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Capstone Project Document

StockFlow Commerce – Unified Warehouse & E-commerce Platform for F&B Supplies with Custom Cup & Packaging Printing

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Finally, while we have put our best effort into this project, there may still be room for improvement regarding technology and business logic. We welcome any feedback and appreciate your continued support as we start our careers.

Sincerely.

Definition and Acronyms

Acronym	Definition
ACID	Atomicity, Consistency, Isolation and Durability
API	Application Program Interface
AWS	Amazon Web Services
BR	Business Rule
CD	Change Directory
CDC	Change Data Capture
CDN	Content Delivery Network
CI	Continuous Integration
CPU	Central Processing Unit
CR	Common Requirement
DTO	Data Transfer Object
FK	Foreign Key

Acronym	Definition
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ID	Identity Document
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
JWT	JSON Web Token
MFA	Multi-factor authentication
NPM	Node Package Manager
OCR	Optical Character Recognition
ORM	Object-Relational Mapping
OTP	One-time Password
PK	Primary Key
RBAC	Role-Based Access Control
RPC	Remote Procedure Call
SQL	Structured Query Language
TDD	Test-Driven Development
TLS	Transport Layer Security
QR	Quick-response
UAT	User Acceptance Test
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
WAL	Write-Ahead Log
2D / 3D	Two-dimensional / Three-dimensional
AP	Accounts Payable

Acronym	Definition
ATP	Available To Promise
AWB	Air Waybill
COD	Cash On Delivery
FEFO	First-Expired-First-Out
FIFO	First-In-First-Out
MOQ	Minimum Order Quantity
PO	Purchase Order
POD	Proof of Delivery
QC	Quality Control
RMA	Return Merchandise Authorization
RTO	Return To Origin
SKU	Stock Keeping Unit
SLA	Service Level Agreement
TO	Transfer Order
UoM	Unit of Measure
WMS	Warehouse Management System

Table 1. Definition and Acronyms

I. Project Introduction

1. Overview

1.1 Project Information

- Project name: StockFlow Commerce – Unified Warehouse & E-commerce Platform for F&B Supplies with Custom Cup & Packaging Printing
- Project code: FA26SE029
- Group name: GFA26SE03
- Software type: Web app, Mobile app

1.2 Project Team

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Table 2. Project Team

2. Product Background

StockFlowCommerce was created in response to the persistent gap between warehouse operations and online selling at businesses dealing in customizable and print-on-demand products. Today these two sides typically run on separate toolsets — stock tracked in spreadsheets, while orders arrive through a storefront or social media channels and are re-entered by hand — leading to three major problems: inventory that is not updated in real time, resulting in overselling; fragmented data that makes end-to-end traceability impossible, from the supplier purchase order through to the parcel handed to the carrier; and warehouse space that has never been digitized, leaving both the storing and the locating of goods dependent on staff memory, which lengthens travel distance and drives up picking errors.

The idea for StockFlowCommerce originated from the observation that existing solutions force businesses to stitch together a warehouse management system (WMS) for the back office and a separate e-commerce platform for sales, synchronized manually through file exports. That seam is precisely where stock discrepancies and broken traceability arise. It pointed to the need for a unified platform that treats inventory as the meeting point between the two sides and covers the full product lifecycle — from product setup, procurement, receiving and storage, through to the customer placing an order and receiving it.

The platform serves three groups: warehouse teams, e-commerce teams, and end customers. What sets it apart is the digitization of warehouse space: the entire facility is modeled down to the individual bin and rendered as an interactive 2D/3D map, allowing staff to locate goods on screen, while a slotting engine recommends the optimal storage position — first eliminating any location that violates a hard constraint, then scoring the rest on distance to the packing area, FIFO/FEFO compliance, capacity, and load balance — with a stated reason behind every recommendation. Together with real-time inventory, document-backed traceability, and a 2D design tool with live 3D preview for printed products, StockFlowCommerce standardizes operations, reduces manual errors, and shortens the time from order placement to delivery.

3. Existing Systems

3.1 Sapo Omnichannel (Sales & E-commerce Management Platform)

- **Description:** Sapo is one of the largest multichannel retail and e-commerce management platforms in Vietnam. It enables businesses to synchronize and manage sales operations across physical stores, independent websites, and various e-commerce marketplaces within a centralized system.
- **Link:** https://www.sapo.vn
- **System Actors:** Customers (B2B/B2C), Store Owners, Inventory Staff, System Administrators.
- **Features:**
    - Multichannel order synchronization and management (Shopee, Lazada, TikTok Shop, Website, Facebook).

    - Basic inventory tracking with automatic stock deduction upon order creation.
    - Customer Relationship Management (CRM) and purchase history retention.
    - Built-in integration with domestic payment gateways and shipping carriers.
- **Pros:** User-friendly interface, rapid deployment, and strong support for fast-paced B2C retail campaigns and order fulfillment.
- **Cons:** Lacks advanced Warehouse Management System (WMS) capabilities for large manufacturing or distribution enterprises; struggles with complex unit-of-measure conversions (e.g., inputting in Kilograms, outputting in Packs/Boxes/Pieces) typical in the packaging industry; limited B2B debt and credit management features.

3.2 Bravo ERP - Inventory & Production Management (ERP & WMS System)

- **Description:** Bravo is a leading Vietnamese Enterprise Resource Planning (ERP) software provider, highly customized for medium to large enterprises. Its system is widely adopted by manufacturing and trading companies in Vietnam to handle complex internal operations, particularly financial accounting, production planning, and deep warehouse management.
- **Link:** https://www.bravo.com.vn
- **System Actors:** Warehouse Managers, Production Planners, Accountants, Procurement Staff, System Administrators.
- **Features:**
    - In-depth inventory control supporting multi-warehouse architecture, bin locations, and lot/expiration date tracking.
    - Manufacturing execution features, including Bill of Materials (BoM), raw material consumption calculation, and production cost accounting.
    - Advanced handling of multiple units of measure (e.g., receiving raw materials in Kilograms and outputting finished goods in Lốc/Thùng/Cái) and complex B2B debt/credit policies.
    - Deeply tailored to comply with Vietnamese Accounting Standards (VAS) and local business workflows.
- **Pros:** Extremely strong core in manufacturing and warehouse operations; highly customizable to fit the specific, complex workflows of the packaging and plastics industry; excellent at managing B2B pricing tiers and detailed material tracking.
- **Cons:** Traditional user interface that can feel heavy and outdated; deployment is time-consuming due to the high level of customization; primarily designed for B2B and internal factory operations, making it lack native, high-speed API synchronization with modern B2C e-commerce marketplaces (like Shopee, TikTok Shop, or Lazada).

3.3 Haravan (Omnichannel E-commerce & Retail Platform)

- **Description:** Haravan is a prominent e-commerce and omnichannel retail platform in Vietnam, empowering merchants to manage sales across independent websites, social media (Facebook, Instagram), and major e-commerce marketplaces from a single dashboard.
- **Link:** https://www.haravan.com
- **System Actors:** Customers (B2C), Merchants/Store Owners, Sales Staff, System Administrators.
- **Features:**

    - Centralized order processing and synchronization across multiple online channels (Shopee, Lazada, TikTok Shop).
    - Integrated social commerce tools (chatbots, automatic comment replies).
    - Basic inventory management that syncs stock levels across all connected sales channels.
    - Built-in marketing, customer loyalty programs, and automated shipping carrier integration.
- **Pros:** Exceptional at handling social commerce and modern B2C e-commerce integrations; highly optimized for marketing, user experience, and rapid retail sales; fast setup and deployment.
- **Cons:** Not designed for manufacturing or complex distribution workflows; lacks the ability to handle intricate warehousing scenarios, such as managing raw materials, multi-level unit conversions (e.g., from plastic resins to finished cups), or complex B2B wholesale pricing matrices.

3.4 Expert ERP (Manufacturing & Enterprise Management System)

- **Description:** Expert ERP is a robust, Vietnam-developed ERP solution specifically engineered for the manufacturing and heavy distribution sectors. It is implemented by many large-scale factories to optimize production lines, manage deep inventory logistics, and control costs.
- **Link:** https://experterp.vn
- **System Actors:** Factory Managers, Warehouse Supervisors, Production Planners, Accountants, System Administrators.
- **Features:**
    - Detailed Bill of Materials (BoM) management and production capacity planning.
    - Strict lot, batch tracking, and advanced inventory valuation methods.
    - Comprehensive B2B contract, purchase order, and debt management.
    - Precise cost calculation for manufacturing processes and material wastage tracking.
- **Pros:** Exceptional at controlling production costs and complex warehouse logistics; highly capable of managing the specialized demands of the packaging industry on a B2B level, ensuring strict material control.
- **Cons:** The architecture is focused heavily on back-office and factory operations, making it cumbersome for fast retail; it lacks native, agile integration with B2C e-commerce platforms; the user interface is strictly functional and requires extensive training for basic staff.

4. Business Opportunity

Businesses selling customizable and print-on-demand products face a structural gap between their online storefront and their warehouse operations. Storefront platforms such as Haravan or Sapo Omnichannel handle catalog and checkout well, while warehouse management systems handle stock and fulfillment, but the two rarely share a single source of product and inventory data. The result is duplicated product records, inventory that drifts out of sync, and overselling—problems made worse for print-on-demand items, where each order carries a customer-approved design that must travel intact from checkout all the way to production and packing. Generic tools were not built for this design-to-delivery workflow, so teams stitch together spreadsheets and manual handoffs that erode accuracy and slow fulfillment.

The opportunity lies in StockFlowCommerce: a unified platform that merges a print-on-demand storefront with a full warehouse management system around one shared inventory core. Unlike stitched-together solutions, StockFlowCommerce declares each product once and reuses it across both the catalog and the warehouse, keeps inventory real-time and location-based to prevent overselling, and preserves an immutable design snapshot on every custom order so production works from exactly what the customer approved. End-to-end traceability links supplier PO, receipt, storage location, customer order, and outbound shipment, while 2D/3D design preview, warehouse mapping, slotting suggestions, and optimized picking routes streamline work for customers, sales staff, and warehouse teams alike.

The timing is favorable. Vietnam's e-commerce market continues to grow rapidly, and demand for personalized, made-to-order goods is rising alongside it, pushing more small and mid-sized sellers into print-on-demand. These sellers increasingly need operational rigor—accurate stock, clean fulfillment, and reliable handling of custom design files—rather than just a checkout page. By unifying storefront and warehouse in one system built specifically for customizable products, StockFlowCommerce positions itself as a scalable, transparent platform that reduces oversell and fulfillment errors while giving both buyers and operators confidence at every step from design to delivery.

5. Software Product Vision

StockFlowCommerce empowers custom-print and made-to-order businesses to run warehouse operations and online sales on a single unified platform. Each product is declared once in the back-office and shared across both warehouse logistics and the e-commerce catalog, giving the entire organization one source of truth. Procurement, warehouse, sales, and coordination staff collaborate digitally through structured workflows for purchasing, receiving, put-away, picking, packing, and shipping.

Customers design their own products in 2D, preview them live in 3D, and confirm the design before payment, while the locked design snapshot travels with the order into fulfillment. Every inventory movement is tied to a source document — purchase order, receipt, pick, adjustment, or transfer — so stock is always location-based, real-time, and fully traceable from supplier PO to delivered parcel.

The vision is to close the gap between warehouse operations and online selling — a platform where every unit is tracked to its exact bin, every order is fulfilled without overselling, and every step from product creation to delivery is verifiable end to end.

6. Project Scope & Limitations

6.1 Major Features

- **Warehouse Staff**
    - Log into the system.
    - Receive goods (Receipt): count, scan, record batch/lot, expiry, and serial, and capture photo evidence.
    - Put away goods (Putaway) following location suggestions; scan to confirm the location.
    - Pick goods (Picking) by the assigned pick list and route.
    - Pack goods (Packing), label them, and attach the design snapshot for print orders.

    - Hand goods over to the carrier (Shipping).
    - Execute stock issue/receipt for warehouse transfers.
- **Warehouse Managers**
    - Log into the system.
    - Monitor warehouse operation KPIs and view the 2D/3D warehouse map.
    - Approve or reject inventory adjustments and over-threshold transfer orders.
    - Resolve discrepancies and exceptions, override slotting suggestions, and run periodic re-slotting analyses.
    - Approve handling of over/under-receipt beyond tolerance.
- **Inventory Planners**
    - Log into the system.
    - Monitor stock levels across warehouses.
    - Propose and create Transfer Orders (TO) to rebalance stock.
- **QC Staff**
    - Log into the system.
    - Inspect the quality of inbound goods upon receipt (accept / reject / quarantine).
    - Inspect the quality of produced/printed goods before fulfillment.
    - Record QC results and move goods into/out of the quarantine area.
- **Customers**
    - Register and verify an account via email/phone (OTP) or social login; guest checkout supported.
    - Log into the system.
    - Browse the catalog, search, filter, and view product detail pages (PDP) with real-time availability.
    - Design print-on-demand products in a 2D design tool and preview the result live on a 3D model (Live 3D Preview); save and confirm the design (design snapshot).
    - Manage the cart, apply discount codes/vouchers, place orders (checkout), and select delivery address and shipping method.
    - Pay online (payment gateway/e-wallet/card/bank transfer), by cash on delivery (COD), or by deposit; receive a payment receipt/invoice.
    - Track order status and history along an order timeline.
    - Submit requests to edit/cancel orders, return/exchange items, and request refunds.
    - Chat directly with sales staff (Sales Chat) for consultation, design confirmation, and post-sale handling.
    - Manage their personal profile: address book, saved payment methods (tokens), and communication preferences; request export/deletion of personal data.
- **Sales Staff**
    - Log into the system.
    - Advise and support customers through a context-aware chat channel (product / design / cart / order).
    - Edit designs on the customer's behalf, verify print feasibility, and confirm the design together with the customer.
    - Create and edit assisted orders on the customer's behalf, apply promotions within limits, and send payment links.
    - Look up orders and handle post-sale requests: order edits/cancellations, returns/exchanges, and delivery complaints.
    - View customer profiles (per permissions) and add internal notes.
- **Order Coordinators**

    - Log into the system.
    - Review and release orders to the warehouse to start fulfillment.
    - Allocate the fulfilling warehouse, merge/split orders, and handle on-hold orders.
    - Monitor consolidated order status and fulfillment SLAs.
- **E-commerce Administrators**
    - Log into the system.
    - Manage the catalog: categories, display order, featured products, PDP content, and SEO.
    - Configure selling prices, promotions/vouchers, and cart policies.
    - Configure design templates, print areas, and technical constraints; run prepress/preflight checks before printing.
    - Manage customer groups/segments, data policies, and duplicate-profile merging.
    - Configure the payment and shipping methods shown on the storefront.
- **Procurement Staff**
    - Log into the system.
    - Create and track supplier Purchase Orders (PO) and send them to suppliers.
    - Perform three-way matching (PO ↔ Receipt ↔ Invoice).
    - Coordinate with suppliers when receipt discrepancies occur.
- **Accountants**
    - Log into the system.
    - Create and process supplier invoices; perform three-way matching and track supplier payables.
    - Reconcile online payments / COD / refunds with the payment gateway and carriers.
    - Approve payments and refunds within limits.
- **System Administrators**
    - Log into the system.
    - Configure the whole system; declare warehouses, zones, and the location hierarchy
    - Create and manage product master data and SKU-generation rules.
    - Configure slotting suggestion rules and weights.
    - Manage access control (RBAC), user accounts, and roles.

- **Production Staff**
    - Log into the system.
    - View the production orders assigned to their own print station.
    - Scan the production order code at the station to start printing (Assigned → In Progress).
    - Print using the approved print-ready file and the locked design snapshot.
    - Report the good quantity and the defective (scrap) quantity, with a reason for each defect.
    - Hand finished goods over to QC for inspection (In Progress → QC Pending).

6.2 Limitations & Exclusions

6.2.1 Limitations

- The system serves a **single legal entity** (one company, one tax code) that owns multiple physical warehouses; inter-warehouse transfers are internal stock movements and do **not** create internal invoices or payables.
- Available-to-promise stock is added only **after Putaway is completed**; goods in the inbound area are considered received but not yet available.
- Relies on **third-party payment gateways and carriers** (integrated via API); interruptions to their services may affect payment and delivery.
- Payments/refunds above thresholds follow **manual approval** — there is no automatic fund transfer.
- Storage location suggestions (slotting) are **rule-based** with a feedback loop; re-slotting requires warehouse-manager approval and is **not** fully automatic.
- Primarily uses **VND** and Vietnamese localization (multi-level VN administrative addresses); procurement may follow the supplier's currency.
- **Live 3D Preview** renders on the client (WebGL) with a static-image fallback; quality depends on the customer's device/browser.
- Personalized products are **not eligible for change-of-mind returns/exchanges** — accepted only for production defects; design-change requests are accepted only until production begins.

6.2.2 Exclusions

- No support for **multi-company / multi-entity** operation.
- No integration with **third-party marketplaces** (Shopee, Lazada, TikTok Shop) — only the system's own storefront.
- No **full accounting general ledger** — only accounts-payable (AP) matching and payment proposals; the ledger resides in an external system.
- No integration with **external chat channels** (Facebook Messenger, Zalo, email, hotline) — only 1-to-1 in-storefront chat.
- No **chatbot / AI assistant** for automated replies — conversations are routed to sales staff.
- No **inventory synchronization from an external ERP** — StockFlowCommerce is the single source of truth for inventory.
- **VAT e-invoices for customers** and an **open B2B partner API** are out of scope (not yet defined).
- No support for **consolidating multiple orders of one customer into a single parcel** (consolidated shipment) — by default, one parcel per order.

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

#	WBS Item	Complexity	Est. Effort (man-days)
1	**Project Initialization**		**33**
1.1	Define project scope	Medium	8
1.2	Define project requirement and objectives	Complex	12
1.3	Research existing solution	Medium	4
1.4	Create Project Introduction document	Simple	4
1.5	Review project with mentor	Simple	5
2	**Project Planning**		**45**
2.1	Create Project Management Plan document	Complex	8
2.2	Schedule project meeting	Simple	16
2.3	Set up project repository	Complex	21
3	**Project Execution**		**353**
3.1	**Product Creation**		**25**
3.1.1	**Product Master Data Management**	Medium	**3**
3.1.1.1	Product draft creation & editing UI	Simple	1
3.1.1.2	Product information fields (name, code, category, brand)	Simple	1
3.1.1.3	Multi-level approval workflow (Draft → Pending → Approved)	Medium	1
3.1.2	**Variant & SKU Management**		**4**
3.1.2.1	Variant attribute configuration (size, color, material)	Medium	1
3.1.2.2	Variant matrix generation UI	Medium	1
3.1.2.3	Auto SKU code generation with rules	Medium	1
3.1.2.4	Manual SKU code input with uniqueness validation	Simple	1
3.1.3	**Logistics Data Management**		**3**
3.1.3.1	Dimensions & weight configuration	Simple	1
3.1.3.2	Packaging specifications	Simple	1
3.1.3.3	Shipping rules & restrictions	Medium	1
3.1.4	**Procurement Data**		**2**
3.1.4.1	Supplier mapping & lead times	Medium	1
3.1.4.2	MOQ & reorder point configuration	Simple	1
3.1.5	**Inventory Control Configuration**		**3**
3.1.5.1	FIFO/FEFO rules setup	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.1.5.2	Batch/lot/serial tracking flags	Simple	1
3.1.5.3	Expiry date management rules	Medium	1
3.1.6	**E-commerce Data Management**		**3**
3.1.6.1	Catalog visibility & publish rules	Medium	1
3.1.6.2	Product image gallery management	Simple	1
3.1.6.3	SEO metadata & descriptions	Simple	1
3.1.7	**Customization/Print Product Configuration**		**3**
3.1.7.1	Print area definition & templates	Complex	1
3.1.7.2	Design constraints (DPI, bleed, safe area)	Medium	1
3.1.7.3	3D model upload & association	Medium	1
3.1.8	**Product Lifecycle Management**		**2**
3.1.8.1	Discontinuation workflow	Medium	1
3.1.8.2	Version control & history	Medium	1
3.1.9	**Business Rules & Validation**		**2**
3.1.9.1	Cross-field validation logic	Medium	1
3.1.9.2	Duplicate detection	Simple	1
3.2	**Procurement**		**20**
3.2.1	**Purchase Order Creation**		**3**
3.2.1.1	PO draft creation UI with line items	Medium	1
3.2.1.2	Supplier selection & contact info	Simple	1
3.2.1.3	Delivery warehouse & date specification	Simple	1
3.2.2	**PO Approval Workflow**	Medium	**2**
3.2.2.1	Approval threshold configuration	Simple	1
3.2.2.2	Multi-level approval routing	Medium	1
3.2.3	**PO Confirmation & Supplier Communication**		**2**
3.2.3.1	Send PO to supplier (email/API)	Medium	1
3.2.3.2	Supplier confirmation tracking	Simple	1
3.2.4	**PO Tracking & Status Management**		**3**
3.2.4.1	Open quantity calculation	Simple	1
3.2.4.2	Expected delivery date tracking	Simple	1
3.2.4.3	PO status lifecycle (Draft → Confirmed → Received → Closed)	Medium	1
3.2.5	**PO Amendment & Revision Control**		**3**
3.2.5.1	PO revision creation with version control	Medium	1
3.2.5.2	Change log & audit trail	Medium	1
3.2.5.3	Supplier notification of changes	Simple	1
3.2.6	**Supplier Master Data**		**3**
3.2.6.1	Supplier profile & contact management	Simple	1
3.2.6.2	Payment terms & lead time defaults	Simple	1
3.2.6.3	Supplier performance tracking	Medium	1
3.2.7	**PO Reporting & Analytics**		**2**

#	WBS Item	Complexity	Est. Effort (man-days)
3.2.7.1	PO status dashboard	Medium	1
3.2.7.2	Supplier spend analysis	Medium	1
3.2.8	**Business Rules & Validation**	Medium	**2**
3.2.8.1	PO line item validation against product master	Simple	1
3.2.8.2	Duplicate PO detection	Simple	1
3.3	**Goods Receipt**		**21**
3.3.1	**Receipt Creation**	Medium	**2**
3.3.1.1	Create receipt from PO reference	Medium	1
3.3.1.2	Blind receipt (no PO) support	Medium	1
3.3.2	**Physical Counting & Scanning**		**3**
3.3.2.1	Barcode/QR scanning for SKU identification	Medium	1
3.3.2.2	Quantity counting UI (mobile-friendly)	Medium	1
3.3.2.3	Discrepancy flagging (vs. PO expected)	Simple	1
3.3.3	**Batch/Lot/Serial Capture**		**4**
3.3.3.1	Batch/lot number entry & validation	Medium	1
3.3.3.2	Expiry date entry with format validation	Simple	1
3.3.3.3	Serial number capture for serialized items	Medium	1
3.3.3.4	Manufacturing date tracking	Simple	1
3.3.4	**Quality Control Integration**		**4**
3.3.4.1	QC task generation for received items	Medium	1
3.3.4.2	QC result capture (Accepted/Quarantine/Rejected)	Medium	1
3.3.4.3	Quarantine location assignment	Simple	1
3.3.4.4	RTV (Return to Vendor) workflow for rejected goods	Medium	1
3.3.5	**Receipt Confirmation**		**2**
3.3.5.1	Receipt finalization & PO update	Medium	1
3.3.5.2	Inventory status update (Inbound → Available/Quarantine)	Medium	1
3.3.6	**Putaway Task Generation**		**2**
3.3.6.1	Auto-generate putaway tasks for accepted goods	Medium	1
3.3.6.2	Putaway task prioritization	Simple	1
3.3.7	**Receipt Documentation**		**2**
3.3.7.1	Receipt document generation (PDF)	Simple	1
3.3.7.2	Photo capture for damaged goods	Simple	1
3.3.8	**Business Rules & Validation**		**2**
3.3.8.1	Over-receipt tolerance checking	Simple	1
3.3.8.2	Expiry date validation against product master	Simple	1
3.4	**Invoice Matching**		**19**
3.4.1	**Invoice Entry**		**2**
3.4.1.1	Manual invoice data entry form	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.4.1.2	OCR-based invoice scanning (optional)	Complex	1
3.4.2	**PO & Receipt Linking**		**3**
3.4.2.1	Link invoice to PO	Simple	1
3.4.2.2	Link invoice to receipts	Simple	1
3.4.2.3	Multi-receipt invoice support	Medium	1
3.4.3	**Three-Way Matching Engine**		**4**
3.4.3.1	Price matching (PO vs. Invoice)	Medium	1
3.4.3.2	Quantity matching (Receipt vs. Invoice)	Medium	1
3.4.3.3	Tax & charges validation	Medium	1
3.4.3.4	Tolerance threshold configuration	Medium	1
3.4.4	**Exception Handling**		**3**
3.4.4.1	Price variance flagging & review UI	Medium	1
3.4.4.2	Quantity variance flagging & review UI	Medium	1
3.4.4.3	Manual override with approval	Medium	1
3.4.5	**Approval for Payment**		**3**
3.4.5.1	Approved invoice queue for accountant	Medium	1
3.4.5.2	Payment status tracking (Pending/Paid)	Simple	1
3.4.5.3	Payment due date reminders	Simple	1
3.4.6	**Credit Note & Debit Note Handling**		**2**
3.4.6.1	Credit note entry & matching	Medium	1
3.4.6.2	Invoice adjustment workflow	Medium	1
3.4.7	**Business Rules & Reporting**		**2**
3.4.7.1	AP aging report	Medium	1
3.4.7.2	Matching exception dashboard	Simple	1
3.5	**Putaway & Slotting**		**19**
3.5.1	**Warehouse Map & Location Master**		**4**
3.5.1.1	2D warehouse map visualization	Medium	1
3.5.1.2	Hierarchical location structure (Zone/Aisle/Rack/Level/Bin)	Medium	1
3.5.1.3	Location barcode generation & printing	Simple	1
3.5.1.4	Capacity configuration per location (weight/volume/slots)	Medium	1
3.5.2	**Slotting Rules Engine**		**4**
3.5.2.1	ABC/XYZ classification configuration	Medium	1
3.5.2.2	Velocity-based slotting rules	Complex	1
3.5.2.3	Hard constraint rules (hazmat, temp, single-SKU)	Medium	1
3.5.2.4	Compatibility matrix (SKUs allowed/forbidden together)	Medium	1
3.5.3	**Slotting Suggestion Algorithm**		**4**
3.5.3.1	Multi-criteria scoring (velocity, golden zone, proximity)	Complex	2

#	WBS Item	Complexity	Est. Effort (man-days)
3.5.3.2	Capacity-aware recommendation	Medium	1
3.5.3.3	Ranked suggestion list generation	Medium	1
3.5.4	**Putaway Task Management**		**3**
3.5.4.1	Putaway task creation & assignment	Medium	1
3.5.4.2	Mobile putaway UI with scanning	Medium	1
3.5.4.3	Putaway confirmation & inventory update	Simple	1
3.5.5	**Re-slotting & Optimization**		**2**
3.5.5.1	Periodic re-slotting analysis	Medium	1
3.5.5.2	Re-slotting task generation	Medium	1
3.5.6	**Location Utilization Monitoring**		**2**
3.5.6.1	Location fill rate heatmap	Medium	1
3.5.6.2	Honeycombing detection & alerts	Medium	1
3.6	**Inventory Tracking**		**20**
3.6.1	**Real-Time Inventory Status**		**3**
3.6.1.1	Location-level inventory view	Medium	1
3.6.1.2	SKU-level aggregated inventory	Medium	1
3.6.1.3	Inventory status categories (Available/Allocated/Reserved/Quarantine/In-Transit)	Medium	1
3.6.2	**Batch/Lot Tracking**		**3**
3.6.2.1	Batch/lot expiry tracking dashboard	Medium	1
3.6.2.2	FEFO enforcement in allocation	Medium	1
3.6.2.3	Batch/lot traceability report	Medium	1
3.6.3	**Serial Number Tracking**		**2**
3.6.3.1	Serial number lifecycle (Receipt → Allocation → Shipment)	Medium	1
3.6.3.2	Serial number lookup & history	Medium	1
3.6.4	**ATP (Available to Promise) Calculation**		**3**
3.6.4.1	ATP calculation engine (Available - Reserved - Allocated)	Complex	1
3.6.4.2	Multi-warehouse ATP aggregation	Medium	1
3.6.4.3	Real-time ATP API for storefront	Medium	1
3.6.5	**Inventory Adjustment**		**3**
3.6.5.1	Manual adjustment entry with reason codes	Medium	1
3.6.5.2	Approval workflow for adjustments	Medium	1
3.6.5.3	Adjustment audit log	Simple	1
3.6.6	**Cycle Count**		**3**
3.6.6.1	Cycle count schedule & task generation	Medium	1
3.6.6.2	Mobile counting UI	Medium	1
3.6.6.3	Discrepancy detection & auto-adjustment	Medium	1
3.6.7	**Inventory Alerts & Notifications**		**3**

#	WBS Item	Complexity	Est. Effort (man-days)
3.6.7.1	Low stock alerts	Simple	1
3.6.7.2	Expiry date warnings	Simple	1
3.6.7.3	Negative inventory alerts	Simple	1
3.7	**Picking**		**19**
3.7.1	**Order Allocation**		**4**
3.7.1.1	Inventory allocation by location & lot (FEFO)	Complex	2
3.7.1.2	Multi-location allocation for single order	Medium	1
3.7.1.3	Allocation conflict resolution	Medium	1
3.7.2	**Wave Planning**		**3**
3.7.2.1	Wave creation rules (time-based, volume-based, priority-based)	Medium	1
3.7.2.2	Order grouping into waves	Medium	1
3.7.2.3	Wave release & picking task generation	Medium	1
3.7.3	**Pick List Generation**		**2**
3.7.3.1	Optimized pick route calculation	Complex	1
3.7.3.2	Pick list document generation (sorted by location)	Medium	1
3.7.4	**Picking Execution**		**4**
3.7.4.1	Mobile picking UI with barcode scanning	Medium	2
3.7.4.2	Location-by-location pick confirmation	Medium	1
3.7.4.3	Quantity verification & scanning	Medium	1
3.7.5	**Short Pick Handling**		**2**
3.7.5.1	Short pick reporting & re-allocation	Medium	1
3.7.5.2	Backorder creation for unfulfilled quantity	Medium	1
3.7.6	**Batch Picking (Multi-Order)**		**2**
3.7.6.1	Batch picking UI (pick once, sort later)	Medium	1
3.7.6.2	Sorting to individual orders after picking	Medium	1
3.7.7	**Staging & Put-Back**		**2**
3.7.7.1	Staging location assignment & transfer	Medium	1
3.7.7.2	Put-back workflow for canceled orders	Medium	1
3.8	**Packing**		**20**
3.8.1	**Packing Station Setup**		**3**
3.8.1.1	Packing station configuration	Simple	1
3.8.1.2	Packer login & task assignment	Simple	1
3.8.1.3	Packing materials inventory	Medium	1
3.8.2	**Order Verification**		**3**
3.8.2.1	Order item list display	Simple	1
3.8.2.2	SKU-by-SKU scanning verification	Medium	1
3.8.2.3	Serial number verification for serialized items	Medium	1
3.8.3	**Design Snapshot Verification (Print Products)**		**3**
3.8.3.1	Display design snapshot for visual comparison	Medium	1
3.8.3.2	Packer confirmation of design match	Simple	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.8.3.3	Verification failure escalation workflow	Medium	1
3.8.4	**Cartonization**		**3**
3.8.4.1	Box type recommendation based on dimensions	Medium	1
3.8.4.2	Multi-box split for large orders	Medium	1
3.8.4.3	Packing material selection	Simple	1
3.8.5	**Weighing & Dimensioning**		**3**
3.8.5.1	Weight capture & validation	Simple	1
3.8.5.2	Dimension capture (L×W×H)	Simple	1
3.8.5.3	Dimensional weight calculation	Medium	1
3.8.6	**Packing Completion**		**3**
3.8.6.1	Packing confirmation & status update	Simple	1
3.8.6.2	Packing slip generation & insertion	Medium	1
3.8.6.3	Box sealing & labeling instructions	Simple	1
3.8.7	**Business Rules**		**2**
3.8.7.1	Fragile item handling rules	Simple	1
3.8.7.2	Gift wrapping option support	Medium	1
3.9	**Shipping**		**21**
3.9.1	**Carrier Integration**		**5**
3.9.1.1	Carrier API configuration (multiple carriers)	Complex	2
3.9.1.2	Rate shopping across carriers	Medium	1
3.9.1.3	Service level mapping	Medium	1
3.9.1.4	Carrier credentials management	Simple	1
3.9.2	**Shipping Label Creation**		**4**
3.9.2.1	API call to carrier for label generation	Medium	1
3.9.2.2	Label download & print	Medium	1
3.9.2.3	Tracking number capture & storage	Simple	1
3.9.2.4	Label error handling & retry	Medium	1
3.9.3	**Address Validation**		**2**
3.9.3.1	Pre-ship address validation API	Medium	1
3.9.3.2	Address correction suggestions	Medium	1
3.9.4	**Shipping Manifest**		**3**
3.9.4.1	Daily manifest generation by carrier	Medium	1
3.9.4.2	Manifest print & handover workflow	Simple	1
3.9.4.3	Handover confirmation & timestamp	Simple	1
3.9.5	**Tracking & Status Updates**		**3**
3.9.5.1	Webhook receiver for carrier status updates	Medium	1
3.9.5.2	Tracking status parsing & normalization	Medium	1
3.9.5.3	Customer notification triggers	Simple	1
3.9.6	**POD (Proof of Delivery)**		**2**
3.9.6.1	POD data capture from carrier	Medium	1
3.9.6.2	POD image/signature storage	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.9.7	**RTO (Return to Origin) Handling**		**2**
3.9.7.1	RTO status detection	Medium	1
3.9.7.2	RTO receipt workflow (re-putaway)	Medium	1
3.10	**Inter-Warehouse Transfer**		**18**
3.10.1	**Transfer Order Creation**		**3**
3.10.1.1	TO creation UI (source/dest warehouse, SKU, qty)	Medium	1
3.10.1.2	Availability check at source warehouse	Medium	1
3.10.1.3	Transfer reason tracking	Simple	1
3.10.2	**TO Approval Workflow**		**2**
3.10.2.1	Approval threshold configuration	Simple	1
3.10.2.2	Multi-level approval for high-value transfers	Medium	1
3.10.3	**Source Warehouse Fulfillment**		**3**
3.10.3.1	Inventory reservation at source	Medium	1
3.10.3.2	Picking workflow (FEFO)	Medium	1
3.10.3.3	Packing & dispatch from source	Medium	1
3.10.4	**In-Transit Inventory Management**		**2**
3.10.4.1	In-Transit inventory status tracking	Medium	1
3.10.4.2	Inventory adjustment (source → In-Transit)	Medium	1
3.10.5	**Destination Warehouse Receipt**		**3**
3.10.5.1	Receipt creation from TO	Medium	1
3.10.5.2	Quantity verification & discrepancy logging	Medium	1
3.10.5.3	Putaway at destination warehouse	Medium	1
3.10.6	**Transfer Discrepancy Handling**		**3**
3.10.6.1	Discrepancy detection & flagging	Medium	1
3.10.6.2	Investigation workflow & resolution	Medium	1
3.10.6.3	Adjustment approval for losses	Medium	1
3.10.7	**TO Completion**		**2**
3.10.7.1	TO closure when all received	Medium	1
3.10.7.2	Transfer history & audit log	Medium	1
3.11	**Intra-Warehouse Transfer (Replenishment)**		**13**
3.11.1	**Replenishment Rule Configuration**		**2**
3.11.1.1	Min/max level setup per location	Medium	1
3.11.1.2	Replenishment trigger thresholds	Medium	1
3.11.2	**Replenishment Task Auto-Generation**		**3**
3.11.2.1	Monitor picking location inventory levels	Medium	1
3.11.2.2	Auto-generate replenishment task when below min	Complex	1
3.11.2.3	Source location selection (reserve zone)	Medium	1
3.11.3	**Replenishment Task Execution**		**3**
3.11.3.1	Mobile replenishment UI	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.11.3.2	Scan source location → SKU → quantity → dest location	Medium	1
3.11.3.3	Inventory update (source → dest within same warehouse)	Simple	1
3.11.4	**Replenishment Task Prioritization**		**2**
3.11.4.1	High-priority replenishment (blocking active picks)	Medium	1
3.11.4.2	Replenishment task queue management	Medium	1
3.11.5	**Discrepancy Handling**		**2**
3.11.5.1	Quantity mismatch detection during move	Medium	1
3.11.5.2	Cycle count trigger for both locations	Medium	1
3.11.6	**Reporting**		**1**
3.11.6.1	Replenishment frequency & performance metrics	Medium	1
3.12	**Product Design (2D/3D Customization)**		**26**
3.12.1	**Design Canvas (2D Editor)**		**6**
3.12.1.1	Canvas initialization with product template	Medium	1
3.12.1.2	Image upload & placement	Medium	1
3.12.1.3	Text tool with font selection	Medium	1
3.12.1.4	Shape & icon library	Medium	1
3.12.1.5	Layer management (bring forward, send back)	Medium	1
3.12.1.6	Zoom, pan, undo/redo	Simple	1
3.12.2	**Live 3D Preview**		**8**
3.12.2.1	3D model loading (GLB/GLTF format)	Complex	2
3.12.2.2	Texture mapping from 2D design to 3D model	Complex	3
3.12.2.3	Real-time preview updates on design changes	Complex	2
3.12.2.4	3D viewer controls (rotate, zoom)	Medium	1
3.12.3	**Design Constraints & Print Area**		**3**
3.12.3.1	Print area boundary visualization	Medium	1
3.12.3.2	Safe area & bleed line indicators	Medium	1
3.12.3.3	Design element snapping to boundaries	Simple	1
3.12.4	**Preflight Validation**		**4**
3.12.4.1	DPI/resolution check for images	Medium	1
3.12.4.2	Color mode validation (RGB → CMYK warning)	Medium	1
3.12.4.3	Bleed & safe area compliance check	Medium	1
3.12.4.4	Font embedding & compatibility check	Medium	1
3.12.5	**Design Snapshot (Artifact)**		**2**
3.12.5.1	Generate immutable design snapshot JSON	Medium	1
3.12.5.2	Snapshot versioning & storage	Medium	1
3.12.6	**Design Draft Management**		**2**
3.12.6.1	Save design as draft (logged-in users)	Medium	1
3.12.6.2	Load & edit saved drafts	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.12.7	**Print-Ready File Generation**		**1**
3.12.7.1	Export design to PDF/PNG for production	Complex	1
3.13	**Catalog Management**		**17**
3.13.1	**Product Listing Page (PLP)**		**3**
3.13.1.1	Grid/list view with pagination	Medium	1
3.13.1.2	Filter by category, brand, price, attributes	Medium	1
3.13.1.3	Sort options (price, popularity, new arrivals)	Medium	1
3.13.2	**Product Detail Page (PDP)**		**4**
3.13.2.1	Product image gallery with zoom	Medium	1
3.13.2.2	Variant selector (size, color, material)	Medium	1
3.13.2.3	Stock availability display (ATP-based)	Medium	1
3.13.2.4	Product description & specifications	Simple	1
3.13.3	**Search Functionality**		**3**
3.13.3.1	Full-text search with autocomplete	Complex	2
3.13.3.2	Search result ranking & relevance tuning	Medium	1
3.13.4	**Category & Navigation Management**		**2**
3.13.4.1	Category hierarchy setup (admin)	Medium	1
3.13.4.2	Mega menu rendering	Medium	1
3.13.5	**Pricing & Promotions Display**		**3**
3.13.5.1	Regular price & sale price display	Simple	1
3.13.5.2	Promotional badge/tag rendering	Simple	1
3.13.5.3	Price tier display (for bulk/segment)	Medium	1
3.13.6	**SEO & Metadata**		**2**
3.13.6.1	Meta title, description, keywords per product	Simple	1
3.13.6.2	Canonical URL & structured data (JSON-LD)	Medium	1
3.14	**Shopping Cart & Checkout**		**20**
3.14.1	**Shopping Cart Management**		**4**
3.14.1.1	Add to cart (standard & customized products)	Medium	1
3.14.1.2	Cart line item display with thumbnails	Medium	1
3.14.1.3	Update quantity & remove items	Simple	1
3.14.1.4	Cart persistence (logged-in & guest)	Medium	1
3.14.2	**Price Calculation**		**4**
3.14.2.1	Line item price calculation (base + customization fee)	Medium	1
3.14.2.2	Subtotal, tax calculation	Medium	1
3.14.2.3	Shipping cost estimation API	Medium	1
3.14.2.4	Discount/voucher application	Medium	1
3.14.3	**Voucher & Promotion Engine**		**4**
3.14.3.1	Voucher code validation	Medium	1
3.14.3.2	Discount calculation (percentage, fixed, BOGO)	Complex	2
3.14.3.3	Promotion rule engine (min order, SKU eligibility)	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.14.4	**Checkout Flow**		**5**
3.14.4.1	Guest vs. logged-in user flow	Medium	1
3.14.4.2	Shipping address entry & validation	Medium	1
3.14.4.3	Shipping method selection	Medium	1
3.14.4.4	Billing address (same as shipping / different)	Simple	1
3.14.4.5	Order review page with final totals	Medium	1
3.14.5	**Inventory Reservation**		**2**
3.14.5.1	Reserve stock at checkout (time-limited)	Medium	1
3.14.5.2	Reservation expiry & auto-release	Medium	1
3.14.6	**Order Creation**		**1**
3.14.6.1	Create order record (Pending Payment status)	Medium	1
3.15	**Payment Processing**		**19**
3.15.1	**Payment Gateway Integration**		**4**
3.15.1.1	Payment gateway API configuration (Stripe, PayPal, etc.)	Complex	2
3.15.1.2	Payment method selection UI (credit card, wallet, QR)	Medium	1
3.15.1.3	PCI-DSS compliance (tokenization, no card storage)	Complex	1
3.15.2	**Payment Execution**		**3**
3.15.2.1	Authorize & capture flow	Medium	1
3.15.2.2	Payment webhook receiver	Medium	1
3.15.2.3	Payment result verification & signature check	Medium	1
3.15.3	**Bank Transfer Payment**		**3**
3.15.3.1	Display bank transfer instructions	Simple	1
3.15.3.2	Manual payment confirmation by accountant	Medium	1
3.15.3.3	Bank statement reconciliation helper	Medium	1
3.15.4	**COD (Cash on Delivery)**		**3**
3.15.4.1	COD eligibility check (customer segment, order value)	Medium	1
3.15.4.2	COD order flagging for carrier	Simple	1
3.15.4.3	COD remittance tracking from carrier	Medium	1
3.15.5	**Deposit / Partial Payment**		**3**
3.15.5.1	Deposit percentage configuration	Simple	1
3.15.5.2	Deposit payment flow	Medium	1
3.15.5.3	Remaining balance payment link generation	Medium	1
3.15.6	**Refund Processing**		**3**
3.15.6.1	Refund request creation (full/partial)	Medium	1
3.15.6.2	Gateway API call for refund	Medium	1
3.15.6.3	Refund status tracking	Simple	1
3.16	**Sales Chat & Messaging**		**18**

#	WBS Item	Complexity	Est. Effort (man-days)
3.16.1	**Real-Time Chat Interface**		**4**
3.16.1.1	WebSocket-based real-time messaging	Complex	2
3.16.1.2	Chat UI (customer & sales staff views)	Medium	1
3.16.1.3	Message history loading & pagination	Medium	1
3.16.2	**Conversation Management**		**3**
3.16.2.1	Conversation creation & context attachment (product, order, design)	Medium	1
3.16.2.2	Conversation status (Open/Resolved/Escalated)	Medium	1
3.16.2.3	Conversation search & filtering (sales staff)	Medium	1
3.16.3	**Routing & Assignment**		**3**
3.16.3.1	Auto-assign conversations to available sales staff	Medium	1
3.16.3.2	Skill-based routing (product expertise)	Medium	1
3.16.3.3	Manual reassignment & escalation	Simple	1
3.16.4	**Rich Message Support**		**3**
3.16.4.1	Image & file attachment	Medium	1
3.16.4.2	Quick reply templates	Simple	1
3.16.4.3	Product card & order card embeds	Medium	1
3.16.5	**Integrated Actions**		**3**
3.16.5.1	Design link sharing & inline edit	Medium	1
3.16.5.2	Assisted order creation (sales creates order for customer)	Medium	1
3.16.5.3	Payment link generation	Simple	1
3.16.6	**SLA & Performance Tracking**		**2**
3.16.6.1	First response time (FRT) tracking	Simple	1
3.16.6.2	SLA violation alerts	Simple	1
3.17	**Order Management**		**21**
3.17.1	**Order Lifecycle Management**		**4**
3.17.1.1	Order status state machine (Pending Payment → Confirmed → Ready to Fulfill → In Production → Picking → Packed → Shipped → Delivered → Completed/Cancelled)	Complex	2
3.17.1.2	Status transition rules & validations	Medium	1
3.17.1.3	Order timeline/history display	Medium	1
3.17.2	**Order Release & Fulfillment Sourcing**		**4**
3.17.2.1	Multi-warehouse fulfillment sourcing logic	Complex	2
3.17.2.2	Order release to warehouse (manual/auto)	Medium	1
3.17.2.3	Split fulfillment (partial shipments from multiple warehouses)	Medium	1
3.17.3	**Order Amendment**		**3**
3.17.3.1	Allow address change before shipment	Medium	1
3.17.3.2	Allow item addition/removal (restricted by status)	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.17.3.3	Amendment audit log	Simple	1
3.17.4	**Order Cancellation**		**2**
3.17.4.1	Customer-initiated cancellation (before shipment)	Medium	1
3.17.4.2	Admin/sales-initiated cancellation with reason	Medium	1
3.17.5	**Return & Exchange (RMA)**		**4**
3.17.5.1	RMA request creation (customer portal)	Medium	1
3.17.5.2	RMA approval workflow (sales/admin)	Medium	1
3.17.5.3	Return shipping label generation	Medium	1
3.17.5.4	Return receipt & refund processing	Medium	1
3.17.6	**Order Hold & Exception Handling**		**2**
3.17.6.1	Place order on hold (payment issue, address issue, inventory issue)	Medium	1
3.17.6.2	Exception resolution workflow	Medium	1
3.17.7	**Customer Order Portal**		**2**
3.17.7.1	Order history display for logged-in customers	Medium	1
3.17.7.2	Order detail view with tracking link	Medium	1
3.18	**Customer Information Management**		**17**
3.18.1	**Customer Profile**		**3**
3.18.1.1	Customer registration & login	Medium	1
3.18.1.2	Profile edit (name, email, phone, password)	Medium	1
3.18.1.3	Guest checkout (no registration required)	Medium	1
3.18.2	**Address Book**		**3**
3.18.2.1	Add/edit/delete multiple addresses	Medium	1
3.18.2.2	Default shipping & billing address	Simple	1
3.18.2.3	Address validation on save	Medium	1
3.18.3	**Customer Segmentation**		**2**
3.18.3.1	Segment definition (VIP, enterprise, new, at-risk)	Medium	1
3.18.3.2	Auto-assignment to segments based on rules	Medium	1
3.18.4	**Order History Integration**		**2**
3.18.4.1	Link customer to all orders (cross-device/session)	Medium	1
3.18.4.2	Customer lifetime value (CLV) calculation	Medium	1
3.18.5	**Marketing Consent & GDPR**		**3**
3.18.5.1	Marketing consent opt-in/opt-out	Medium	1
3.18.5.2	Data export request (GDPR)	Medium	1
3.18.5.3	Right to be forgotten (anonymization)	Medium	1
3.18.6	**Risk & Fraud Management**		**2**
3.18.6.1	Blacklist management (high-risk customers)	Medium	1
3.18.6.2	Fraud flag & manual review workflow	Medium	1
3.18.7	**Merge & Deduplication**		**2**
3.18.7.1	Duplicate customer detection	Medium	1
3.18.7.2	Customer profile merge with conflict resolution	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.19.1	**User Management & RBAC**		**4**
3.19.1.1	Role-based access control (10 roles)	Complex	2
3.19.1.2	User creation & role assignment	Medium	1
3.19.1.3	Permission management UI	Medium	1
3.19.2	**Notification Engine**		**3**
3.19.2.1	Email notification templates	Medium	1
3.19.2.2	SMS notification integration	Medium	1
3.19.2.3	In-app notification center	Medium	1
3.19.3	**Audit Log & Activity Tracking**		**3**
3.19.3.1	Comprehensive audit log for all entities	Complex	2
3.19.3.2	Activity search & filtering	Medium	1
3.19.4	**System Configuration**		**3**
3.19.4.1	Company & warehouse setup	Medium	1
3.19.4.2	Tax rate & currency configuration	Medium	1
3.19.4.3	Business hours & holiday calendar	Simple	1
3.19.5	**API & Webhooks**		**3**
3.19.5.1	RESTful API for external integrations	Complex	2
3.19.5.2	Webhook configuration & delivery	Medium	1
3.19.6	**File Storage & CDN Integration**		**2**
3.19.6.1	Product image upload & optimization	Medium	1
3.19.6.2	Design artifact storage	Medium	1
3.19.7	**Multi-Warehouse Orchestration**		**2**
3.19.7.1	Warehouse master data management	Medium	1
3.19.7.2	Cross-warehouse inventory visibility	Medium	1
3.20.1	**Warehouse KPI Dashboard**		**4**
3.20.1.1	Receipt volume & lead time	Medium	1
3.20.1.2	Picking productivity & accuracy	Medium	1
3.20.1.3	Packing throughput	Medium	1
3.20.1.4	Shipping on-time rate	Medium	1
3.20.2	**Inventory Reports**		**4**
3.20.2.1	Stock level by warehouse/location	Medium	1
3.20.2.2	Aging inventory report	Medium	1
3.20.2.3	Expiry date report	Medium	1
3.20.2.4	Inventory turnover analysis	Medium	1
3.20.3	**E-commerce Analytics**		**3**
3.20.3.1	Sales by product/category/period	Medium	1
3.20.3.2	Conversion funnel (cart → order → paid)	Medium	1
3.20.3.3	Customer acquisition & retention metrics	Medium	1
3.20.4	**Financial Reports**		**3**
3.20.4.1	Revenue by channel/product	Medium	1
3.20.4.2	AP aging (supplier payables)	Medium	1

#	WBS Item	Complexity	Est. Effort (man-days)
3.20.4.3	Payment reconciliation report	Medium	1
3.21.1	**Unit Testing**		**23**
3.21.1.1	Backend unit tests (80% coverage target)	Complex	14
3.21.1.2	Frontend unit tests (React components)	Medium	9
3.21.2	**Integration Testing**		**17**
3.21.2.1	Module-to-module integration tests	Complex	10
3.21.2.2	External API integration tests (carrier, payment gateway)	Medium	7
3.21.3	**End-to-End Testing**		**13**
3.21.3.1	Critical user journeys (purchase flow, fulfillment flow)	Complex	9
3.21.3.2	Browser compatibility testing	Medium	4
3.21.4	**Performance & Load Testing**		**5**
3.21.4.1	Load testing (concurrent users, inventory updates)	Complex	3
3.21.4.2	Performance optimization based on results	Medium	2
3.21.5	**User Acceptance Testing (UAT)**		**3**
3.21.5.1	UAT scenario preparation & execution	Medium	2
3.21.5.2	Bug fix & re-test cycle	Simple	1
3.22.1	**CI/CD Pipeline Setup**		**2**
3.22.1.1	GitHub Actions / GitLab CI configuration	Medium	1
3.22.1.2	Automated build, test, deploy workflow	Medium	1
3.22.2	**Cloud Infrastructure Provisioning**		**4**
3.22.2.1	AWS/GCP/Azure setup (VPC, subnets, security groups)	Complex	2
3.22.2.2	Database setup (RDS/Cloud SQL with replication)	Medium	1
3.22.2.3	CDN & static asset hosting	Medium	1
3.22.3	**Application Deployment**		**2**
3.22.3.1	Docker containerization	Medium	1
3.22.3.2	Kubernetes orchestration (if applicable)	Medium	1
3.22.4	**Monitoring & Logging**		**2**
3.22.4.1	Application monitoring (Datadog, New Relic)	Medium	1
3.22.4.2	Centralized logging (ELK/CloudWatch)	Medium	1
3.22.5	**Security Hardening**		**3**
3.22.5.1	SSL/TLS certificate setup	Simple	1
3.22.5.2	Firewall rules & intrusion detection	Medium	1
3.22.5.3	Security audit & penetration testing	Medium	1
3.23	**Testing Phrase**		**42**
3.23.1	Write Software Test document	Complex	14
3.23.2	Write, execute and write unit test's documents	Complex	14
3.23.3	Perform and write system test's documents	Complex	14

#	WBS Item	Complexity	Est. Effort (man-days)
3.24	**Deployment Phrase**		**9**
3.24.1	Deploy to server	Complex	9
4	**Project Monitoring & Controlling**		**12**
4.1	Monitor and control risk	Complex	12
5	**Project Closing**		**16**
5.1	Write Software User Guide's document	Complex	8
5.2	Write Final Project Report	Complex	8
	Total Estimated Effort (man-days)		**560**

Table 3. Scope and Estimation

1.2 Project Objectives

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Reviewing	100% of documents	0	0	Review of all requirements , design, and code documents.
3	Integration Test	>= 90% of module integration	0	0	
4	System Test	100% of critical flow	0	0	

Table 4. Project Objectives

- **Milestone Timeliness (%): The project must be finished before (24 December 2026)**
- **Allocated Effort (man-days): 5 (members) * 16(weeks) * 7(days/weeks) = 560 man-days**

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	Conflicts between team members	Medium	Medium	Organize team meetings to address and resolve conflicts
2	Solution does not meet real-world requirements	High	Medium	Extensive business requirements analysis, researching existing solution
3	Limited technical knowledge of relevant frameworks.	Medium	Low	Allocate time for technical training

4	Data security vulnerabilities	Medium	Medium	Data security vulnerabilities
5	Task deadline is too tight	Medium	Medium	Prioritize key tasks, delegate when possible, and extend the deadline.

Table 5. Project Risks

2. Management Approach

2.1 Project Process

Our software development lifecycle methodology is Agile, which is an iterative and flexible framework for developing, delivering, and maintaining products. Agile emphasizes adaptability, collaboration, and continuous delivery of value to ensure that the product can evolve alongside changing requirements and user needs. The reasons for adopting Agile in our project include:

- **Iteration planning**: This allows the team to break down the project into smaller, manageable tasks, prioritize them effectively, and set clear goals for each sprint. It helps ensure that the most important features are always being worked on first.
- **Daily Stand-ups**: These short meetings keep everyone aligned, encourage collaboration, and help quickly identify any blockers. By maintaining constant communication, the team can adapt and resolve issues before they grow into bigger problems.
- **Sprint reviews and demo**: At the end of each sprint, the team showcases the completed work to stakeholders. This not only provides transparency but also invites early feedback, making it easier to adjust direction if needed and ensuring the product evolves in line with expectations.

This iterative approach enables rapid response to changes and continuous improvement while keeping the project on track.

2.2 Quality Management

The following quality management measures will be applied to ensure that FoodFund fulfills its quality objectives:

- Defect Prevention:
    - Focus on applying coding best practices and conducting peer reviews to identify potential issues at an early stage. Maintaining clear documentation along with regular knowledge-sharing sessions will further help prevent defects.
- Reviewing:
    - Perform both formal and informal reviews of code, design, and requirements throughout the development process. In addition, holding regular sprint retrospectives enables the team to improve practices and proactively address areas needing enhancement.
- Unit Testing:
    - Unit testing is carried out continuously during development, especially under practices like Test-Driven Development (TDD) or Continuous Integration (CI). Developers create tests alongside their code to verify that each feature functions properly from the beginning.
    - Integration testing is checked whether various parts of the system interact and function correctly together. It usually takes place early in the development process,

right after unit testing, and continues as new features are added and integrated into the system.

- System Testing:
    - System testing is performed at the end of each sprint to verify the complete functionality of the system in a real-world setting. Its goal is to ensure that both the new features and the overall system meet user requirements through end-to-end testing that mirrors real usage scenarios.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criterias
Java Spring	HungNL, PhuongNH, TuUT	23/08/2026	Mandatory
React Native	HungNL, PhuongNH, TuUT	23/08/2026	Mandatory
Elasticsearch	TuUT, PhuongNH	23/08/2026	Mandatory
VPS	TuUT, PhuongNH	23/08/2026	Mandatory
NextJS	ThaiVHM, TuUT, PhuongNH	23/08/2026	Mandatory

Table 6. Training Plan

3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Project Introduction (Report 1)	31/08/2026	Project Introduction Document
2	Project Management Plan (Report 2)	07/8/2026	Project Management Plan Document
3	Software Requirement Specification (Report 3)	21/10/2026	Context Diagram, Use case diagram, ...
4	Software Design Documentation (Report 4)	10/11/2026	System architecture, class diagram, sequence diagram, ….
5	Software Testing Documentation (Report 5)	19/11/2026	Test strategy, test plan, test case, ...
6	Software User Guides (Report 6)	20/11/2026	Installation guides, user manual guides, ...
7	Final Report (Report 7)	7/12/2026	Include all project documentations
8	Code Package Back-end	10/12/2026	Code & Unit Test, Integration Test
9	Code Package Front-end	13/12/2026	Code & Unit Test, Integration Test

#	Deliverable	Due Date	Notes
10	Code Package Mobile	10/12/2026	Code & Unit Test, Integration Test

Table 7. Project Deliverables

4. Responsibility Assignments

D~Do; R~Review; S~Support; I~Informed; <blank>- Omitted

Responsibility	Phuong NH	TuUT	HungNL	ThaiVHM	Minh VH
Project Introduction (Report 1)	D	R	S	S	D
Project Management Plan (Report 2)	D	R	D	D	S
Software Requirement Specification (Report 3)	R	D	S	S	D
Software Design Documentation (Report 4)	D	R	S	S	D
Software Testing Documentation (Report 5)	S	D	R	S	R
Software User Guides (Report 6)	R	S	S	D	D
Final Report (Report 7)	D	R	D	D	D
Code Package Back-end	D	D	I	I	D
Code Package Front-end	S	S	D	D	S
Code Package Mobile	S	R	D	D	S

Table 8. Responsibility Assignments

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Supervisor (Lecturer)	Mr. Nguyen Minh Sang	- Provide document template - Instruct the project team - Review deliverables - Supervise project status - Receive report project	Weekly	Zalo
Meeting	Team members	- Collaborate on mutual task - Planning	Daily	Google Meets, Discord, Zalo
Messages	Team members	- Share daily progress made on project tasks - Share current and potential	Daily	Zalo, Discord

		roadblocks		

Table 9. Project Communications

6. Configuration Management

6.1 Document Management

- Document tools: Google Docs, Google Sheets
- File Management: Google Drive

6.2 Source Code Management

- Our team uses GitHub to manage the source code version

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	NextJS (Frontend), React Native (Mobile app), JavaSpring (Backend), ThreeJS (Frontend)
Database	PostgreSQL, Redis, Elasticsearch
IDEs/Editors	Visual Studio Code, IntelliJ
Diagramming	DrawIO, Visual Paradigm, Mermaid Chart
Documentation	Google Docs/Sheets/Slides
Version Control	GitHub (Source Codes), Google Drive (Documents)
Deployment server	VPS, Vercel
Project management	Google Sheet, Jira, Github

Table 10. Tools & Infrastructure

III. Software Requirement Specification

1. Product Overview

FoodFund is a web and mobile-based crowdfunding and operations management platform designed specifically for charitable food preparation and distribution campaigns. The system centralizes donation collection, transparent fund disbursement, ingredient management, kitchen workflows, and delivery tracking into a single integrated solution.

Currently, food charity initiatives in Vietnam are often managed manually or via fragmented social media groups, leading to inefficiencies, lack of transparency, and low donor trust. FoodFund addresses these challenges by digitizing and standardizing the entire workflow of a fundraisingcampaign—from receiving donations to preparing and delivering meals.

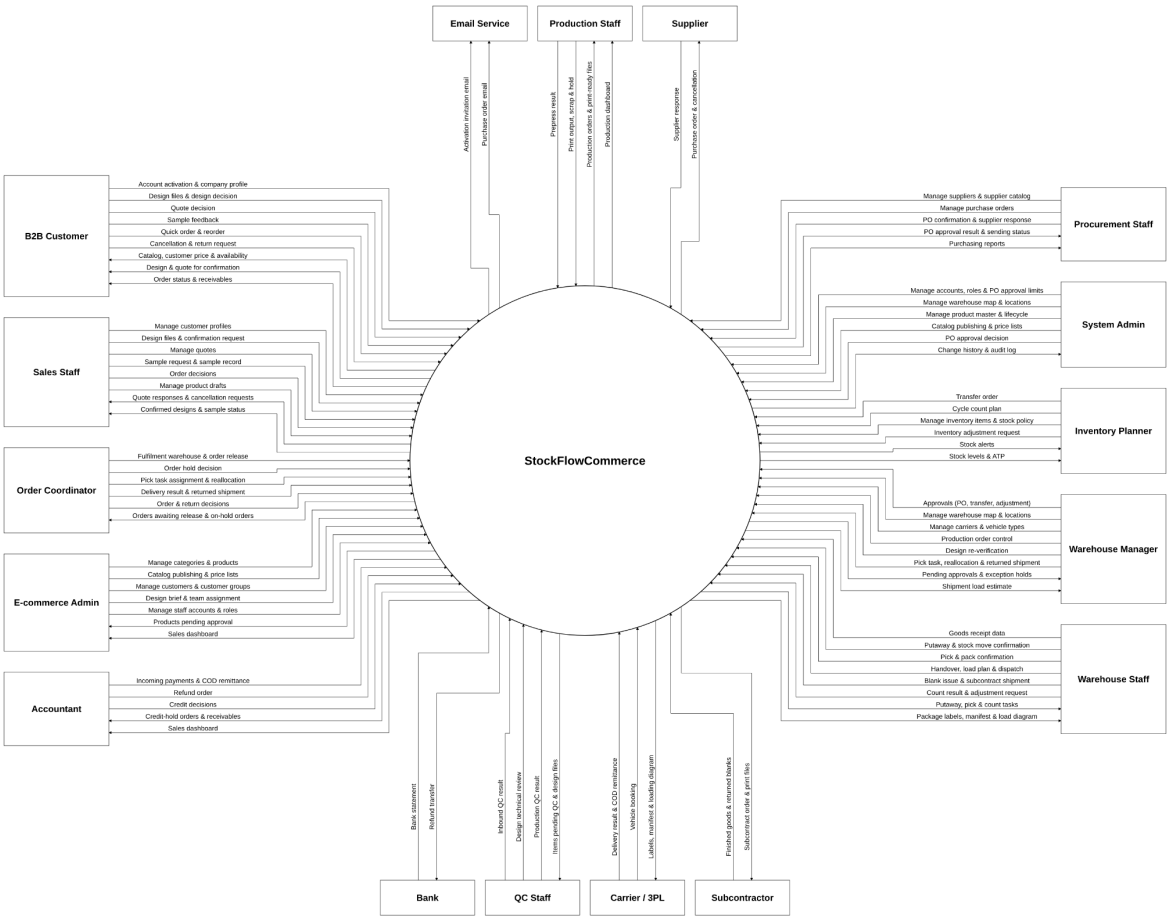


Figure 1. StockFlow Commerce Context Diagram

2. User Requirements

2.1 Actors

#	Actor	Description
1	User	An abstract actor representing any signed-in user; all other actors inherit its functions to log in, log out, change password, manage login sessions and view in-app notifications.
2	B2B Customer	A wholesale business customer, invited by Sales, who uses the customer portal with functions to browse the catalog with customer-specific prices, confirm designs, accept quotes, approve samples, place and track orders, request cancellations or returns, and view receivables and credit position.
3	Sales Staff	A sales user who manages customer relationships, with functions to create customer profiles, design products together with customers, prepare and send quotes, request samples, convert accepted quotes into orders, and decide on cancellations and partial deliveries.
4	Order Coordinator	A user who controls order fulfilment, with functions to select the fulfilment warehouse and delivery method, release or hold orders, assign pick tasks, reallocate short-picked lines, record delivery results and returned shipments, and decide on return requests.

#	Actor	Description
5	E-commerce Admin	A storefront administrator with functions to maintain product master data, approve products, publish the catalog, manage prices, manage customer status and groups, assign design work, and administer staff accounts and roles within their own permissions.
6	System Admin	A privileged user who holds every permission of the platform and has functions to manage staff accounts, roles and PO approval limits, configure warehouse maps, govern product master data and lifecycle, and review the change history and audit log.
7	Accountant	A finance user with functions to record and allocate incoming payments, record carrier remittances, create refunds, approve payment terms and credit limits, decide on credit-hold orders, and view order details in read-only mode.
8	Procurement Staff	A purchasing user with functions to manage suppliers and supplier catalogs, create and submit purchase orders, confirm and send POs to suppliers, record supplier responses, close or cancel POs, and view purchasing reports.
9	Inventory Planner	A planning user who keeps stock balanced across warehouses, with functions to create transfer orders, plan cycle counts, maintain stock policies, request inventory adjustments and acknowledge stock alerts.
10	Warehouse Manager	A supervisor of a warehouse who inherits all functions of Warehouse Staff, Inventory Planner and Production Staff, and additionally approves POs, transfers and inventory adjustments, configures the warehouse map, controls production orders and handles operational exceptions.
11	Warehouse Staff	A warehouse operator with functions to receive and put away goods, pick and pack orders, hand over shipments and load trucks, dispatch transfers, issue blank materials to production, and perform cycle counts.
12	Production Staff	A print-shop operator with functions to check print files before production, print and report good and scrap quantities, put production orders on hold, and view the production dashboard.
13	QC Staff	A quality inspector with functions to inspect inbound goods, technically review design files, record production QC results, and download design files for comparison.
14	Supplier	An external organization that supplies goods; it receives confirmed purchase orders and cancellation notices from the system by email or API, and its responses are recorded by Procurement Staff.
15	Email Service	An external email service used by the system to send activation invitations, purchase orders to suppliers and alert notifications, returning the delivery status of each email.

#	Actor	Description
16	Bank	An external financial institution whose bank statements provide customer transfers and carrier remittances for the Accountant to record, and which executes refund transfers to customers.
17	Carrier / 3PL	An external shipping partner that receives package labels, handover manifests and loading diagrams, provides vehicles for full-truckload delivery, and reports delivery results and COD remittances back to staff.
18	Subcontractor	An external print shop that receives subcontract orders, print files and blank materials when in-house capacity is insufficient, and returns finished goods, unused blanks and scrap for reconciliation.

Table 11. Actors

2.2 Use Case

2.2.1 Use Case Diagram(s)

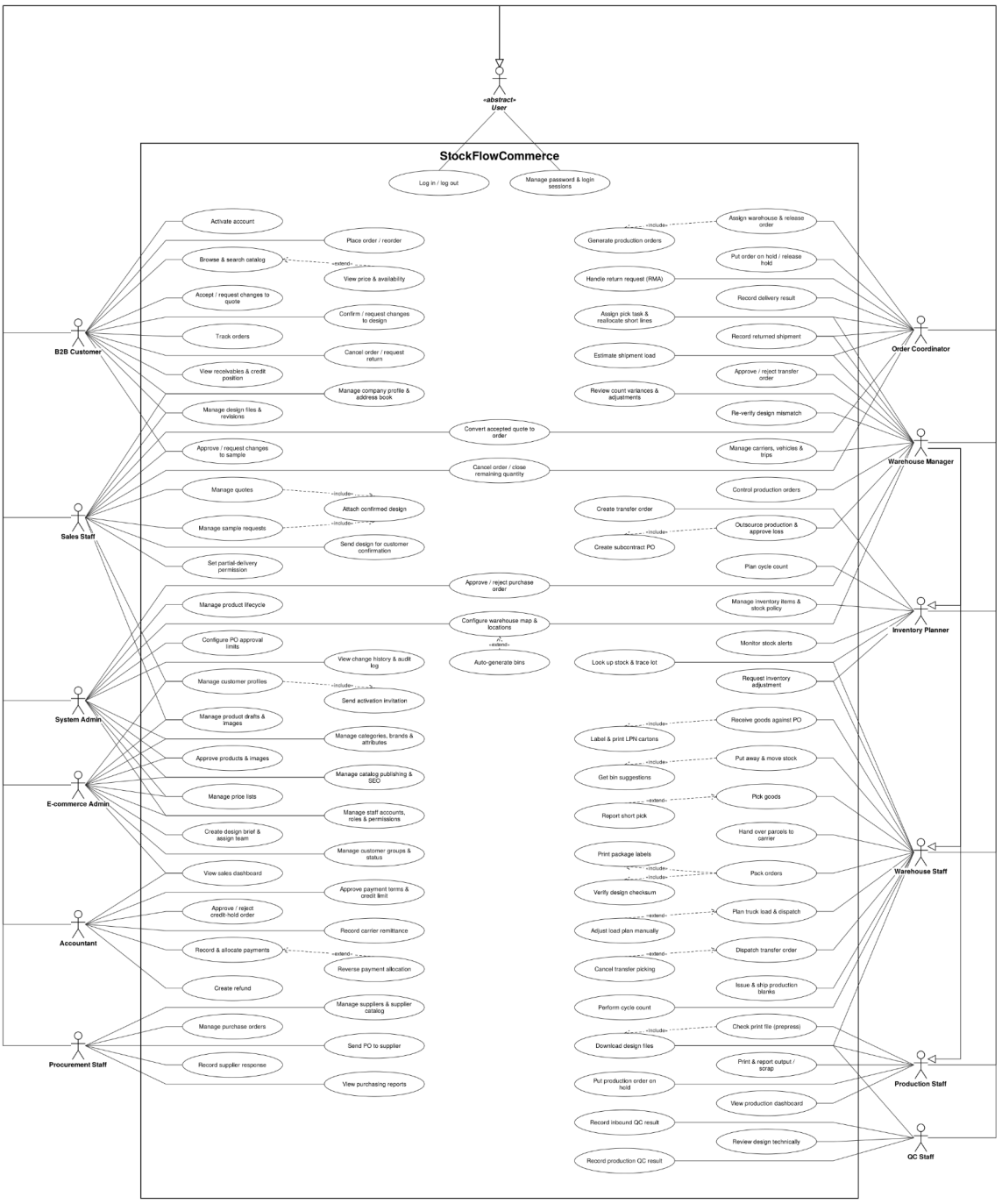


Figure 2. Use Case Diagram

2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
01	Log in / log out	User	Allows any user to sign in with the account issued to them and to sign out of the current session. Five consecutive failed sign-in attempts lock the account for 15 minutes.
02	Manage password & login sessions	User	Allows a user to change their password by entering the current one, view their active login sessions and revoke any of them, optionally signing out all other devices.
03	Activate account	B2B Customer	Allows the customer contact to open the invitation link sent by the system (valid for 72 hours) and set a password to activate the B2B account.
04	Browse & search catalog	B2B Customer	Allows the B2B Customer to browse the category tree and to search, filter and sort published products in the catalog.
05	View price & availability	B2B Customer	«extend» of "Browse & search catalog". Shows the price applied to the customer and a stock indicator (in stock, low stock, out of stock) without revealing actual quantities.
06	Place order / reorder	B2B Customer	Allows the B2B Customer to quickly order in-stock items by SKU and quantity, or to reorder from a previous order; reordering a quote-based order creates a new draft quote for Sales.
07	Accept / request changes to quote	B2B Customer	Allows the B2B Customer to accept a quote by choosing the quantity of each line, or to request changes so that Sales can revise it.
08	Confirm / request changes to design	B2B Customer	Allows the B2B Customer to confirm a design, which creates an immutable snapshot used for printing, or to request changes with a reason.
09	Track orders	B2B Customer	Allows the B2B Customer to view their orders and follow a simplified status timeline of each order.
10	Cancel order / request return	B2B Customer	Allows the B2B Customer to cancel an order that has not been released, request cancellation of a released order that is not yet packed, or request a return of defective goods within the policy period.
11	View receivables & credit position	B2B Customer	Allows the B2B Customer to view outstanding receivables, credit limit, overdue items and the bank transfer instructions for payment.

ID	Use Case	Actors	Use Case Description
12	Manage company profile & address book	B2B Customer, Sales Staff	Allows the B2B Customer, or Sales Staff on the customer's behalf, to maintain the company name, phone number and the delivery and billing address book.
13	Manage design files & revisions	B2B Customer, Sales Staff	Allows Sales Staff (designer) to upload design files and specifications and the B2B Customer to upload reference files; a pending confirmation request can be withdrawn and a new design version created from a confirmed snapshot.
14	Approve / request changes to sample	B2B Customer, Sales Staff	Allows the B2B Customer, or Sales Staff on the customer's behalf, to approve a sample round, which locks the design and blank, or to request changes with feedback.
15	Manage quotes	Sales Staff	Allows Sales Staff to create a quote with blanks, price tiers, validity and payment terms, revise it on request, send it to the customer and cancel a quote that is no longer valid.
16	Attach confirmed design	Sales Staff	«include» of "Manage quotes" and "Manage sample requests". Attaches a confirmed design snapshot to the quote or sample request after checking the customer, the blank product and the file checksum.
17	Send design for customer confirmation	Sales Staff	Allows Sales Staff (designer) to send a design that has passed technical review to the customer for confirmation.
18	Manage sample requests	Sales Staff	Allows Sales Staff to create and submit a sample request to the workshop, which reserves blanks and generates a sample production order, and to record when the sample is sent to the customer.
19	Set partial-delivery permission	Sales Staff	Allows Sales Staff to switch the partial-delivery permission of an order on or off.
20	Manage customer profiles	Sales Staff, E-commerce Admin	Allows Sales Staff or the E-commerce Admin to create a wholesale customer profile with initial payment terms (prepaid or deposit) and to resend the activation invitation, which revokes the previous link.

ID	Use Case	Actors	Use Case Description
21	Send activation invitation	Sales Staff, E-commerce Admin	«include» of "Manage customer profiles". The system emails an activation invitation to the customer contact when the profile is created.
22	Manage product drafts & images	Sales Staff, E-commerce Admin	Allows Sales Staff or the E-commerce Admin to create draft products, declare variant axes, generate SKUs, upload virus-scanned images and submit the product for approval.
23	Convert accepted quote to order	Sales Staff, Order Coordinator	Allows Sales Staff or the Order Coordinator to convert an accepted quote into a sales order with the agreed quantities, prices and terms.
24	Cancel order / close remaining quantity	Sales Staff, Order Coordinator	Allows Sales Staff or the Order Coordinator to approve or reject a customer cancellation request, cancel an order with a reason code (charging a production fee if printing has started) and close the undelivered remainder of a partial order.
25	Assign warehouse & release order	Order Coordinator	Allows the Order Coordinator to choose the fulfilment warehouse and delivery method (parcel or full truckload) and release a confirmed order to the warehouse; stock lines go to picking and printed lines go to production.
26	Generate production orders	Order Coordinator	«include» of "Assign warehouse & release order". The system generates one production order for each printed line of the released order.
27	Put order on hold / release hold	Order Coordinator	Allows the Order Coordinator to put an order on hold with a reason and release it from hold with a note.
28	Handle return request (RMA)	Order Coordinator	Allows the Order Coordinator to review a customer return request and either approve it by issuing an RMA or reject it.
29	Record delivery result	Order Coordinator	Allows the Order Coordinator to record whether a delivery succeeded or failed, with the reason reported by the carrier.
30	Assign pick task & reallocate short lines	Order Coordinator, Warehouse Manager	Allows the Order Coordinator or Warehouse Manager to create pick tasks for an order, assign them to warehouse staff and reallocate a short-picked line to another bin in the same warehouse.

ID	Use Case	Actors	Use Case Description
31	Record returned shipment	Order Coordinator, Warehouse Manager	Allows the Order Coordinator or Warehouse Manager to record a shipment that the carrier has returned to the warehouse (RTO).
32	Estimate shipment load	Order Coordinator, Warehouse Manager	Allows the Order Coordinator or Warehouse Manager to estimate the number of cartons and vehicles needed for an order or transfer before packing.
33	Create design brief & assign team	E-commerce Admin	Allows the E-commerce Admin to create a design brief for a customer and product and assign the designer and the QC reviewer.
34	Manage customer groups & status	E-commerce Admin	Allows the E-commerce Admin to block, unblock, close or reopen a customer with a reason, and to create customer groups and assign customers to them.
35	Manage categories, brands & attributes	E-commerce Admin, System Admin	Allows the E-commerce Admin or System Admin to maintain the shared category tree, brands and attribute dictionary used by products.
36	Approve products & images	E-commerce Admin, System Admin	Allows the E-commerce Admin or System Admin to approve or reject a submitted product and publish images uploaded by another user (four-eyes rule).
37	Manage catalog publishing & SEO	E-commerce Admin, System Admin	Allows the E-commerce Admin or System Admin to publish an approved product to the catalog or unlist it, and to edit its slug, SEO title and SEO description.
38	Manage price lists	E-commerce Admin, System Admin	Allows the E-commerce Admin or System Admin to maintain base SKU prices, customer-specific tiered prices and customer-group prices.
39	Manage staff accounts, roles & permissions	E-commerce Admin, System Admin	Allows the E-commerce Admin or System Admin to create, lock and disable staff accounts, reset passwords, and manage roles, permission matrices and data scopes.
40	View sales dashboard	E-commerce Admin, Accountant	Allows the E-commerce Admin or Accountant to view daily orders and revenue, top-selling SKUs and orders by status.
41	Manage product lifecycle	System Admin	Allows the System Admin to suspend, resume or discontinue a product, giving a mandatory reason.

ID	Use Case	Actors	Use Case Description
42	Configure PO approval limits	System Admin	Allows the System Admin to configure the purchase-order approval limit of each role.
43	View change history & audit log	System Admin	Allows the System Admin to view the change history of master data and permissions and the audit log of warehouse layout changes.
44	Approve / reject purchase order	System Admin, Warehouse Manager	Allows the Warehouse Manager or System Admin to approve or reject a pending purchase order; the approver must be different from the submitter.
45	Configure warehouse map & locations	System Admin, Warehouse Manager	Allows the Warehouse Manager or System Admin to configure the warehouse, zones, shelves, levels, bins, areas, walls and doors, and to change location status.
46	Auto-generate bins	System Admin, Warehouse Manager	«extend» of "Configure warehouse map & locations". Generates bins automatically by rows and columns for the levels of a shelf while configuring the warehouse map.
47	Approve payment terms & credit limit	Accountant	Allows the Accountant to approve a customer's payment terms, credit limit, credit term and deposit rate.
48	Approve / reject credit-hold order	Accountant	Allows the Accountant to approve or reject an order held because the customer exceeded the credit limit or has overdue debt.
49	Record & allocate payments	Accountant	Allows the Accountant to record incoming bank transfers from the bank statement and allocate them to orders or receivables.
50	Reverse payment allocation	Accountant	«extend» of "Record & allocate payments". Reverses a wrong payment allocation and returns the money to the customer's surplus balance.
51	Record carrier remittance	Accountant	Allows the Accountant to record the cash-on-delivery money collected and remitted by the carrier.
52	Create refund	Accountant	Allows the Accountant to create a refund order that returns a customer's surplus balance by bank transfer.
53	Manage suppliers & supplier catalog	Procurement Staff	Allows Procurement Staff to maintain supplier profiles and supplier catalogs, including price, MOQ, pack size and lead time.

ID	Use Case	Actors	Use Case Description
54	Manage purchase orders	Procurement Staff	Allows Procurement Staff to create and edit purchase orders validated against MOQ, pack size and catalog price, submit them for approval (auto-approved within the submitter's limit), and close or cancel them.
55	Send PO to supplier	Procurement Staff	Allows Procurement Staff to confirm an approved purchase order and send it to the supplier by email or API.
56	Record supplier response	Procurement Staff	Allows Procurement Staff to record the supplier's confirmation or rejection of a purchase order.
57	View purchasing reports	Procurement Staff	Allows Procurement Staff to view purchase orders by status, spending by supplier and supplier performance.
58	Create transfer order	Inventory Planner	Allows the Inventory Planner to create a transfer order with source and destination warehouses, delivery method and SKU lines, and submit it for approval.
59	Plan cycle count	Inventory Planner	Allows the Inventory Planner to plan a cycle count by scope, choose blind counting and assign the counter.
60	Manage inventory items & stock policy	Inventory Planner	Allows the Inventory Planner to maintain the logistics data, unit conversions and stock policy (reorder and safety levels) of an inventory item.
61	Monitor stock alerts	Inventory Planner	Allows the Inventory Planner to monitor reorder, safety-stock, expiring and expired stock alerts and acknowledge them.
62	Look up stock & trace lot	Inventory Planner, Warehouse Staff	Allows the Inventory Planner or Warehouse Staff to look up stock and ATP by warehouse, location, lot and status, and to trace a lot to its origin, reservations and customer orders.
63	Request inventory adjustment	Inventory Planner, Warehouse Staff	Allows the Inventory Planner or Warehouse Staff to request a stock increase or decrease with a reason, pending the Warehouse Manager's approval.
64	Approve / reject transfer order	Warehouse Manager	Allows the Warehouse Manager to approve or reject a transfer order that exceeds the approval threshold.

ID	Use Case	Actors	Use Case Description
65	Review count variances & adjustments	Warehouse Manager	Allows the Warehouse Manager to review count variances, approve or reject inventory adjustments, request a recount or cancel the count.
66	Re-verify design mismatch	Warehouse Manager	Allows the Warehouse Manager to re-verify a packing session that was held because the design did not match the confirmed snapshot.
67	Manage carriers, vehicles & trips	Warehouse Manager	Allows the Warehouse Manager to maintain the carrier list and vehicle types and cancel a dispatch trip that has not left the warehouse.
68	Control production orders	Warehouse Manager	Allows the Warehouse Manager to reopen a production order from hold, cancel it before printing or stop it during printing or QC.
69	Outsource production & approve loss	Warehouse Manager	Allows the Warehouse Manager to split or move a production order to a subcontractor, top it up or bring it back in-house, and approve subcontracting losses above the allowed rate.
70	Create subcontract PO	Warehouse Manager	«include» of "Outsource production & approve loss". The system creates a draft subcontract purchase order for the outsourced production quantity.
71	Receive goods against PO	Warehouse Staff	Allows Warehouse Staff to create a goods receipt against a purchase order, count the goods with lot and expiry data, confirm it and move lines that require QC to the QC area.
72	Label & print LPN cartons	Warehouse Staff	«include» of "Receive goods against PO". Labels full cartons with LPN codes during receiving and prints or reprints the labels.
73	Put away & move stock	Warehouse Staff	Allows Warehouse Staff to put away goods by scanning the carton or quantity and the destination bin, move stock between locations, view the warehouse map and print location labels.
74	Get bin suggestions	Warehouse Staff	«include» of "Put away & move stock". The system suggests up to ten valid destination bins, ranked, during put-away.

ID	Use Case	Actors	Use Case Description
75	Pick goods	Warehouse Staff	Allows Warehouse Staff to start a pick task and confirm each line by scanning the location, SKU or LPN and quantity, and to put back picked goods of a cancelled order.
76	Report short pick	Warehouse Staff	«extend» of "Pick goods". Reports a shortage at the bin during picking; the system holds the order line and opens a blind count of that bin.
77	Pack orders	Warehouse Staff	Allows Warehouse Staff to open packages, record dimensions and weight, seal them and complete packing of an order.
78	Print package labels	Warehouse Staff	«include» of "Pack orders". Prints package labels when packing is completed and reprints outdated labels.
79	Verify design checksum	Warehouse Staff	«include» of "Pack orders". The system compares the design file checksum with the snapshot confirmed by the customer before packing is completed.
80	Hand over parcels to carrier	Warehouse Staff	Allows Warehouse Staff to scan parcels onto a trip and hand them over to the carrier with a signed manifest.
81	Plan truck load & dispatch	Warehouse Staff	Allows Warehouse Staff to calculate, review and confirm a 3D truck load plan, record vehicle bookings, and load, scan and dispatch each truck.
82	Adjust load plan manually	Warehouse Staff	«extend» of "Plan truck load & dispatch". Lets staff drag packages in the 3D model and recalculate a single vehicle when the automatic plan needs adjustment.
83	Dispatch transfer order	Warehouse Staff	Allows Warehouse Staff to pick and pack transfer goods at the source warehouse and dispatch them to the destination.
84	Cancel transfer picking	Warehouse Staff	«extend» of "Dispatch transfer order". Cancels transfer picking while the goods have not yet left the source warehouse.
85	Issue & ship production blanks	Warehouse Staff	Allows Warehouse Staff to issue reserved blanks to the production area, ship blanks to a subcontractor and reconcile the returned blanks and scrap.
86	Perform cycle count	Warehouse Staff	Allows Warehouse Staff to perform an assigned cycle count; in blind counts the book quantity is hidden.

ID	Use Case	Actors	Use Case Description
87	Download design files	Warehouse Staff, Production Staff, QC Staff	Allows Warehouse Staff, Production Staff or QC Staff to download the design files they are permitted to see within the scope of their task.
88	Check print file (prepress)	Production Staff	Allows Production Staff to check the print-ready file before production; a rejected file puts the production order on hold.
89	Print & report output / scrap	Production Staff	Allows Production Staff to start printing and report printed and scrap quantities with reasons.
90	Put production order on hold	Production Staff	Allows Production Staff to put a production order on hold with a reason.
91	View production dashboard	Production Staff	Allows Production Staff to view open and overdue production orders, the scrap rate and samples awaiting customer feedback.
92	Record inbound QC result	QC Staff	Allows QC Staff to record the QC result of inbound goods as accepted, quarantined or rejected with a reason.
93	Review design technically	QC Staff	Allows the assigned QC reviewer to pass or fail the current design files with comments before they go to the customer.
94	Record production QC result	QC Staff	Allows QC Staff to record the passed and failed quantities of each production QC round.

Table 12. Use Case Diagram Descriptions

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

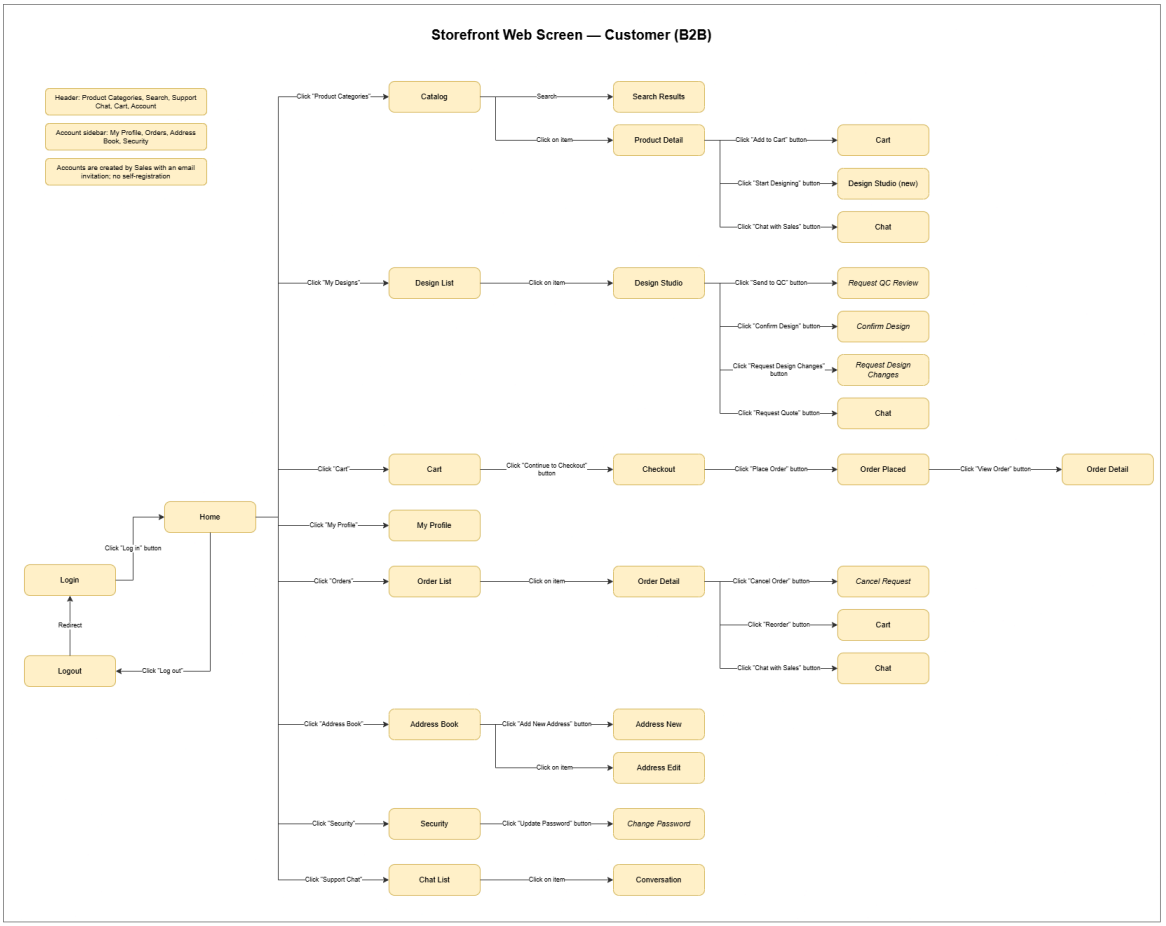


Figure 3. Storefront Web Screen - Customer

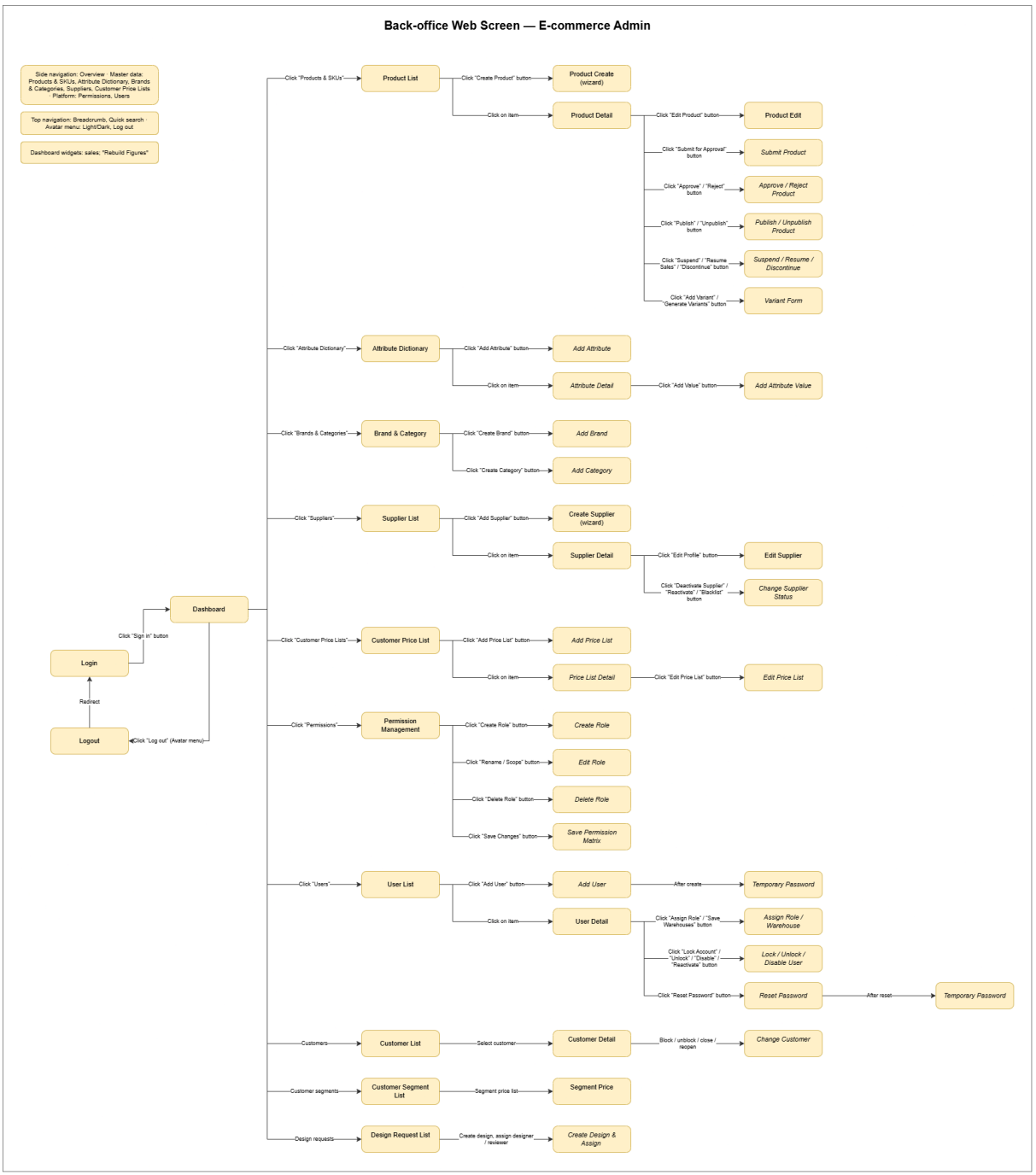


Figure 4. Back-office Web Screen - E-commerce Admin

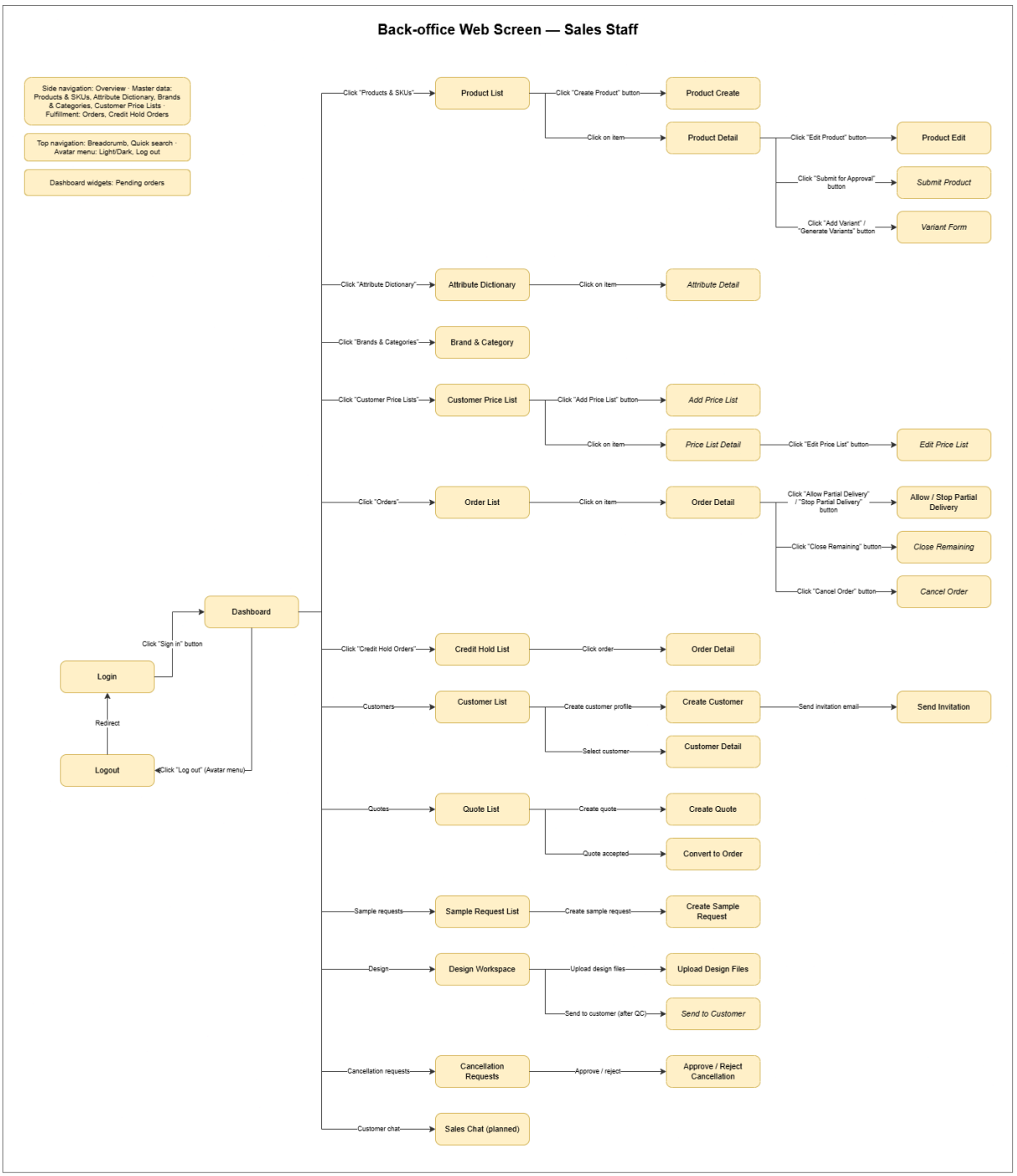


Figure 5. Back-office Web Screen - Sales Staff

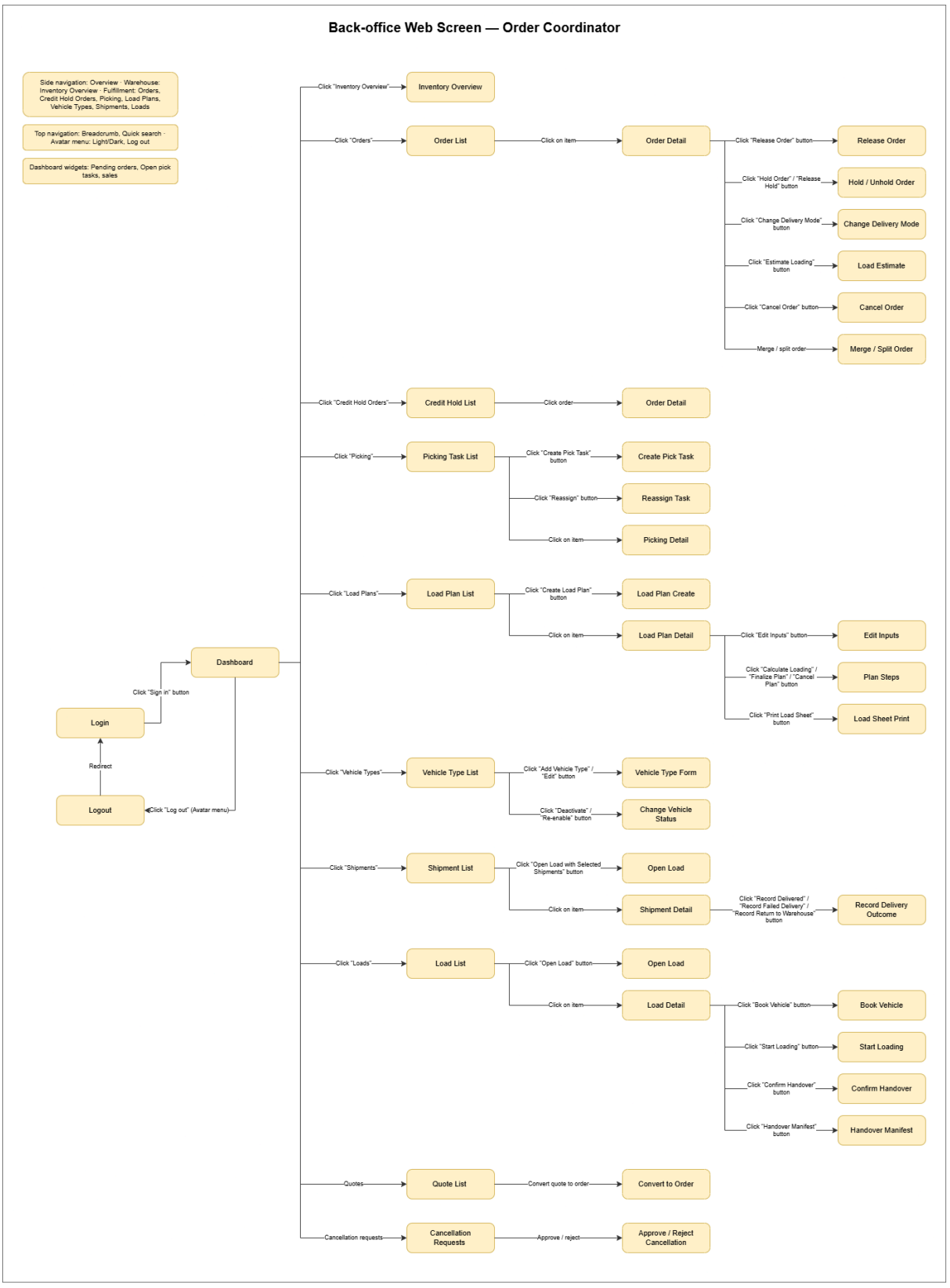


Figure 6. Back-office Web Screen - Order Coordinator

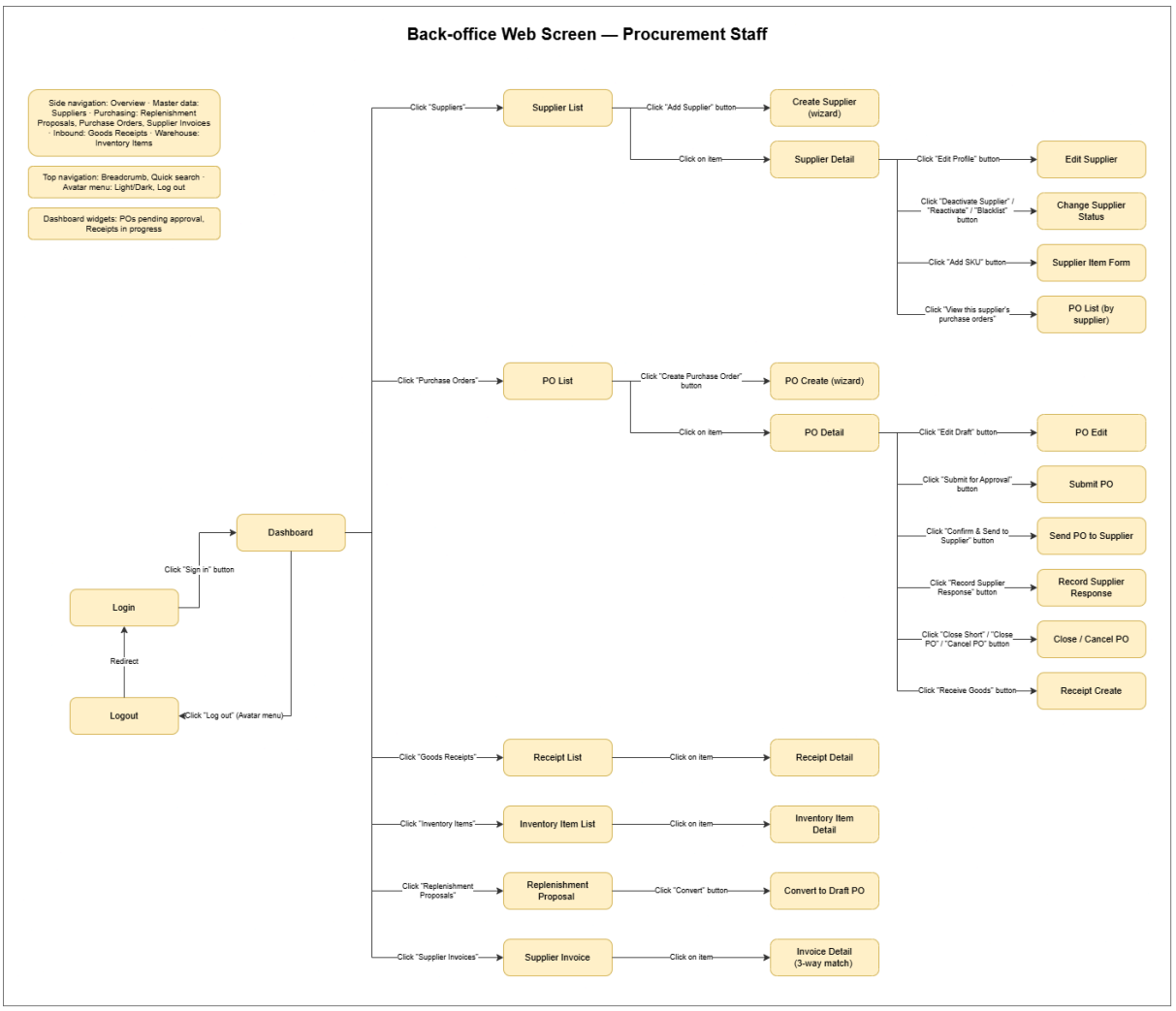


Figure 7. Back-office Web Screen - Procurement Staff

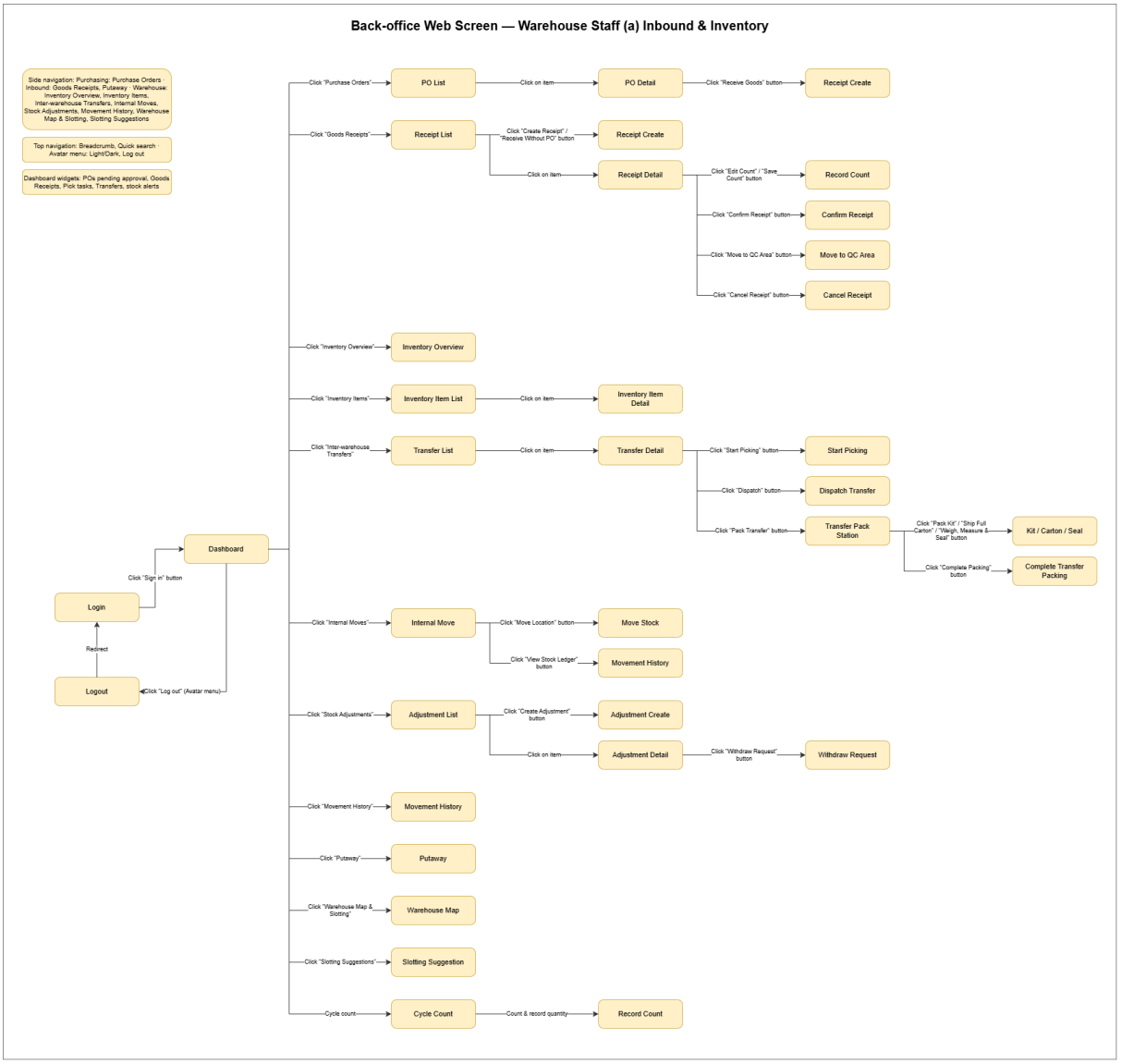


Figure 8. Back-office Web Screen - Warehouse Staff (a) Inbound & Inventory

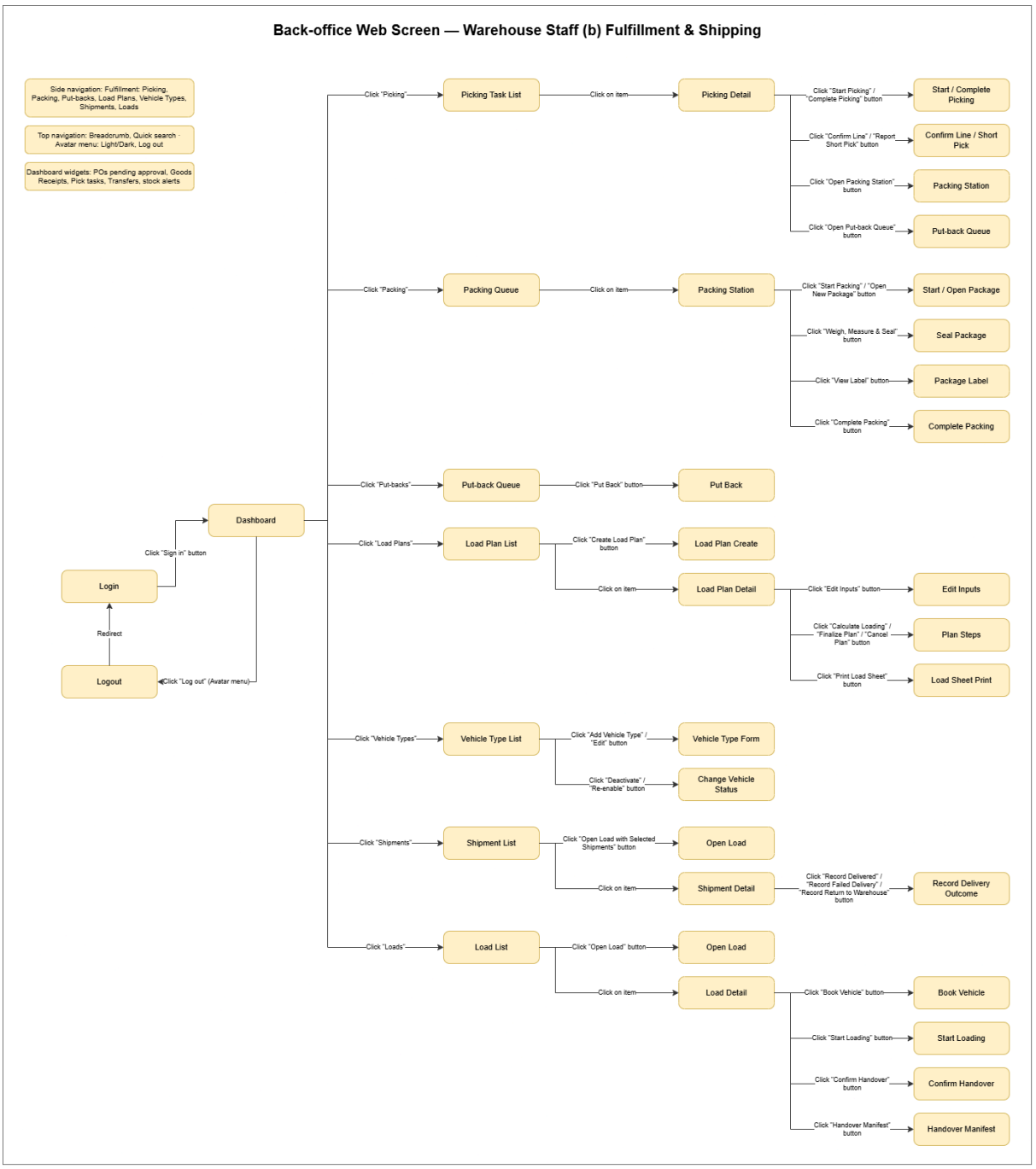


Figure 9. Back-office Web Screen - Warehouse Staff (b) Fulfillment & Shipping

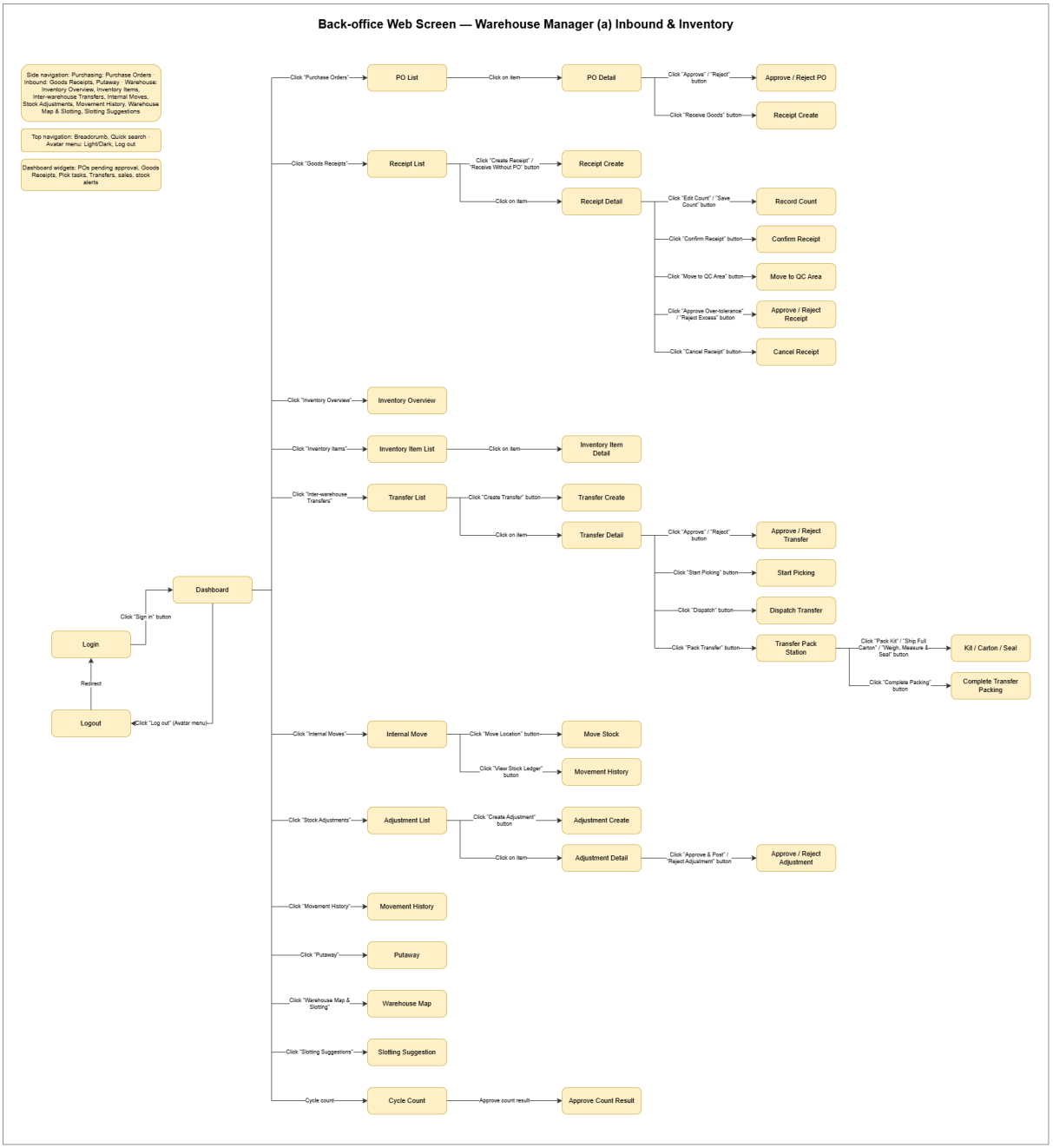


Figure 10. Back-office Web Screen - Warehouse Manager (a) Inbound & Inventory

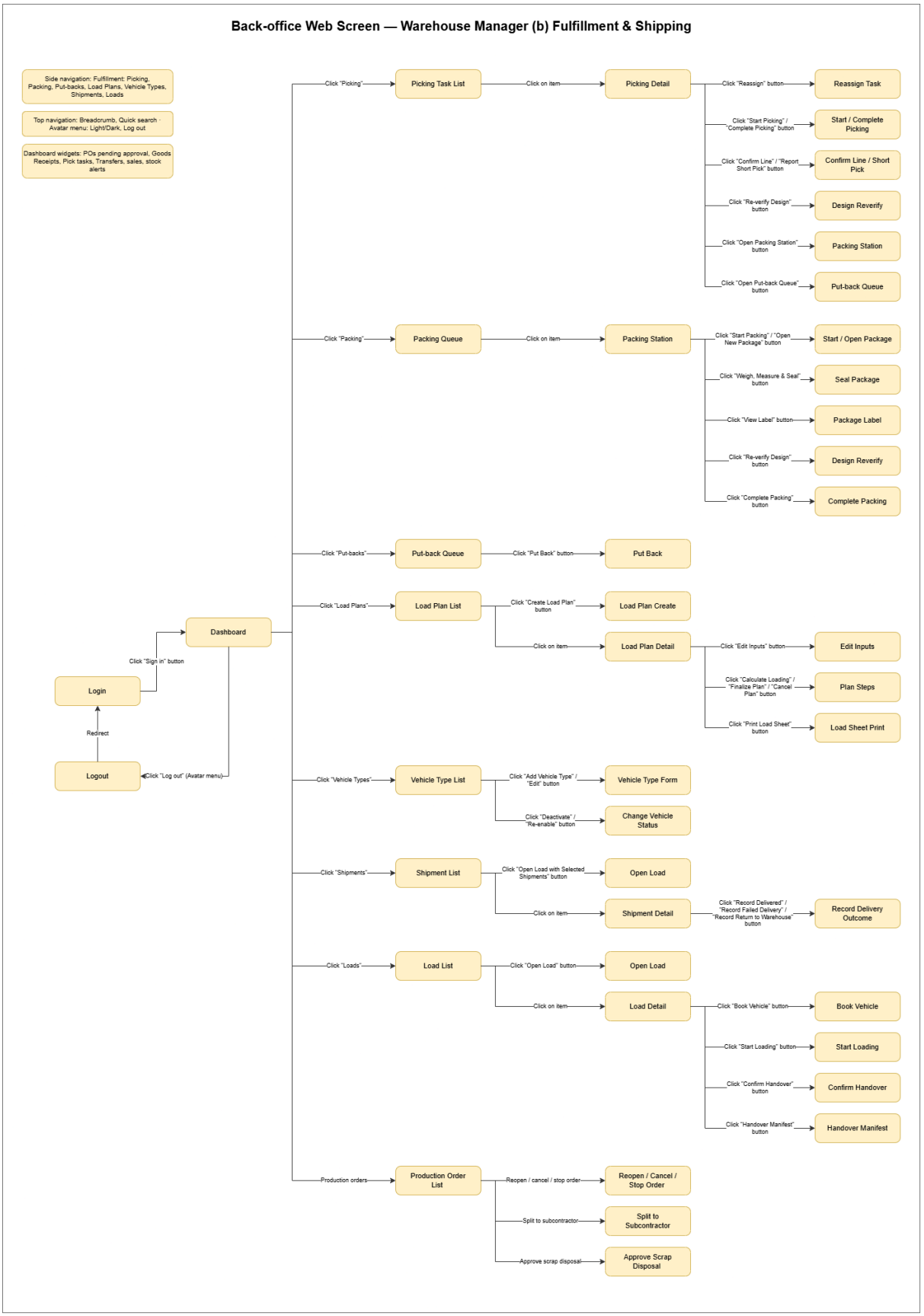


Figure 11. Back-office Web Screen - Warehouse Manager (b) Fulfillment & Shipping

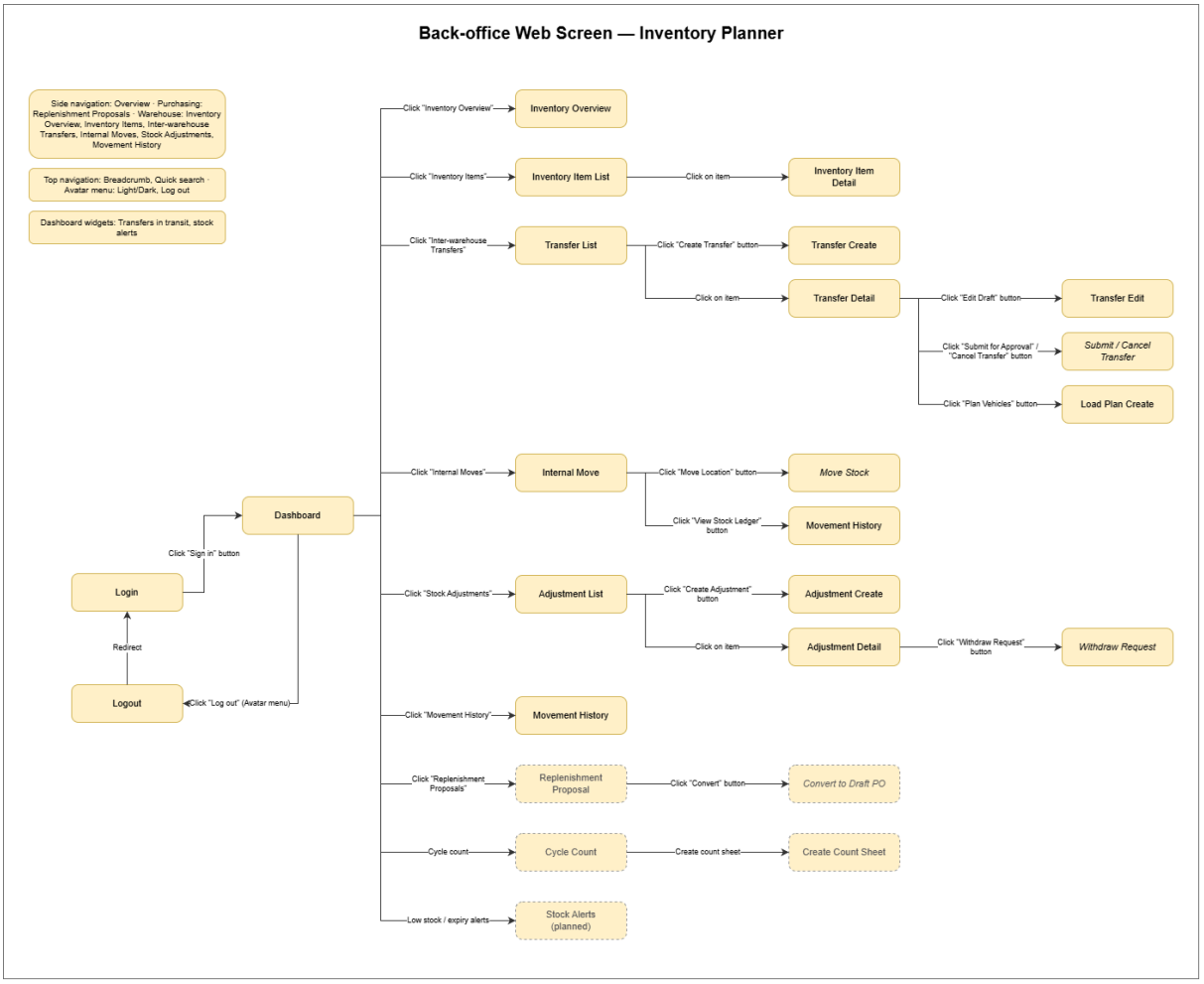


Figure 12. Back-office Web Screen - Inventory Planner

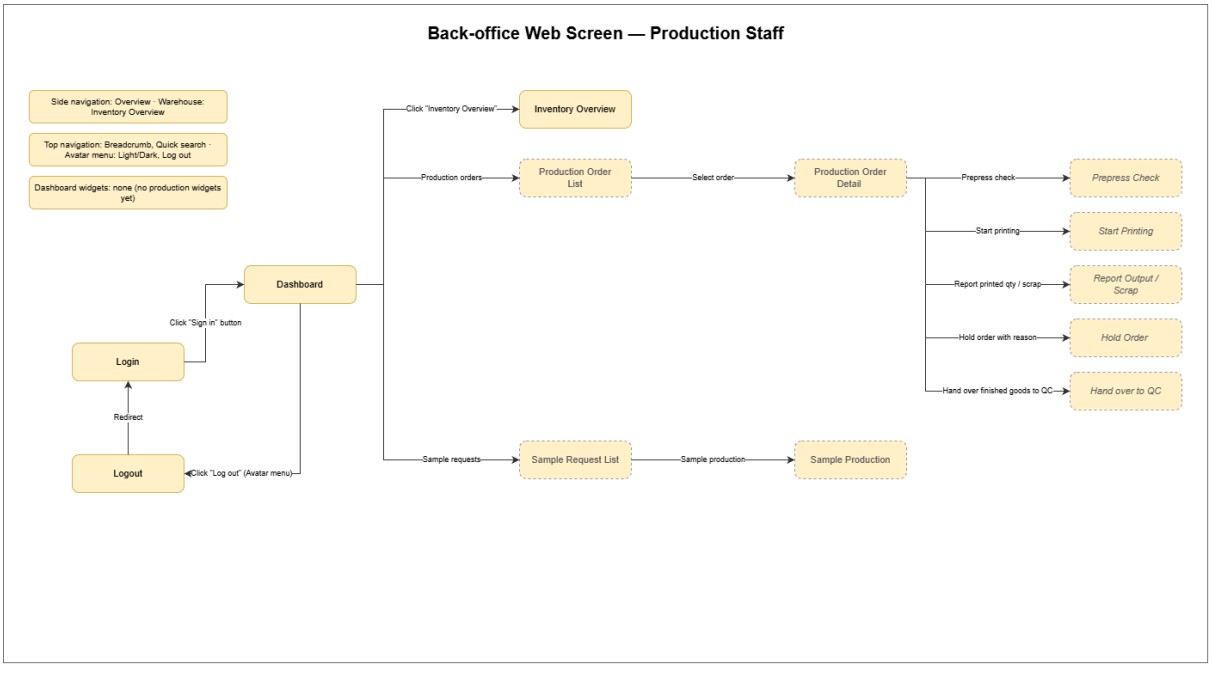


Figure 13. Back-office Web Screen - Production Staff

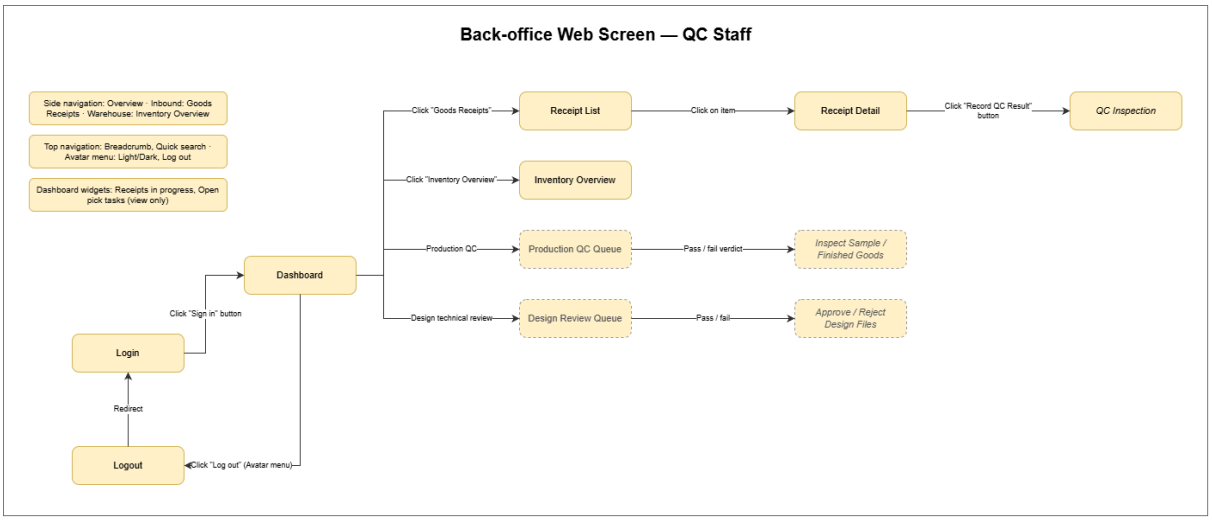


Figure 14. Back-office Web Screen - QC Staff

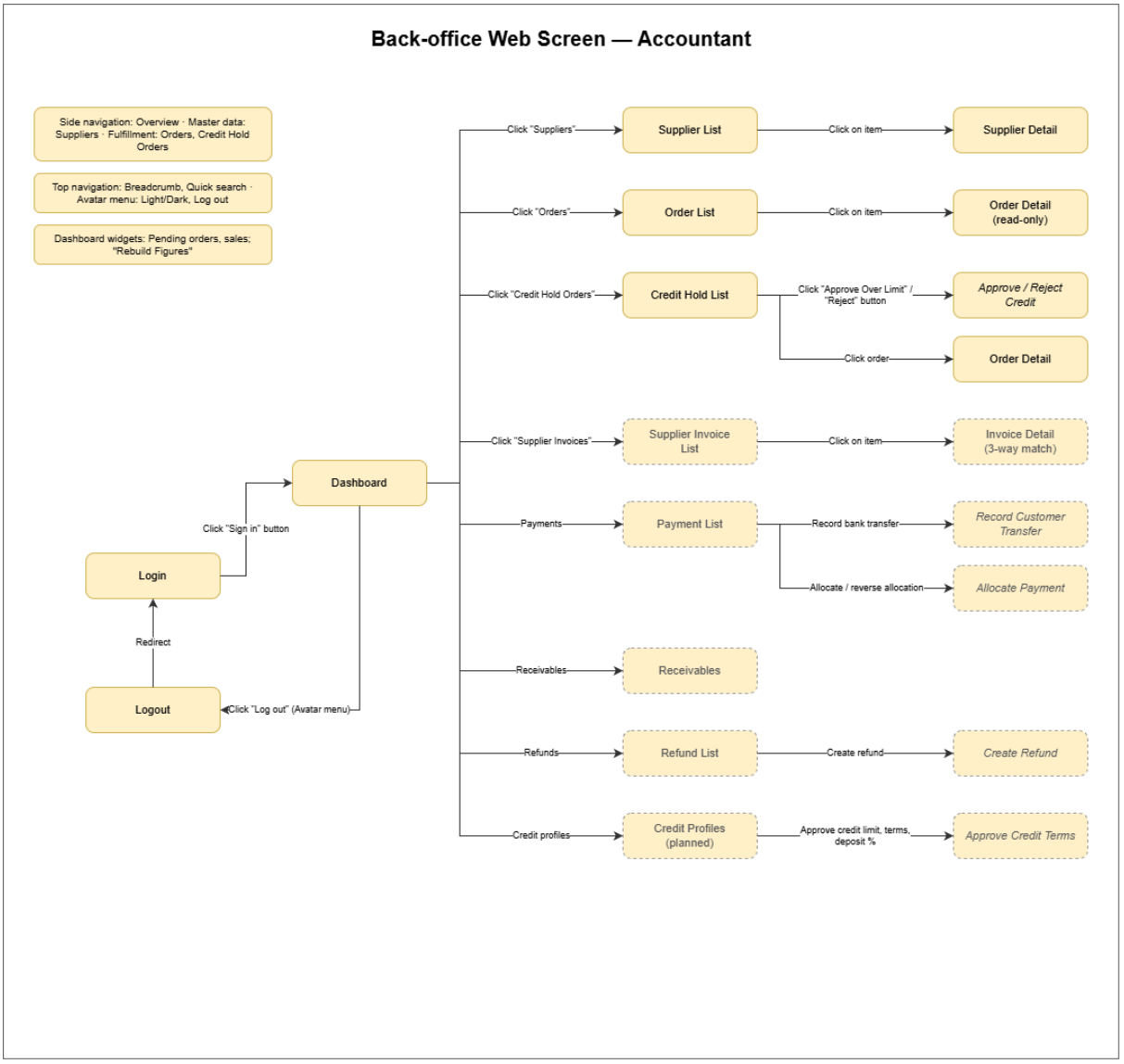


Figure 15. Back-office Web Screen - Accountant

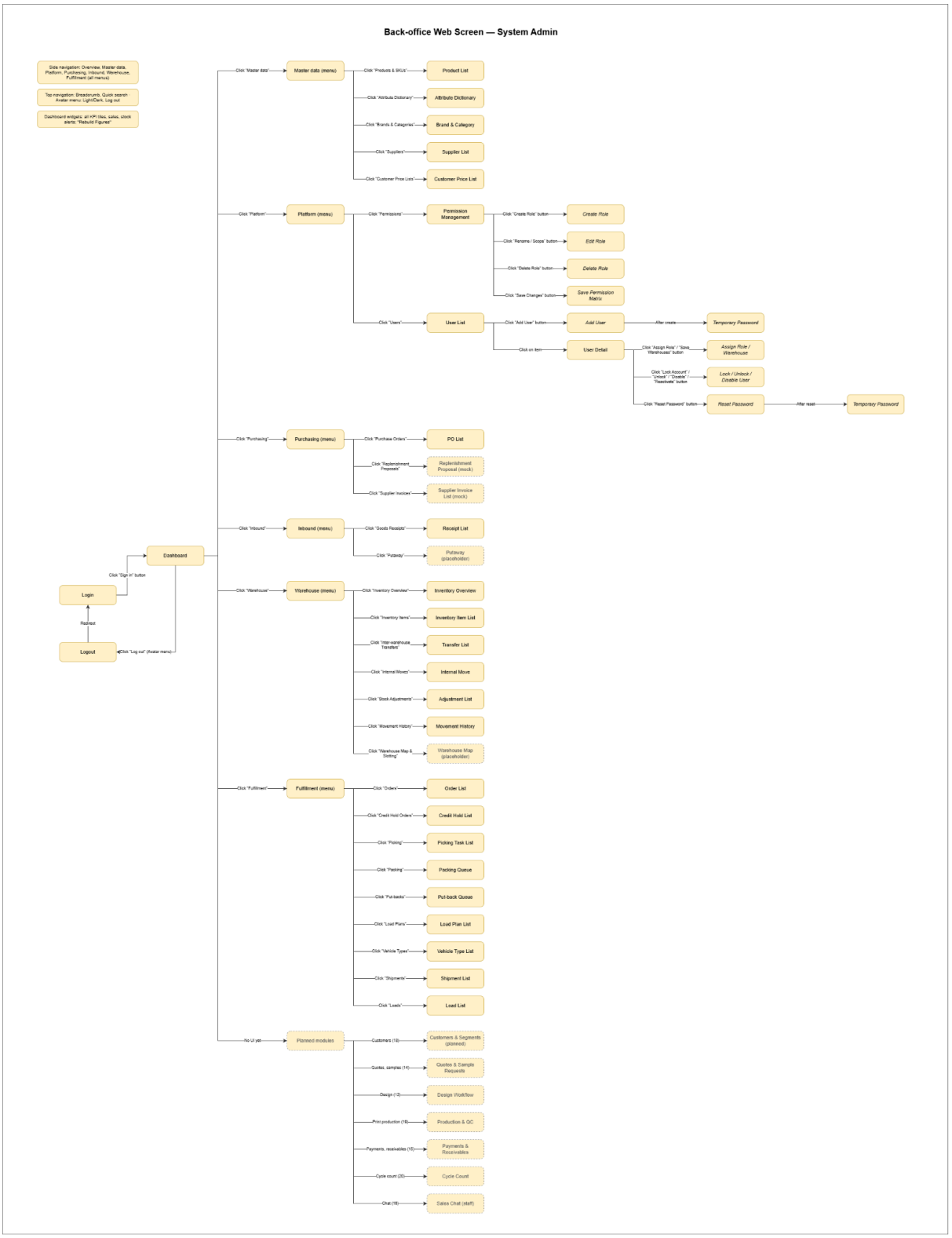


Figure 16. Back-office Web Screen - System Admin

3.1.2 Screen Descriptions

3.1.2.1 Storefront (Customer portal)

#	Feature	Screen	Description
1	Login	Storefront Login	Screen allows customers to log in to the storefront with their email and password; staff accounts are refused.
2	Logout	Header, Account	Screen in the header and account section allows customers to log out of the storefront.
3	Change password	Account security	Screen allows customers to change their password; it opens first when an administrator has set the password.
4	View home page	Home	Screen allows all users to view the store introduction, product categories, featured products and design steps.
5	View product catalog	Catalog	Screen allows customers to browse products with their own prices and stock, filter by category and sort the list.
6	Search products	Search	Screen allows customers to search products by keyword with suggestions while typing.
7	View product detail	Product detail	Screen allows customers to view product images, variants (SKU), price tiers, stock and minimum order quantity.
8	Add product to cart	Product detail	Screen allows customers to choose a SKU and quantity and add it to their cart.
9	View cart	Cart	Screen allows customers to review cart items, change quantities, remove items and see live price and stock.
10	Checkout	Checkout	Screen allows customers to choose a shipping address and payment term (prepaid or credit) and place an order.
11	View my orders	My orders	Screen allows customers to view their order list and filter it by status.
12	View order detail	Order detail	Screen allows customers to follow an order's status, timeline, items, production progress and payment.
13	Cancel order	Order detail	Screen allows customers to cancel an order, or request cancellation from Sales once processing has started.
14	Reorder	Order detail	Screen allows customers to put the items of a past order back into the cart at today's prices.
15	View profile	Account	Screen allows customers to view their company and contact information.
16	Edit profile	Account	Screen allows customers to update their contact information.
17	View credit and receivables	Account	Screen allows customers to view their credit limit, amount owed, payment terms and outstanding receivables.
18	View address book	Address book	Screen allows customers to view saved shipping and billing addresses and set the default one.
19	Add address	New address	Screen allows customers to add a new address with province, ward and street details.

#	Feature	Screen	Description
20	Edit address	Edit address	Screen allows customers to update a saved address.
21	Delete address	Address book	Screen allows customers to delete an address they no longer use.
22	View my designs	My designs	Screen allows customers to view their print designs grouped by status.
23	Start new design	Design launcher	Screen allows customers to choose a print template for a product variant and start a new design.
24	Design product	Design studio	Screen allows customers to design print faces with text, images, brush and background tools, layers and a 3D preview.
25	Submit design for QC review	Design studio	Screen allows customers to run the print preflight check and send the design to QC for review.
26	Confirm design	Design studio	Screen allows customers to finalize their design, or confirm the version Sales sent, accepting any print warnings.
27	Request design changes	Design studio	Screen allows customers to request changes to a design Sales sent, or withdraw it to edit again, with a reason.
28	Chat with Sales	Sales chat	Screen allows customers to view conversations and message Sales to adjust their design.

3.1.2.2 Back-office (Staff portal)

#	Feature	Screen	Description
1	Login	Back-office Login	Screen allows staff to log in to the back-office with their account and password; customer accounts are refused.
2	Logout	Header	Screen in the header allows staff to log out of the back-office.
3	View notifications	Notification bell	Screen allows staff to view notifications, mark them as read and open the related record.
4	View dashboard	Dashboard	Screen allows staff to view key indicators (orders, purchase orders, receipts, transfers, picking) and stock alerts.
5	View sales report	Dashboard	Screen allows authorized staff to view daily revenue, top SKUs and orders by status.
6	Rebuild sales report	Dashboard	Screen allows authorized staff to recalculate the sales report data.
7	View product list	Product list	Screen allows authorized staff to search and filter products by status, brand and category.
8	View product detail	Product detail	Screen allows authorized staff to view product information, variants, images, prices and history.
9	Create product	Create product	Screen allows authorized staff to create a draft product step by step (information, variants, images, prices).
10	Edit product	Edit product	Screen allows authorized staff to edit a draft product and submit it for approval.
11	Delete draft product	Product detail	Screen allows authorized staff to delete a product that is still a draft.

#	Feature	Screen	Description
12	Approve product	Product detail	Screen allows authorized staff to approve or reject a product submitted for approval.
13	Publish product	Product detail	Screen allows authorized staff to publish an approved product to the storefront or unpublish it.
14	Suspend or discontinue product	Product detail	Screen allows the System Admin to suspend, resume or discontinue a product.
15	Manage product variants	Product detail	Screen allows authorized staff to set variant axes, generate the variant matrix and add or block variants.
16	Activate variant	Product detail	Screen allows authorized staff to activate a variant or stop using it.
17	View attribute dictionary	Attribute dictionary	Screen allows authorized staff to view product attributes and their values.
18	Create attribute	Attribute dictionary	Screen allows authorized staff to add a new attribute to the dictionary.
19	Edit attribute	Attribute dictionary	Screen allows authorized staff to edit an attribute, manage its values and assign attributes to categories.
20	Delete attribute	Attribute dictionary	Screen allows authorized staff to delete an attribute that is no longer used.
21	View brands and categories	Brands & categories	Screen allows authorized staff to view the brand list and the category tree.
22	Create brand or category	Brands & categories	Screen allows authorized staff to add a new brand or category.
23	Edit brand or category	Brands & categories	Screen allows authorized staff to update a brand or category.
24	View supplier list	Supplier list	Screen allows authorized staff to search and filter suppliers by status.
25	View supplier detail	Supplier detail	Screen allows authorized staff to view supplier information, contacts, payment terms and purchase orders.
26	Create supplier	Create supplier	Screen allows authorized staff to create a new supplier.
27	Edit supplier	Edit supplier	Screen allows authorized staff to update supplier information.
28	Change supplier status	Supplier detail	Screen allows authorized staff to reactivate a supplier, stop working with it or put it on the blacklist.
29	View supplier items	Supplier detail	Screen allows authorized staff to view the SKUs a supplier provides with purchase price and lead time.
30	Manage supplier items	Supplier detail	Screen allows authorized staff to add, edit or remove SKUs in a supplier's catalog.
31	View customer prices	Customer prices	Screen allows authorized staff to choose a customer and view their own price list per SKU.
32	Set customer price	Customer prices	Screen allows authorized staff to add or update a customer's price for a SKU.

#	Feature	Screen	Description
33	View permission matrix	Permissions	Screen allows authorized staff to view roles and the permissions granted to each role.
34	Edit role permissions	Permissions	Screen allows authorized staff to grant or revoke permissions of a role and set its data scope.
35	Create role	Permissions	Screen allows authorized staff to create a custom role, optionally copying permissions from another role.
36	Edit role	Permissions	Screen allows authorized staff to rename a custom role and change its description.
37	Delete role	Permissions	Screen allows authorized staff to delete a custom role that no one holds.
38	View user list	Users	Screen allows authorized staff to search staff accounts and filter them by role and status.
39	View user detail	User detail	Screen allows authorized staff to view a staff account's profile, roles, warehouses and status.
40	Create user	Users	Screen allows authorized staff to create a staff account.
41	Edit user	User detail	Screen allows authorized staff to edit a staff profile and assign roles and working warehouses.
42	Lock or disable user	User detail	Screen allows authorized staff to lock, unlock, disable or re-enable an account and sign out its sessions.
43	Reset user password	User detail	Screen allows authorized staff to reset a staff member's password.
44	View purchase order list	Purchase order list	Screen allows authorized staff to search and filter purchase orders by status, supplier and date.
45	View purchase order detail	Purchase order detail	Screen allows authorized staff to view order lines, amounts, approval and supplier delivery history.
46	Create purchase order	Create purchase order	Screen allows authorized staff to create a purchase order with a supplier, lines, prices and delivery date.
47	Edit purchase order	Edit purchase order	Screen allows authorized staff to edit a draft purchase order and submit it for approval.
48	Approve purchase order	Purchase order detail	Screen allows authorized staff to approve a purchase order or return it to draft.
49	Send purchase order to supplier	Purchase order detail	Screen allows authorized staff to send an approved order to the supplier and record the supplier's response.
50	Resend purchase order	Purchase order detail	Screen allows authorized staff to resend an order whose delivery to the supplier failed.
51	Close or cancel purchase order	Purchase order detail	Screen allows authorized staff to close an order short, close it or cancel it.
52	View goods receipt list	Goods receipt list	Screen allows authorized staff to search and filter goods receipts by status, supplier and warehouse.
53	View goods receipt detail	Goods receipt detail	Screen allows authorized staff to view received lines, counted quantities and QC results.
54	Create goods receipt	Create goods receipt	Screen allows authorized staff to create a receipt for goods arriving against a purchase order.

#	Feature	Screen	Description
55	Create receipt without PO	Create goods receipt	Screen allows authorized staff to record goods that arrive without a purchase order.
56	Count and confirm receipt	Goods receipt detail	Screen allows authorized staff to count received goods, move them to the QC area, confirm or cancel the receipt.
57	Record QC result	Goods receipt detail	Screen allows QC staff to record the inspection result (pass or fail) of received lines.
58	Approve over-receipt	Goods receipt detail	Screen allows authorized staff to approve quantities over tolerance or reject the excess.
59	View inventory overview	Inventory overview	Screen allows authorized staff to view stock on hand, reserved and available quantities by SKU and warehouse.
60	View stock alerts	Stock alerts	Screen allows authorized staff to view low-stock and out-of-stock alerts.
61	View inventory item list	Inventory items	Screen allows authorized staff to view the logistics profile and stock policy of each SKU.
62	View inventory item detail	Inventory item detail	Screen allows authorized staff to view a SKU's dimensions, weight, packaging and reorder settings.
63	Edit inventory item	Inventory item detail	Screen allows authorized staff to update a SKU's logistics data and stock policy (min, max, reorder point).
64	View transfer order list	Transfer order list	Screen allows authorized staff to view transfer orders between warehouses and filter them by status.
65	View transfer order detail	Transfer order detail	Screen allows authorized staff to view transfer lines, approval, picking and dispatch progress.
66	Create transfer order	Create transfer order	Screen allows authorized staff to create a transfer order from a source to a destination warehouse.
67	Edit transfer order	Edit transfer order	Screen allows authorized staff to edit a draft transfer order, submit it for approval or cancel it.
68	Approve transfer order	Transfer order detail	Screen allows the Warehouse Manager to approve a transfer order or return it to draft.
69	Pick and dispatch transfer	Transfer order detail	Screen allows authorized staff to start picking at the source warehouse and dispatch the goods.
70	Pack transfer order	Transfer packing station	Screen allows authorized staff to pack a full-truck transfer into sealed packages before load planning.
71	Move stock internally	Internal move	Screen allows authorized staff to move stock from one bin location to another in the same warehouse.
72	View stock adjustment list	Stock adjustment list	Screen allows authorized staff to view stock adjustment requests and filter them by status.
73	View stock adjustment detail	Stock adjustment detail	Screen allows authorized staff to view adjusted quantities, reason and approval history.
74	Create stock adjustment	Create stock adjustment	Screen allows authorized staff to request an increase or decrease of stock with a reason, or withdraw it.
75	Approve stock adjustment	Stock adjustment detail	Screen allows authorized staff to approve or reject a stock adjustment (four-eyes rule).

#	Feature	Screen	Description
76	View stock movement history	Stock movements	Screen allows authorized staff to view the stock ledger of every inbound, outbound and internal movement.
77	View order list	Order list	Screen allows authorized staff to search and filter sales orders by status, customer and date.
78	View order detail	Order detail	Screen allows authorized staff to view order lines, payment, credit position, production and shipments.
79	Release order	Order detail	Screen allows authorized staff to release an order to the warehouse or production, hold it or remove a hold.
80	Cancel order	Order detail	Screen allows authorized staff to cancel an order or close the remaining quantity of a partly shipped order.
81	Allow partial shipment	Order detail	Screen allows authorized staff to allow or stop partial shipment of an order.
82	Choose delivery mode	Order detail	Screen allows authorized staff to choose or change how an order is delivered.
83	Estimate vehicle load	Order detail	Screen allows authorized staff to estimate the vehicles needed for an order before packing.
84	View credit-hold orders	Credit holds	Screen allows authorized staff to view orders held because the customer exceeded the credit limit.
85	Approve credit hold	Credit holds	Screen allows the Accountant to approve or reject an order over the credit limit.
86	View picking tasks	Picking list	Screen allows authorized staff to view picking tasks and filter them by status, warehouse and assignee.
87	Create and assign picking task	Picking list	Screen allows authorized staff to create a picking task for a released order and assign it to warehouse staff.
88	Pick goods	Picking detail	Screen allows warehouse staff to start picking, confirm each line by bin, report shortages and complete the task.
89	Put back goods	Put-back queue	Screen allows warehouse staff to view goods waiting to go back to the shelf and return them to a bin.
90	View packing queue	Packing list	Screen allows authorized staff to view orders that are picked and waiting to be packed.
91	Pack order	Packing station	Screen allows authorized staff to open, scan, seal and void packages and create the shipment.
92	Take over packing session	Packing station	Screen allows the Warehouse Manager to take over a packing session started by another packer.
93	Re-verify design	Packing station	Screen allows authorized staff to re-check a print design whose checksum did not match and release the hold.
94	View load plan list	Load plan list	Screen allows authorized staff to view vehicle load plans and filter them by status.
95	Create load plan	Create load plan	Screen allows authorized staff to create a load plan from orders or a transfer order and choose vehicle types.

#	Feature	Screen	Description
96	View load plan detail	Load plan detail	Screen allows authorized staff to view the 3D loading layout, vehicles and unloaded packages.
97	Calculate and confirm load plan	Load plan detail	Screen allows authorized staff to calculate, edit placements, confirm, reopen or cancel a load plan.
98	Print load sheet	Load sheet print	Screen allows authorized staff to print the loading sheet of one vehicle.
99	View vehicle types	Vehicle types	Screen allows authorized staff to view vehicle types with their cargo dimensions and payload.
100	Manage vehicle types	Vehicle types	Screen allows authorized staff to add or edit a vehicle type and activate or deactivate it.
101	View shipment list	Shipment list	Screen allows authorized staff to view shipments waiting to leave and shipments already delivered.
102	View shipment detail	Shipment detail	Screen allows authorized staff to view the packages, carrier and delivery status of a shipment.
103	Assign carrier	Shipment detail	Screen allows authorized staff to choose or clear the carrier of a shipment.
104	Record delivery outcome	Shipment detail	Screen allows authorized staff to record that a shipment was returned to the warehouse.
105	View load list	Load list	Screen allows authorized staff to view vehicle loads (trips) and their status.
106	Load and hand over	Load detail	Screen allows authorized staff to open a load, add shipments, scan packages onto the vehicle and hand it to the carrier.
107	Cancel load	Load detail	Screen allows authorized staff to cancel a load before the vehicle leaves the warehouse.
108	View load manifest	Load manifest	Screen allows authorized staff to view and print the manifest of packages on a vehicle.

3.1.3 Screen Authorization

3.1.3.1 Storefront (Customer portal)

Screen	Guest	Customer
Login	X	X
Logout		X
Change password		X
View home page	X	X
View product catalog		X
Search products		X
View product detail		X
Add product to cart		X
View cart		X
Checkout		X
View my orders		X
View order detail		X
Cancel order		X
Reorder		X
View profile		X
Edit profile		X
View credit and receivables		X

Screen	Guest	Customer
View address book		X
Add address		X
Edit address		X
Delete address		X
View my designs		X
Start new design		X
Design product		X
Submit design for QC review		X
Confirm design		X
Request design changes		X
Chat with Sales		X

3.1.3.2 Back-office (Staff portal)

Screen	System Admin	E-commerce Admin	Warehouse Manager	Warehouse Staff	QC Staff	Inventory Planner	Procurement Staff	Sales Staff	Order Coordinator	Accountant	Production Staff
Login	X	X	X	X	X	X	X	X	X	X	X
Logout	X	X	X	X	X	X	X	X	X	X	X
View notifications	X	X	X	X	X	X	X	X	X	X	X
View dashboard	X	X	X	X	X	X	X	X	X	X	X
View sales report	X	X	X						X	X	
Rebuild sales report	X	X								X	
View product list	X	X						X			
View product detail	X	X						X			
Create product	X	X						X			
Edit product	X	X						X			
Delete draft product	X	X									
Approve product	X	X									
Publish product	X	X									
Suspend or discontinue product	X										
Manage product variants	X	X						X			
Activate variant	X	X									
View attribute dictionary	X	X						X			
Create attribute	X	X									
Edit attribute	X	X									
Delete attribute	X	X									
View brands and categories	X	X						X			
Create brand or category	X	X						X			
Edit brand or category	X	X						X			
View supplier list	X	X					X			X	
View supplier detail	X	X					X			X	
Create supplier	X	X					X				
Edit supplier	X	X					X				

Screen	System Admin	E-commerce Admin	Warehouse Manager	Warehouse Staff	QC Staff	Inventory Planner	Procurement Staff	Sales Staff	Order Coordinator	Accountant	Production Staff
Change supplier status	X	X					X				
View supplier items	X						X				
Manage supplier items	X						X				
View customer prices	X	X						X			
Set customer price	X	X									
View permission matrix	X	X									
Edit role permissions	X	X									
Create role	X	X									
Edit role	X	X									
Delete role	X										
View user list	X	X									
View user detail	X	X									
Create user	X	X									
Edit user	X	X									
Lock or disable user	X	X									
Reset user password	X	X									
View purchase order list	X		X	X			X				
View purchase order detail	X		X	X			X				
Create purchase order	X						X				
Edit purchase order	X						X				
Approve purchase order	X		X								
Send purchase order to supplier	X						X				
Resend purchase order	X		X								
Close or cancel purchase order	X						X				
View goods receipt list	X		X	X	X		X				
View goods receipt detail	X		X	X	X		X				
Create goods receipt	X		X	X			X				
Create receipt without PO	X		X	X			X				
Count and confirm receipt	X		X	X			X				
Record QC result	X				X						
Approve over-receipt	X		X								
View inventory overview	X		X	X	X	X			X		X
View stock alerts	X		X	X		X					
View inventory item list	X		X	X		X	X				
View inventory item detail	X		X	X		X	X				
Edit inventory item	X		X			X					
View transfer order list	X		X	X		X					
View transfer order detail	X		X	X		X					
Create transfer order	X		X			X					
Edit transfer order	X		X			X					
Approve transfer order	X		X								
Pick and dispatch transfer	X		X	X		X					

Screen	System Admin	E-commerce Admin	Warehouse Manager	Warehouse Staff	QC Staff	Inventory Planner	Procurement Staff	Sales Staff	Order Coordinator	Accountant	Production Staff
Pack transfer order	X		X	X							
Move stock internally	X		X	X		X					
View stock adjustment list	X		X	X		X					
View stock adjustment detail	X		X	X		X					
Create stock adjustment	X		X	X		X					
Approve stock adjustment	X		X								
View stock movement history	X		X	X		X					
View order list	X							X	X	X	
View order detail	X							X	X	X	
Release order	X								X		
Cancel order	X							X	X		
Allow partial shipment	X							X			
Choose delivery mode	X								X		
Estimate vehicle load	X		X						X		
View credit-hold orders	X							X	X	X	
Approve credit hold	X									X	
View picking tasks	X		X	X					X		
Create and assign picking task	X		X						X		
Pick goods	X		X	X							
Put back goods	X		X	X							
View packing queue	X		X	X							
Pack order	X		X	X							
Take over packing session	X		X								
Re-verify design	X		X								
View load plan list	X		X	X					X		
Create load plan	X		X	X							
View load plan detail	X		X	X					X		
Calculate and confirm load plan	X		X	X							
Print load sheet	X		X	X					X		
View vehicle types	X		X	X					X		
Manage vehicle types	X		X								
View shipment list	X		X	X					X		
View shipment detail	X		X	X					X		
Assign carrier	X		X	X							
Record delivery outcome	X		X						X		
View load list	X		X	X					X		
Load and hand over	X		X	X							
Cancel load	X		X								
View load manifest	X		X	X					X		

3.1.4 Non-screen Functions

#	Feature	System Function	Description
1	Inventory Management	Release Expired Stock Reservations	Runs every 1 minute. Finds stock reservations whose hold time (30 minutes, BR-014) has passed and releases them in bounded batches, so stock held by abandoned checkouts returns to Available-to-Promise.
2	Inventory Management	Evaluate Stock Alerts	Runs every 30 seconds. Re-evaluates reorder-point and safety-stock alerts only for SKUs whose stock changed since the last run (changes are queued by database triggers on stock items). A SKU is also evaluated immediately after its thresholds are saved.
3	Inventory Management	Reconcile Stock Alerts	Runs daily at 01:30. Re-evaluates every SKU that is low on stock or still has an open alert, as a safety net for missed changes (e.g., data restored or thresholds edited directly).
4	Inventory Management	Expiry Sweep	Runs daily at 02:00. Moves expired stock to EXPIRED status and raises Expiring / Expired alerts for the affected warehouse.
5	Inventory Management	Create Putaway Tasks	When received goods are confirmed (line without QC) or accepted by QC, automatically creates putaway tasks for the warehouse: one task per carton plus one task for loose quantity.
6	Inventory Management	Release Stock on Payment Failure	When a payment fails, automatically releases the stock that the order was holding.
7	Goods Receipt	Close Goods Receipt	When a putaway task is completed, checks whether all lines of the goods receipt have been put away and, if so, changes the receipt from IN_PUTAWAY to CLOSED.
8	Order Management	Cancel Unpaid Order	When a stock reservation expires before the order is paid, the system automatically cancels the order.
9	Order Management	Update Order on Payment Result	When a payment is captured, reversed or failed, the system moves the order to the next status or cancels it accordingly.
10	Order Management	Record Production Progress	When a sample is approved or a production order is completed, the system records the result on the related order so it can continue to fulfillment.
11	Order Management	Expire Quotations	Runs every 5 minutes. Changes sent quotations whose validity date has passed from SENT to EXPIRED.
12	Order Management	Expire Return Requests	Runs hourly. Changes approved returns whose goods did not arrive within the deadline (14 days) from AWAITING_RETURN to EXPIRED.

#	Feature	System Function	Description
13	Order Management	Settle Orders	Runs daily at 00:30. Completes and closes orders that are fully delivered and paid and have no open return; orders not yet eligible are checked again the next night.
14	Payment	Create Receivable	When a credit (debt) order is delivered, the system automatically creates a receivable for the customer.
15	Payment	Mark Overdue Receivables	Runs daily at 00:15. Changes receivables past their due date to OVERDUE.
16	Payment	Create Refund Requests	When an order is cancelled or its remaining quantity is closed, cancels the amounts still awaited and creates PENDING refund requests for the money received in excess, for the Accountant to process.
17	Production	Create Production Orders	When the Order Coordinator releases order lines that need printing, the system automatically creates the corresponding production orders.
18	Production	Cancel Production Orders	When an order is cancelled or its remainder is closed, the related production orders are cancelled/stopped automatically.
19	Production	Hold Subcontracted Goods	When a goods receipt for a subcontract purchase order is confirmed, the received goods are held for the related order so they are not put up for sale.
20	Fulfillment	Stop Warehouse Work on Cancellation	When an order is cancelled before shipping, cancels its picking and packing tasks, flags picked goods for put-back, voids unshipped packages and labels, and cancels the shipment waiting for dispatch.
21	Notification	Send Customer Emails	Sends emails to the customer when an order is placed, handed to the carrier, delivered, not delivered, cancelled or partly closed, and when a return request changes status. Emails are sent via Resend/SMTP and logged.
22	Notification	Send Purchase Order to Supplier	When a purchase order is sent, amended or cancelled, the system emails the purchase order document to the supplier.
23	Notification	Retry Failed Emails	Runs every 1 minute. Resends customer and supplier emails whose last attempt failed, after their back-off time has passed.
24	Notification	Receive Email Delivery Status	Webhook endpoint that receives delivery reports from the email provider (delivered, bounced, complained, delayed, failed), verifies the signature and updates the delivery log.
25	Notification	Send In-app Notifications to Staff	Creates in-app notifications for the responsible staff when stock alerts are raised, damage is reported, a production order is held, a customer acts on a quotation, or an order cancellation needs manual handling.

#	Feature	System Function	Description
26	Notification	Purge Inbox Messages	Runs daily at 03:45. Deletes read in-app messages older than 90 days; unread messages are kept.
27	Catalog	Sync Storefront Catalog	When product listing data changes or a product is discontinued, updates the published catalog shown on the storefront; failed updates are retried every 30 seconds and repaired on application start.
28	Reporting	Update Sales Statistics	When an order is placed or cancelled, or a return is resolved, updates the sales summary data used by the dashboard. On first start after deployment, rebuilds statistics from historical orders.
29	System	Purge Audit Logs	Runs daily at 03:30. Deletes audit log entries whose retention period has passed, in small batches.
30	System	Purge Expired Sessions	Runs hourly. Deletes login sessions one day after their token expired.
31	System	Handle Idempotent Requests	Stores the result of requests sent with an Idempotency-Key so a repeated request returns the same result instead of being processed twice; records older than 24 hours are purged hourly.
32	System	Lock Scheduled Jobs	Uses a database lock (ShedLock) so each scheduled job runs on only one server instance at a time.

Table 13. Non-Screen Functions

3.1.5 Entity Relationship Diagram

3.1.5.1 Entity Relationship Diagram Purchase Order

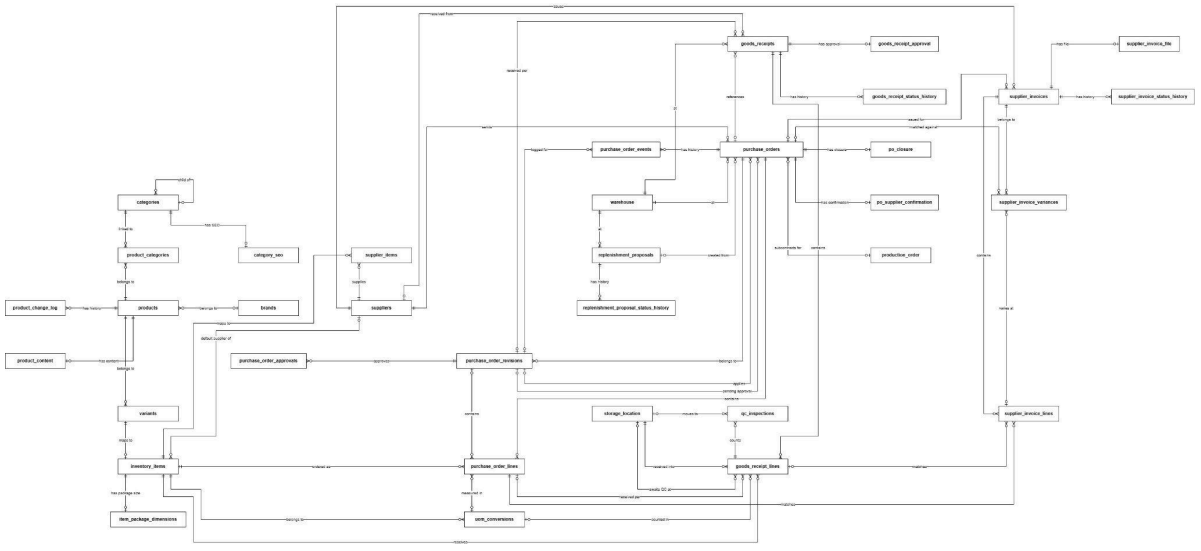


Figure 17. Entity Relationship Diagram Purchase Order

Entities Description

#	Entity	Description
1	categories	Hierarchical classification of products. Each category can have a parent category (self-reference "child of"), forming a multi-level category tree. A product must be placed in at least one category before it can be submitted for approval.
2	product_categories	Junction table linking products to categories, allowing one product to belong to many categories and one category to contain many products. Exactly one link per product is flagged as its primary category.
3	products	Master record of a sellable product (name, description, status). Each product belongs to a brand, is linked to categories and has one or more variants. It is either standard or customizable (printed to the customer's design) and must be approved before it can be published.
4	brands	Brand or manufacturer under which products are marketed. One brand can own many products. A brand is required before a product is submitted for approval, and a deactivated brand cannot be chosen for new products.
5	variants	A specific version of a product (e.g. size, color, packaging), identified by a system-wide unique SKU that is frozen once the variant leaves Draft. Each variant maps to an inventory item, created automatically with it, so its stock can be tracked.
6	inventory_items	Warehouse-side record of a SKU, created automatically when a variant is created. It holds what the warehouse needs to store and restock the item: unit of measure, dimensions, lot / expiry tracking, QC requirement, reorder point, removal strategy (FIFO / FEFO) and default supplier.
7	uom_conversions	Packaging units of an inventory item: one base unit plus larger packs (e.g. box, carton) with their conversion factor and dimensions. They let purchase order lines, goods receipt counts and sales be expressed in packs instead of single units.
8	suppliers	Vendor from whom goods are bought or printing is outsourced (print subcontractor). Stores payment terms, lead time, tolerances, bank details and the channel for sending purchase orders (email or API). Only active suppliers can receive new purchase orders.

#	Entity	Description
9	supplier_items	The supplier's catalogue: which inventory items a supplier sells, with current price, currency, minimum order quantity (MOQ), pack size and lead time. Every purchase order line must come from this catalogue.
10	replenishment_proposals	Proposal to restock a warehouse, generated by an MRP run or created manually. After review it is either rejected or converted into a purchase order, which keeps a reference to the proposal it was created from.
11	purchase_orders	Order sent to a supplier to buy goods into a warehouse, with totals and payment terms. It moves from Draft through approval and supplier confirmation to Received and Closed, or is Cancelled. A PO is either standard or a subcontract order for a production order.
12	purchase_order_revisions	Append-only snapshot of a purchase order, taken each time it is submitted for approval or amended (initial or amendment). The PO points to its currently active revision ("applies") and to the one awaiting approval ("pending approval").
13	purchase_order_lines	One item ordered on a purchase order: inventory item, quantity, unit price, tax and discount rates and line totals. A line can be ordered in packs through a unit conversion and tracks its own receiving status, from Open to Received or Closed.
14	purchase_order_approvals	Approval decision recorded against a purchase order revision: approved, auto-approved when the amount is within the submitter's approval limit, rejected or withdrawn. A manual approver must be a different person from the one who submitted the PO.
15	goods_receipts	Goods receipt note for a supplier delivery arriving at a warehouse, normally against a PO revision; a receipt without a PO names the supplier and a reason. Over-tolerance or PO-less receipts need warehouse manager approval before QC and putaway.
16	goods_receipt_lines	One counted line on a goods receipt: the inventory item, the PO line it fulfils, the receiving location, the quantity received, lot number and expiry date. Lines that require QC are moved to a QC location to await inspection.

#	Entity	Description
17	qc_inspections	Quality-check result for part of a goods receipt line: accepted, quarantined or rejected, with the quantity, the destination location and the reason. The inspected quantities of a line add up to the quantity sent to QC.
18	supplier_invoices	Invoice issued by a supplier for a purchase order, entered with its scanned file. It is matched three ways against the PO and the goods received, ending as Matched or Exception before payment; one entered by mistake is cancelled.
19	supplier_invoice_lines	One billed line on a supplier invoice, tied to the PO line it bills and optionally to the goods receipt line it covers, with invoiced quantity, unit price, tax rate and line total.
20	supplier_invoice_variances	Discrepancy found by the three-way match, one row per issue: quantity billed above what was received, goods still awaiting QC, a price or tax rate that differs from the PO, or a subcontract bill above its settlement. These rows feed the review queue for invoice exceptions.
21	warehouse	Physical warehouse where goods are received and stored. Purchase orders, goods receipts and replenishment proposals are all placed at a warehouse. Shown here as a reference only.
22	storage_location	Any location inside a warehouse (a bin or an area), identified by a unique location code. Goods receipt lines record where goods were received and where they await QC; QC inspections record where checked goods are moved.
23	production_order	Order to produce printed cups or packaging, owned by the production module. A subcontract purchase order references the production order whose printing it outsources. Shown here as a reference only.

Table 14. Entities Descriptions Purchase Order

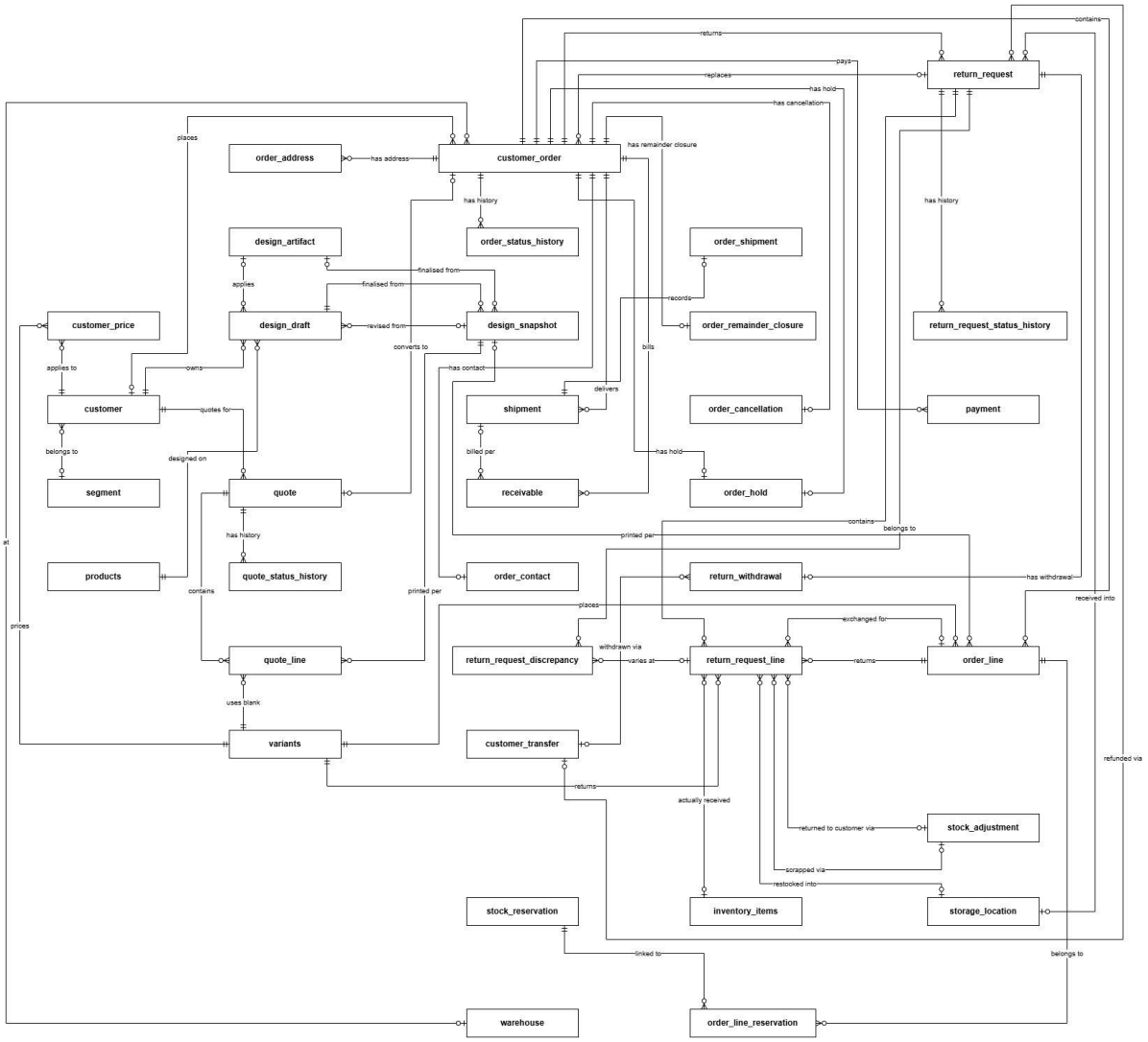


Figure 18. Entity Relationship Diagram Order

Entities Description

#	Entity	Description
1	customer	Wholesale (B2B) customer who buys printed cups and packaging: name, email, phone, tax code and status (pending activation, active, inactive, blocked). A customer belongs to a segment and may be linked to a login account for the customer portal.
2	segment	Customer group used for group pricing and reporting, identified by a fixed code. A customer without its own price list pays the price of its segment.

#	Entity	Description
3	customer_price	A customer's own negotiated price for one SKU, set by quantity tiers; the first tier is that customer's minimum order quantity. Quotes start from this price list, and a customer without one falls back to the segment or base price.
4	design_draft	Editable design a customer prepares on a customizable product, with its specification and files. It goes from Draft to Submitted, then Approved or Rejected after preflight checks and staff review, and can be revised from an earlier confirmed snapshot.
5	design_artifact	Append-only file revision of a design draft (artwork, print file). The draft points to its current artifact, and a confirmed snapshot references the exact revision that was approved.
6	design_snapshot	Immutable, confirmed version of a design, finalised from a draft and its files and identified by a checksum. Quote lines and order lines reference the snapshot to print, so what is printed is exactly what the customer approved.
7	quote	B2B quotation Sales sends to a customer: printed lines with price breaks, a payment term, a deposit and a validity date. It moves Draft → Sent → Accepted → Converted (or Changes requested, Expired, Cancelled), and an accepted quote becomes exactly one sales order.
8	quote_line	One quoted line: the blank variant to print on, the confirmed design snapshot, the print specification and the quantity, with price tiers whose lowest tier is the MOQ. The quantity and unit price the customer accepted are fixed on the line.
9	quote_status_history	Append-only log of a quote's status changes per revision, with the reason, who changed it and when. The offer that was sent is kept with each entry, so what the customer accepted survives later revisions.
10	customer_order	Wholesale sales order of a customer, placed directly or converted from a quote and fulfilled from a warehouse. It holds totals, payment term (prepaid, deposit or credit), amounts paid and billed, delivery mode and status, from Draft through payment, production and delivery to Completed. A replacement order points to the return request it replaces.

#	Entity	Description
11	order_line	One SKU ordered on a sales order, with its quantity and the agreed unit price, copied at ordering so later price changes do not affect it. A printed line references the design snapshot and its checksum; cancelled and short-closed quantities are tracked on the line.
12	order_line_reservation	Junction table linking an order line to the stock reservations that hold its goods. A line drawn from several lots holds several reservations, all of which are released if the order is cancelled.
13	order_contact	Contact person of an order (name, email, phone), copied when the order is placed so that later profile changes do not alter it. One-to-one with the order.
14	order_address	Address snapshot of an order (recipient, phone, street, ward, province, country), one row per address kind: shipping or billing. Copied at checkout so later edits to the customer's addresses cannot change the order.
15	order_hold	Hold placed on an order: the reason (credit, design mismatch, payment, address, inventory, customer, fraud, other), the status to return to and a note. Credit and design-mismatch holds are raised by the system; the others by the Order Coordinator.
16	order_cancellation	Cancellation details of an order: reason code and text, plus any fee agreed with the customer when printing had already started and the amount kept from what was paid. The rest is refunded; an unpaid fee becomes a receivable.
17	order_remainder_closure	Closure of the unfulfilled remainder of a partially delivered order: the amount refunded for the units that will no longer ship, and any fee charged, retained or paid for closing it.
18	order_status_history	Append-only log of every status change of an order: from and to status, reason, actor and time.
19	order_shipment	The order's view of one shipment: the value of the goods it carried, its status and delivery outcome (delivered, or failed with a reason), the amount billed and the COD amount the carrier collects. An order may be delivered in several shipments.

#	Entity	Description
20	payment	Money paid by a customer against an order: amount, method (card, bank transfer, COD, e-wallet, deposit), status and provider reference.
21	receivable	Amount owed on an order: a delivered credit order's bill, a cancellation fee, or a deposit balance collected by the carrier. It has a due date, tracks how much has been paid and who pays it (customer or carrier), and may be tied to the shipment it bills.
22	customer_transfer	Incoming money recorded by the accountant from the bank statement: a customer's transfer, a carrier's COD remittance, or the credit a resolved return gives the customer. It is allocated to orders and receivables; what is not allocated stays as the customer's credit.
23	return_request	Return request (RMA) a customer opens on a delivered order within the return window, with reason, requested resolution and status (Requested → Awaiting return → Received → Inspected → Resolved, or Rejected / Cancelled / Expired). It records where returned goods are received and how the claim was settled.
24	return_request_line	One order line being returned: quantity requested, approved, received, accepted or rejected by QC, and what happened to the units (restocked, scrapped or sent back to the customer). Each accepted unit is either replaced through a new order line or credited.
25	return_request_discrepancy	Difference between what a returned parcel held and what was approved: short, excess, wrong item or unexpected item. Recorded at receipt for Sales to follow up; the credited quantity is decided at inspection.
26	return_withdrawal	Withdrawal of the free replacement order of a resolved return in exchange for credit: proposed, then approved or rejected by another person. The replaced value is set off against receivables or kept as the customer's credit.
27	return_request_status_history	Append-only log of every status change of a return request: from and to status, reason, actor and time.
28	products	Product master record from the product module. A design draft is created on a customizable product. Shown here as a reference only.

#	Entity	Description
29	variants	Sellable SKU of a product. Order lines, quote lines (as the blank to print on), customer prices and return lines all refer to a variant. Shown here as a reference only.
30	inventory_items	Warehouse-side record of a SKU. A return line records the SKU actually received when it differs from the one ordered. Shown here as a reference only.
31	stock_reservation	Stock set aside in the inventory module for an order line, so goods are held before the customer pays and the order is picked. Shown here as a reference only.
32	stock_adjustment	Inventory adjustment document. Returned units that are scrapped or sent back to the customer leave stock through an adjustment. Shown here as a reference only.
33	warehouse	Warehouse that fulfils an order. Shown here as a reference only.
34	storage_location	Location in a warehouse. A return request records where returned parcels are received, and a return line where accepted units are restocked. Shown here as a reference only.
35	shipment	Shipment created by the fulfillment module to deliver an order, identified by a shipment code. Shown here as a reference only.

Table 15. Entities Descriptions Order

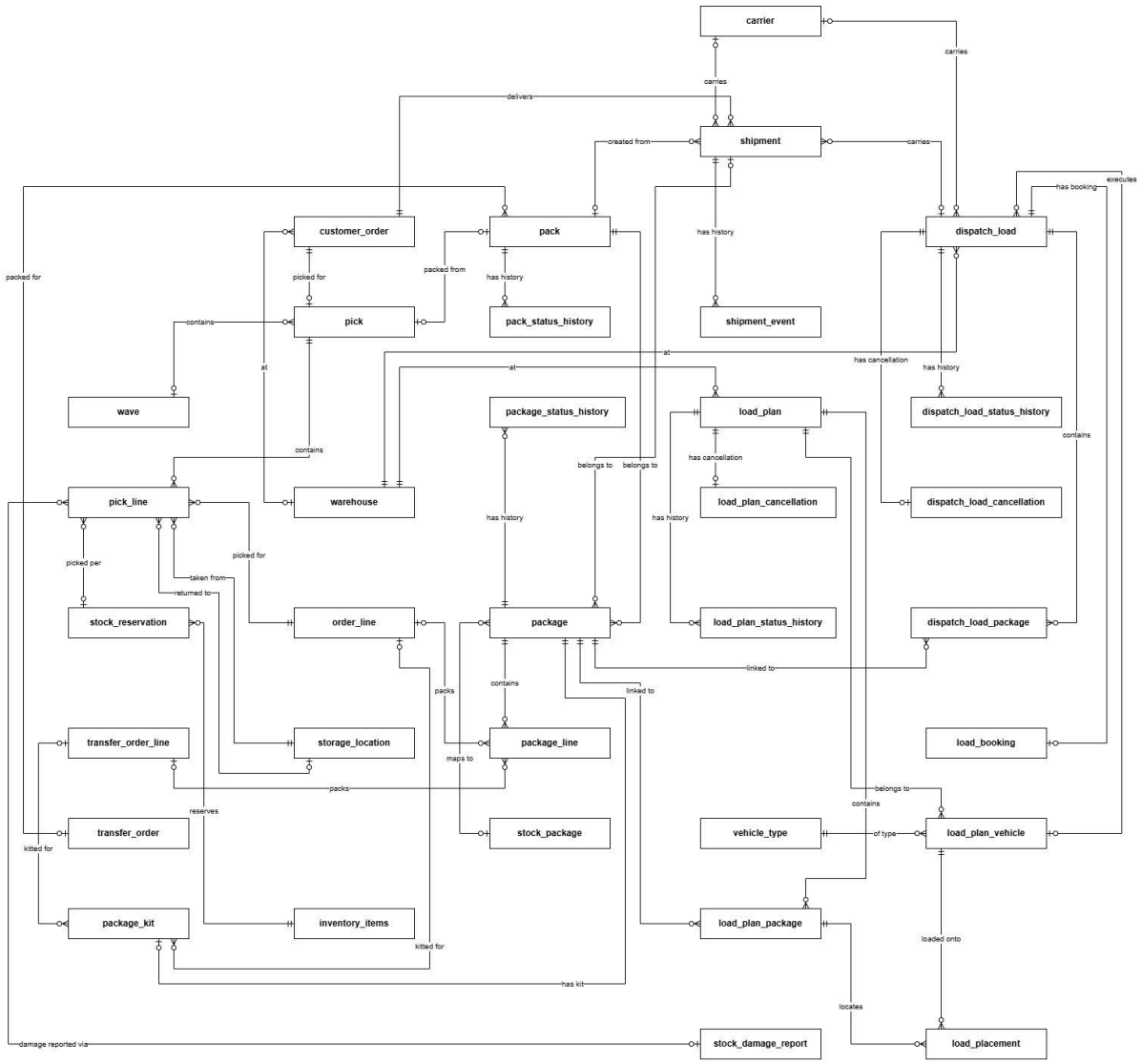


Figure 19. Entity Relationship Diagram Purchase Fulfillment

Entities Description

#	Entity	Description
1	warehouse	Warehouse where customer orders are fulfilled and where load plans and dispatch loads are performed.
2	customer_order	Sales order to be fulfilled from a warehouse; it is picked, packed and delivered by shipments.
3	order_line	Line of a customer order (item, quantity) that is picked, packed and kitted.
4	transfer_order	Inter-warehouse transfer order whose goods are packed for shipment.
5	transfer_order_line	Line of a transfer order that is packed into package lines and kitted.
6	inventory_items	Stock-keeping item that is reserved for picking.

#	Entity	Description
7	stock_reservation	Stock allocated to an order line at a storage location; pick lines are picked per reservation.
8	storage_location	Location from which goods are taken when picking and to which they are returned if not used.
9	wave	Group of picks released together to optimize picking work in the warehouse.
10	pick	Picking task for a customer order, containing pick lines; may belong to a wave.
11	pick_line	Line of a pick: item and quantity taken from a storage location per stock reservation for an order line; leftover goods can be returned and damages reported.
12	stock_damage_report	Report of goods found damaged during picking.
13	pack	Packing session that packs picked goods for a customer order or transfer order and produces packages.
14	pack_status_history	Log of status changes of a pack.
15	package	Physical package (carton) produced from a pack and belonging to a shipment; it contains package lines and is linked to load plans and dispatch loads.
16	package_status_history	Log of status changes of a package.
17	package_line	Content line of a package: quantity of an order line or transfer order line packed.
18	package_kit	Kit (set of items assembled together) inside a package, kitted for an order line or transfer order line.
19	stock_package	Inventory handling unit that a package maps to.
20	carrier	Shipping carrier / logistics provider that carries shipments and dispatch loads.
21	shipment	Delivery of packages to a customer for a customer order, created from a pack and carried by a carrier.
22	shipment_event	Tracking event of a shipment (picked up, in transit, delivered, failed...).
23	vehicle_type	Type of vehicle with its capacity (weight, volume, dimensions).
24	load_plan	Plan for loading packages onto vehicles at a warehouse before dispatch.
25	load_plan_status_history	Log of status changes of a load plan.
26	load_plan_cancellation	Cancellation record of a load plan (reason, time, user).
27	load_plan_vehicle	Vehicle of a given type assigned to a load plan.
28	load_plan_package	Package included in a load plan.

#	Entity	Description
29	load_placement	Position where a load plan package is placed inside a load plan vehicle.
30	dispatch_load	Actual dispatch of a loaded vehicle from a warehouse, executing a load plan vehicle and carrying shipments with a carrier.
31	dispatch_load_status_history	Log of status changes of a dispatch load.
32	dispatch_load_cancellation	Cancellation record of a dispatch load.
33	dispatch_load_package	Package actually loaded in a dispatch load.
34	load_booking	Booking of a carrier / vehicle slot for a dispatch load.

Table 16. Entities Descriptions Fulfillment

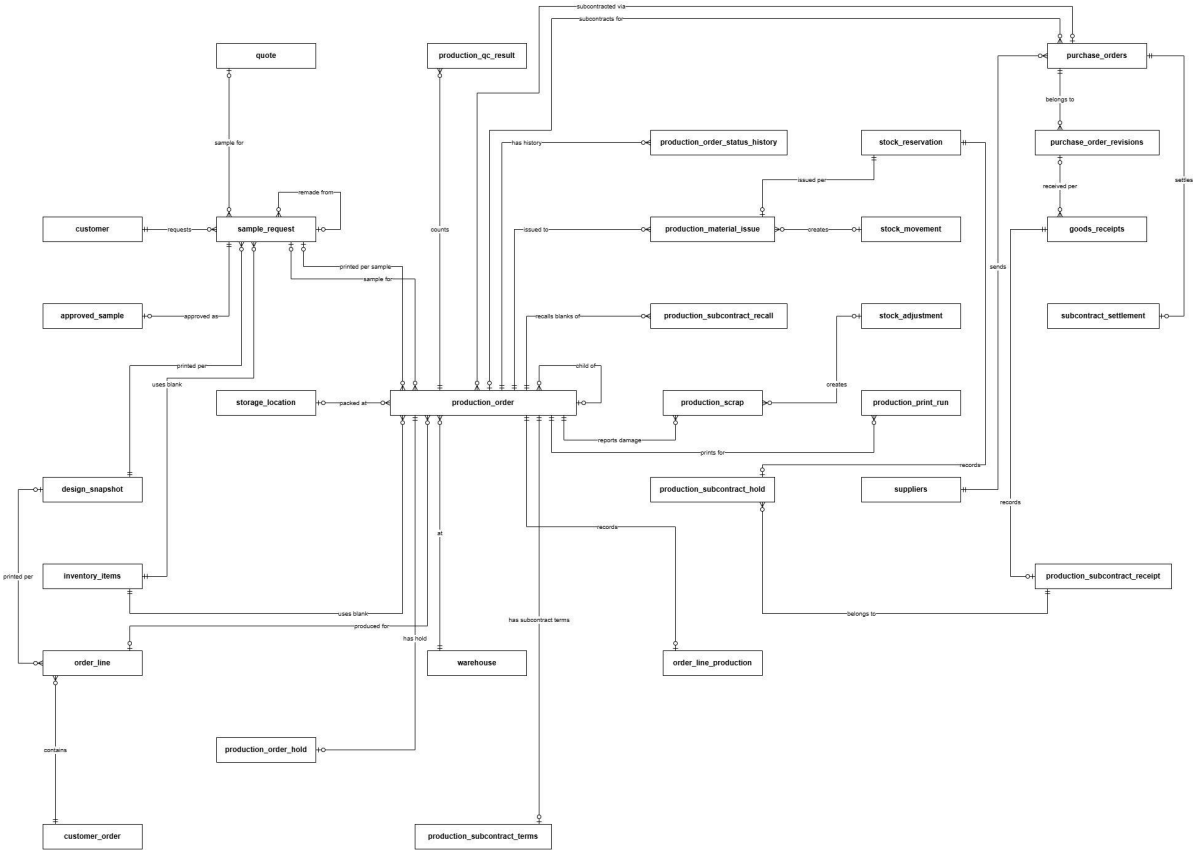


Figure 20. Entity Relationship Diagram Production

Entities Description

#	Entity	Description
1	customer	Wholesale (B2B) customer who buys printed cups and packaging: name, email, phone, tax code and status (pending activation, active, inactive, blocked). A customer belongs to a segment and may be linked to a login account for the customer portal.
2	segment	Customer group used for group pricing and reporting, identified by a fixed code. A customer without its own price list pays the price of its segment.
3	customer_price	A customer's own negotiated price for one SKU, set by quantity tiers; the first tier is that customer's minimum order quantity. Quotes start from this price list, and a customer without one falls back to the segment or base price.
4	design_draft	Editable design a customer prepares on a customizable product, with its specification and files. It goes from Draft to Submitted, then Approved or Rejected after preflight checks and staff review, and can be revised from an earlier confirmed snapshot.
5	design_artifact	Append-only file revision of a design draft (artwork, print file). The draft points to its current artifact, and a confirmed snapshot references the exact revision that was approved.
6	design_snapshot	Immutable, confirmed version of a design, finalised from a draft and its files and identified by a checksum. Quote lines and order lines reference the snapshot to print, so what is printed is exactly what the customer approved.
7	quote	B2B quotation Sales sends to a customer: printed lines with price breaks, a payment term, a deposit and a validity date. It moves Draft → Sent → Accepted → Converted (or Changes requested, Expired, Cancelled), and an accepted quote becomes exactly one sales order.
8	quote_line	One quoted line: the blank variant to print on, the confirmed design snapshot, the print specification and the quantity, with price tiers whose lowest tier is the MOQ. The quantity and unit price the customer accepted are fixed on the line.

#	Entity	Description
9	quote_status_history	Append-only log of a quote's status changes per revision, with the reason, who changed it and when. The offer that was sent is kept with each entry, so what the customer accepted survives later revisions.
10	customer_order	Wholesale sales order of a customer, placed directly or converted from a quote and fulfilled from a warehouse. It holds totals, payment term (prepaid, deposit or credit), amounts paid and billed, delivery mode and status, from Draft through payment, production and delivery to Completed. A replacement order points to the return request it replaces.
11	order_line	One SKU ordered on a sales order, with its quantity and the agreed unit price, copied at ordering so later price changes do not affect it. A printed line references the design snapshot and its checksum; cancelled and short-closed quantities are tracked on the line.
12	order_line_reservation	Junction table linking an order line to the stock reservations that hold its goods. A line drawn from several lots holds several reservations, all of which are released if the order is cancelled.
13	order_contact	Contact person of an order (name, email, phone), copied when the order is placed so that later profile changes do not alter it. One-to-one with the order.
14	order_address	Address snapshot of an order (recipient, phone, street, ward, province, country), one row per address kind: shipping or billing. Copied at checkout so later edits to the customer's addresses cannot change the order.
15	order_hold	Hold placed on an order: the reason (credit, design mismatch, payment, address, inventory, customer, fraud, other), the status to return to and a note. Credit and design-mismatch holds are raised by the system; the others by the Order Coordinator.
16	order_cancellation	Cancellation details of an order: reason code and text, plus any fee agreed with the customer when printing had already started and the amount kept from what was paid. The rest is refunded; an unpaid fee becomes a receivable.

#	Entity	Description
17	order_remainder_closure	Closure of the unfulfilled remainder of a partially delivered order: the amount refunded for the units that will no longer ship, and any fee charged, retained or paid for closing it.
18	order_status_history	Append-only log of every status change of an order: from and to status, reason, actor and time.
19	order_shipment	The order's view of one shipment: the value of the goods it carried, its status and delivery outcome (delivered, or failed with a reason), the amount billed and the COD amount the carrier collects. An order may be delivered in several shipments.
20	payment	Money paid by a customer against an order: amount, method (card, bank transfer, COD, e-wallet, deposit), status and provider reference.
21	receivable	Amount owed on an order: a delivered credit order's bill, a cancellation fee, or a deposit balance collected by the carrier. It has a due date, tracks how much has been paid and who pays it (customer or carrier), and may be tied to the shipment it bills.
22	customer_transfer	Incoming money recorded by the accountant from the bank statement: a customer's transfer, a carrier's COD remittance, or the credit a resolved return gives the customer. It is allocated to orders and receivables; what is not allocated stays as the customer's credit.
23	return_request	Return request (RMA) a customer opens on a delivered order within the return window, with reason, requested resolution and status (Requested → Awaiting return → Received → Inspected → Resolved, or Rejected / Cancelled / Expired). It records where returned goods are received and how the claim was settled.
24	return_request_line	One order line being returned: quantity requested, approved, received, accepted or rejected by QC, and what happened to the units (restocked, scrapped or sent back to the customer). Each accepted unit is either replaced through a new order line or credited.
25	return_request_discrepancy	Difference between what a returned parcel held and what was approved: short, excess, wrong item or unexpected item. Recorded at receipt for Sales to follow up; the credited quantity is decided at inspection.

#	Entity	Description
26	return_withdrawal	Withdrawal of the free replacement order of a resolved return in exchange for credit: proposed, then approved or rejected by another person. The replaced value is set off against receivables or kept as the customer's credit.
27	return_request_status_history	Append-only log of every status change of a return request: from and to status, reason, actor and time.
28	products	Product master record from the product module. A design draft is created on a customizable product. Shown here as a reference only.
29	variants	Sellable SKU of a product. Order lines, quote lines (as the blank to print on), customer prices and return lines all refer to a variant. Shown here as a reference only.
30	inventory_items	Warehouse-side record of a SKU. A return line records the SKU actually received when it differs from the one ordered. Shown here as a reference only.
31	stock_reservation	Stock set aside in the inventory module for an order line, so goods are held before the customer pays and the order is picked. Shown here as a reference only.
32	stock_adjustment	Inventory adjustment document. Returned units that are scrapped or sent back to the customer leave stock through an adjustment. Shown here as a reference only.
33	warehouse	Warehouse that fulfils an order. Shown here as a reference only.
34	storage_location	Location in a warehouse. A return request records where returned parcels are received, and a return line where accepted units are restocked. Shown here as a reference only.
35	shipment	Shipment created by the fulfillment module to deliver an order, identified by a shipment code. Shown here as a reference only.

Table 17. Entities Descriptions Production

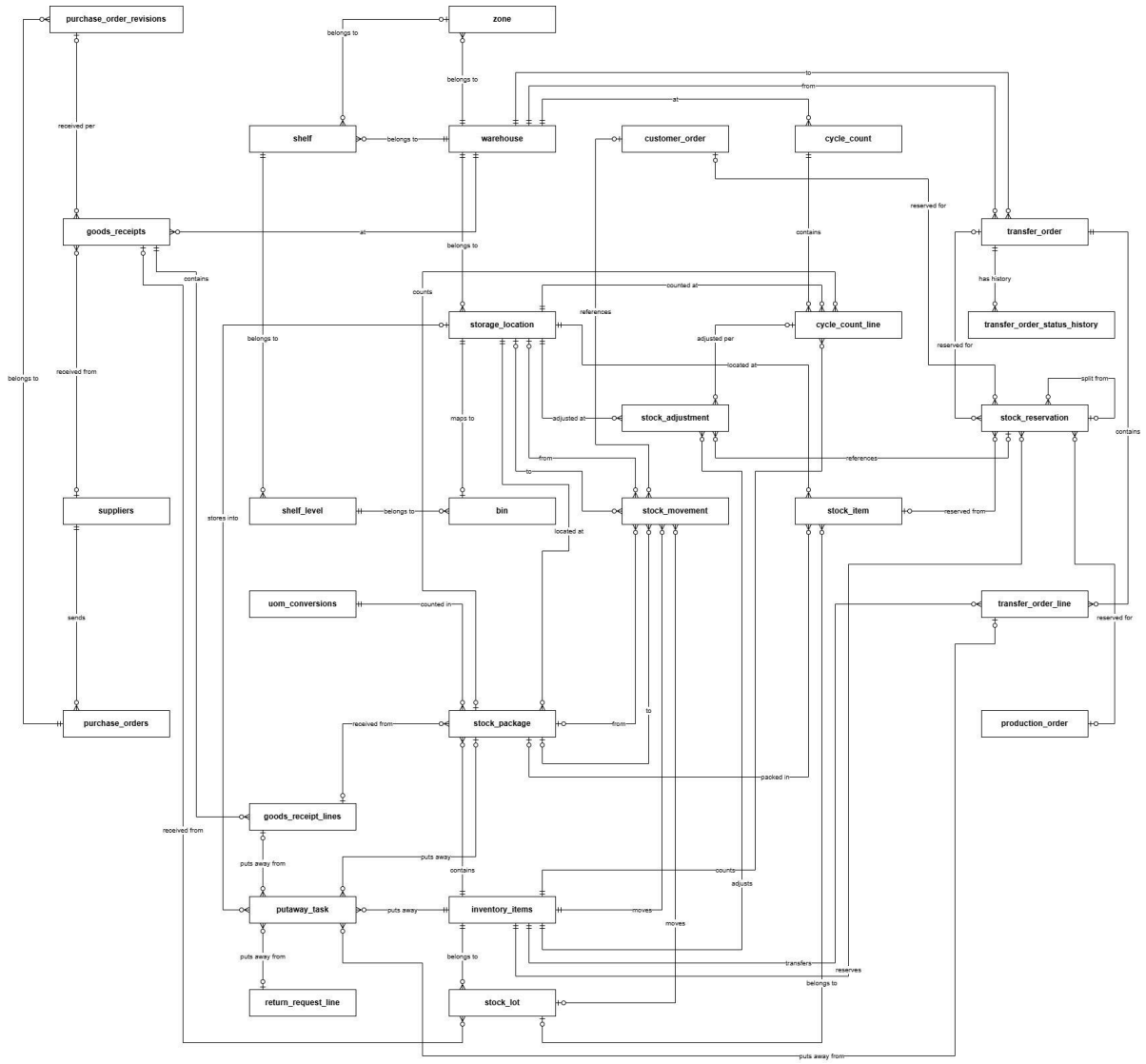


Figure 21. Entity Relationship Diagram Warehouse

Entities Description

#	Entity	Description
1	warehouse	Physical warehouse that owns zones, shelves and storage locations. Cycle counts, goods receipts and transfer orders (from / to) are performed at a warehouse.
2	zone	Functional area inside a warehouse (e.g. receiving, storage, picking, QC). A zone contains many shelves.
3	shelf	Shelving unit (rack) that belongs to a zone of a warehouse. A shelf is divided into shelf levels.
4	shelf_level	A horizontal level (tier) of a shelf. Each shelf level contains one or more bins.

#	Entity	Description
5	bin	Smallest physical storage slot on a shelf level. Each bin maps to a storage location used by inventory transactions.
6	storage_location	Logical, addressable location in a warehouse where stock is held. Referenced by stock packages, stock items, stock movements (from / to), putaway tasks, cycle count lines and stock adjustments.
7	suppliers	Vendors that send purchase orders and from which goods receipts are received.
8	purchase_orders	Order issued to a supplier for goods. Each purchase order has one or more revisions.
9	purchase_order_revisions	Approved version of a purchase order against which goods are actually received.
10	goods_receipts	Header document recording goods received from a supplier at a warehouse per purchase order revision. Contains goods receipt lines and creates the stock lots received.
11	goods_receipt_lines	Line of a goods receipt (item, received quantity). Received goods are recorded as stock packages and become the source of putaway tasks.
12	return_request_line	Line of a customer return request. Returned goods are put back into storage through putaway tasks.
13	putaway_task	Warehouse task to move received, returned or transferred goods (from goods receipt lines, return request lines or transfer order lines) into a target storage location, specifying the stock package and inventory item put away.
14	inventory_items	Stock-keeping item managed in inventory. Has stock lots, is contained in stock packages, and is moved, counted, adjusted, reserved and transferred.
15	uom_conversions	Unit-of-measure conversion rules (e.g. carton, box, piece) in which stock packages are counted.
16	stock_package	Physical handling unit (carton, pallet, box) containing an inventory item, located at a storage location. It is the unit that is put away, moved (from / to) and counted.
17	stock_lot	Batch / lot of an inventory item received from a goods receipt, used for traceability (lot number, production date, expiry date).
18	stock_item	Individual stock unit belonging to a stock lot, packed in a stock package and located at a storage location. It can be moved, counted and reserved.
19	stock_movement	Record of every stock transfer between storage locations or stock packages (from / to), including the

#	Entity	Description
		inventory item, lot or item moved and the reference document (e.g. customer order).
20	cycle_count	Periodic stock-counting session performed at a warehouse. Contains many cycle count lines.
21	cycle_count_line	Line of a cycle count recording the counted quantity of an inventory item / stock package at a storage location compared with the system quantity.
22	stock_adjustment	Correction of on-hand quantity of an inventory item at a storage location, created per cycle count line (count difference) or for other reasons, and may reference a stock reservation.
23	customer_order	Sales order placed by a customer, for which stock is reserved and referenced by stock movements.
24	production_order	Production order for which raw materials / blanks are reserved.
25	transfer_order	Order to transfer stock from one warehouse to another (from / to). Contains transfer order lines and can have stock reserved for it.
26	transfer_order_line	Line of a transfer order specifying the inventory item and quantity to transfer; received goods are put away through putaway tasks.
27	stock_reservation	Allocation of available stock (inventory item / stock item) to a customer order, transfer order or production order so it cannot be used elsewhere. A reservation can be split from another reservation.

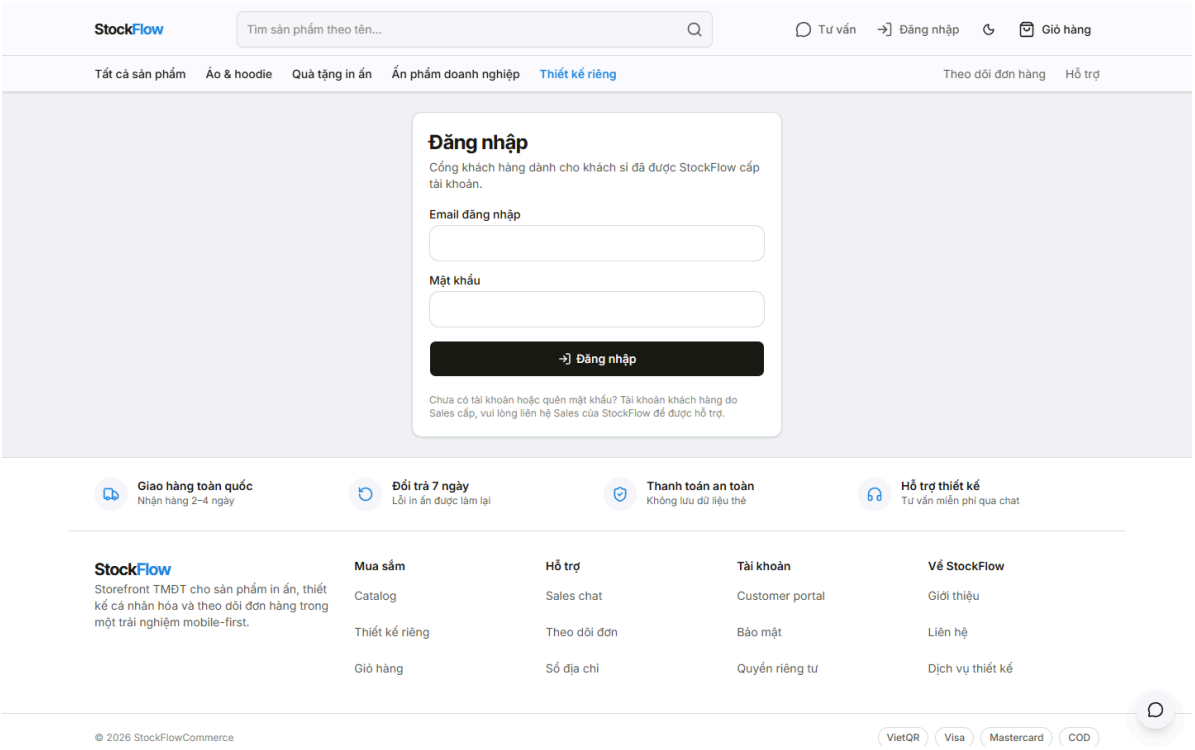
Table 18. Entities Descriptions Warehouse

3.2 Storefront (Customer portal)

3.2.1 Authentication

3.2.1.1 Sign In

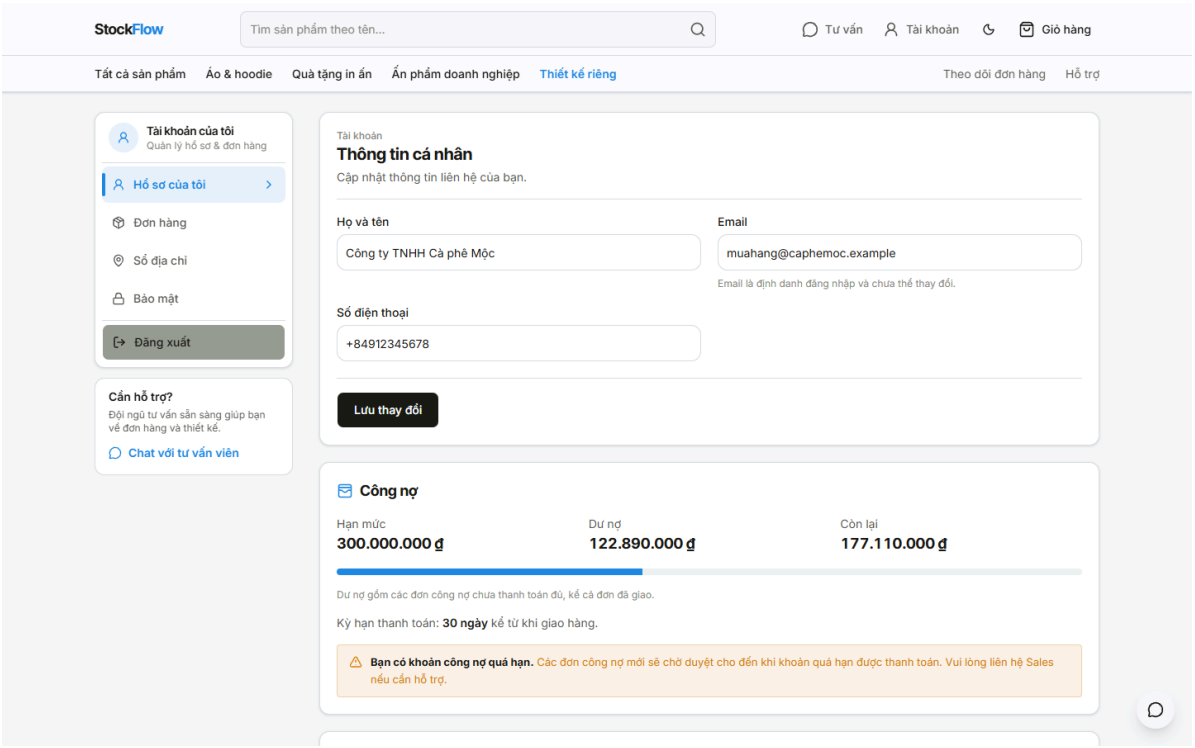
- **Function trigger:** Users click “Đăng nhập” in the header, or open the login page from any screen that asks them to sign in.
- **Function description:**
  - **Actors:** Guest, Customer
  - **Purpose:** Sign in to the customer storefront.



- **Function details:**
  - **Data requirement:** Email (login), password
  - **Validation:** Email required (trimmed); password required (not trimmed); BE: both @NotBlank; BE allows 5 sign-in attempts per minute per IP and temporarily locks the account after repeated wrong passwords.
  - **Business rules:** BR-01, BR-01a, BR-06 (18 Customer Information)

3.2.1.2 Logout

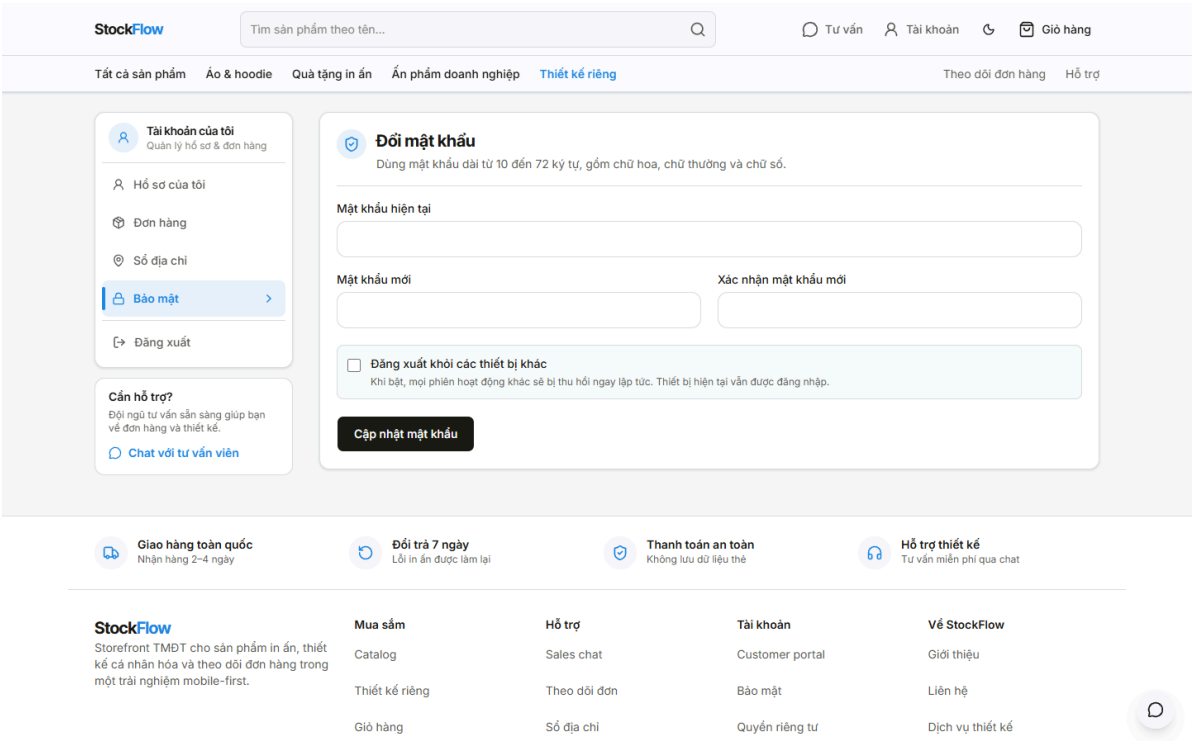
- **Function trigger:** Users click “Đăng xuất” in the account sidebar (“Tài khoản của tôi”) or the account tab strip on mobile.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Sign out of the storefront.



- **Function details:**
  - **Data requirement:** None
  - **Validation:** None
  - **Business rules:** None

3.2.1.3 Change Password

- **Function trigger:** Users open “Bảo mật” in the account menu, fill the form and click “Cập nhật mật khẩu”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Change the account password.

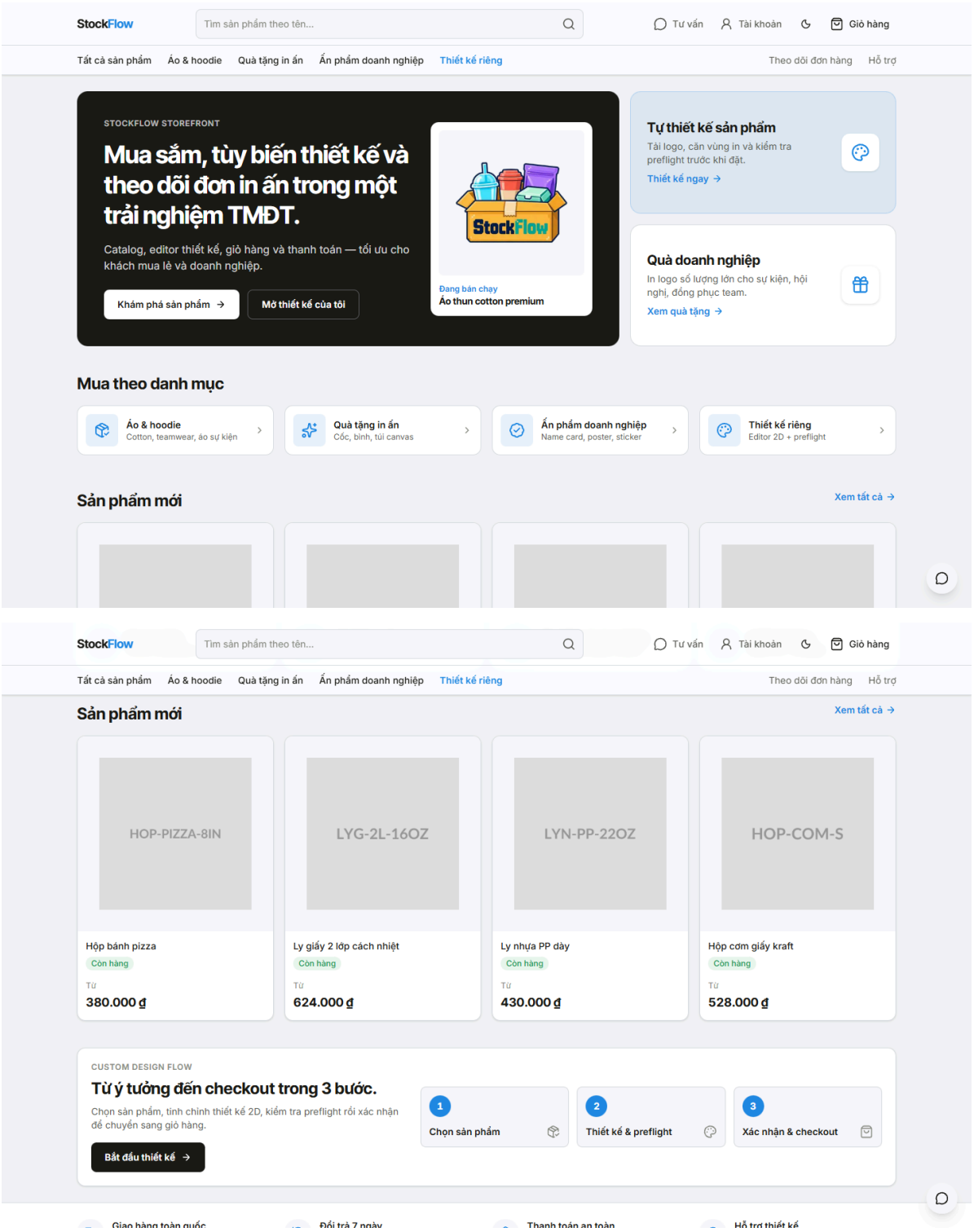


- **Function details:**
  - **Data requirement:** Current password, new password, confirm new password, “Đăng xuất khỏi các thiết bị khác” option
  - **Validation:** Current password required; new password 10–72 characters with an upper-case letter, a lower-case letter and a digit; confirmation must match; BE rejects a wrong current password or an unchanged password.
  - **Business rules:** None

3.2.2 Home & Catalog

3.2.2.1 View Home Page

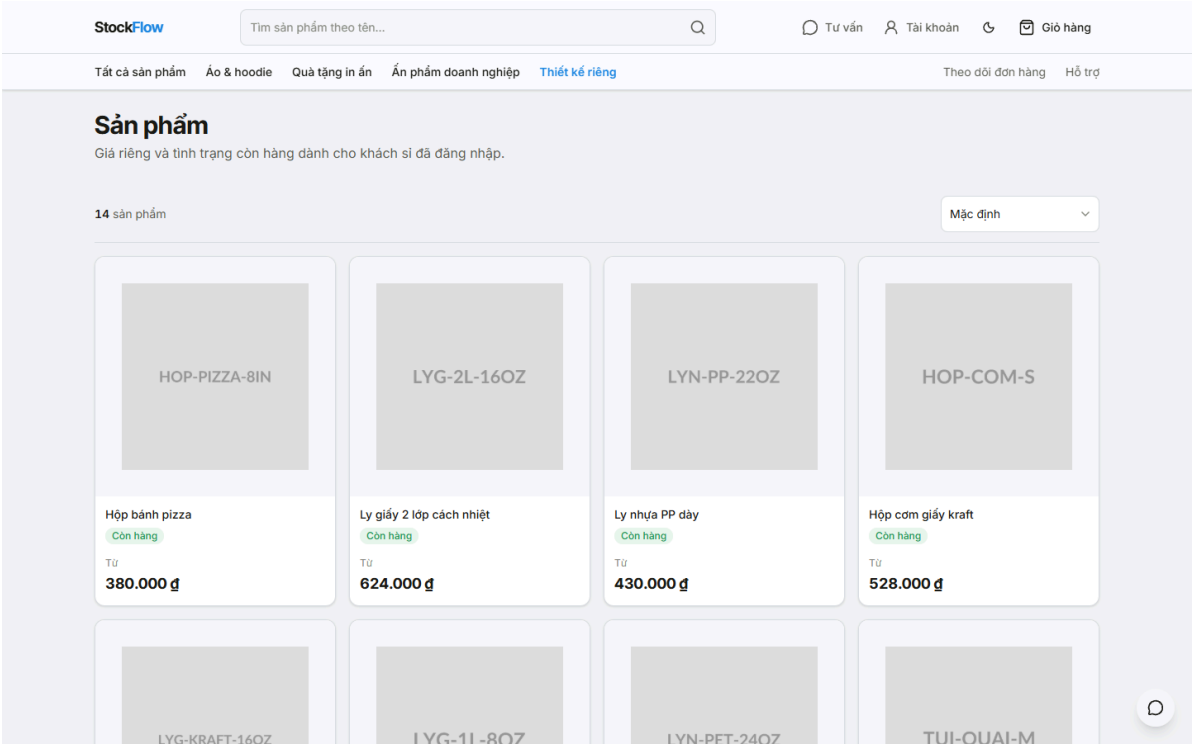
- **Function trigger:** Users open the storefront address or click the “StockFlow” logo in the header.
- **Function description:**
  - **Actors:** Guest, Customer
  - **Purpose:** Introduce the store and link to products and design.



- **Function details:**
  - **Data requirement:** Hero banners, “Mua theo danh mục” tiles, “Sản phẩm mới” (4 newest products), three design steps
  - **Validation:** None
  - **Business rules:** BR-01 (13 Catalog)

3.2.2.2 View Product Catalog

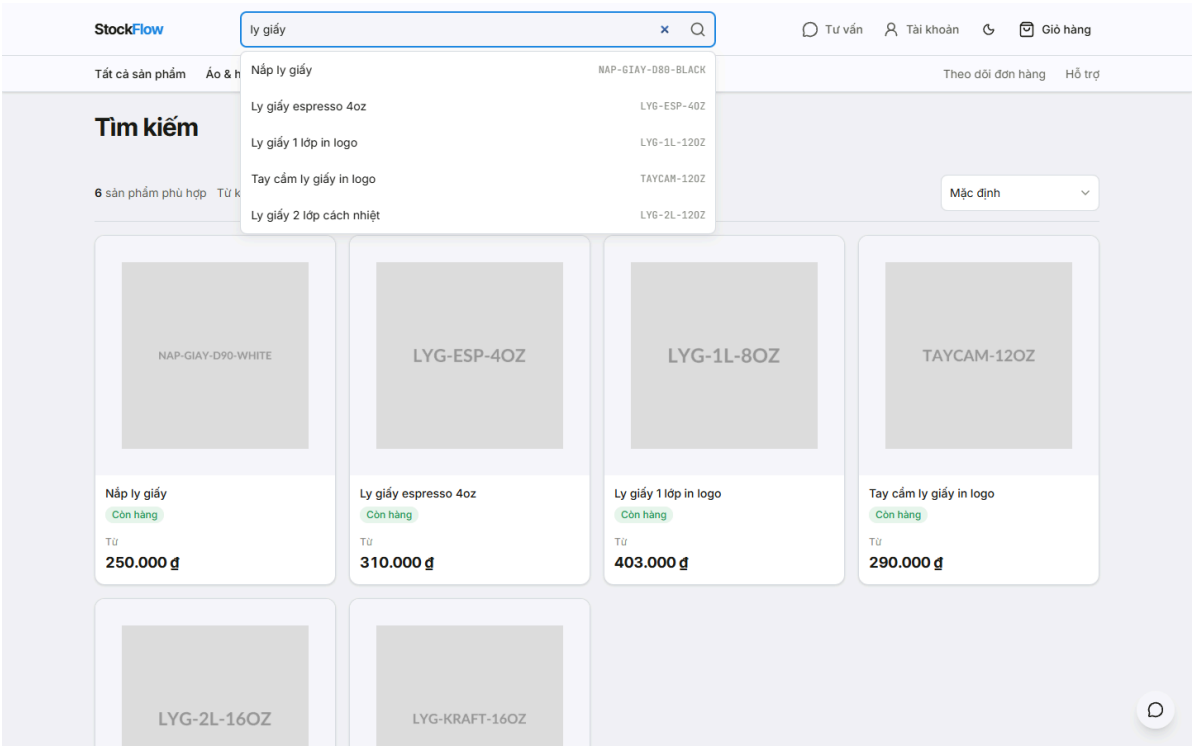
- **Function trigger:** Users click “Tất cả sản phẩm” in the header category bar or “Khám phá sản phẩm” on the home page.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Browse published products with the customer's own prices and stock.



- **Function details:**
  - **Data requirement:** Product cards (image, name, lowest “Từ” price, stock badge), sort, page
  - **Validation:** Sort limited to “Mặc định”, “Mới nhất”, “Tên A–Z”; page ≥ 1; 12 products per page; results capped at 500.
  - **Business rules:** BR-01, BR-02, BR-03, BR-06 (13 Catalog)

3.2.2.3 Search Products

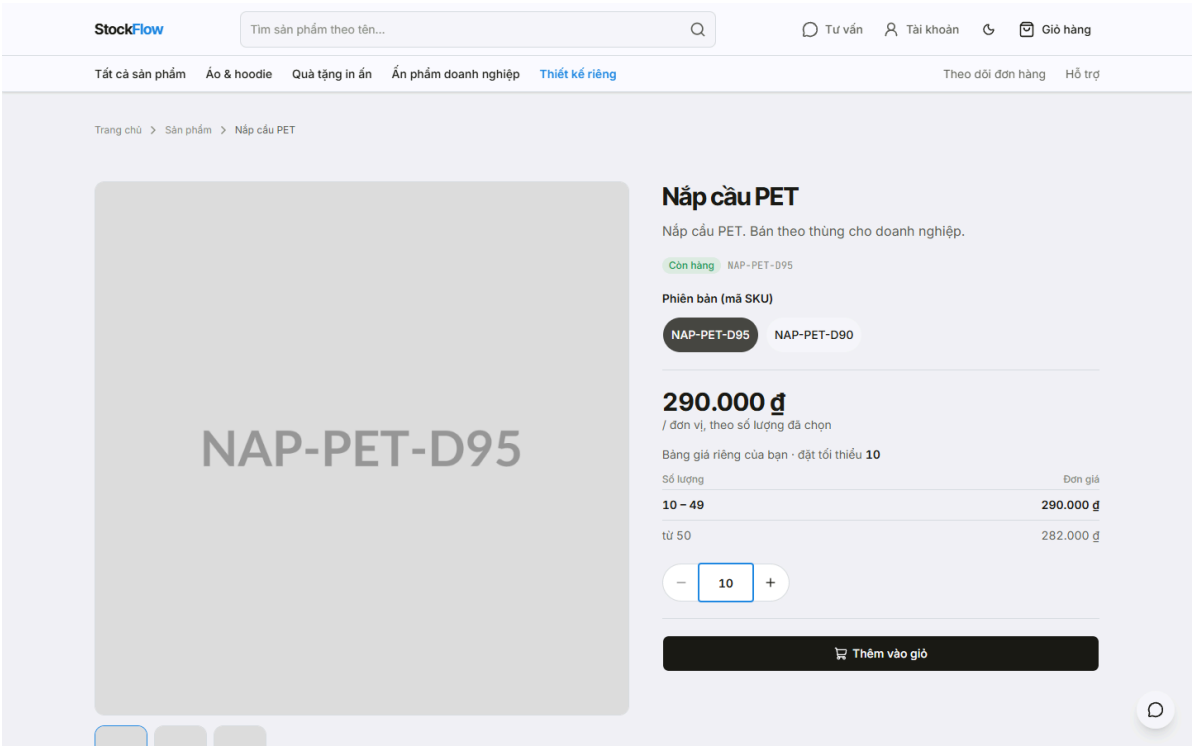
- **Function trigger:** Users type in the “Tìm sản phẩm theo tên…” box in the header, then press Enter or pick a suggestion.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Find products by name or SKU.



- **Function details:**
  - **Data requirement:** Keyword, suggestions (name, SKU), result cards, sort including “Liên quan nhất”
  - **Validation:** Keyword ≤ 100 characters, accent-insensitive; suggestions need ≥ 2 letters or digits, up to 5 shown, fetched 250 ms after typing stops; empty keyword shows a prompt instead of results.
  - **Business rules:** BR-01, BR-02, BR-06 (13 Catalog)

3.2.2.4 View Product Detail

- **Function trigger:** Users click a product card in the catalog, search results or “Sản phẩm mới”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** View a product's images, variants, price tiers and stock.

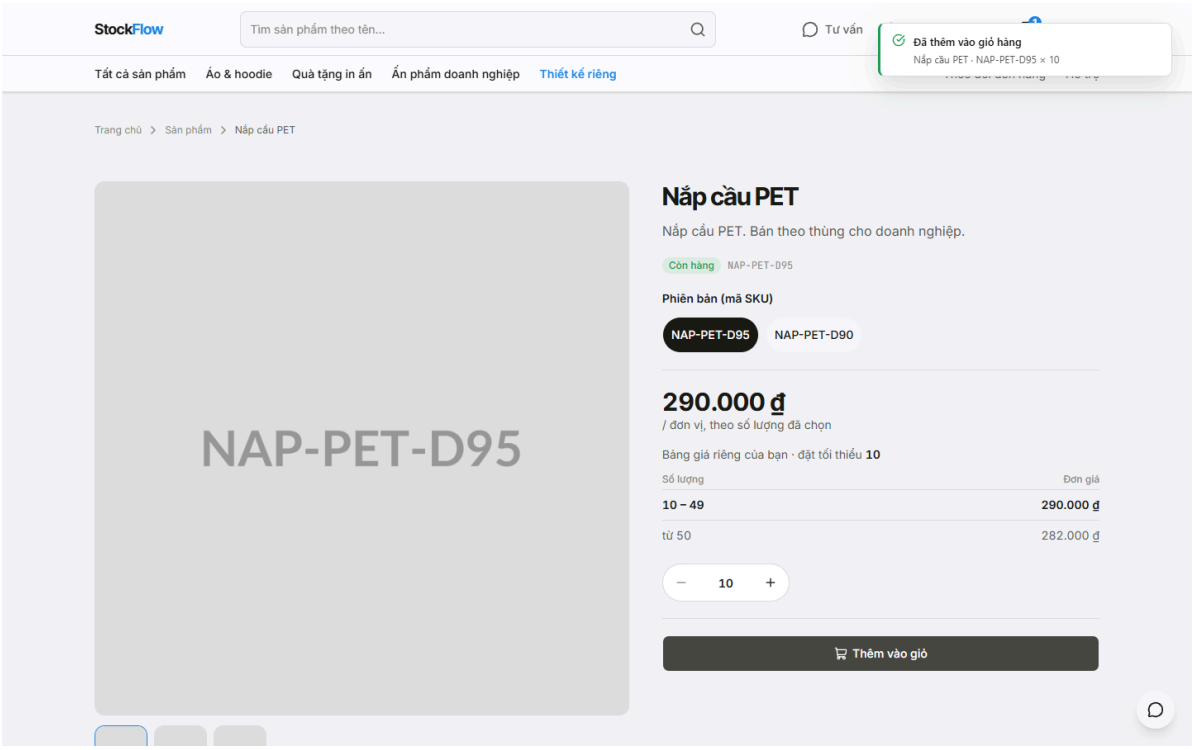


- **Function details:**
  - **Data requirement:** Images, name, description, stock badge, “Phiên bản (mã SKU)”, unit price, price tiers, minimum order quantity
  - **Validation:** None (view); out-of-stock SKUs are disabled in the variant selector.
  - **Business rules:** BR-01, BR-02, BR-03, BR-04, BR-06, BR-10 (13 Catalog)

3.2.3 Cart & Checkout

3.2.3.1 Add Product to Cart

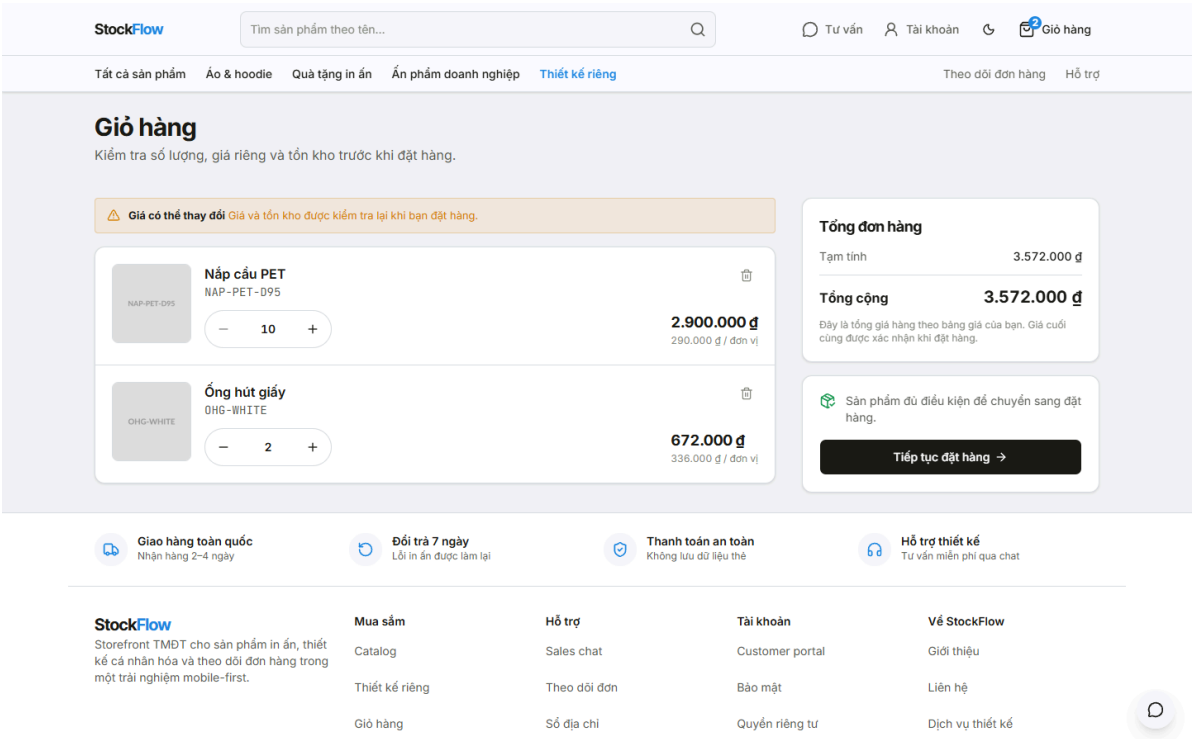
- **Function trigger:** Users choose a SKU and quantity on the product detail page and click “Thêm vào giỏ”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Add a stock product to the cart.



- **Function details:**
  - **Data requirement:** SKU, quantity
  - **Validation:** Quantity is a whole number from the MOQ (first price tier) to 1,000,000; button disabled when the SKU is out of stock, has no price or is below MOQ.
  - **Business rules:** BR-04, BR-07, BR-10 (13 Catalog); BR-11, BR-12 (14 Cart & Order)

3.2.3.2 View Cart

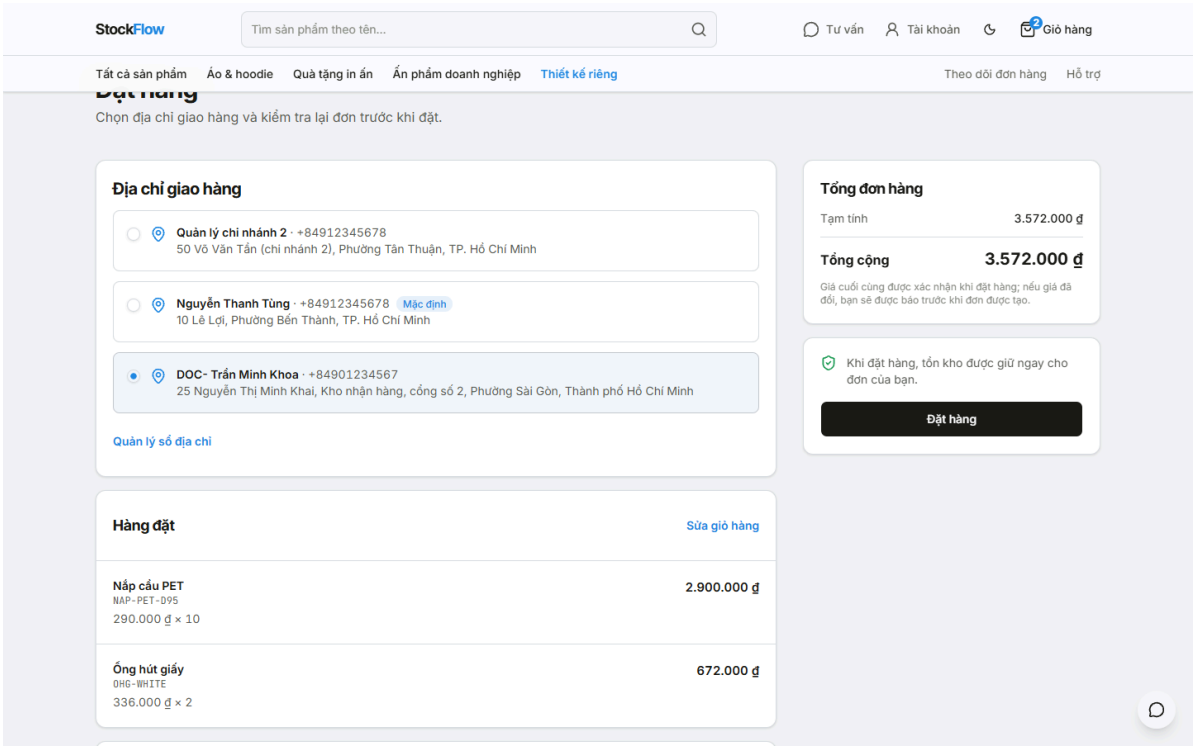
- **Function trigger:** Users click “Giỏ hàng” in the header.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Review cart lines with live prices and stock before ordering.

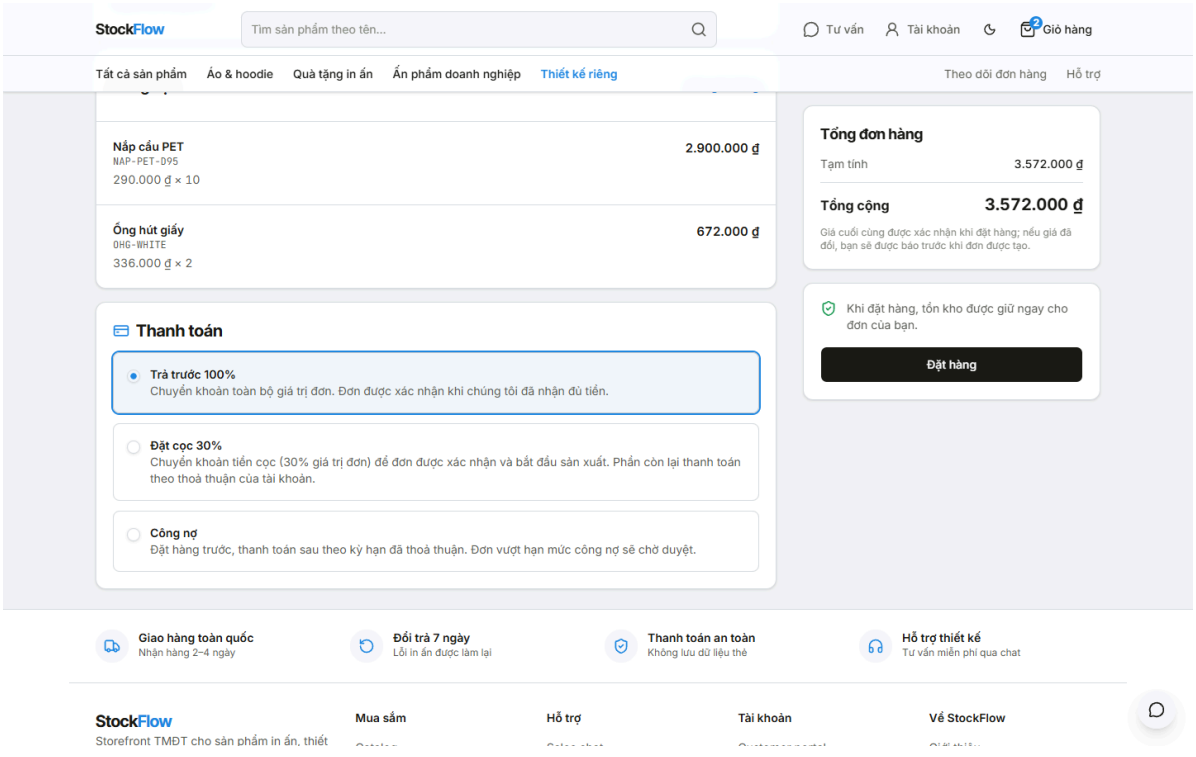


- **Function details:**
  - **Data requirement:** Lines (image, name, SKU, quantity, unit price, line total), subtotal, line warnings
  - **Validation:** Quantity 1–1,000,000; “Tiếp tục đặt hàng” is blocked while a line is out of stock, has no price, is below MOQ or has a non-whole price.
  - **Business rules:** BR-02, BR-04, BR-06, BR-07 (13 Catalog); BR-04 (14 Cart & Order)

3.2.3.3 Checkout

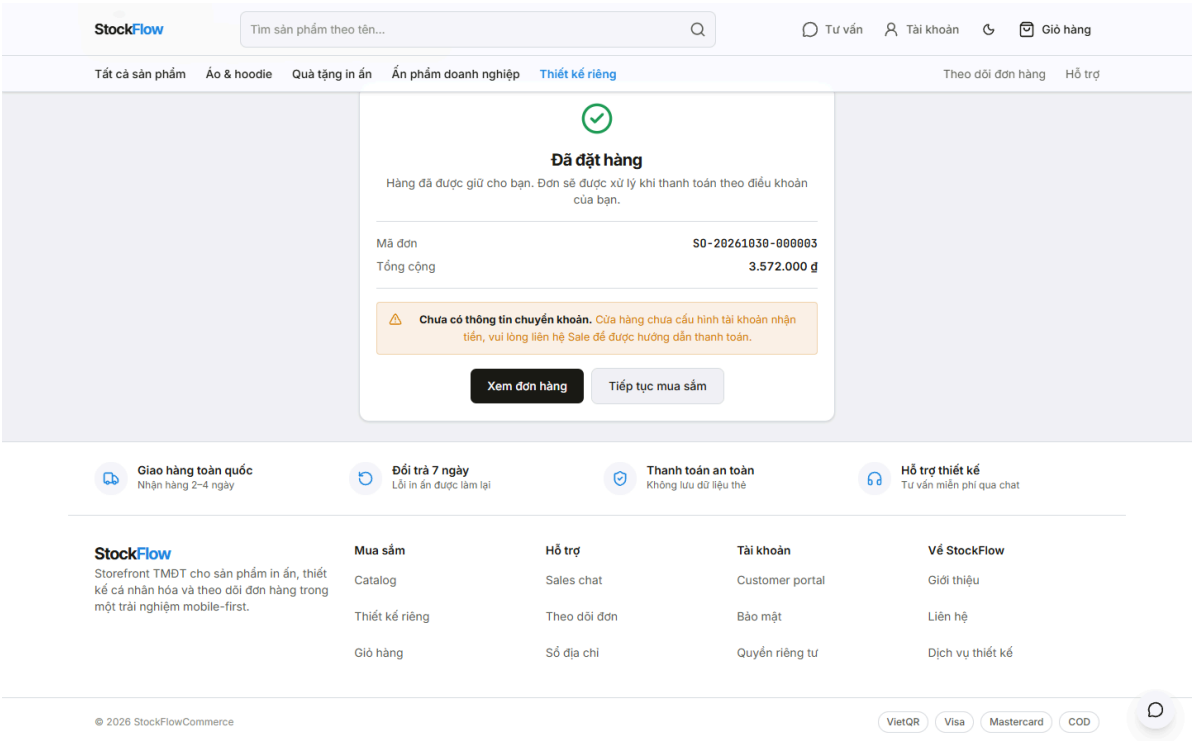
- **Function trigger:** Users click “Tiếp tục đặt hàng” in the cart, then “Đặt hàng” on the checkout page.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Place an order for the cart.


StockFlow
Tìm sản phẩm theo tên...
Tư vấn
Tài khoản
Giỏ hàng
Tất cả sản phẩm
Áo & hoodie
Quà tặng in ấn
Ấn phẩm doanh nghiệp
Thiết kế riêng
Theo dõi đơn hàng
Hỗ trợ
Chọn địa chỉ giao hàng và kiểm tra lại đơn trước khi đặt.
Địa chỉ giao hàng
Quản lý chi nhánh 2 · +84912345678
50 Võ Văn Tần (chi nhánh 2), Phường Tân Thuận, TP. Hồ Chí Minh
Nguyễn Thanh Tùng · +84912345678 Mặc định
10 Lê Lợi, Phường Bến Thành, TP. Hồ Chí Minh
DOC- Trần Minh Khoa · +84901234567
25 Nguyễn Thị Minh Khai, Kho nhận hàng, cổng số 2, Phường Sài Gòn, Thành phố Hồ Chí Minh
Quản lý sổ địa chỉ
Tổng đơn hàng
Tạm tính
3.572.000 ₫
Tổng cộng
3.572.000 ₫
Giá cuối cùng được xác nhận khi đặt hàng; nếu giá đã đổi, bạn sẽ được báo trước khi đơn được tạo.
Khi đặt hàng, tồn kho được giữ ngay cho đơn của bạn.
Đặt hàng
Hàng đặt
Sửa giỏ hàng
Nắp cầu PET
NAP-PET-D95
290.000 ₫ × 10
2.900.000 ₫
Ống hút giấy
OHG-WHITE
336.000 ₫ × 2
672.000 ₫


StockFlow
Tìm sản phẩm theo tên...
Tư vấn
Tài khoản
Giỏ hàng
Tất cả sản phẩm
Áo & hoodie
Quà tặng in ấn
Ấn phẩm doanh nghiệp
Thiết kế riêng
Theo dõi đơn hàng
Hỗ trợ
Nắp cầu PET
NAP-PET-D95
290.000 ₫ × 10
2.900.000 ₫
Ống hút giấy
OHG-WHITE
336.000 ₫ × 2
672.000 ₫
Tổng đơn hàng
Tạm tính
3.572.000 ₫
Tổng cộng
3.572.000 ₫
Giá cuối cùng được xác nhận khi đặt hàng; nếu giá đã đổi, bạn sẽ được báo trước khi đơn được tạo.
Khi đặt hàng, tồn kho được giữ ngay cho đơn của bạn.
Đặt hàng
Thanh toán
Trả trước 100%
Chuyển khoản toàn bộ giá trị đơn. Đơn được xác nhận khi chúng tôi đã nhận đủ tiền.
Đặt cọc 30%
Chuyển khoản tiền cọc (30% giá trị đơn) để đơn được xác nhận và bắt đầu sản xuất. Phần còn lại thanh toán theo thoả thuận của tài khoản.
Công nợ
Đặt hàng trước, thanh toán sau theo kỳ hạn đã thoả thuận. Đơn vượt hạn mức công nợ sẽ chờ duyệt.

Giao hàng toàn quốc
Nhận hàng 2–4 ngày
Đổi trả 7 ngày
Lỗi in ấn được làm lại
Thanh toán an toàn
Không lưu dữ liệu thẻ
Hỗ trợ thiết kế
Tư vấn miễn phí qua chat

StockFlow
Storefront TMĐT cho sản phẩm in ấn, thiết
Mua sắm
Hỗ trợ
Tài khoản
Về StockFlow

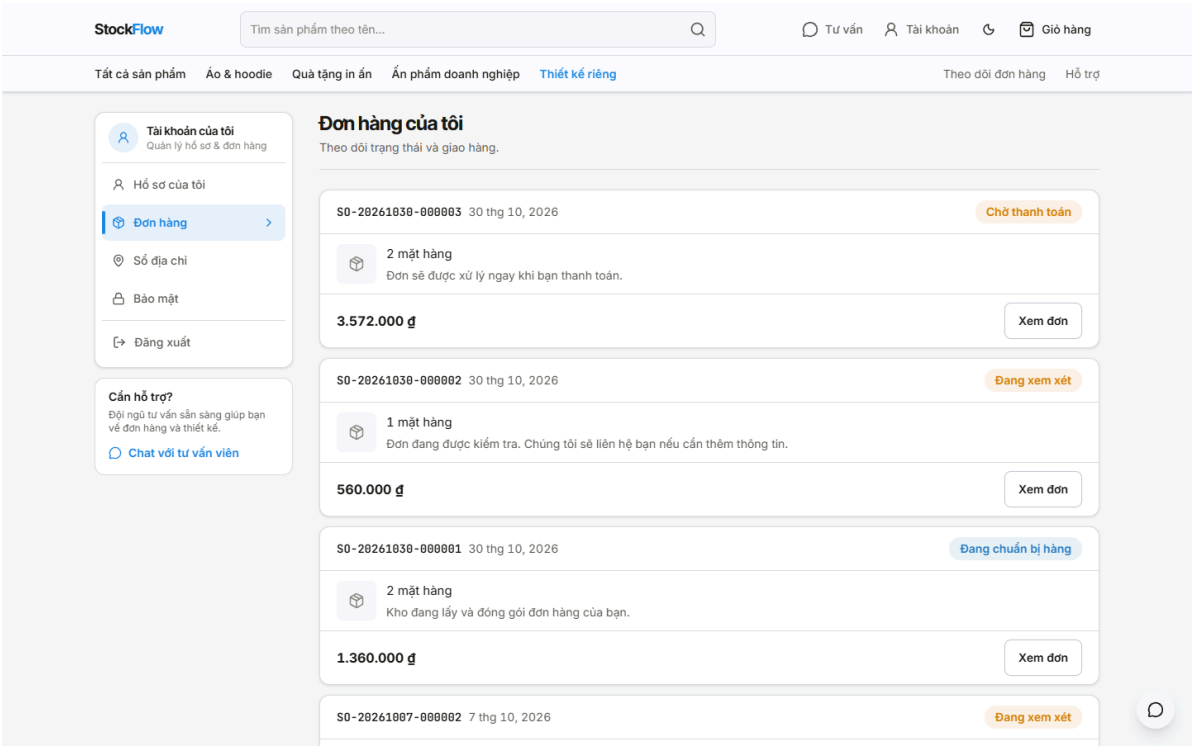


- **Function details:**
  - **Data requirement:** Shipping address, payment term (“Trả trước 100%”, “Đặt cọc”, “Công nợ”), deposit balance option, order lines, total
  - **Validation:** Shipping address required; at least one line; quantity whole number ≥ 1; whole-VND unit price; only allowed terms; balance option for deposit only; changed price or suspected duplicate must be confirmed.
  - **Business rules:** BR-01, BR-03, BR-04, BR-05, BR-06, BR-07, BR-12 (14 Cart & Order); BR-01, BR-10, BR-12 (15 Payment)

3.2.4 My Orders

3.2.4.1 View My Orders

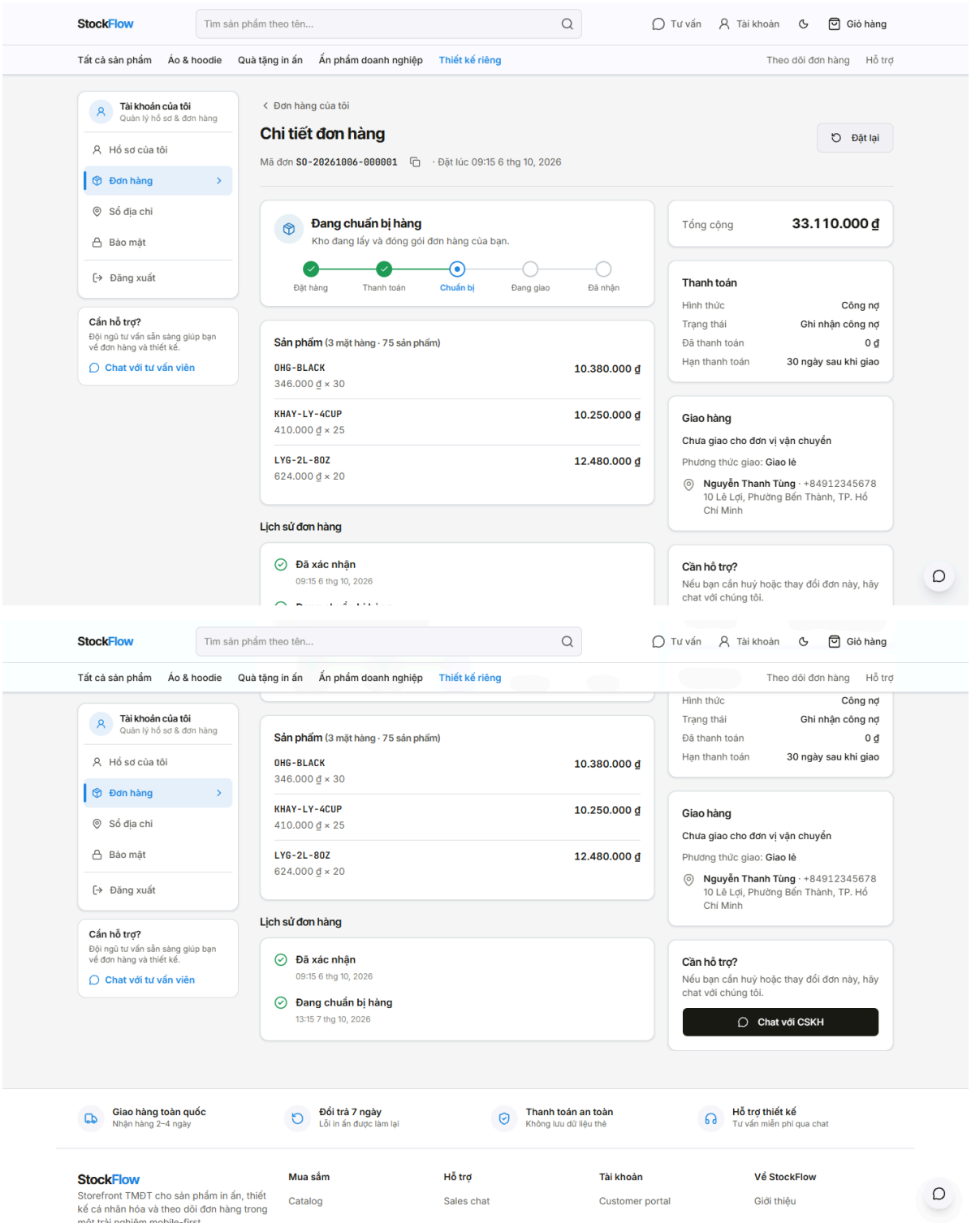
- **Function trigger:** Users open “Đơn hàng” in the account menu or click “Theo dõi đơn hàng” in the header.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** List the customer's own orders.



- **Function details:**
  - **Data requirement:** Order cards: order number, order date, status, number of items, total
  - **Validation:** None
  - **Business rules:** BR-01 (17 Order Management)

3.2.4.2 View Order Detail

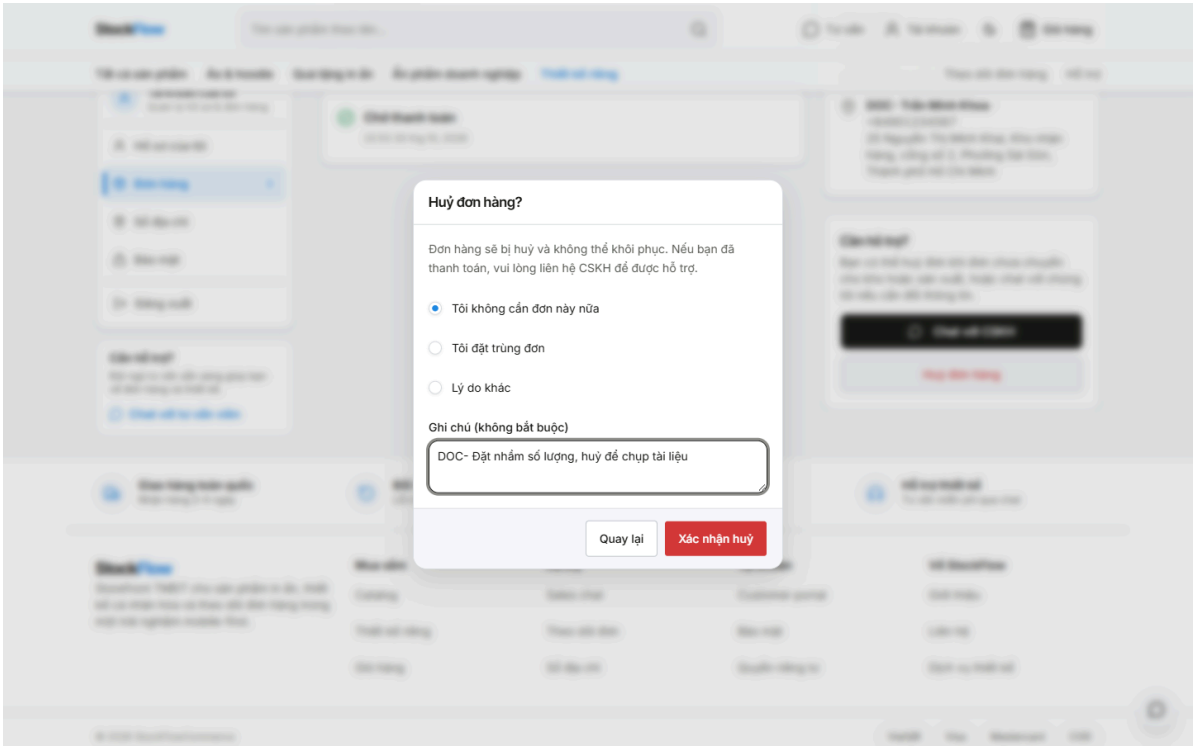
- **Function trigger:** Users click “Xem đơn” on an order card in “Đơn hàng của tôi”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Follow one order's status, items, production and payment.

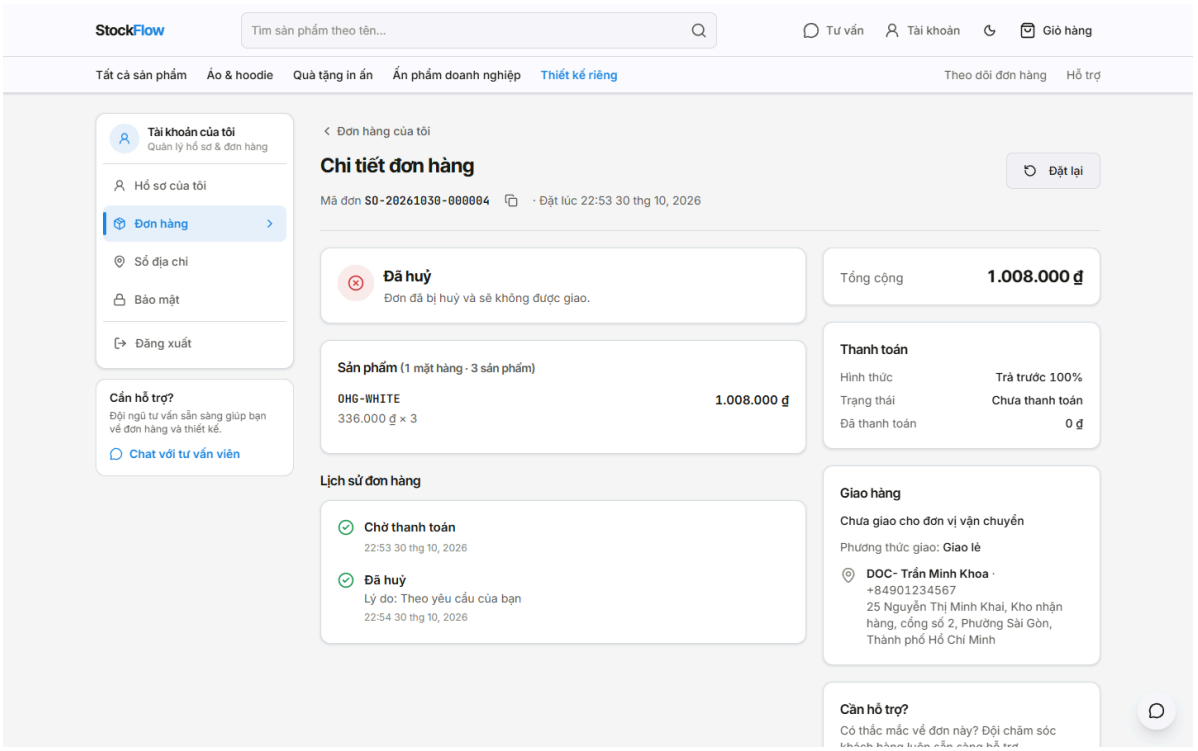


- **Function details:**
  - **Data requirement:** Order number, order time, status and hold reason, lines with production progress, timeline, total, payment (term, status, paid, due, transfer details), delivery address and mode
  - **Validation:** None
  - **Business rules:** BR-01 (17 Order Management); BR-04 (15 Payment)

3.2.4.3 Cancel Order

- **Function trigger:** Users click “Huỷ đơn hàng”, or “Gửi yêu cầu huỷ” once the order is in production or the warehouse, on the order detail page.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Cancel an order or ask Sales to cancel it.



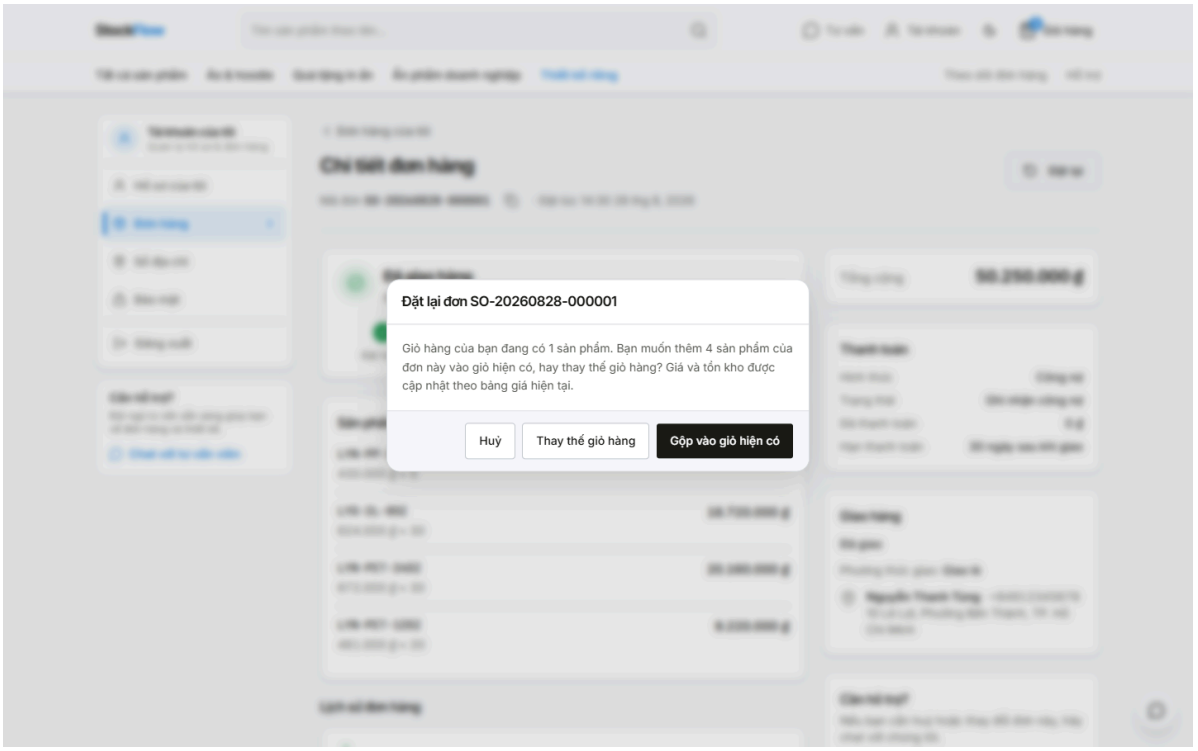


- **Function details:**
  - **Data requirement:** Reason (“Lý do huỷ”), note

- **Validation:** Reason is “Tôi không cần đơn này nữa”, “Tôi đặt trùng đơn” or “Lý do khác”; note ≤ 1,000 characters, required for “Lý do khác”.
- **Business rules:** BR-10, BR-11, BR-12, BR-14 (17 Order Management)

3.2.4.4 Reorder

- **Function trigger:** Users click “Đặt lại” on the order detail page.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Order a past order's items again.

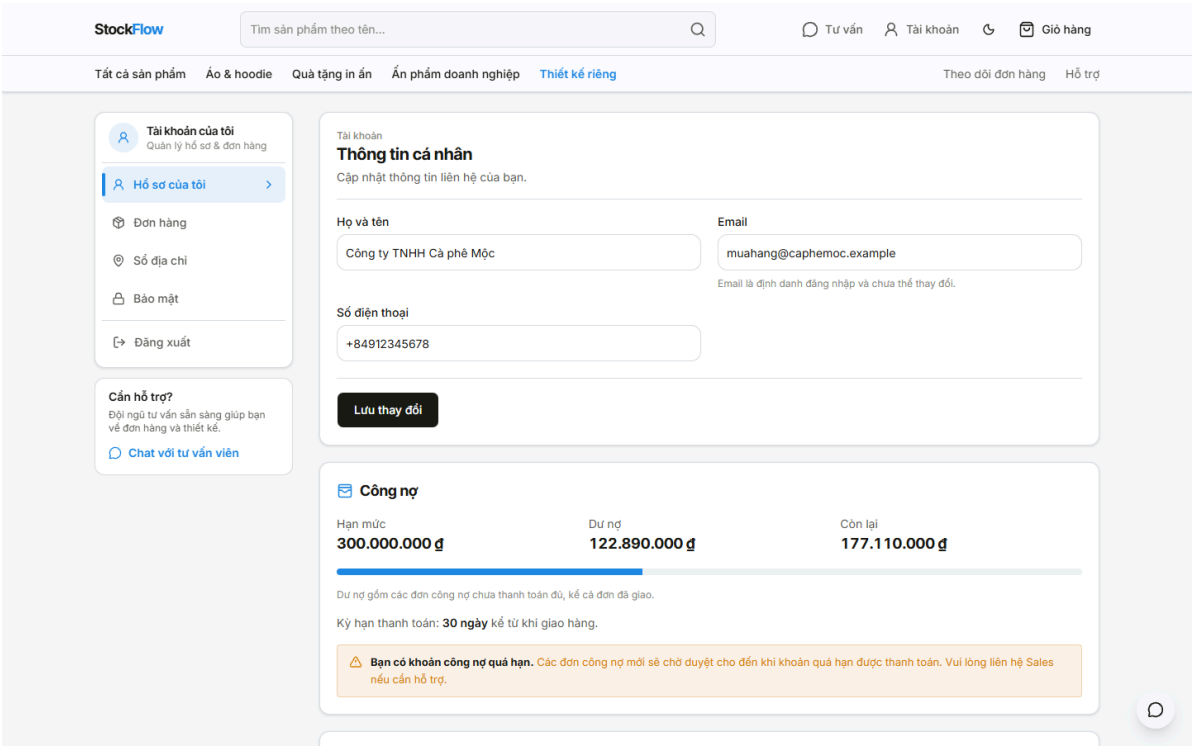


- **Function details:**
  - **Data requirement:** Past order lines; choice “Gộp vào giỏ hiện có” or “Thay thế giỏ hàng” when the cart is not empty
  - **Validation:** Design lines and SKUs no longer on sale are skipped and listed; quantities capped at 1,000,000.
  - **Business rules:** BR-03, BR-12, BR-15 (14 Cart & Order)

3.2.5 Account & Address Book

3.2.5.1 View Profile

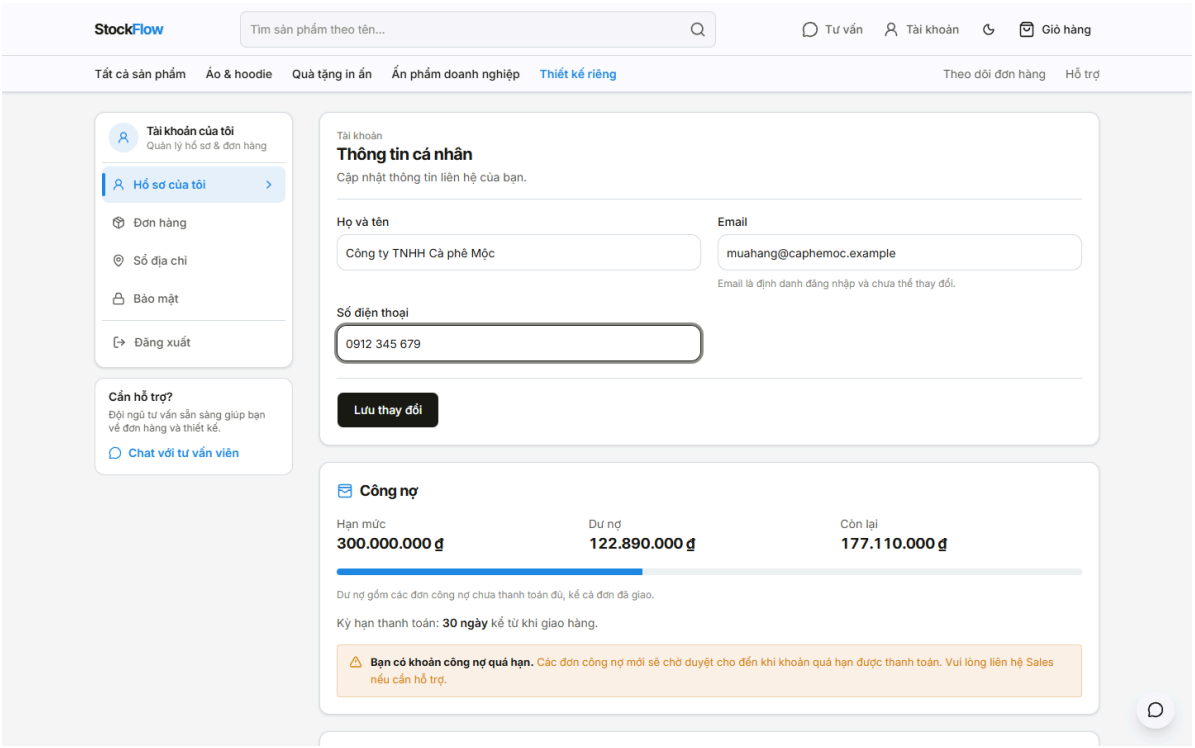
- **Function trigger:** Users click “Tài khoản” in the header, which opens “Hồ sơ của tôi”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** View the account's contact information.



- **Function details:**
  - **Data requirement:** Name (“Họ và tên”), email, phone
  - **Validation:** None
  - **Business rules:** BR-03 (18 Customer Information)

3.2.5.2 Edit Profile

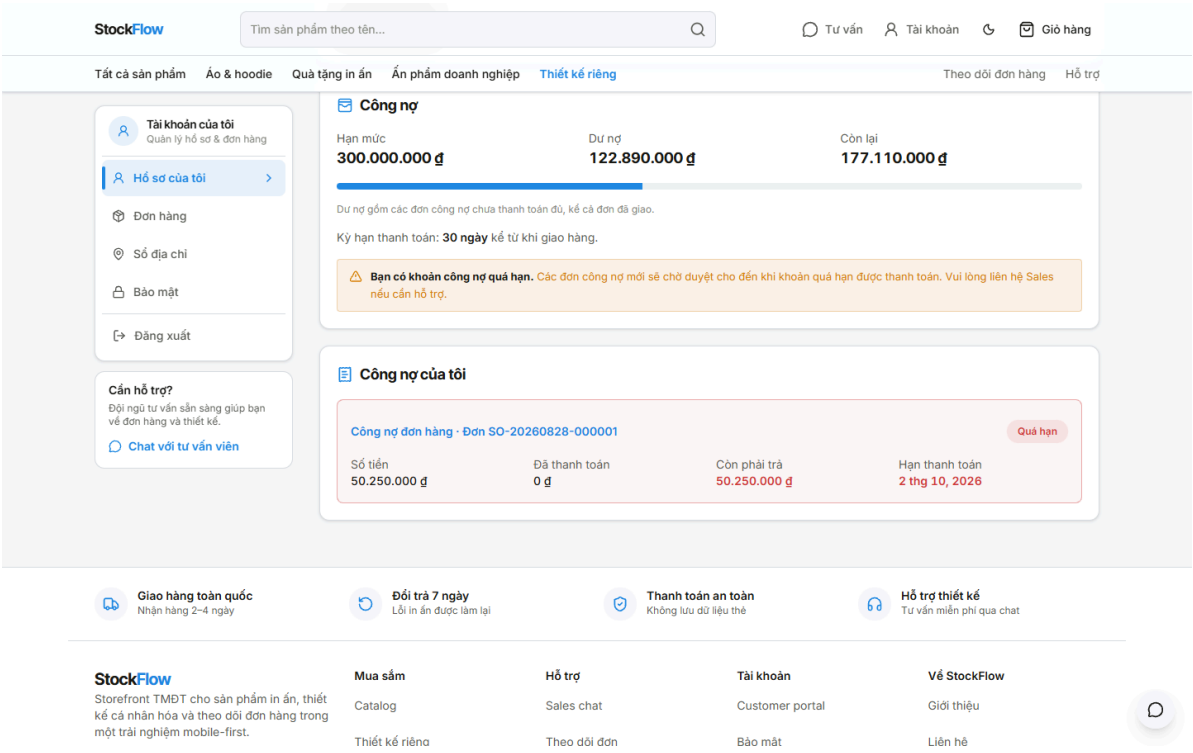
- **Function trigger:** Users edit “Thông tin cá nhân” on “Hồ sơ của tôi” and click “Lưu thay đổi”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Update the account's name and phone number.



- **Function details:**
  - **Data requirement:** Name, phone
  - **Validation:** Name required, ≤ 200 characters; phone optional, valid Vietnamese number; email read-only; a stale version is rejected and the customer must reload.
  - **Business rules:** BR-03, BR-07 (18 Customer Information)

3.2.5.3 View Credit and Receivables

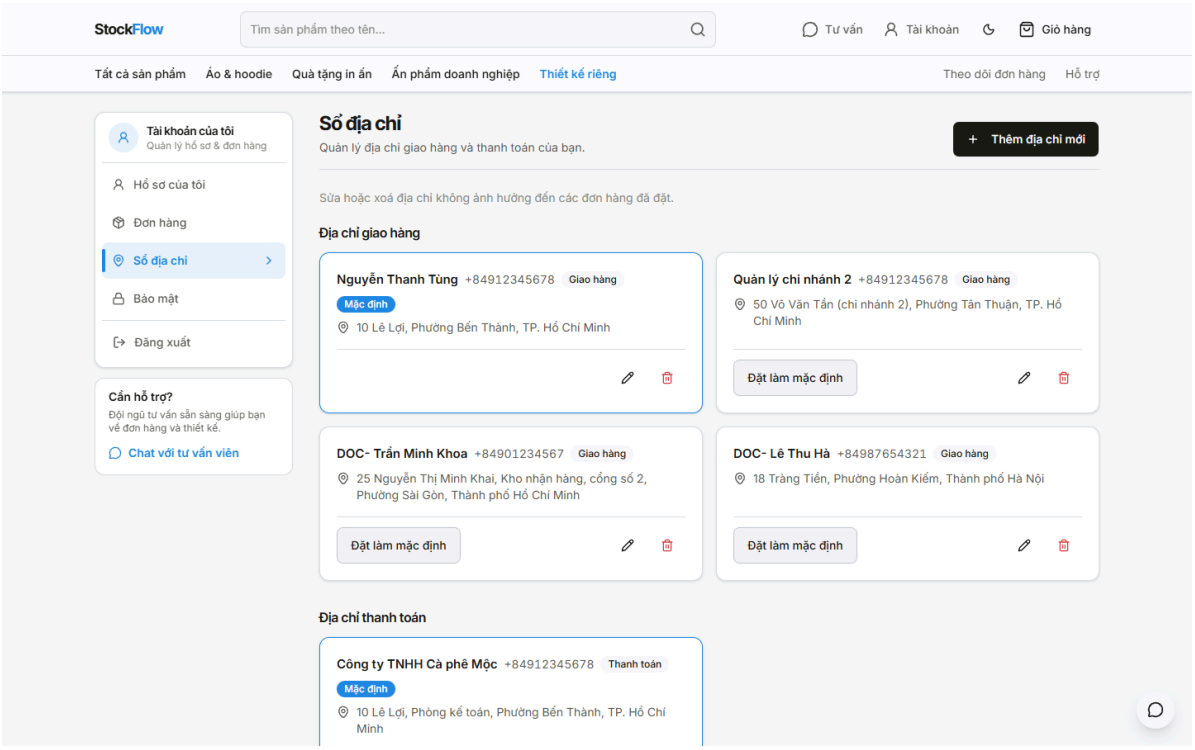
- **Function trigger:** Users open “Hồ sơ của tôi” and scroll to the “Công nợ” and “Công nợ của tôi” sections.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** View credit position and outstanding receivables.



- **Function details:**
  - **Data requirement:** Credit limit, amount owed, remaining limit, payment term days, overdue warning; receivables (type, order, amount, paid, outstanding, due date, status)
  - **Validation:** None
  - **Business rules:** BR-03, BR-13, BR-14 (15 Payment); BR-03 (18 Customer Information)

3.2.5.4 View Address Book

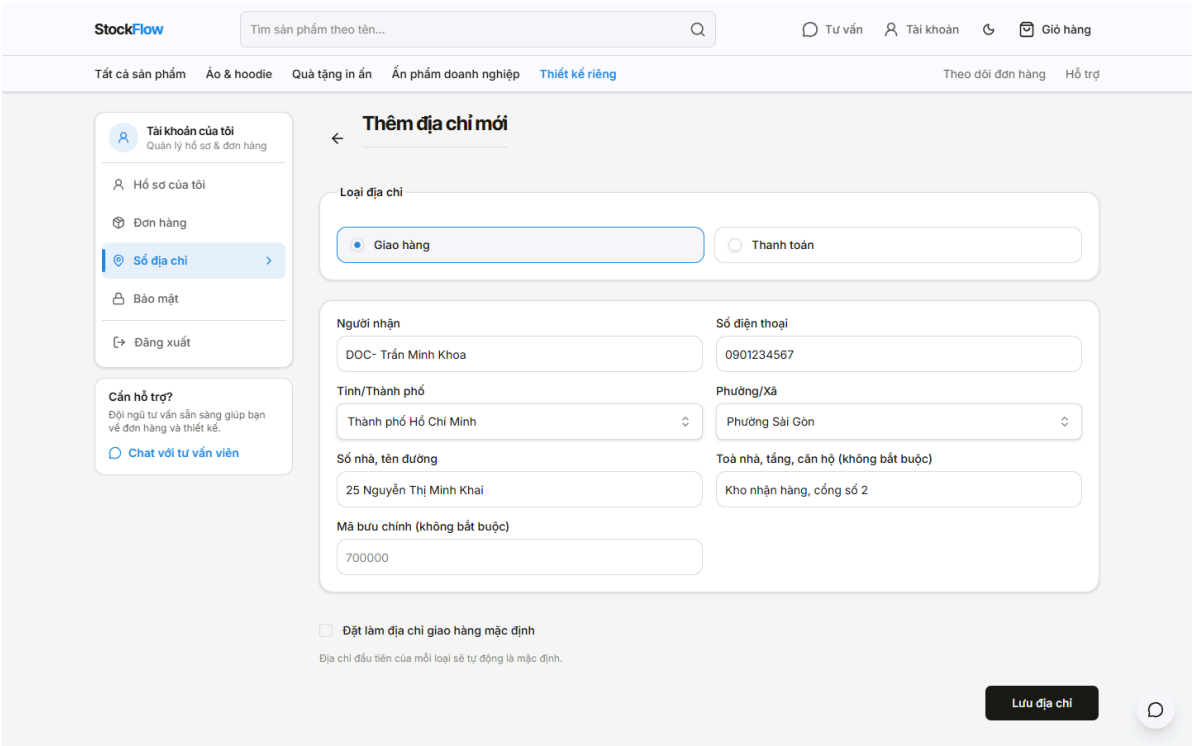
- **Function trigger:** Users open “Sổ địa chỉ” in the account menu.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** View saved addresses and choose the default one.



- **Function details:**
  - **Data requirement:** Addresses grouped by type (“Giao hàng”, “Thanh toán”): recipient, phone, address, “Mặc định” badge
  - **Validation:** None
  - **Business rules:** BR-03 (18 Customer Information)

3.2.5.5 Add Address

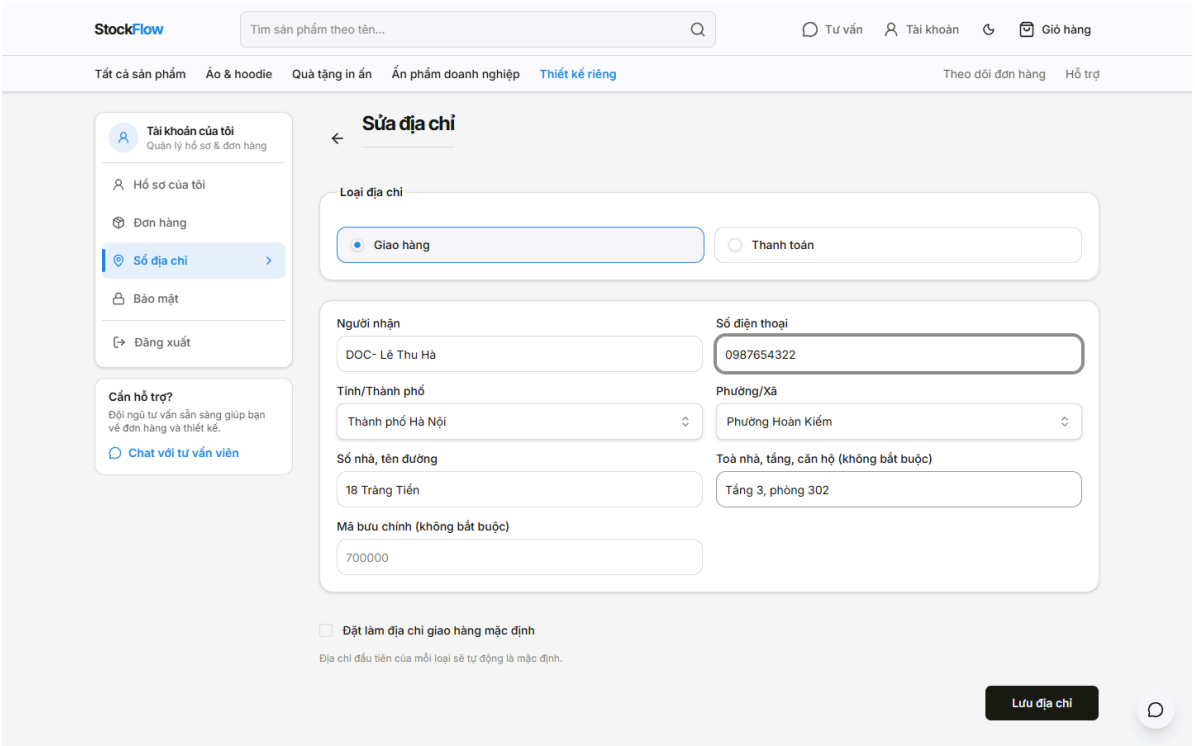
- **Function trigger:** Users click “Thêm địa chỉ mới” in “Sổ địa chỉ”, fill the form and click “Lưu địa chỉ”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Add a shipping or billing address.



- **Function details:**
  - **Data requirement:** Address type, recipient, phone, province/city, ward, street address, building (optional), postal code (optional), default flag
  - **Validation:** Recipient required, ≤ 200; valid Vietnamese phone required; street required, ≤ 255; building ≤ 255; province and ward required from the list; postal code ≤ 20; Vietnam only.
  - **Business rules:** None

3.2.5.6 Edit Address

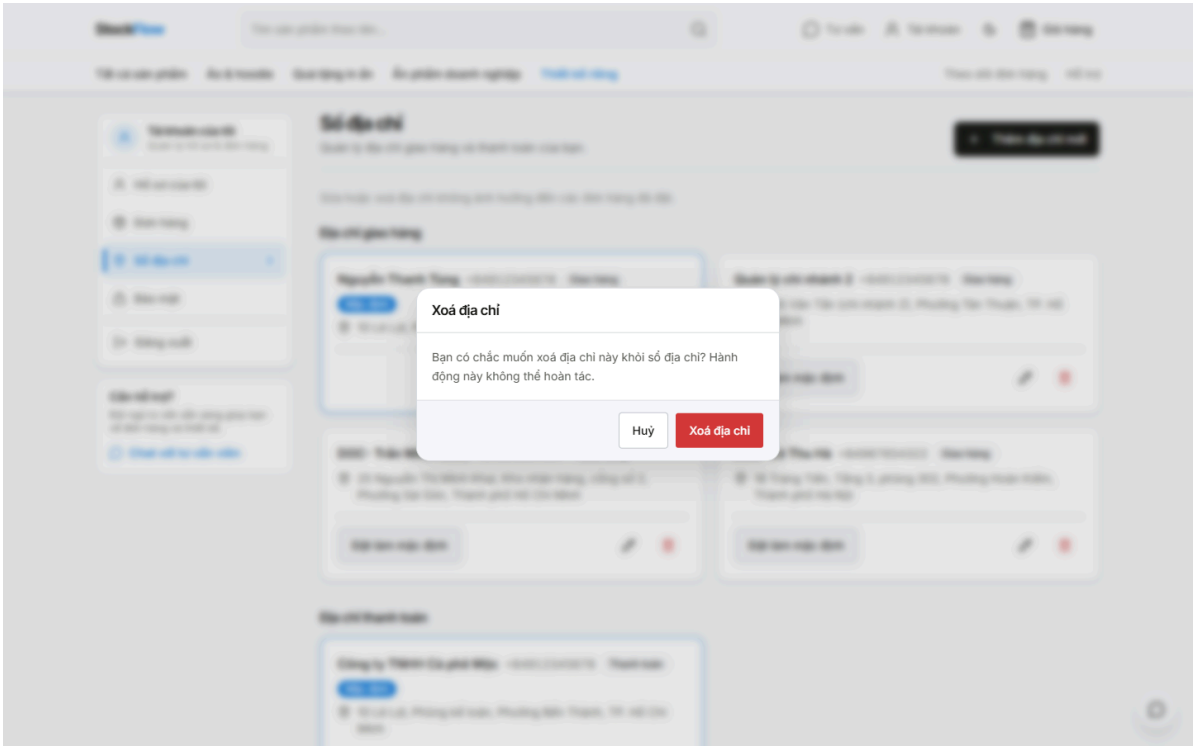
- **Function trigger:** Users click the edit icon (“Sửa địa chỉ”) on an address card, change fields and click “Lưu địa chỉ”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Update a saved address.



- **Function details:**
  - **Data requirement:** Same fields as adding an address
  - **Validation:** Same rules as adding an address; a stale version is rejected and the latest address is reloaded.
  - **Business rules:** None

3.2.5.7 Delete Address

- **Function trigger:** Users click the delete icon (“Xoá địa chỉ”) on an address card and confirm “Xoá địa chỉ”.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Remove an address from the address book.

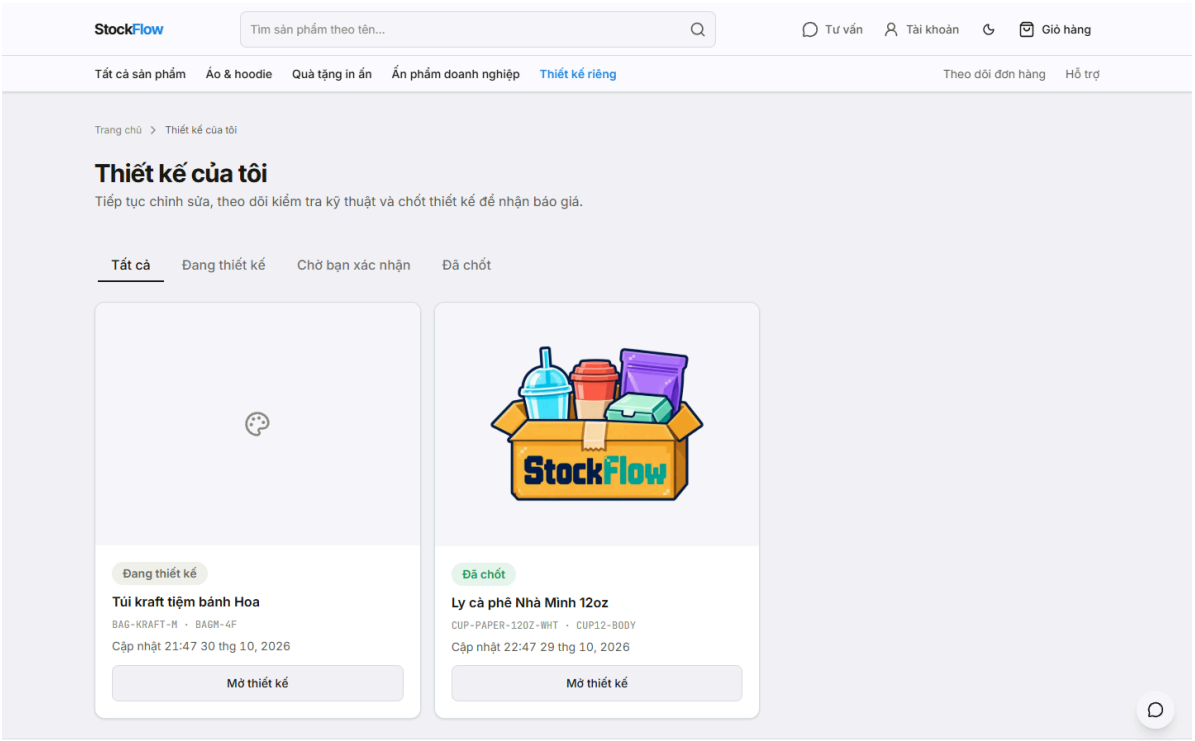


- **Function details:**
  - **Data requirement:** None (selected address)
  - **Validation:** Confirmation required.
  - **Business rules:** None

3.2.6 Product Design

3.2.6.1 View My Designs

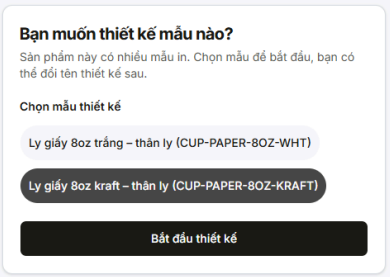
- **Function trigger:** Users click “Thiết kế riêng” in the header category bar or “Mở thiết kế của tôi” on the home page.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Under development — List the customer's print designs.



- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.2.6.2 Start New Design

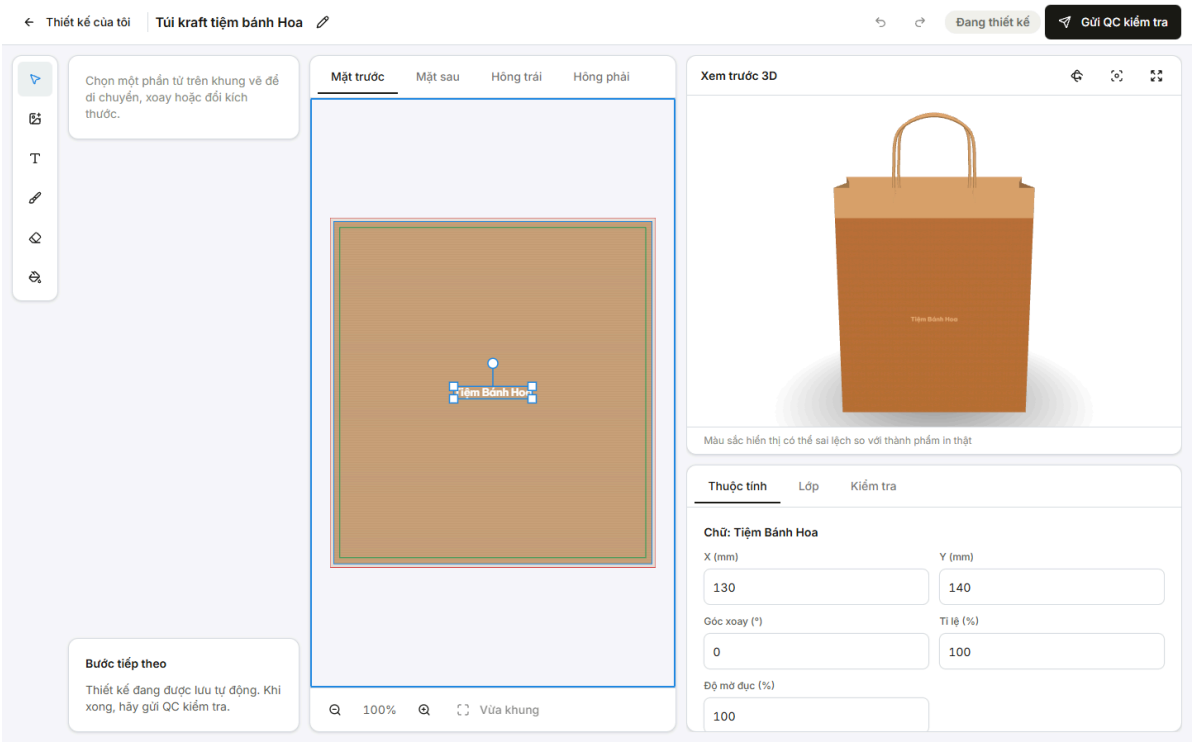
- **Function trigger:** Users click “Bắt đầu thiết kế” on a customizable product's detail page, then pick a template (“Chọn mẫu thiết kế”) if several exist.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Under development — Create a new design from a print template.



- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.2.6.3 Design Product

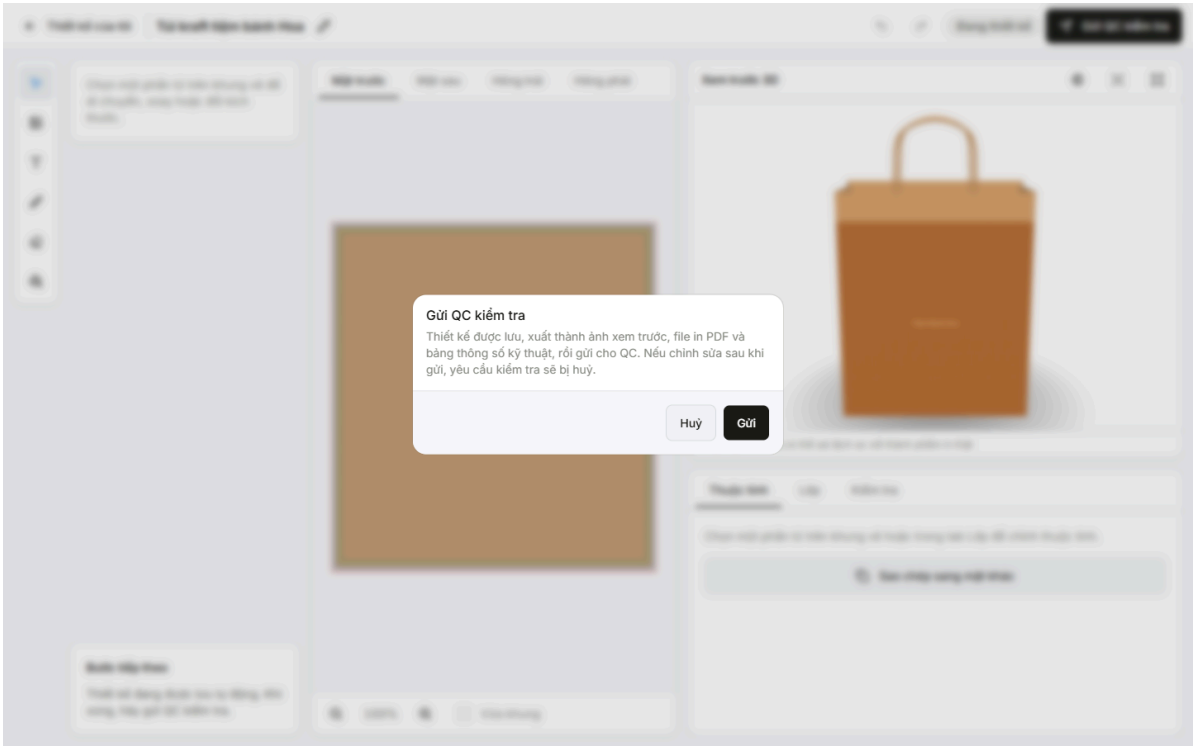
- **Function trigger:** Users click “Mở thiết kế” on a design card in “Thiết kế của tôi” to open the design studio.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Under development — Design print faces with images, text, drawing and backgrounds.

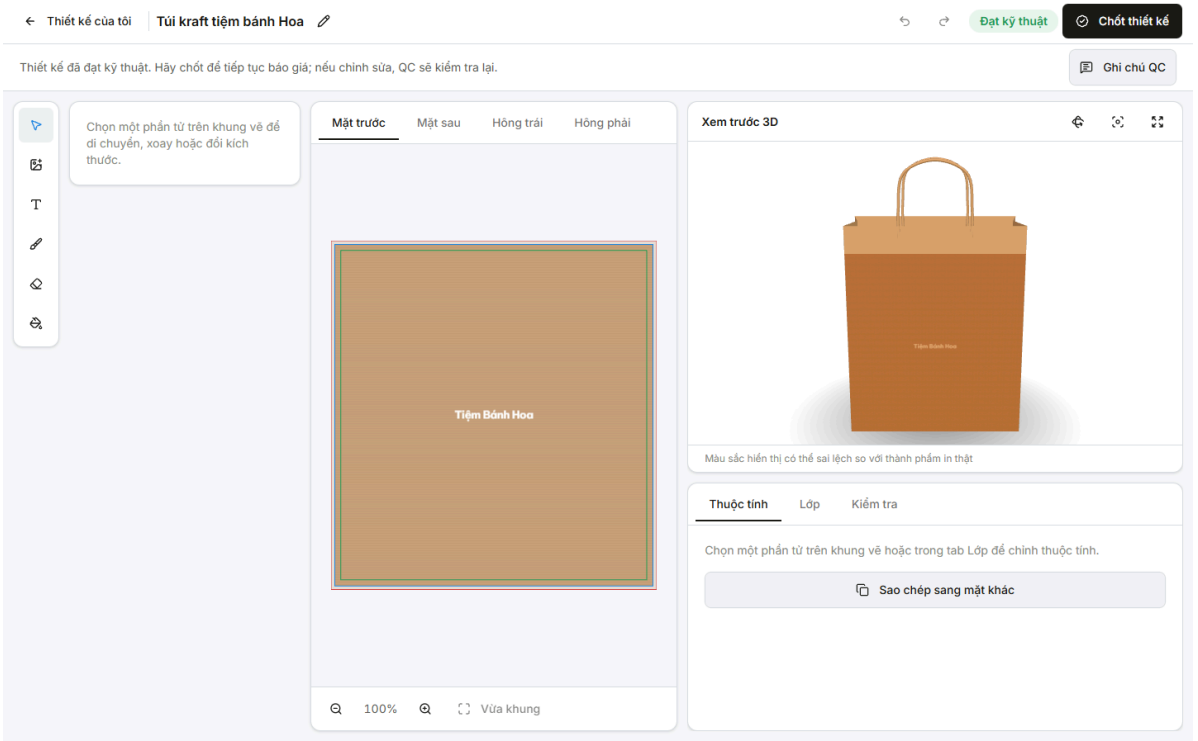


- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.2.6.4 Submit Design for QC Review

- **Function trigger:** Users click “Gửi QC kiểm tra” in the design studio header.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Under development — Run the print preflight check and send the design to QC.





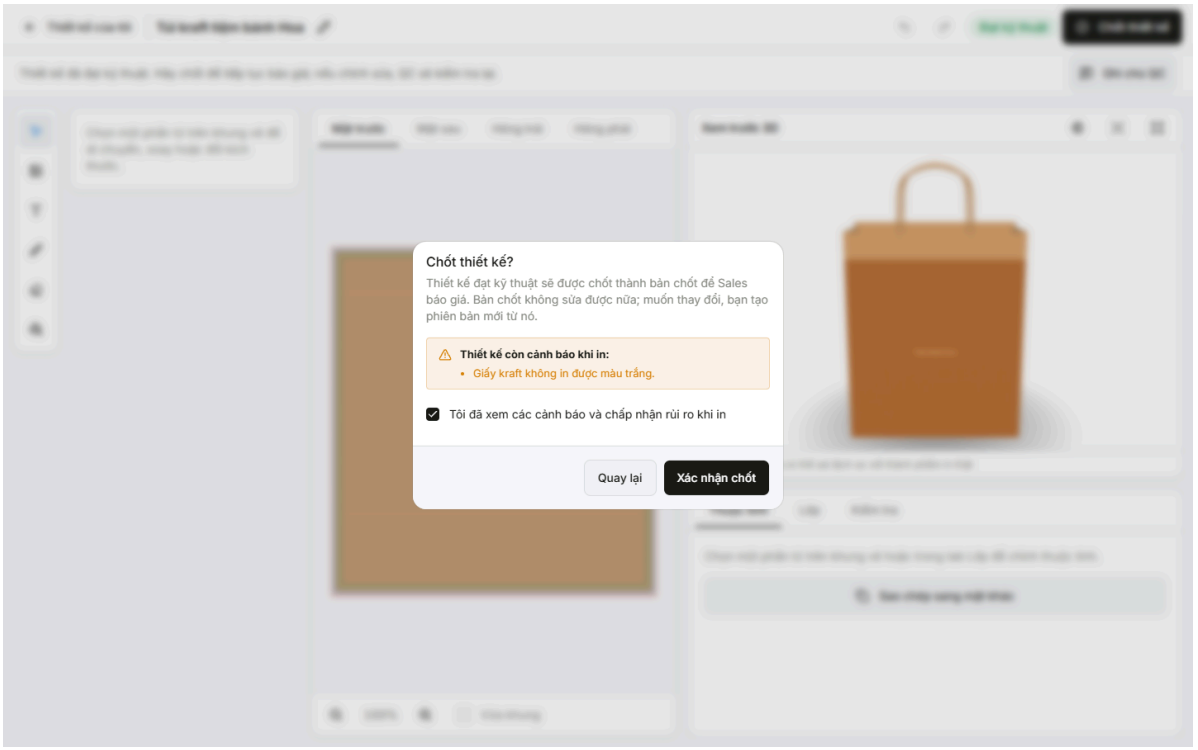
- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

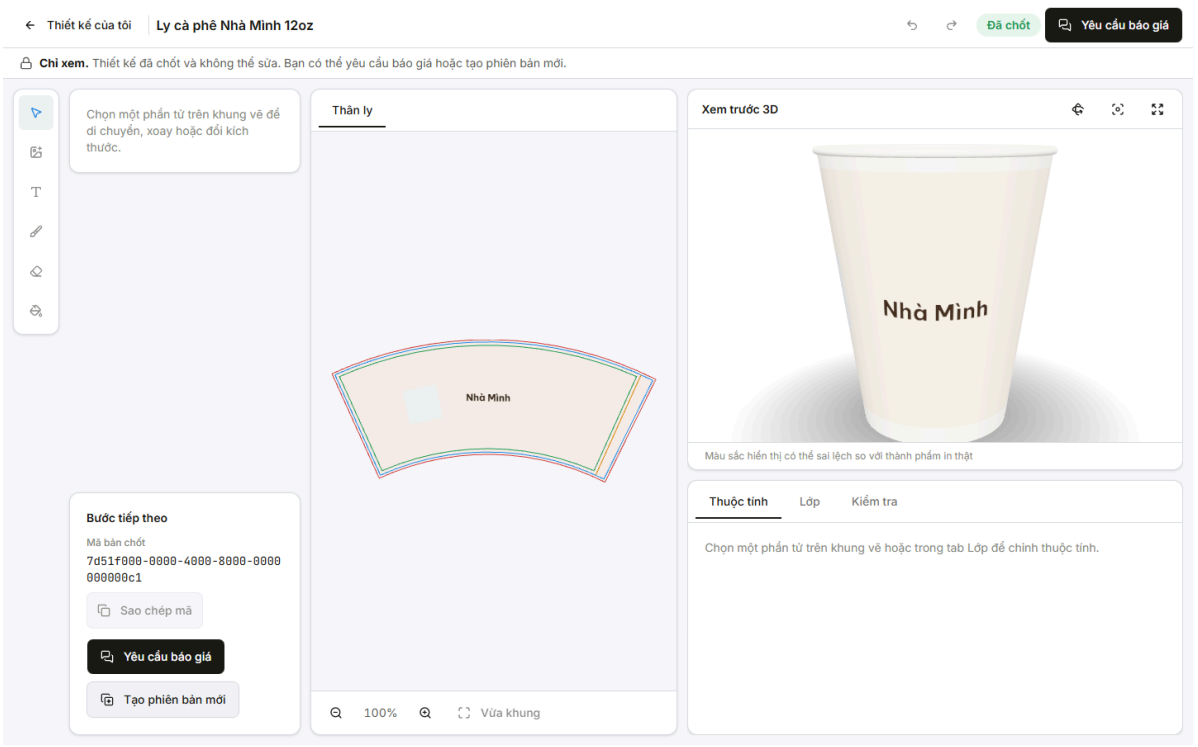
3.2.6.5 Confirm Design

- **Function trigger:** Users click “Chốt thiết kế” in the design studio header once QC has passed (tick “Tôi đã xem các cảnh báo và chấp nhận rủi ro khi in” when print

warnings remain, then “Xác nhận chốt”), or “Xác nhận thiết kế” on a design Sales sent (status “Chờ bạn xác nhận”).

- **Function description:**
    - **Actors:** Customer
    - **Purpose:** Confirm a design and freeze it as the final version.



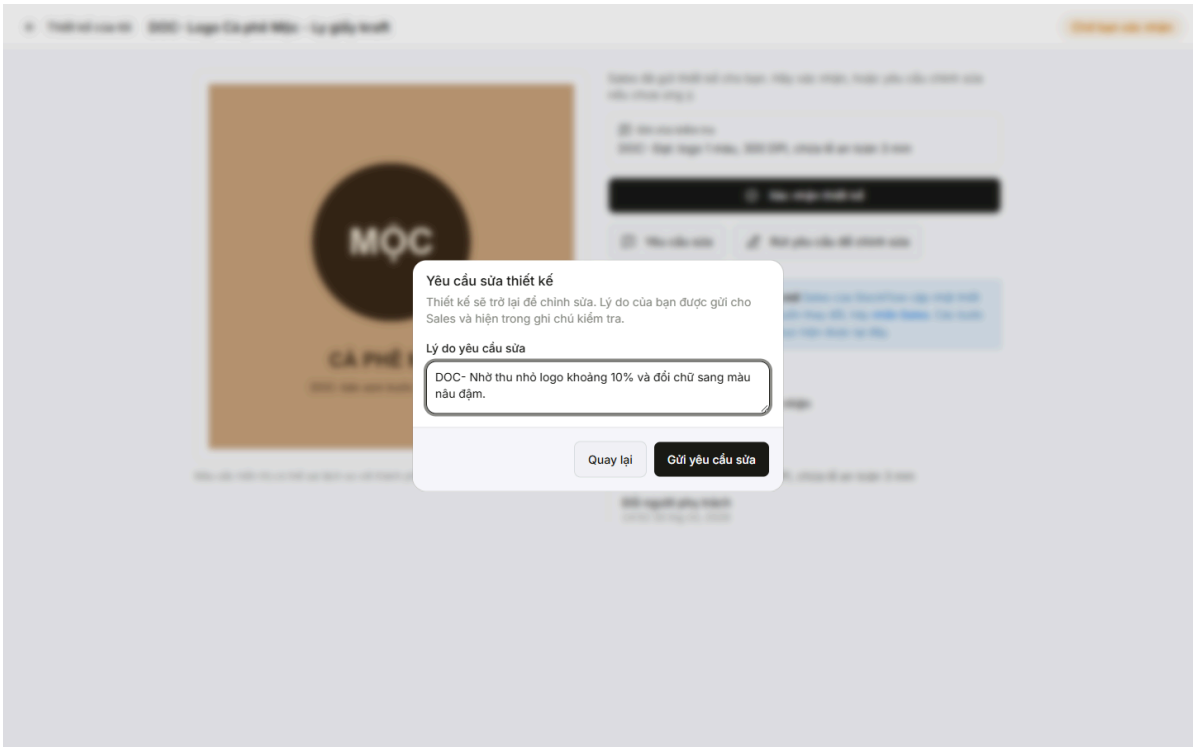


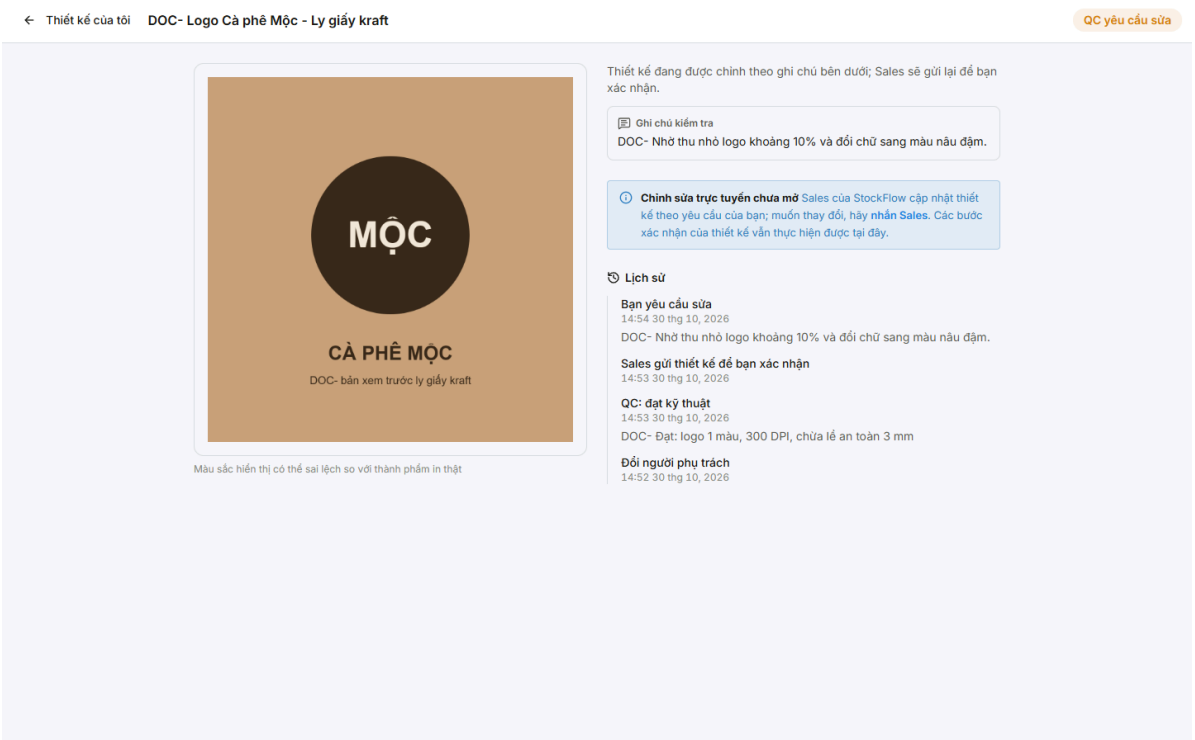
- **Function details:**
    - **Data requirement:** Print warnings, risk-acceptance checkbox

- **Validation:** No preflight error allowed; when warnings exist, “Tôi đã xem các cảnh báo và chấp nhận rủi ro khi in” must be ticked.
- **Business rules:** BR-07, BR-08, BR-10, BR-12 (12 Product Design)

3.2.6.6 Request Design Changes

- **Function trigger:** Users click “Yêu cầu sửa” or “Rút yêu cầu để chỉnh sửa” on a design awaiting their confirmation.
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Send a design back for changes, with a reason.



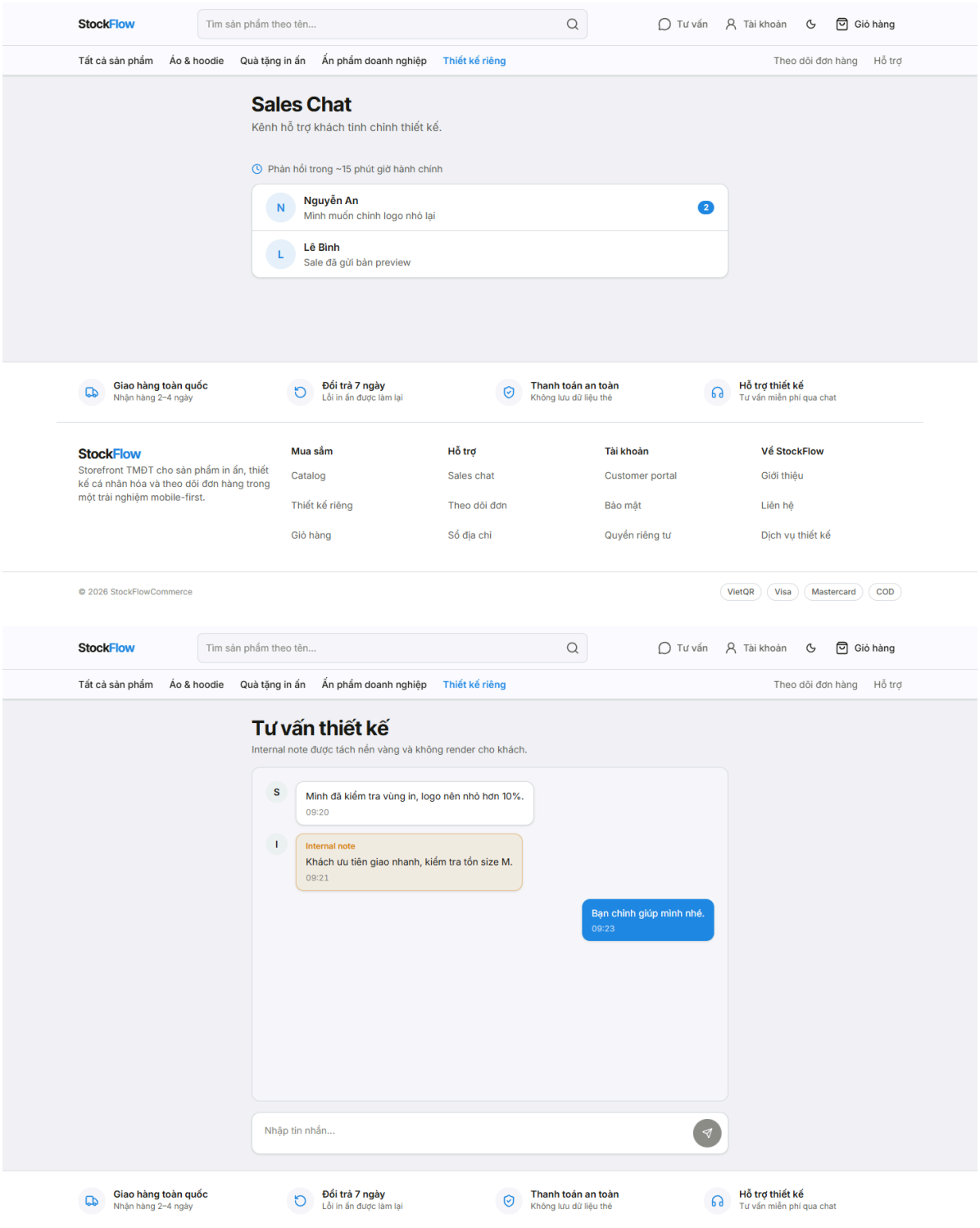


- **Function details:**
  - **Data requirement:** Reason
  - **Validation:** Reason required, ≤ 2,000 characters (BE @NotBlank @Size(max = 2000)).
  - **Business rules:** BR-06, BR-10, BR-12 (12 Product Design)

3.2.7 Sales Chat

3.2.7.1 Chat With Sales

- **Function trigger:** Users click “Tư vấn” in the header (or the floating chat button).
- **Function description:**
  - **Actors:** Customer
  - **Purpose:** Under development — Chat with Sales about designs and orders.



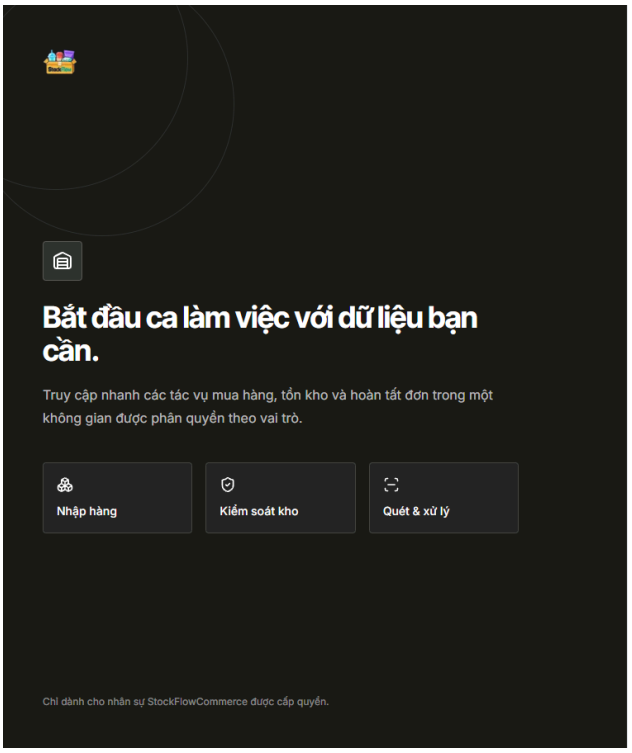
- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

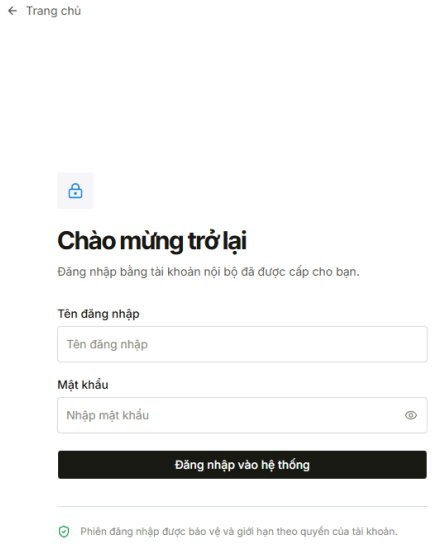
3.3 Back-office (Staff portal)

3.3.1 Authentication & Dashboard

3.3.1.1 Sign In

- **Function trigger:** Users open the login page, enter their credentials and click “Đăng nhập vào hệ thống”.
- **Function description:**
    - **Actors:** System Admin, E-commerce Admin, Warehouse Manager, Warehouse Staff, QC Staff, Inventory Planner, Procurement Staff, Sales Staff, Order Coordinator, Accountant, Production Staff
    - **Purpose:** Authenticate a staff member and open a back-office session.

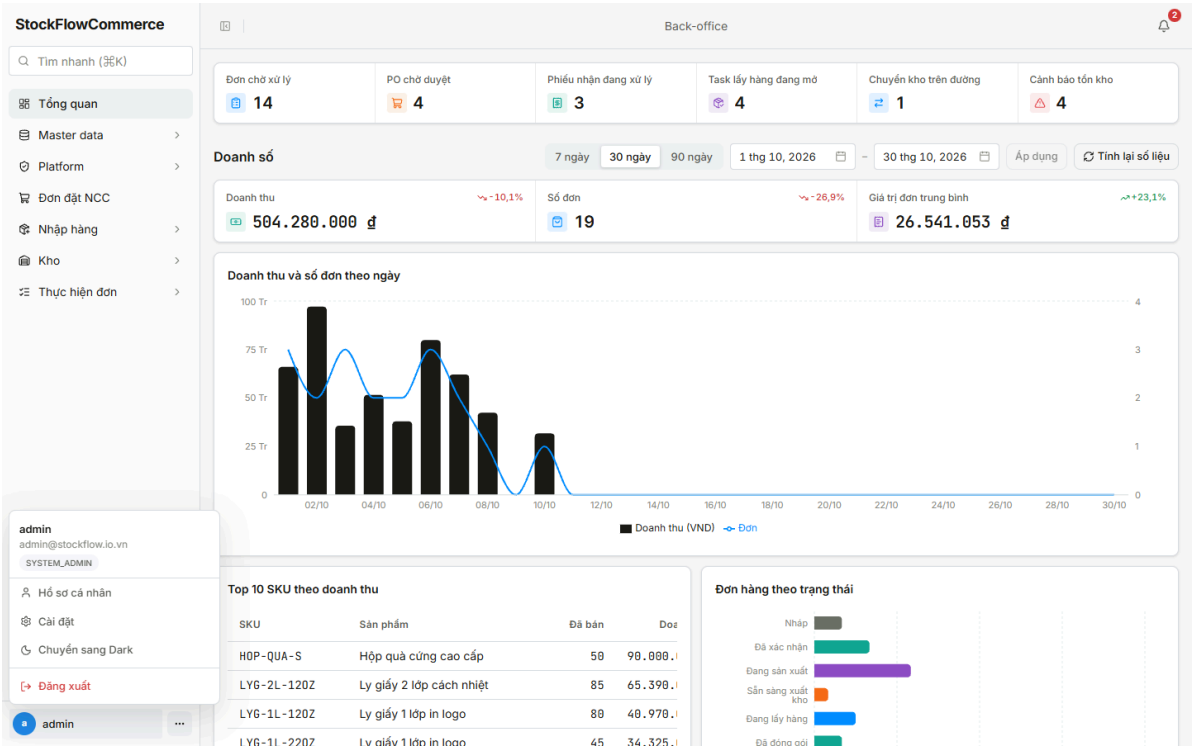




- **Function details:**
    - **Data requirement:** Username, password.
    - **Validation:** Username and password required; one generic error for unknown user or wrong password; 5 consecutive failures lock the account for 15 minutes; max 5 attempts per minute per IP; locked or disabled accounts refused.
    - **Business rules:** §2.5.1, §2.5.2 (00 System Overview)

3.3.1.2 Logout

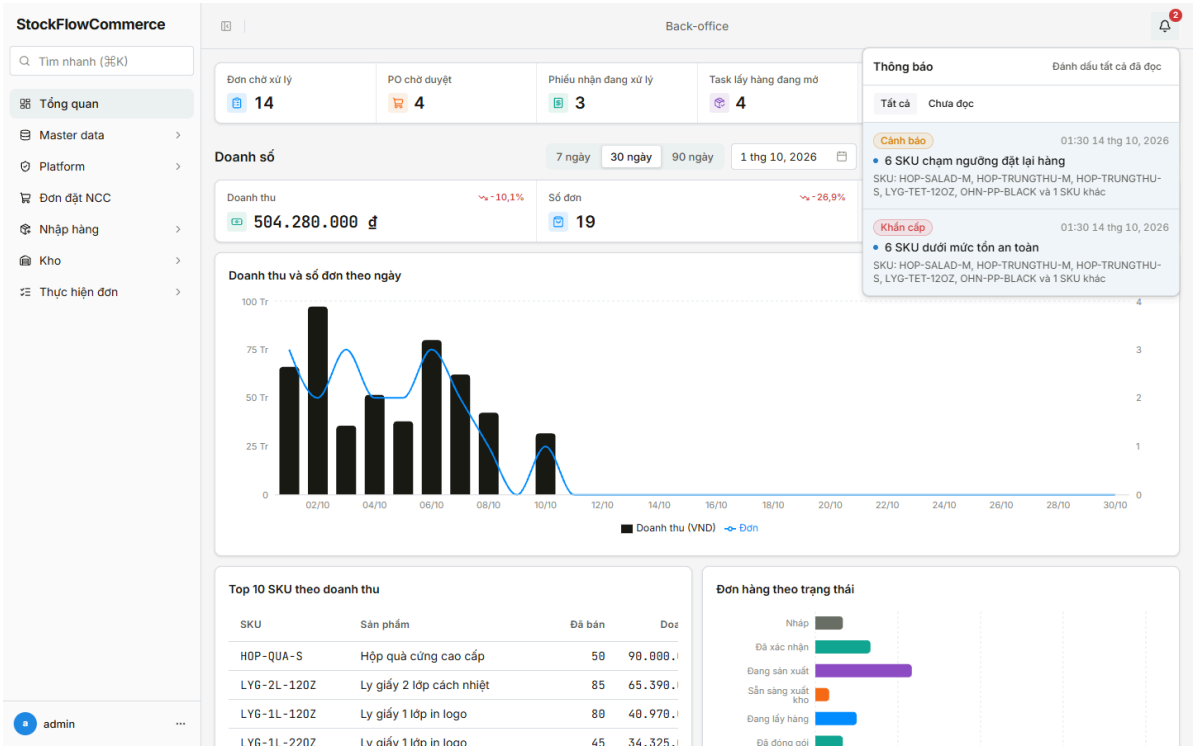
- **Function trigger:** Users open the account menu at the bottom of the sidebar and click “Đăng xuất”.
- **Function description:**
    - **Actors:** System Admin, E-commerce Admin, Warehouse Manager, Warehouse Staff, QC Staff, Inventory Planner, Procurement Staff, Sales Staff, Order Coordinator, Accountant, Production Staff
    - **Purpose:** End the current session and return to the login page.



- **Function details:**
  - **Data requirement:** None.
  - **Validation:** None.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.1.3 View Notifications

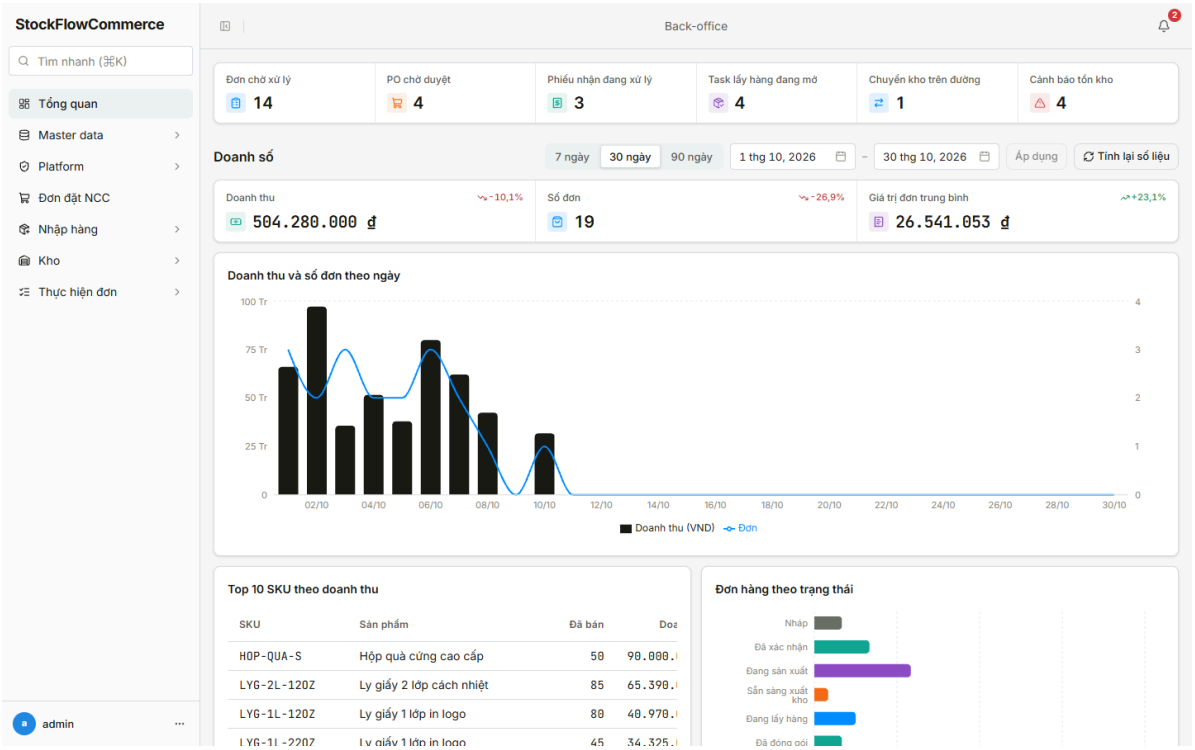
- **Function trigger:** Users click the “Thông báo” bell icon in the top bar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Warehouse Manager, Warehouse Staff, QC Staff, Inventory Planner, Procurement Staff, Sales Staff, Order Coordinator, Accountant, Production Staff
  - **Purpose:** Review in-app notifications and jump to the related record.



- **Function details:**
  - **Data requirement:** Severity, title, content, time, read state; filter “Tất cả” / “Chưa đọc”.
  - **Validation:** None.
  - **Business rules:** §6 In-app notifications (00 System Overview)

3.3.1.4 View Dashboard

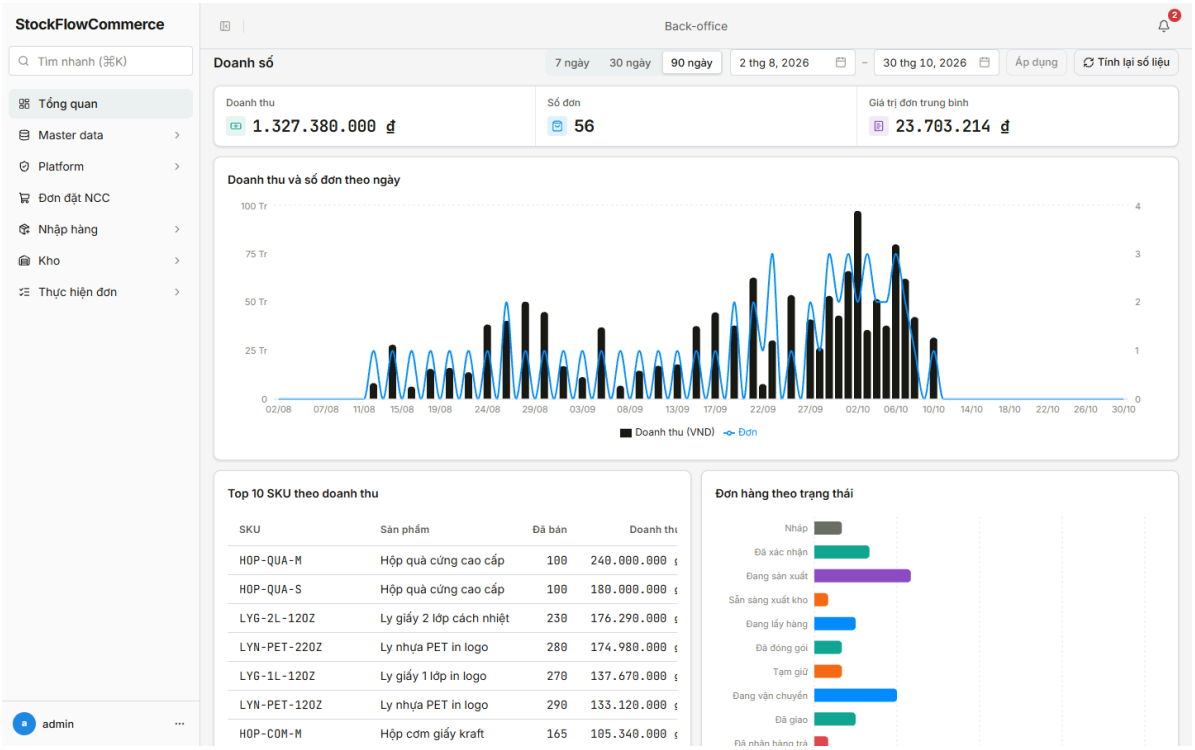
- **Function trigger:** Users open “Tổng quan” in the sidebar (the default page after login).
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Warehouse Manager, Warehouse Staff, QC Staff, Inventory Planner, Procurement Staff, Sales Staff, Order Coordinator, Accountant, Production Staff
  - **Purpose:** Show key operational counts and open stock alerts.

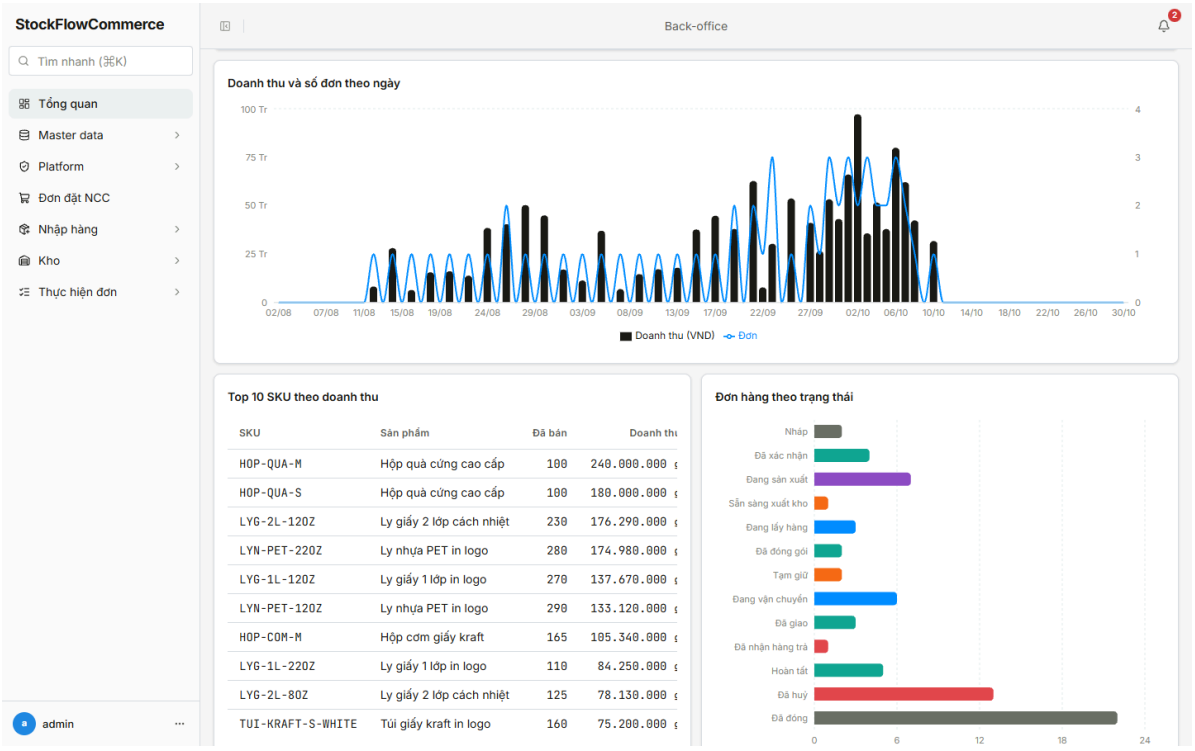


- **Function details:**
  - **Data requirement:** Pending orders, POs awaiting approval, receipts in progress, open pick tasks, transfers in transit; latest open stock alerts (type, SKU, warehouse, on hand, threshold, lot, expiry).
  - **Validation:** None.
  - **Business rules:** None

3.3.1.5 View Sales Report

- **Function trigger:** Users open “Tổng quan” and use the “Doanh số” section, picking “7 ngày”, “30 ngày” or “90 ngày”, or setting a from/to date and clicking “Áp dụng”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Warehouse Manager, Order Coordinator, Accountant
  - **Purpose:** Show sales figures for a chosen date range.

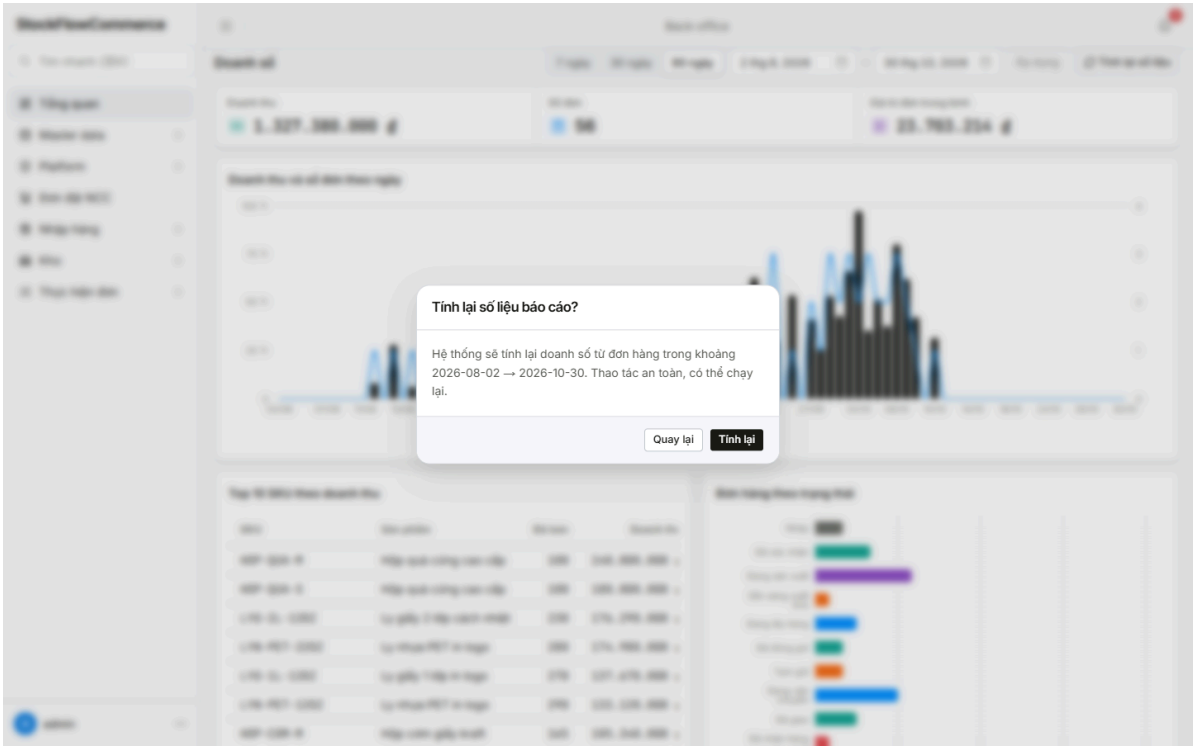




- **Function details:**
  - **Data requirement:** Date range (from, to); shows revenue, order count, average order value versus previous period, daily chart, top 10 SKUs, orders by status.
  - **Validation:** From date must not be after to date; range at most 366 days; defaults to the last 30 days.
  - **Business rules:** None

3.3.1.6 Rebuild Sales Report

- **Function trigger:** Users click “Tính lại số liệu” in the “Doanh số” section and confirm with “Tính lại”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Accountant
  - **Purpose:** Recalculate the sales report figures from the orders.

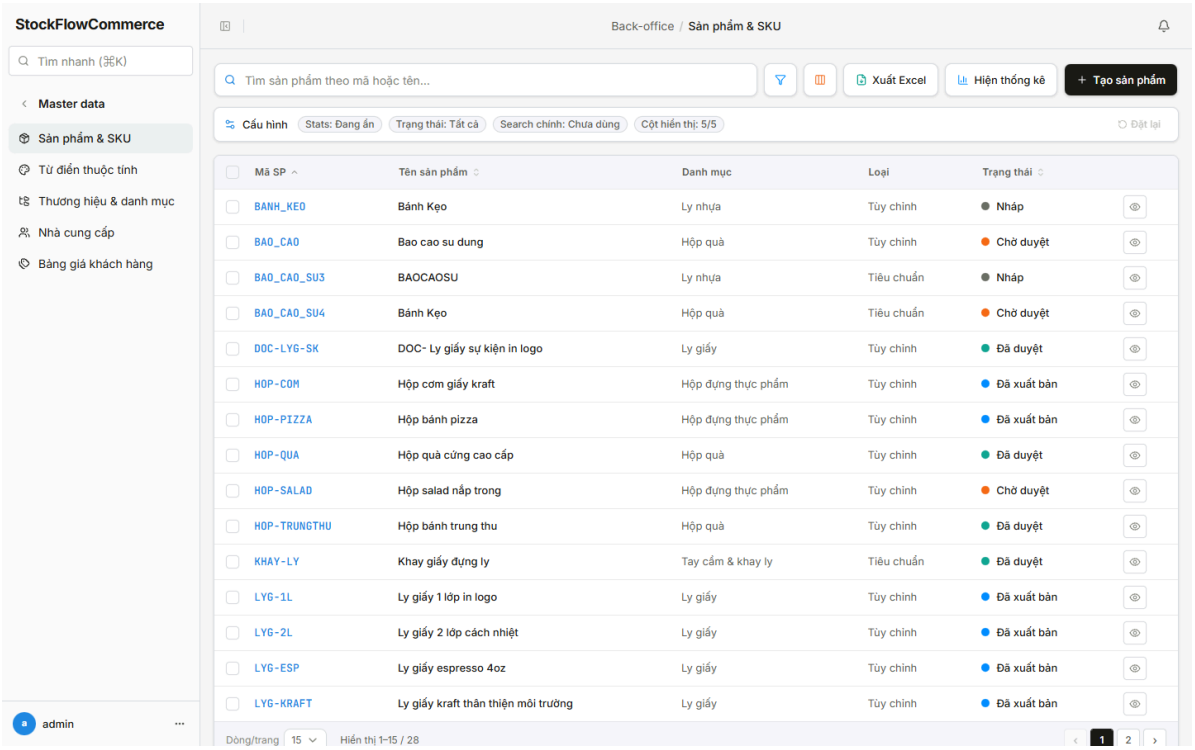


- **Function details:**
  - **Data requirement:** Currently selected date range.
  - **Validation:** Range at most 366 days and not inverted; requires reporting-reports:EXPORT.
  - **Business rules:** None

3.3.2 Product Management

3.3.2.1 View Product List

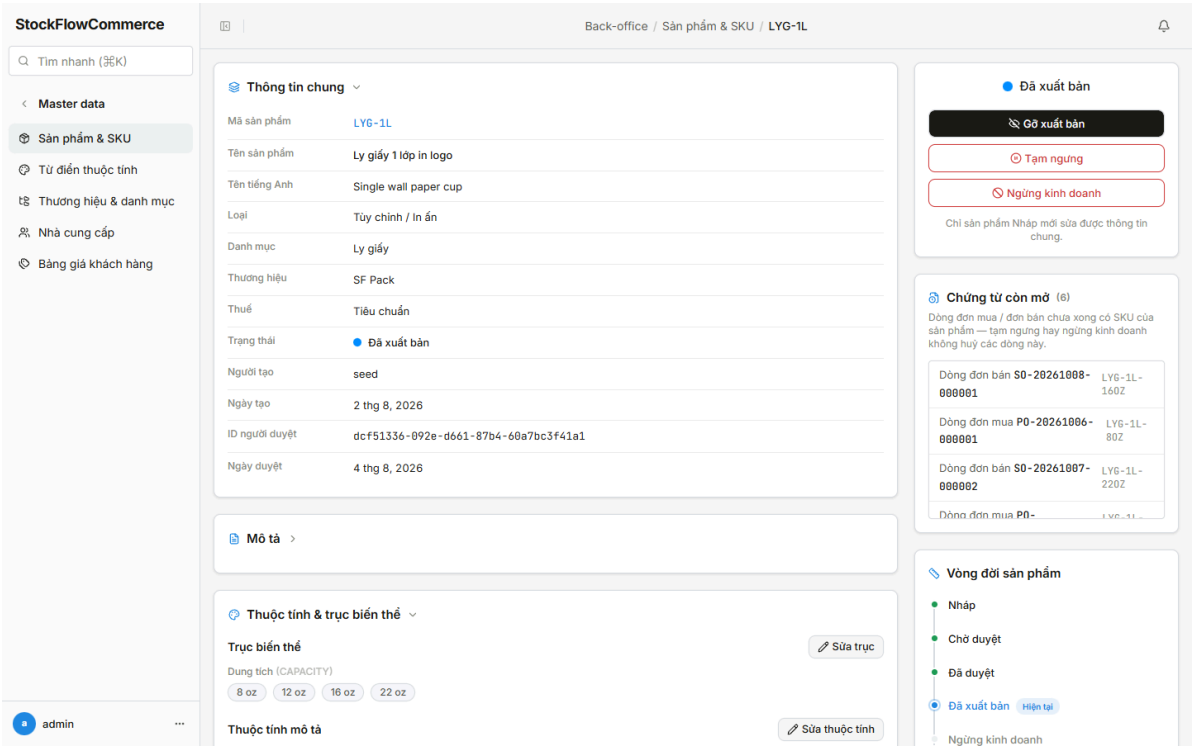
- **Function trigger:** Users open “Sản phẩm & SKU” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Browse, search and filter products.



- **Function details:**
  - **Data requirement:** Keyword (product code or name), status filter; columns product code, name, category, type, status.
  - **Validation:** None.
  - **Business rules:** None

3.3.2.2 View Product Detail

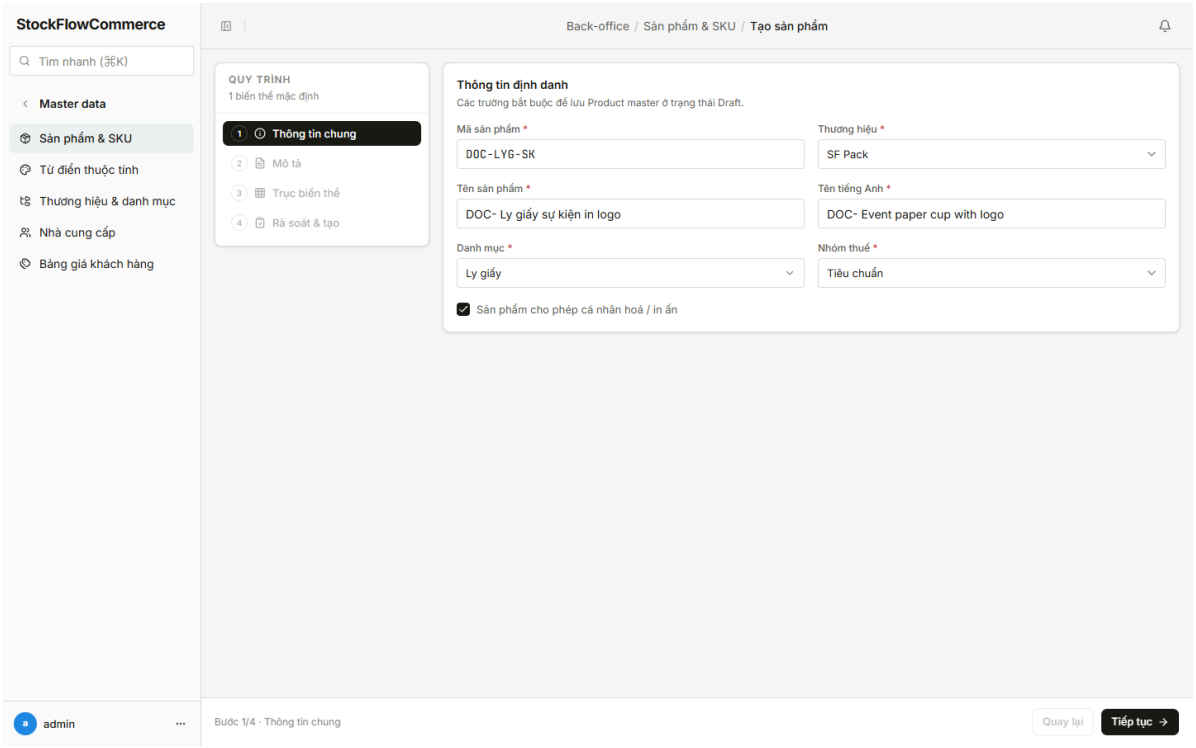
- **Function trigger:** Users click a product row or “Xem chi tiết” on the product list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Show all information and the lifecycle of one product.

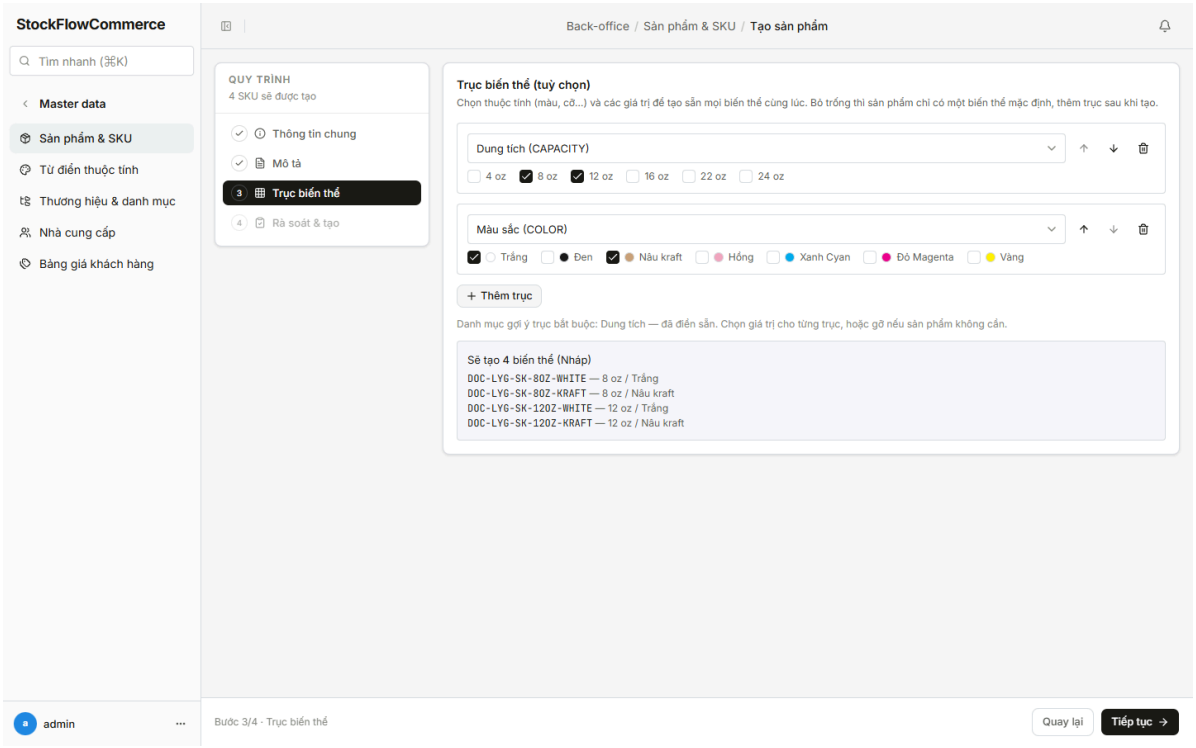


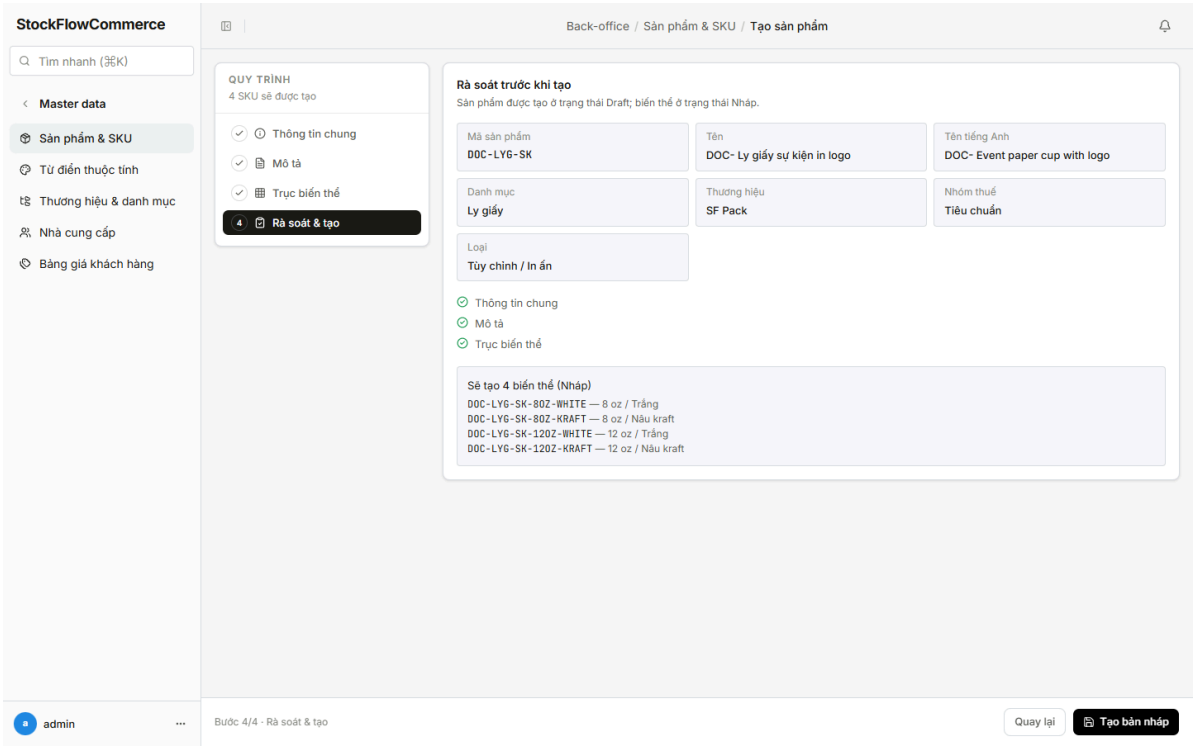
- **Function details:**
  - **Data requirement:** General info, description, attributes and axes, variants (SKU, status, base price, images, inventory, purchasing), store info (slug, SEO), change history, open references, lifecycle.
  - **Validation:** None.
  - **Business rules:** BR-15 (01 Product Creation)

3.3.2.3 Create Product

- **Function trigger:** Users click “Tạo sản phẩm” on the product list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Create a draft product, optionally with its variant matrix.


StockFlowCommerce
Tìm nhanh (⌘K)
Master data
Sản phẩm & SKU
Từ điển thuộc tính
Thương hiệu & danh mục
Nhà cung cấp
Bảng giá khách hàng
admin
Back-office / Sản phẩm & SKU / Tạo sản phẩm
QUY TRÌNH
1 biến thể mặc định
1 Thông tin chung
2 Mô tả
3 Trục biến thể
4 Rà soát & tạo
Thông tin định danh
Các trường bắt buộc để lưu Product master ở trạng thái Draft.
Mã sản phẩm *
DOC-LYG-SK
Thương hiệu *
SF Pack
Tên sản phẩm *
DOC- Ly giấy sự kiện in logo
Tên tiếng Anh *
DOC- Event paper cup with logo
Danh mục *
Ly giấy
Nhóm thuế *
Tiêu chuẩn
Sản phẩm cho phép cá nhân hoá / in ấn
Bước 1/4 · Thông tin chung
Quay lại
Tiếp tục →

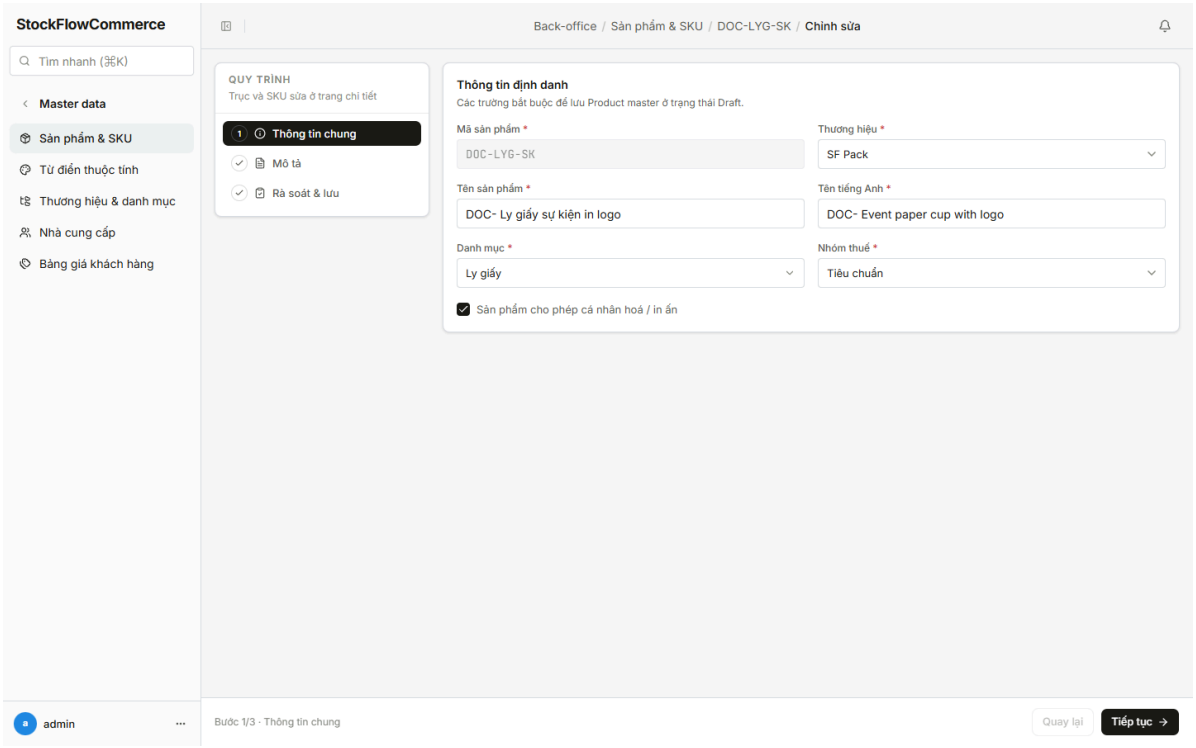

StockFlowCommerce
Tìm nhanh (⌘K)
Master data
Sản phẩm & SKU
Từ điển thuộc tính
Thương hiệu & danh mục
Nhà cung cấp
Bảng giá khách hàng
admin
Back-office / Sản phẩm & SKU / Tạo sản phẩm
QUY TRÌNH
4 SKU sẽ được tạo
Thông tin chung
Mô tả
3 Trục biến thể
4 Rà soát & tạo
Trục biến thể (tuỳ chọn)
Chọn thuộc tính (màu, cỡ...) và các giá trị để tạo sẵn mọi biến thể cùng lúc. Bỏ trống thì sản phẩm chỉ có một biến thể mặc định, thêm trục sau khi tạo.
Dung tích (CAPACITY)
4 oz
8 oz
12 oz
16 oz
22 oz
24 oz
Màu sắc (COLOR)
Trắng
Đen
Nâu kraft
Hồng
Xanh Cyan
Đỏ Magenta
Vàng
+ Thêm trục
Danh mục gợi ý trục bắt buộc: Dung tích — đã điền sẵn. Chọn giá trị cho từng trục, hoặc gỡ nếu sản phẩm không cần.
Sẽ tạo 4 biến thể (Nháp)
DOC-LYG-SK-8OZ-WHITE — 8 oz / Trắng
DOC-LYG-SK-8OZ-KRAFT — 8 oz / Nâu kraft
DOC-LYG-SK-12OZ-WHITE — 12 oz / Trắng
DOC-LYG-SK-12OZ-KRAFT — 12 oz / Nâu kraft
Bước 3/4 · Trục biến thể
Quay lại
Tiếp tục →



- **Function details:**
  - **Data requirement:** Product code, Vietnamese name, English name, brand, category, tax class, customizable flag, short description, Vietnamese/English descriptions, variant axes and values.
  - **Validation:** Code required, 1–50 letters, digits, _ or -, unique; name ≤255; English name ≤300; brand, category, tax class required; each axis ≥1 value; ≤100 combinations; generated SKUs ≤64 characters and unique.
  - **Business rules:** BR-01, BR-07 (01 Product Creation)

3.3.2.4 Edit Product

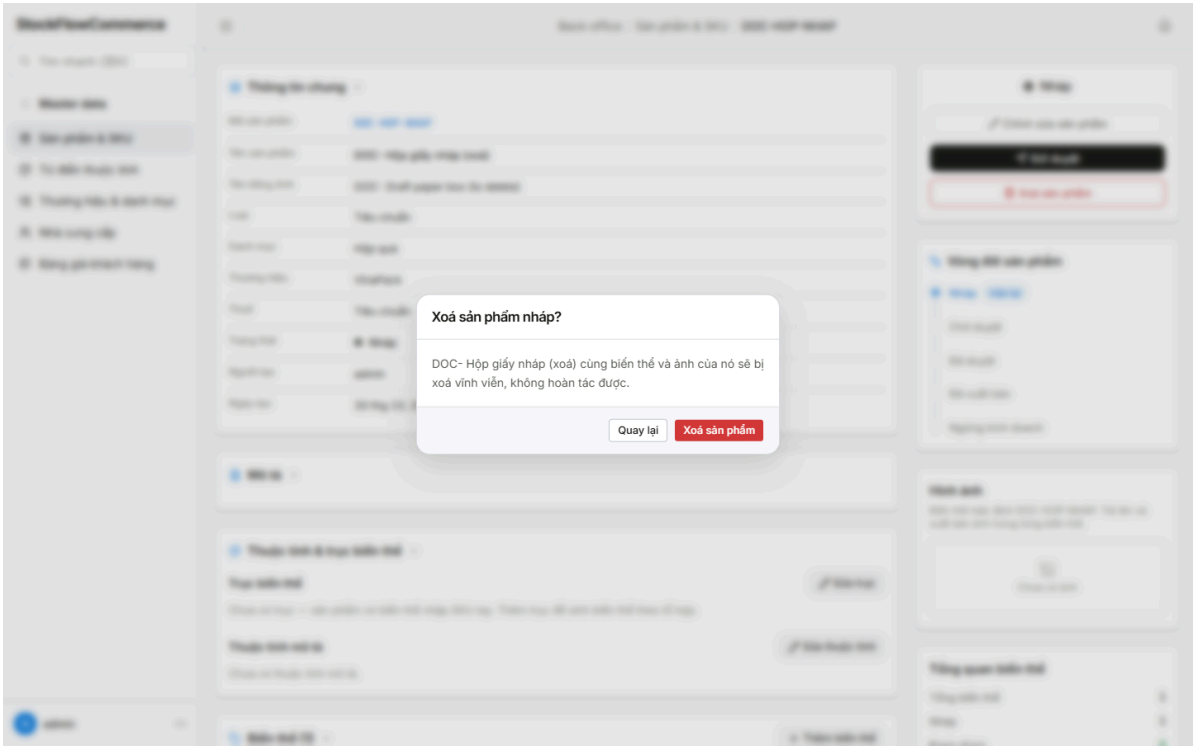
- **Function trigger:** Users click “Chỉnh sửa sản phẩm” on a Draft product's detail page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Update the main information of a draft product.



- **Function details:**
  - **Data requirement:** Vietnamese name, English name, brand, category, tax class, customizable flag, short description, Vietnamese/English descriptions.
  - **Validation:** Draft products only; name ≤255; English name ≤300; brand, category, tax class required; version must match (optimistic lock), otherwise the changed fields are shown.
  - **Business rules:** BR-13, BR-15 (01 Product Creation)

3.3.2.5 Delete Draft Product

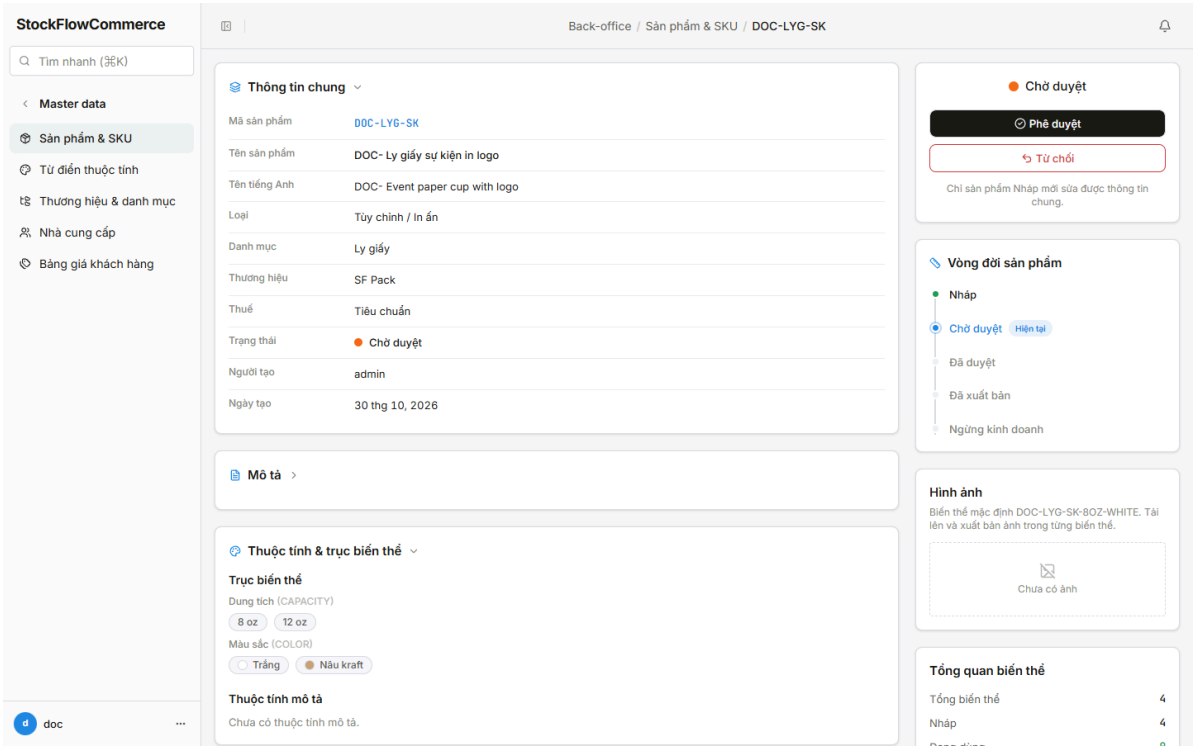
- **Function trigger:** Users click “Xoá sản phẩm” on a Draft product's detail page and confirm.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Permanently remove an unused draft product.

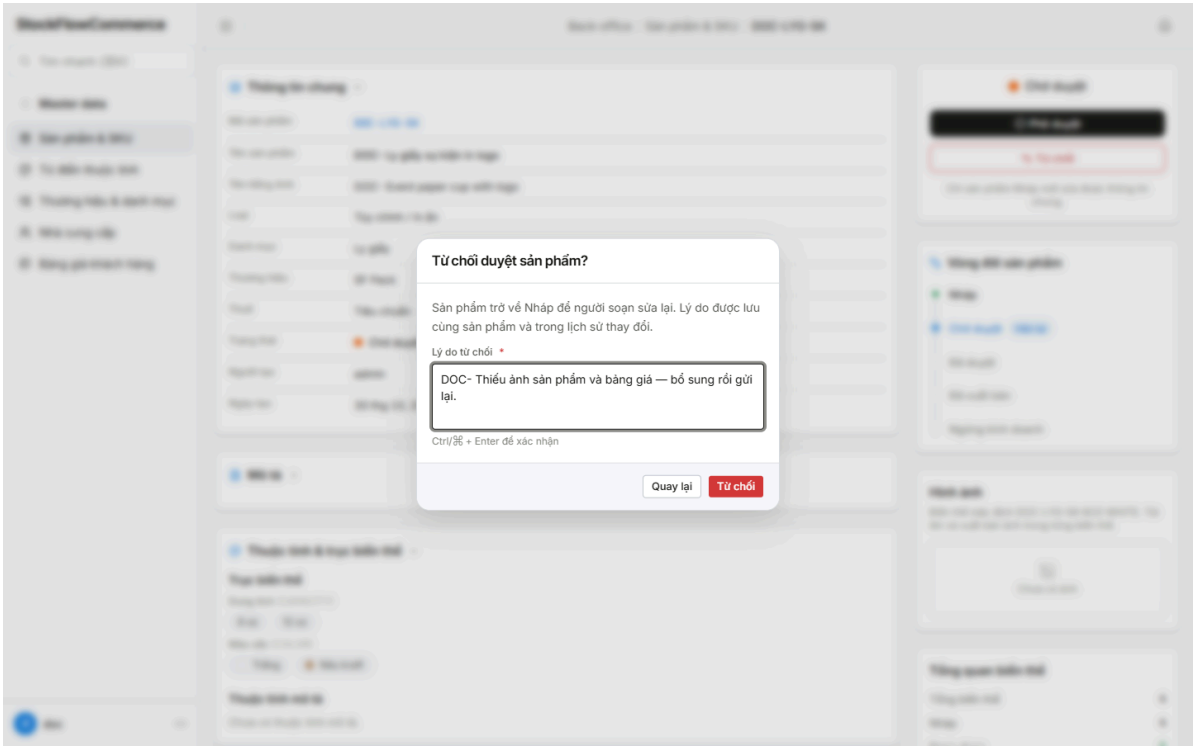


- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Product must be Draft; no SKU may be referenced by any document; requires product delete permission.
  - **Business rules:** BR-02 (01 Product Creation)

3.3.2.6 Approve Product

- **Function trigger:** Users click “Phê duyệt” or “Từ chối” on a Pending Approval product's detail page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Approve or reject a product submitted for approval.

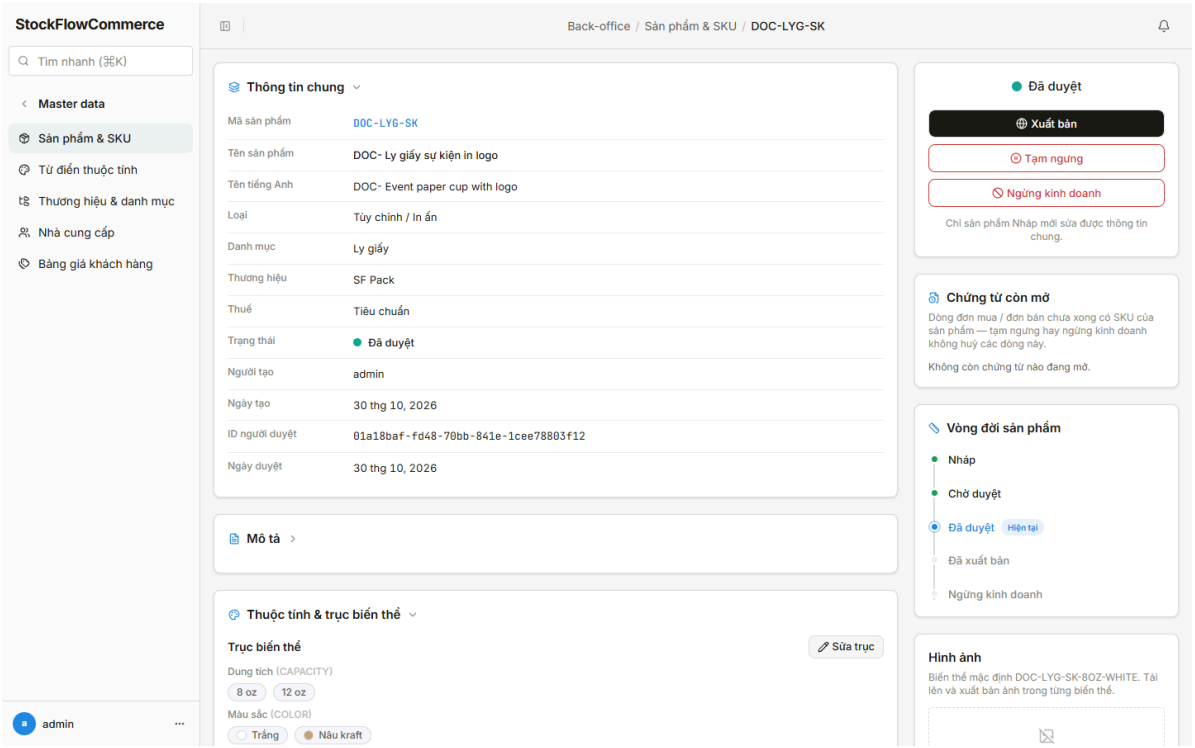




- **Function details:**
  - **Data requirement:** Rejection reason (reject only).
  - **Validation:** Product must be Pending Approval; approver must differ from submitter; rejection reason required.
  - **Business rules:** BR-12, BR-13 (01 Product Creation)

3.3.2.7 Publish Product

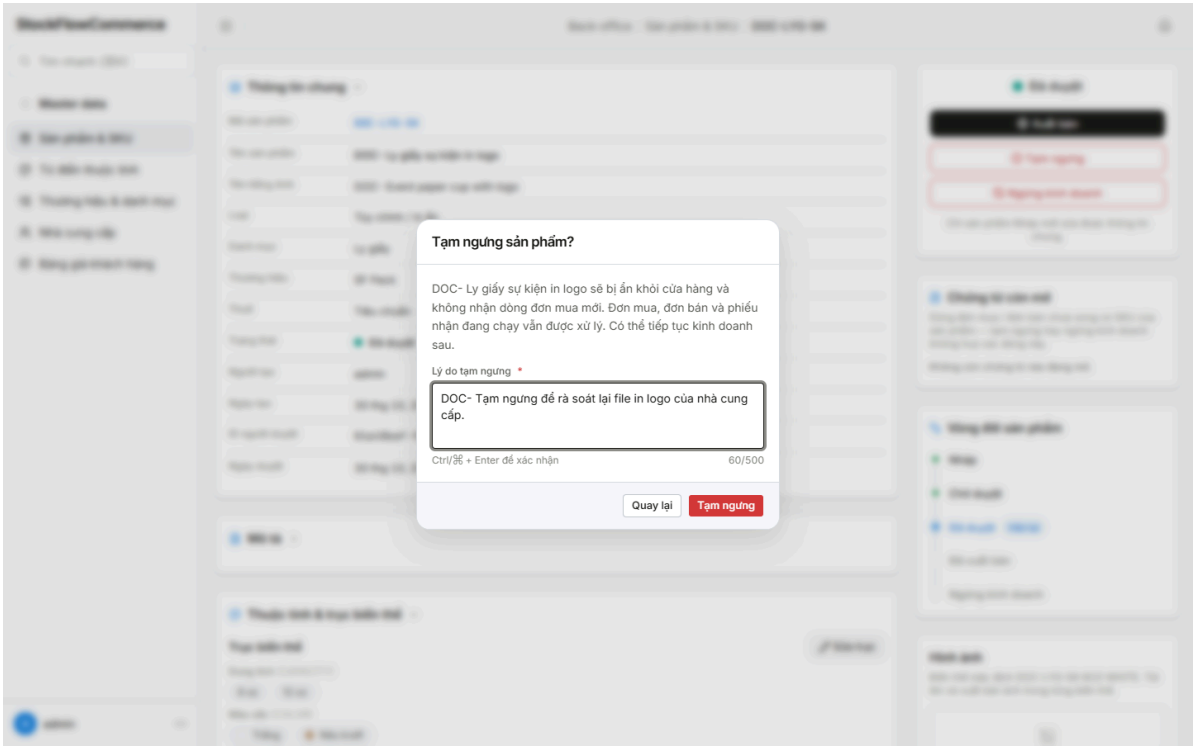
- **Function trigger:** Users click “Xuất bản” on an Approved product or “Gỡ xuất bản” on a Published product.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Show the product on, or hide it from, the customer catalog.

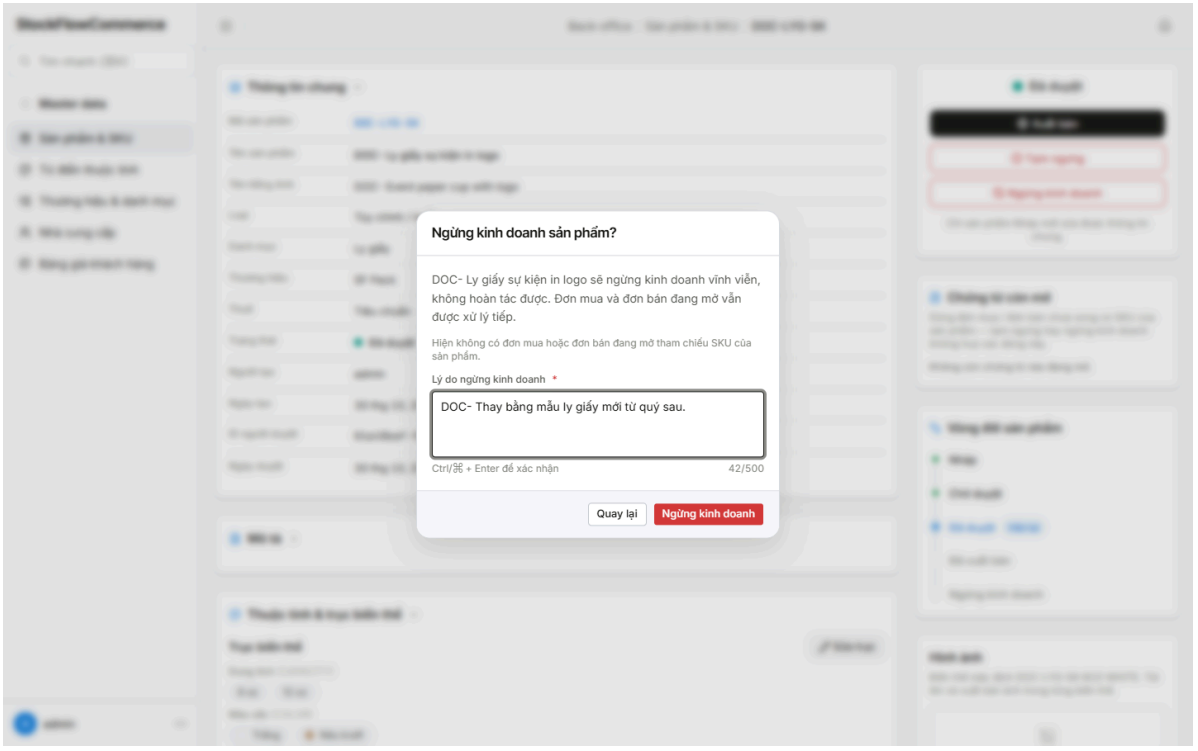


- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Store listing must exist; product Approved; category assigned; at least one active SKU; every SKU priced; at least one published image.
  - **Business rules:** BR-04, BR-05 (01 Product Creation); BR-01, BR-08 (13 Catalog)

3.3.2.8 Suspend or Discontinue Product

- **Function trigger:** Users click “Tạm ngưng”, “Tiếp tục kinh doanh” or “Ngừng kinh doanh” on the product detail page and confirm.
- **Function description:**
  - **Actors:** System Admin
  - **Purpose:** Suspend, resume or permanently discontinue a product.

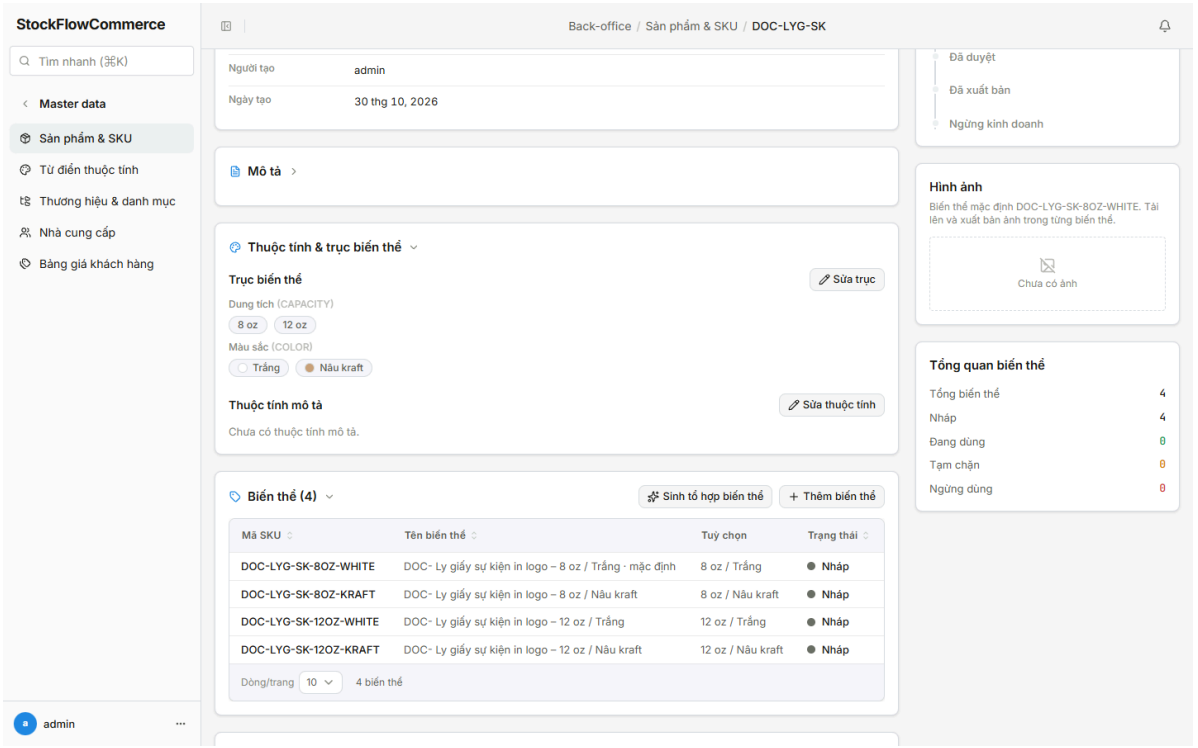


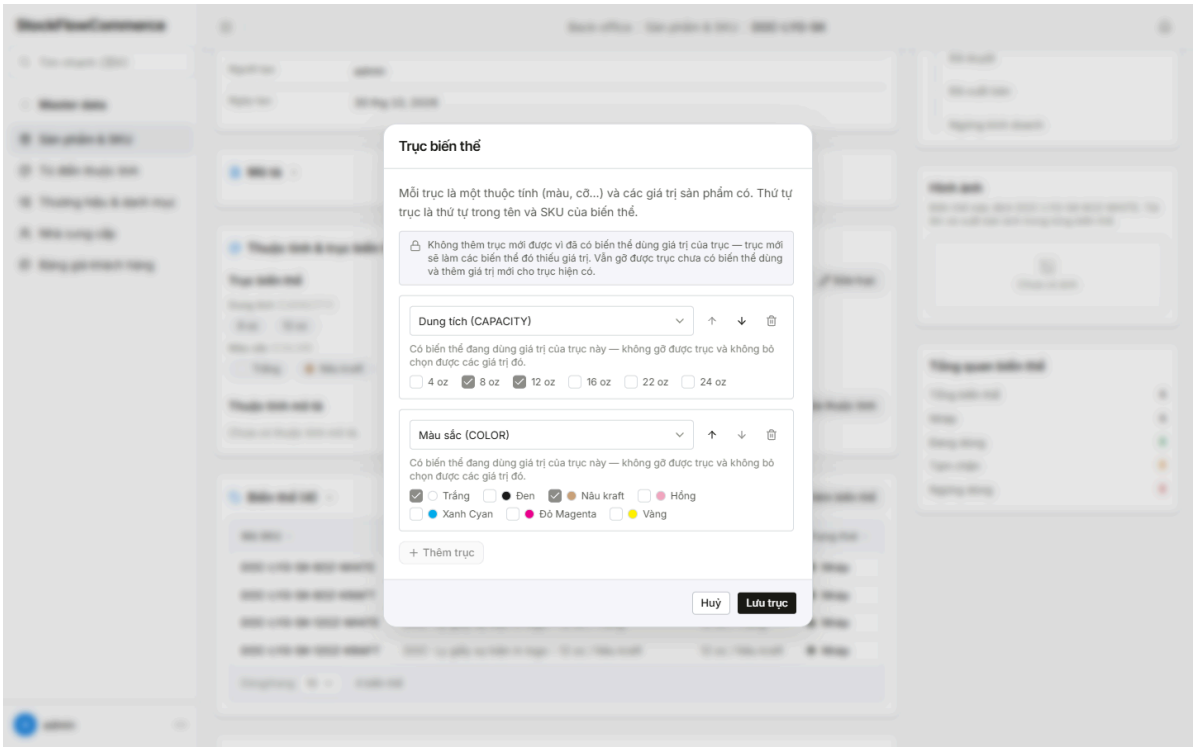


- **Function details:**
  - **Data requirement:** Reason (suspend and discontinue).
  - **Validation:** Suspend: Approved/Published only; Resume: Inactive only; Discontinue: Approved/Published/Inactive; reason required, ≤500 characters; version must match.
  - **Business rules:** BR-14, BR-15 (01 Product Creation)

3.3.2.9 Manage Product Variants

- **Function trigger:** Users click “Sửa trục”, “Sinh tổ hợp biến thể” or “Thêm biến thể” on the product detail page, or “Tạm chặn” in a variant panel.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Define variant axes, generate or add variants, and block variants.



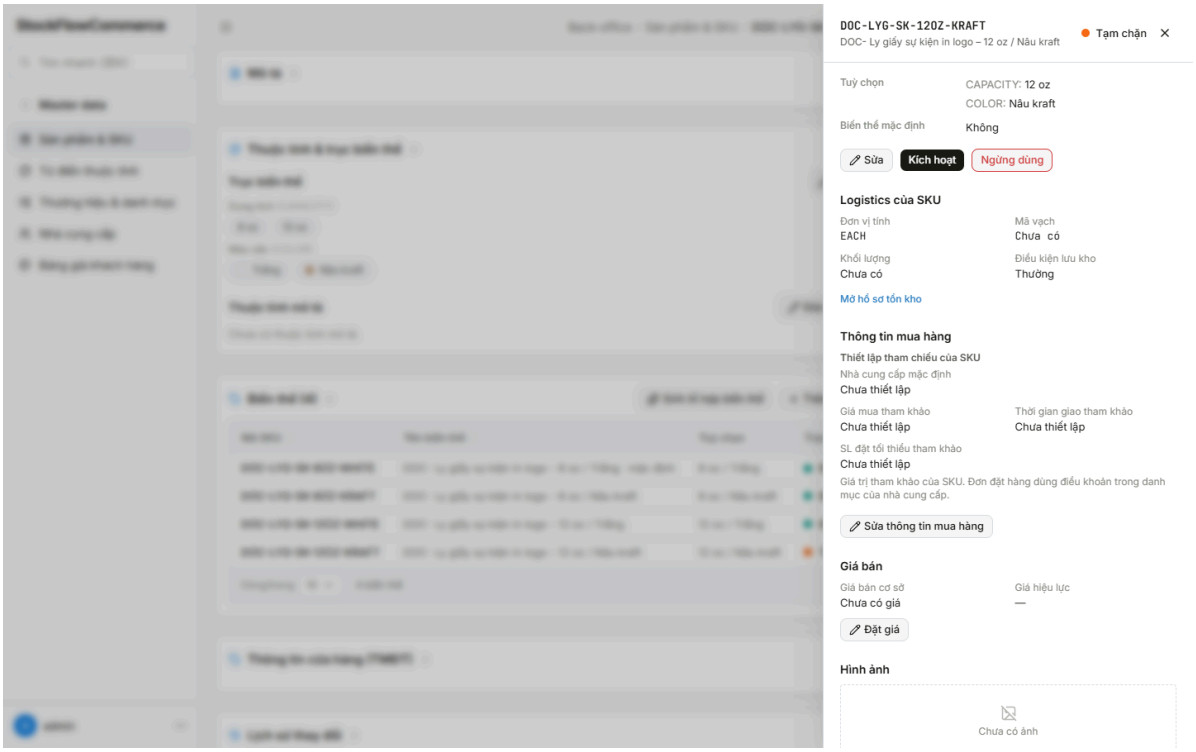


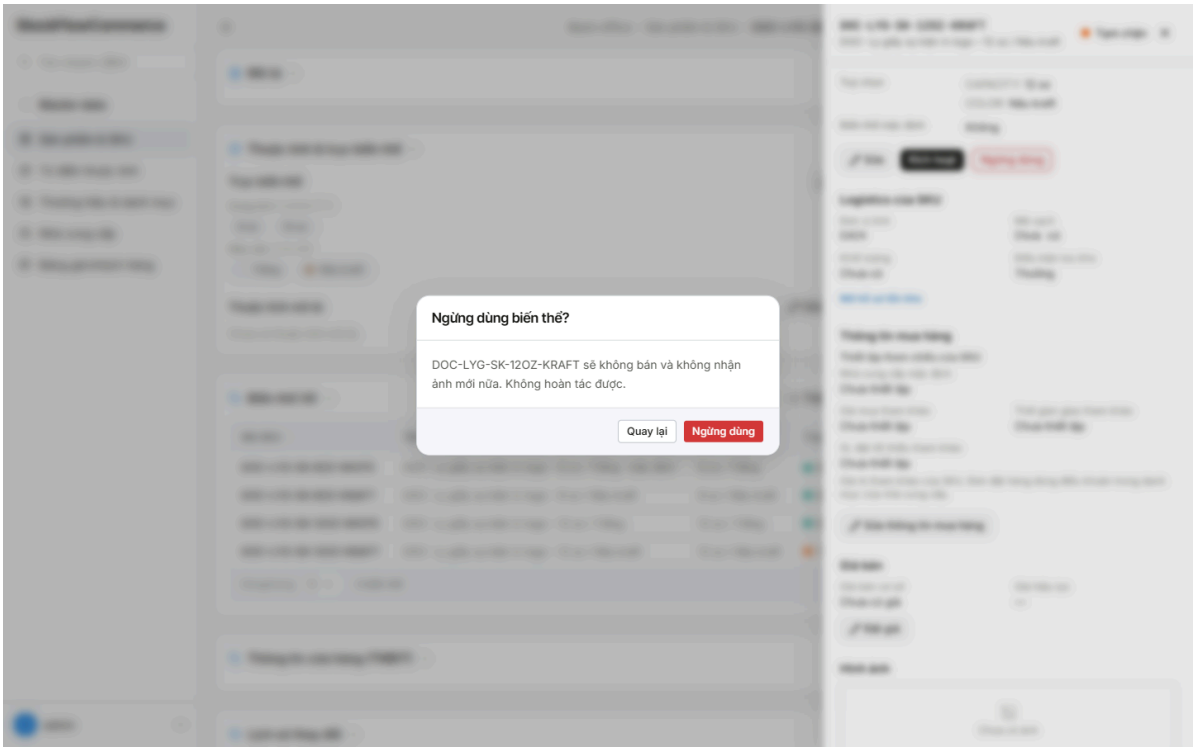
- **Function details:**

- **Data requirement:** Axis attributes and values, SKU, variant name, one value per axis.
- **Validation:** Axes only SELECT/COLOR, each attribute once, ≥1 value; adding axes only in Draft; ≤100 combinations per run; SKU ≤64 characters, unique; without axes SKU and name (≤255) required.
- **Business rules:** BR-01, BR-07 (01 Product Creation)

3.3.2.10 Activate Variant

- **Function trigger:** Users open a variant from the product's variant list and click “Kích hoạt” or “Ngừng dùng”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Activate a variant or retire it permanently.



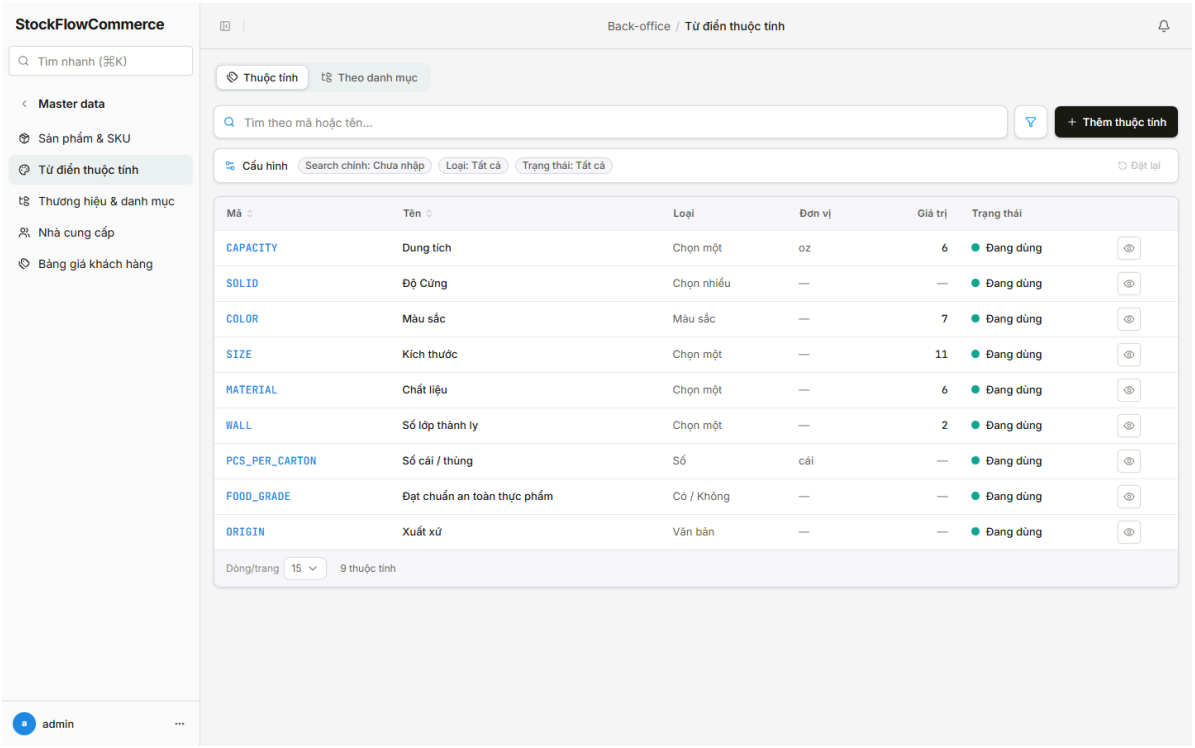


- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Activate: variant Draft or Blocked and product Approved/Published; Obsolete: from Draft, Active or Blocked; the default variant cannot be made obsolete.
  - **Business rules:** BR-01, BR-04 (01 Product Creation)

3.3.3 Attribute Dictionary

3.3.3.1 View Attribute Dictionary

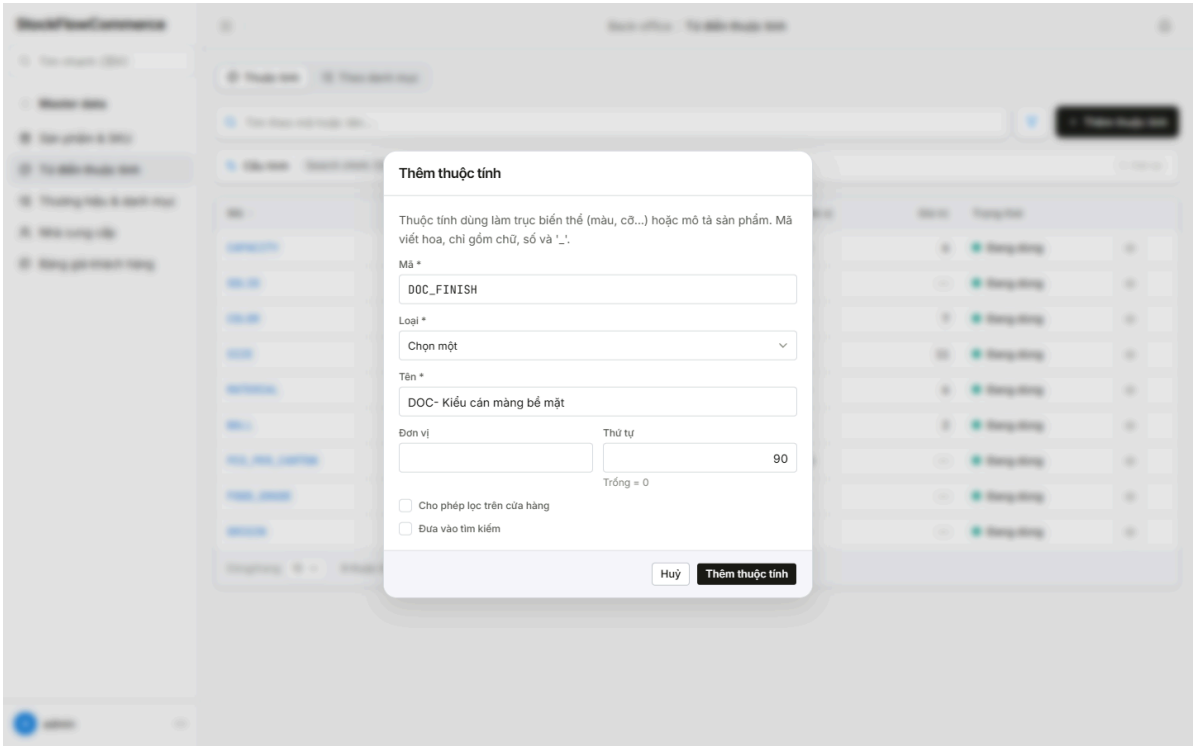
- **Function trigger:** Users open “Từ điển thuộc tính” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Browse product attributes and their values.



- **Function details:**
  - **Data requirement:** Keyword (code or name), type and status filters; code, name, type, unit, values, status; “Theo danh mục” tab shows attributes per category.
  - **Validation:** None.
  - **Business rules:** §4.1 (01 Product Creation)

3.3.3.2 Create Attribute

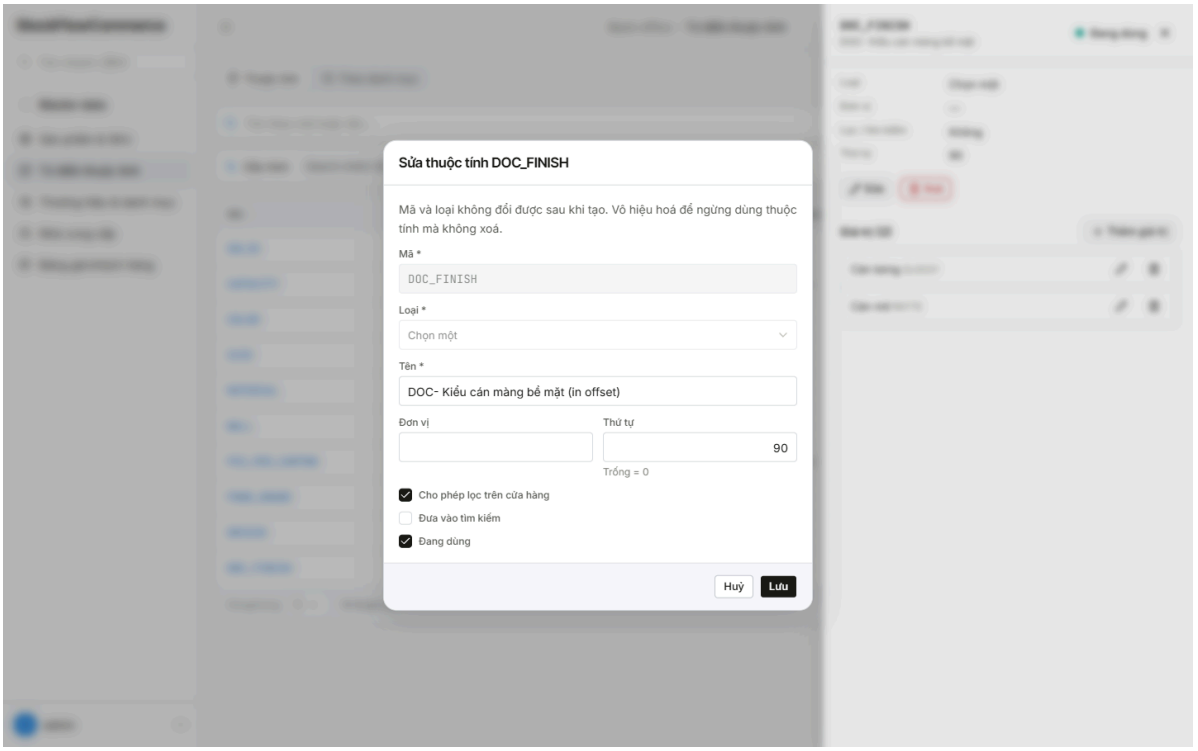
- **Function trigger:** Users click “Thêm thuộc tính” on the attribute dictionary.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Add a new attribute to the dictionary.

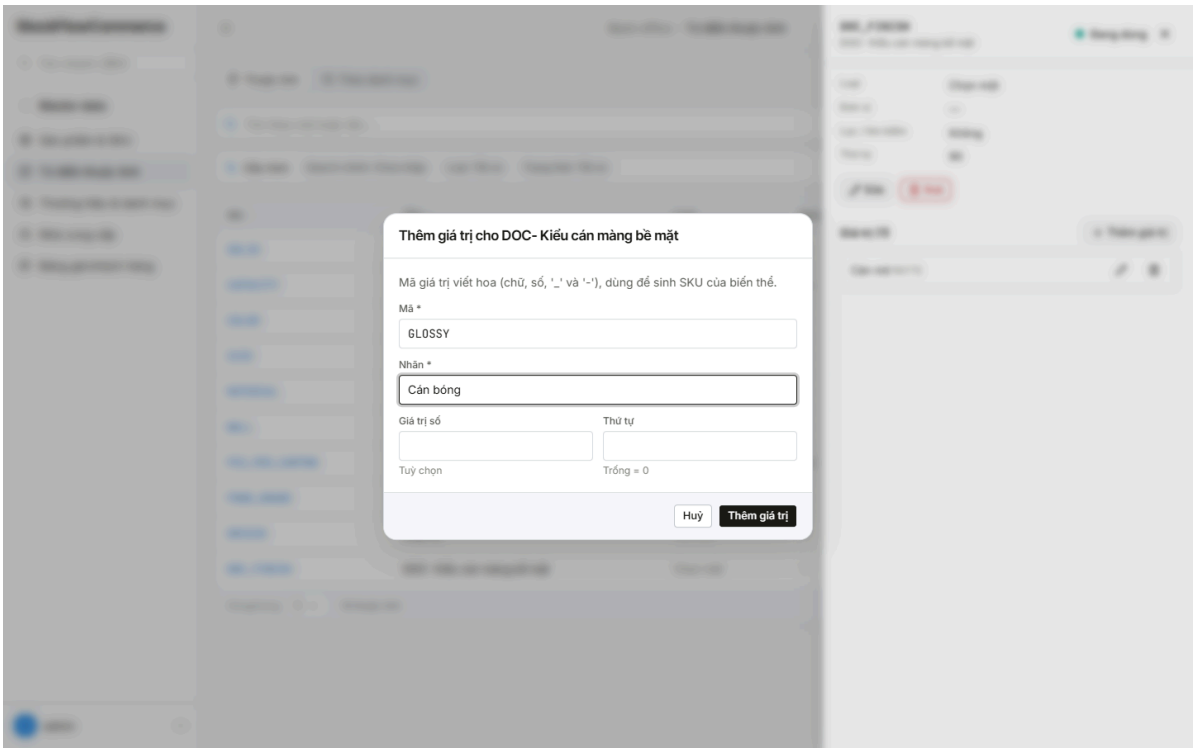


- **Function details:**
  - **Data requirement:** Code, data type, name, unit, filterable, searchable, sort order, active.
  - **Validation:** Code required, letters/digits/_ up to 50, starts with letter or digit, unique; data type required; name required ≤255; unit ≤20; sort order integer.
  - **Business rules:** §4.1 (01 Product Creation)

3.3.3.3 Edit Attribute

- **Function trigger:** Users open an attribute and click “Sửa” or “Thêm giá trị”, or edit a category in the “Theo danh mục” tab and click “Lưu bộ thuộc tính”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Update an attribute, its values and category attribute templates.

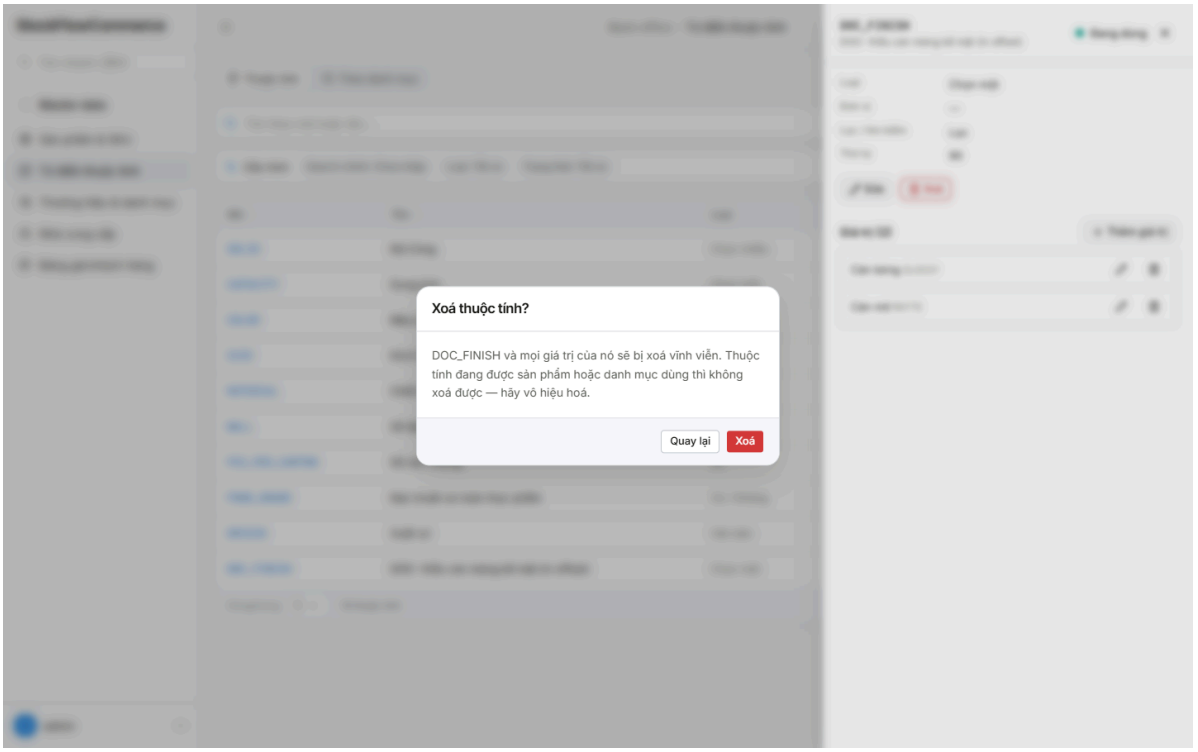




- **Function details:**
  - **Data requirement:** Attribute name, unit, flags, sort order, active; value code, label, number, hex colour; category attributes with role, required flag, order.
  - **Validation:** Name ≤255; unit ≤20; value code ≤50 (letters, digits, _, -); label ≤255; COLOR needs #RRGGBB; number ≤14 integer, 4 decimal digits; attribute once per category.
  - **Business rules:** BR-02 (01 Product Creation)

3.3.3.4 Delete Attribute

- **Function trigger:** Users open an attribute and click “Xoá”, then confirm.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Permanently delete an attribute that is not in use.

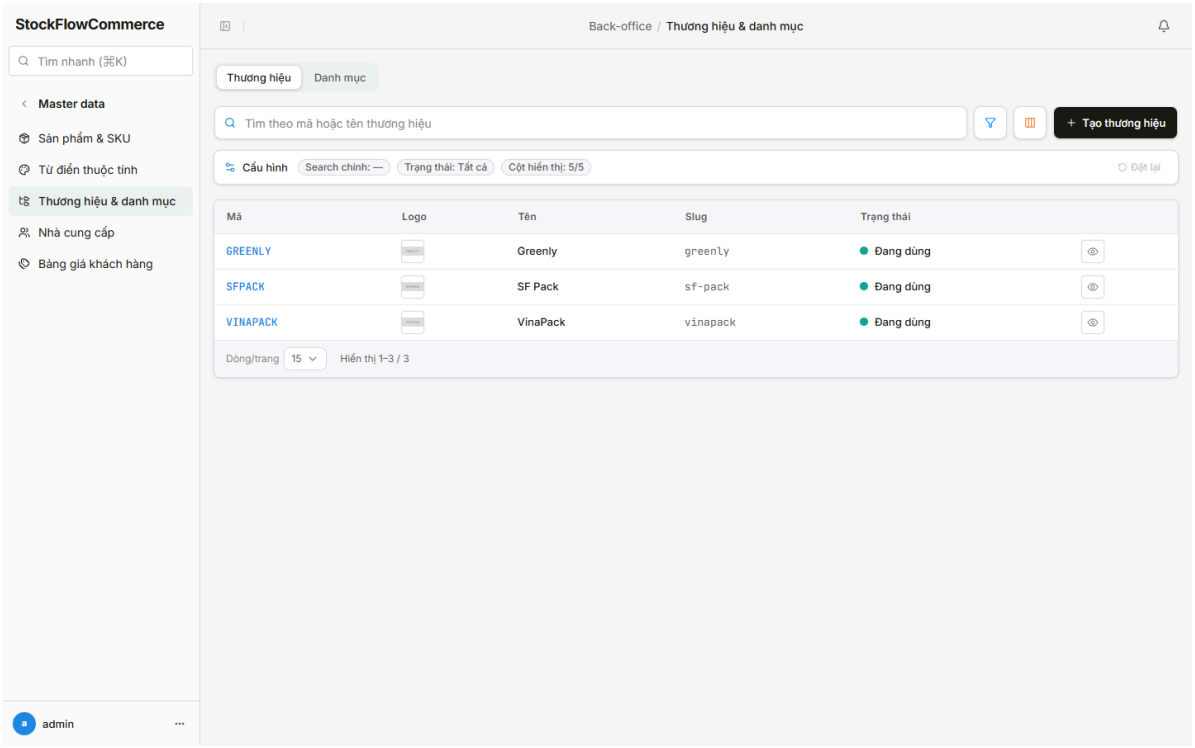


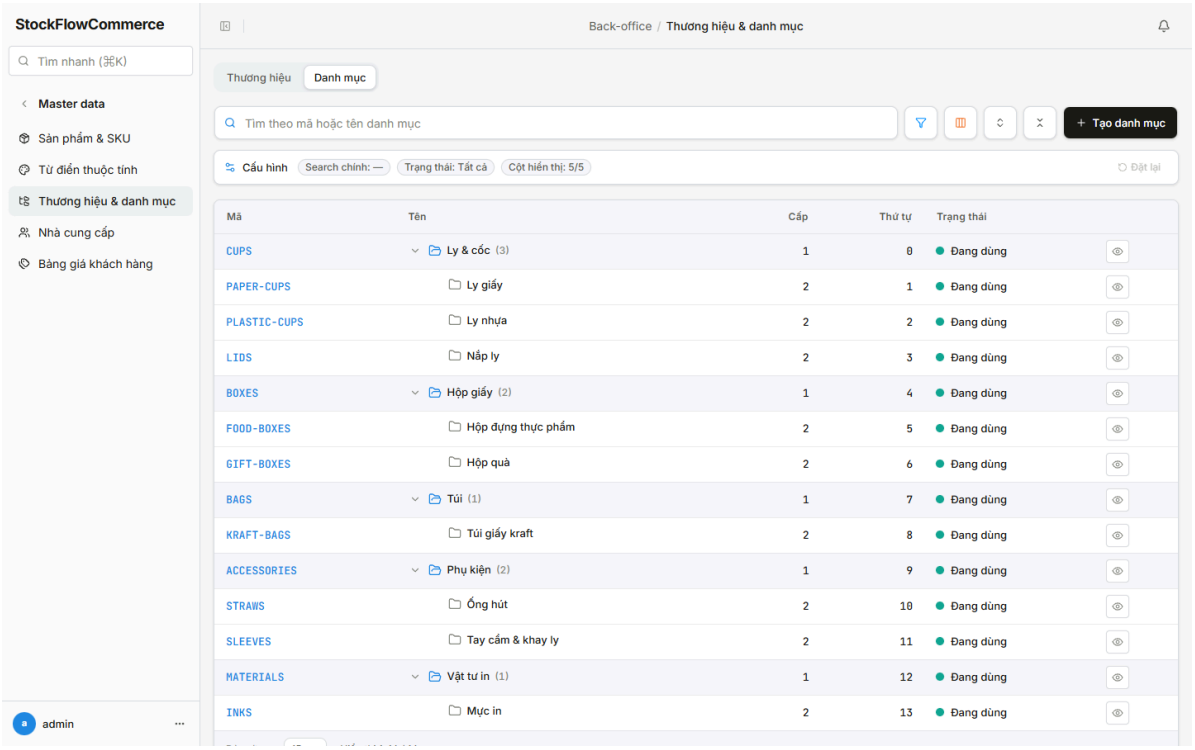
- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Attribute must not be used by any product or category template.
  - **Business rules:** BR-02 (01 Product Creation)

3.3.4 Brands & Categories

3.3.4.1 View Brands and Categories

- **Function trigger:** Users open “Thương hiệu & danh mục” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Browse brands and the category tree.

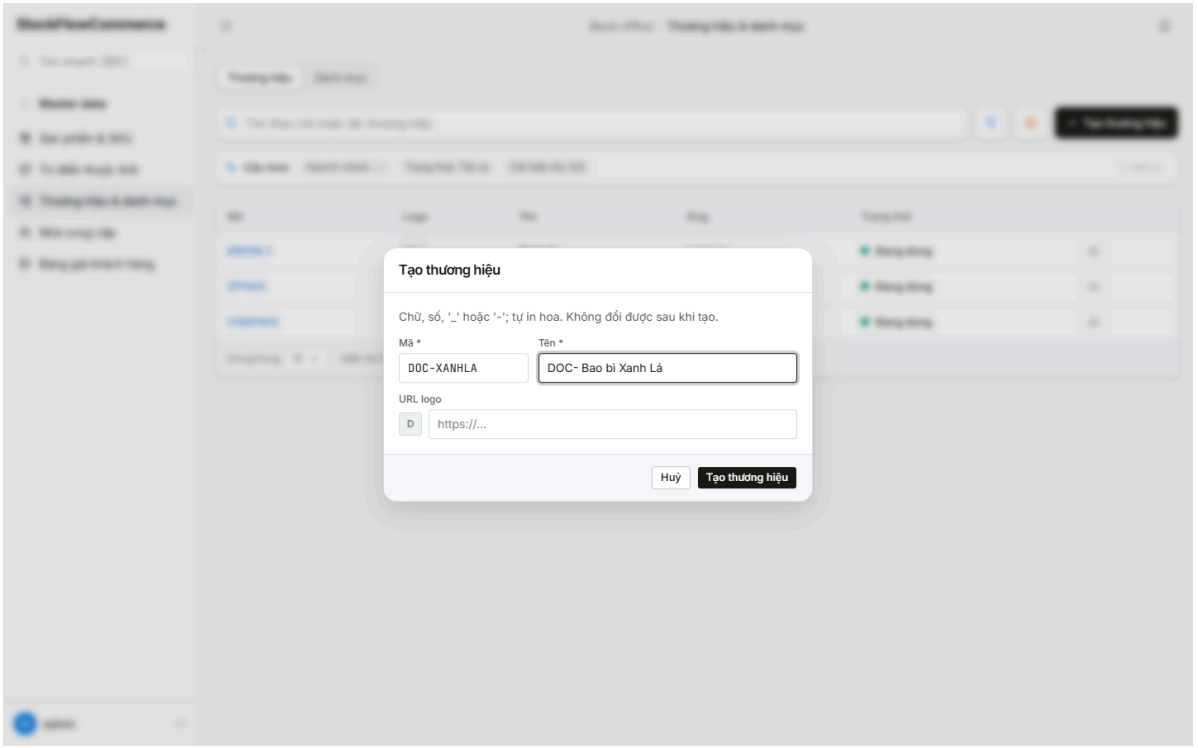


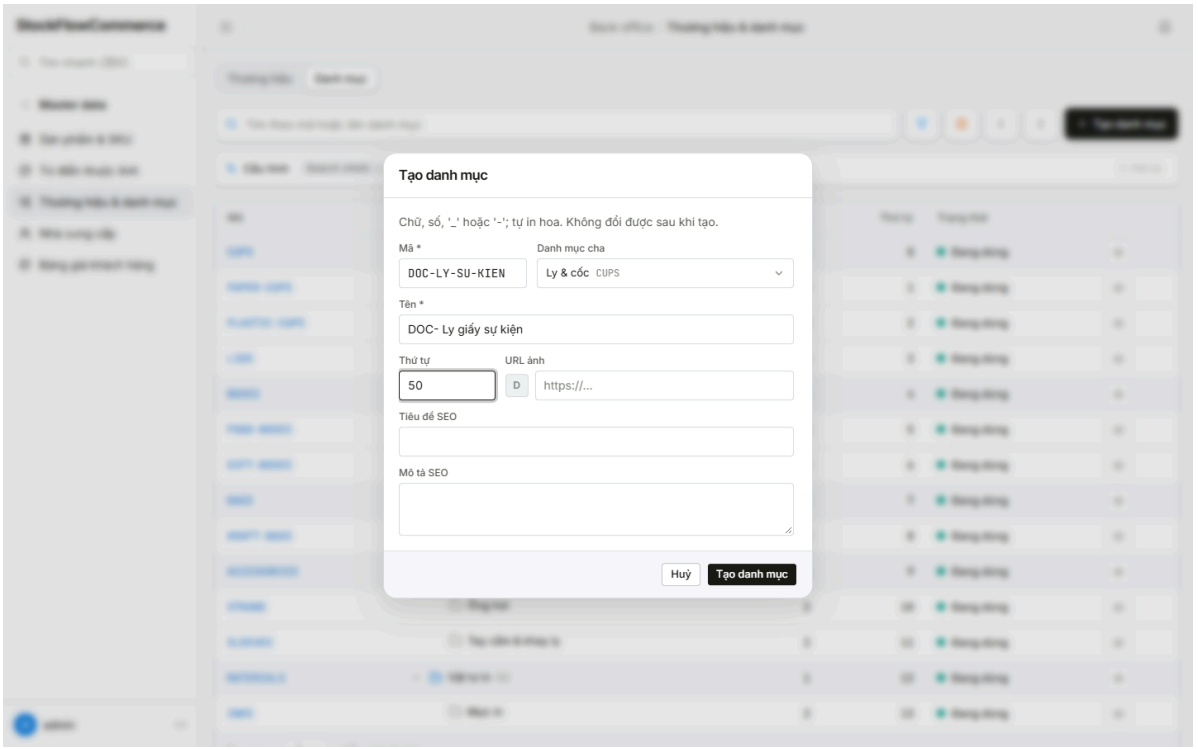


- **Function details:**
  - **Data requirement:** Tabs “Thương hiệu” / “Danh mục”; keyword (code or name), status filter; code, name, slug, logo or image, parent, level, status.
  - **Validation:** None.
  - **Business rules:** §4.1 (01 Product Creation)

3.3.4.2 Create Brand or Category

- **Function trigger:** Users click “Tạo thương hiệu” or “Tạo danh mục” on the brands & categories page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Add a new brand or category.



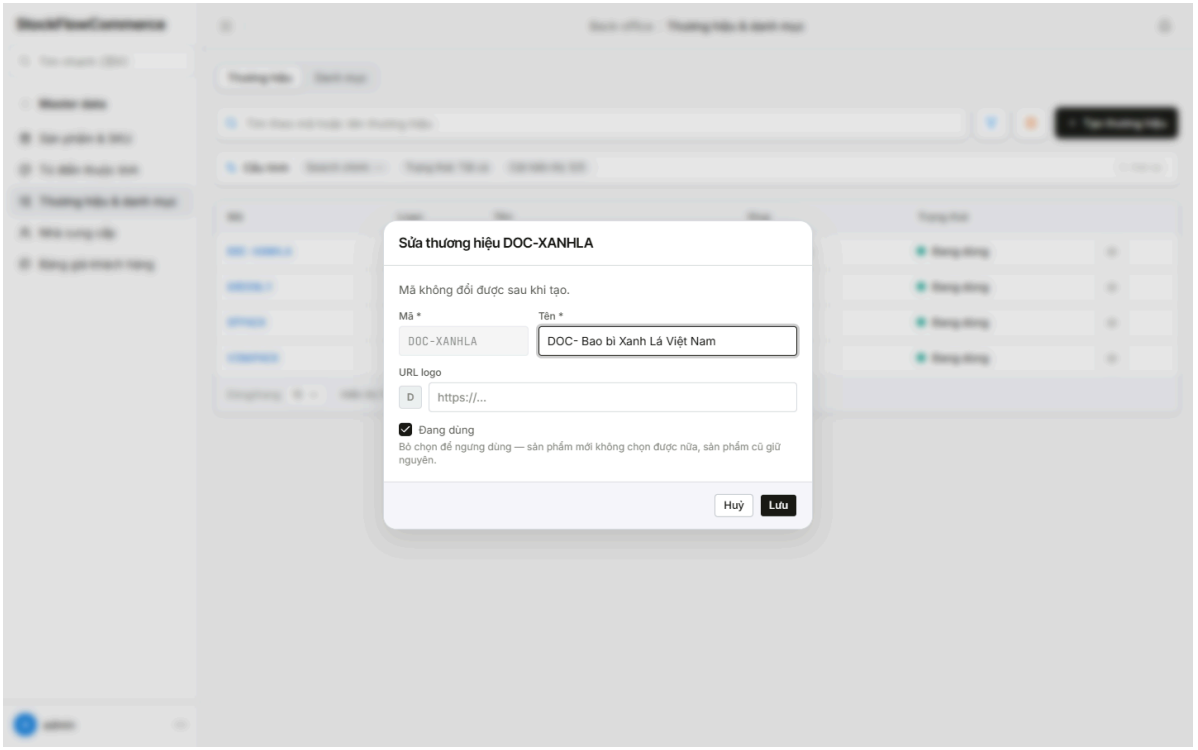


- **Function details:**

- **Data requirement:** Brand: code, name, logo URL. Category: code, parent category, name, sort order, image URL, SEO title, SEO description.
- **Validation:** Code required, letters/digits/_/- up to 50, auto upper-cased, unique (slug too); name required ≤255; sort order integer; SEO title ≤255; tree at most 11 levels.
- **Business rules:** §4.1 (01 Product Creation)

3.3.4.3 Edit Brand or Category

- **Function trigger:** Users click a brand or category row (or “Xem”) and save with “Lưu”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Update a brand or category, including deactivating it.

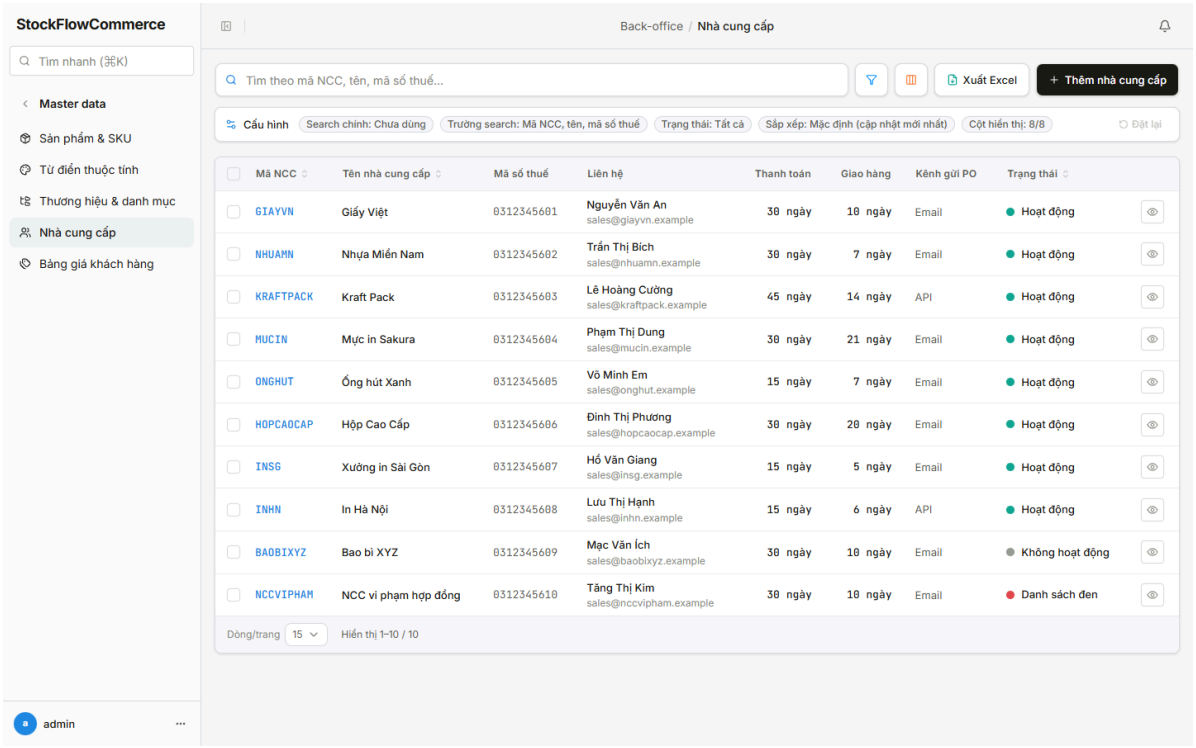


- **Function details:**
  - **Data requirement:** Brand: name, logo URL, active. Category: name, sort order, image URL, SEO title, SEO description, active.
  - **Validation:** Name required ≤255; sort order integer; SEO title ≤255; data changed by someone else is rejected.
  - **Business rules:** §4.1 (01 Product Creation)

3.3.5 Supplier Management

3.3.5.1 View Supplier List

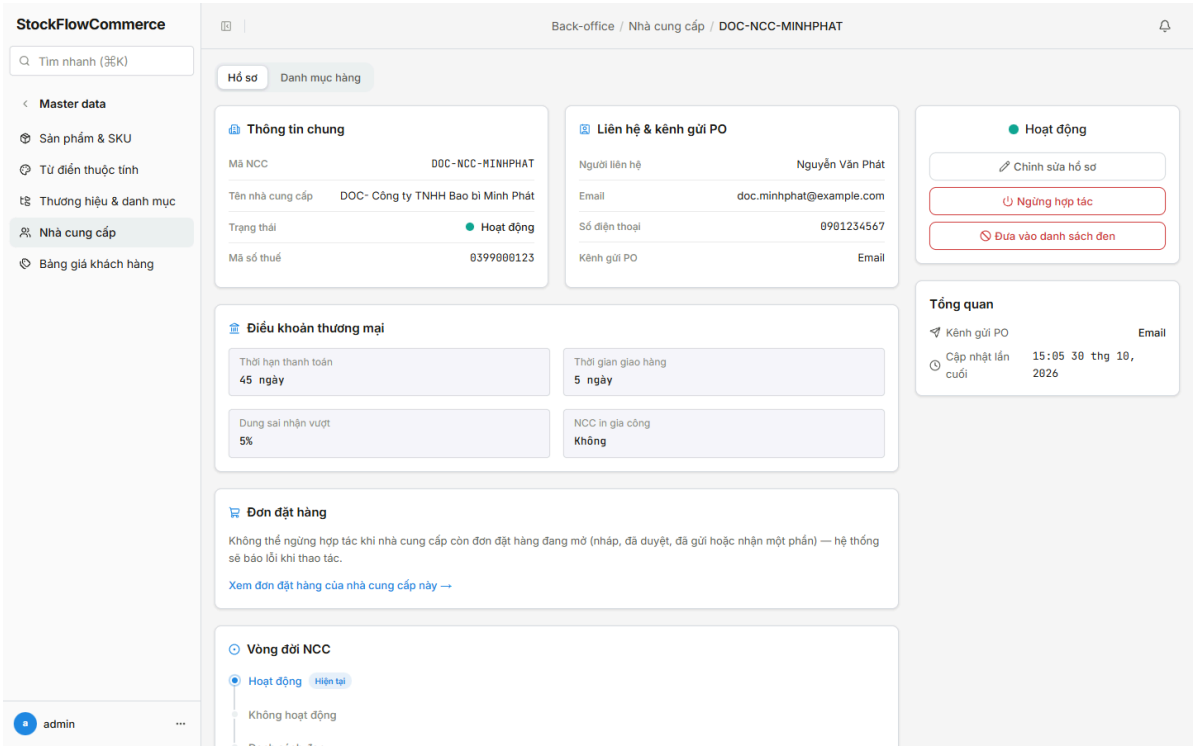
- **Function trigger:** Users open “Nhà cung cấp” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Procurement Staff, Accountant
  - **Purpose:** Search and filter suppliers.



- **Function details:**
  - **Data requirement:** Keyword (supplier code, name, tax code), status filter; code, name, tax code, contact, payment term, lead time, PO channel, status.
  - **Validation:** None.
  - **Business rules:** None

3.3.5.2 View Supplier Detail

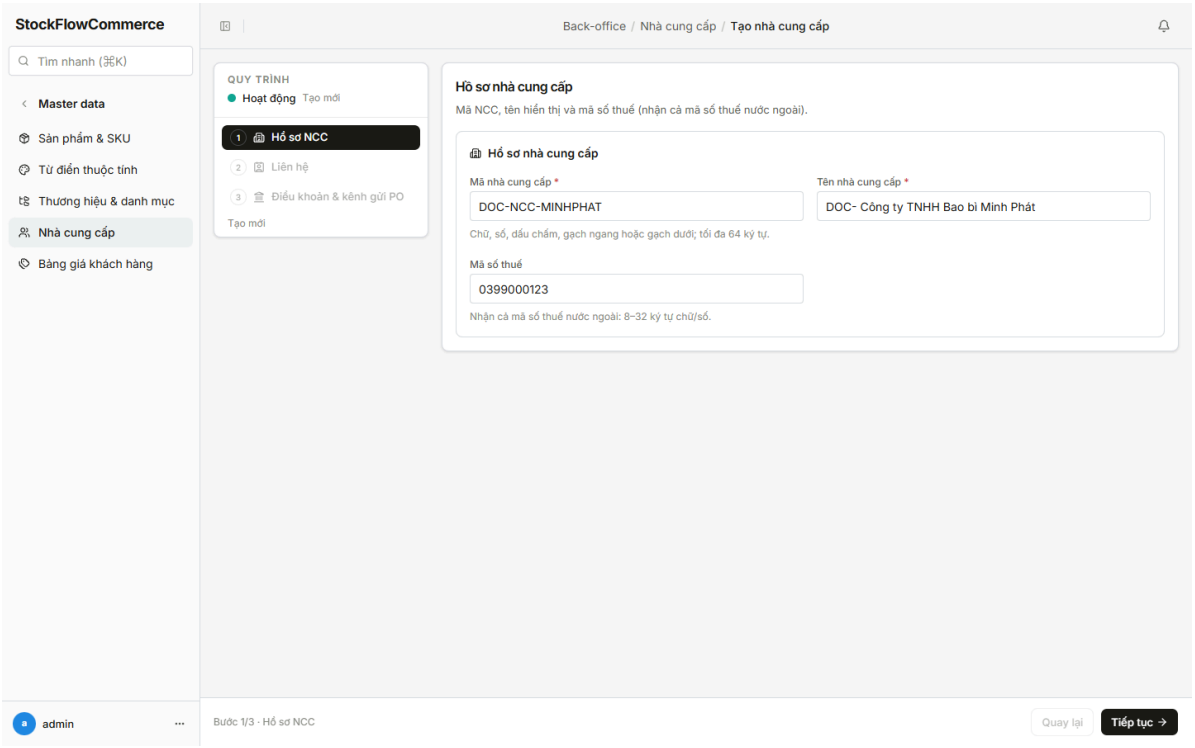
- **Function trigger:** Users click a supplier row or “Xem chi tiết” on the supplier list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Procurement Staff, Accountant
  - **Purpose:** Show a supplier's profile and status.

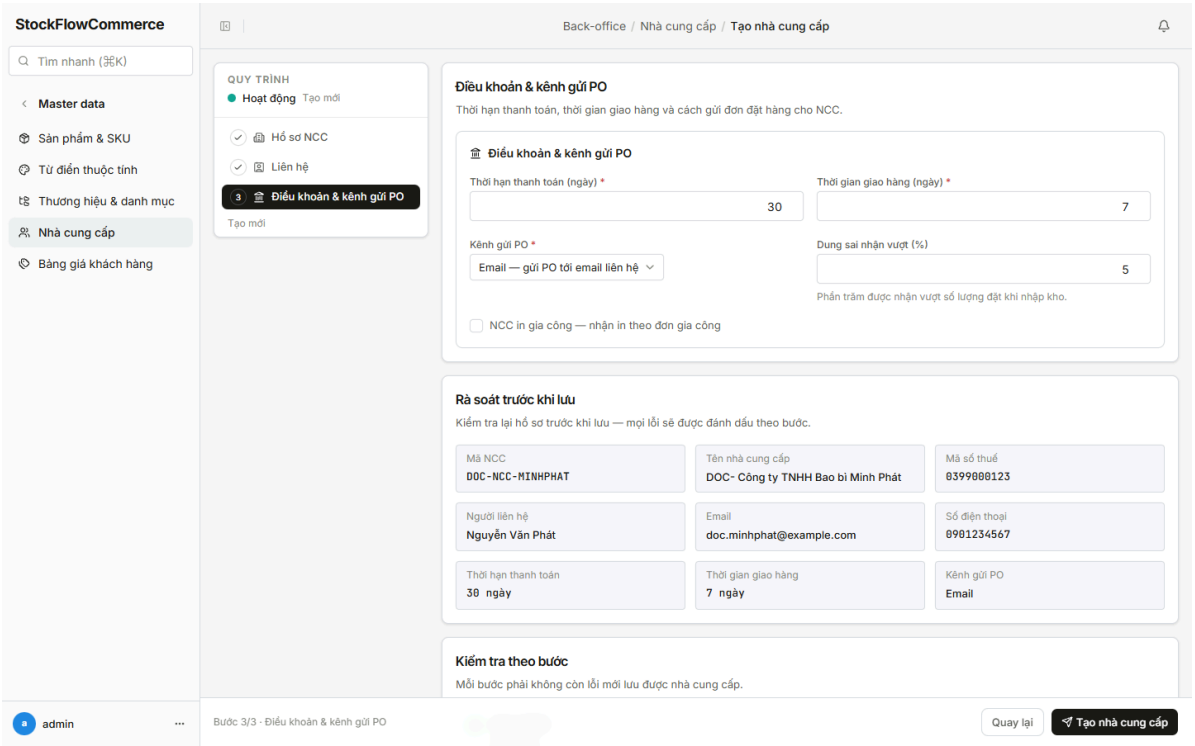


- **Function details:**
  - **Data requirement:** Code, name, status, tax code, contact name, email, phone, PO channel, API endpoint, payment term, lead time, over-receipt tolerance, print subcontractor flag, lifecycle.
  - **Validation:** None.
  - **Business rules:** BR-11 (02 Purchase Order)

3.3.5.3 Create Supplier

- **Function trigger:** Users click “Thêm nhà cung cấp” on the supplier list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Procurement Staff
  - **Purpose:** Create a new supplier profile.

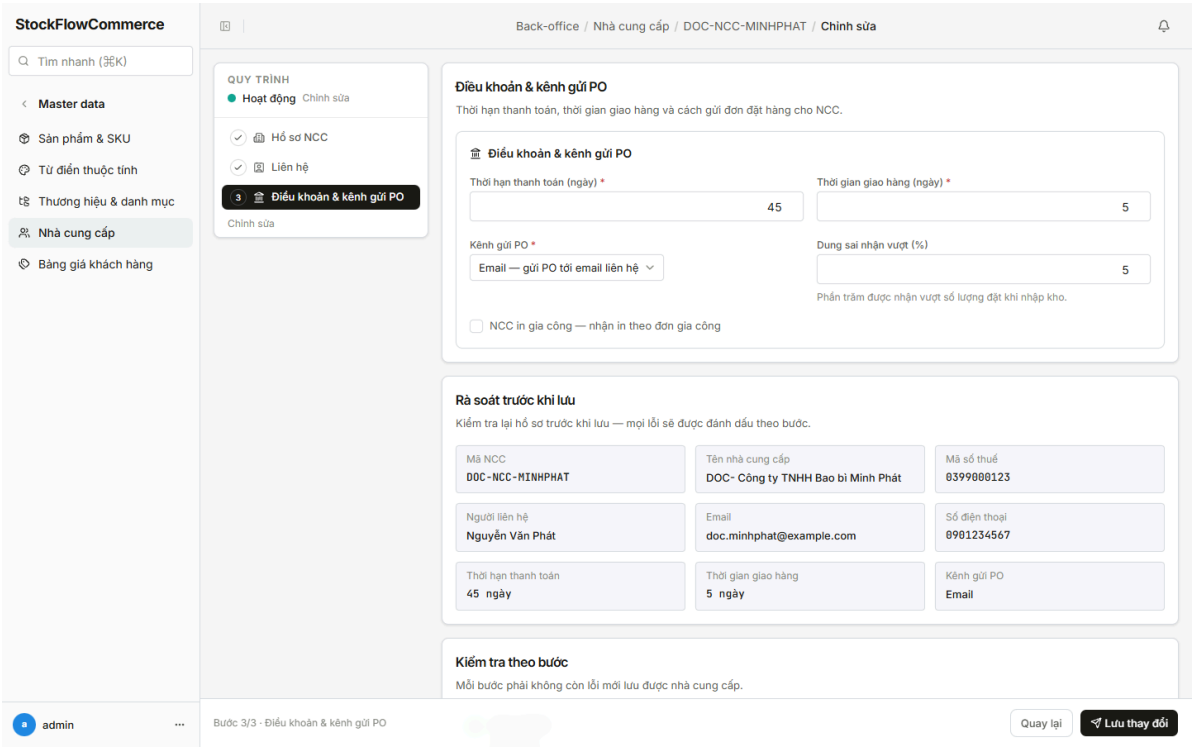




- **Function details:**
  - **Data requirement:** Code, name, tax code, contact name, email, phone, payment term days, lead time days, PO channel, API endpoint, over-receipt tolerance %, print subcontractor flag, loss tolerance %.
  - **Validation:** Code ≤30, unique; name ≤200; tax code 8–30 characters, unique; valid email ≤150; phone 8–15 digits; terms 0–365 days; percentages 0–100; EMAIL channel needs email; API needs HTTPS endpoint.
  - **Business rules:** BR-11 (02 Purchase Order)

3.3.5.4 Edit Supplier

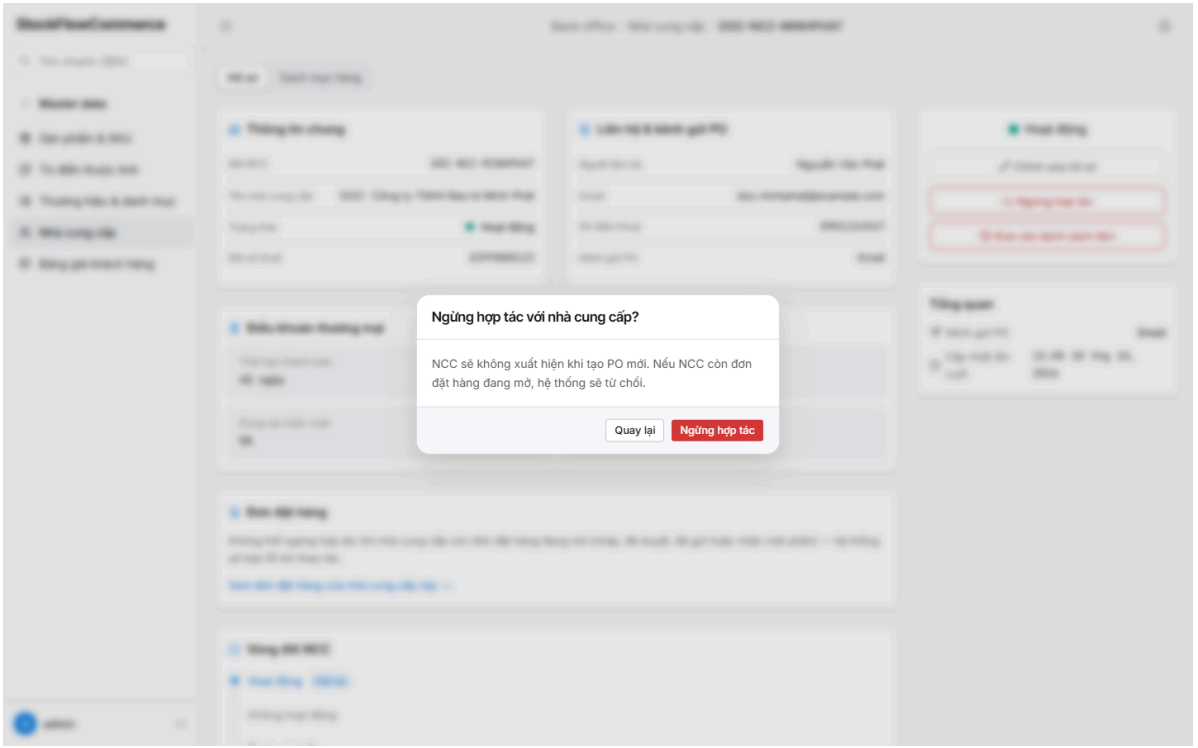
- **Function trigger:** Users click “Chỉnh sửa hồ sơ” on the supplier detail page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Procurement Staff
  - **Purpose:** Update an existing supplier profile.

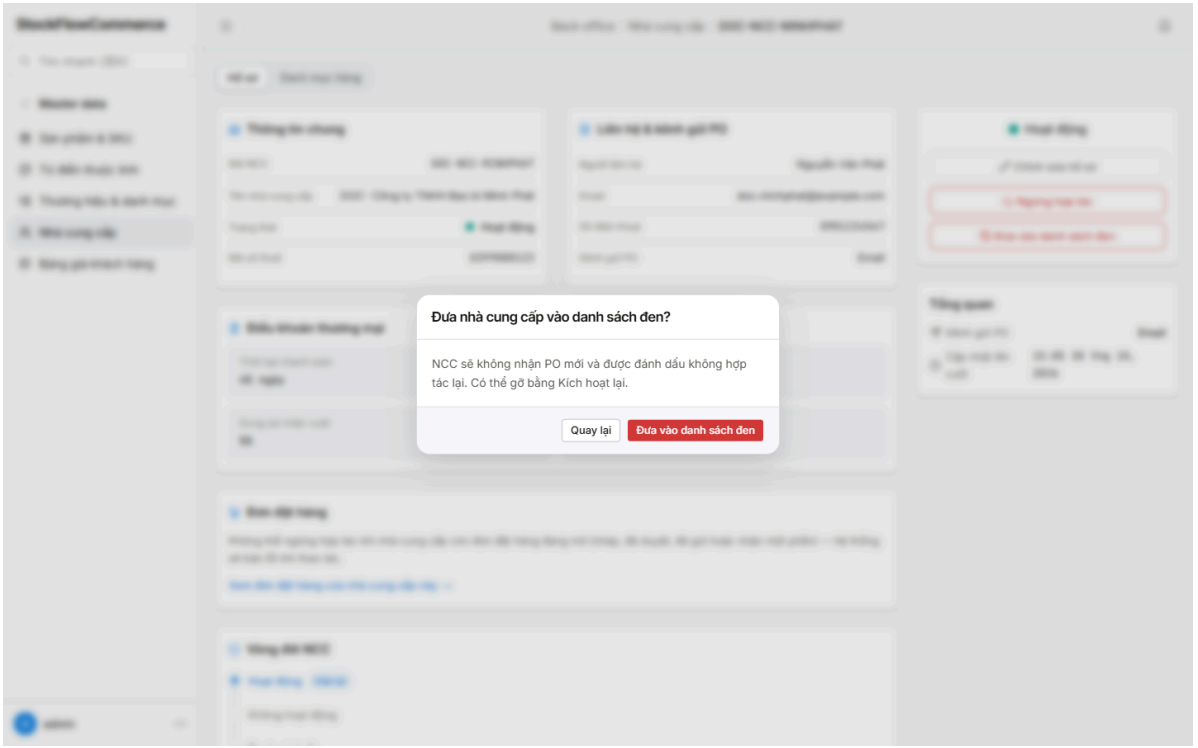


- **Function details:**
  - **Data requirement:** Same fields as creation except the supplier code.
  - **Validation:** Same as creation: name ≤200; tax code format and unique; email ≤150; phone 8–15 digits; terms 0–365; percentages 0–100; channel contact required.
  - **Business rules:** BR-11 (02 Purchase Order)

3.3.5.5 Change Supplier Status

- **Function trigger:** Users click “Kích hoạt lại”, “Ngừng hợp tác” or “Đưa vào danh sách đen” on the supplier detail page and confirm.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Procurement Staff
  - **Purpose:** Change a supplier's status.

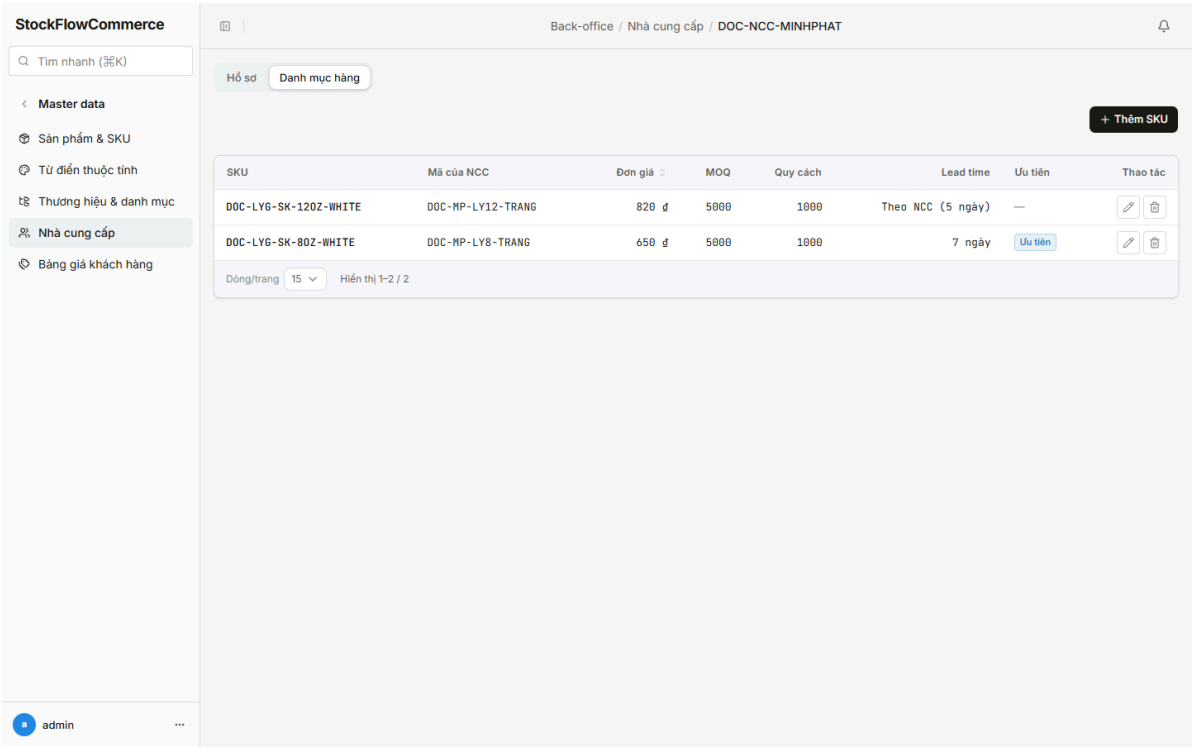




- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Allowed: Active→Inactive/Blacklisted, Inactive→Active/Blacklisted, Blacklisted→Active; cannot leave Active while open POs exist.
  - **Business rules:** BR-11 (02 Purchase Order)

3.3.5.6 View Supplier Items

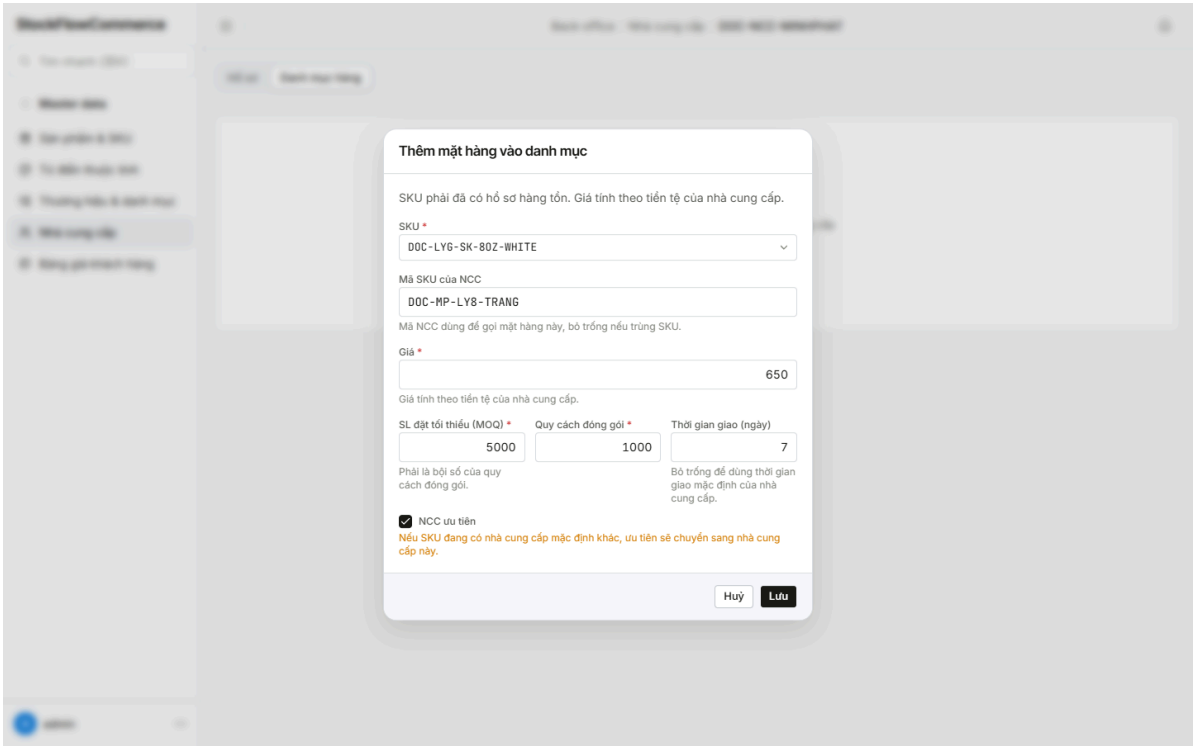
- **Function trigger:** Users open the “Danh mục hàng” tab on the supplier detail page.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff
  - **Purpose:** Show the SKUs a supplier sells and their terms.

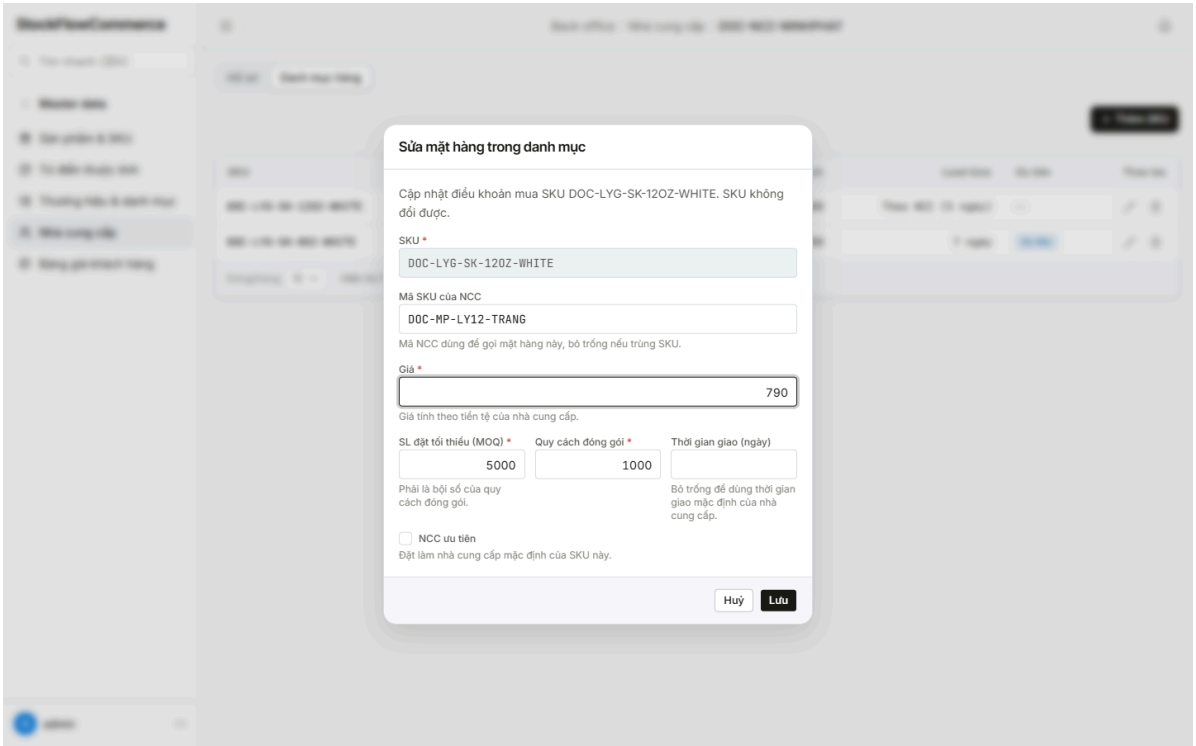


- **Function details:**
  - **Data requirement:** SKU, supplier SKU code, price, MOQ, pack size, lead time, preferred flag.
  - **Validation:** None.
  - **Business rules:** BR-07, BR-08 (02 Purchase Order)

3.3.5.7 Manage Supplier Items

- **Function trigger:** Users click “Thêm SKU”, or “Sửa” / “Xoá” on a row, in the “Danh mục hàng” tab.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff
  - **Purpose:** Maintain the SKUs in a supplier's catalog.



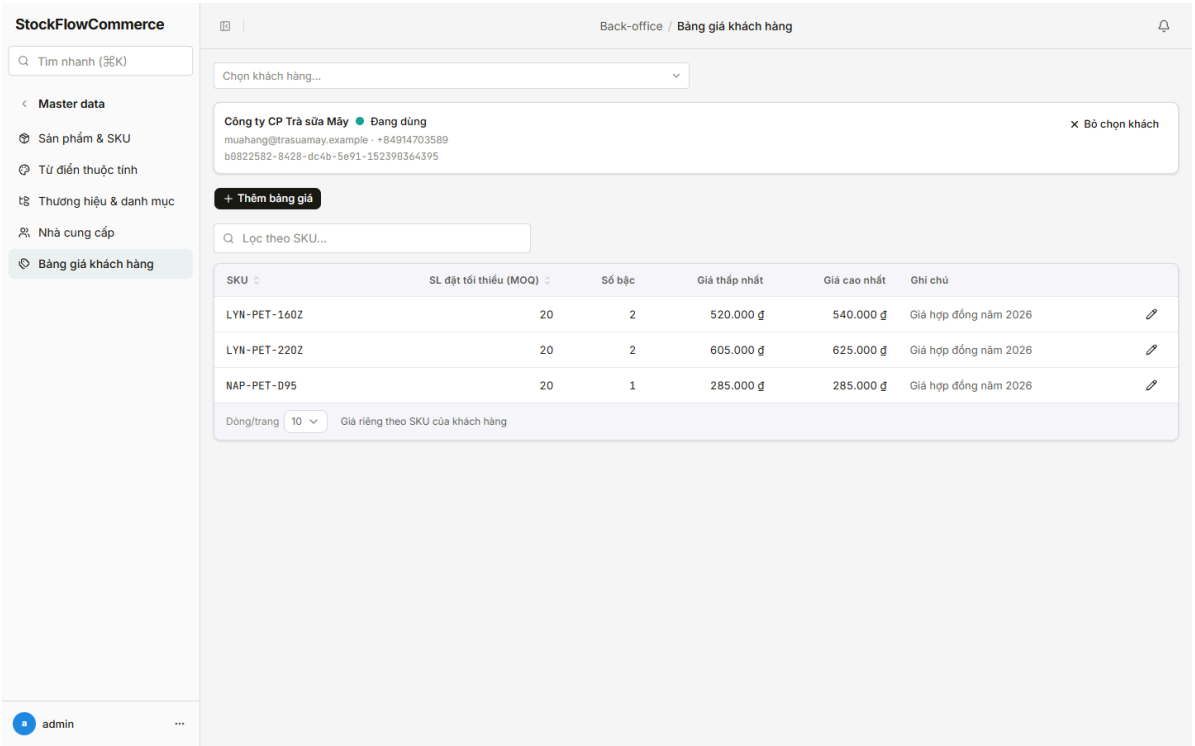


- **Function details:**
  - **Data requirement:** SKU, supplier SKU code, price, MOQ, pack size, lead time days, preferred supplier flag.
  - **Validation:** SKU valid and purchasable, one row per supplier and SKU; supplier code ≤50; price >0 within currency decimals; MOQ and pack size ≥1; MOQ multiple of pack size; lead time 0–365.
  - **Business rules:** BR-01, BR-08, BR-11 (02 Purchase Order)

3.3.6 Customer Prices

3.3.6.1 View Customer Prices

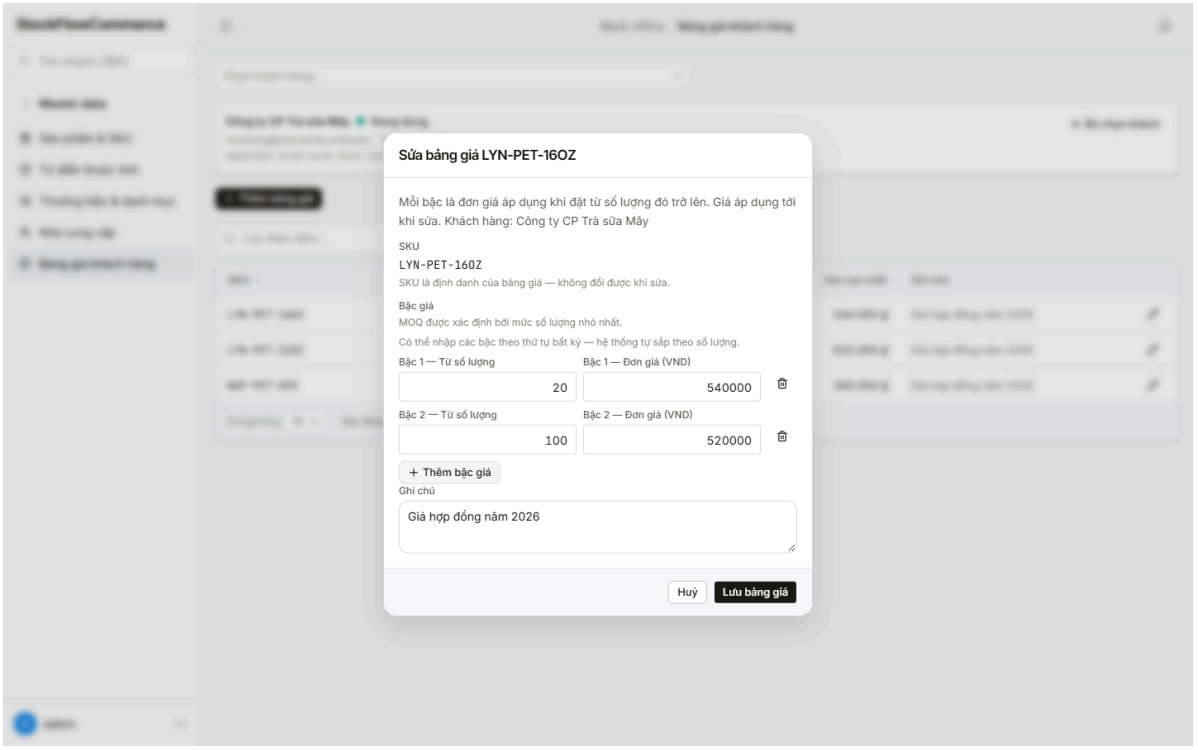
- **Function trigger:** Users open “Bảng giá khách hàng” in the sidebar and pick a customer.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin, Sales Staff
  - **Purpose:** Show a customer's own price list per SKU.

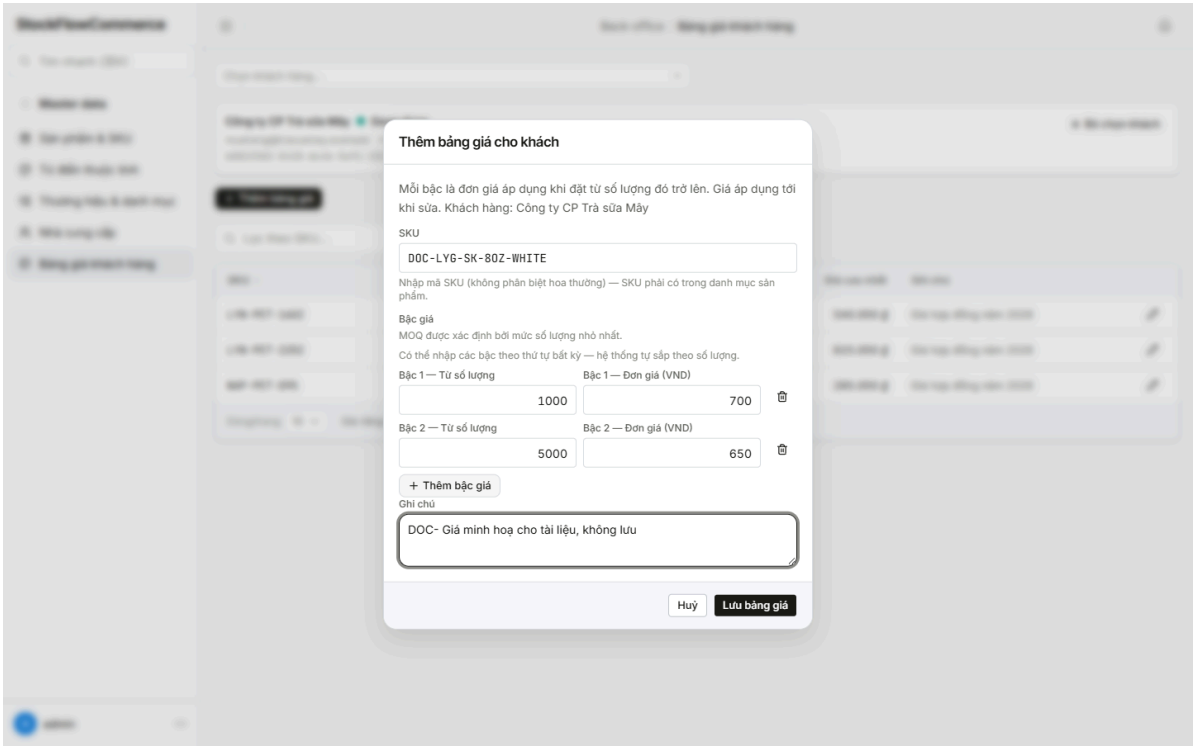


- **Function details:**
  - **Data requirement:** Customer (search by name, email or phone); SKU filter; SKU, MOQ, number of tiers, lowest and highest price, note.
  - **Validation:** Customer ID in the URL must be valid.
  - **Business rules:** BR-03, BR-04, BR-09 (13 Catalog)

3.3.6.2 Set Customer Price

- **Function trigger:** Users click “Thêm bảng giá” or “Sửa bảng giá” for a SKU on the customer price page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Create or update a customer's tiered price for one SKU.



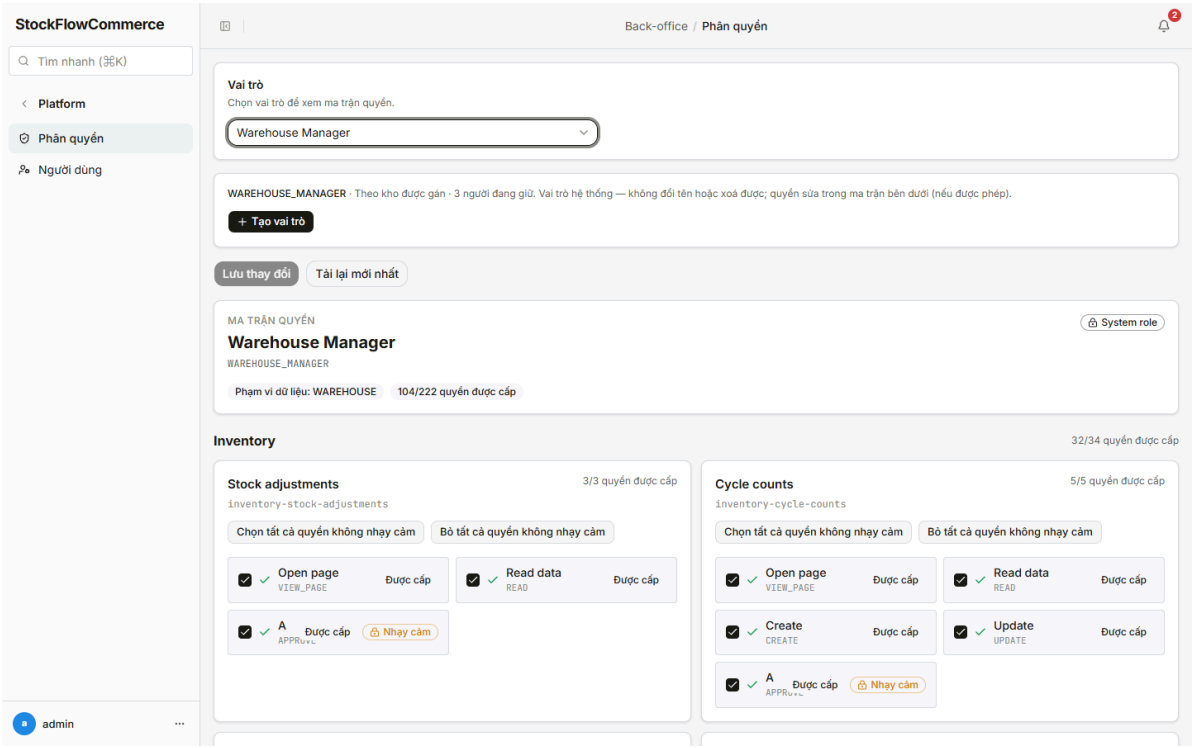


- **Function details:**
  - **Data requirement:** SKU, price tiers (from quantity, unit price VND), note.
  - **Validation:** SKU must exist; 1–20 tiers; from-quantity integer ≥1, no duplicates; unit price whole VND >0, ≤16 digits; note ≤500; one price per customer and SKU; version must match on update.
  - **Business rules:** BR-03, BR-04, BR-05, BR-09 (13 Catalog)

3.3.7 Permission Management

3.3.7.1 View Permission Matrix

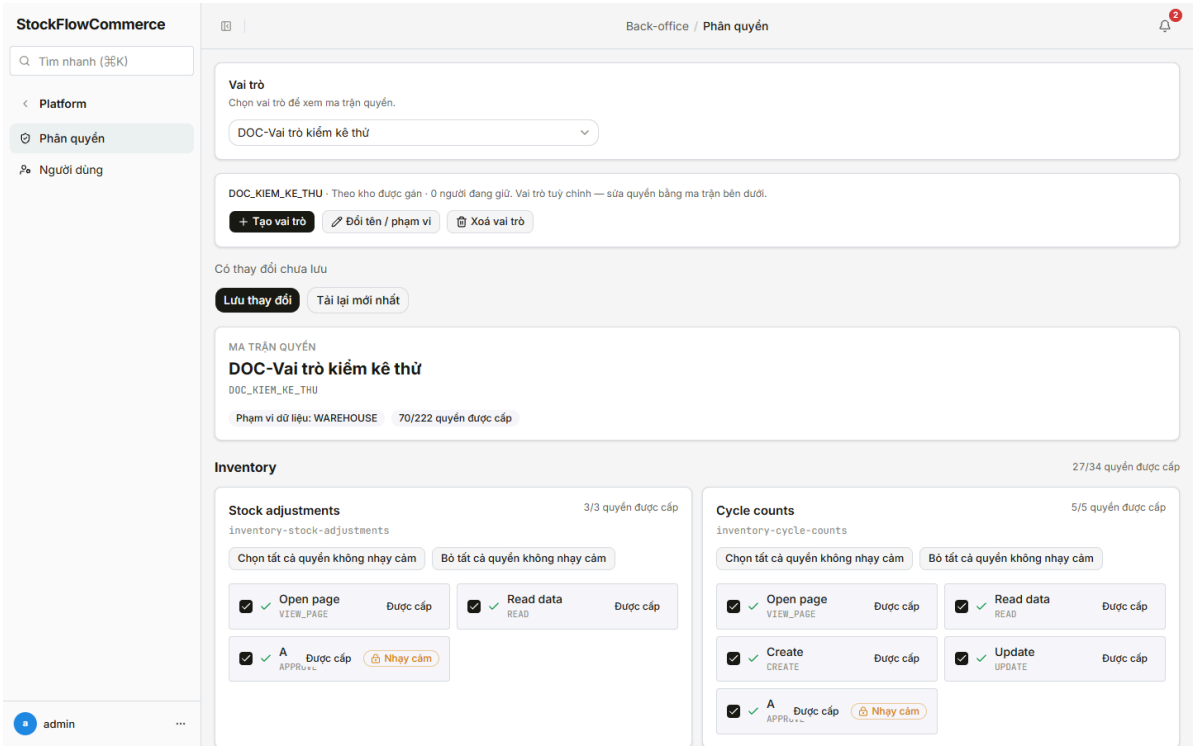
- **Function trigger:** Users open “Phân quyền” in the sidebar and choose a role.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Show roles and the permissions each role holds.



- **Function details:**
  - **Data requirement:** Role; role code, data scope, holder count; permission matrix of resources × actions.
  - **Validation:** None.
  - **Business rules:** §2.5.3 (00 System Overview)

3.3.7.2 Edit Role Permissions

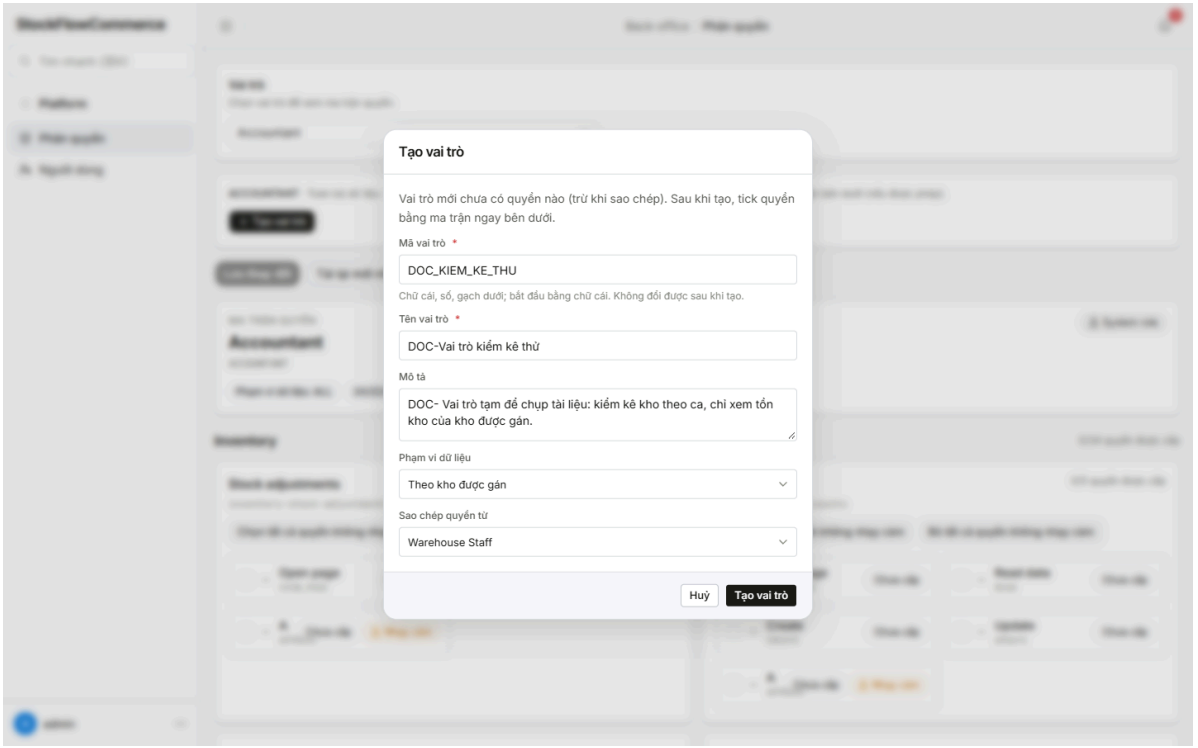
- **Function trigger:** Users tick or untick permissions in the matrix for a role and click “Lưu thay đổi”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Grant or revoke permissions of a role.



- **Function details:**
  - **Data requirement:** Permission selections (resource and action).
  - **Validation:** Requires identity-rbac:APPROVE; at most 2000 permissions; only permissions the editor holds; at least one role keeps permission management; version must match; read-only roles cannot change.
  - **Business rules:** §2.5.1, §2.5.3 (00 System Overview)

3.3.7.3 Create Role

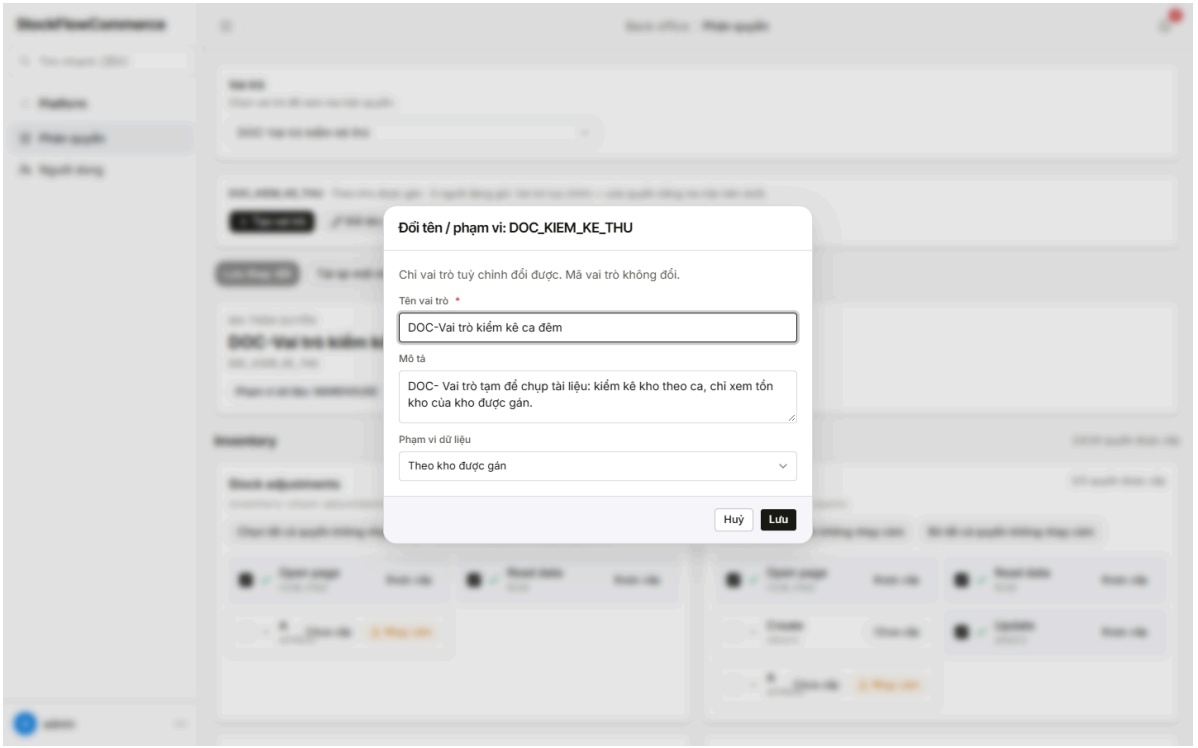
- **Function trigger:** Users click “Tạo vai trò” on the permissions page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Create a custom role, optionally copying another role's permissions.



- **Function details:**
  - **Data requirement:** Role code, role name, description, data scope, copy permissions from.
  - **Validation:** Code 2–64 characters, letters/digits/_, starts with a letter, unique (case-insensitive); name required ≤200; description ≤1000; scope OWN, WAREHOUSE or ALL.
  - **Business rules:** §2.5.1, §2.5.3 (00 System Overview)

3.3.7.4 Edit Role

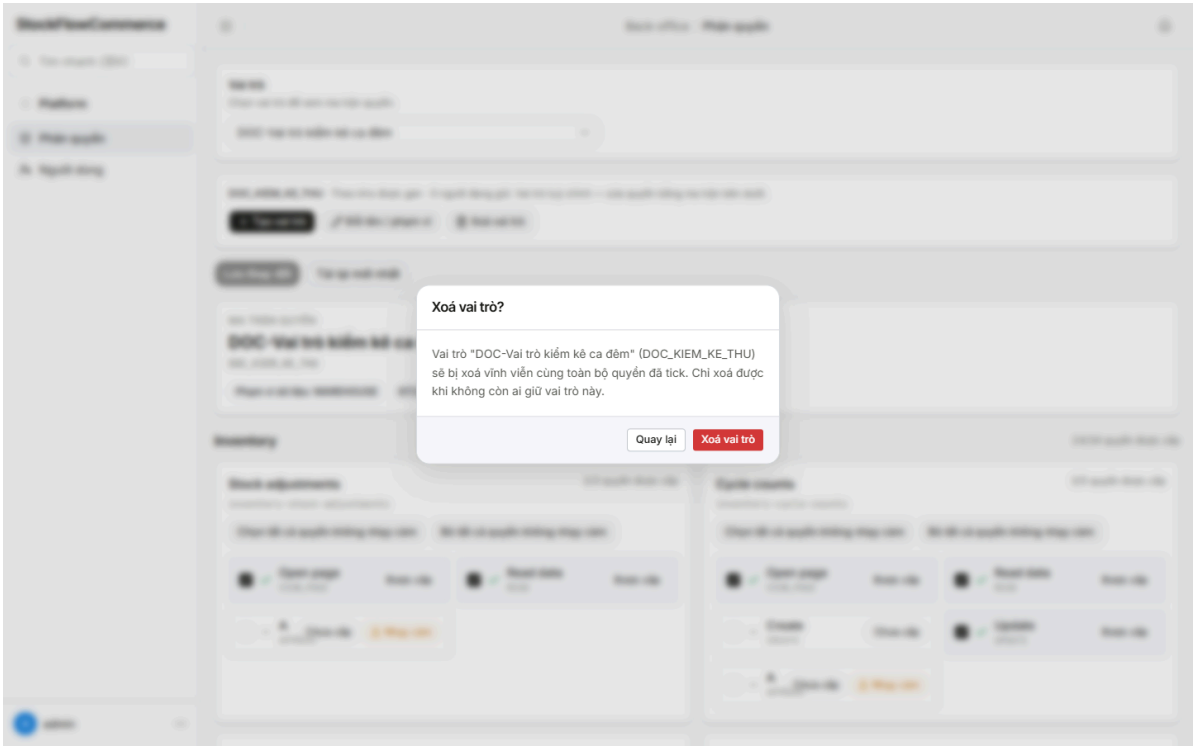
- **Function trigger:** Users select a custom role and click “Đổi tên / phạm vi”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Rename a custom role and change its description or data scope.



- **Function details:**
  - **Data requirement:** Role name, description, data scope.
  - **Validation:** Custom roles only; name required ≤200; description ≤1000; scope OWN, WAREHOUSE or ALL; version must match.
  - **Business rules:** §2.5.3 (00 System Overview)

3.3.7.5 Delete Role

- **Function trigger:** Users select a custom role, click “Xoá vai trò” and confirm.
- **Function description:**
  - **Actors:** System Admin
  - **Purpose:** Delete a custom role that no one holds.

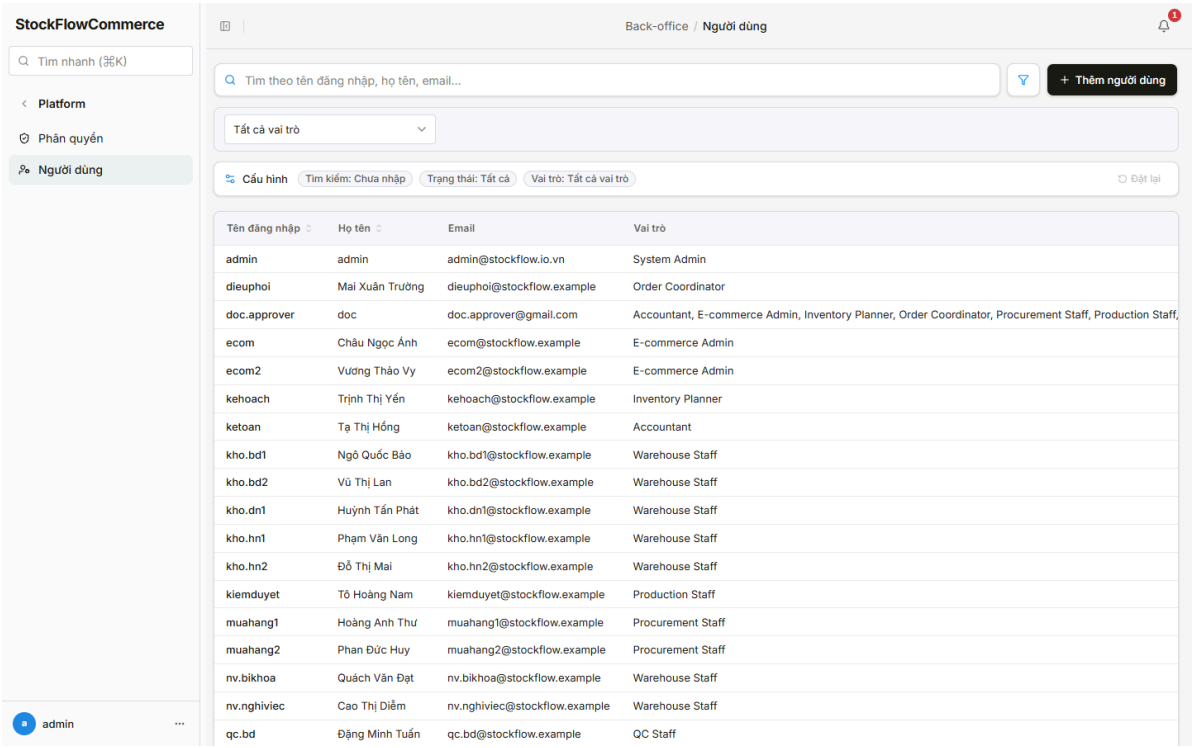


- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Custom role only; no holders (button disabled otherwise); requires identity-roles:DELETE.
  - **Business rules:** §2.5.3 (00 System Overview)

3.3.8 User Management

3.3.8.1 View User List

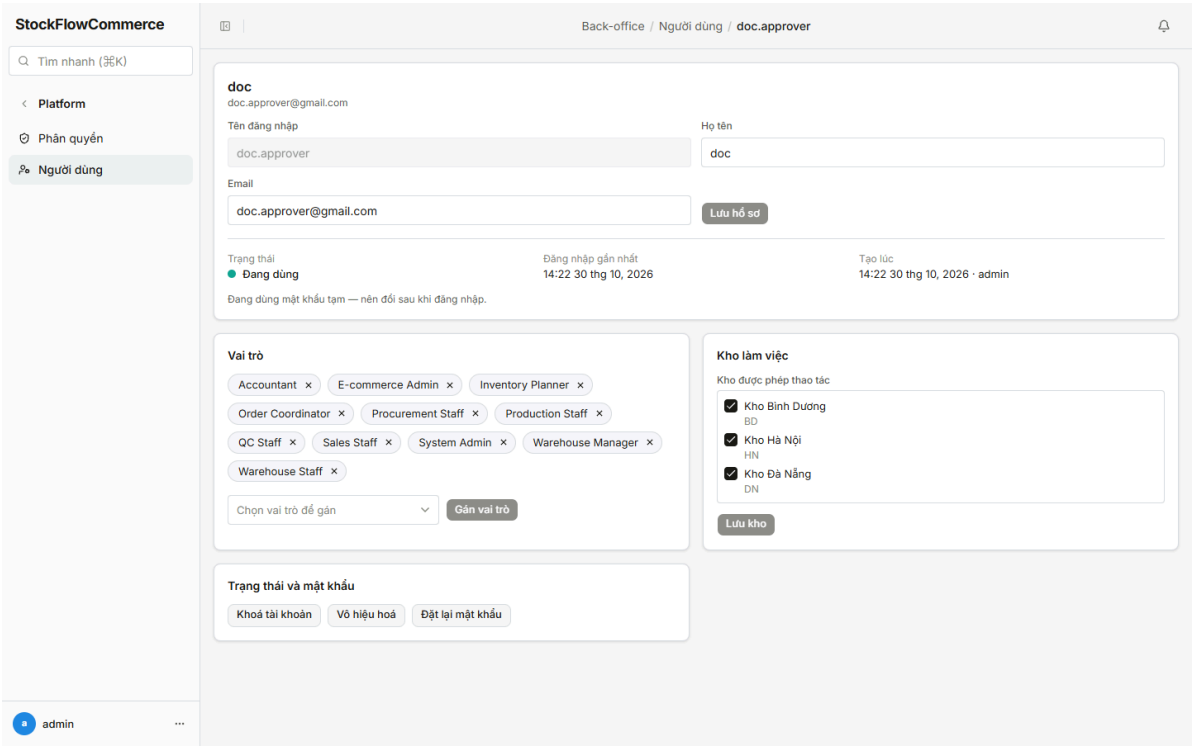
- **Function trigger:** Users open “Người dùng” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Search staff accounts and filter them by role and status.



- **Function details:**
  - **Data requirement:** Keyword (username, full name, email), role filter, status filter; username, full name, email, roles, warehouses, status, last login.
  - **Validation:** None.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.8.2 View User Detail

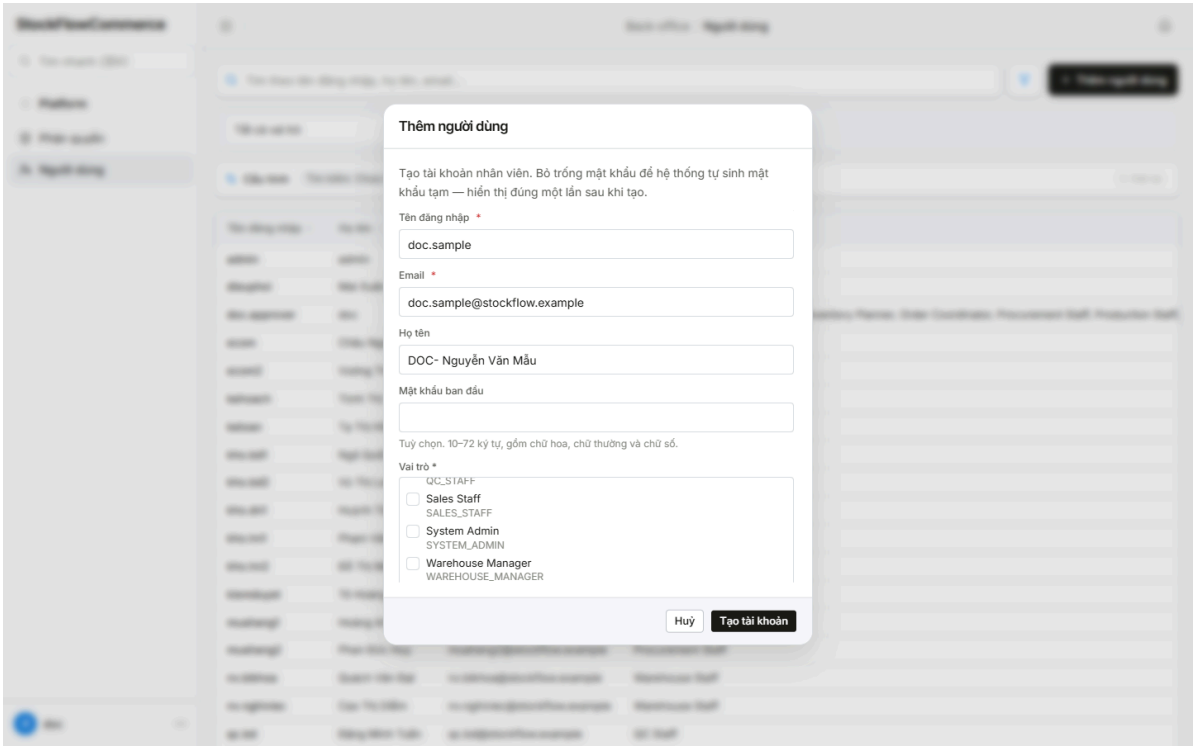
- **Function trigger:** Users click a username on the user list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Show a staff account's profile, roles, warehouses and status.

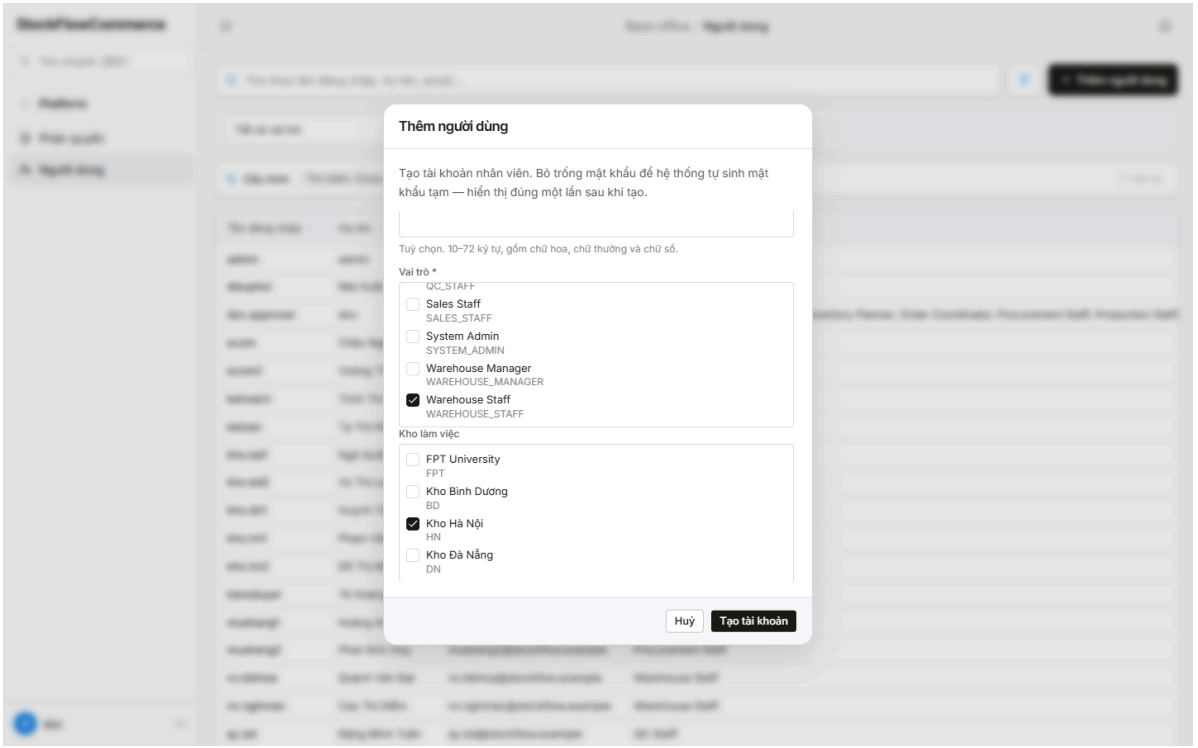


- **Function details:**
  - **Data requirement:** Username, full name, email, status, last login, created at, roles, working warehouses, temporary-lock and must-change-password flags.
  - **Validation:** None.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.8.3 Create User

- **Function trigger:** Users click “Thêm người dùng” on the user list.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Create a staff account.

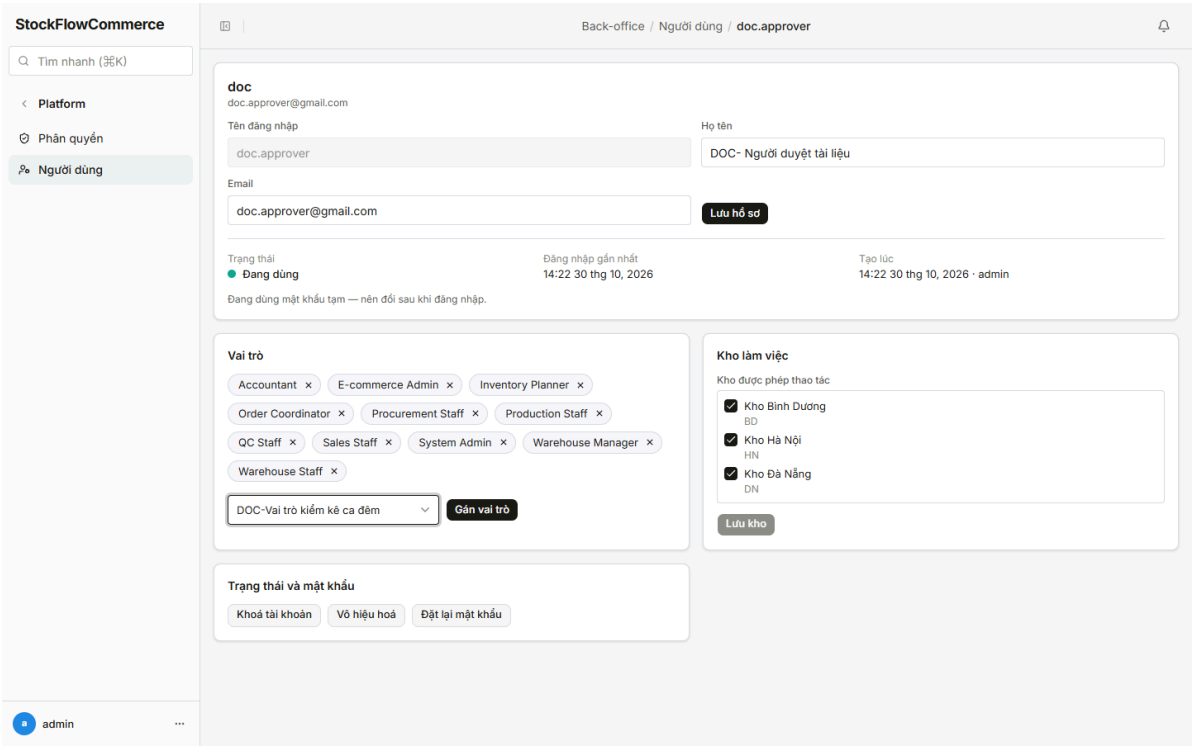




- **Function details:**
  - **Data requirement:** Username, email, full name, initial password, roles, working warehouses.
  - **Validation:** Username 3–100 letters/digits/._-, unique; email required, valid, ≤320, unique; full name ≤200; optional password 10–72 with upper, lower, digit; 1–20 roles, CUSTOMER not allowed; ≤50 warehouses.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.8.4 Edit User

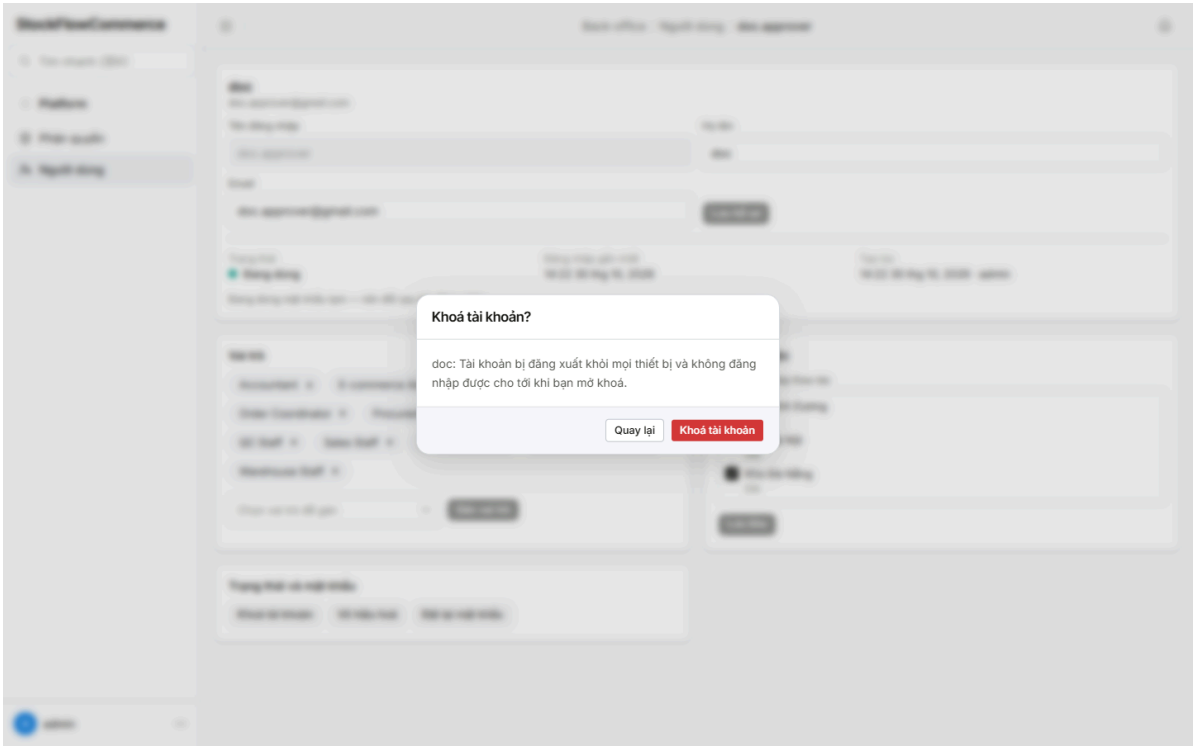
- **Function trigger:** Users edit fields on the user detail page and click “Lưu hồ sơ”, “Gán vai trò” or “Lưu kho”.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Edit a staff profile and assign roles and working warehouses.

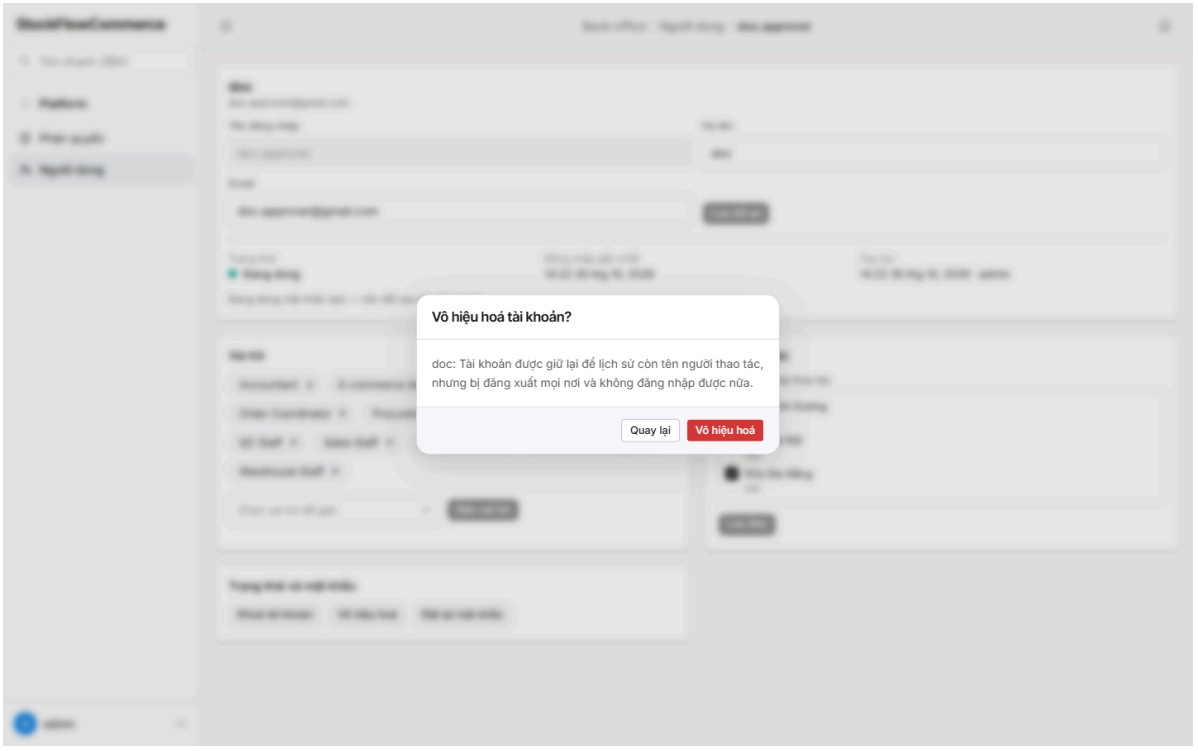


- **Function details:**
  - **Data requirement:** Full name, email, role to assign or remove, working warehouses.
  - **Validation:** Email required, valid, ≤320, unique; full name ≤200; ≤50 warehouses; cannot manage own account; only roles held by the editor; CUSTOMER not assignable; keep one active System Admin; version must match.
  - **Business rules:** §2.5.1, §2.5.3 (00 System Overview)

3.3.8.5 Lock or Disable User

- **Function trigger:** Users click “Khoá tài khoản”, “Mở khoá”, “Vô hiệu hoá” or “Kích hoạt lại” on the user detail page and confirm.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Lock, unlock, disable or re-enable an account.

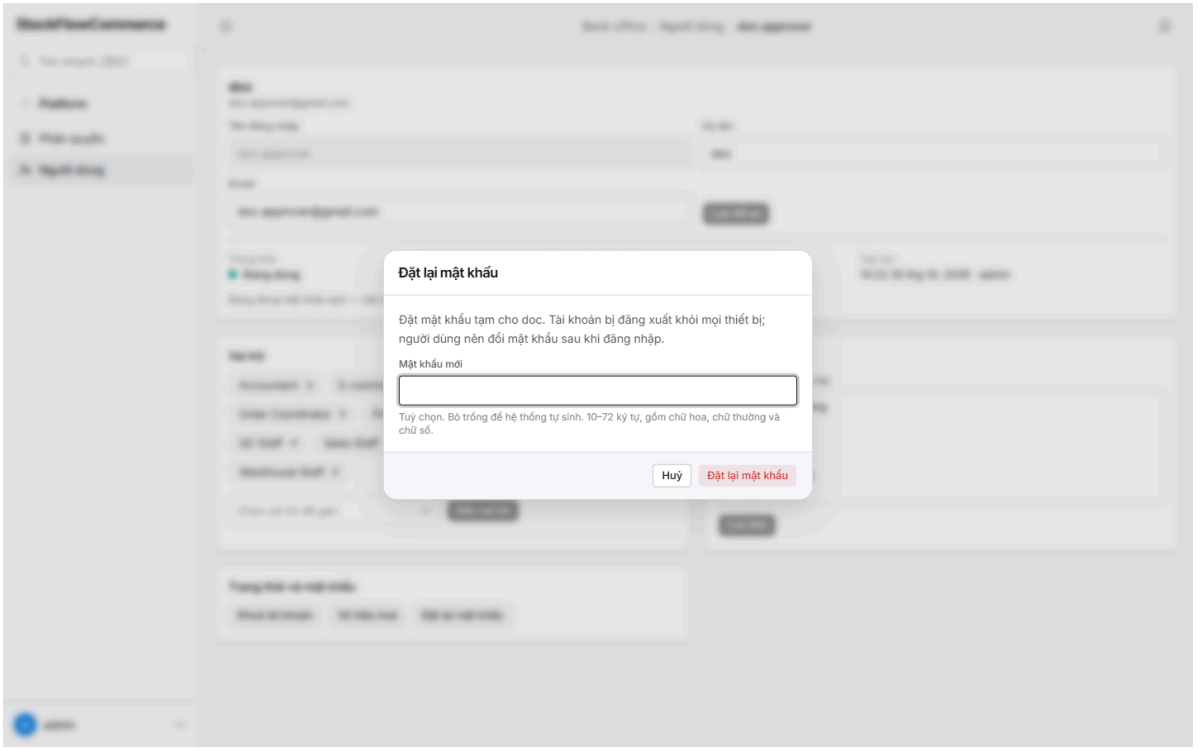




- **Function details:**
  - **Data requirement:** None.
  - **Validation:** Active→Locked/Disabled; Locked→Active/Disabled; Disabled→Active; cannot act on own account; cannot lock or disable the last active System Admin.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.8.6 Reset User Password

- **Function trigger:** Users click “Đặt lại mật khẩu” on the user detail page.
- **Function description:**
  - **Actors:** System Admin, E-commerce Admin
  - **Purpose:** Set a new temporary password for a staff member.

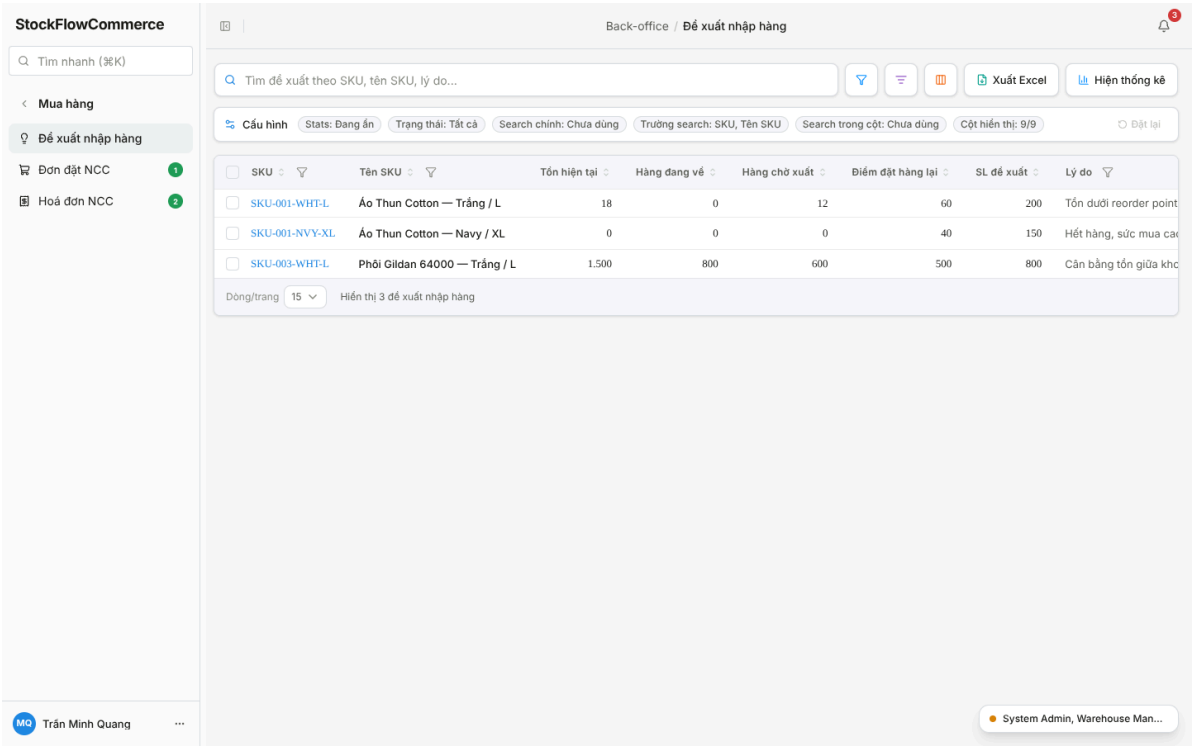


- **Function details:**
  - **Data requirement:** New password (optional).
  - **Validation:** If entered: 10–72 characters with upper-case, lower-case and digit; cannot reset own account.
  - **Business rules:** §2.5.1 (00 System Overview)

3.3.9 Replenishment

3.3.9.1 View Replenishment Suggestions

- **Function trigger:** Users click “Đề xuất nhập hàng” in the “Mua hàng” sidebar group (the item is shown only in mock mode).
- **Function description:**
  - **Actors:** Inventory Planner, Procurement Staff
  - **Purpose:** Under development — Review SKUs below their reorder point with a suggested purchase quantity and convert a reviewed proposal into a draft purchase order.

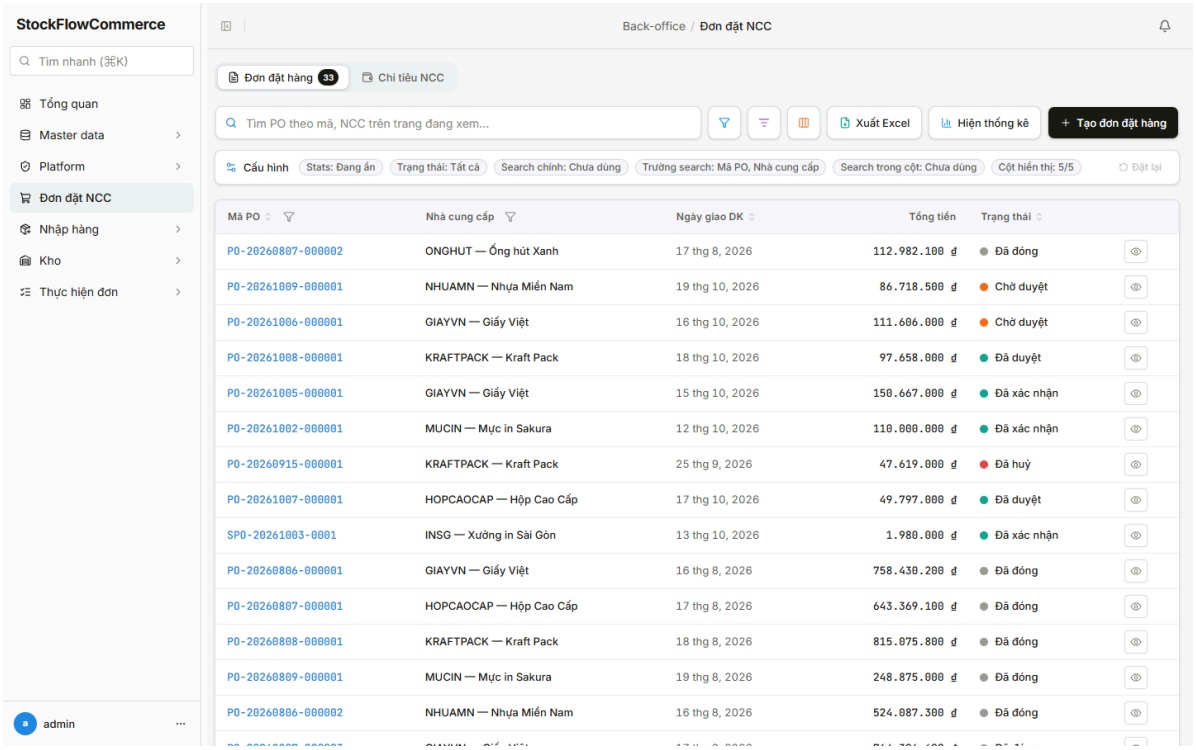


- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.3.10 Purchase Orders

3.3.10.1 View Purchase Order List

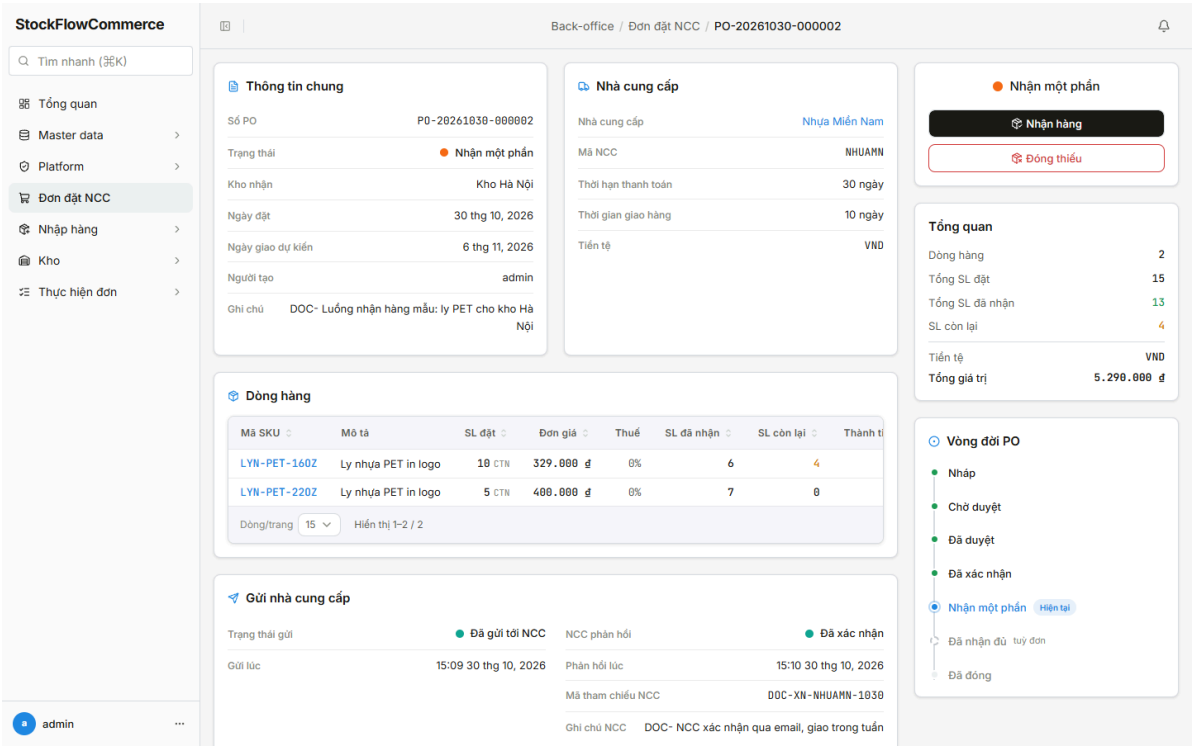
- **Function trigger:** Users click “Đơn đặt NCC” in the sidebar.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Procurement Staff
  - **Purpose:** Browse, search and filter purchase orders to find one to work on.

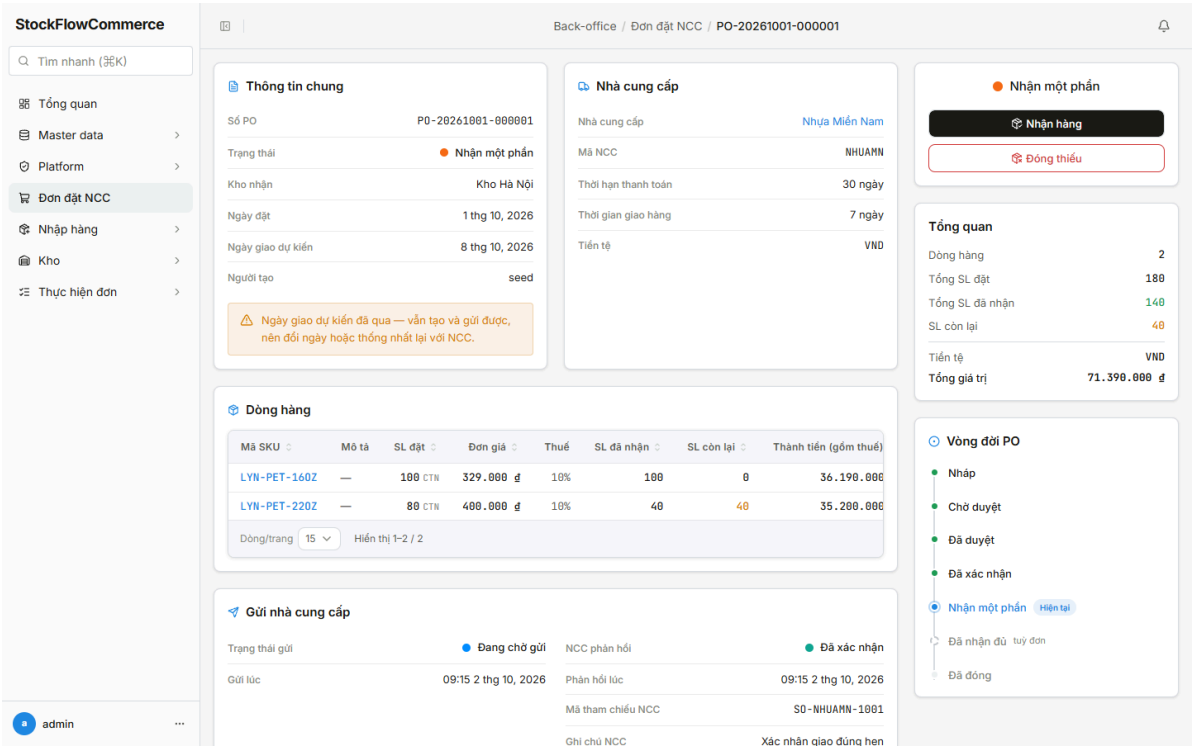


- **Function details:**
  - **Data requirement:** Status filter, keyword (PO number, supplier), PO number and supplier column filters, sort by PO number/expected date/status; shows PO number, supplier, expected delivery date, total amount, status.
  - **Validation:** None
  - **Business rules:** BR-14 (02 Purchase Order)

3.3.10.2 View Purchase Order Detail

- **Function trigger:** Users click a row or the view icon (“Xem chi tiết …”) on the purchase order list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Procurement Staff
  - **Purpose:** View a purchase order's full content, progress and supplier delivery history.

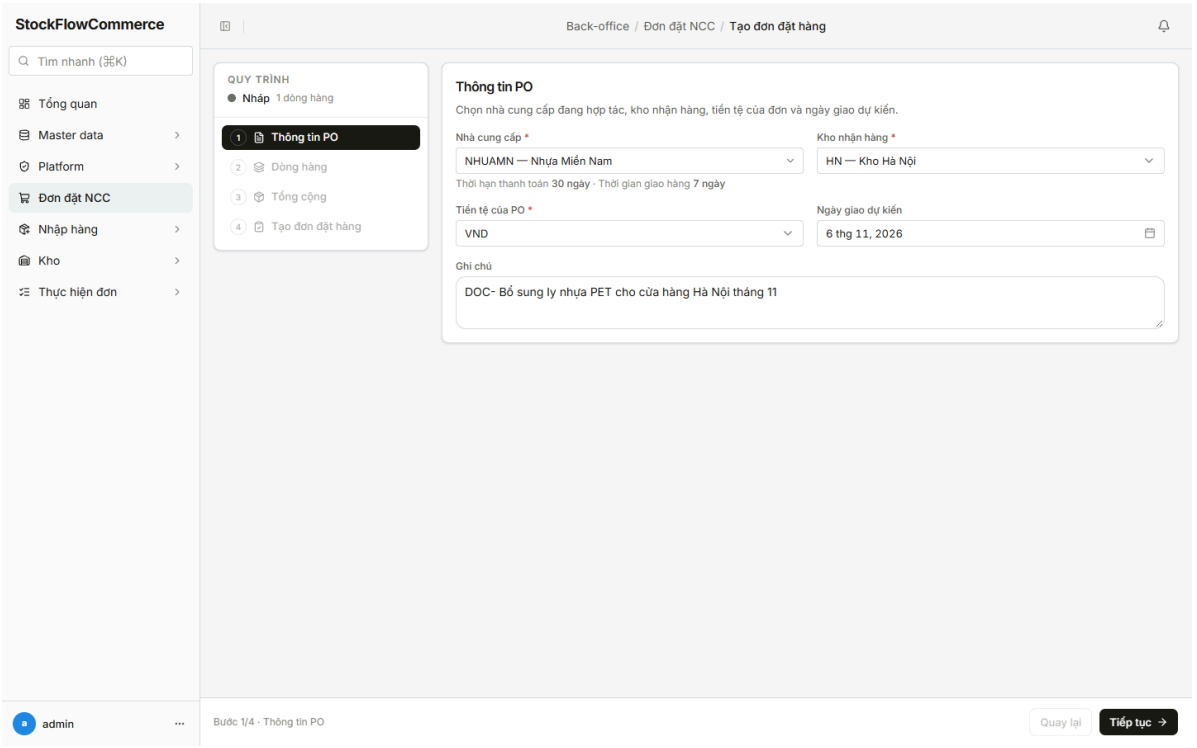


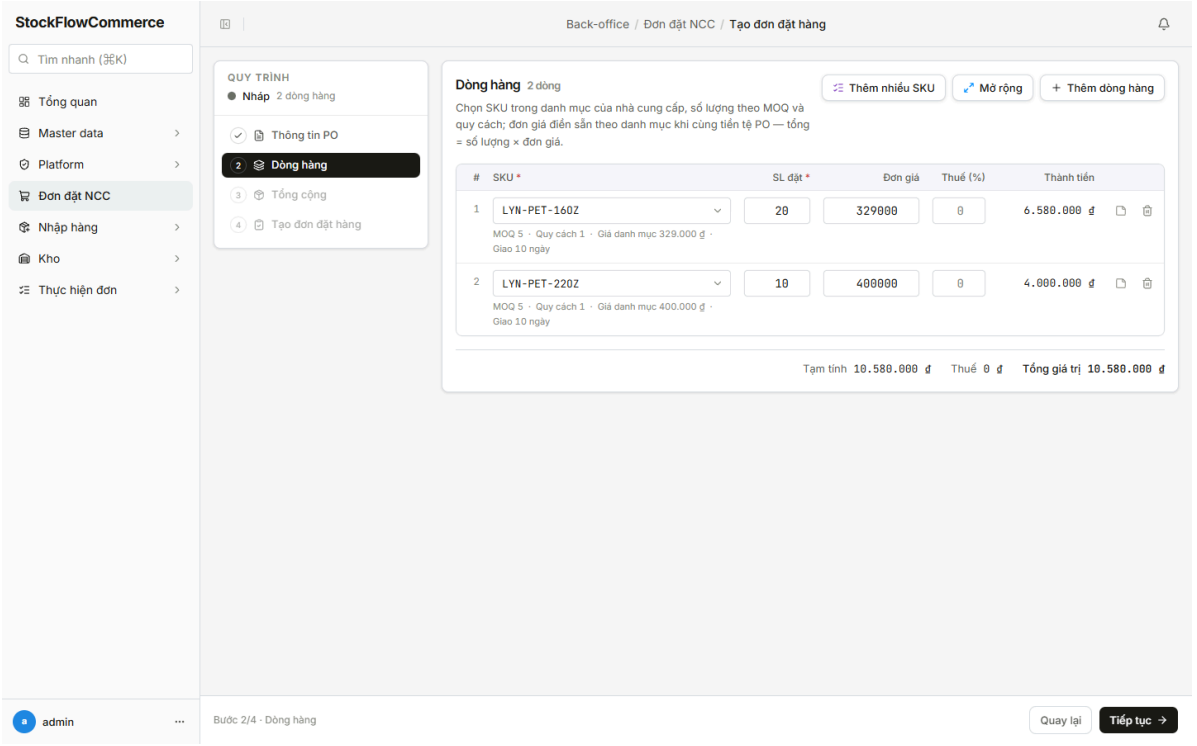


- **Function details:**
  - **Data requirement:** PO number, status, supplier, receiving warehouse, currency, expected date, lines (SKU, ordered/received/open quantity, unit price, tax, line total), totals, supplier response and reference, delivery attempts and decisions, lifecycle timeline.
  - **Validation:** None
  - **Business rules:** BR-12, BR-13, BR-14 (02 Purchase Order)

3.3.10.3 Create Purchase Order

- **Function trigger:** Users click “Tạo đơn đặt hàng” button on the purchase order list.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff
  - **Purpose:** Create a new draft purchase order for one supplier and one receiving warehouse.



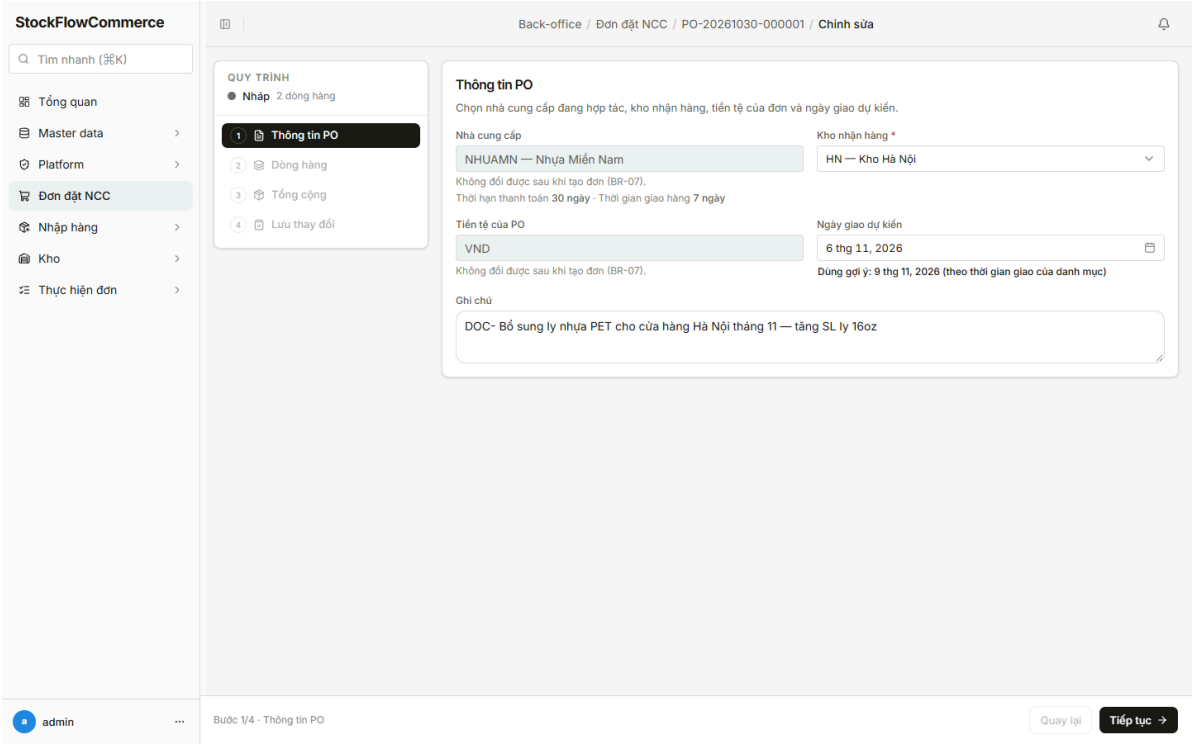


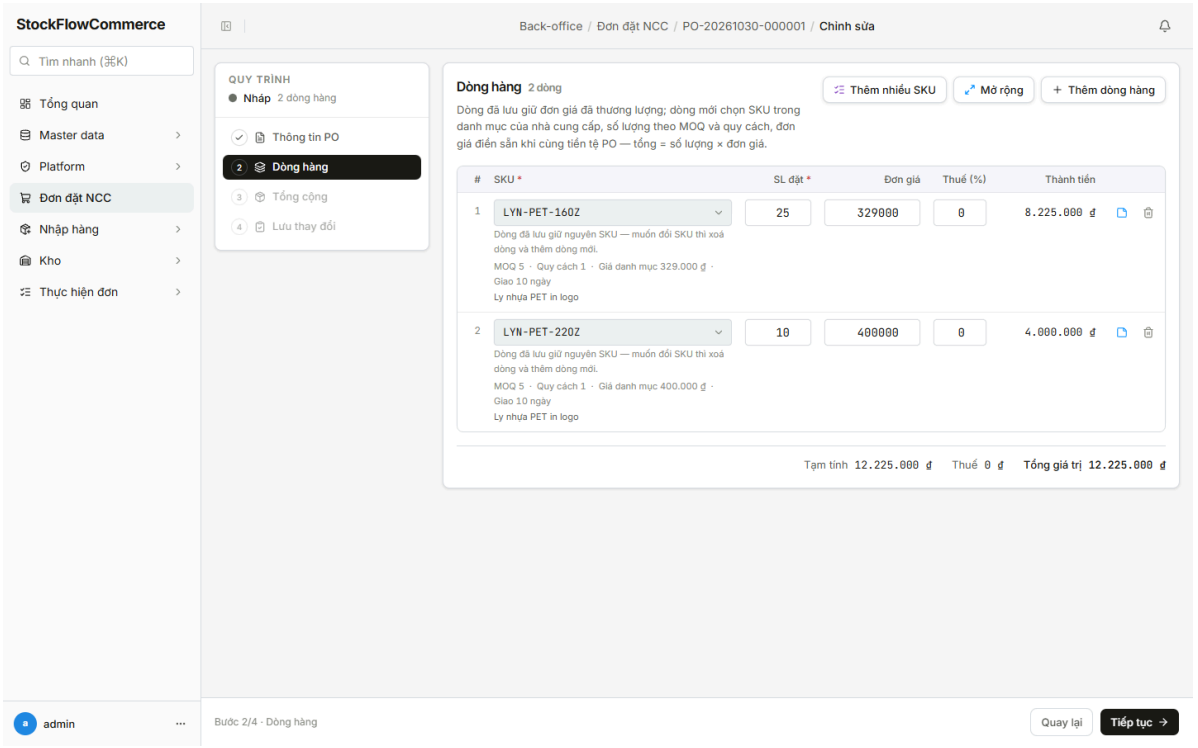
- **Function details:**

- **Data requirement:** Supplier, receiving warehouse, currency (VND/USD/CNY), expected delivery date, note; lines: SKU from the supplier catalog, ordered quantity, unit price, tax rate (%), description.
- **Validation:** Supplier and warehouse required; at least 1 line; SKU from supplier catalog, unique per PO; quantity integer 1–1,000,000, ≥ MOQ and a pack-size multiple; unit price >0 (blank = catalog price if same currency; VND whole numbers); tax 0–100%; description ≤255; note ≤2000 chars.
- **Business rules:** BR-01, BR-06, BR-07, BR-08, BR-11, BR-14 (02 Purchase Order)

3.3.10.4 Edit Purchase Order

- **Function trigger:** Users click “Sửa nháp” on a Draft purchase order's detail page and save with “Lưu thay đổi”, then click “Gửi duyệt”.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff
  - **Purpose:** Revise a draft standard purchase order and submit it for approval.

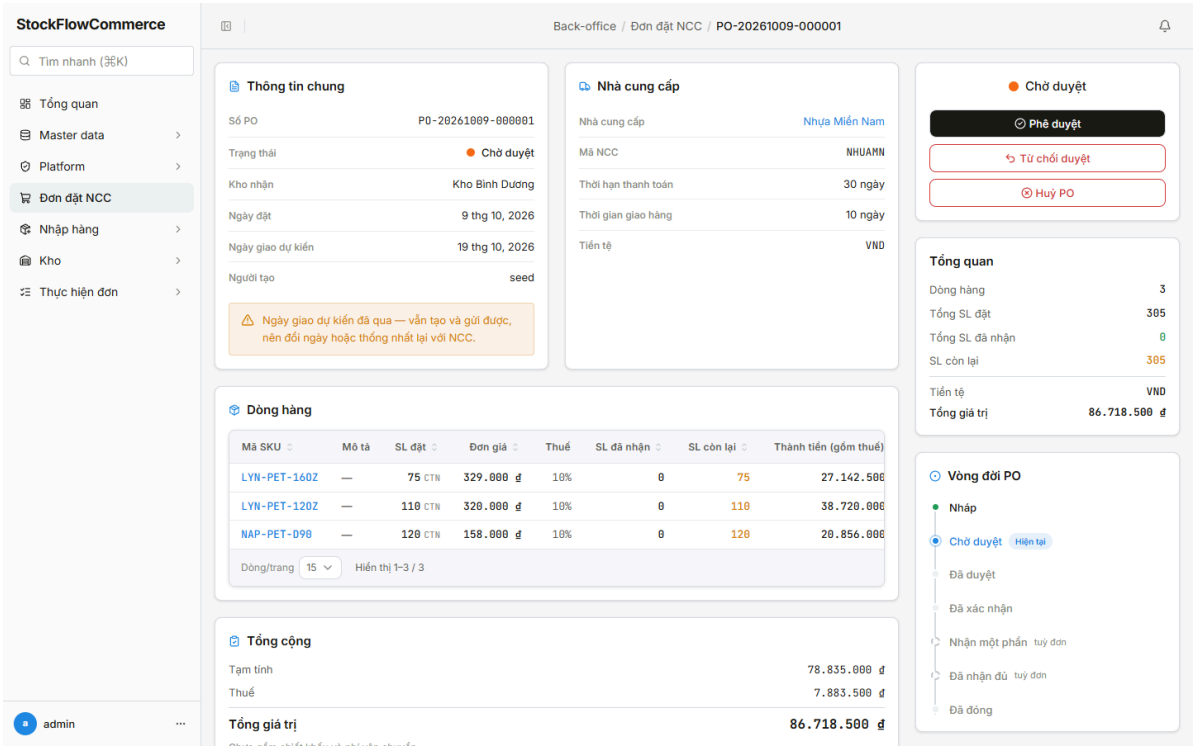


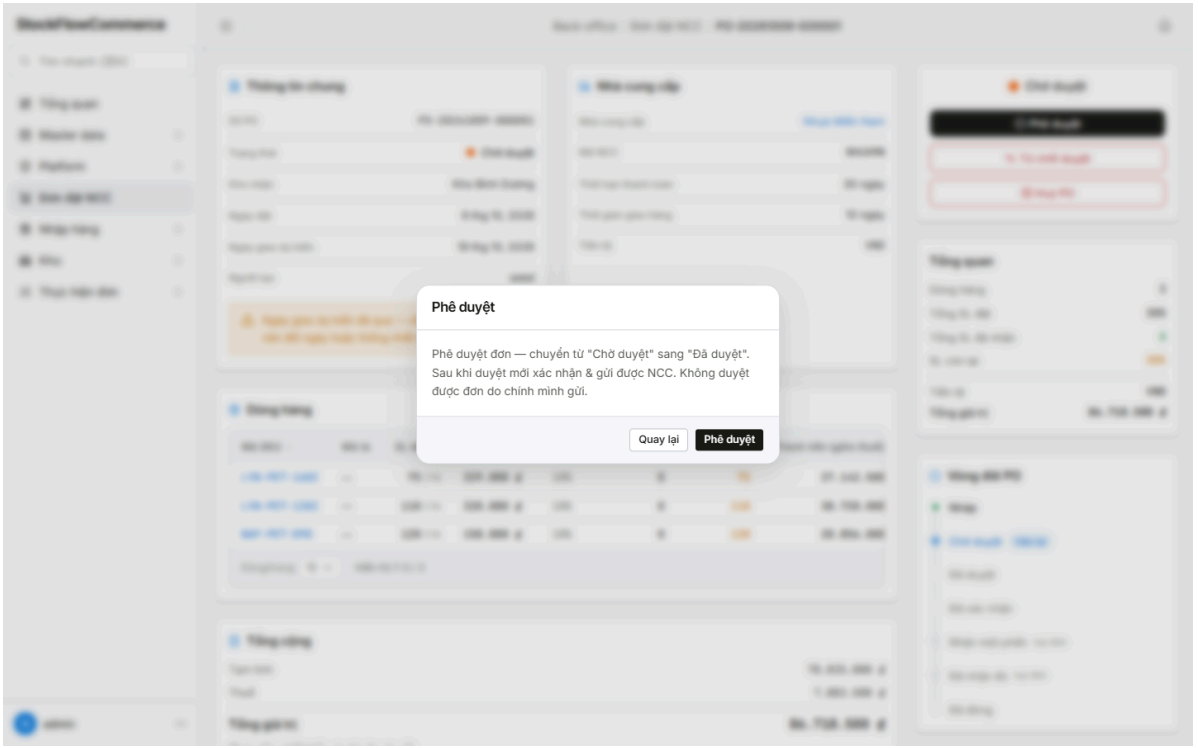


- **Function details:**
  - **Data requirement:** Receiving warehouse, expected delivery date, note, lines (add/remove; quantity, unit price, tax rate, description); supplier and currency shown read-only.
  - **Validation:** Only Draft STANDARD PO; warehouse and expected date required, date ≥ order date; 1–200 lines; saved lines keep their SKU and must keep a unit price; line rules as on create; stale version rejected (409 OPTIMISTIC_LOCK).
  - **Business rules:** BR-01, BR-02, BR-06, BR-07, BR-08 (02 Purchase Order)

3.3.10.5 Approve Purchase Order

- **Function trigger:** Users click “Phê duyệt” or “Từ chối duyệt” on a Pending Approval purchase order's detail page.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Approve a submitted purchase order or return it to Draft with a reason.

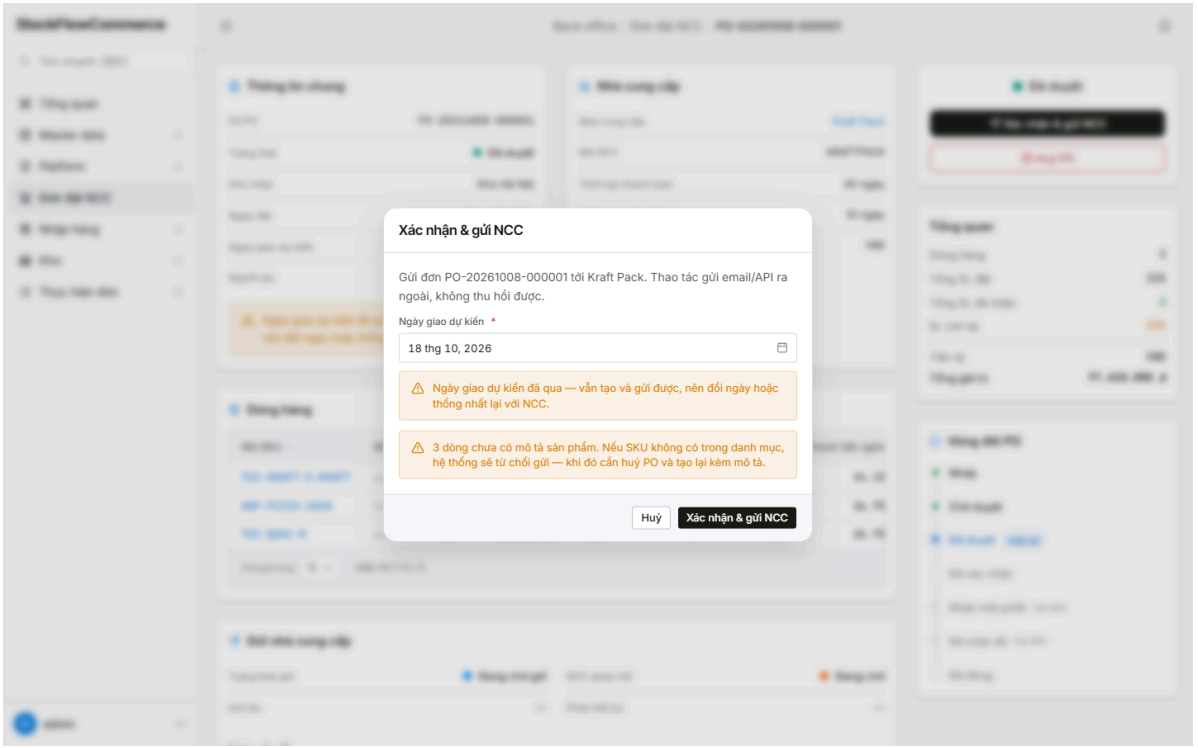


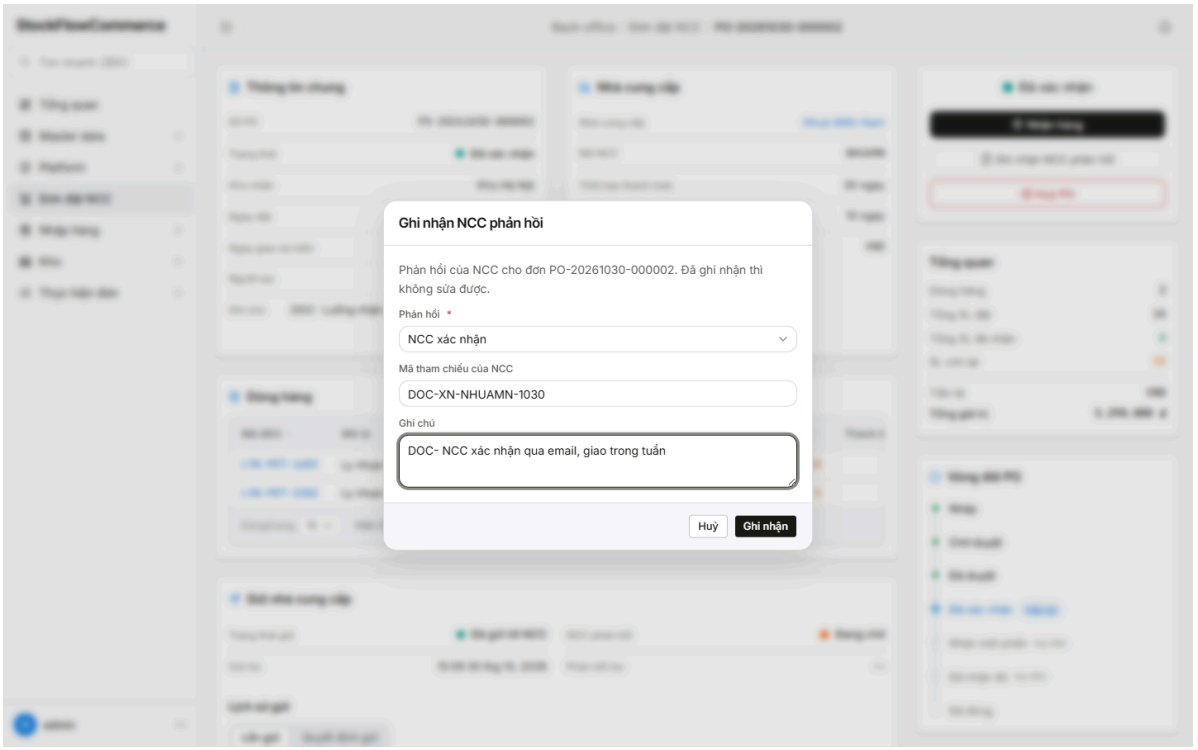


- **Function details:**
  - **Data requirement:** Rejection reason (reject only).
  - **Validation:** Only Pending Approval; rejection reason required, ≤500 chars; approver must differ from the submitter (backend 409 SELF_APPROVAL_NOT_ALLOWED); requires the PO approve permission.
  - **Business rules:** BR-01, BR-09 (02 Purchase Order)

3.3.10.6 Send Purchase Order to Supplier

- **Function trigger:** Users click “Xác nhận & gửi NCC” on an Approved purchase order, and later “Ghi nhận NCC phản hồi” to record the supplier's reply.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff
  - **Purpose:** Confirm the purchase order and send it to the supplier, then record the supplier's confirmation or rejection.



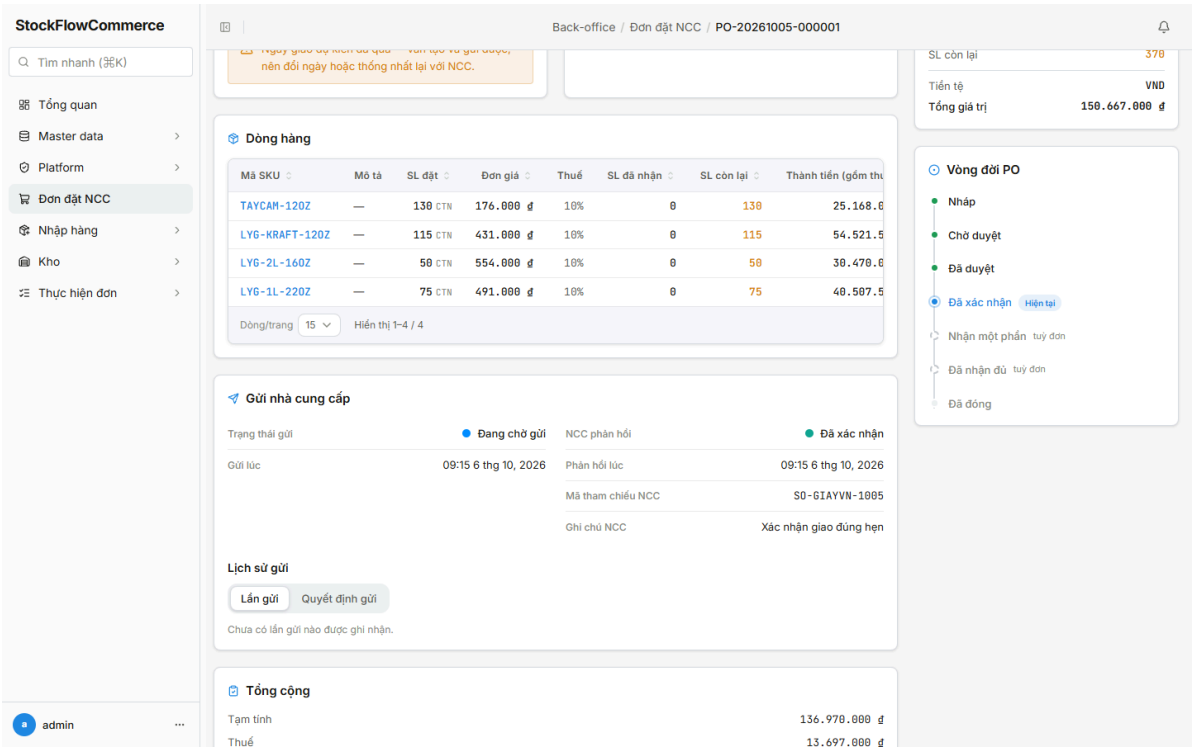


- **Function details:**

- **Data requirement:** Delivery date, reason for changing it; supplier response ("NCC xác nhận" / "NCC từ chối"), supplier reference, note.
- **Validation:** Delivery date required, ≥ order date (past date only warns); reason required if date changed, ≤1000; reference ≤100; note ≤1000 and required on rejection; rejection only while PO is Confirmed (nothing received); response recorded once.
- **Business rules:** BR-06, BR-12, BR-13 (02 Purchase Order)

3.3.10.7 Resend Purchase Order

- **Function trigger:** Users click "Khôi phục gửi NCC" on a Confirmed purchase order whose last delivery failed, then "Gửi lại".
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Resend a purchase order whose delivery to the supplier failed.

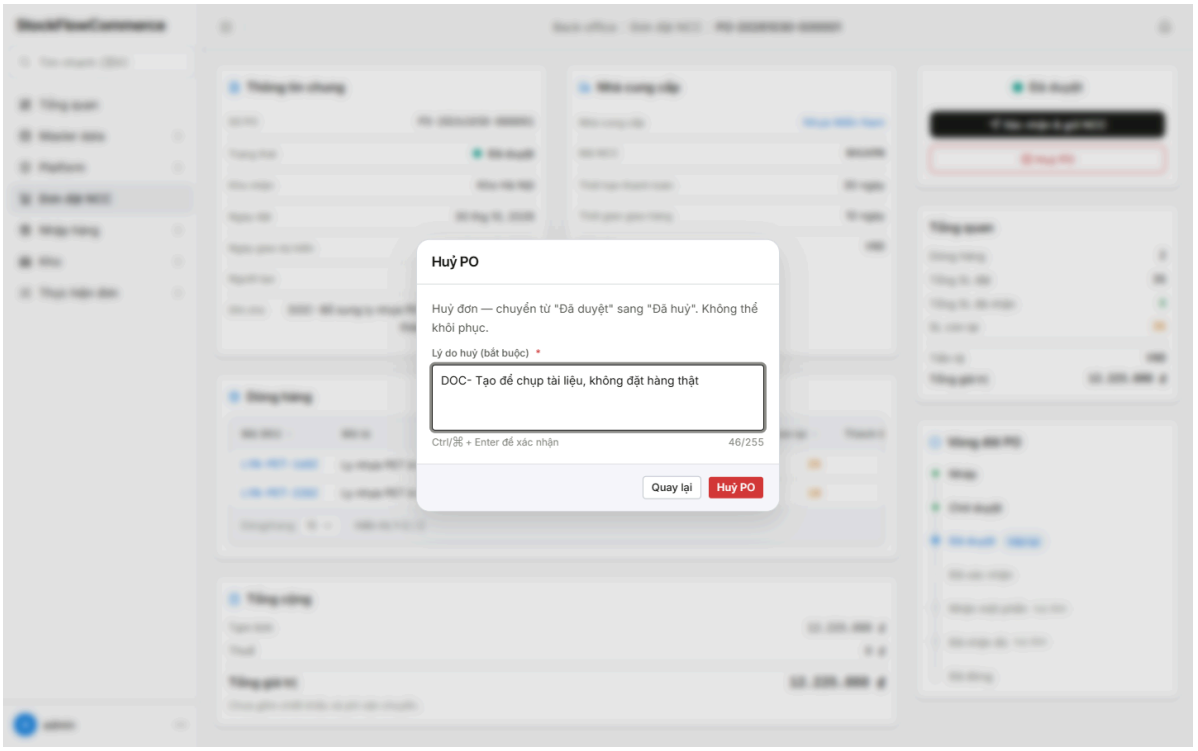


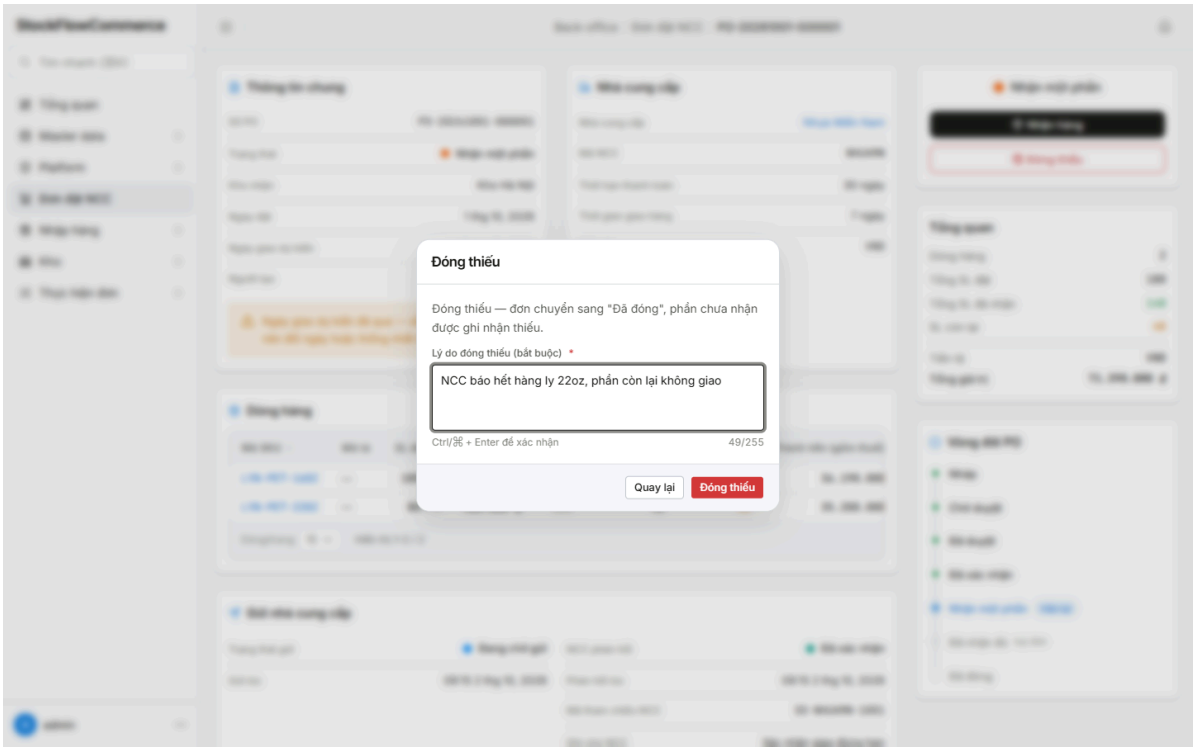
- **Function details:**
  - **Data requirement:** Reason, confirmation of reconciliation with the previous attempt, past-due delivery date acknowledgement.
  - **Validation:** Only Confirmed PO with supplier response Pending and last delivery Failed; reason required, ≤1000 chars; reconciliation checkbox required; past-due acknowledgement required when delivery date is past or missing; requires the PO approve permission.
  - **Business rules:** BR-12 (02 Purchase Order)

3.3.10.8 Close or Cancel Purchase Order

- **Function trigger:** Users click "Đóng thiếu", "Đóng đơn" or "Huỷ PO" on the purchase order detail page.
- **Function description:**
  - **Actors:** System Admin, Procurement Staff

- **Purpose:** End a purchase order: close it short, close it when fully received, or cancel it before any receipt.



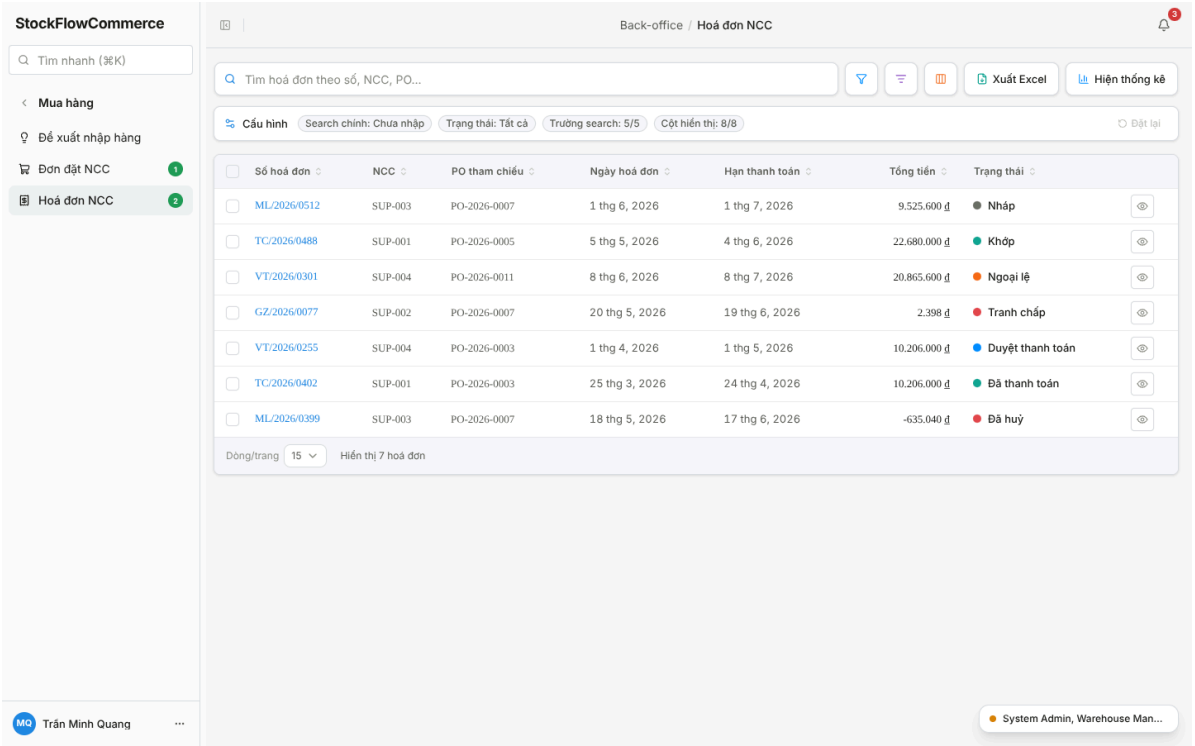


- **Function details:**
  - **Data requirement:** Reason (close short and cancel).
  - **Validation:** Close short only when Partially Received; close only when Received; cancel only from Draft, Pending Approval, Approved or Confirmed; reason required, ≤255 chars, for close short and cancel.
  - **Business rules:** BR-05 (02 Purchase Order)

3.3.11 Supplier Invoices

3.3.11.1 View Supplier Invoices

- **Function trigger:** Users click “Hoá đơn NCC” in the “Mua hàng” sidebar group (the item is shown only in mock mode).
- **Function description:**
  - **Actors:** Accountant
  - **Purpose:** Under development — View supplier invoices and their three-way match against purchase orders and goods receipts.

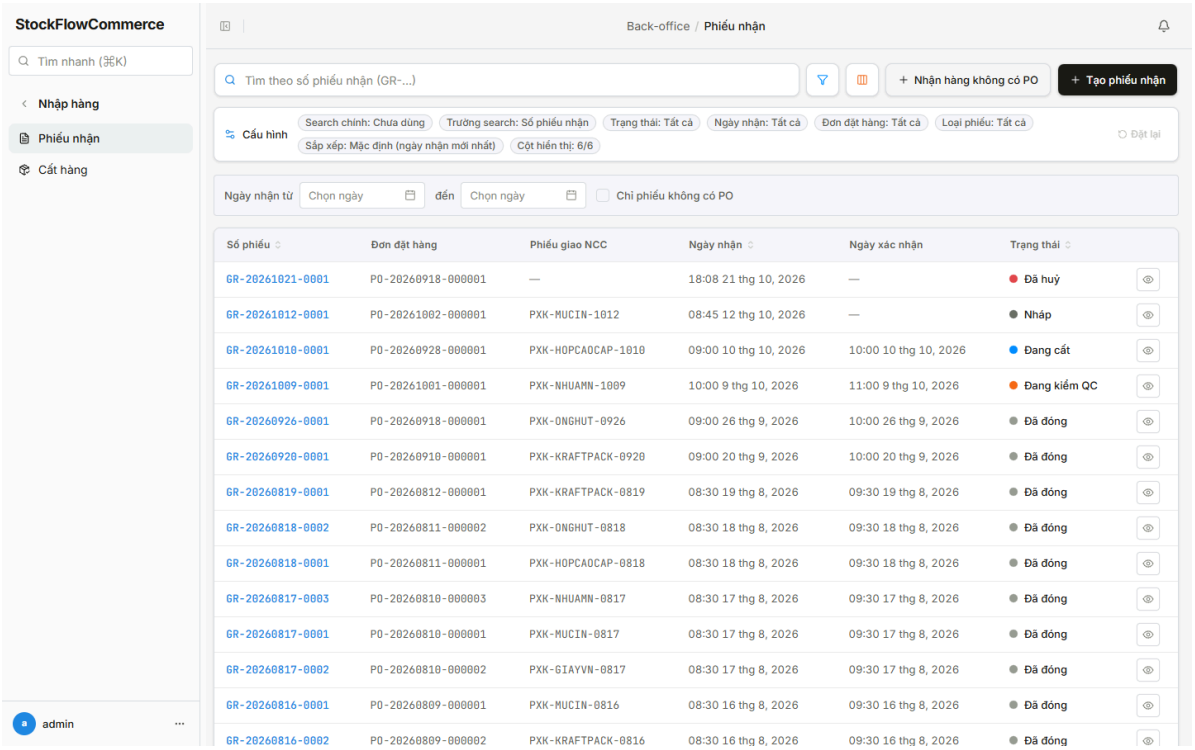


- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.3.12 Goods Receipt

3.3.12.1 View Goods Receipt List

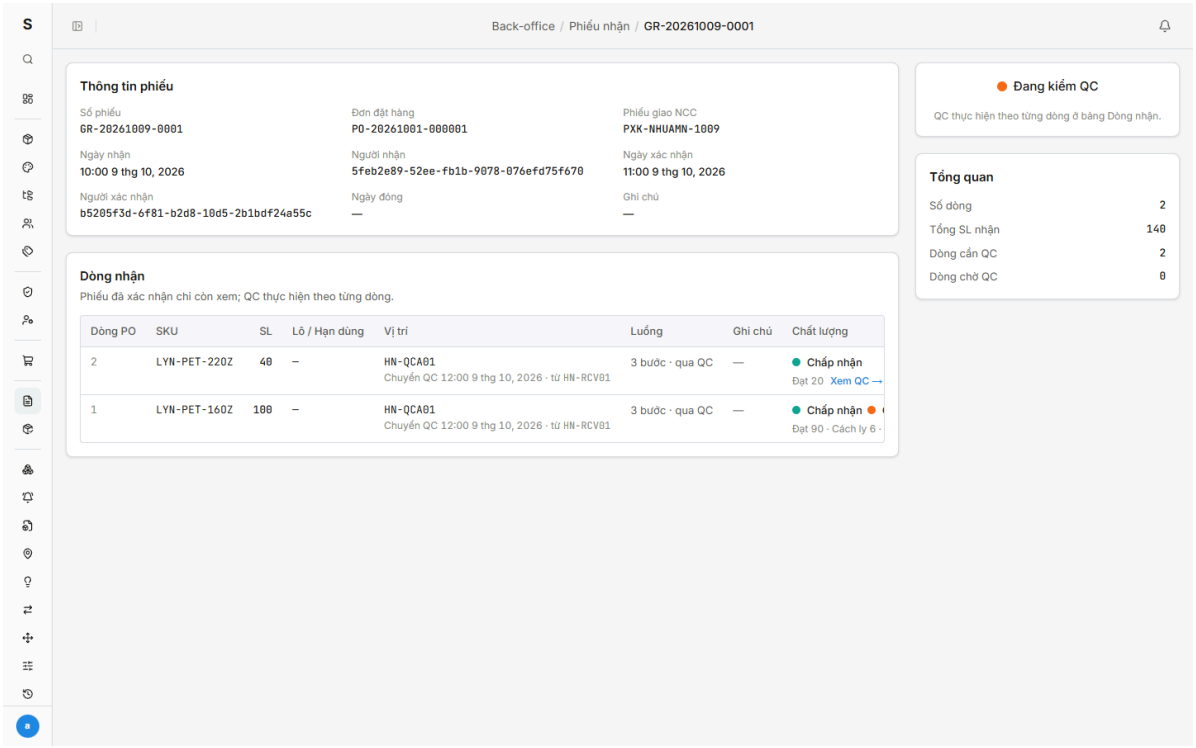
- **Function trigger:** Users click “Phiếu nhận” in the “Nhập hàng” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, QC Staff, Procurement Staff
  - **Purpose:** Find goods receipts and track their status.

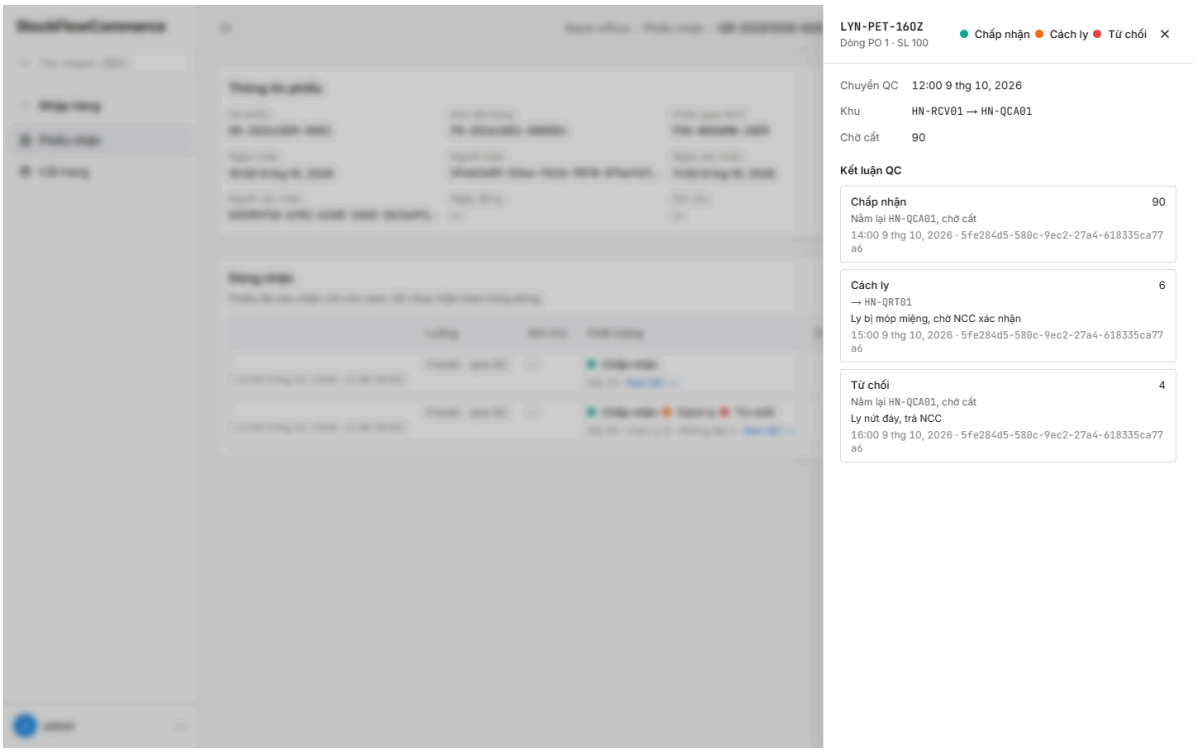


- **Function details:**
  - **Data requirement:** Status filter, receipt-number search, received-date range (“Ngày nhận từ … đến”), “Chỉ phiếu không có PO”, PO filter (via link); shows receipt number, PO, supplier delivery note, received date, confirmed date, status.
  - **Validation:** Start date after end date shows a warning and matches nothing.
  - **Business rules:** None

3.3.12.2 View Goods Receipt Detail

- **Function trigger:** Users click a row or the view icon (“Xem chi tiết …”) on the goods receipt list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, QC Staff, Procurement Staff
  - **Purpose:** View a receipt's counted lines, QC progress and approval decision.

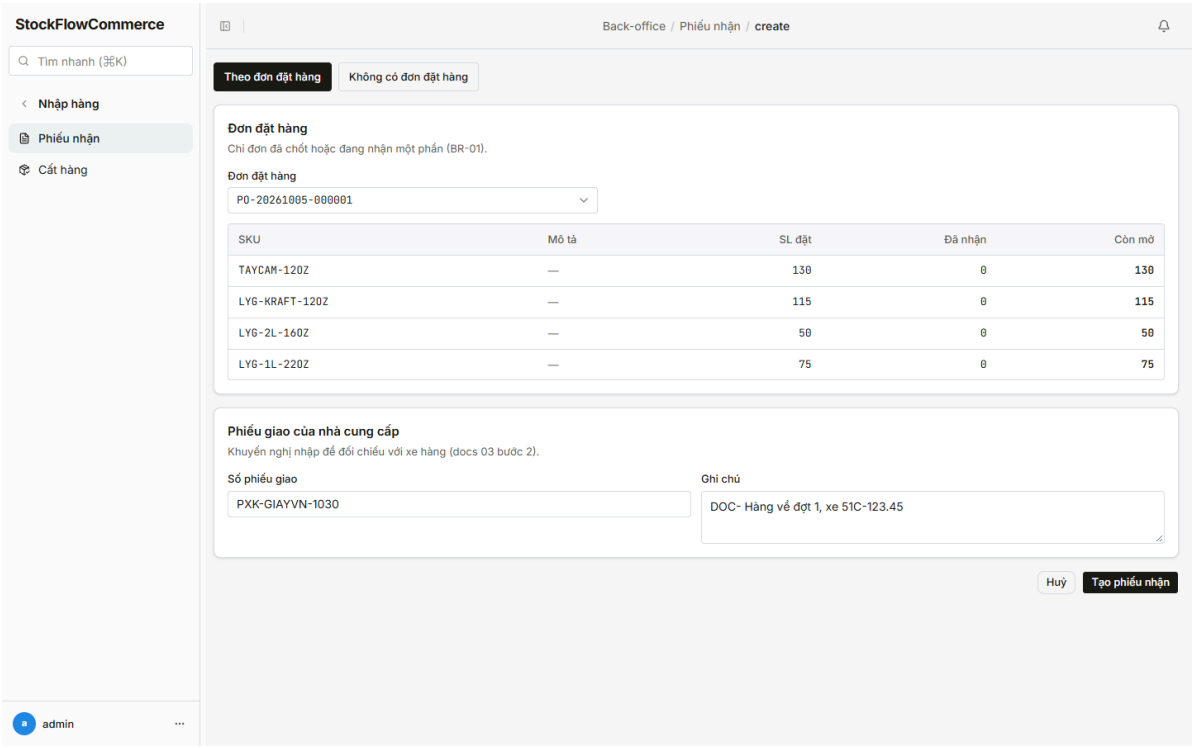


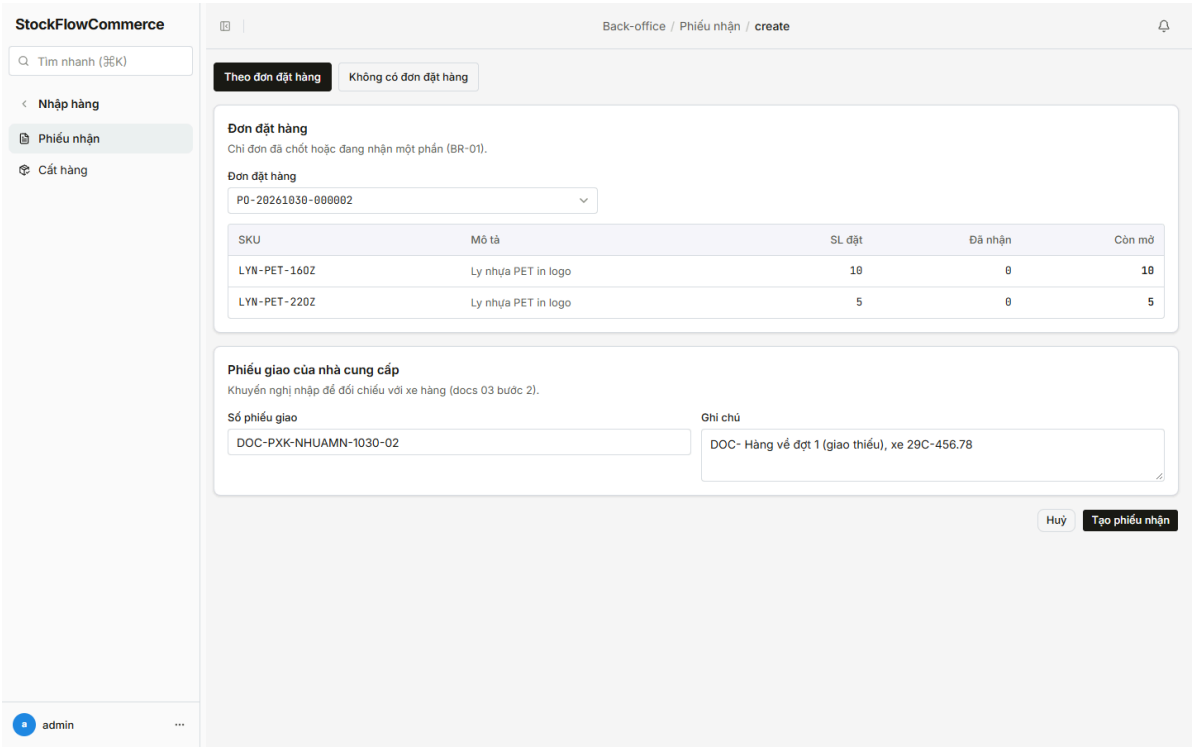


- **Function details:**
  - **Data requirement:** Receipt number, PO (or “Không có PO”), delivery note, status, received/confirmed time; lines: SKU, quantity, lot, expiry, location, flow (2-step or 3-step with QC), quality result; approval requester, decision, reason.
  - **Validation:** None
  - **Business rules:** BR-04, BR-07, BR-09 (03 Receipt)

3.3.12.3 Create Goods Receipt

- **Function trigger:** Users click “Tạo phiếu nhận” on the goods receipt list, or “Nhận hàng” on a Confirmed purchase order.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Procurement Staff
  - **Purpose:** Open a Draft goods receipt for goods arriving against one purchase order.

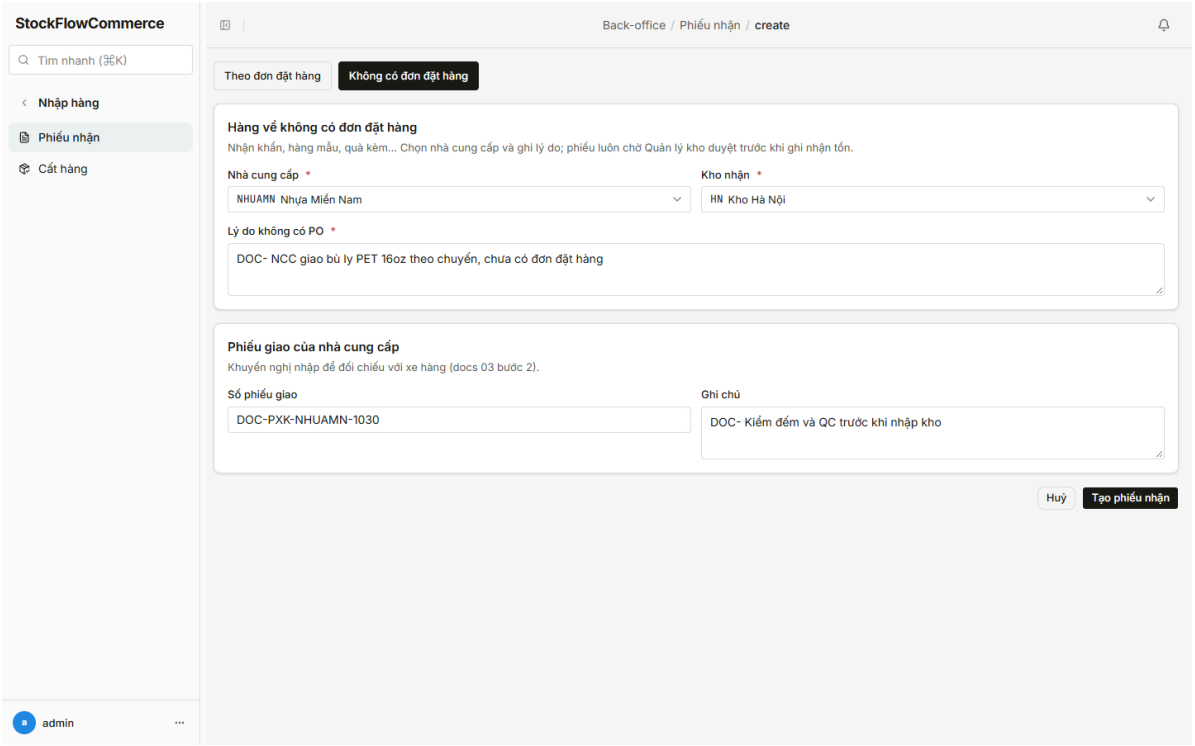




- **Function details:**
    - **Data requirement:** Purchase order, supplier delivery note number (“Số phiếu giao”), note.
    - **Validation:** PO required, only Confirmed or Partially Received and not rejected by the supplier; delivery note ≤100 chars; note ≤2000 chars; an existing Draft receipt for the PO is flagged to avoid double counting.
    - **Business rules:** BR-01 (03 Receipt); BR-03 (02 Purchase Order)

3.3.12.4 Create Receipt Without PO

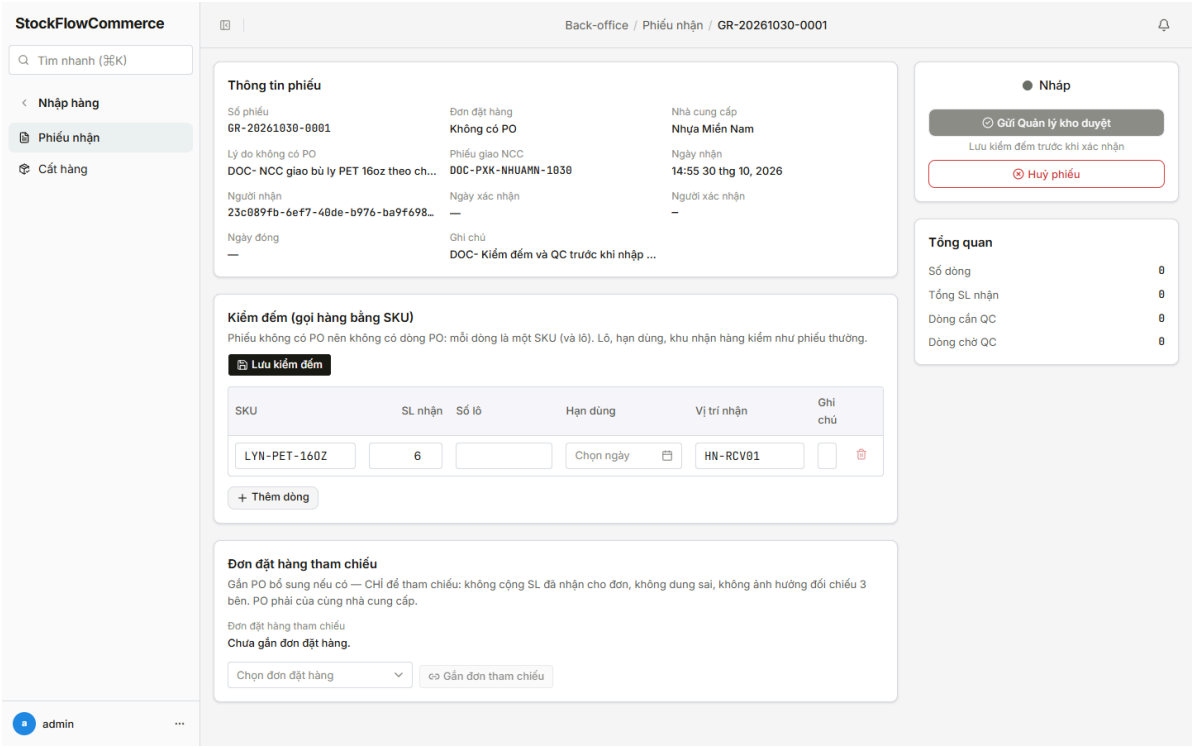
- **Function trigger:** Users click “Nhận hàng không có PO” on the goods receipt list (or the “Không có đơn đặt hàng” option on the create screen).
- **Function description:**
    - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Procurement Staff
    - **Purpose:** Record urgent, sample or bonus goods that arrive without a purchase order.

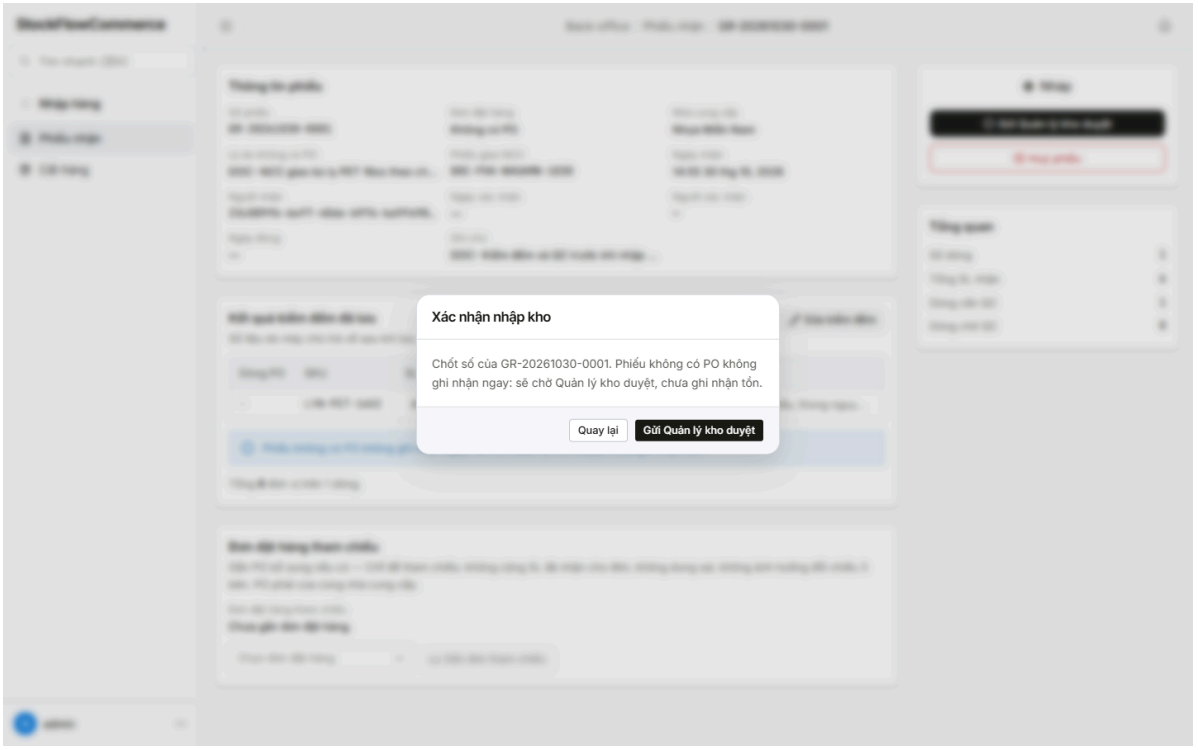


- **Function details:**
    - **Data requirement:** Supplier, receiving warehouse, reason, delivery note number, note; count lines by SKU (quantity, lot, expiry, location, note); optional reference PO.
    - **Validation:** Supplier, warehouse and reason required (reason ≤500); delivery note ≤100; note ≤2000; SKU required ≤64; quantity integer >0; one line per SKU and lot; expiry after receipt date; always needs Warehouse Manager approval.
    - **Business rules:** BR-RCP-002 (backend code, SCRUM-478); BR-06 (03 Receipt)

3.3.12.5 Count and Confirm Receipt

- **Function trigger:** Users enter counts on a Draft receipt and click “Lưu kiểm đếm”, then “Xác nhận nhập kho” (the button reads “Gửi Quản lý kho duyệt” on receipts without a PO or over tolerance); QC lines use “Chuyển sang khu QC”; “Huỷ phiếu” cancels.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Procurement Staff
  - **Purpose:** Record actual received quantities, post the receipt to stock and move QC-required lines to the QC area.

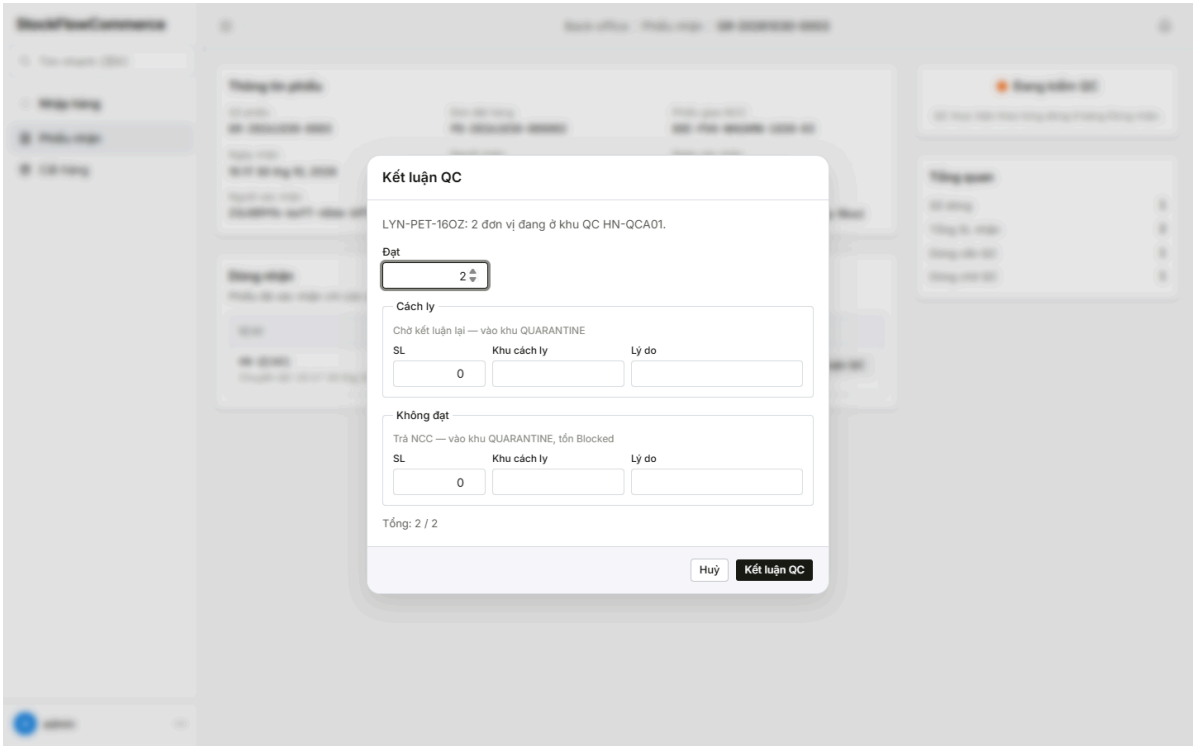




- **Function details:**
  - **Data requirement:** Per line: received quantity, lot, expiry date, receiving location code, note; QC area location code.
  - **Validation:** 1–200 lines; quantity integer >0; one line per PO line and lot; lot/expiry required for tracked SKUs; expiry after receipt date; location required ≤64; lot ≤64; note ≤255; confirm needs ≥1 line; cancel only Draft or Pending Approval.
  - **Business rules:** BR-02, BR-03, BR-05, BR-06, BR-07, BR-09, BR-12 (03 Receipt)

3.3.12.6 Record QC Result

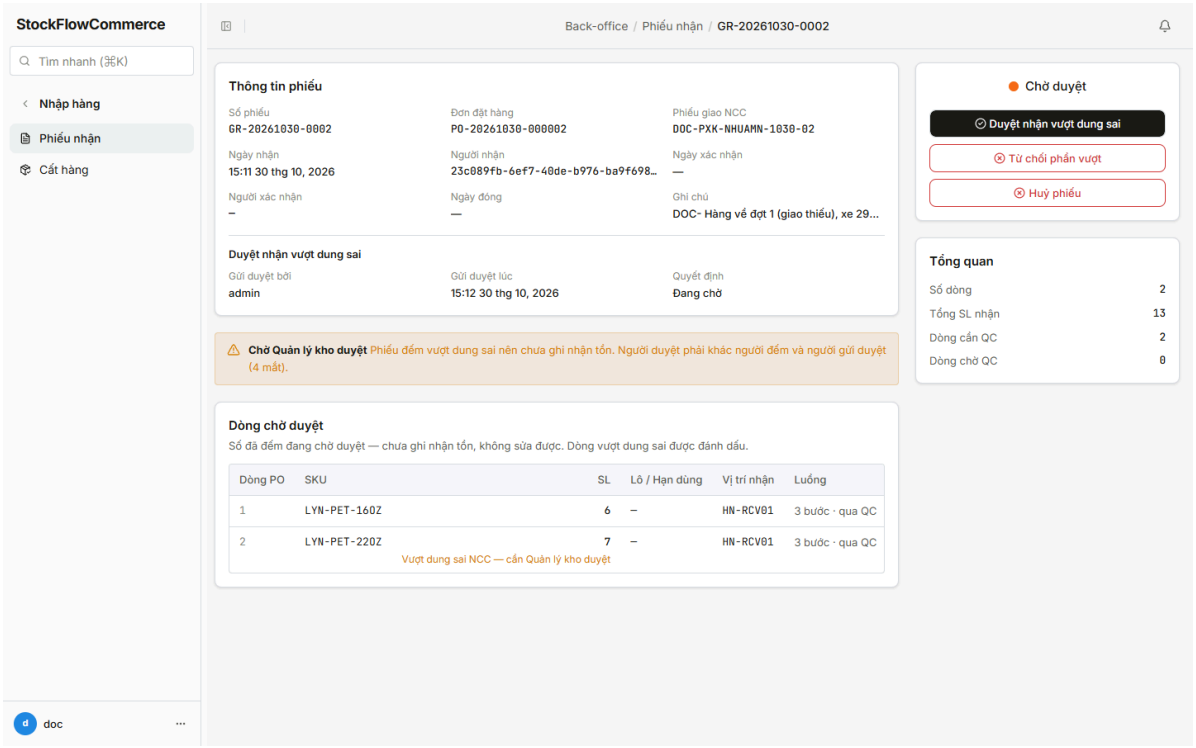
- **Function trigger:** QC staff click “Kết luận QC” on a line already in the QC area of an In QC receipt.
- **Function description:**
  - **Actors:** System Admin, QC Staff
  - **Purpose:** Record the QC result of a received line as accepted, quarantined and rejected quantities.

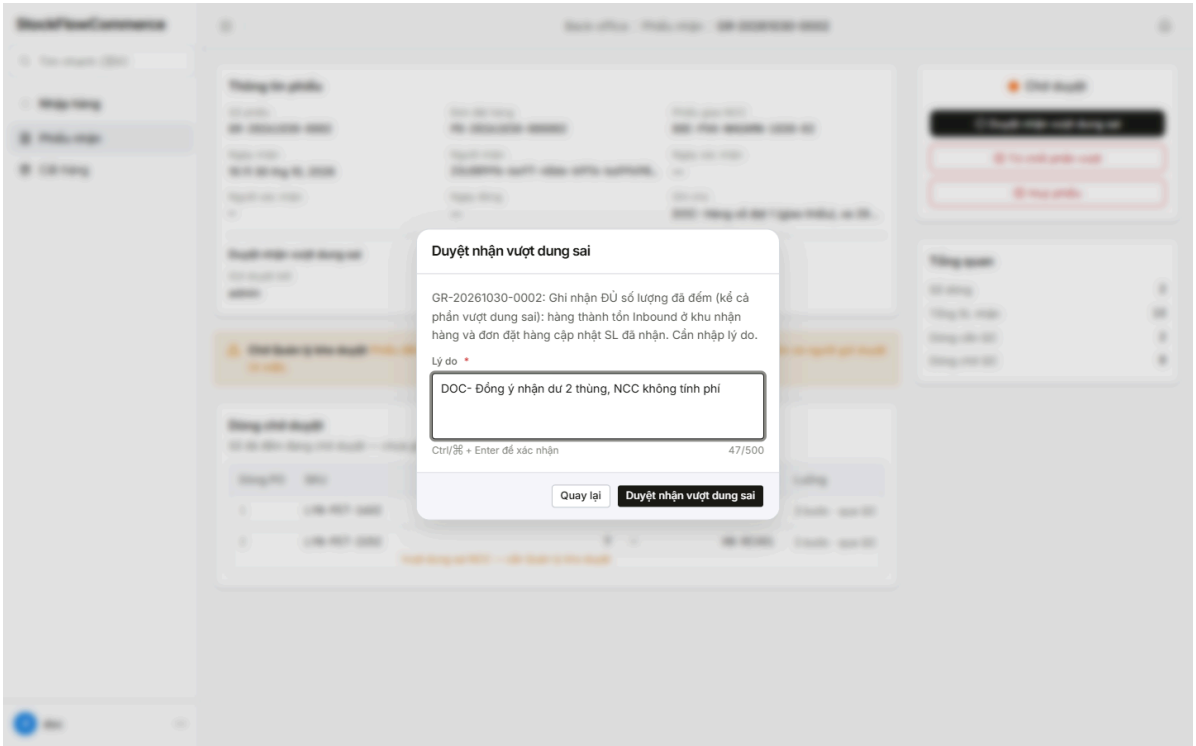


- **Function details:**
  - **Data requirement:** Accepted quantity; quarantined quantity, quarantine location, reason; rejected quantity, quarantine location, reason.
  - **Validation:** Quantities integer ≥0; accepted + quarantined + rejected = quantity moved to QC; quarantined/rejected parts need a location and a reason (≤500); the two quarantine locations must differ; one decision per line; requires QC approve permission.
  - **Business rules:** BR-07, BR-08 (03 Receipt)

3.3.12.7 Approve Over-Receipt

- **Function trigger:** Users click “Duyệt nhận vượt dung sai” or “Từ chối phần vượt” on a Pending Approval receipt (“Duyệt nhận hàng không có PO” / “Từ chối phiếu” for no-PO receipts).
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Decide on a receipt counted beyond supplier tolerance (or without PO): accept the full count or post only up to tolerance.



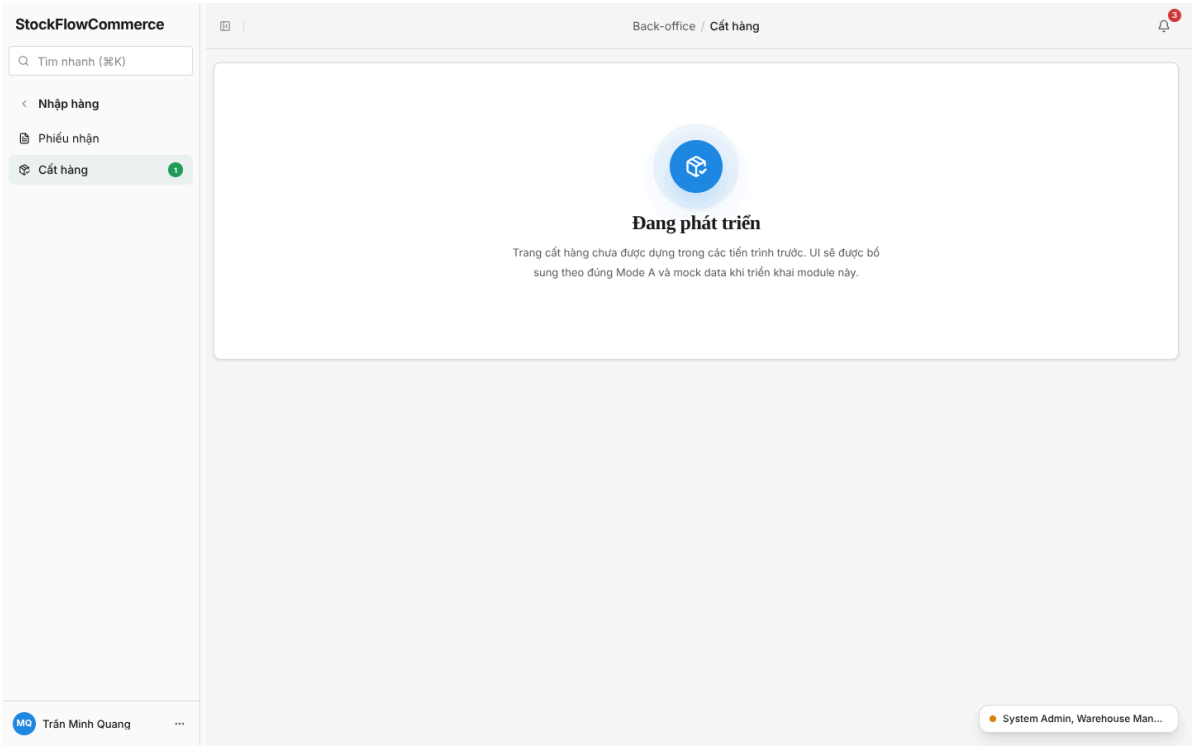


- **Function details:**
  - **Data requirement:** Decision reason.
  - **Validation:** Reason required, ≤500 chars; approver must differ from the counter and the submitter (backend RECEIPT_SELF_APPROVAL); requires receipt approve permission; rejecting the excess cancels the receipt if nothing fits within tolerance.
  - **Business rules:** BR-RCP-001 (backend code, SCRUM-477); BR-02 (03 Receipt)

3.3.13 Putaway

3.3.13.1 Put Away Goods

- **Function trigger:** Users click “Cất hàng” in the “Nhập hàng” sidebar group.
- **Function description:**
  - **Actors:** Warehouse Staff, Warehouse Manager
  - **Purpose:** Under development — Put received goods away from the receiving or QC area into storage bins through putaway tasks.

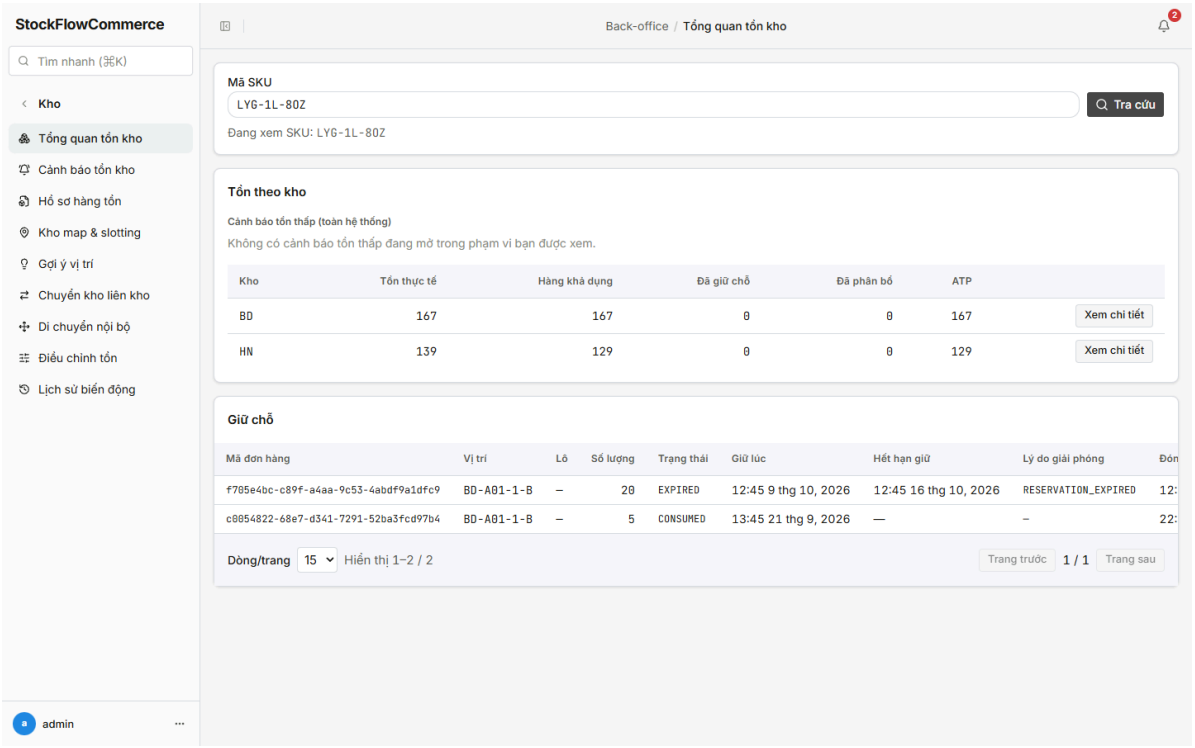


- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.3.14 Inventory

3.3.14.1 View Inventory Overview

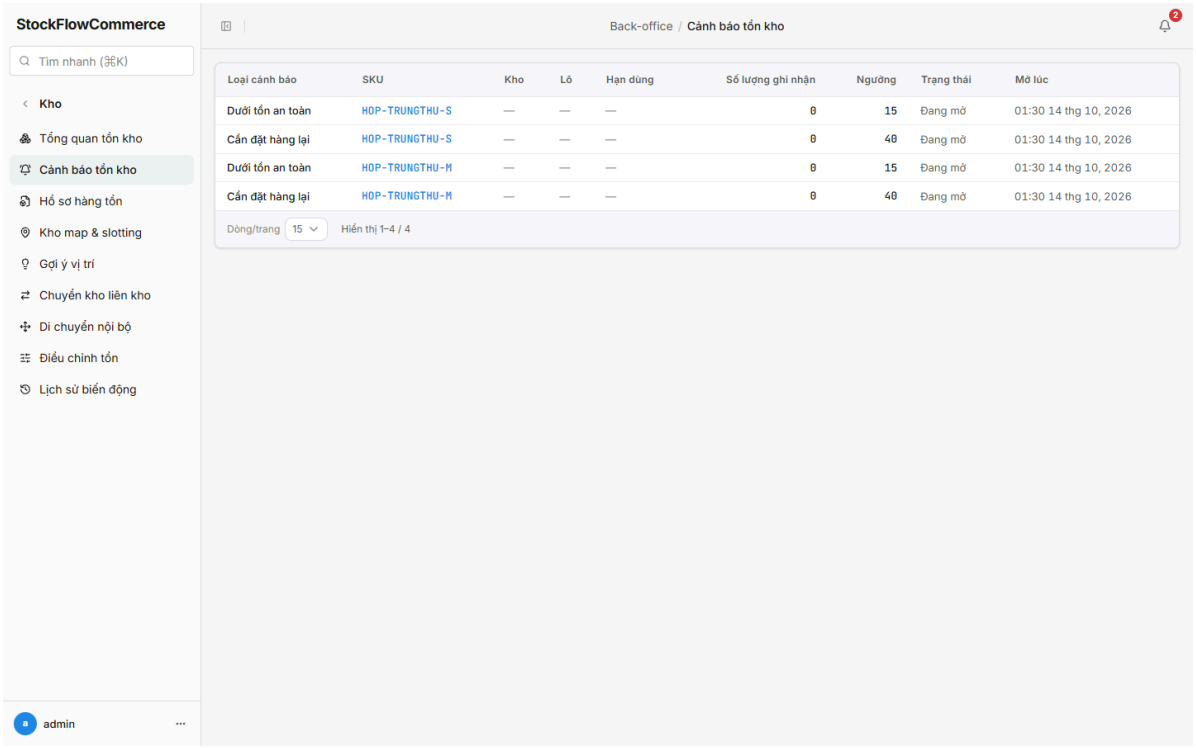
- **Function trigger:** Users click “Tổng quan tồn kho” in the “Kho” sidebar group, enter an SKU code and click “Tra cứu”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, QC Staff, Inventory Planner, Order Coordinator, Production Staff
  - **Purpose:** Look up one SKU's stock, ATP and reservations across warehouses.



- **Function details:**
  - **Data requirement:** SKU code; shows per warehouse on hand, available, reserved, allocated, ATP; stock by location and lot with expiry and received time; reservations; low-stock and expiry alerts.
  - **Validation:** SKU required, 1–64 letters, digits, '.', '_' or '-', starting with a letter or digit; upper-cased; exact match only.
  - **Business rules:** BR-01, BR-02, BR-07, BR-19 (20 Inventory Control)

3.3.14.2 View Stock Alerts

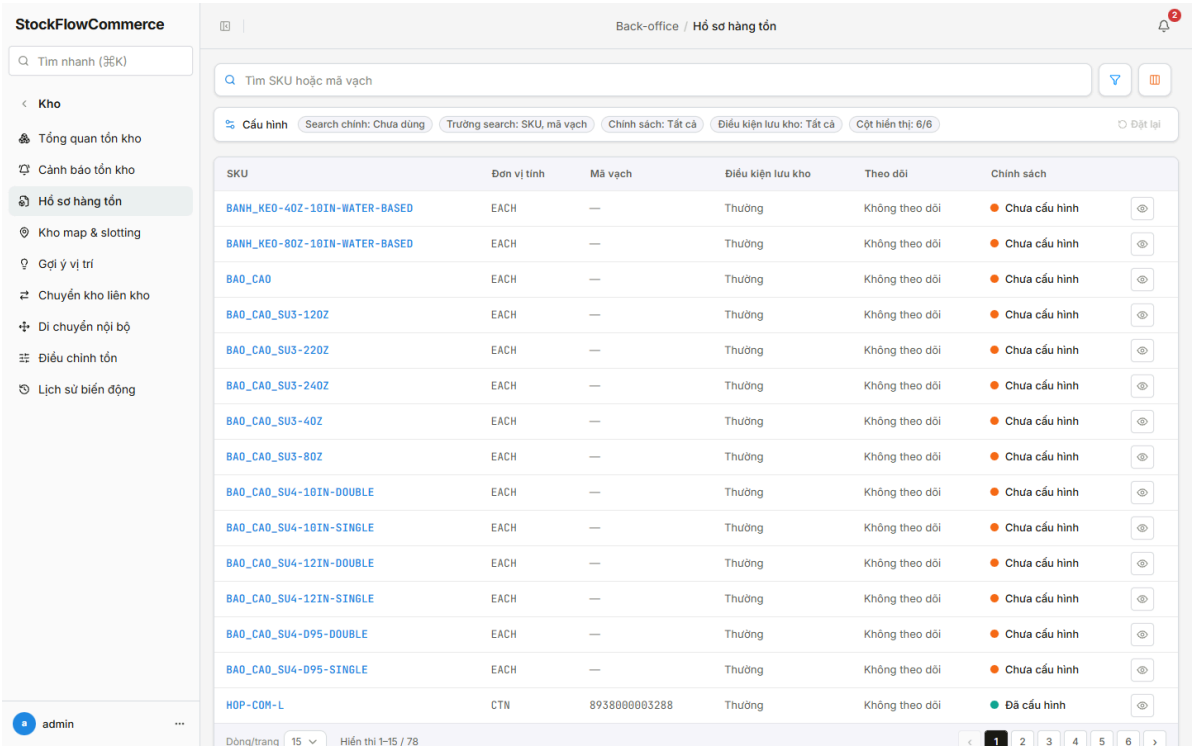
- **Function trigger:** Users click “Cảnh báo tồn kho” in the “Kho” sidebar group, or open an inventory-alert notification link.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Monitor open low-stock and expiry alerts.



- **Function details:**
  - **Data requirement:** Alert kind and warehouse filters (from the URL); shows kind (“Cần đặt hàng lại”, “Dưới tồn an toàn”, “Sắp hết hạn”, “Đã quá hạn”), SKU, warehouse, lot, expiry, observed quantity, threshold, status, opened at.
  - **Validation:** Invalid URL filter parameters are ignored with a warning.
  - **Business rules:** BR-13, BR-14, BR-15, BR-17 (20 Inventory Control)

3.3.14.3 View Inventory Item List

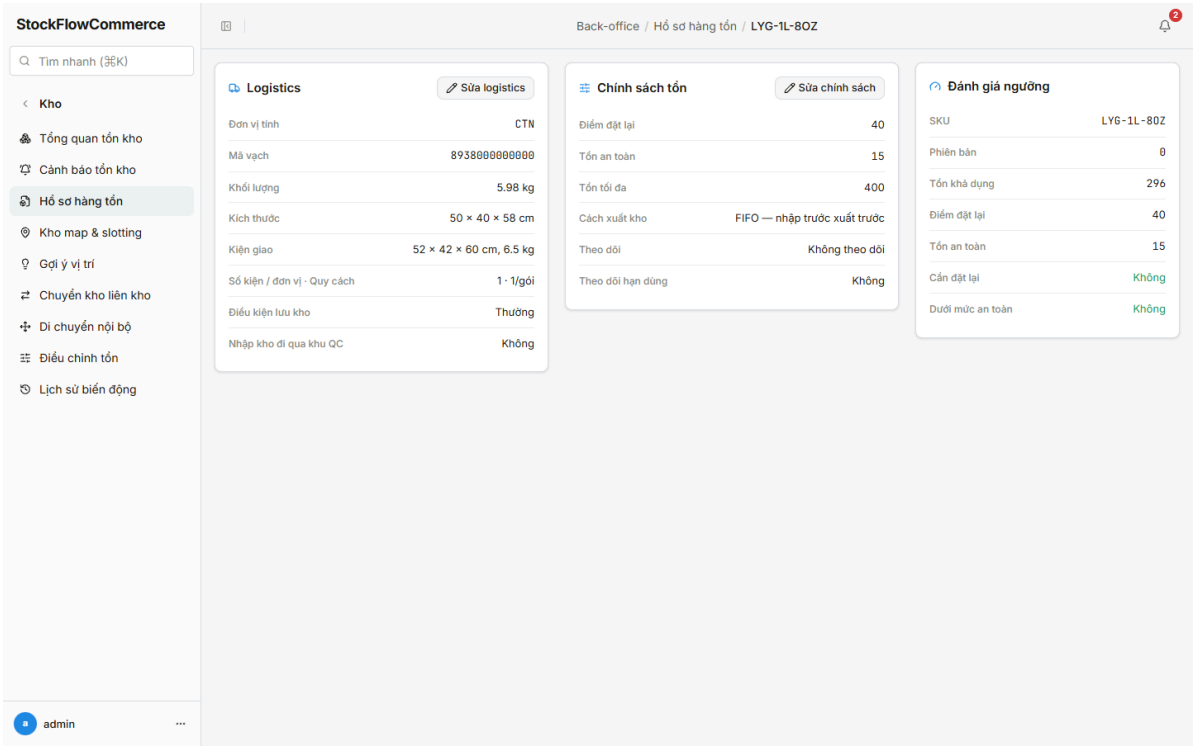
- **Function trigger:** Users click “Hồ sơ hàng tồn” in the “Kho” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner, Procurement Staff
  - **Purpose:** Browse the logistics profile and stock-policy status of every SKU.



- **Function details:**
  - **Data requirement:** Search by SKU or barcode, storage-class and policy-configured filters; shows SKU, unit of measure, barcode, storage class, tracking mode, policy status (rows without a policy are flagged).
  - **Validation:** None
  - **Business rules:** BR-07, BR-08 (01 Product Creation)

3.3.14.4 View Inventory Item Detail

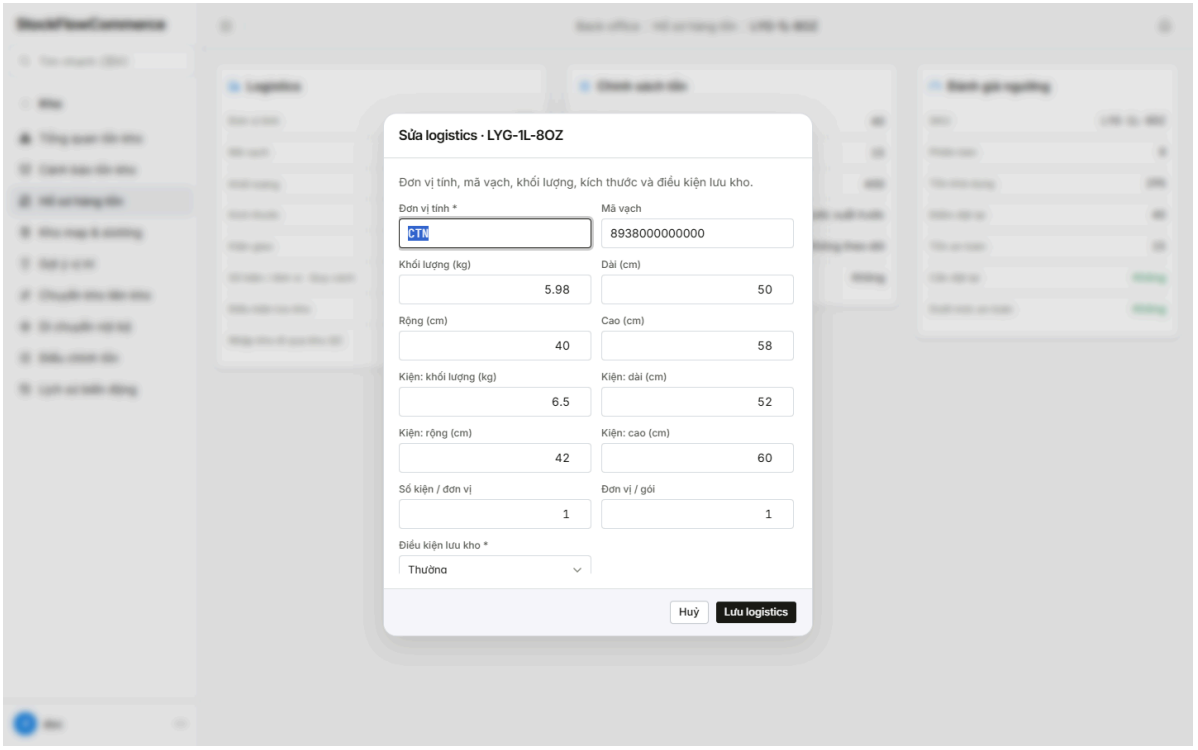
- **Function trigger:** Users click a row or the view icon on the inventory item list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner, Procurement Staff
  - **Purpose:** View one SKU's logistics data, stock policy and threshold evaluation.

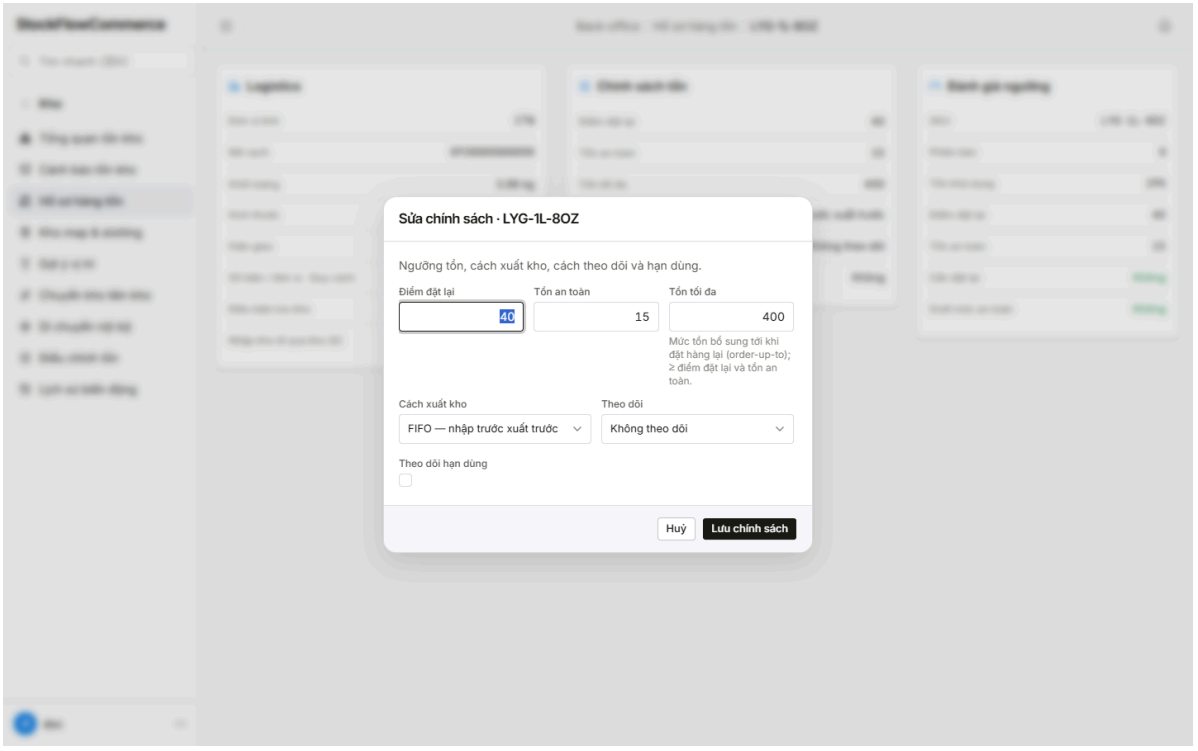


- **Function details:**
  - **Data requirement:** SKU, version; unit of measure, barcode, weight, dimensions, package data, pack size, storage class, QC required, rotatable, shipping restriction; reorder point, safety stock, max quantity, removal strategy, tracking, expiry, max shelf life; usable on hand, reorder/safety flags.
  - **Validation:** None
  - **Business rules:** BR-07, BR-08 (01 Product Creation); BR-13 (20 Inventory Control)

3.3.14.5 Edit Inventory Item

- **Function trigger:** Users click “Sửa logistics” or “Sửa chính sách” on the inventory item detail, then “Lưu logistics” or “Lưu chính sách”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Inventory Planner
  - **Purpose:** Update a SKU's logistics data or stock policy (thresholds, tracking, removal strategy).





- **Function details:**
  - **Data requirement:** Logistics: unit of measure, barcode, weight, dimensions, package data, pack size, storage class, adult signature, shipping note, QC required, rotatable. Policy: reorder point, safety stock, max quantity, removal strategy, tracking mode, expiry tracked, max shelf life.
  - **Validation:** Unit of measure [A-Z0-9_]{1,16}; barcode ≤64; weights/dimensions >0 if set; package count, pack size integers ≥1; shipping note ≤500; thresholds integers ≥0 with safety ≤ reorder ≤ max;

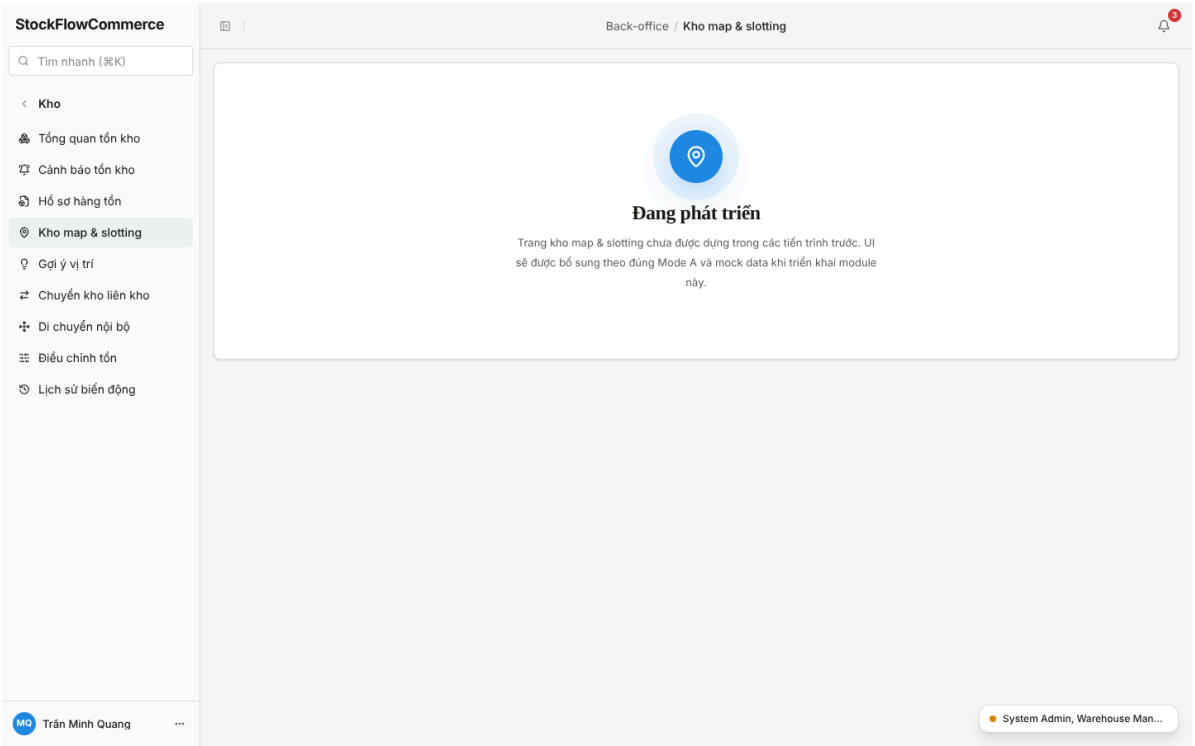
expiry tracking needs LOT/LOT_SERIAL and FEFO; shelf life 1–36,500 days; stale version 409.

- **Business rules:** BR-03, BR-09, BR-15, BR-17 (01 Product Creation); BR-13 (20 Inventory Control)

3.3.15 Warehouse Map & Slotting

3.3.15.1 View Warehouse Map

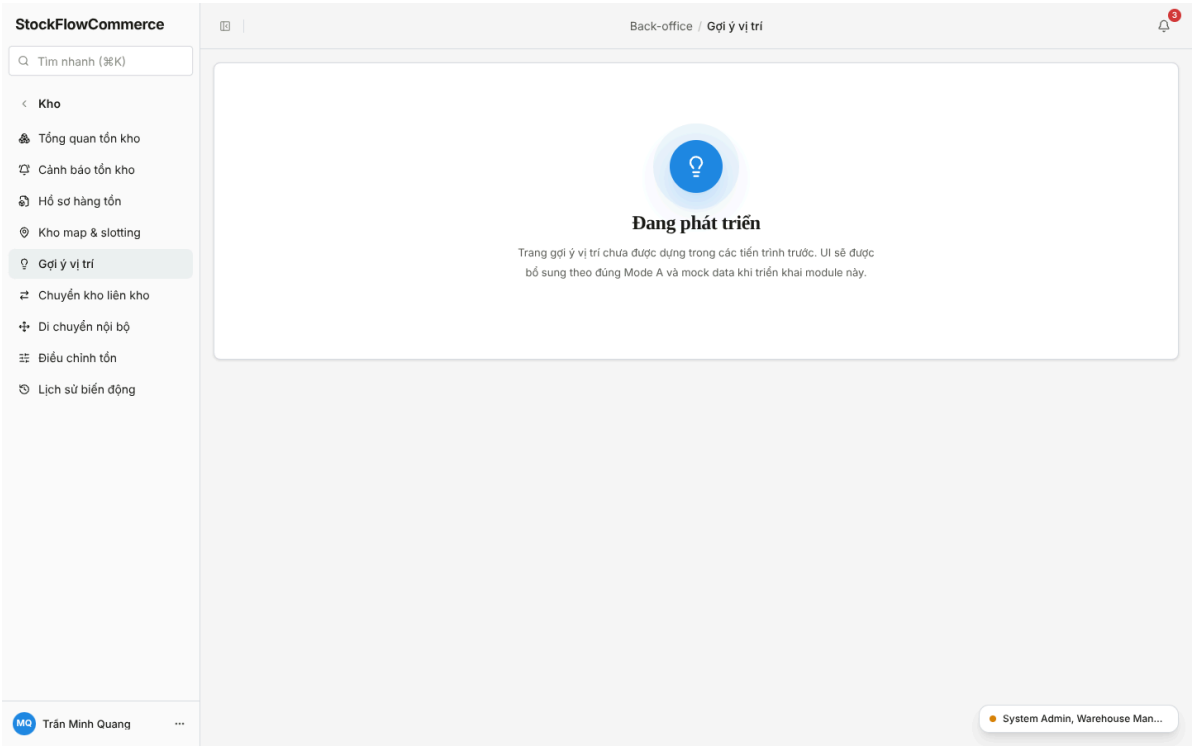
- **Function trigger:** Users click “Kho map & slotting” in the “Kho” sidebar group.
- **Function description:**
  - **Actors:** Warehouse Manager, Warehouse Staff
  - **Purpose:** Under development — View the warehouse layout of areas, shelves and bins as a map.



- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.3.15.2 View Slotting Suggestions

- **Function trigger:** Users click “Gợi ý vị trí” in the “Kho” sidebar group.
- **Function description:**
  - **Actors:** Warehouse Manager, Warehouse Staff
  - **Purpose:** Under development — Suggest suitable bin locations for goods to be stored.

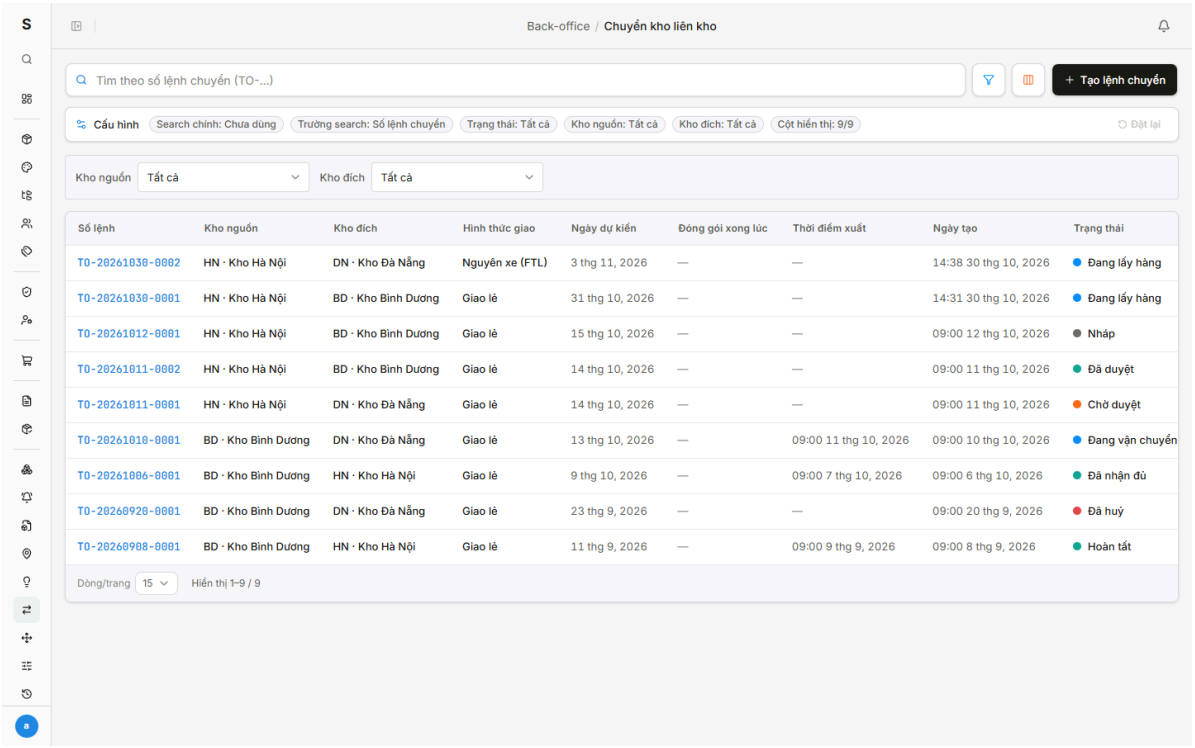


- **Function details:**
  - **Data requirement:** Under development
  - **Validation:** Under development
  - **Business rules:** Under development

3.3.16 Transfer Orders

3.3.16.1 View Transfer Order List

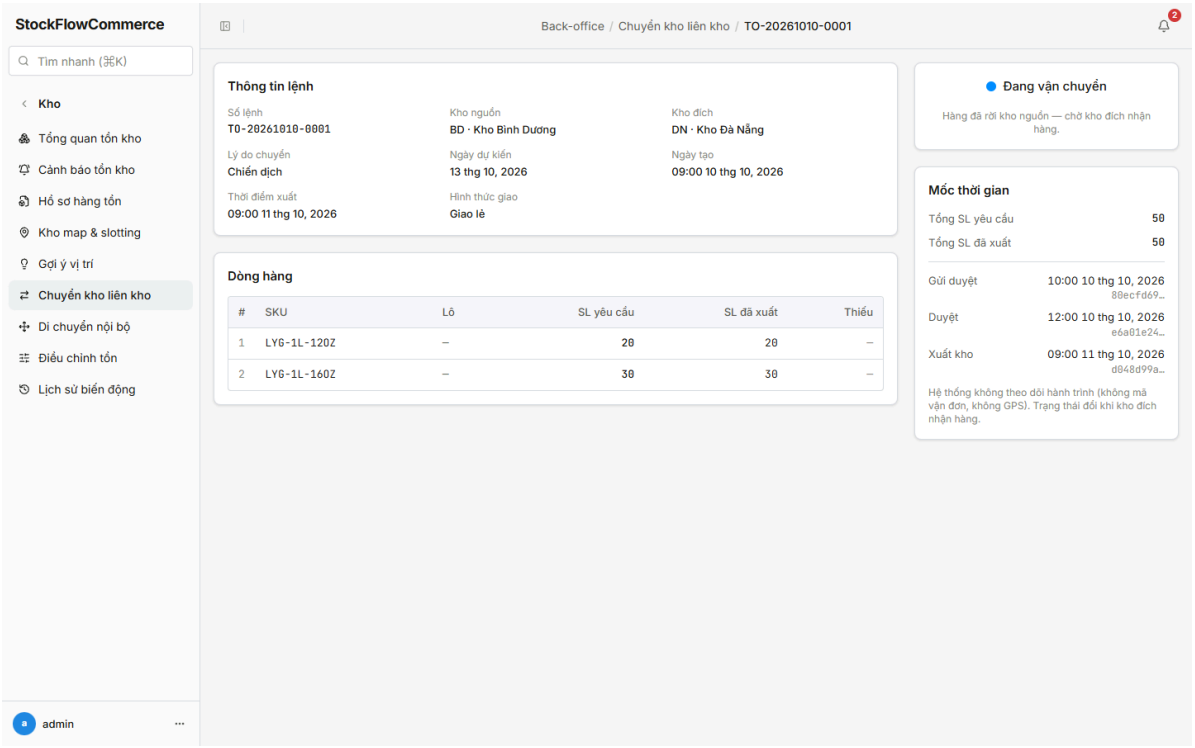
- **Function trigger:** Users click “Chuyển kho liên kho” in the “Kho” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Browse and filter transfer orders between warehouses.

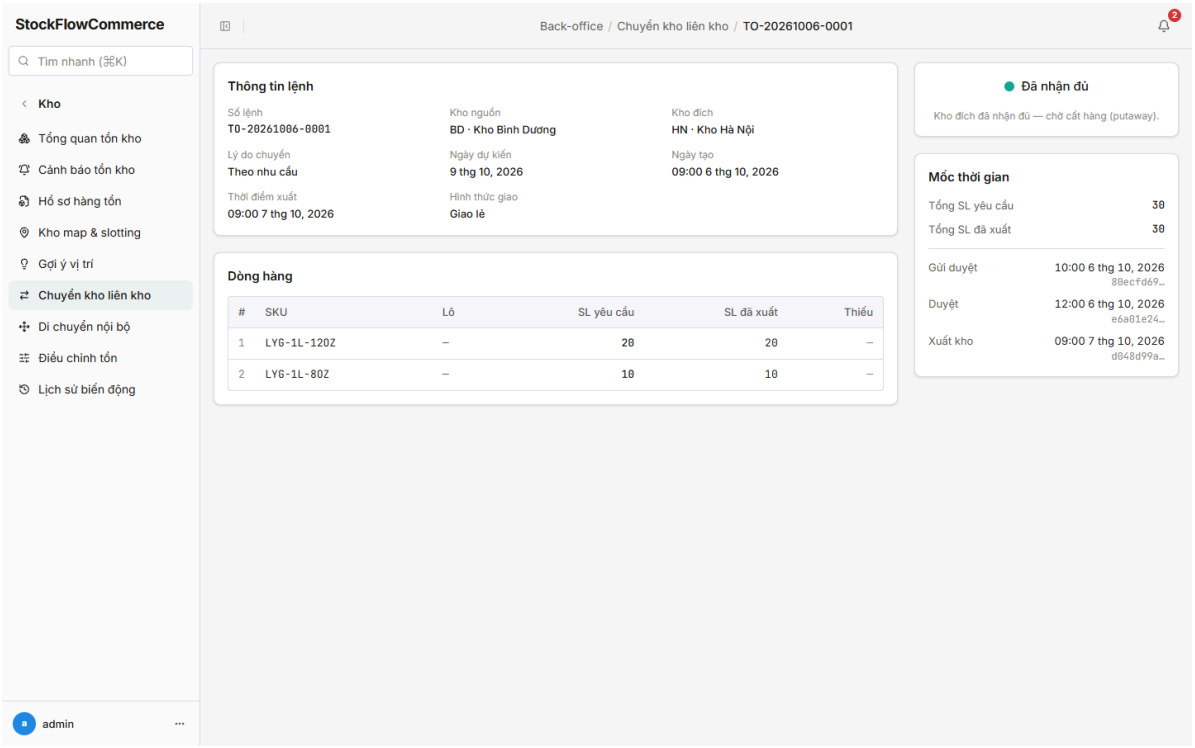


- **Function details:**
  - **Data requirement:** Transfer-number search, status, source warehouse and destination warehouse filters; shows number, source, destination, delivery mode, expected date, packed at, dispatched at, created at, status.
  - **Validation:** None
  - **Business rules:** None

3.3.16.2 View Transfer Order Detail

- **Function trigger:** Users click a row or the view icon on the transfer order list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** View a transfer order's lines and its approval, picking, packing and dispatch progress.

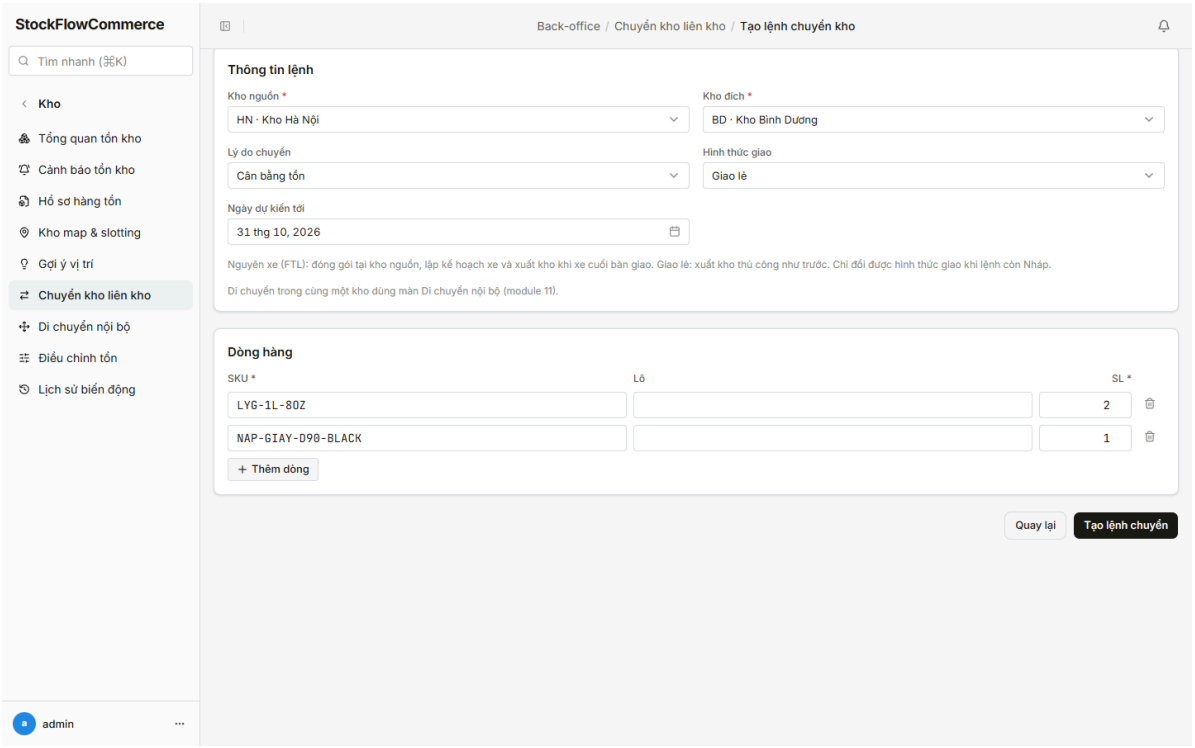




- **Function details:**
  - **Data requirement:** Number, status, source and destination warehouse, reason, expected date, delivery mode, packed at, goods left at; lines (SKU, lot, requested, shipped); rejection reason; timeline; next-step hint.
  - **Validation:** None
  - **Business rules:** BR-03, BR-04, BR-09 (10 Inter-warehouse Transfer)

3.3.16.3 Create Transfer Order

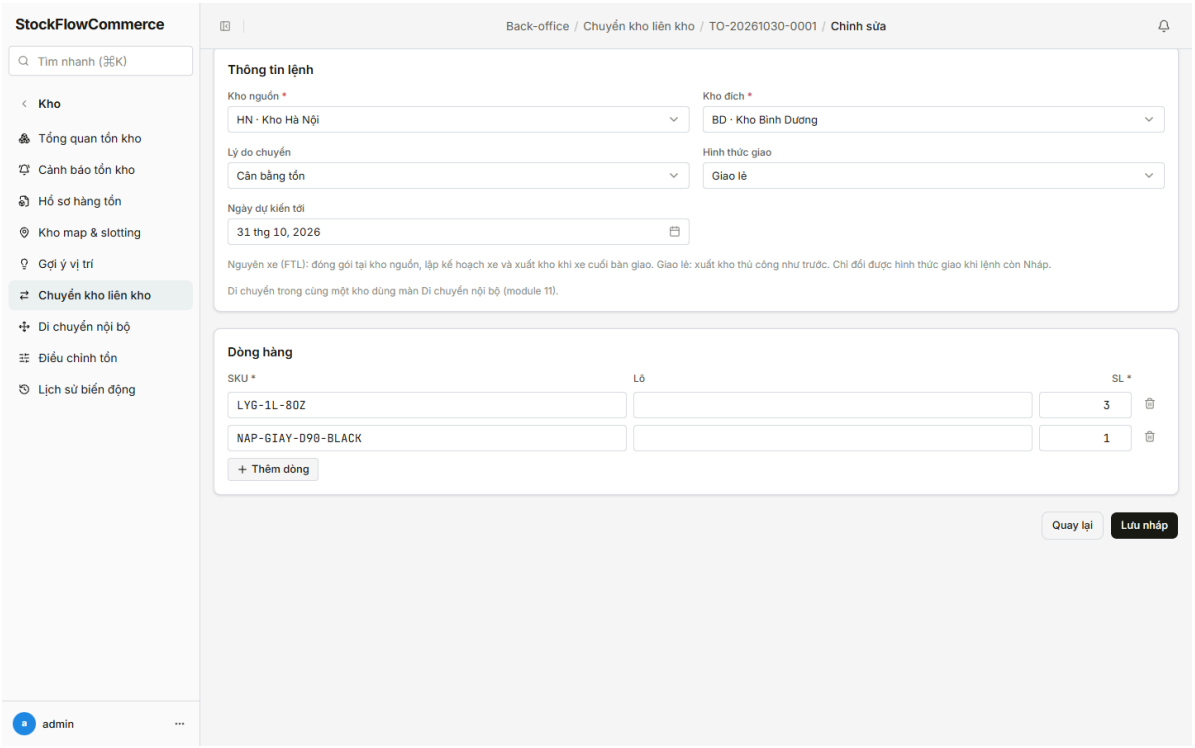
- **Function trigger:** Users click “Tạo lệnh chuyển” button on the transfer order list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Inventory Planner
  - **Purpose:** Create a Draft transfer order moving stock from a source warehouse to a destination warehouse.

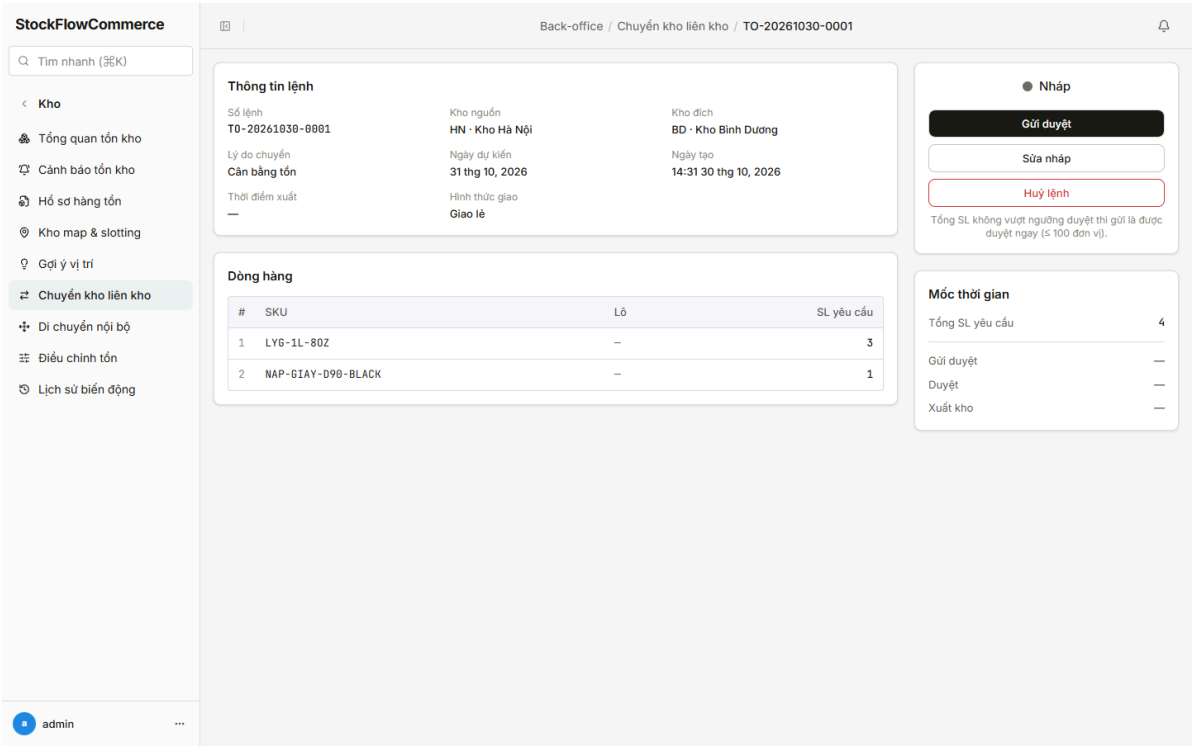


- **Function details:**
  - **Data requirement:** Source warehouse, destination warehouse, transfer reason, expected arrival date, delivery mode (“Nguyên xe (FTL)” / “Giao lẻ”); lines: SKU, lot, quantity.
  - **Validation:** Both warehouses required and different; 1–200 lines; SKU 1–64 valid characters; lot ≤64; quantity integer 1–1,000,000; one line per SKU and lot; backend checks each line has enough unreserved Available stock at the source.
  - **Business rules:** BR-01, BR-02, BR-09 (10 Inter-warehouse Transfer)

3.3.16.4 Edit Transfer Order

- **Function trigger:** Users click “Sửa nháp” on a Draft transfer order and save with “Lưu nháp”; “Gửi duyệt” and “Huỷ lệnh” are on the detail page.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Inventory Planner
  - **Purpose:** Revise a draft transfer order, submit it for approval or cancel it.

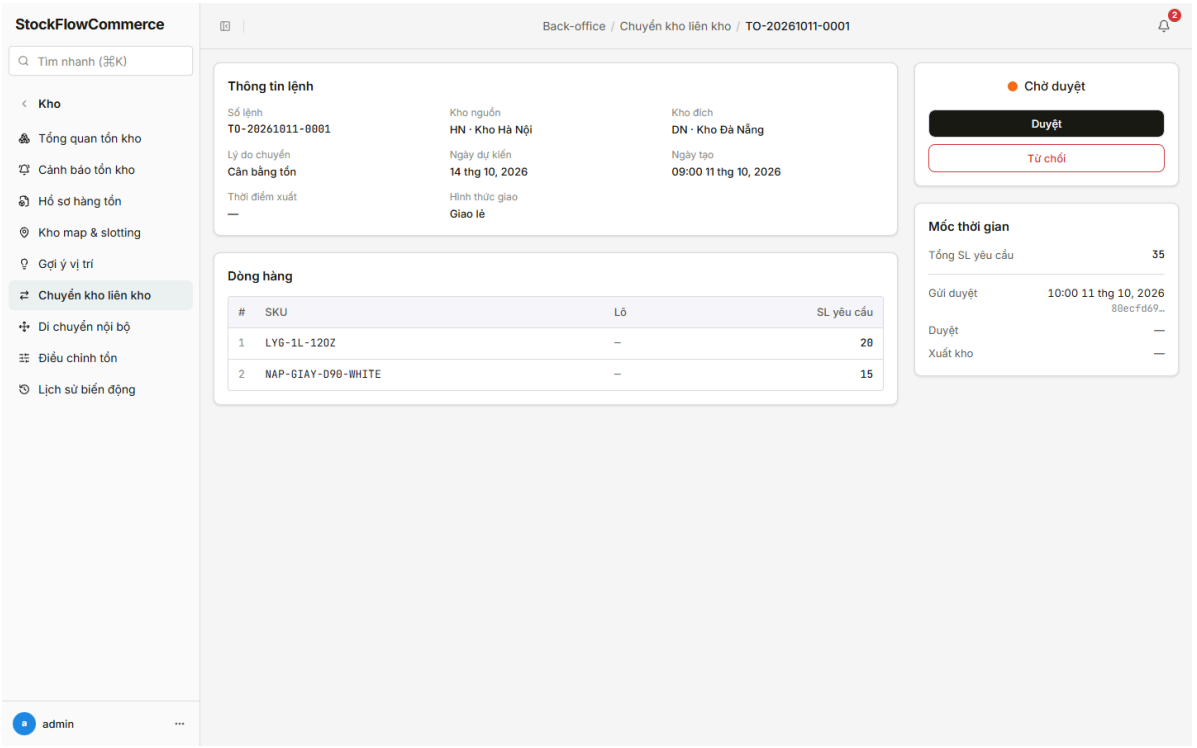




- **Function details:**
  - **Data requirement:** Same fields as create (warehouses, reason, expected date, delivery mode, lines); cancellation reason.
  - **Validation:** Same rules as create; only Draft is editable; stale version rejected (409); cancel only from Draft or Approved, reason required ≤500 chars.
  - **Business rules:** BR-01, BR-02, BR-04, BR-08 (10 Inter-warehouse Transfer)

3.3.16.5 Approve Transfer Order

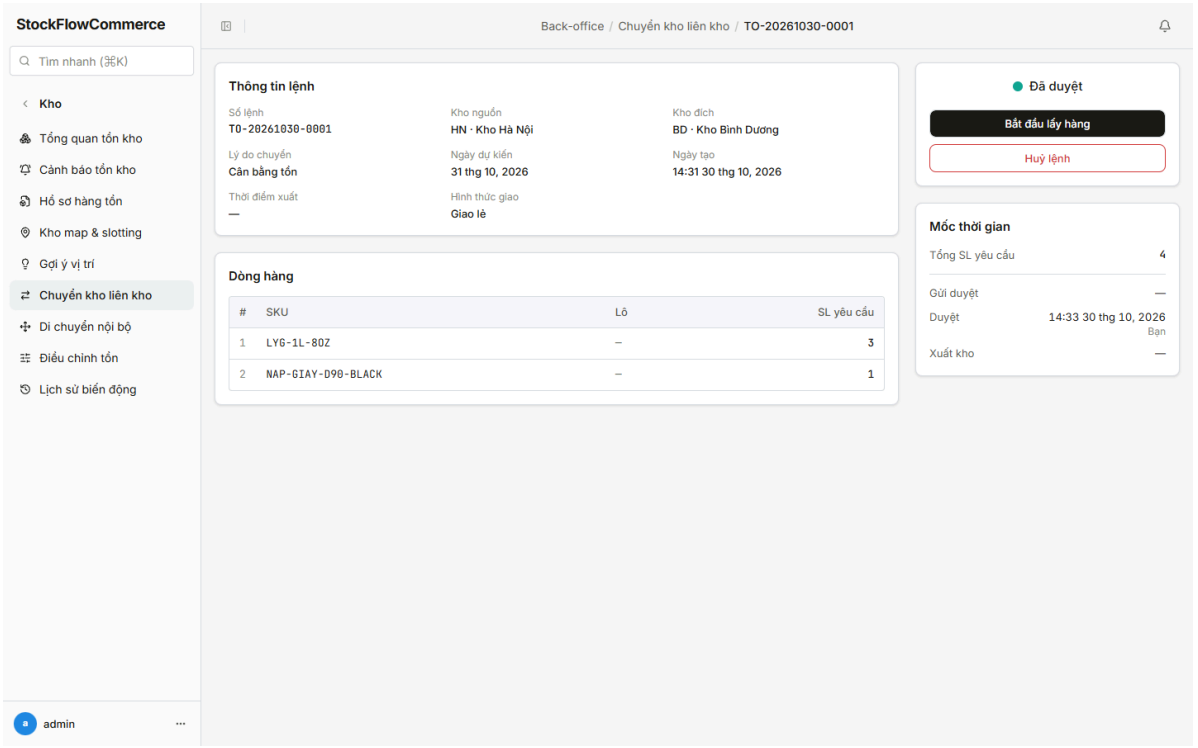
- **Function trigger:** Users click “Duyệt” or “Từ chối” on a Pending Approval transfer order's detail page.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Approve a submitted transfer order or return it to Draft with a reason.

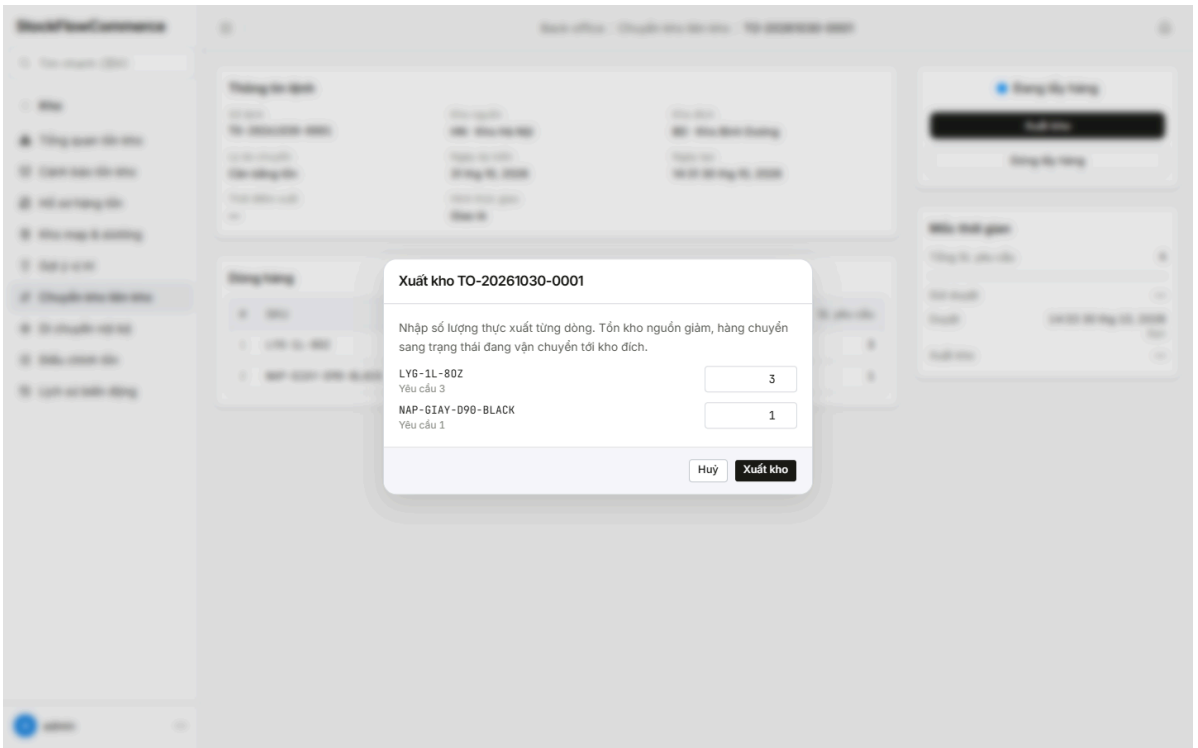


- **Function details:**
  - **Data requirement:** Rejection reason (reject only).
  - **Validation:** Only Pending Approval; rejection reason required, ≤500 chars; the submitter sees both buttons disabled (“Bạn là người gửi duyệt — cần người khác duyệt.”); requires the transfer approve permission.
  - **Business rules:** BR-08 (10 Inter-warehouse Transfer)

3.3.16.6 Pick and Dispatch Transfer

- **Function trigger:** Users click “Bắt đầu lấy hàng” on an Approved transfer order, then “Xuất kho” on a “Giao lẻ” transfer (or “Dừng lấy hàng” to stop picking); a “Nguyên xe (FTL)” transfer has no “Xuất kho” button because it is dispatched through its load plan.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Start picking at the source warehouse and confirm shipped quantities so the goods go in transit.

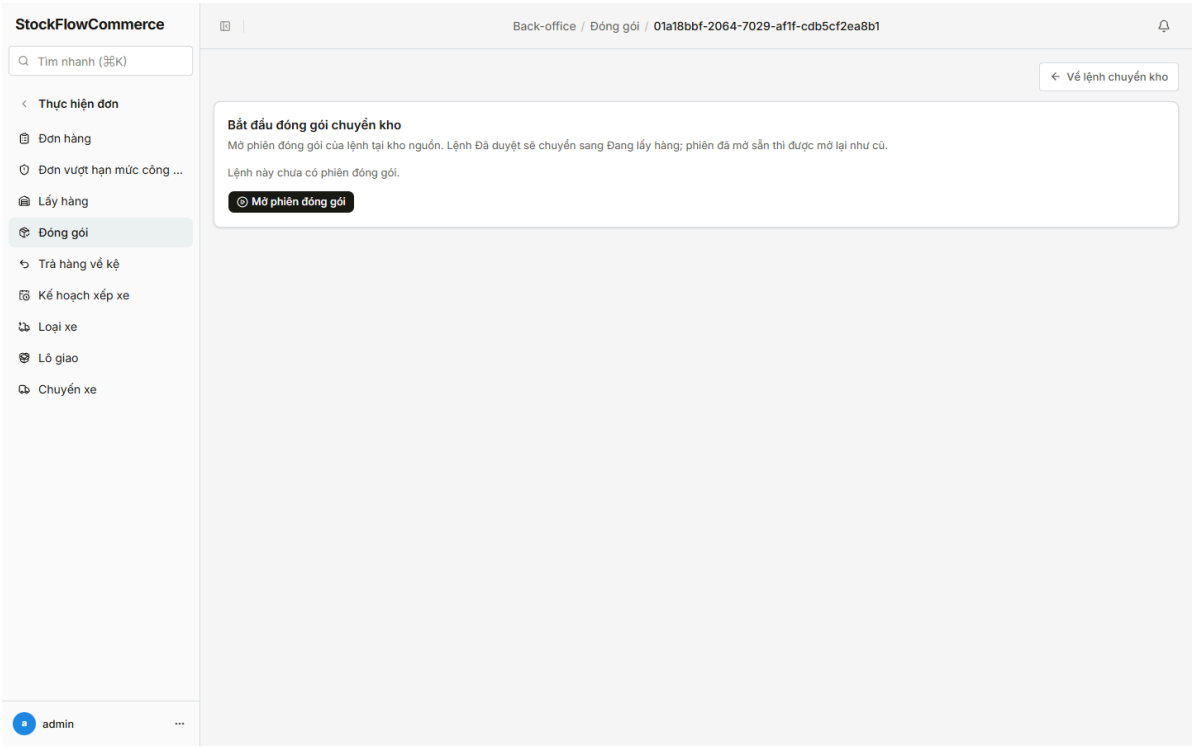


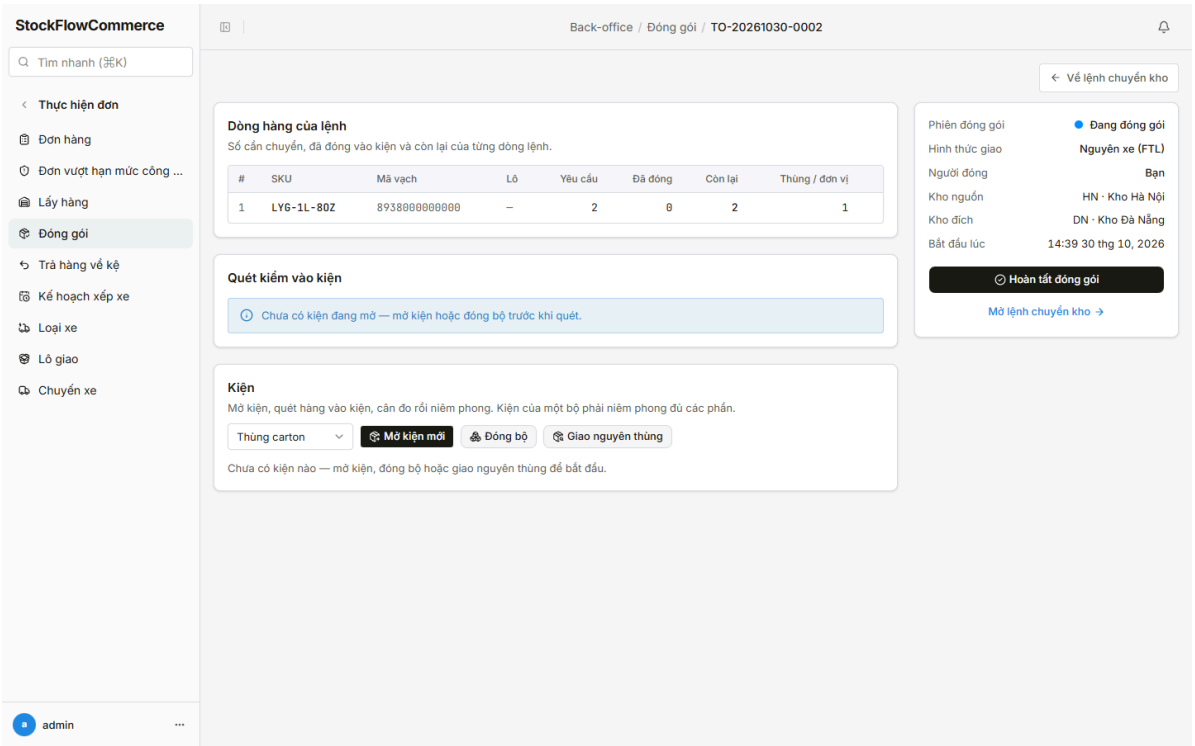


- **Function details:**
  - **Data requirement:** Shipped quantity per line.
  - **Validation:** Start only from Approved; dispatch only for Picking orders in "Giao lẻ" mode; each line 0 to requested quantity, total ≥1; stop picking blocked once goods have left the source.
  - **Business rules:** BR-04, BR-09, BR-11 (10 Inter-warehouse Transfer)

3.3.16.7 Pack Transfer Order

- **Function trigger:** Users click “Đóng gói chuyển kho” on an Approved or Picking full-truckload transfer order, then “Mở phiên đóng gói” on the packing station.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Pack a full-truckload transfer into sealed, labelled packages before load planning.





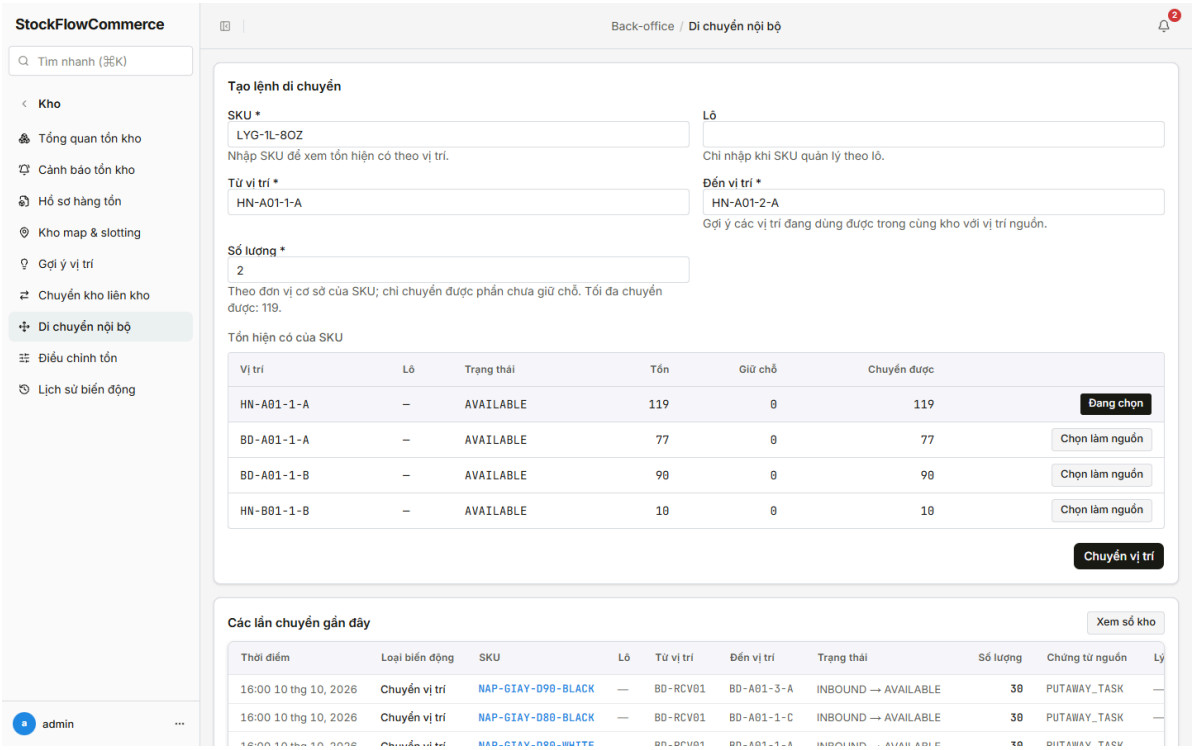
- **Function details:**

- **Data requirement:** Scanned item code and quantity (line/lot when ambiguous); package weight, dimensions, package type, rotatable, fragile; kit part count; carton LPN; completion mode (full/partial) and note.
- **Validation:** Scan quantity integer 1–1,000,000; weight >0, ≤9,999,999.999 kg (3 decimals); dimensions >0, ≤100,000 cm (2 decimals); LPN required ≤30; kit parts 2–99 or blank; kit-force reason ≤500; partial completion needs a note ≤500.
- **Business rules:** BR-10, BR-12 (10 Inter-warehouse Transfer)

3.3.17 Stock Control

3.3.17.1 Move Stock Internally

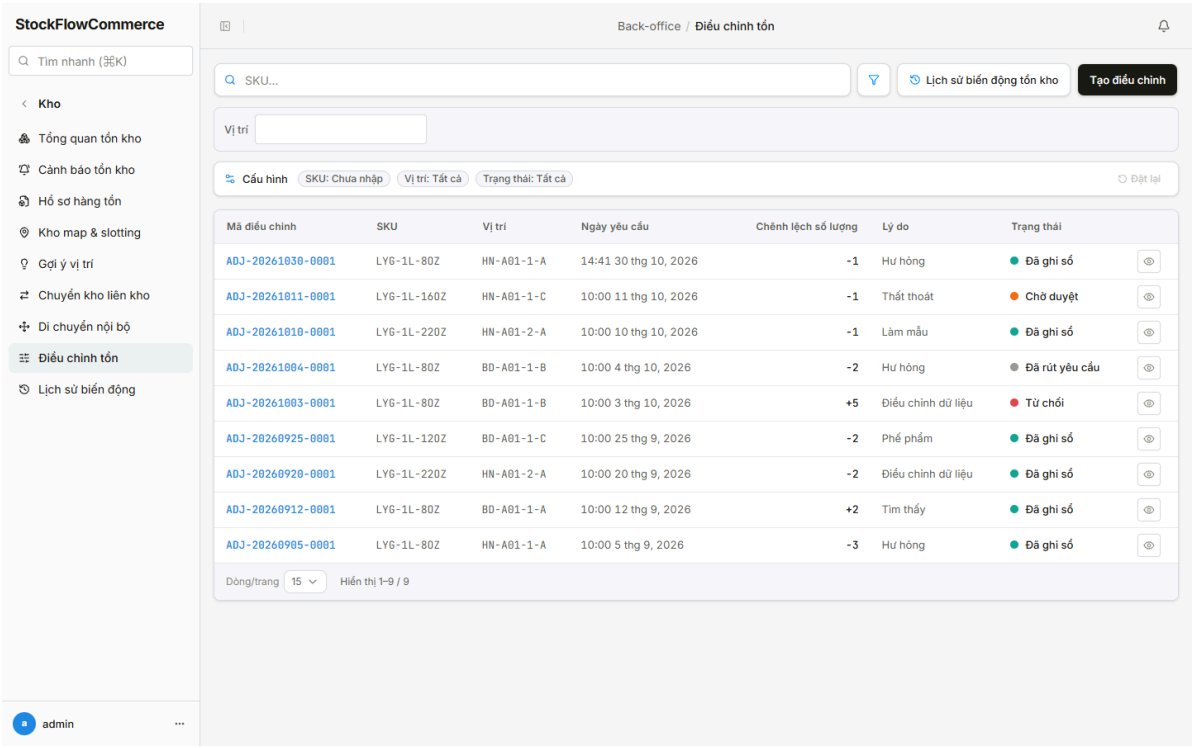
- **Function trigger:** Users click “Di chuyển nội bộ” in the “Kho” sidebar group, pick a source row (“Chọn làm nguồn”), then click “Chuyển vị trí”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Move unreserved stock of one SKU and lot between two locations in the same warehouse.



- **Function details:**
  - **Data requirement:** SKU, lot, from location, to location, quantity.
  - **Validation:** SKU 1–64 valid characters; lot ≤64; both locations required ≤64, different and in the same warehouse; quantity integer 1–1,000,000, ≤ unreserved quantity at source; no move out of or into a slot under cycle count.
  - **Business rules:** BR-01, BR-02, BR-03, BR-04, BR-05, BR-06, BR-07 (11 Intra-warehouse Transfer)

3.3.17.2 View Stock Adjustment List

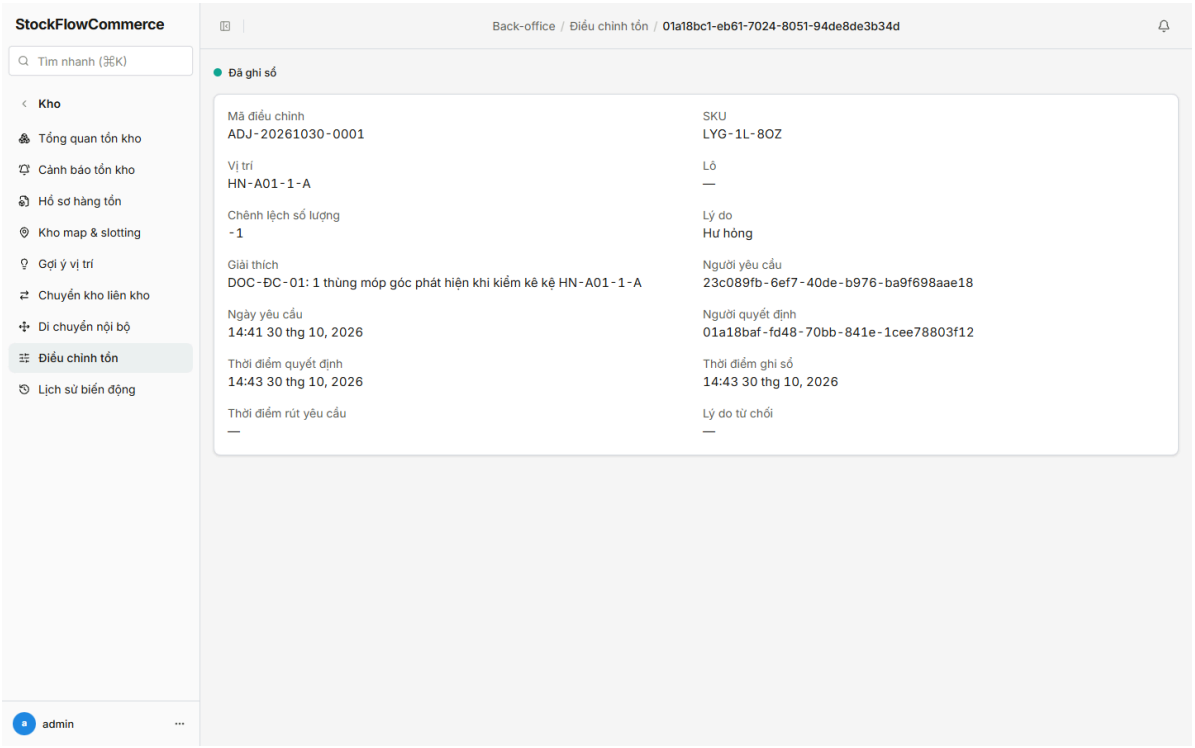
- **Function trigger:** Users click “Điều chỉnh tồn” in the “Kho” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Browse stock adjustment requests and find those pending approval.



- **Function details:**
  - **Data requirement:** SKU, location and status filters; shows adjustment number, SKU, location, requested date, quantity delta, reason, status (pending rows flagged).
  - **Validation:** None
  - **Business rules:** BR-19 (20 Inventory Control)

3.3.17.3 View Stock Adjustment Detail

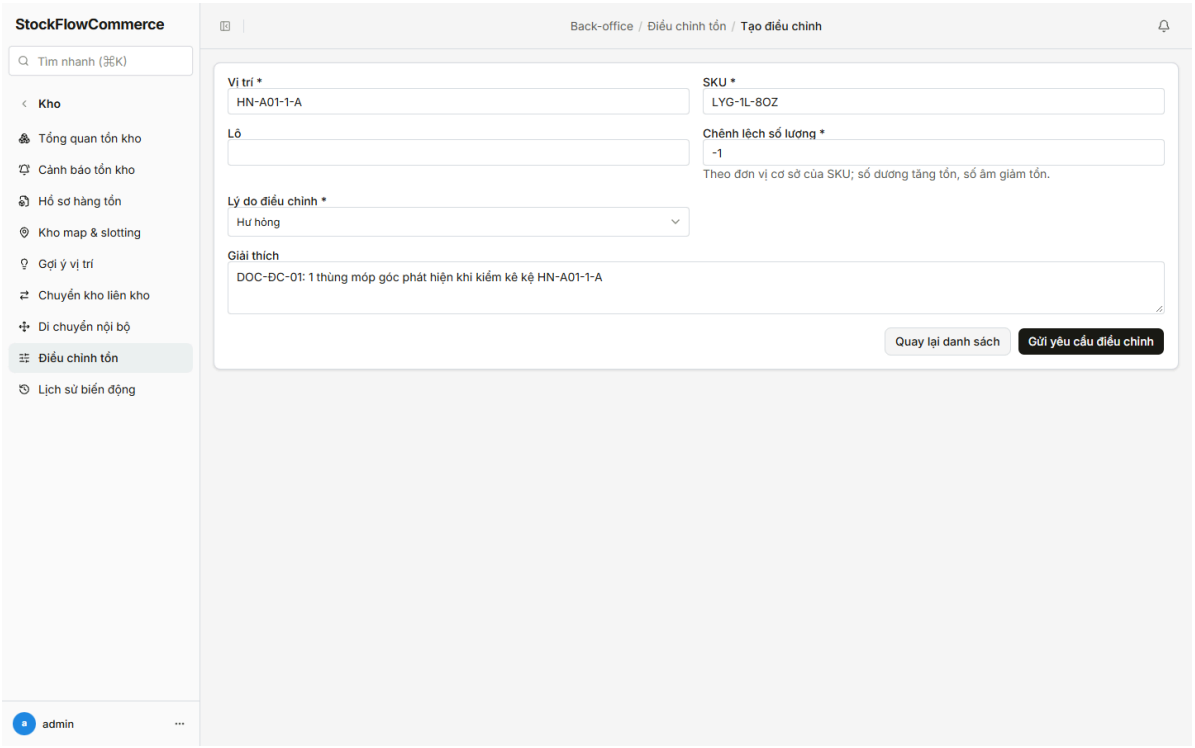
- **Function trigger:** Users click the adjustment number or the view icon (“Xem …”) on the stock adjustment list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** View an adjustment's quantities, reason and decision history.

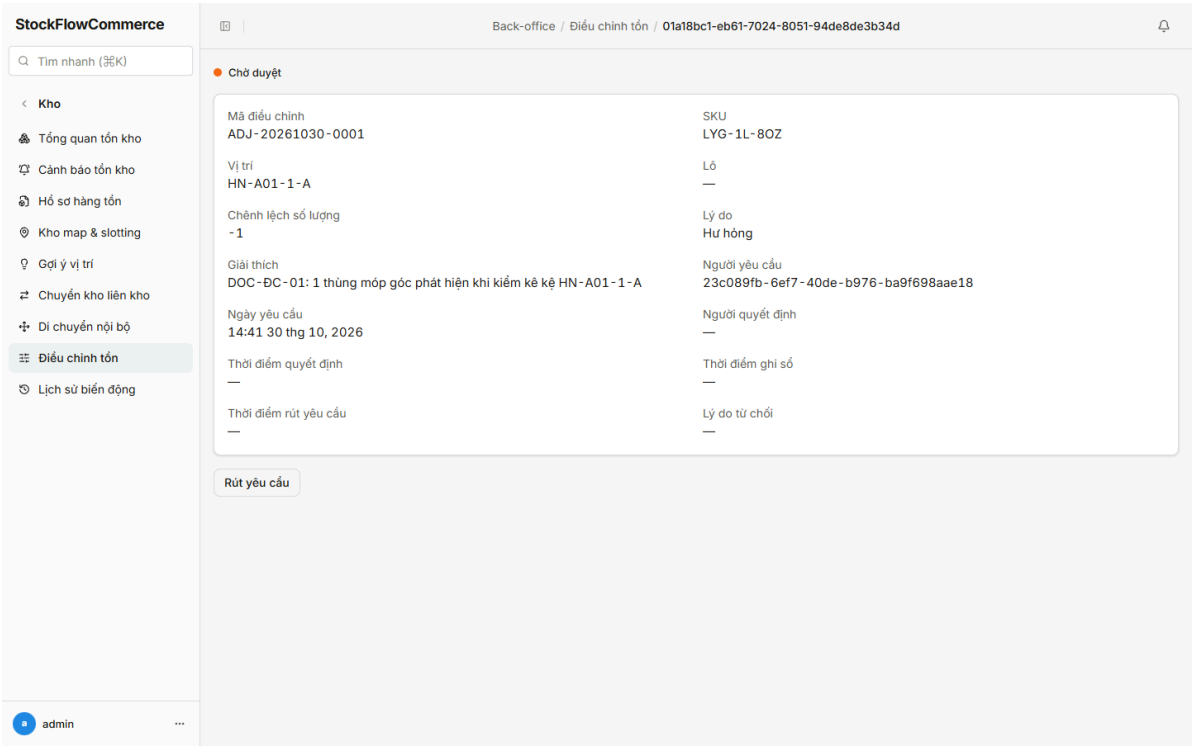


- **Function details:**
  - **Data requirement:** Number, location, SKU, lot, quantity delta, reason, note, status, requested by/at, decided by/at, rejection reason, posted at, withdrawn at.
  - **Validation:** None
  - **Business rules:** BR-05, BR-06 (20 Inventory Control)

3.3.17.4 Create Stock Adjustment

- **Function trigger:** Users click “Tạo điều chỉnh” on the stock adjustment list, then “Gửi yêu cầu điều chỉnh”; the requester can later click “Rút yêu cầu”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** Request a stock increase or decrease with a reason for another person to approve, or withdraw the request.

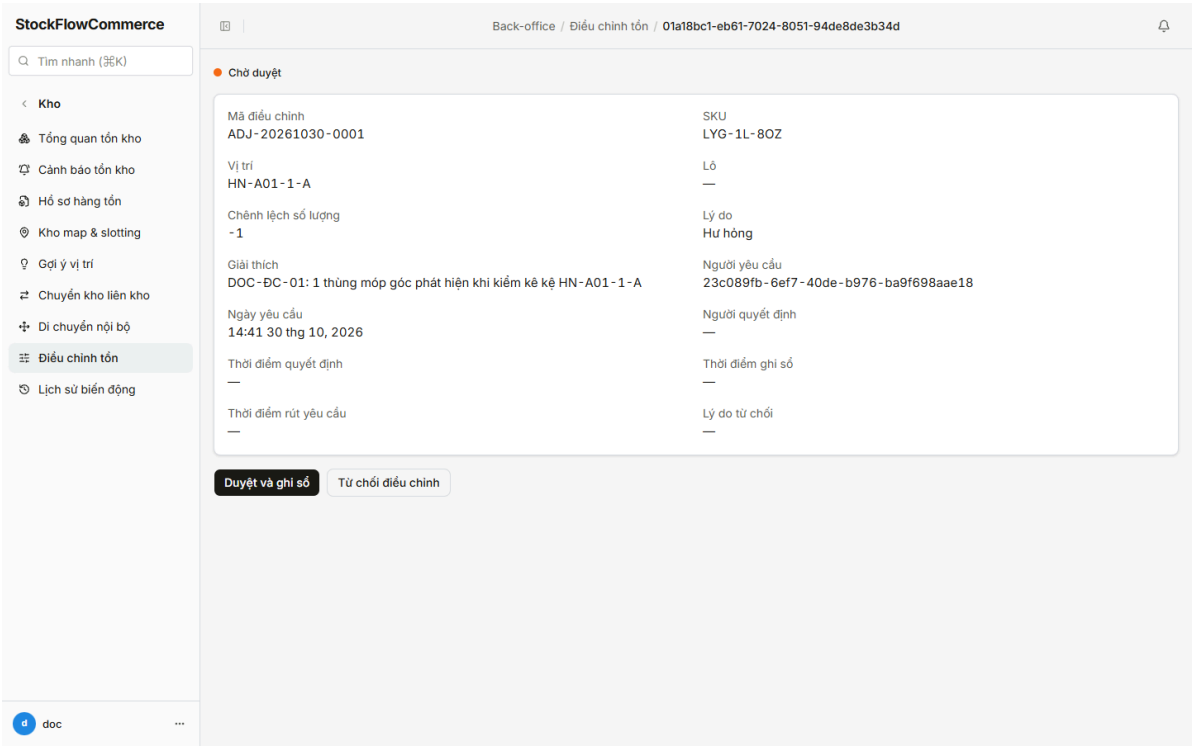


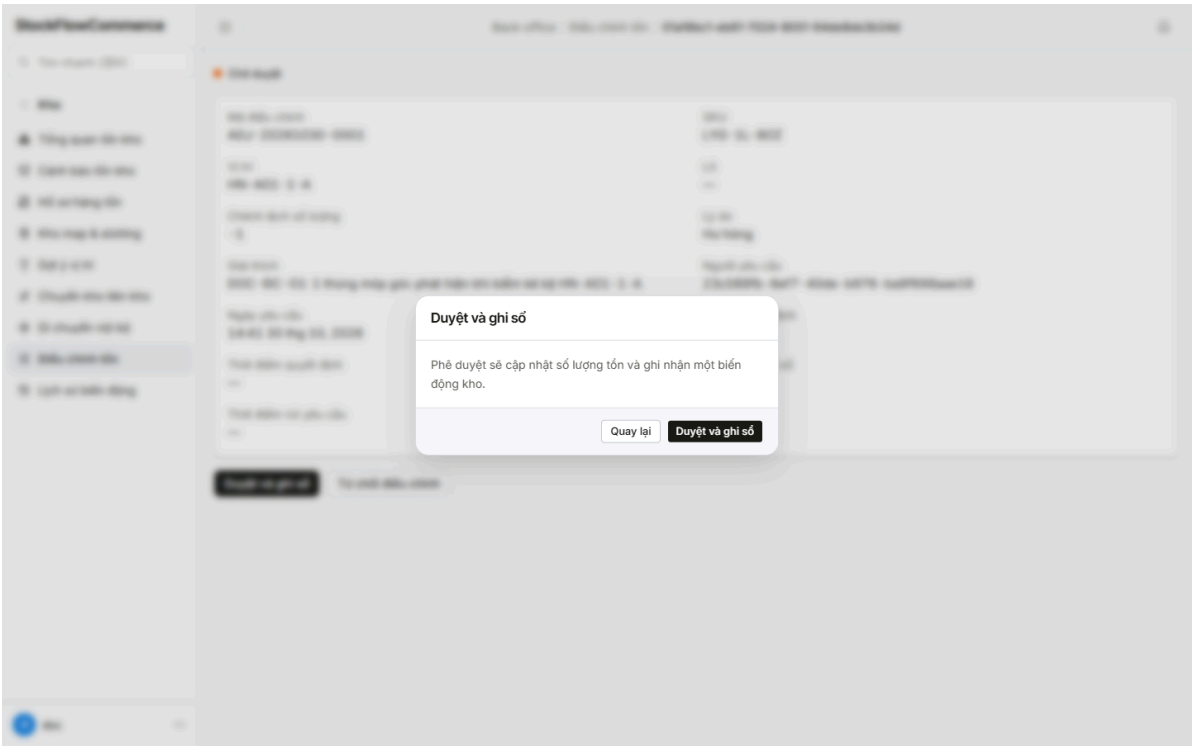


- **Function details:**
  - **Data requirement:** Location, SKU, lot, quantity delta (positive increases, negative decreases, base units), reason, note.
  - **Validation:** Location required ≤64; SKU 1–64 valid characters; lot ≤64; delta integer, non-zero, −1,000,000 to 1,000,000; SCRAP/SAMPLE only decrease; OTHER requires a note; note ≤1000; withdraw only by the requester while Pending Approval.
  - **Business rules:** BR-03, BR-04, BR-06 (20 Inventory Control)

3.3.17.5 Approve Stock Adjustment

- **Function trigger:** Users click “Duyệt và ghi sổ” or “Từ chối điều chỉnh” on a Pending Approval adjustment's detail page.
- **Function description:**
    - **Actors:** System Admin, Warehouse Manager
    - **Purpose:** Approve and post, or reject, another user's stock adjustment request.



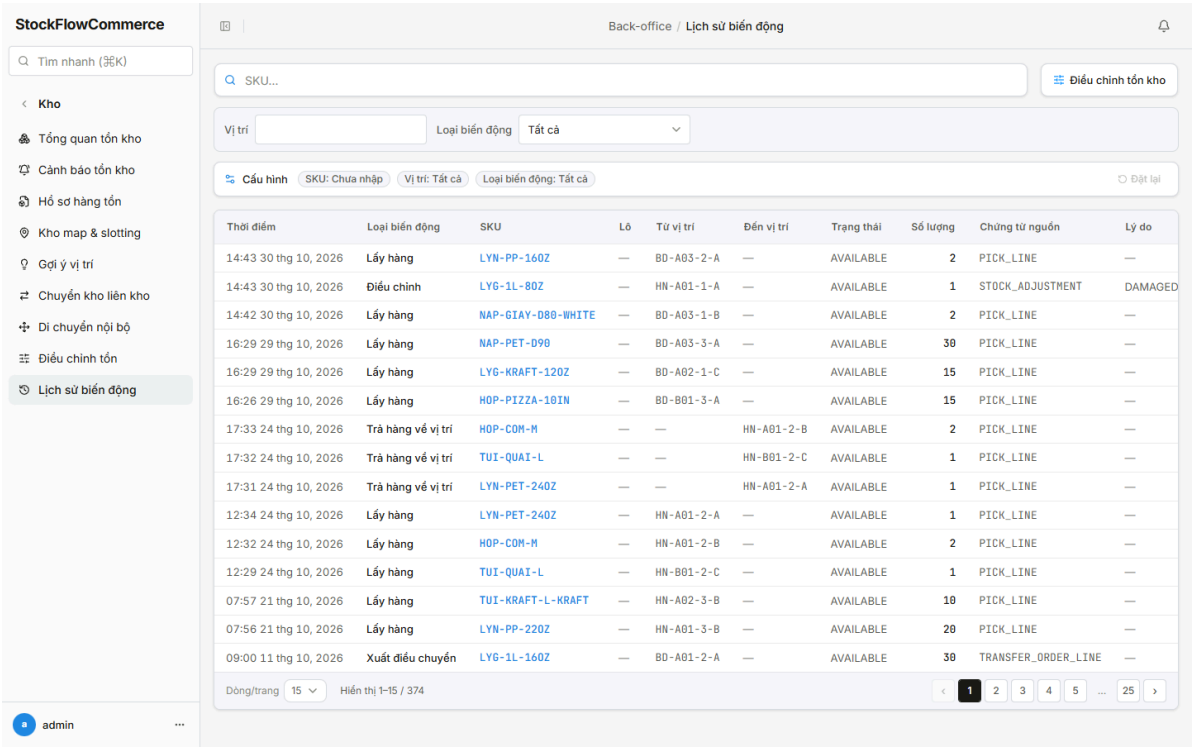


- **Function details:**

- **Data requirement:** Rejection reason (reject only).
- **Validation:** Only Pending Approval; the requester cannot approve or reject their own request (buttons hidden, backend re-checks); rejection reason required ≤500; posting fails if stock would fall below the reserved quantity.
- **Business rules:** BR-02, BR-05 (20 Inventory Control)

3.3.17.6 View Stock Movement History

- **Function trigger:** Users click “Lịch sử biến động” in the “Kho” sidebar group (or “Xem sổ kho” on the internal move screen).
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Inventory Planner
  - **Purpose:** View the read-only stock ledger of every stock movement.

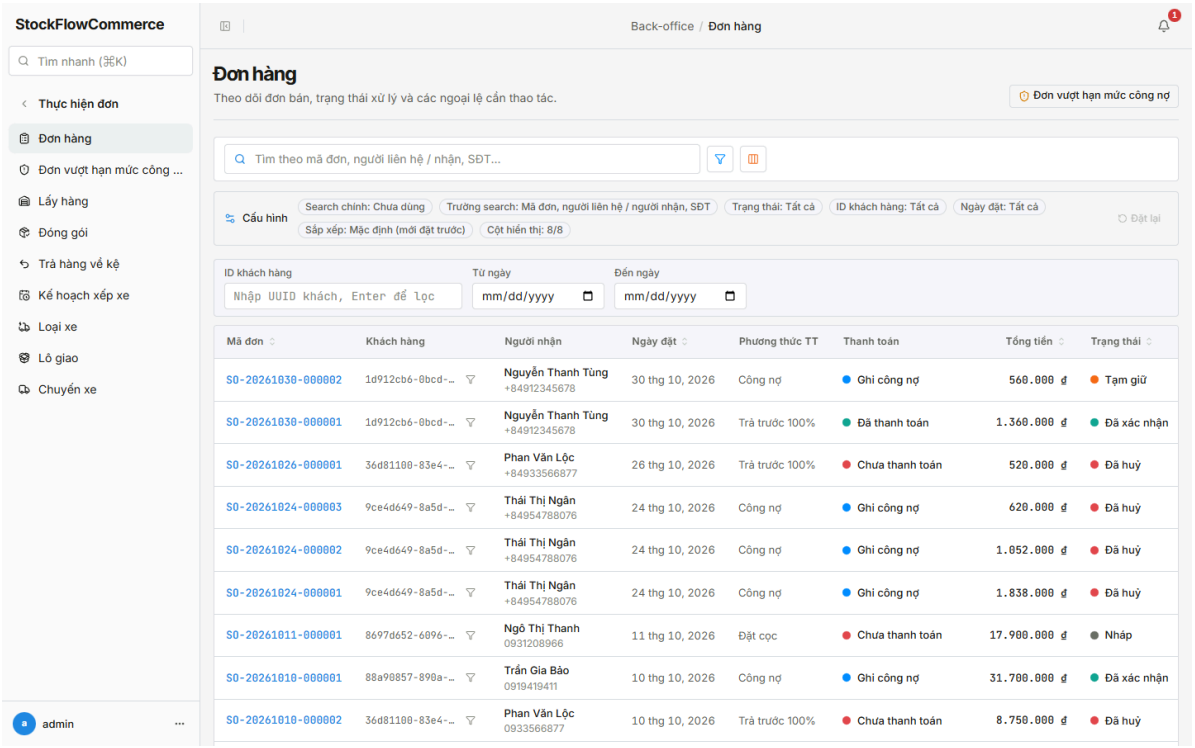


- **Function details:**
  - **Data requirement:** SKU, location and movement-type filters; shows time, movement type, SKU, lot, from/to location, stock status, quantity, source document type, reason.
  - **Validation:** None
  - **Business rules:** BR-16 (20 Inventory Control); BR-09 (03 Receipt); BR-06 (11 Intra-warehouse Transfer)

3.3.18 Sales Orders

3.3.18.1 View Order List

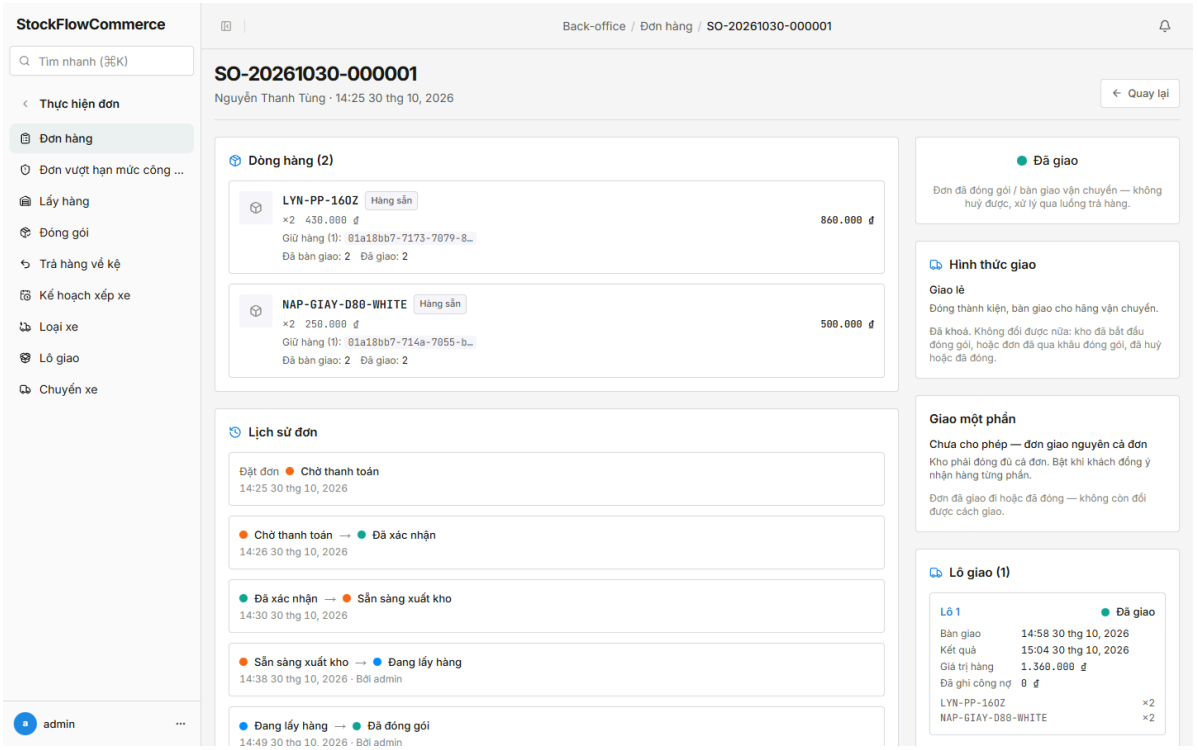
- **Function trigger:** Users click “Đơn hàng” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Sales Staff, Order Coordinator, Accountant
  - **Purpose:** Find and browse sales orders.



- **Function details:**
  - **Data requirement:** Search keyword, status, customer ID, placed-from date, placed-to date; list shows order number, customer, recipient, placed date, payment term, payment status, total amount, status.
  - **Validation:** Customer ID must be a valid UUID (“ID khách hàng phải là UUID hợp lệ.”); end date must be on or after start date; server sort only by order number, placed date, total amount, status.
  - **Business rules:** BR-01 (17 Order Management)

3.3.18.2 View Order Detail

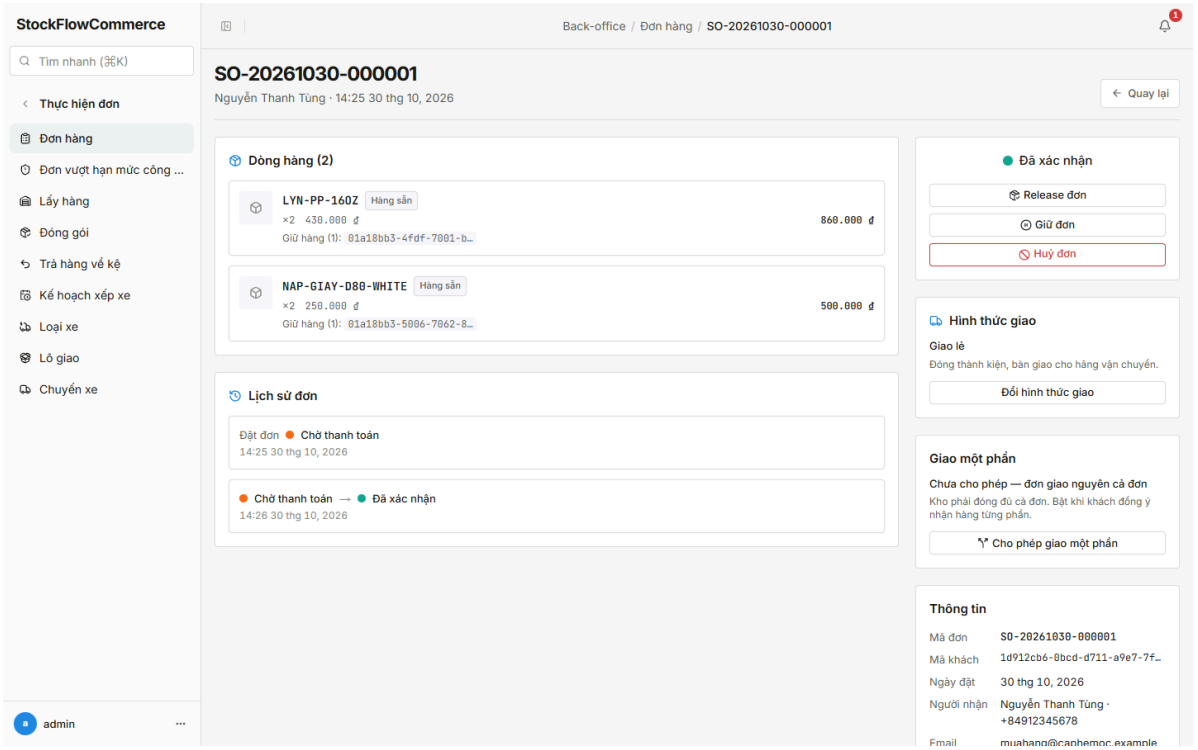
- **Function trigger:** Users click an order number (“Mã đơn”) or the view icon in the order list.
- **Function description:**
  - **Actors:** System Admin, Sales Staff, Order Coordinator, Accountant
  - **Purpose:** Review all information of one order.

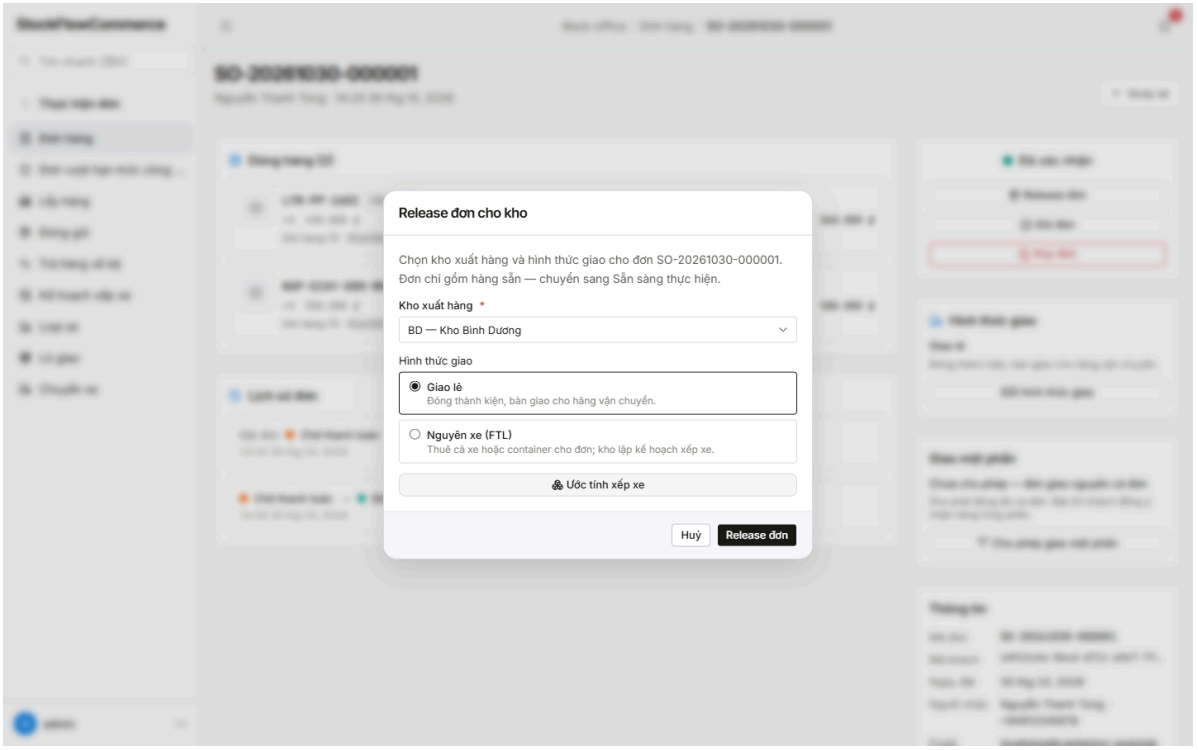


- **Function details:**
  - **Data requirement:** Order lines, production orders per print line, order history, delivery mode, partial-shipment flag, shipments, info (customer, recipient, address, warehouse, release time), payment (term, status, paid, deposit, balance policy, amounts due, credit).
  - **Validation:** None
  - **Business rules:** BR-01, BR-05 (17 Order Management)

3.3.18.3 Release Order

- **Function trigger:** Users click “Release đơn”, “Giữ đơn” or “Gỡ giữ” in the action card of the order detail.
- **Function description:**
  - **Actors:** System Admin, Order Coordinator
  - **Purpose:** Hand a confirmed order to a warehouse or production, or pause and resume it.

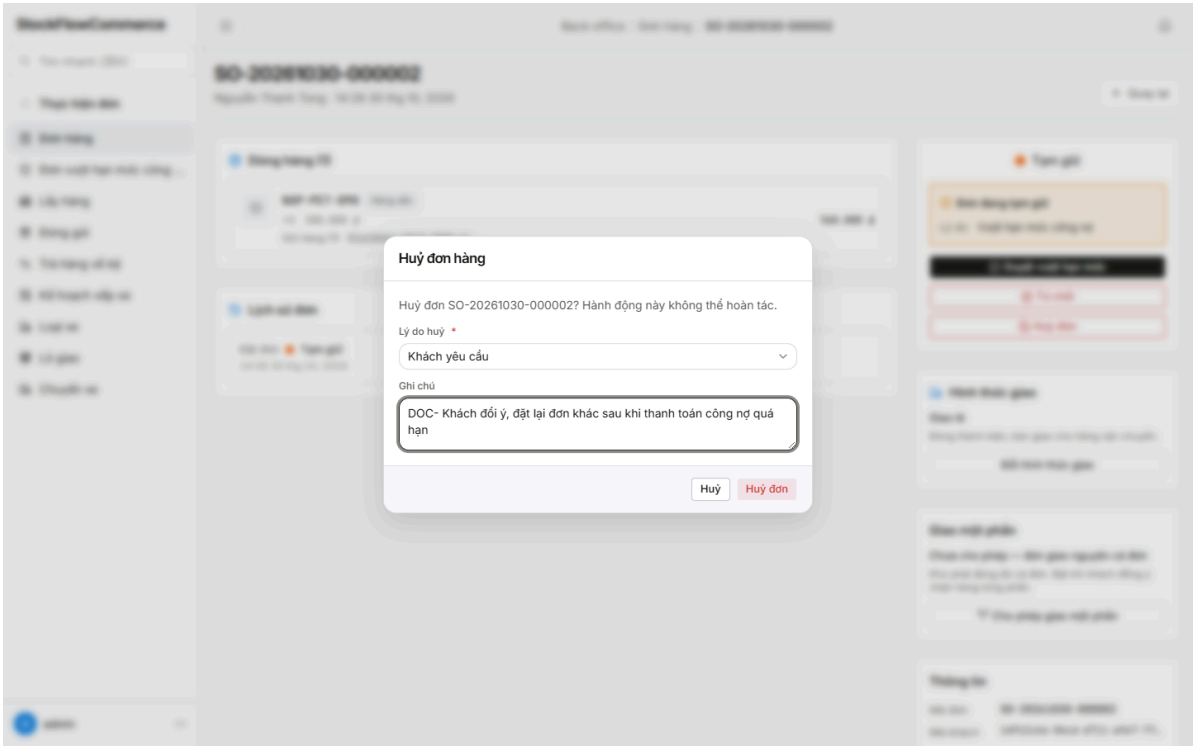




- **Function details:**
  - **Data requirement:** Release: warehouse (“Kho xuất hàng”), delivery mode. Hold: hold reason, note. Resolve hold: resolution note (“Cách đã xử lý”).
  - **Validation:** Release only from Confirmed; warehouse required; server refuses unpaid deposit, unapproved design sample, overdue credit, FTL with COD. Hold reason required (manual reasons only), note ≤1000, required for “Lý do khác”; resolve note required, ≤1000.
  - **Business rules:** BR-02, BR-09, BR-13, BR-15 (17 Order Management); BR-02 (15 Payment)

3.3.18.4 Cancel Order

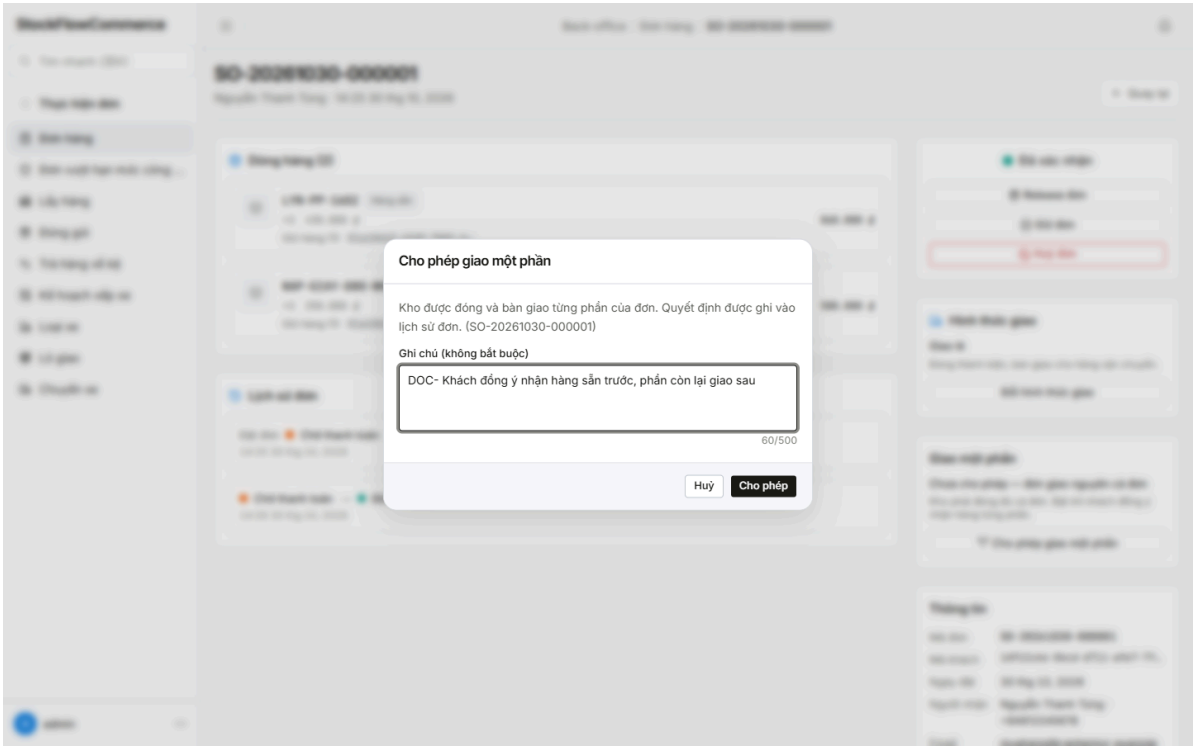
- **Function trigger:** Users click “Huỷ đơn” (or “Đóng phần còn lại” for a partly shipped order) on the order detail.
- **Function description:**
  - **Actors:** System Admin, Sales Staff, Order Coordinator
  - **Purpose:** Cancel an order, or stop delivering its undelivered remainder.



- **Function details:**
  - **Data requirement:** Cancel: reason code, note. Close remainder: reason code, note, optional agreed fee (amount, fee reason, agreed detail).
  - **Validation:** Reason required; note ≤1000, mandatory for “Lý do khác”; cancel only Draft–Picking/On Hold and before any package is packed (except PACK-EARLY); remainder closure only when Partially Fulfilled; fee amount integer >0, reason ≤500, detail ≤2000.
  - **Business rules:** BR-03, BR-04, BR-10, BR-11, BR-12 (17 Order Management); BR-17 (15 Payment)

3.3.18.5 Allow Partial Shipment

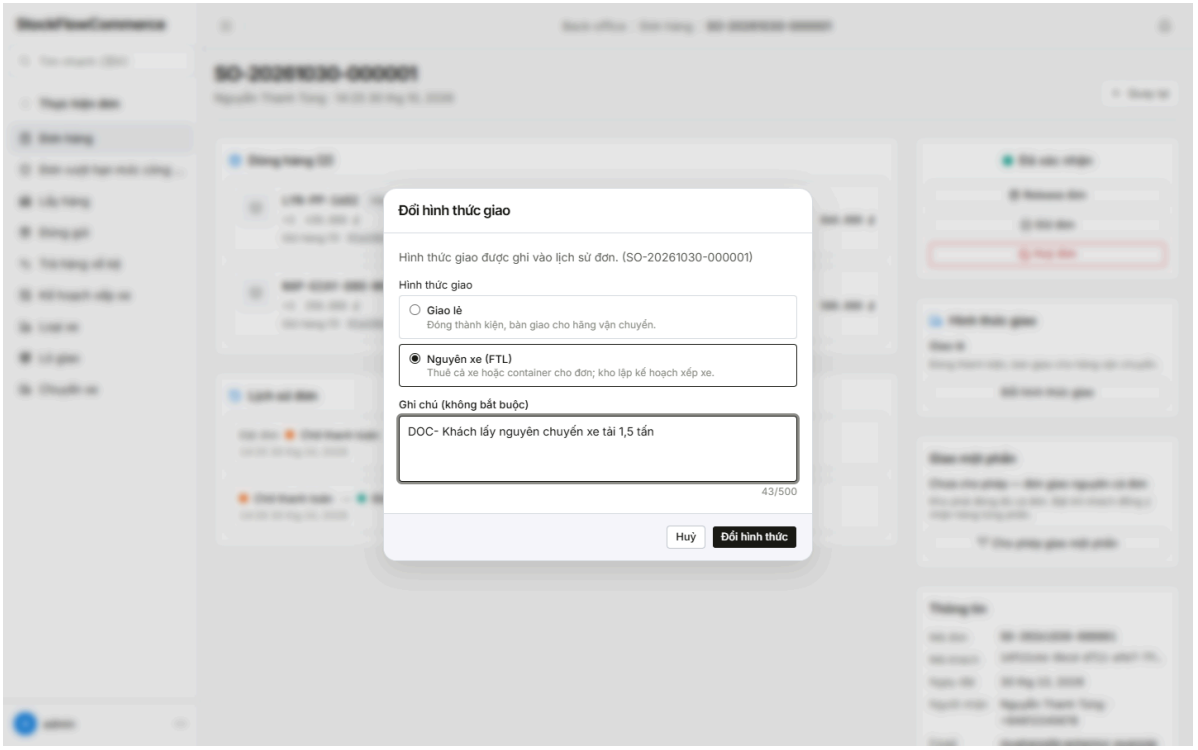
- **Function trigger:** Users click “Cho phép giao một phần” or “Dừng giao một phần” on the “Giao một phần” card of the order detail.
- **Function description:**
  - **Actors:** System Admin, Sales Staff
  - **Purpose:** Decide whether the warehouse may ship the order in several parts.



- **Function details:**
  - **Data requirement:** Allowed flag (toggle), note (optional).
  - **Validation:** Note ≤500 characters; locked from Shipped onward (or closed/cancelled); cannot stop while ready stock already left ahead of print lines (ORDER_PARTIAL_SHIPMENT_LOCKED).
  - **Business rules:** BR-06 (17 Order Management); BR-03 (09 Shipping); BR-01 (08 Packing)

3.3.18.6 Choose Delivery Mode

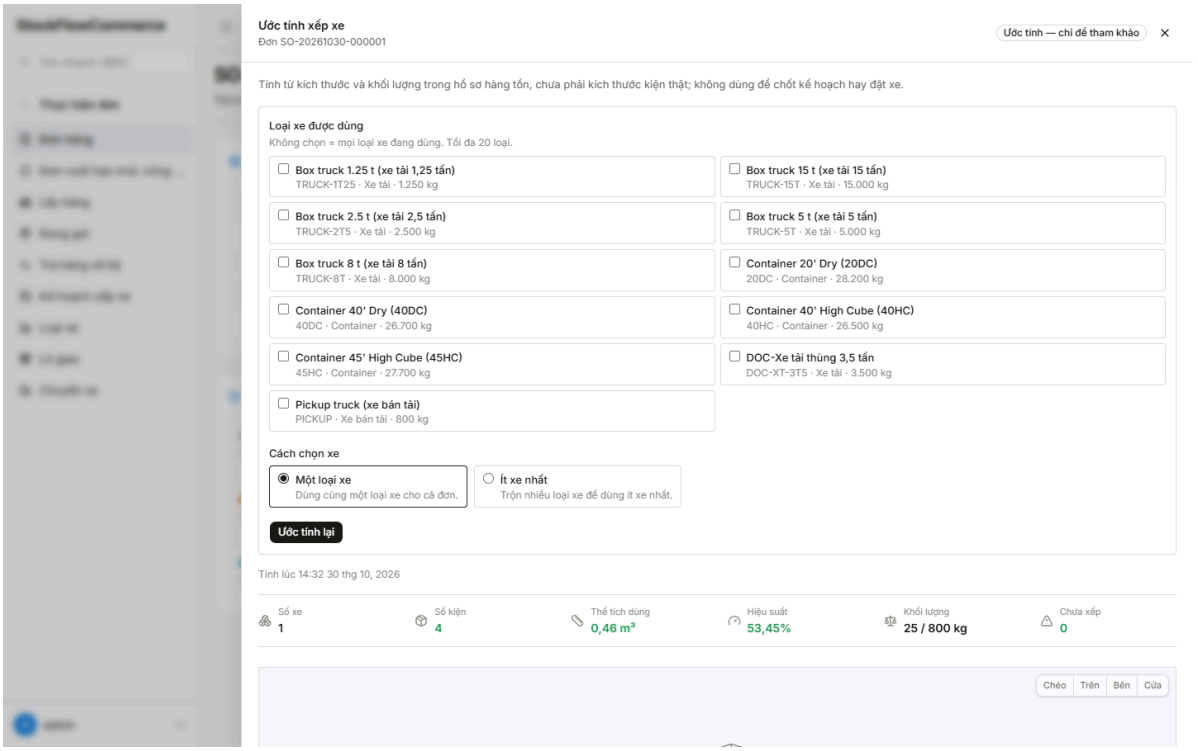
- **Function trigger:** Users click “Đổi hình thức giao” on the “Hình thức giao” card of the order detail (or choose it in the release dialog).
- **Function description:**
  - **Actors:** System Admin, Order Coordinator
  - **Purpose:** Set the order to parcel or full-truckload delivery.

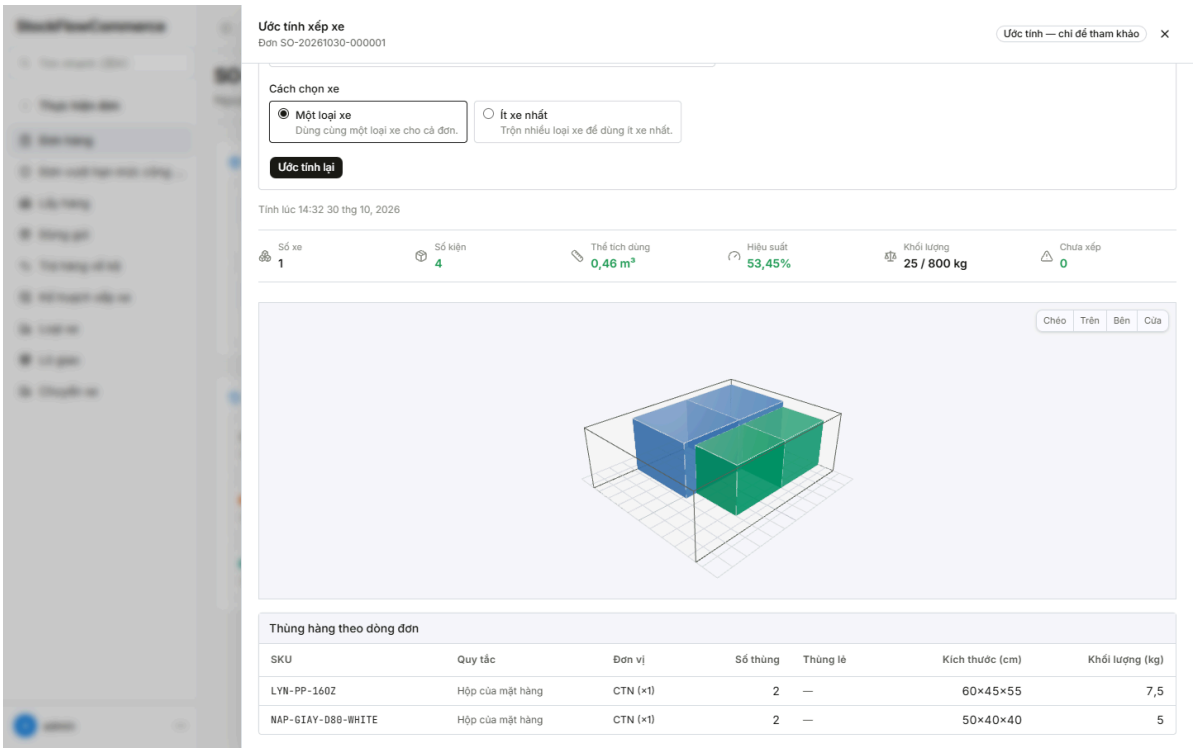


- **Function details:**
  - **Data requirement:** Delivery mode (Giao lẻ / Nguyên xe (FTL)), note (optional).
  - **Validation:** Mode required and must differ from current; note ≤500; locked once the first package is opened or the order passed packing/cancelled; FTL refused for deposit orders collecting the balance on delivery (DELIVERY_MODE_FTL_NO_COD).
  - **Business rules:** BR-13 (17 Order Management); BR-04, BR-08, BR-21 (09 Shipping); BR-18 (15 Payment)

3.3.18.7 Estimate Vehicle Load

- **Function trigger:** Users click “Ước tính xếp xe” on the “Hình thức giao” card (released order) or in the release dialog (confirmed order), then click “Ước tính”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Order Coordinator
  - **Purpose:** Preview how many vehicles an order would need.

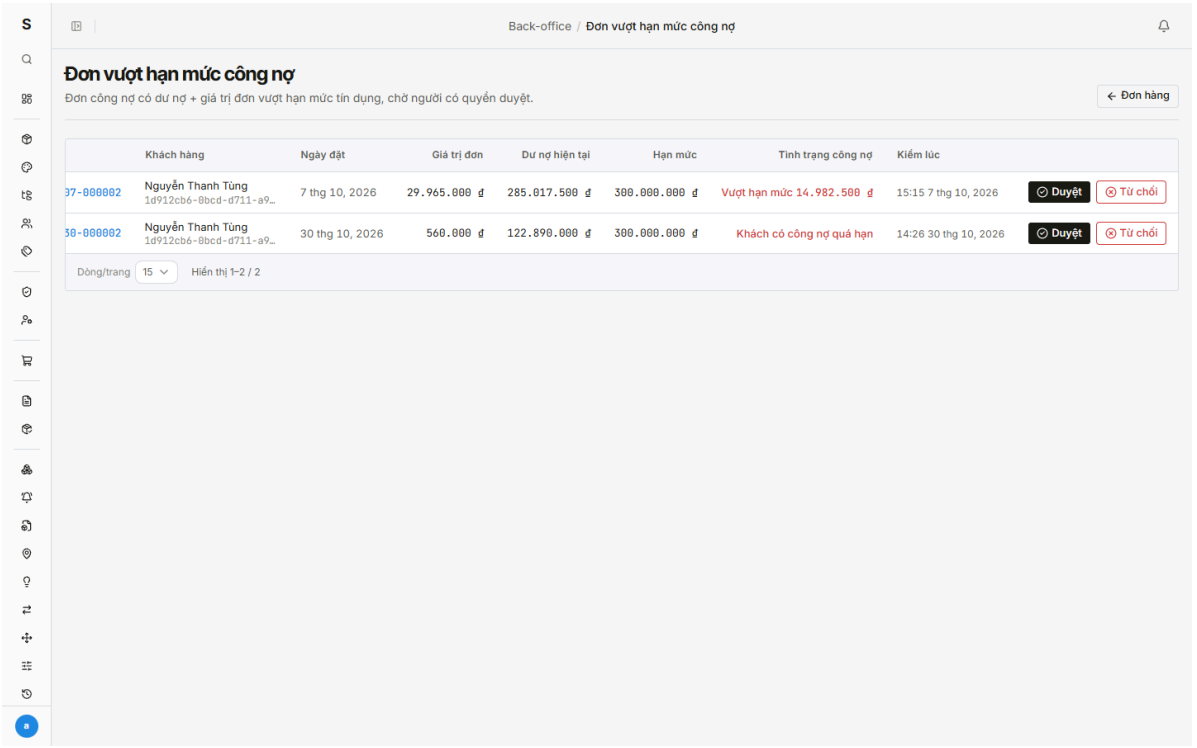




- **Function details:**
  - **Data requirement:** Vehicle types (“Loại xe được dùng”, optional), selection mode (Một loại xe / Ít xe nhất); result shows vehicles, cartons per line, unplaced items, warnings.
  - **Validation:** At most 20 vehicle types (none = all active); warehouse required for unreleased orders; inactive or unknown vehicle types rejected; order must be in an estimable status.
  - **Business rules:** BR-20 (09 Shipping)

3.3.18.8 View Credit-Hold Orders

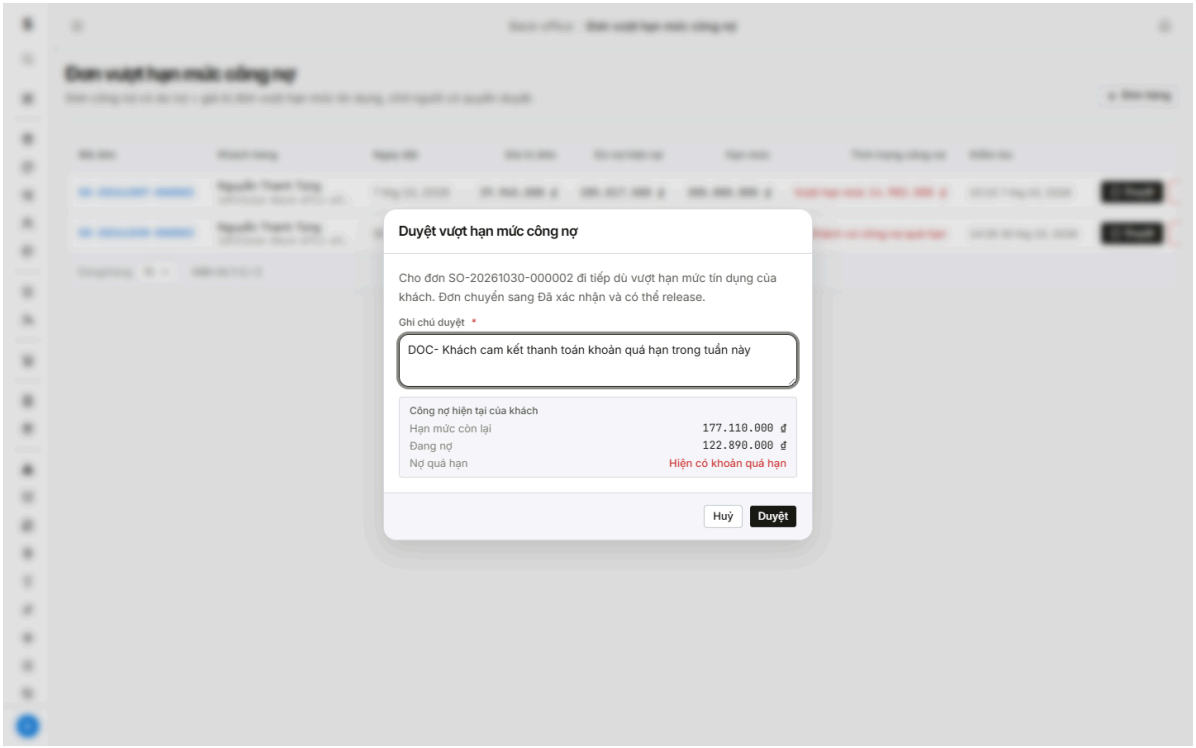
- **Function trigger:** Users click “Đơn vượt hạn mức công nợ” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Sales Staff, Order Coordinator, Accountant
  - **Purpose:** Review credit orders held for exceeding the credit limit.

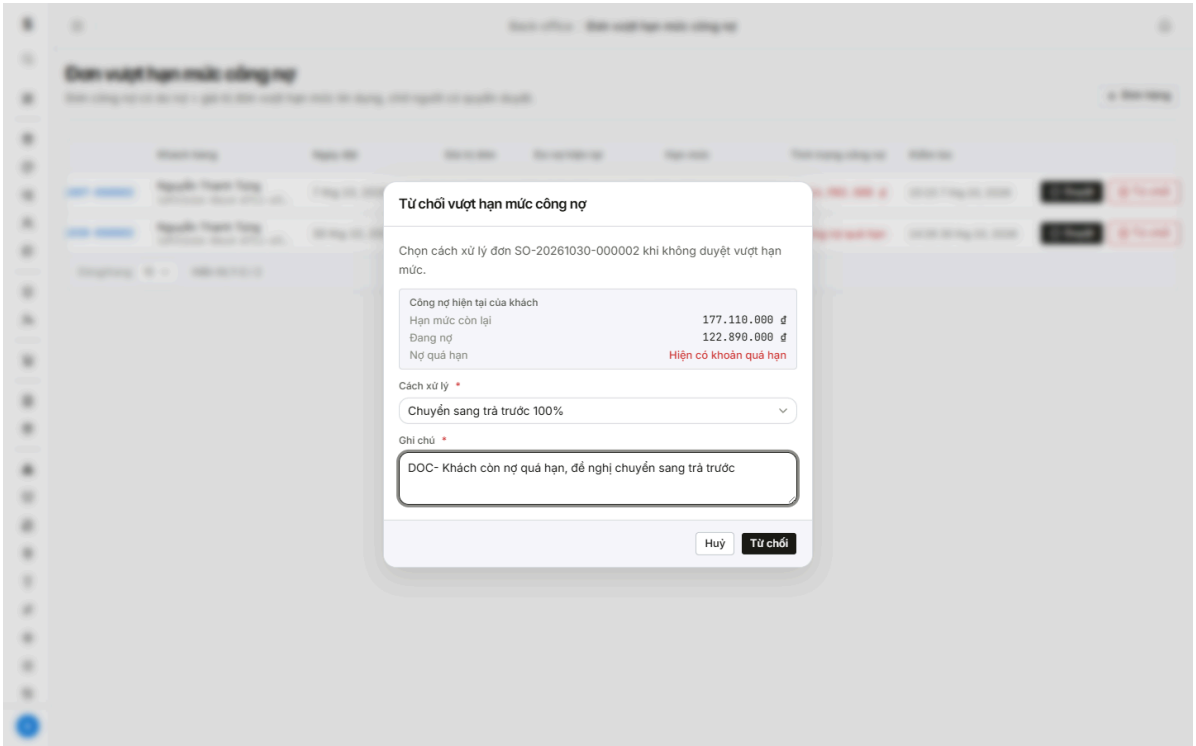


- **Function details:**
  - **Data requirement:** Order number, customer, placed date, order amount, current exposure (“Dư nợ hiện tại”), credit limit, credit status, checked-at time.
  - **Validation:** None
  - **Business rules:** BR-06 (14 Cart & Order); BR-03, BR-13 (15 Payment)

3.3.18.9 Approve Credit Hold

- **Function trigger:** Accountant clicks “Duyệt” or “Từ chối” on a row of “Đơn vượt hạn mức công nợ” (or “Duyệt vượt hạn mức” / “Từ chối” on the order detail).
- **Function description:**
  - **Actors:** System Admin, Accountant
  - **Purpose:** Let an over-limit credit order proceed, or resolve it otherwise.



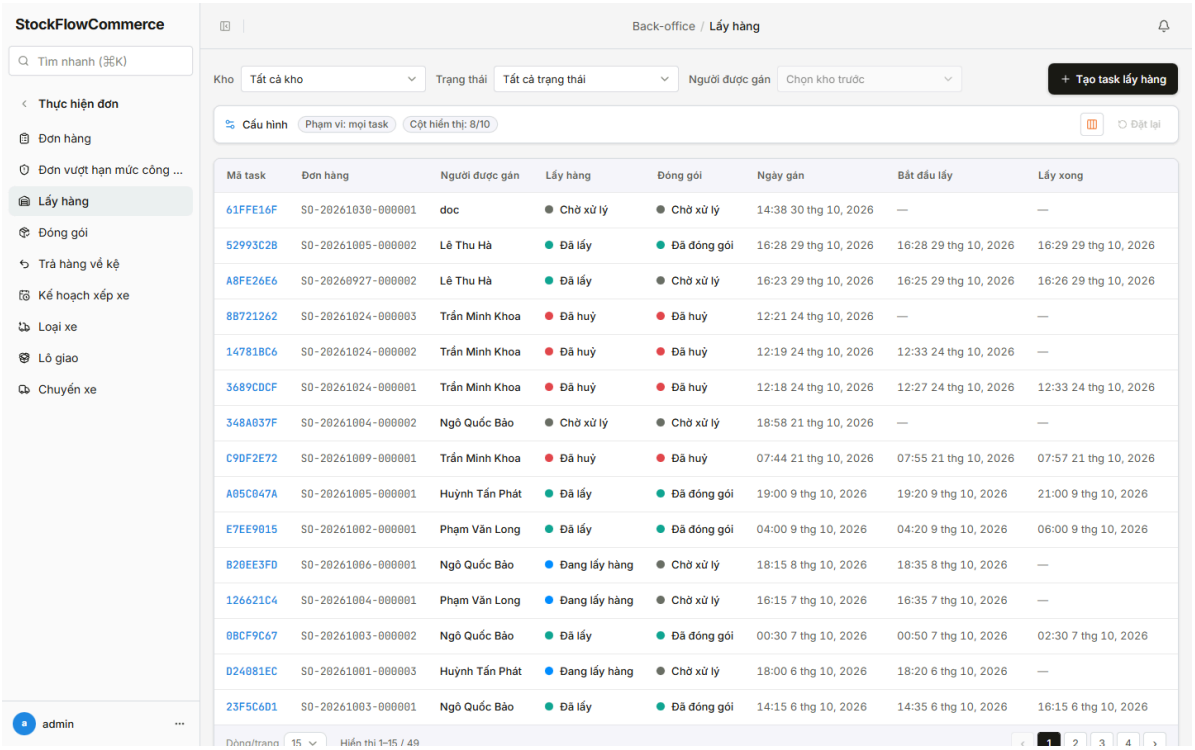


- **Function details:**
  - **Data requirement:** Approve: note (“Ghi chú duyệt”). Reject: handling (“Cách xử lý”: cancel, switch to prepaid, switch to deposit), deposit percent (optional), note.
  - **Validation:** Note required, ≤1000; handling required and limited to terms allowed for the customer; deposit percent 0 < x < 100 (blank = customer default); only On Hold orders with CREDIT hold reason and credit terms.
  - **Business rules:** BR-06 (14 Cart & Order); BR-03 (15 Payment); BR-15 (17 Order Management)

3.3.19 Picking

3.3.19.1 View Picking Tasks

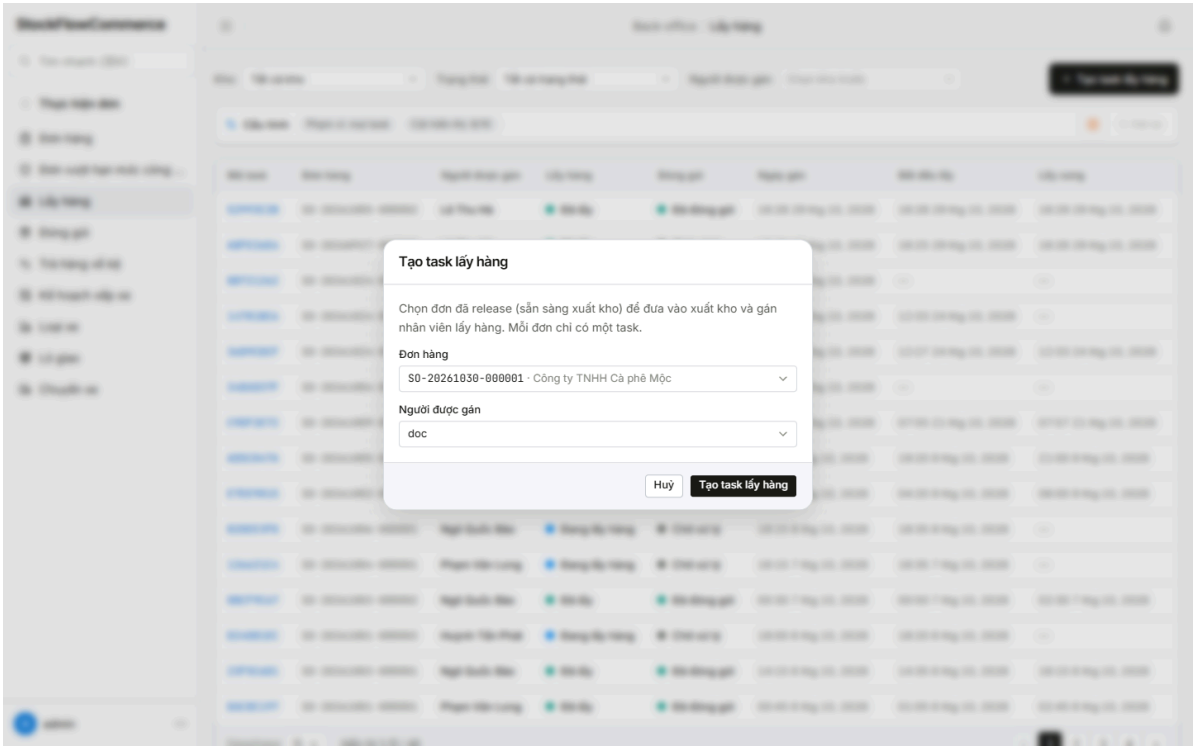
- **Function trigger:** Users click “Lấy hàng” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Track picking tasks.



- **Function details:**
  - **Data requirement:** Filters warehouse (“Kho”), status (“Trạng thái”), assignee (“Người được gán”); list shows task code, order, assignee, pick status, pack status, assigned at, picking started, picked at.
  - **Validation:** Assignee filter enabled only after choosing a warehouse; warehouses outside the user's data scope are refused.
  - **Business rules:** BR-01, BR-07 (07 Picking)

3.3.19.2 Create and Assign Picking Task

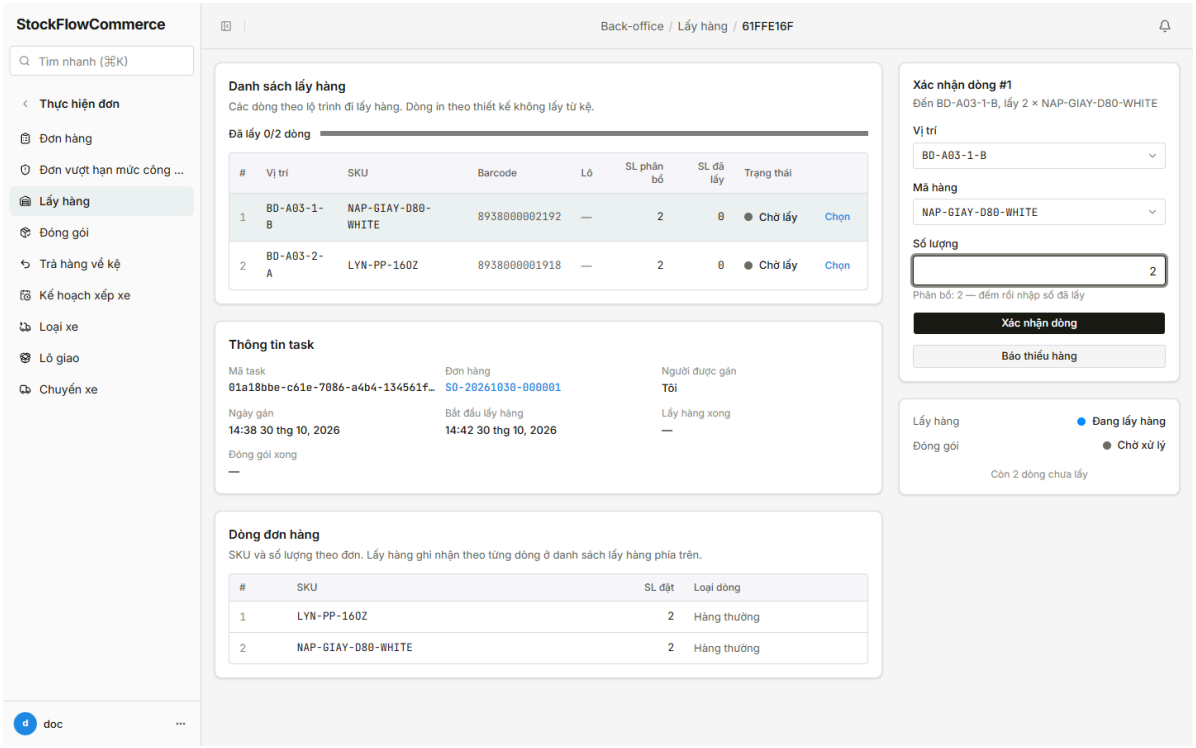
- **Function trigger:** Users click “Tạo task lấy hàng” on the picking list (or “Gán lại” on a task row or detail to reassign).
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Order Coordinator
  - **Purpose:** Create one picking task per released order and assign a picker.

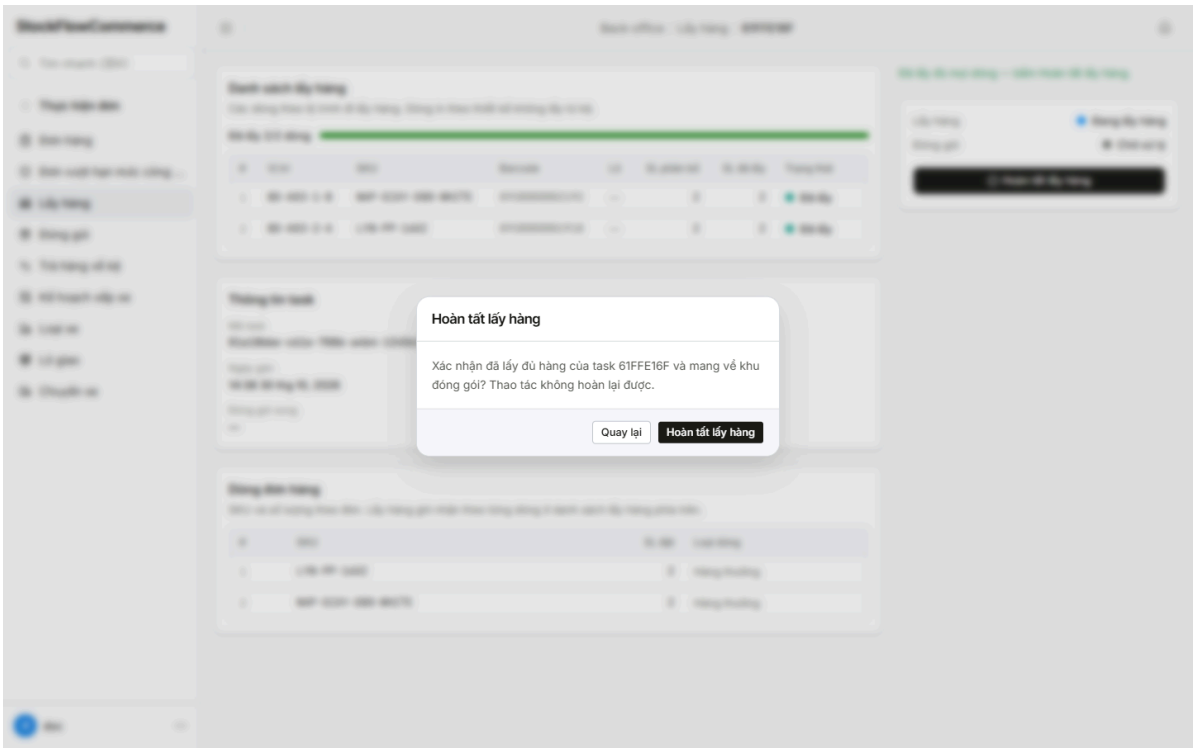


- **Function details:**
  - **Data requirement:** Order (“Đơn hàng”), assignee (“Người được gán”).
  - **Validation:** Order and assignee required; only released orders without a task are listed; assignee must be an active warehouse staff/manager of that warehouse; reassign only while pick is Pending and to a different person.
  - **Business rules:** BR-01, BR-07, BR-08 (07 Picking)

3.3.19.3 Pick Goods

- **Function trigger:** The assigned picker clicks “Bắt đầu lấy hàng” on the picking detail, confirms each line with “Xác nhận dòng”, then clicks “Hoàn tất lấy hàng”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Pick allocated stock line by line.

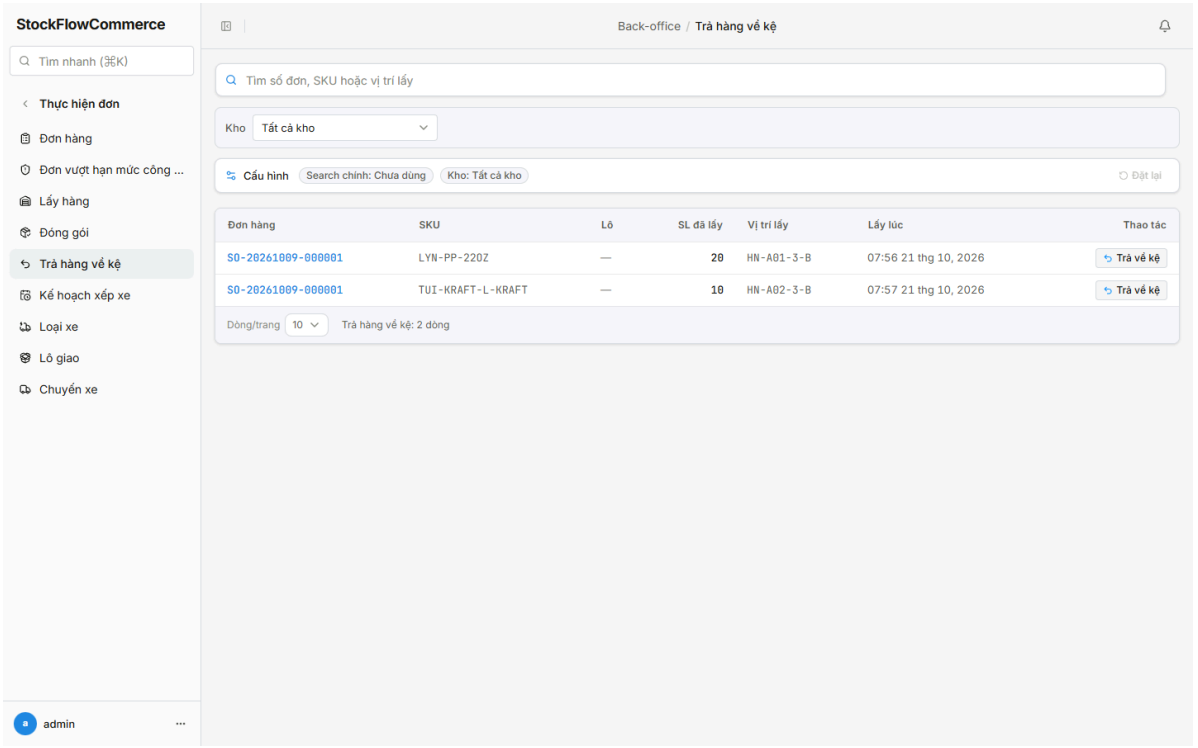


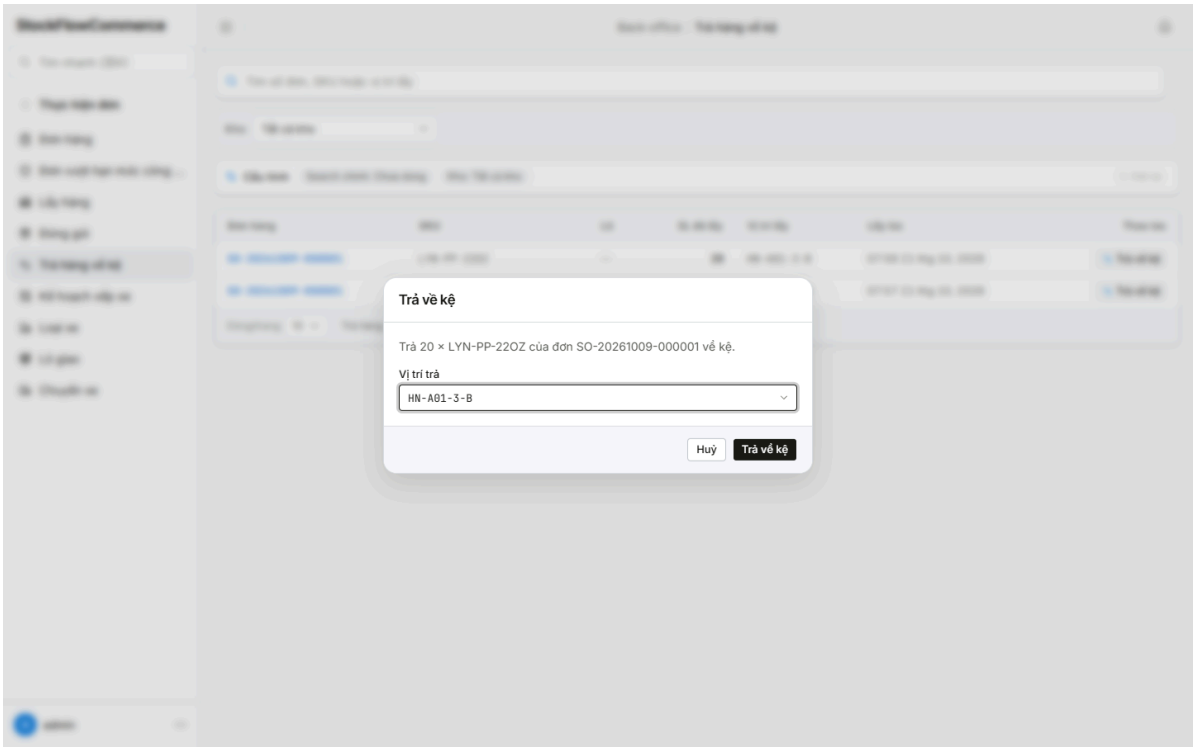


- **Function details:**
  - **Data requirement:** Location (“Vị trí”), scanned SKU/barcode or carton LPN (“Mã hàng”), quantity; short pick: location, found quantity (“Số tìm thấy”).
  - **Validation:** Location and code required, ≤64 chars; quantity whole number >0 equal to allocated; wrong bin/item rejected; found quantity 0 to allocated−1; only the assignee acts; completion requires all lines picked or cancelled.
  - **Business rules:** BR-02, BR-03, BR-04, BR-08, BR-09, BR-10 (07 Picking)

3.3.19.4 Put Back Goods

- **Function trigger:** Users click “Trả hàng về kệ” in the sidebar, then “Trả về kệ” on a queue row.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Return picked goods of cancelled orders to a shelf bin.





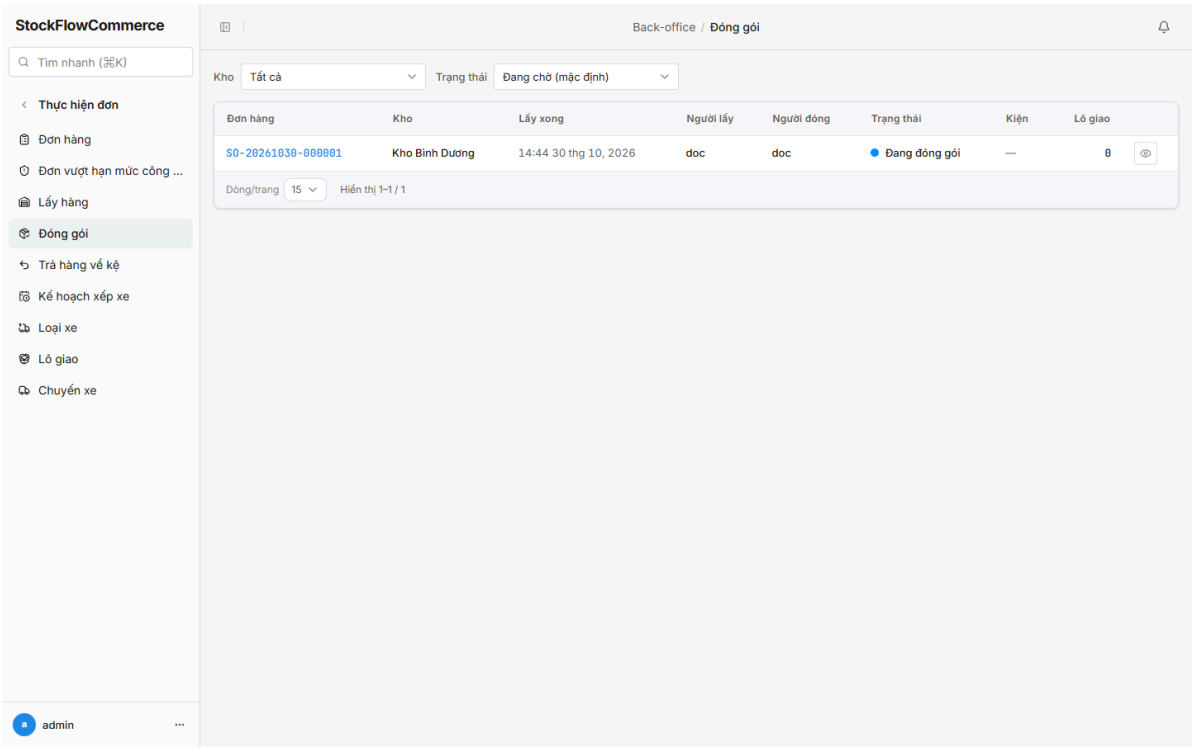
- **Function details:**

- **Data requirement:** Search (order, SKU, pick location), warehouse filter; return location ("Vị trí trả"); rows show order, SKU, lot, picked quantity, pick location, picked at.
- **Validation:** Location required, ≤64 chars; must be an active bin of the same warehouse (server check).
- **Business rules:** BR-05 (07 Picking); BR-03 (17 Order Management)

3.3.20 Packing

3.3.20.1 View Packing Queue

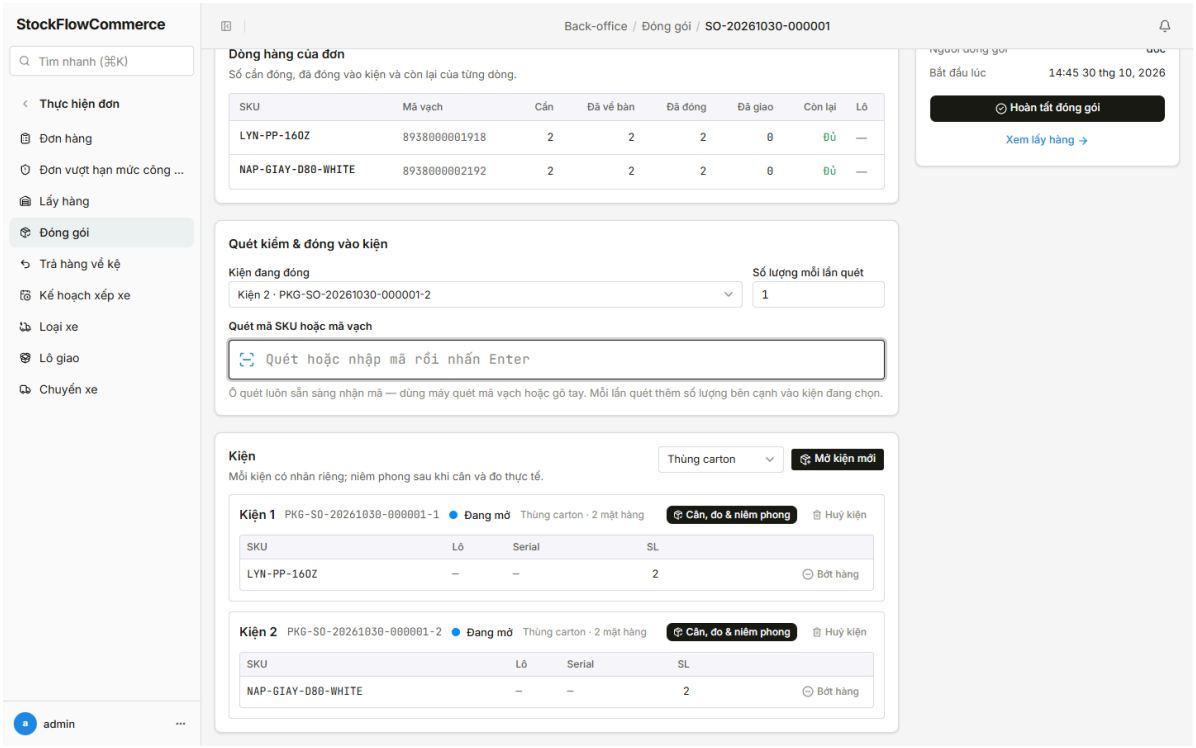
- **Function trigger:** Users click "Đóng gói" in the "Thực hiện đơn" sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** See picked orders waiting to be packed.

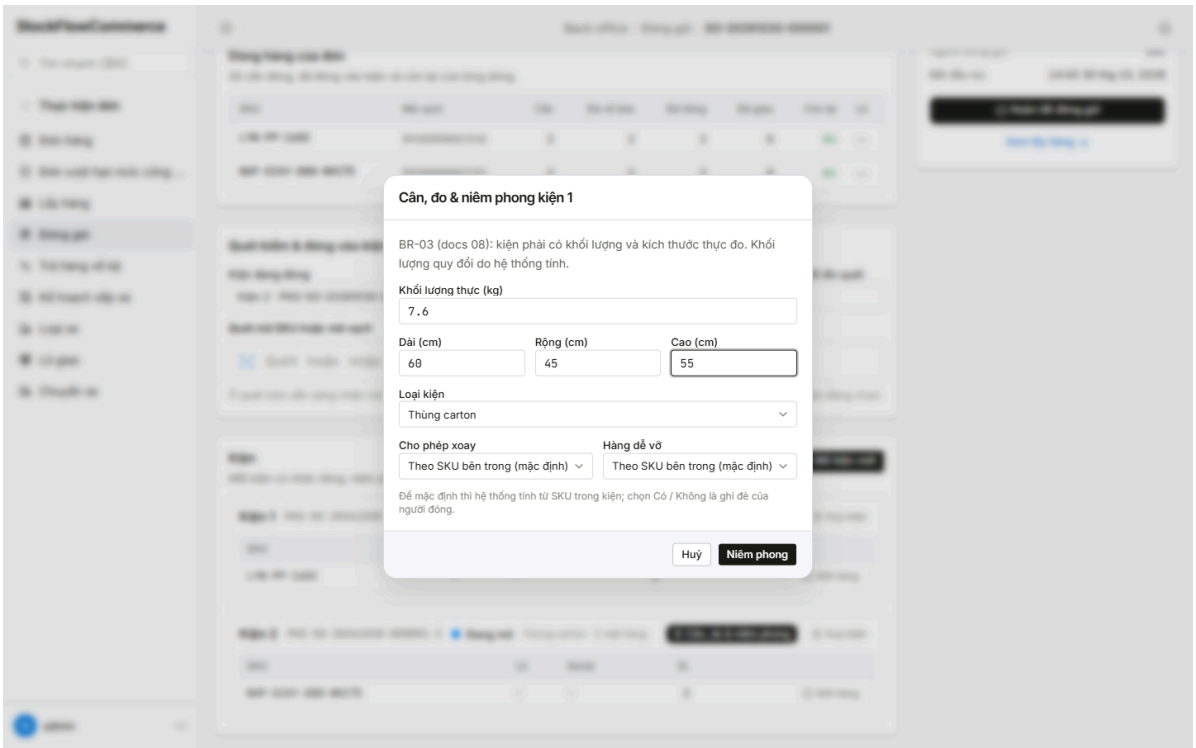


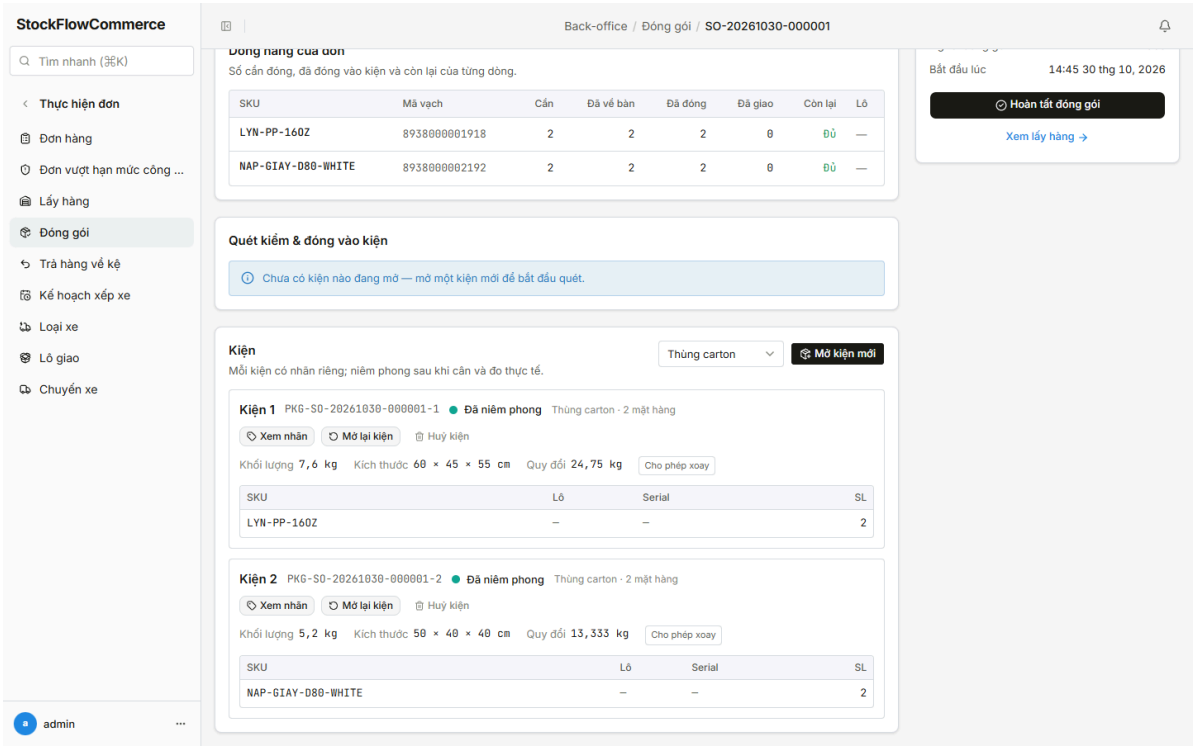
- **Function details:**
  - **Data requirement:** Filters warehouse, status (waiting, packing, on hold); list shows order, warehouse, picked at, picker, packer, status, sealed/total packages, shipments.
  - **Validation:** None
  - **Business rules:** BR-01 (08 Packing)

3.3.20.2 Pack Order

- **Function trigger:** Users click the eye icon ("Mở trạm") on a row of the "Đóng gói" queue, then "Bắt đầu đóng gói", "Mở kiện mới", scan items, "Cân, đo & niêm phong" (confirm with "Niêm phong") and "Hoàn tất đóng gói".
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Pack picked goods into sealed, labelled packages and create the shipment.


StockFlowCommerce
Tìm nhanh (⌘K)
Thực hiện đơn
Đơn hàng
Đơn vượt hạn mức công ...
Lấy hàng
Đóng gói
Trả hàng về kệ
Kế hoạch xếp xe
Loại xe
Lô giao
Chuyến xe
admin
Back-office / Đóng gói / SO-20261030-000001
Dòng hàng của đơn
Số cần đóng, đã đóng vào kiện và còn lại của từng dòng.
SKU | Mã vạch | Cần | Đã về bàn | Đã đóng | Đã giao | Còn lại | Lô
LYN-PP-16OZ | 8938000001918 | 2 | 2 | 2 | 0 | Đủ | —
NAP-GIAY-D80-WHITE | 8938000002192 | 2 | 2 | 2 | 0 | Đủ | —
Bắt đầu lúc 14:45 30 thg 10, 2026
Hoàn tất đóng gói
Xem lấy hàng →
Quét kiểm & đóng vào kiện
Kiện đang đóng
Kiện 2 · PKG-SO-20261030-000001-2
Số lượng mỗi lần quét
1
Quét mã SKU hoặc mã vạch
Quét hoặc nhập mã rồi nhấn Enter
Ô quét luôn sẵn sàng nhận mã — dùng máy quét mã vạch hoặc gõ tay. Mỗi lần quét thêm số lượng bên cạnh vào kiện đang chọn.
Kiện
Mỗi kiện có nhãn riêng; niêm phong sau khi cân và đo thực tế.
Thùng carton
Mở kiện mới
Kiện 1 PKG-SO-20261030-000001-1 Đang mở Thùng carton · 2 mặt hàng
Cân, đo & niêm phong
Huỷ kiện
SKU | Lô | Serial | SL
LYN-PP-16OZ | – | – | 2 | Bớt hàng
Kiện 2 PKG-SO-20261030-000001-2 Đang mở Thùng carton · 2 mặt hàng
Cân, đo & niêm phong
Huỷ kiện
SKU | Lô | Serial | SL
NAP-GIAY-D80-WHITE | – | – | 2 | Bớt hàng

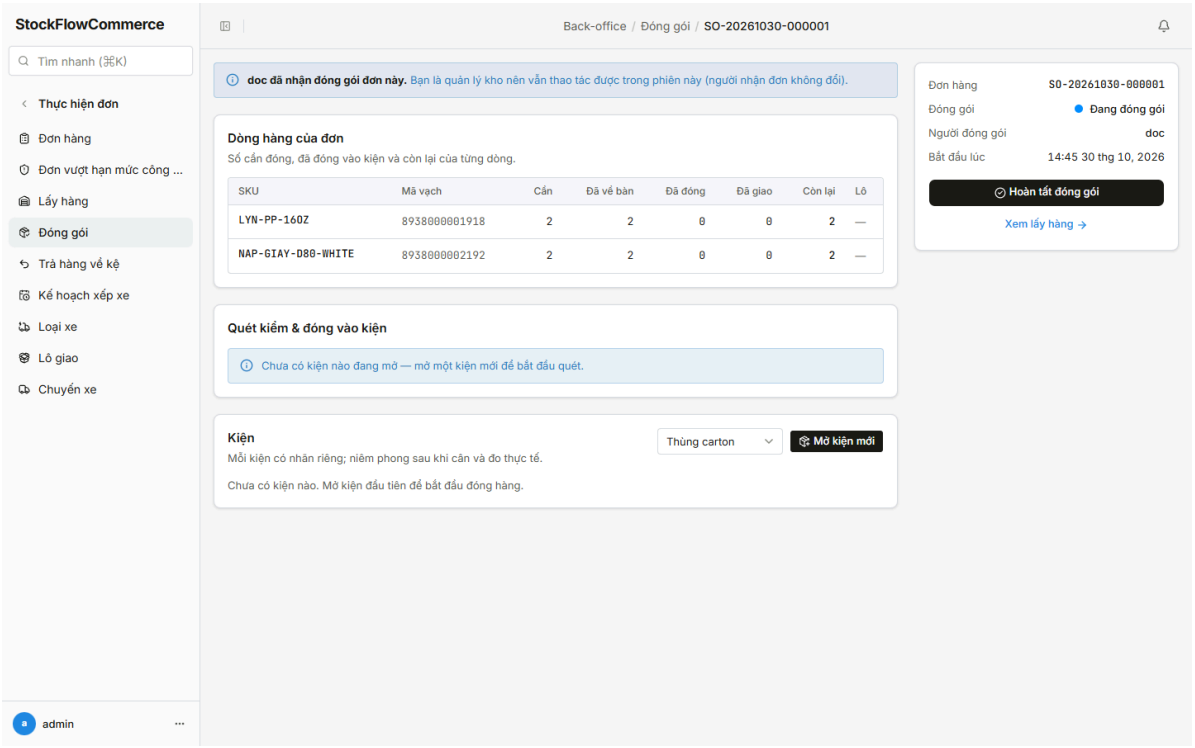

Cân, đo & niêm phong kiện 1
BR-03 (docs 08): kiện phải có khối lượng và kích thước thực đo. Khối lượng quy đổi do hệ thống tính.
Khối lượng thực (kg)
7.6
Dài (cm)
60
Rộng (cm)
45
Cao (cm)
55
Loại kiện
Thùng carton
Cho phép xoay
Theo SKU bên trong (mặc định)
Hàng dễ vỡ
Theo SKU bên trong (mặc định)
Để mặc định thì hệ thống tính từ SKU trong kiện; chọn Có / Không là ghi đè của người đóng.
Huỷ
Niêm phong



- **Function details:**
  - **Data requirement:** Scanned code, quantity, order line/lot/serial when needed; seal: weight (kg), length/width/height (cm), package type, rotatable, fragile; completion mode (full/partial), partial reason.
  - **Validation:** Code ≤64; quantity 1–1,000,000; lot/serial required when tracked; weight >0 (3 decimals); dimensions >0, ≤100,000 cm (2 decimals); open packages must be sealed or voided; partial needs reason ≤500 and Sales-allowed partial shipment.
  - **Business rules:** BR-01, BR-03, BR-04, BR-06, BR-08, BR-09, BR-10 (08 Packing); BR-06 (17 Order Management)

3.3.20.3 Take Over Packing Session

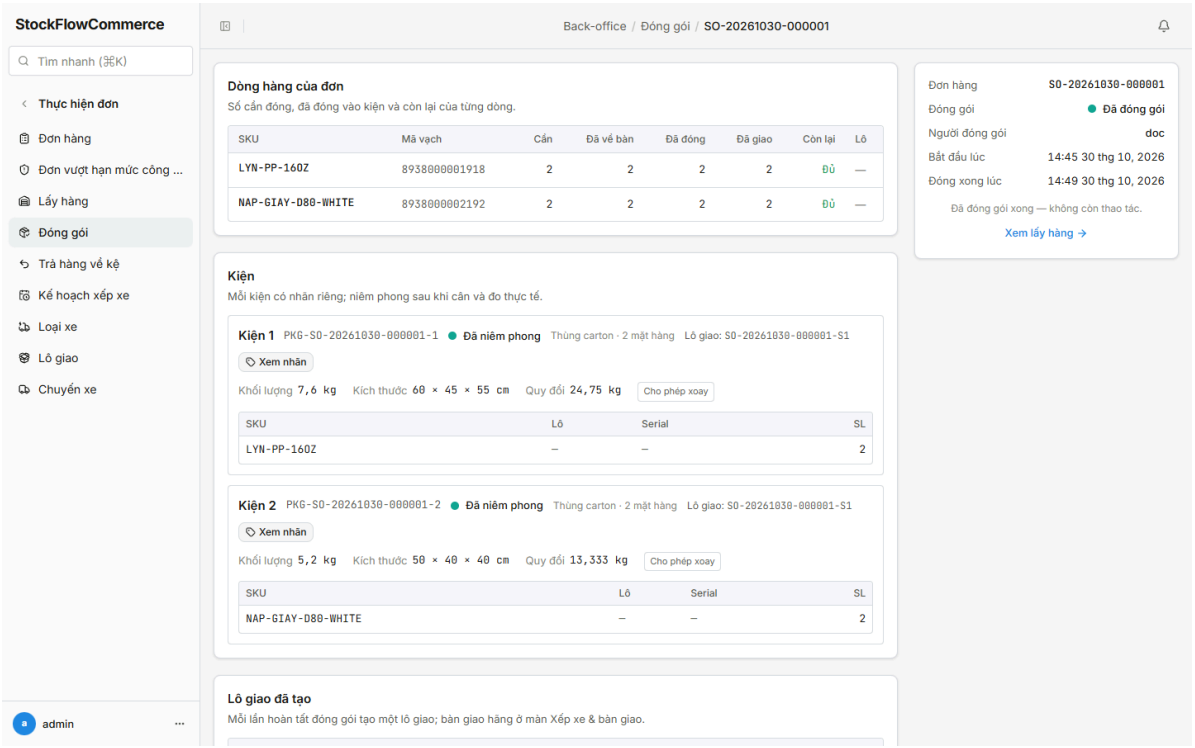
- **Function trigger:** The Warehouse Manager clicks the eye icon (“Mở trạm”) on an order another packer started; the station shows “<tên> đã nhận đóng gói đơn này.” and keeps actions enabled.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Let a manager continue or assist another packer's session.



- **Function details:**
  - **Data requirement:** None (same packing actions as the owner).
  - **Validation:** Session must be Packing; needs fulfillment-packing:APPROVE, otherwise the station is read-only and the server returns PACKING_TAKEN (“Đơn này đang được nhân viên khác đóng gói.”).
  - **Business rules:** BR-07 (08 Packing)

3.3.20.4 Re-Verify Design

- **Function trigger:** Users click “Kiểm tra lại thiết kế” on the packing station of a session held for design mismatch.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Re-check print files against the locked design checksum and release the hold.

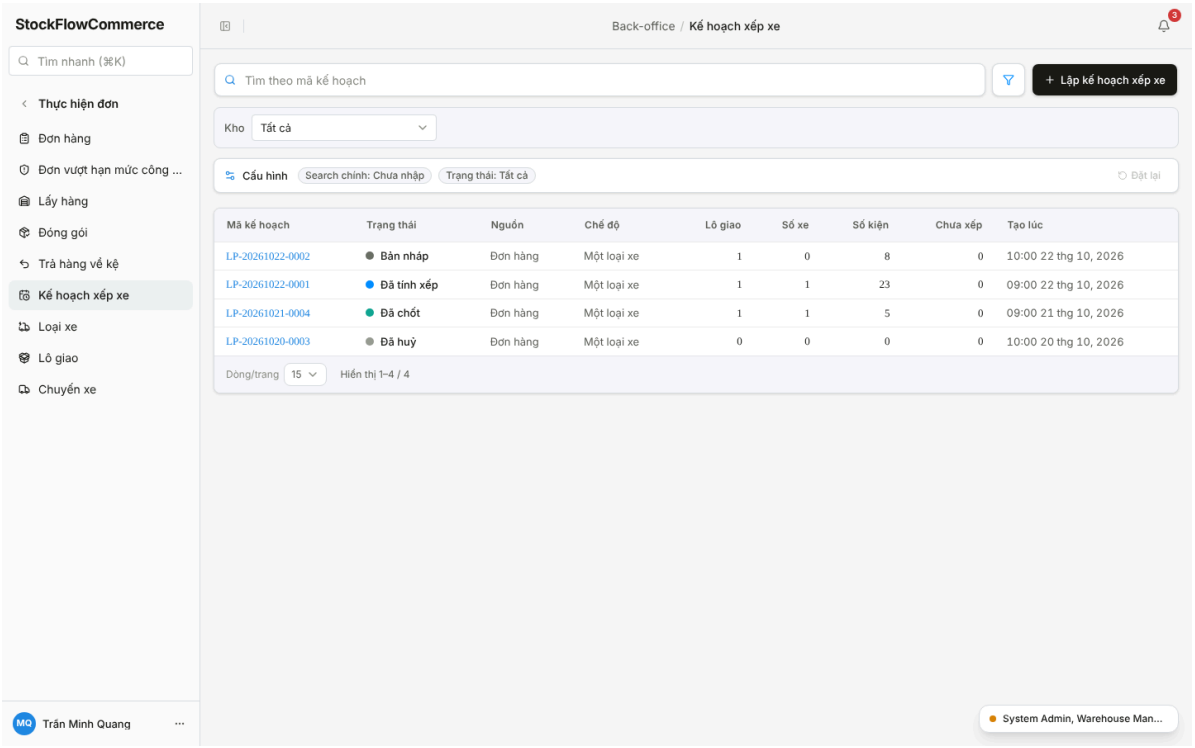


- **Function details:**
  - **Data requirement:** Resolution note (“Cách đã xử lý”).
  - **Validation:** Note required, ≤1000 characters; only when pack status is On Hold; no skip option.
  - **Business rules:** BR-02 (08 Packing); BR-15 (17 Order Management)

3.3.21 Load Planning

3.3.21.1 View Load Plan List

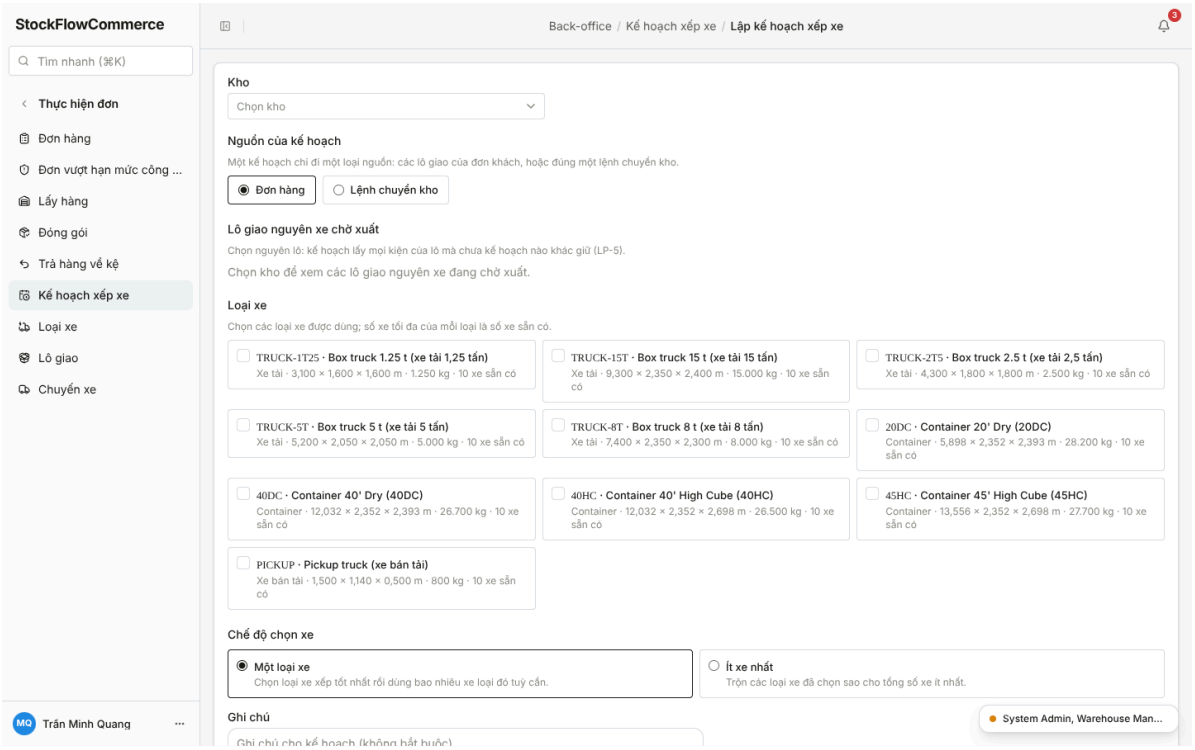
- **Function trigger:** Users click “Kế hoạch xếp xe” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Browse vehicle load plans.



- **Function details:**
  - **Data requirement:** Search by plan code, status, warehouse; list shows plan code, source, status, selection mode, shipments, vehicles, packages, unplaced, created at.
  - **Validation:** None
  - **Business rules:** BR-08 (09 Shipping)

3.3.21.2 Create Load Plan

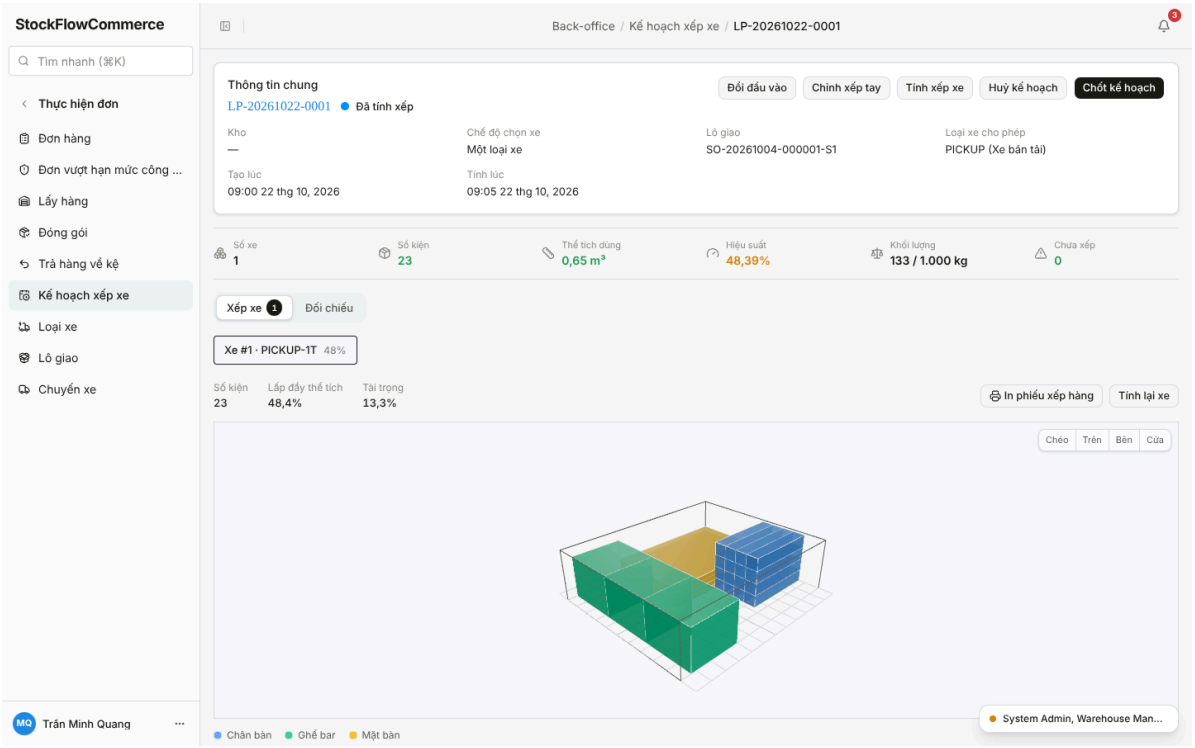
- **Function trigger:** Users click “Lập kế hoạch xếp xe” on the load plan list, fill the form and click “Tạo và tính xếp xe”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Create a load plan and run its first calculation.

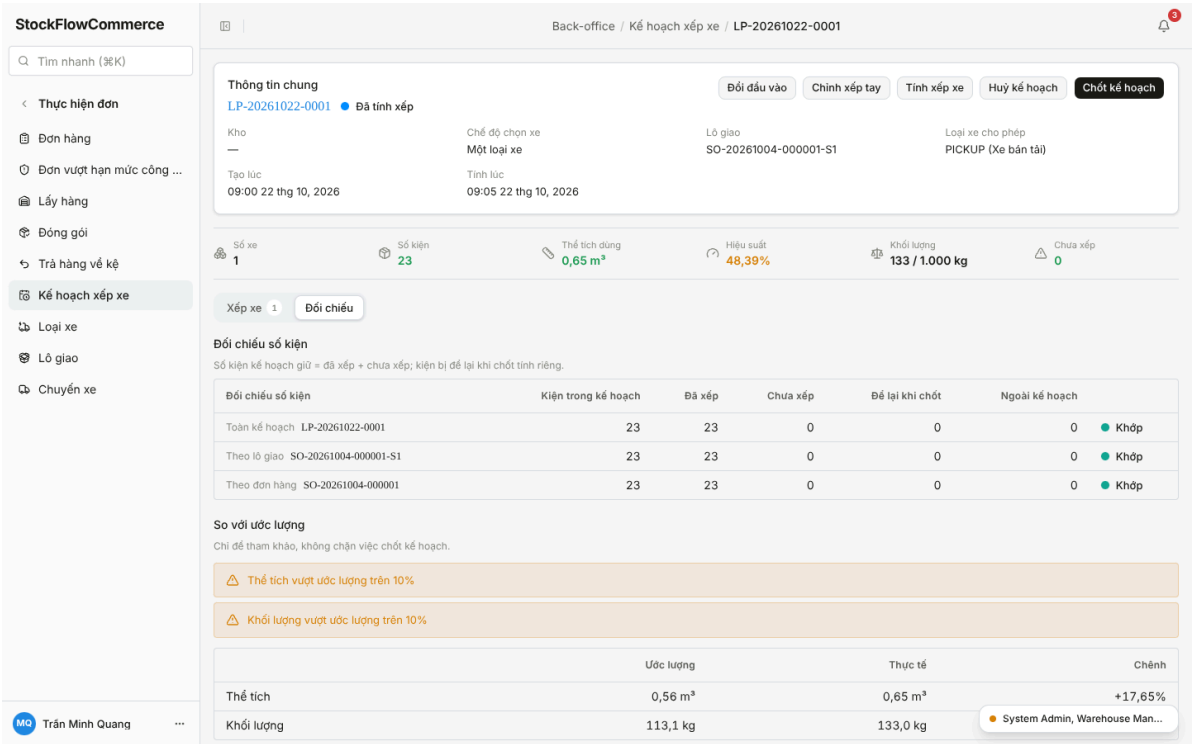


- **Function details:**
  - **Data requirement:** Source (orders / transfer order), warehouse, full-truckload shipments or one transfer order, vehicle types, selection mode (Một loại xe / Ít xe nhất), note.
  - **Validation:** Warehouse required; 1–200 shipments or exactly one packed FTL transfer order; 1–50 vehicle types; note ≤500; packages must not be in another plan; inactive vehicle types and non-FTL shipments rejected.
  - **Business rules:** BR-08, BR-09, BR-12 (09 Shipping)

3.3.21.3 View Load Plan Detail

- **Function trigger:** Users click a plan code (“Mã kế hoạch”) in the load plan list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Inspect a plan's 3D loading layout and results.

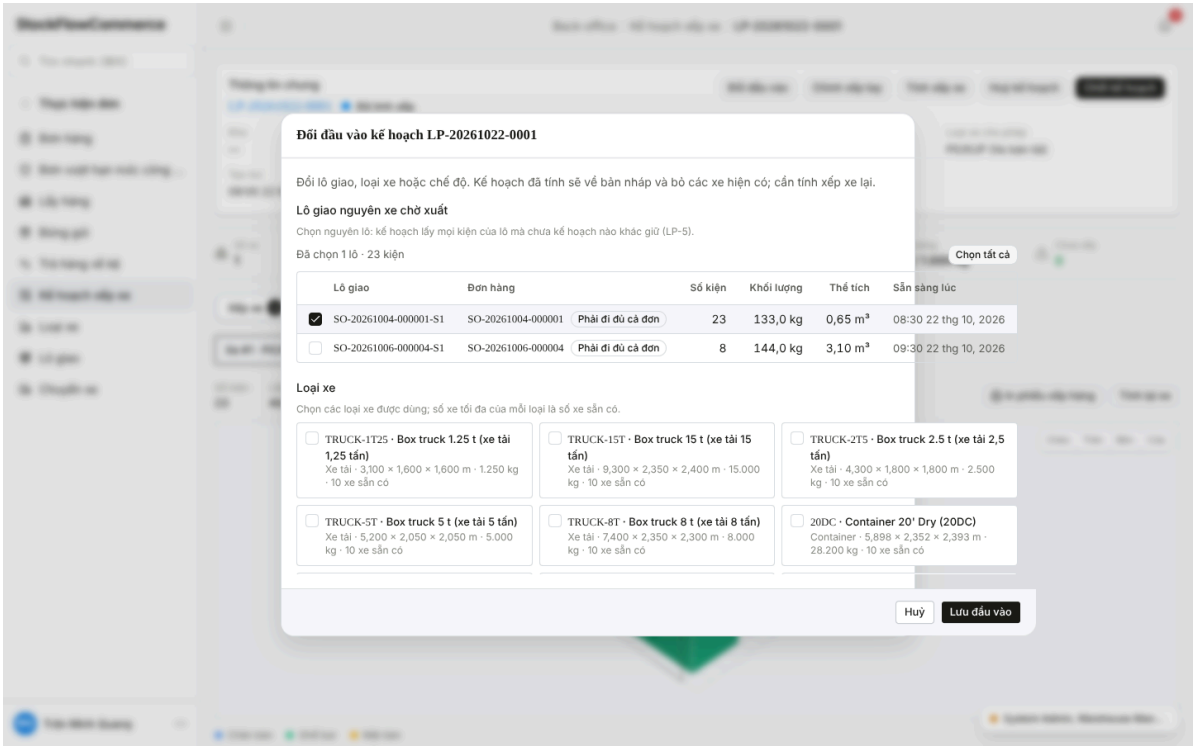


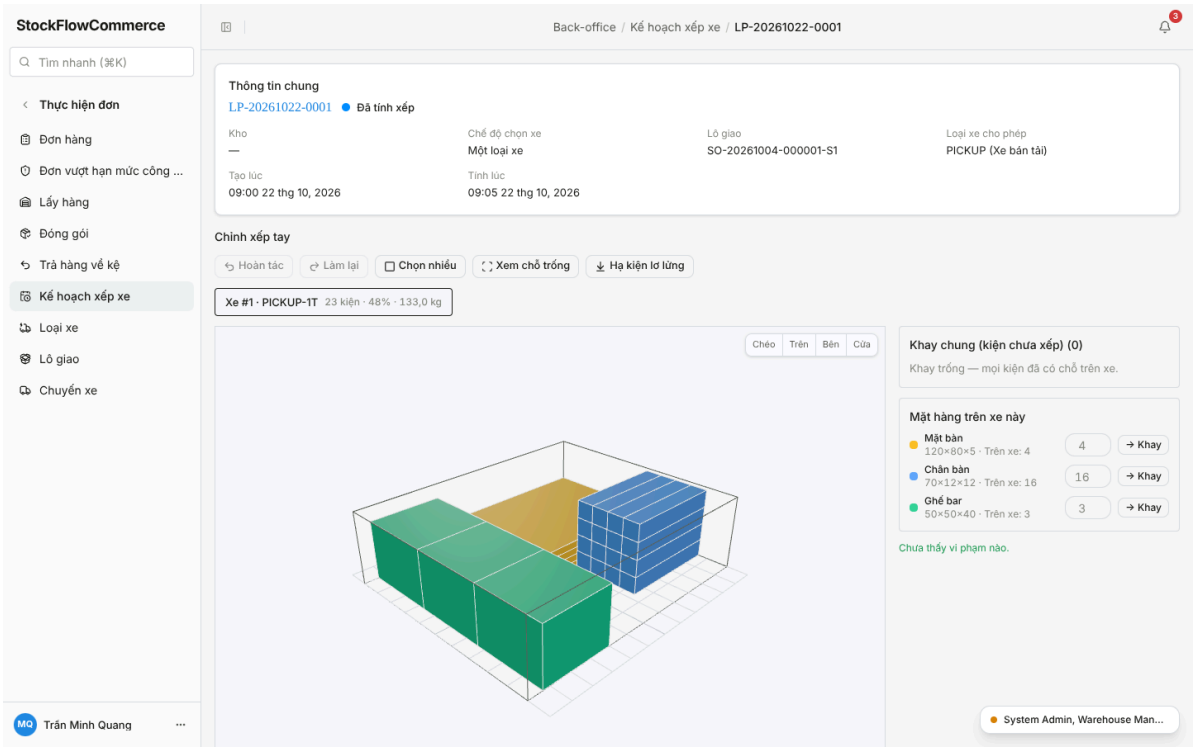


- **Function details:**
  - **Data requirement:** Plan code, status, warehouse, mode, shipments or transfer, allowed vehicle types, timestamps; per vehicle 3D view, volume and weight fill, parcels; unplaced packages with reasons; “Đối chiếu” piece check and estimate comparison.
  - **Validation:** None
  - **Business rules:** BR-10, BR-13, BR-20 (09 Shipping)

3.3.21.4 Calculate and Confirm Load Plan

- **Function trigger:** Users click “Tính xếp xe”, “Chỉnh xếp tay”, “Để lại các kiện chưa xếp”, “Chốt kế hoạch”, “Mở lại” or “Huỷ kế hoạch” on the load plan detail.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Calculate, adjust, lock or withdraw a load plan.



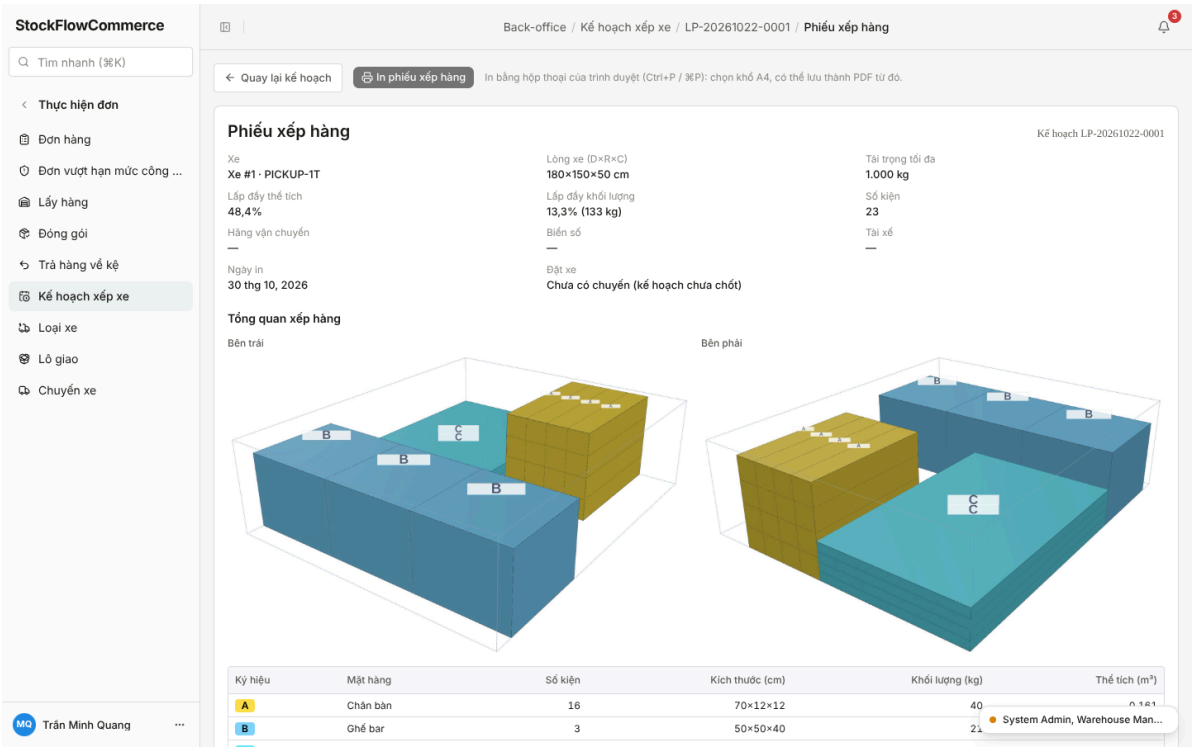


- **Function details:**

- **Data requirement:** Manual placements, changed inputs (“Đổi đầu vào”), packages returned from the tray, reopen/cancel reason.
- **Validation:** Reason required, ≤500; confirm only when Planned, with vehicles, none awaiting recalculation and unplaced packages acknowledged (transfer plans: none left); placements must pass loading rules; reopen/cancel only before loading starts.
- **Business rules:** BR-10, BR-11, BR-13, BR-14, BR-17 (09 Shipping)

3.3.21.5 Print Load Sheet

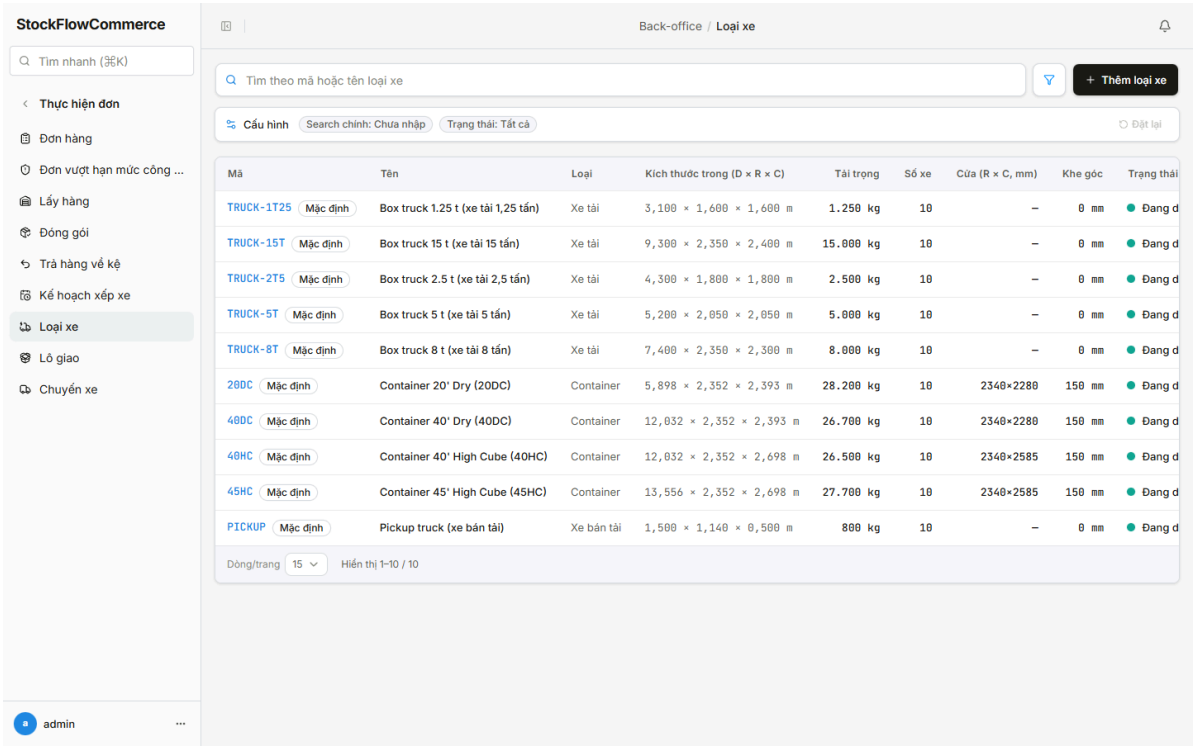
- **Function trigger:** Users click “In phiếu xếp hàng” on a vehicle panel of the load plan detail, then print with the browser (Ctrl+P / ⌘P).
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Print the loading sheet of one vehicle.



- **Function details:**
  - **Data requirement:** Shown: plan, vehicle, inner size, payload, volume/weight fill, booking (carrier, plate, driver), contents table, step-by-step 3D images, parcel list, unplaced packages, signature boxes.
  - **Validation:** Vehicle sequence must exist in the plan.
  - **Business rules:** BR-15 (09 Shipping)

3.3.21.6 View Vehicle Types

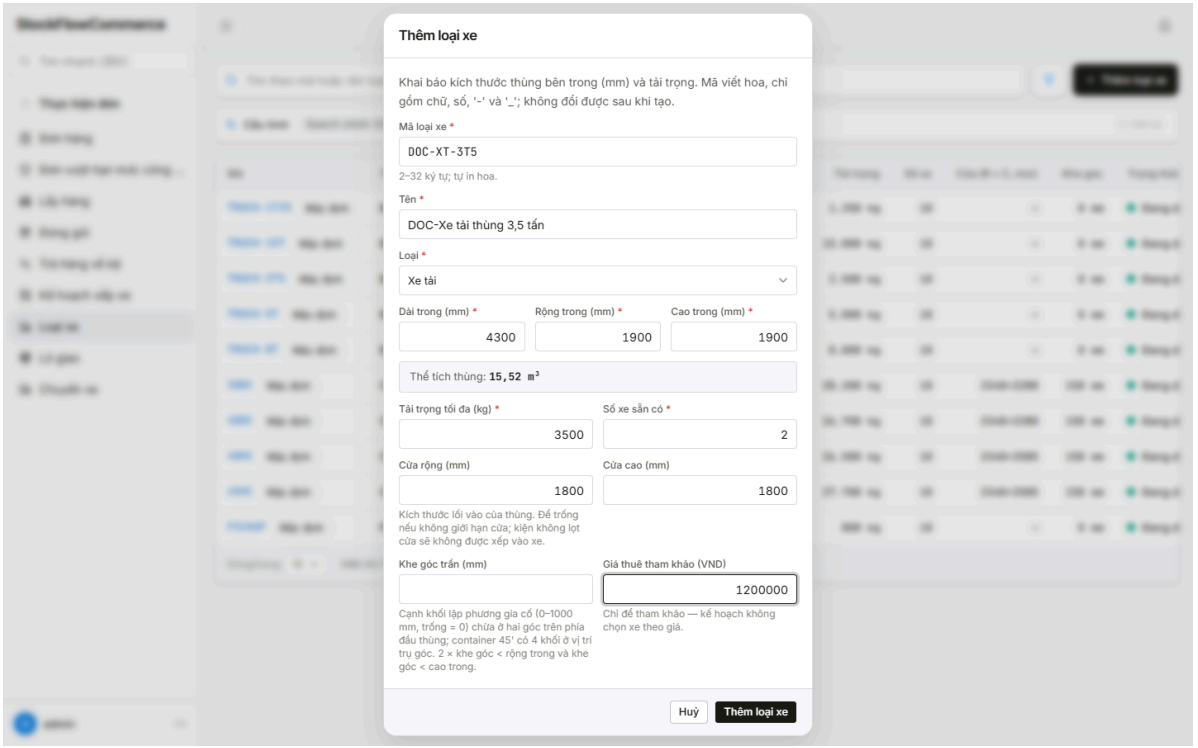
- **Function trigger:** Users click “Loại xe” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Browse the vehicle type catalogue.

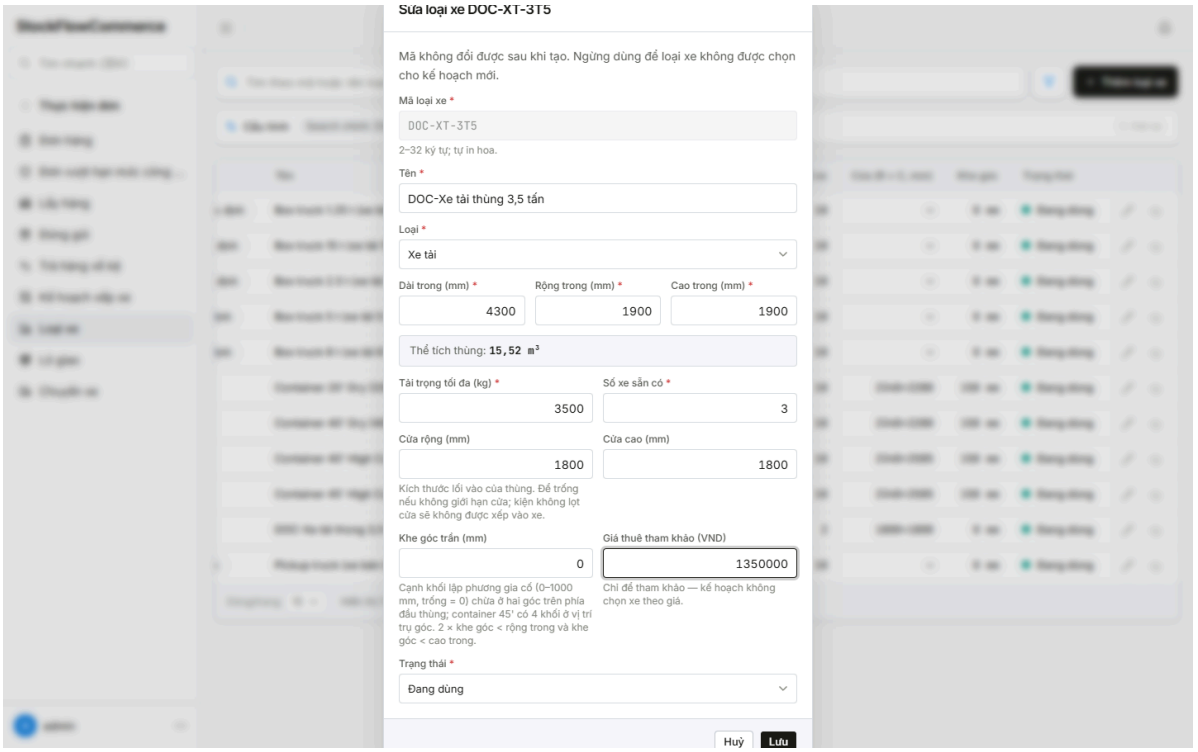


- **Function details:**
  - **Data requirement:** Search by code or name, status filter; list shows code, name, kind, inner size, payload, available count, door size, gusset, status.
  - **Validation:** None
  - **Business rules:** BR-12 (09 Shipping)

3.3.21.7 Manage Vehicle Types

- **Function trigger:** Users click “Thêm loại xe”, the row “Sửa” icon, or “Ngừng dùng” / “Dùng lại” on the vehicle type list.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Maintain vehicle types used for load planning.



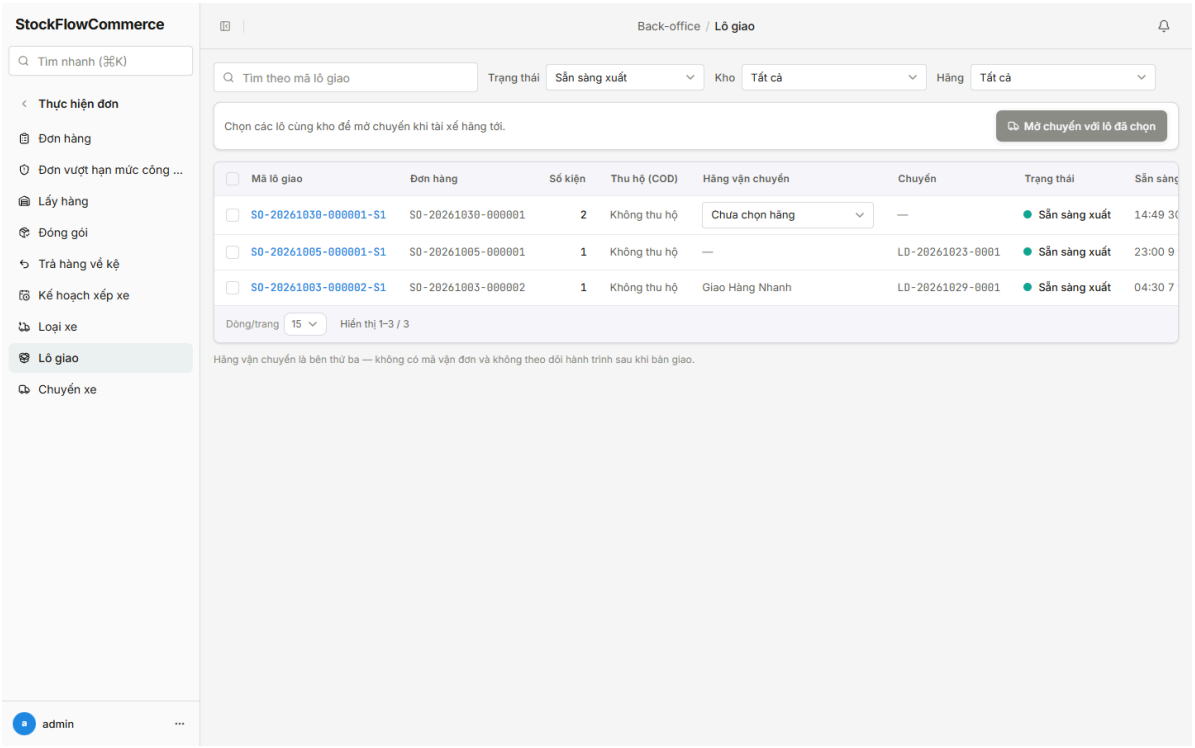


- **Function details:**
  - **Data requirement:** Code, name, kind, inner length/width/height (mm), gusset, door width/height, max payload (kg), available count, reference rental price, status.
  - **Validation:** Code 2–32 chars [A-Z0-9_-], immutable; name ≤120; dimensions 1–30000 mm; gusset 0–1000 and fitting the box; door ≤ inner width/height; payload >0, ≤100000 kg (3 decimals); count 0–10000; price ≥0.
  - **Business rules:** BR-10, BR-12 (09 Shipping)

3.3.22 Shipping

3.3.22.1 View Shipment List

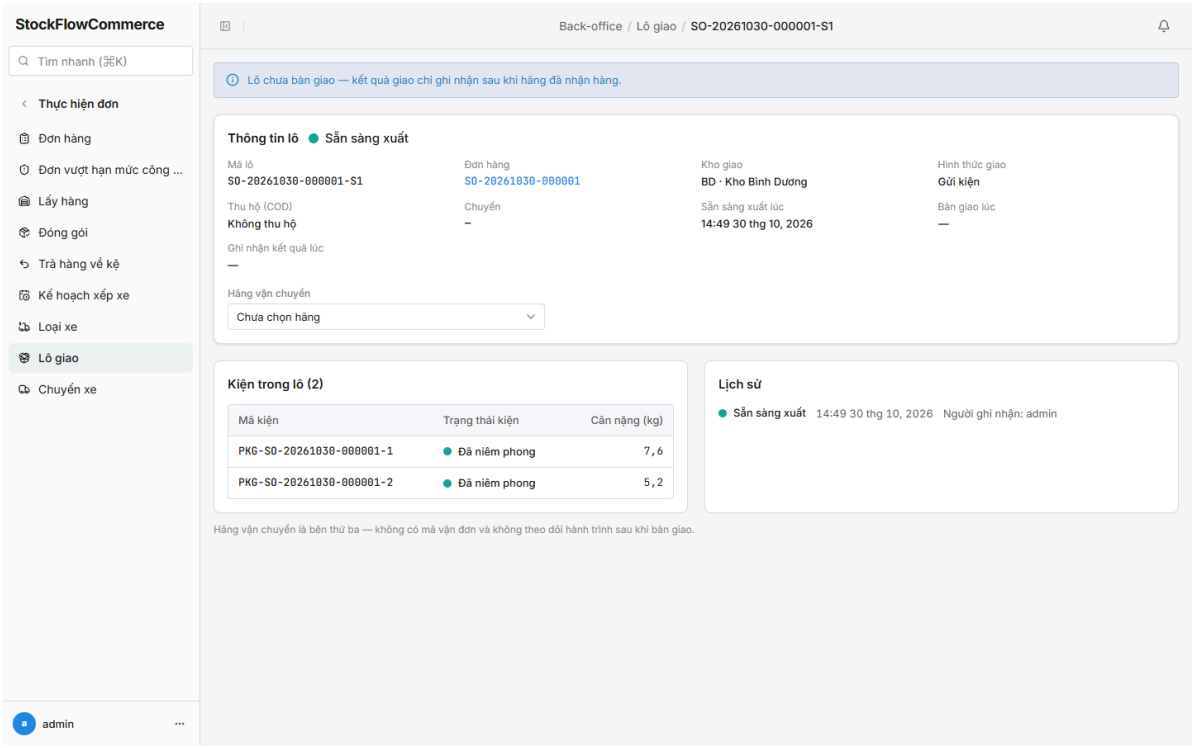
- **Function trigger:** Users click “Lô giao” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Monitor shipments awaiting handover and after it.



- **Function details:**
  - **Data requirement:** Search by shipment code, status (default Ready to Dispatch), warehouse, carrier; list shows shipment code, order, packages, COD, carrier, load, status.
  - **Validation:** Shipments selected for “Mở chuyến với lô đã chọn” must belong to one warehouse.
  - **Business rules:** BR-01, BR-04 (09 Shipping)

3.3.22.2 View Shipment Detail

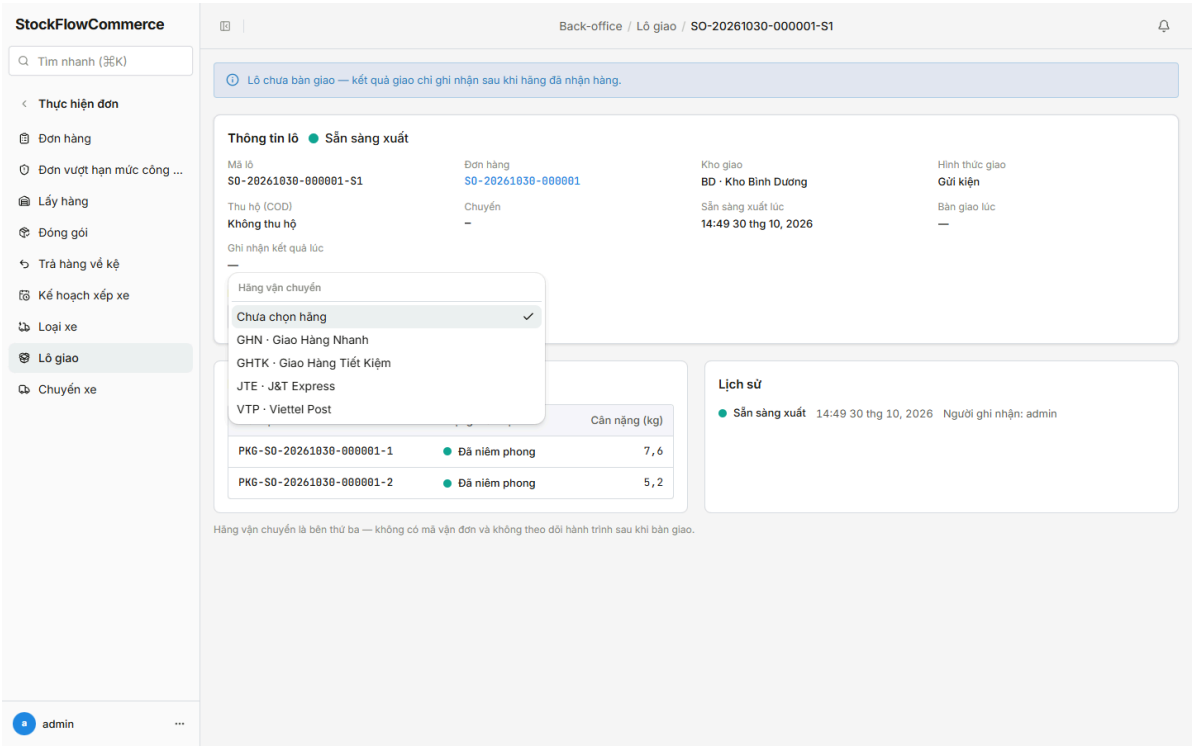
- **Function trigger:** Users click a shipment code (“Mã lô giao”) in the shipment list or “Xem lô giao” on the order detail.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** View one shipment in detail.



- **Function details:**
  - **Data requirement:** Shipment code, order, delivery mode, ready/handed-over/outcome times, failure reason, internal note, load, carrier, packages and their status, history timeline.
  - **Validation:** None
  - **Business rules:** BR-02, BR-04 (09 Shipping)

3.3.22.3 Assign Carrier

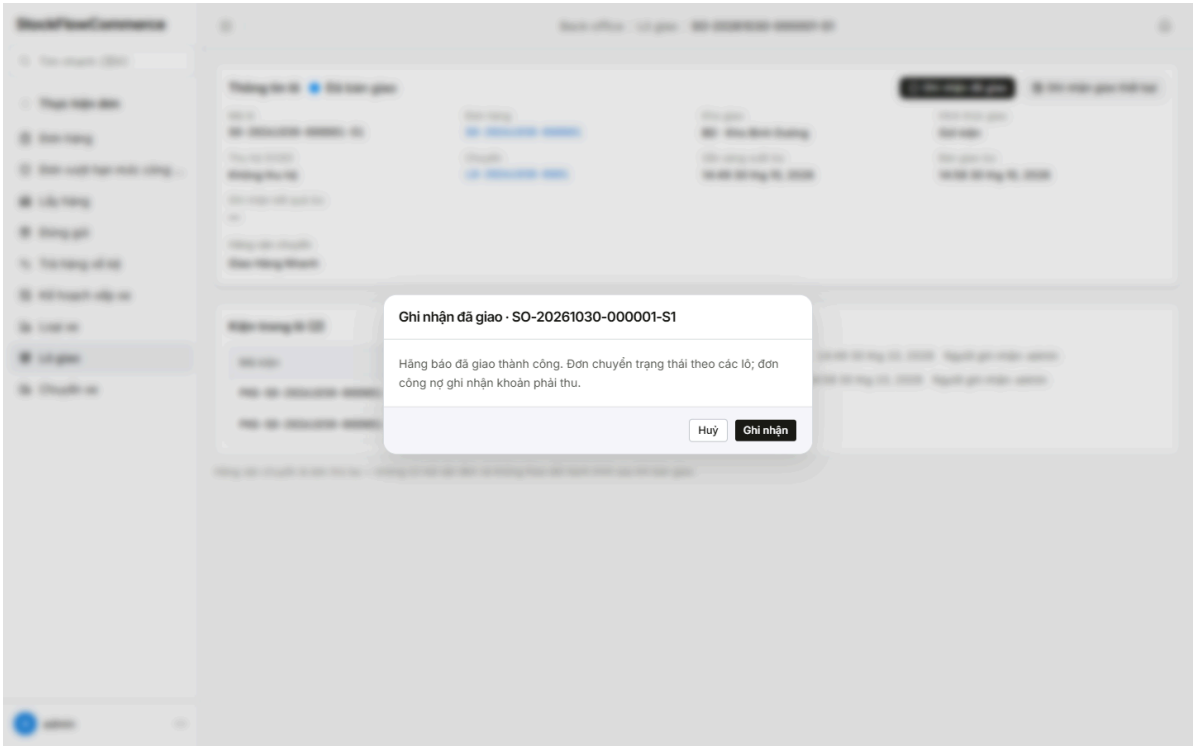
- **Function trigger:** Users pick a carrier in the “Hãng vận chuyển” select of the shipment detail (or in the “Hãng vận chuyển” column of a shipment-list row); “Chưa chọn hãng” clears it.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Assign or clear a shipment's carrier.

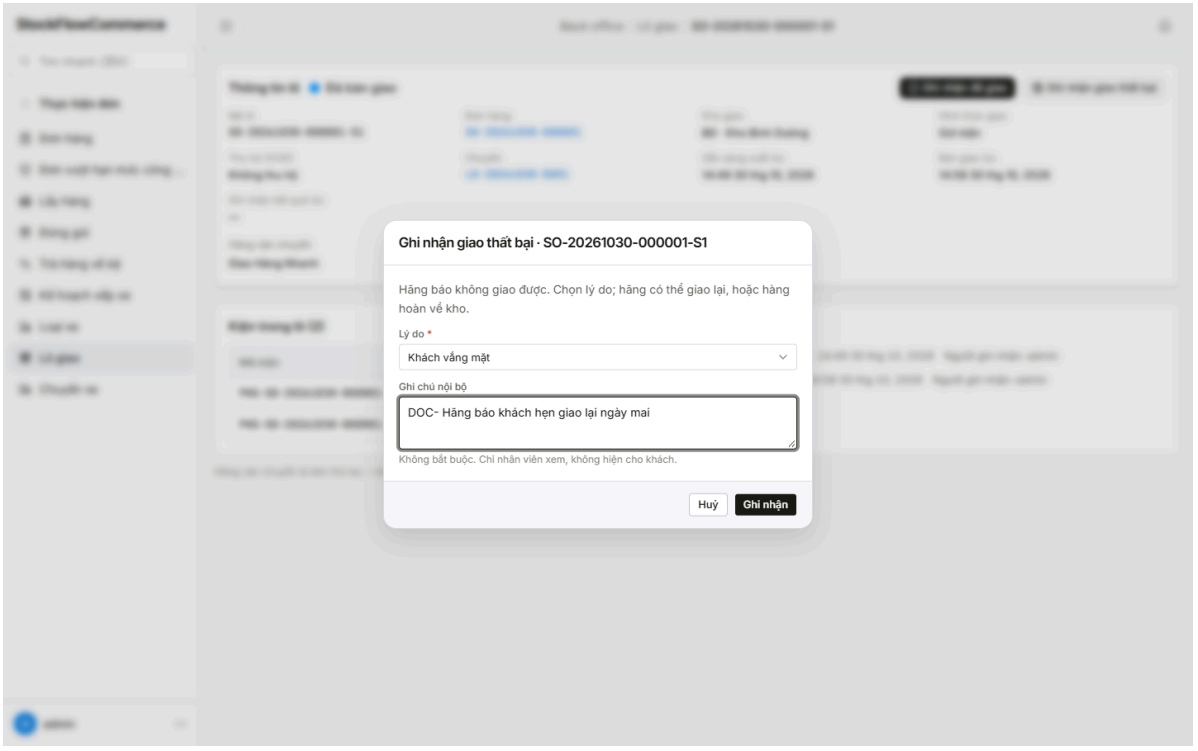


- **Function details:**
  - **Data requirement:** Carrier.
  - **Validation:** Only Ready to Dispatch shipments not yet on a load; carrier must be active (CARRIER_INACTIVE).
  - **Business rules:** BR-16 (09 Shipping)

3.3.22.4 Record Delivery Outcome

- **Function trigger:** Users click “Ghi nhận đã giao” or “Ghi nhận giao thất bại” (or “Ghi nhận hàng hoàn về kho” after a failed delivery) on the shipment detail, then confirm with “Ghi nhận”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Order Coordinator
  - **Purpose:** Record what the carrier reported after handover.

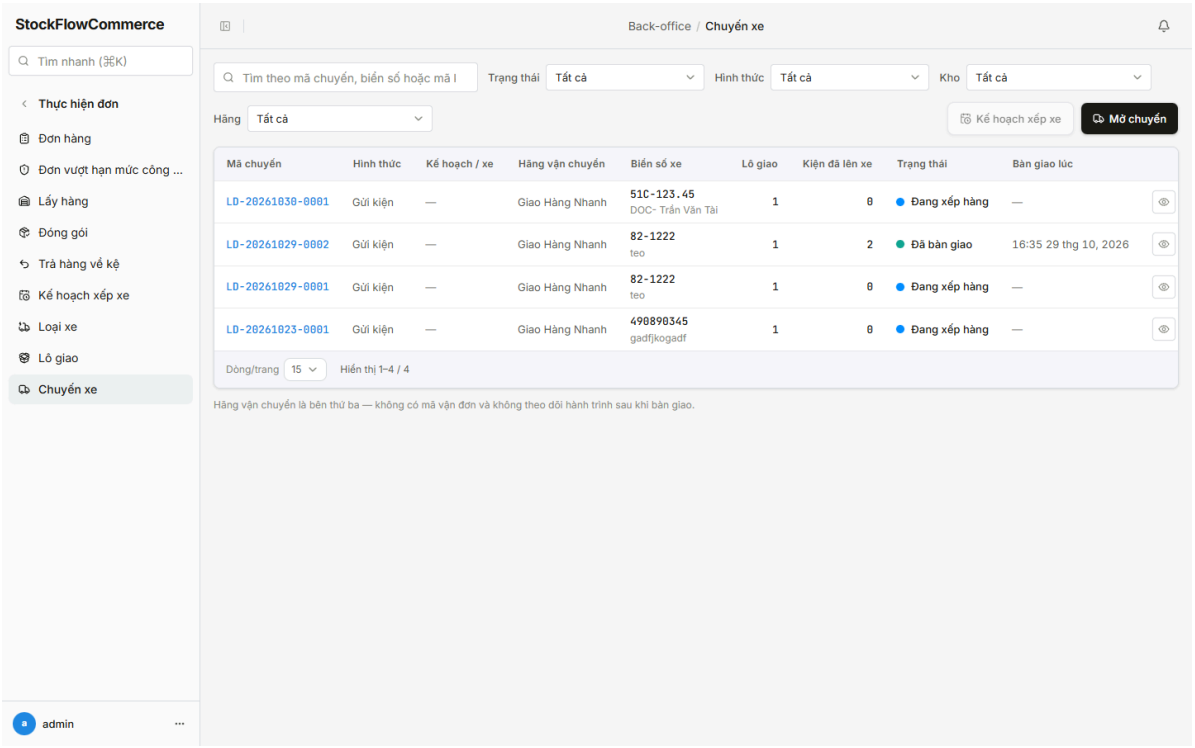




- **Function details:**
  - **Data requirement:** Outcome, failure reason (failed delivery only), internal note.
  - **Validation:** Failed delivery requires a reason; “Lý do khác” requires a note; note ≤500; only outcomes allowed from the current shipment status.
  - **Business rules:** BR-02, BR-06 (09 Shipping)

3.3.22.5 View Load List

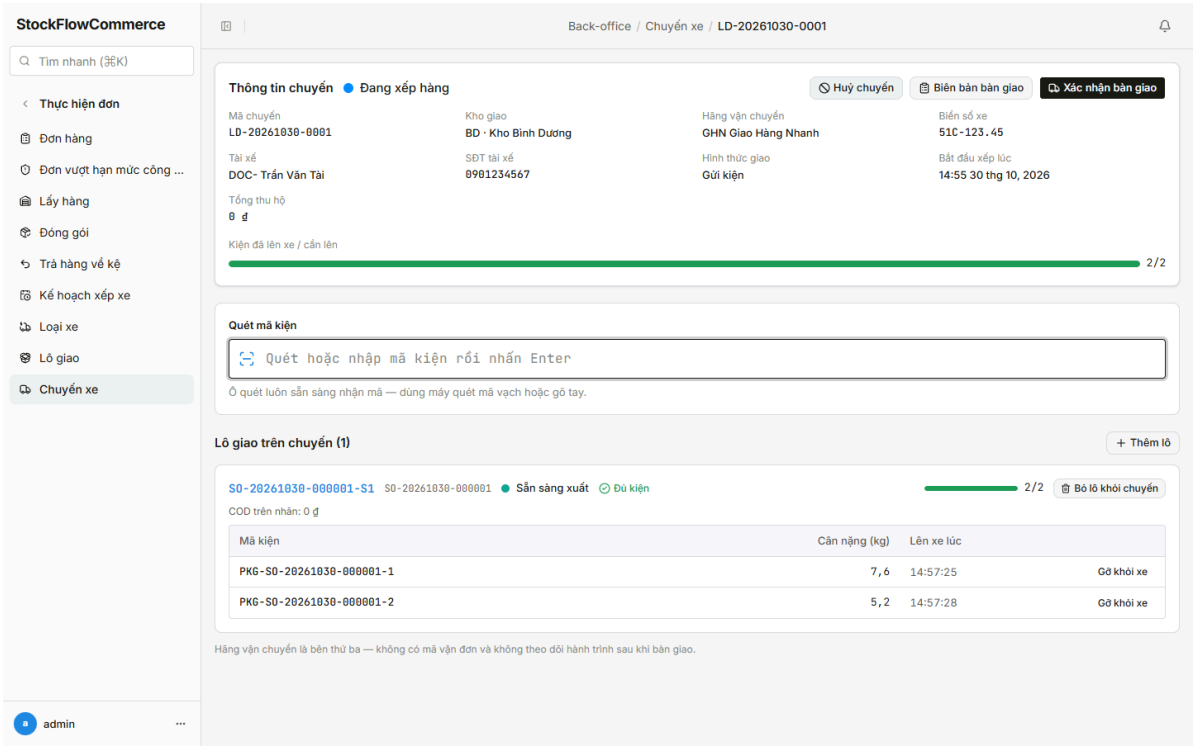
- **Function trigger:** Users click “Chuyến xe” in the “Thực hiện đơn” sidebar group.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Browse dispatch loads (trips).

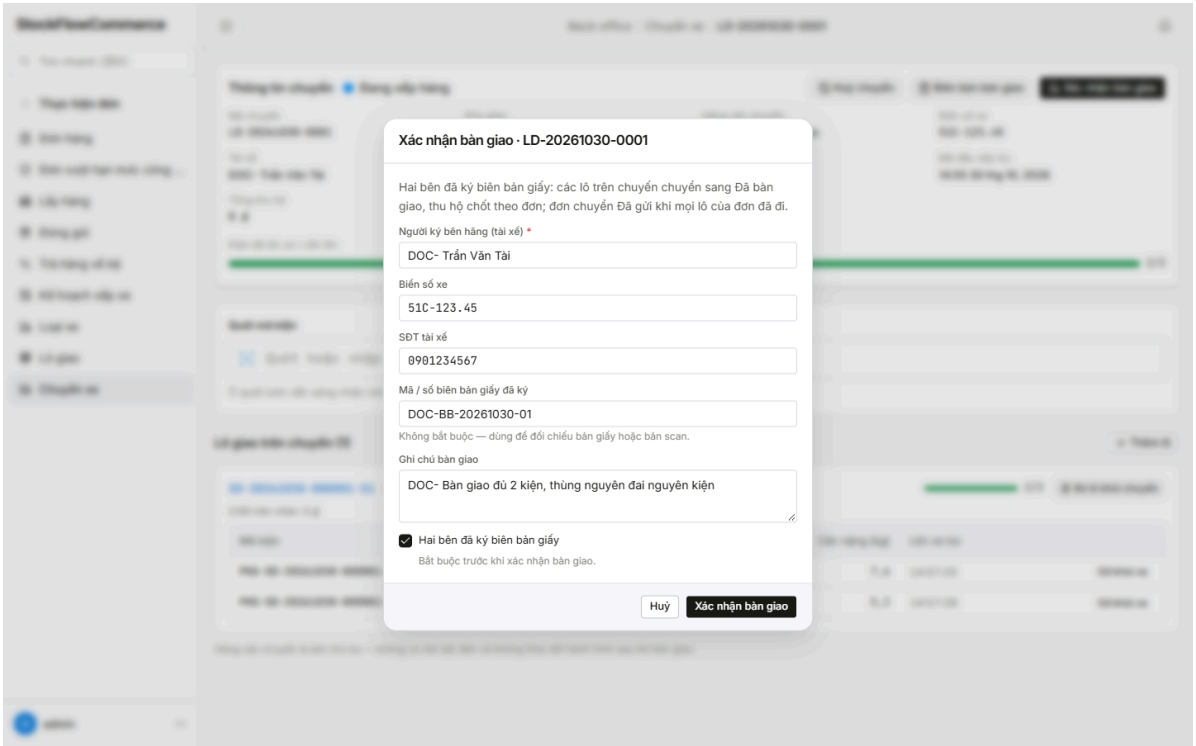


- **Function details:**
  - **Data requirement:** Search by load code, plate or plan code; filters status, delivery mode, warehouse; list shows load code, mode, plan/vehicle, carrier, plate/driver, shipments, loaded packages, status, dispatched at.
  - **Validation:** None
  - **Business rules:** BR-15, BR-16 (09 Shipping)

3.3.22.6 Load and Hand Over

- **Function trigger:** Users click “Mở chuyến” (load list, or “Mở chuyến với lô đã chọn” on shipments), scan in “Quét mã kiện”, then click “Xác nhận bàn giao”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff
  - **Purpose:** Load packages onto a vehicle and hand them to the carrier.



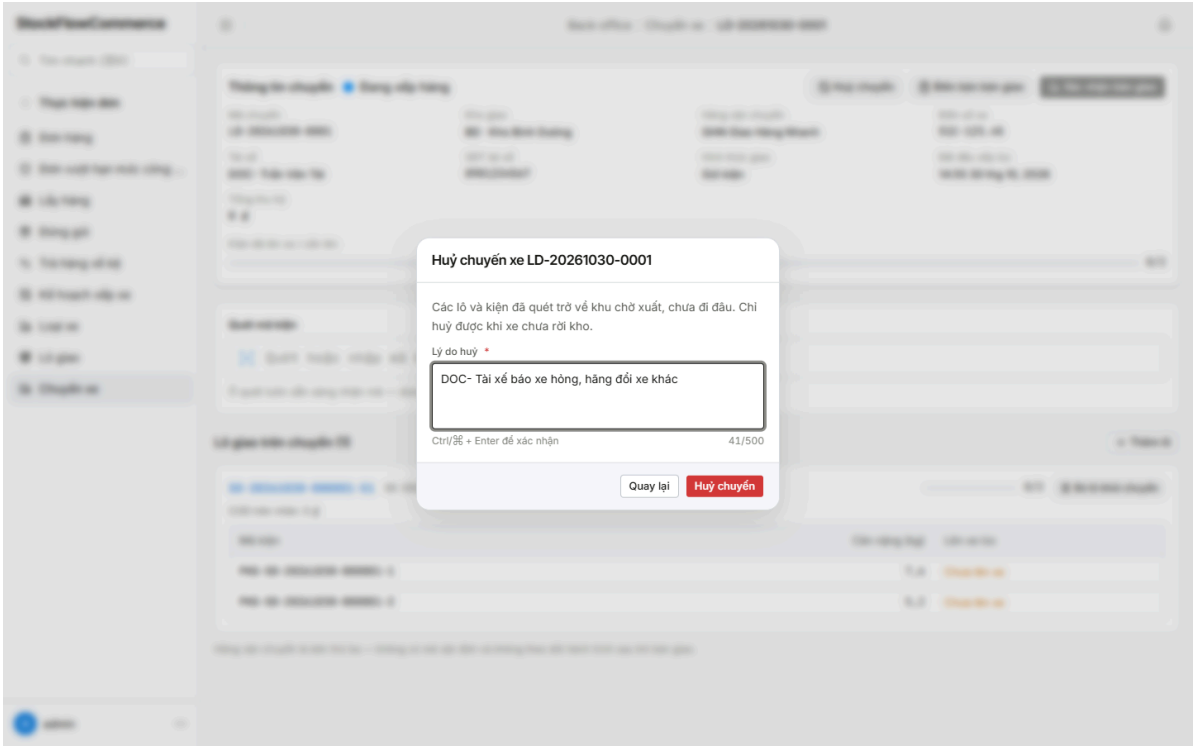


- **Function details:**
  - **Data requirement:** Warehouse, carrier, shipments, plate, driver name/phone; scanned package code; handover: carrier representative, plate, driver phone, signed-manifest ref, note, signed confirmation.
  - **Validation:** Warehouse and carrier required; ≤200 shipments, one warehouse and carrier; code ≤48 printable ASCII; handover needs representative ≤120, plate ≤20 (unless recorded), signed checkbox, every package loaded, orders paid and not held.

- **Business rules:** BR-01, BR-03, BR-04, BR-16, BR-17, BR-18 (09 Shipping); BR-09 (08 Packing); BR-11 (15 Payment)

3.3.22.7 Cancel Load

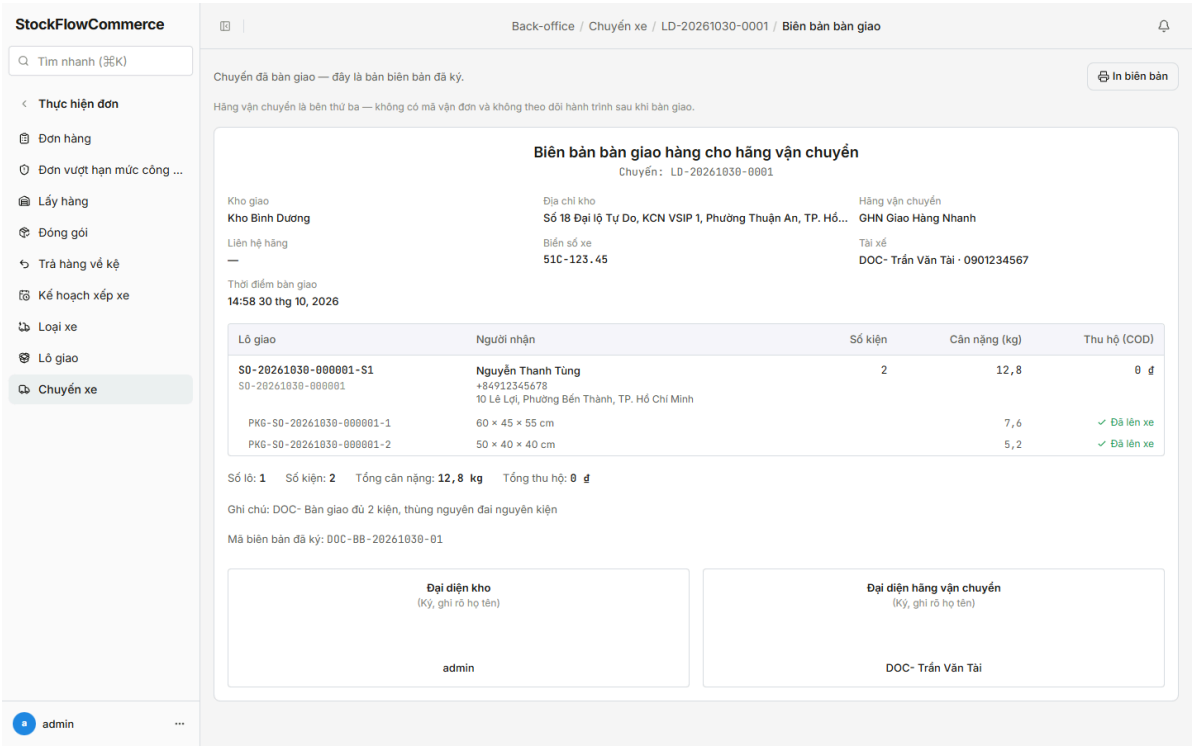
- **Function trigger:** Users click “Huỷ chuyến” on the load detail and confirm with a reason.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager
  - **Purpose:** Cancel a parcel load before the vehicle leaves.



- **Function details:**
  - **Data requirement:** Cancel reason (“Lý do huỷ”).
  - **Validation:** Reason required, ≤500; only parcel loads in Planned/Vehicle Booked/Loading; needs fulfillment-load-cancellations:UPDATE.
  - **Business rules:** BR-19 (09 Shipping)

3.3.22.8 View Load Manifest

- **Function trigger:** Users click “Biên bản bàn giao” on the load detail, then “In biên bản”.
- **Function description:**
  - **Actors:** System Admin, Warehouse Manager, Warehouse Staff, Order Coordinator
  - **Purpose:** Reconcile and print the handover manifest for signing.



- **Function details:**
  - **Data requirement:** Shown: load, warehouse and address, carrier contact, vehicle, shipments (order, recipient, address, packages, weight, COD), parcel codes, totals, signature blocks.
  - **Validation:** “In biên bản” is disabled until every package of every shipment is loaded; until then a draft banner is shown.
  - **Business rules:** BR-04, BR-15, BR-18 (09 Shipping)

4. Non-Functional Requirements

4.1 External Interfaces

- The application shall operate entirely in the Vietnamese language to ensure accessibility for local users.
- The system shall maintain compatibility with the latest stable releases of major browsers, including Chrome, Firefox, and Safari.
- The interface design shall be responsive, adapting smoothly to various screen sizes and device orientations.
- The system shall provide clear, concise, and user-friendly error messages whenever issues occur during external service interactions.
- All icons shall be easily recognizable and accurately represent their intended functions.
- Text elements must use legible fonts that support readability across different display resolutions.

4.2 Quality Attributes

4.2.1 Usability

- The platform shall be designed to be user-friendly and easy to navigate.
- The user interface shall follow modern web design principles to ensure a consistent and intuitive experience.

- Navigation and layout shall follow familiar and predictable patterns to enhance usability.

4.2.2 Reliability

- The system shall maintain an uptime of at least **99%**, excluding scheduled maintenance periods.
- Data integrity must be ensured to prevent the loss or corruption of donation and transaction records.

4.2.3 Performance

- It shall be capable of processing multiple simultaneous donations or campaign submissions efficiently.
- The platform shall handle a high number of concurrent users during peak fundraising periods without performance degradation.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	A staff username never changes once the account is created. Each staff account and each customer contact has a unique username and a unique email address.
BR-02	When an administrator creates a staff account without a password, or resets a password, the system generates a 16-character temporary password that is shown only once, and the user must change it at the next sign-in. A password reset also ends every session of the account.
BR-03	A password must be 10 to 72 characters long and contain at least one upper-case letter, one lower-case letter and one digit. Changing a password requires the current password, and the new password must differ from it.
BR-04	Five consecutive failed sign-in attempts lock the account for 15 minutes, after which it unlocks automatically; during the lock the system answers exactly as for a wrong password. Each IP address may attempt to sign in at most 5 times per minute.
BR-05	Each sign-in creates a session that lasts 8 hours and is not refreshed. Users can see their own sessions, sign out, and sign out their other devices.
BR-06	All sessions of an account end immediately when its roles change, when it is locked or disabled, when an administrator resets its password, or when the user changes the password and chooses to sign out other devices. A change of assigned warehouses takes effect from the next action without signing in again.
BR-07	A staff account is Active, Locked or Disabled. Only an account administrator locks, unlocks, disables or re-enables an account; unlocking also clears an automatic sign-in lock, and a disabled account is kept for history.
BR-08	An administrator can grant only roles and permissions they fully hold, can manage only accounts that hold no permission beyond their own (so only a System Admin can grant the System Admin role), and cannot manage their own account.
BR-09	At least one active System Admin must always remain, and at least one role must keep the right to edit permissions.

BR-10	The Customer role is never given to a staff account, and a customer account holds no other role.
BR-11	A permission is a resource combined with an action: View page, Read, Create, Update, Delete, Approve or Export. Delete, Approve and Export are sensitive actions.
BR-12	Each role has a data scope: Warehouse (only the warehouses assigned to the account; with none assigned the user sees nothing), All (the whole system) or Own (the customer's own data). A user with several roles gets the union of their permissions and the widest scope.
BR-13	The 12 system roles cannot be renamed or deleted. A custom role can be created (optionally copied from another role), renamed and re-scoped, and can be deleted only by a System Admin and only when no account holds it.
BR-14	Editing a role's permission matrix requires the permission-approval right; the matrices of the Customer and System Admin roles are fixed. A change takes effect on the holders' next action without a new sign-in.
BR-15	B2B customers cannot register themselves and there is no guest checkout. Only Sales Staff or the E-commerce Admin creates a customer profile, which starts as Pending Activation with one login account for its contact; the email is the login name.
BR-16	An activation invitation is valid for 72 hours and can be used only once. Each account has at most one open invitation; resending revokes the previous link and is allowed at least 1 minute after the previous sending.
BR-17	A tax code is optional. When entered it must be 10 digits, 10-3 digits or 12 digits (spaces and dots are removed and 13 digits become 10-3), it must be unique across all customer profiles including closed ones, and once set it cannot be removed.
BR-18	Only a customer whose profile is Active can see prices, receive quotes and place orders. A Pending Activation customer can sign in but sees no price or stock.
BR-19	A Blocked or Inactive customer cannot sign in or place new orders: its accounts are disabled and their sessions end, while orders already running are processed to the end.
BR-20	Every customer status change is recorded with a reason; unblocking and reopening require a reason, and nobody can set Pending Activation by hand. A closed customer is reopened by the E-commerce Admin instead of being created again.
BR-21	Credit limits, payment terms, deposit rates, the balance collection policy and the allowed payment terms are approved only by the Accountant, with the approver and time recorded. When creating a customer, Sales can only choose prepaid or deposit terms (deposit rate above 0% and below 100%).
BR-22	Every change of a customer's company name or tax code is recorded with the old value, new value, who and when; staff can read this history, the customer cannot. Tax codes are masked in lists and customer-facing screens.
BR-23	A customer group code never changes after creation, and each customer belongs to at most one group.

BR-24	A customer keeps separate shipping and billing addresses with at most one default per type; the first address of a type becomes the default, and deleting the default promotes another address of the same type. Customers access only their own profile and addresses.
BR-25	Only Published products are shown in the catalog and only Active SKUs are listed. The catalog is available only to signed-in users.
BR-26	The price applied to a customer is resolved as customer price, then customer-group price, then base price. Group and base prices have a single level; two rules of the same priority for the same SKU are reported as a configuration error, never chosen automatically.
BR-27	A customer price has 1 to 20 quantity tiers and its first tier is the customer's MOQ. The tier is chosen by the total quantity of the SKU in the order or quote; below the MOQ there is no price and no fallback, so the customer must ask Sales for a quote.
BR-28	All prices are positive whole VND amounts per base unit, with no validity period and no price approval. The price is fixed on the order when it is created; later price changes never alter placed orders, and a customer whose price changed during checkout must confirm the new price.
BR-29	Availability is computed from system-wide ATP and shown only as in stock, low stock or out of stock; the ATP quantity is never shown. There is no backorder: an out-of-stock SKU cannot be ordered.
BR-30	A product's slug is locked after its first publication, and the catalog content of a Discontinued product cannot be edited.
BR-31	Only the E-commerce Admin and System Admin change prices, customer groups and publication; Sales Staff and the Order Coordinator can only view customer and group prices.
BR-32	Lines with a print design enter an order only through a quote; customers cannot order print lines directly on the portal.
BR-33	A product code never changes. A SKU and its axis values are locked once the variant leaves Draft. A SKU is unique across the system, including discontinued products, and has at most 64 characters made of upper-case letters, digits, '.', '_' and '-'.
BR-34	Only a Draft product whose SKUs are not referenced by any document can be deleted; any other product can only be suspended or discontinued. An attribute in use can only be deactivated.
BR-35	Only Active SKUs of Approved or Published products can be used on new purchase orders, quotes and orders; this is checked at every step and each failing line is named.
BR-36	Publishing a product requires catalog content, a category, at least one Active SKU, a price for every Active SKU and at least one published image.
BR-37	Each SKU has exactly one inventory item, created automatically with the SKU, which holds only physical, stock-control and reference purchasing data and no sales content. Combos and bundles are not sold.

BR-38	Stock and every warehouse document are kept in the SKU's base unit. A conversion factor is a whole number of at least 1, and no new SKU is created per pack size. Once a unit is used by a document its factor cannot change: a new unit is declared and the old one retired.
BR-39	A variant's sales units are chosen from its inventory item's units. The base unit is sold at the catalog base price; only other sales units, such as a carton, have their own price.
BR-40	Four-eyes rule: the person who approves a product must not be the person who submitted it, and the person who publishes an image must not be the person who uploaded it.
BR-41	A product can be submitted for approval only with a category, a brand, an English name and a tax class. Its main information can be edited only in Draft, and a product Pending Approval cannot be edited at all.
BR-42	Only the System Admin suspends or discontinues a product, always with a reason of 1 to 500 characters. Discontinuing makes every variant obsolete and warns about open documents without blocking them.
BR-43	Every product change carries the version the user is viewing (optimistic locking), and every change is kept in a history that is never edited or deleted. A possible duplicate product is only warned about, never blocked.
BR-44	A SKU's base unit cannot change while the SKU has stock, a stock policy that conflicts with existing stock is refused, and expiry tracking cannot be switched off once stock with expiry dates exists.
BR-45	Product images must be JPEG, PNG or WebP files of at most 10 MB and are virus-scanned; when the scanner is unavailable an image is not accepted. A variant holds at most 20 images with exactly one cover image.
BR-46	Logistics data (weight and dimensions) is optional, but when entered it must be positive. A deactivated brand or category cannot be chosen for a product.
BR-47	Each design belongs to exactly one customer and one product, and its owner must be the active customer account of that customer.
BR-48	A design's designer, if any, is an active Sales Staff or E-commerce Admin user. Its technical reviewer is mandatory and must be an active QC Staff user who is neither the owner nor the designer.
BR-49	A design is visible only to its owner, designer and reviewer; to anyone else it does not exist. Confirmed designs can be looked up by Sales per customer to prepare quotes and samples.
BR-50	Design files are only added, never edited or deleted, and only a Draft design accepts files. Files must be JPEG, PNG, WebP or PDF of at most 50 MB and pass a malware scan.
BR-51	A design cannot be submitted for review while the print preflight fails: image resolution below the minimum DPI, artwork outside the safe area, or a font that is not embedded.
BR-52	Technical review is independent (four-eyes): only the assigned reviewer can review, the reviewer must not be the owner or the last editor, and a review is tied to the current preview file and carries a comment.

BR-53	Any change clears the review result, including uploading a file, editing the specification, reassigning people, a customer's change request or a withdrawal; the design must be reviewed again before it is sent to the customer.
BR-54	Customer confirmation does not replace technical review: a design is sent to the customer only after its current files passed review, only the owner can confirm, and the customer confirms exactly the preview that is waiting; if the files changed, the design must be sent again.
BR-55	A design snapshot is immutable from the moment the customer confirms it. It holds the files, the SHA-256 checksum of each file and of the whole set, and the specification; a confirmed design accepts no files, specification edits or reassignment.
BR-56	Quotes, samples and orders accept only snapshots of the same customer and of the same product as the SKU. At every use (quote, order, prepress, download, packing) the stored files must match the confirmed checksum; a mismatch is refused or held, and an unconfirmed file is never printed.
BR-57	Every write to a design carries the version the user is viewing; if someone changed it first, the action is refused and the user reloads. Every decision is recorded with the person, file, content or reason, version and time.
BR-58	Orders, samples and quotes keep the snapshot attached to them; later revisions do not affect them. When staff edit a design on a customer's behalf, a new version is created and the customer must confirm it before it can be ordered.
BR-59	A print line needs a design snapshot confirmed by the customer, of the same customer and the same product as the blank SKU, with matching checksums. For a new design, a quote can be accepted only when an Approved sample exists.
BR-60	Each line quantity is a whole number of at least 1 in the base unit and at least the MOQ of the quote or price list; the price tier is chosen by the total quantity of the blank across all lines of the quote. The same quantity is used for reservation, allocation and production orders.
BR-61	An order may use only the payment terms allowed for the customer (prepaid, deposit or credit). A deposit rate exists only on deposit quotes and must be above 0% and below 100%.
BR-62	A credit order is confirmed only when the customer has no overdue receivable and the outstanding balance plus the order value is within the credit limit; otherwise it is put On Hold until the Accountant approves it, and if the Accountant rejects it the order is cancelled or switched to prepaid or deposit.
BR-63	Stock is held for an order awaiting prepayment or deposit for a limited time (30 minutes by default, configurable). When the hold expires unpaid, the stock is released and the order is cancelled; an order whose printing has started is referred to staff instead.
BR-64	An order total equals subtotal plus shipping fee plus tax minus discount, and every amount is shown to the customer before they confirm.
BR-65	Only the customer accepts a quote or requests changes to it. Only staff (Sales Staff or the Order Coordinator) convert an accepted quote into an order, and converting it again returns the same order.

BR-66	A quote can be sent only when every line has a price, the payment term is allowed for the customer and the validity date is in the future. An expired quote cannot be accepted, and a quote holds its prices during its validity.
BR-67	Every quote action carries the version the user is viewing, and each reopening after a change request creates a new revision; every version sent is kept.
BR-68	A SKU must still be sellable when a quote is drafted, edited, sent, accepted and converted; the system names each failing line.
BR-69	A draft quote can be cancelled with a reason. An accepted quote can be cancelled only by Sales, with a reason, when one of its SKUs is no longer sellable.
BR-70	Reordering a quote-based order creates a new draft quote with the same blanks, designs and quantities, priced from the current price list and never from the old negotiated price.
BR-71	An order with the same items as one the customer placed a moment ago is placed only after the customer confirms it. A Confirmed order's content (SKU, quantity, address) changes only through Order Management, never freely.
BR-72	A prepaid order is Confirmed only when it is fully paid. A deposit order is Confirmed, and can be released to production, only when the full deposit has been received.
BR-73	Money is recorded only when the Accountant matches it against the bank statement, never on the customer's word, and each statement reference is recorded only once, including carrier remittances.
BR-74	A refund never exceeds the amount collected, and a refund of surplus money never exceeds the unallocated surplus. A refund above the approver's limit must be escalated.
BR-75	The payment ledger is append-only: an incoming payment never changes its amount and mistakes are corrected with reversal lines. A payment's amount equals its net allocations plus refunds plus the unallocated remainder.
BR-76	A payment on credit is allocated to the earliest due receivable first unless an order or receivable is named; any remainder is kept as the customer's credit balance.
BR-77	Payment terms and the balance collection policy are copied from the customer when the order is placed and can only be tightened (to before delivery): collection on delivery is allowed only when it is the customer's policy, and balance on credit only when the customer has credit.
BR-78	When the amount due before delivery has not been fully paid, handover to the carrier is blocked. The amount the carrier collects on delivery equals what the order still owes after the deposit, is fixed at handover from the ledger and never collects more than is owed.
BR-79	A deposit order with the balance on credit is refused when the customer is overdue or over the credit limit, showing the remaining limit; when converted from a quote it switches to paying the balance before delivery.

BR-80	A customer's exposure is the value of undelivered credit orders plus undelivered balances on credit plus unpaid customer receivables; the remaining limit is the credit limit minus exposure and is never negative. Amounts collected by a carrier are not counted in exposure or overdue, are not shown to the customer and are not paid by customer transfers.
BR-81	A delivered credit order becomes a receivable due on the delivery day plus the customer's payment term, and a partial payment does not clear an overdue receivable. An order delivered in several shipments opens one receivable per delivery, with prepaid money applied to the earliest deliveries first, so the customer never pays more than the goods received.
BR-82	Reversing an allocation that leaves a confirmed order short of money requires putting the order on hold for payment; the order is never silently returned to unpaid.
BR-83	A full-truckload order is never collected on delivery and must be fully paid before handover.
BR-84	An order's status is unified from the modules that execute it and changes only along the order lifecycle; both the application and the database refuse other transitions.
BR-85	An order cannot be released while its payment condition is not met, the customer has an overdue receivable (credit or balance on credit), the order is On Hold, a new-design line has no approved sample, or there is not enough stock to allocate. An order is released to one warehouse; print lines go to production and stock lines to picking.
BR-86	Cancelling an order after warehouse work has started reverses that work in a controlled way (put-back of stock, voiding of labels, stopping of production orders), so no stock or money is left hanging.
BR-87	Every order change that affects money creates matching lines in Payment; total collected minus total refunded always equals the net amount received for the order.
BR-88	A Completed or Closed order cannot be changed; later complaints are handled with new documents.
BR-89	Partial shipment is off by default. A mixed order is shipped in parts only when Sales allows it for that order, and the decision is locked once the order is Shipped.
BR-90	Returned goods always re-enter stock through a goods receipt before stock is increased or intact goods are refunded. Products printed to a design are exchanged or returned only for production defects, and returns are accepted only within 7 days after delivery.
BR-91	Packing is the cancellation cut-off: the customer cancels directly only while the order is Draft, Pending Payment or Confirmed; after that, until packing is finished, the customer submits a cancellation request; once a package is packed or handed over the order cannot be cancelled, only returned.
BR-92	Each order has at most one pending cancellation request, which Sales or the Order Coordinator approves or rejects; a request that can no longer be decided expires automatically.

BR-93	There is no cancellation fee before printing starts and the order is refunded in full. After printing has started only staff can cancel, and a fee may be kept: a share of the money collected or an agreed amount not above the order total, which requires a reason and details.
BR-94	Every cancellation has a reason code; customers can use only Customer request, Duplicate order or Other, and Other requires a note.
BR-95	The delivery mode (parcel or full truckload) can be changed only until the first package is opened. A full-truckload order cannot be combined with a deposit whose balance is collected on delivery.
BR-96	A manual hold requires a reason code (Other with a note) and releasing it requires a note; the order returns to the status it was held from. Credit and design holds are released only through their own flows, and a payment hold only when the money is sufficient again.
BR-97	Only the undelivered remainder of a partially fulfilled order can be closed; money received above the order's new value is refunded.
BR-98	A delivered order is completed automatically 7 days after delivery when nothing is owed and no return is requested, and closed automatically 30 days later; the job runs every night.
BR-99	Every consequential action taken from a chat (placing an order on the customer's behalf, applying a discount, sending a payment link, raising a refund request) is logged with its thread and its actor; change, cancellation and refund requests go through Order Management, and a chat never changes an order directly.
BR-100	Internal notes in a chat are never visible to the customer, and sensitive data such as card numbers or passwords is never requested in chat and is masked when typed.
BR-101	A standard purchase order contains only purchasable SKUs (Active variants of Approved or Published products) that are listed in the supplier's catalog; this is checked on create, edit, submit and approve, and each failing line is named.
BR-102	A purchase order line is in the SKU's stock unit, at least the supplier's MOQ and a multiple of the supplier's pack size; in the supplier catalog the MOQ is a multiple of the pack size, the price is above zero and the lead time is 0 to 365 days.
BR-103	A purchase order's currency is fixed for the whole order, and supplier catalog prices are in the supplier's currency. An expected delivery date in the past only raises a warning.
BR-104	Each submission of a purchase order records an immutable revision, and a Confirmed standard purchase order cannot be edited. Two purchase orders to the same supplier for the same SKU and delivery date are flagged as possible duplicates.
BR-105	Four-eyes rule: the approver of a purchase order must not be its submitter, and an approver cannot approve a purchase order above their own approval limit. A purchase order within the submitter's limit is approved automatically on submission; one that was ever rejected is always approved by a person.
BR-106	To prevent order splitting, purchase orders approved automatically for the same person to the same supplier in the previous 7 days are added to the new order's total when checking the limit.

BR-107	An approver who rejects a purchase order must give a reason, and the order returns to draft.
BR-108	Editing purchase order approval limits is a separate permission: by default only the System Admin can edit them, while Procurement Staff, the Warehouse Manager and the Accountant can view them.
BR-109	A supplier code never changes, and supplier codes and tax codes are unique. A supplier cannot leave Active while it has an open purchase order, and only Active suppliers receive new purchase orders and new catalog items.
BR-110	Confirming a purchase order means sending it to the supplier, which requires the expected delivery date, the buyer's company details and the receiving address. Changing an agreed delivery date requires a reason. A failed sending is retried automatically up to 5 times and can then be resent only by the Warehouse Manager, with a reason and a confirmation that it was reconciled.
BR-111	The supplier's response is recorded once; a rejection requires a note and is allowed only before any goods are received.
BR-112	Goods are received only against a purchase order that is Confirmed or Partially Received and has not been rejected by the supplier. A purchase order can be cancelled only when nothing has been received and no draft receipt exists, and closing it short requires a reason.
BR-113	A subcontract purchase order is placed only with a supplier flagged as a print subcontractor and is always linked to exactly one child production order. Its quantity changes only through the production split and resend flows, and a supplement after approval waits for four-eyes approval within the limit, one at a time.
BR-114	Warehouse-bound users see and act only on purchase orders of their assigned warehouses, and purchasing reports are refused to them.
BR-115	Each goods receipt belongs to exactly one purchase order. The total received for a purchase order line, over every receipt that is not cancelled, may exceed the ordered quantity by at most the supplier's over-receipt tolerance, rounded down (0% when the supplier has none).
BR-116	A receipt counted beyond the tolerance is not posted: it waits for a Warehouse Manager, who neither counted it nor sent it for approval, to approve the whole count or reject the excess, always with a reason.
BR-117	A goods receipt without a purchase order requires the supplier and a reason, and its stock is counted in only after a Warehouse Manager approves it. A purchase order attached afterwards is a reference only.
BR-118	A lot number and an expiry date are mandatory for SKUs tracked by lot or expiry and not allowed for other SKUs. The expiry date must be after the receiving date and not later than the receiving date plus the SKU's maximum shelf life; an expiry within 30 days of receipt raises a near-expiry warning.
BR-119	Only a Draft receipt can be cancelled, which voids its cartons. A confirmed receipt is never deleted or cancelled; mistakes are corrected with an approved stock adjustment.

BR-120	A receipt line of a SKU that requires QC cannot be put away before QC concludes, and goods are put away only from the QC area, never straight from the receiving area to a bin.
BR-121	QC concludes only lines already moved to the QC area, once per line. Accepted plus quarantined plus rejected must equal the quantity received, rejected goods need a reason, and quarantined and rejected goods go to two different quarantine areas.
BR-122	Every movement in receiving (receiving area to QC area to quarantine area or bin) writes a stock ledger line whose source is the receipt; stock is never edited directly.
BR-123	Cartons are labelled with a unique LPN on Draft receipt lines and become In Stock when the receipt is confirmed. A carton always moves whole through QC, and a line that has cartons cannot be removed; at most 500 carton labels are printed at a time.
BR-124	A warehouse must have a QC area before it can confirm a receipt that has QC lines or a subcontract receipt. A subcontract receipt has no cartons, no warehouse QC and no putaway: it closes on confirmation and its finished goods go to production QC.
BR-125	Only goods that passed QC, or goods of SKUs that do not require QC, get putaway tasks. Goods waiting for QC, Quarantine and Blocked goods are never put into storage.
BR-126	Stock is always at a location, either a bin or a storage area; there is no stock without a location except goods in transit on a transfer order.
BR-127	Goods are put away only into putaway-target bins of the same warehouse whose storage class matches the SKU and whose bin, shelf and warehouse are Active. A bin's capacity units are never exceeded, nor its maximum weight when both the bin limit and the SKU weight are known; a bin may hold several SKUs and lots.
BR-128	Bin suggestions are ranked same lot, then same SKU, then empty bin, then bin with room, and more free room first; at most 10 suggestions are given, each with a reason. FEFO is guaranteed at allocation for picking, not at putaway.
BR-129	A bin other than the suggested one may be chosen without a reason as long as it meets every hard constraint; the constraints are checked again on the scanned bin at confirmation, and putaways into the same bin are processed one at a time.
BR-130	A carton is put away whole by scanning its own LPN. Loose goods may be put away in parts, each not above the quantity left on the task.
BR-131	Every putaway writes a stock ledger line whose source is the task, and the quantity left is computed from the ledger. A Done task is never edited: later moves use an intra-warehouse transfer and discrepancies use a count or an approved adjustment.
BR-132	Available-to-promise increases only when goods are put away into a bin; goods in the receiving, QC and quarantine areas cannot be sold.
BR-133	Shelves, bins, areas and zones are never deleted, only made Inactive. The map is a visual tool and is not the source of truth for stock, which is the stock recorded per location.
BR-134	Shelves, areas and both ends of walls and doors must lie inside the map frame (touching the edge counts as inside), and the frame can shrink only to the extent of the layout.

BR-135	Shelves and areas that are not Inactive must not overlap (touching is allowed). Bins lie inside their shelf and do not overlap other bins on the same level, a shelf cannot shrink below its bins, and walls are not checked for overlap. Rotations are only 0, 90, 180 or 270 degrees.
BR-136	A shelf with pickable bins must have at least one pick face.
BR-137	Location codes are built by the system and unique across the system: a bin is warehouse prefix-shelf-level-bin (for example HN-A01-2-03), an area is prefix-area code and a subcontractor location is prefix-SUB-supplier code. Each part uses only A-Z and 0-9 (lower case is converted), the prefix has at most 10 characters, shelf, bin and area codes at most 20, and the full code at most 64.
BR-138	A warehouse prefix is unique and never changes, and it is used in location codes and in the codes of documents created at that warehouse. The prefix, shelf code, level number, bin code, area code and map unit never change; a location created by mistake is made Inactive and its code is never reused.
BR-139	Stock is kept only in bins, storage areas (receiving, QC, quarantine, production, packing, dispatch, overflow) or subcontractor locations, never in a non-storage area, and a storage area cannot become non-storage.
BR-140	Picking allocation uses only pickable, usable locations, and putaway suggestions return only bins.
BR-141	Bins are generated only on levels that have no bin yet, with at least 1 row and 1 column, at most 200 bins per level and at most 20 levels per run; a shelf has at most 20 levels. Generated bins share the same properties, and pickable bins require a shelf with a pick face.
BR-142	A location's effective status is the narrowest of the warehouse, shelf and location statuses and is computed when read. Location labels are printed at most 2,000 per request, in walking order, skipping Inactive locations.
BR-143	Only released orders have pick tasks, with one task per order, and the stock held for an order must be in the order's warehouse. Payment is checked before release and at handover, not during picking.
BR-144	Pickers take only the stock allocated to the line (a specific bin and lot), never Blocked, Quarantine or another order's stock. Only the assigned picker works on a task, and it can be reassigned only while Pending.
BR-145	Allocation and reallocation use only pickable, usable bins of the same warehouse and never expired stock: FEFO or FIFO comes first, then within the same expiry or receipt date whole cartons (largest first), opened cartons, loose stock and finally opening a new carton.
BR-146	The scanned quantity must equal the line's allocation: a shortfall is reported as a short pick, and picking more requires a reallocation. A carton line is confirmed only by scanning that carton's LPN; another LPN, the SKU or a barcode is refused.
BR-147	A short pick always puts the order on hold and opens a blind count of that location. The free stock there is blocked from sale until the variance is decided, and reallocation is done by the Order Coordinator or Warehouse Manager, never automatically.

BR-148	Picked goods of a cancelled order are put back in a controlled way into a bin of the same warehouse. Print lines have no pick lines: blanks are issued to production by scanning, and QC-passed finished goods go straight to the packing area.
BR-149	Packing starts only when the task's picking is Picked. Packing can be completed with goods missing only when the order allows partial shipment, with a note.
BR-150	For printed products the design files must match the checksum of the snapshot confirmed by the customer. A mismatch puts the packing session and the order on hold (design mismatch), only the Warehouse Manager can re-verify and release it, and no shipment is created while it remains.
BR-151	A package must have a weight and dimensions above zero when it is sealed, and belongs to exactly one order. Volumetric weight is length × width × height in cm divided by 6,000 (configurable).
BR-152	Only the packer who started a packing session works on it; the Warehouse Manager may act in their place. A package can be reopened only while it is not in a shipment or load plan, and voided only before delivery.
BR-153	A kit shipped as several packages is complete only when all its packages are sealed, voiding one package voids the whole kit, and packing a kit item as an ordinary package requires a reason.
BR-154	Labels are printed once a package is sealed. An outdated label (the package changed shipment or the amount to collect changed) must be reprinted before loading or handover.
BR-155	When an order is cancelled before handover, its shipment is cancelled and its packages and labels are voided. Packing does not deduct printed goods again, because they leave stock when their production order is completed.
BR-156	Only sealed packages with weight and dimensions, of shipments Ready to Dispatch and with current labels, are handed over or put into a load plan.
BR-157	The system does not track parcels after handover: Delivered and Delivery Failed are recorded only by the Order Coordinator from the carrier's report, Returned by the Order Coordinator or Warehouse Manager, each with who and when. A failed delivery needs a reason (absent, refused, wrong address or lost).
BR-158	An order becomes Shipped only when all its packages are handed over; otherwise it is Partially Fulfilled, which happens only when Sales allowed partial shipment.
BR-159	The amount a carrier collects is fixed per shipment at handover and must match the package label. Shipping fees charged to the customer and freight paid to the carrier are separate amounts reconciled separately.
BR-160	Load planning applies only to full-truckload deliveries (customer orders assigned full truckload and truck or container transfers). Only whole shipments with dimensions and weights enter a plan, a package belongs to at most one plan that is not cancelled, and a plan holds either customer orders or one transfer order.
BR-161	Loading rules are hard constraints: each package base must be at least 98% supported with no overhang, the total weight on a vehicle must not exceed its payload, a package may be rotated in its 6 orientations only when rotation is allowed, the reinforced ceiling corners stay free and packages are placed parallel to the walls.

BR-162	Manual changes on the 3D model must still satisfy the loading rules, a violating position is refused, and the changed vehicle is recalculated. Vehicles of a type never exceed the available count, and Inactive vehicle types are not proposed.
BR-163	A load plan is Confirmed only when every package is placed or explicitly left out and no vehicle awaits recalculation; parts of an order that does not allow partial shipment, or of a transfer order, cannot be left out. The parts of a kit always travel in one vehicle.
BR-164	After confirmation a load plan's packages and layout are locked. It can be reopened with a reason only before any vehicle starts loading, which cancels its trips; once loading has started it can no longer be reopened or cancelled.
BR-165	Each vehicle of a plan is one handover trip with its own manifest, and the plan is Dispatched only when every vehicle has been signed for and has left. A trip carries one Active carrier and one warehouse, and a shipment's carrier can be set only while it is on no trip.
BR-166	Handover is all or nothing: every package of the trip must be scanned, the manifest signed with the carrier's representative and plate number, and every shipment allowed to leave (nothing due before delivery, not on hold, not missing parts when partial shipment is not allowed).
BR-167	A parcel trip can be cancelled only by the Warehouse Manager, with a reason, before it leaves; trips of a load plan change only through the plan. A load estimate is for reference only and neither confirms a plan nor blocks packing.
BR-168	A transfer order's source and destination warehouses must differ and both be active; moves inside one warehouse use an intra-warehouse transfer. Only Available stock that is not held or allocated can be transferred, and each line must have enough stock at the source when saved.
BR-169	A transfer order whose total requested quantity is at most the approval threshold (100 base units by default) is approved automatically on submission; above it, approval follows the four-eyes rule and the approver must not be the submitter.
BR-170	In-transit stock belongs to the transfer order and cannot be sold in any warehouse until it is received and put away at the destination, which always goes through a receipt rather than adding the dispatched quantity directly.
BR-171	A transfer order can be cancelled, with a reason, only before goods leave the source; after it is In Transit it can only be closed with the real goods handled. Discrepancies are recorded and adjusted with a reason through an approved count or adjustment, and overdue in-transit stock raises a warning.
BR-172	A full-truckload transfer is dispatched only through its load plan, which carries that single transfer order and no customer orders, travels whole, and lets the goods leave the source when the last vehicle is signed; a parcel transfer is dispatched by manual confirmation.
BR-173	Transfer picking can be cancelled only while no goods have left and no vehicle has started loading, which releases the holds and voids the packages, labels and plans. Packing less than requested requires a note, and only the packed part is dispatched.

BR-174	An internal move has different source and destination locations in the same warehouse, and the destination must be usable. The warehouse total does not change, and status, lot, expiry and receipt date are kept.
BR-175	Only free loose stock is moved; stock that is held or allocated, and stock in LPN cartons, is not moved this way, and nothing moves into or out of a location held by a cycle count for that SKU and lot.
BR-176	Every internal move writes a stock ledger line (source, destination, SKU, lot, quantity, person, time), resubmitting the same request does not move stock twice, and users move stock only in their assigned warehouses.
BR-177	Each custom order line or sample request has exactly one production order, and recreating it does not duplicate it (except child orders split for subcontracting). A production order is always tied to a customer-confirmed, checksummed design snapshot, which together with its print file cannot be edited.
BR-178	Nothing is printed before prepress passes and while the file checksum does not match the locked design. A production order can be assigned only to a warehouse that has a production area.
BR-179	Blanks are issued only against a production order and never above its remaining need, and blanks held for an order still awaiting payment cannot be issued. An in-house order starts Printing only when all its blanks are issued, and each print run cannot exceed the issued blanks not yet printed.
BR-180	Goods awaiting QC equal printed minus scrap. Scrap always has a reason (Other with a note), leaves the hold at once and is written off through a four-eyes approved adjustment; the production order does not wait for that approval.
BR-181	A production order is Completed only after QC: an order run is checked against the customer-approved sample, a sample run against the print file, and each QC round checks everything waiting (passed plus failed equals waiting). A reprint issues blanks only for the shortfall, using issued but unprinted blanks first.
BR-182	An order leaves In Production only when all its production orders, parent and child, are Completed or their lines cancelled. Finished goods of an order run leave stock when the production order completes.
BR-183	The design cannot change once a production order is Printing (a child order: Subcontracted); before that, changing the design means cancelling the line and creating it again.
BR-184	An order line with a new design cannot be released unless an Approved sample round exists for that design and blank, or the design was already produced for another order. A deposit order is released to production only after the deposit arrives.
BR-185	A production order can be cancelled only before printing (Pending Prepress, Ready or Material Issued); while Printing or in QC it can only be stopped, and its printed goods become scrap. Only the Warehouse Manager cancels or stops, with a reason, and the system never cancels a printed order automatically.
BR-186	An order cancelled after printing is charged a production fee, and its printed goods that cannot be delivered are written off as scrap through an approved adjustment, so stock is deducted only once.

BR-187	A production order is put on hold by Production Staff or the Warehouse Manager with a reason (Other with a note); only the Warehouse Manager resumes it, and it returns to the state it was held from.
BR-188	Blank holds for production never expire, and submitting a sample request requires enough available blanks at that moment. A sample request can be submitted only for an Active customer with a confirmed design snapshot.
BR-189	A sample round answered with Changes Requested is final: a new round is a new request numbered one higher and linked to the previous one, each round has at most one next round, and customers answer only their own sample rounds. An Approved sample locks the design and blank pair and becomes the QC reference.
BR-190	Only order production orders can be subcontracted; sample orders are always printed in-house because the sample is the QC reference for later orders.
BR-191	Only the not-yet-issued part of an in-house parent order in Pending Prepress, Ready or On Hold can be split off, with a split quantity above 0 and below the unissued quantity; the whole order is moved only when nothing has been issued or printed.
BR-192	The subcontractor must be an Active supplier flagged as a print subcontractor, and a child order is sent only after its subcontract purchase order is Approved. The file sent is the prepress-checked print-ready file with a matching checksum, after which the design is locked as when printing.
BR-193	Blanks sent to a subcontractor remain company stock at the subcontractor location, never count as available, and leave only through a receipt, a reconciliation or an approved subcontract-loss write-off.
BR-194	A child order is Completed only after its receipt is confirmed, the blank reconciliation is done and QC passes the full quantity. Loss above the allowed tolerance needs Warehouse Manager approval with a deduction per blank, QC-failed units are not paid, and a subcontract cancellation fee above zero requires a reason.
BR-195	Available-to-promise (ATP) is the Available, unexpired stock that is not held (no allocation, production hold or transfer hold) minus the active SKU reservations; it can be viewed system-wide or per warehouse. Inbound, Quarantine, Blocked, Expired, In Production and In-Transit stock and stock at subcontractor locations are not part of ATP.
BR-196	For every stock record the quantity on hand is at least the total held on it, and stock is never negative. Stock can be reserved only when ATP is at least the requested quantity.
BR-197	Outside the business flows (receiving, putaway, picking, transfer, production) stock quantities change only through an approved adjustment, and every stock change has a source document.
BR-198	An adjustment reason is Damaged, Lost, Found, Expired, Data correction or Other (with a mandatory note) for increases or decreases; Scrap, Sample and Subcontract loss only decrease stock; Count variance is created only by cycle counts and cannot be chosen by users.

BR-199	Four-eyes rule: an adjustment is approved or rejected only by someone other than the requester, and approval posts the stock immediately. Only the requester can withdraw an adjustment, and only before a decision; a count variance cannot be withdrawn.
BR-200	In a blind count the counter does not see the book quantity or the count notes; only users who can approve counts see them, and counts opened from a short pick are always blind.
BR-201	Submitting a cycle count never posts stock directly: it creates count-variance adjustments pending four-eyes approval. The variance is computed from the book quantity read at counting time without freezing the warehouse, and a submission is refused when a shortage exceeds the current stock, so the line must be recounted.
BR-202	A count line (location, SKU and lot, or a carton) is busy while it is on a Planned or In Progress count, or on a Completed count whose variance is still pending. A busy line cannot be added to a new count, added by hand or recounted, and manual moves into or out of it are refused.
BR-203	The free stock of a short-picked line is excluded from ATP and allocation until every adjustment of that line is decided, and any pending decrease adjustment blocks that free stock as well.
BR-204	A count has at most 500 lines, 50 zones and 200 SKUs. The counter must be an active warehouse staff member or manager of that warehouse and can be changed only before counting starts; a rejected line can be recounted once.
BR-205	Stock alerts compare usable stock (Available, unexpired, not reduced by holds): Reorder at or below the reorder point and Safety below the safety stock; an empty threshold is not tracked. The safety stock must not exceed the reorder point, the maximum quantity must be at least both, and safety stock is never deducted from ATP.
BR-206	Reorder and Safety alerts exist only for SKUs that can still be purchased; when a SKU can no longer be purchased its open alerts are closed. A condition has at most one open alert, and a condition that returns within 6 hours reopens its alert without notifying again.
BR-207	Stock past its expiry date is moved to Expired every night and drops out of ATP, but stock held for an order is never silently dropped: the alert of the lot names the affected order goods for someone to handle.
BR-208	Stock ledger lines (receipt, move, adjustment, transfer out, status change, pick, put-back, consumption) are never edited or deleted.
BR-209	An LPN carton carries no quantity of its own: its goods are stock at the carton's location, a carton is at one location only, a carton variance is one adjustment for the whole carton, and an increase refills the same carton without creating a new lot.
BR-210	Warehouse-bound users see and act only on the stock, counts, adjustments, alerts and traces of their assigned warehouses.
BR-211	When a customer accepts a quote, asks for changes or asks to reorder, the Sales user who sent the quote is notified in-app and by email; if that account is no longer active, every Sales user is notified for one to take over.

BR-212	When printing has started and a payment fails or does not arrive, every Sales user and Order Coordinator is warned to decide on the cancellation. A production hold notifies Sales (file rejected or waiting for the customer), Procurement (blanks short) or the Warehouse Manager of that warehouse (other reasons).
BR-213	Stock alerts notify the Inventory Planner and System Admin (reorder and safety) and the Warehouse Manager, Inventory Planner and System Admin (near expiry and expired), within their warehouses, at most once per type, per warehouse, per day.

Table 19. Business Rules

5.2 Common Requirements

- CR-01: Every user must sign in before using the system. Only the storefront home page, the sign-in pages and the account activation page are available without signing in.
- CR-02: B2B customers use the customer portal (storefront) and staff use the back-office. Each portal accepts only its own kind of account: a customer account cannot sign in to the back-office and a staff account cannot sign in to the storefront.
- CR-03: Every function that reads or changes business data requires a permission (a resource and an action). Permissions are granted to roles and are checked by the server on every request, so a change of role or permission takes effect from the user's next action.
- CR-04: A data scope limits what each user can see and act on: warehouse roles work only in the warehouses assigned to them, other staff roles see the whole system, and a customer sees only the data of their own company.
- CR-05: The system serves one legal entity that owns several physical warehouses. Moving goods between warehouses is an internal stock movement and creates no internal invoice or receivable.
- CR-06: Every stock quantity is recorded in the SKU's base unit and always belongs to a location (a bin or a storage area) or to a transfer in transit. Every stock change carries a source document such as a receipt, putaway, pick, adjustment, transfer or production order.
- CR-07: Document statuses move only along their lifecycle. Cancellations and reversals are controlled actions that are logged, and approval steps follow the four-eyes rule: the person who submits or requests cannot approve.
- CR-08: When two users change the same record at the same time, the later save is refused and the user is asked to reload the record, so that no change is lost silently. A data-changing request can be resubmitted (double click, network retry) without being processed twice.
- CR-09: Every error is returned in one standard format: a message for the user, field errors that the form shows under the matching input, and a correlation id that support staff use to trace the request.
- CR-10: The system is presented in Vietnamese by default and is also available in English; every system message exists in both languages.
- CR-11: All list screens support paging, sorting and filtering.
- CR-12: Each account has an in-app notification inbox with a severity (info, warning or critical), a title, content, a link to the related screen and a read flag. Recipients are chosen by role and warehouse when the notification is sent, each person receives one copy, and the screen refreshes the unread count periodically.
- CR-13: Key business actions (approvals, rejections, cancellations, stock adjustments, account and permission changes) are recorded in an audit log stating who did what and when.

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In red, under the text box	A required field is left empty.	{field} is required.
2	MSG02	In red, under the text box	A text field is shorter or longer than allowed.	{field} must be between {min} and {max} characters.
3	MSG03	In red, under the text box	An email field does not hold a valid email address.	{field} must be a valid email address.
4	MSG04	In red, under the text box	A phone field does not hold a valid Vietnamese phone number.	{field} must be a valid Vietnamese phone number.
5	MSG05	Toast message	A form is submitted while some fields are invalid.	Please check the form again.
6	MSG06	In line	A search or filter returns no record.	No results found.
7	MSG07	In line	The screen cannot reach the server.	Cannot connect to the server. Check your network connection and try again.
8	MSG08	In line	The user opens a screen whose data they are not allowed to read.	You do not have permission to view this data.
9	MSG09	Toast message	The user calls a function their roles do not grant.	You do not have permission to do this.
10	MSG10	Toast message	The user opens a record outside their data scope, for example another warehouse.	You do not have access to this data.
11	MSG11	Toast message	The record was saved by another user after it was opened.	Someone else changed this record. Please reload and try again.
12	MSG12	Toast message	The same request is submitted again while the first one is still running.	An identical request is still being processed.
13	MSG13	Toast message	The user uploads a file above the size limit.	The file is too large.
14	MSG14	Toast message	The user uploads a file whose type is not accepted.	This file type is not accepted.
15	MSG15	Toast message	The user sends too many requests in a short time.	Too many requests. Please slow down.

16	MSG16	Toast message	An unexpected server error occurs.	Something went wrong on our side.
17	MSG17	Toast message	The user signs in with a wrong username or password, or while the account is temporarily locked.	Invalid username or password.
18	MSG18	Toast message	A locked, disabled or not yet activated account tries to sign in.	This account cannot sign in right now.
19	MSG19	Toast message	The session has expired or was ended.	You are not signed in.
20	MSG20	In red, under the text box	A new password does not meet the password policy.	Password must be 10-72 characters and contain upper-case, lower-case and digit characters.
21	MSG21	In red, under the text box	The user enters a wrong current password when changing the password.	The current password is incorrect.
22	MSG22	In red, under the text box	The new password is the same as the current one.	The new password must differ from the current one.
23	MSG23	Toast message	An administrator creates a staff account.	Account created.
24	MSG24	Toast message	An administrator saves a staff profile.	Profile saved.
25	MSG25	Toast message	An administrator changes a user's working warehouses.	Working warehouses updated.
26	MSG26	Toast message	An administrator locks an account.	Account locked and all its sessions signed out.
27	MSG27	Toast message	An administrator disables an account.	Account disabled and all its sessions signed out.
28	MSG28	Toast message	An administrator resets a user's password.	Password reset.
29	MSG29	Toast message	An administrator saves a role and its permissions.	Role saved.
30	MSG30	Toast message	An administrator grants a permission or role they do not hold.	You cannot grant or manage permissions you do not hold yourself.
31	MSG31	Toast message	An administrator tries to change their own account from user management.	You cannot change your own account from user management.

32	MSG32	Toast message	The change would leave no active System Admin.	At least one active System Admin must remain.
33	MSG33	Toast message	A role that is still assigned is deleted.	This role is still assigned to users.
34	MSG34	Email	Sales creates a B2B customer profile or resends the invitation.	Activate the customer portal account of {company}
35	MSG35	Toast message	The customer opens an expired, used or replaced invitation link.	This portal invitation is invalid or no longer available. Contact Sales for a new invitation.
36	MSG36	Toast message	Someone tries to register on the portal.	Public registration is unavailable. Contact Sales to activate your business account.
37	MSG37	Toast message	Sales creates a customer whose tax code is already registered.	A customer with this tax code already exists.
38	MSG38	Toast message	The customer orders less than the minimum quantity of their price list.	The quantity is below your price list minimum for this product; ask Sales for a quote.
39	MSG39	Toast message	An administrator creates a category.	Category created.
40	MSG40	Toast message	The submitter tries to approve their own product.	A product cannot be approved by the person who submitted it.
41	MSG41	Toast message	A product is submitted for approval without an English name or a tax class.	An English name and a tax class are required before submitting for approval.
42	MSG42	Toast message	A product is published without an approved image gallery.	An approved non-empty product gallery is required.
43	MSG43	Toast message	The user tries to delete a product that is no longer a draft.	Only a draft product can be deleted.
44	MSG44	Toast message	A quote or order line uses a SKU that is no longer active.	The SKU is no longer active; change the line to an active SKU.
45	MSG45	Toast message	The customer accepts a quote after its validity date.	This quote is no longer valid.
46	MSG46	Toast message	An order line with a new design is released before its sample is approved.	A design on this order has no approved sample yet.

47	MSG47	Toast message	The customer places an order without a shipping address.	Add a shipping address before placing an order.
48	MSG48	Toast message	The requested quantity is more than the available stock.	Not enough stock available.
49	MSG49	Toast message	The selling price changed between viewing the cart and placing the order.	The selling price changed. Refresh the basket and confirm the new price.
50	MSG50	Confirmation dialog	The customer places an order with the same items as an order placed a moment ago.	An order with the same items was placed recently; confirm to place another.
51	MSG51	In line	A placed order waits for the customer's bank transfer.	Transfer this amount to confirm your order; an unpaid order is cancelled when its stock hold runs out.
52	MSG52	Toast message	The Order Coordinator releases an order.	Order released to the warehouse.
53	MSG53	Toast message	Staff cancel an order.	Order cancelled.
54	MSG54	Toast message	The Accountant approves an order held over the credit limit.	Over-limit credit approved.
55	MSG55	Toast message	The customer cancels an order that is already packed.	This order can no longer be cancelled; raise a return instead.
56	MSG56	Toast message	The customer cancels an order whose printing has started.	Printing has started: staff decide this order's cancellation.
57	MSG57	Toast message	A credit order is placed or released for a customer with overdue receivables.	The customer has overdue receivables.
58	MSG58	Toast message	A deposit order is released to production before the deposit arrives.	The deposit for this order has not arrived yet.
59	MSG59	Toast message	Part of an order is packed or handed over while partial shipment is not allowed.	This order does not allow partial shipments.
60	MSG60	In-app notification	The customer accepts a quote.	Quote {quote} was accepted by the customer: convert it into an order

61	MSG61	In-app notification	Printing has started and the payment failed or did not arrive.	Order {order} needs a decision: printing started, the system did not cancel it
62	MSG62	Toast message	Procurement Staff create a supplier.	Supplier created.
63	MSG63	Toast message	Procurement Staff add a supplier to the blacklist.	Supplier added to the blacklist.
64	MSG64	Toast message	A supplier with open purchase orders is deactivated.	Close or cancel all open purchase orders before deactivating this supplier.
65	MSG65	Toast message	Procurement Staff add an item to a supplier catalog.	Item added to the catalog.
66	MSG66	Toast message	Procurement Staff create a purchase order.	Purchase order created.
67	MSG67	Toast message	A purchase order looks like another one to the same supplier.	Possible duplicate purchase order.
68	MSG68	Toast message	Procurement Staff submit a purchase order.	Purchase order submitted for approval.
69	MSG69	Toast message	An approver approves a purchase order.	Purchase order approved.
70	MSG70	Toast message	Procurement Staff confirm a purchase order.	Purchase order confirmed and queued for sending to the supplier.
71	MSG71	Toast message	Sending a purchase order to the supplier fails.	Sending to the supplier failed; check the sending history.
72	MSG72	Toast message	Procurement Staff record the supplier's response.	Supplier response recorded.
73	MSG73	Toast message	A purchase order line quantity is below the supplier's MOQ.	The quantity is below the supplier's minimum order quantity.
74	MSG74	Toast message	A purchase order line quantity is not a multiple of the pack size.	The quantity must be a multiple of the supplier's pack size.
75	MSG75	Toast message	The approver's limit is lower than the purchase order total.	The purchase order total is above your approval limit.
76	MSG76	Toast message	Warehouse Staff create a goods receipt.	Goods receipt created.
77	MSG77	Toast message	Warehouse Staff move receipt lines to the QC area.	Goods moved to the QC area.

78	MSG78	Toast message	QC Staff record an inspection result.	QC result recorded.
79	MSG79	Toast message	Warehouse Staff confirm a goods receipt.	Goods receipt confirmed into stock.
80	MSG80	Toast message	A receipt is confirmed with a count beyond the tolerance.	The receipt exceeds the tolerance and was sent to the Warehouse Manager for approval; no stock has been recorded yet.
81	MSG81	Toast message	The Warehouse Manager rejects the excess of a receipt.	Excess rejected; only the quantity within tolerance was recorded.
82	MSG82	Toast message	A goods receipt without a purchase order is created without a reason.	A receipt without a purchase order needs a reason.
83	MSG83	In red, under the text box	A lot-tracked item is counted without a lot number.	This item is tracked by lot: enter the lot number.
84	MSG84	In red, under the text box	An expiry date is later than the receiving date plus the maximum shelf life.	The expiry date is later than the receiving date plus the maximum shelf life of the item.
85	MSG85	Toast message	The scanned carton is not the carton of the putaway task.	The scanned carton is not the one this putaway task is for.
86	MSG86	Toast message	The putaway bin cannot hold the quantity or weight.	This bin cannot take that many units or that much weight.
87	MSG87	Toast message	A shelf or area is placed over another object on the warehouse map.	This would overlap something already on the map.
88	MSG88	Toast message	Stock is moved to another warehouse without a transfer order.	Stock moves to another warehouse only through a transfer order.
89	MSG89	Toast message	The requester tries to approve their own stock adjustment.	A stock adjustment cannot be decided by the person who requested it.
90	MSG90	Toast message	Stock is moved in or out of a slot that is being counted.	Stock cannot be moved out of or into a slot that is under a cycle count until the count is done.
91	MSG91	Toast message	Stock changed after a count line was counted.	Stock changed since it was counted - recount the line.
92	MSG92	Toast message	An Inventory Planner saves the stock policy of an item.	Stock policy saved.

93	MSG93	In-app notification	One or more SKUs fall to or below their reorder point.	{count} SKU(s) at or below the reorder point
94	MSG94	In-app notification	Lots in a warehouse are close to their expiry date.	{count} lot(s) near expiry at warehouse {warehouse}
95	MSG95	Toast message	An Inventory Planner submits a transfer order.	Transfer order submitted for approval.
96	MSG96	Toast message	A submitted transfer order is within the approval threshold.	The transfer is within the approval threshold and was approved immediately.
97	MSG97	Toast message	The submitter tries to approve their own transfer order.	A transfer order cannot be decided by the person who submitted it.
98	MSG98	Toast message	Warehouse Staff dispatch a transfer order.	Goods dispatched; the transfer is in transit.
99	MSG99	Toast message	A pick task is created for an order.	Pick task created.
100	MSG100	Toast message	The picker scans a location that is not the pick line's location.	The scanned location is not this line's.
101	MSG101	Toast message	The picker confirms fewer units than allocated.	Fewer than allocated - report a short pick.
102	MSG102	Toast message	The picker reports a short pick.	Short pick reported: {quantity} × {sku}.
103	MSG103	Toast message	The Order Coordinator reallocates a short line.	Short line reallocated.
104	MSG104	Toast message	Warehouse Staff put goods of a cancelled order back on a shelf.	Returned {quantity} × {sku} to {bin}.
105	MSG105	Toast message	A second packer opens an order that is already being packed.	Another packer is packing this order.
106	MSG106	Toast message	The packer seals a package.	Package sealed.
107	MSG107	Toast message	A printed design does not match the confirmed snapshot at packing.	Design checksum mismatch: packing is on hold until the design is re-verified.
108	MSG108	Toast message	The Warehouse Manager re-verifies a held design and it matches.	The design matches: the hold is released and the packer can complete again.
109	MSG109	Toast message	The packer completes packing of the whole order.	The whole order is packed.

110	MSG110	Toast message	A package label shows an amount to collect that is no longer correct.	The package label shows another COD amount than the order now fixes; reprint it.
111	MSG111	Toast message	A manual placement breaks a loading rule in the 3D load plan.	A placement breaks a loading rule.
112	MSG112	Toast message	Warehouse Staff confirm a load plan.	Load plan confirmed.
113	MSG113	Toast message	The load is handed over before both sides sign the manifest.	Both sides must sign the manifest.
114	MSG114	Toast message	The load is handed over before every package on its manifest is scanned onto the vehicle.	Some packages on the manifest are not scanned yet.
115	MSG115	Toast message	Warehouse Staff hand a load over to the carrier.	Load handed over to the carrier.
116	MSG116	Toast message	The Order Coordinator records a delivery result.	Delivery result recorded.
117	MSG117	Toast message	At prepress, the print file does not match the design snapshot locked for the order.	The print file does not match the locked design.
118	MSG118	Toast message	Warehouse Staff issue more blanks than the production order still needs.	More blanks than the production order still needs.
119	MSG119	In-app notification	A production order is put on hold.	Production order {order} on hold: {reason}

Table 20. Application Messages List

5.4 Other Requirements

Additional requirements include:

- Audit and data retention: Routine audit records are kept for 180 days and security records (sign-ins, permission grants, exports and deletions) for 730 days. Read in-app notifications are deleted after 90 days; unread ones are kept.
- File upload safety: Every uploaded file is checked against an allow-list of file types by its content and scanned for viruses, and an upload is refused while the scanner is unavailable. Private file download links expire after 10 minutes.
- Rate limiting: Sensitive operations are rate limited: sign-in to 5 attempts per minute per IP address, password change to 5 attempts per 5 minutes per user, portal account activation to 10 requests per minute per IP address and customer creation to 30 requests per minute per user.

- Label printing: Location labels carry a Code 128 barcode and a QR code of the location code and are printed on A4 sheets of 24 labels (70 × 37 mm) or on 60 × 40 mm thermal labels. Carton (LPN) labels and package labels are printed from the receipt and packing screens.
- Background jobs: Scheduled jobs, such as releasing expired stock reservations every minute, the nightly stock alert reconciliation at 01:30, the expiry sweep at 02:00 and the nightly order completion, run on only one server instance at a time.
- Performance: One 3D load plan calculation is limited to 5 seconds and returns the best plan found within that time.
- Observability: Every request carries a correlation id that appears in the logs and in error responses. Logs are written as structured JSON outside development, and requests are traced end to end.
- Scalability and maintainability: The backend is a modular monolith of fourteen business modules, each owning its own database schema. Module boundaries are verified by automated tests on every pull request, so a module can later be moved out into a separate service.
- Deployment: The system runs in Docker containers (application, PostgreSQL 16, Redis, MinIO object storage and a mail server), with configuration supplied through environment variables and no credentials stored in the source code.
- Later phases: Self-service password reset by email, two-factor authentication, session refresh, notifications to customers (order placed, payment failed, shipped or delivered) and supplier invoice matching are planned for later phases; until then a user who forgets a password asks an administrator to reset it.

IV. Software Design Description

1. System Design

1.1 System Architecture

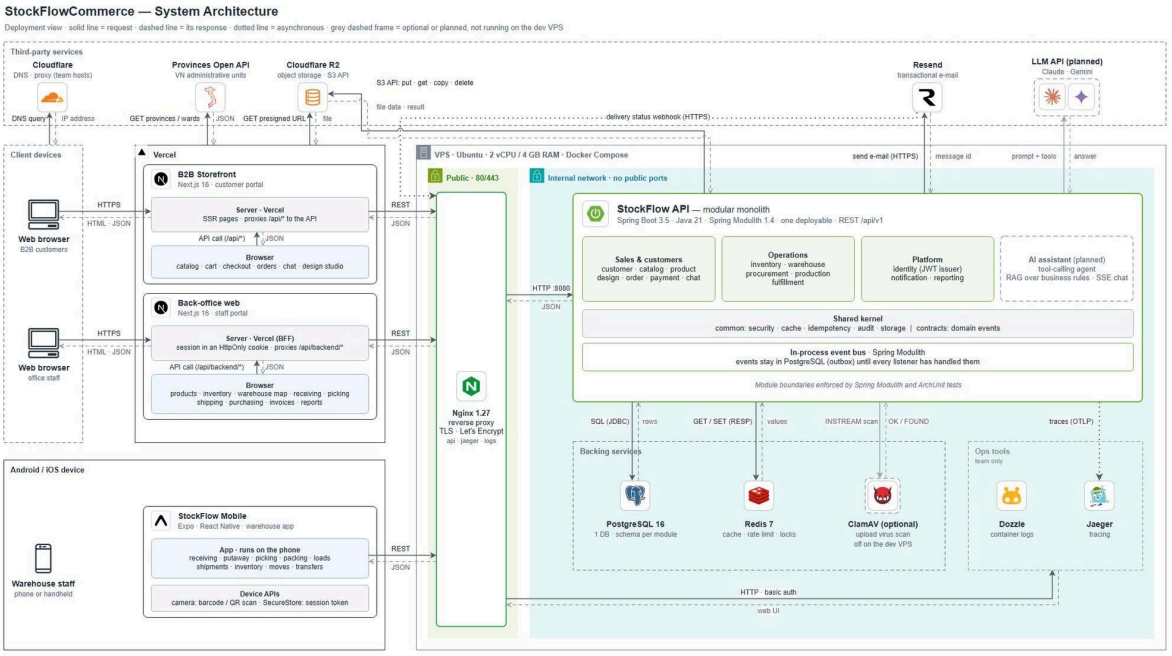


Figure 22. Backend Package Diagram

1.2 Package Diagram

1.2.1 Backend Package Diagram

Figure 23. Backend Package Diagram

Package Descriptions

No	Package	Description
01	Web Layer	Entry point for client requests: Spring MVC DispatcherServlet, Security Filter Chain (JWT), GlobalExceptionHandler and correlation-id / request-logging filters.
02	Business Modules	Container of the 15 application modules declared with @ApplicationModule; each module owns one database schema and exposes only its api package.

No	Package	Description
03	catalog	Storefront listings, product detail pages, search and SEO.
04	customer	B2B customer profiles, addresses, segments and data-privacy requests.
05	order	Cart, checkout, order lifecycle, amendments and returns (RMA).
06	payment	Cash on delivery, deposits, refunds, receivables and reconciliation.
07	chat	Sales chat, conversation routing and assisted ordering.
08	warehouse	Warehouse map, storage locations, slotting and putaway.
09	inventory	Location-level stock, available-to-promise, reservations, allocations, counts and transfers.
10	procurement	Suppliers, purchase orders, goods receipt, quality control and invoice matching.
11	fulfillment	Picking, packing, shipping and proof of delivery.
12	reporting	Read models for warehouse KPIs, inventory and finance reports.
13	product	Product master data, variants, SKUs and print configuration.
14	design	2D design canvas, 3D preview, preflight checks and design snapshots.
15	production	Print production: sample requests and production orders.
16	identity	Authentication, users, roles and the permission matrix.
17	notification	Templated email and in-app notifications.
18	api	Public interface of a module (@NamedInterface): service interfaces, commands, results and read views; the only package other modules may import.
19	controller	Presentation layer of a module: REST controllers and request/response DTOs.
20	service	Application layer of a module: service implementations, event listeners and scheduled jobs; the transaction boundary.

No	Package	Description
21	domain	Domain layer of a module: aggregates, value objects and repository ports, free of Spring and JPA.
22	repository	Infrastructure layer of a module: Spring Data repositories, port adapters and entity-to-domain mappers.
23	entity	Persistence model of a module: JPA entities that map database tables only.
24	Shared Kernel Layer	Open modules that every business module may use: common and contracts.
25	Cross-Cutting (common)	Security, auditing, logging, i18n, validation, error model, API envelope, web filters and shared domain types.
26	Events (contracts)	Domain-event records (e.g. OrderPlaced, PaymentFailed, StockReserved) published after commit for asynchronous communication between modules.
27	Infrastructures (common)	Persistence base classes, caching, distributed locks, idempotency, rate limiting, file storage, HTTP clients and configuration.
28	Externals	External systems the backend depends on: PostgreSQL, Redis, MinIO (S3), SMTP / Resend, OpenTelemetry with Jaeger, and Prometheus.

Table 21. Backend Package Description

1.2.2 Frontend Back-office Package Diagram

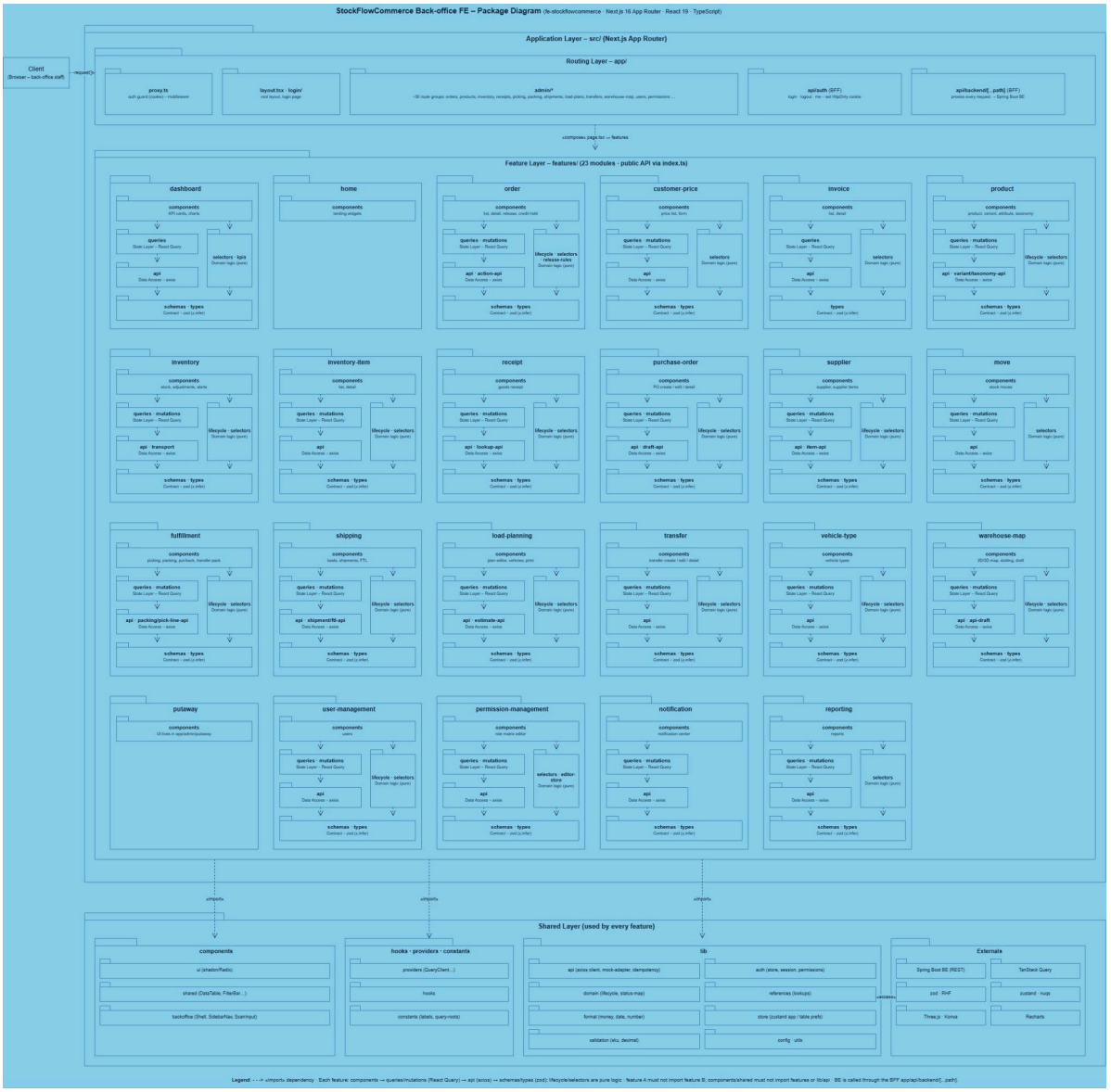


Figure 24. Frontend Back-office Package Diagram

Package Descriptions

No	Package	Description
01	Routing Layer (app/)	Next.js App Router layer that maps URLs to pages; pages only compose features and components.
02	proxy.ts	Route guard that redirects unauthenticated users to the login page based on the session cookie.
03	layout.tsx · login/	Root layout of the application and the login page.

No	Package	Description
04	admin/*	About 30 route groups for back-office screens: orders, products, inventory, receipts, picking, packing, shipments, load plans, transfers, warehouse map, users and permissions.
05	api/auth (BFF)	Backend-for-frontend route handlers for login, logout and current user; they set the HttpOnly session cookie.
06	api/backend/[...path] (BFF)	Backend-for-frontend proxy that forwards every API request to the Spring Boot backend.
07	Feature Layer (features/)	Business features of the back-office; each feature exposes its public API through index.ts and must not import another feature.
08	dashboard	KPI cards and charts on the home dashboard.
09	home	Landing widgets of the back-office.
10	order	Order list and detail, release to warehouse, credit holds and partial shipments.
11	customer-price	Customer-specific price lists.
12	invoice	Invoice list and detail.
13	product	Products, variants, attributes, taxonomy and purchasing data.
14	inventory	Stock levels, stock adjustments and stock alerts.
15	inventory-item	Inventory item list and detail.
16	receipt	Goods receipt against purchase orders.
17	purchase-order	Purchase order creation, editing, approval and tracking.
18	supplier	Suppliers and the items each supplier provides.
19	move	Stock moves between locations.
20	fulfillment	Picking, packing, put-back and transfer packing.
21	shipping	Loads, shipments and full-truckload shipping.

No	Package	Description
22	load-planning	Load plan editor, vehicle assignment, load estimates and printed load sheets.
23	transfer	Warehouse-to-warehouse transfer orders.
24	vehicle-type	Vehicle types used for load planning.
25	warehouse-map	2D/3D warehouse map, racks, slotting and draft layout editing.
26	putaway	Placeholder feature; the putaway screen lives in app/admin/putaway.
27	user-management	Back-office users and their warehouse assignments.
28	permission-management	Roles and the permission matrix editor.
29	notification	Notification center for back-office users.
30	reporting	Operational and financial reports.
31	components	Presentation layer of a feature: page-level React components.
32	queries · mutations	State layer of a feature: TanStack Query hooks for reading and changing server data, with query keys and cache invalidation.
33	api	Data access layer of a feature: axios calls to the backend.
34	schemas · types	Contract layer of a feature: zod schemas that validate API data and form input, with types inferred from them.
35	lifecycle · selectors	Pure domain logic of a feature: status transition tables, allowed actions per role and derived values.
36	Shared Layer	Code shared by every feature.
37	components	Shared UI: shadcn/Radix primitives, generic components (DataTable, FilterBar) and the back-office shell (sidebar, scan input).
38	hooks · providers · constants	App-wide providers (React Query client), reusable hooks and constants such as labels and query roots.

No	Package	Description
39	lib	Shared infrastructure: axios client and mock adapter, auth session and permissions, domain status maps, reference lookups, formatting, zustand stores and validation helpers.
40	Externals	Spring Boot backend REST API and third-party libraries: TanStack Query, zod, React Hook Form, zustand, nuqs, Three.js, Konva and Recharts.

Table 22. Frontend Package Description

1.2.3 Frontend Store-Front Package Diagram

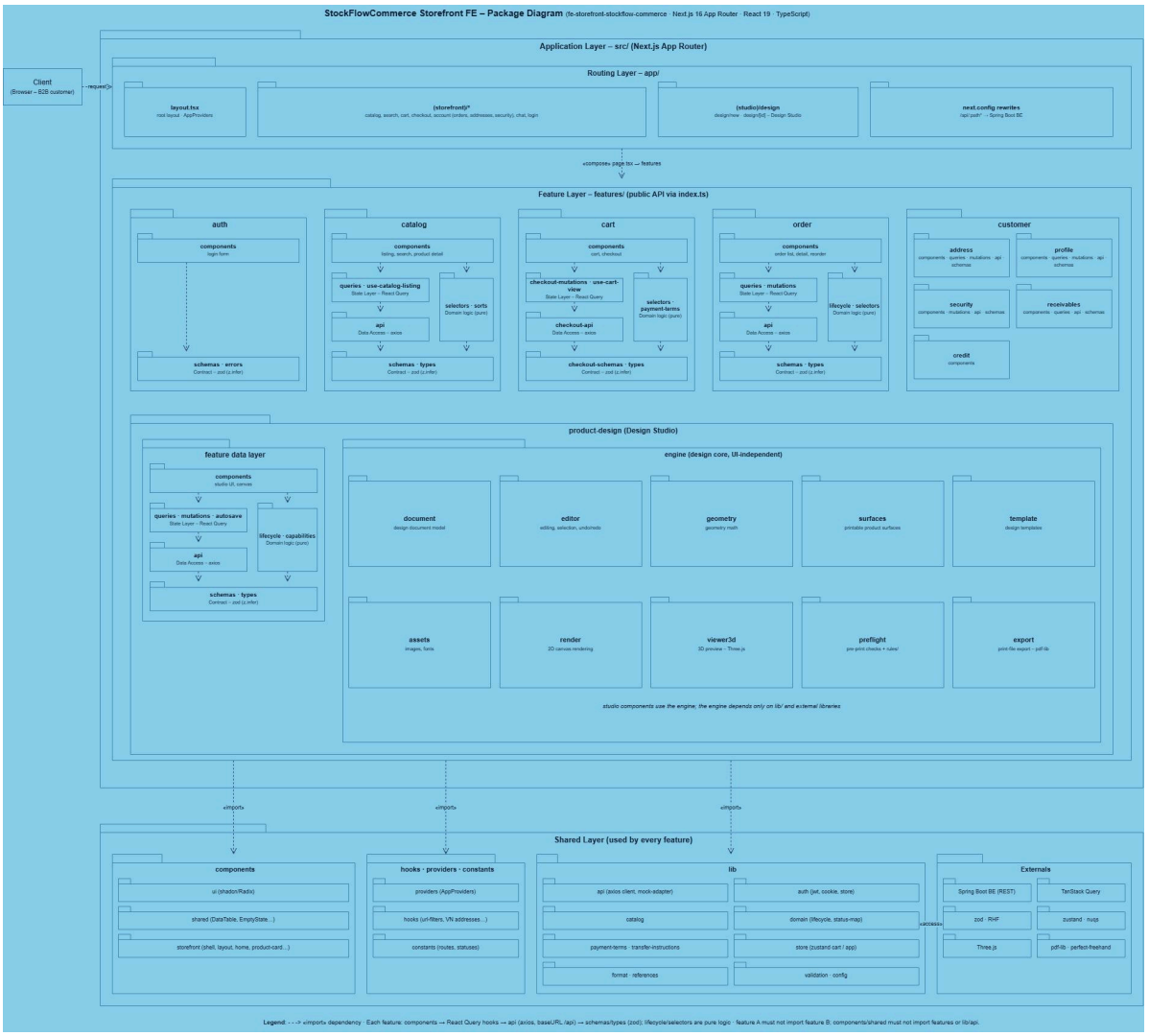


Figure 25. Frontend Store-Front Package Diagram

Package Descriptions

No	Package	Description
01	Routing Layer (app/)	Next.js App Router layer that maps URLs to pages; pages only compose features and components.
02	layout.tsx	Root layout that wraps every page with the application providers.
03	(storefront)/*	Customer-facing routes: catalog, search, cart, checkout, account (orders, addresses, security), chat and login.
04	(studio)/design	Design Studio routes for creating a new design and editing an existing one.
05	next.config rewrites	Rewrites /api/* requests to the Spring Boot backend.
06	Feature Layer (features/)	Business features of the storefront; each feature exposes its public API through index.ts and must not import another feature.
07	auth	Customer login form and its validation.
08	catalog	Product listing, search, sorting and product detail.
09	cart	Shopping cart and checkout, including payment terms.
10	order	Customer order list, order detail and reorder.
11	customer	Customer account area, split into address, profile, security, receivables and credit sub-features.
12	product-design	Design Studio feature: studio UI, design data access, autosave and design lifecycle.
13	engine	Design core independent of the UI, made of the document, editor, geometry, surfaces, template, assets, render, viewer3d, preflight and export packages.
14	document	Data model of a design document.
15	editor	Editing operations, selection and undo/redo.
16	geometry	Geometry calculations for shapes and transforms.
17	surfaces	Printable surfaces of each product.

No	Package	Description
18	template	Ready-made design templates.
19	assets	Images and fonts used in designs.
20	render	2D canvas rendering of a design.
21	viewer3d	3D preview of the product with Three.js.
22	preflight	Pre-print checks and their rules.
23	export	Export of print-ready files with pdf-lib.
24	components	Presentation layer of a feature: page-level React components.
25	queries · mutations	State layer of a feature: TanStack Query hooks for reading and changing server data.
26	api	Data access layer of a feature: axios calls to the backend through /api.
27	schemas · types	Contract layer of a feature: zod schemas that validate API data and form input, with types inferred from them.
28	lifecycle · selectors	Pure domain logic of a feature: status transitions, allowed actions and derived values.
29	Shared Layer	Code shared by every feature.
30	components	Shared UI: shadcn/Radix primitives, generic components and storefront building blocks (shell, layout, home, product card).
31	hooks · providers · constants	App-wide providers, reusable hooks (URL filters, Vietnamese address lookup) and constants such as routes and statuses.
32	lib	Shared infrastructure: axios client and mock adapter, JWT and cookie auth, catalog, domain status maps, payment terms, transfer instructions, zustand stores, formatting and validation.
33	Externals	Spring Boot backend REST API and third-party libraries: TanStack Query, zod, React Hook Form, zustand, nuqs, Three.js, pdf-lib and perfect-freehand.

Table 23. Frontend Package Description

2. Database Design

2.1 Database Design for Booking Products

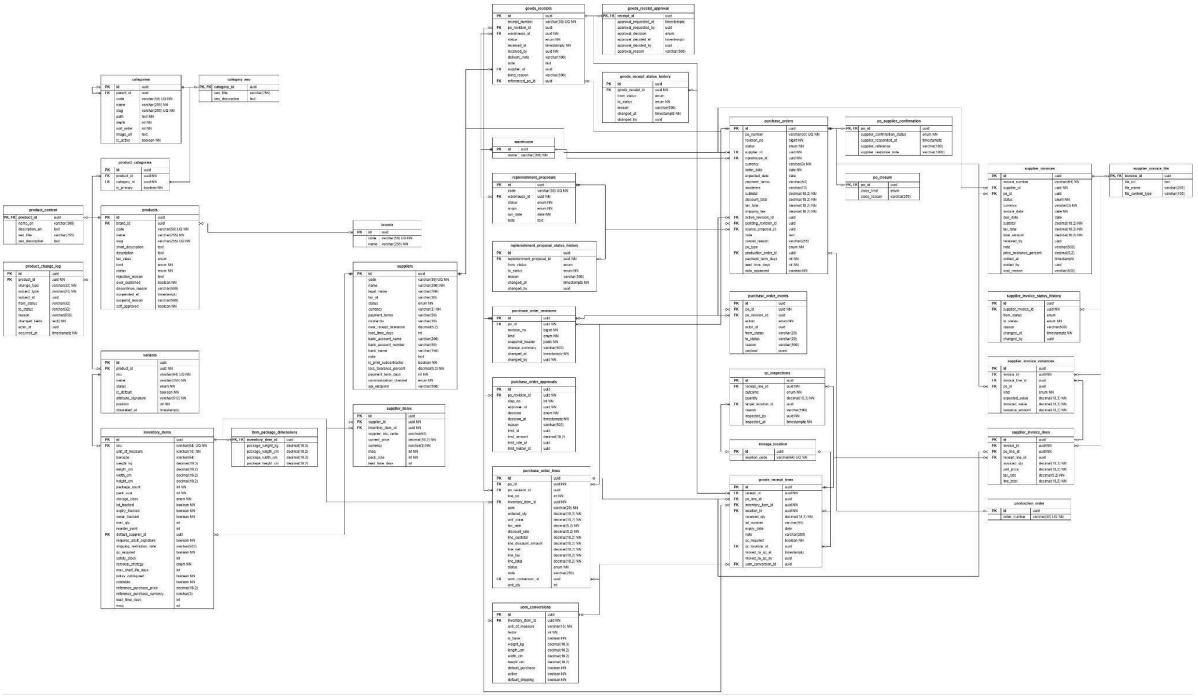


Figure 26. Database Design for Booking Products

Table Descriptions:

No	Table	Description
01	product_categories	Maps products to their assigned categories.
02	categories	Stores hierarchical categories for product classification.
03	replenishment_proposals	Stores warehouse replenishment proposals and their statuses.
04	supplier_items	Maps suppliers to inventory items with supplier-specific pricing and SKUs.
05	production_order	Stores production orders and their related warehouse, material, and subcontractor references.
06	products	Stores basic product information, types, and statuses.
07	variants	Stores product variants with unique SKUs and attributes.
08	warehouse	Stores warehouse information.
09	purchase_orders	Stores purchase orders, supplier details, financial totals, and revision references.
10	suppliers	Stores supplier profiles, payment details, and procurement settings.
11	inventory_items	Stores inventory item specifications, tracking settings, and reference pricing.
12	purchase_order_revisions	Tracks purchase order revisions and their revision numbers.

No	Table	Description
13	goods_receipts	Records goods receipt transactions associated with purchase orders and warehouses.
14	goods_receipt_lines	Records received item quantities, batches, expiration dates, and storage locations.
15	purchase_order_lines	Stores ordered items, quantities, unit prices, discounts, and taxes for each purchase order.
16	supplier_invoices	Stores supplier invoices, related purchase orders, and financial totals.
17	purchase_order_approvals	Records approval decisions, approval steps, and authorization limits for purchase order revisions.
18	uom_conversions	Stores item-specific units of measurement, weights, and dimensions.
19	qc_inspections	Records quality inspection quantities, reasons, and target locations for received goods.
20	supplier_invoice_lines	Stores invoiced items, quantities, prices, and references to purchase order and receipt lines.
21	storage_location	Stores storage locations and their associated warehouses.

Table 24. Database Design for Booking Products

2.2 Database Design for Mass Ordering

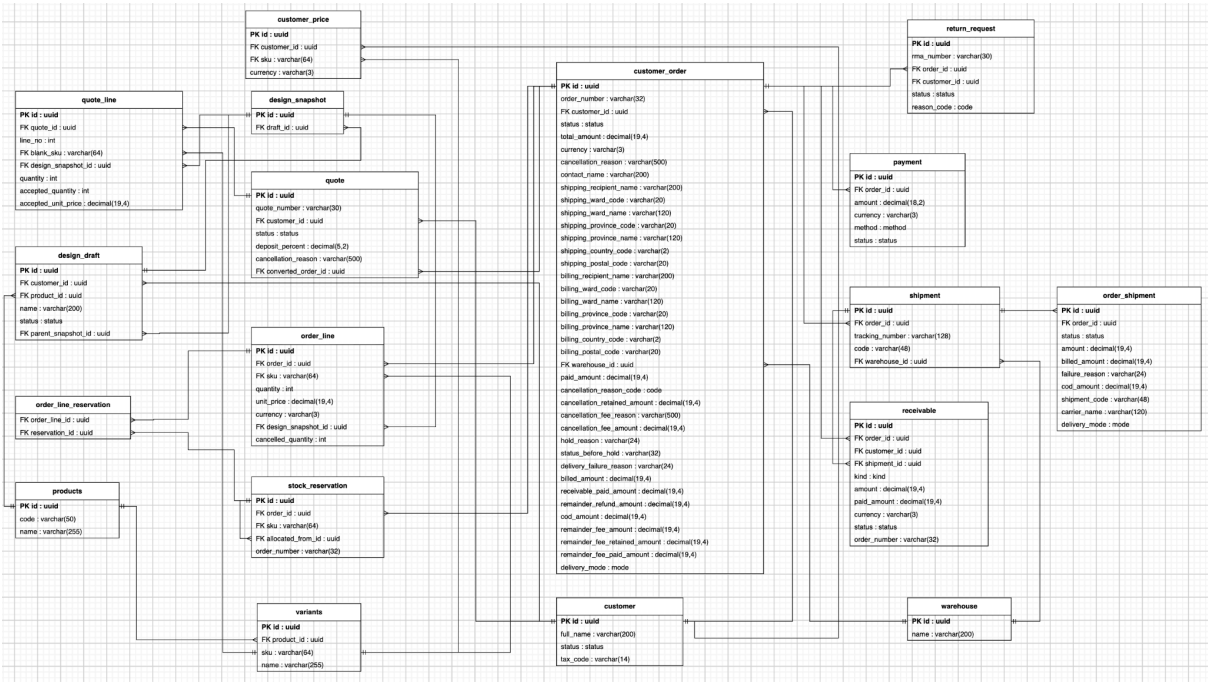


Figure 27. Database Design for Mass Ordering

Table Descriptions:

No	Table	Description
01	customer_price	Associates customers with product SKUs and currencies for customer-specific pricing.

02	quote_line	Stores quoted items, quantities, accepted prices, and design references.
03	design_snapshot	Stores snapshots associated with customer design drafts.
04	design_draft	Stores customer design drafts, related products, and their statuses.
05	quote	Stores customer quotations, deposit requirements, statuses, and converted order references.
06	order_line	Stores ordered items, quantities, unit prices, and design references.
07	order_line_reservation	Links customer order lines to their stock reservations.
08	products	Stores basic product information, including codes and names.
09	stock_reservation	Tracks inventory reservations and allocations for customer orders.
10	variants	Stores product variants with unique SKUs and names.
11	customer_order	Stores customer orders, billing and shipping details, payment amounts, and fulfillment statuses.
12	customer	Stores customer profiles, statuses, and tax information.
13	return_request	Records customer return requests, reasons, and processing statuses.
14	payment	Records order payments, including amounts, methods, and transaction statuses.
15	shipment	Stores shipment records, tracking numbers, and associated warehouses.
16	order_shipment	Tracks order shipment statuses, delivery details, billing amounts, and cash-on-delivery information.
17	receivable	Tracks outstanding amounts, received payments, and receivable statuses for customer orders.
18	warehouse	Stores warehouse information.

Table 25. Database Design for Mass Ordering

2.3 Database Design for Fulfilment

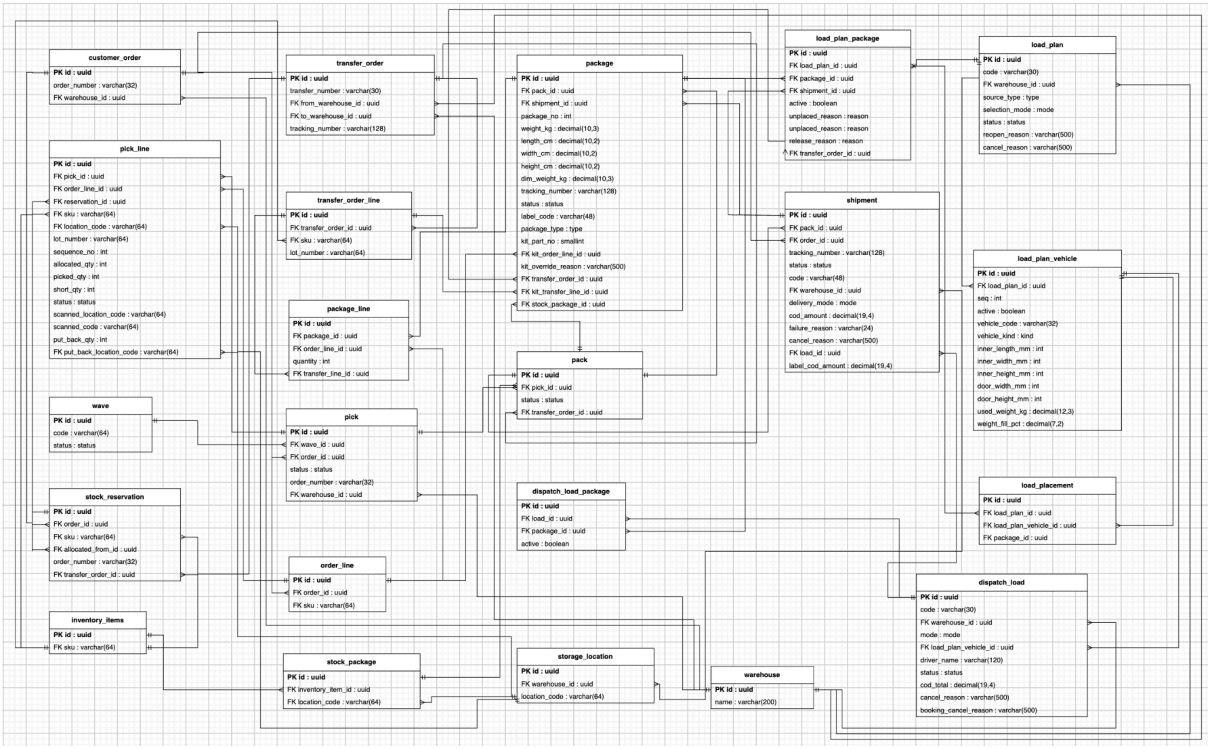


Figure 28. Database Design for Fulfilment

Table Descriptions:

No	Table	Description
01	customer_order	Stores customer order references and assigned fulfillment warehouses.
02	transfer_order	Records inventory transfers between warehouses, including transfer numbers and tracking information.
03	package	Stores package details, dimensions, tracking information, and associated packing or shipment records.
04	load_plan_package	Links packages and shipments to load plans, including allocation status and removal reasons.
05	load_plan	Stores warehouse loading plans, planning methods, statuses, and cancellation reasons.
06	pick_line	Records item picking activities, including allocated quantities, picked quantities, locations, and discrepancies.
07	transfer_order_line	Stores inventory items and lot numbers associated with warehouse transfer orders.

08	shipment	Stores shipment records, tracking information, delivery modes, and associated orders and loads.
09	load_plan_vehicle	Stores planned vehicles, cargo dimensions, weight utilization, and loading capacities.
10	package_line	Records items and quantities packed into packages for customer orders or warehouse transfers.
11	pack	Tracks packing operations and their associated picking tasks or transfer orders.
12	wave	Groups warehouse picking operations into processing waves.
13	pick	Tracks order picking tasks, assigned warehouses, processing waves, and picking statuses.
14	dispatch_load_package	Links packages to dispatch loads and tracks their active allocation status.
15	load_placement	Records package placements within vehicles assigned to loading plans.
16	stock_reservation	Tracks inventory reservations and allocations for customer orders and warehouse transfers.
17	order_line	Stores ordered items and their associated customer orders.
18	dispatch_load	Records dispatch loads, assigned vehicles, drivers, delivery statuses, and cash-on-delivery totals.
19	inventory_items	Stores inventory item identifiers and SKUs.
20	stock_package	Associates inventory items with their physical storage locations.
21	storage_location	Stores warehouse storage locations and their location codes.
22	warehouse	Stores warehouse information.

Table 26. Database Design for Fulfillment

2.4 Database Design for Warehouse Management

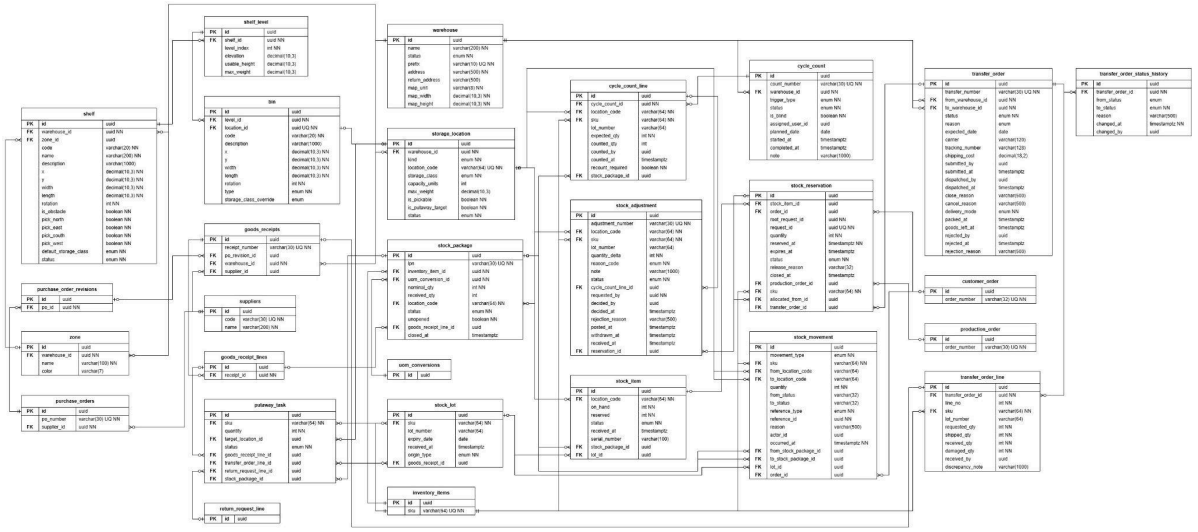


Figure 29. Database Design for Warehouse Management

Table Descriptions:

No	Table	Description
01	shelf_level	Stores shelf levels, including elevation, usable height, and maximum weight capacity.
02	purchase_orders	Stores purchase order references and their associated suppliers.
03	purchase_order_revisions	Tracks purchase order revisions referenced by goods receipts.
04	shelf	Stores warehouse shelves, including dimensions, positions, storage classes, and picking directions.
05	zone	Stores warehouse zones used to organize storage areas.
06	bin	Stores storage bins, including shelf levels, physical dimensions, positions, and storage classifications.
07	goods_receipts	Records incoming goods receipts associated with purchase order revisions, suppliers, and warehouses.
08	suppliers	Stores supplier identifiers, codes, and names.
09	goods_receipt_lines	Stores individual goods receipt line references used in inventory receiving and putaway operations.
10	putaway_task	Tracks inventory placement tasks, quantities, target locations, source documents, and processing statuses.

11	return_request_line	Stores return request line references used in inventory putaway operations.
12	warehouse	Stores warehouse information, operational statuses, addresses, and layout dimensions.
13	storage_location	Stores warehouse storage locations, capacity limits, storage classes, and picking eligibility.
14	stock_package	Tracks inventory packages, quantities, unit conversions, storage locations, and processing statuses.
15	uom_conversions	Stores unit-of-measure conversion references associated with inventory packages.
16	stock_lot	Stores inventory lots, batch numbers, expiration dates, and receiving information.
17	inventory_items	Stores inventory item identifiers and unique SKUs.
18	cycle_count_line	Records expected and counted quantities for inventory items, lots, and storage locations.
19	cycle_count	Manages warehouse inventory counting sessions, assignments, counting methods, and processing statuses.
20	stock_adjustment	Records inventory quantity adjustments, approval decisions, reasons, and posting information.
21	stock_item	Tracks inventory balances, reserved quantities, and stock statuses by location, package, and lot.
22	stock_movement	Records inventory movements, source and destination locations, quantities, transaction references, and audit details.
23	stock_reservation	Manages inventory reservations, allocated quantities, expiration times, and associated customer, production, or transfer orders.
24	transfer_order	Manages inventory transfers between warehouses, including approval, dispatch, receipt, and rejection statuses.
25	customer_order	Stores customer order references associated with inventory movements and reservations.

26	transfer_order_line	Tracks transferred items, requested quantities, shipped quantities, received quantities, and discrepancies.
27	production_order	Stores production order references associated with inventory reservations.

Table 27. Database Design for Warehouse Management

2.5 Database Design for Customized Production

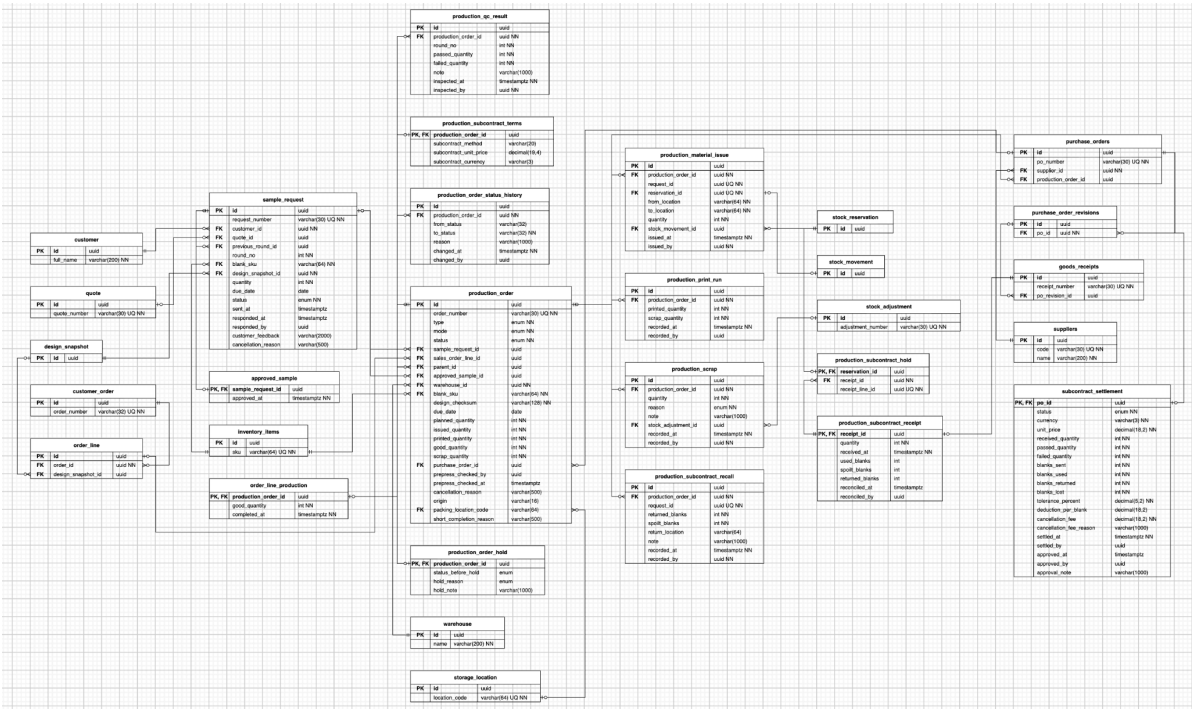


Figure 30. Database Design for Customized Production

Table Descriptions:

No	Table	Description
01	customer	Stores customer identifiers and names.
02	quote	Stores quotation references associated with sample requests.
03	design_snapshot	Stores design snapshot references used in sample requests and order lines.
04	customer_order	Stores customer order references associated with production orders.

05	order_line	Stores customer order line references and associated design snapshots.
06	sample_request	Manages customer sample requests, design references, requested quantities, feedback, and approval statuses.
07	approved_sample	Records approved customer samples and their approval timestamps.
08	inventory_items	Stores inventory item identifiers and unique SKUs used in production.
09	order_line_production	Links production orders to customer order lines and tracks completed quantities.
10	production_qc_result	Records production quality inspections, including passed quantities, failed quantities, and inspection rounds.
11	production_subcontract_terms	Stores subcontracting methods, unit prices, and currencies for production orders.
12	production_order_status_history	Tracks production order status changes, reasons, timestamps, and responsible users.
13	production_order	Manages production orders, quantities, materials, processing statuses, quality checks, and related procurement records.
14	production_order_hold	Records production order hold statuses, reasons, and additional notes.
15	warehouse	Stores warehouse information associated with production operations.
16	storage_location	Stores storage location references used in production operations.
17	production_material_issue	Records material issuance transactions, quantities, inventory reservations, and source and destination locations.
18	production_print_run	Tracks production printing runs, including printed quantities, scrap quantities, and recording details.
19	production_scrap	Records production waste quantities, scrap reasons, and associated inventory adjustments.

20	production_subcontract_recall	Tracks recalled subcontracted materials, including returned quantities, spoiled quantities, and return locations.
21	stock_reservation	Stores inventory reservation references associated with production material issuance.
22	stock_movement	Stores inventory movement references associated with production material issuance.
23	stock_adjustment	Stores inventory adjustment references associated with production scrap records.
24	production_subcontract_hold	Links subcontracted production materials to inventory reservations and goods receipt lines.
25	production_subcontract_receipt	Records subcontracted production receipts, received quantities, material usage, losses, returns, and reconciliation details.
26	purchase_orders	Stores purchase order references associated with suppliers and production orders.
27	purchase_order_revisions	Tracks purchase order revisions associated with goods receipts.
28	goods_receipts	Records incoming goods receipts associated with purchase order revisions.
29	suppliers	Stores supplier identifiers, codes, and names.
30	subcontract_settlement	Manages subcontractor settlements, received quantities, material losses, deductions, cancellation fees, and approval details.

Table 28. Database Design for Customized Production

3. Detailed Design

3.1 Authentication & Session Management

3.1.1 Class Diagram

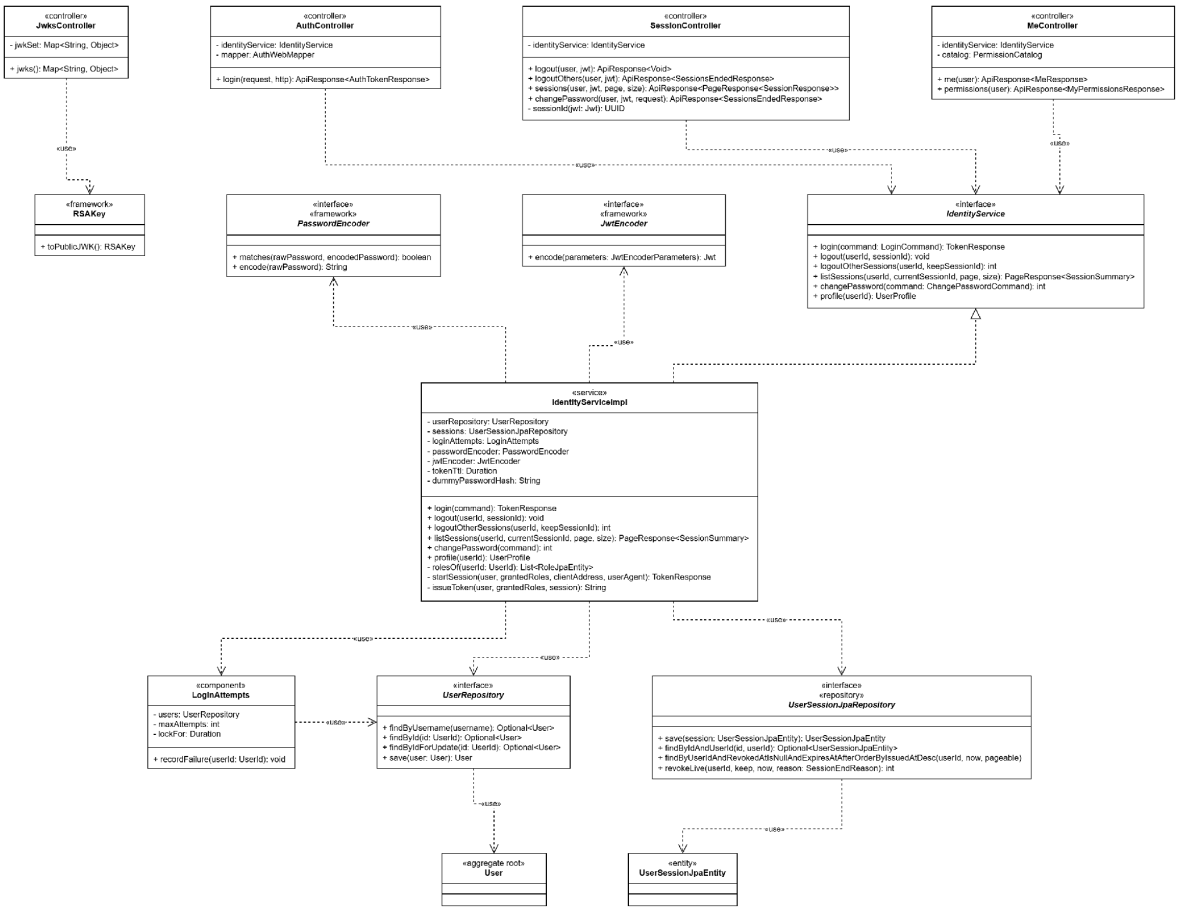


Figure 31. Authentication and Session Management Class Diagram (1/4): Sign-in and Sessions

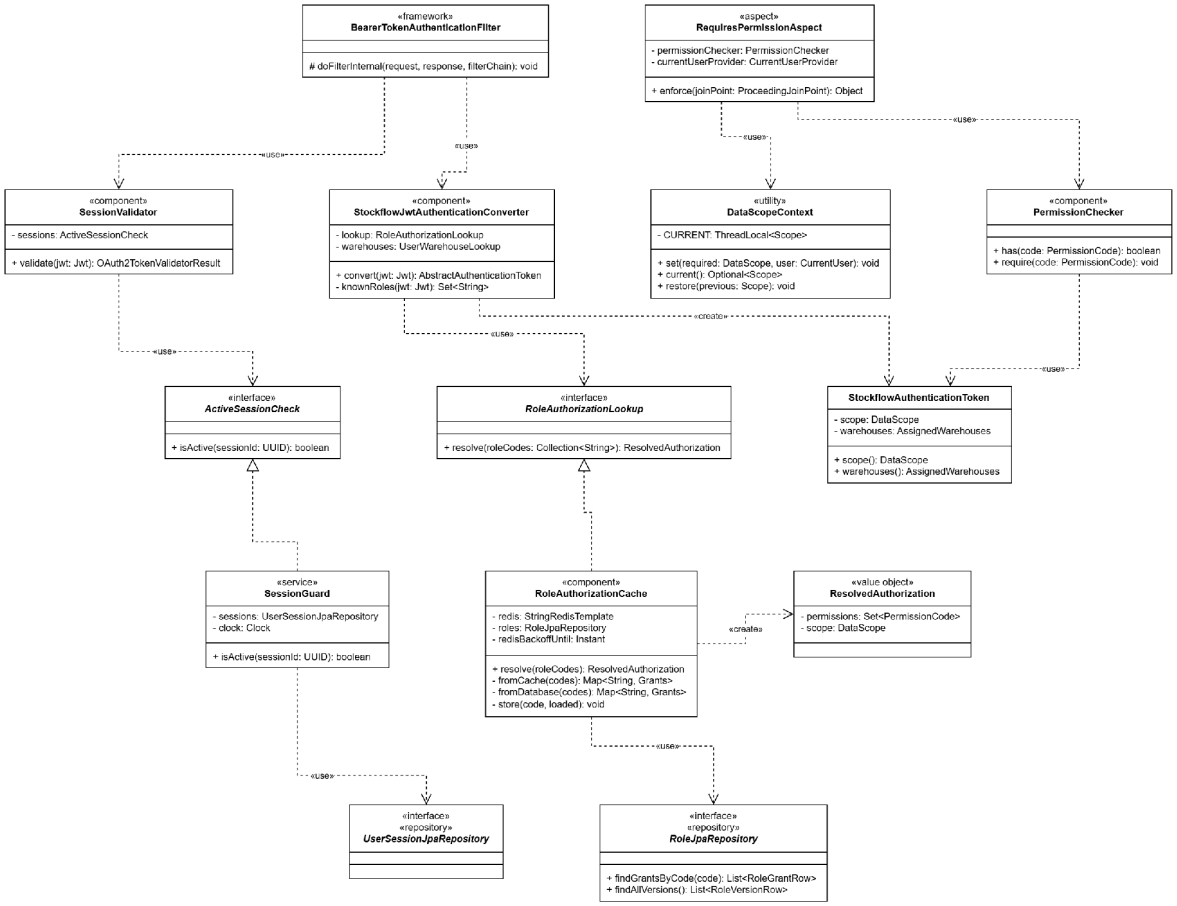


Figure 32. Authentication and Session Management Class Diagram (2/4): Request Authorization

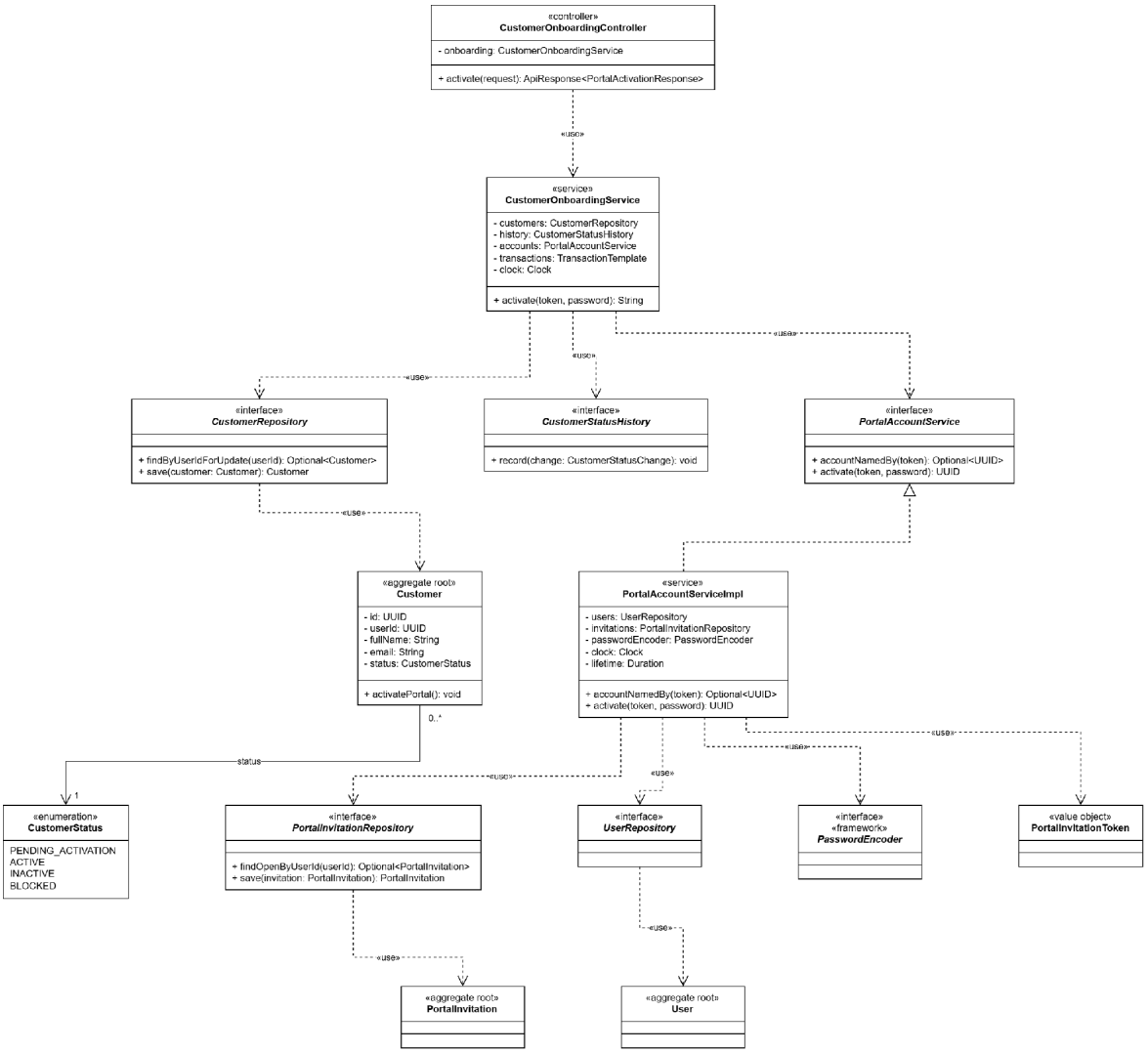


Figure 33. Authentication and Session Management Class Diagram (3/4): Portal Account Activation

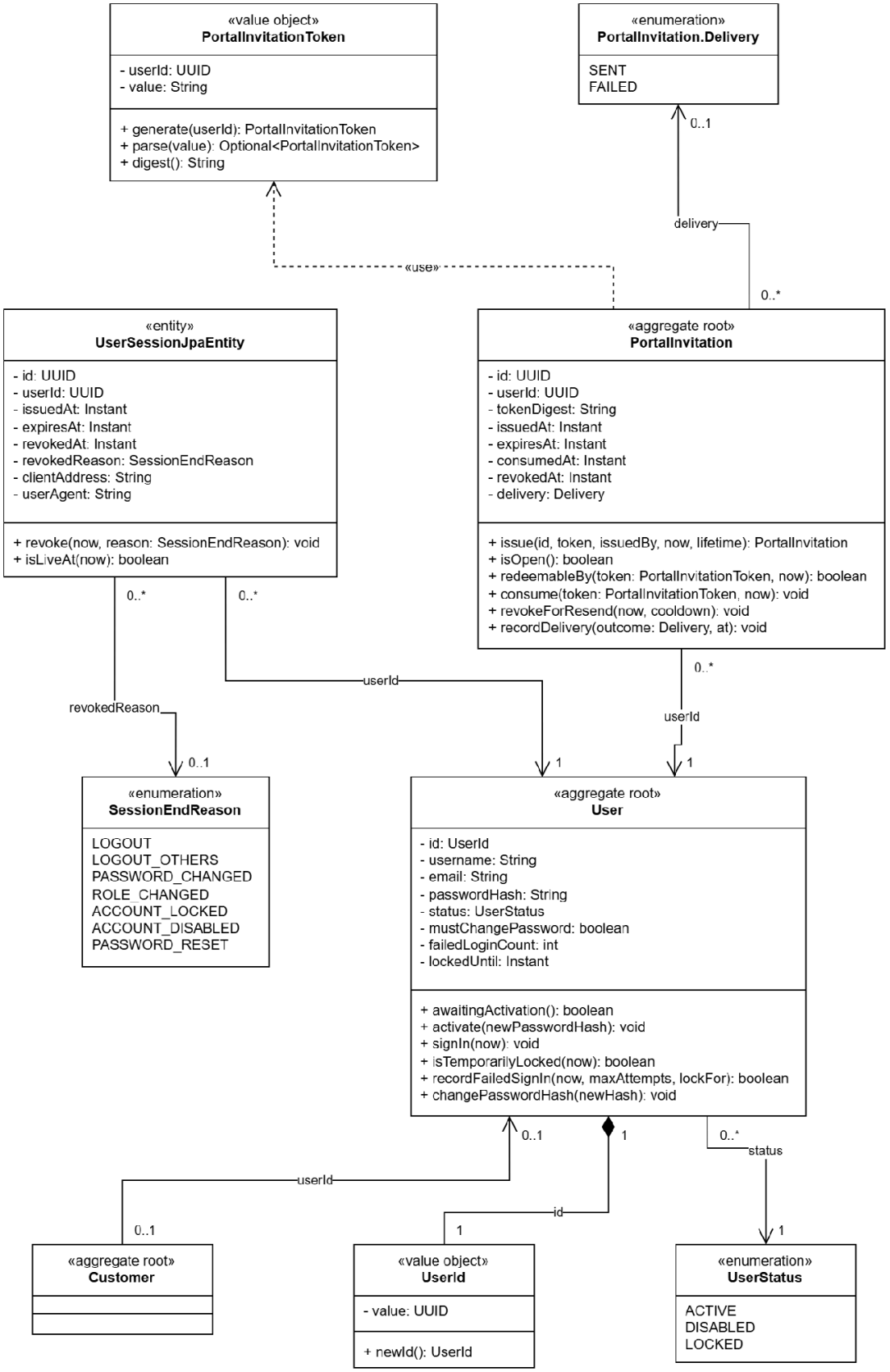


Figure 34. Authentication and Session Management Class Diagram (4/4): Domain Model

3.1.2 Class Specification

AuthController

No	Method	Description
1	login(request, http)	POST /api/v1/identity/auth/login: Signs a staff user (username) or customer (e-mail) in and returns a bearer JWT for a new session; anonymous, rate-limited to 5 per minute per IP; an unknown name, a not-yet-activated account and a wrong password all answer the same 401 UNAUTHORIZED, and the 5th consecutive wrong password locks the account for 15 minutes.

Table 29. AuthController Class Specification

SessionController

No	Method	Description
1	logout(user, jwt)	POST /api/v1/identity/auth/logout: Revokes the session named by the token's sid (reason LOGOUT) so this token is refused from the next request; a token without sid is a no-op.
2	logoutOthers(user, jwt)	POST /api/v1/identity/auth/logout-others: Revokes every other live session of the caller's own account in one UPDATE, keeping the current one, and returns how many were ended.
3	sessions(user, jwt, page, size)	GET /api/v1/identity/auth/sessions: Pages the caller's live sessions (not revoked, not expired), newest first, with client address and user agent, flagging the current one.
4	changePassword(user, jwt, request)	PUT /api/v1/identity/auth/password: Changes the caller's own password after verifying the current one and the password policy, optionally ending all other sessions; 400 INVALID_CURRENT_PASSWORD / PASSWORD_UNCHANGED, rate-limited to 5 per 5 minutes per user.

Table 30. SessionController Class Specification

MeController

No	Method	Description
1	me(user)	GET /api/v1/identity/me: Returns the signed-in user's profile (username, e-mail, full name, status, roles, last login, mustChangePassword); the login response itself carries only the token.

2	permissions(user)	GET /api/v1/identity/me/permissions: Returns the caller's roles, effective permissions and data scope exactly as resolved server-side for this request (ADR-0008); with security switched off it returns the whole catalog instead of 401.

Table 31. MeController Class Specification

JwksController

No	Method	Description
1	jwks()	GET /oauth2/jwks: Publishes the public half of the RSA signing key as a standard JWK Set (RFC 7517) used by the resource server to verify tokens; anonymous and deliberately not wrapped in ApiResponse.

Table 32. JwksController Class Specification

3.1.3 Log In Sequence Diagram

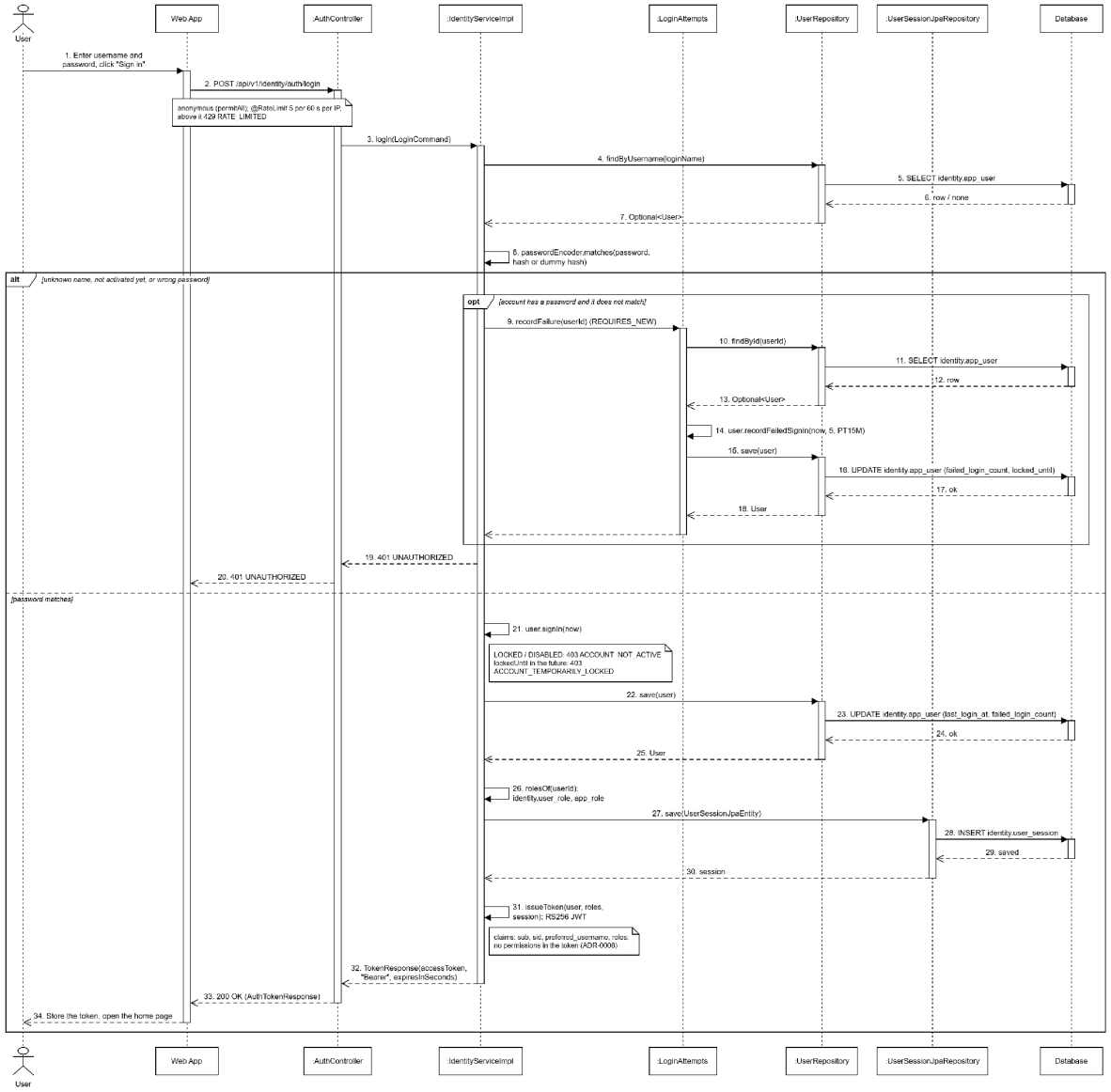


Figure 35. Log In Sequence Diagram

3.1.4 Authorize Request Sequence Diagram

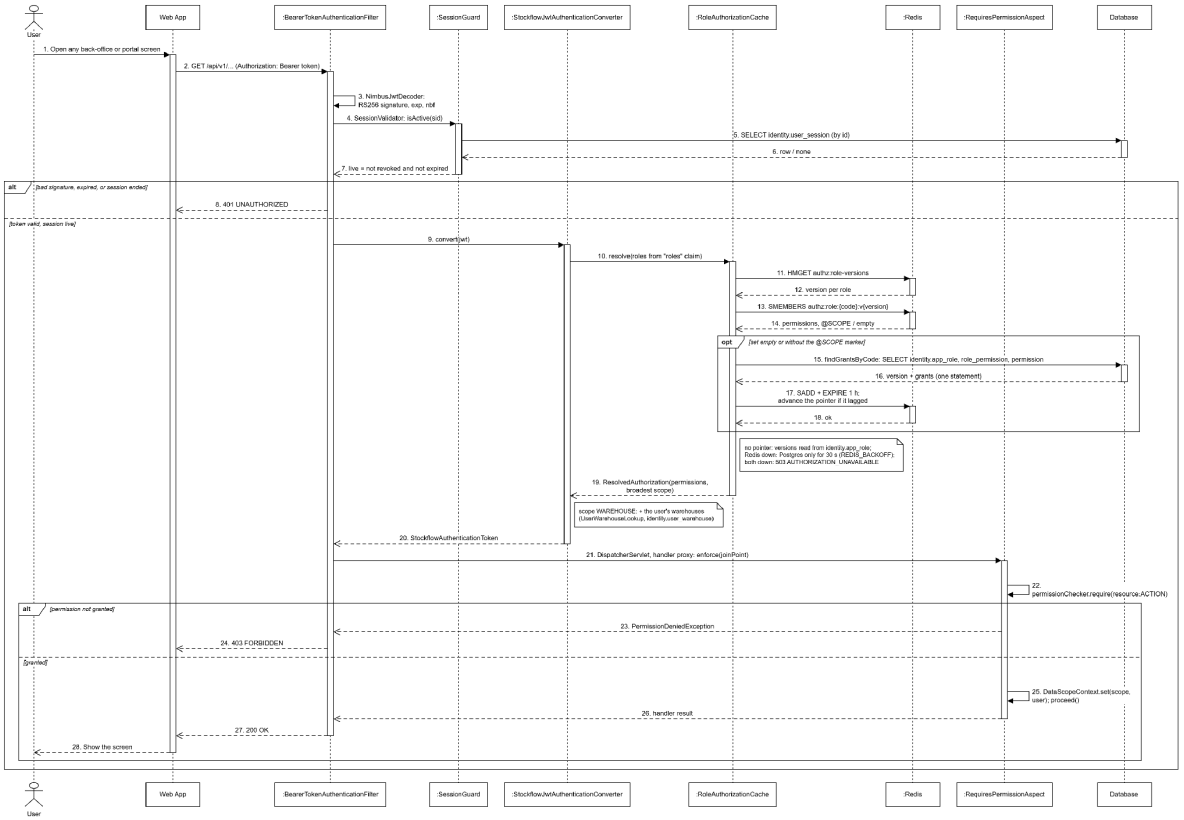


Figure 36. Authorize Request Sequence Diagram

3.1.5 Log Out Sequence Diagram

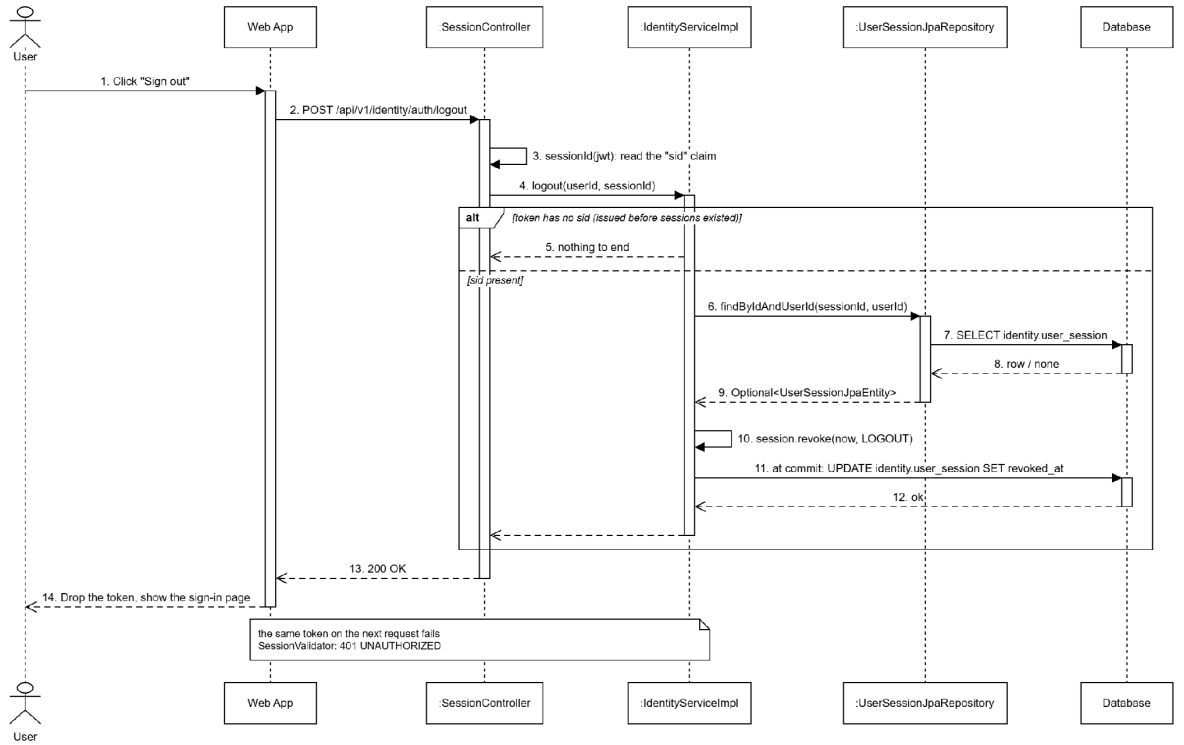


Figure 37. Log Out Sequence Diagram

3.1.6 Manage Login Sessions Sequence Diagram

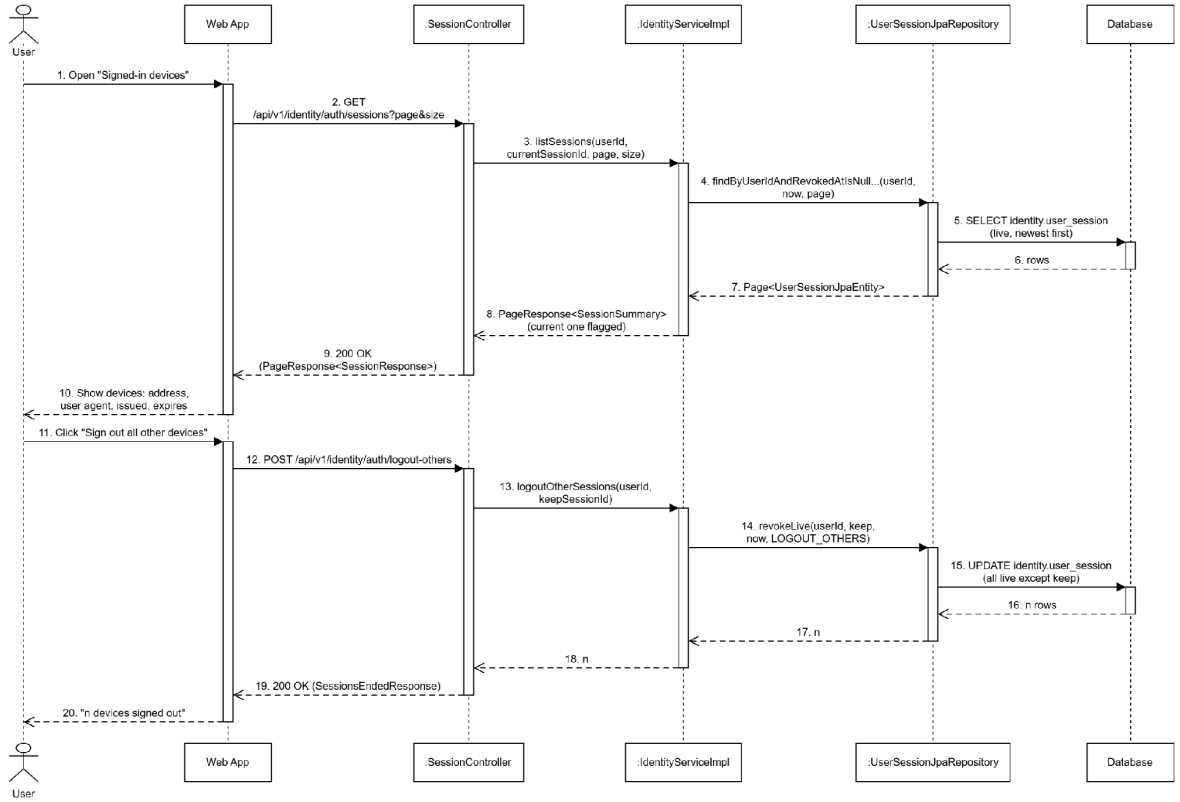


Figure 38. Manage Login Sessions Sequence Diagram

3.1.7 Change Password Sequence Diagram

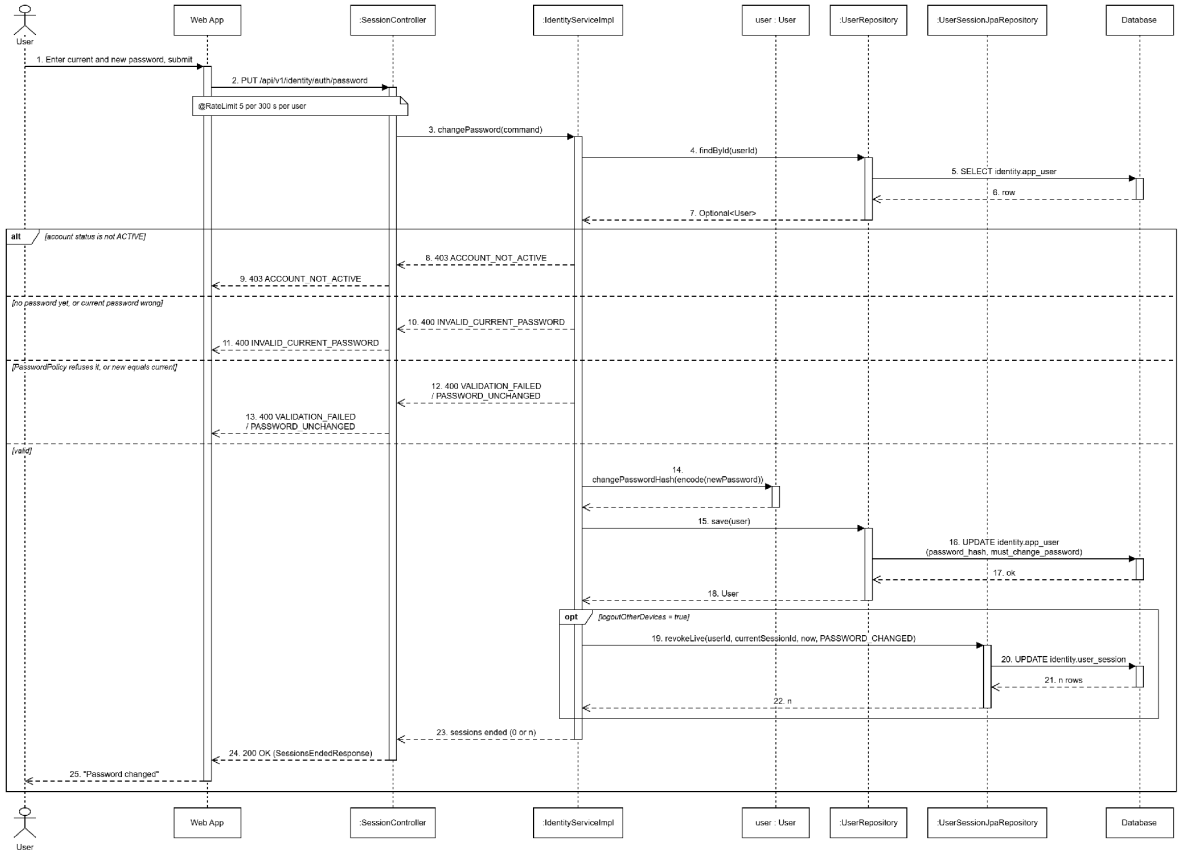


Figure 39. Change Password Sequence Diagram

3.1.8 Activate Account From Invitation Sequence Diagram

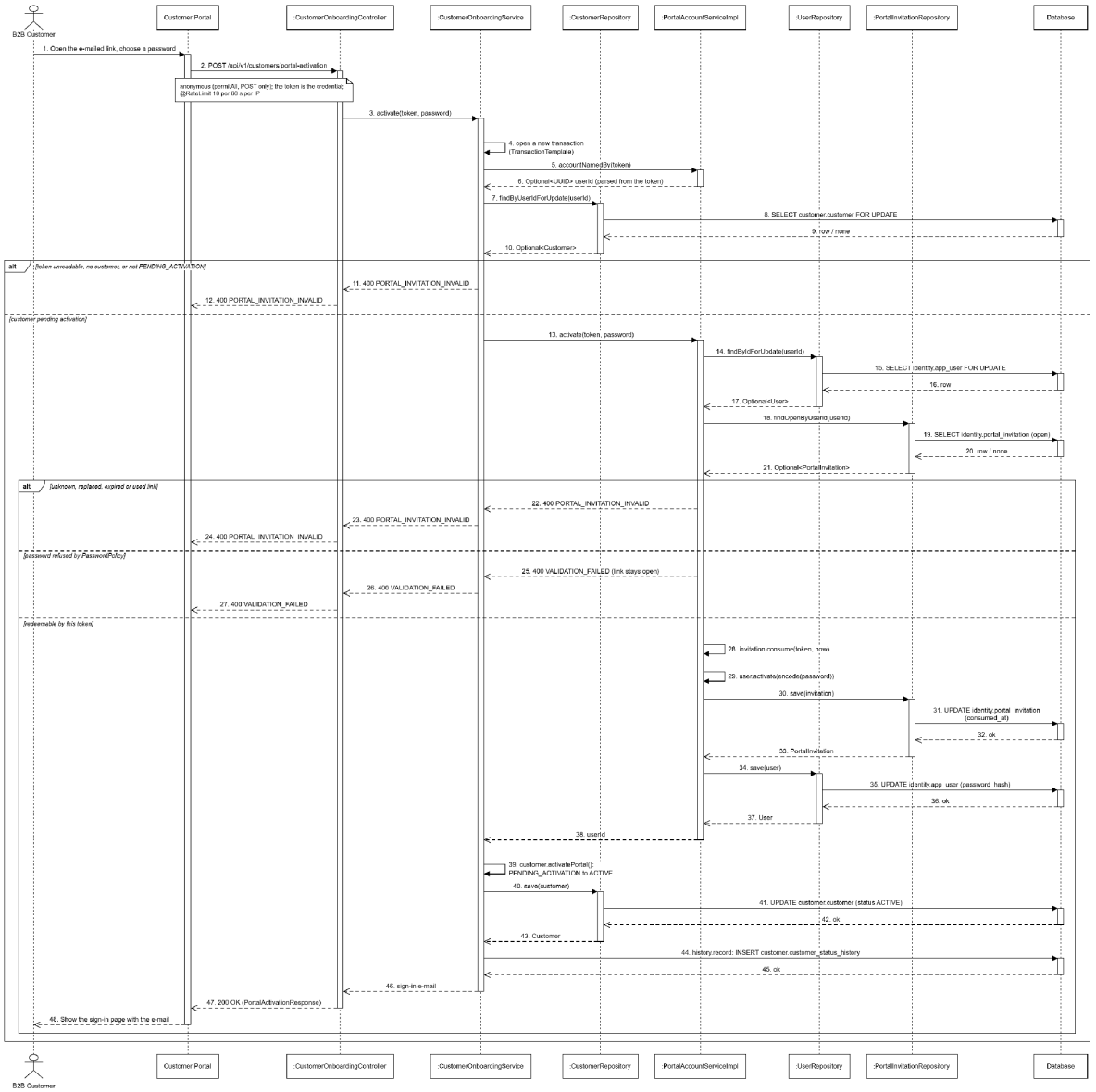


Figure 40. Activate Account From Invitation Sequence Diagram

3.2 Staff Account, Role & Permission Management

3.2.1 Class Diagram

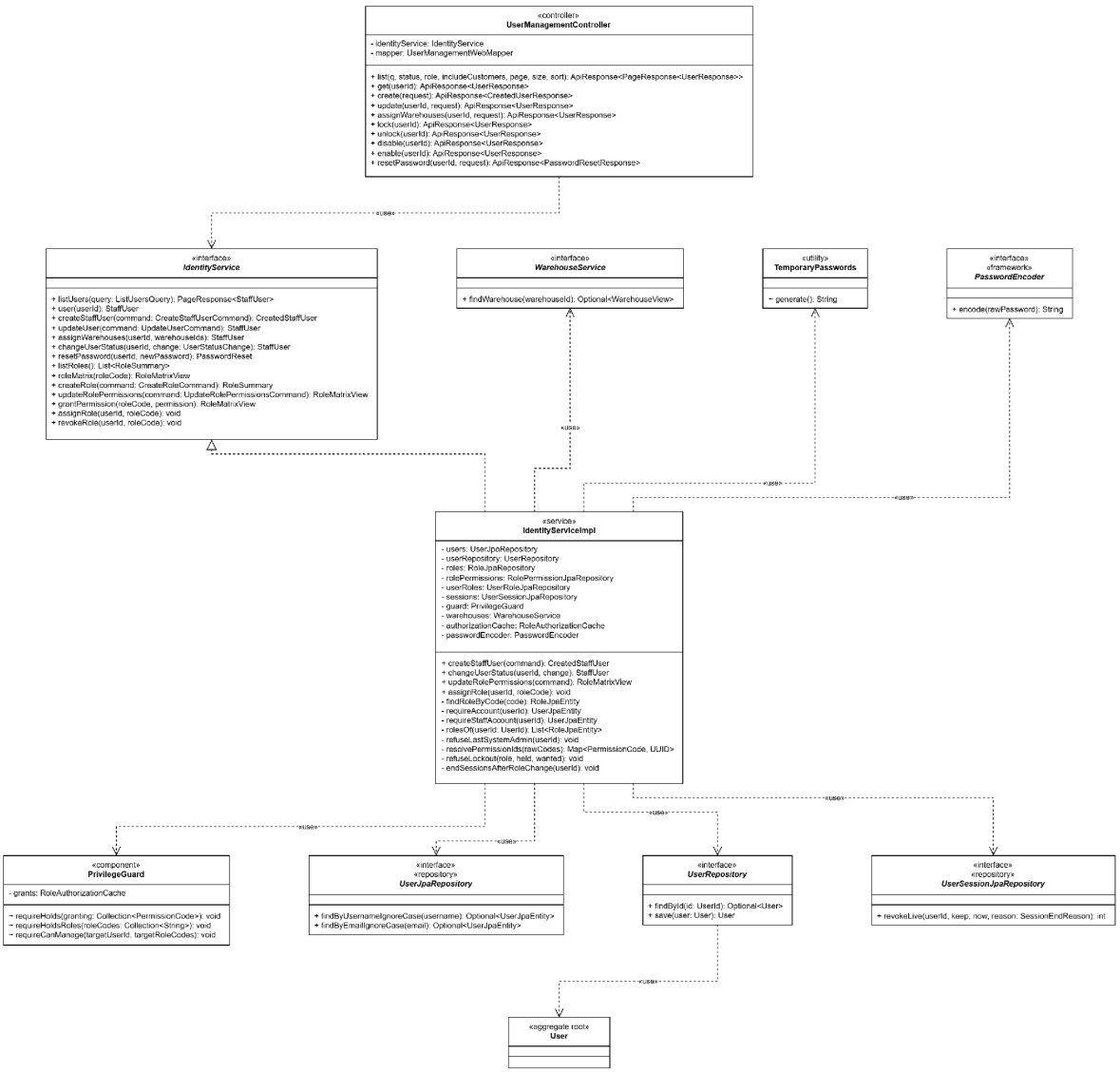


Figure 41. Staff Account, Role and Permission Management Class Diagram (1/3): Staff Accounts

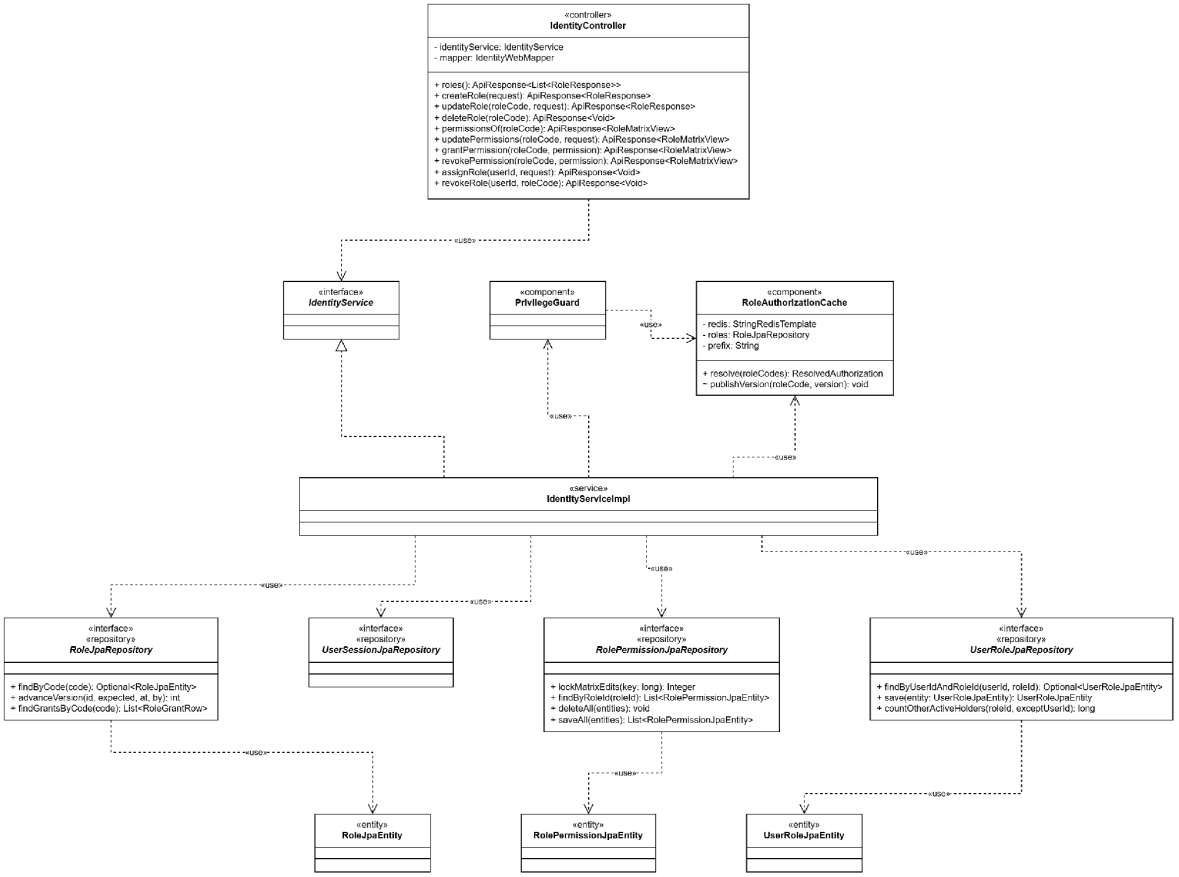


Figure 42. Staff Account, Role and Permission Management Class Diagram (2/3): Roles and Permissions

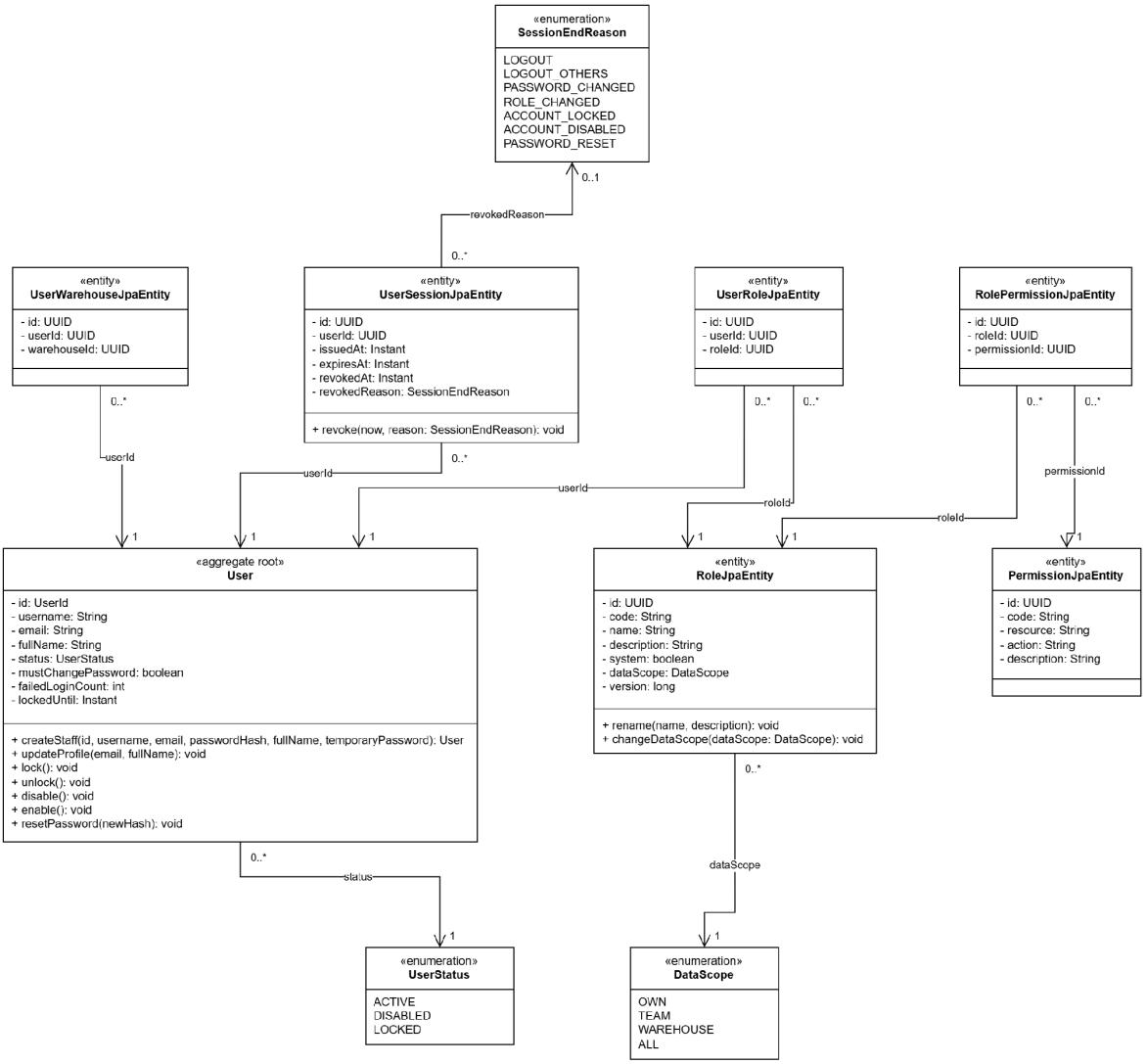


Figure 43. Staff Account, Role and Permission Management Class Diagram (3/3): Domain Model

3.2.2 Class Specification

UserManagementController

No	Method	Description
1	list(q, status, role, includeCustomers, page, size, sort)	GET /api/v1/identity/users: Pages staff accounts filtered by text, status and role with a whitelisted sort; customer accounts are left out unless includeCustomers; needs identity-users:READ.
2	get(userId)	GET /api/v1/identity/users/{userId}: Returns one account with its roles and assigned warehouses; 404 USER_NOT_FOUND.

3	create(request)	POST /api/v1/identity/users: Creates an ACTIVE staff account with roles and warehouses; without a password a temporary one is generated and returned once (must be changed); refuses the CUSTOMER role, a taken username or e-mail (409) and roles granting more than the caller holds (403); needs identity-users:CREATE.
4	update(userId, request)	PATCH /api/v1/identity/users/{userId}: Edits e-mail or full name of a staff account the caller may manage, with the loaded version (409 OPTIMISTIC_LOCK) and a unique e-mail.
5	assignWarehouses(userId, request)	PUT /api/v1/identity/users/{userId}/warehouses: Replaces the warehouses a warehouse-scoped account may act on, writing only the difference; every id must exist (404 WAREHOUSE_NOT_FOUND); takes effect on the user's next request.
6	lock(userId)	POST /api/v1/identity/users/{userId}/lock: ACTIVE → LOCKED and ends every session; refuses the last active System Admin (409 LAST_SYSTEM_ADMIN) and a DISABLED account (409 INVALID_USER_STATUS_TRANSITION).
7	unlock(userId)	POST /api/v1/identity/users/{userId}/unlock: LOCKED → ACTIVE, also ending a running wrong-password lockout; refused for a DISABLED account.
8	disable(userId)	POST /api/v1/identity/users/{userId}/disable: Retires the account (kept for its history) and ends every session; refuses the last active System Admin.
9	enable(userId)	POST /api/v1/identity/users/{userId}/enable: DISABLED → ACTIVE; a LOCKED account must be unlocked instead (409 INVALID_USER_STATUS_TRANSITION).
10	resetPassword(userId, request)	POST /api/v1/identity/users/{userId}/password-reset: Sets a temporary password (generated if none is given, else policy-checked), forces a change at next use, clears the lockout and ends every session of the account.

Table 33. UserManagementController Class Specification

IdentityController

No	Method	Description
1	roles()	GET /api/v1/identity/roles: Lists every role with its holder count, system flag, data scope and version; needs identity-roles:READ.

2	createRole(request)	POST /api/v1/identity/roles: Creates a custom role, empty or copying another role's grants (the caller must hold every copied permission); 409 ROLE_CODE_ALREADY_EXISTS; needs identity-roles:CREATE.
3	updateRole(roleCode, request)	PATCH /api/v1/identity/roles/{roleCode}: Renames a custom role or changes its data scope (OWN, WAREHOUSE, ALL) against the loaded version; system roles answer 409 SYSTEM_ROLE_IMMUTABLE.
4	deleteRole(roleCode)	DELETE /api/v1/identity/roles/{roleCode}: Deletes a custom role nobody holds (409 ROLE_IN_USE) with its grants, refusing to remove the last holder of identity-rbac:APPROVE.
5	permissionsOf(roleCode)	GET /api/v1/identity/roles/{roleCode}/permissions: Returns the role's permission matrix (RoleMatrixView, grouped by resource with its version) for the permission screen; needs identity-rbac:READ.
6	updatePermissions(roleCode, request)	PUT /api/v1/identity/roles/{roleCode}/permissions: Replaces the role's declared grants under an advisory lock and a version compare-and-set (409 ROLE_PERMISSIONS_CHANGED); refuses CUSTOMER and SYSTEM_ADMIN (409 ROLE_NOT_EDITABLE), a lockout (409 RBAC_LOCKOUT) and grants the caller lacks; needs identity-rbac:APPROVE.
7	grantPermission(roleCode, permission)	POST /api/v1/identity/roles/{roleCode}/permissions/{permission}: Ticks one resource:ACTION onto a role (the caller must hold it); idempotent, advances the role version.
8	revokePermission(roleCode, permission)	DELETE /api/v1/identity/roles/{roleCode}/permissions/{permission}: Unticks one permission; idempotent, refuses to remove the last identity-rbac:APPROVE grant.
9	assignRole(userId, request)	POST /api/v1/identity/users/{userId}/roles: Gives a staff account a role (never CUSTOMER, 409 ROLE_NOT_ASSIGNABLE) only if the caller holds everything it grants, then ends the account's sessions; needs identity-users:UPDATE.
10	revokeRole(userId, roleCode)	DELETE /api/v1/identity/users/{userId}/roles/{roleCode}: Takes a role away and ends the account's sessions; refuses CUSTOMER and the last active System Admin.

Table 34. IdentityController Class Specification

3.2.3 Manage Staff Accounts (Create Account) Sequence Diagram

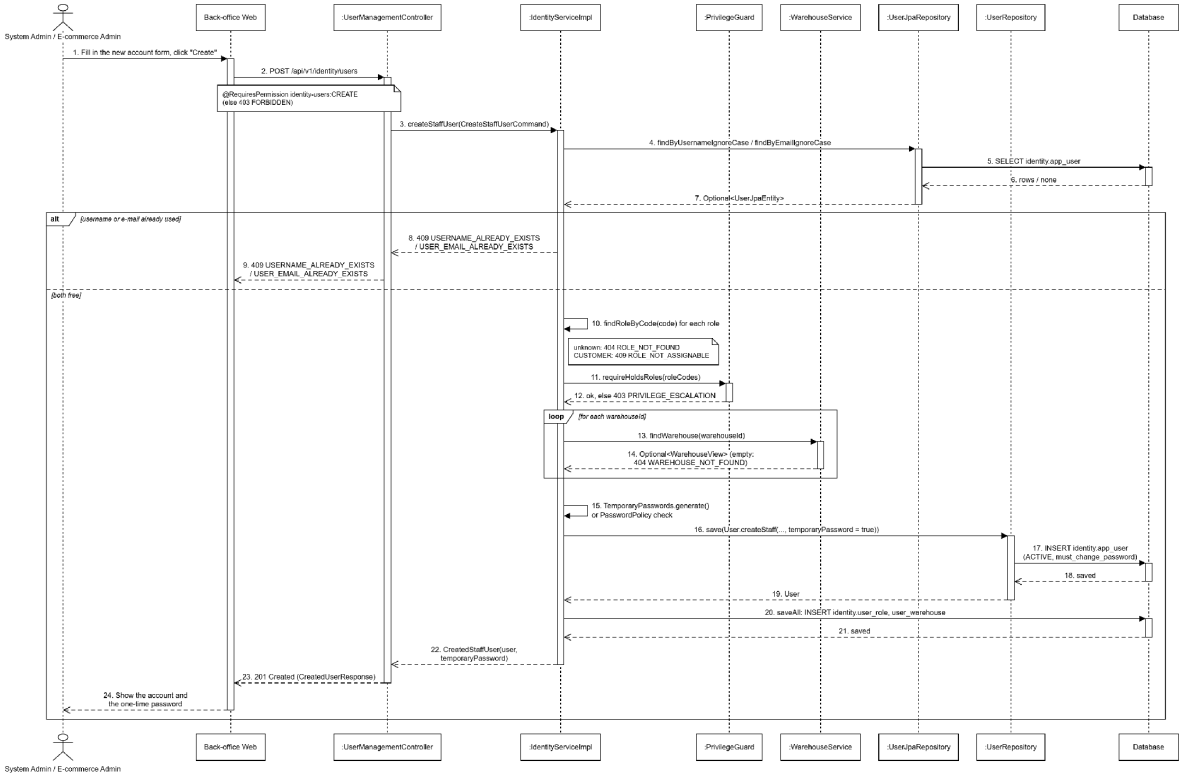


Figure 44. Manage Staff Accounts (Create Account) Sequence Diagram

3.2.4 Manage Staff Accounts (Lock Account) Sequence Diagram

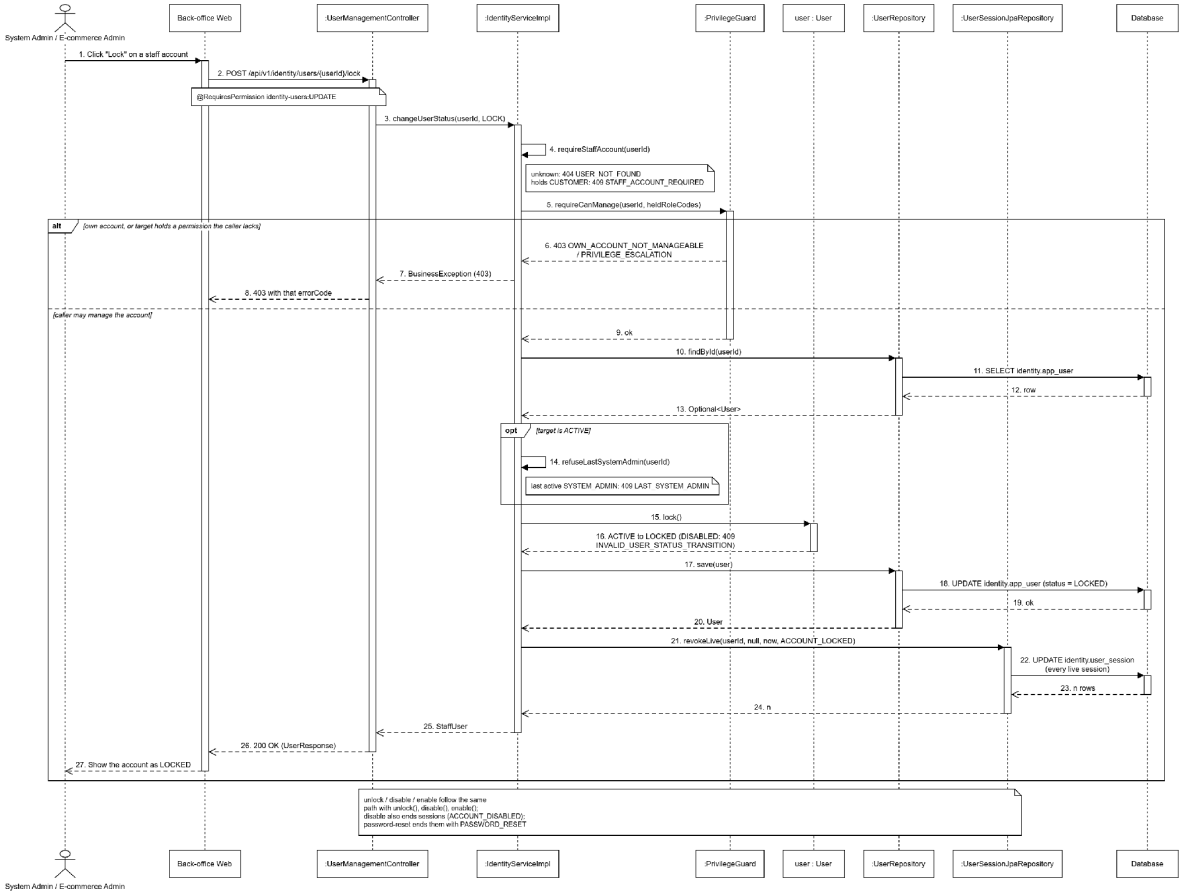


Figure 45. Manage Staff Accounts (Lock Account) Sequence Diagram

3.2.5 Manage Roles & Permissions (Edit Permission Matrix) Sequence Diagram

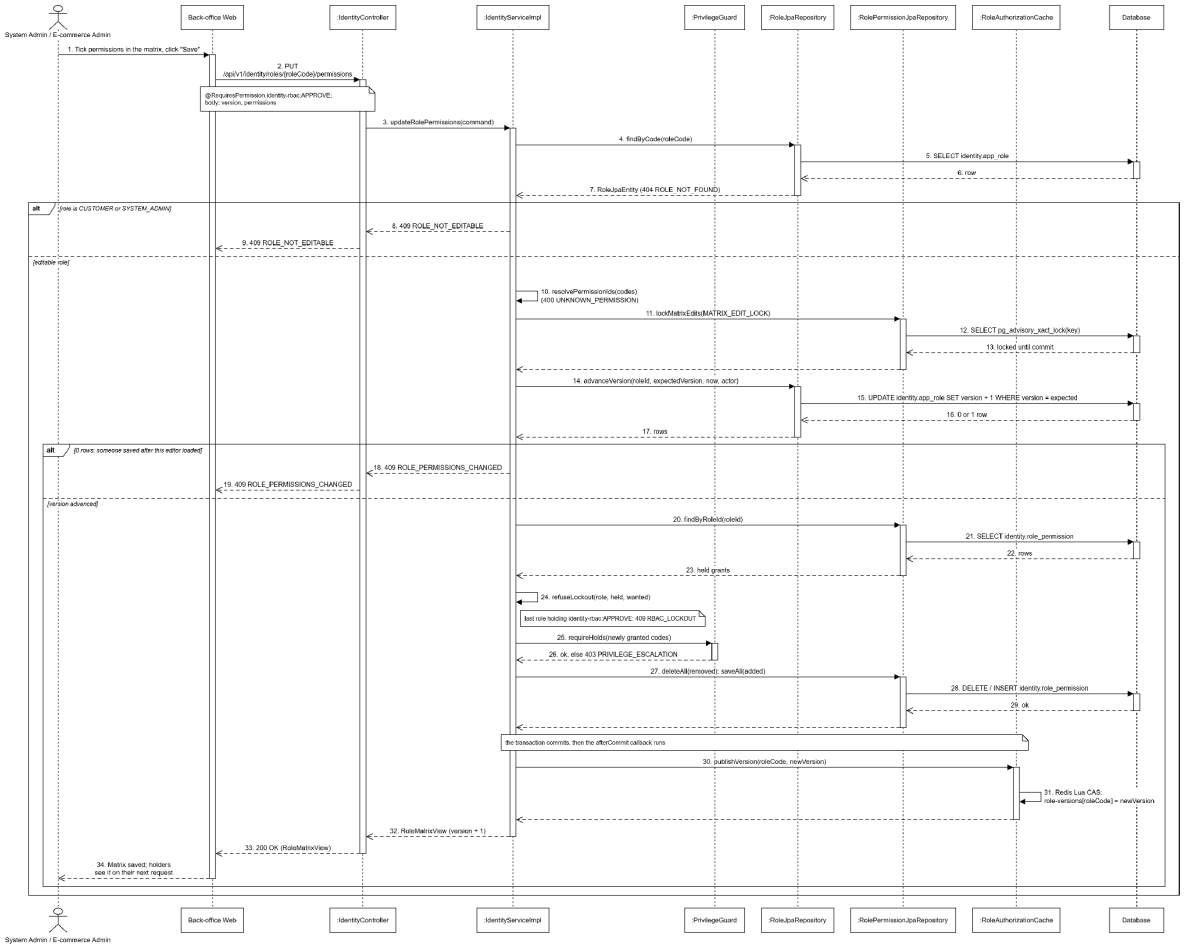


Figure 46. Manage Roles & Permissions (Edit Permission Matrix) Sequence Diagram

3.2.6 Manage Roles & Permissions (Assign Role) Sequence Diagram

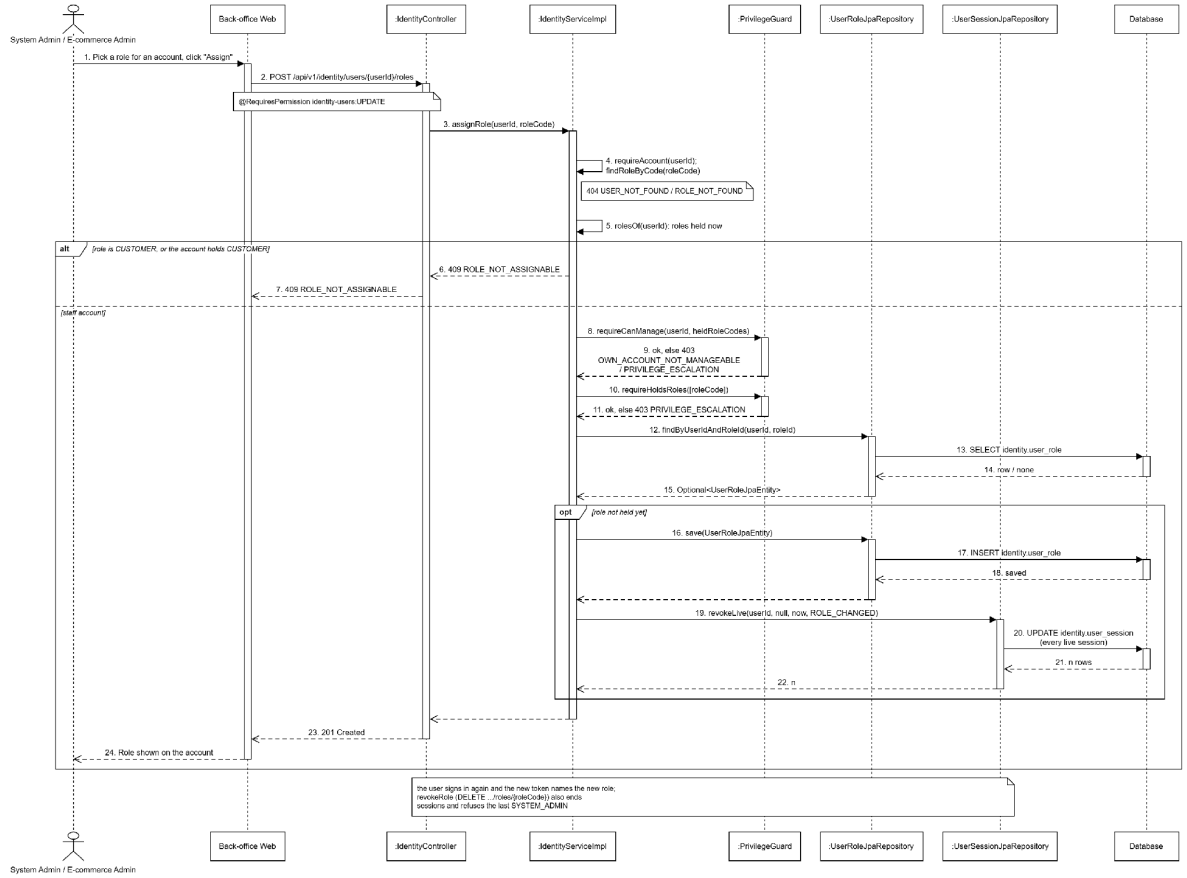


Figure 47. Manage Roles & Permissions (Assign Role) Sequence Diagram

3.2.7 User Account State Diagram

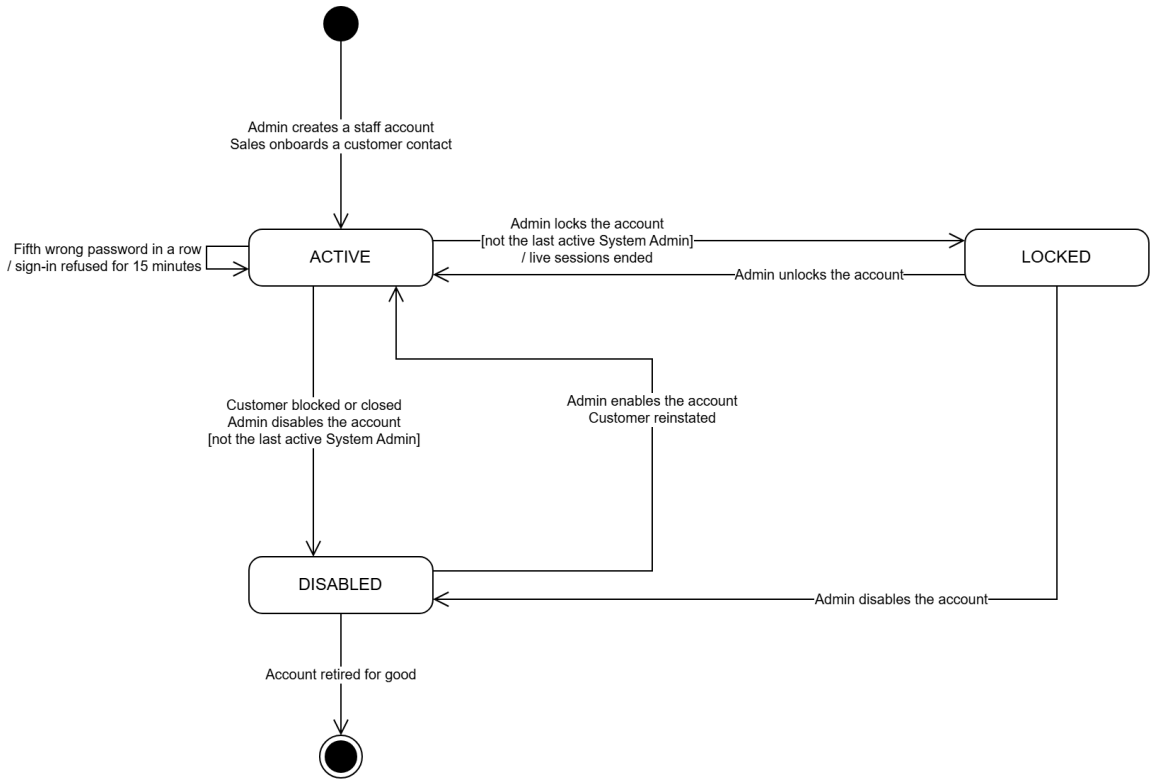


Figure 48. User Account State Diagram

3.3 Notification

3.3.1 Class Diagram

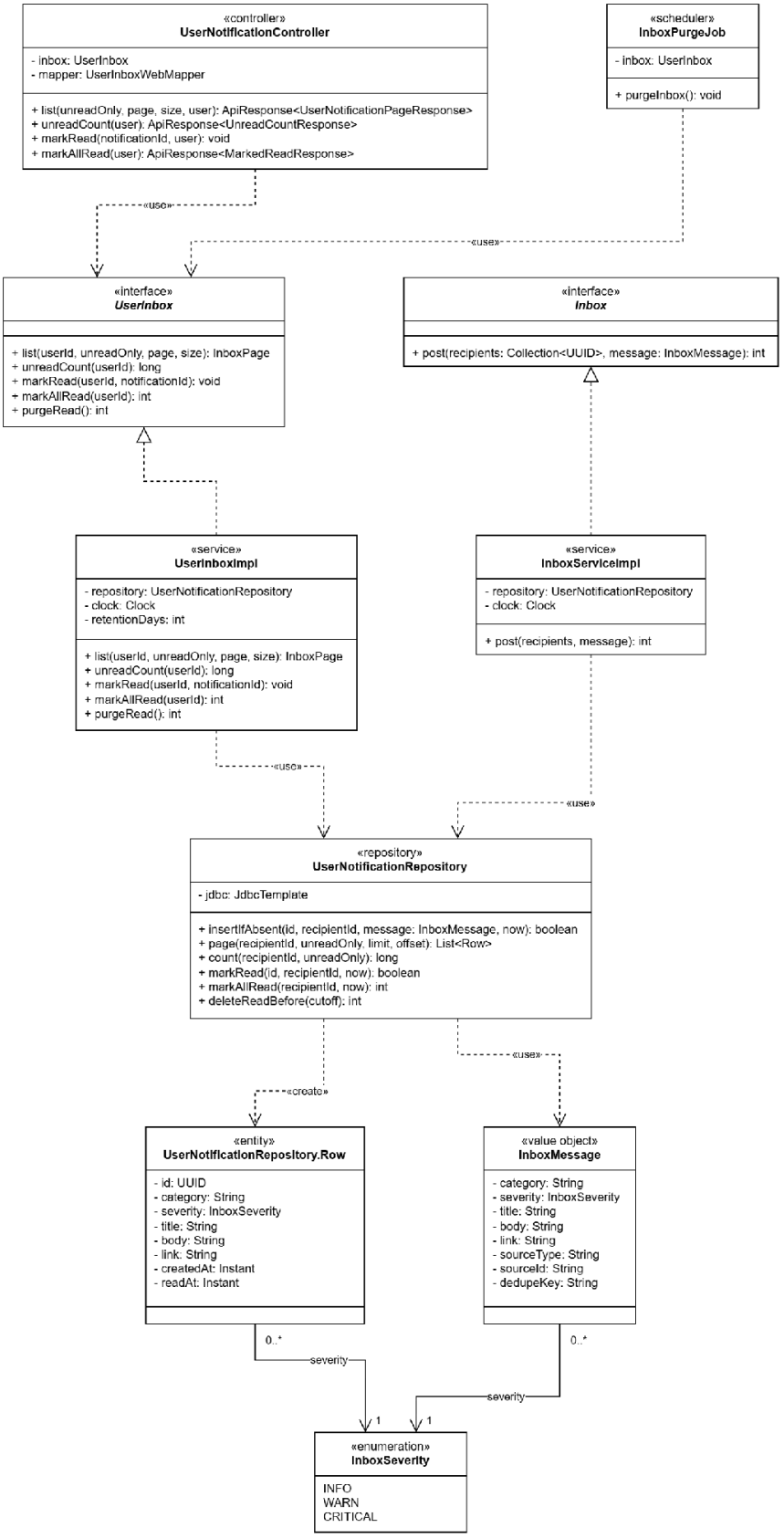


Figure 49. Notification Class Diagram (1/4): In-App Inbox

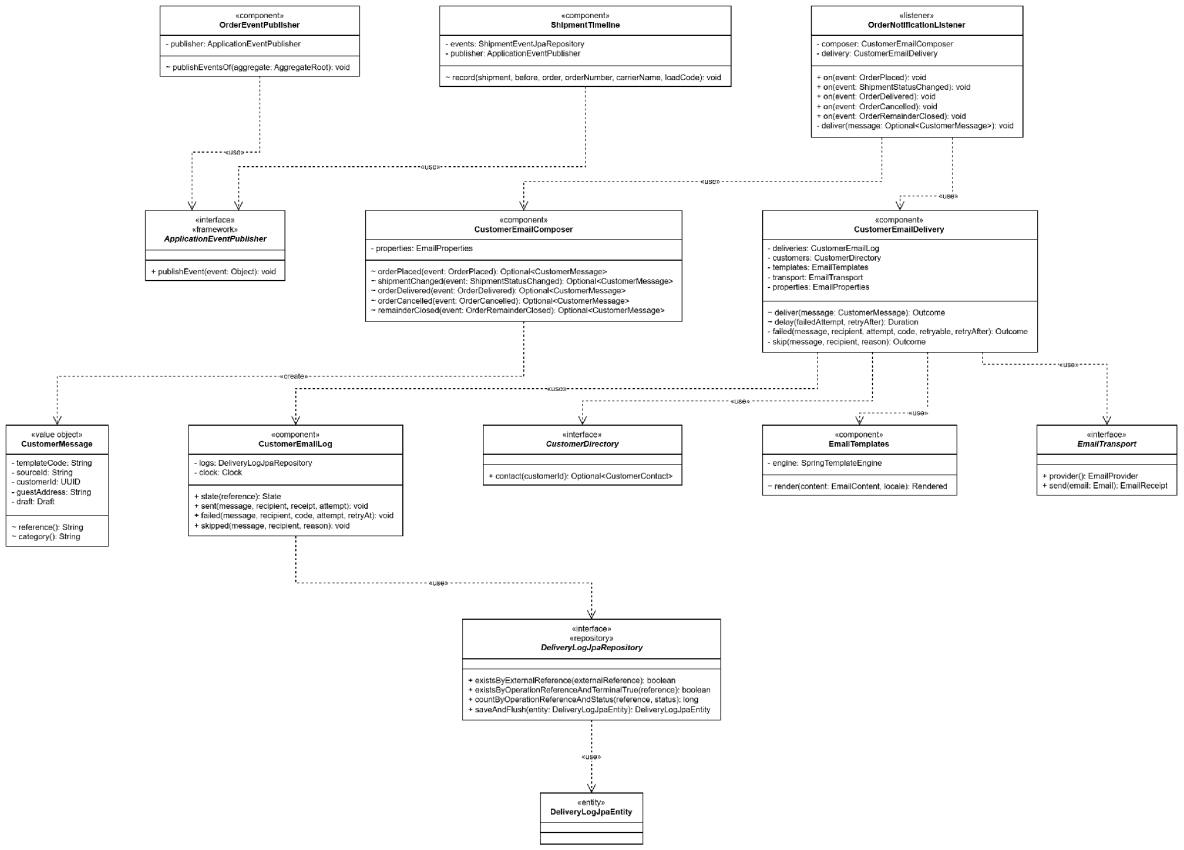


Figure 50. Notification Class Diagram (2/4): Customer Order E-mails

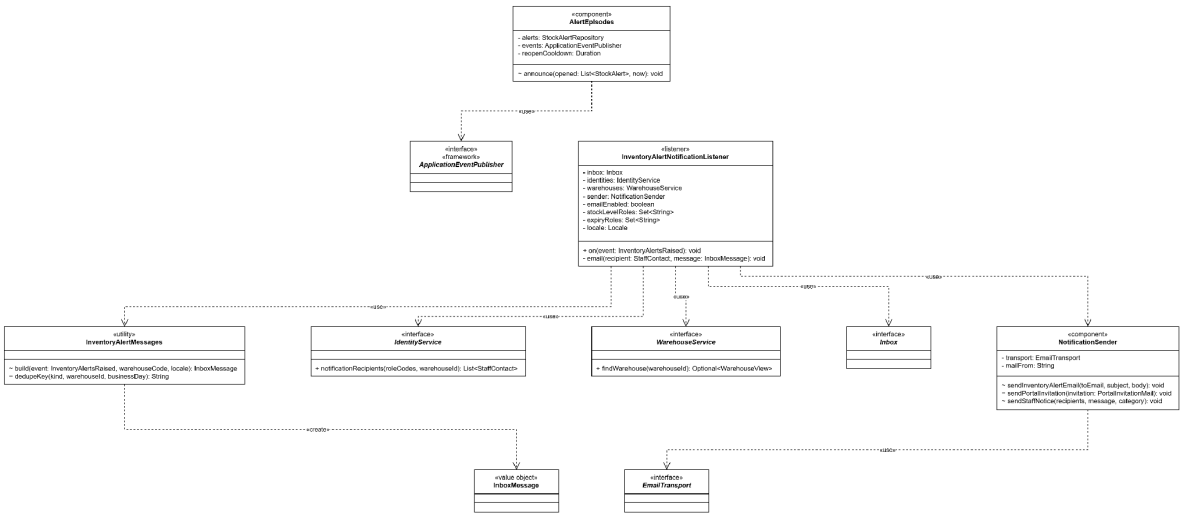


Figure 51. Notification Class Diagram (3/4): Stock Alerts

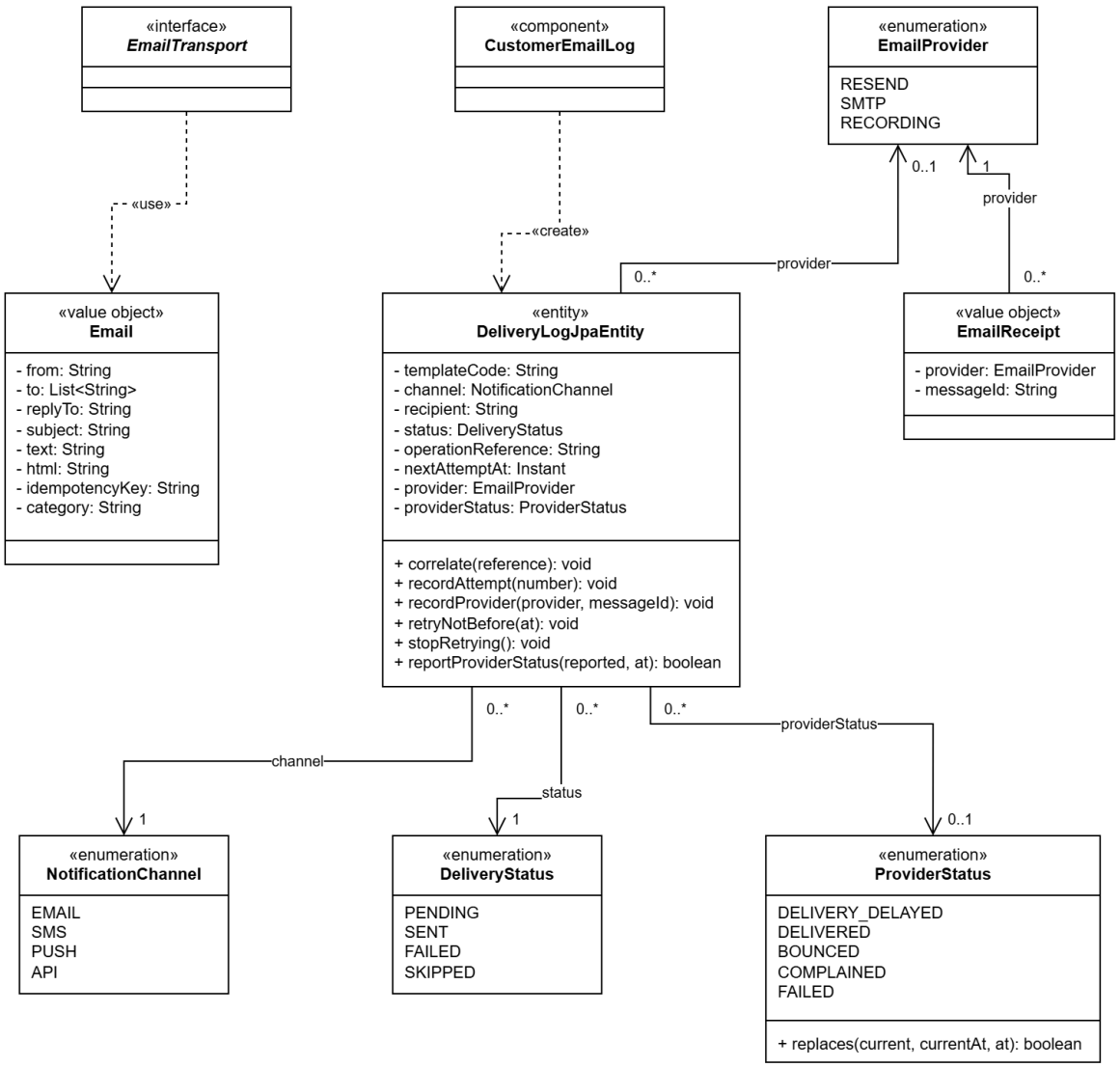


Figure 52. Notification Class Diagram (4/4): E-mail Delivery Model

3.3.2 Class Specification

UserNotificationController

No	Method	Description
1	list(unreadOnly, page, size, user)	GET /api/v1/notifications/me: Pages the caller's own messages newest first (optionally unread only) with the total unread count; needs notification-notifications:READ (scope OWN), rows filtered by the token's user id.
2	unreadCount(user)	GET /api/v1/notifications/me/unread-count: Returns how many of the caller's messages are unread, for polling the bell badge.

3	markRead(notificationId, user)	POST /api/v1/notifications/me/{notificationId}/read: Marks one of the caller's messages read (the first read time is kept) and answers 204; an unknown id or another user's message answers 404 NOTIFICATION_NOT_FOUND; needs notification-notifications:UPDATE.
4	markAllRead(user)	POST /api/v1/notifications/me/read-all: Marks every unread message of the caller read and returns how many were changed.

Table 35. UserNotificationController Class Specification

3.3.3 View Notifications Sequence Diagram

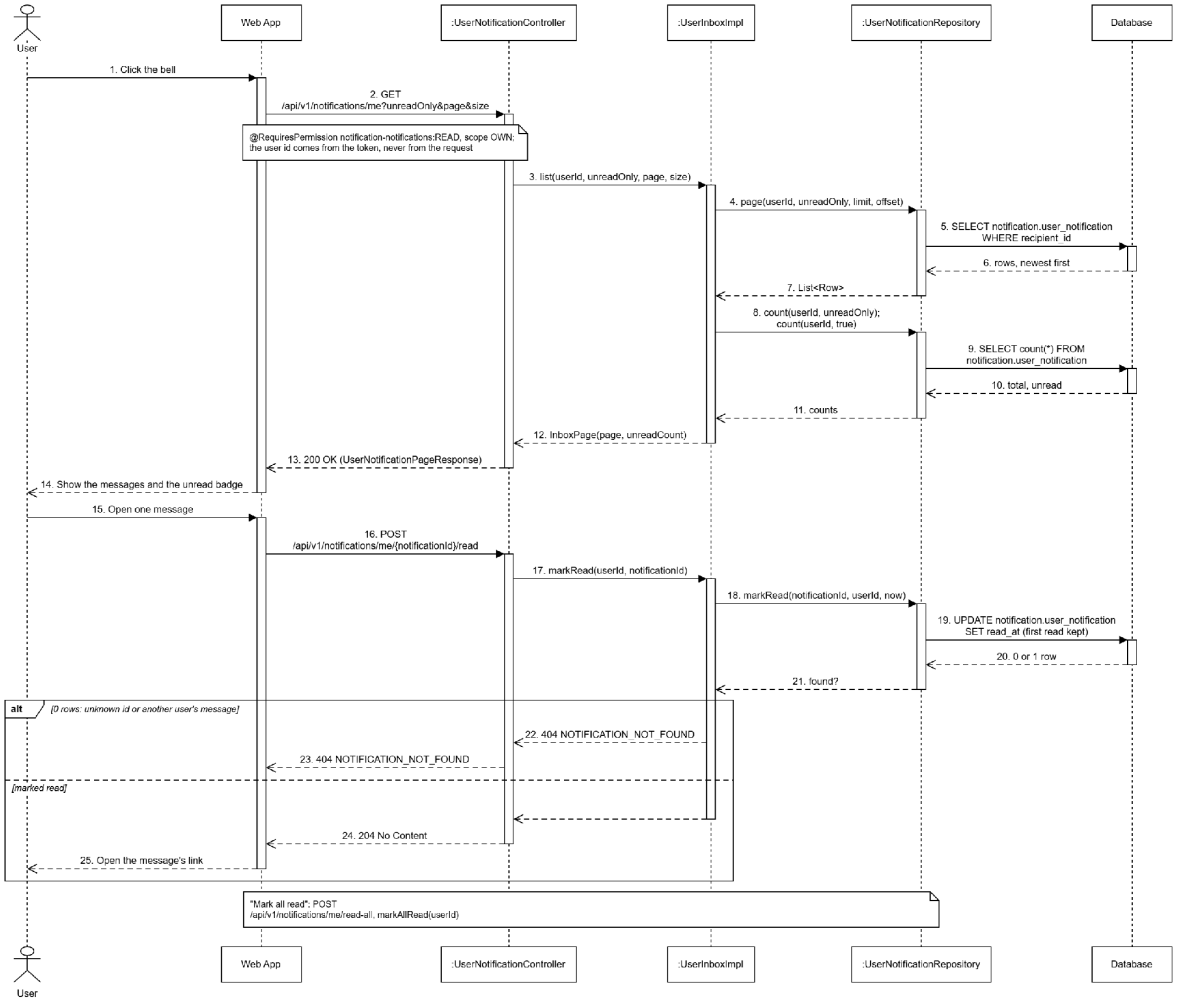


Figure 53. View Notifications Sequence Diagram

3.3.4 Order Milestone Notification Sequence Diagram

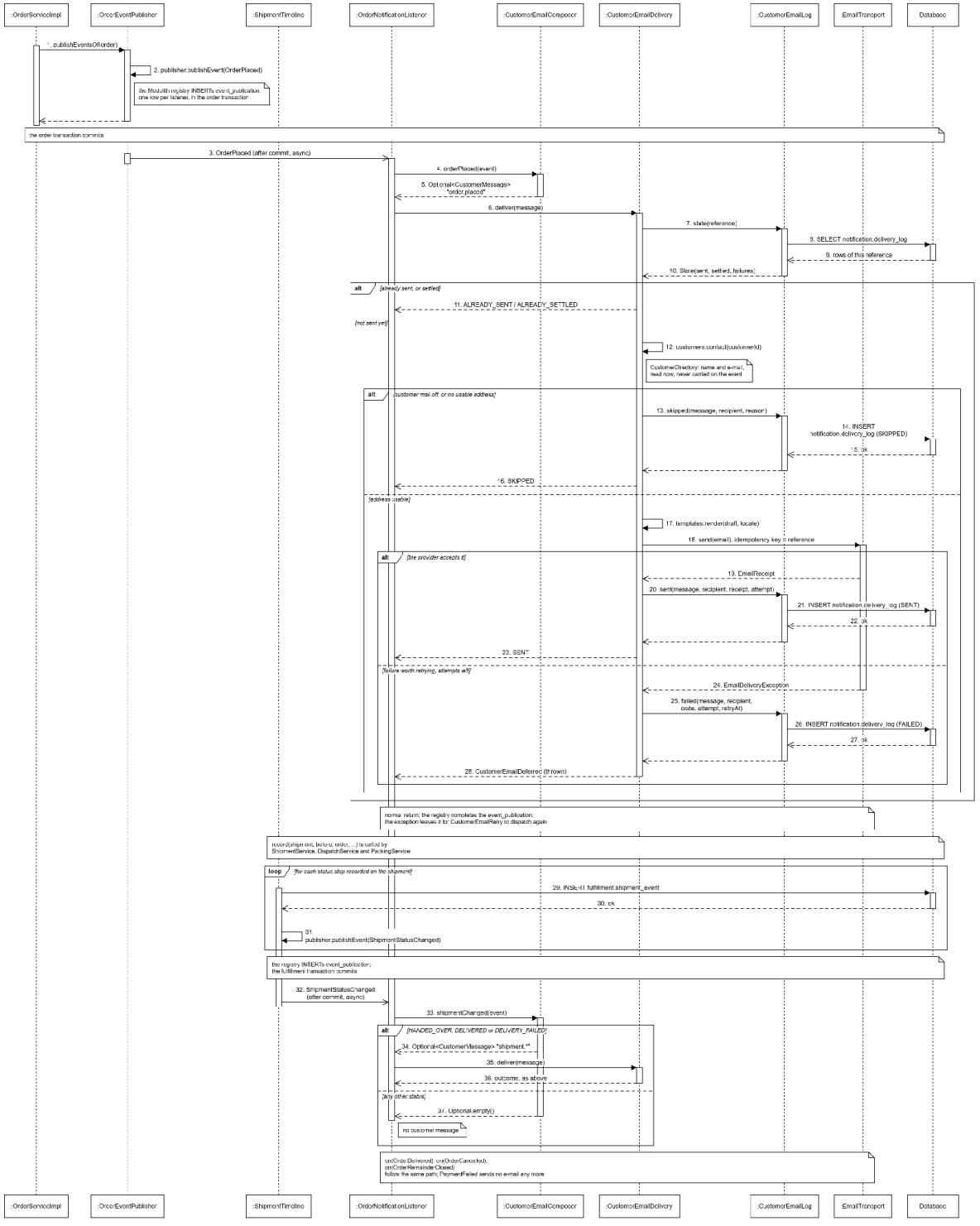


Figure 54. Order Milestone Notification Sequence Diagram

3.3.5 Stock Alert Notification Sequence Diagram

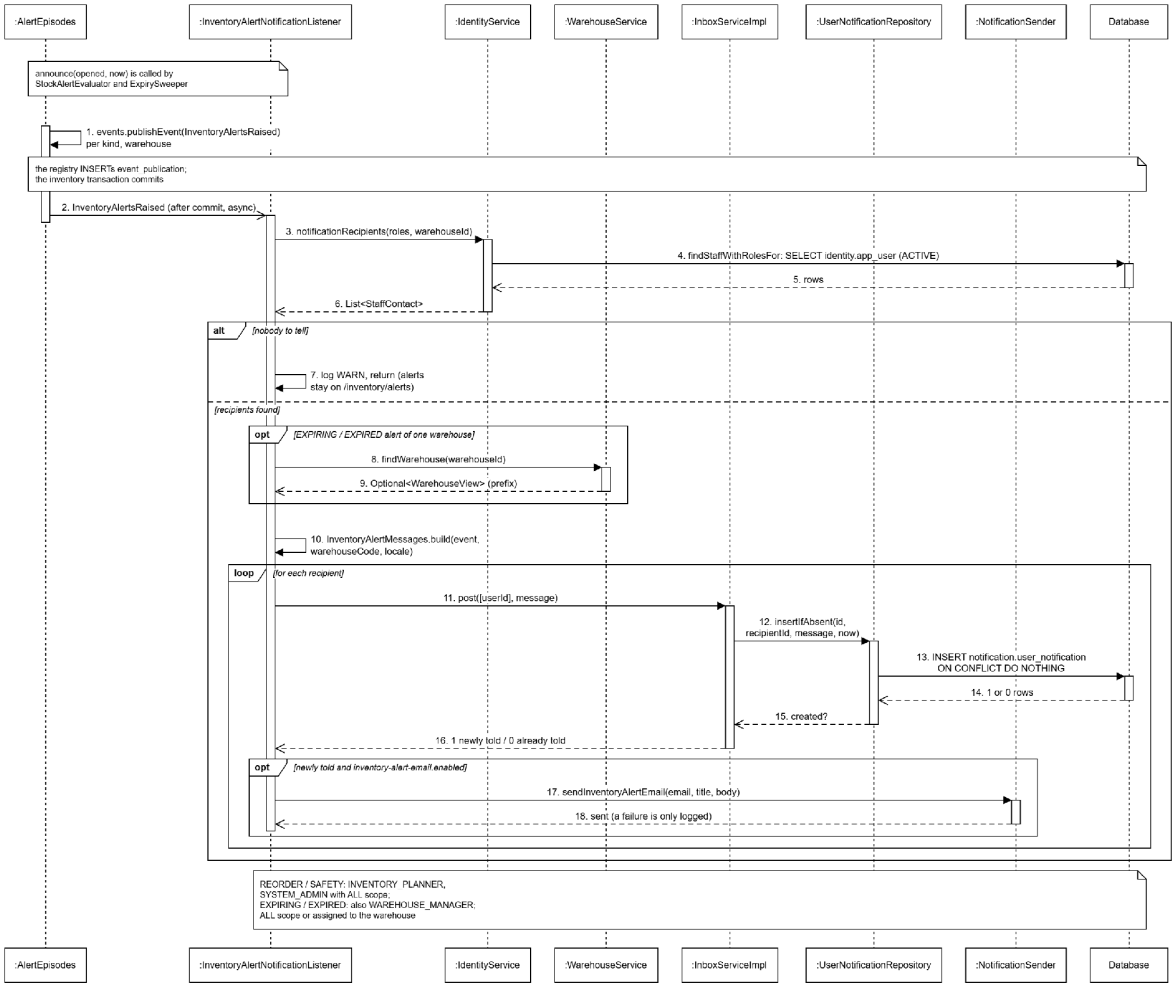


Figure 55. Stock Alert Notification Sequence Diagram

3.4 Customer Management

3.4.1 Class Diagram

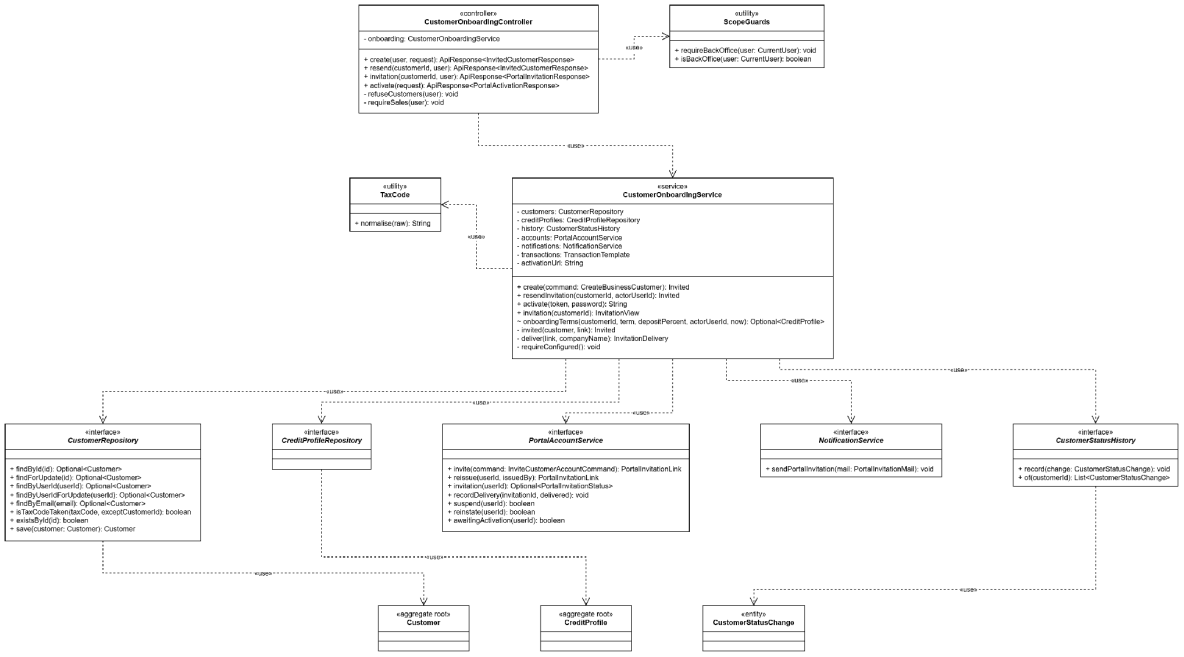


Figure 56. Customer Management Class Diagram (1/4): Onboarding and Portal Invitations

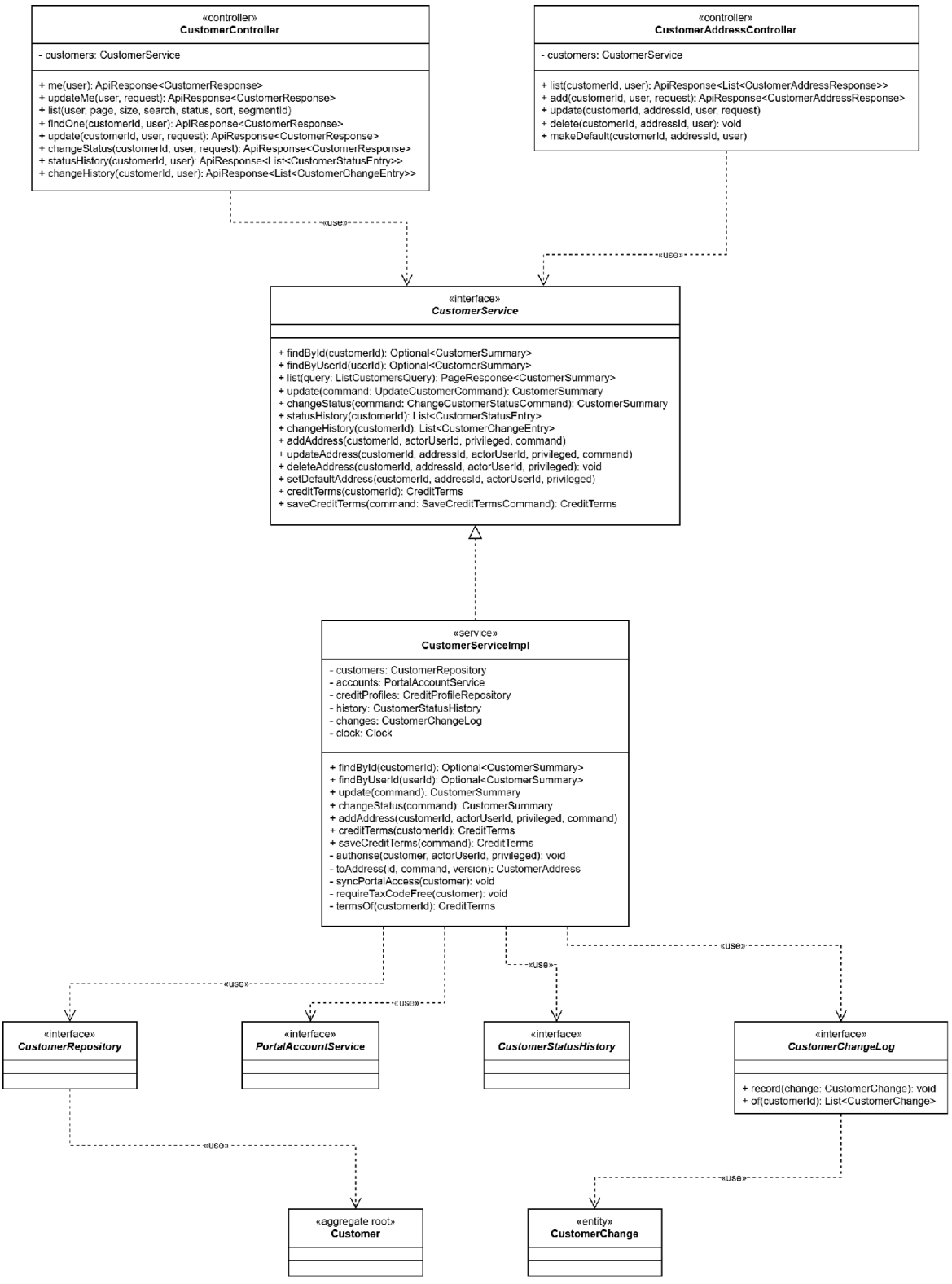


Figure 57. Customer Management Class Diagram (2/4): Profile, Address Book and Status

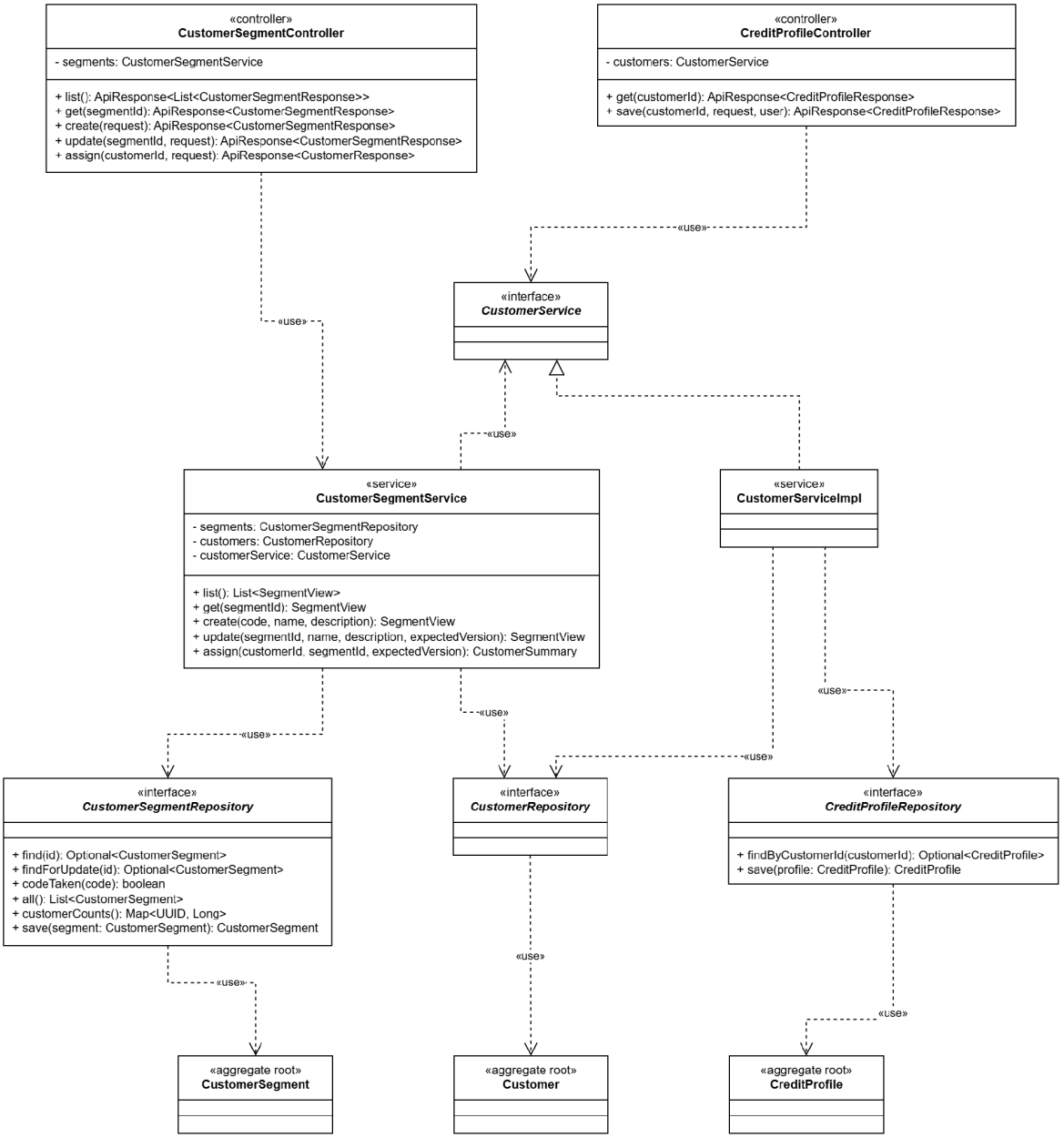


Figure 58. Customer Management Class Diagram (3/4): Customer Groups and Credit Terms

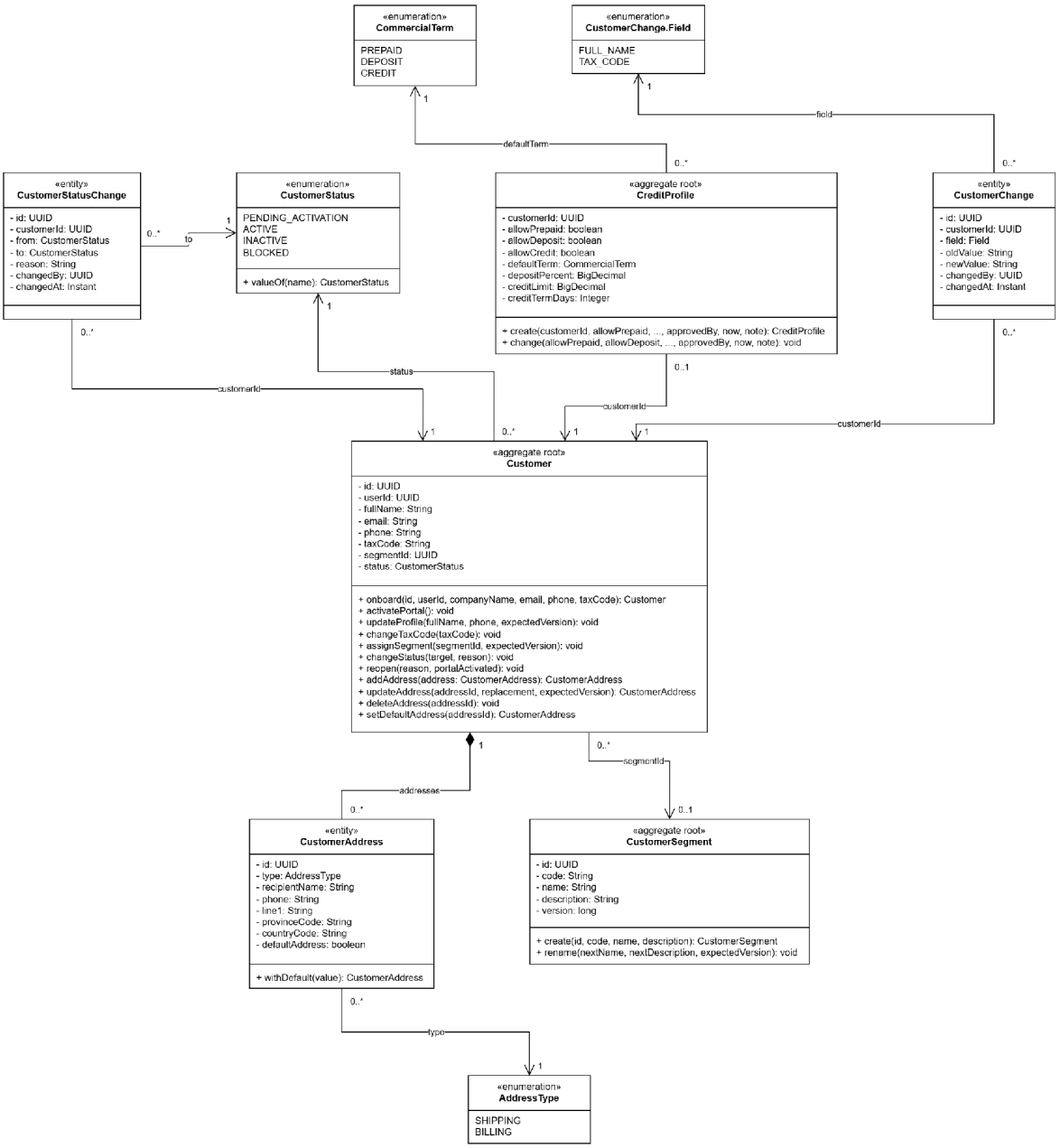


Figure 59. Customer Management Class Diagram (4/4): Domain Model

3.4.2 Class Specification

CustomerOnboardingController

No	Method	Description
1	create(user, request)	POST /api/v1/customers: Sales creates a B2B customer (PENDING_ACTIVATION) plus a password-less CUSTOMER account and e-mails the activation link after commit; needs customer-customers:CREATE, refused to CUSTOMER callers, 409 on a duplicate tax code or e-mail.

2	resend(customerId, user)	POST /api/v1/customers/{customerId}/portal-invitation: Issues a new invitation, which revokes the previous link; only for a customer still pending activation, 429 RATE_LIMITED within the 1-minute cooldown, 409 PORTAL_ACCOUNT_ALREADY_ACTIVE once activated.
3	invitation(customerId, user)	GET /api/v1/customers/{customerId}/portal-invitation: Shows back-office staff the invitation state (PENDING, EXPIRED, ACTIVATED, NO_ACCOUNT, UNAVAILABLE), its expiry and whether the e-mail went out.
4	activate(request)	POST /api/v1/customers/portal-activation: The invited contact redeems the token and chooses a password; the account gets its password and the customer becomes ACTIVE in one transaction; anonymous (the token is the credential), 10 per minute per IP, every unusable link answers 400 PORTAL_INVITATION_INVALID.

Table 36. CustomerOnboardingController Class Specification

CustomerController

No	Method	Description
1	me(user)	GET /api/v1/customers/me: The caller's own profile, masked; 404 CUSTOMER_NOT_FOUND for an account without one (customer-customers:READ). (:50)
2	updateMe(user, request)	PUT /api/v1/customers/me: The portal contact changes company name and phone against the version read (409 OPTIMISTIC_LOCK); a name change is kept in the change history. (:58)
3	list(user, page, size, search, status, sort, segmentId)	GET /api/v1/customers: Back-office page of customers filtered by text q, status and group; a portal caller gets 403 FORBIDDEN (ScopeGuards.requireBackOffice). (:69)
4	findOne(customerId, user)	GET /api/v1/customers/{customerId}: One customer, unmasked, back office only. (:85)
5	update(customerId, user, request)	PUT /api/v1/customers/{customerId}: Staff edit of name and phone, plus correction of the tax code, which stays unique across all statuses (409 CUSTOMER_TAX_CODE_ALREADY_EXISTS); name and tax-code changes are logged. (:94)

6	changeStatus(customerId, user, request)	POST /api/v1/customers/{customerId}/status: Blocks, unblocks, closes or reopens a customer (customer-status:APPROVE); unblock and reopen need a reason, block and close do not; the portal account is suspended or reinstated in the same transaction. (:106)
7	statusHistory(customerId, user)	GET /api/v1/customers/{customerId}/status-history: Status changes with reason, actor and time, back office only. (:115)
8	changeHistory(customerId, user)	GET /api/v1/customers/{customerId}/change-history: Every change of company name and tax code (before, after, who, when), back office only. (:124)

Table 37. CustomerController Class Specification

CustomerAddressController

No	Method	Description
1	list(customerId, user)	GET /api/v1/customers/{customerId}/addresses: The address book; a portal user only its own customer's (404 NOT_FOUND otherwise) (customer-addresses:READ). (:38)
2	add(customerId, user, request)	POST /api/v1/customers/{customerId}/addresses: Adds a DELIVERY or BILLING address, Vietnamese only (400 VALIDATION_FAILED); the first of a type becomes its default (201). (:47)
3	update(customerId, addressId, user, request)	PUT /api/v1/customers/{customerId}/addresses/{addressId} : Replaces an address against its version (409 OPTIMISTIC_LOCK); each type keeps exactly one default. (:56)
4	delete(customerId, addressId, user)	DELETE /api/v1/customers/{customerId}/addresses/{addressId} : Removes an address (204); if it was the default, another of its type becomes default. (:67)
5	makeDefault(customerId, addressId, user)	POST /api/v1/customers/{customerId}/addresses/{addressId} /default: Makes the address its type's default and clears the previous one. (:74)

Table 38. CustomerAddressController Class Specification

CreditProfileController

No	Method	Description

1	get(customerId)	GET /api/v1/customers/{customerId}/credit-profile: Allowed payment terms, default term, deposit %, credit limit, days to pay and version; "prepaid only" when none is set (customer-credit-profiles:READ). (:48)
2	save(customerId, request, user)	PUT /api/v1/customers/{customerId}/credit-profile: Sets or changes the terms (customer-credit-profiles:APPROVE: Accountant, System Admin, V20261013000400:111); expectedVersion required once terms exist (400), 409 OPTIMISTIC_LOCK when stale, 400 CREDIT_PROFILE_INVALID for inconsistent terms; records approver and time. (:55)

Table 39. CreditProfileController Class Specification

CustomerSegmentController

No	Method	Description
1	list()	GET /api/v1/customer-segments: All customer groups with member counts (customer-segments:READ: E-commerce Admin, Sales, Order Coordinator, System Admin). (:47)
2	get(segmentId)	GET /api/v1/customer-segments/{segmentId}: One group with its member count; 404 SEGMENT_NOT_FOUND. (:53)
3	create(request)	POST /api/v1/customer-segments: Creates a group whose upper-case code is unique (409 CONFLICT if taken) (customer-segments:CREATE: E-commerce Admin, System Admin). (:60)
4	update(segmentId, request)	PUT /api/v1/customer-segments/{segmentId}: Renames a group and its description against its version (409 OPTIMISTIC_LOCK); the code never changes. (:66)
5	assign(customerId, request)	PUT /api/v1/customers/{customerId}/segment: Puts a customer in a group, or in none (segmentId null), against the customer's version (customer-segments:UPDATE). (:75)

Table 40. CustomerSegmentController Class Specification

3.4.3 Create B2B Customer Profile Sequence Diagram

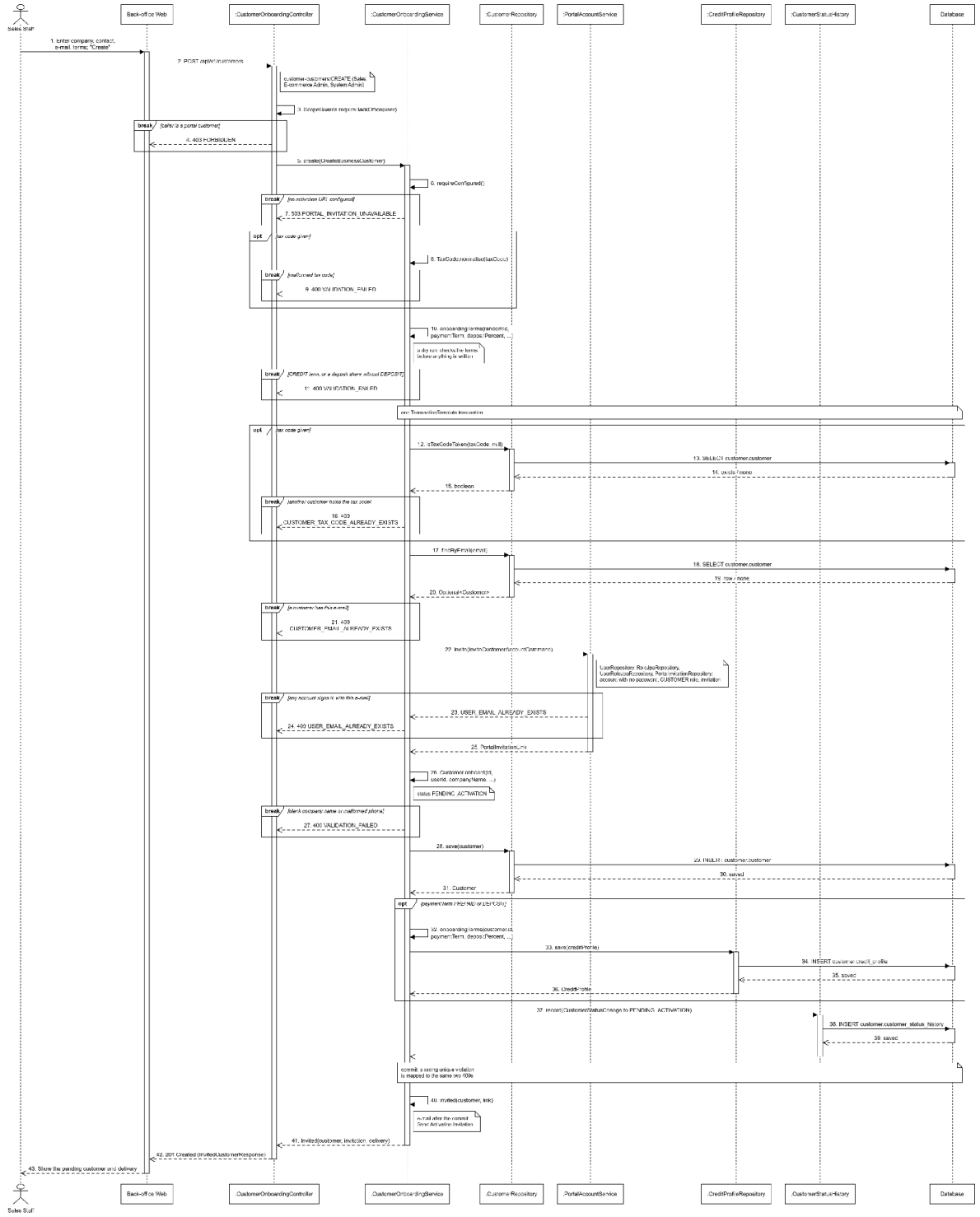


Figure 60. Create B2B Customer Profile Sequence Diagram

3.4.4 Send Activation Invitation Sequence Diagram

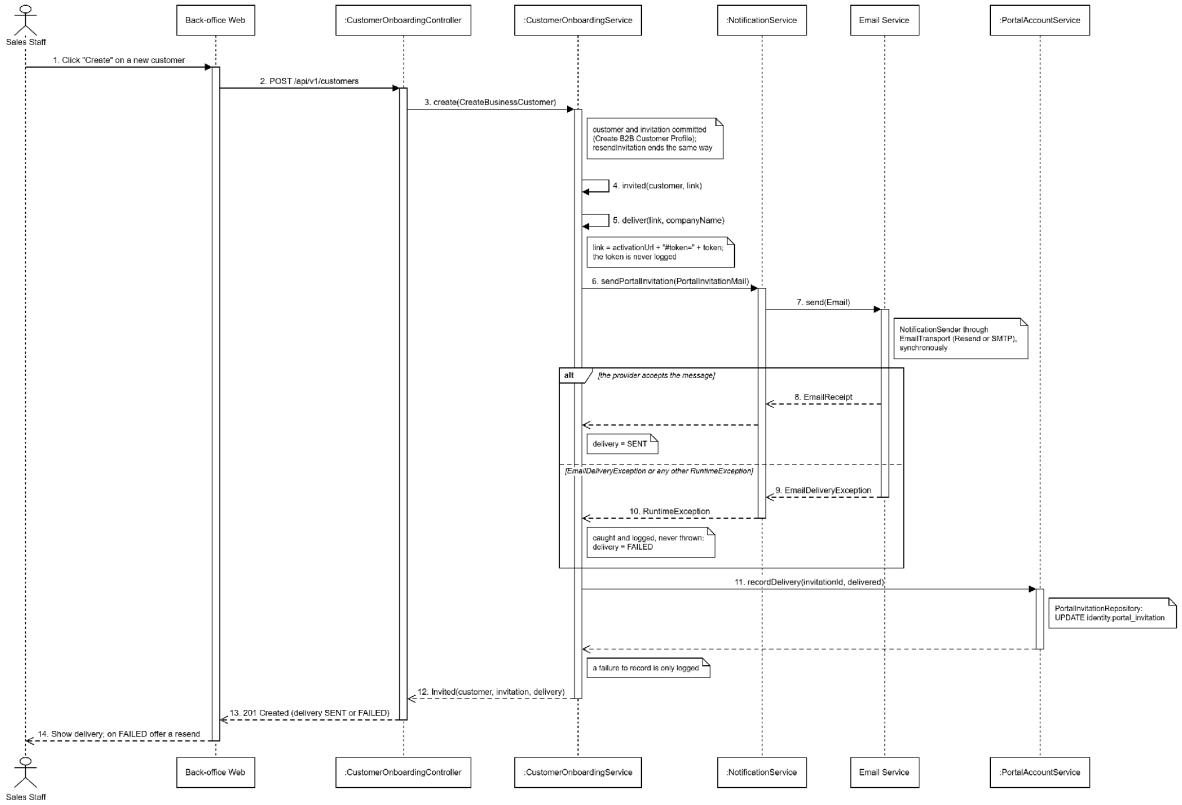


Figure 61. Send Activation Invitation Sequence Diagram

3.4.5 Resend Activation Invitation Sequence Diagram

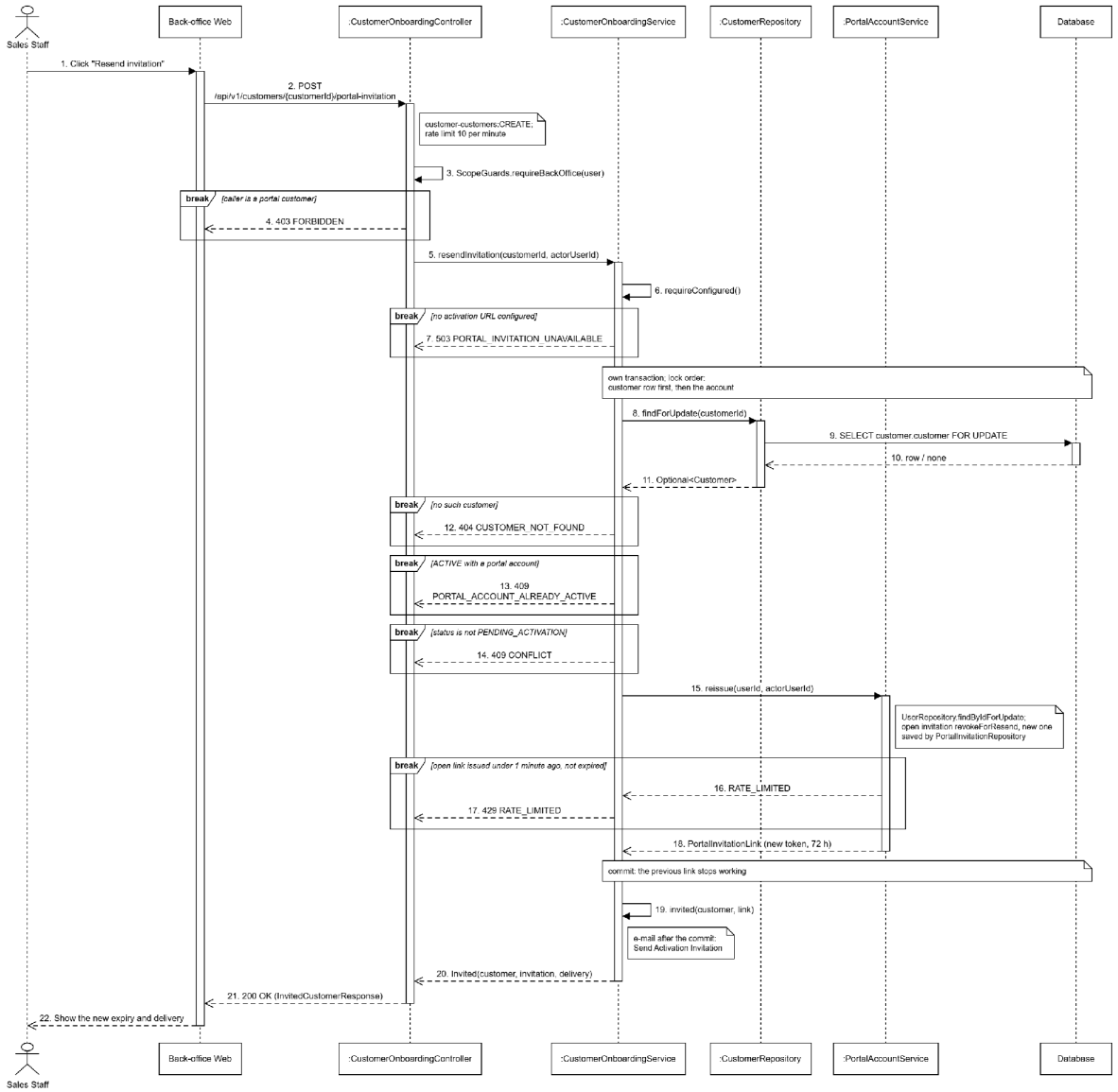


Figure 62. Resend Activation Invitation Sequence Diagram

3.4.6 Manage Company Profile & Address Book Sequence Diagram

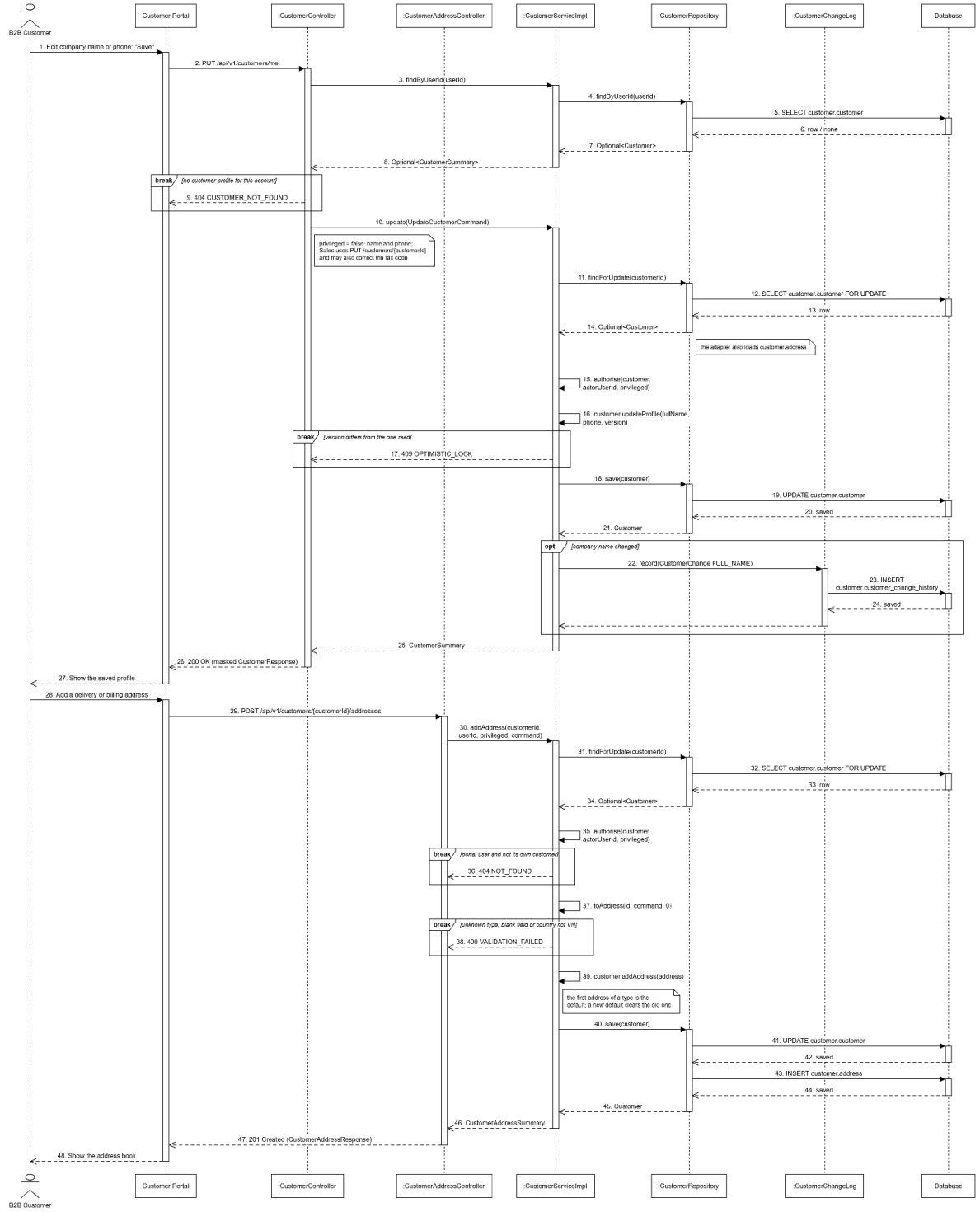


Figure 63. Manage Company Profile & Address Book Sequence Diagram

3.4.7 Change Customer Status Sequence Diagram

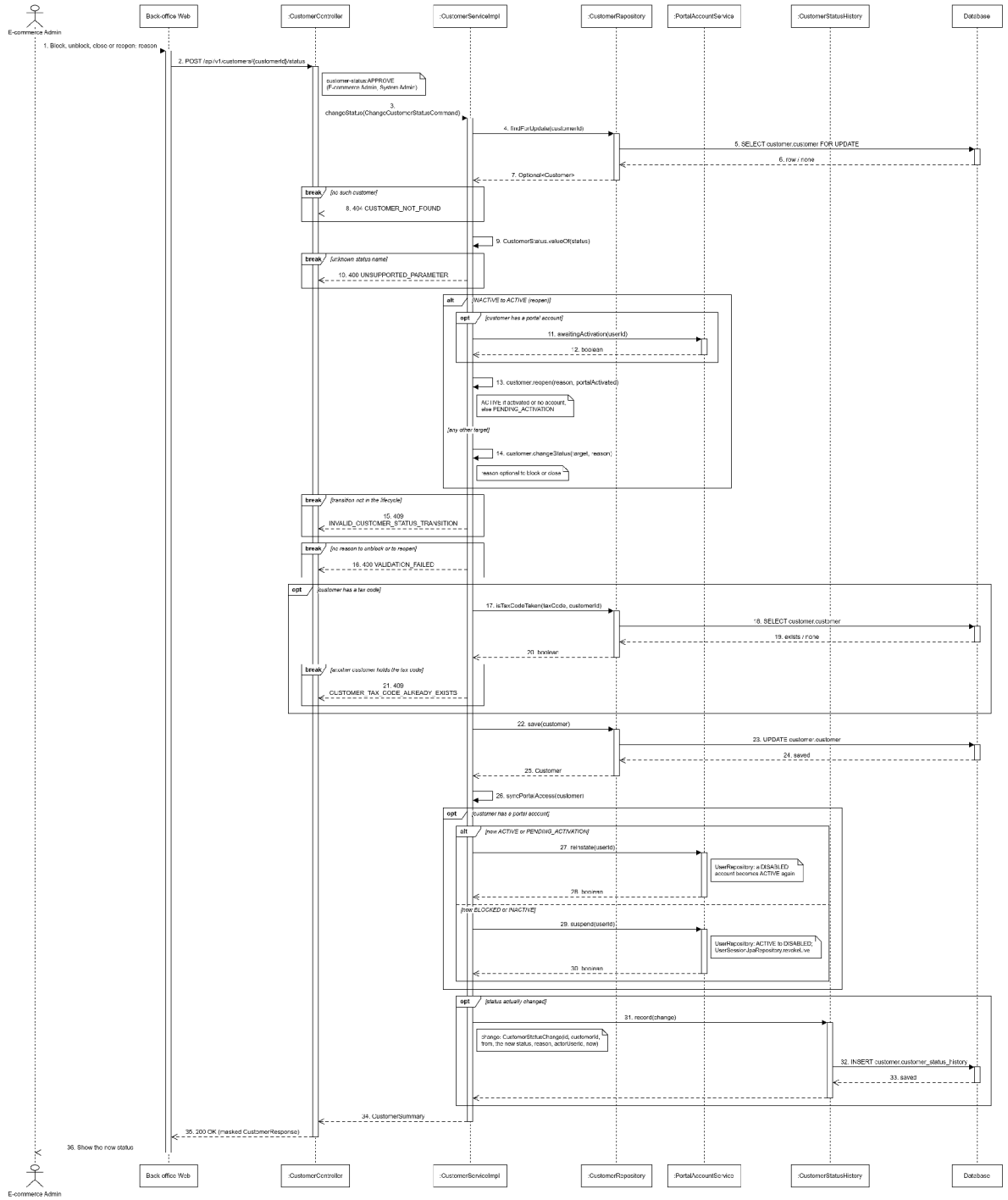


Figure 64. Change Customer Status Sequence Diagram

3.4.8 Manage Customer Groups Sequence Diagram

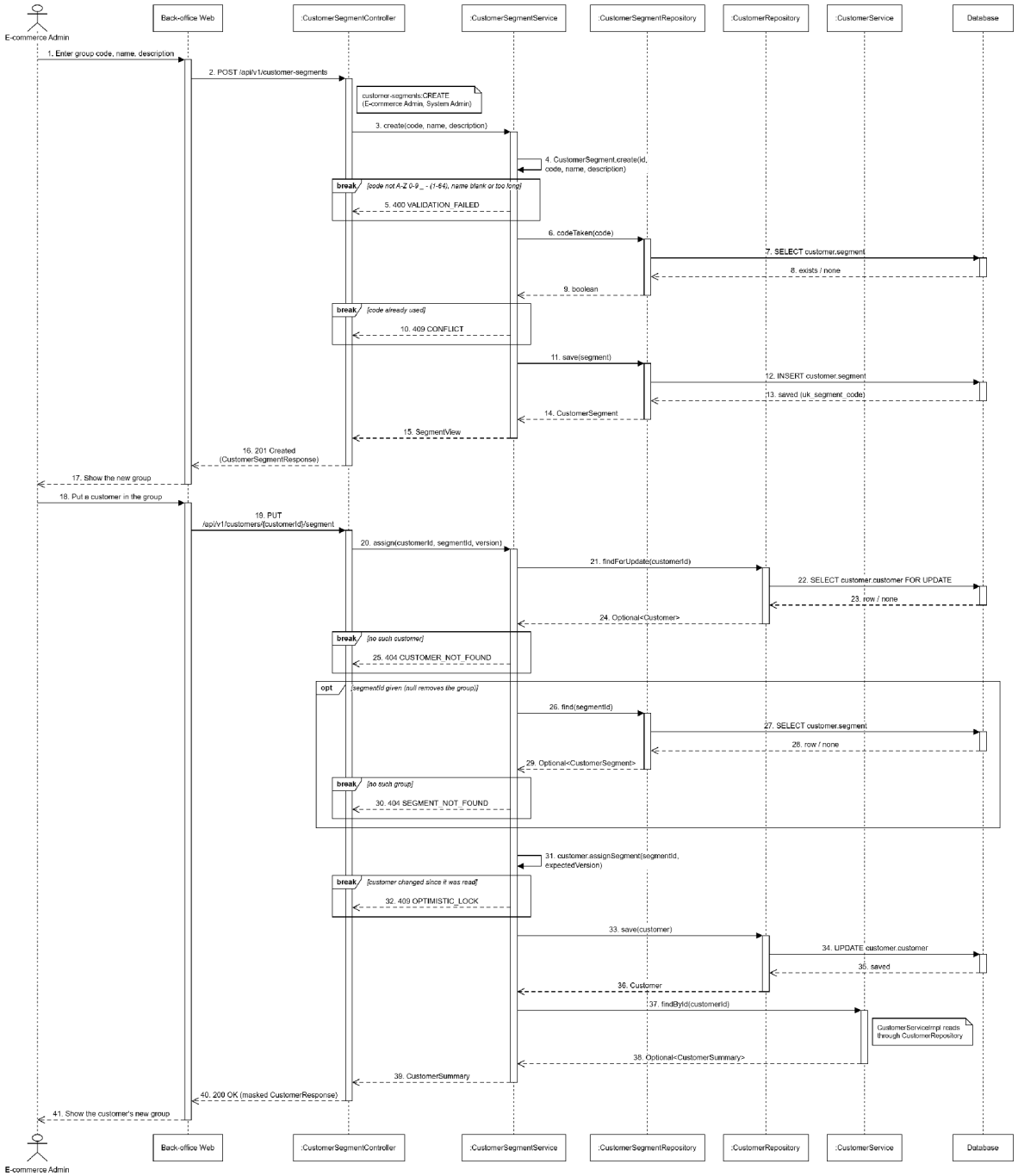


Figure 65. Manage Customer Groups Sequence Diagram

3.4.9 Approve Payment Terms & Credit Limit Sequence Diagram

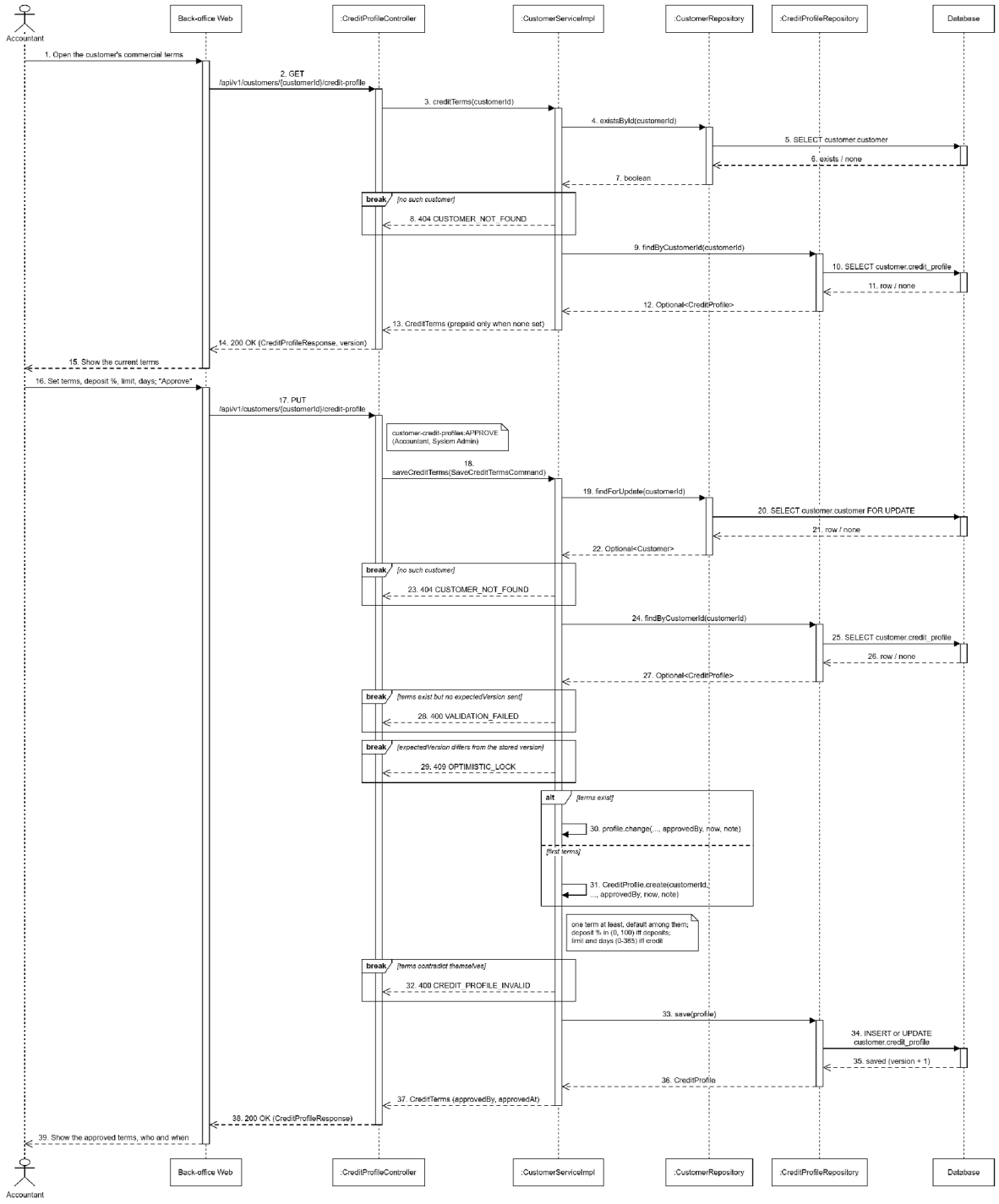


Figure 66. Approve Payment Terms & Credit Limit Sequence Diagram

3.4.10 Customer State Diagram

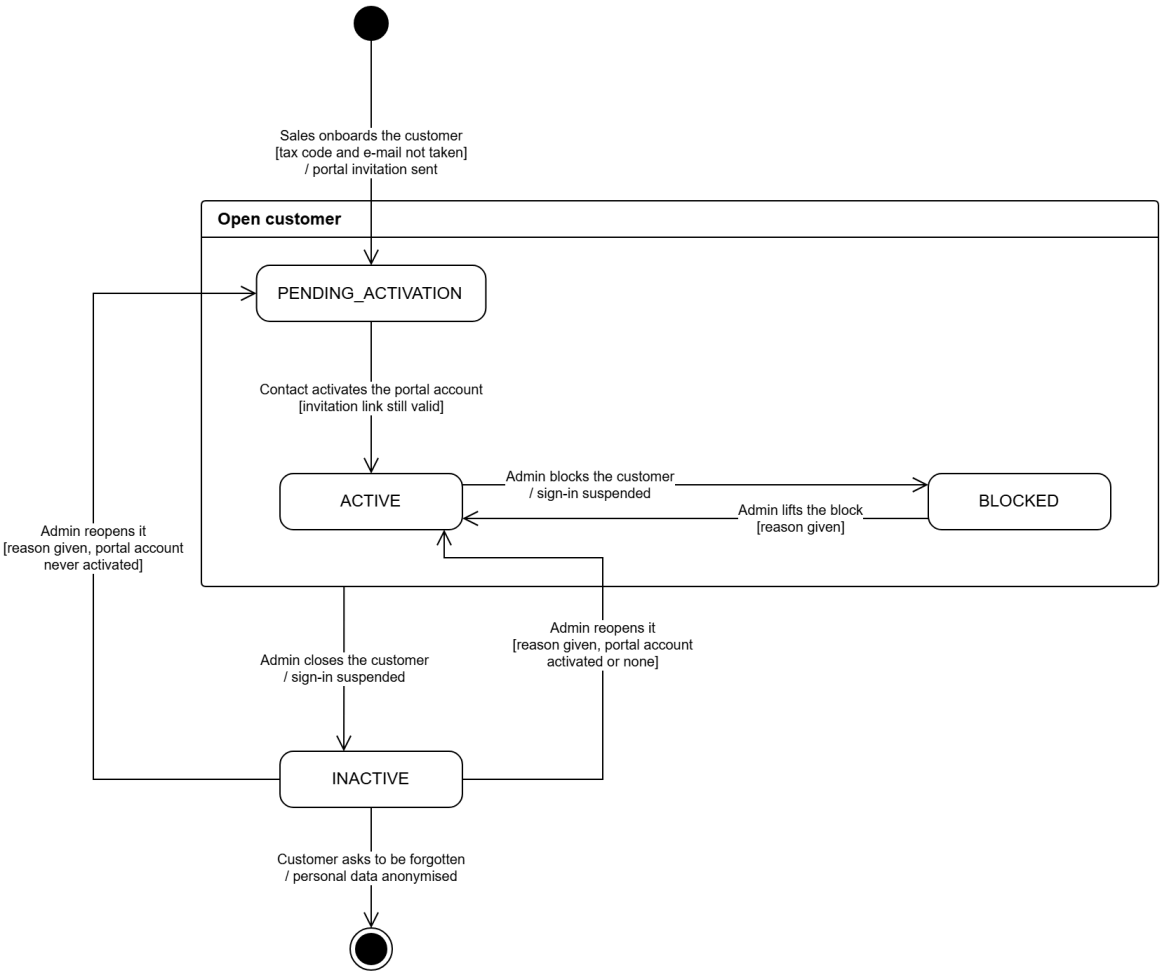


Figure 67. Customer State Diagram

3.5 Product Management

3.5.1 Class Diagram

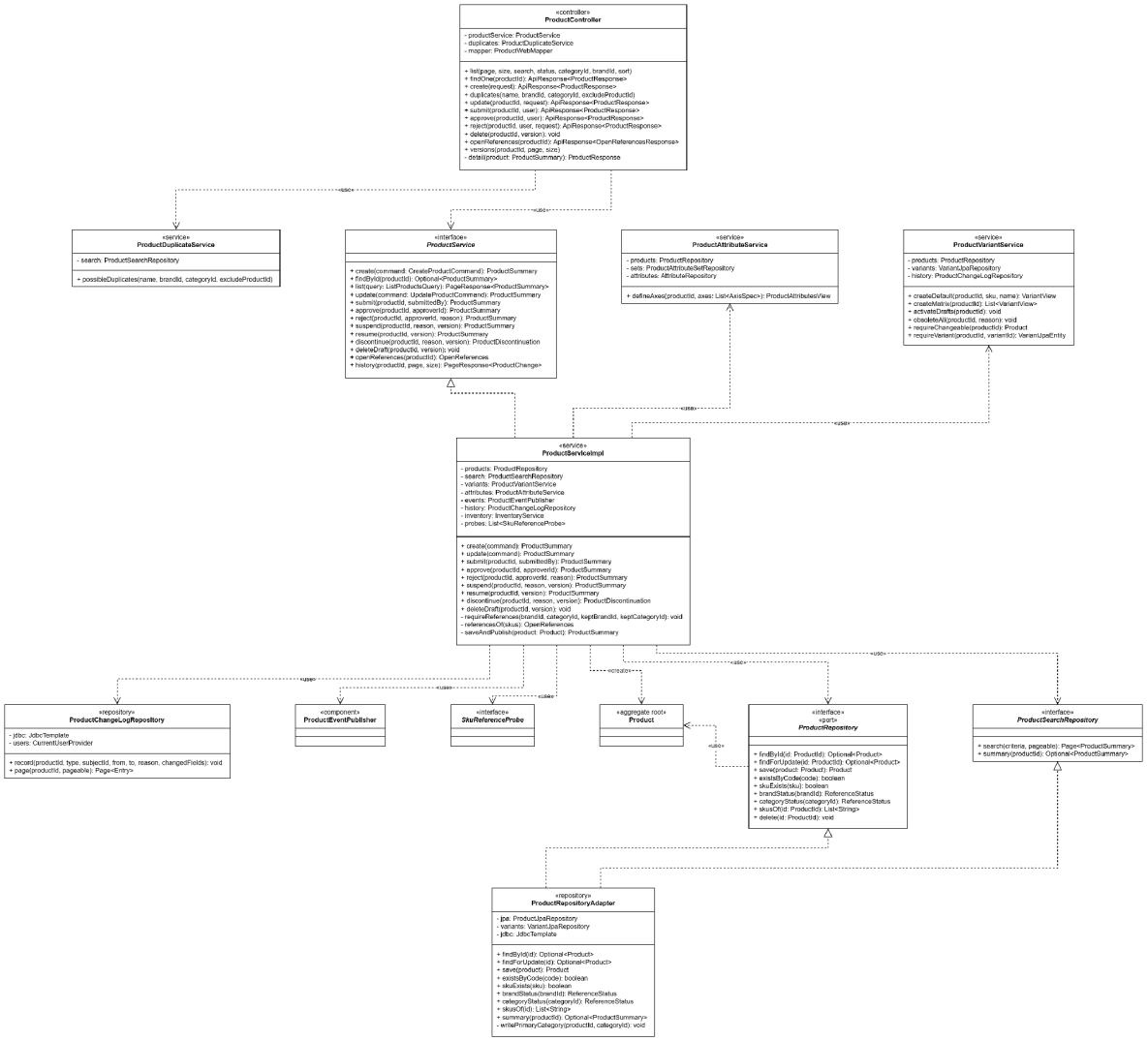


Figure 68. Product Management Class Diagram (1/6): Create, Edit and Approve Products

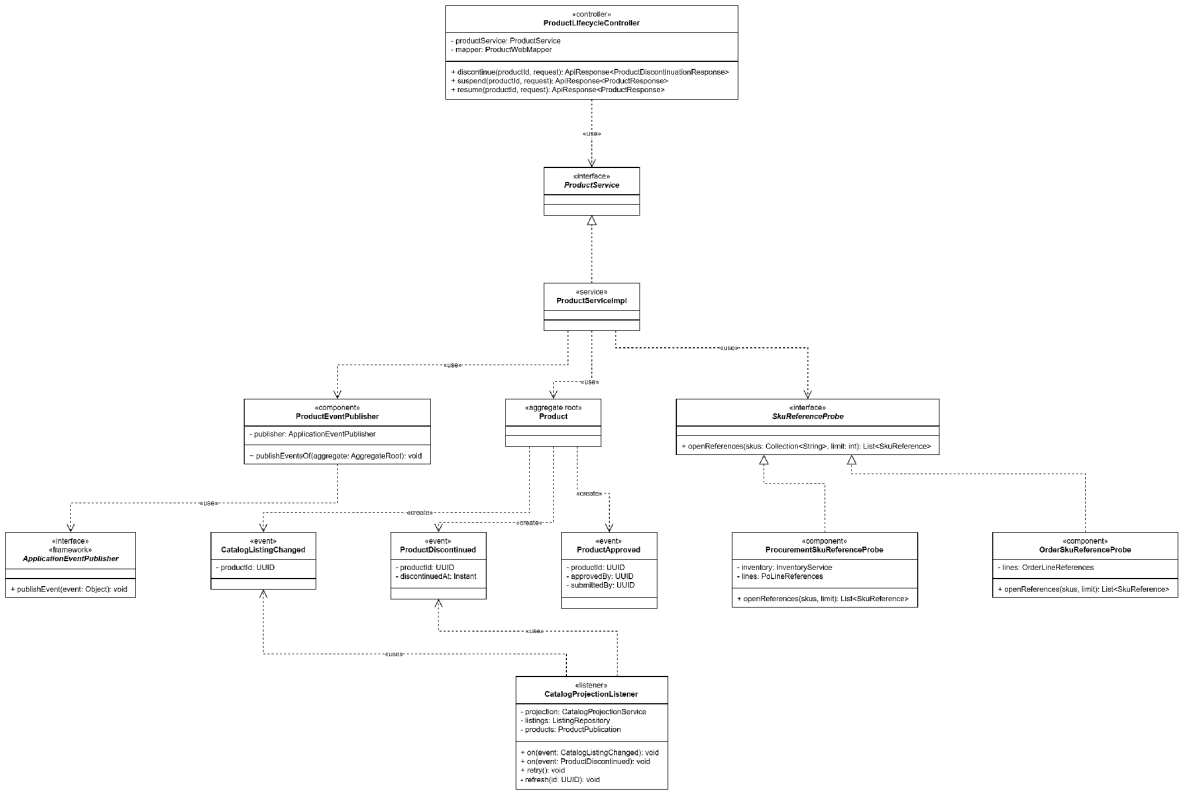


Figure 69. Product Management Class Diagram (2/6): Suspend, Resume, Discontinue and Catalog Events

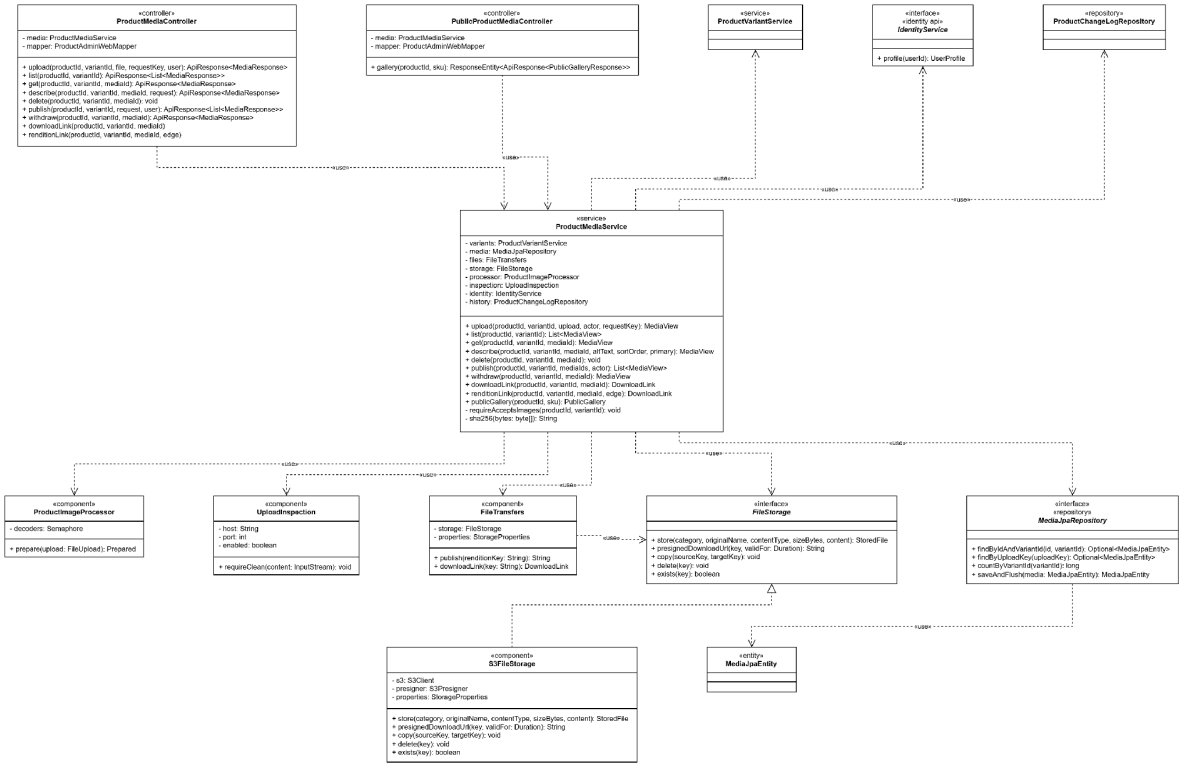


Figure 70. Product Management Class Diagram (3/6): Product Images

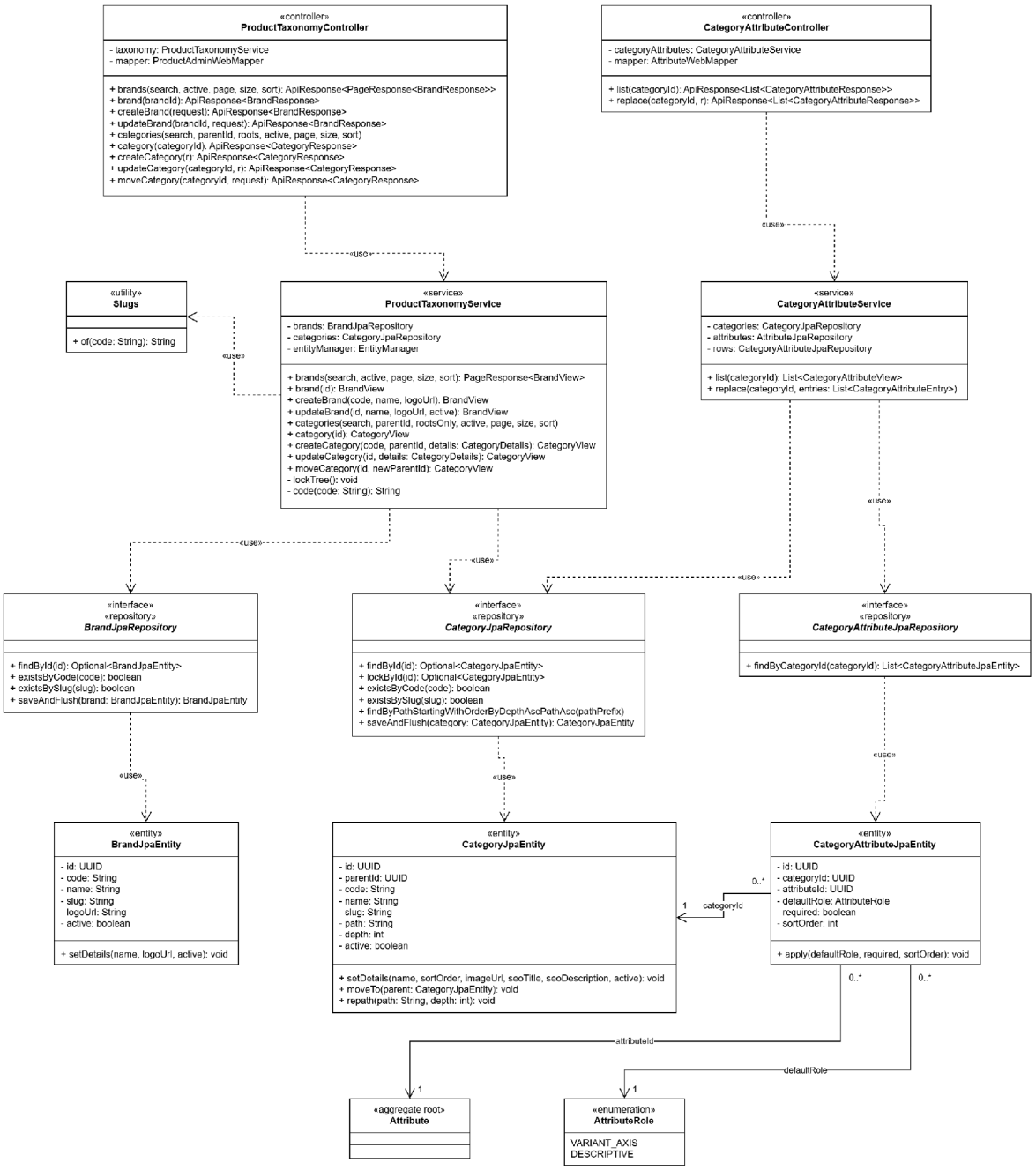


Figure 71. Product Management Class Diagram (4/6): Categories and Brands

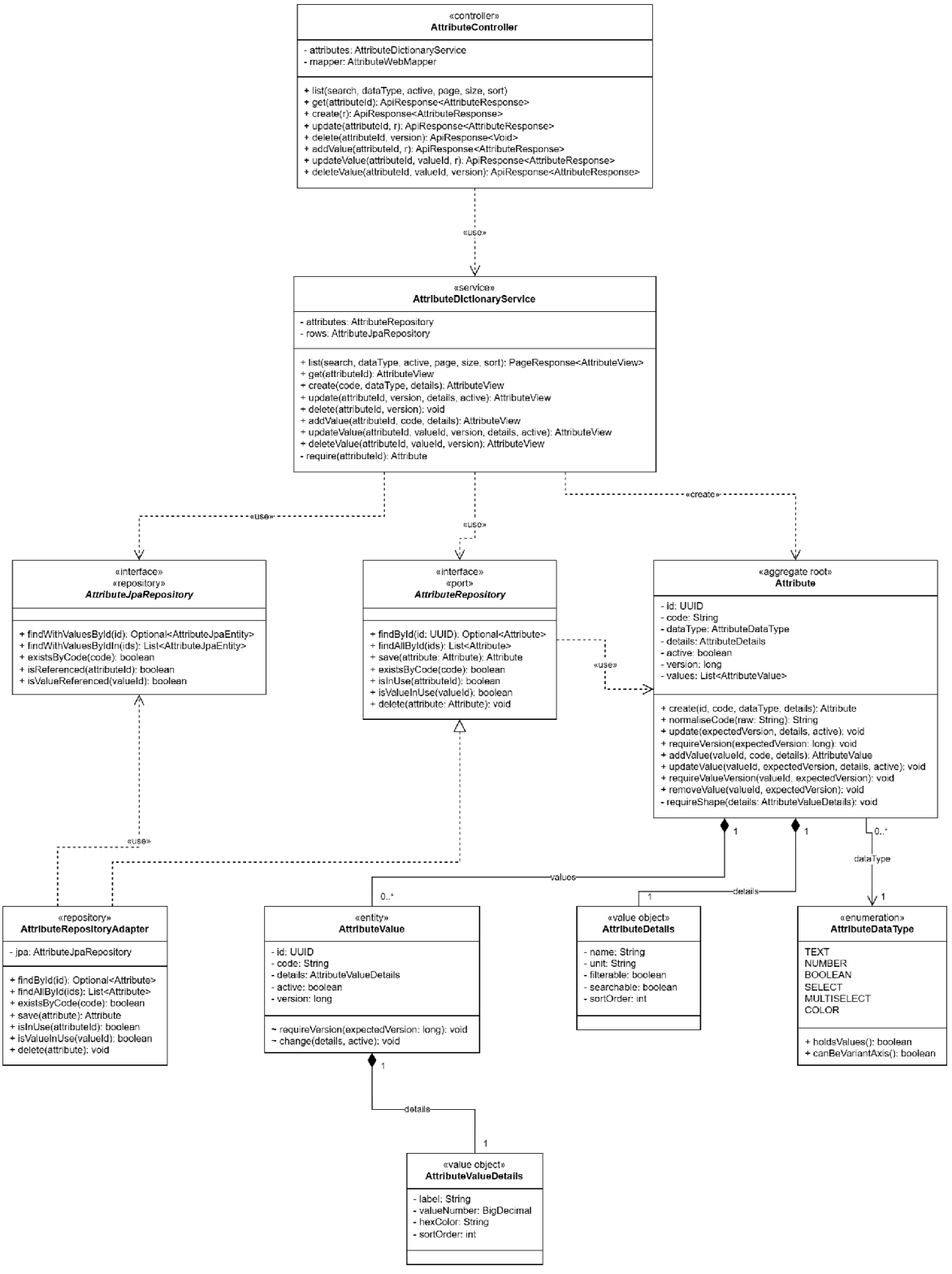


Figure 72. Product Management Class Diagram (5/6): Attribute Dictionary

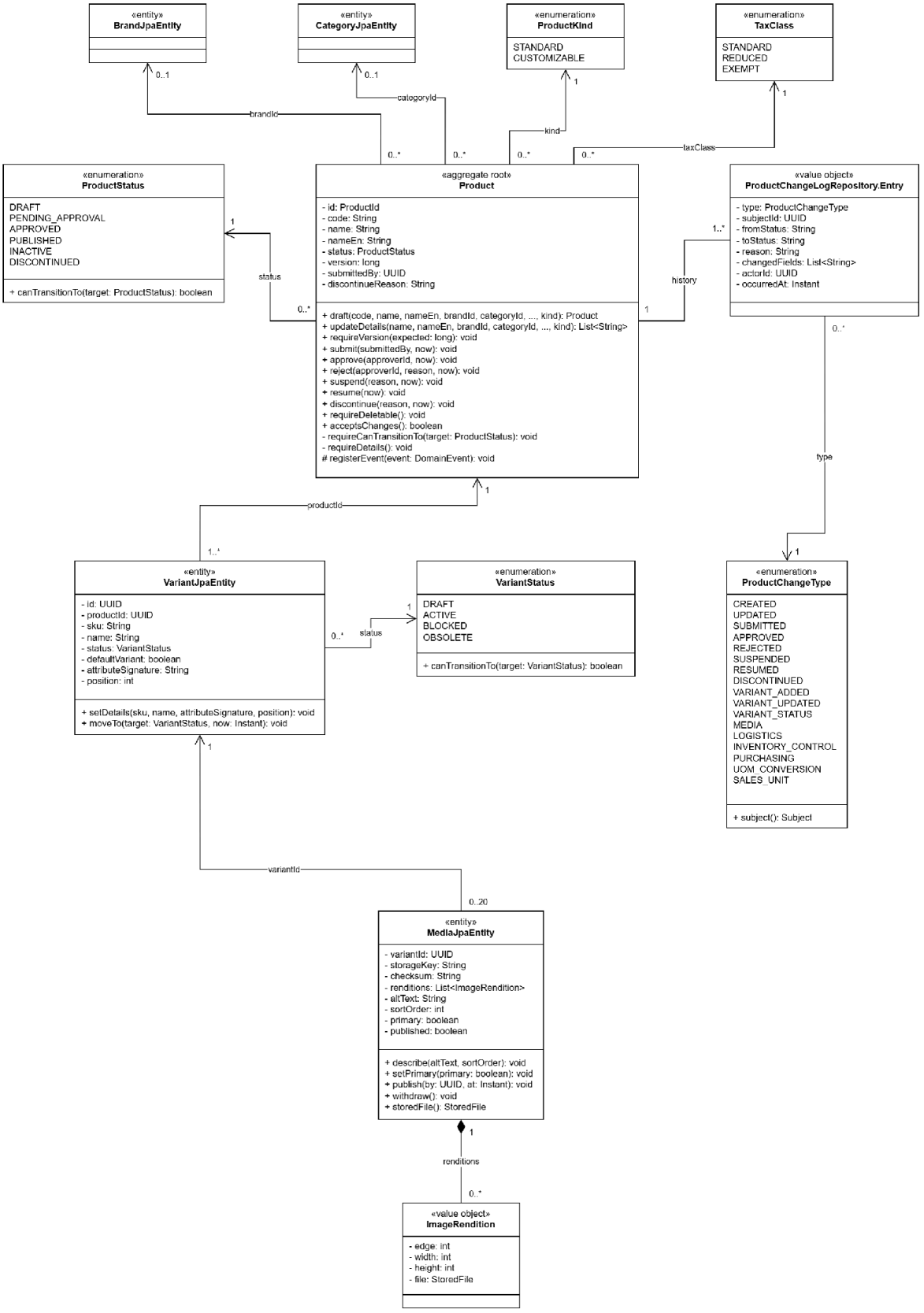


Figure 73. Product Management Class Diagram (6/6): Domain Model

3.5.2 Class Specification

ProductController

No	Method	Description
1	list(page, size, search, status, categoryId, brandId, sort)	GET /api/v1/products: Pages products filtered by text (q, name or code), statuses, category and brand; sorted by a whitelist (default lastModifiedAt desc); rows carry no duplicate warning.
2	findOne(productId)	GET /api/v1/products/{productId}: Returns one product with the products it may duplicate (same brand and primary category, same name key); 404 PRODUCT_NOT_FOUND.
3	create(request)	POST /api/v1/products: Creates a DRAFT product with one default variant (SKU = code) or, with axes, every combination of their active options (at most 100); a taken code is 409 PRODUCT_CODE_ALREADY_EXISTS, an inactive brand or category 409 BRAND_INACTIVE / CATEGORY_INACTIVE; answers 201 with possible duplicates.
4	duplicates(name, brandId, categoryId, excludeProductId)	GET /api/v1/products/duplicates: Lists products of the same brand and primary category whose name has the same key (case, accents, spaces ignored), before anything is saved; a warning, never a refusal, empty without brand or category.
5	update(productId, request)	PUT /api/v1/products/{productId}: Replaces the details of a DRAFT product under a row lock; the quoted version must match (409 OPTIMISTIC_LOCK), any other status is 409 INVALID_PRODUCT_STATUS_TRANSITION.
6	submit(productId, user)	POST /api/v1/products/{productId}/submission: Moves DRAFT to PENDING_APPROVAL when the product has a category, a brand, an English name and a tax class (409 INVALID_PRODUCT_STATUS_TRANSITION / PRODUCT_BRAND_REQUIRED / PRODUCT_DETAILS_REQUIRED); takes no version.
7	approve(productId, user)	POST /api/v1/products/{productId}/approval: Approves a PENDING_APPROVAL product and activates its DRAFT variants; the submitter cannot approve (409 SELF_APPROVAL_NOT_ALLOWED); takes no version.
8	reject(productId, user, request)	POST /api/v1/products/{productId}/rejection: Sends a PENDING_APPROVAL product back to DRAFT with a mandatory reason and clears the submission; takes no version.

9	delete(productId, version)	DELETE /api/v1/products/{productId}?version=: Deletes a DRAFT product with its variants, inventory items and images (files after commit); refused while an open document or any stock, pricing or purchasing history refers to a SKU (409 PRODUCT_IN_USE), other statuses 409 PRODUCT_NOT_DELETABLE.
10	openReferences(productId)	GET /api/v1/products/{productId}/open-references: Lists up to 20 open purchase order lines and unshipped order lines that still refer to the product's SKUs.
11	versions(productId, page, size)	GET /api/v1/products/{productId}/versions: Pages the product's history newest first: status moves with reason, and variant, image, unit, sales-unit and purchasing changes, with the actor's name.

Table 41. ProductController Class Specification

ProductLifecycleController

No	Method	Description
1	discontinue(productId, request)	POST /api/v1/products/{productId}/discontinuation: System Admin only (product-lifecycle:APPROVE): APPROVED, PUBLISHED or INACTIVE to DISCONTINUED with a 1-500 character reason and the read version; every variant becomes OBSOLETE; open PO lines and unshipped order lines come back as warnings, never a refusal.
2	suspend(productId, request)	POST /api/v1/products/{productId}/suspension: System Admin only: APPROVED or PUBLISHED to INACTIVE with a reason and version (409 OPTIMISTIC_LOCK); the catalog hides it and no new PO line may use it.
3	resume(productId, request)	POST /api/v1/products/{productId}/resumption: System Admin only: INACTIVE back to APPROVED (not PUBLISHED; it must be published again) with the read version.

Table 42. ProductLifecycleController Class Specification

ProductMediaController

No	Method	Description

1	upload(productId, variantId, file, requestKey, user)	POST /api/v1/products/{productId}/variants/{variantId}/media: Uploads an image (multipart): decoded, resized to 256/768/1600 px renditions, virus-scanned by clamd, stored privately in MinIO and attached unpublished; at most 20 per variant, not while the product is PENDING_APPROVAL or DISCONTINUED nor on an OBSOLETE variant; an Idempotency-Key replays the same upload.
2	list(productId, variantId)	GET …/media: Lists all images of the variant (at most 20) by sort order.
3	get(productId, variantId, mediaId)	GET …/media/{mediaId}: Returns one image of the variant; 404 MEDIA_NOT_FOUND.
4	describe(productId, variantId, mediaId, request)	PUT …/media/{mediaId}: Changes alt text (max 255), position and cover flag; making one image the cover takes the flag from the previous one; not while the product is under review or discontinued.
5	delete(productId, variantId, mediaId)	DELETE …/media/{mediaId}: Removes the image row (the next image becomes cover if needed); stored objects are left to the bucket lifecycle rule because the CDN may still cache them.
6	publish(productId, variantId, request, user)	POST …/media/publication: Publishes the named images (four-eyes: not by their uploader, 409 SELF_APPROVAL_NOT_ALLOWED) of an APPROVED or PUBLISHED product (409 PRODUCT_NOT_APPROVED); copies each rendition to the public prefix in the same transaction.
7	withdraw(productId, variantId, mediaId)	POST …/media/{mediaId}/withdrawal: Takes an image off the storefront (is_published false); the product stays published even with no image left.
8	downloadLink(productId, variantId, mediaId)	GET …/media/{mediaId}/download-url: Returns a short-lived presigned MinIO URL of the private original (Cache-Control no-store); 409 CONFLICT for an external-URL image.
9	renditionLink(productId, variantId, mediaId, edge)	GET …/media/{mediaId}/renditions/{edge}/download-url: Returns a presigned URL of one display rendition (no-store); 404 NOT_FOUND for an unknown edge.

Table 43. ProductMediaController Class Specification

PublicProductMediaController

No	Method	Description

1	gallery(productId, sku)	GET /api/v1/public/products/{productId}/gallery: Anonymous storefront gallery of a PUBLISHED product: the published images of the ACTIVE variant named by sku (or the default one), falling back to the default variant's; public rendition URLs, cached publicly 5 minutes; 404 otherwise.

Table 44. PublicProductMediaController Class Specification

ProductTaxonomyController

No	Method	Description
1	brands(search, active, page, size, sort)	GET /api/v1/brands: Pages brands filtered by text and active flag.
2	brand(brandId)	GET /api/v1/brands/{brandId}: Returns one brand; 404 BRAND_NOT_FOUND.
3	createBrand(request)	POST /api/v1/brands: Creates an active brand; code upper-cased and pattern-checked, slug from the code; a taken code or slug is 409 BRAND_CODE_ALREADY_EXISTS.
4	updateBrand(brandId, request)	PUT /api/v1/brands/{brandId}: Changes name, logo and active flag (retiring, never deleting); no version.
5	categories(search, parentId, roots, active, page, size, sort)	GET /api/v1/categories: Pages categories: all, the roots, or the children of parentId.
6	category(categoryId)	GET /api/v1/categories/{categoryId}: Returns one category; 404 CATEGORY_NOT_FOUND.
7	createCategory(r)	POST /api/v1/categories: Creates a category under the tree's advisory lock, at most 11 levels deep (400 VALIDATION_FAILED); a taken code or slug is 409 CATEGORY_CODE_ALREADY_EXISTS.
8	updateCategory(categoryId, r)	PUT /api/v1/categories/{categoryId}: Changes name, order, image, SEO texts and active flag; no version.
9	moveCategory(categoryId, request)	PUT /api/v1/categories/{categoryId}/parent: Moves a category and its subtree under another parent (or to the root), rewriting path and depth parents first; under itself or its own subtree, or deeper than 11 levels, is 400 VALIDATION_FAILED.

Table 45. ProductTaxonomyController Class Specification

AttributeController

No	Method	Description
1	list(search, dataType, active, page, size, sort)	GET /api/v1/attributes: Pages the attribute dictionary with each attribute's values.

2	get(attributeId)	GET /api/v1/attributes/{attributeId}: Returns one attribute with its values; 404 ATTRIBUTE_NOT_FOUND.
3	create(r)	POST /api/v1/attributes: Creates an attribute (code, data type, name, unit, filter/search flags); a taken code is 409 ATTRIBUTE_CODE_ALREADY_EXISTS.
4	update(attributeId, r)	PUT /api/v1/attributes/{attributeId}: Changes details and active flag; the data type is fixed; a stale version is 409 OPTIMISTIC_LOCK.
5	delete(attributeId, version)	DELETE /api/v1/attributes/{attributeId}?version=: Deletes an attribute and its values for good; 409 OPTIMISTIC_LOCK when stale, 409 ATTRIBUTE_IN_USE when a product or category uses it.
6	addValue(attributeId, r)	POST /api/v1/attributes/{attributeId}/values: Adds a value to a SELECT, MULTISELECT or COLOR attribute (a colour needs #RRGGBB); a code already in the attribute is 409 ATTRIBUTE_CODE_ALREADY_EXISTS.
7	updateValue(attributeId, valueId, r)	PUT /api/v1/attributes/{attributeId}/values/{valueId}: Changes a value's label, number, colour, order and active flag under its own version.
8	deleteValue(attributeId, valueId, version)	DELETE /api/v1/attributes/{attributeId}/values/{valueId}?version=: Removes a value no product option or descriptive value uses (409 ATTRIBUTE_IN_USE); answers the attribute without it.

Table 46. AttributeController Class Specification

CategoryAttributeController

No	Method	Description
1	list(categoryId)	GET /api/v1/categories/{categoryId}/attributes: Lists the attributes a category suggests, with their default role and required flag.
2	replace(categoryId, r)	PUT /api/v1/categories/{categoryId}/attributes: Replaces the category's suggested attribute set whole (empty list clears it), under a lock on the category; a duplicate, an inactive attribute newly added, or an axis role for a type that cannot be an axis is 400 VALIDATION_FAILED.

Table 47. CategoryAttributeController Class Specification

3.5.3 Create / Edit Product & Variants (Create Product) Sequence Diagram

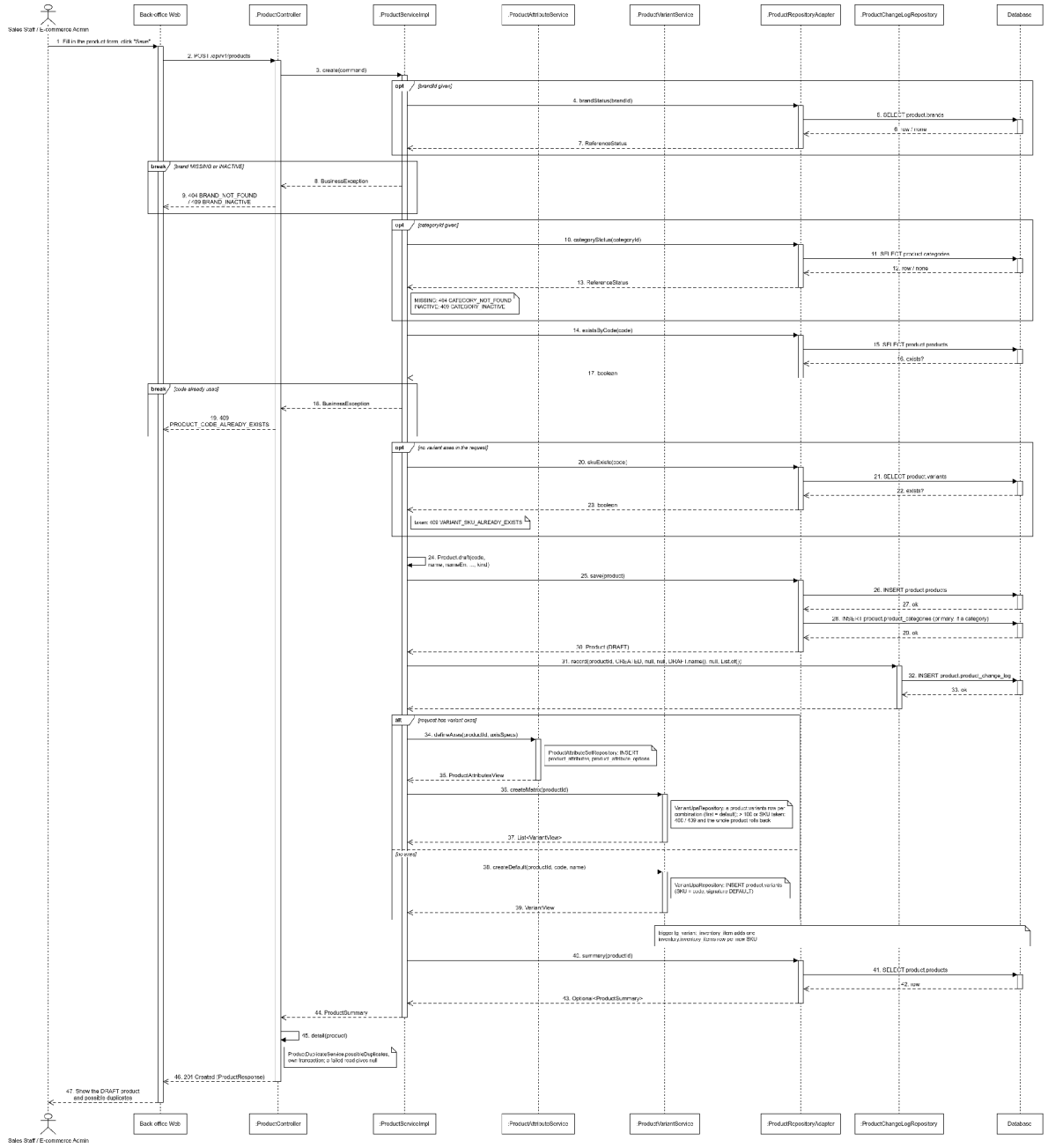


Figure 74. Create / Edit Product & Variants (Create Product) Sequence Diagram

3.5.4 Create / Edit Product & Variants (Edit Product) Sequence Diagram

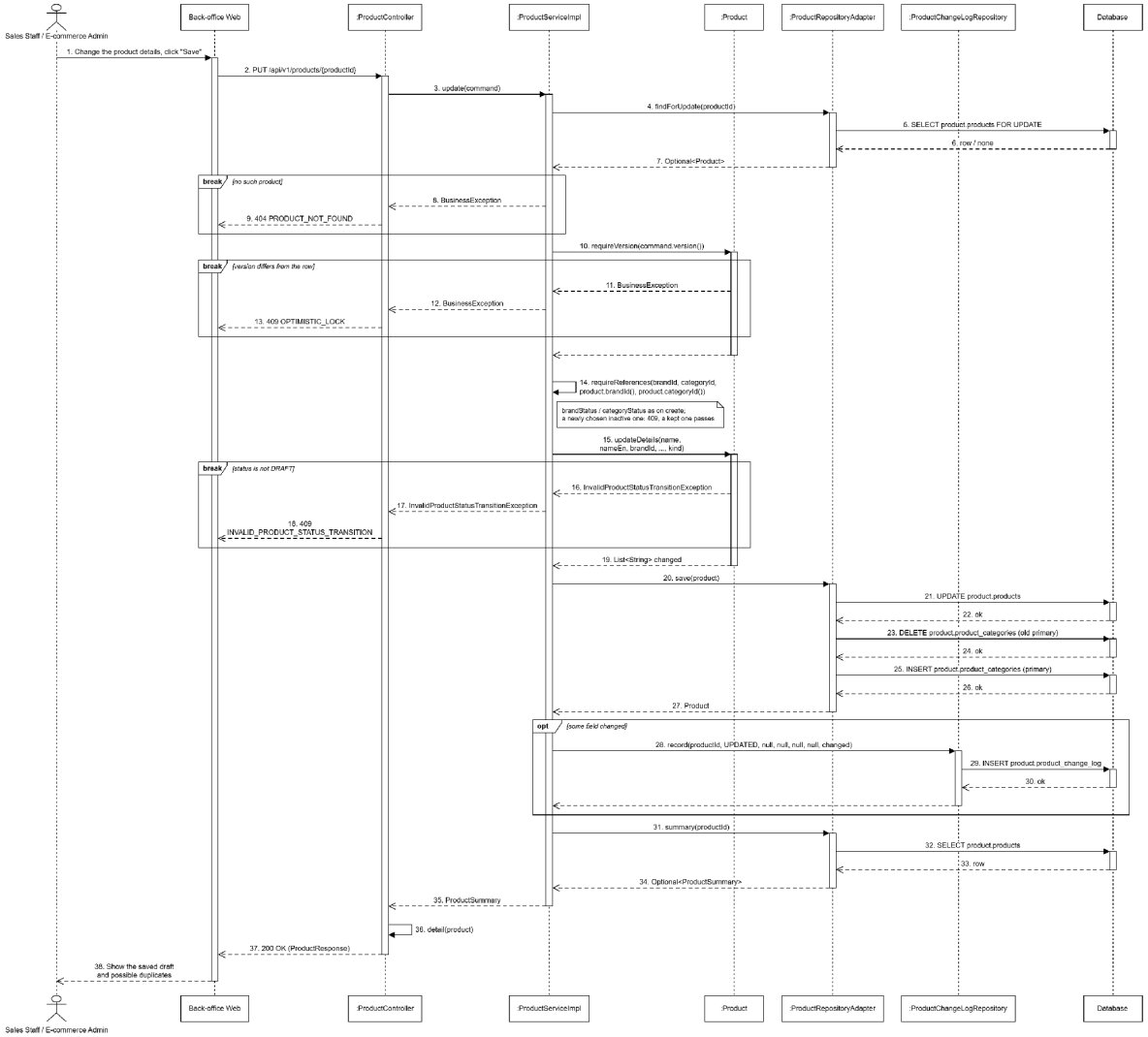


Figure 75. Create / Edit Product & Variants (Edit Product) Sequence Diagram

3.5.5 Upload Product Images (Upload Image) Sequence Diagram

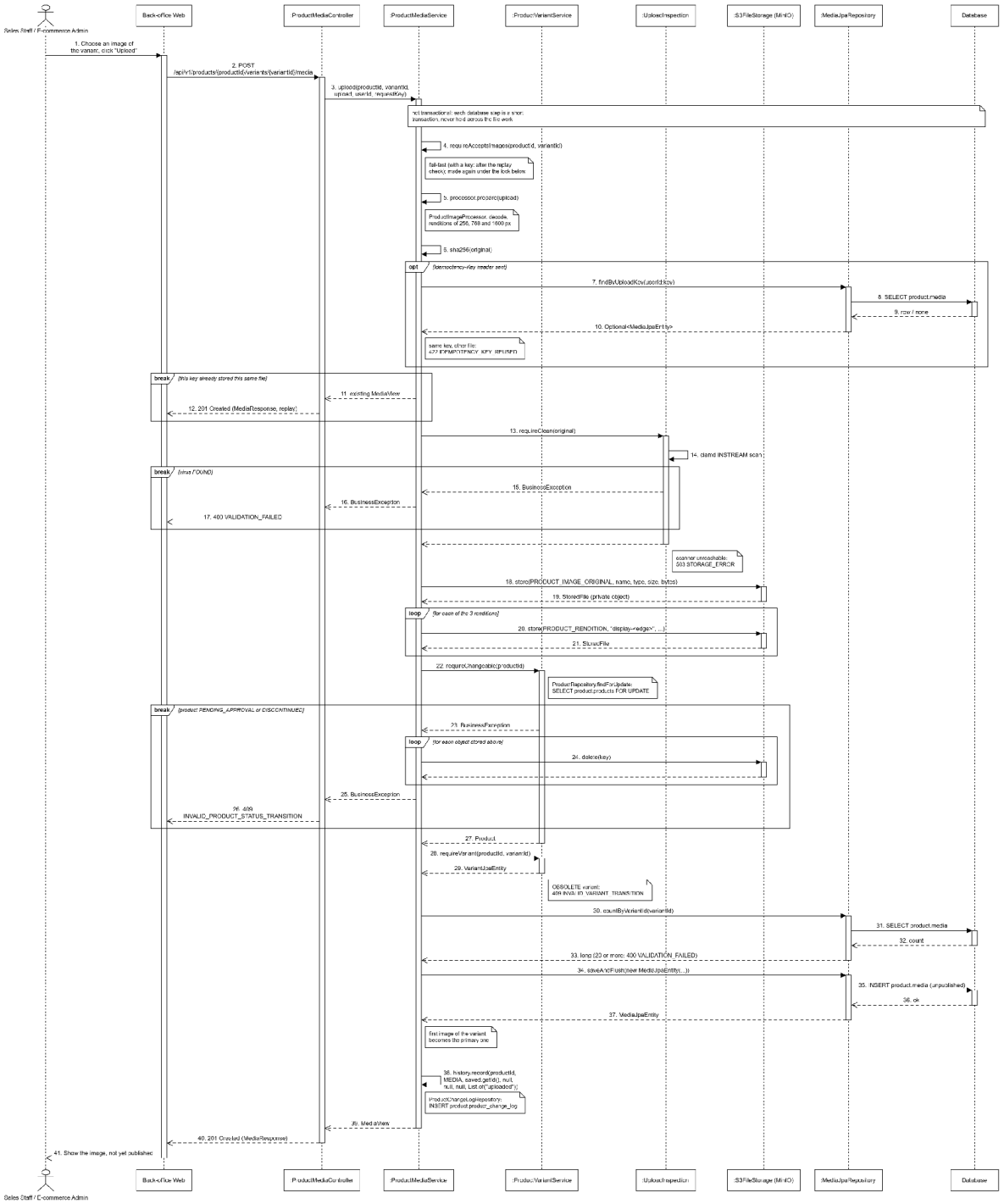


Figure 76. Upload Product Images (Upload Image) Sequence Diagram

3.5.6 Upload Product Images (Preview Image by Presigned URL) Sequence Diagram

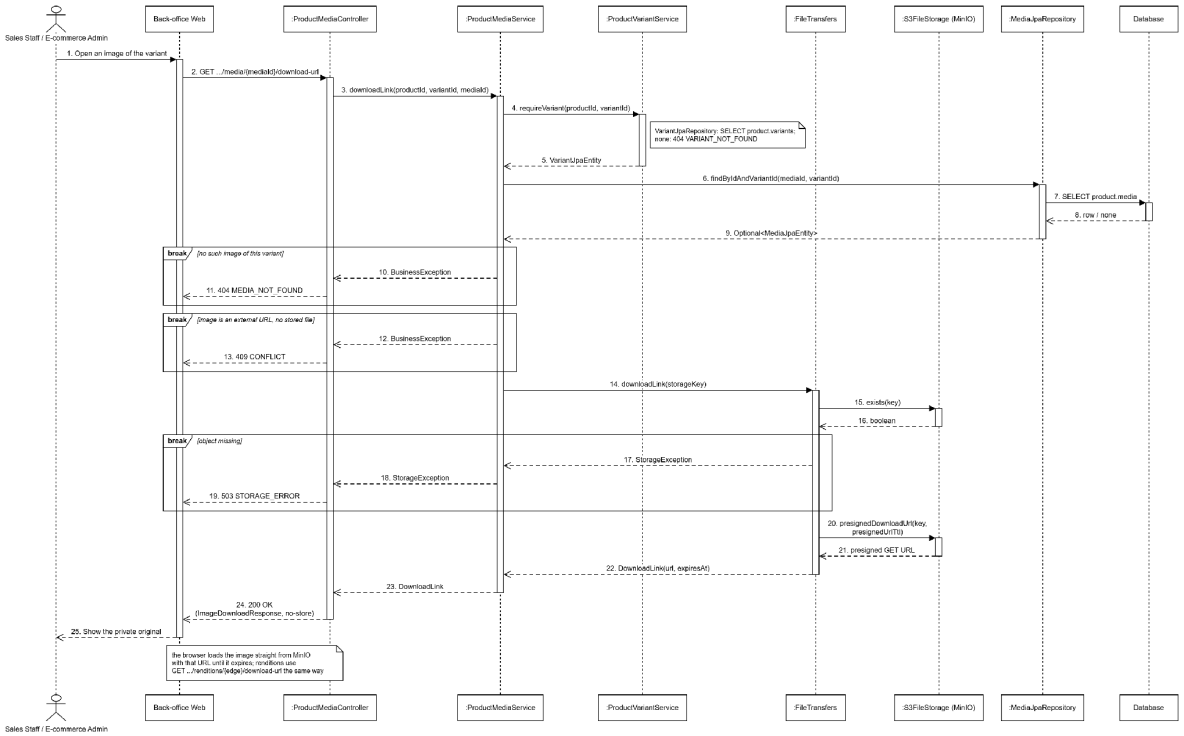


Figure 77. Upload Product Images (Preview Image by Presigned URL) Sequence Diagram

3.5.7 Submit Product for Approval Sequence Diagram

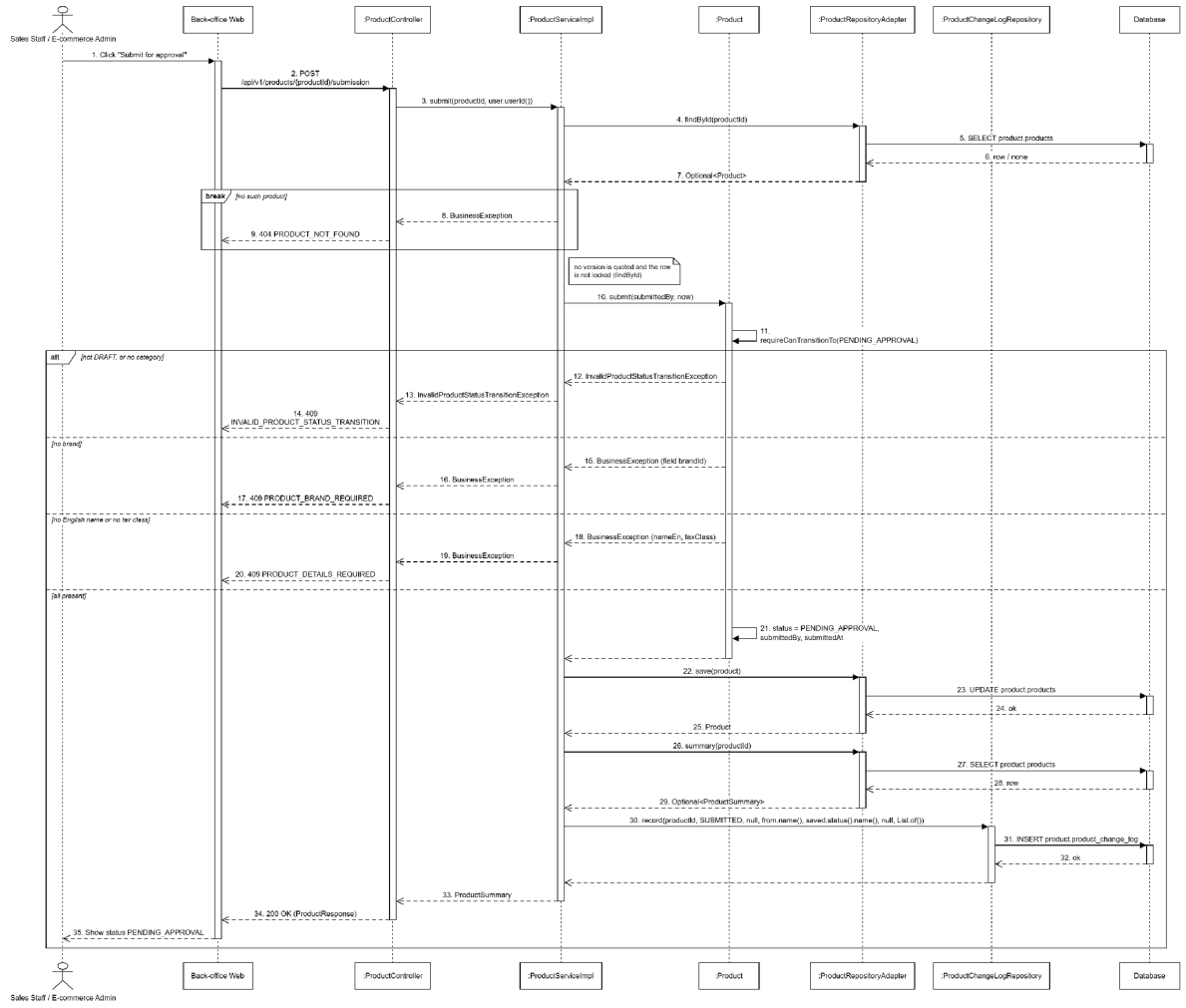


Figure 78. Submit Product for Approval Sequence Diagram

3.5.8 Approve Product & Publish Images (Approve or Reject Product) Sequence Diagram

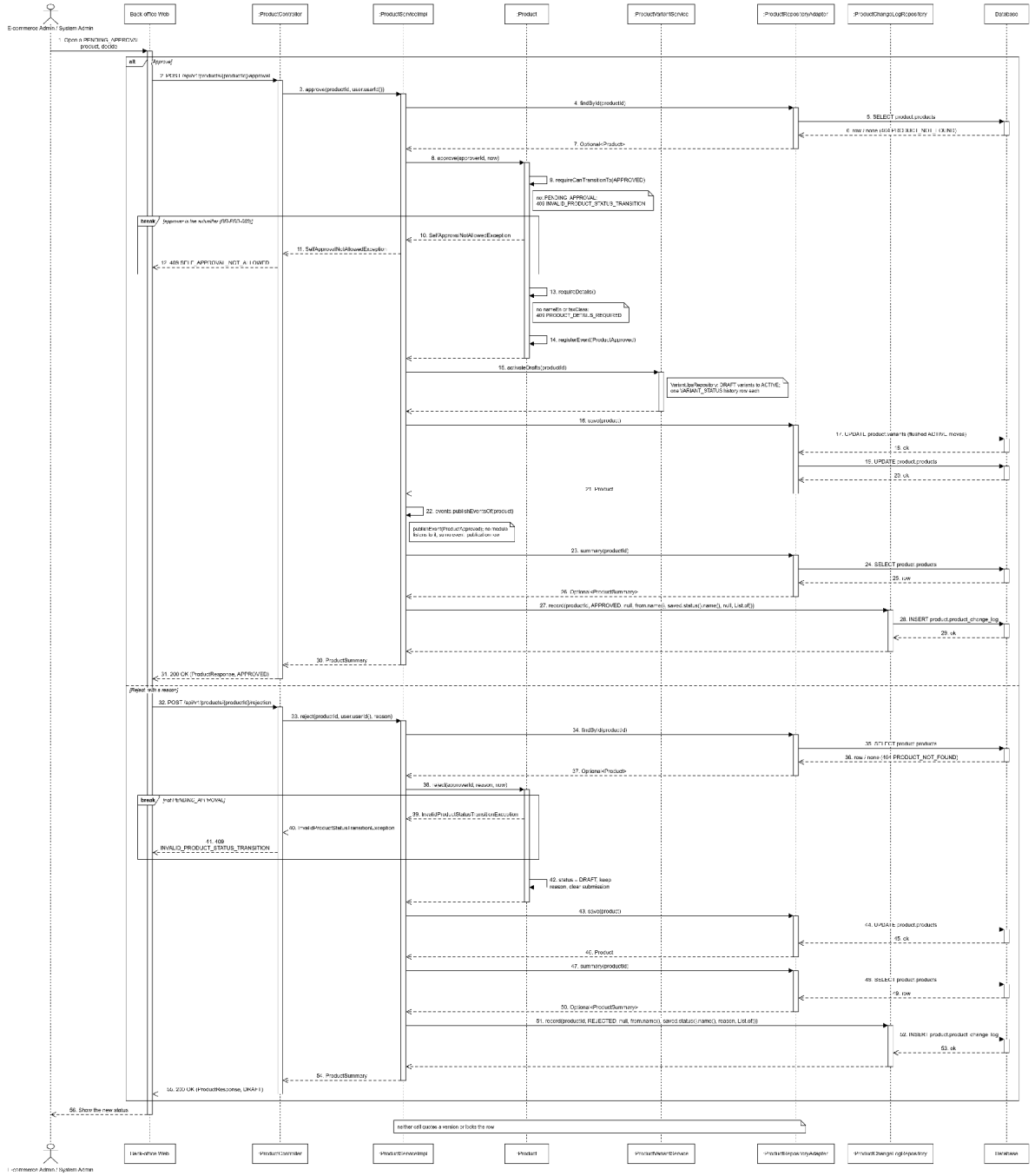


Figure 79. Approve Product & Publish Images (Approve or Reject Product) Sequence Diagram

3.5.9 Approve Product & Publish Images (Publish Images) Sequence Diagram

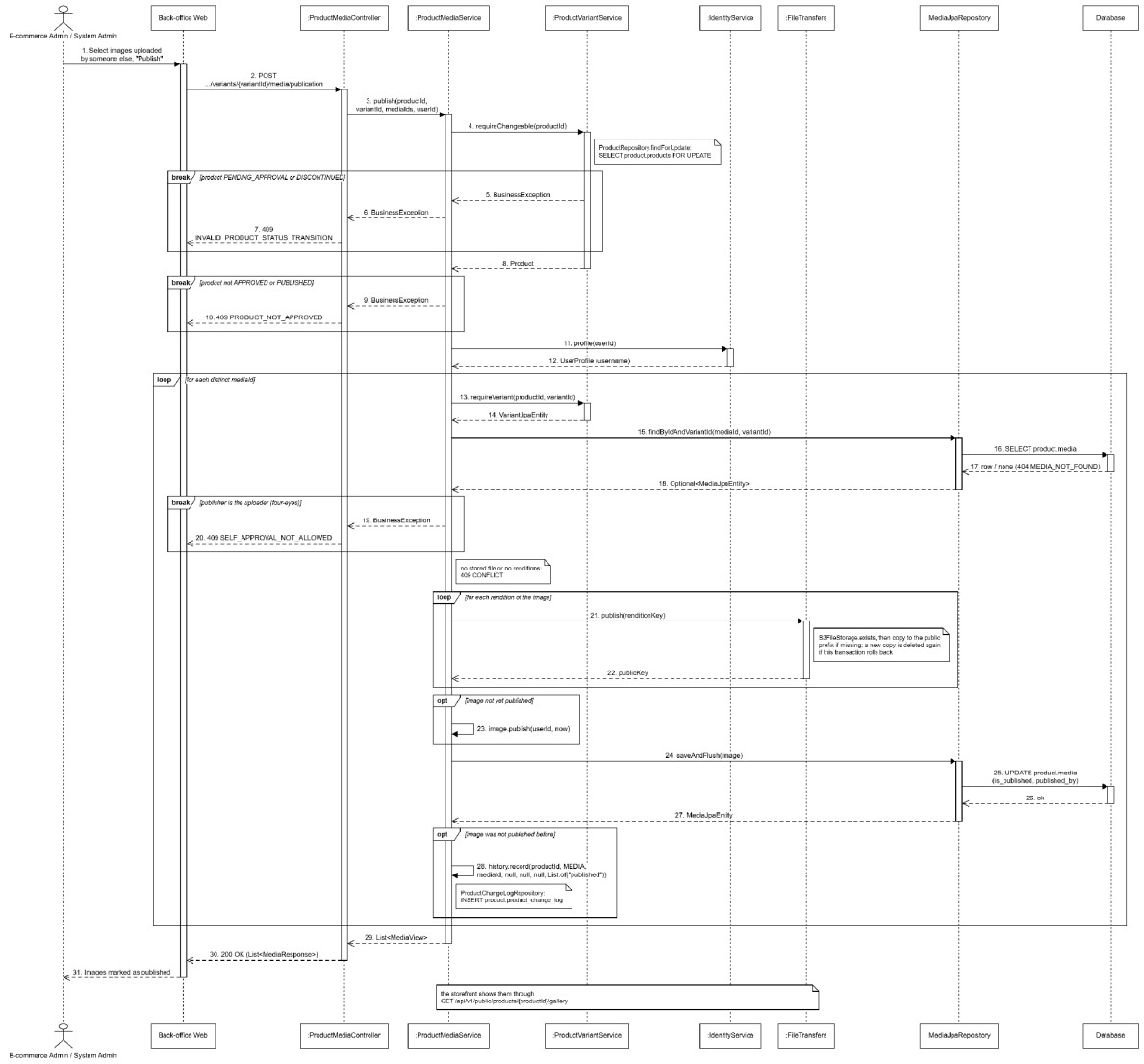


Figure 80. Approve Product & Publish Images (Publish Images) Sequence Diagram

3.5.10 Suspend / Resume / Discontinue Product (Suspend or Resume) Sequence Diagram

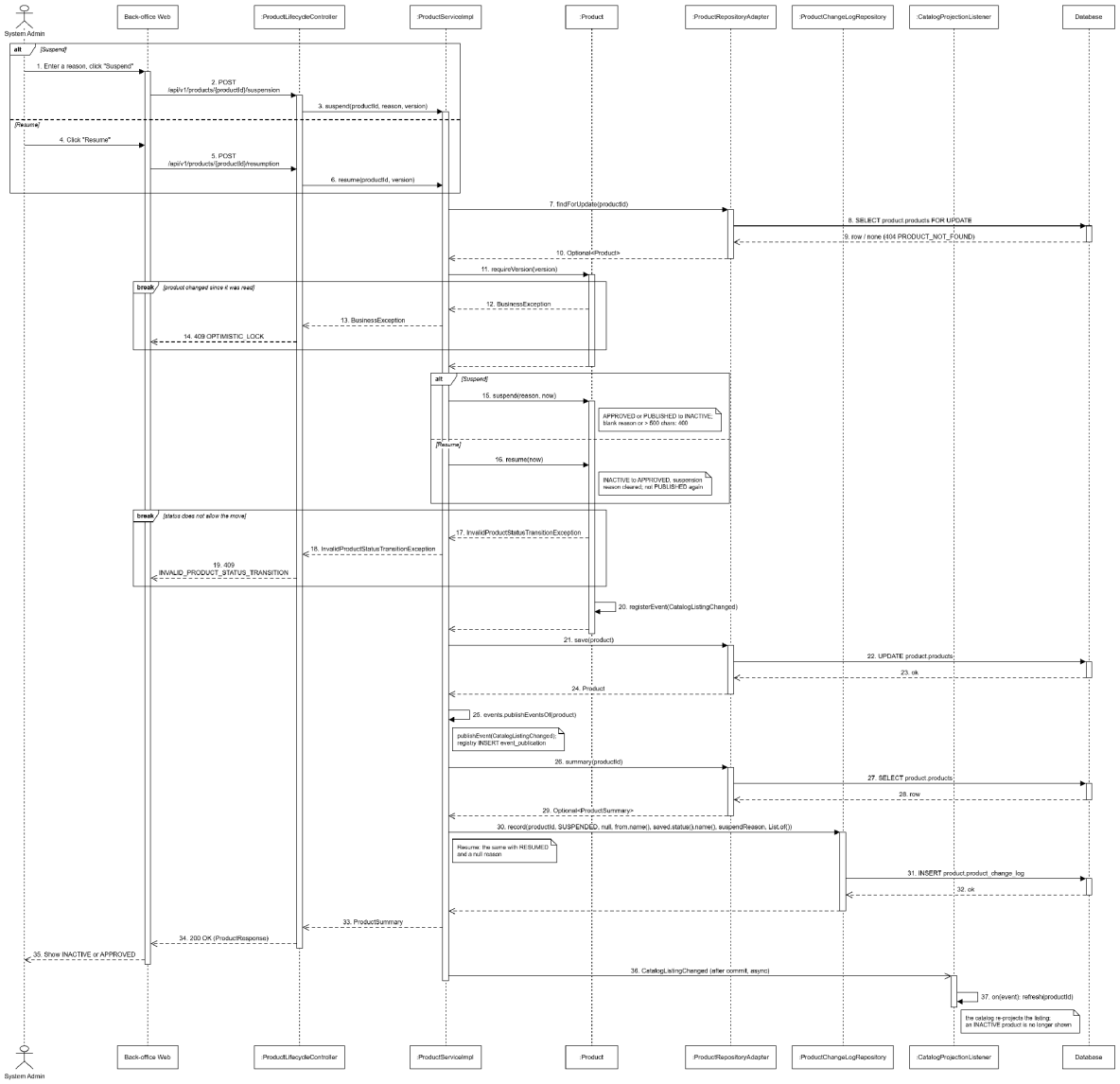


Figure 81. Suspend / Resume / Discontinue Product (Suspend or Resume) Sequence Diagram

3.5.11 Suspend / Resume / Discontinue Product (Discontinue) Sequence Diagram

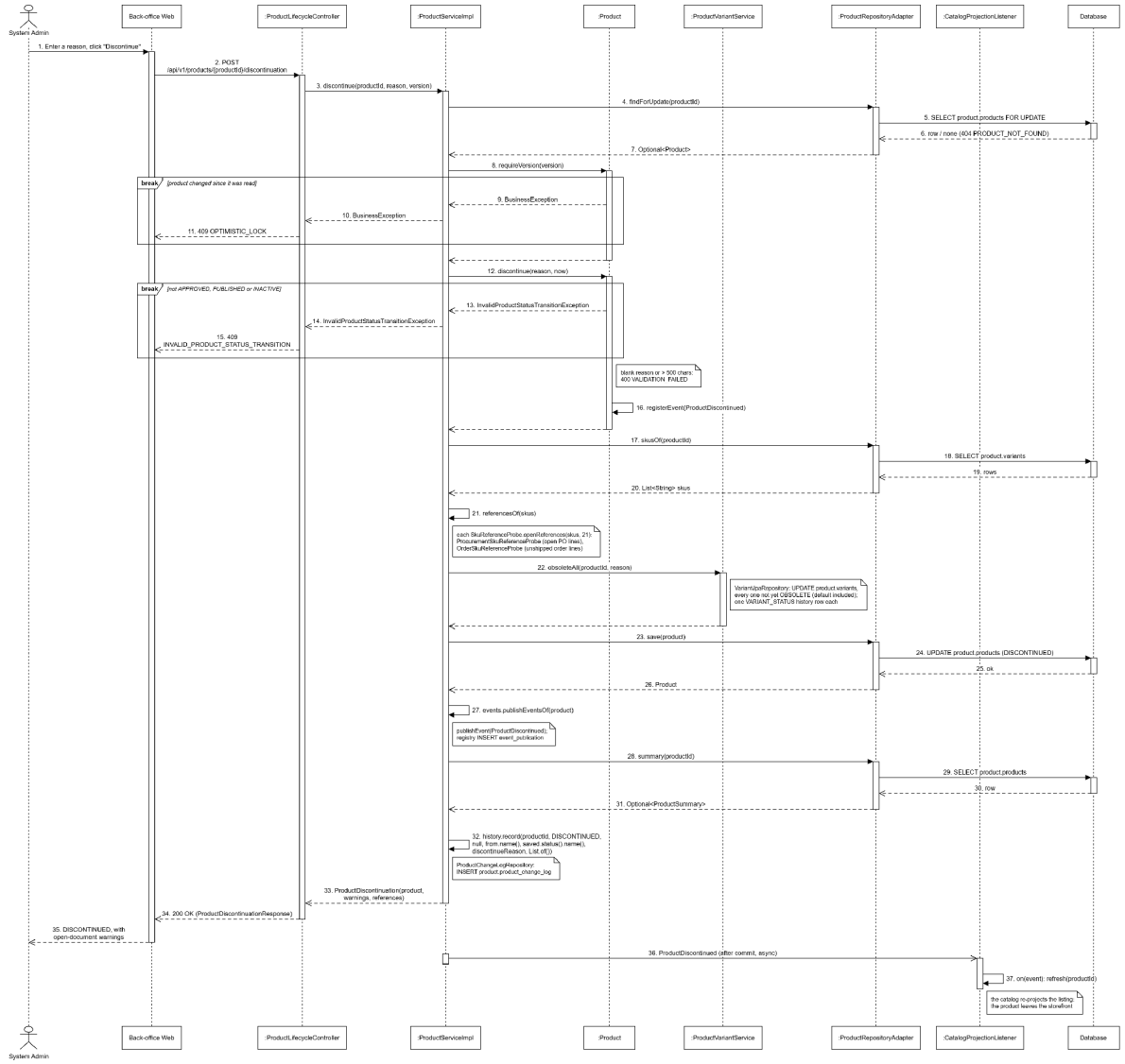


Figure 82. Suspend / Resume / Discontinue Product (Discontinue) Sequence Diagram

3.5.12 Manage Categories, Brands & Attributes (Categories) Sequence Diagram

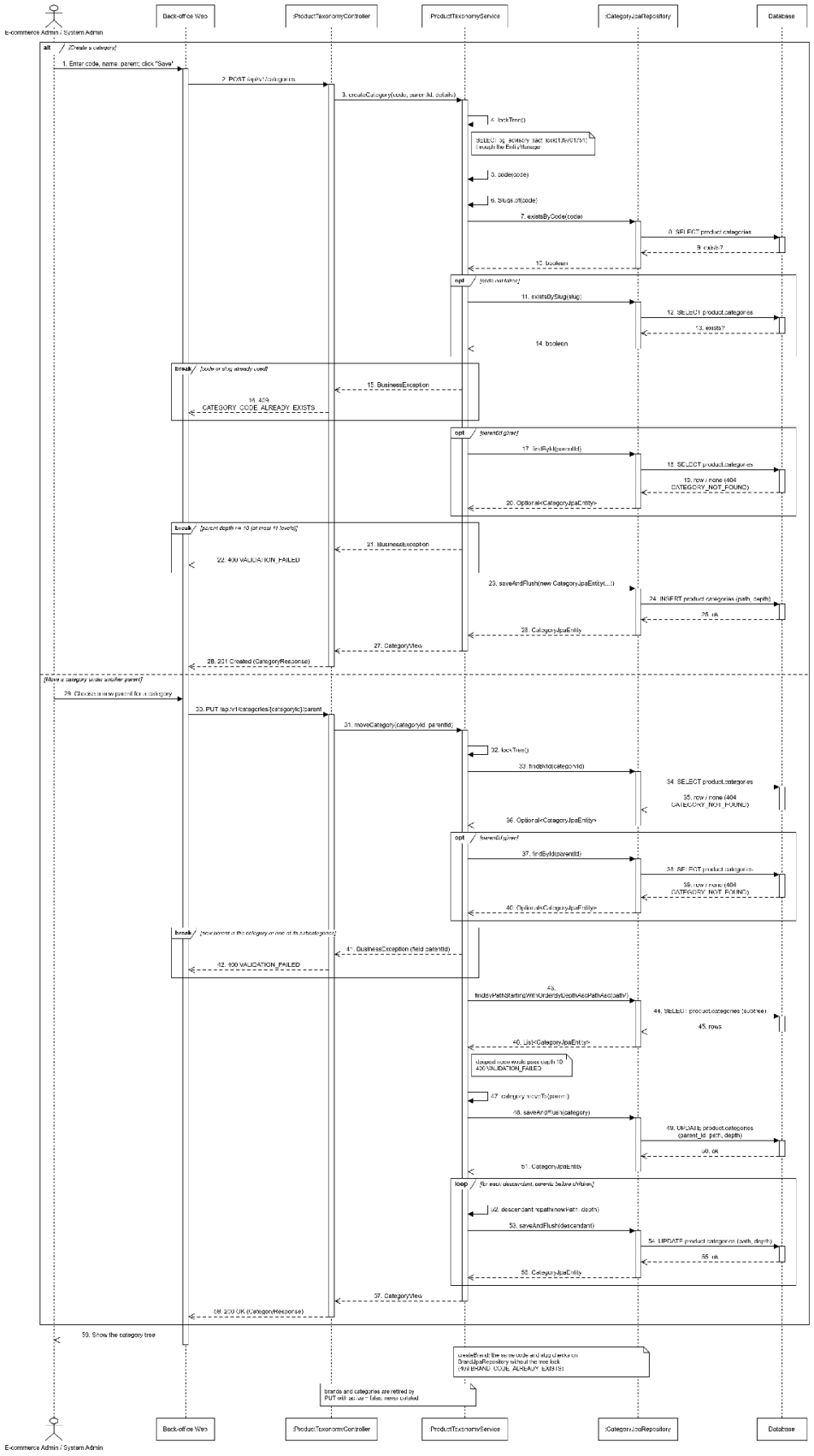


Figure 83. Manage Categories, Brands & Attributes (Categories) Sequence Diagram

3.5.13 Manage Categories, Brands & Attributes (Attribute Dictionary) Sequence Diagram

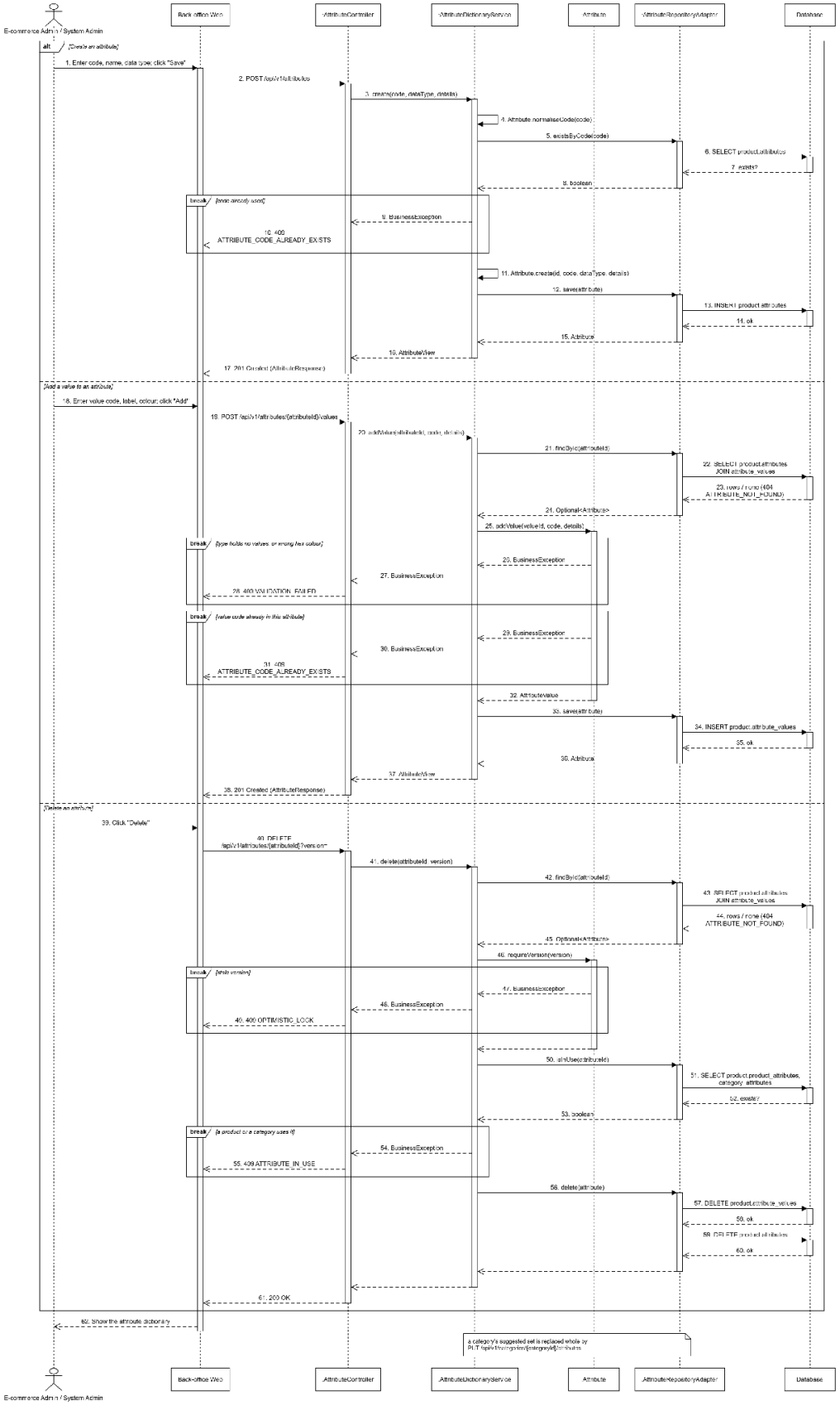


Figure 84. Manage Categories, Brands & Attributes (Attribute Dictionary) Sequence Diagram

3.5.14 Product State Diagram

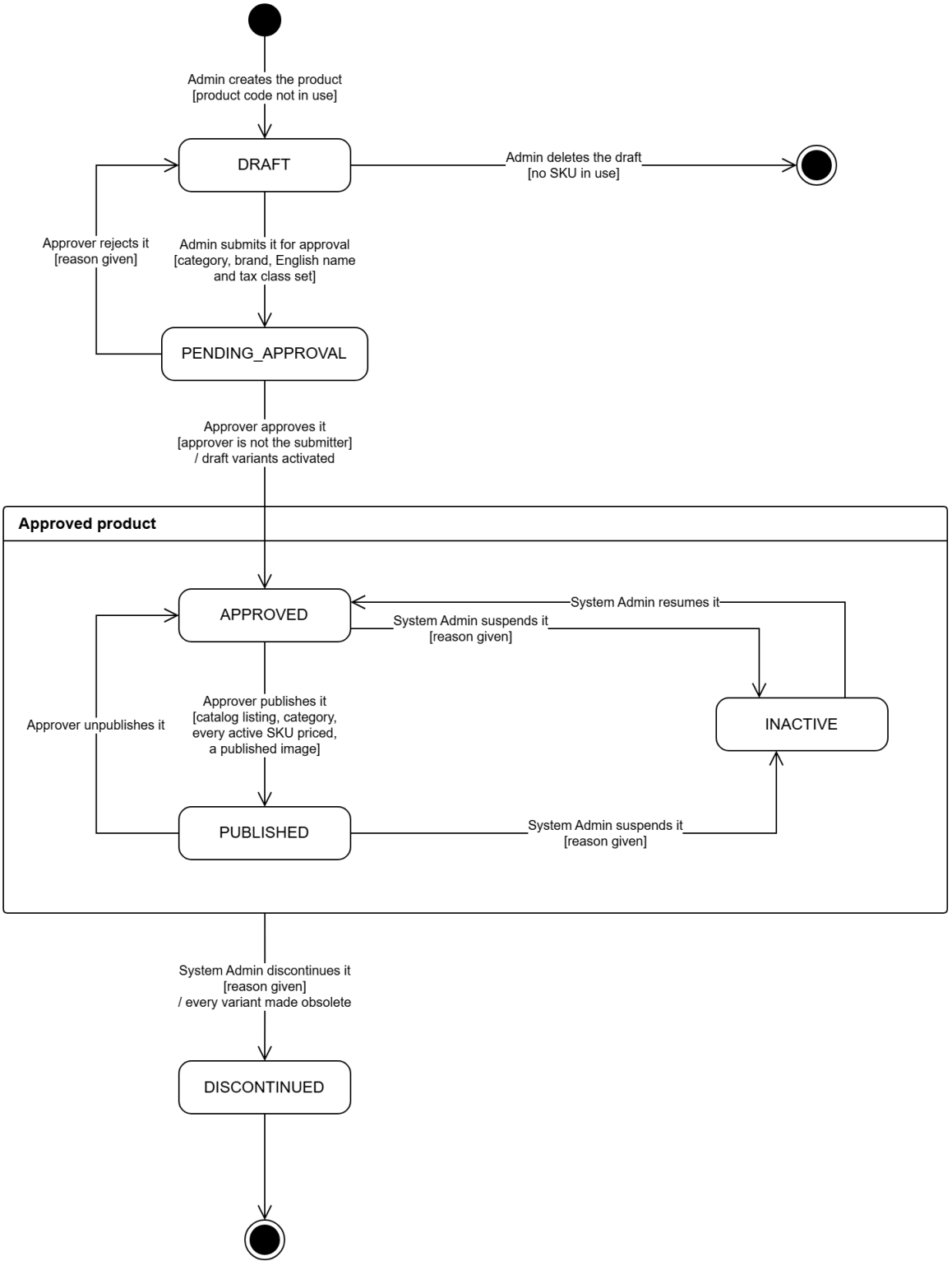


Figure 85. Product State Diagram

3.5.15 Product Image State Diagram (values of product.media.is_published)

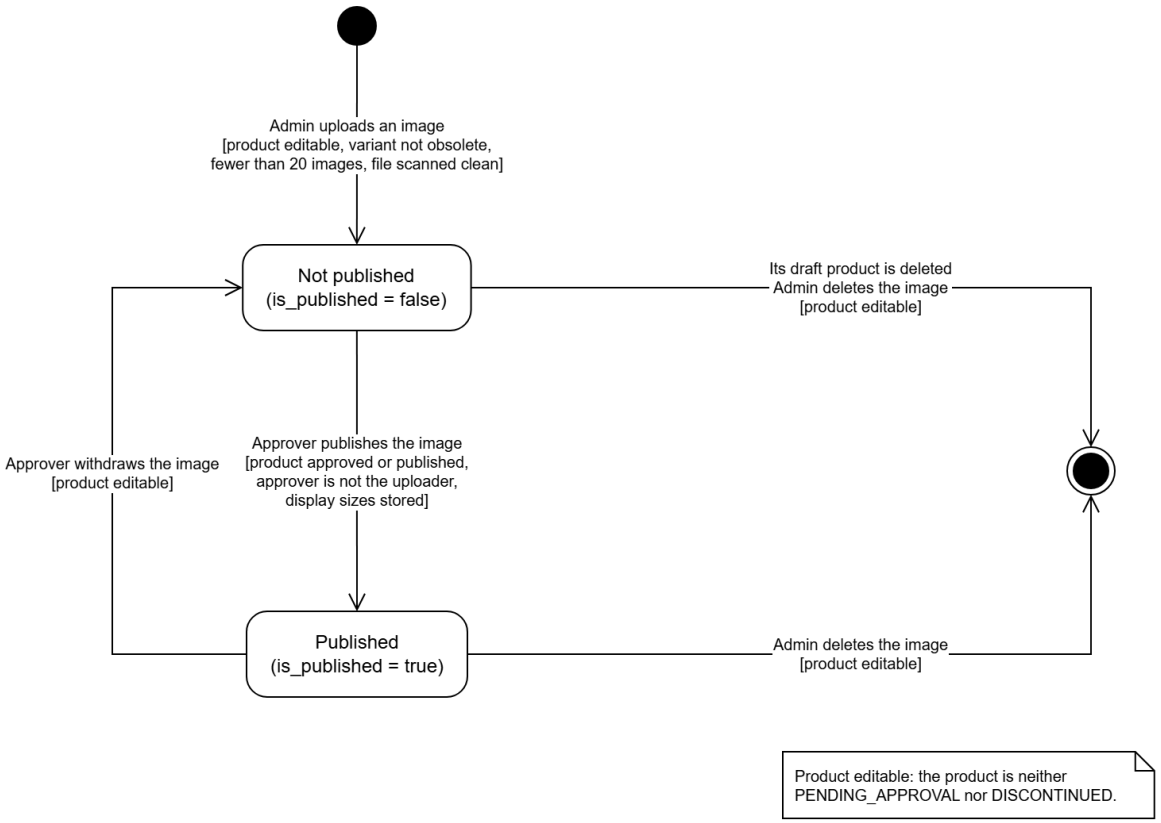


Figure 86. Product Image State Diagram (values of product.media.is_published)

3.6 Product Master Data

3.6.1 Class Diagram

Figure 87. Product Master Data Class Diagram (1/4): Variants

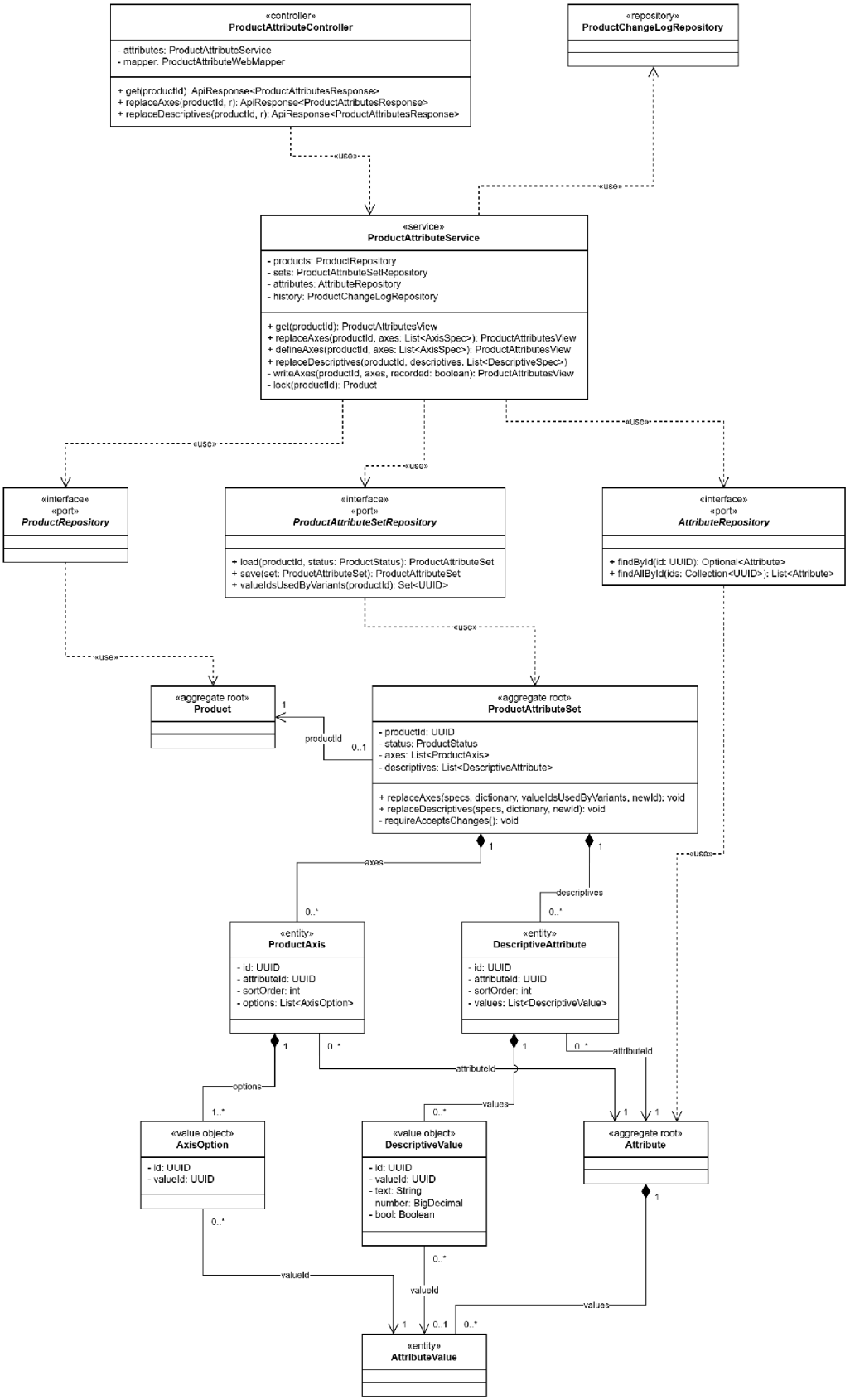


Figure 88. Product Master Data Class Diagram (2/4): Variant Axes and Descriptive Attributes

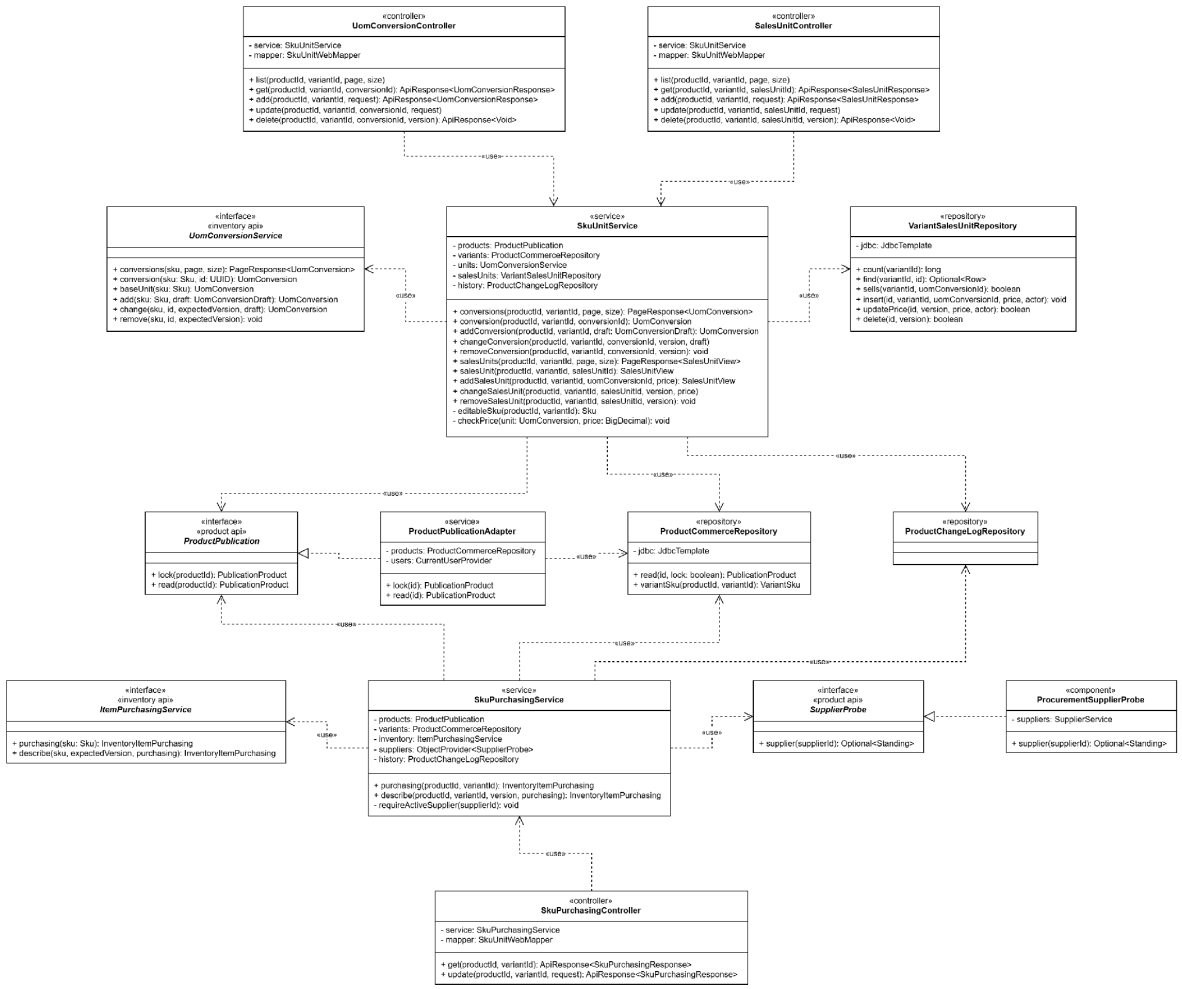


Figure 89. Product Master Data Class Diagram (3/4): Packing Units, Sales Units and Purchasing

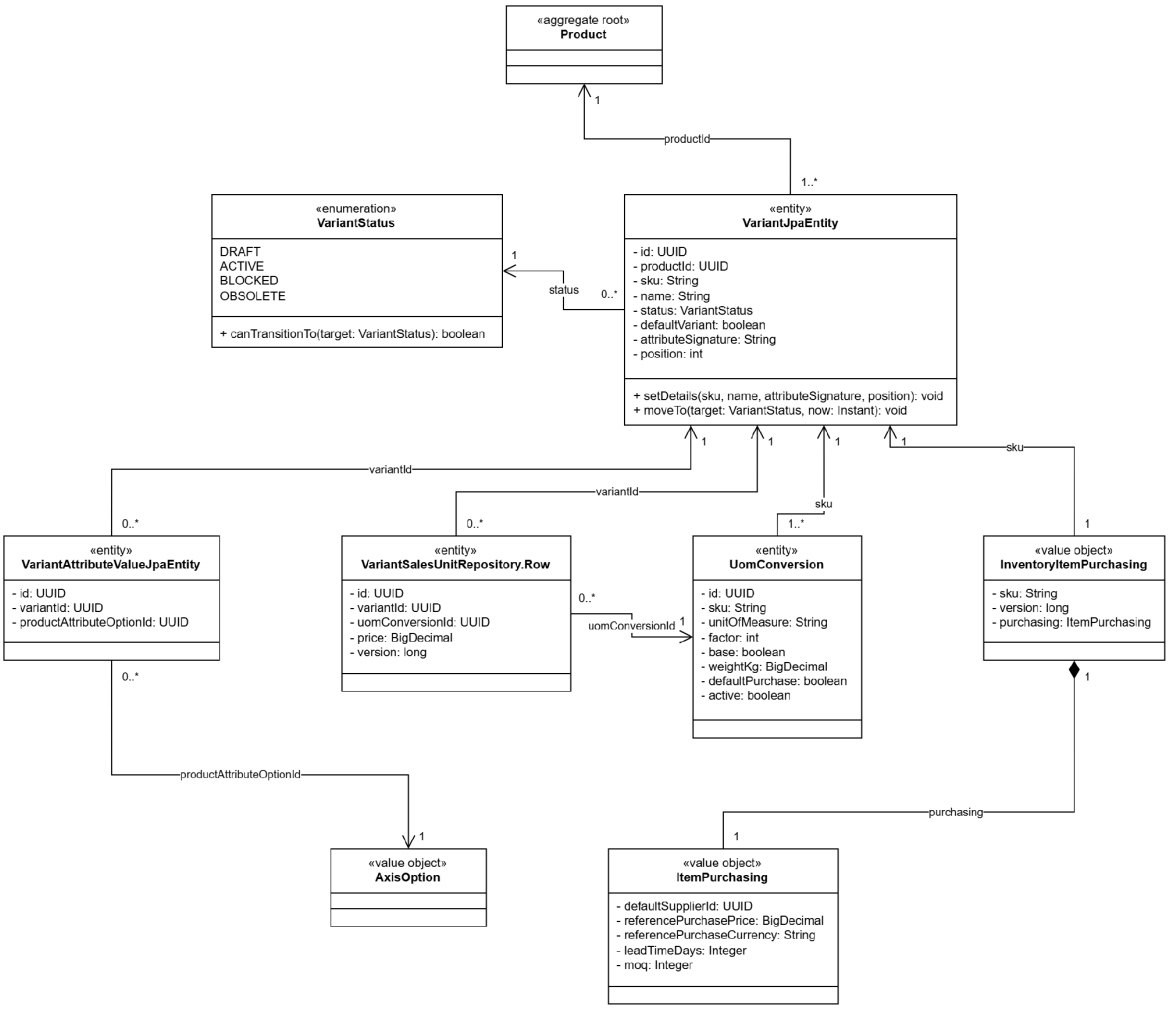


Figure 90. Product Master Data Class Diagram (4/4): Domain Model

3.6.2 Class Specification

ProductVariantController

No	Method	Description
1	list(productId, page, size)	GET /api/v1/products/{productId}/variants: Pages the product's variants by position with their options and each SKU's inventory item id.
2	get(productId, variantId)	GET …/variants/{variantId}: Returns one variant with its options and inventory item id; 404 VARIANT_NOT_FOUND.
3	add(productId, request)	POST …/variants: Adds a DRAFT variant for one active option per axis (a SKU alone on a product without axes); the matrix-first combination is given to the DRAFT default instead of a new row; taken SKU or combination 409 VARIANT_SKU_ALREADY_EXISTS, inactive value 400 VARIANT_OPTIONS_INVALID; not while the product is under review or discontinued.

4	generate(productId, request)	POST …/variants/generation: Creates the missing combinations of the option matrix (all options, or those named) as DRAFT variants with SKUs by rule, skipping existing ones; more than 100 is 400 VARIANT_MATRIX_TOO_LARGE; idempotent.
5	update(productId, variantId, request)	PUT …/variants/{variantId}: Partial update: name and position always, SKU and options only while the variant is DRAFT (409 INVALID_VARIANT_TRANSITION otherwise or when the SKU is already referenced).
6	activate(productId, variantId)	POST …/variants/{variantId}/activation: Puts a DRAFT or BLOCKED variant on sale (ACTIVE) once the product is APPROVED or PUBLISHED (409 INVALID_VARIANT_TRANSITION).
7	block(productId, variantId)	POST …/variants/{variantId}/blocking: Takes an ACTIVE variant off sale for now (BLOCKED).
8	obsolete(productId, variantId)	POST …/variants/{variantId}/obsoletion: Retires a variant for good (OBSOLETE); the default variant is refused (409 INVALID_VARIANT_TRANSITION).

Table 48. ProductVariantController Class Specification

ProductAttributeController

No	Method	Description
1	get(productId)	GET /api/v1/products/{productId}/attributes: Returns the product's variant axes with their options (dictionary order) and its descriptive attributes.
2	replaceAxes(productId, r)	PUT /api/v1/products/{productId}/axes: Replaces the axes whole; not while the product is PENDING_APPROVAL or DISCONTINUED (409 INVALID_PRODUCT_STATUS_TRANSITION); adding or removing an axis needs a DRAFT product, and adding one also no variant with options (409 PRODUCT_AXES_LOCKED); removing an option or axis a variant uses is 409 ATTRIBUTE_IN_USE; options of kept axes may change in DRAFT, APPROVED, PUBLISHED and INACTIVE.
3	replaceDescriptives(productId, r)	PUT /api/v1/products/{productId}/descriptive-attributes: Replaces the descriptive attribute values whole (empty list clears them); guarded by product-products:UPDATE; not while the product is PENDING_APPROVAL or DISCONTINUED (409 INVALID_PRODUCT_STATUS_TRANSITION).

Table 49. ProductAttributeController Class Specification

UomConversionController

No	Method	Description
1	list(productId, variantId, page, size)	GET …/uom-conversions: Pages the SKU's units of measure: the base unit first, then the packing units by code.
2	get(productId, variantId, conversionId)	GET …/uom-conversions/{conversionId}: Returns one unit of the SKU; 404 UOM_CONVERSION_NOT_FOUND.
3	add(productId, variantId, request)	POST …/uom-conversions: Declares a packing unit (factor to base, size, weight, default purchase / shipping flags) on the SKU's inventory item; a taken code is 409 UOM_CONVERSION_DUPLICATE; not while the product is PENDING_APPROVAL or DISCONTINUED (409 CONFLICT) or the variant OBSOLETE (409 VARIANT_OBSOLETE).
4	update(productId, variantId, conversionId, request)	PUT …/uom-conversions/{conversionId}: Replaces a unit (on the base unit only defaultPurchase); retiring a unit the variant sells in, or changing the factor or code of a unit a document uses, is 409 UOM_CONVERSION_IN_USE; stale version 409 OPTIMISTIC_LOCK.
5	delete(productId, variantId, conversionId, version)	DELETE …/uom-conversions/{conversionId}?version=: Deletes a packing unit; the base unit, a unit the variant sells in or one a document uses is 409 UOM_CONVERSION_IN_USE.

Table 50. UomConversionController Class Specification

SalesUnitController

No	Method	Description
1	list(productId, variantId, page, size)	GET …/sales-units: Pages the variant's sales units; with none listed, one implicit entry for the base unit at the catalog base price.
2	get(productId, variantId, salesUnitId)	GET …/sales-units/{salesUnitId}: Returns one sales unit; 404 SALES_UNIT_NOT_FOUND.
3	add(productId, variantId, request)	POST …/sales-units: Sells the variant in one active unit of its own inventory item (404 UOM_CONVERSION_NOT_FOUND otherwise); a packing unit needs a positive whole-VND price, the base unit none (400 VALIDATION_FAILED); already listed 409 SALES_UNIT_DUPLICATE.
4	update(productId, variantId, salesUnitId, request)	PUT …/sales-units/{salesUnitId}: Reprices a sales unit under its version (409 OPTIMISTIC_LOCK).

5	delete(productId, variantId, salesUnitId, version)	DELETE …/sales-units/{salesUnitId}?version=: Stops selling in a unit; removing the last one brings back the base-unit default.

Table 51. SalesUnitController Class Specification

SkuPurchasingController

No	Method	Description
1	get(productId, variantId)	GET …/purchasing: Returns the SKU's default supplier, reference purchase price, lead time and MOQ, held on its inventory item.
2	update(productId, variantId, request)	PUT …/purchasing: Replaces them under the item's version (409 OPTIMISTIC_LOCK); a newly chosen default supplier must exist (404 SUPPLIER_NOT_FOUND) and be ACTIVE (409 SUPPLIER_INACTIVE); not while the product is PENDING_APPROVAL or DISCONTINUED (409 CONFLICT).

Table 52. SkuPurchasingController Class Specification

3.6.3 Create / Edit Product & Variants (Generate Variants) Sequence Diagram

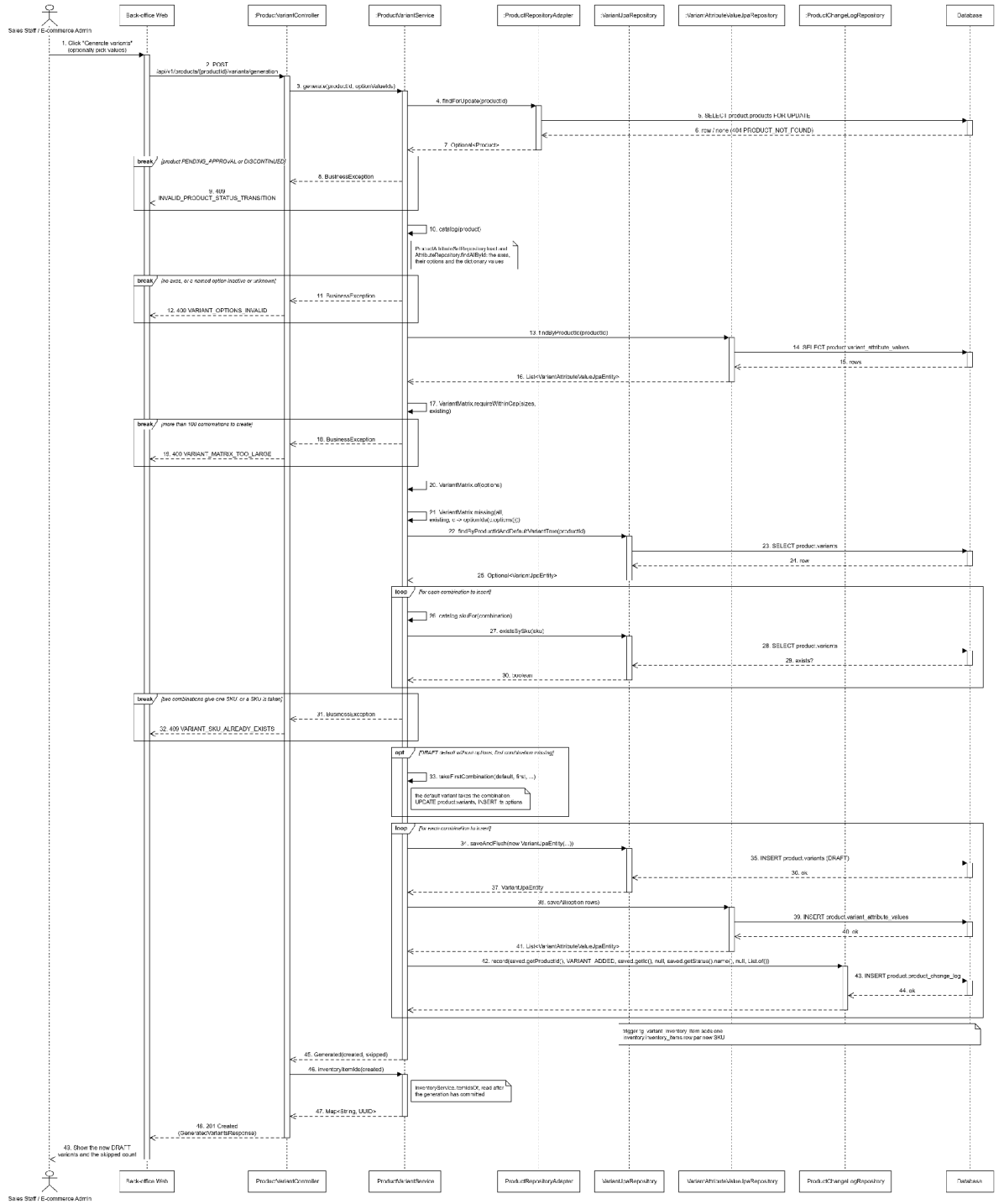


Figure 91. Create / Edit Product & Variants (Generate Variants) Sequence Diagram

3.6.4 Create / Edit Product & Variants (Declare Packing Unit) Sequence Diagram

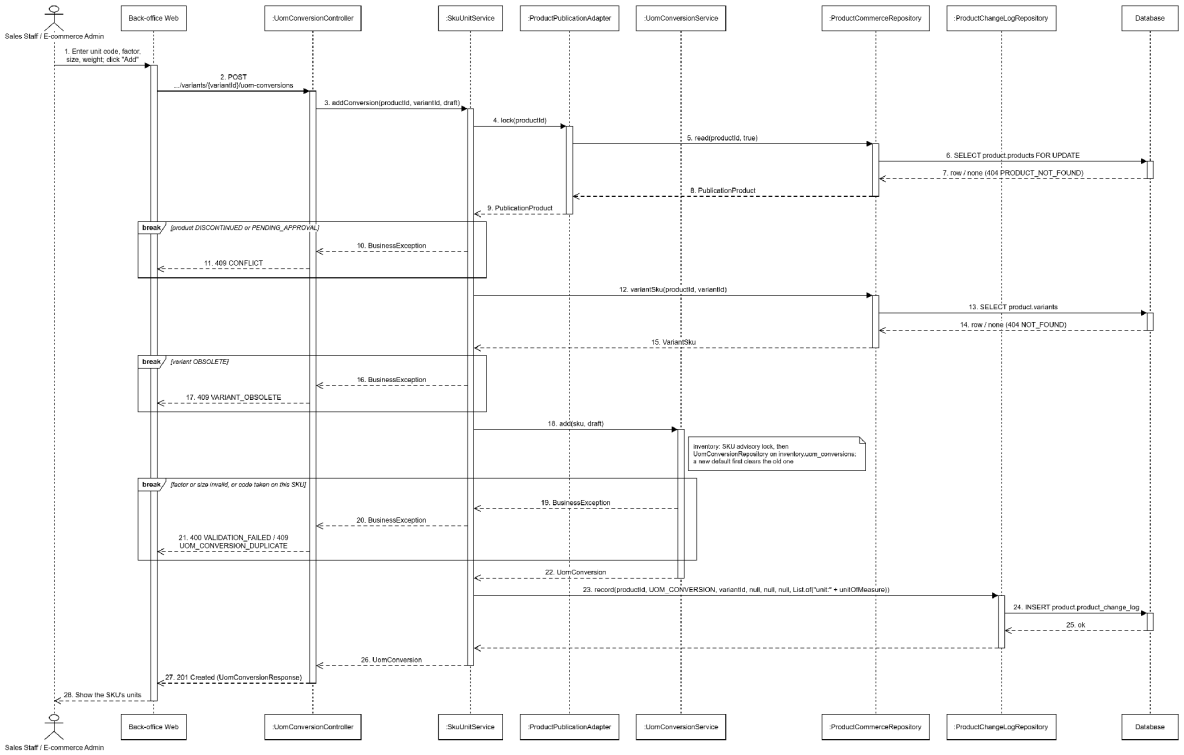


Figure 92. Create / Edit Product & Variants (Declare Packing Unit) Sequence Diagram

3.6.5 Create / Edit Product & Variants (Add Sales Unit) Sequence Diagram

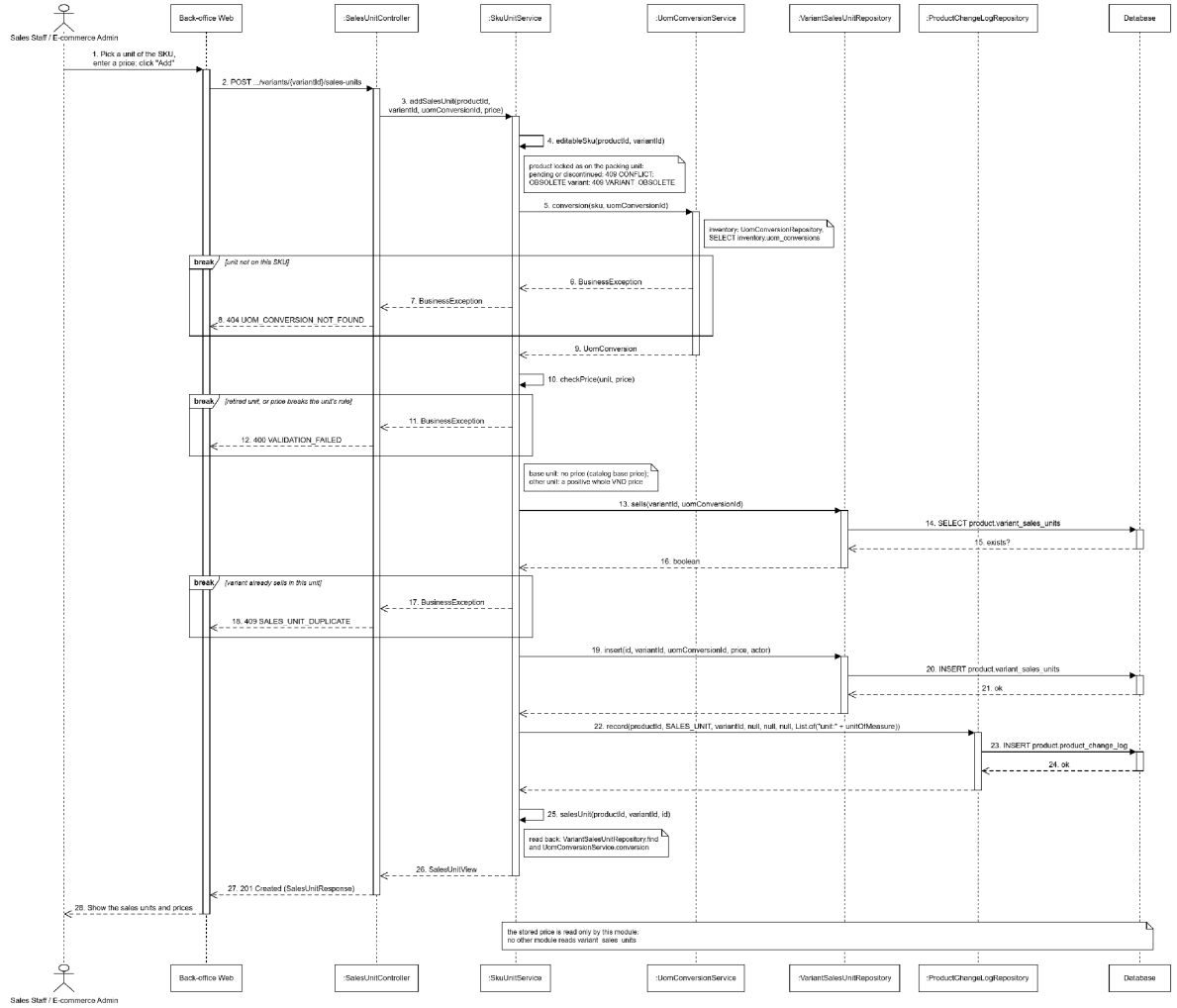


Figure 93. Create / Edit Product & Variants (Add Sales Unit) Sequence Diagram

3.6.6 Create / Edit Product & Variants (Edit SKU Purchasing Information) Sequence Diagram

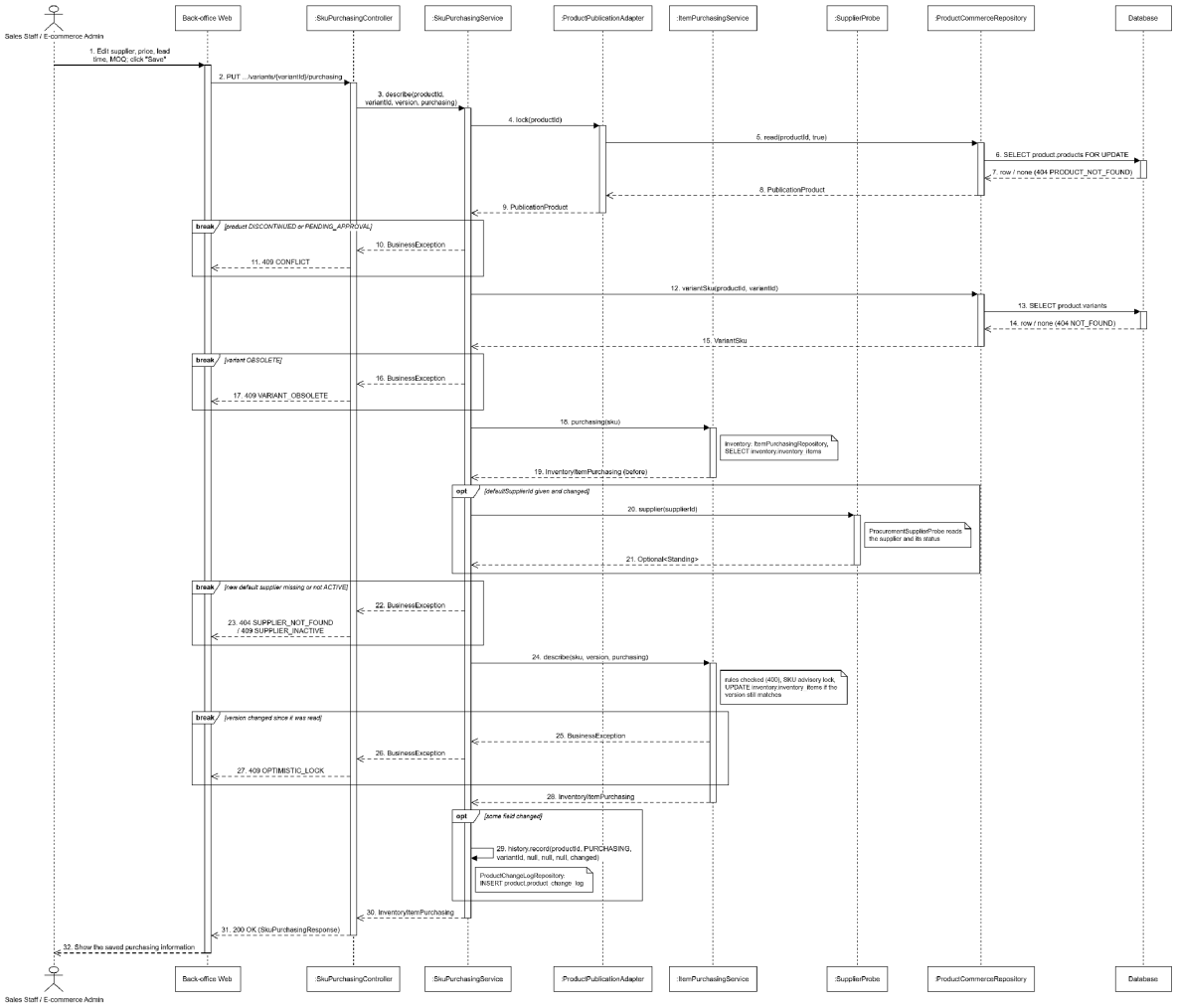


Figure 94. Create / Edit Product & Variants (Edit SKU Purchasing Information) Sequence Diagram

3.6.7 Product Variant (SKU) State Diagram

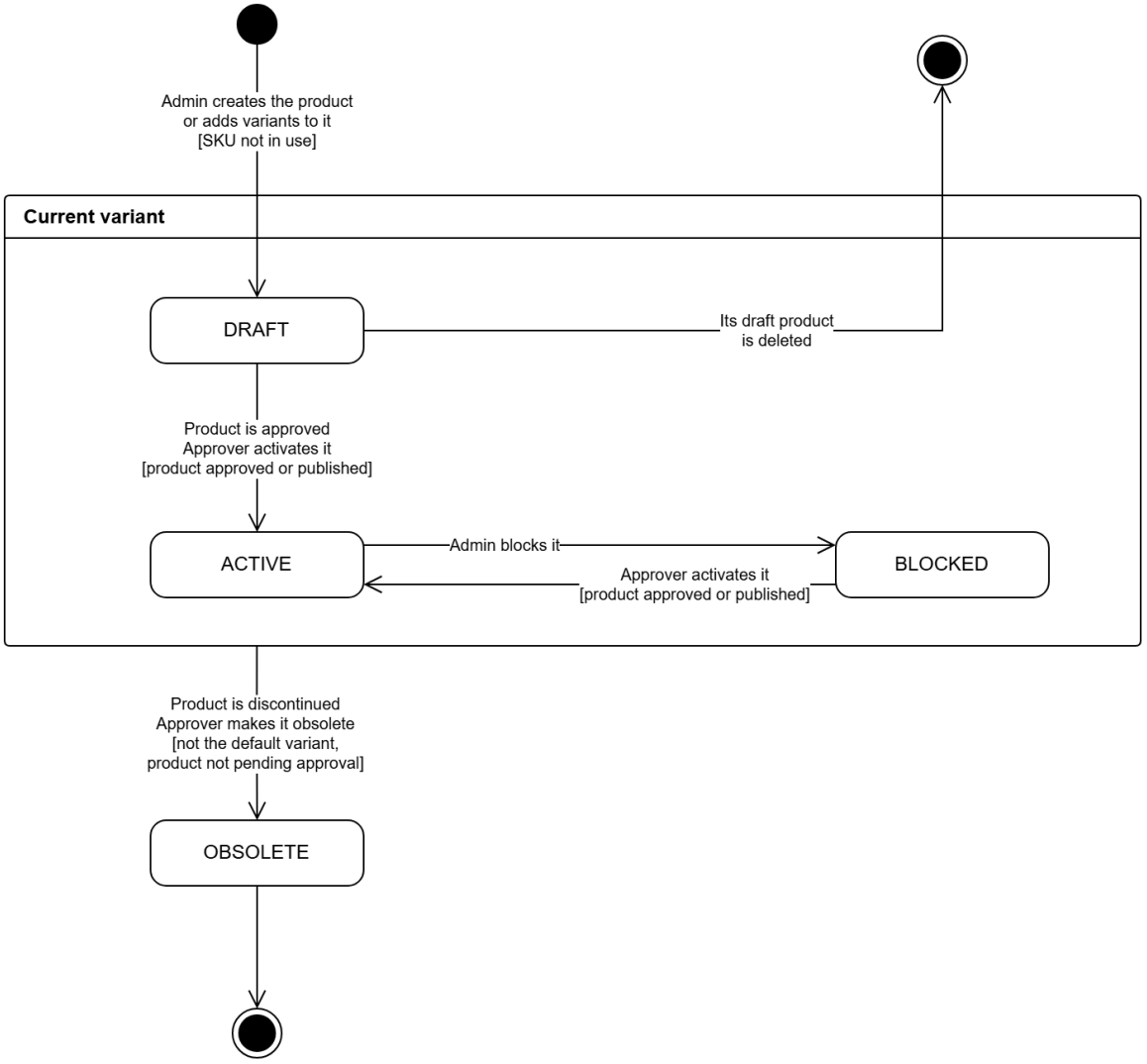


Figure 95. Product Variant (SKU) State Diagram

3.7 Catalog & Pricing

3.7.1 Class Diagram

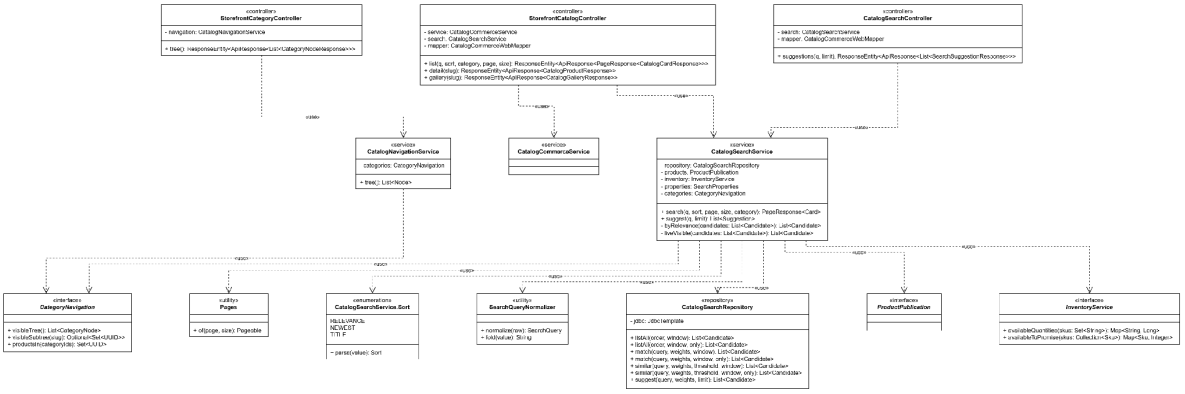


Figure 96. Catalog and Pricing Class Diagram (1/5): Storefront Browsing and Search

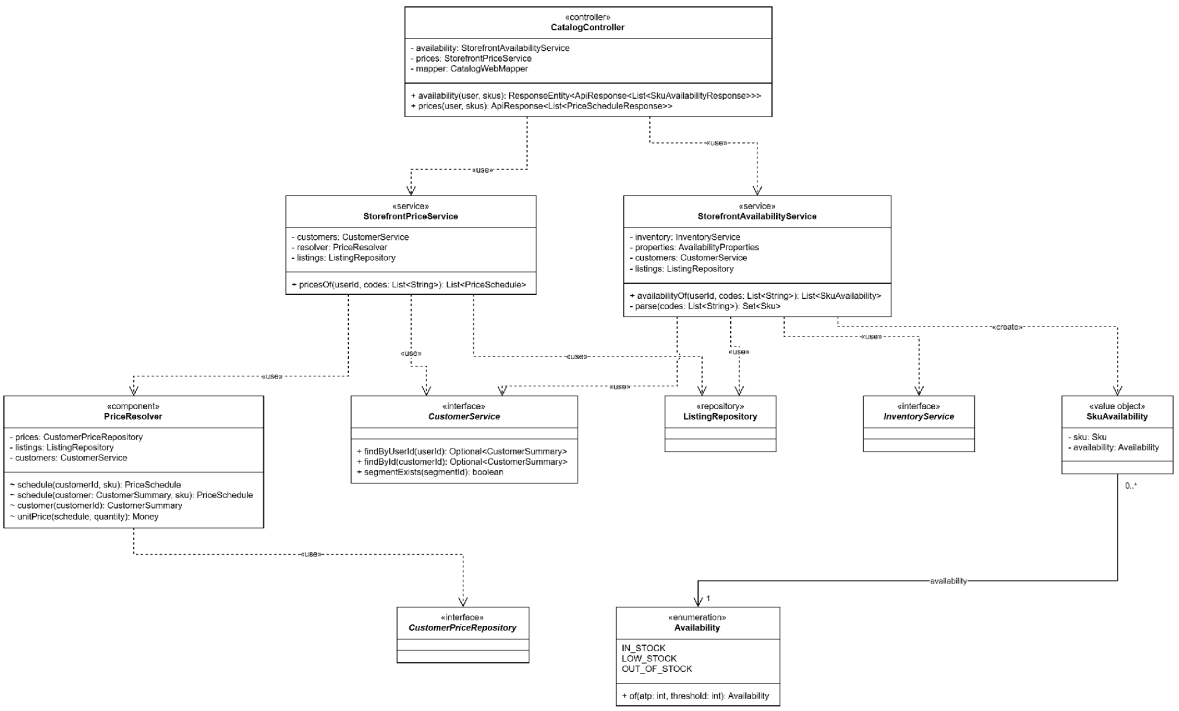


Figure 97. Catalog and Pricing Class Diagram (2/5): Storefront Prices and Availability

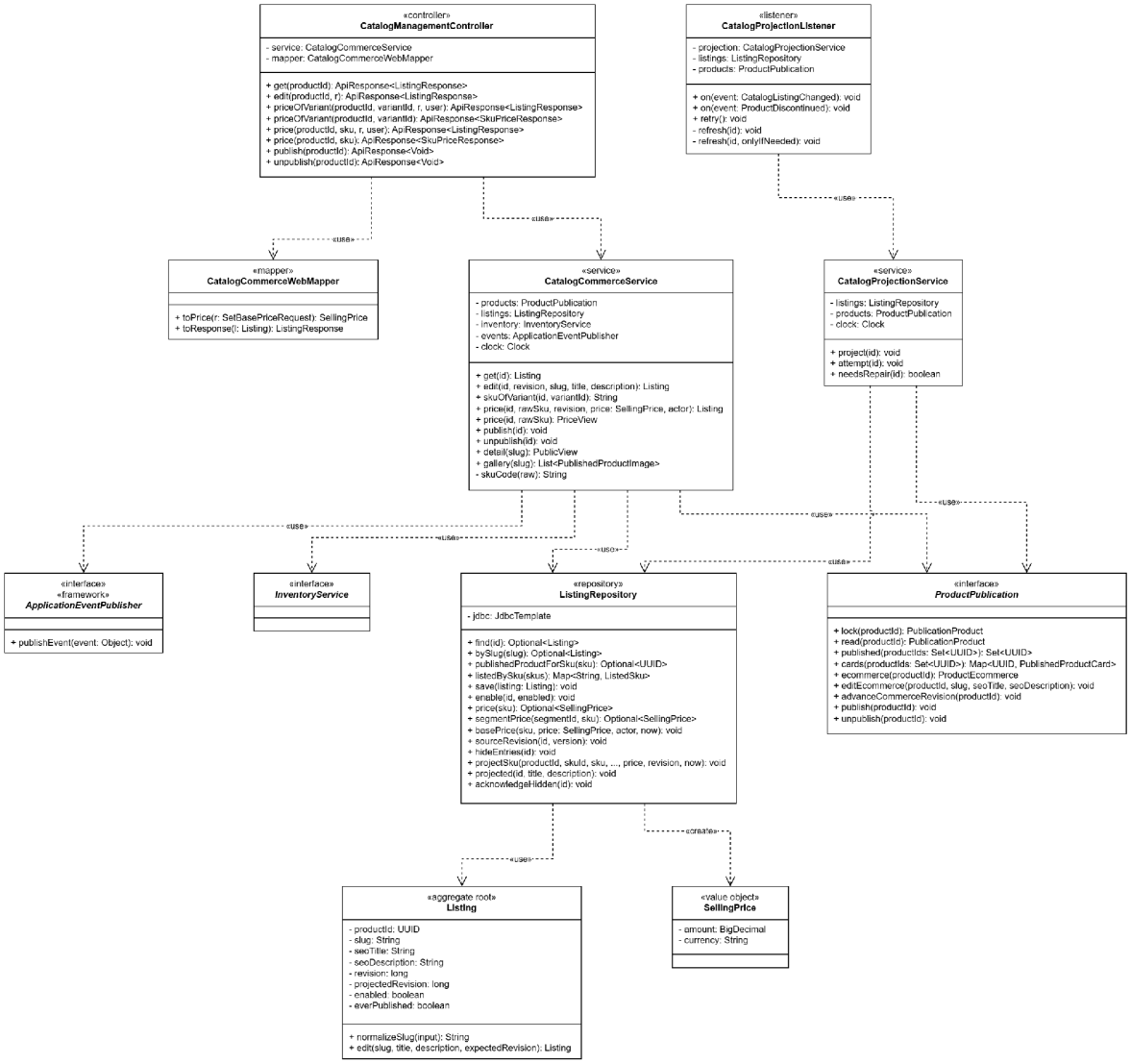


Figure 98. Catalog and Pricing Class Diagram (3/5): Catalog Content, Publication and Base Price

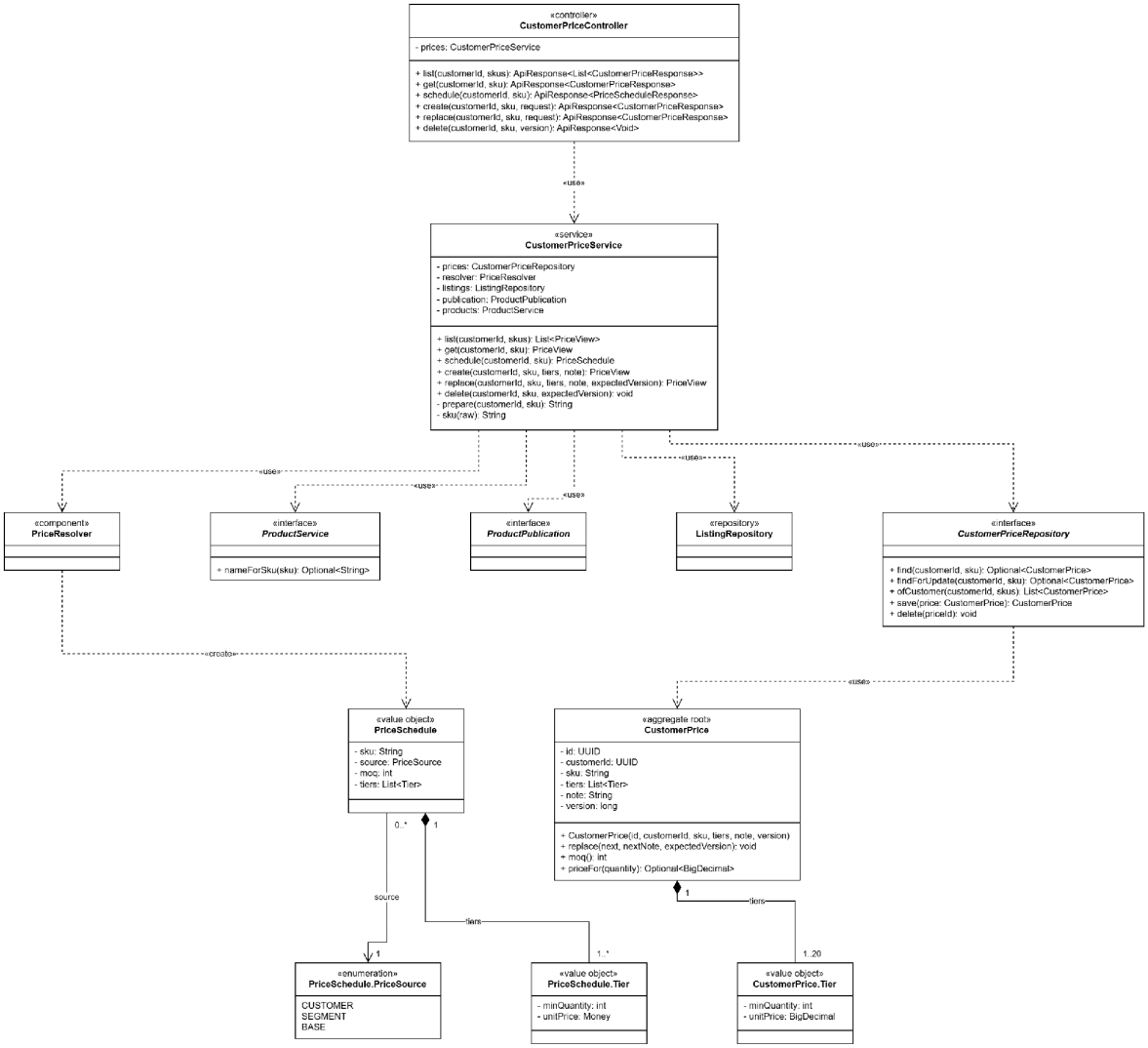


Figure 99. Catalog and Pricing Class Diagram (4/5): Customer Prices

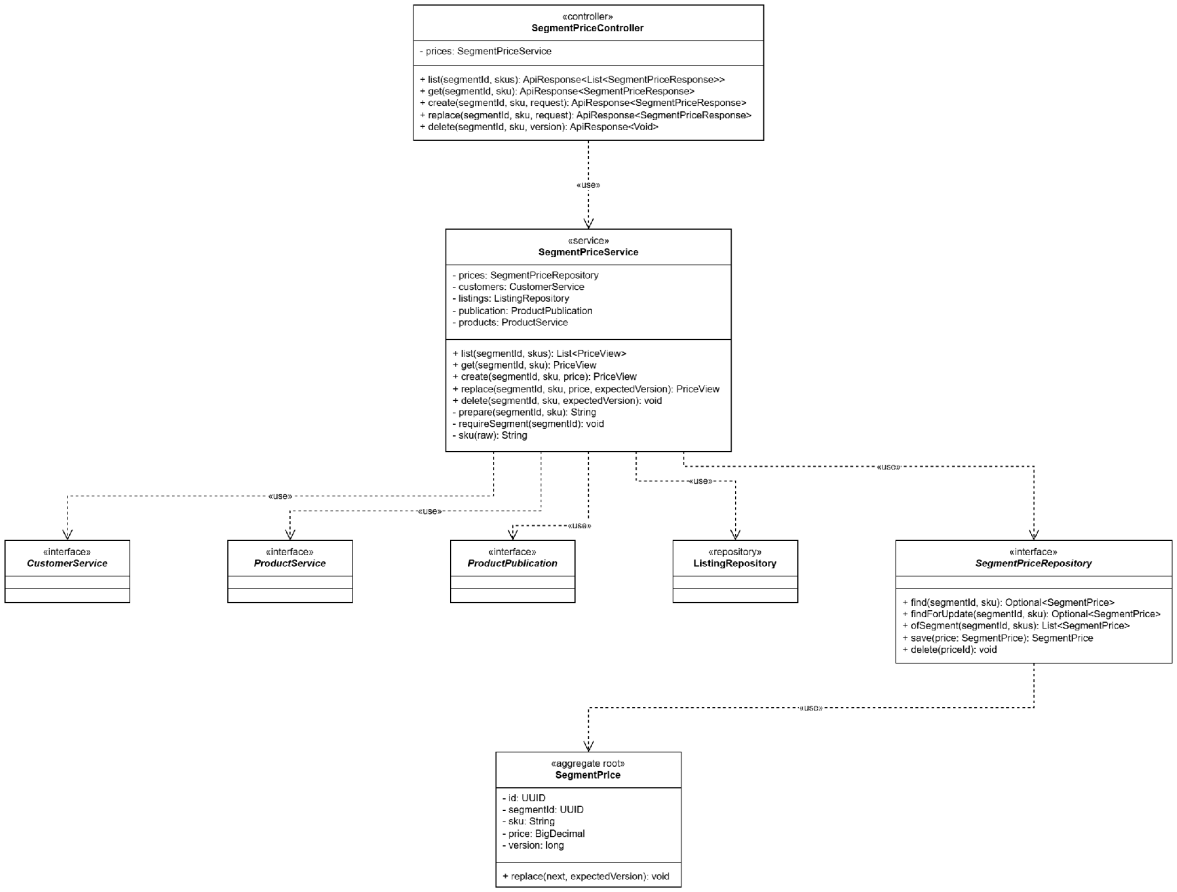


Figure 100. Catalog and Pricing Class Diagram (5/5): Customer Group Prices

3.7.2 Class Specification

StorefrontCatalogController

No	Method	Description
1	list(q, sort, category, page, size)	GET /api/v1/catalog/products: The published catalogue, searched with q (full text, then typo-tolerant), within a category subtree; sort relevance/newest/title (anything else 400 UNSUPPORTED_PARAMETER); cards carry no price or stock; no @RequiresPermission (any signed-in account), no-store. (:39)
2	detail(slug)	GET /api/v1/catalog/products/{slug}: Product page of an enabled, projected listing of a PUBLISHED product with each SKU's base price and IN_STOCK / OUT_OF_STOCK; no permission and no ACTIVE check (any signed-in account); 404 NOT_FOUND otherwise. (:51)
3	gallery(slug)	GET /api/v1/catalog/products/{slug}/gallery: Published images of the listing's product; 404 NOT_FOUND for an unknown slug; any signed-in account. (:58)

Table 53. StorefrontCatalogController Class Specification

StorefrontCategoryController

No	Method	Description
1	tree()	GET /api/v1/catalog/categories: The visible category tree, nested; an inactive category hides its subtree; any signed-in account, no-store. (:41)

Table 54. StorefrontCategoryController Class Specification

CatalogSearchController

No	Method	Description
1	suggestions(q, limit)	GET /api/v1/catalog/search/suggestions: Title completions for the search box: q of 2-100 characters (400 VALIDATION_FAILED), limit 1-10 (400 UNSUPPORTED_PARAMETER); 30 calls per 10 s per user; any signed-in account. (:44)

Table 55. CatalogSearchController Class Specification

CatalogController

No	Method	Description
1	availability(user, skus)	GET /api/v1/catalog/products/availability: IN_STOCK / LOW_STOCK / OUT_OF_STOCK per listed SKU, never a quantity, for the caller's own ACTIVE customer (403 FORBIDDEN otherwise); 1-50 SKUs; catalog-storefront:READ scope OWN (CUSTOMER role); 120 calls/min; cached privately 10 s. (:73)
2	prices(user, skus)	GET /api/v1/catalog/products/prices: The caller's own price per published SKU: its tiers and MOQ, else its group's price, else the base price; unpriced SKUs left out; ACTIVE customer only (403); 1-50 SKUs. (:90)

Table 56. CatalogController Class Specification

CatalogManagementController

No	Method	Description
1	get(productId)	GET /api/v1/products/{productId}/ecommerce: Listing view: slug, SEO, revision, projection progress, enabled / everPublished and the storefront copy (product-products:READ). (:33)
2	edit(productId, r)	PUT /api/v1/products/{productId}/ecommerce: Edits slug, SEO title and SEO description against the revision read; slug locked after the first publish (409 CONFLICT) and unique (409 CATALOG_SLUG_EXISTS); refused for DISCONTINUED (product-products:UPDATE). (:39)

3	priceOfVariant(productId, variantId, r, user)	PUT /api/v1/products/{productId}/variants/{variantId}/base-price: Sets the base price of an ACTIVE variant's SKU (positive whole VND) against the revision; needs a listing (409 CATALOG_LISTING_REQUIRED); rolled back when two rules tie (409 CATALOG_PRICE_AMBIGUOUS) (product-products:UPDATE). (:54)
4	priceOfVariant(productId, variantId)	GET /api/v1/products/{productId}/variants/{variantId}/base-price: Base and effective price of the variant's SKU, with the revision (product-products:READ). (:65)
5	price(productId, sku, r, user)	PUT /api/v1/products/{productId}/skus/{sku}/base-price: Deprecated alias of the variant route, by SKU code in any case. (:75)
6	price(productId, sku)	GET /api/v1/products/{productId}/skus/{sku}/base-price: Deprecated alias: base and effective price by SKU code. (:97)
7	publish(productId)	POST /api/v1/products/{productId}/publication: Publishes an APPROVED product: needs a listing, a category, SKUs, a selling price per SKU and an image approved by another user; the product becomes PUBLISHED and the storefront is projected after commit (product-products:APPROVE: E-commerce Admin, System Admin). (:104)
8	unpublish(productId)	POST /api/v1/products/{productId}/unpublication: Unlists a PUBLISHED product back to APPROVED (no-op when already APPROVED, 409 CONFLICT for other statuses) and hides its catalogue entries (product-products:APPROVE). (:111)

Table 57. CatalogManagementController Class Specification

CustomerPriceController

No	Method	Description
1	list(customerId, skus)	GET /api/v1/catalog/customer-prices?customerId=&sku=: A customer's own price lists, optionally for some SKUs (catalog-customer-prices:READ: E-commerce Admin, Sales, Order Coordinator, System Admin). (:53)
2	get(customerId, sku)	GET /api/v1/catalog/customer-prices/{customerId}/{sku}: One own price with tiers, MOQ and version; 404 CUSTOMER_PRICE_NOT_FOUND. (:61)

3	schedule(customerId, sku)	GET /api/v1/catalog/customer-prices/{customerId}/{sku}/schedule: What the customer pays now (own tiers, else group price, else base price): the default of a quote line. (:68)
4	create(customerId, sku, request)	POST /api/v1/catalog/customer-prices/{customerId}/{sku}: Creates 1-20 quantity tiers, first tier = MOQ, whole VND; 409 CONFLICT if one exists; locks the published product first (CREATE: E-commerce Admin, System Admin). (:75)
5	replace(customerId, sku, request)	PUT /api/v1/catalog/customer-prices/{customerId}/{sku}: Replaces the tiers against the version (400 without one, 409 OPTIMISTIC_LOCK when stale) (UPDATE). (:82)
6	delete(customerId, sku, version)	DELETE /api/v1/catalog/customer-prices/{customerId}/{sku}?version=: Deletes the own price against its version (DELETE). (:93)

Table 58. CustomerPriceController Class Specification

SegmentPriceController

No	Method	Description
1	list(segmentId, skus)	GET /api/v1/catalog/segment-prices?segmentId=&sku=: A customer group's prices; 404 SEGMENT_NOT_FOUND (catalog-segment-prices:READ). (:50)
2	get(segmentId, sku)	GET /api/v1/catalog/segment-prices/{segmentId}/{sku}: One group price with version; 404 SEGMENT_PRICE_NOT_FOUND. (:58)
3	create(segmentId, sku, request)	POST /api/v1/catalog/segment-prices/{segmentId}/{sku}: One price (no tiers) per group and SKU, positive whole VND; 409 CONFLICT if it exists (CREATE: E-commerce Admin, System Admin). (:65)
4	replace(segmentId, sku, request)	PUT /api/v1/catalog/segment-prices/{segmentId}/{sku}: Changes the price against the version (400 without one, 409 OPTIMISTIC_LOCK when stale) (UPDATE). (:72)

5	delete(segmentId, sku, version)	DELETE /api/v1/catalog/segment-prices/{segmentId}/{sku}?version=: Removes the group price against its version (DELETE). (:82)

Table 59. SegmentPriceController Class Specification

3.7.3 Browse & Search Catalog Sequence Diagram (1/2)

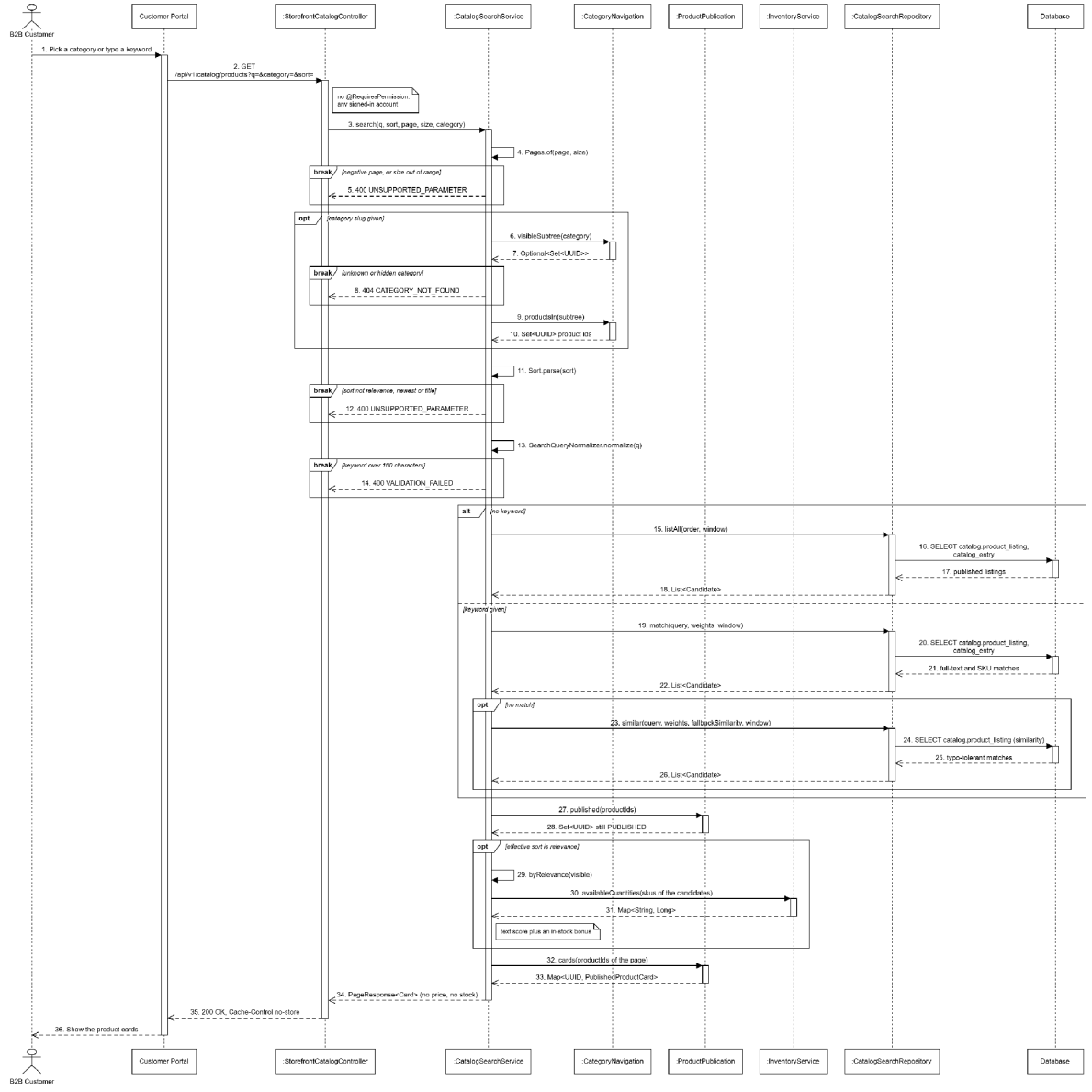


Figure 101. Browse & Search Catalog Sequence Diagram (1/2)

3.7.4 Browse & Search Catalog Sequence Diagram (2/2)

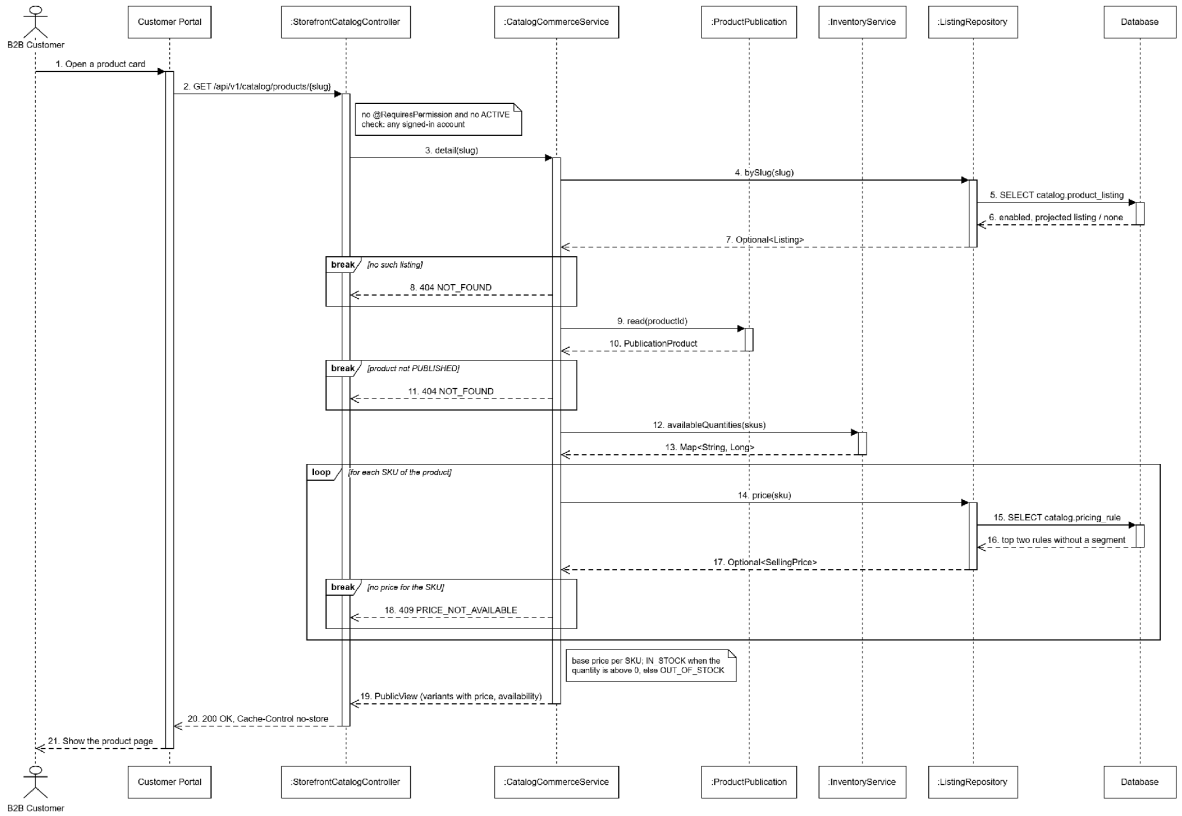


Figure 102. Browse & Search Catalog Sequence Diagram (2/2)

3.7.5 View Product Price & Availability Sequence Diagram (1/2)

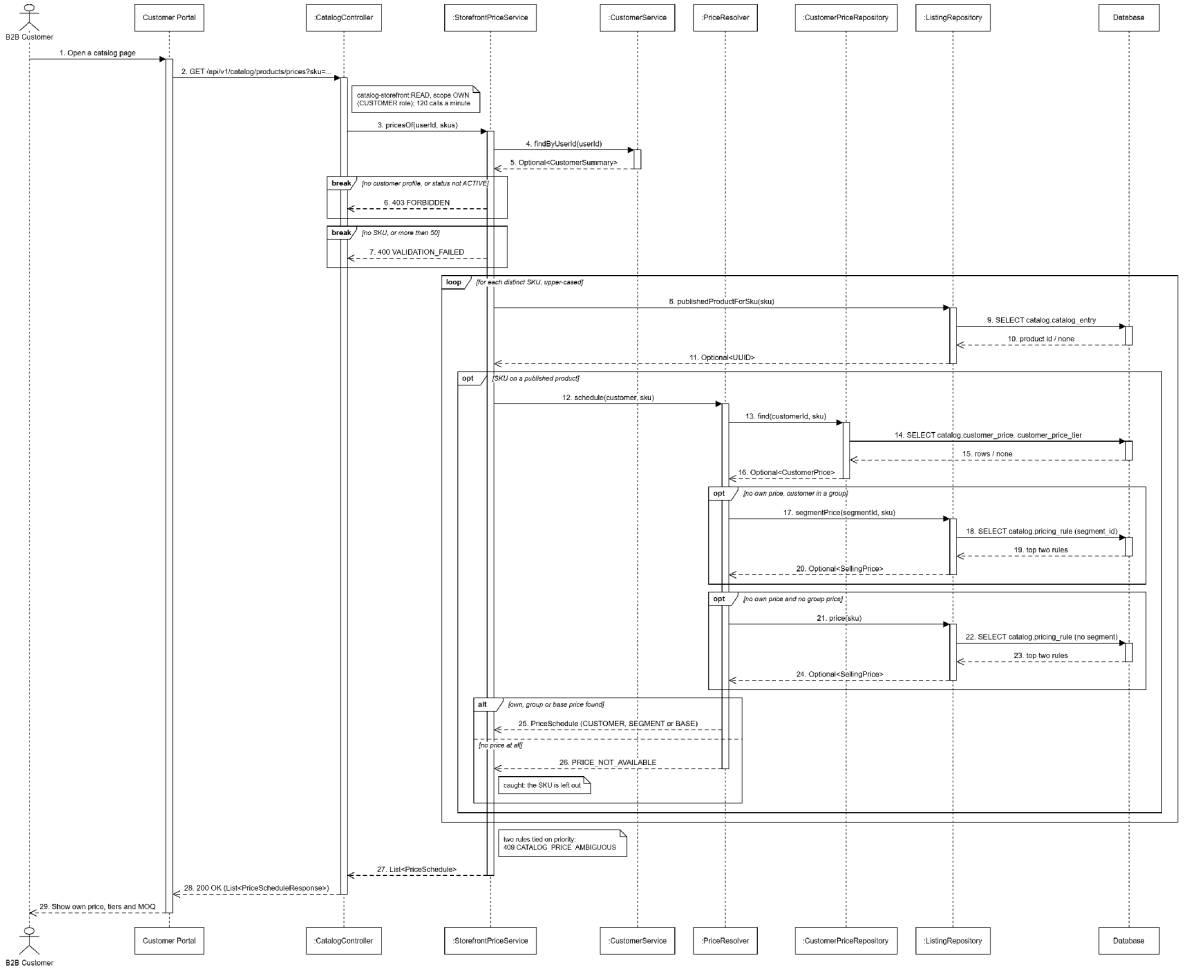


Figure 103. View Product Price & Availability Sequence Diagram (1/2)

3.7.6 View Product Price & Availability Sequence Diagram (2/2)

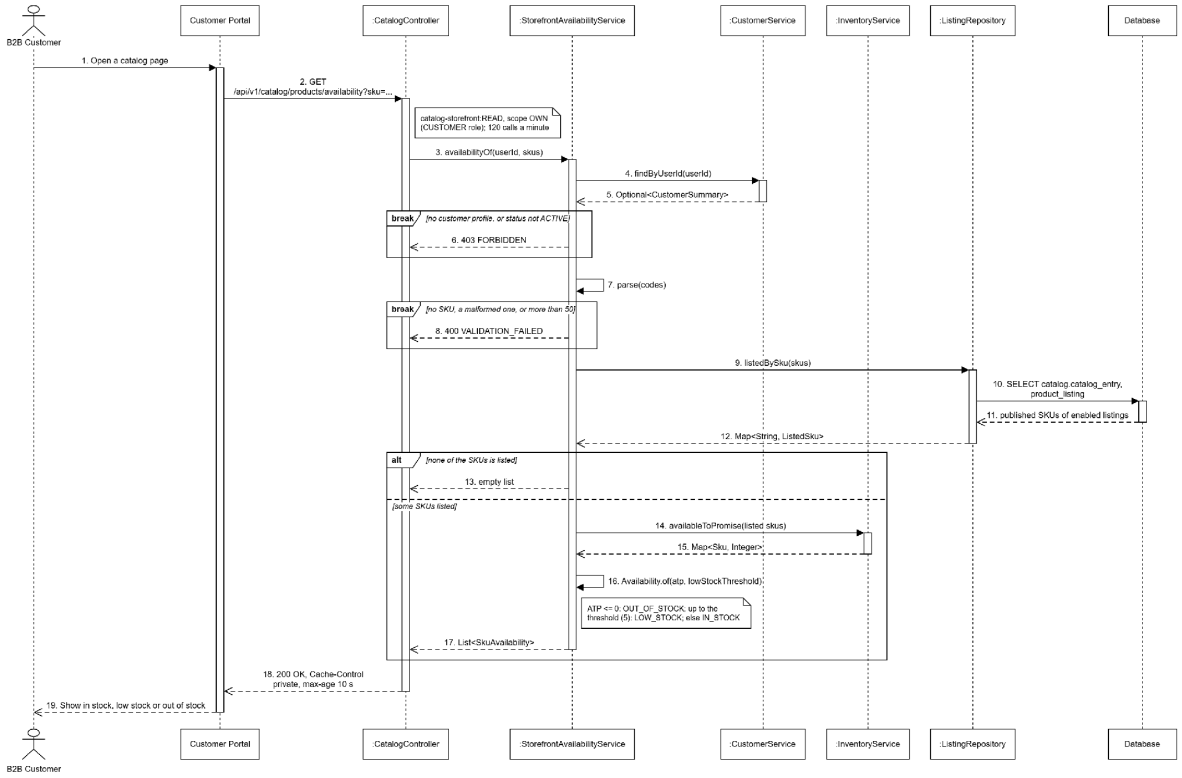


Figure 104. View Product Price & Availability Sequence Diagram (2/2)

3.7.7 Publish / Unlist Product Sequence Diagram (1/2)

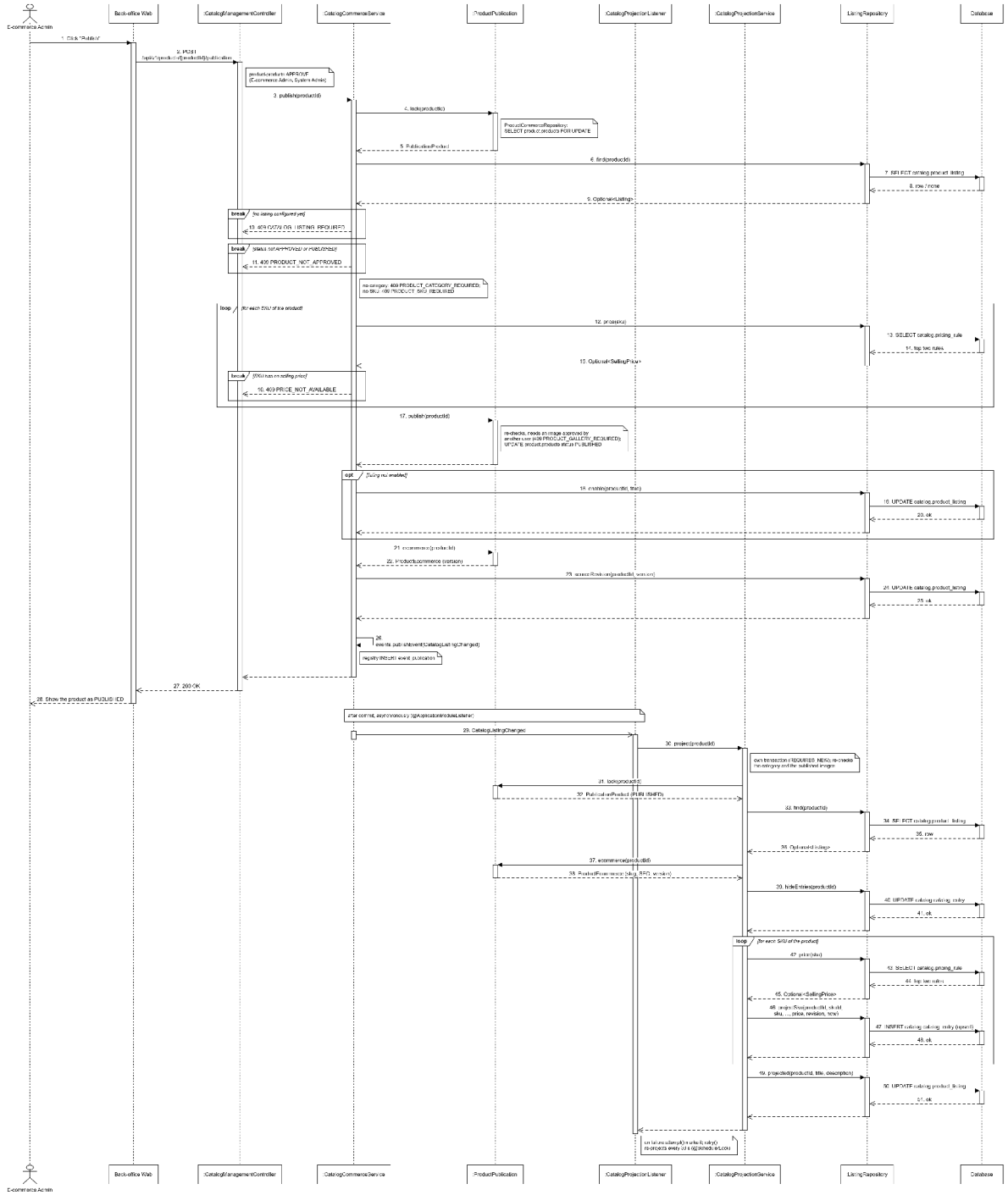


Figure 105. Publish / Unlist Product Sequence Diagram (1/2)

3.7.8 Publish / Unlist Product Sequence Diagram (2/2)

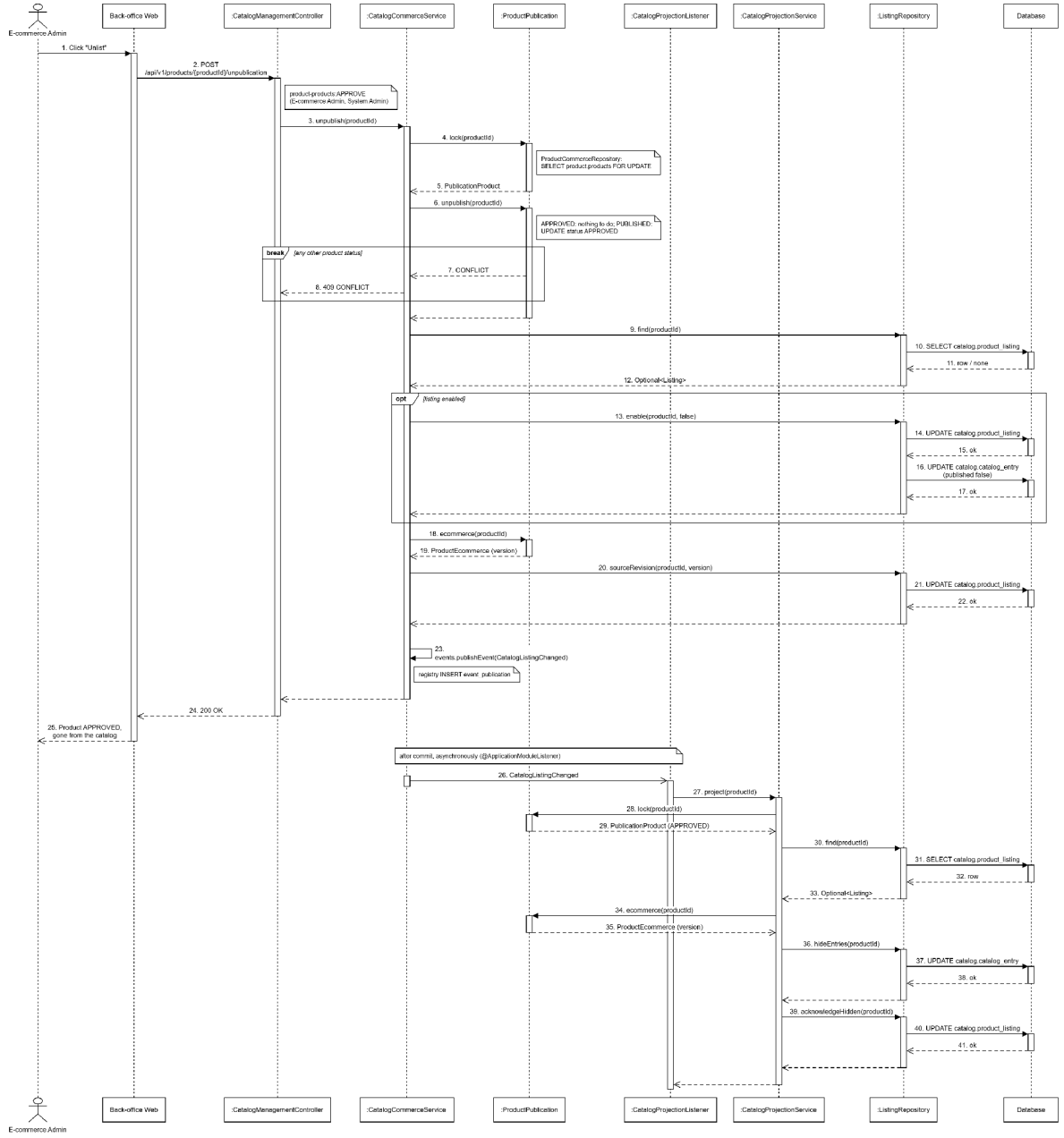


Figure 106. Publish / Unlist Product Sequence Diagram (2/2)

3.7.9 Edit Catalog Content (SEO & Slug) Sequence Diagram

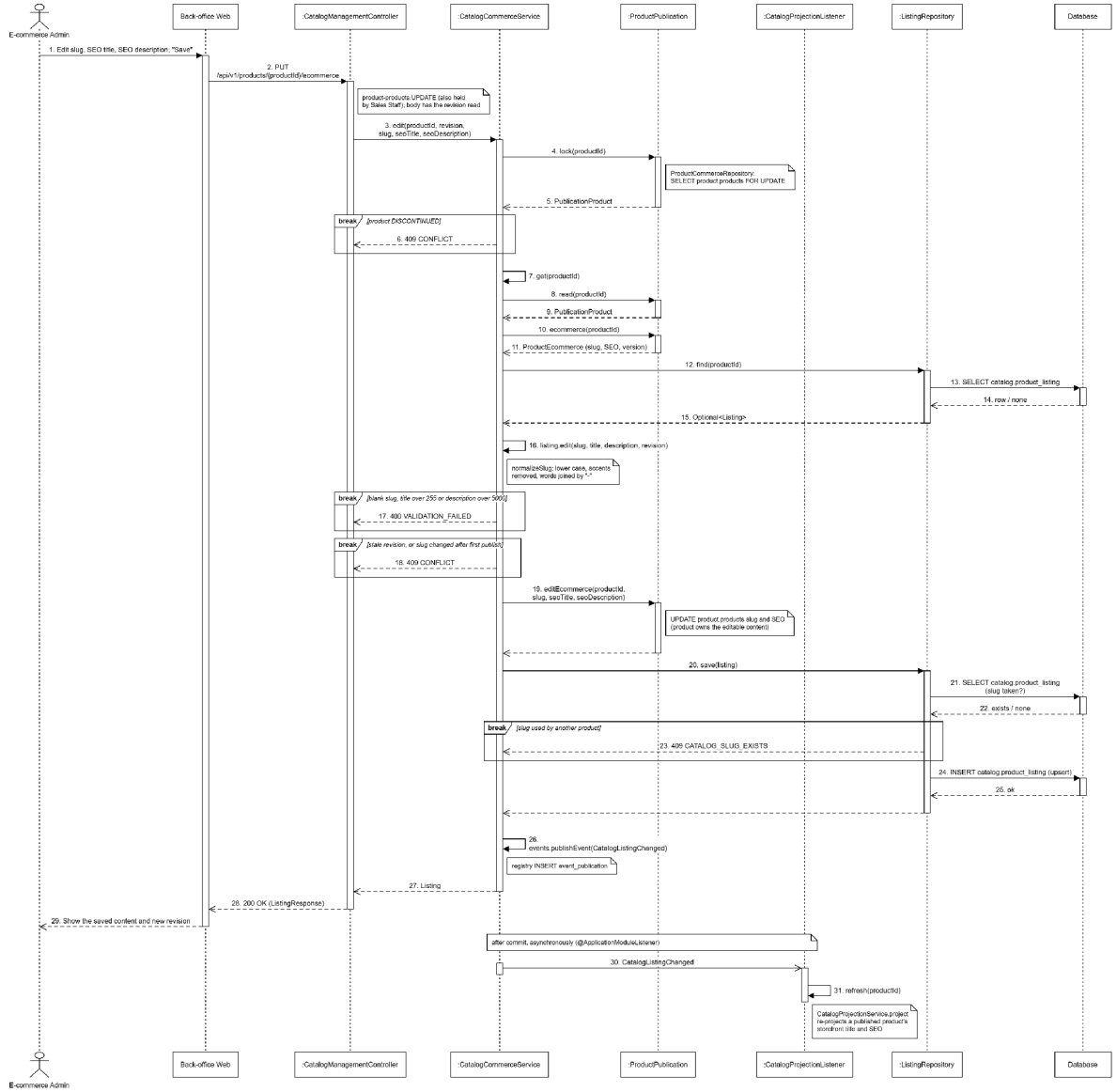


Figure 107. Edit Catalog Content (SEO & Slug) Sequence Diagram

3.7.10 Manage Base, Customer & Group Prices Sequence Diagram (1/3)

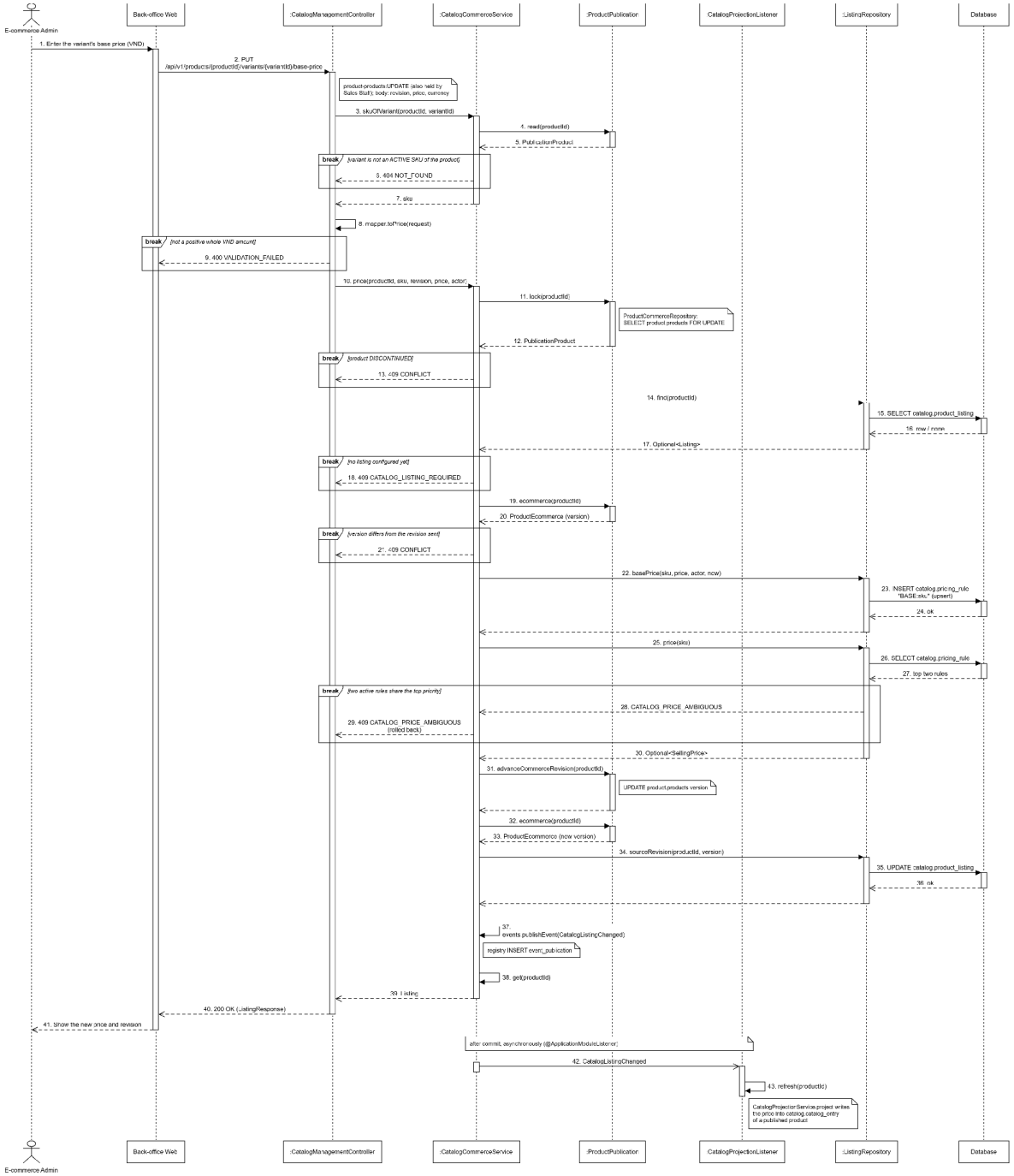


Figure 108. Manage Base, Customer & Group Prices Sequence Diagram (1/3)

3.7.11 Manage Base, Customer & Group Prices Sequence Diagram (2/3)

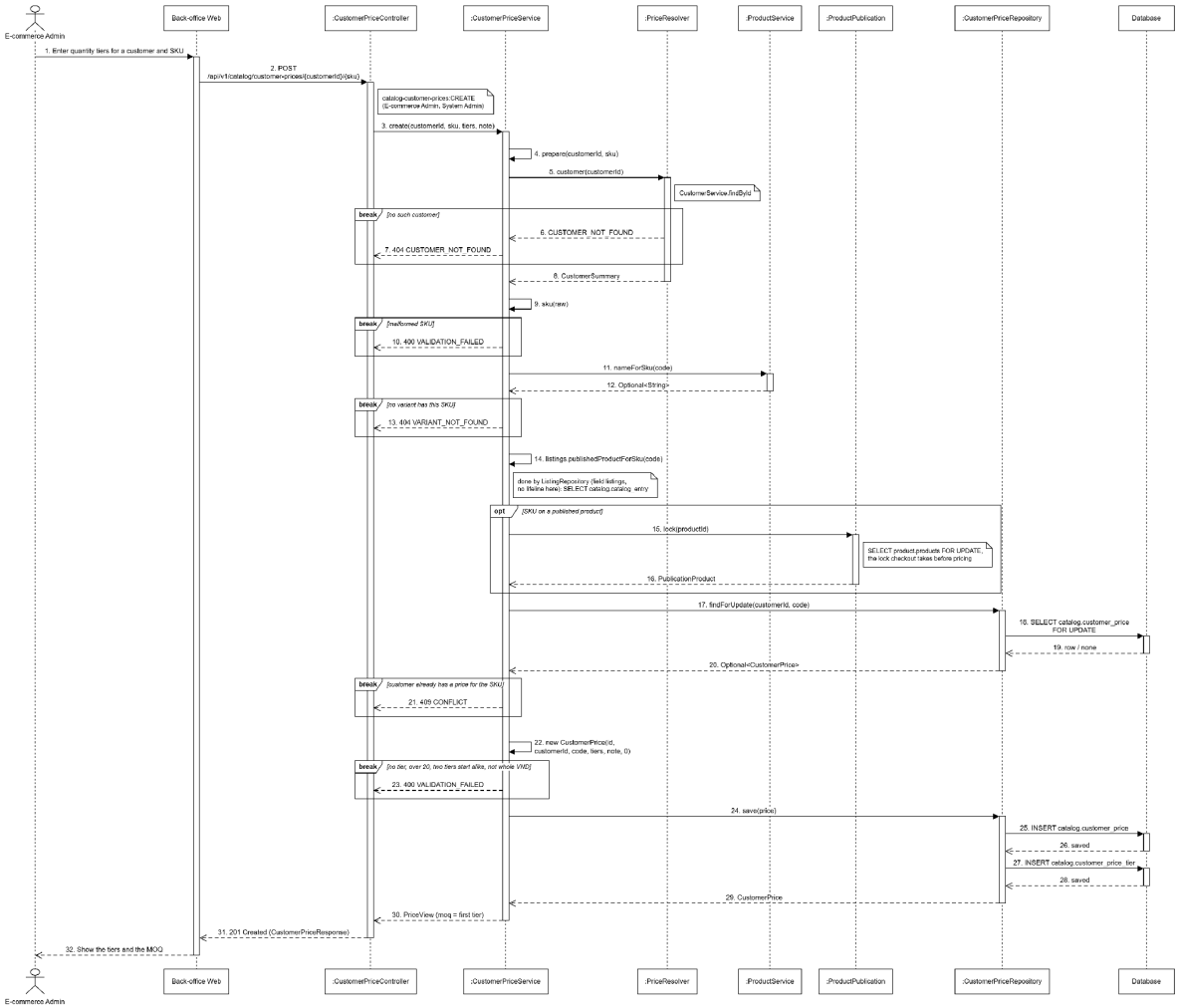


Figure 109. Manage Base, Customer & Group Prices Sequence Diagram (2/3)

3.7.12 Manage Base, Customer & Group Prices Sequence Diagram (3/3)

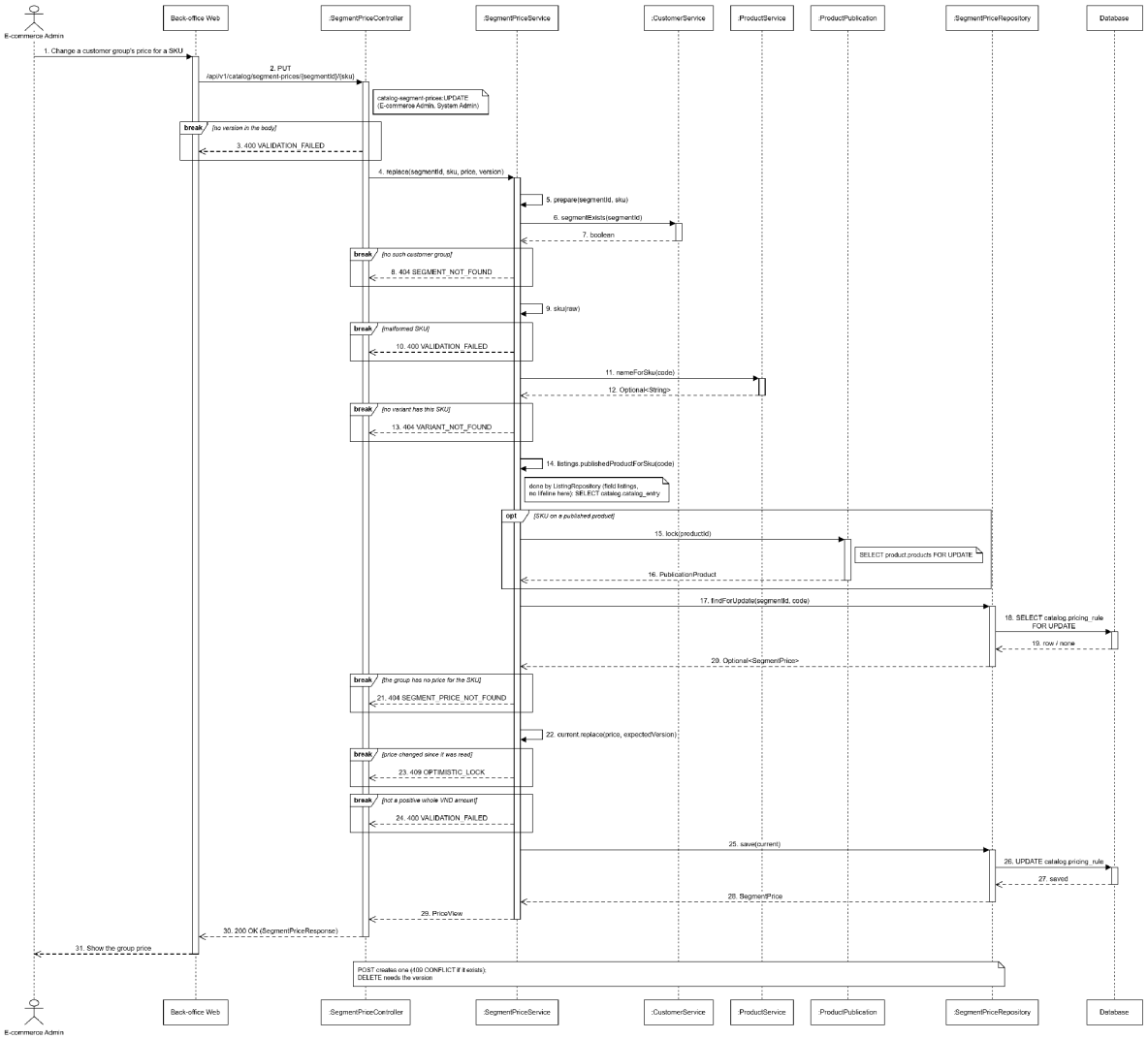


Figure 110. Manage Base, Customer & Group Prices Sequence Diagram (3/3)

3.8 Design Management

3.8.1 Class Diagram

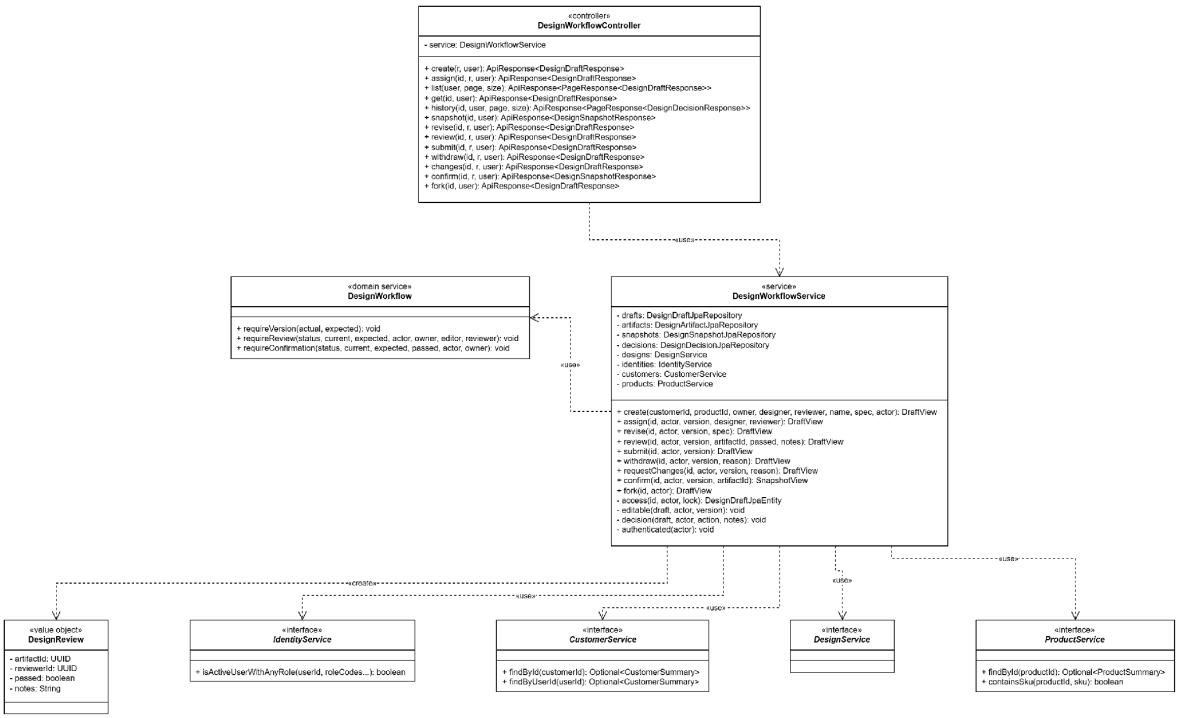


Figure 111. Design Management Class Diagram (1/8): Design Workflow

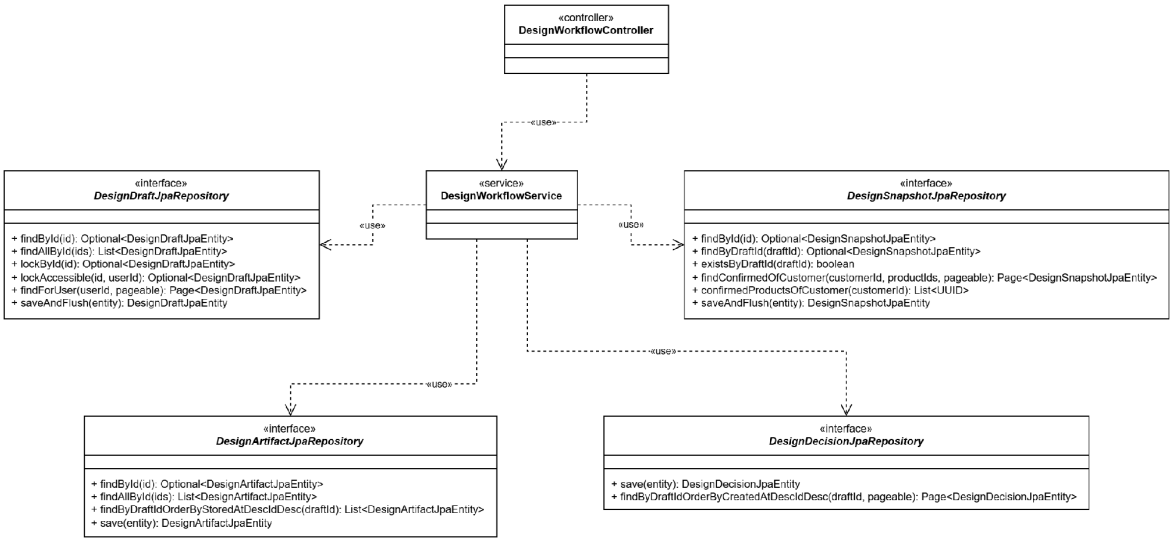


Figure 112. Design Management Class Diagram (2/8): Design Persistence

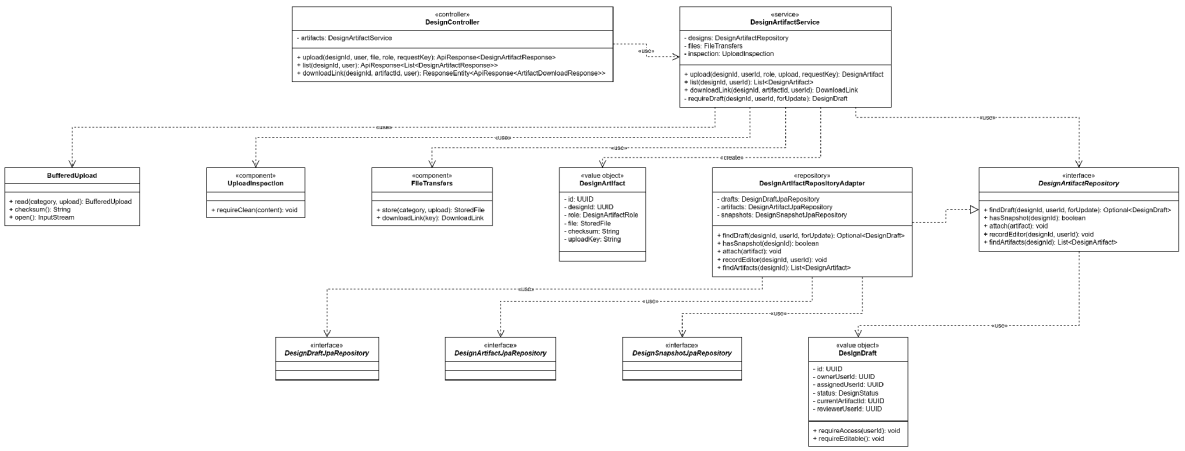


Figure 113. Design Management Class Diagram (3/8): Design Files

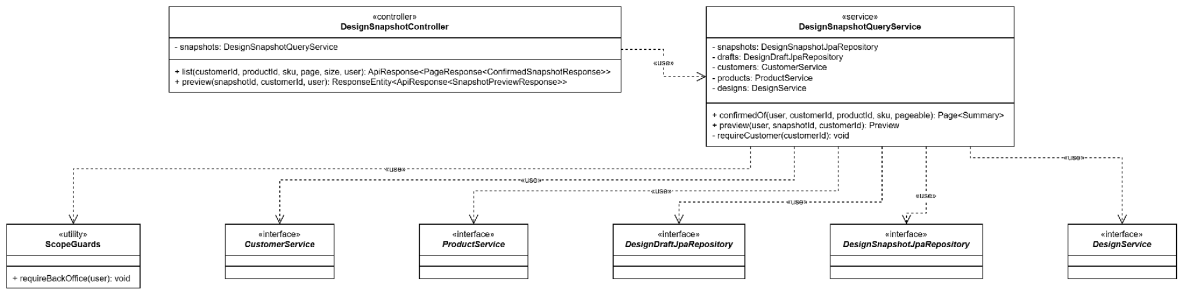


Figure 114. Design Management Class Diagram (4/8): Confirmed Snapshots

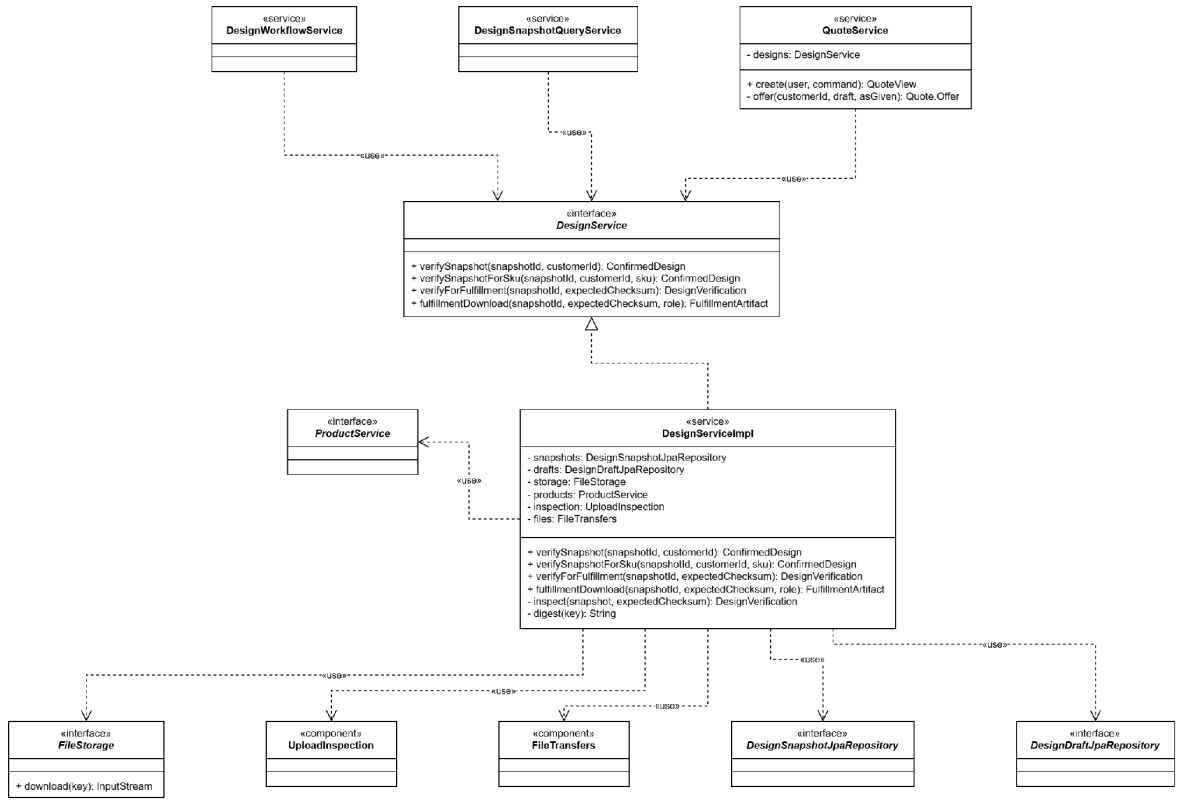


Figure 115. Design Management Class Diagram (5/8): Design Integrity Verification

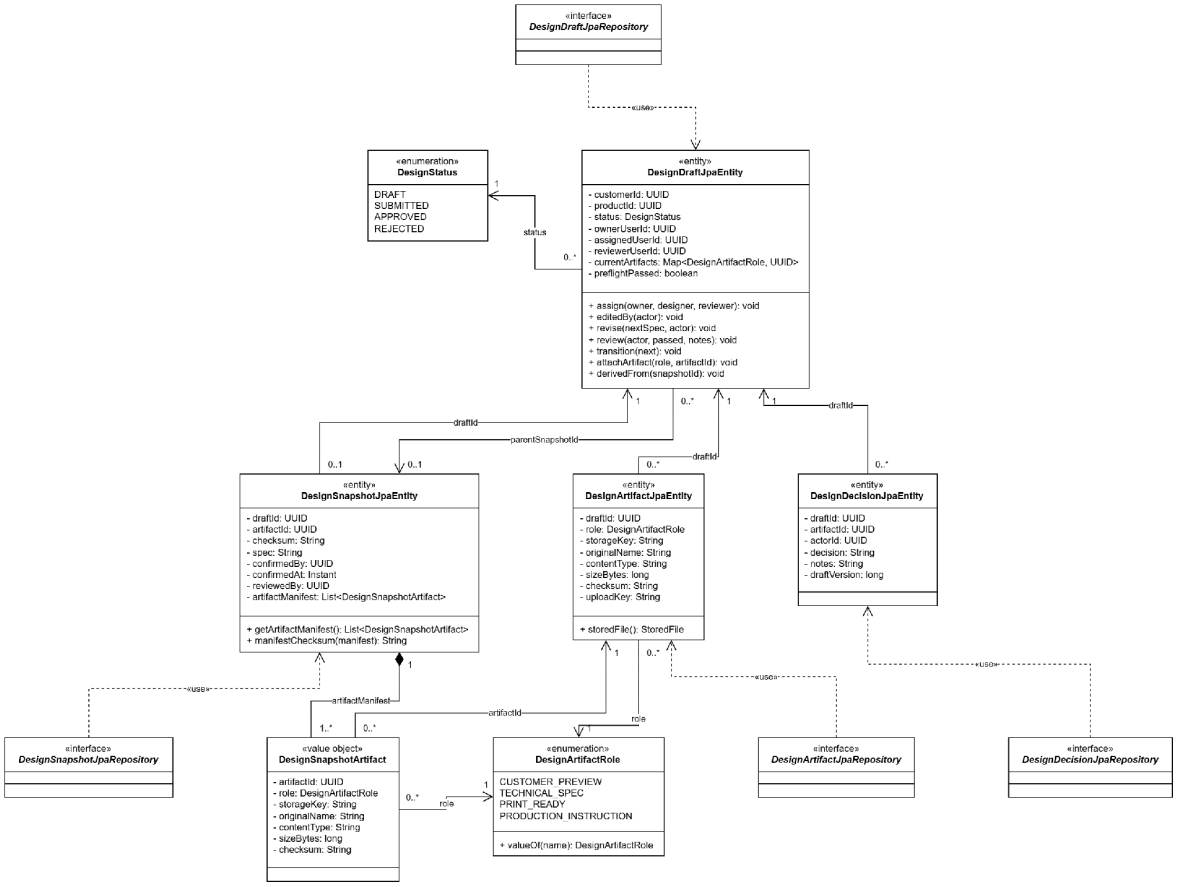


Figure 116. Design Management Class Diagram (6/8): Design Domain Model

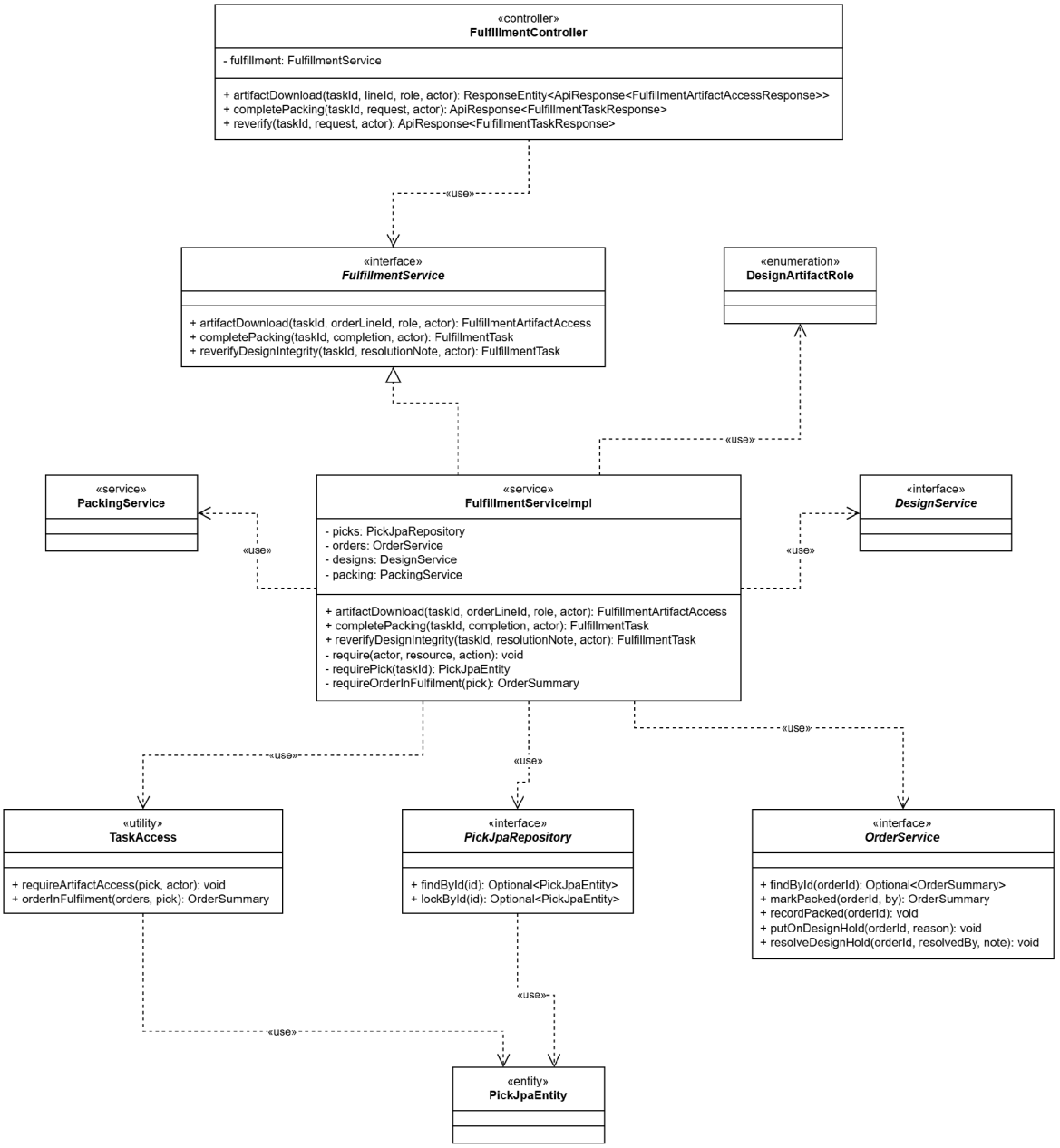


Figure 117. Design Management Class Diagram (7/8): Design Files in Fulfillment

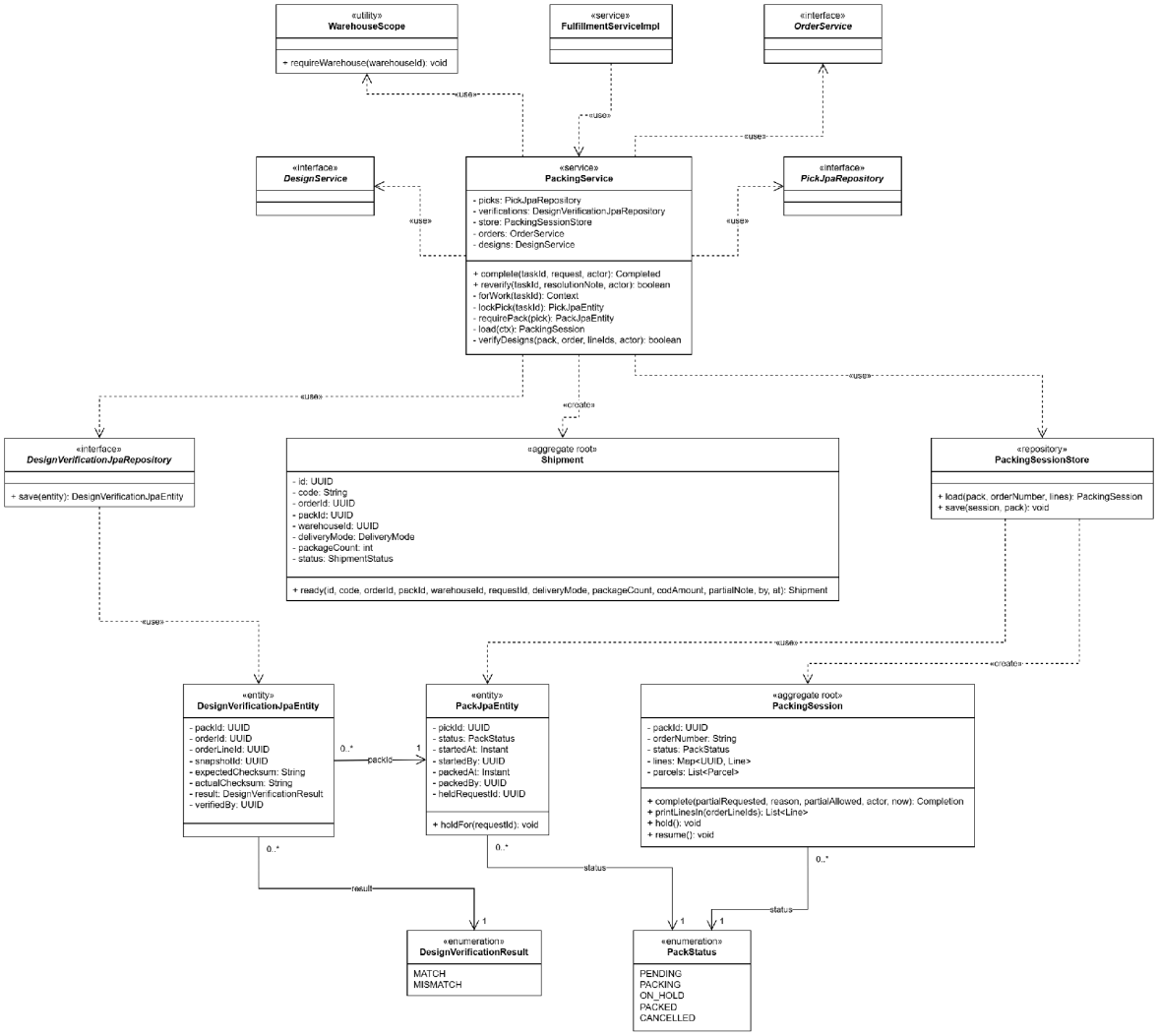


Figure 118. Design Management Class Diagram (8/8): Design Check at Packing

3.8.2 Class Specification

DesignWorkflowController

No	Method	Description
1	assign(id, r, user)	PUT /api/v1/designs/{id}/assignment: design-administration:UPDATE (ECOMMERCE_ADMIN). Replaces the designer and QC reviewer, owner kept; reviewer must be an active QC_STAFF other than owner and designer (400), a confirmed draft is refused (409); resets the draft to DRAFT and clears the QC verdict. Locks the draft by id with no owner/designer/reviewer check and returns the full draft. (:56)

2	create(r, user)	POST /api/v1/designs: design-administration:CREATE. Creates a DRAFT design brief for a customer and product: owner = an active CUSTOMER login of that customer (400 DESIGN_OWNER_CUSTOMER_MISMATCH), optional designer (SALES_STAFF or ECOMMERCE_ADMIN), independent QC reviewer (400); product must exist (404); answers 200. (:63)
3	list(user, page, size)	GET /api/v1/designs: Drafts the caller owns, designs or reviews, newest first; an oversized page is 400. (:70)
4	get(id, user)	GET /api/v1/designs/{id}: One draft; 404 unless the caller is its owner, designer or reviewer. (:76)
5	history(id, user, page, size)	GET /api/v1/designs/{id}/history: The draft's append-only decisions (design.design_decision), newest first; same 404 rule. (:81)
6	snapshot(id, user)	GET /api/v1/designs/{id}/snapshot: The draft's confirmed snapshot with its file manifest; 404 when it has none. (:87)
7	withdraw(id, r, user)	POST /api/v1/designs/{id}/confirmation-withdrawals: Owner or designer withdraws a pending confirmation request with a reason: SUBMITTED → DRAFT and the QC verdict is cleared; 409 when nothing is pending. (:92)
8	revise(id, r, user)	PUT /api/v1/designs/{id}/specification: Owner or designer replaces the specification of a DRAFT (stale version, SUBMITTED or confirmed: 409); clears the QC verdict. (:98)
9	review(id, r, user)	POST /api/v1/designs/{id}/technical-reviews: design-designs:APPROVE. The assigned QC reviewer, never the owner or last editor (403), passes or fails the current CUSTOMER_PREVIEW file of a DRAFT with notes; sets preflight_passed (other file or status: 409). (:104)
10	submit(id, r, user)	POST /api/v1/designs/{id}/confirmation-requests: Owner or designer sends a DRAFT whose current file passed QC review to the customer: DRAFT → SUBMITTED (409 "Technical review is required"). (:110)
11	changes(id, r, user)	POST /api/v1/designs/{id}/change-requests: The owner (customer) asks for changes with a reason: SUBMITTED → DRAFT, QC verdict cleared; anyone else 403, not SUBMITTED 409. (:116)

12	confirm(id, r, user)	POST /api/v1/designs/{id}/confirmations: The owner confirms the current file of a SUBMITTED, reviewed draft: snapshot of the current bundle, stored bytes re-hashed, draft → APPROVED; the same artifact again returns the same snapshot. (:122)
13	fork(id, user)	POST /api/v1/designs/{id}/revisions: Owner or designer starts a new DRAFT from an APPROVED draft (same customer, product, spec and team; parent_snapshot_id set; no file copied); 409 if not APPROVED. (:128)

Table 60. DesignWorkflowController Class Specification

DesignController

No	Method	Description
1	upload(designId, user, file, role, requestKey)	POST /api/v1/designs/{designId}/artifacts (multipart): Stores a new file revision with a role (default CUSTOMER_PREVIEW) on a DRAFT the caller owns or designs, after a malware scan (400 if infected); idempotent per Idempotency-Key (422 when reused for another file); clears the QC verdict; 201. (:51)
2	list(designId, user)	GET /api/v1/designs/{designId}/artifacts: The draft's file revisions, newest first, for its owner, designer or reviewer. (:66)
3	downloadLink(designId, artifactId, user)	GET /api/v1/designs/{designId}/artifacts/{artifactId}/download-url: A short-lived storage link to one revision for the owner, designer or reviewer, Cache-Control: no-store; the file is not re-hashed. (:74)

Table 61. DesignController Class Specification

DesignSnapshotController

No	Method	Description
1	list(customerId, productId, sku, page, size, user)	GET /api/v1/design-snapshots: Back office only: a customer's confirmed snapshots, newest first, optionally only for one product or for products holding a SKU; 404 CUSTOMER_NOT_FOUND. (:48)
2	preview(snapshotId, customerId, user)	GET /api/v1/design-snapshots/{snapshotId}/preview: Back office only: short-lived link to the CUSTOMER_PREVIEW the customer confirmed; another customer's snapshot is 404, files re-hashed first (409 if changed); no-store. (:64)

Table 62. DesignSnapshotController Class Specification

3.8.3 Create Design Brief & Assign Team Sequence Diagram

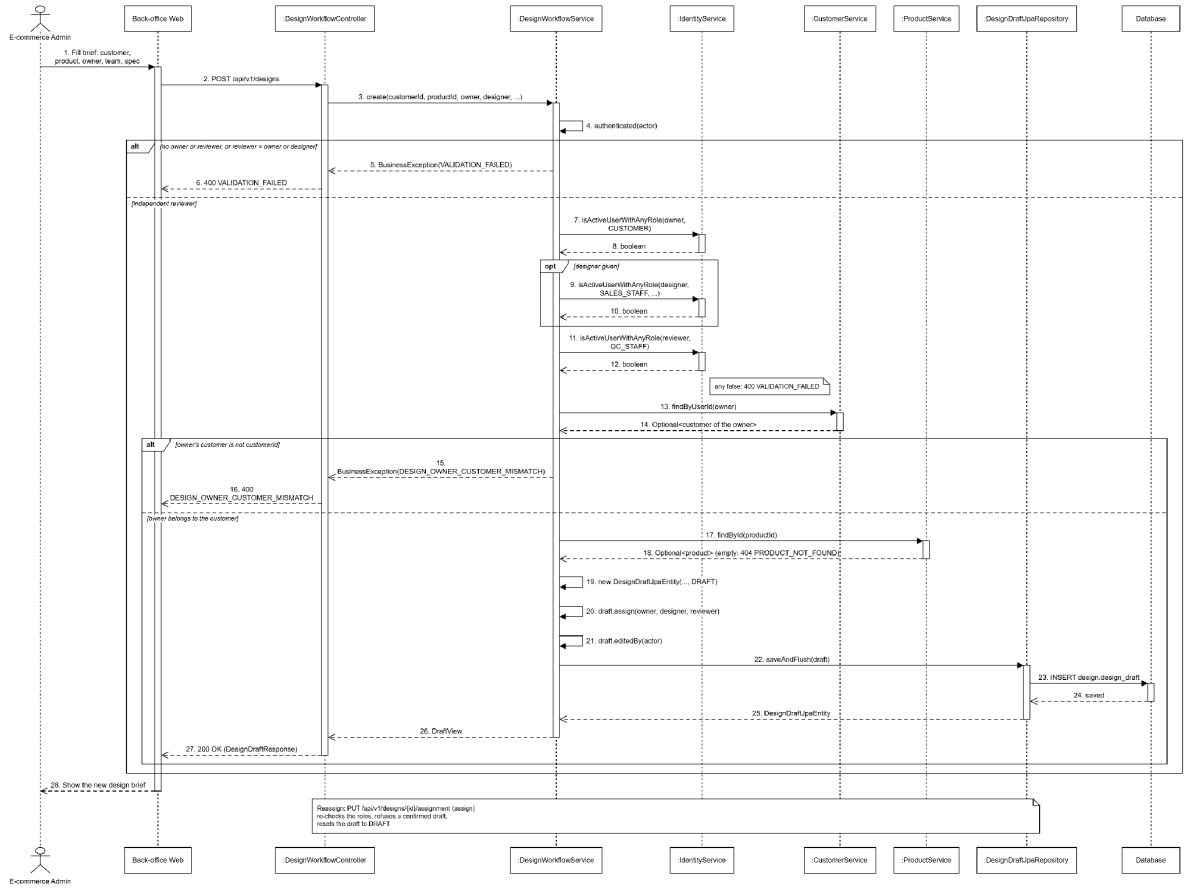


Figure 119. Create Design Brief & Assign Team Sequence Diagram

3.8.4 Upload Design Files Sequence Diagram

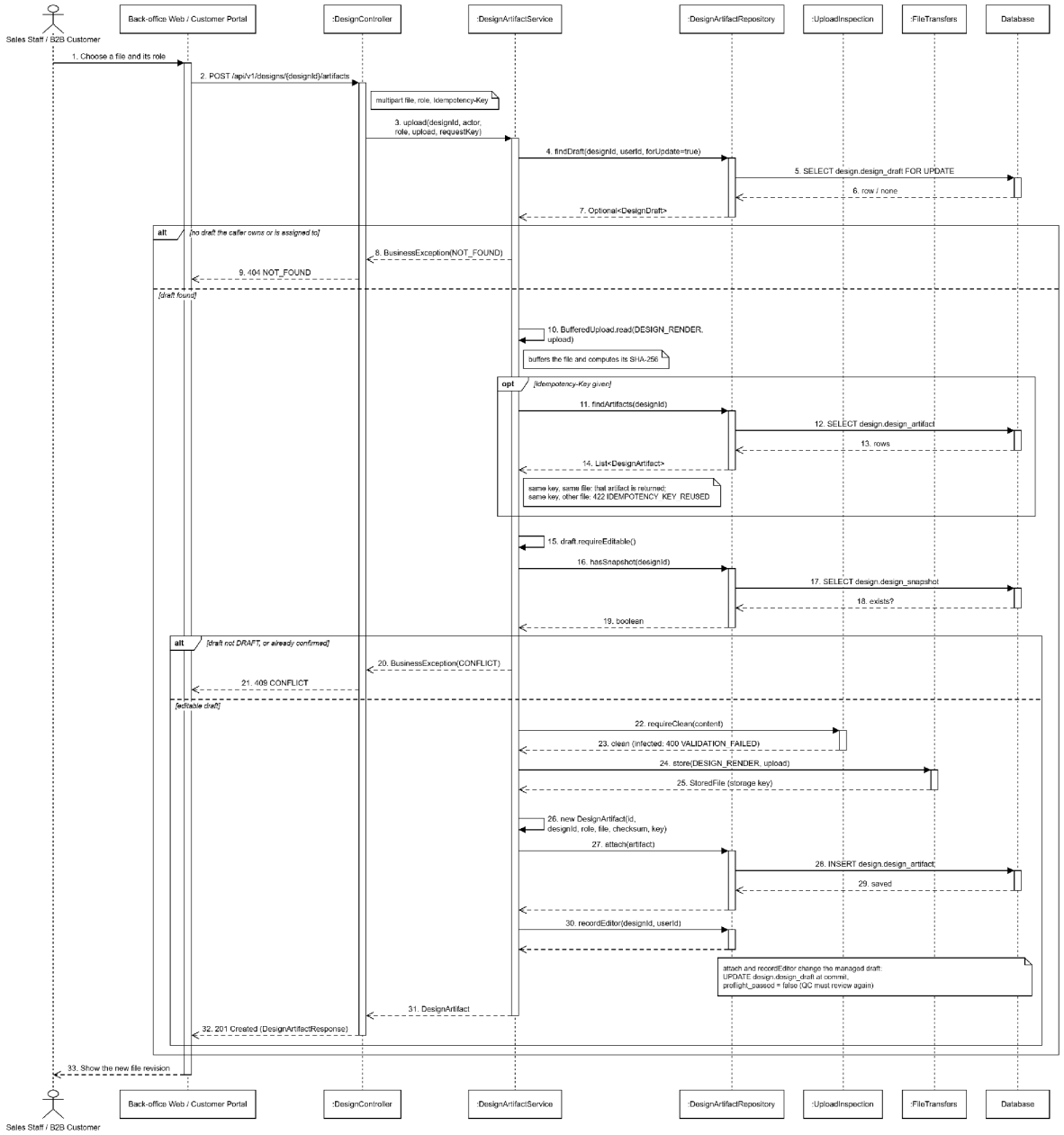


Figure 120. Upload Design Files Sequence Diagram

3.8.5 Create Design Revision Sequence Diagram

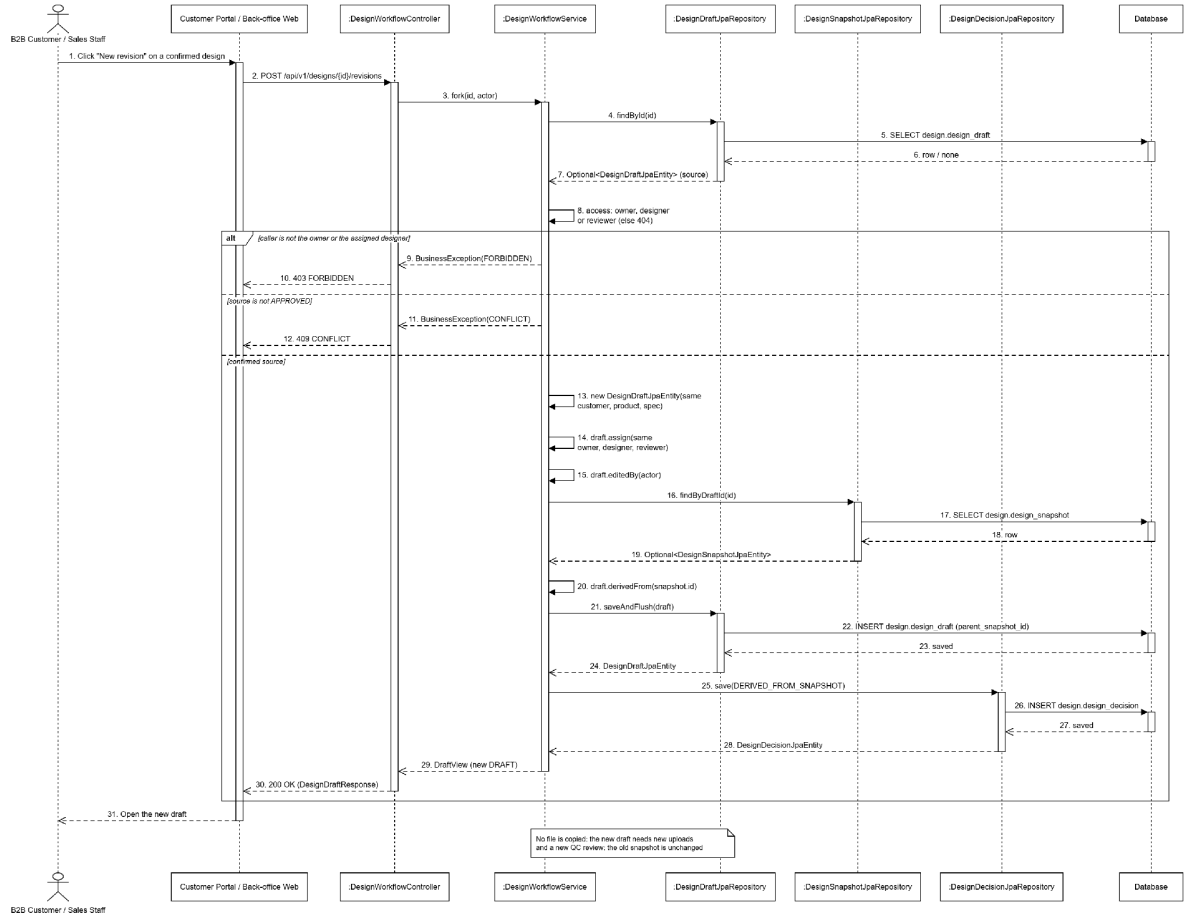


Figure 121. Create Design Revision Sequence Diagram

3.8.6 Review Design Technically Sequence Diagram

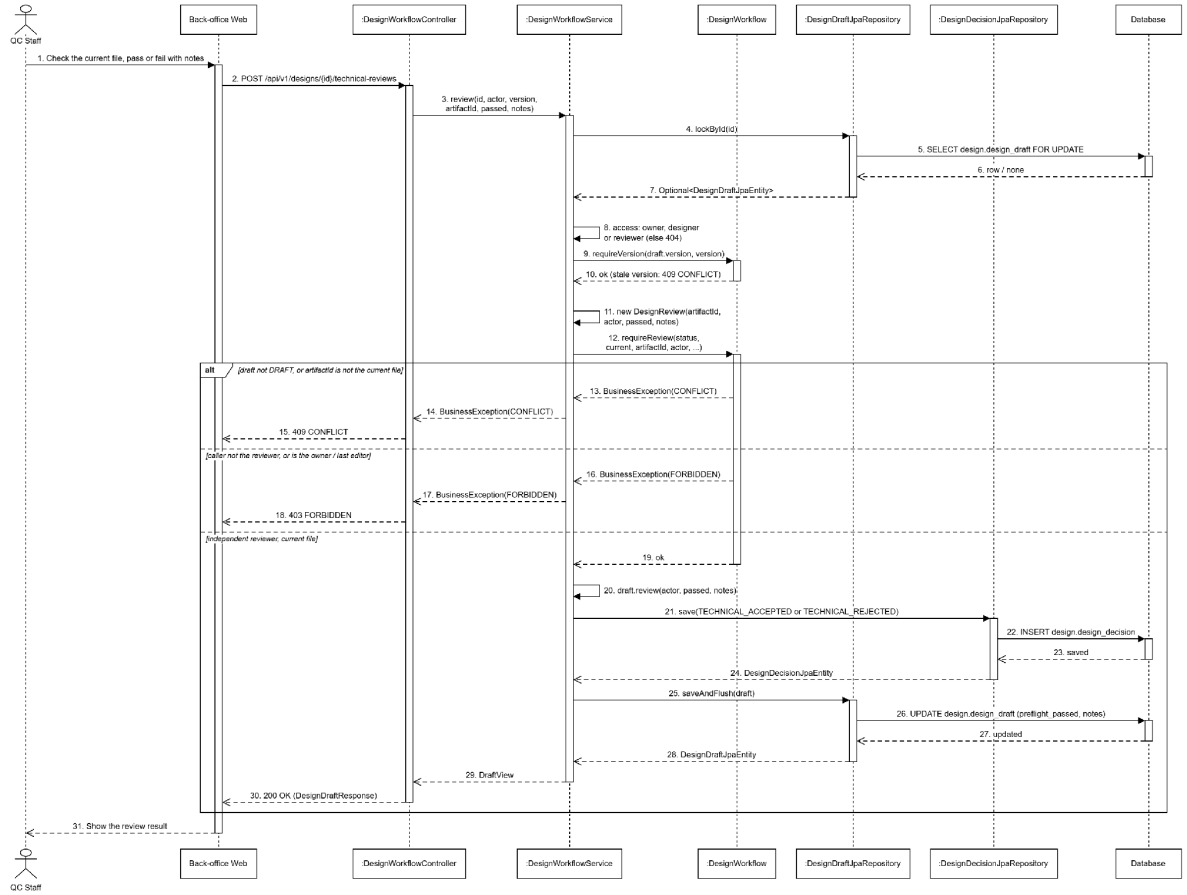


Figure 122. Review Design Technically Sequence Diagram

3.8.7 Send Design for Customer Confirmation Sequence Diagram

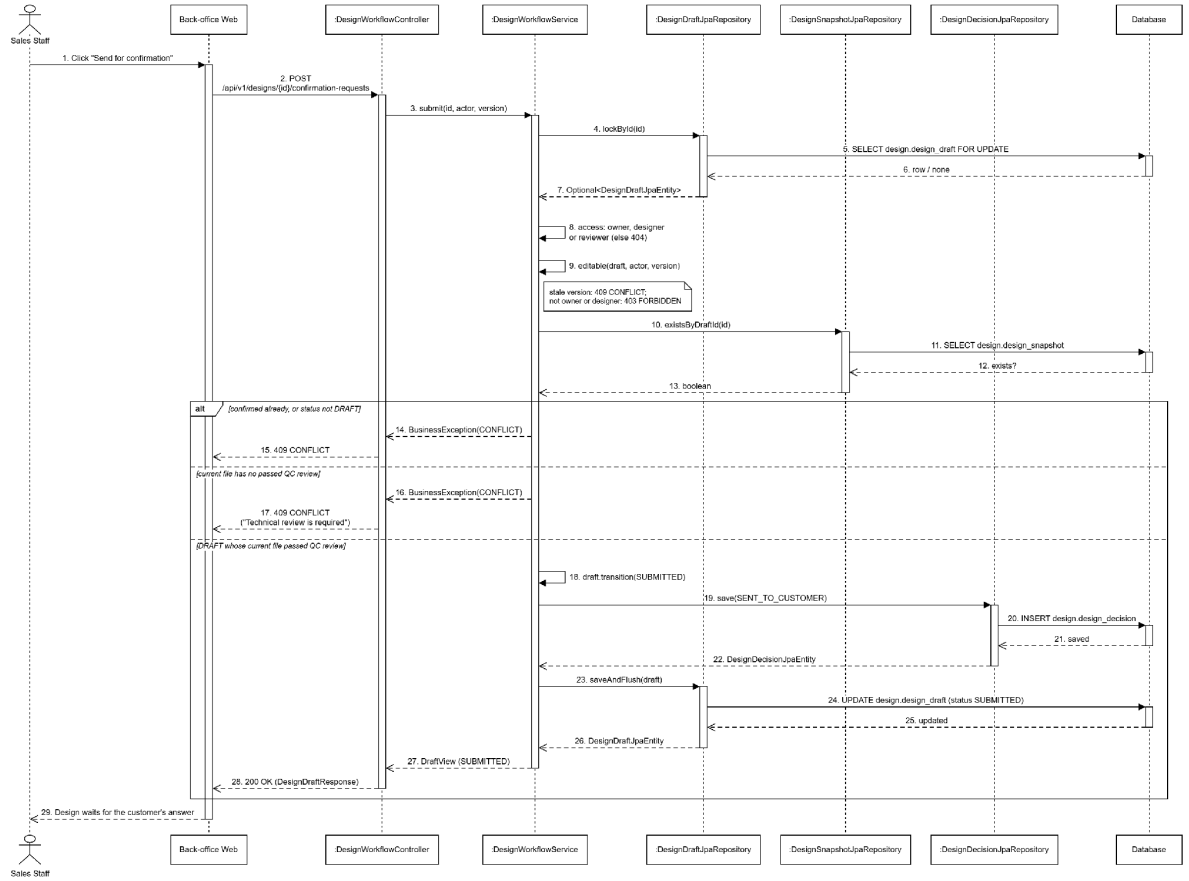


Figure 123. Send Design for Customer Confirmation Sequence Diagram

3.8.8 Withdraw Design Confirmation Request Sequence Diagram

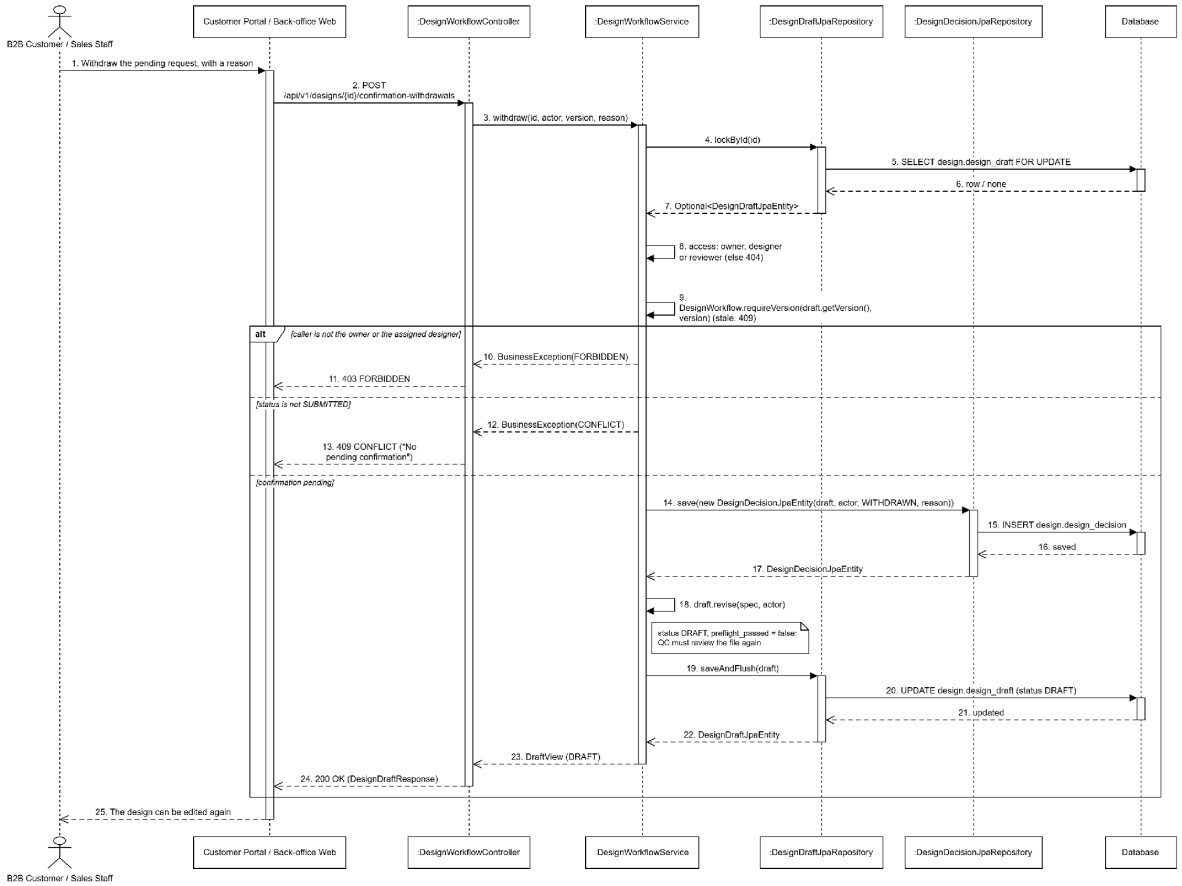


Figure 124. Withdraw Design Confirmation Request Sequence Diagram

3.8.9 Confirm / Request Changes to Design Sequence Diagram

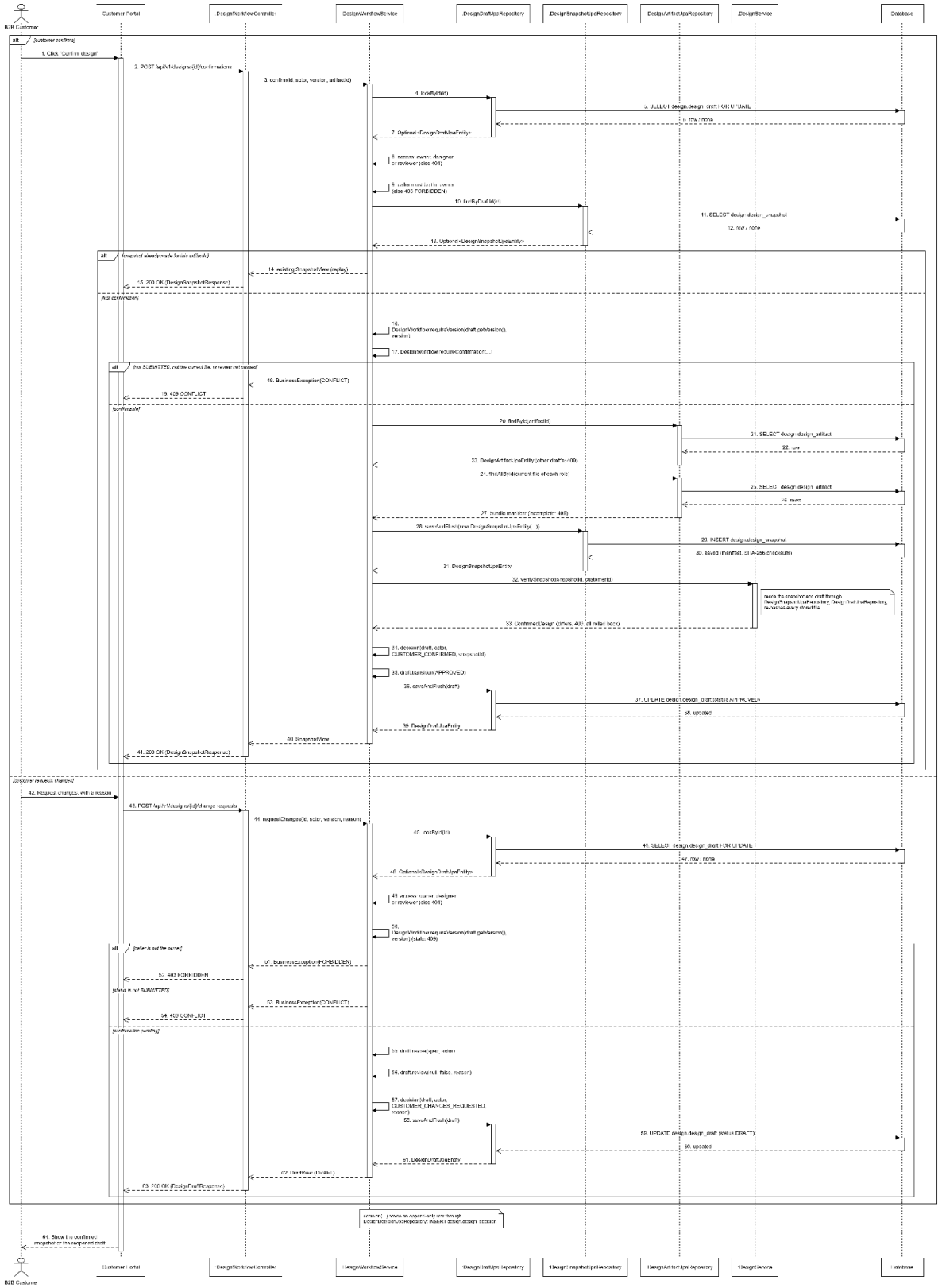


Figure 125. Confirm / Request Changes to Design Sequence Diagram

3.8.10 Attach Confirmed Design Sequence Diagram

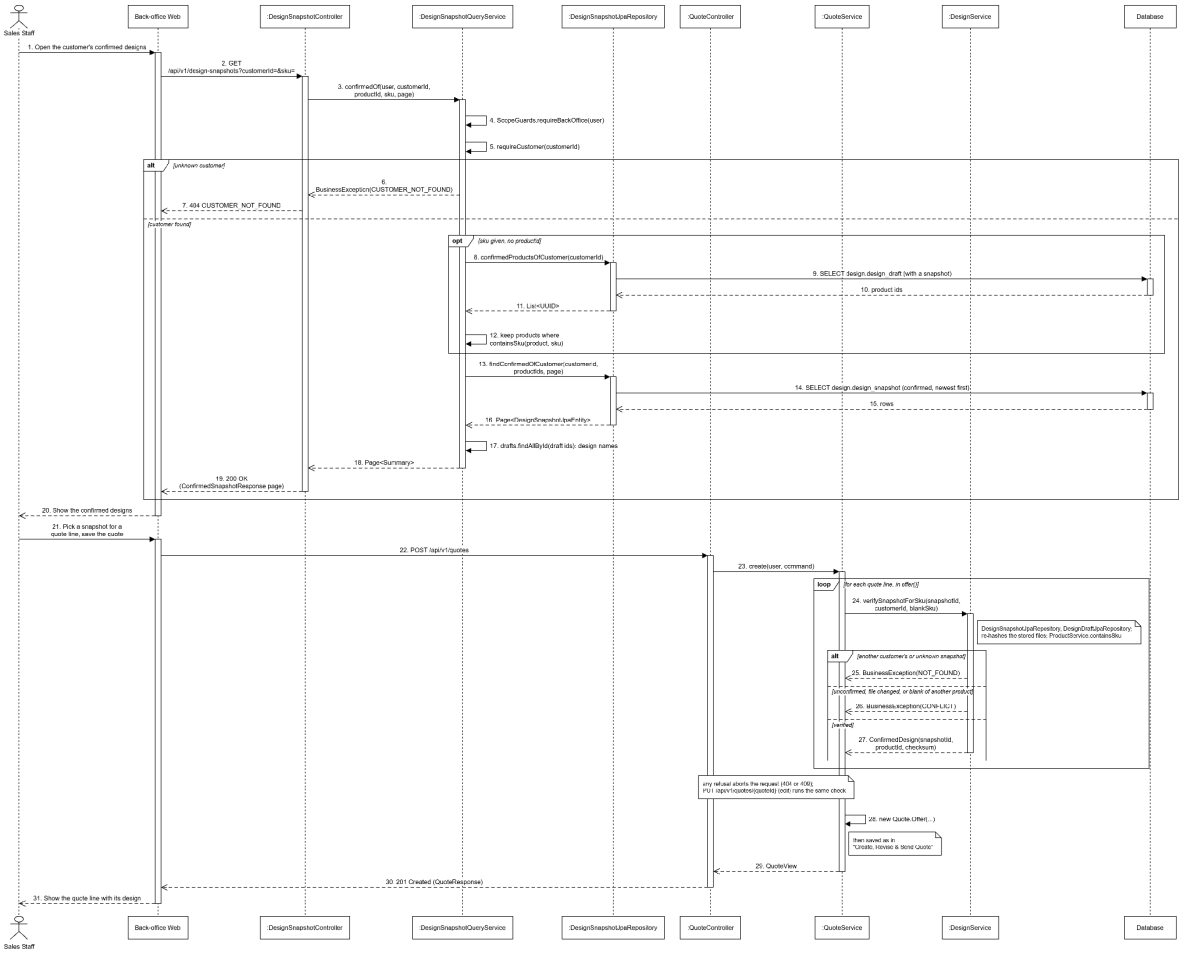


Figure 126. Attach Confirmed Design Sequence Diagram

3.8.11 Download Design Files Sequence Diagram

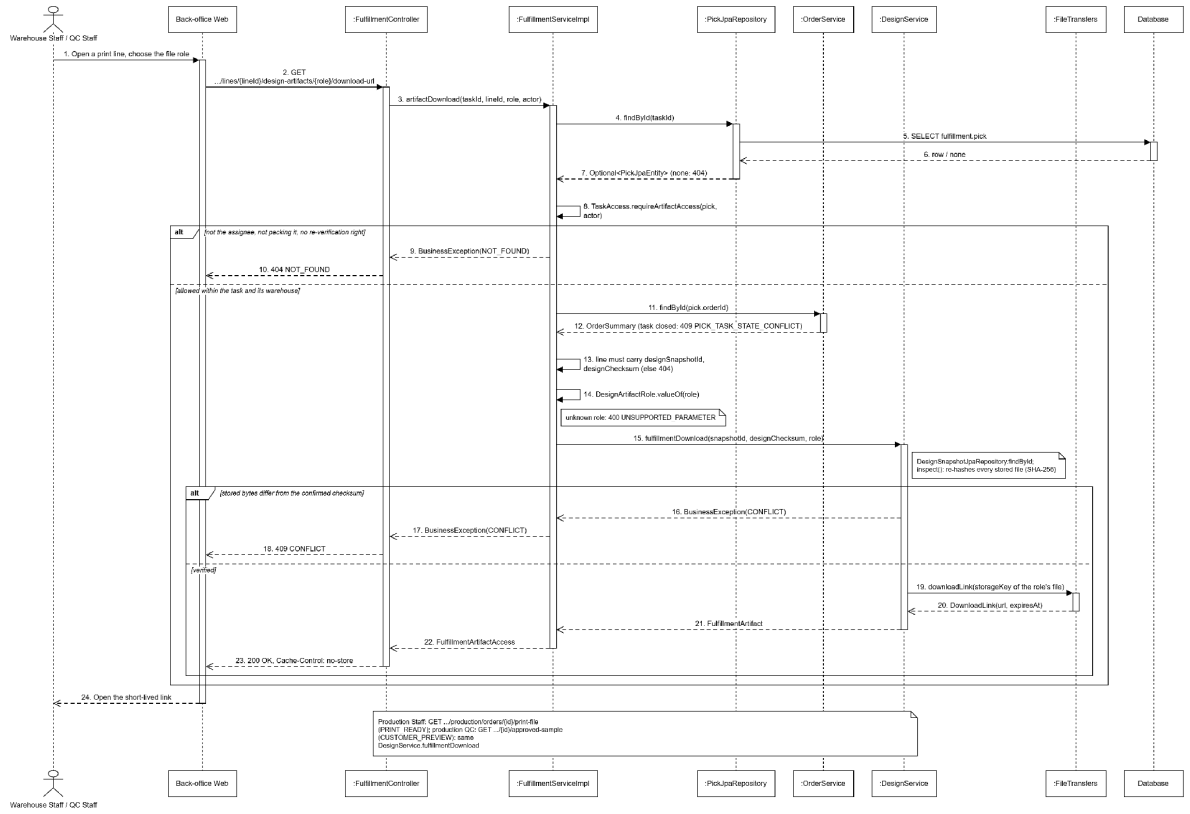


Figure 127. Download Design Files Sequence Diagram

3.8.12 Verify Design Checksum Sequence Diagram

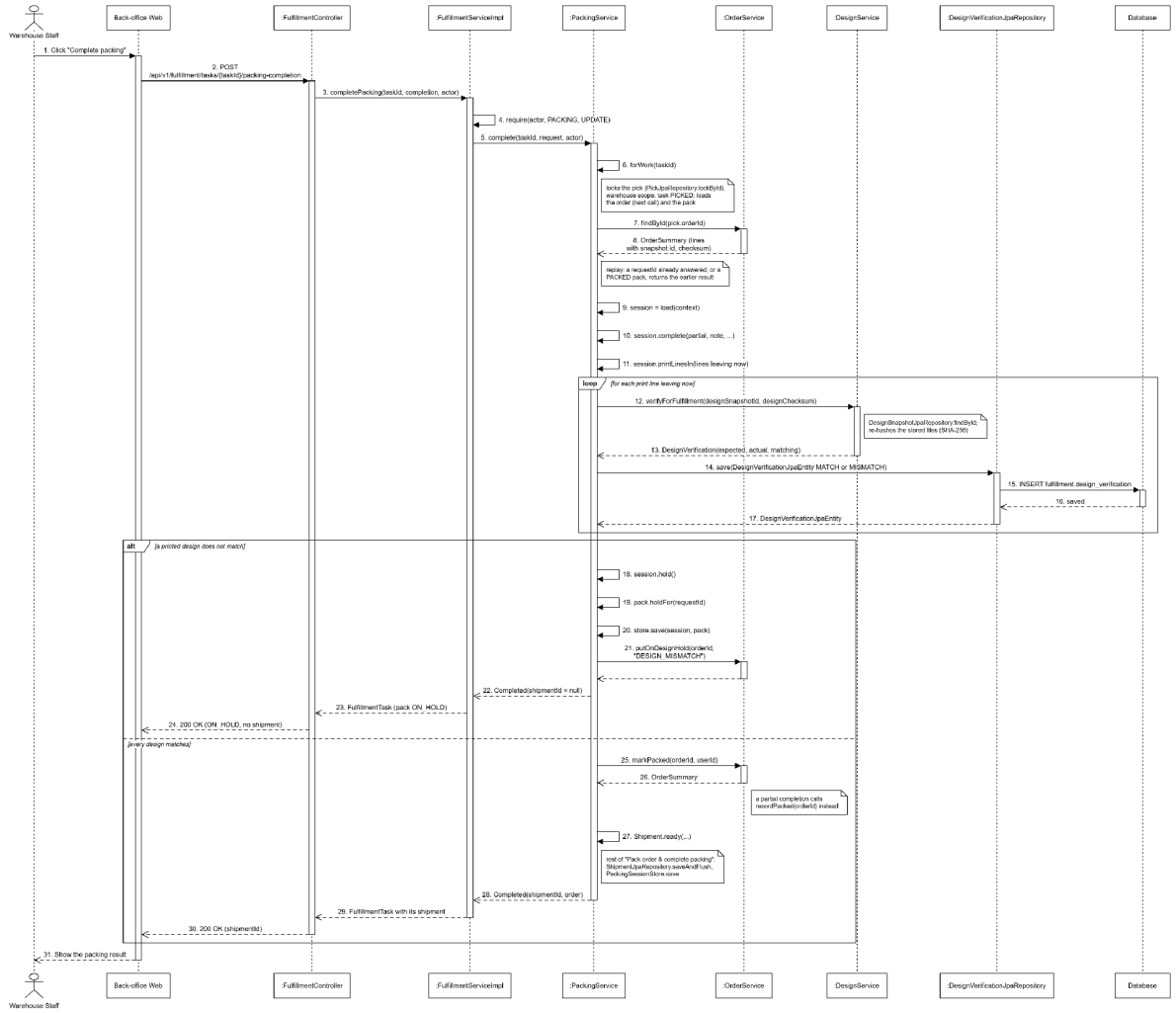


Figure 128. Verify Design Checksum Sequence Diagram

3.8.13 Re-verify Design Mismatch Sequence Diagram

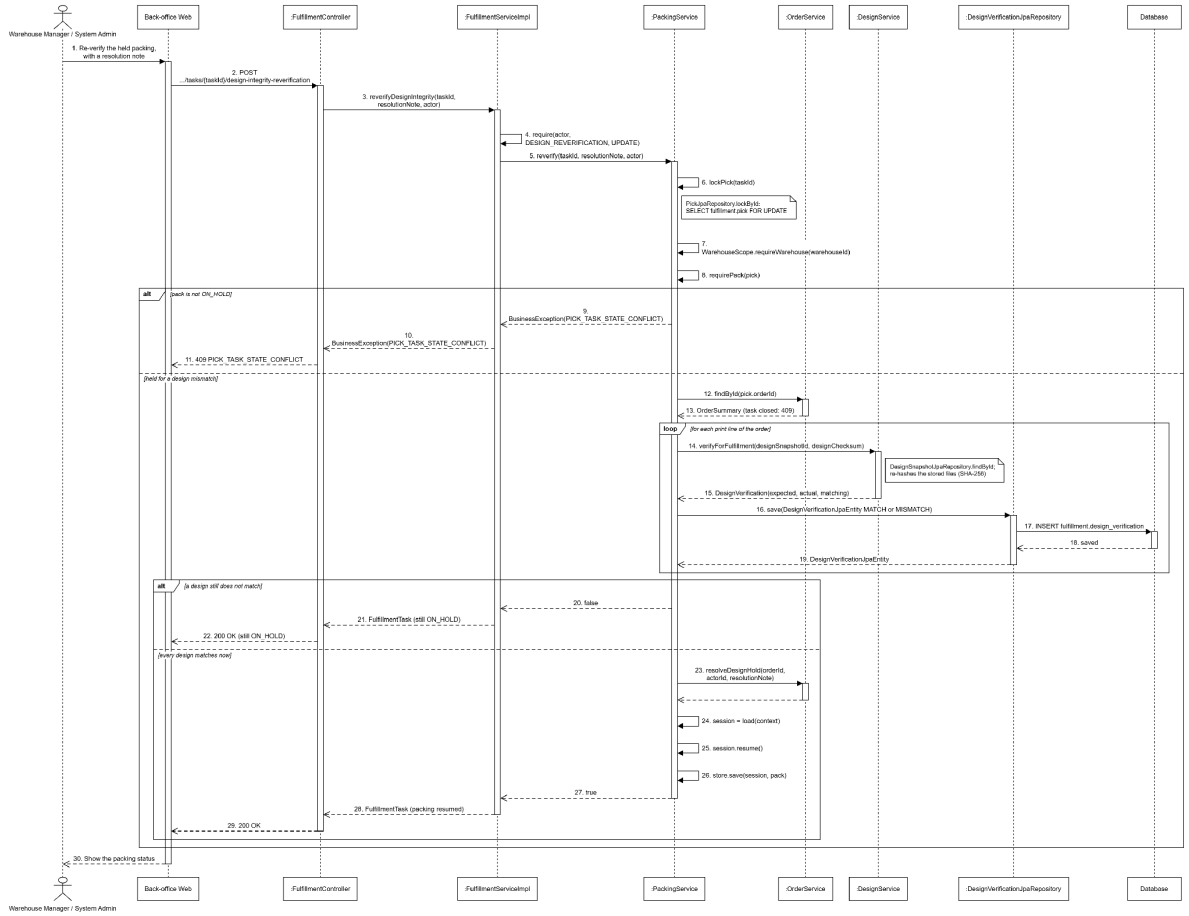


Figure 129. Re-verify Design Mismatch Sequence Diagram

3.8.14 Design Draft State Diagram

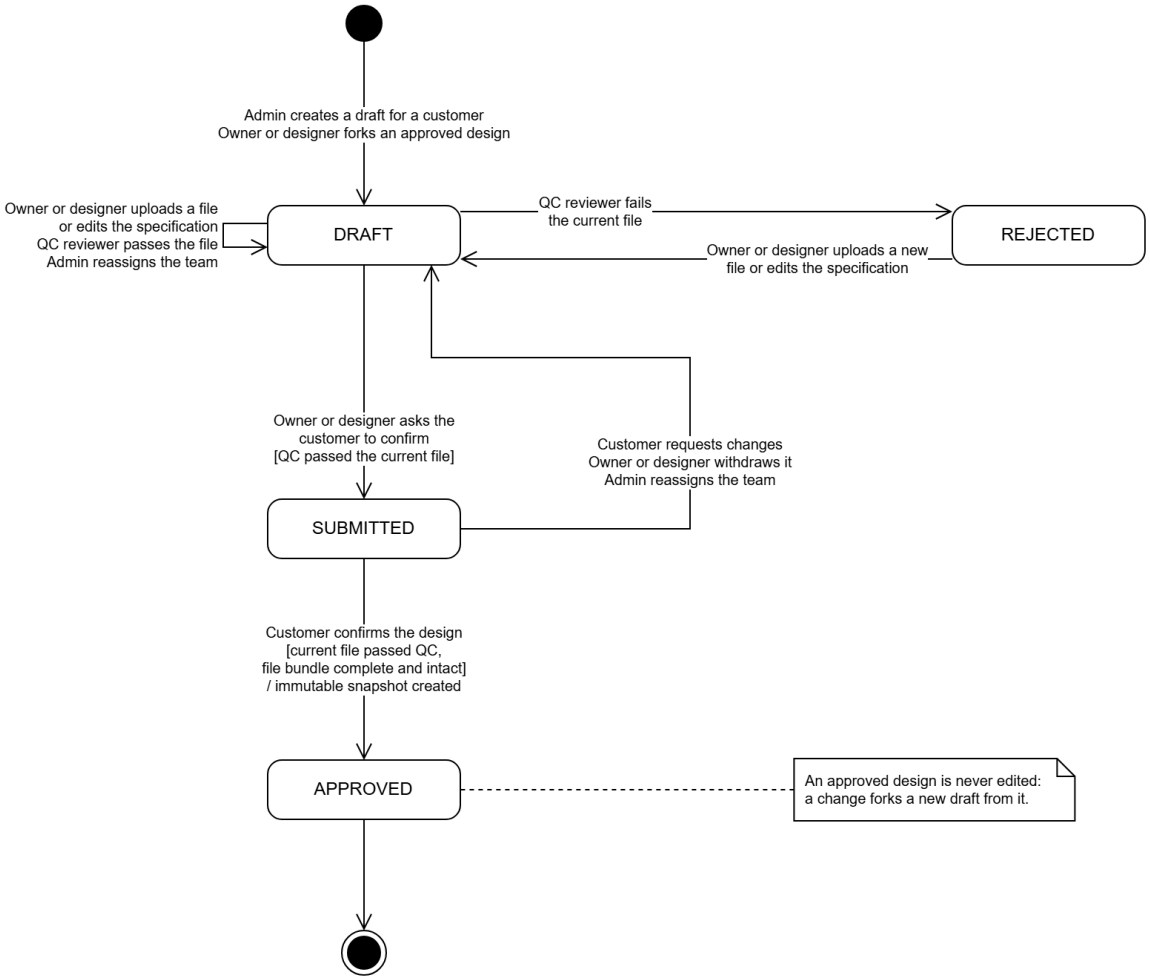


Figure 130. Design Draft State Diagram

3.9 Quote & Sample Management

3.9.1 Class Diagram

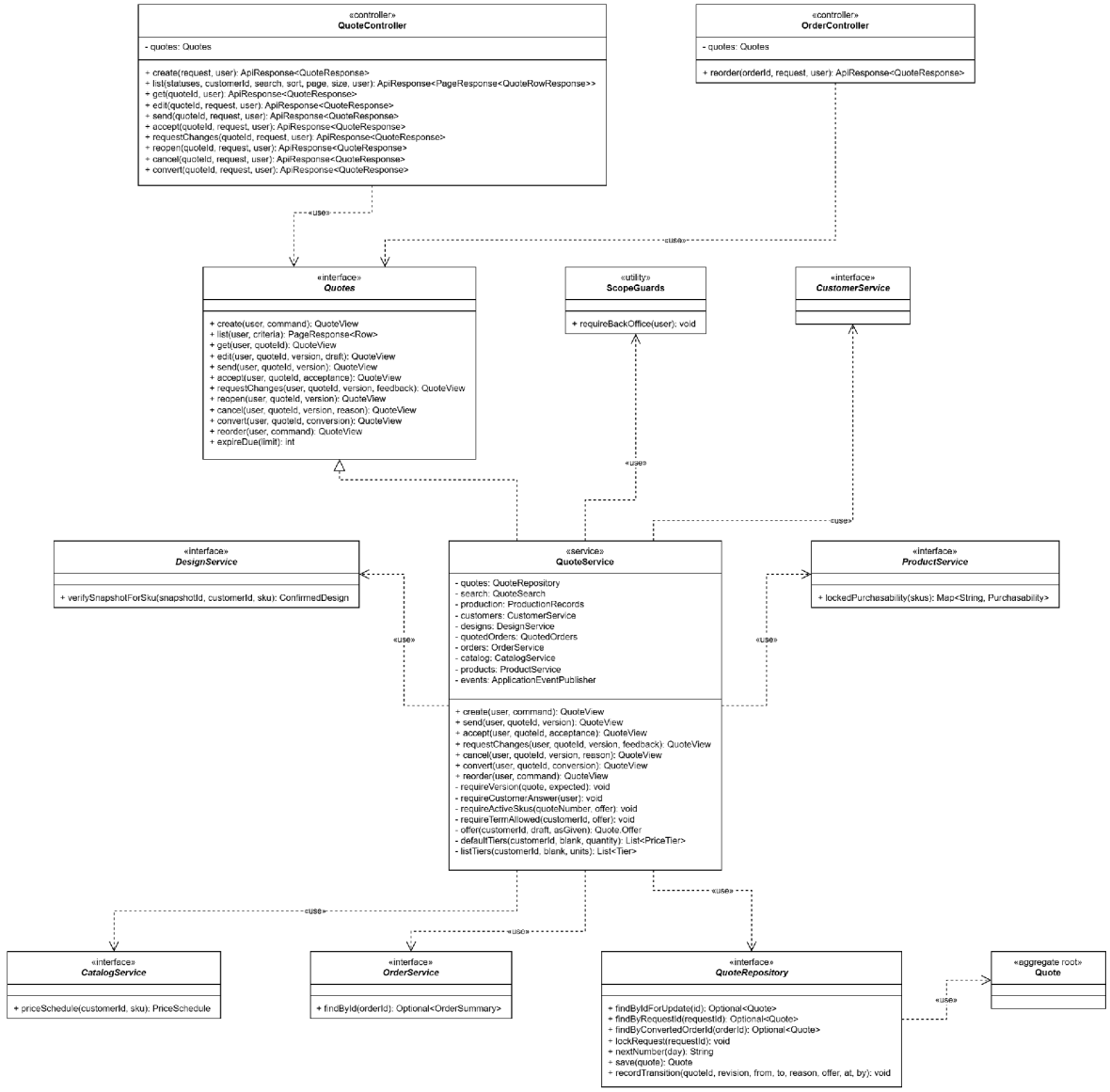


Figure 131. Quote & Sample Management Class Diagram (1/7): Quote Workflow

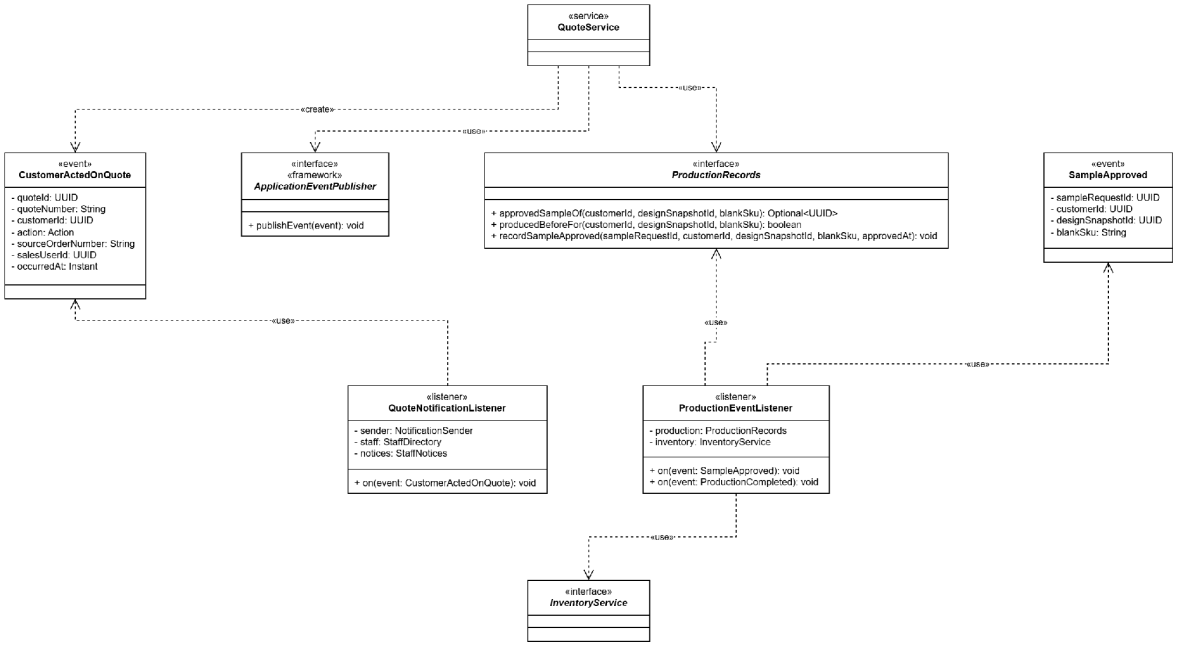


Figure 132. Quote & Sample Management Class Diagram (2/7): Customer Answers and the Sample Gate

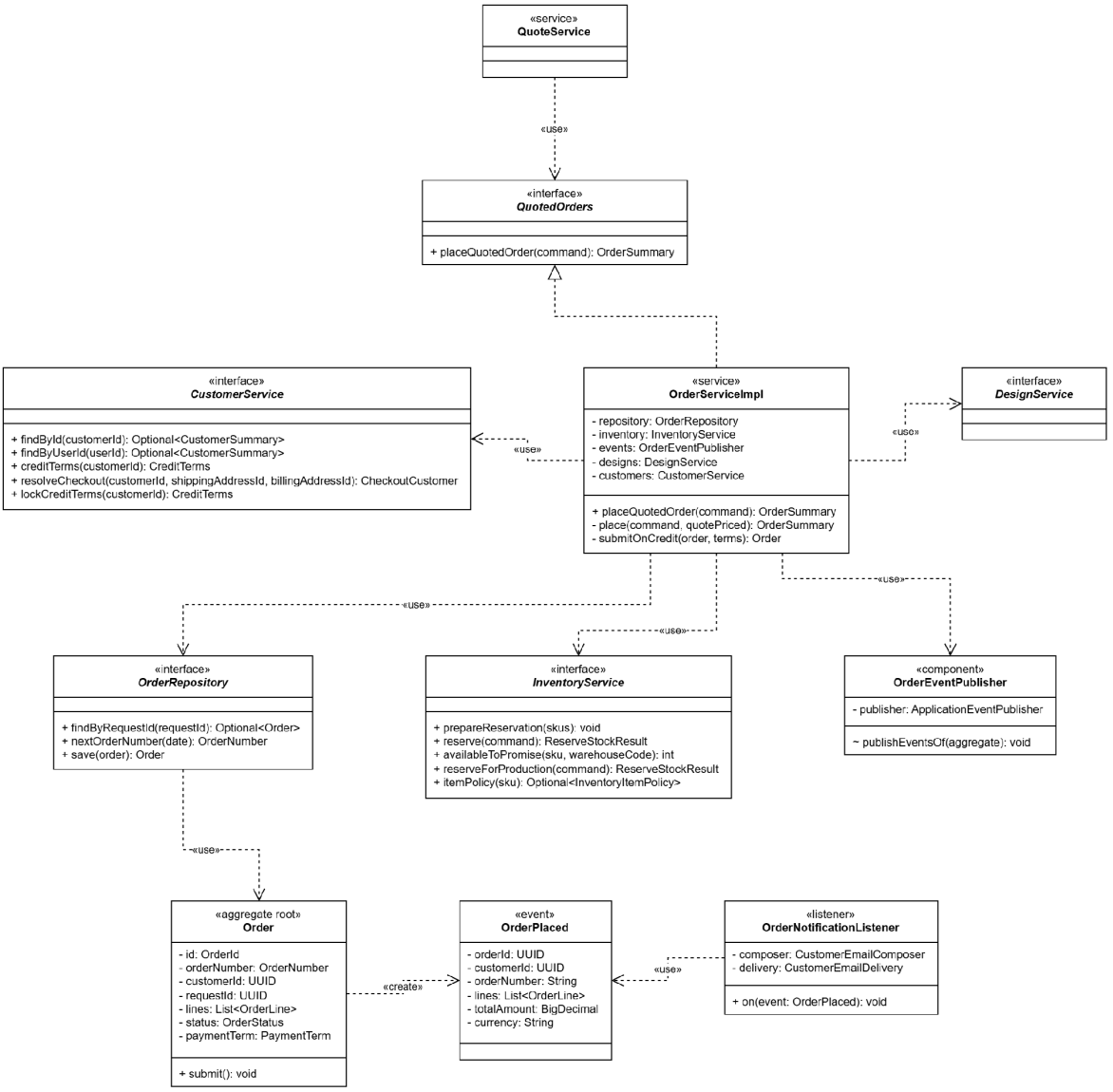


Figure 133. Quote & Sample Management Class Diagram (3/7): Quote Conversion

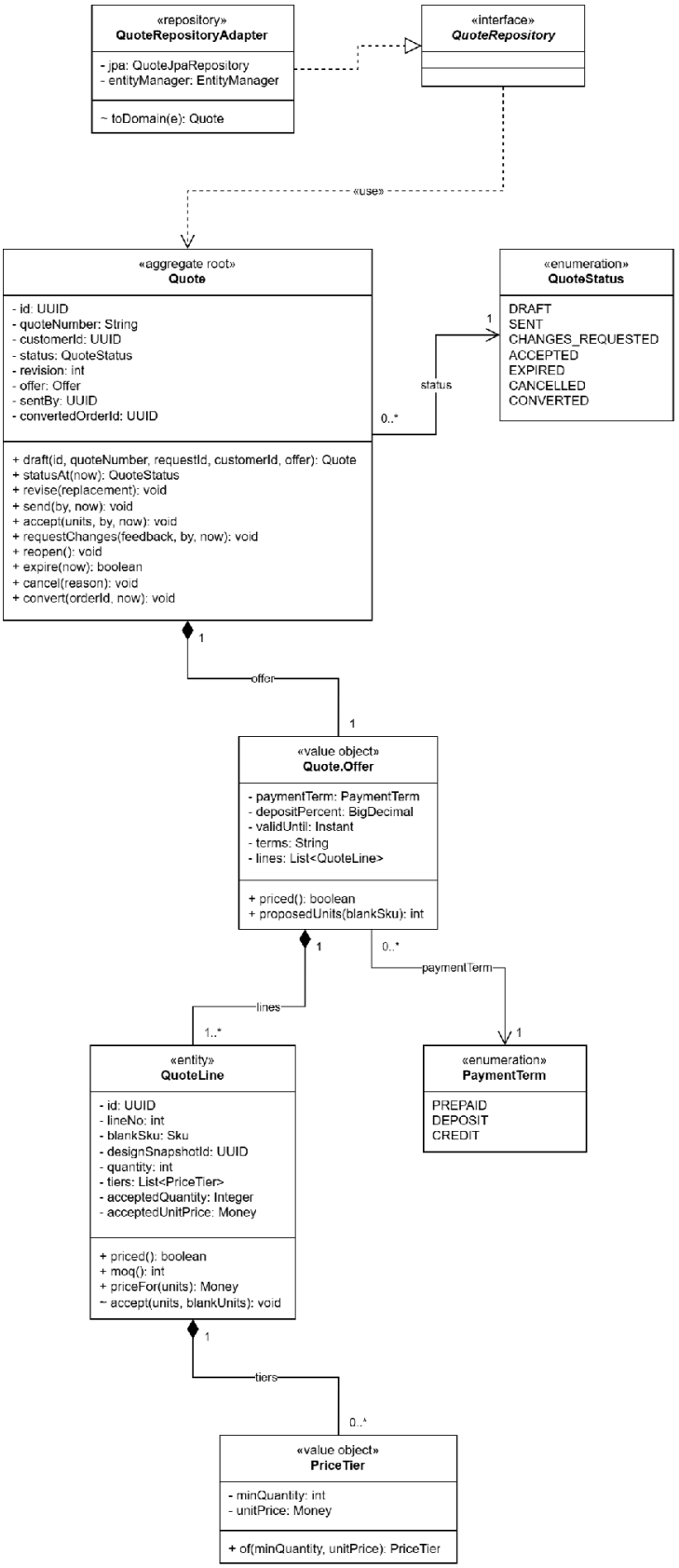


Figure 134. Quote & Sample Management Class Diagram (4/7): Quote Domain Model

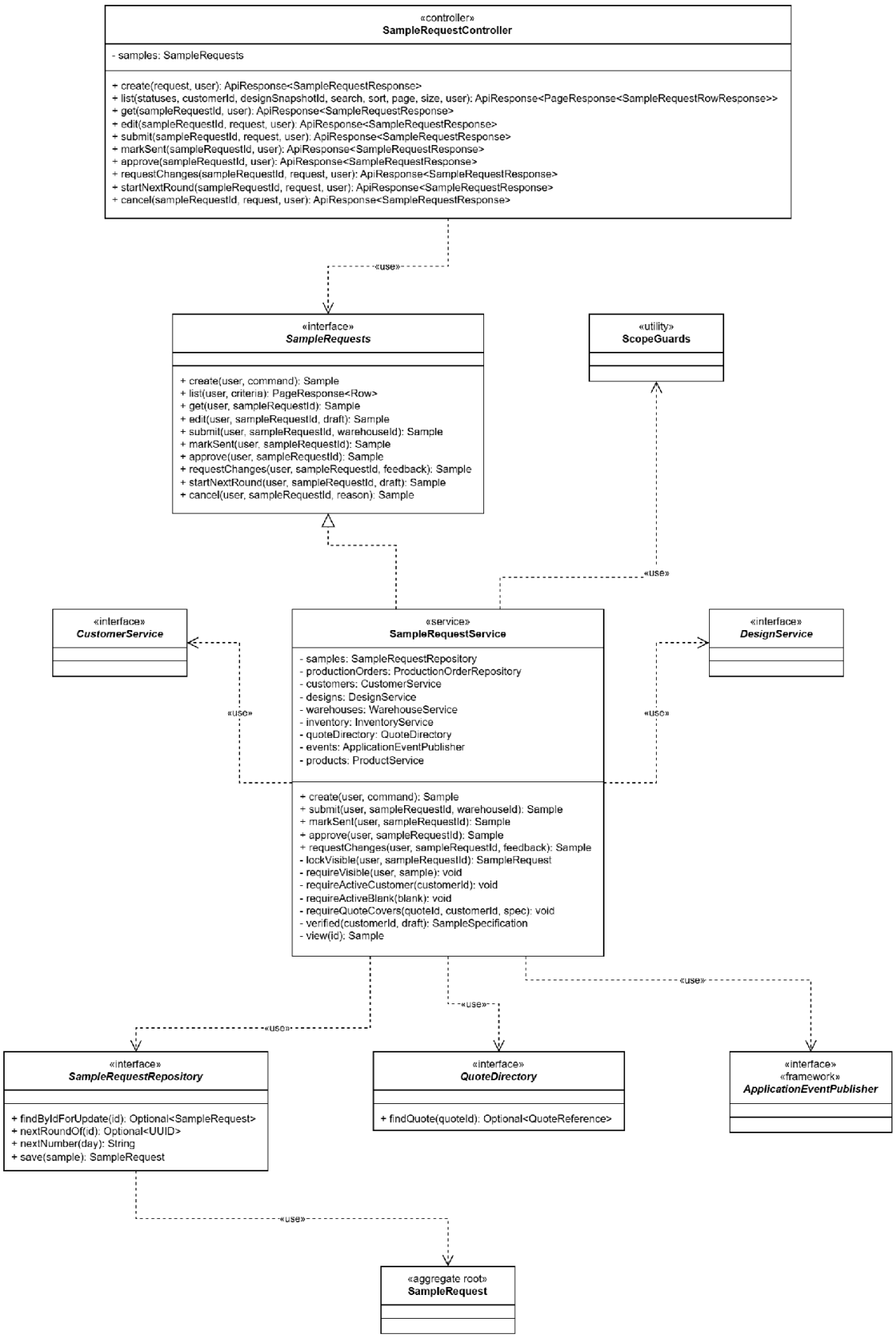


Figure 135. Quote & Sample Management Class Diagram (5/7): Sample Requests

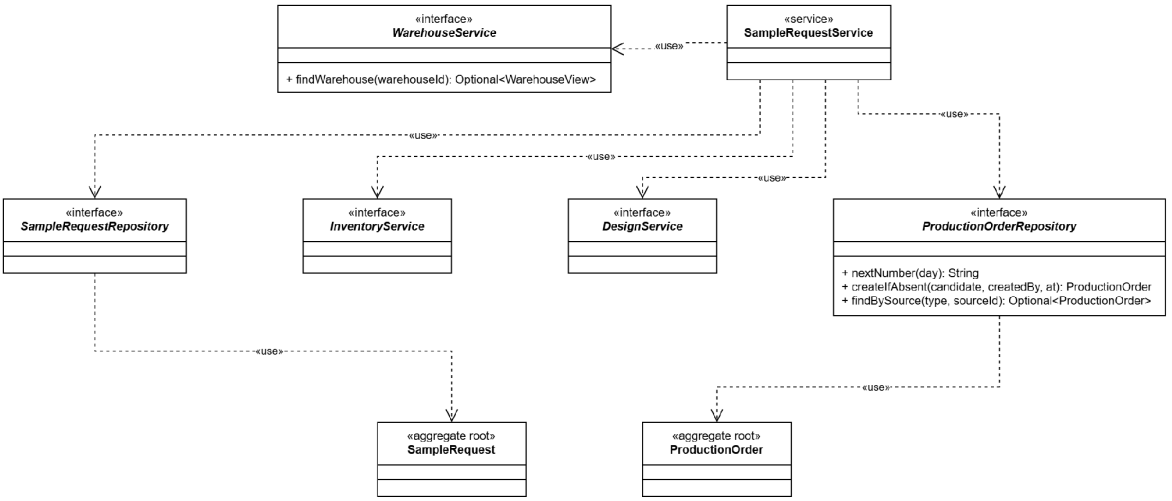


Figure 136. Quote & Sample Management Class Diagram (6/7): Sample Submission to the Workshop

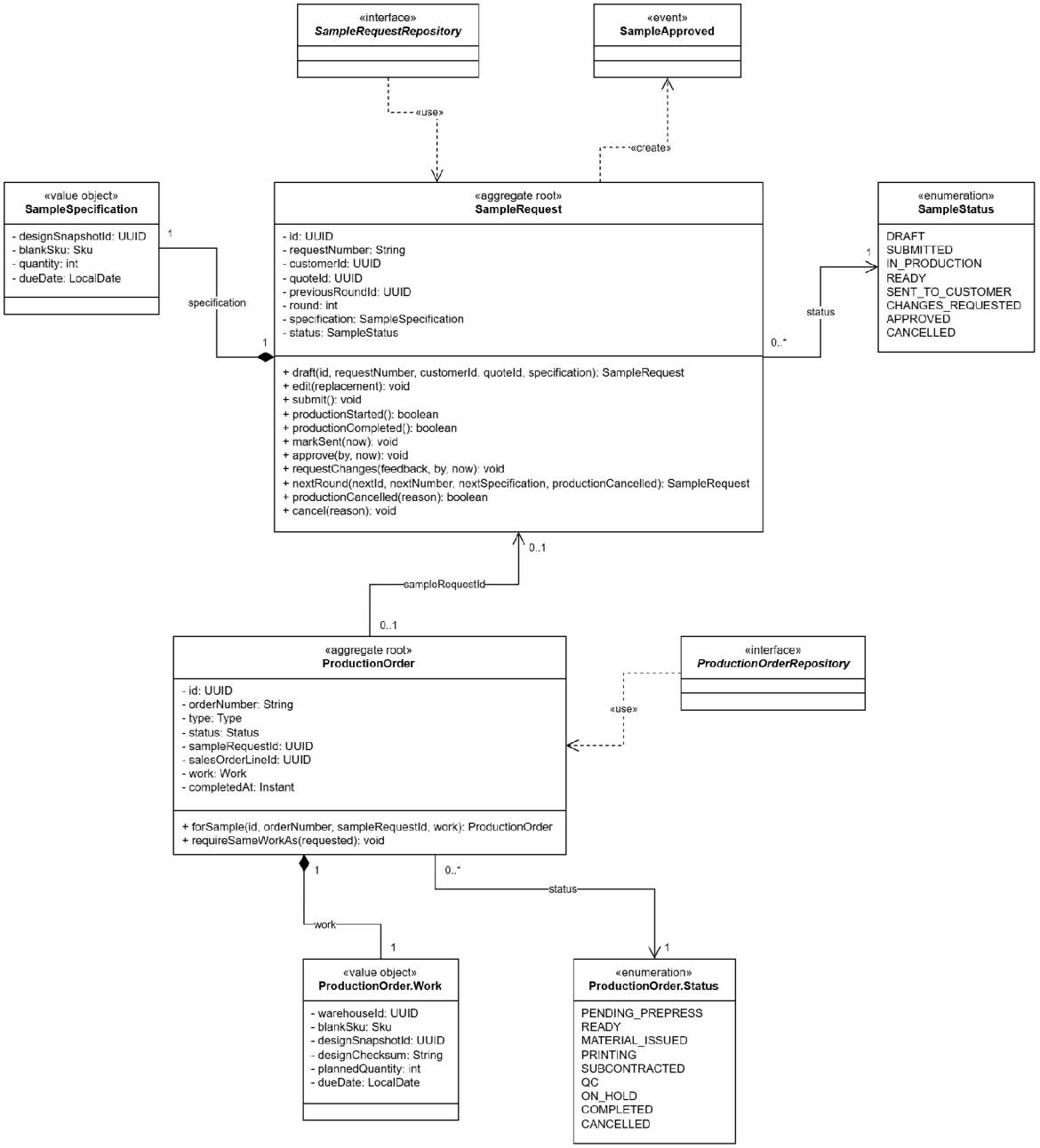


Figure 137. Quote & Sample Management Class Diagram (7/7): Sample Domain Model

3.9.2 Class Specification

SampleRequestController

No	Method	Description
1	create(request, user)	POST /api/v1/production/sample-requests: Drafts the first round of a sample for a customer's confirmed design on a blank.
2	list(statuses, customerId, designSnapshotId, search, sort, page, size, user)	GET /api/v1/production/sample-requests: Pages sample rounds, newest first; a portal customer sees only their own.

3	get(sampleRequestId, user)	GET /api/v1/production/sample-requests/{sampleRequestId}: Returns one round with every round of its request and its sample production order.
4	edit(sampleRequestId, request, user)	PUT /api/v1/production/sample-requests/{sampleRequestId}: Changes a DRAFT round's design, blank, quantity or due date.
5	submit(sampleRequestId, request, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/submission: Sends a DRAFT to the workshop of a warehouse: SUBMITTED, with its SAMPLE production order in PENDING_PREPRESS.
6	markSent(sampleRequestId, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/sending: Records that a READY sample was handed over or couriered to the customer.
7	approve(sampleRequestId, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/approval: The customer approves the sample: design and blank locked, SampleApproved announced.
8	requestChanges(sampleRequestId, request, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/change-request: The customer asks for changes; the feedback stays on this round.
9	startNextRound(sampleRequestId, request, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/next-round: Starts the next round after CHANGES_REQUESTED as a new DRAFT request, round + 1.
10	cancel(sampleRequestId, request, user)	POST /api/v1/production/sample-requests/{sampleRequestId}/cancellation: Cancels a DRAFT round with a reason.

Table 63. SampleRequestController Class Specification

QuoteController

No	Method	Description
1	create(request, user)	POST /api/v1/quotes: Back office drafts a quote (revision 1) for an ACTIVE customer; each line's design must be that customer's confirmed snapshot for the blank, omitted tiers come from the price list, MOQ is checked on the blank's total over all lines; idempotent on requestId; 201. (:67)

2	list(statuses, customerId, search, sort, page, size, user)	GET /api/v1/quotes: Quotes newest first; a portal customer sees only their own once sent; a SENT quote past its validity is shown EXPIRED. (:77)
3	get(quoteId, user)	GET /api/v1/quotes/{quoteId}: One quote with lines, tiers, sample status per line and history; another customer's or an unsent quote is 404 to a portal user. (:92)
4	edit(quoteId, request, user)	PUT /api/v1/quotes/{quoteId}: Replaces the offer of a DRAFT quote with the same design, tier, MOQ, payment-term and active-SKU checks as create (stale version: 409 OPTIMISTIC_LOCK). (:99)
5	send(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/sending: DRAFT → SENT: every line priced (409 QUOTE_NOT_PRICED), validity ahead, customer ACTIVE, term allowed, every blank SKU active; the offer is kept on the history row. (:109)
6	accept(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/acceptance: The portal customer accepts within validity (409 QUOTE_EXPIRED), choosing units per line; every line needs a sample they approved or a reorder (409 DESIGN_SAMPLE_NOT_APPROVED); prices fixed at each blank's break; notifies Sales. (:117)
7	requestChanges(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/change-request: The portal customer asks for changes with feedback within validity: SENT → CHANGES_REQUESTED; notifies Sales. (:131)
8	reopen(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/revision: Sales reopens a CHANGES_REQUESTED quote as the next DRAFT revision (revision + 1, acceptance cleared). (:141)
9	cancel(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/cancellation: Sales cancels with a reason (1–500 chars): a DRAFT always; an ACCEPTED one only when a blank SKU on it is no longer active; SENT or CHANGES_REQUESTED: 409 INVALID_QUOTE_TRANSITION. (:149)
10	convert(quoteId, request, user)	POST /api/v1/quotes/{quoteId}/conversion: sales-quote-conversions:CREATE (Sales, coordinator). ACCEPTED → CONVERTED: places the one order at accepted quantities and prices in the same transaction (request id derived from the quote); a repeat returns the same order. (:162)

Table 64. QuoteController Class Specification

3.9.3 Create, Revise & Send Quote (1: Create) Sequence Diagram

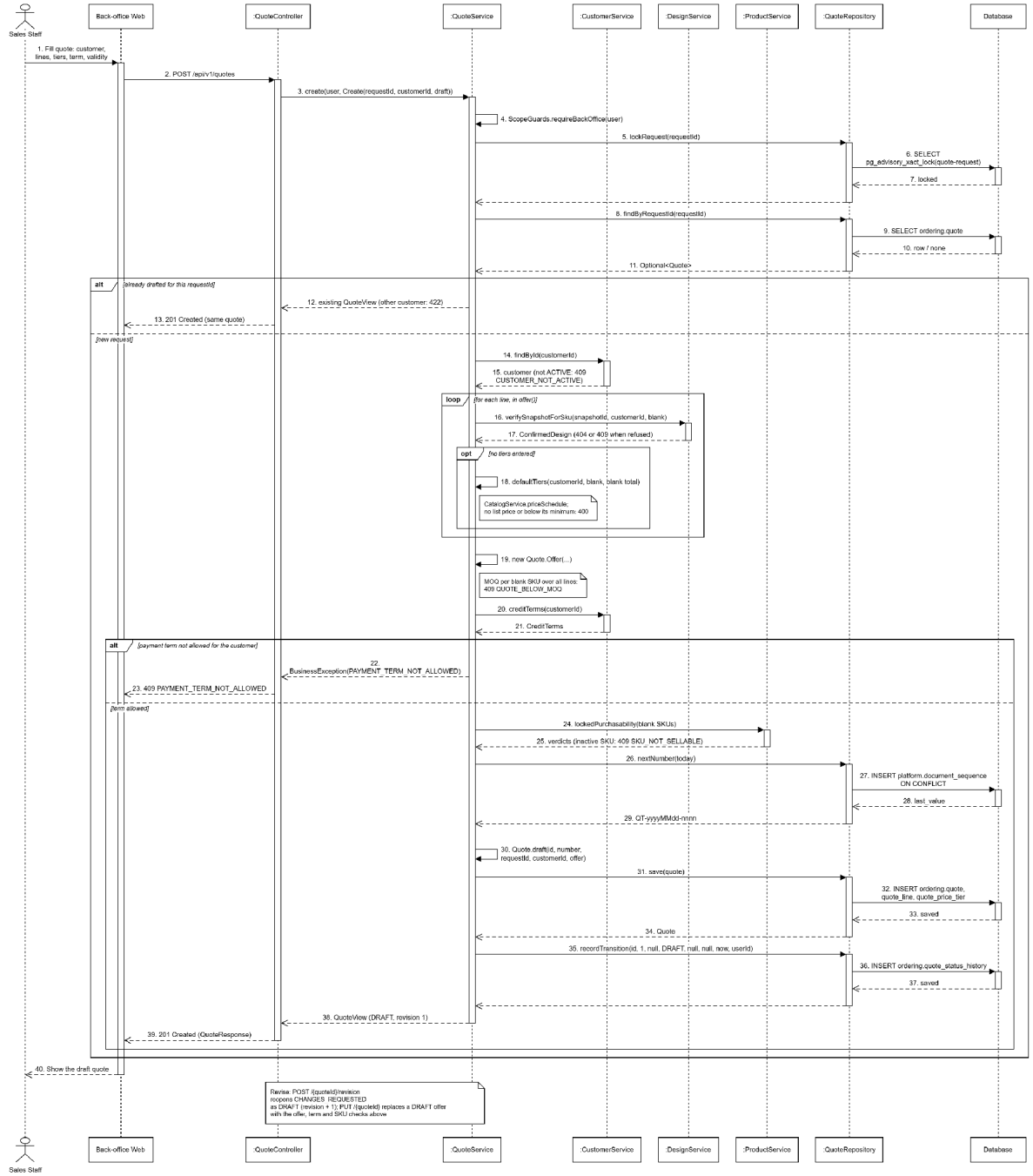


Figure 138. Create, Revise & Send Quote (1: Create) Sequence Diagram

3.9.4 Create, Revise & Send Quote (2: Send) Sequence Diagram

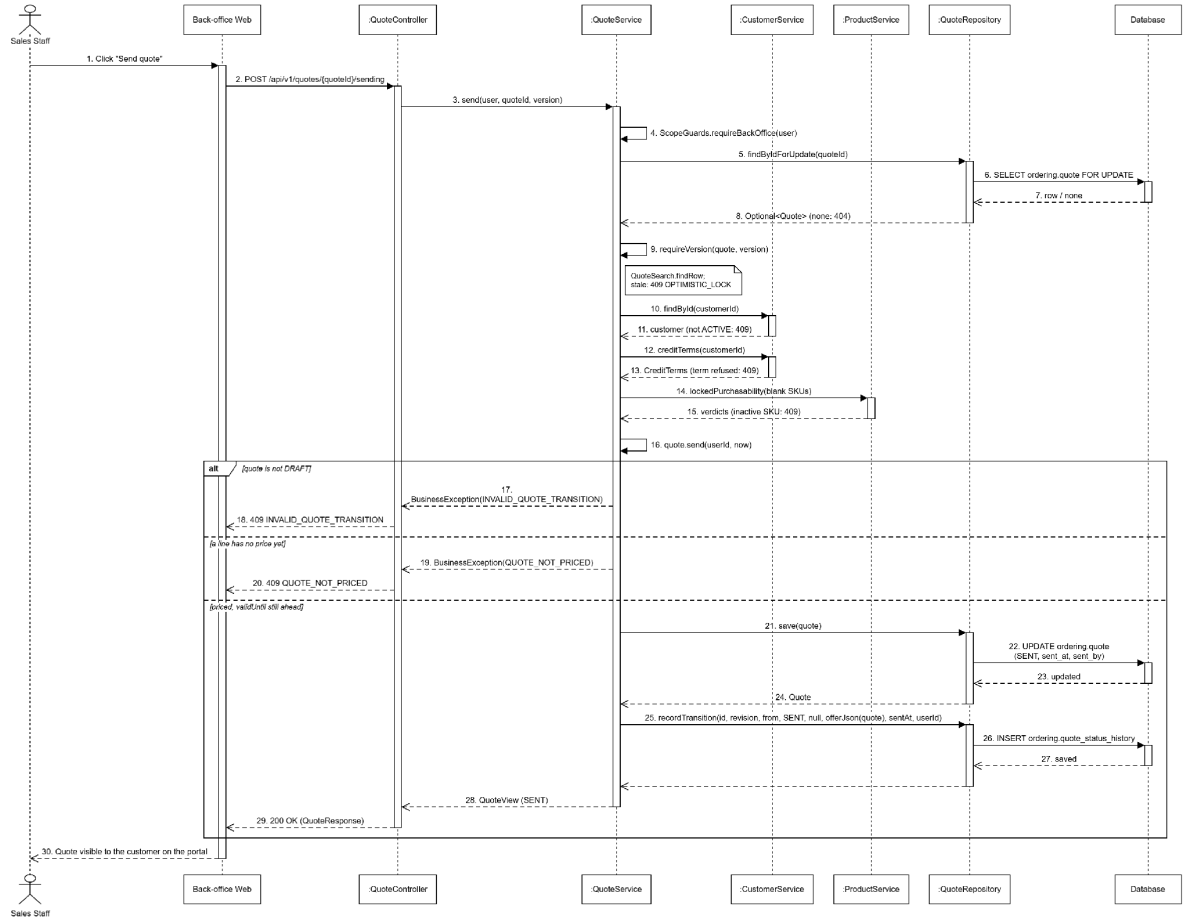


Figure 139. Create, Revise & Send Quote (2: Send) Sequence Diagram

3.9.5 Accept / Request Changes to Quote Sequence Diagram

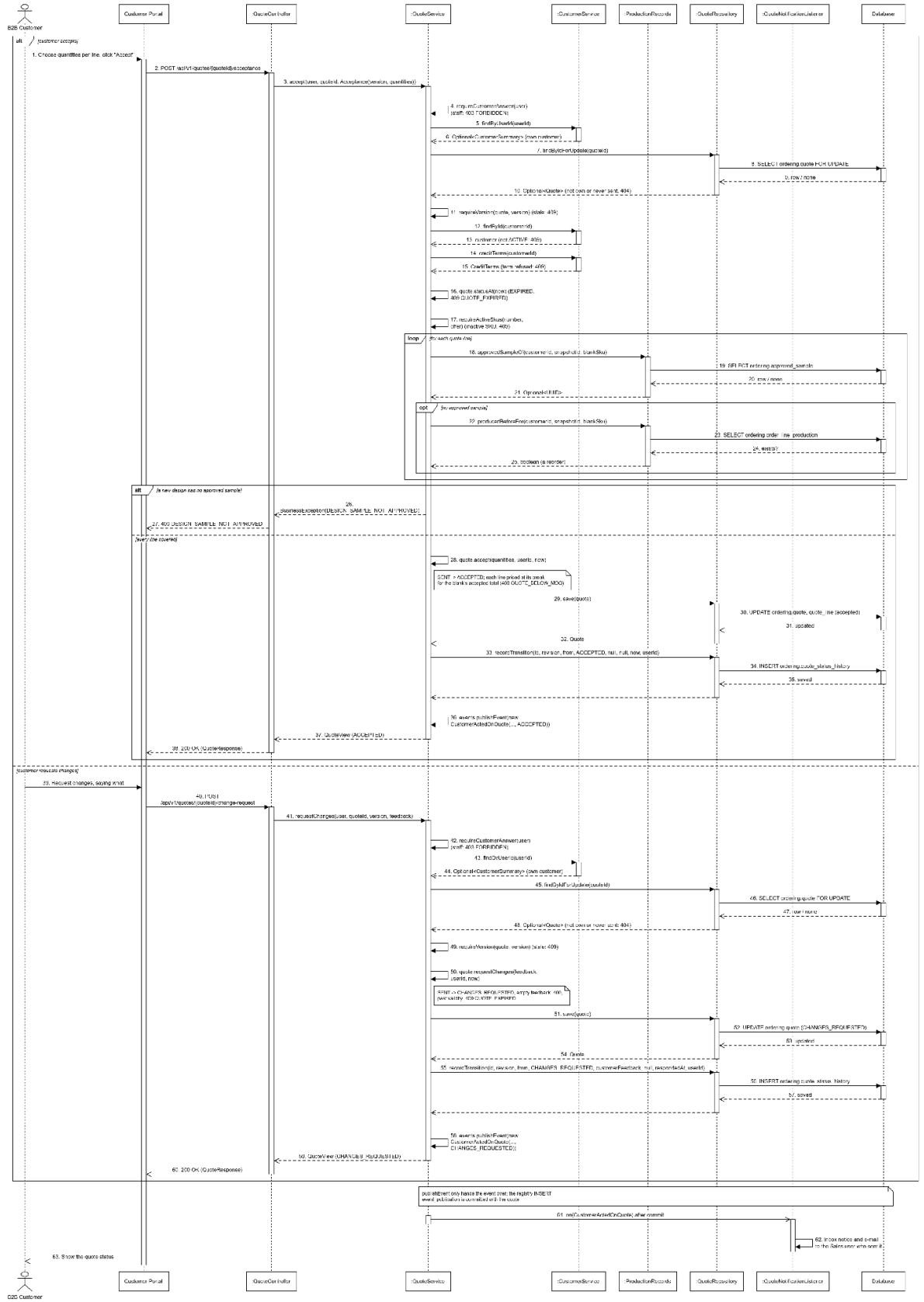


Figure 140. Accept / Request Changes to Quote Sequence Diagram

3.9.6 Cancel Quote Sequence Diagram

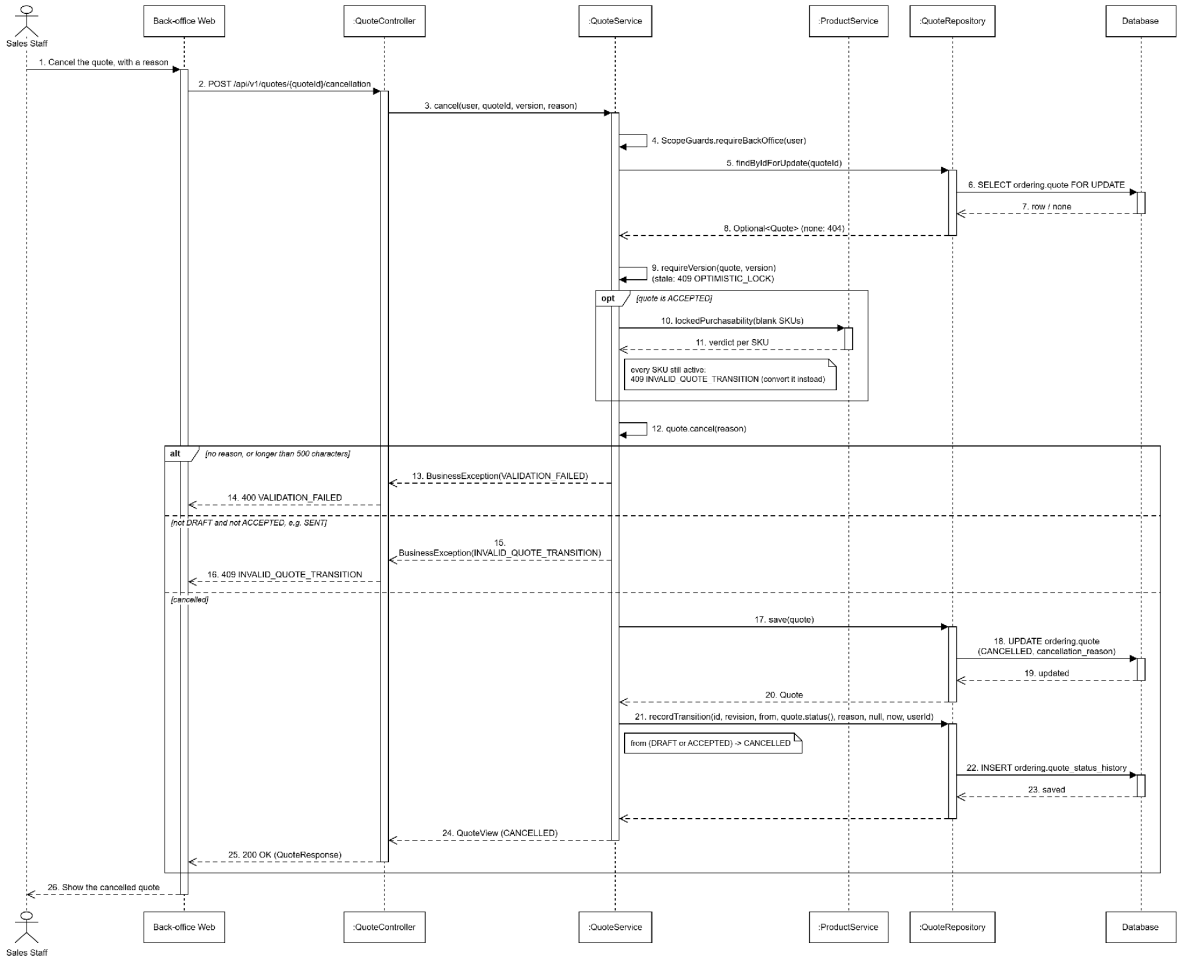


Figure 141. Cancel Quote Sequence Diagram

3.9.7 Convert Accepted Quote to Order Sequence Diagram

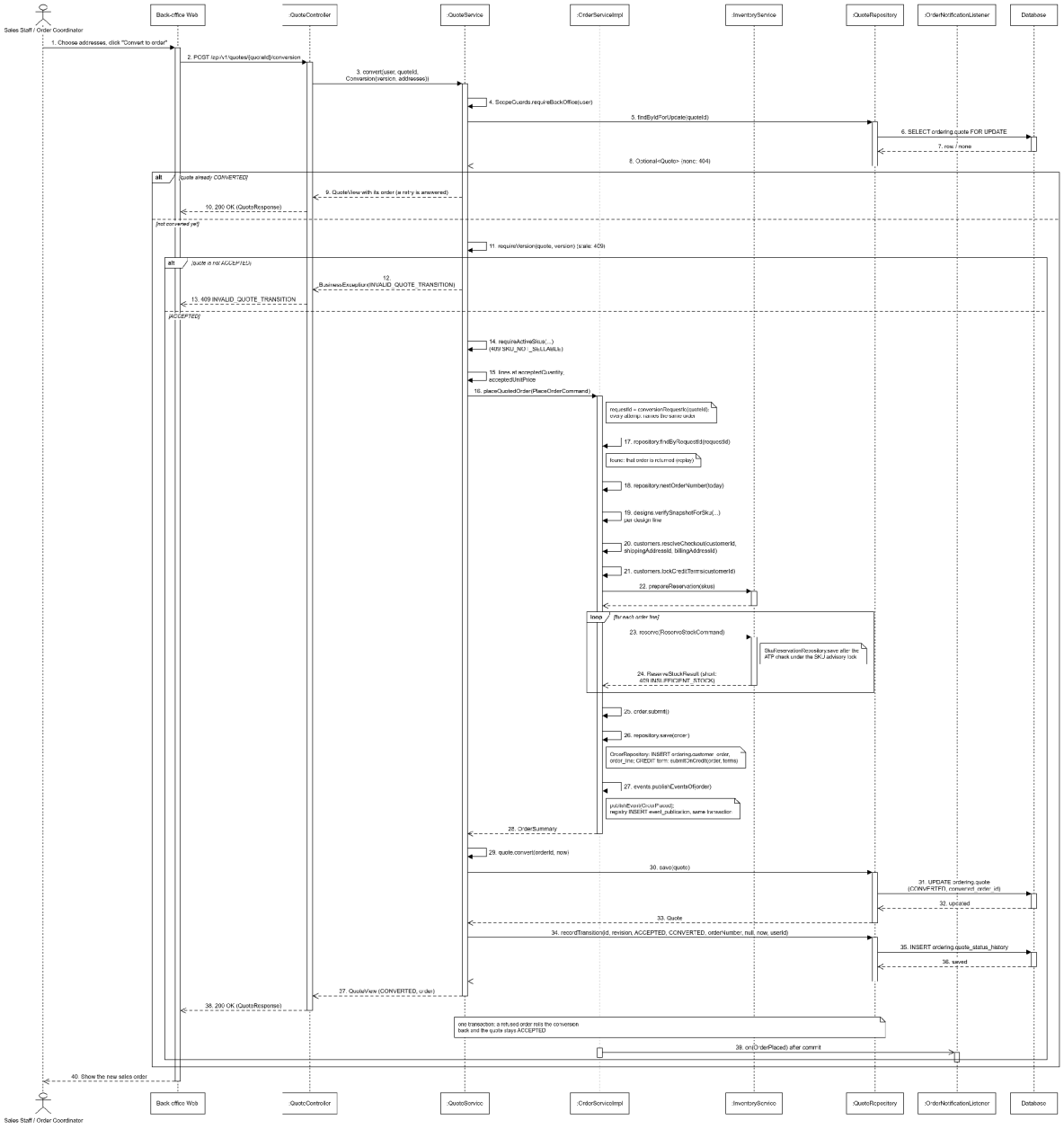


Figure 142. Convert Accepted Quote to Order Sequence Diagram

3.9.8 Reorder from Previous Order Sequence Diagram

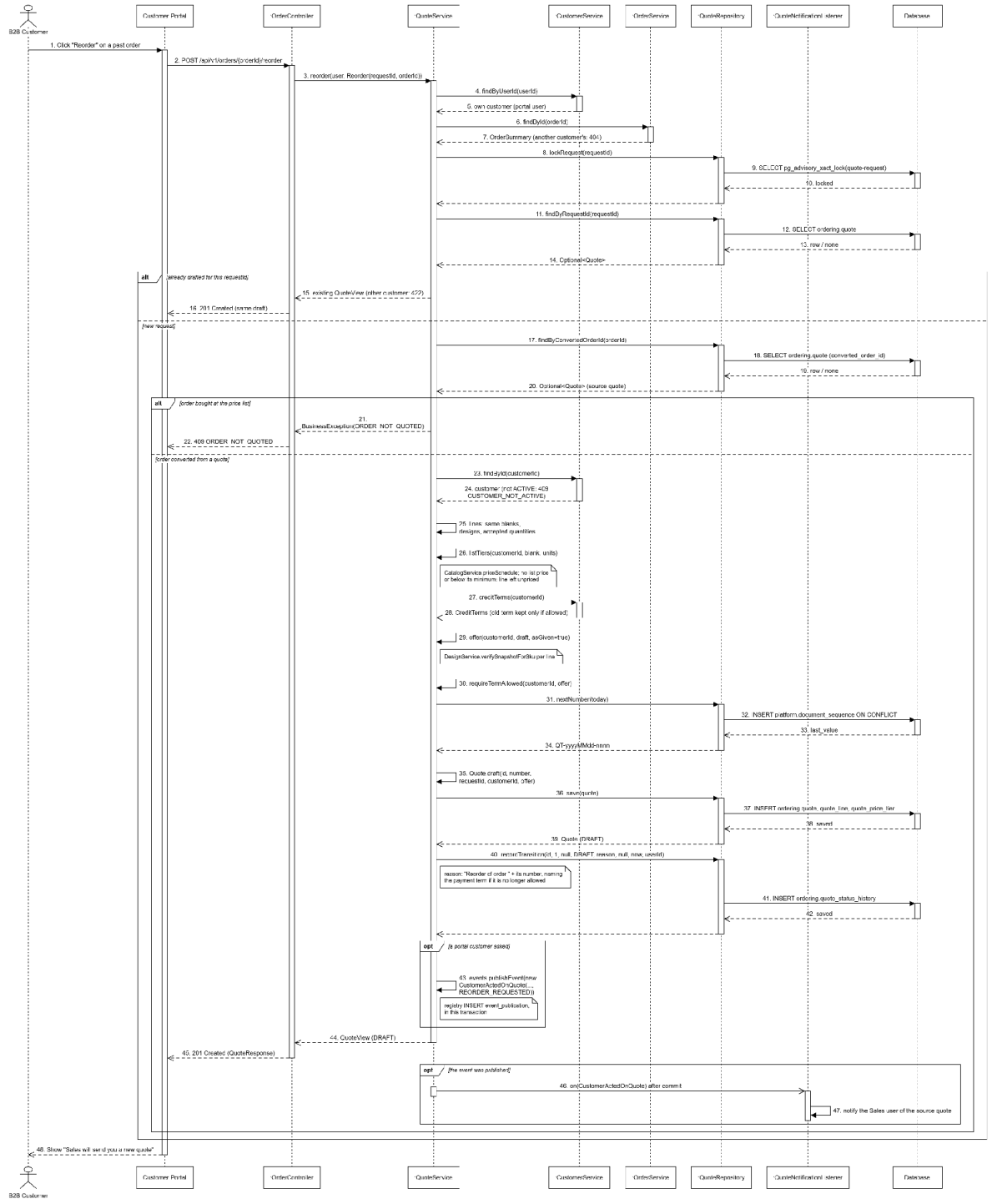


Figure 143. Reorder from Previous Order Sequence Diagram

3.9.9 Create & Submit Sample Request (1: Create) Sequence Diagram

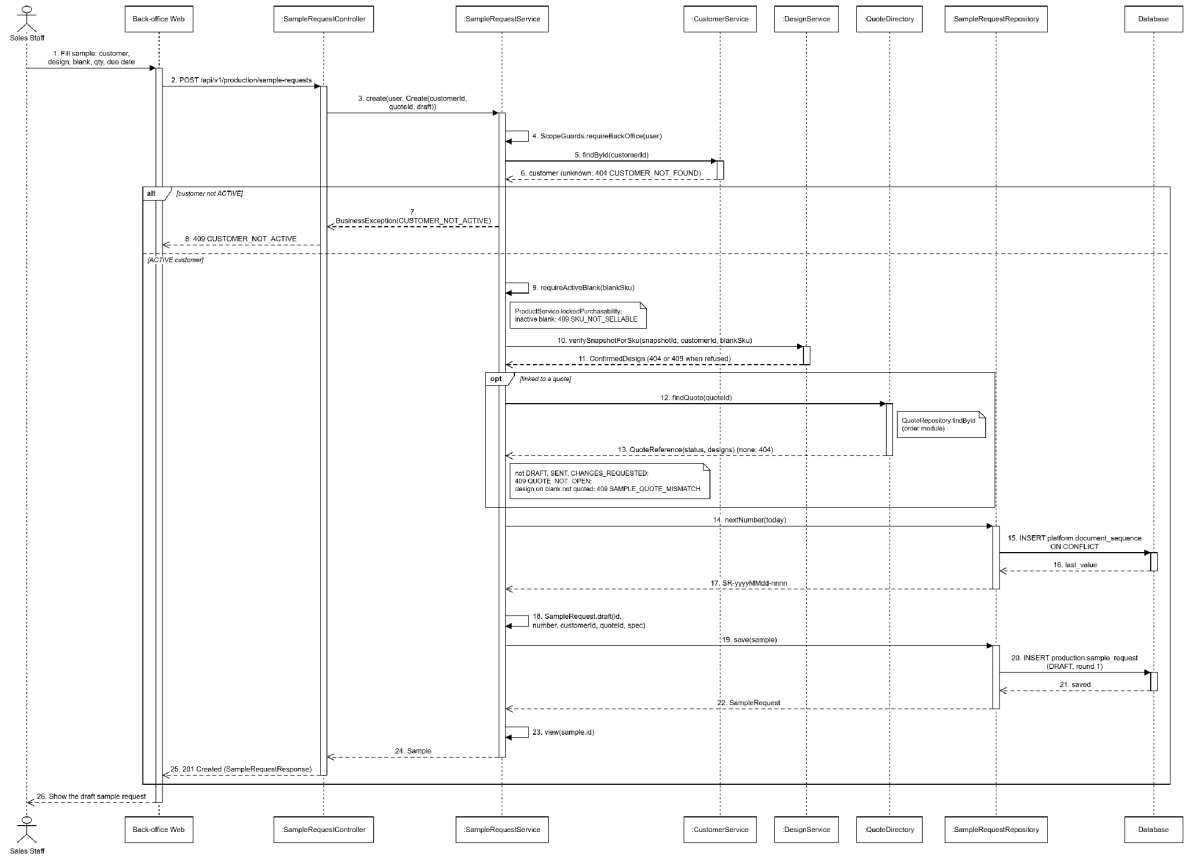


Figure 144. Create & Submit Sample Request (1: Create) Sequence Diagram

3.9.10 Create & Submit Sample Request (2: Submit) Sequence Diagram

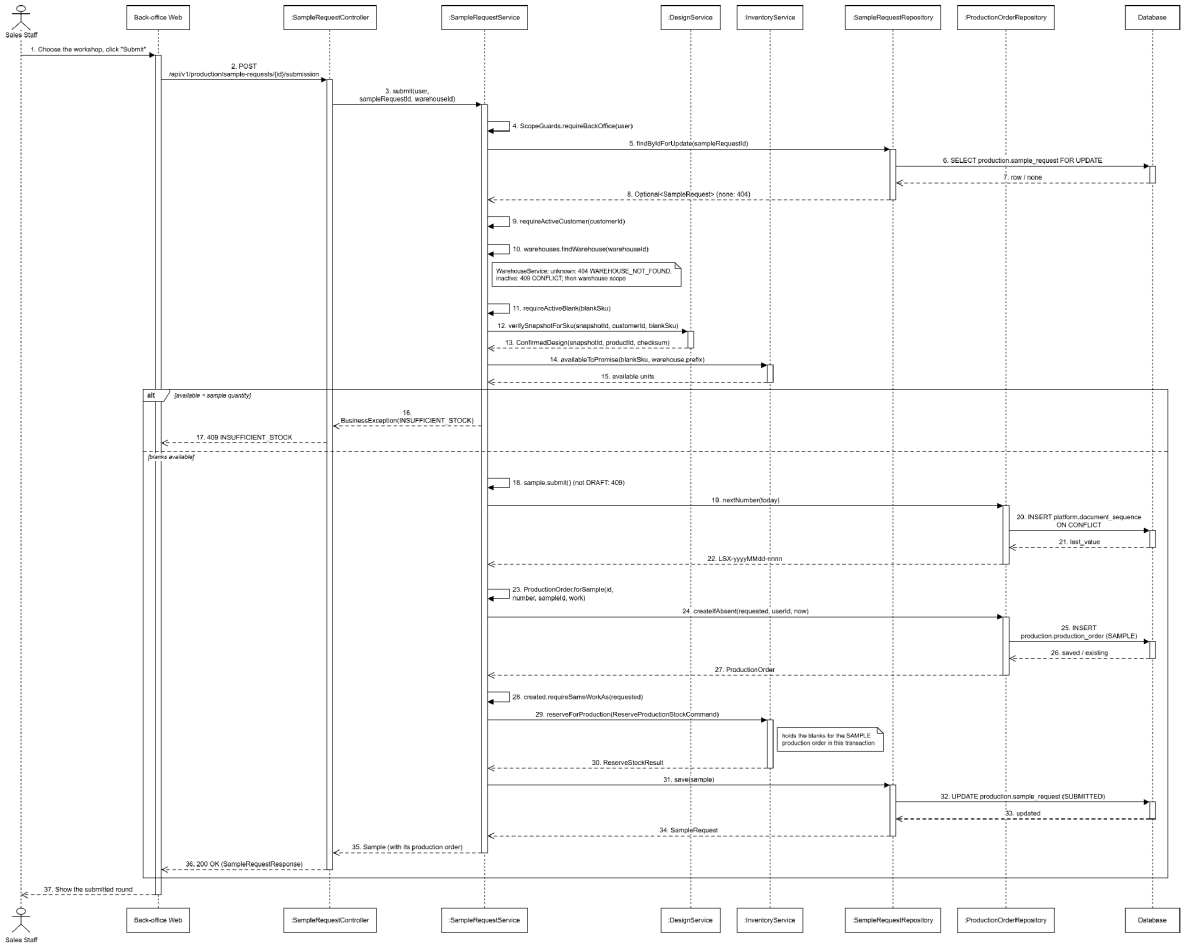


Figure 145. Create & Submit Sample Request (2: Submit) Sequence Diagram

3.9.11 Record Sample Sent to Customer Sequence Diagram

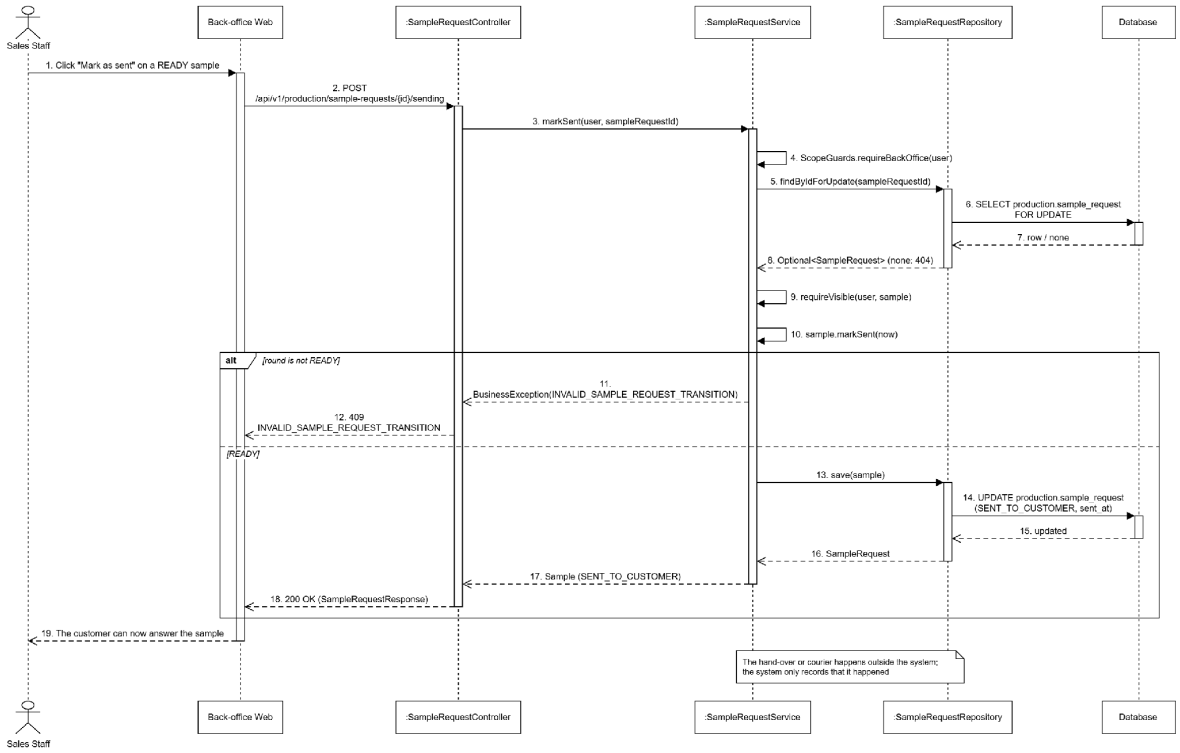


Figure 146. Record Sample Sent to Customer Sequence Diagram

3.9.12 Approve / Request Changes to Sample Sequence Diagram

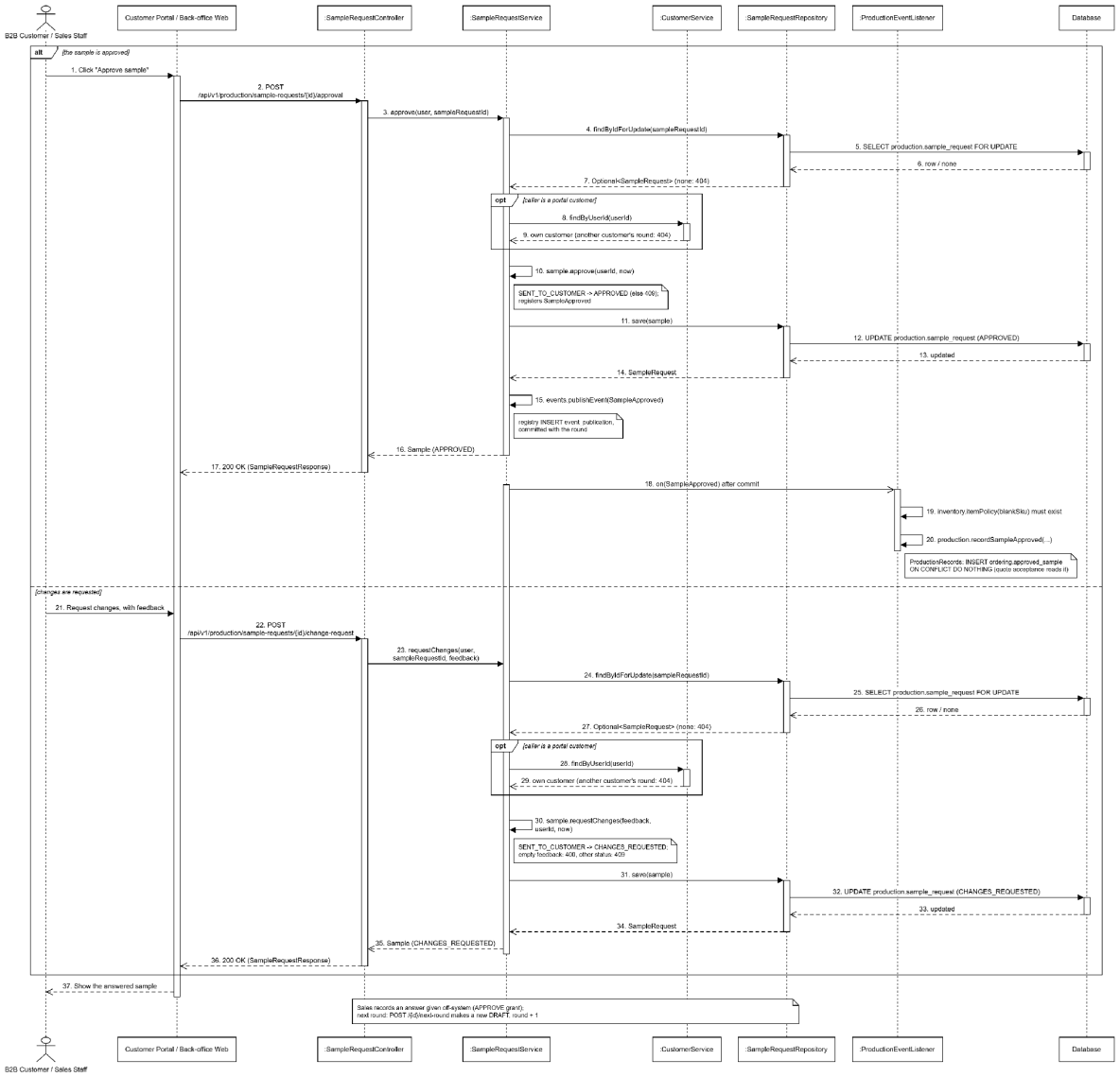


Figure 147. Approve / Request Changes to Sample Sequence Diagram

3.9.13 Quote State Diagram

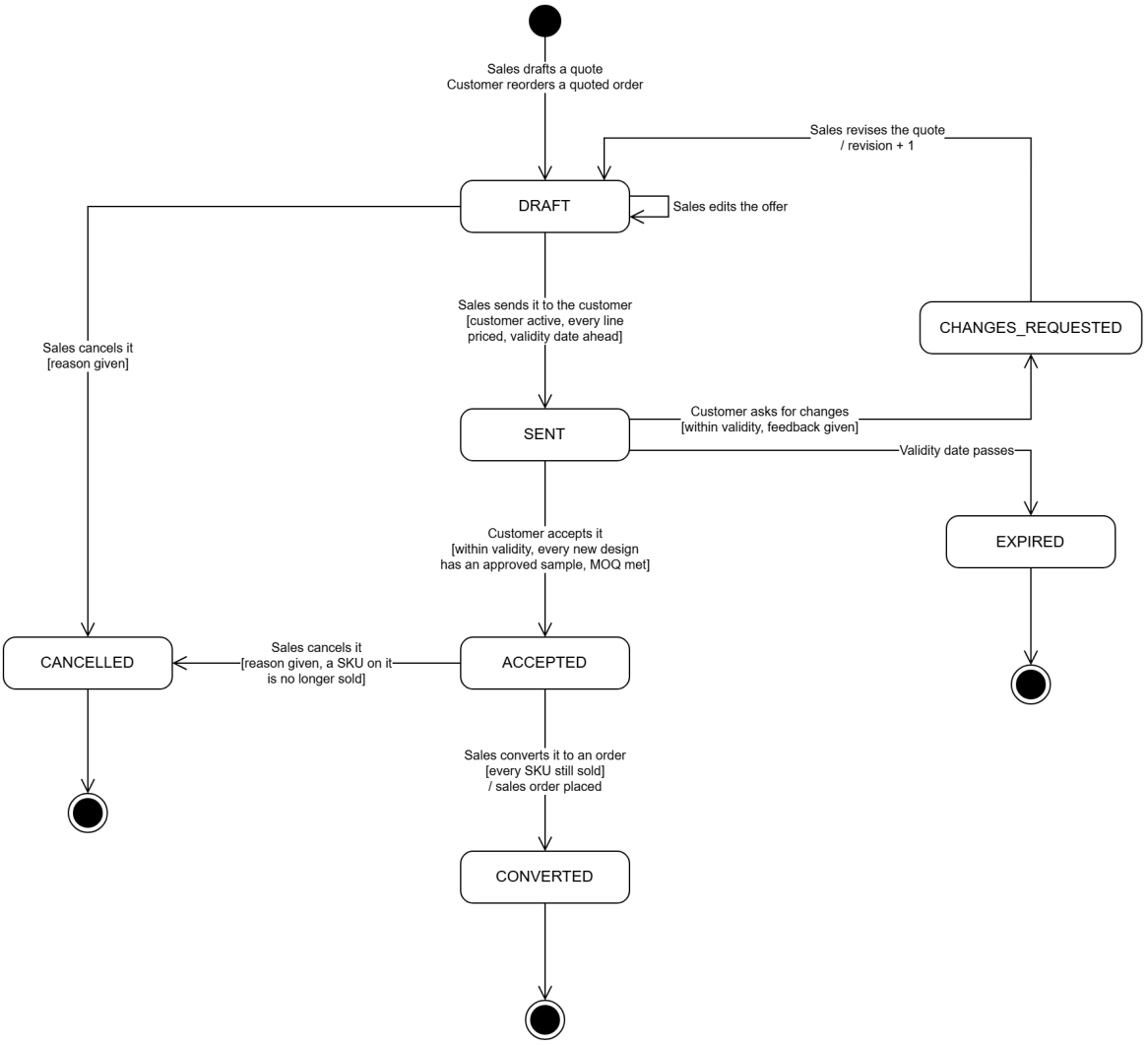


Figure 148. Quote State Diagram

3.9.14 Sample Request State Diagram

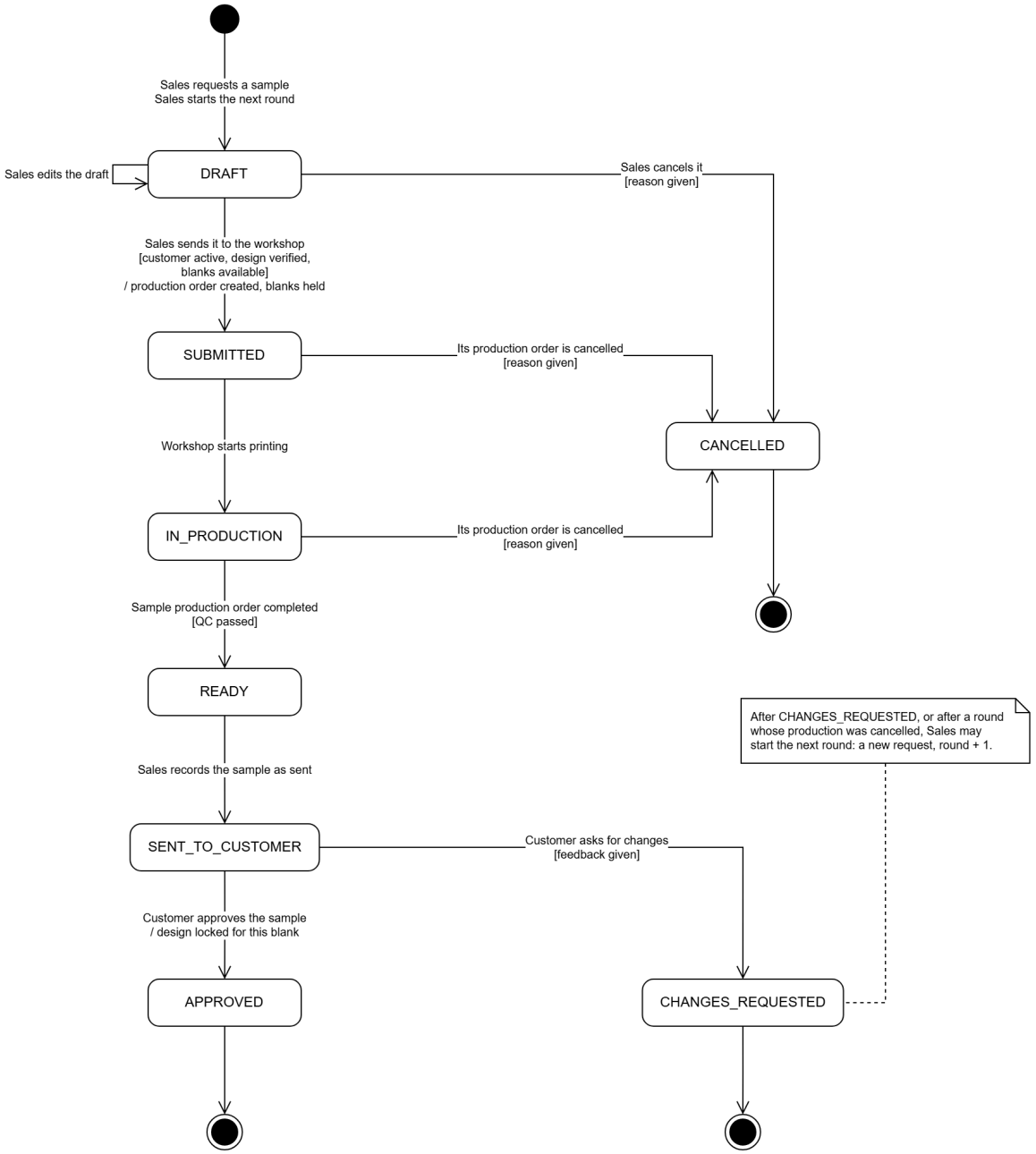


Figure 149. Sample Request State Diagram

3.10 Order Management

3.10.1 Class Diagram

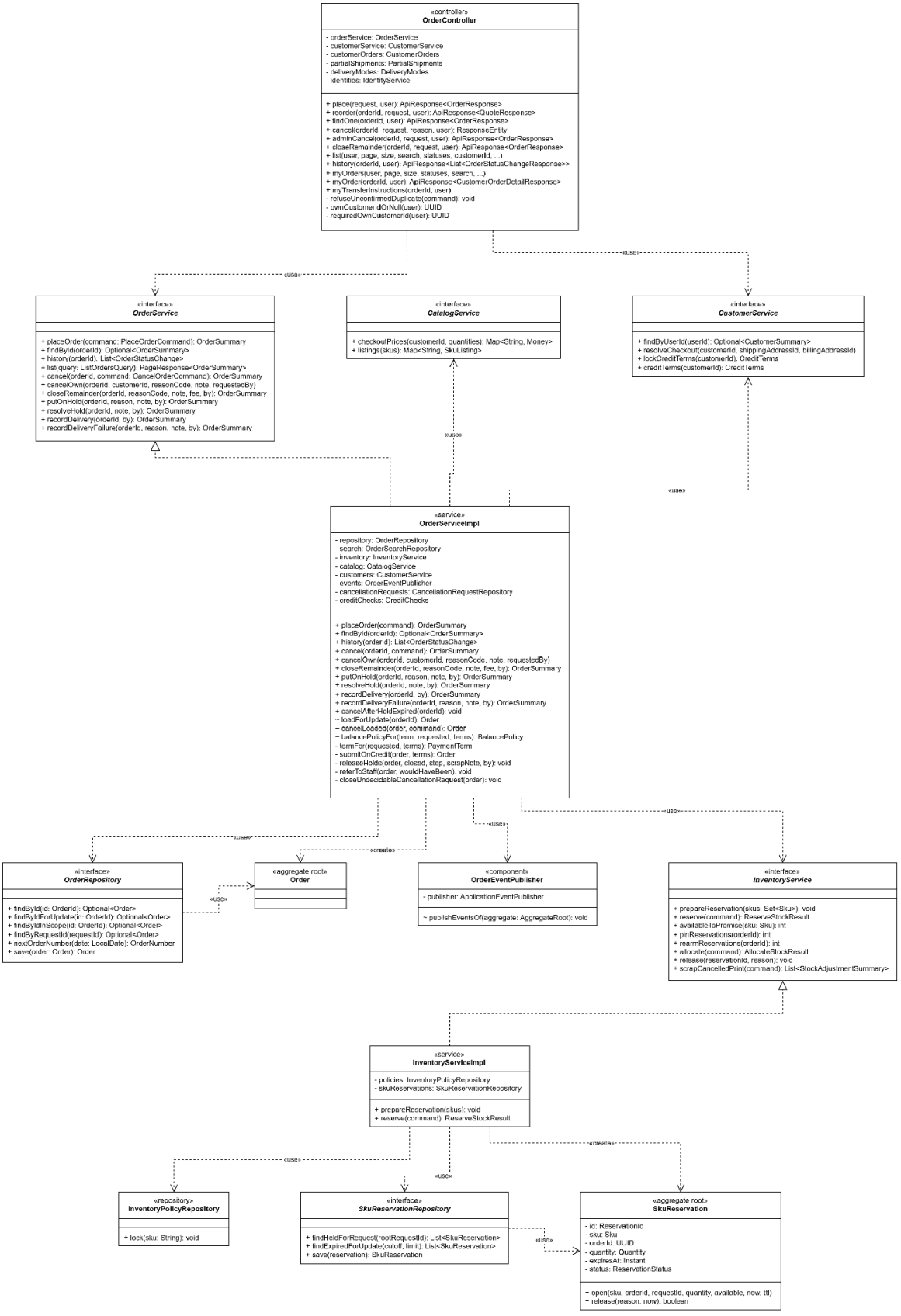


Figure 150. Order Management Class Diagram (1/10): Checkout

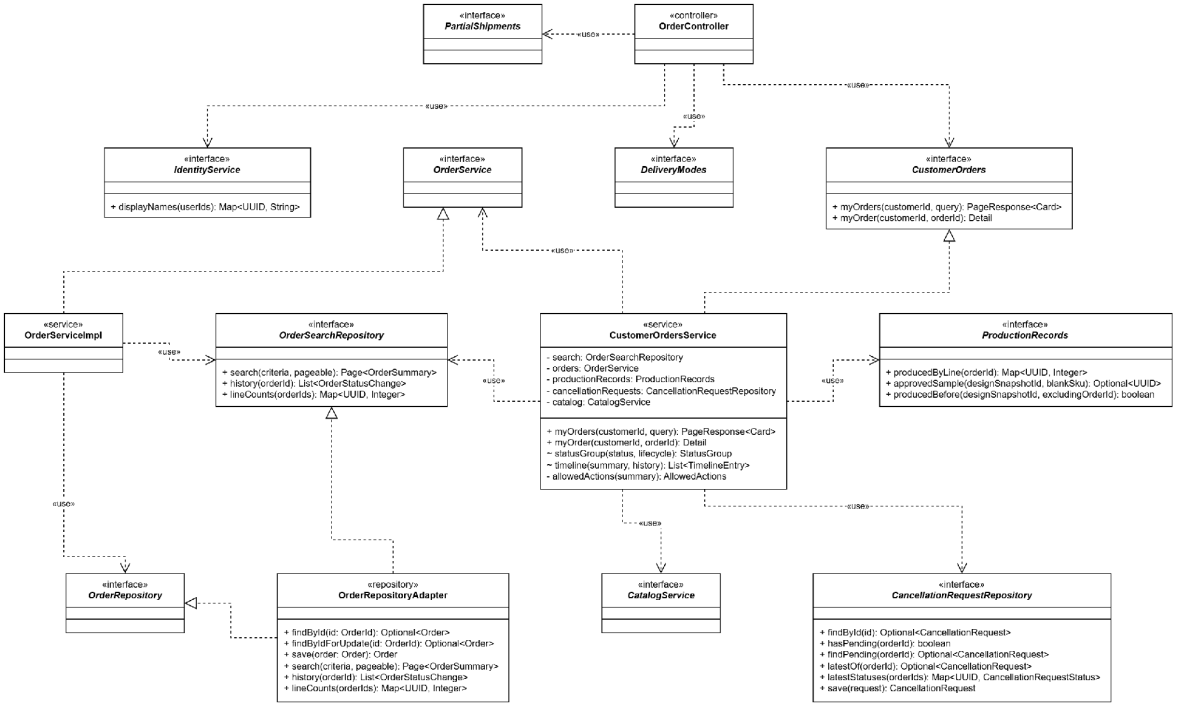


Figure 151. Order Management Class Diagram (2/10): Tracking and Viewing Orders

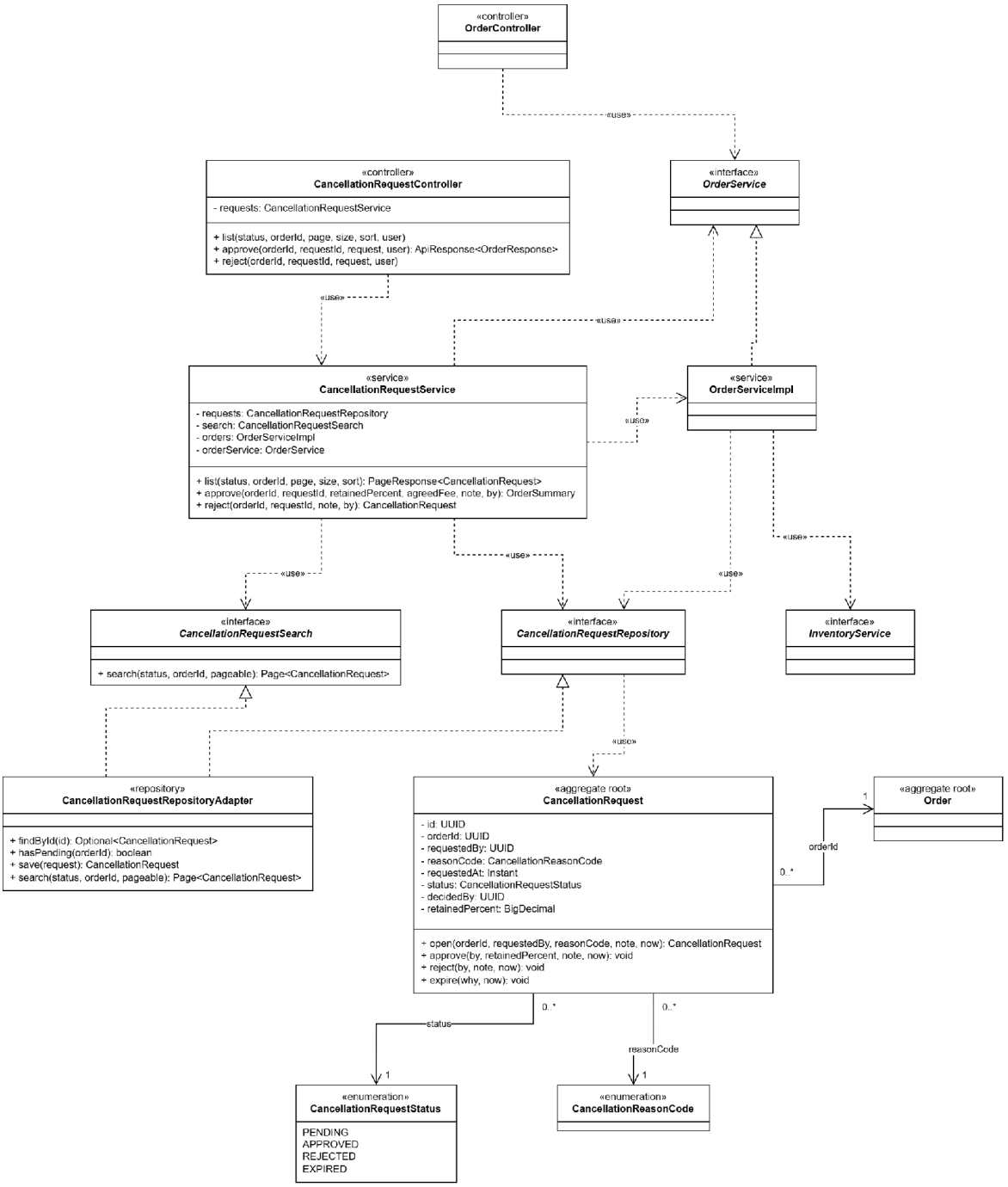


Figure 152. Order Management Class Diagram (3/10): Cancellation

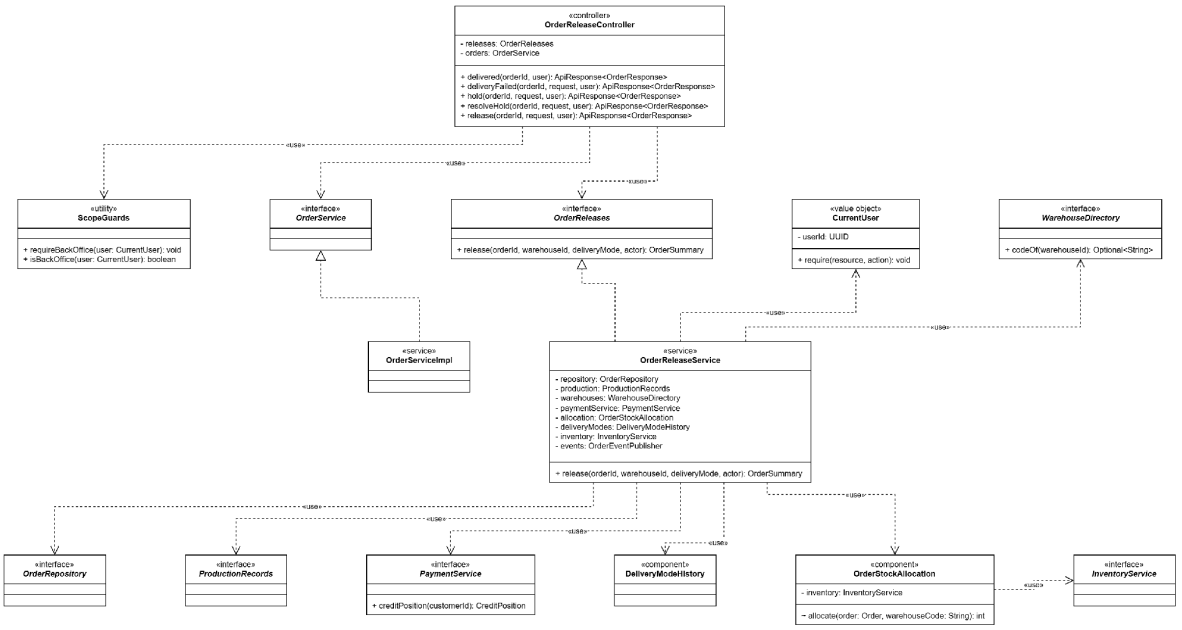


Figure 153. Order Management Class Diagram (4/10): Release, Hold and Delivery Result

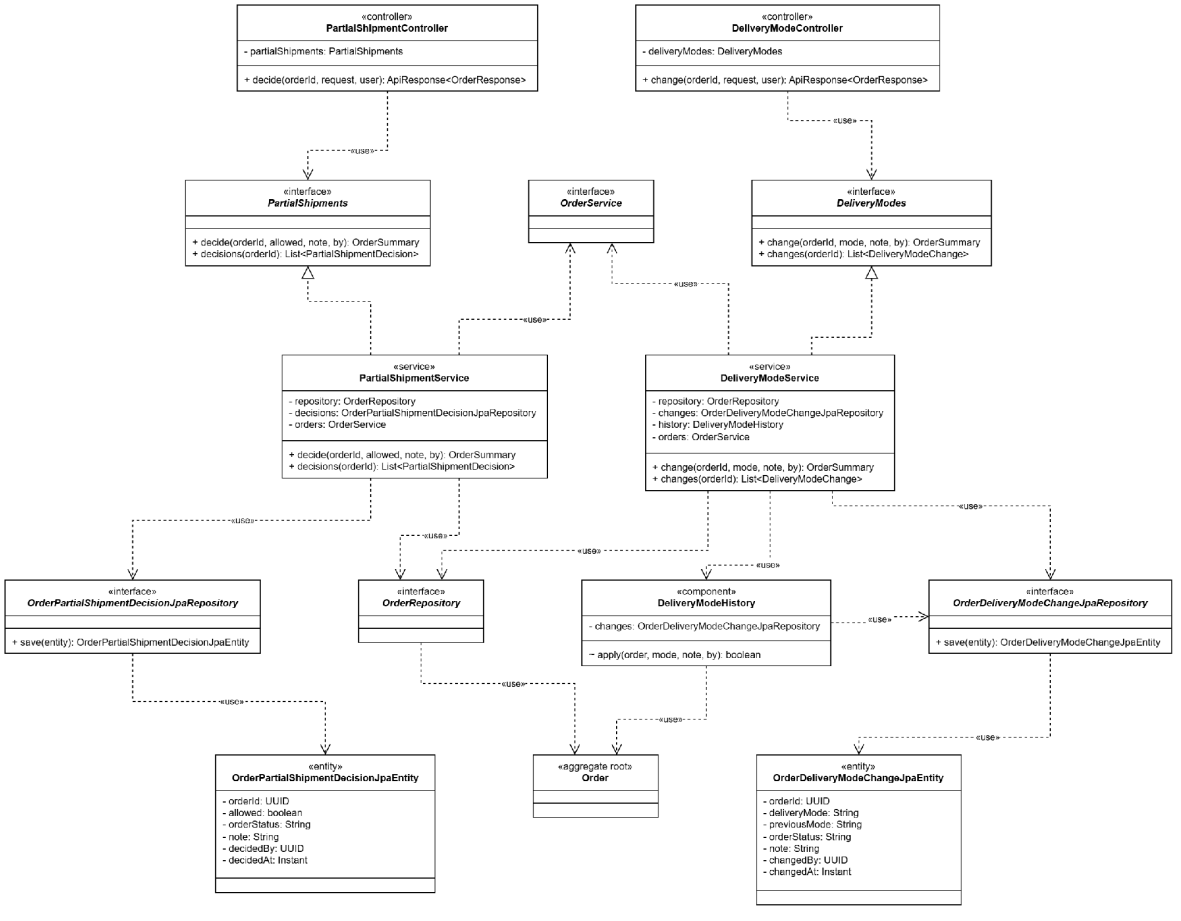


Figure 154. Order Management Class Diagram (5/10): Partial Shipment and Delivery Mode

Figure 155. Order Management Class Diagram (6/10): Credit Holds

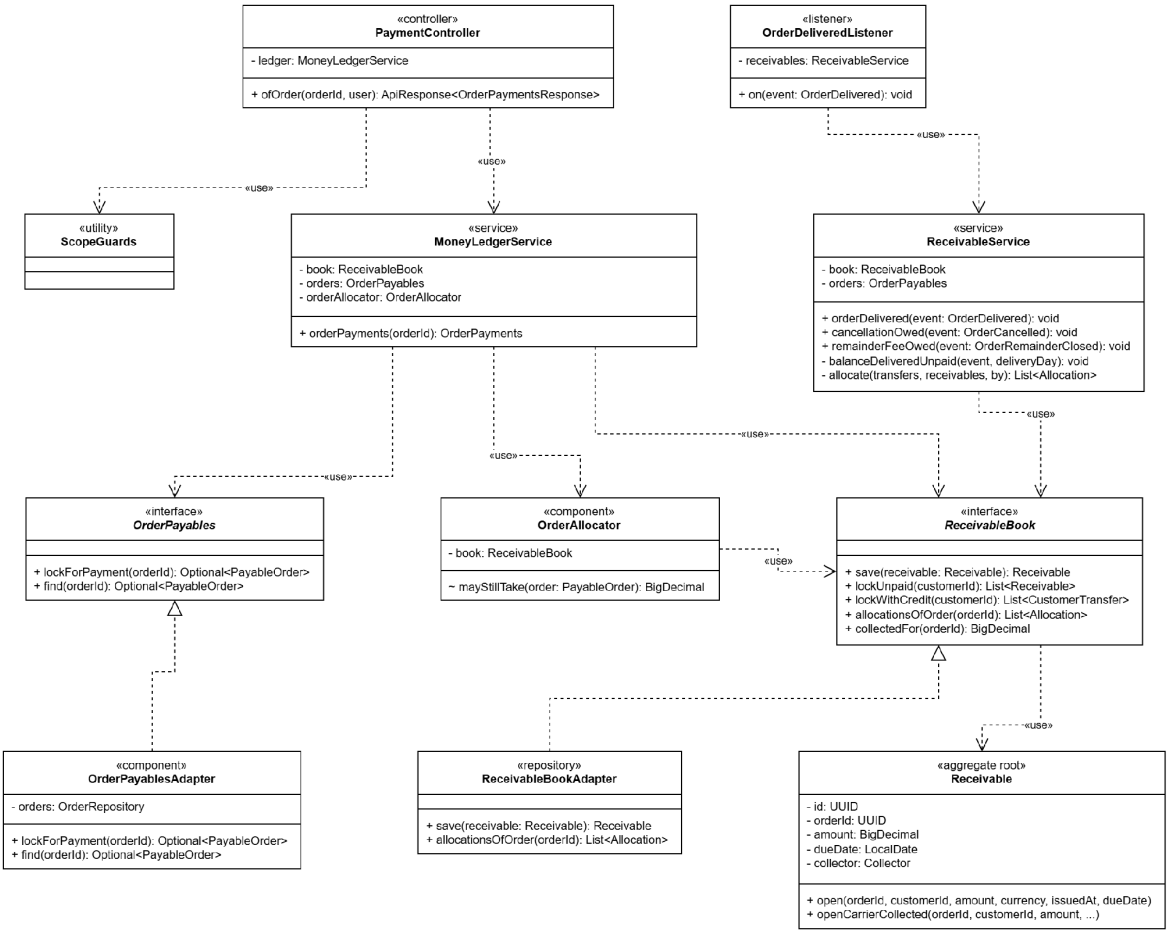


Figure 156. Order Management Class Diagram (7/10): Payments and Receivables of an Order

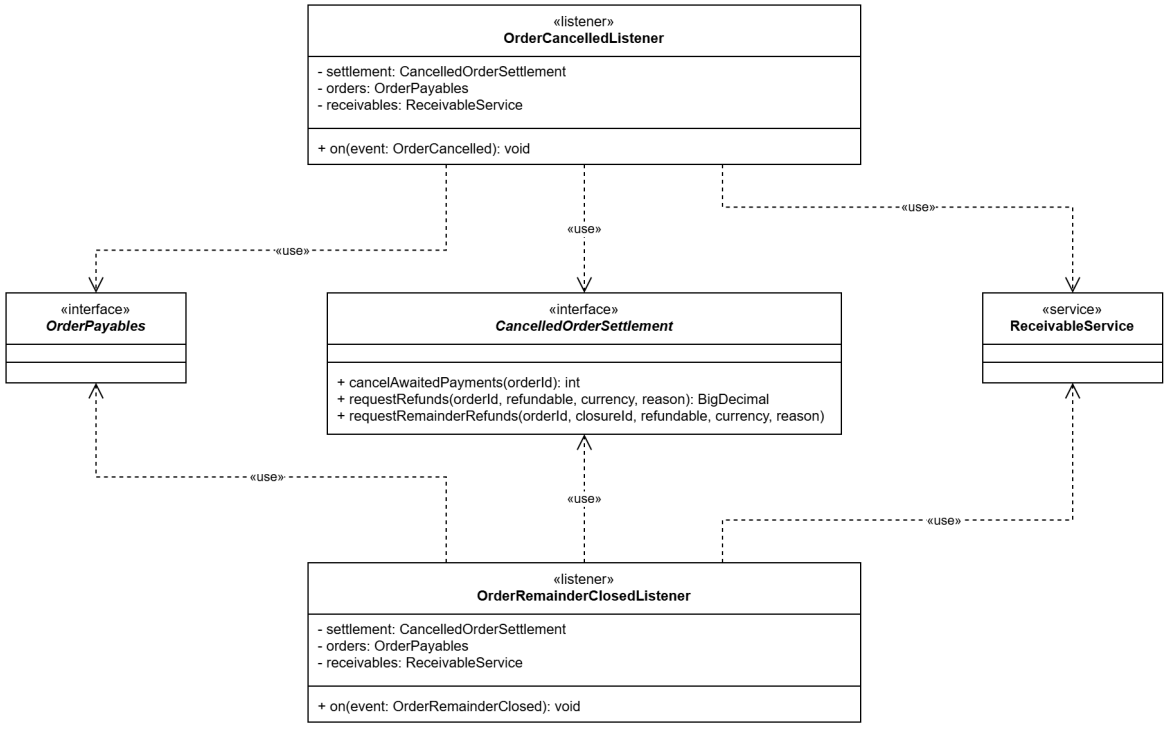


Figure 157. Order Management Class Diagram (8/10): Settling Cancelled and Closed Orders

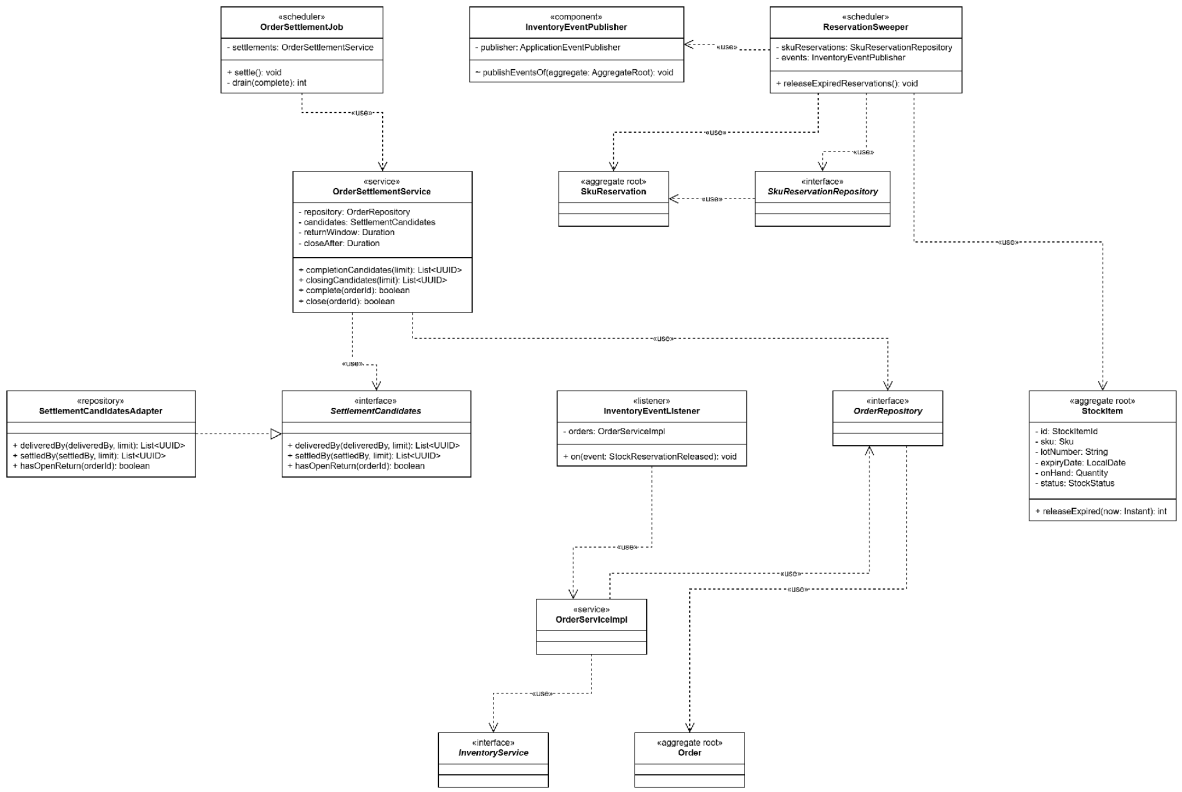


Figure 158. Order Management Class Diagram (9/10): Reservation Expiry and Order Settlement Jobs

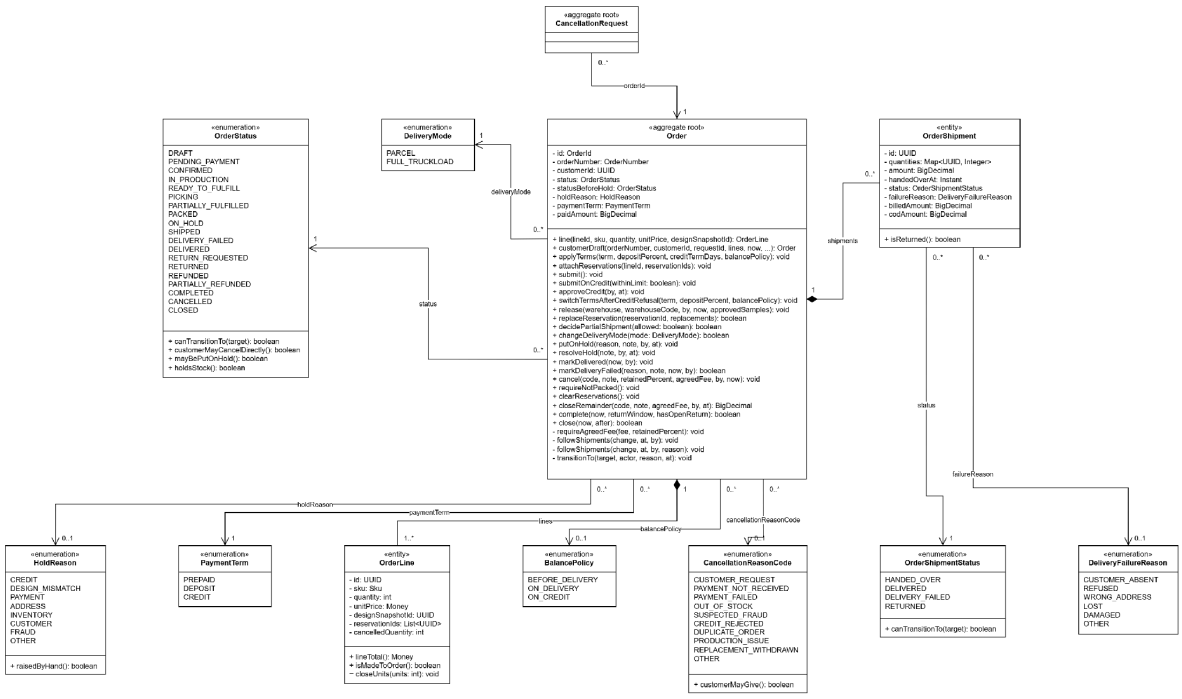


Figure 159. Order Management Class Diagram (10/10): Domain Model

3.10.2 Class Specification

OrderController

No	Method	Description

1	reorder(orderId, request, user)	POST /api/v1/orders/{orderId}/reorder: sales-orders:CREATE scope OWN, rate-limited 10/min. Drafts a new quote with the same blanks, designs and quantities of a quote-based order, priced from today's list (unpriced lines left for Sales); a list-price order is 409 ORDER_NOT_QUOTED; a customer's request notifies Sales; 201. (:175)
2	place(request, user)	POST /api/v1/orders: Places an order and reserves its stock in one transaction (201); a portal account may only order for its own customer (404 otherwise) and is asked to confirm a repeat of a recent identical order (409 DUPLICATE_ORDER_SUSPECTED); 409 INSUFFICIENT_STOCK rolls everything back. (:116)
3	findOne(orderId, user)	GET /api/v1/orders/{orderId}: Returns one order; a customer sees only their own (404 otherwise), staff also see the contact fields. (:186)
4	cancel(orderId, request, reason, user)	POST /api/v1/orders/{orderId}/cancellation: Customer cancellation: DRAFT/PENDING_PAYMENT/CONFIRMED is cancelled at once (200), a released order not yet packed gets a cancellation request (202), packed or later is 409 ORDER_NOT_CANCELLABLE; staff on this path cancel immediately. (:218)
5	adminCancel(orderId, request, user)	POST /api/v1/orders/{orderId}/admin-cancellation: Staff cancel any cancellable order with a reason code, optionally keeping a percentage or an agreed fee; a fee before printing starts is 409 CANCELLATION_FEE_BEFORE_PRINTING. (:262)
6	closeRemainder(orderId, request, user)	POST /api/v1/orders/{orderId}/remainder-closure: Closes the unshipped units of a PARTIALLY_FULFILLED order (else 409 ORDER_INVALID_TRANSITION), releases or scraps their holds and refunds what was paid for them, optionally keeping an agreed fee. (:290)
7	list(user, page, size, search, statuses, customerId, placedFrom, placedTo, sort)	GET /api/v1/orders: Back-office paged search of every customer's orders by text, status, customer and placed date; refused for portal accounts. (:306)
8	history(orderId, user)	GET /api/v1/orders/{orderId}/history: Status history oldest first; staff also get partial-shipment decisions, delivery-mode changes and actor names, a customer only the cancellation reason. (:328)
9	myOrders(user, page, size, statuses, search, placedFrom, placedTo, sort)	GET /api/v1/orders/me: The signed-in customer's order cards with simplified status group, delivery state and latest cancellation-request status. (:384)

10	myOrder(orderId, user)	GET /api/v1/orders/me/{orderId}: One of the customer's own orders with lines, production progress, timeline, payment, shipments and allowed actions (404 for another customer's). (:411)
11	myTransferInstructions(orderId, user)	GET /api/v1/orders/me/{orderId}/transfer-instructions: Company bank account, the order number as transfer content and the amount due now (deposit or balance); 503 TRANSFER_INSTRUCTIONS_UNAVAILABLE when no bank is configured. (:428)

Table 65. OrderController Class Specification

OrderReleaseController

No	Method	Description
1	delivered(orderId, user)	POST /api/v1/orders/{orderId}/delivery: Records that the carrier delivered a whole order (SHIPPED or DELIVERY_FAILED to DELIVERED) and publishes OrderDelivered, which opens a receivable for credit/deposit balances; 409 ORDER_DELIVERY_VIA_SHIPMENT for an order with shipments. (:57)
2	deliveryFailed(orderId, request, user)	POST /api/v1/orders/{orderId}/delivery-failure: Records a failed delivery with its reason code (OTHER needs a note), SHIPPED to DELIVERY_FAILED; 409 ORDER_DELIVERY_VIA_SHIPMENT for an order with shipments. (:67)
3	hold(orderId, request, user)	POST /api/v1/orders/{orderId}/hold: Puts an order on hold for PAYMENT, ADDRESS, INVENTORY, CUSTOMER, FRAUD or OTHER (CREDIT and DESIGN_MISMATCH are system holds: 400); only CONFIRMED to PACKED states may be held (409 ORDER_INVALID_TRANSITION). (:82)
4	resolveHold(orderId, request, user)	POST /api/v1/orders/{orderId}/hold/resolution: Resolves a hold with a note, back to the status it was held from; 409 ORDER_NOT_ON_HOLD, ORDER_CREDIT_HOLD_USE_APPROVAL, ORDER_DESIGN_HOLD_USE_REVERIFICATION, or ORDER_BALANCE_NOT_PAID for a PAYMENT hold whose terms are still unmet. (:100)

5	release(orderId, request, user)	POST /api/v1/orders/{orderId}/release: Releases a CONFIRMED order from a chosen warehouse (to IN_PRODUCTION with print lines, else READY_TO_FULFILL), allocating its SKU reservations to bins of that warehouse; refused for an unpaid deposit, a new design without an approved sample, an overdue credit customer or a shortage; another delivery mode needs sales-order-delivery-modes:UPDATE. (:113)

Table 66. OrderReleaseController Class Specification

CancellationRequestController

No	Method	Description
1	list(status, orderId, page, size, sort, user)	GET /api/v1/orders/cancellation-requests: Back-office queue of cancellation requests, newest first, filterable by status (PENDING = to decide) and order. (:53)
2	approve(orderId, requestId, request, user)	POST /api/v1/orders/{orderId}/cancellation-requests/{requestId}/approval: Approves a PENDING request and cancels the order in the same transaction, keeping a retained percentage or agreed fee; 409 ORDER_CANCELLATION_REQUEST_NOT_PENDING if already decided. (:69)
3	reject(orderId, requestId, request, user)	POST /api/v1/orders/{orderId}/cancellation-requests/{requestId}/rejection: Rejects a PENDING request with a mandatory note; the order goes on. (:81)

Table 67. CancellationRequestController Class Specification

CreditHoldController

No	Method	Description
1	myTerms(user)	GET /api/v1/orders/payment-terms: The signed-in customer's payment terms, credit limit, exposure, credit left, overdue amount and allowed balance policies, for checkout. (:65)
2	position(customerId)	GET /api/v1/orders/credit-position/{customerId}: The same credit position for any customer, for staff (customer-credit-profiles:READ). (:75)
3	holds(page, size)	GET /api/v1/orders/credit-holds: Orders waiting on a CREDIT hold, oldest first, each with its latest credit check (limit, exposure, order amount). (:82)

4	approve(orderId, request, user)	POST /api/v1/orders/{orderId}/credit-approval: Approves a credit-hold order with a mandatory note: ON_HOLD to CONFIRMED, the check and the hold resolution recorded; 409 ORDER_NOT_ON_CREDIT_HOLD otherwise. (:96)
5	refuse(orderId, request, user)	POST /api/v1/orders/{orderId}/credit-rejection: Refuses credit with a note: CANCEL cancels the order, SWITCH_TO_PREPAID / SWITCH_TO_DEPOSIT re-terms it to PENDING_PAYMENT and re-arms its stock holds (409 PAYMENT_TERM_NOT_ALLOWED for terms the customer may not use). (:105)

Table 68. CreditHoldController Class Specification

DeliveryModeController

No	Method	Description
1	change(orderId, request, user)	PUT /api/v1/orders/{orderId}/delivery-mode: Changes how an order travels (PARCEL / FULL_TRUCKLOAD) and records the change in history; 409 DELIVERY_MODE_LOCKED once packing started or past PICKING, 409 DELIVERY_MODE_FTL_NO_COD for a truckload whose balance is collected on delivery. (:56)

Table 69. DeliveryModeController Class Specification

PartialShipmentController

No	Method	Description
1	decide(orderId, request, user)	PUT /api/v1/orders/{orderId}/partial-shipment: Allows or stops partial shipments of an order and records the decision with its note; 409 ORDER_PARTIAL_SHIPMENT_LOCKED once shipped, cancelled or closed, or when turning it off after goods already left. (:54)

Table 70. PartialShipmentController Class Specification

3.10.3 Place Quick Order Sequence Diagram

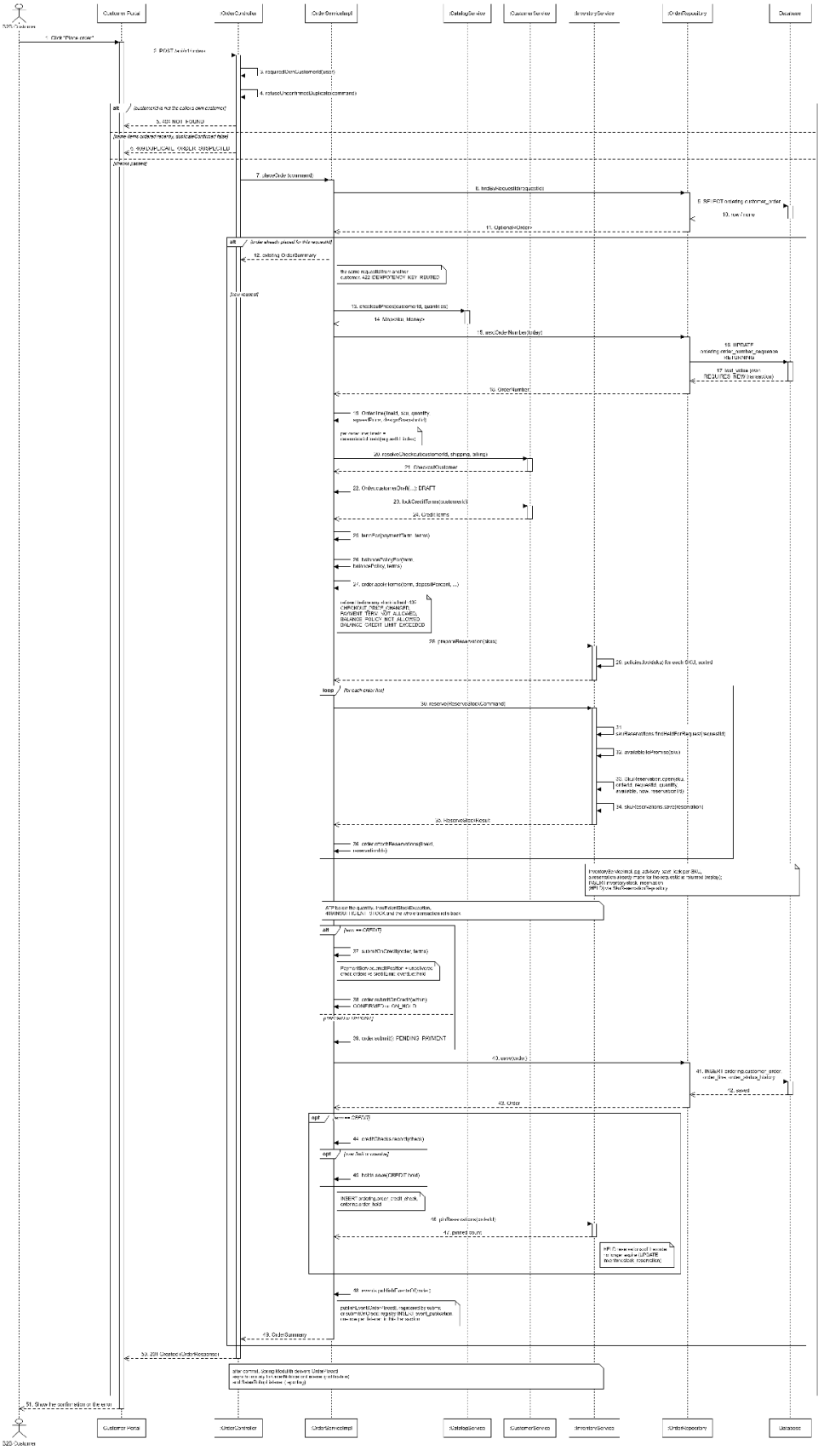


Figure 160. Place Quick Order Sequence Diagram

3.10.4 Track Orders Sequence Diagram

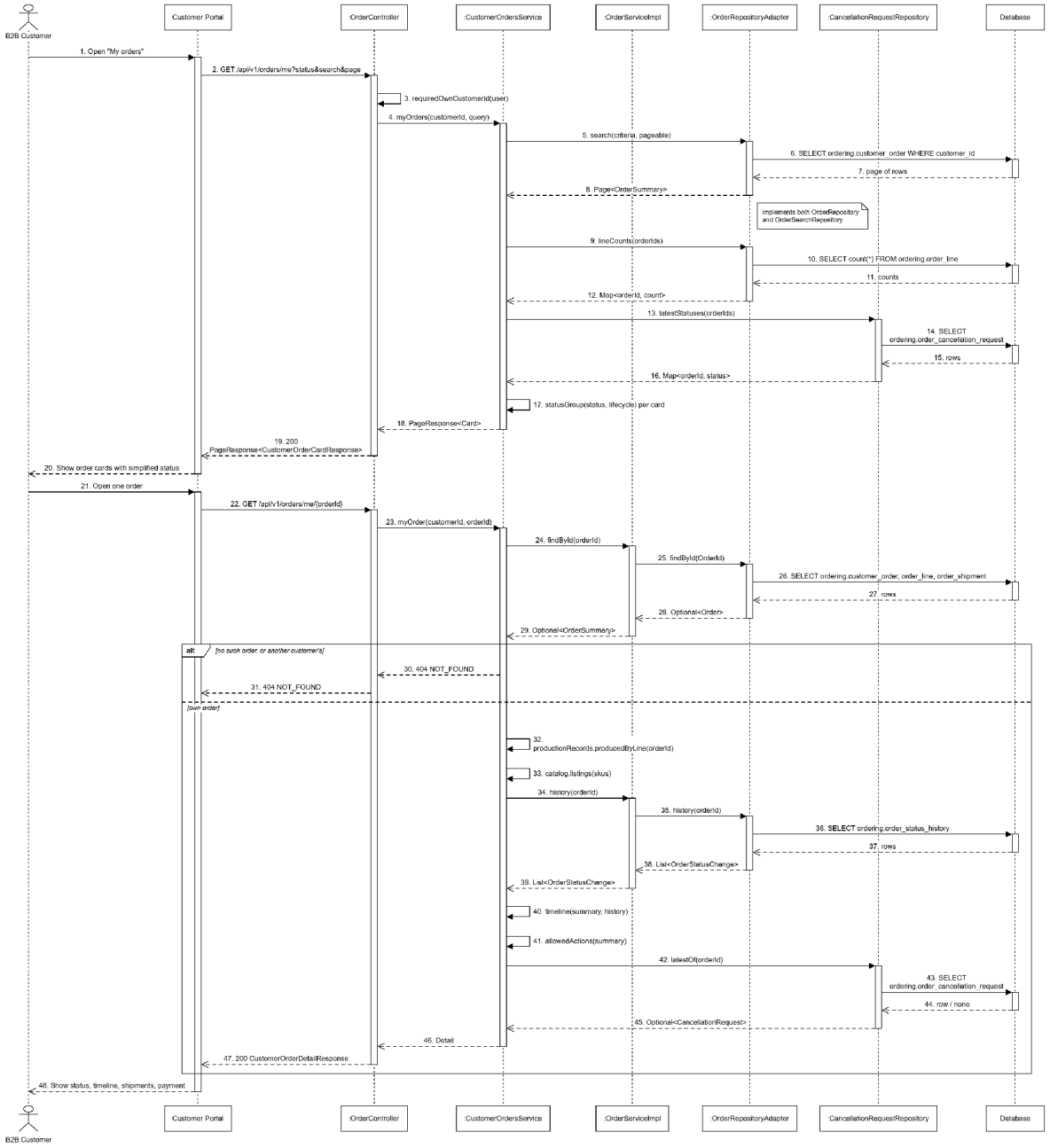


Figure 161. Track Orders Sequence Diagram

3.10.5 View Order Details Sequence Diagram

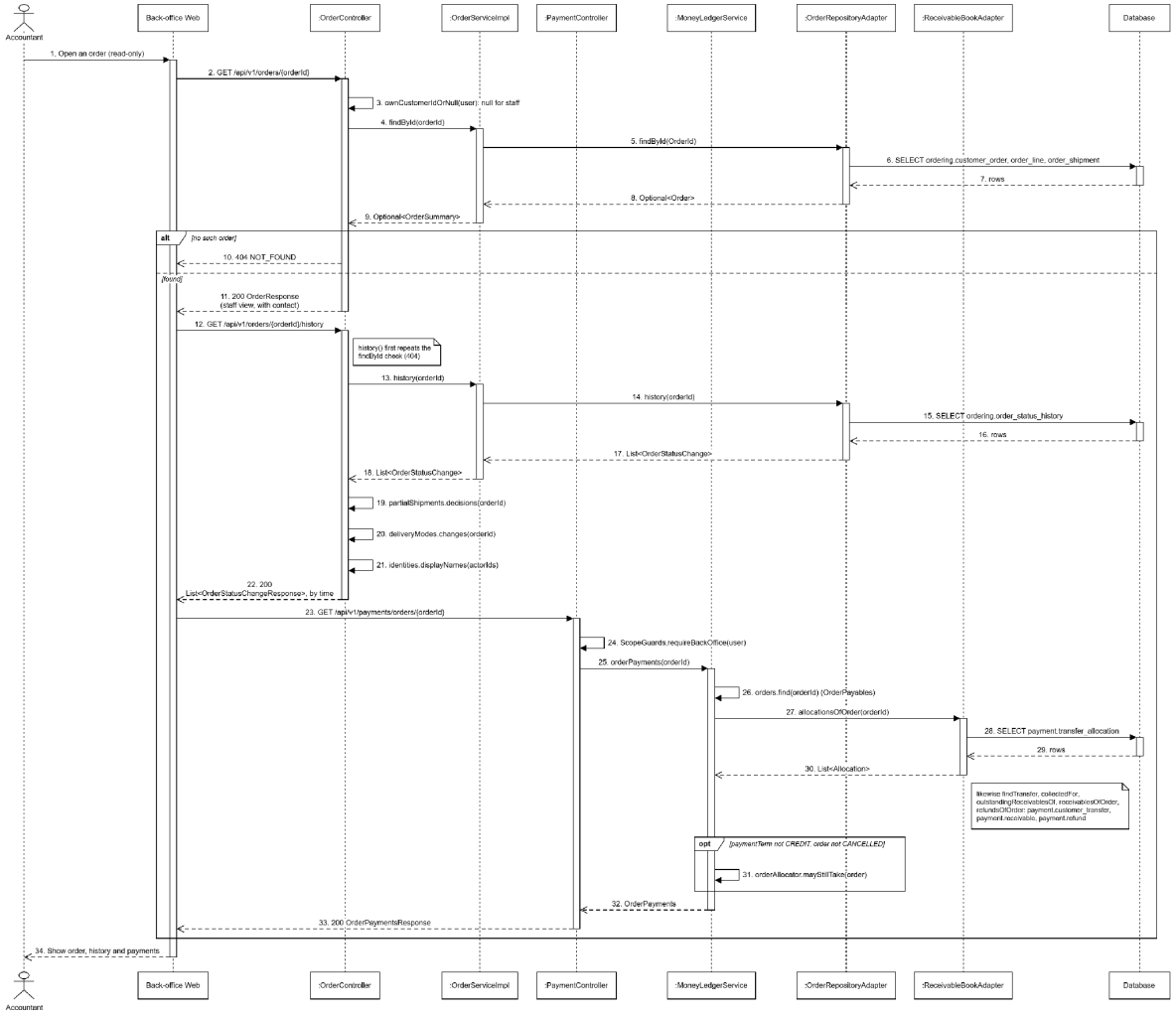


Figure 162. View Order Details Sequence Diagram

3.10.6 Cancel Order / Request Cancellation Sequence Diagram

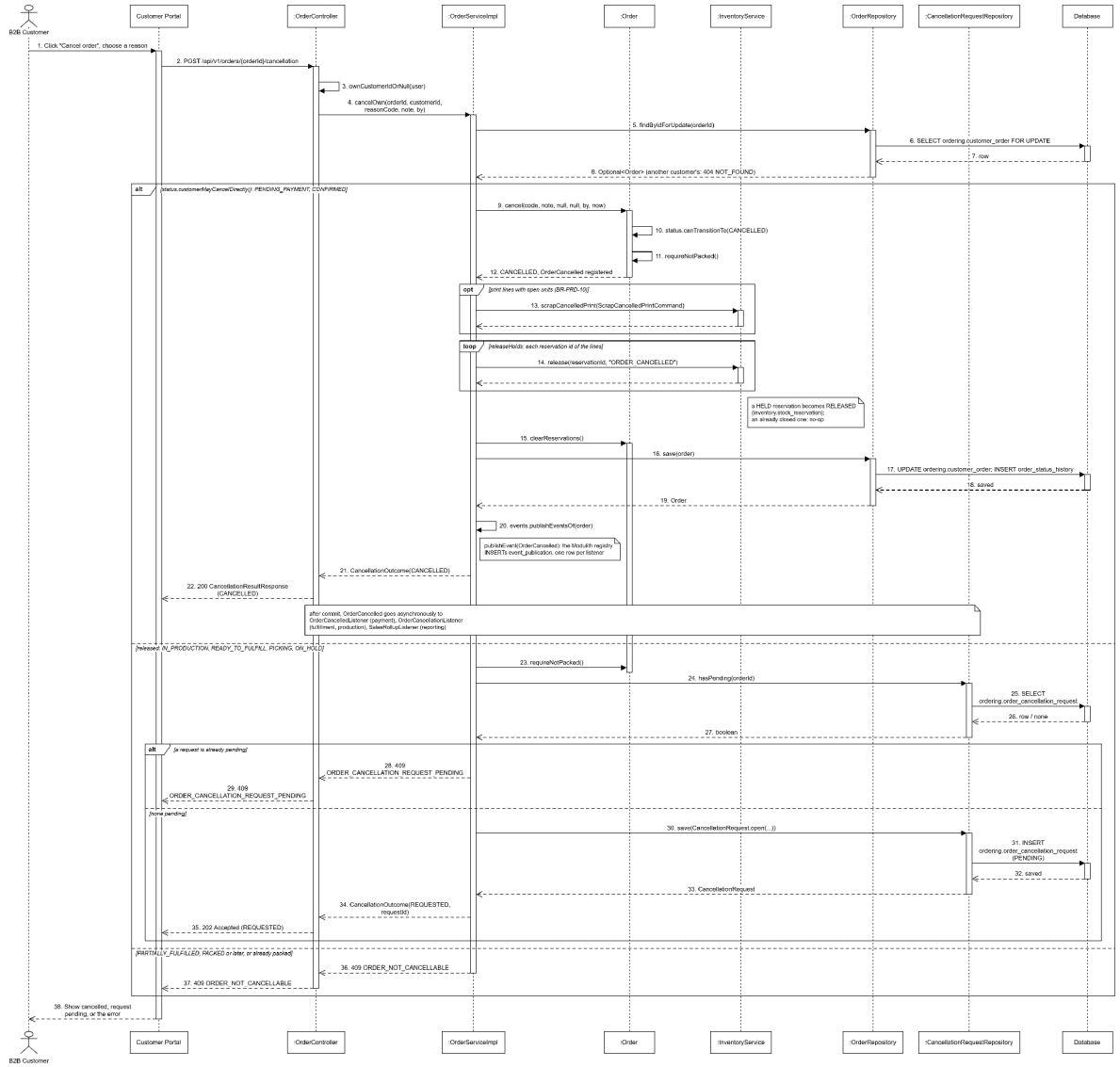


Figure 163. Cancel Order / Request Cancellation Sequence Diagram

3.10.7 Approve / Reject Cancellation Request Sequence Diagram

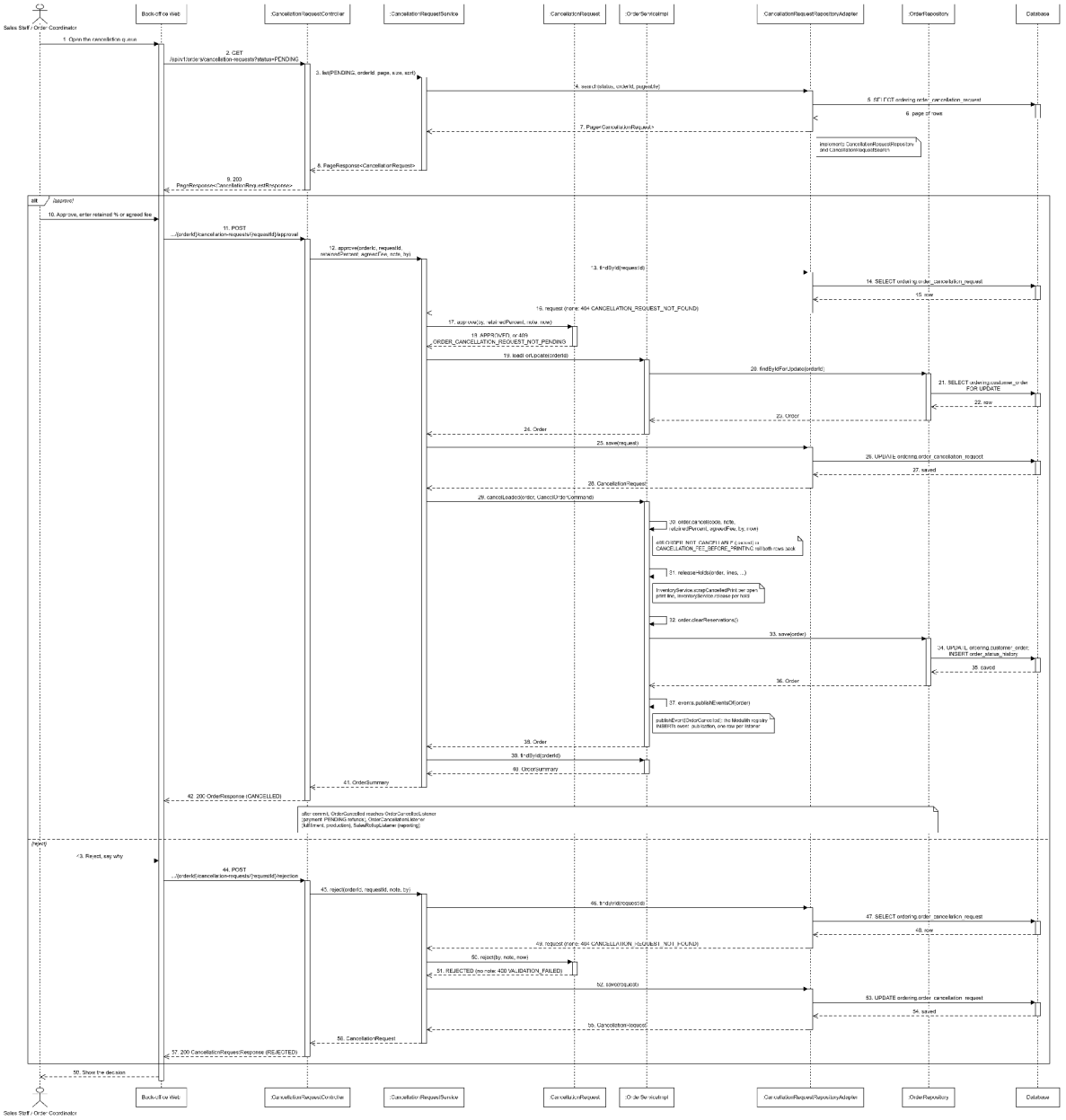


Figure 164. Approve / Reject Cancellation Request Sequence Diagram

3.10.8 Cancel Order (with Fee) Sequence Diagram

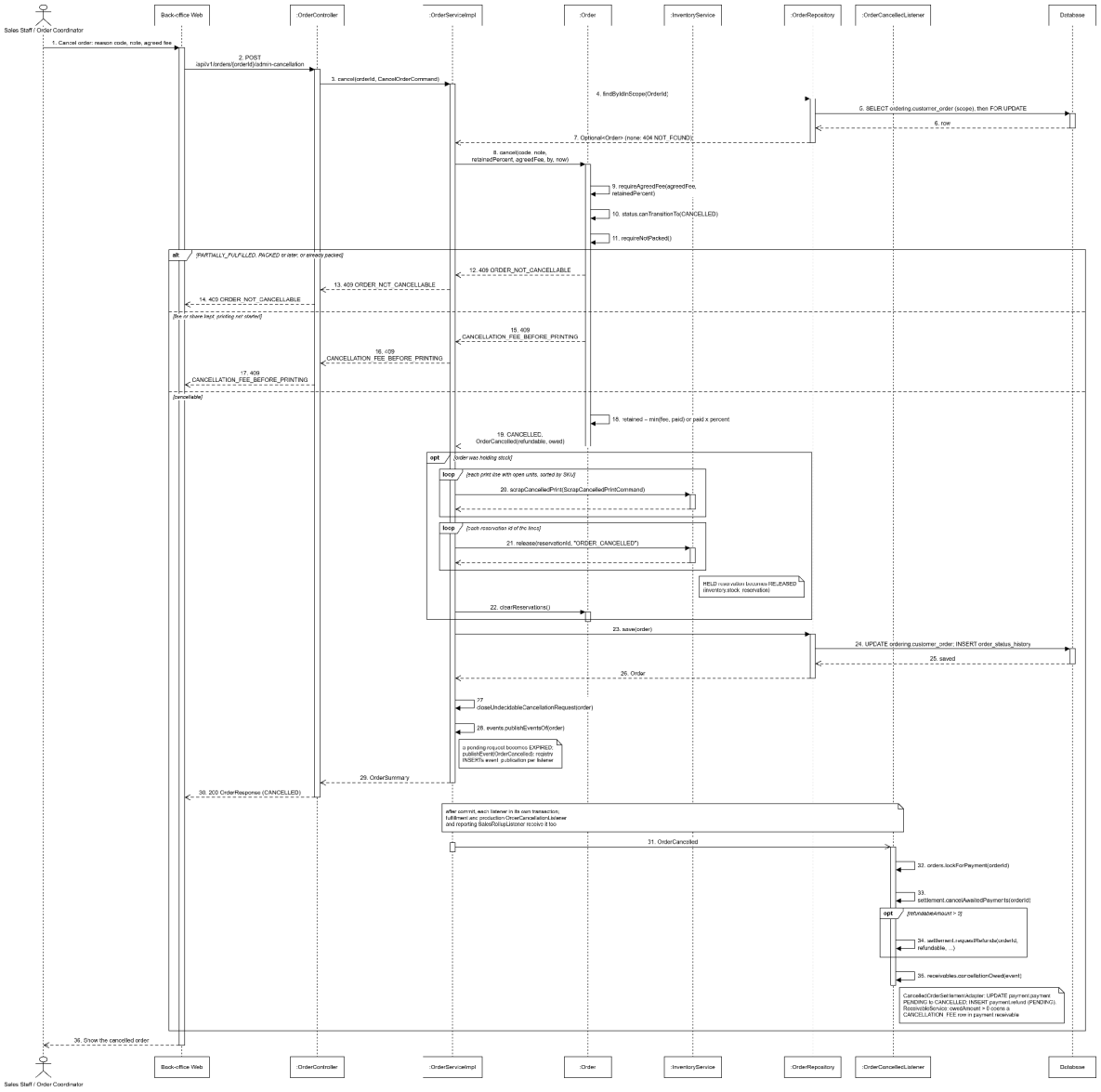


Figure 165. Cancel Order (with Fee) Sequence Diagram

3.10.9 Close Remaining Quantity Sequence Diagram

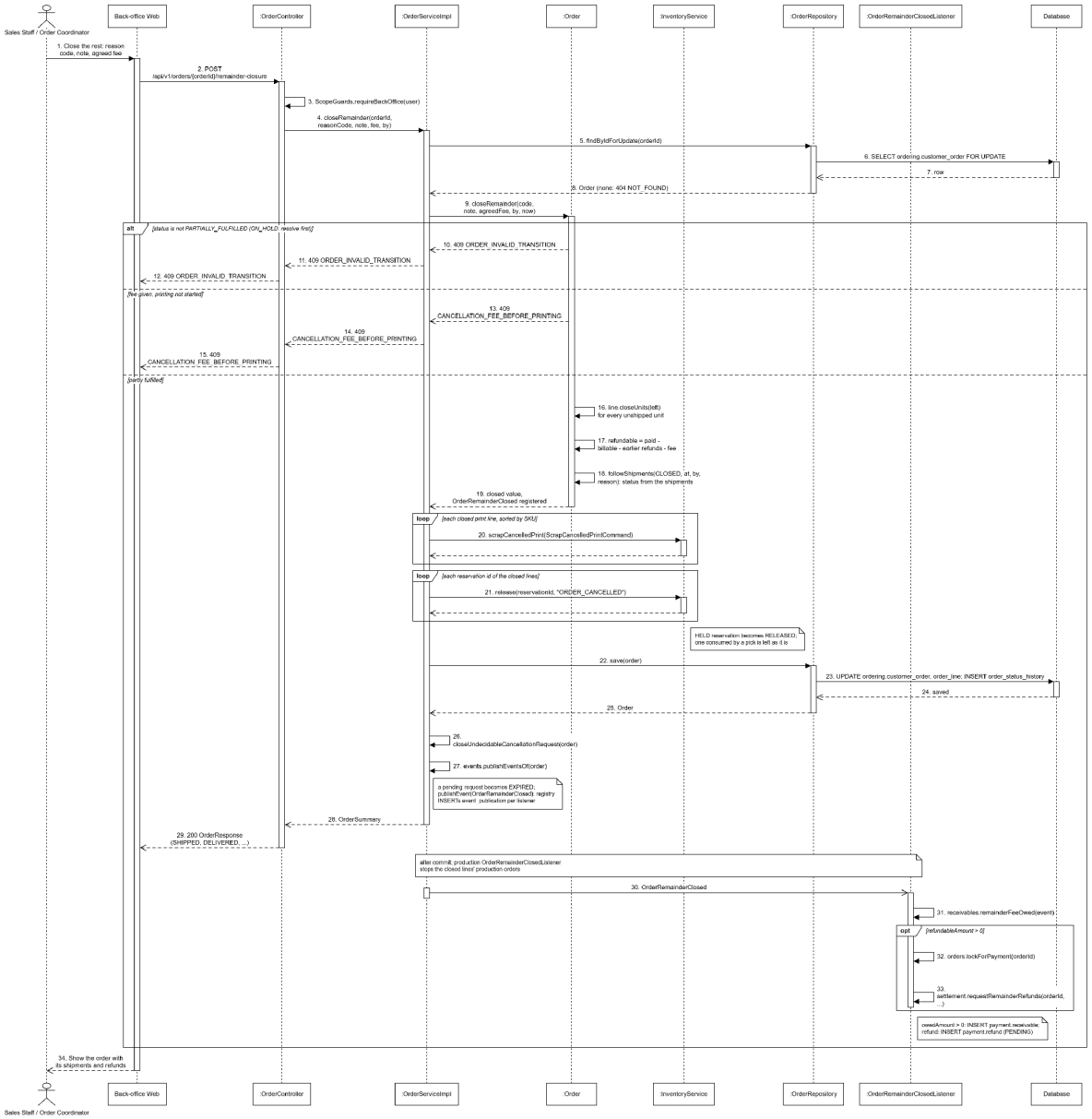


Figure 166. Close Remaining Quantity Sequence Diagram

3.10.10 Set Partial-Delivery Permission Sequence Diagram

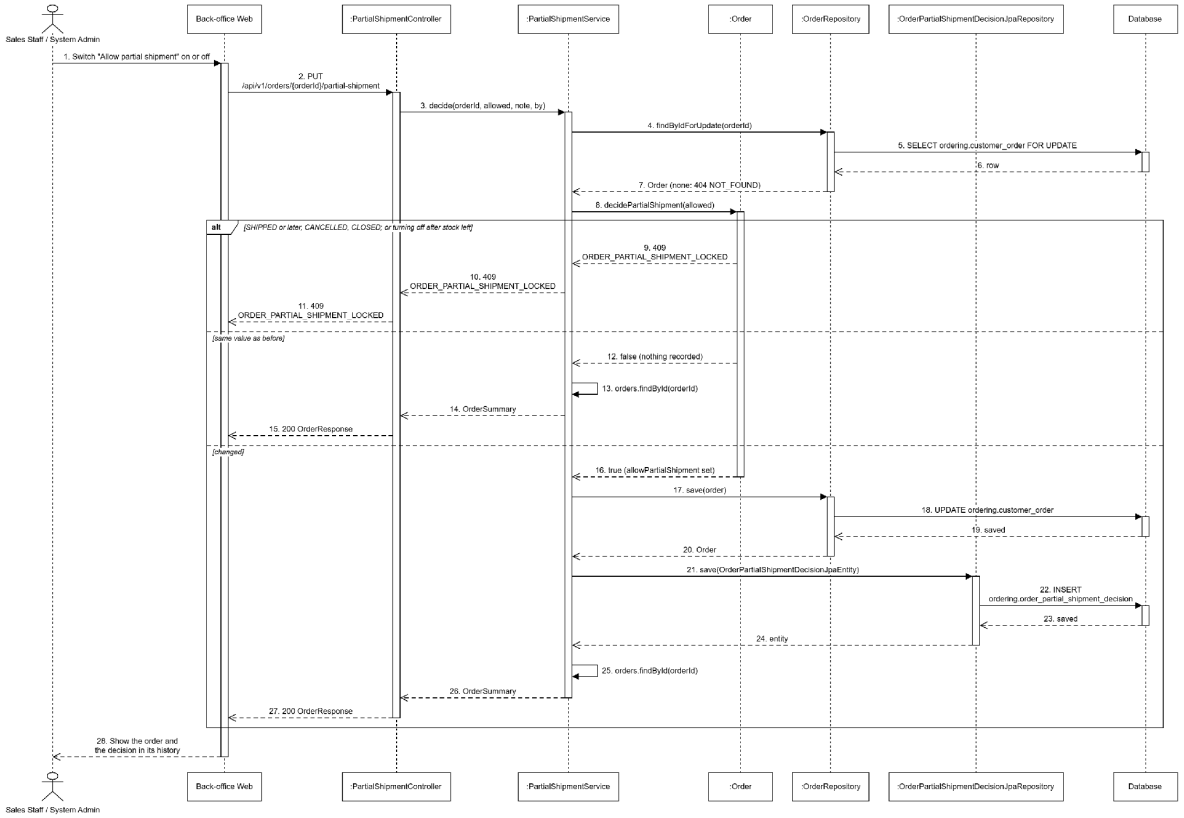


Figure 167. Set Partial-Delivery Permission Sequence Diagram

3.10.11 Select Fulfilment Warehouse & Delivery Method (Change Delivery Mode) Sequence Diagram

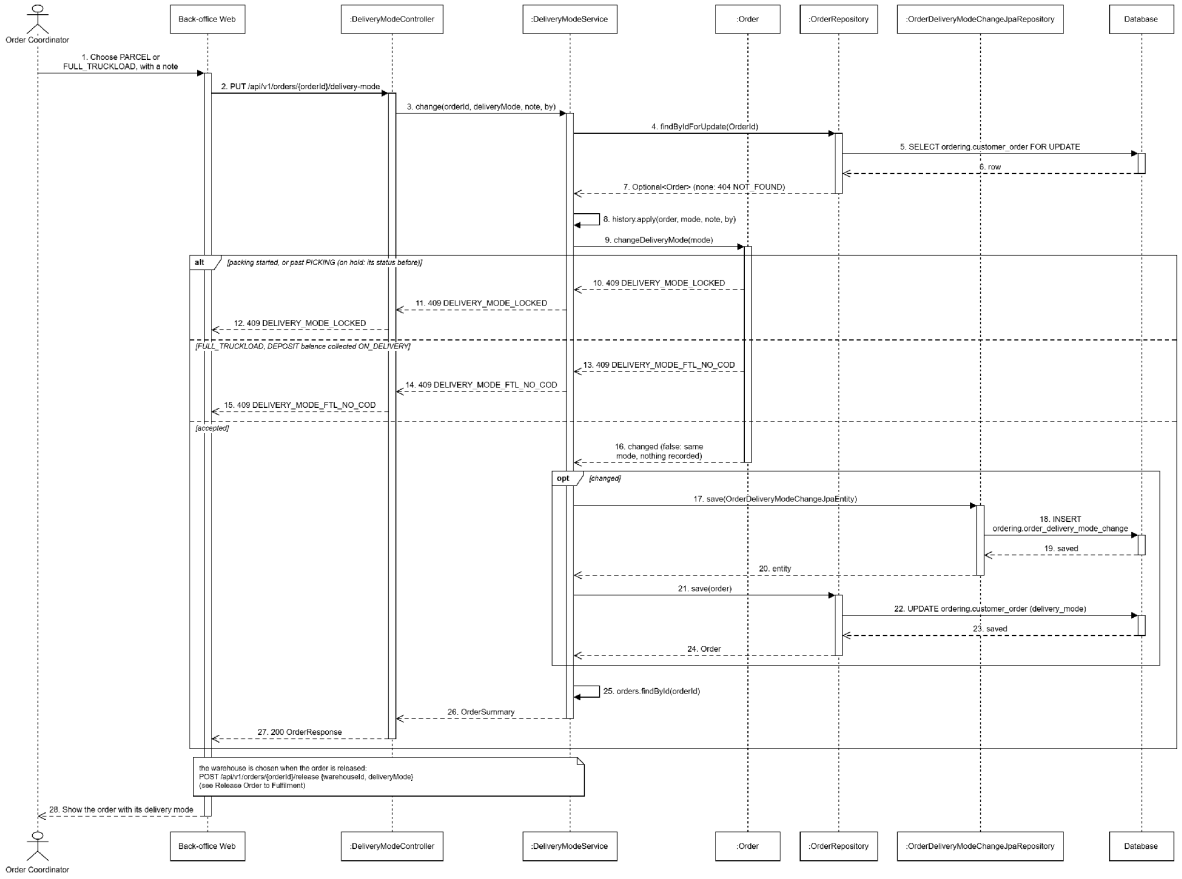


Figure 168. Select Fulfilment Warehouse & Delivery Method (Change Delivery Mode) Sequence Diagram

3.10.12 Release Order to Fulfilment Sequence Diagram

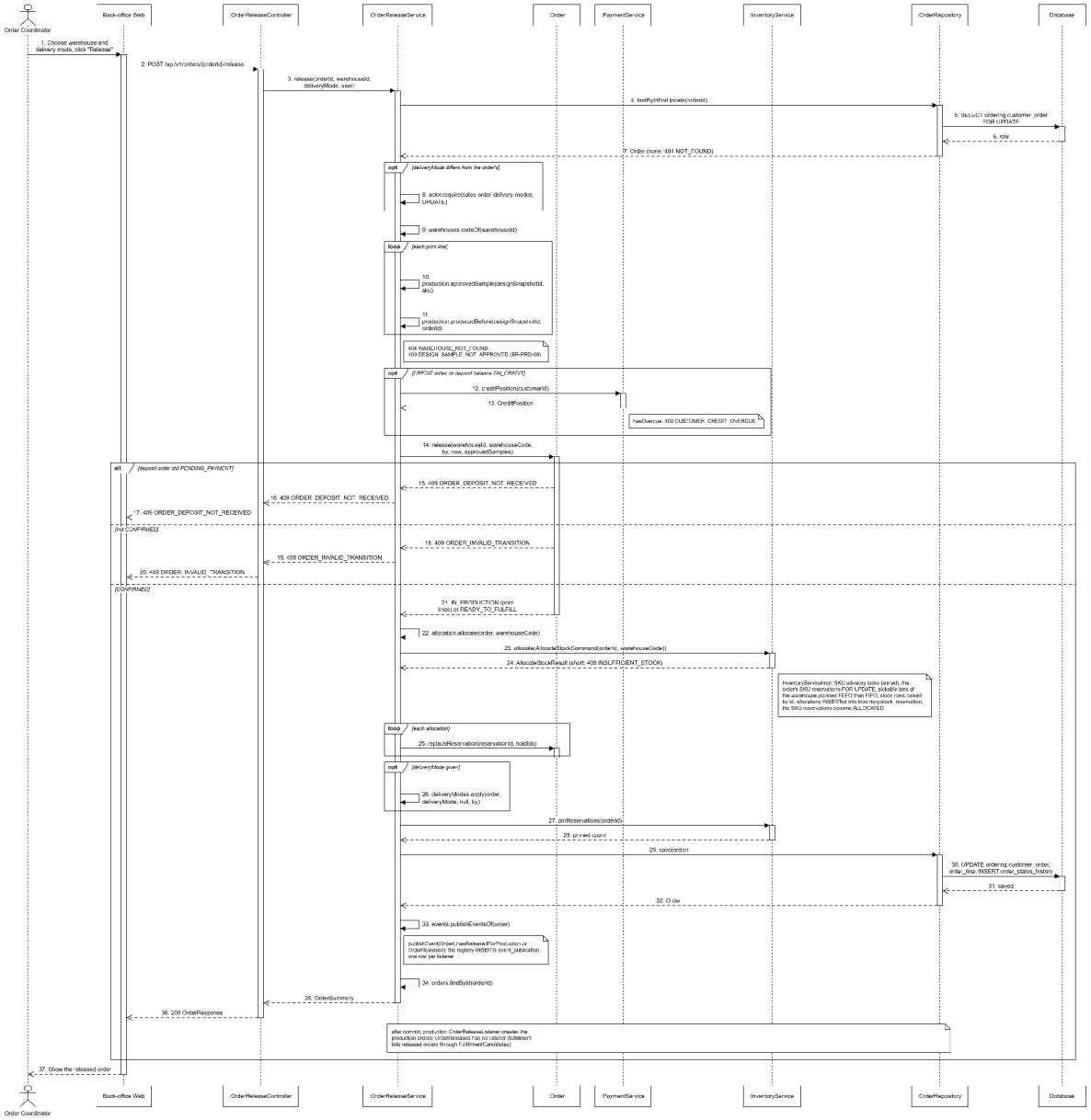


Figure 169. Release Order to Fulfilment Sequence Diagram

3.10.13 Put Order on Hold / Release Hold Sequence Diagram

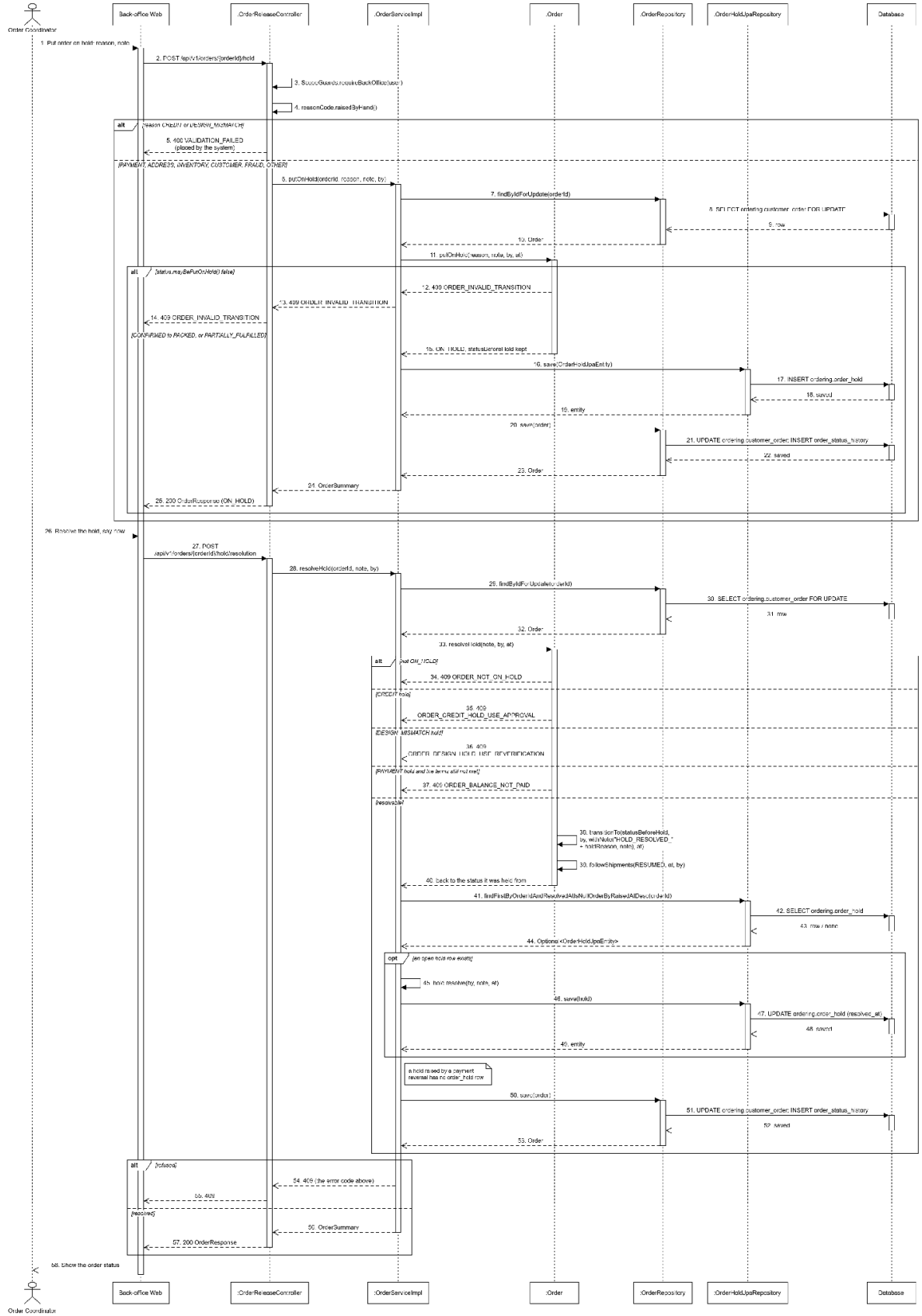


Figure 170. Put Order on Hold / Release Hold Sequence Diagram

3.10.14 Record Delivery Result Sequence Diagram

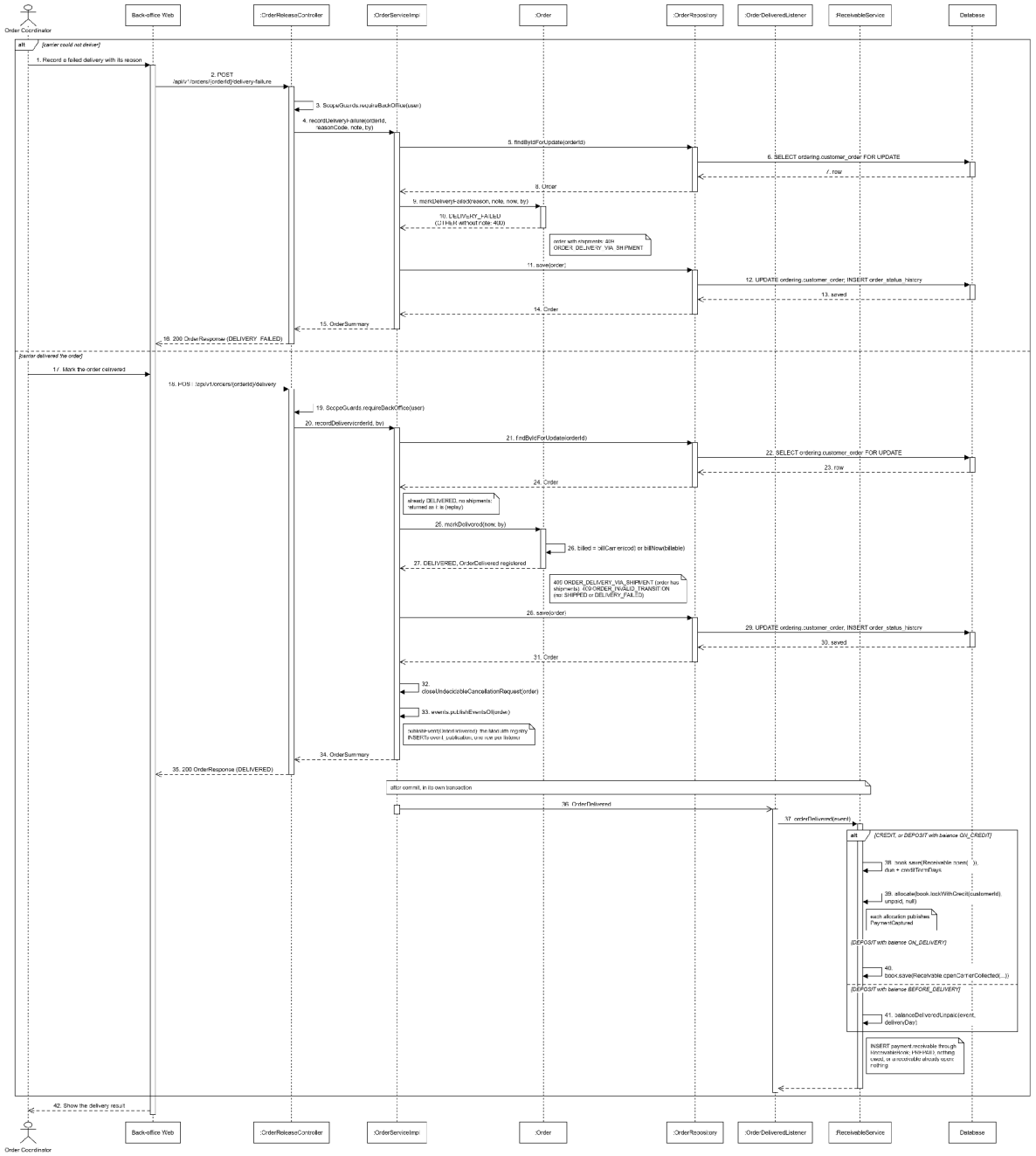


Figure 171. Record Delivery Result Sequence Diagram

3.10.15 Approve / Reject Credit-Hold Order Sequence Diagram

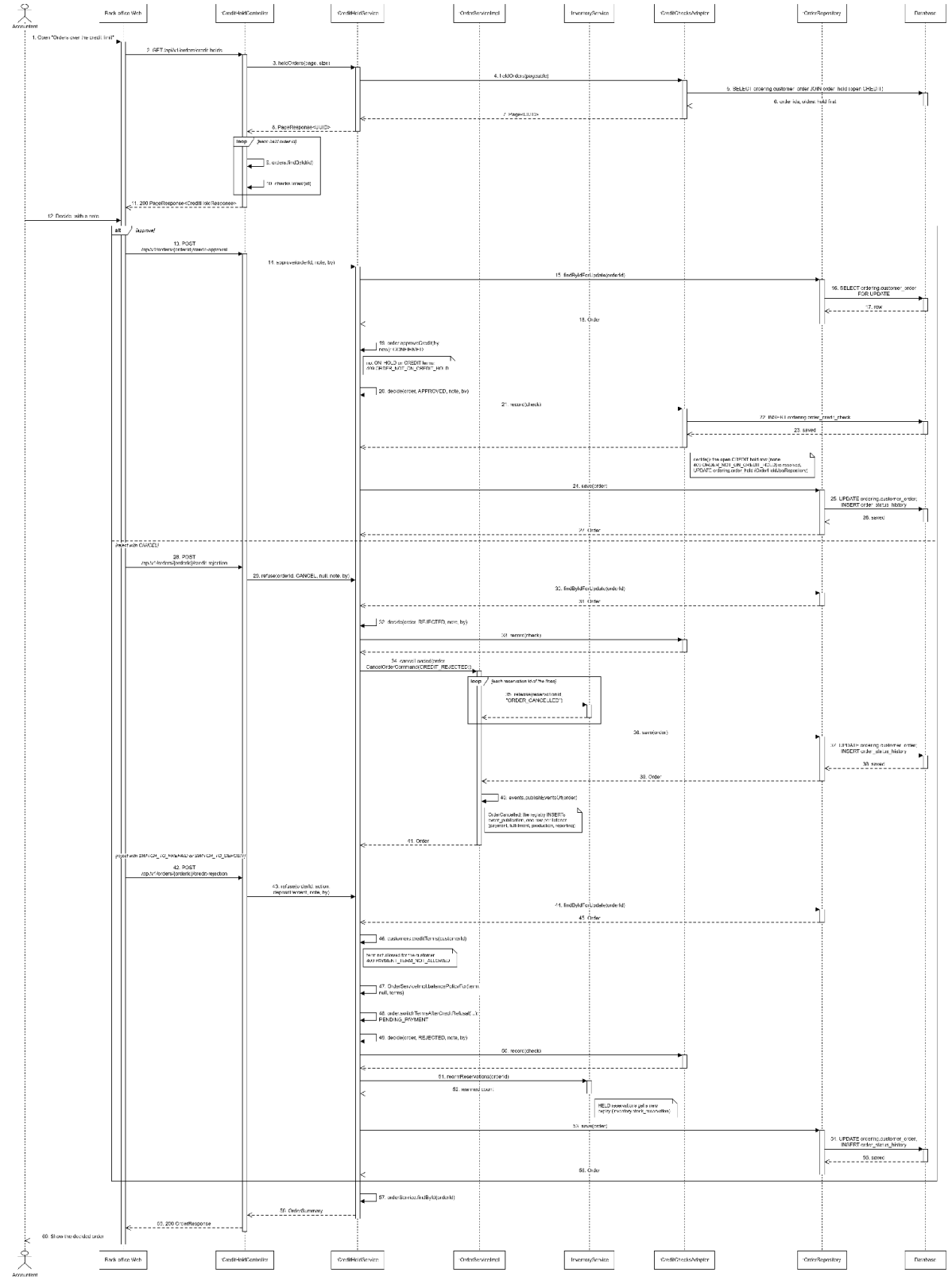


Figure 172. Approve / Reject Credit-Hold Order Sequence Diagram

3.10.16 Reservation Expiry Sweep Sequence Diagram

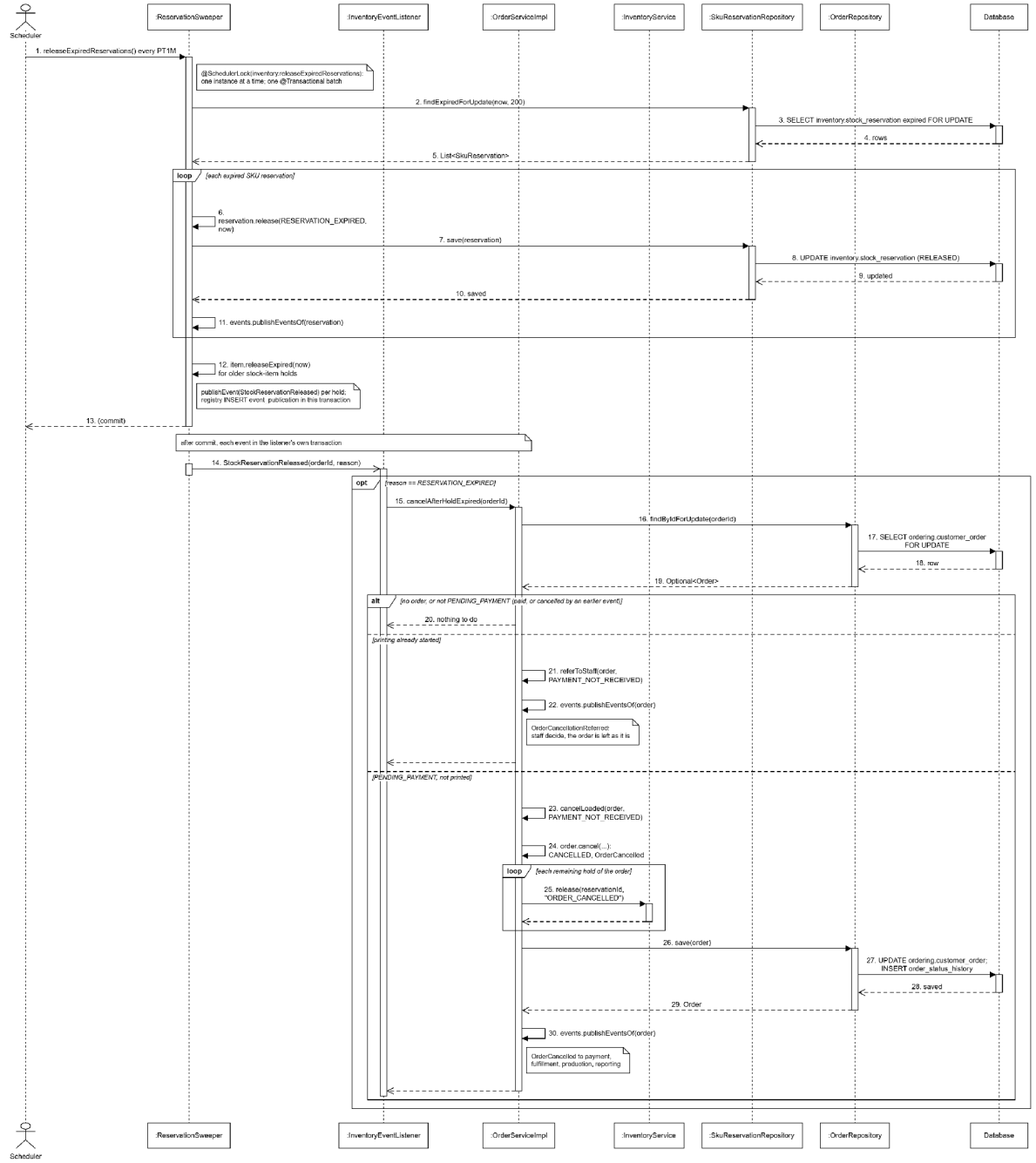


Figure 173. Reservation Expiry Sweep Sequence Diagram

3.10.17 Order Settlement Job Sequence Diagram

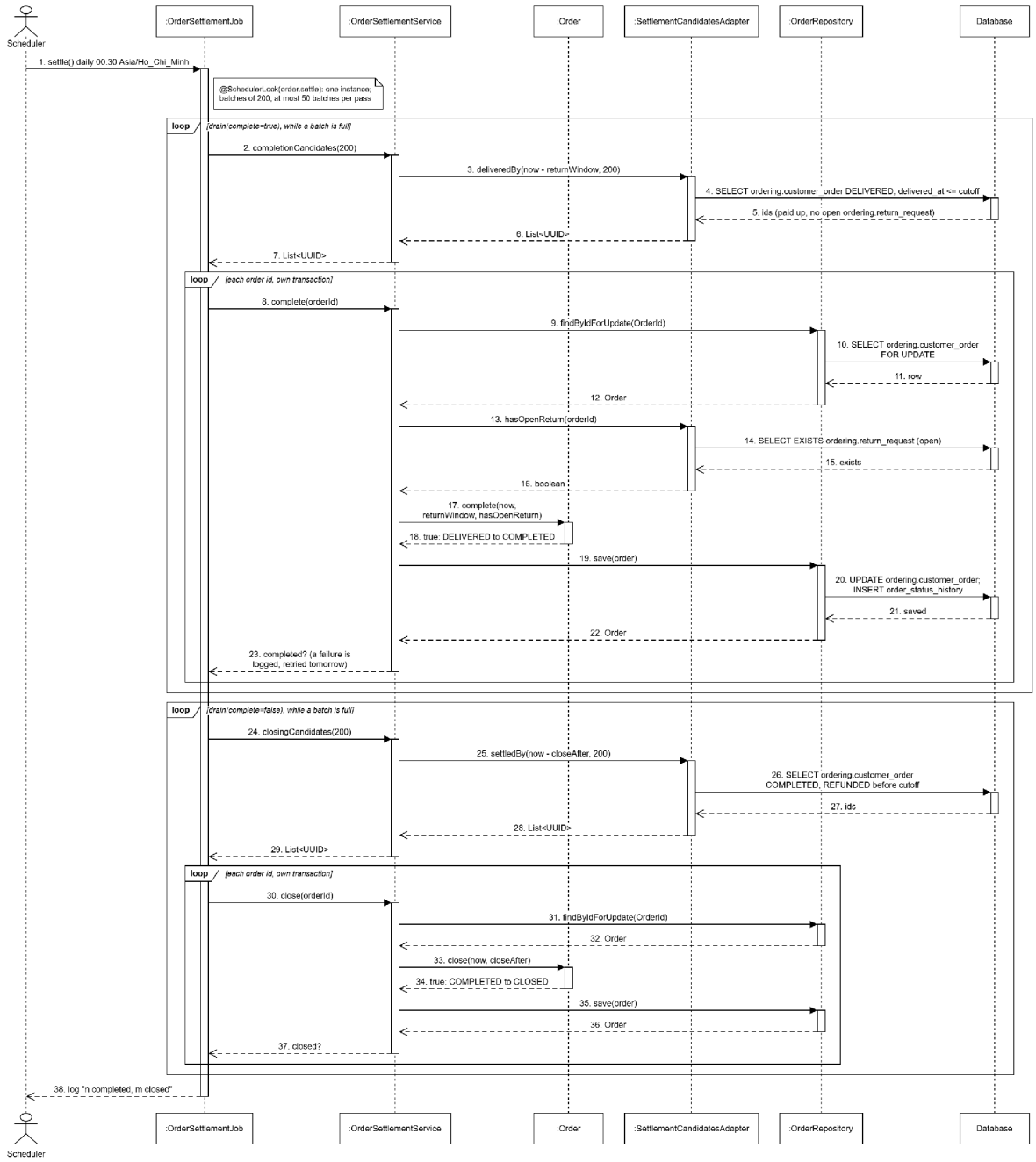


Figure 174. Order Settlement Job Sequence Diagram

3.10.18 Order State Diagram

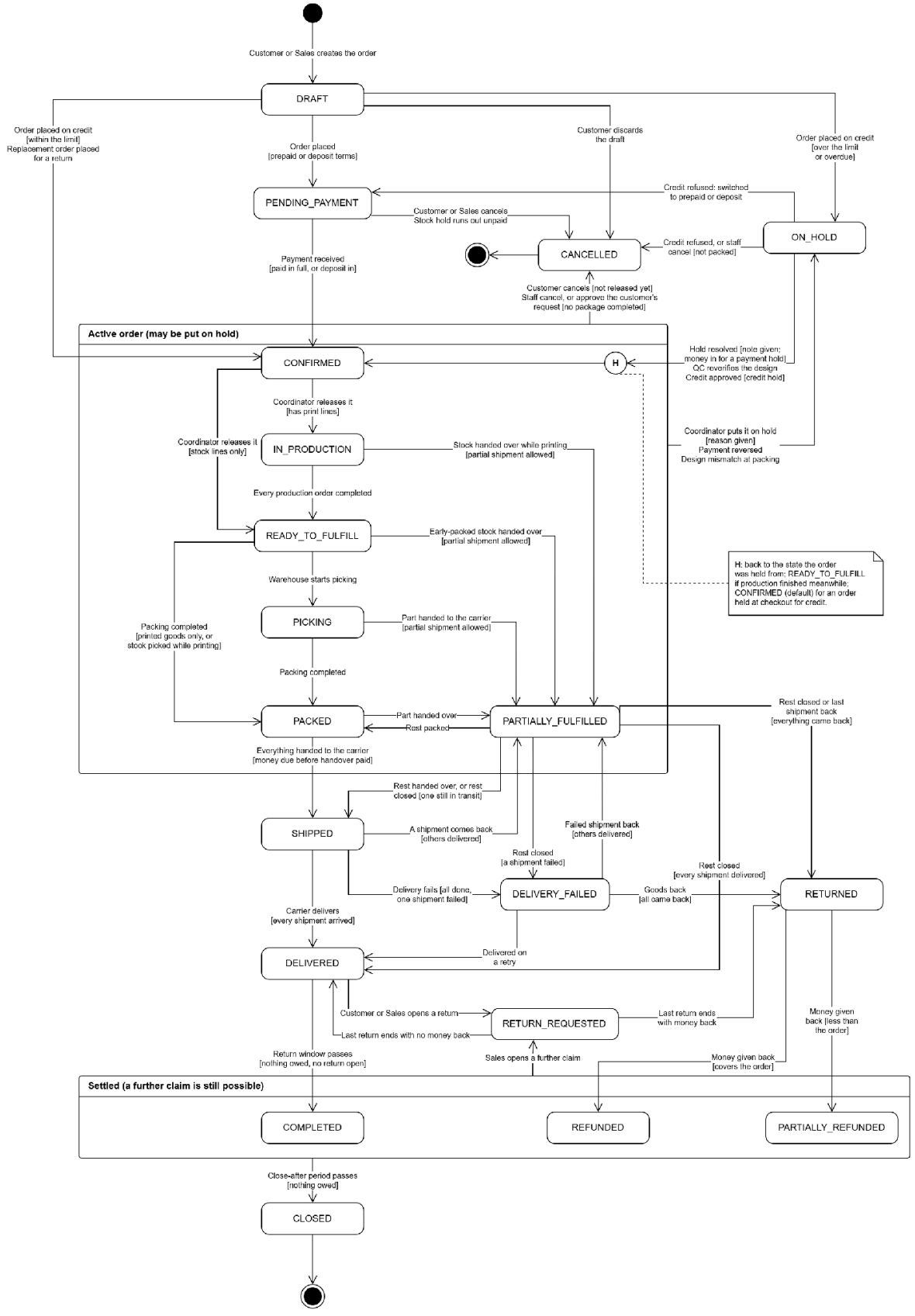


Figure 175. Order State Diagram

3.10.19 Cancellation Request State Diagram

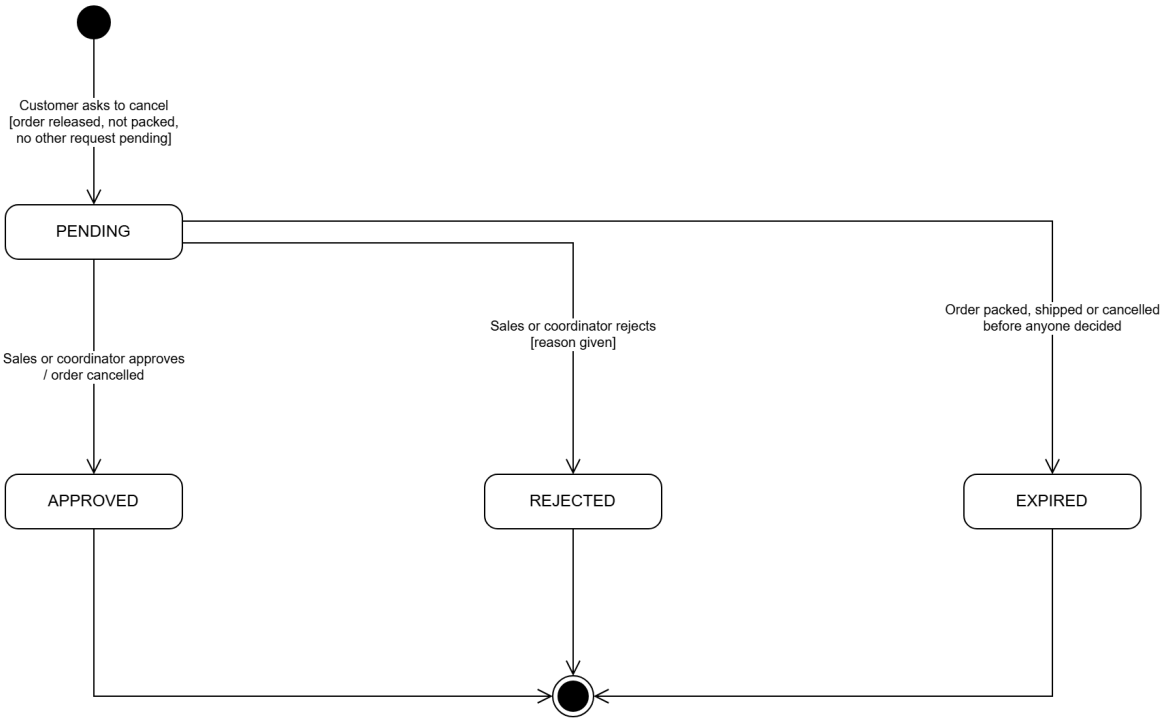


Figure 176. Cancellation Request State Diagram

3.11 Payment & Receivables

3.11.1 Class Diagram

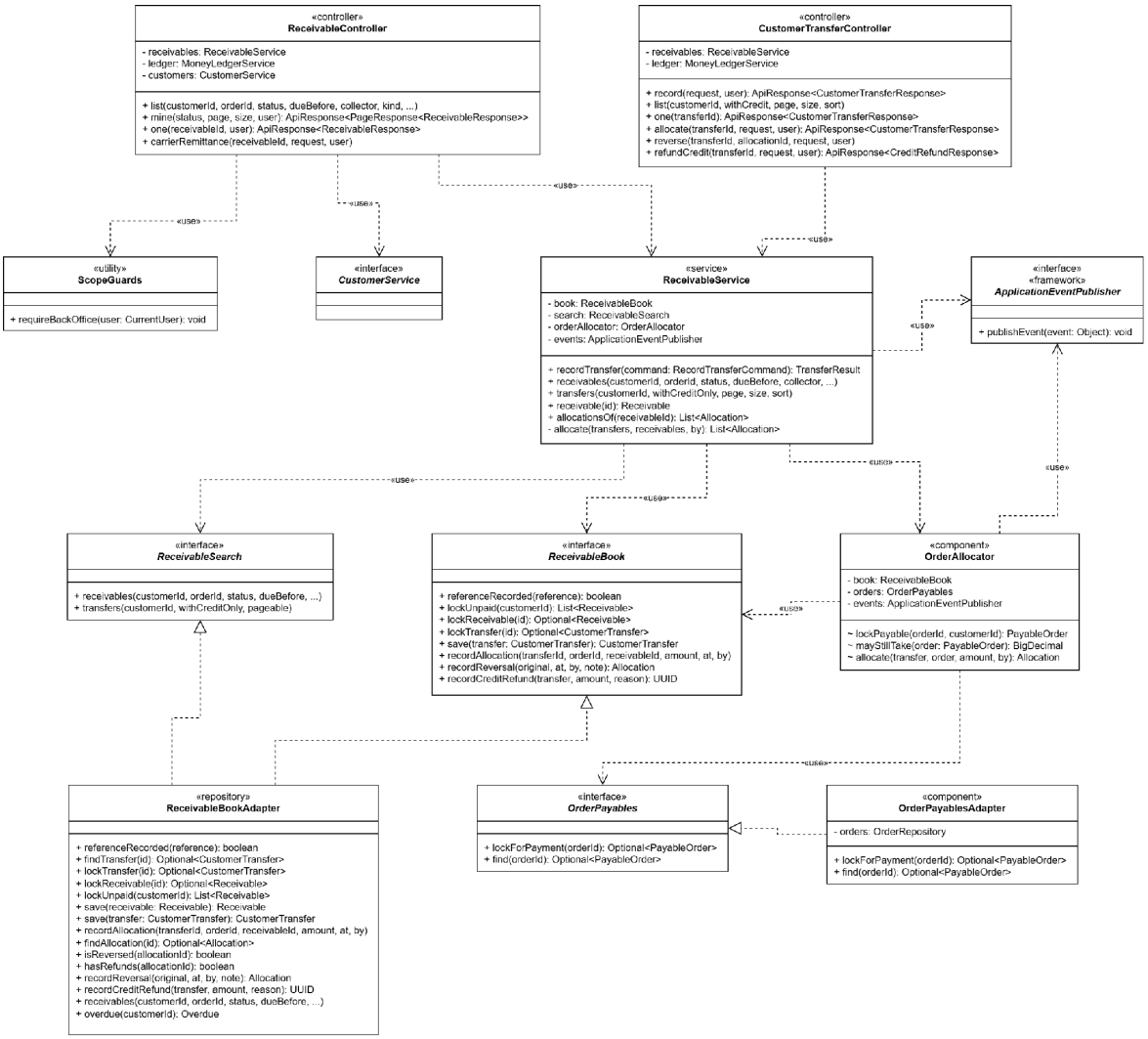


Figure 177. Payment & Receivables Class Diagram (1/4): Recording Transfers and Paying Receivables

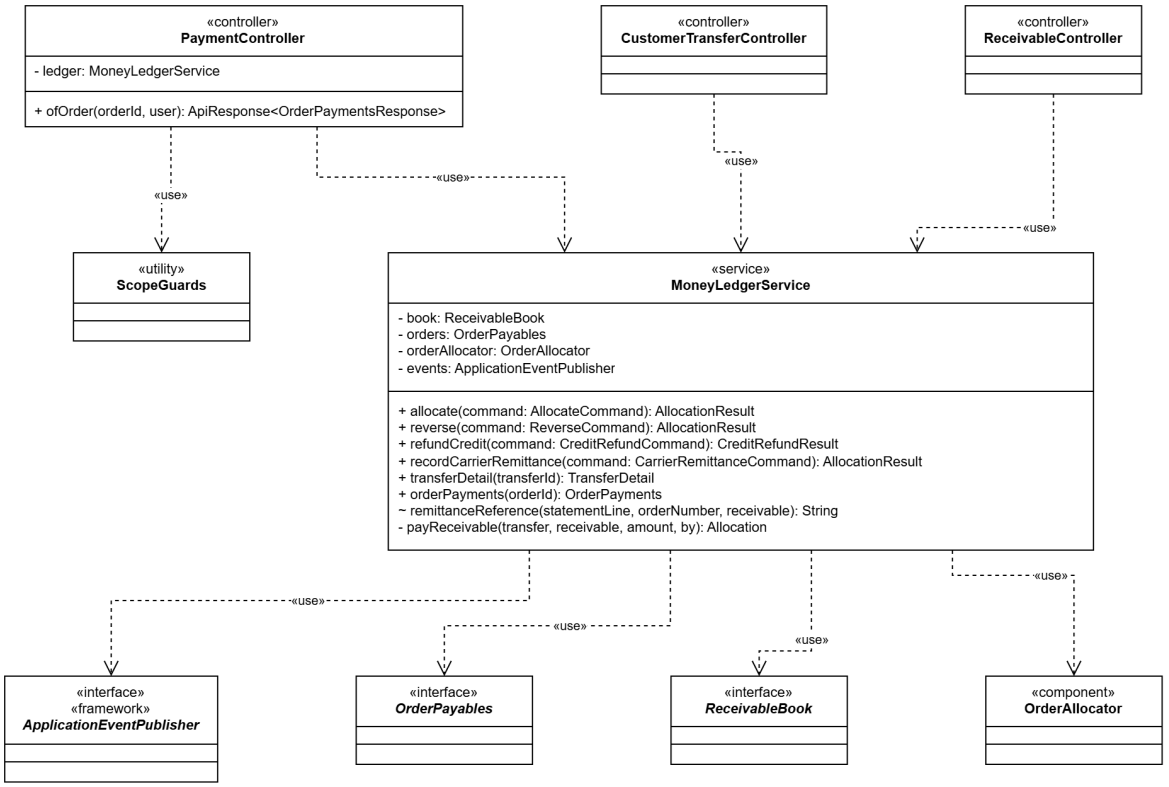


Figure 178. Payment & Receivables Class Diagram (2/4): Allocating, Reversing and Refunding Money

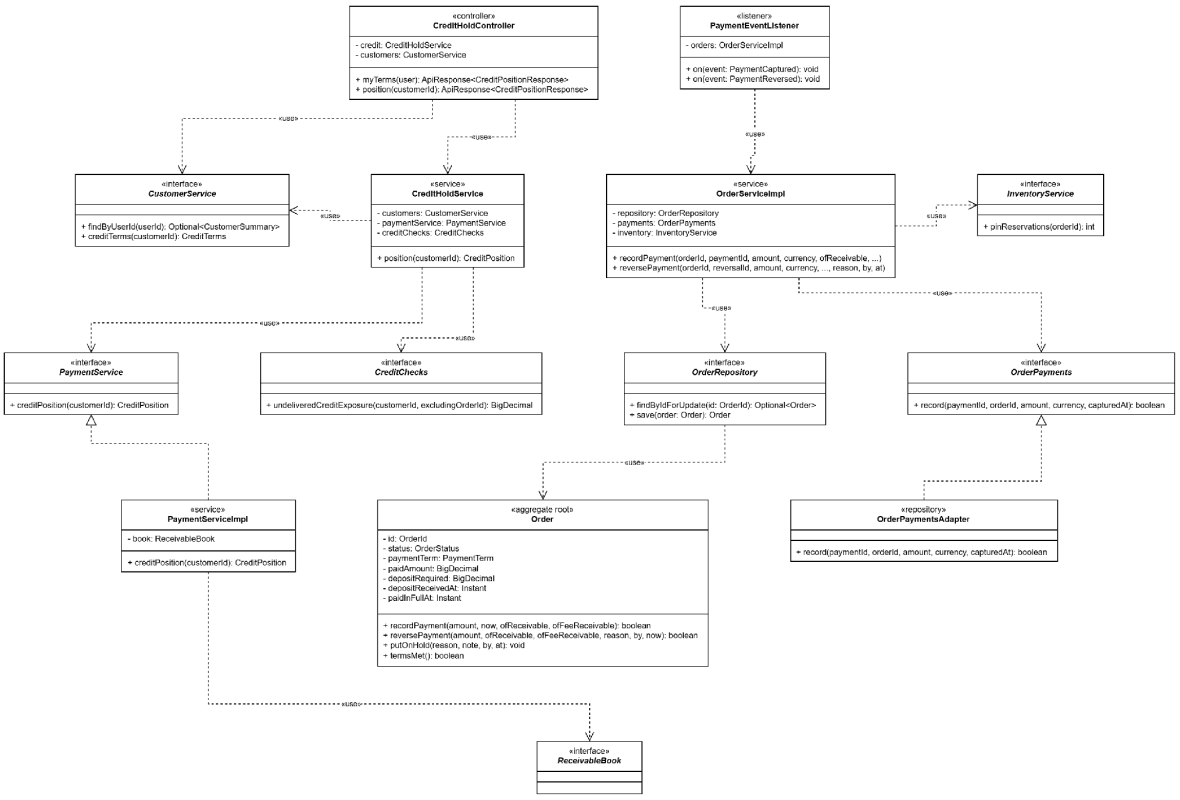


Figure 179. Payment & Receivables Class Diagram (3/4): Applying Payments to Orders and the Credit Position

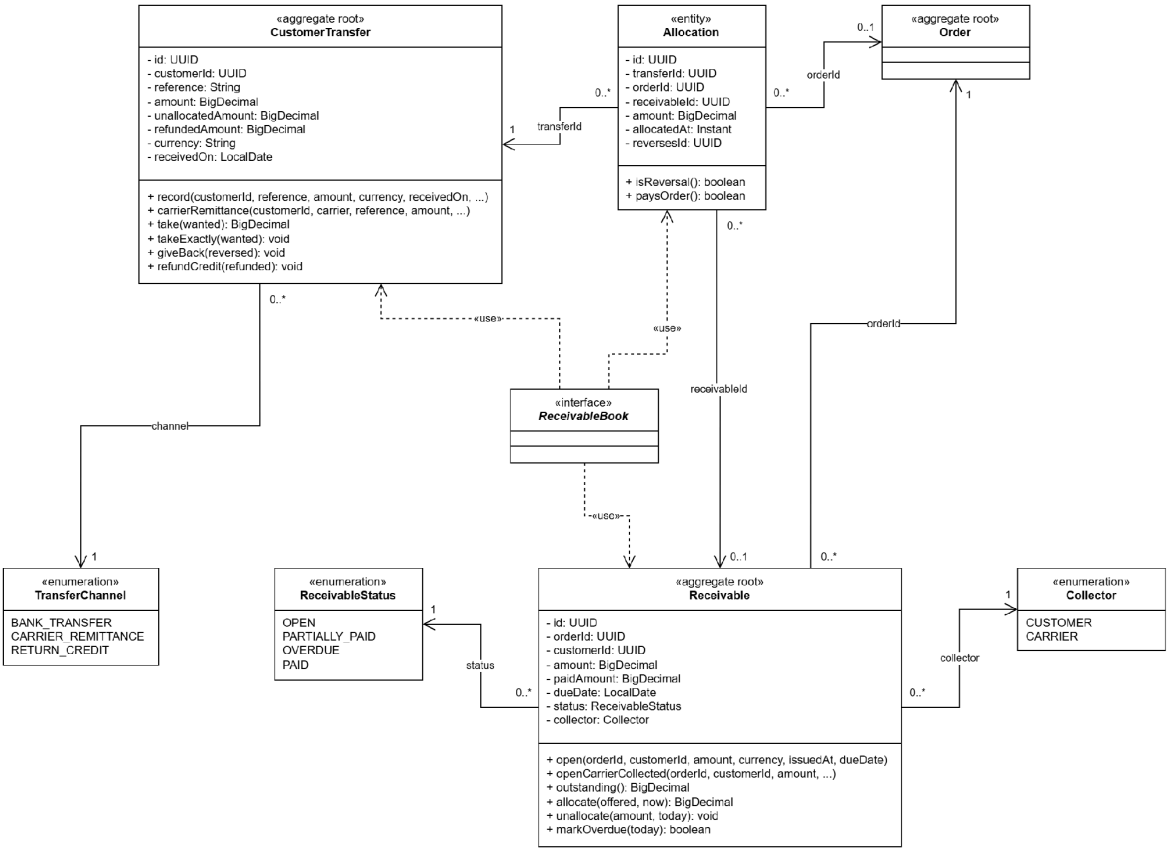


Figure 180. Payment & Receivables Class Diagram (4/4): Domain Model

3.11.2 Class Specification

CustomerTransferController

No	Method	Description
1	record(request, user)	POST /api/v1/customer-transfers: Records a bank-statement transfer (201) and allocates it to the orders named (each the amount given), then to the receivables named or, if none, the earliest due; the rest stays as the customer's credit; a statement reference already recorded is 409 TRANSFER_REFERENCE_ALREADY_RECORDED. (:63)
2	list(customerId, withCredit, page, size, sort)	GET /api/v1/customer-transfers: Recorded transfers, newest first; withCredit=true lists only transfers with unallocated money. (:75)
3	one(transferId)	GET /api/v1/customer-transfers/{transferId}: One transfer with every allocation and reversal of its money; 404 TRANSFER_NOT_FOUND. (:89)

4	allocate(transferId, request, user)	POST /api/v1/customer-transfers/{transferId}/allocations: Pays part of a transfer's unallocated money to one order (PREPAID/DEPOSIT, up to what it may still take) or one receivable of the same customer (201); 409 ALLOCATION_EXCEEDS_AMOUNT_DUE / ALLOCATION_EXCEEDS_TRANSFER / ALLOCATION_ORDER_NOT_PAYABLE. (:98)
5	reverse(transferId, allocationId, request, user)	POST /api/v1/customer-transfers/{transferId}/allocations/{allocationId}/reversal: Takes an allocation back with a reason (a reversal row), the money becomes credit again; refused when already reversed, refunded or the order is settled, and needs holdOrder=true when the order would fall short of its terms (409 PAYMENT_REVERSAL_NEEDS_HOLD). (:111)
6	refundCredit(transferId, request, user)	POST /api/v1/customer-transfers/{transferId}/credit-refunds: Creates a PENDING refund of part of a transfer's unallocated money with a reason (201); more than the unallocated amount is 409 ALLOCATION_EXCEEDS_TRANSFER. (:123)

Table 71. CustomerTransferController Class Specification

ReceivableController

No	Method	Description
1	list(customerId, orderId, status, dueBefore, collector, kind, page, size, sort, user)	GET /api/v1/receivables: Back-office list of receivables, earliest due first, filtered by customer, order, status, due date, collector and kind. (:70)
2	mine(status, page, size, user)	GET /api/v1/receivables/me: What the signed-in customer owes (collector CUSTOMER only: balances the carrier collected are not the customer's to pay). (:91)
3	one(receivableId, user)	GET /api/v1/receivables/{receivableId}: One receivable with the allocations that paid it; 404 RECEIVABLE_NOT_FOUND. (:107)
4	carrierRemittance(receivableId, request, user)	POST /api/v1/receivables/{receivableId}/carrier-remittances: Records a carrier's remittance for a balance it collected on delivery as a CARRIER_REMITTANCE transfer and pays the receivable from it (201); 409 RECEIVABLE_NOT_CARRIER_COLLECTED / RECEIVABLE_SETTLED / TRANSFER_REFERENCE_ALREADY_RECORDED. (:123)

Table 72. ReceivableController Class Specification

PaymentController

No	Method	Description
1	ofOrder(orderId, user)	GET /api/v1/payments/orders/{orderId}: An order's payment history for staff: allocations and reversals, collected and outstanding amounts, receivables, refunds, and what it may still take by transfer (zero for CREDIT or CANCELLED). (:44)

Table 73. PaymentController Class Specification

3.11.3 Record & Allocate Incoming Payment (Record Transfer) Sequence Diagram

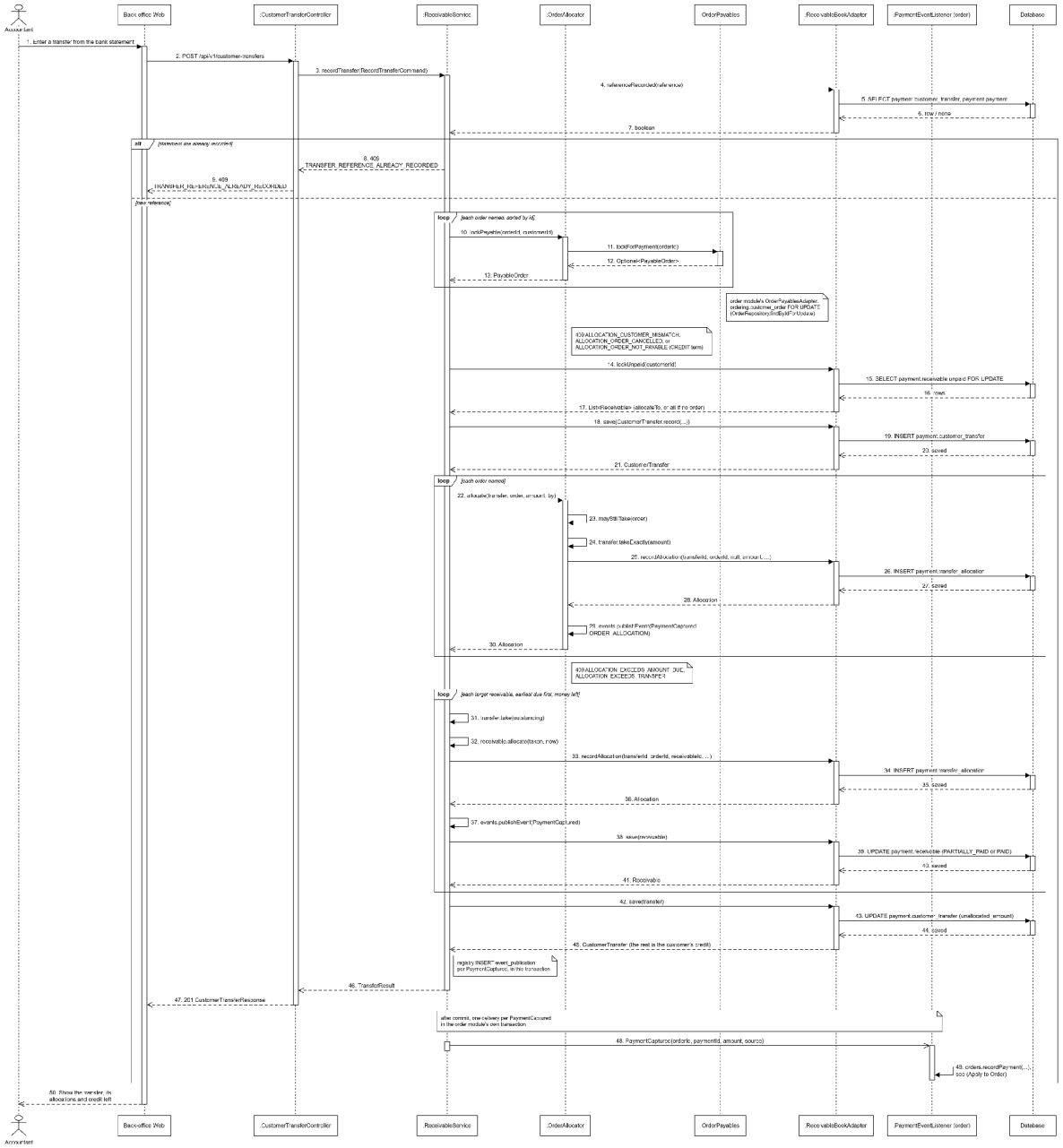


Figure 181. Record & Allocate Incoming Payment (Record Transfer) Sequence Diagram

3.11.4 Record & Allocate Incoming Payment (Allocate Transfer) Sequence Diagram

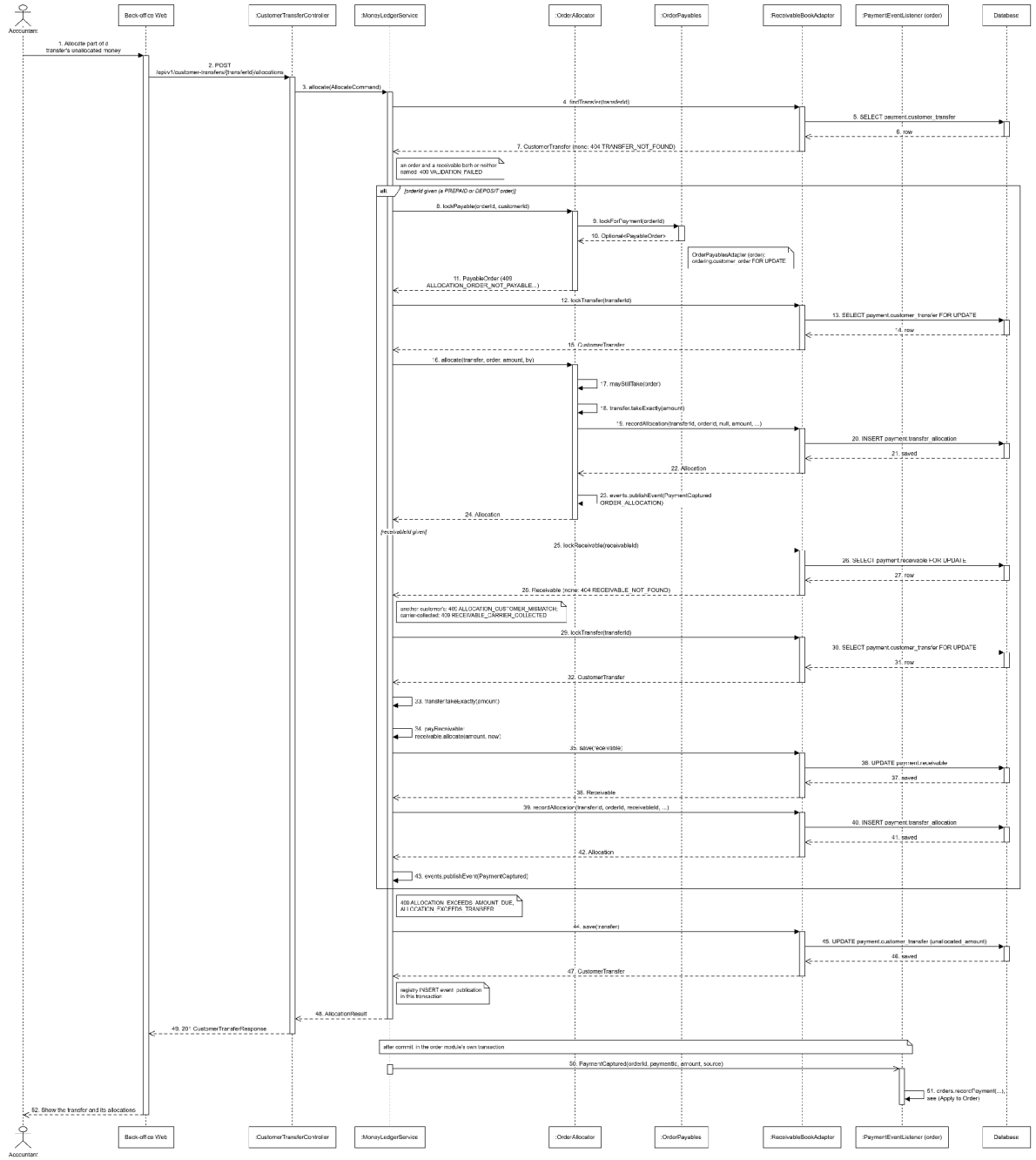


Figure 182. Record & Allocate Incoming Payment (Allocate Transfer) Sequence Diagram

3.11.5 Record & Allocate Incoming Payment (Apply to Order) Sequence Diagram

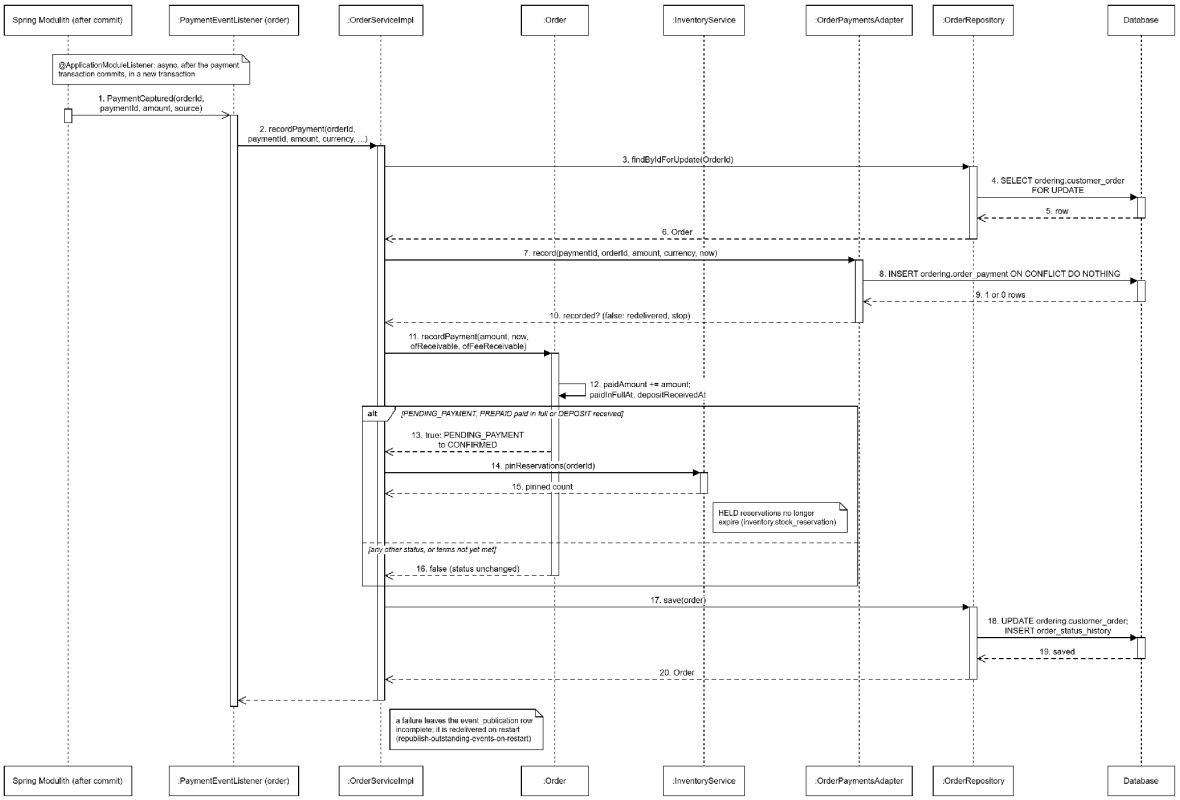


Figure 183. Record & Allocate Incoming Payment (Apply to Order) Sequence Diagram

3.11.6 Reverse Payment Allocation (Apply to Order) Sequence Diagram

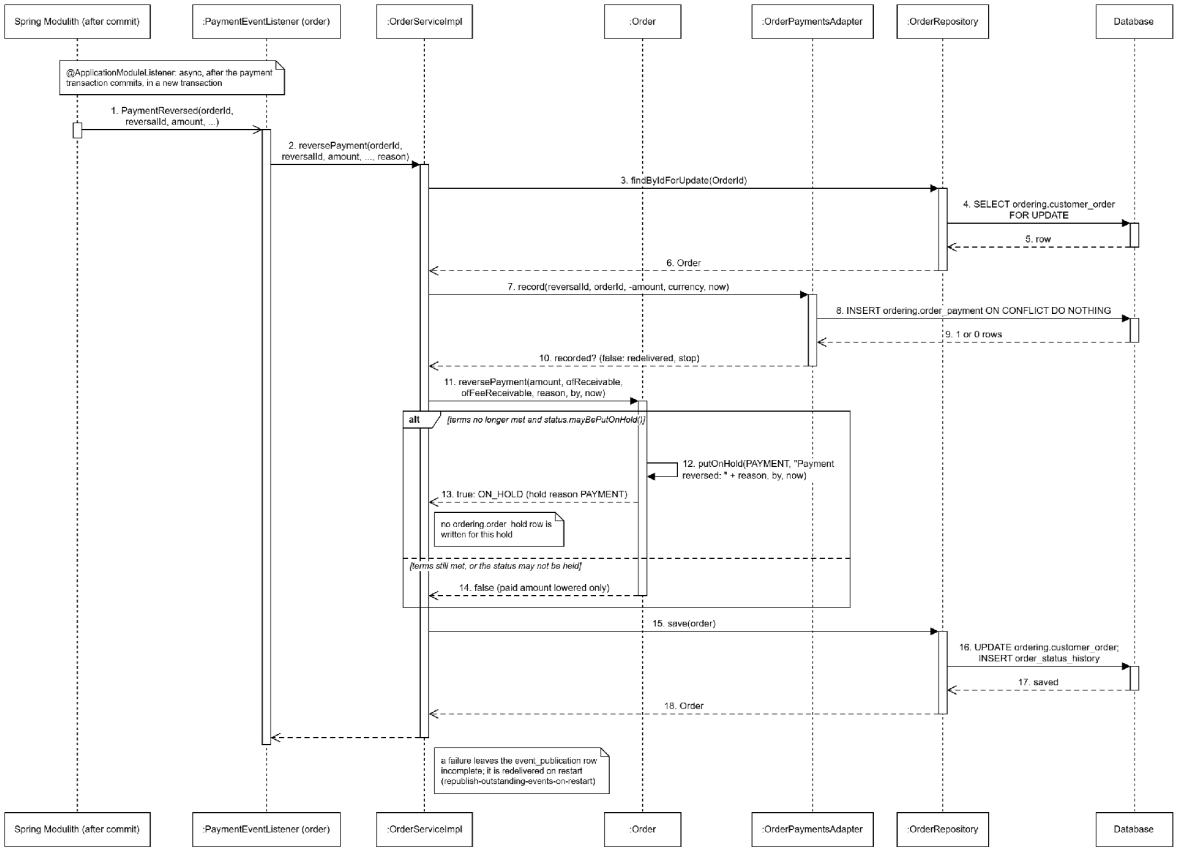


Figure 184. Reverse Payment Allocation (Apply to Order) Sequence Diagram

3.11.7 Reverse Payment Allocation Sequence Diagram

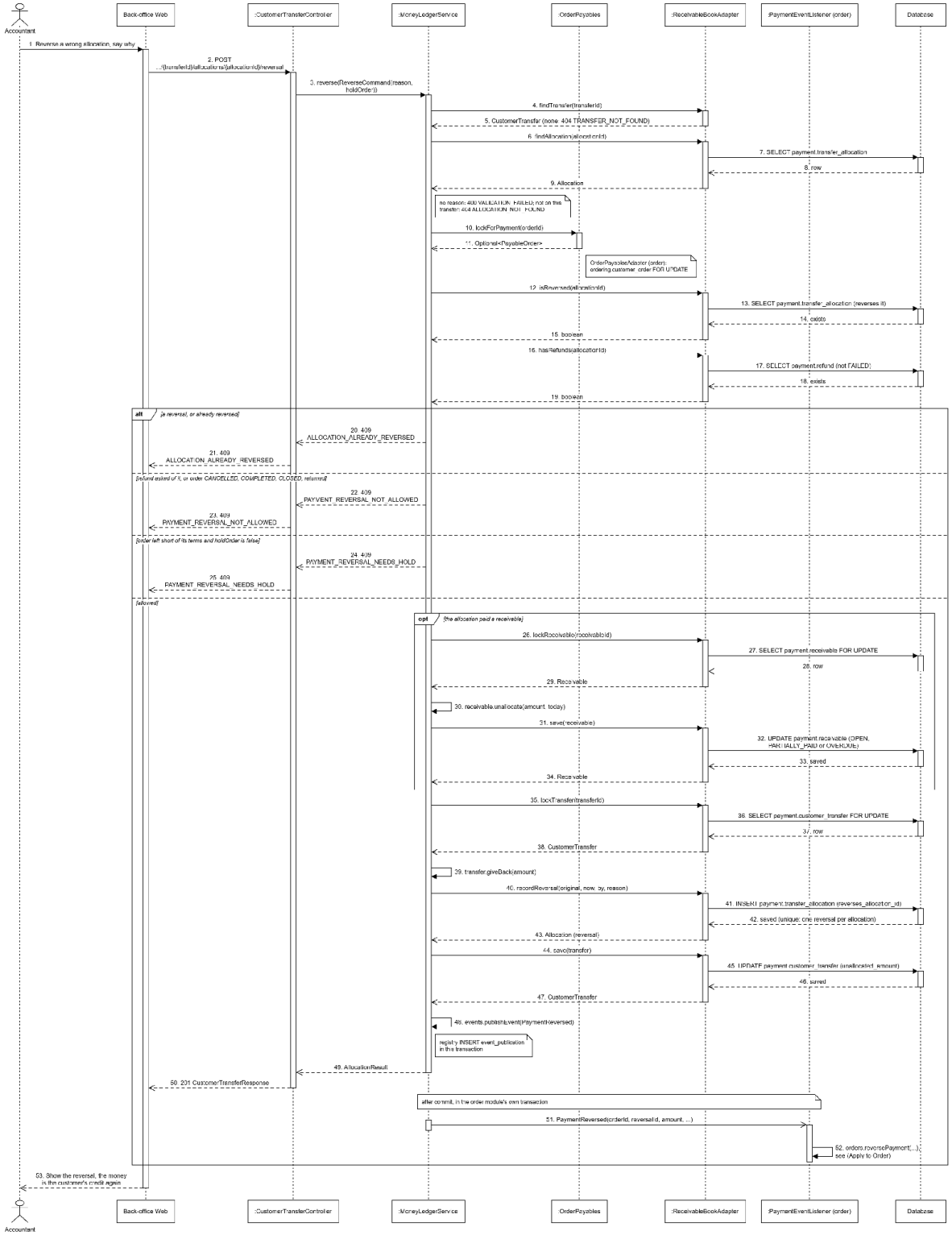


Figure 185. Reverse Payment Allocation Sequence Diagram

3.11.8 Record Carrier Remittance Sequence Diagram

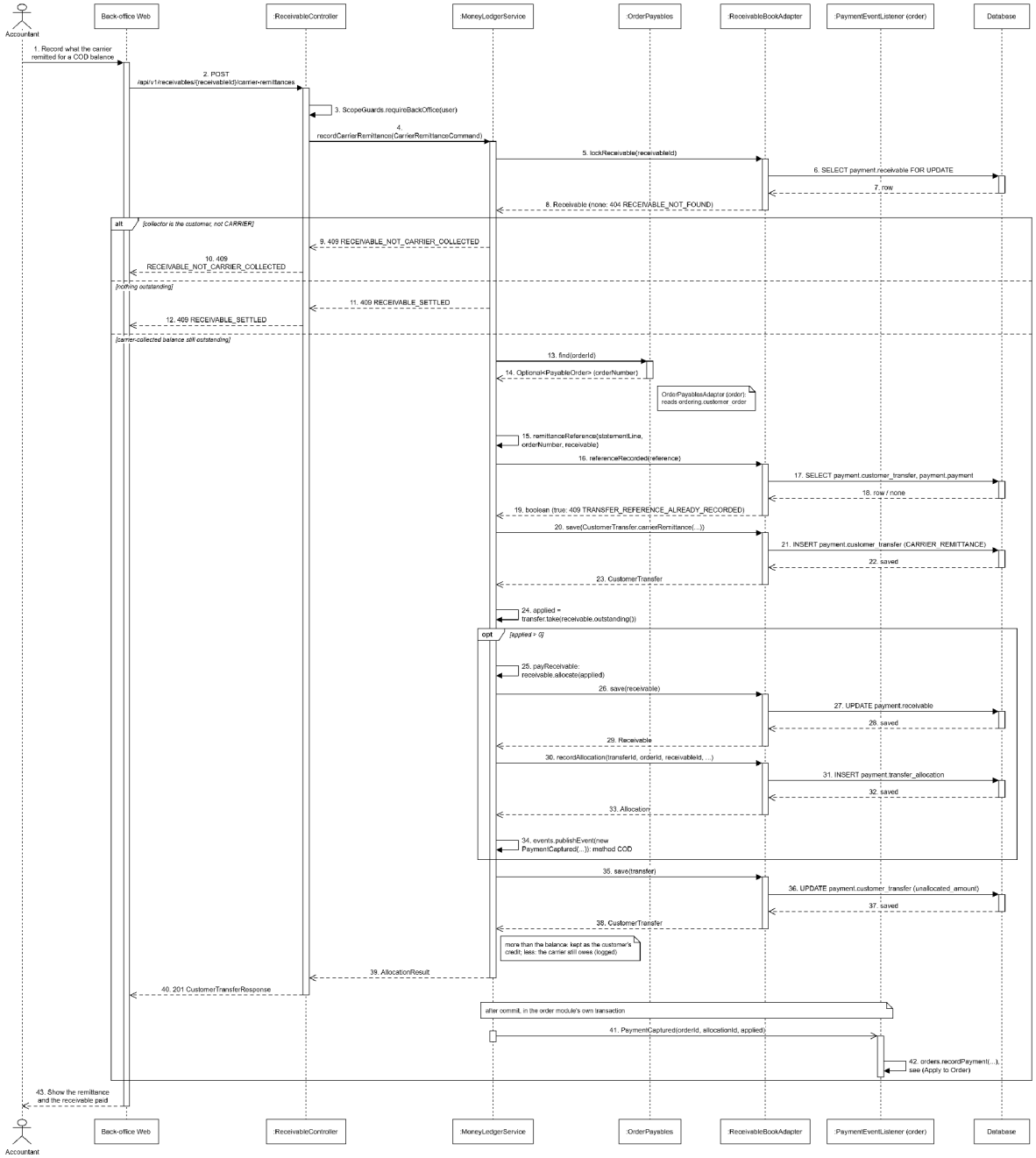


Figure 186. Record Carrier Remittance Sequence Diagram

3.11.9 Create Refund Sequence Diagram

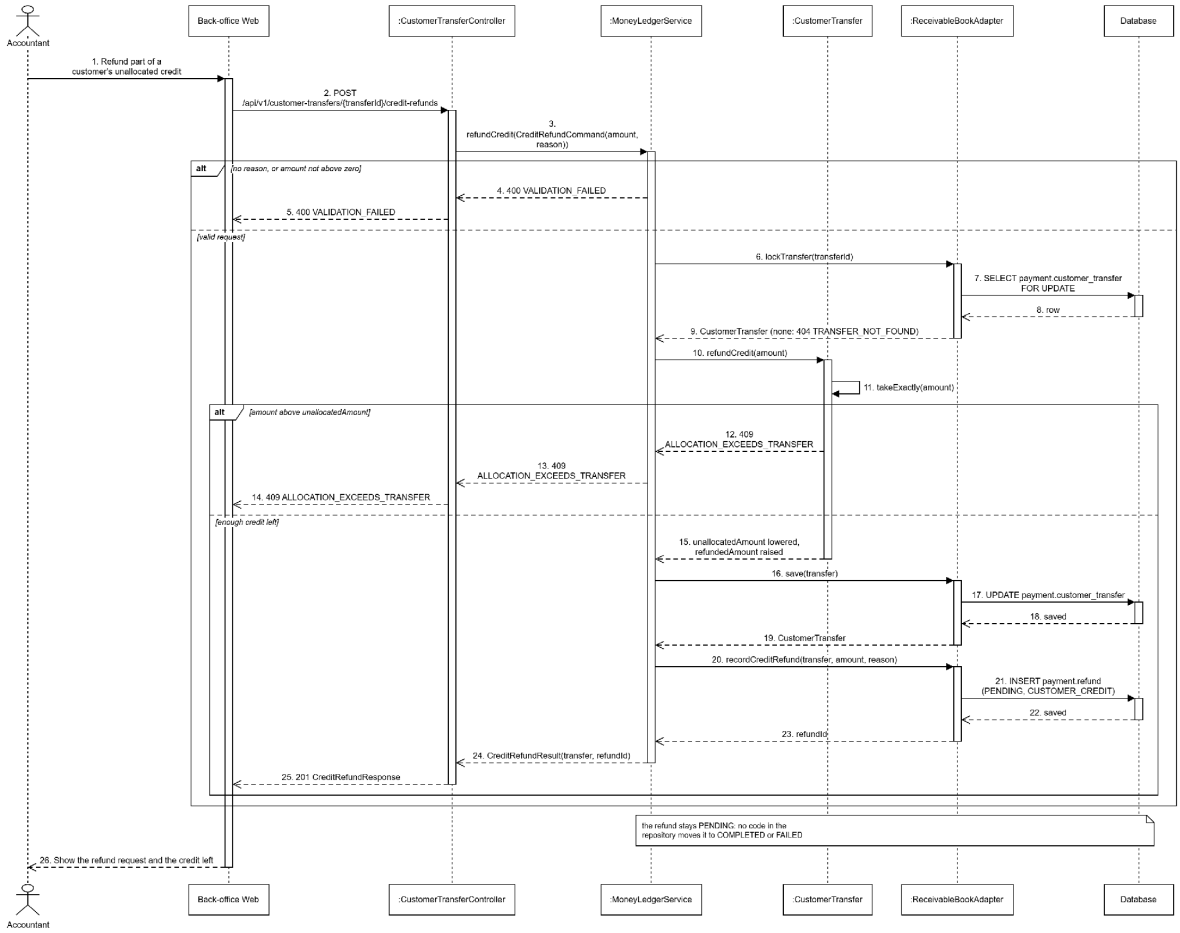


Figure 187. Create Refund Sequence Diagram

3.11.10 View Receivables & Credit Position Sequence Diagram

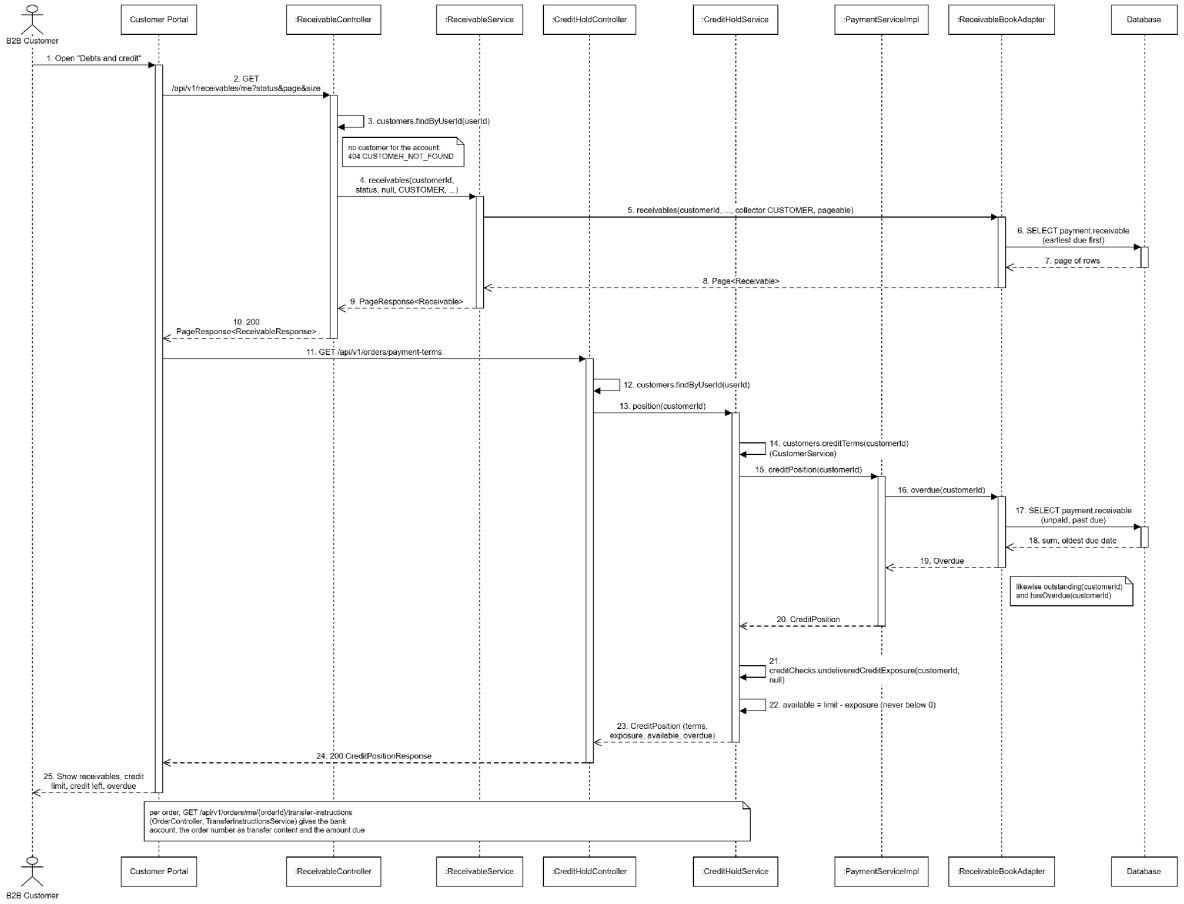


Figure 188. View Receivables & Credit Position Sequence Diagram

3.11.11 Receivable State Diagram

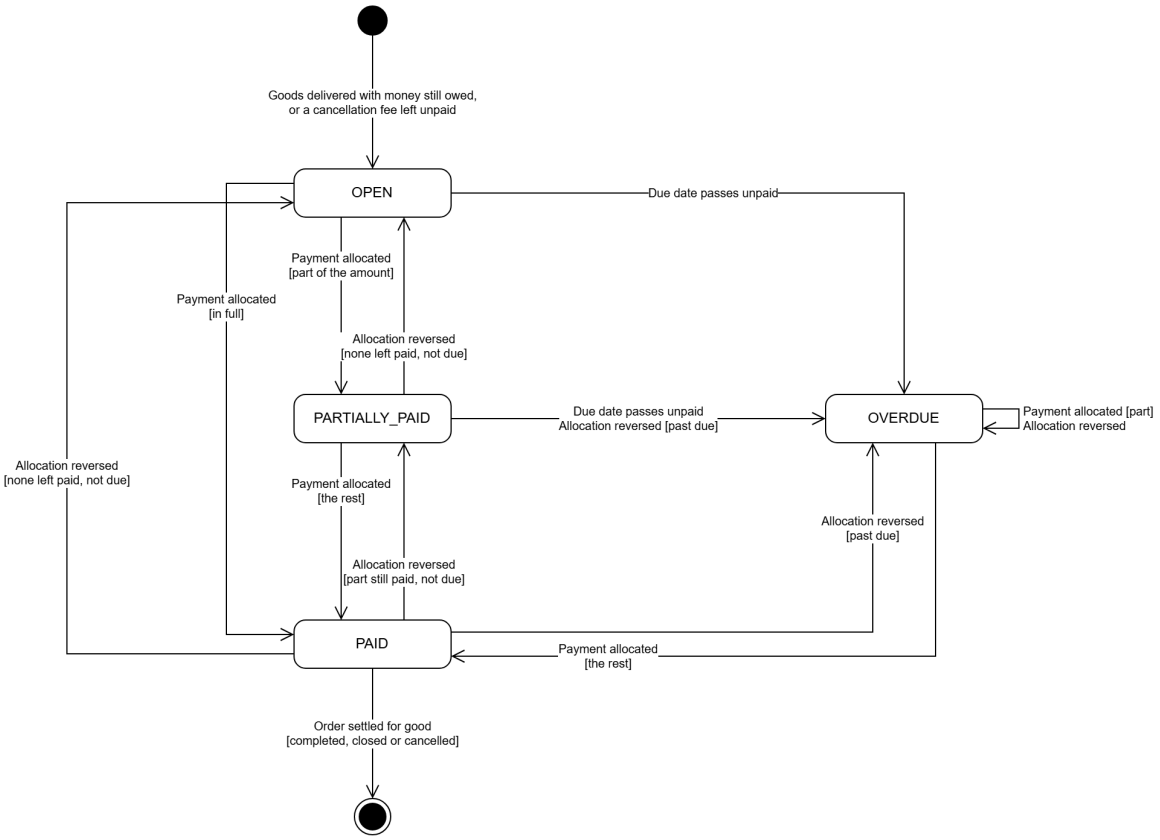


Figure 189. Receivable State Diagram

3.12 Production Management

3.12.1 Class Diagram

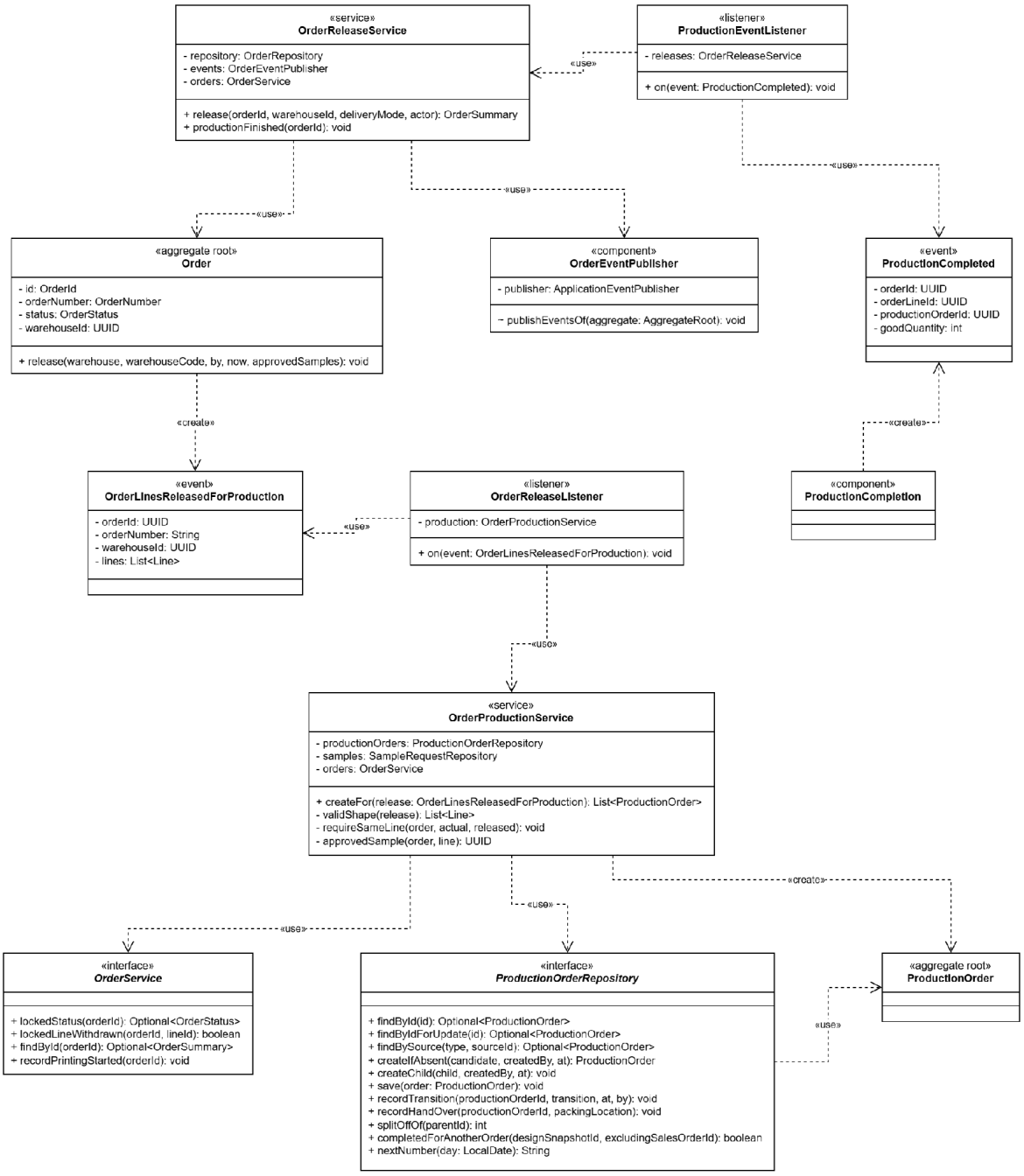


Figure 190. Production Management Class Diagram (1/7): Production Orders from a Release

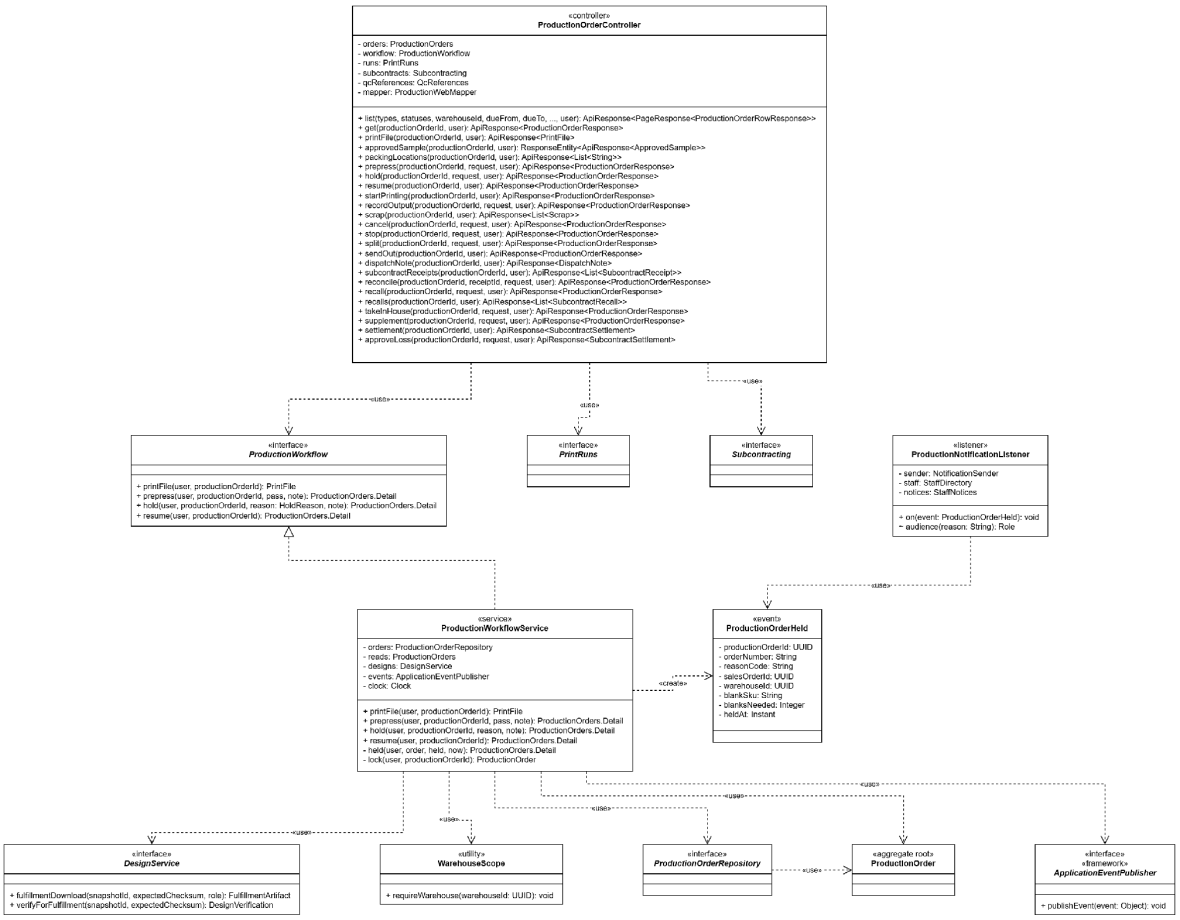


Figure 191. Production Management Class Diagram (2/7): Prepress, Hold and Resume

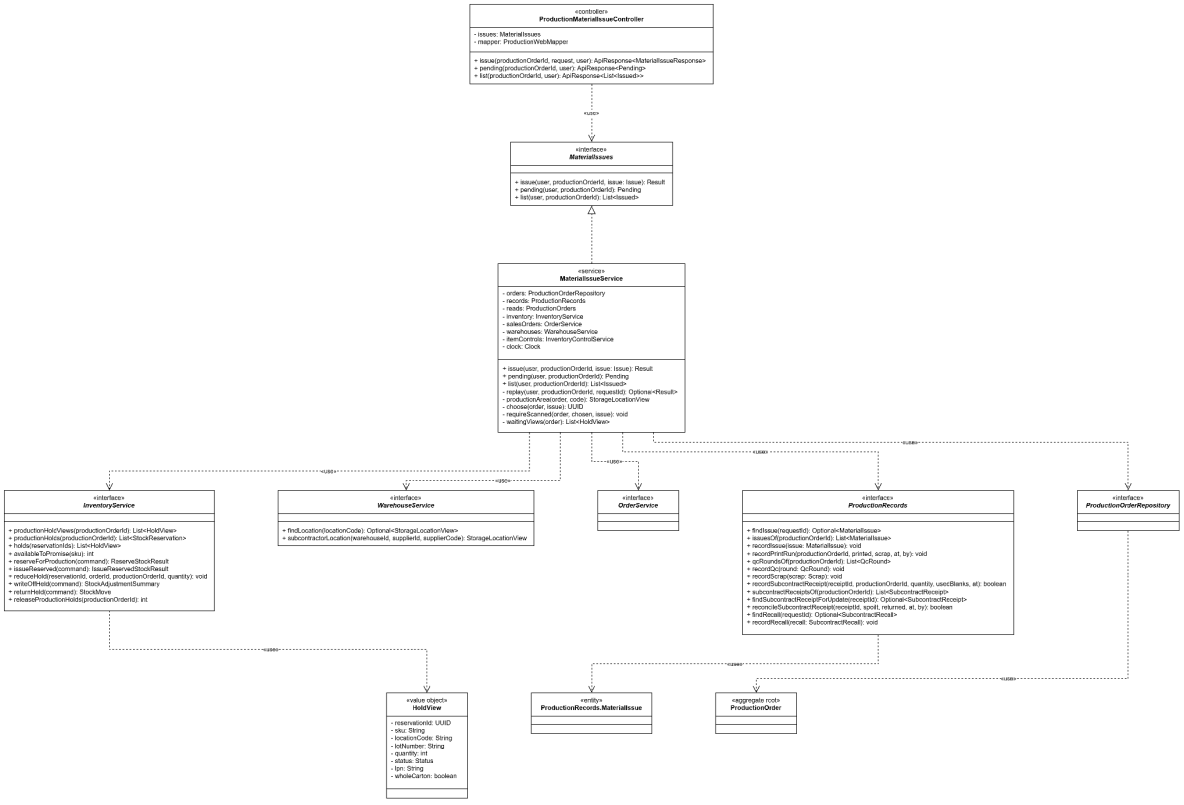


Figure 192. Production Management Class Diagram (3/7): Issuing Blanks

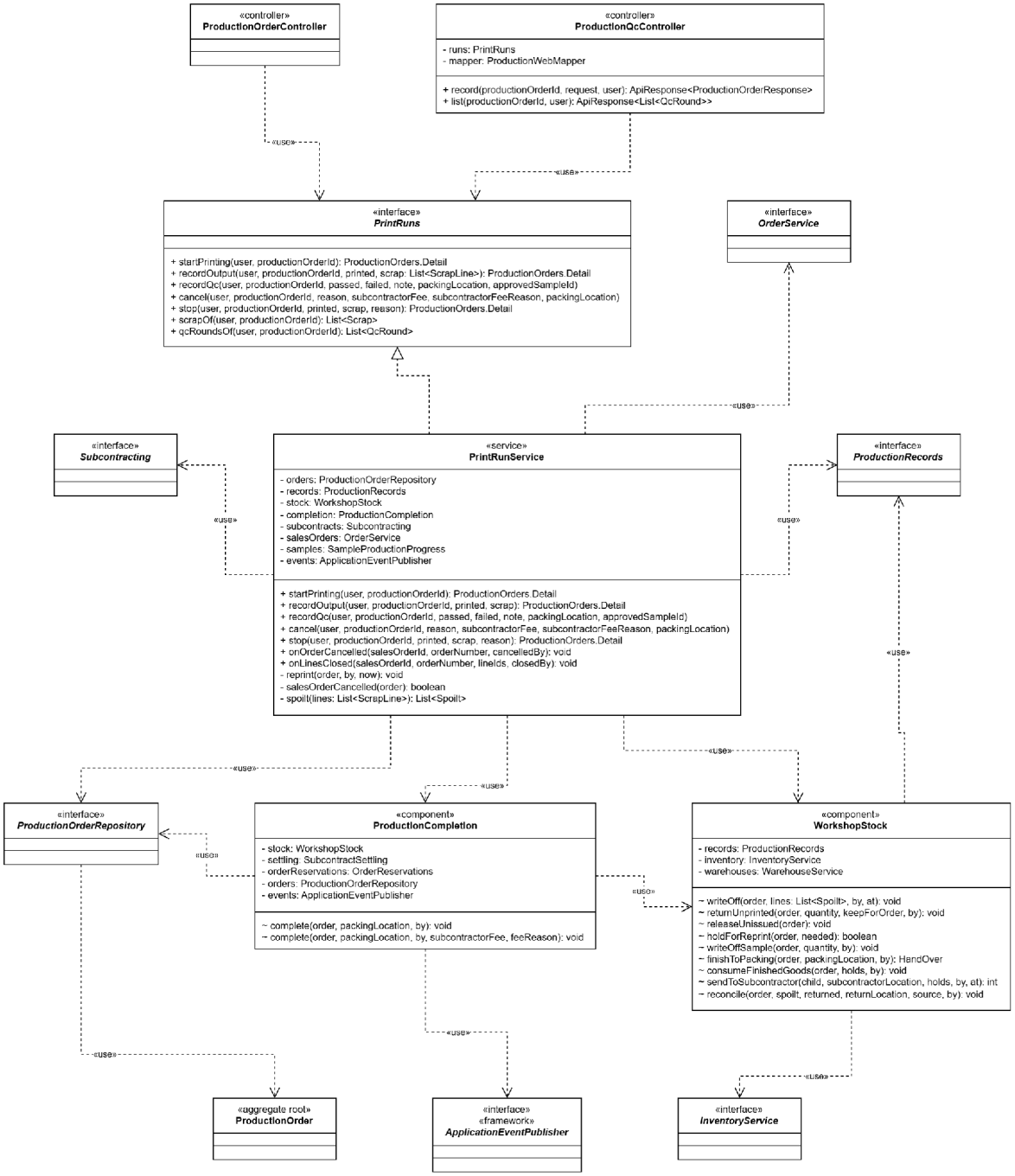


Figure 193. Production Management Class Diagram (4/7): Printing, QC, Cancellation and Stop

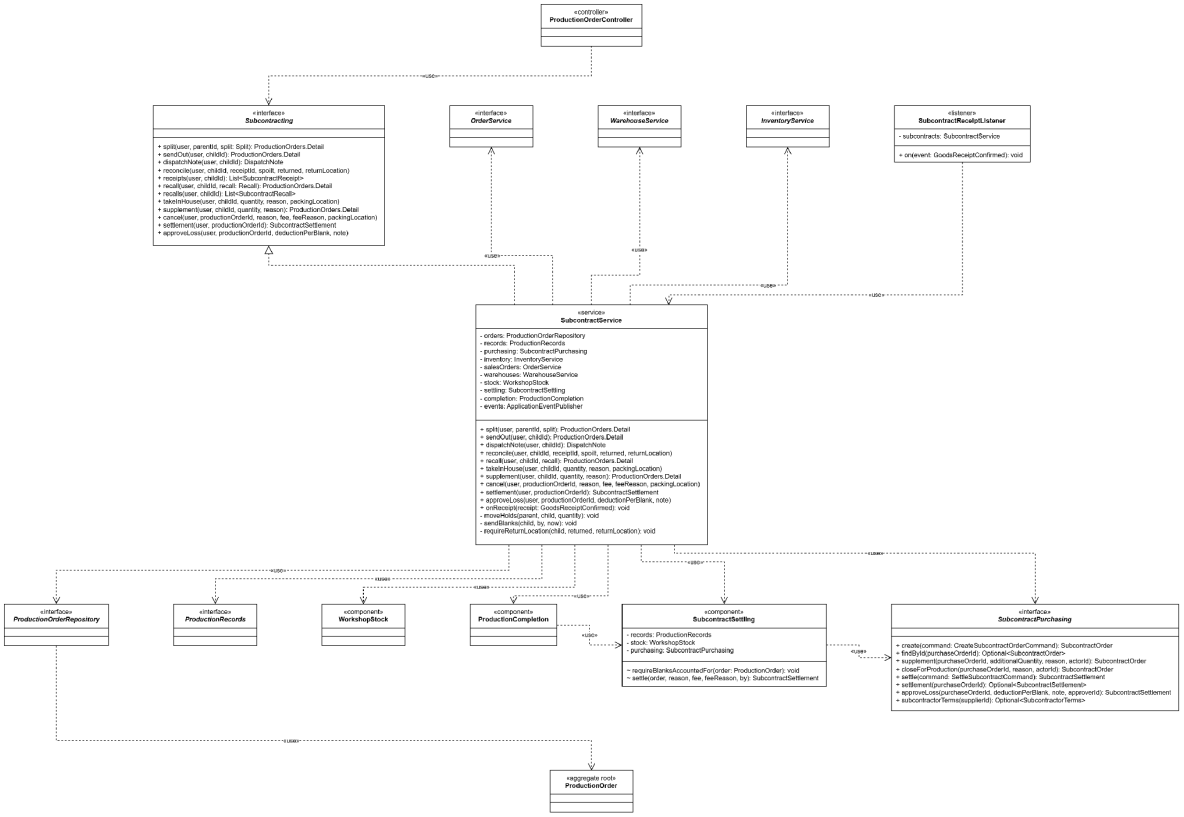


Figure 194. Production Management Class Diagram (5/7): Subcontracting

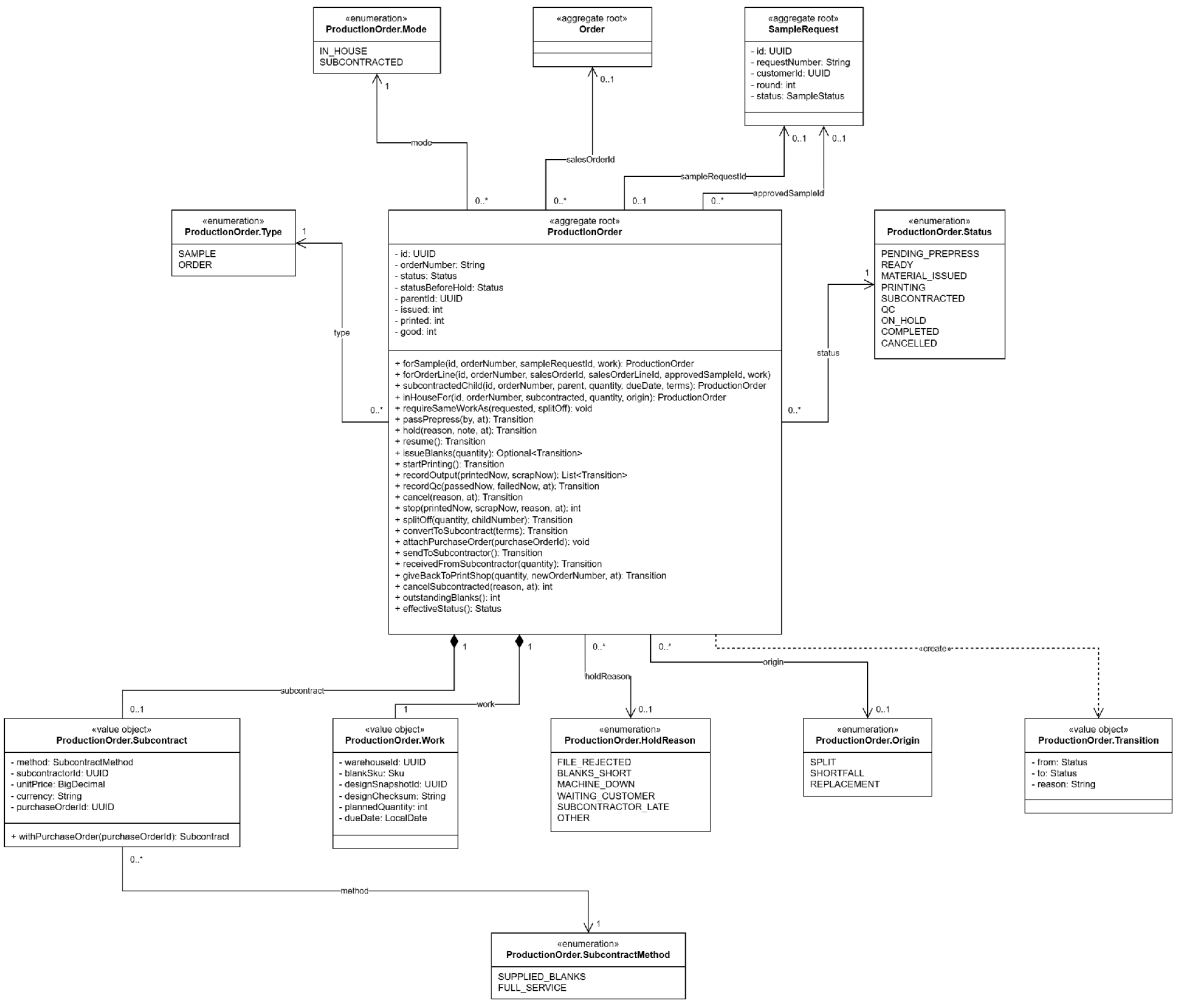


Figure 195. Production Management Class Diagram (6/7): Production Order Domain Model

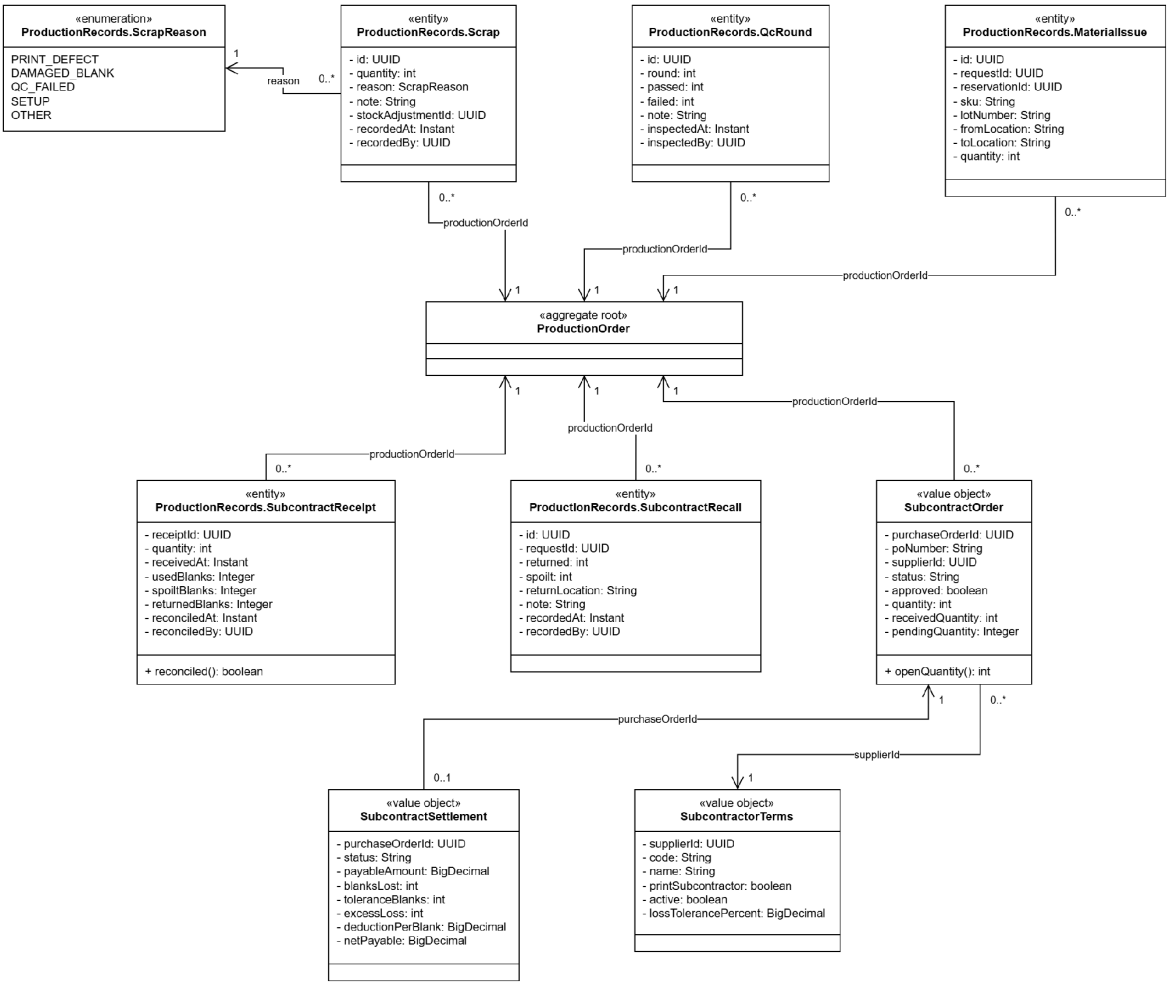


Figure 196. Production Management Class Diagram (7/7): Workshop Records and Subcontract Settlement

3.12.2 Class Specification

ProductionOrderController

No	Method	Description
1	list(types, statuses, warehouseId, dueFrom, dueTo, sampleRequestId, salesOrderId, search, sort, page, size, user)	GET /api/v1/production/orders: Pages production orders filtered by type, status, warehouse, due dates, sample request, sales order and number, newest first, only in the caller's warehouses (production-orders:READ).
2	get(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}: Returns one production order with its quantities, subcontract terms and status history; 404 when unknown, refused outside the caller's warehouses.
3	printFile(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/print-file: Returns a short-lived link to the PRINT_READY file of the locked design, fetched by the design module against the order's checksum.

4	approvedSample(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/approved-sample: Returns the APPROVED sample round an ORDER is QC'd against (BR-PRD-05) with a CUSTOMER_PREVIEW link, no-store; 404 for a SAMPLE order or a repeat design released without a sample, 409 CONFLICT if the round is no longer APPROVED for that design.
5	packingLocations(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/packing-locations: Lists the usable PACKING areas of the order's warehouse a completing QC round may send the goods to (production-qc-results:READ).
6	prepress(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/prepress: Records the prepress verdict: PASS re-verifies the stored file (409 PRINT_FILE_MISMATCH) and moves PENDING_PREPRESS -> READY; FAIL needs a note and puts the order ON_HOLD (FILE_REJECTED), Sales notified.
7	hold(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/hold: Puts a live order (any status before COMPLETED, not already held) ON_HOLD with a reason; OTHER needs a note; publishes ProductionOrderHeld.
8	startPrinting(productionOrderId, user)	POST /api/v1/production/orders/{productionOrderId}/printing: MATERIAL_ISSUED -> PRINTING; for an ORDER tells the sales order printing started, for a SAMPLE moves its round to IN_PRODUCTION.
9	recordOutput(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/output: Records a run (printed, scrap lines with reasons, BR-PRD-04): PRINTING -> QC, spoilt units leave the holds with a SCRAP write-off asked; printed above the unprinted blanks is 400.
10	scrap(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/scrap: Lists what was spoilt, each line with the stock adjustment asked for it.

11	split(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-split: Sends unissued quantity to a print subcontractor (production-orders:APPROVE): part becomes a SUBCONTRACTED child, the whole unissued plan converts the order itself; raises the DRAFT SUBCONTRACT PO (UC94); 409 SUPPLIER_NOT_SUBCONTRACTOR.
12	sendOut(productionOrderId, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-dispatch: READY -> SUBCONTRACTED once the PO is approved and covers what is owed (BR-PRD-13), sending SUPPLIED_BLANKS blanks to the subcontractor's location (production-material-issues:CREATE).
13	reconcile(productionOrderId, receiptId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-receipts/{receiptId}/reconciliation: Reconciles the blanks of one subcontract goods receipt: spoilt written off as SUBCONTRACT_LOSS, returned put back; a second reconciliation with other figures is 409.
14	recall(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-recall: Records blanks brought back from the subcontractor without a receipt (BR-PRD-15), returned or spoilt; idempotent per requestId.
15	recalls(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/subcontract-recalls: Lists the recalls of a subcontracted order.
16	takeInHouse(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-in-house: Moves what the subcontractor still owes to a new in-house order (SHORTFALL, or REPLACEMENT when nothing was received); the subcontracted order shrinks, completes or is cancelled and settled.
17	supplement(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-supplement: Adds quantity to the SUBCONTRACT PO for a resend: a DRAFT is resized, an approved PO gets an amendment (approved at once within the requester's limit).

18	settlement(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/subcontract-settlement: Returns the settlement of the order's SUBCONTRACT PO (QC-passed units paid, excess loss deducted, BR-PRD-17); 404 before the order ends.
19	approveLoss(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/subcontract-settlement/loss-approval: Approves blanks lost above the subcontractor's tolerance with the agreed deduction per blank and a note: AWAITING_APPROVAL -> FINAL; 409 CONFLICT if already FINAL with another deduction.
20	dispatchNote(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/subcontract-dispatch-note: Returns what goes out with the work: PO, subcontractor, PRINT_READY link, approved sample, blanks sent; 409 before prepress passed (BR-PRD-14).
21	subcontractReceipts(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/subcontract-receipts: Lists the goods receipts of a subcontracted order with their blank reconciliation.
22	cancel(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/cancellation: Cancels an in-house order before printing (PENDING_PREPRESS, READY, MATERIAL_ISSUED, or held from one), returning blanks; a subcontracted order is cancelled with its PO and settled, or replaced in house while its sales order is live.
23	stop(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/stop: Stops a job in PRINTING or QC (Q2): the run's output is recorded, good units become scrap and are written off, unprinted blanks go back; -> CANCELLED.
24	resume(productionOrderId, user)	POST /api/v1/production/orders/{productionOrderId}/resume: Returns a held order to the status it was held in (production-orders:APPROVE, Warehouse Manager); 409 when not ON_HOLD.

Table 74. ProductionOrderController Class Specification

ProductionMaterialIssueController

No	Method	Description

1	issue(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/material-issues: Issues one held lot of blanks to a usable PRODUCTION area of the order's warehouse; a carton hold needs its LPN scanned; READY -> MATERIAL_ISSUED when nothing is outstanding; above the need 409 ISSUE_EXCEEDS_REQUIREMENT; idempotent per requestId.
2	pending(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/material-issues/pending: Shows blanks required, issued and outstanding and each hold still to scan with location, lot, quantity, barcode or carton LPN.
3	list(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/material-issues: Lists the blanks issued to the order, oldest first.

Table 75. ProductionMaterialIssueController Class Specification

ProductionQcController

No	Method	Description
1	record(productionOrderId, request, user)	POST /api/v1/production/orders/{productionOrderId}/qc-results: Records a QC round over every unit awaiting QC: enough good -> COMPLETED (goods to packing and consumed, or the sample round READY), short -> READY for a reprint; refused while a subcontract receipt is unreconciled or the sales order is cancelled.
2	list(productionOrderId, user)	GET /api/v1/production/orders/{productionOrderId}/qc-results: Lists the QC rounds of the order, first round first.

Table 76. ProductionQcController Class Specification

3.12.3 Generate Production Orders Sequence Diagram

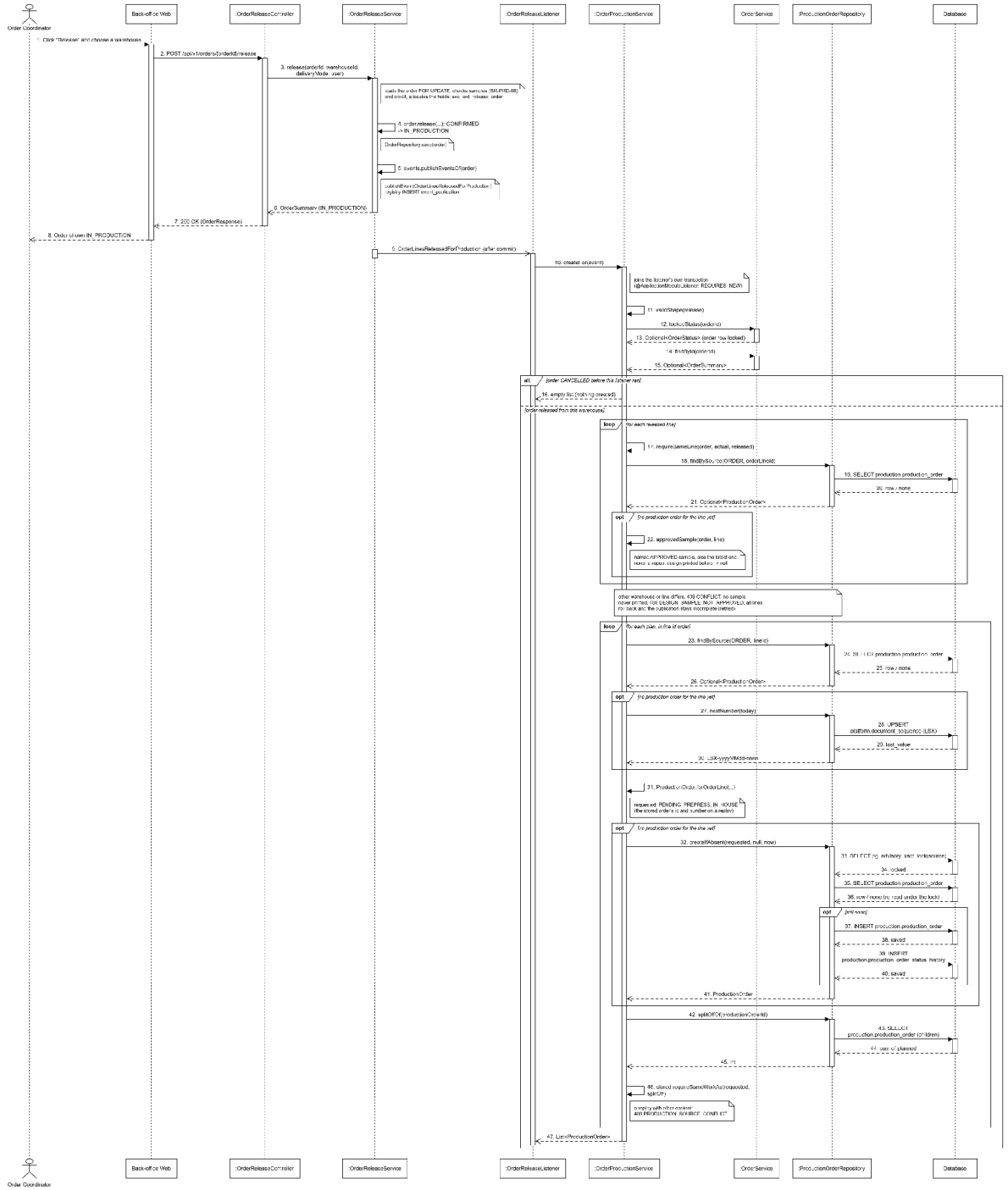


Figure 197. Generate Production Orders Sequence Diagram

3.12.4 Check Print File (Prepress) Sequence Diagram

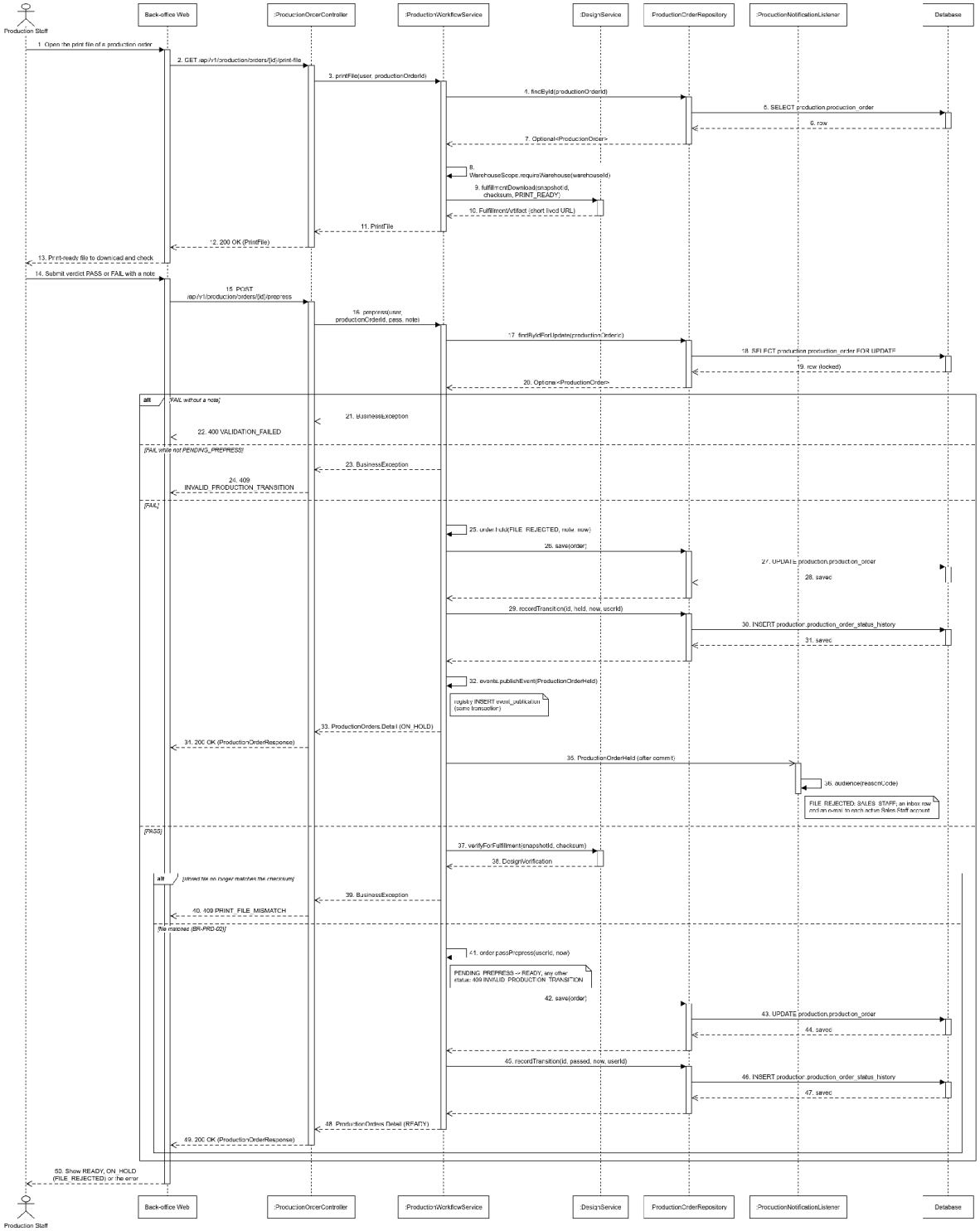


Figure 198. Check Print File (Prepress) Sequence Diagram

3.12.5 Issue Blanks to Production Sequence Diagram

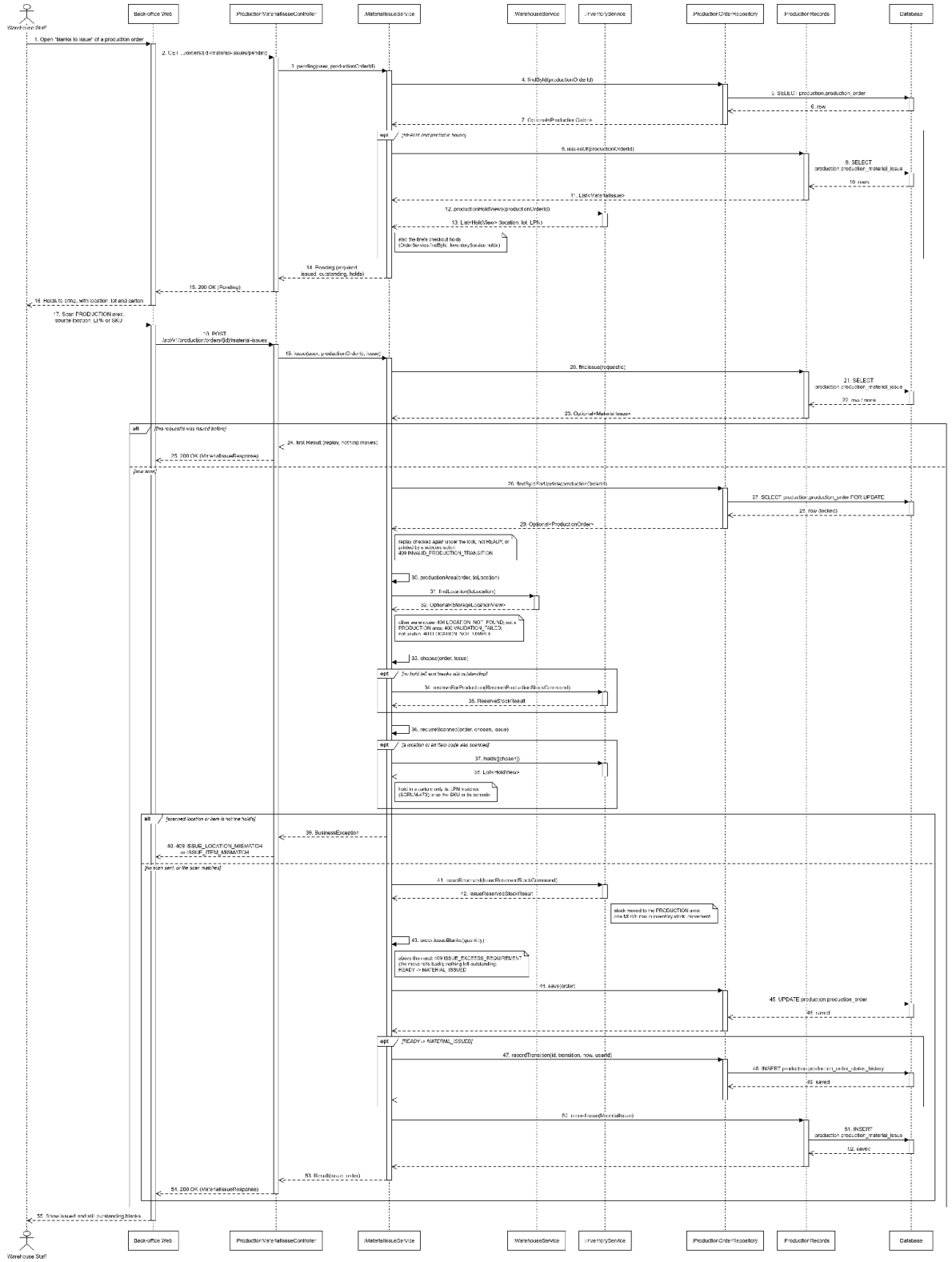


Figure 199. Issue Blanks to Production Sequence Diagram

3.12.6 Print & Report Output / Scrap Sequence Diagram

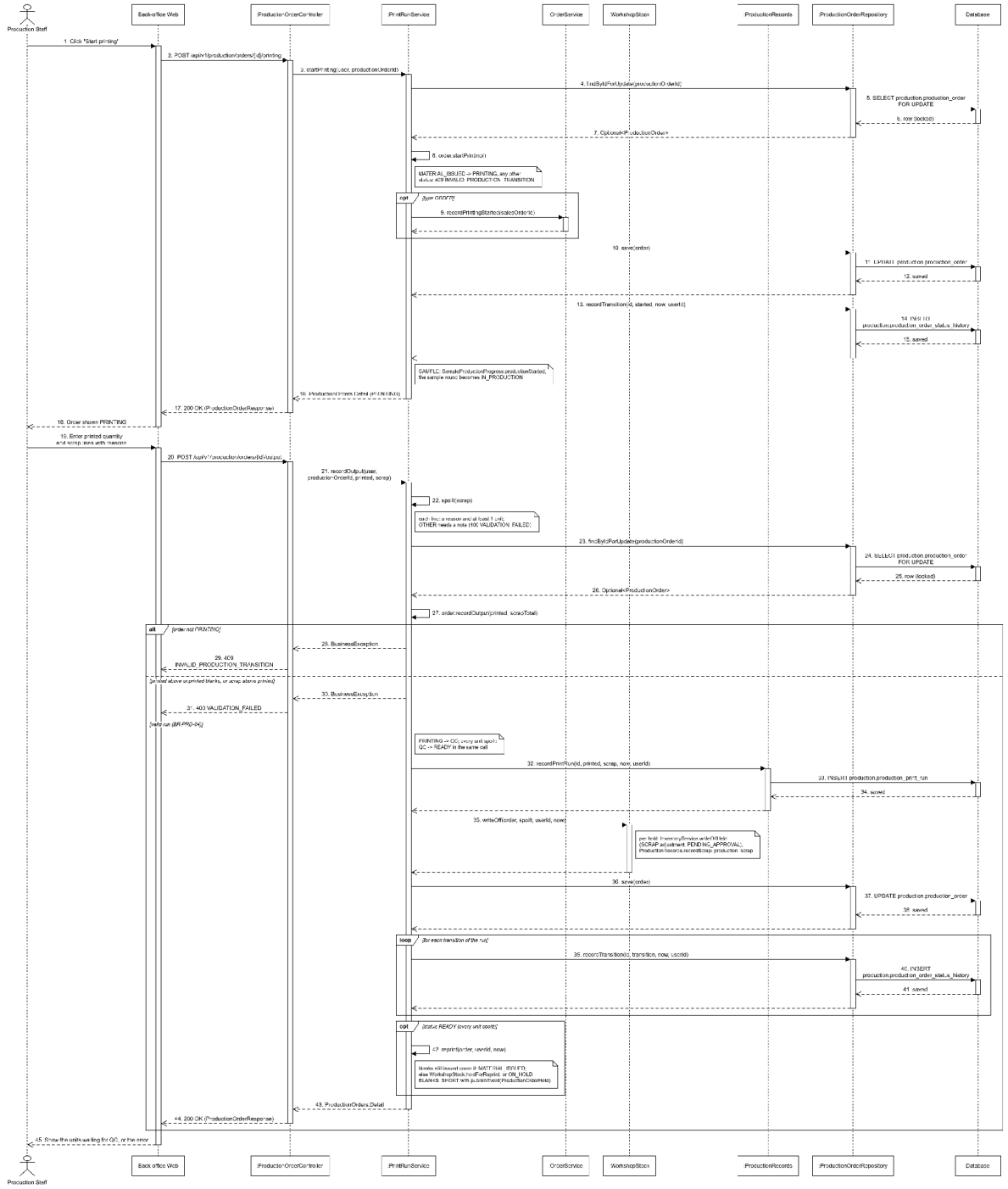


Figure 200. Print & Report Output / Scrap Sequence Diagram

3.12.7 Record Production QC Result Sequence Diagram

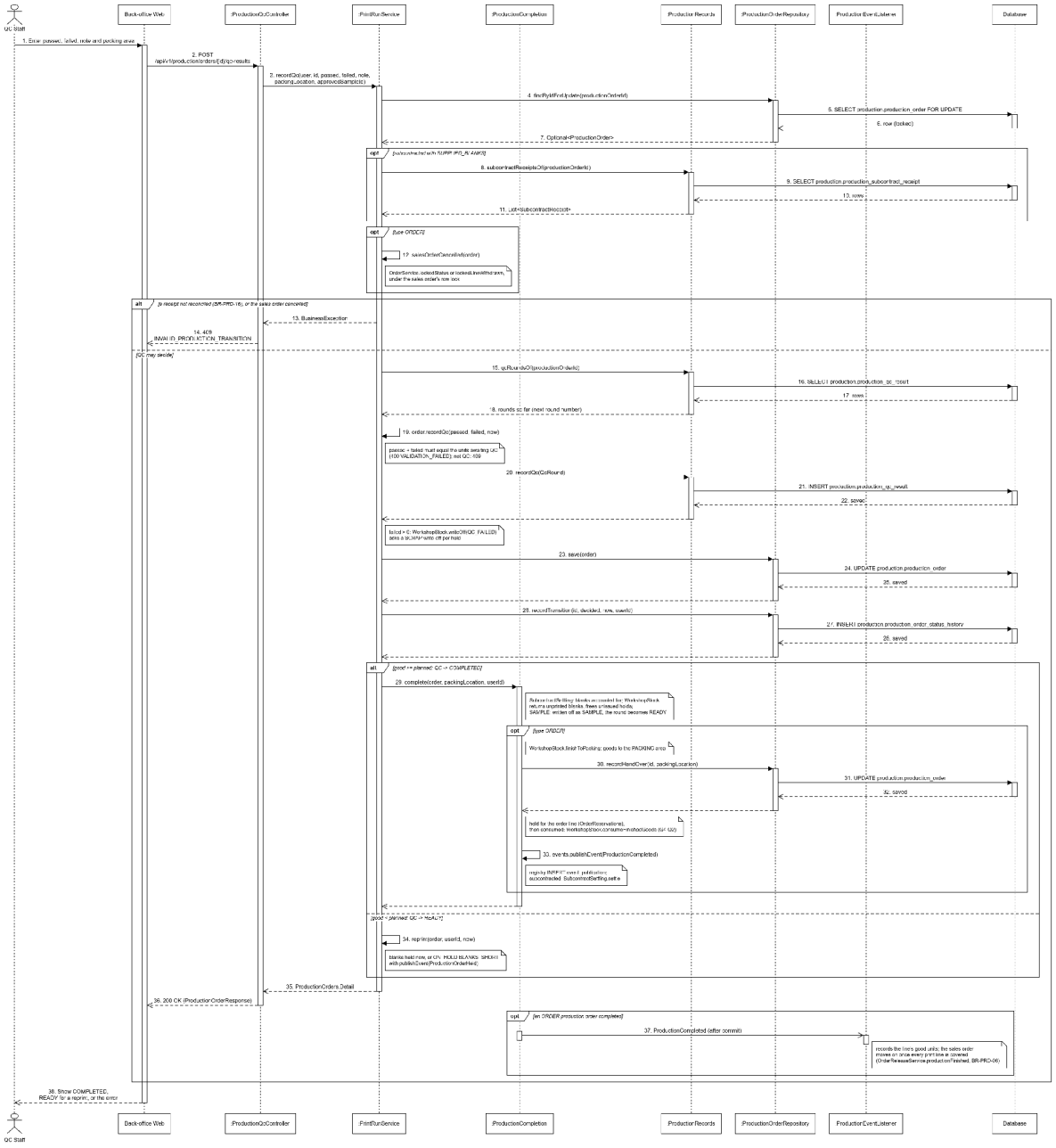


Figure 201. Record Production QC Result Sequence Diagram

3.12.8 Put Production Order on Hold Sequence Diagram

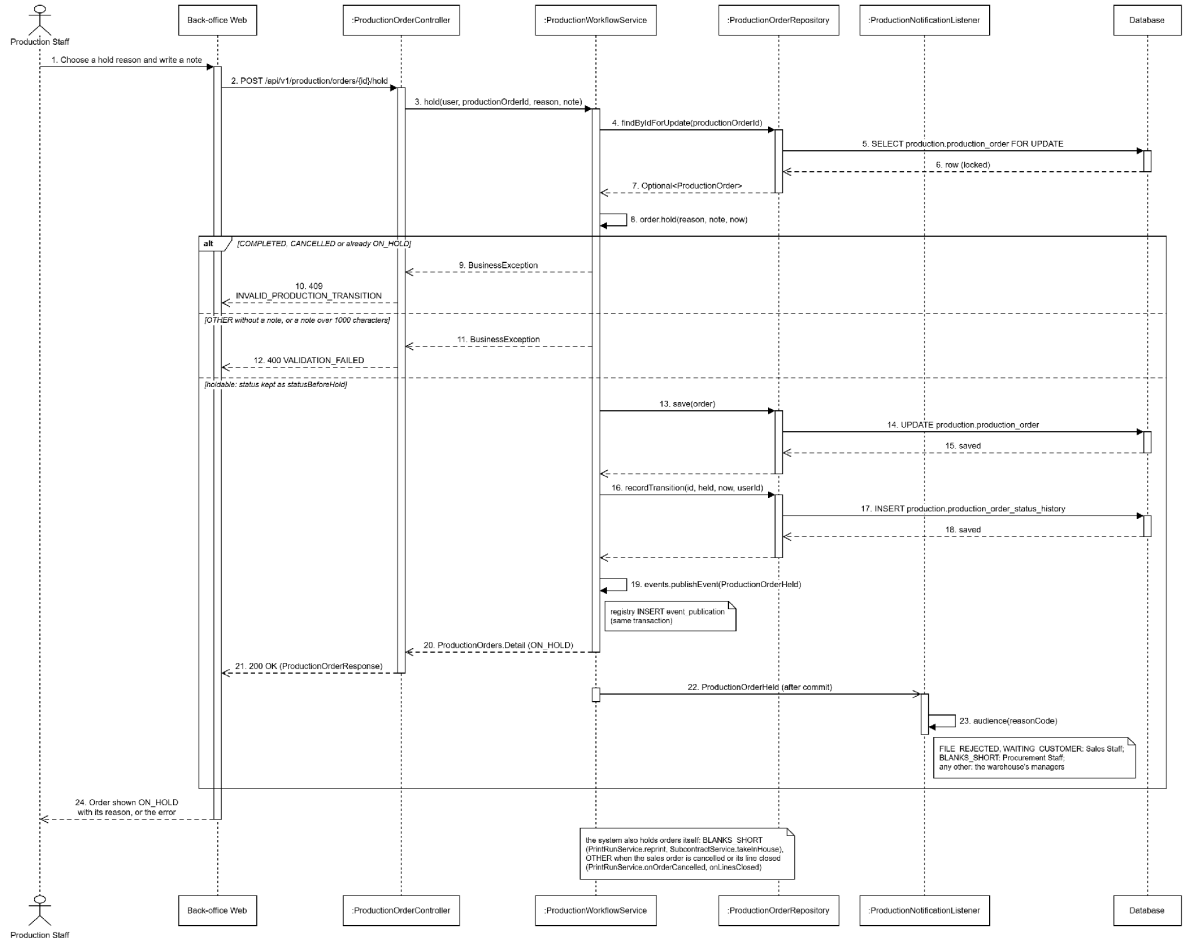


Figure 202. Put Production Order on Hold Sequence Diagram

3.12.9 Reopen / Cancel / Stop Production Order Sequence Diagram (1/3): Reopen

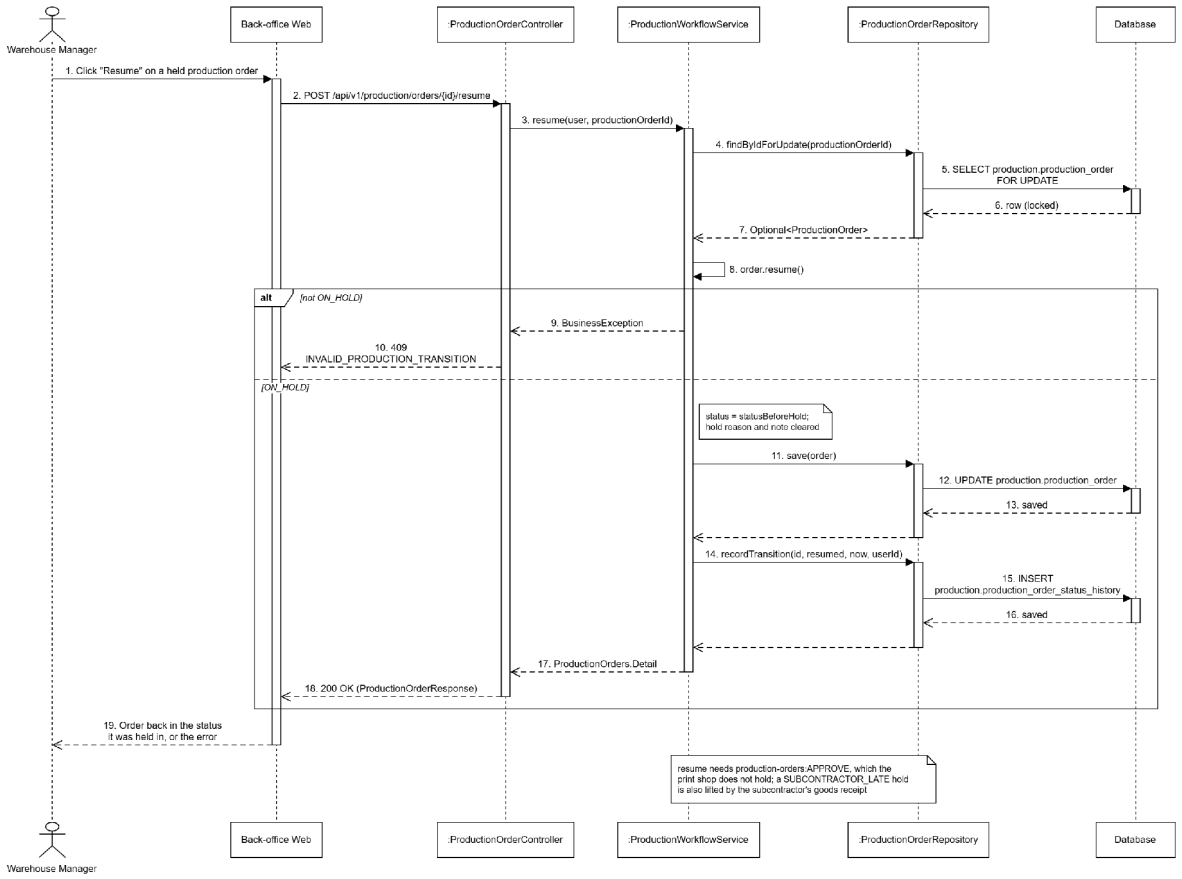


Figure 203. Reopen / Cancel / Stop Production Order Sequence Diagram (1/3): Reopen

3.12.10 Reopen / Cancel / Stop Production Order Sequence Diagram (2/3): Cancel

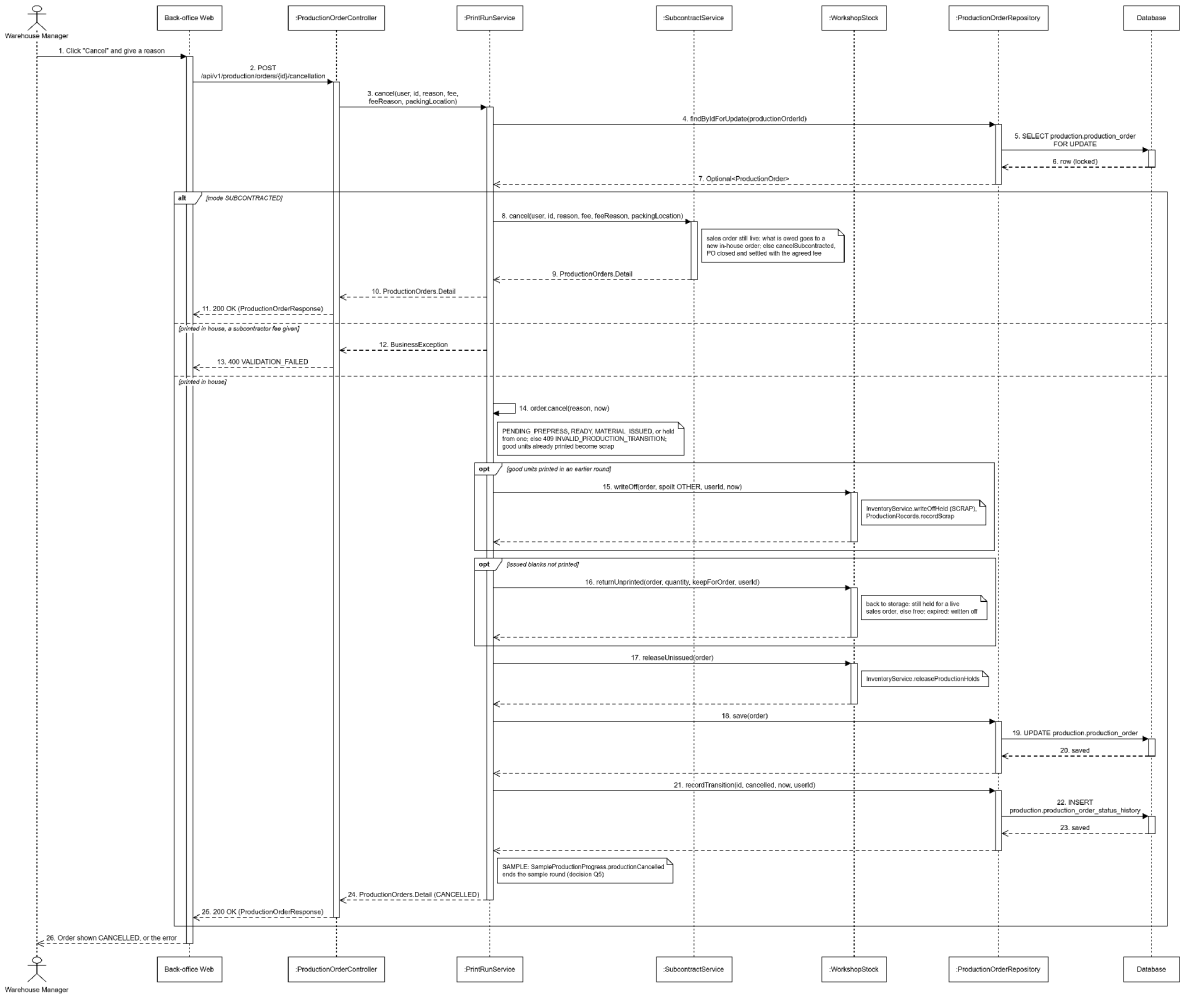


Figure 204. Reopen / Cancel / Stop Production Order Sequence Diagram (2/3): Cancel

3.12.11 Reopen / Cancel / Stop Production Order Sequence Diagram (3/3): Stop

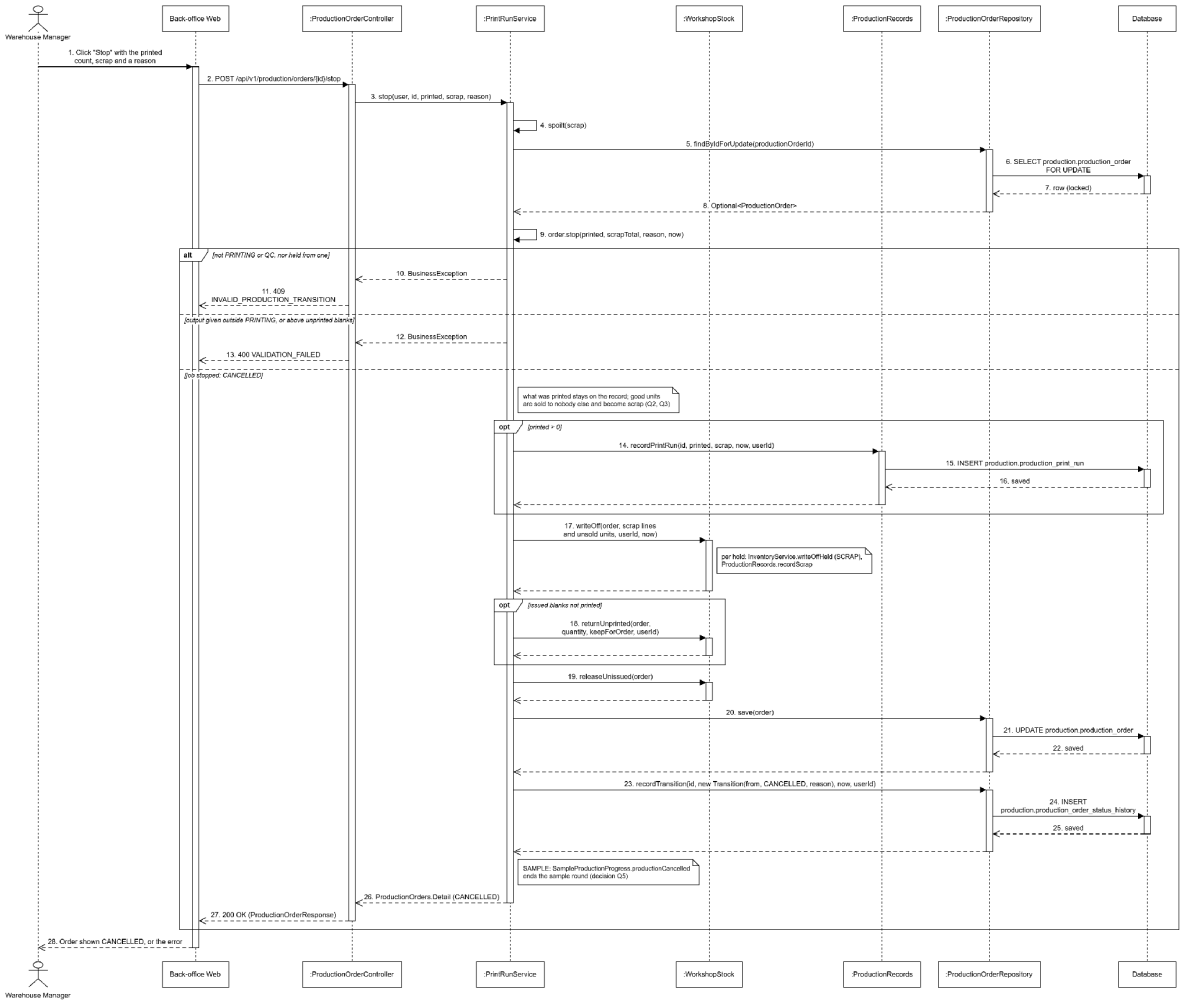


Figure 205. Reopen / Cancel / Stop Production Order Sequence Diagram (3/3): Stop

3.12.12 Outsource Production Order and Create Subcontract PO Sequence Diagram (1/2)

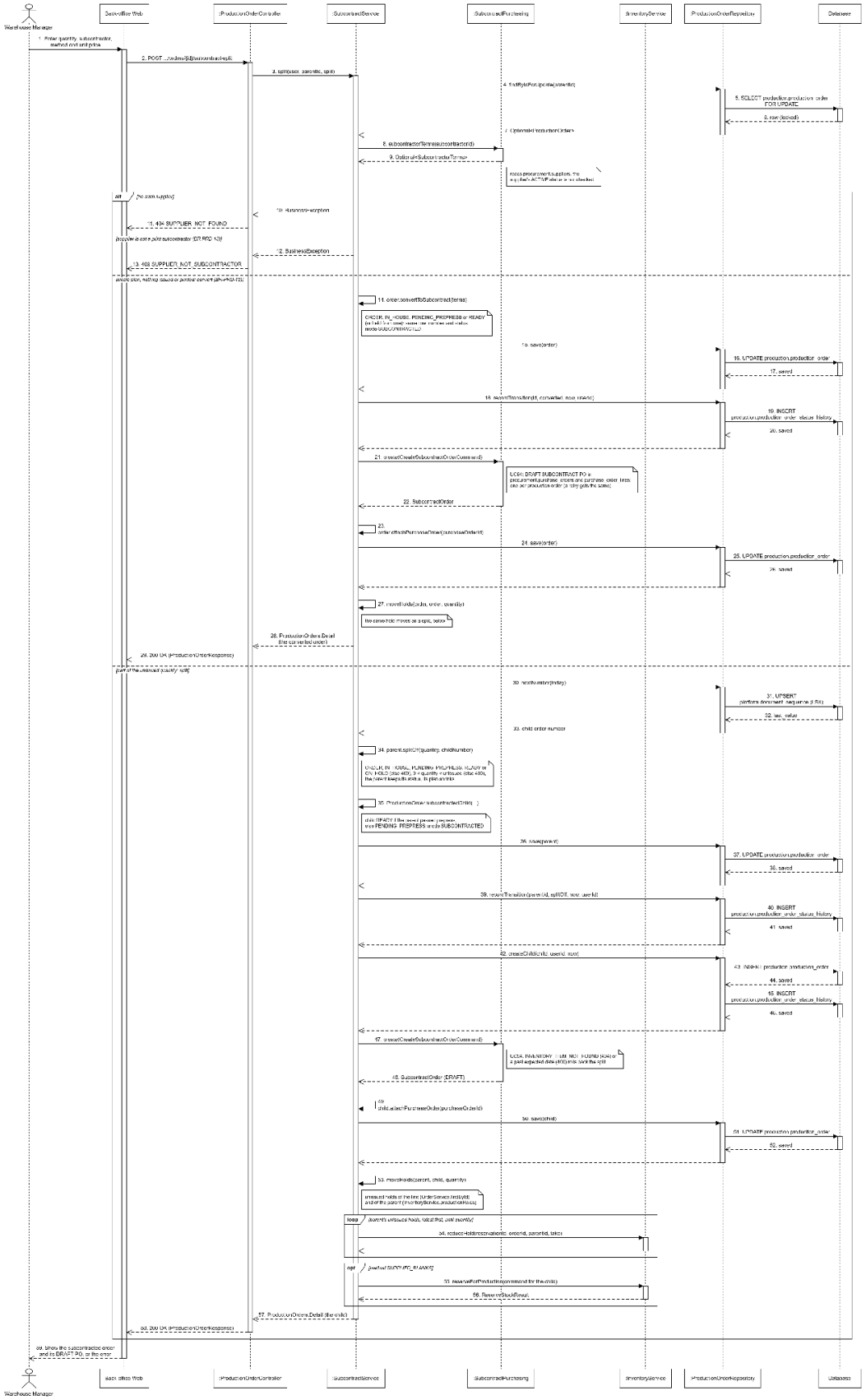


Figure 206. Outsource Production Order and Create Subcontract PO Sequence Diagram (1/2)

3.12.13 Outsource Production Order Sequence Diagram (2/2): Supplement PO or Take Work Back

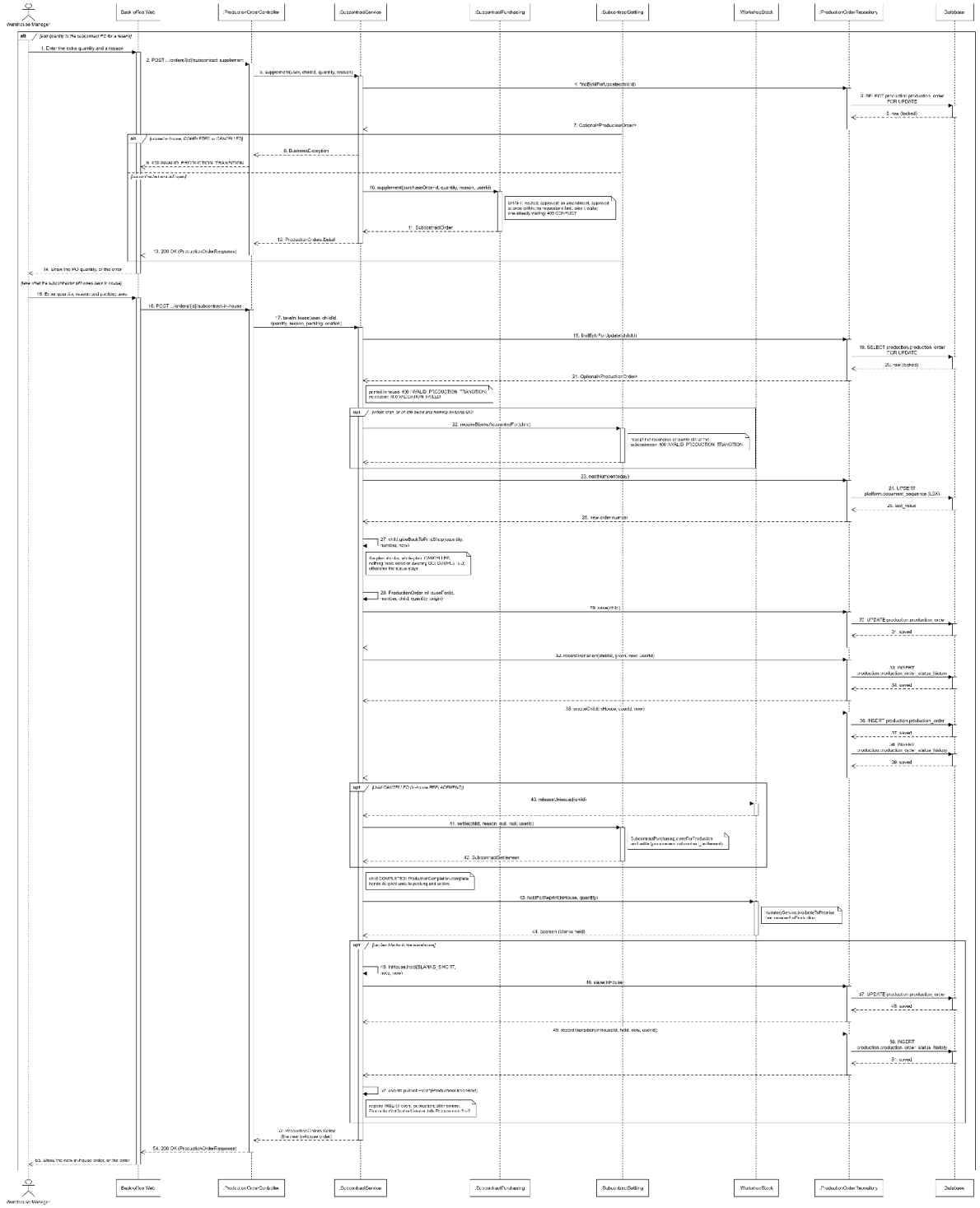


Figure 207. Outsource Production Order Sequence Diagram (2/2): Supplement PO or Take Work Back

3.12.14 Ship & Reconcile Subcontract Blanks Sequence Diagram (1/2): Ship

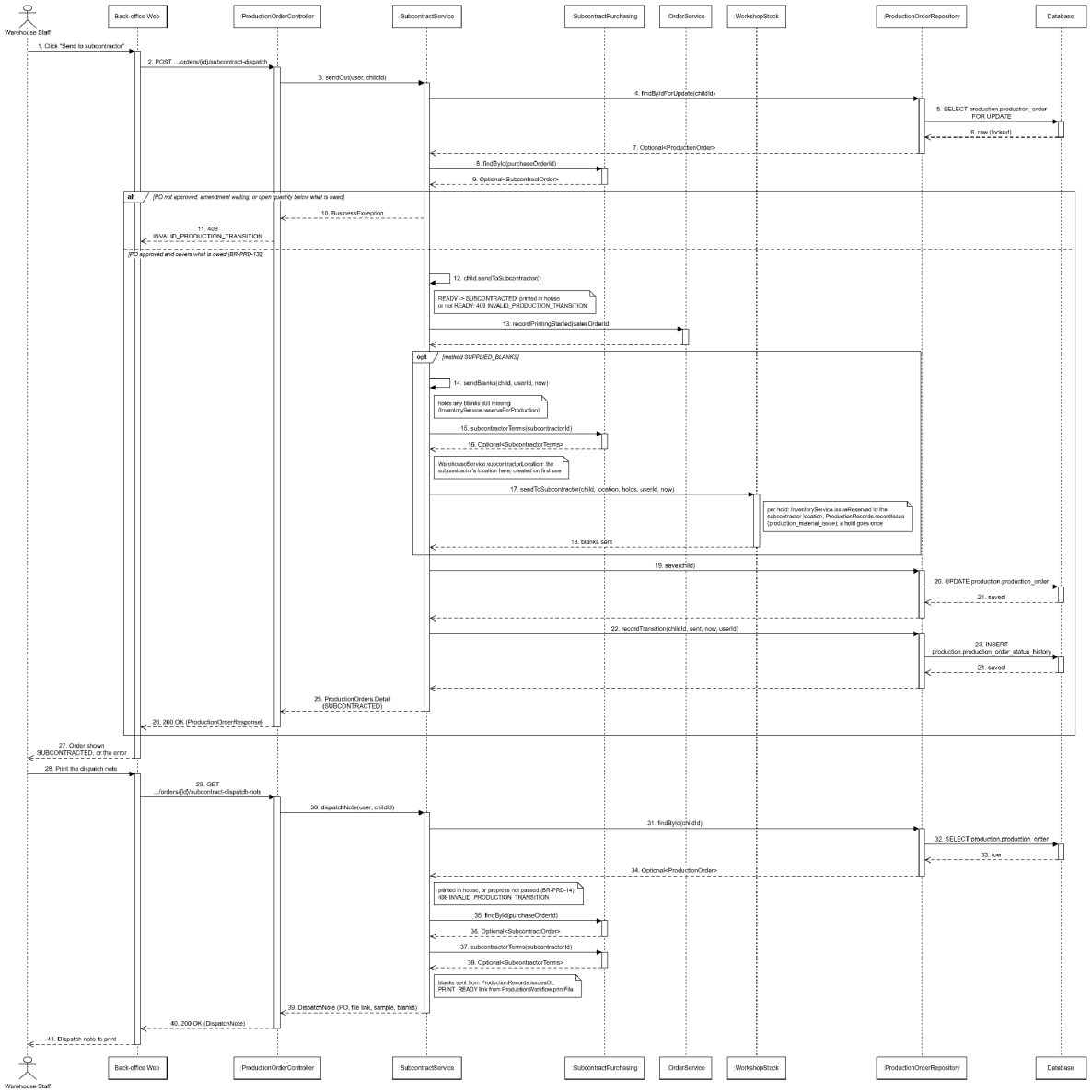


Figure 208. Ship & Reconcile Subcontract Blanks Sequence Diagram (1/2): Ship

3.12.15 Ship & Reconcile Subcontract Blanks Sequence Diagram (2/2): Reconcile

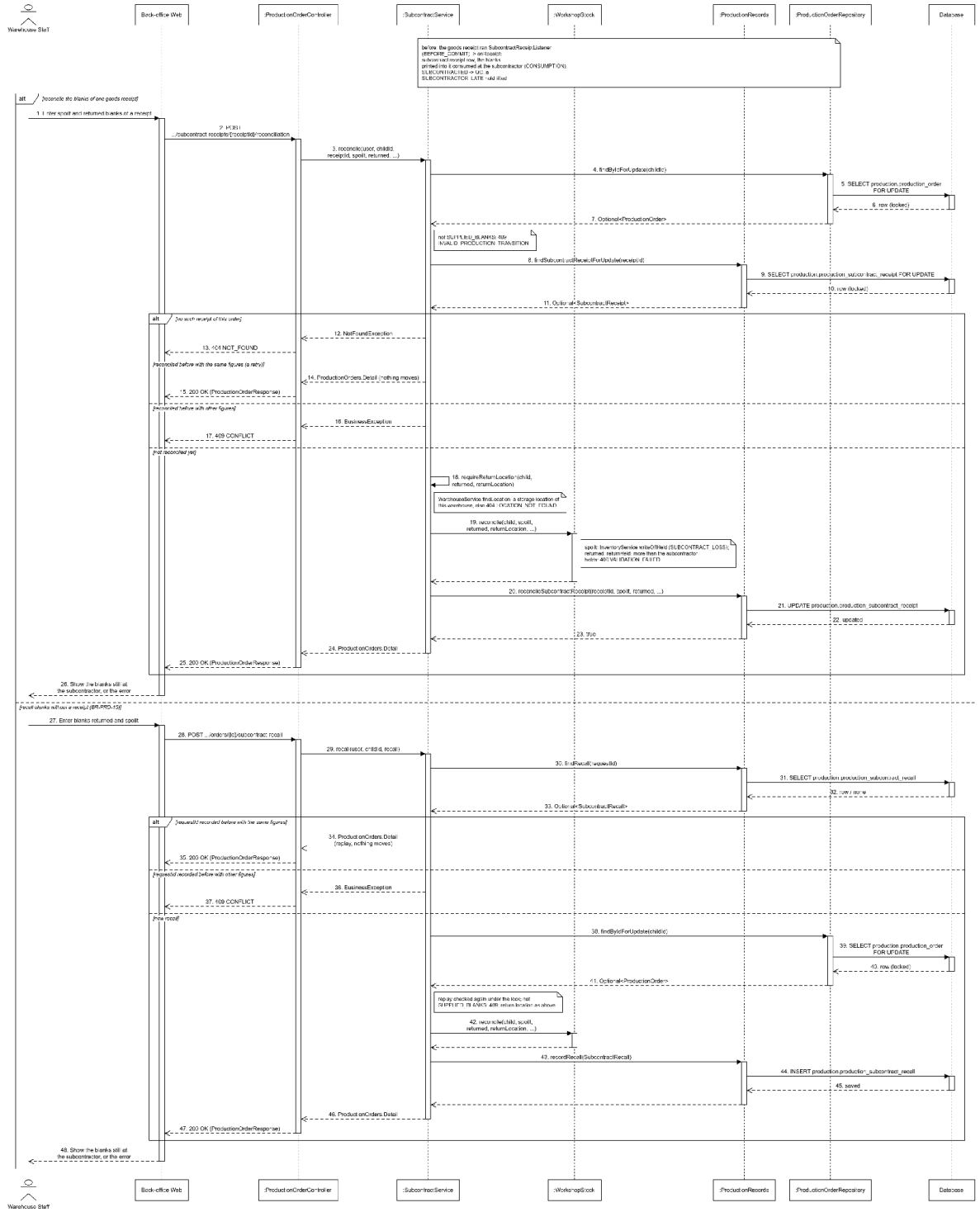


Figure 209. Ship & Reconcile Subcontract Blanks Sequence Diagram (2/2): Reconcile

3.12.16 Approve Subcontract Loss Sequence Diagram

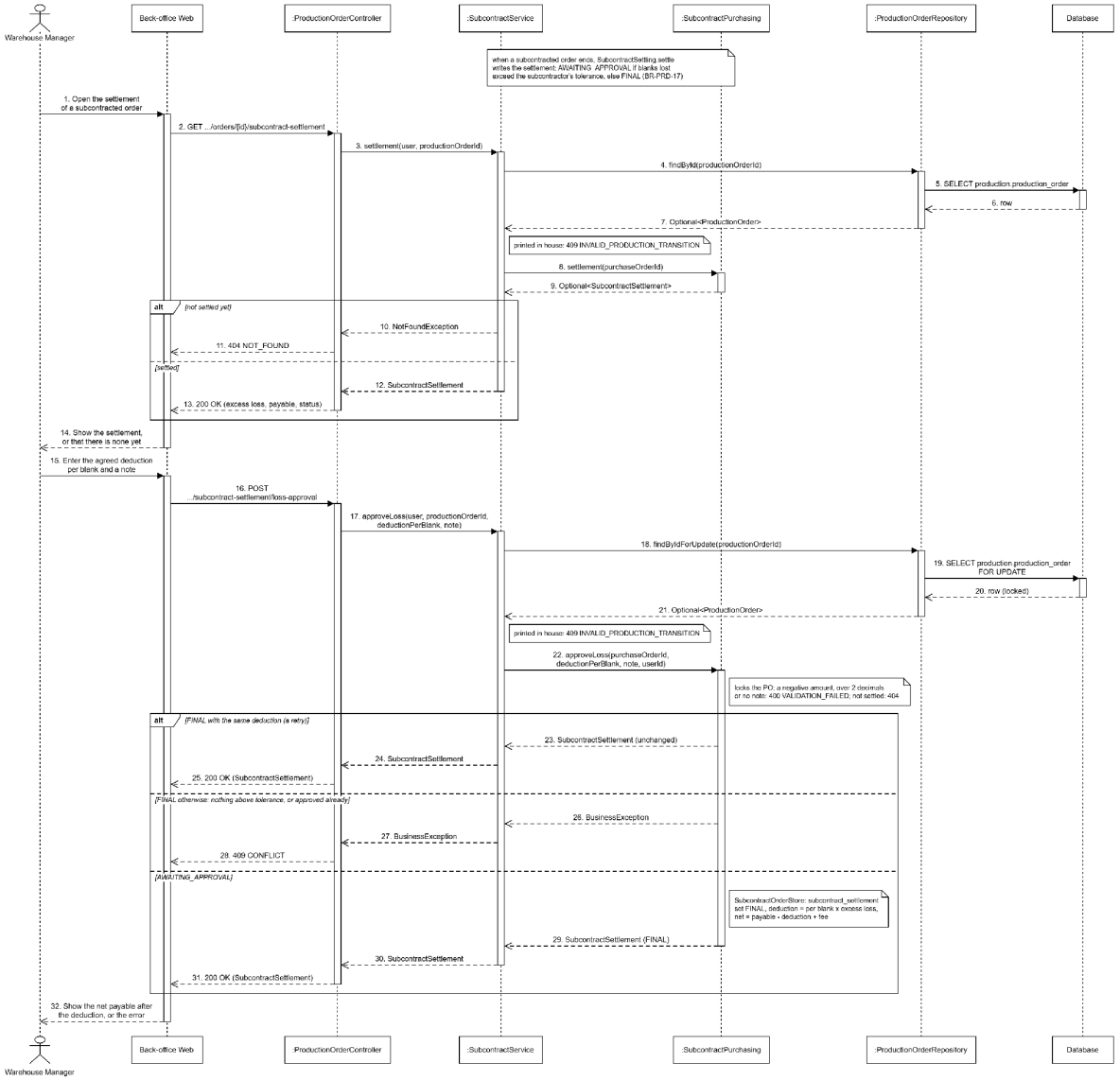


Figure 210. Approve Subcontract Loss Sequence Diagram

3.12.17 Production Order State Diagram

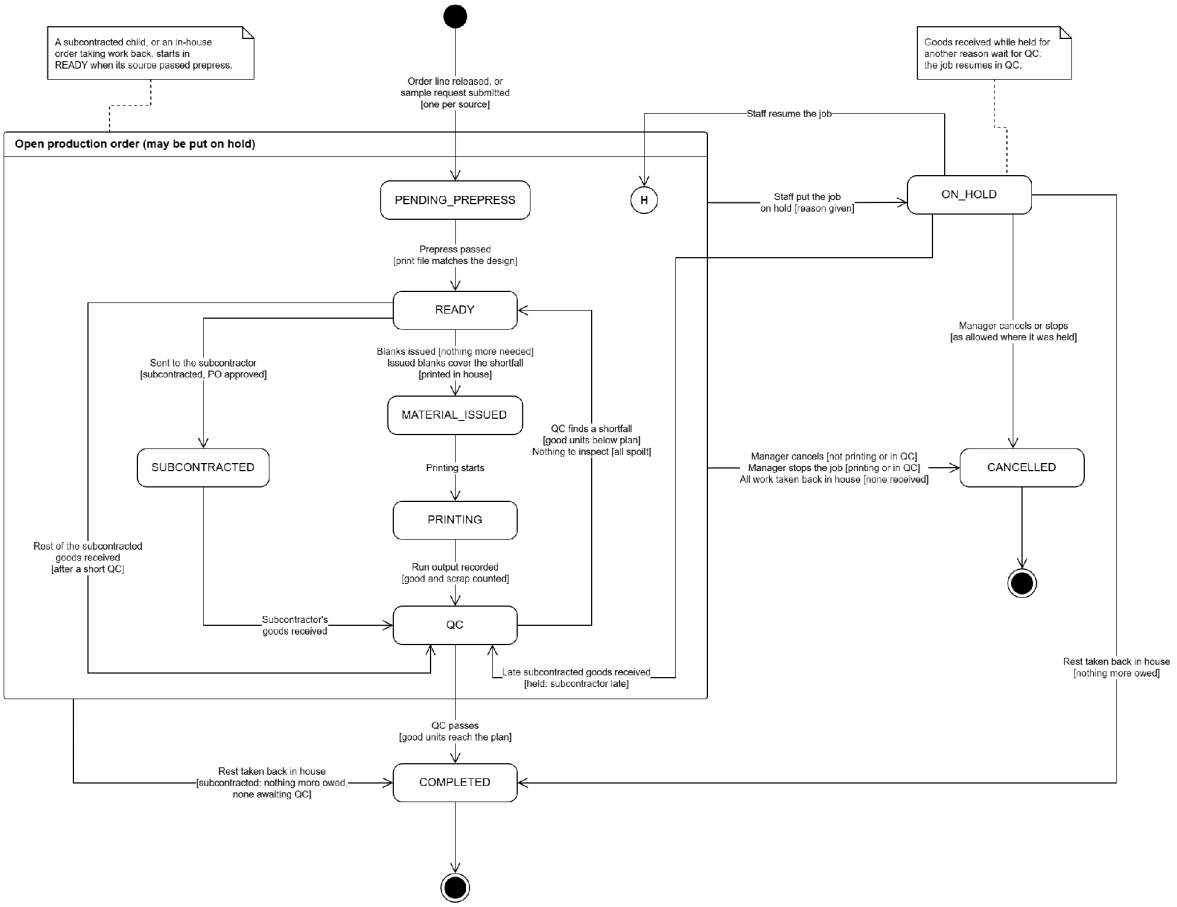


Figure 211. Production Order State Diagram

3.12.18 Subcontract Settlement State Diagram

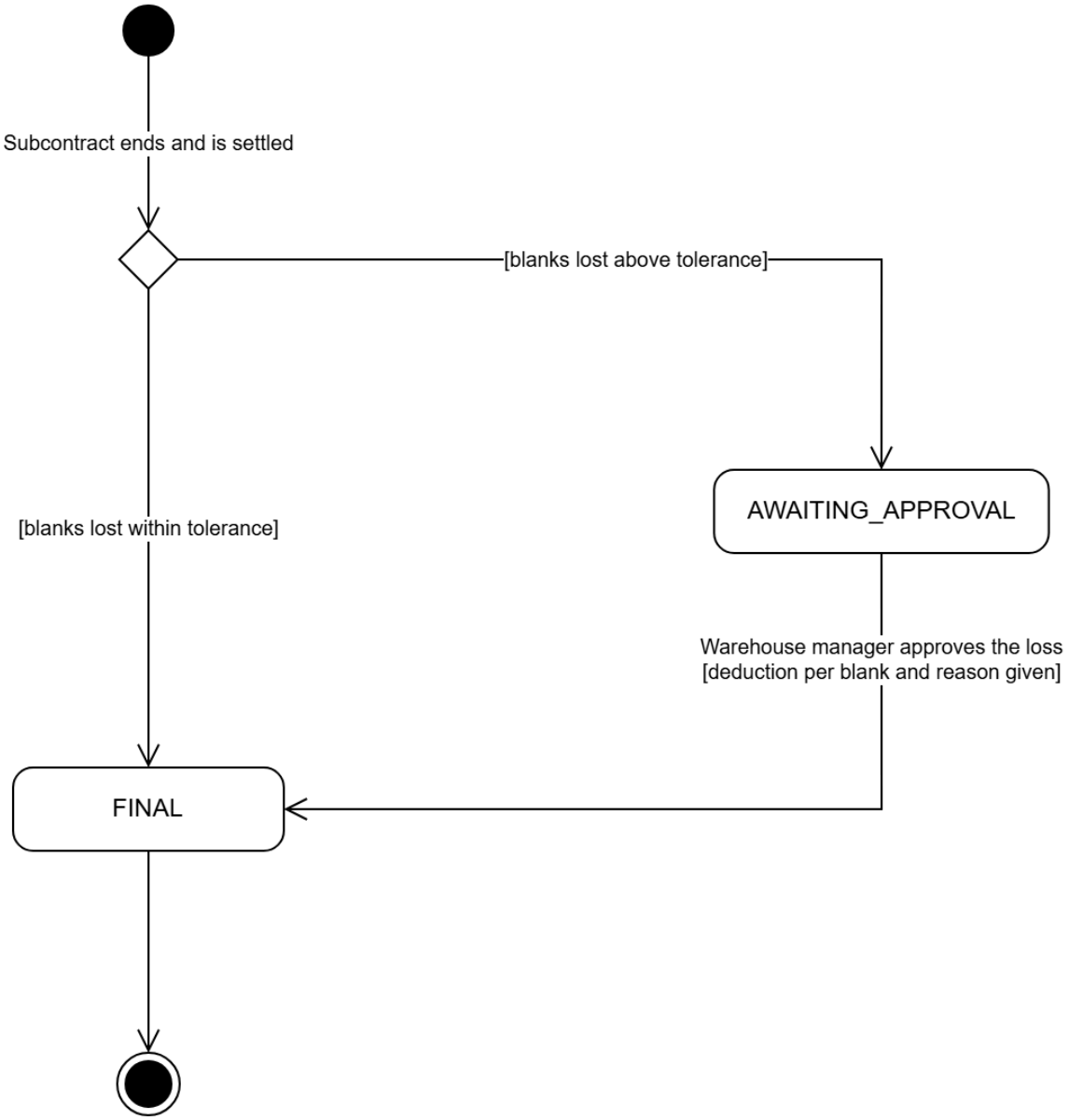


Figure 212. Subcontract Settlement State Diagram

3.13 Warehouse Layout Management

3.13.1 Class Diagram

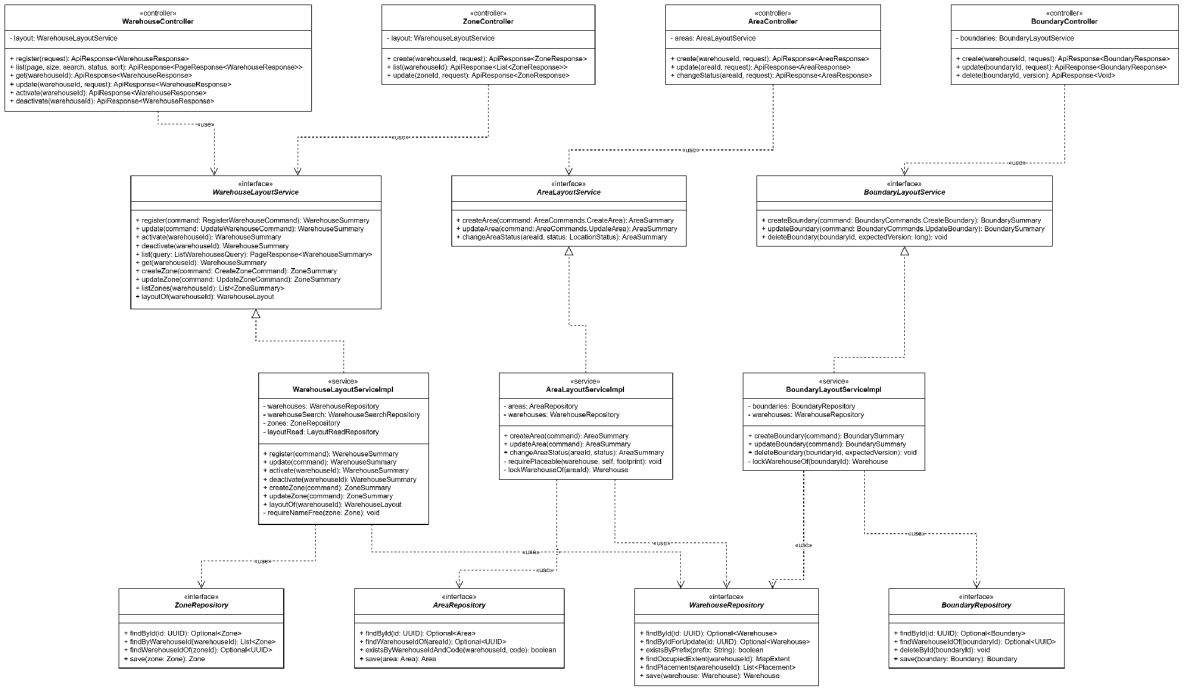


Figure 213. Warehouse Layout Management Class Diagram (1/4): Warehouses, Zones, Areas and Boundaries

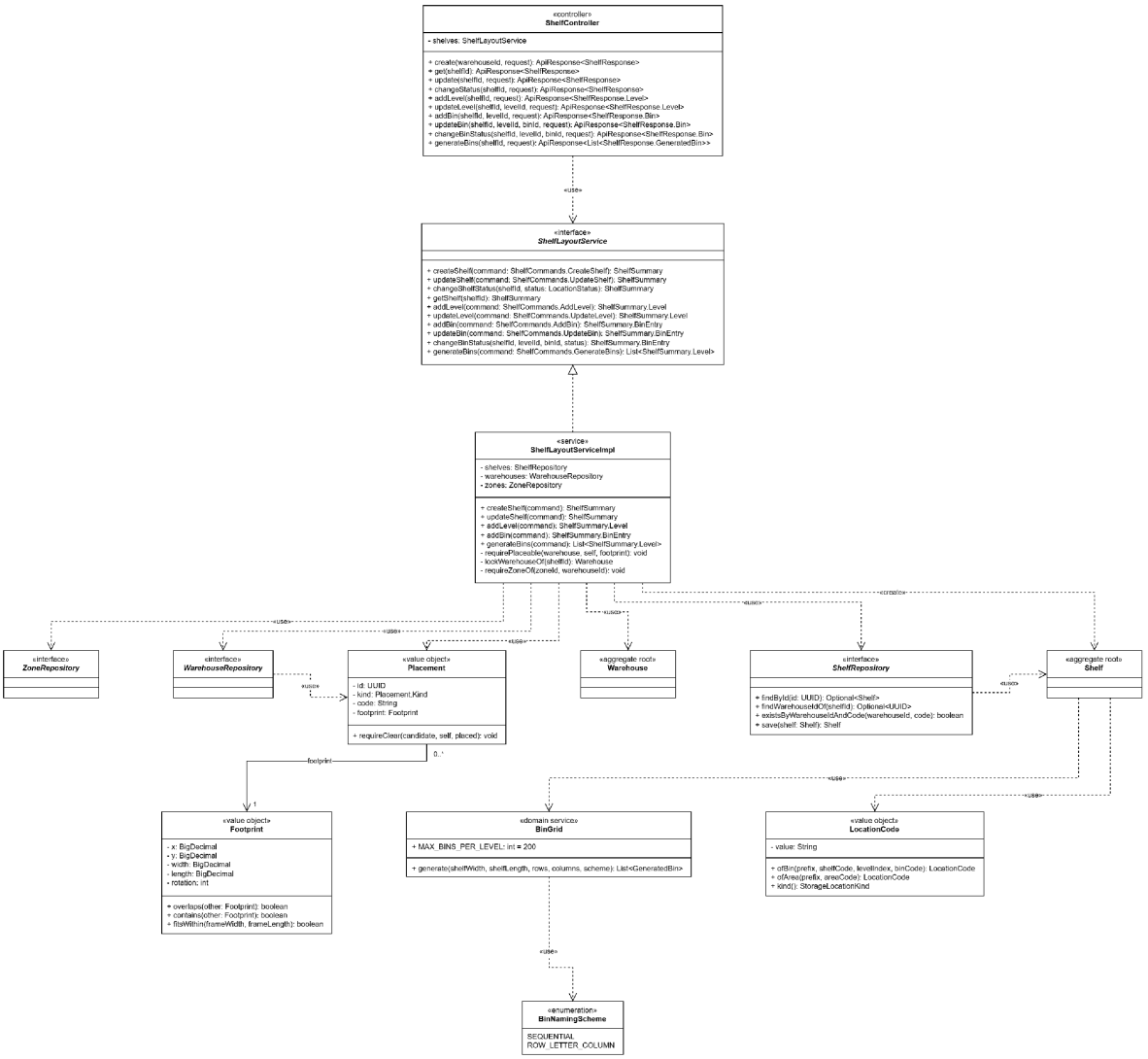


Figure 214. Warehouse Layout Management Class Diagram (2/4): Shelves and Bins

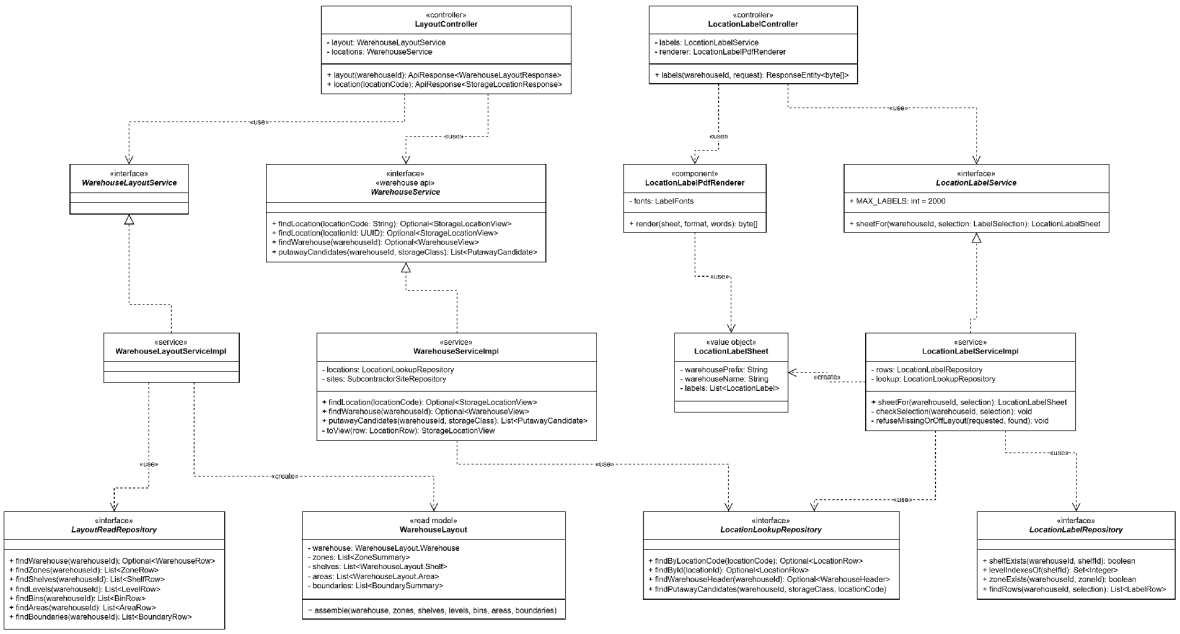


Figure 215. Warehouse Layout Management Class Diagram (3/4): Map View and Location Labels

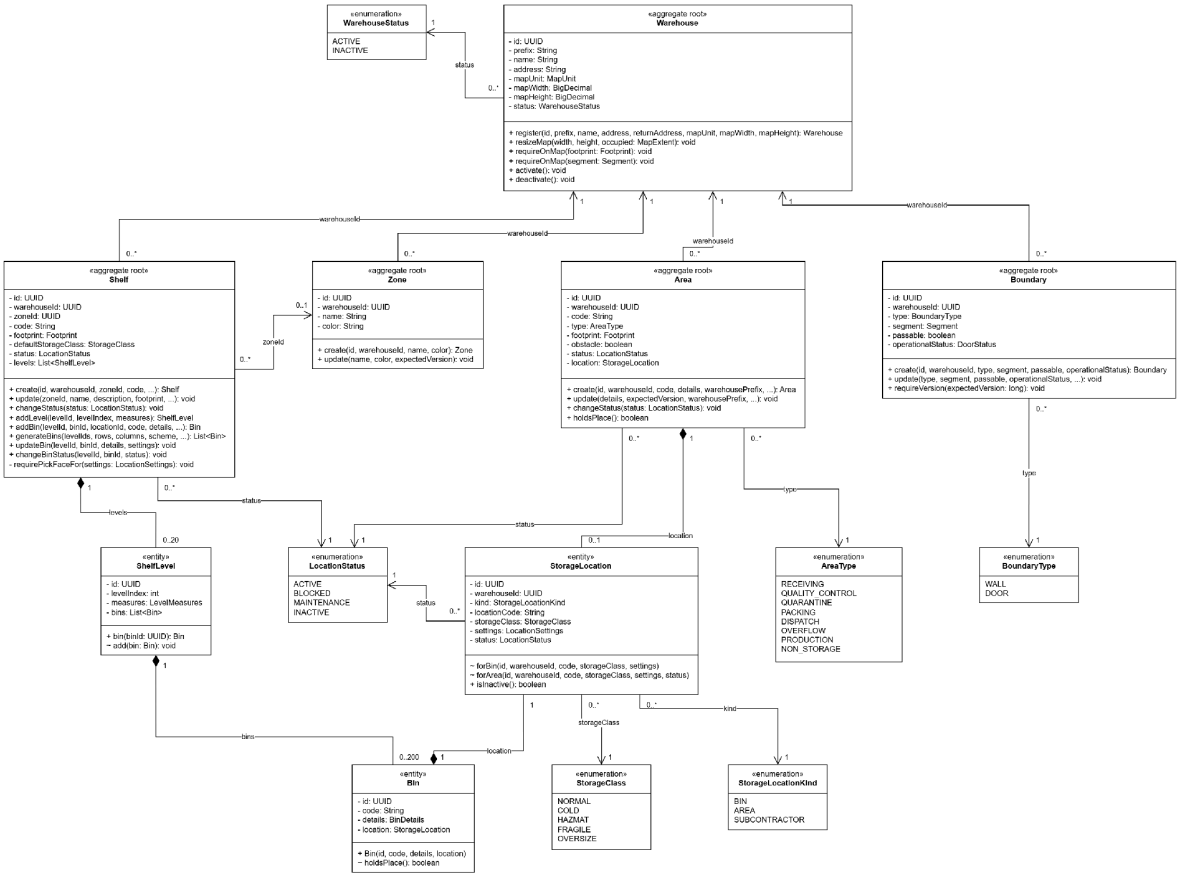


Figure 216. Warehouse Layout Management Class Diagram (4/4): Domain Model

3.13.2 Class Specification

WarehouseController

No	Method	Description

1	register(request)	POST /api/v1/warehouses: Registers an ACTIVE warehouse with its map frame; 409 WAREHOUSE_PREFIX_ALREADY_EXISTS for a prefix in use.
2	list(page, size, search, status, sort)	GET /api/v1/warehouses: Lists warehouses, paginated; the search text matches prefix or name.
3	get(warehouseId)	GET /api/v1/warehouses/{warehouseId}: Returns one warehouse; 404 WAREHOUSE_NOT_FOUND.
4	update(warehouseId, request)	PUT /api/v1/warehouses/{warehouseId}: Edits name and addresses and resizes the map under the warehouse lock; refused if a shelf or area would fall outside the new frame (409 LAYOUT_OUT_OF_BOUNDS).
5	activate(warehouseId)	POST /api/v1/warehouses/{warehouseId}/activation: Sets the warehouse ACTIVE.
6	deactivate(warehouseId)	POST /api/v1/warehouses/{warehouseId}/deactivation: Sets the warehouse INACTIVE; no cascade and no stock check (Warehouse.deactivate, issue #18).

Table 77. WarehouseController Class Specification

ZoneController

No	Method	Description
1	create(warehouseId, request)	POST /api/v1/warehouses/{warehouseId}/zones: Creates a zone (name, colour) under the warehouse lock; the name must be unique in the warehouse.
2	list(warehouseId)	GET /api/v1/warehouses/{warehouseId}/zones: Lists the zones of a warehouse by name.
3	update(zoneId, request)	PUT /api/v1/zones/{zoneId}: Renames or recolours a zone (name still unique, 409 ZONE_NAME_ALREADY_EXISTS).

Table 78. ZoneController Class Specification

AreaController

No	Method	Description
1	create(warehouseId, request)	POST /api/v1/warehouses/{warehouseId}/areas: Places an area (RECEIVING, QUALITY_CONTROL, QUARANTINE, ..., NON_STORAGE) on the map; a storage area gets a storage location; refused off the map (BR-06) or overlapping (BR-07).
2	update(areaId, request)	PUT /api/v1/areas/{areaId}: Moves, resizes or retypes an area; a storage area never becomes NON_STORAGE.

3	changeStatus(areaId, request)	PUT /api/v1/areas/{areaId}/status: Changes an area's status (and its location's).

Table 79. AreaController Class Specification

BoundaryController

No	Method	Description
1	create(warehouseId, request)	POST /api/v1/warehouses/{warehouseId}/boundaries: Draws a wall or door segment; it must lie on the map (409 LAYOUT_OUT_OF_BOUNDS).
2	update(boundaryId, request)	PUT /api/v1/boundaries/{boundaryId}: Redraws a boundary.
3	delete(boundaryId, version)	DELETE /api/v1/boundaries/{boundaryId}: Deletes a boundary for good, against the version the client read.

Table 80. BoundaryController Class Specification

ShelfController

No	Method	Description
1	create(warehouseId, request)	POST /api/v1/warehouses/{warehouseId}/shelves: Places a shelf on the map; 409 SHELF_CODE_ALREADY_EXISTS, LAYOUT_OUT_OF_BOUNDS or LAYOUT_OVERLAP.
2	get(shelfId)	GET /api/v1/shelves/{shelfId}: Returns one shelf with its levels and bins.
3	update(shelfId, request)	PUT /api/v1/shelves/{shelfId}: Moves, resizes or edits a shelf; it must still fit the map and its bins.
4	changeStatus(shelfId, request)	PUT /api/v1/shelves/{shelfId}/status: Changes a shelf's status (ACTIVE, BLOCKED, MAINTENANCE, INACTIVE).
5	addLevel(shelfId, request)	POST /api/v1/shelves/{shelfId}/levels: Adds a level; at most 20 levels (409 SHELF_CAPACITY_EXCEEDED).
6	updateLevel(shelfId, levelId, request)	PUT /api/v1/shelves/{shelfId}/levels/{levelId}: Edits a level's measures.
7	addBin(shelfId, levelId, request)	POST /api/v1/shelves/{shelfId}/levels/{levelId}/bins: Adds one bin and its storage location to a level; it must fit the level and not overlap another bin.
8	updateBin(shelfId, levelId, binId, request)	PUT /api/v1/shelves/{shelfId}/levels/{levelId}/bins/{binId}: Edits a bin and its storage-location settings (pickable, putaway target, capacity).

9	changeBinStatus(shelfId, levelId, binId, request)	PUT /api/v1/shelves/{shelfId}/levels/{levelId}/bins/{binId}/status: Changes a bin's location status; a bin coming back from INACTIVE must still be placeable.
10	generateBins(shelfId, request)	POST /api/v1/shelves/{shelfId}/bin-generation: Fills empty levels with a rows x columns grid of equal bins (BR-14), all levels or none: 404 SHELF_LEVEL_NOT_FOUND, 409 SHELF_LEVEL_HAS_BINS, above 200 bins a level 409 SHELF_CAPACITY_EXCEEDED.

Table 81. ShelfController Class Specification

LayoutController

No	Method	Description
1	layout(warehouseId)	GET /api/v1/warehouses/{warehouseId}/layout: Returns the whole map (zones, shelves, levels, bins, areas, boundaries) with each place's effective status.
2	location(locationCode)	GET /api/v1/locations/{locationCode}: Looks up a storage location by its scanned code; 404 LOCATION_NOT_FOUND.

Table 82. LayoutController Class Specification

LocationLabelController

No	Method	Description
1	labels(warehouseId, request)	POST /api/v1/warehouses/{warehouseId}/location-labels: Prints location labels (text, QR, Code 128) as a PDF for a shelf, zone, list of codes or the whole warehouse; 422 LOCATION_LABELS_EMPTY, more than 2000 labels 422 LOCATION_LABELS_TOO_MANY.

Table 83. LocationLabelController Class Specification

3.13.3 Configure Warehouse Map & Locations (Warehouse and Area) Sequence Diagram

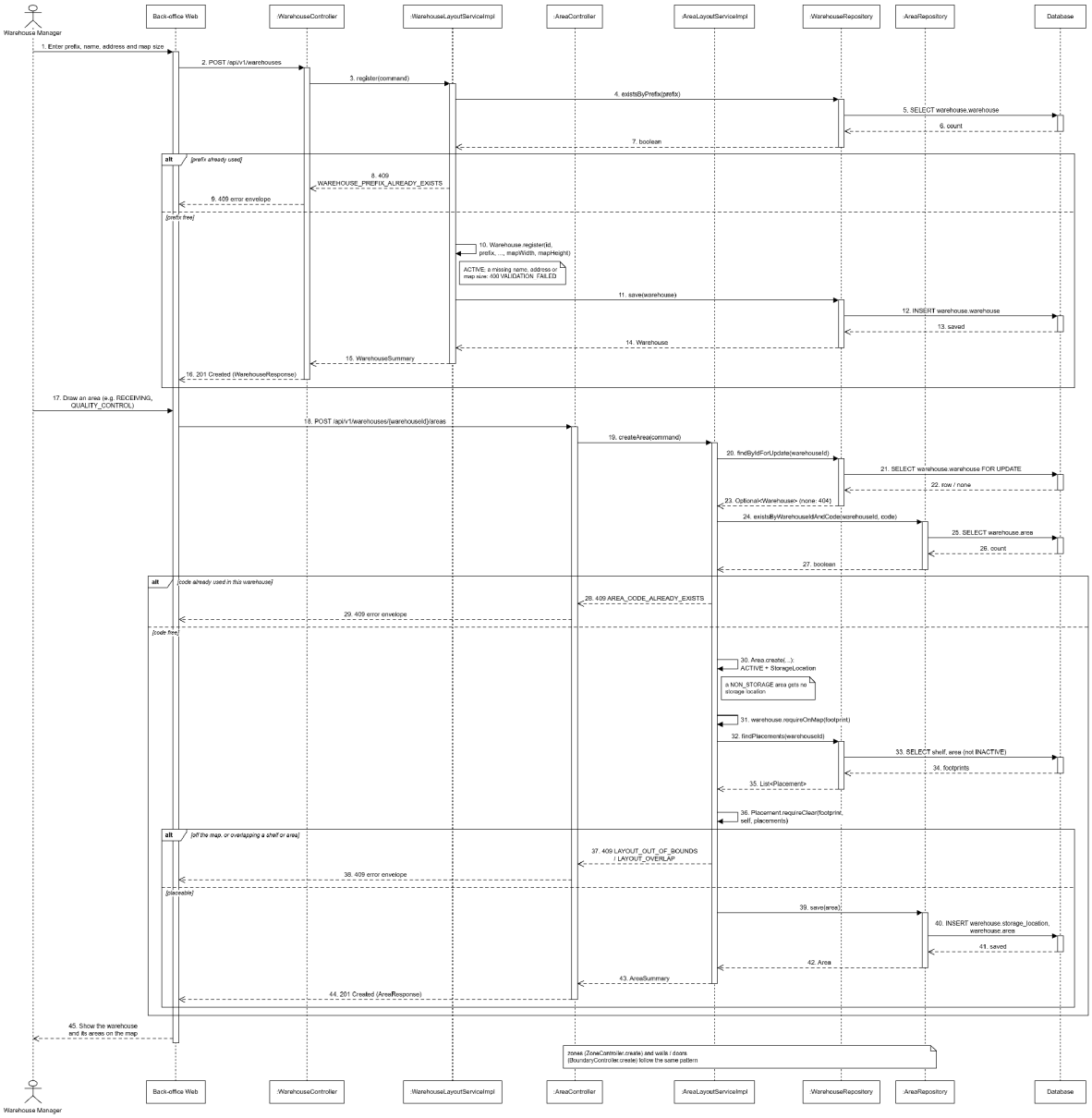


Figure 217. Configure Warehouse Map & Locations (Warehouse and Area) Sequence Diagram

3.13.4 Configure Warehouse Map & Locations (Shelf) Sequence Diagram

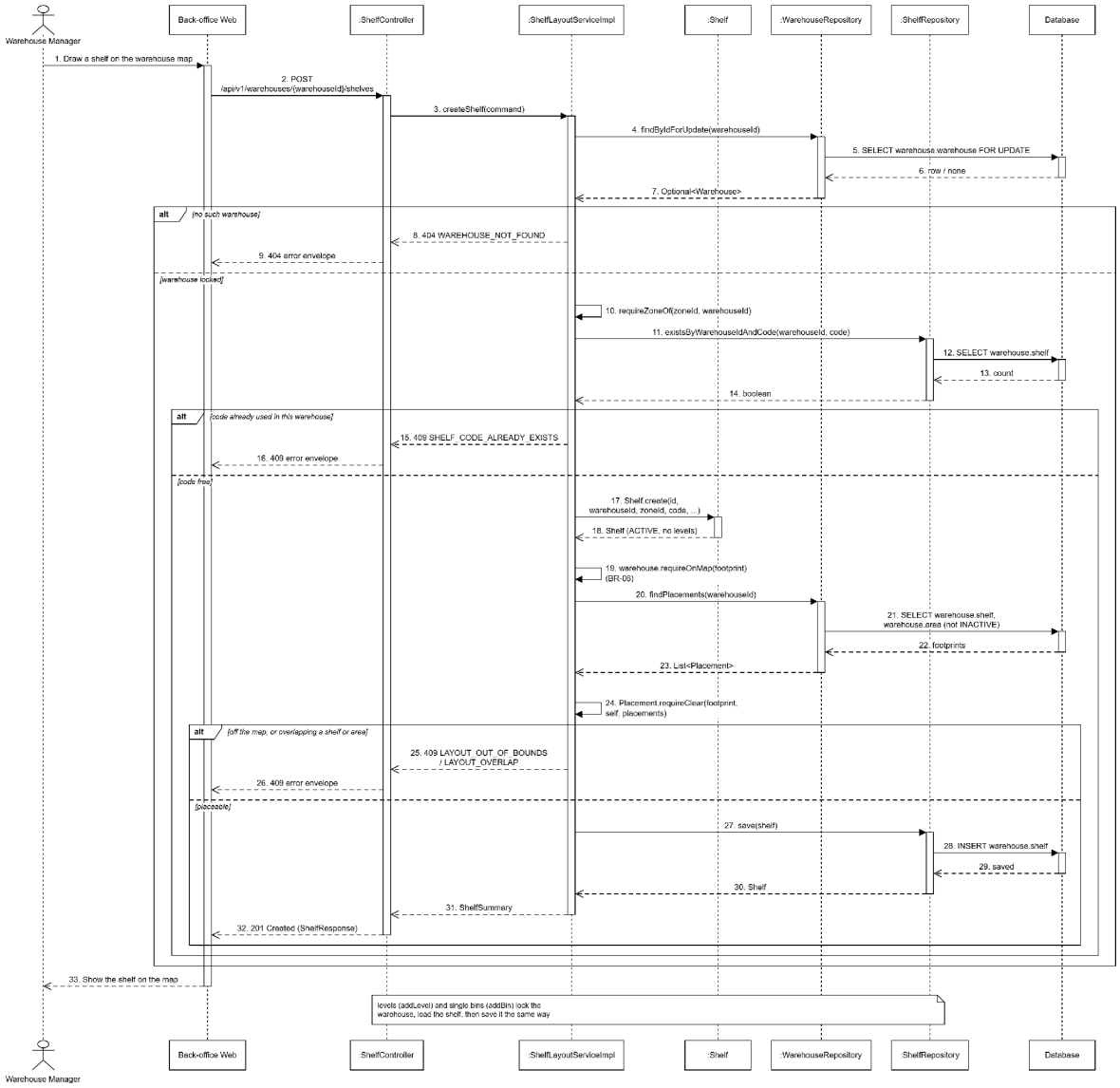


Figure 218. Configure Warehouse Map & Locations (Shelf) Sequence Diagram

3.13.5 Auto-generate Bins Sequence Diagram

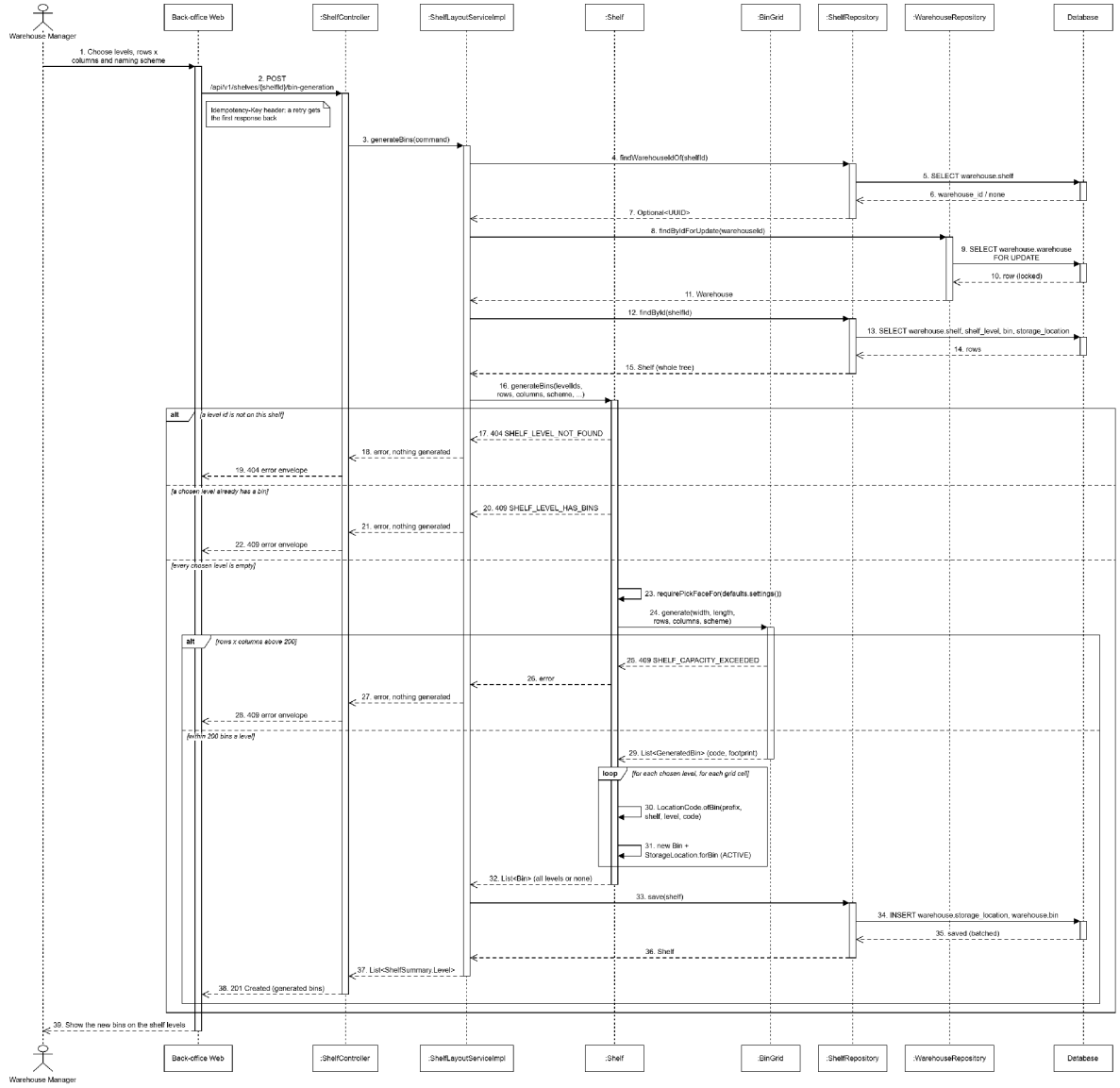


Figure 219. Auto-generate Bins Sequence Diagram

3.13.6 View Map & Print Location Labels (View Map) Sequence Diagram

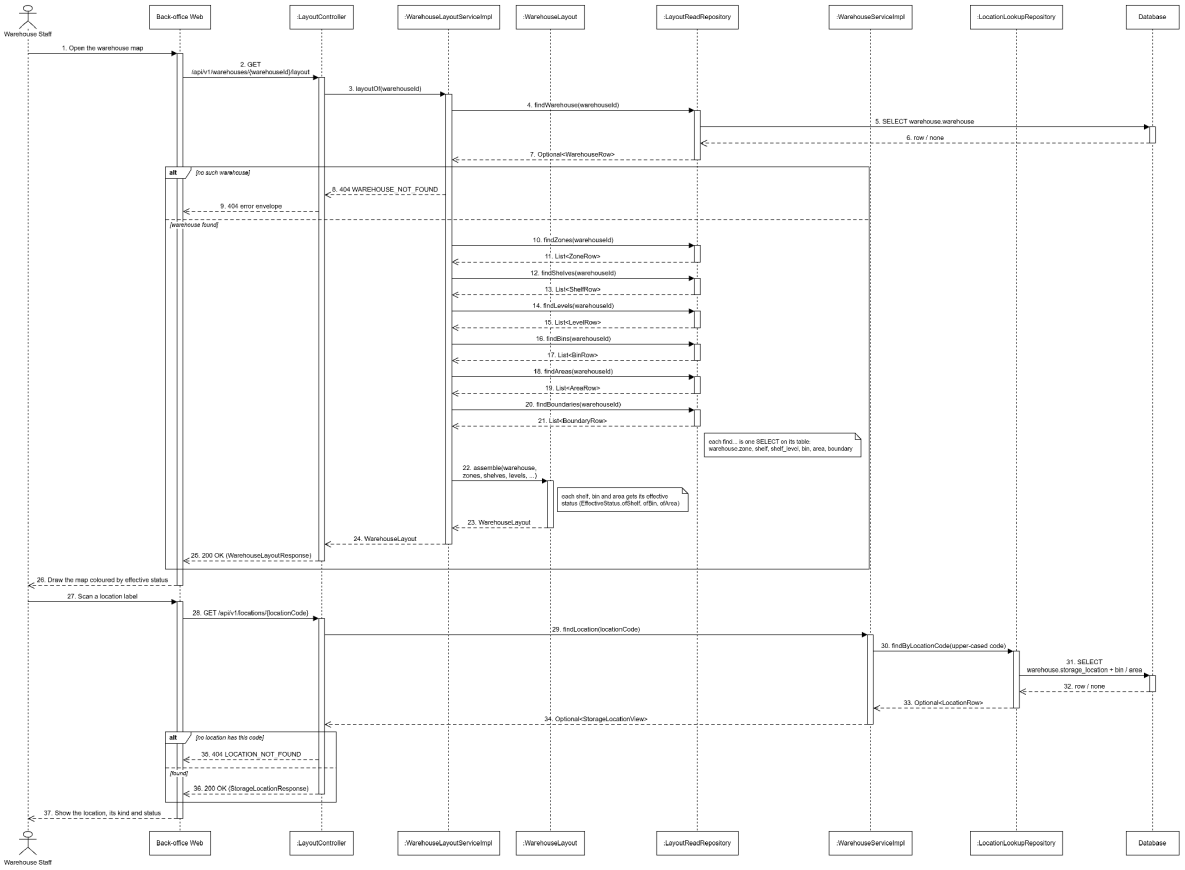


Figure 220. View Map & Print Location Labels (View Map) Sequence Diagram

3.13.7 View Map & Print Location Labels (Print Labels) Sequence Diagram

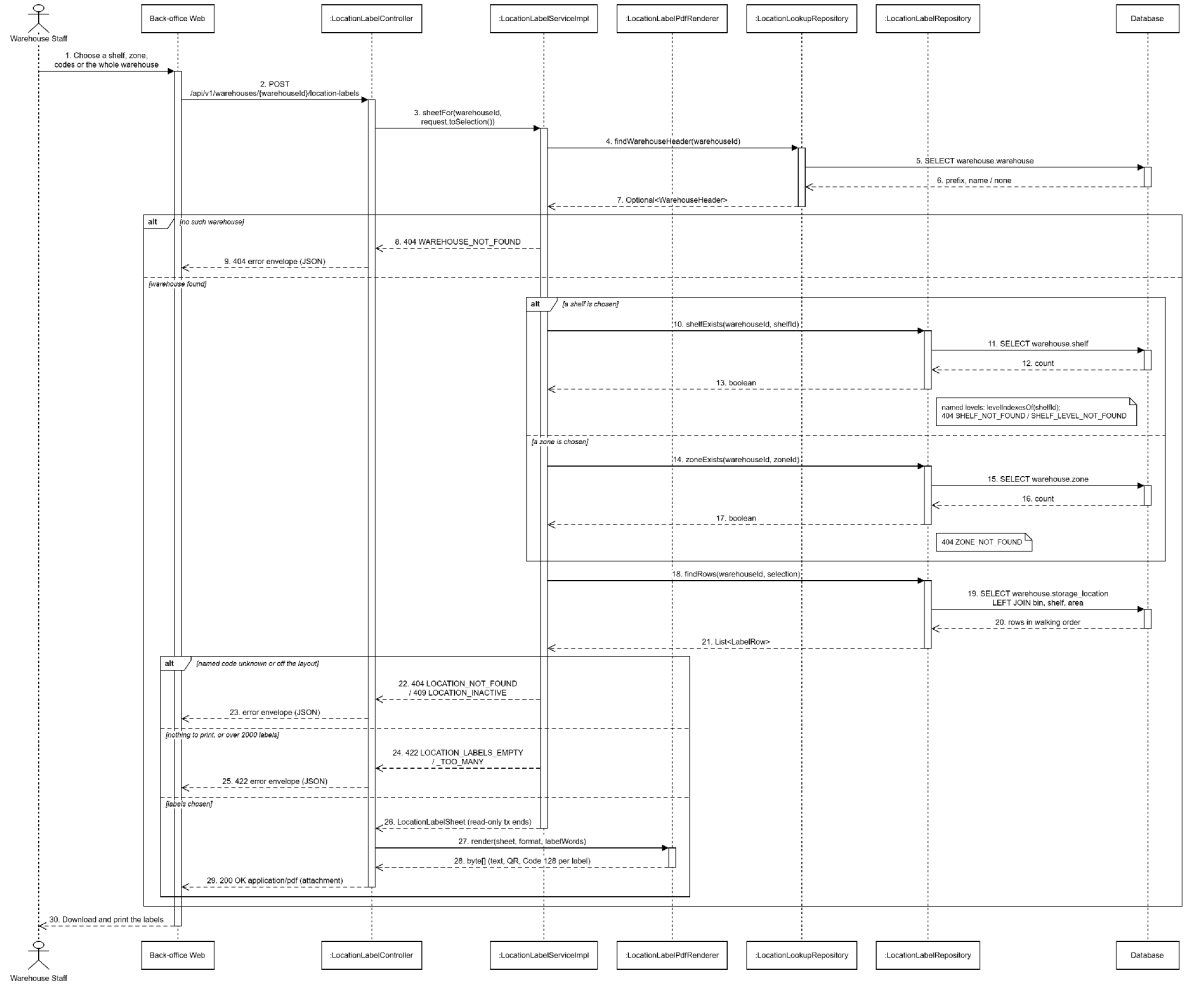


Figure 221. View Map & Print Location Labels (Print Labels) Sequence Diagram

3.14 Procurement

3.14.1 Class Diagram

Figure 222. Procurement Class Diagram (1/5): Suppliers and Supplier Catalog

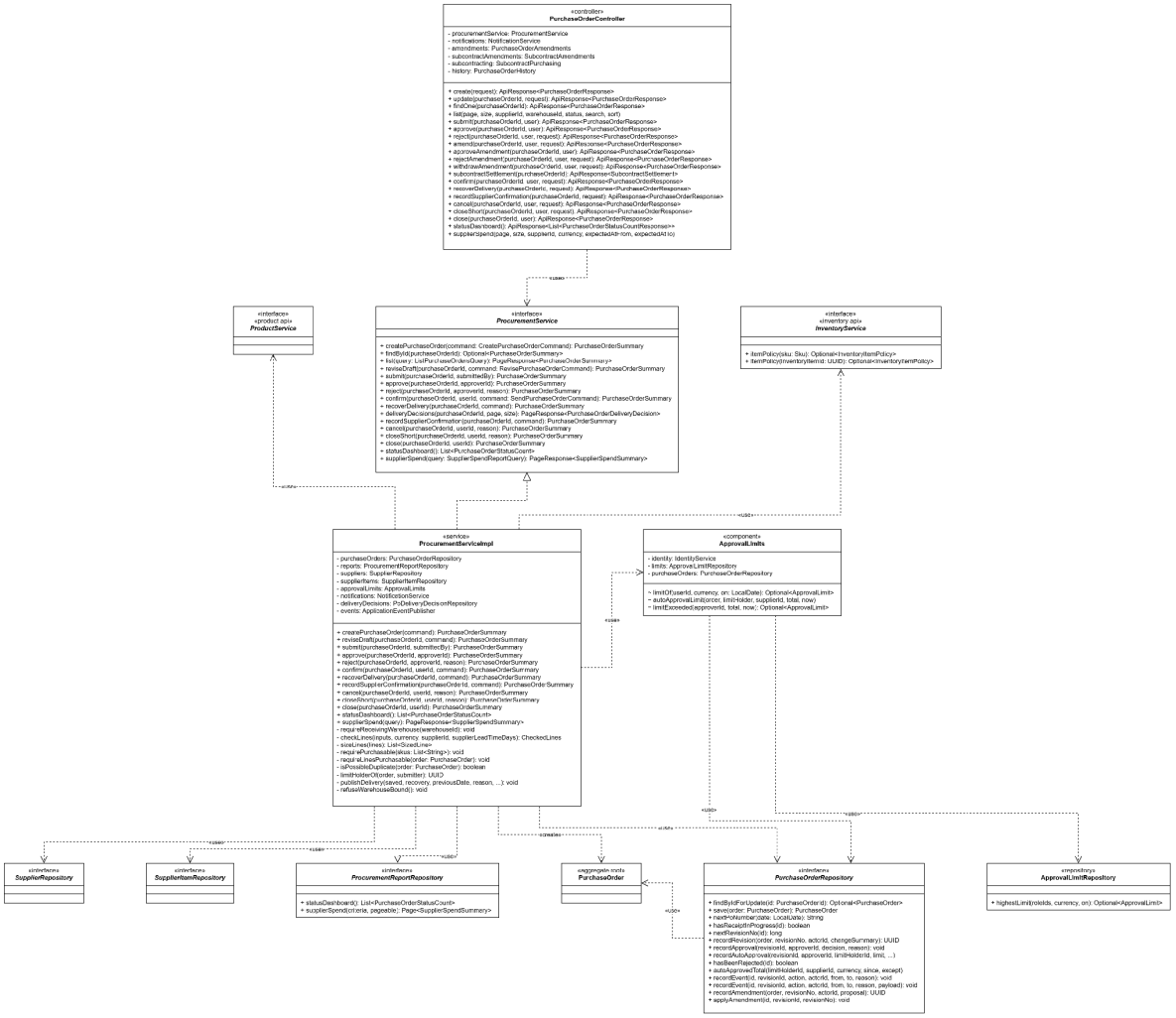


Figure 223. Procurement Class Diagram (2/5): Purchase Order Workflow

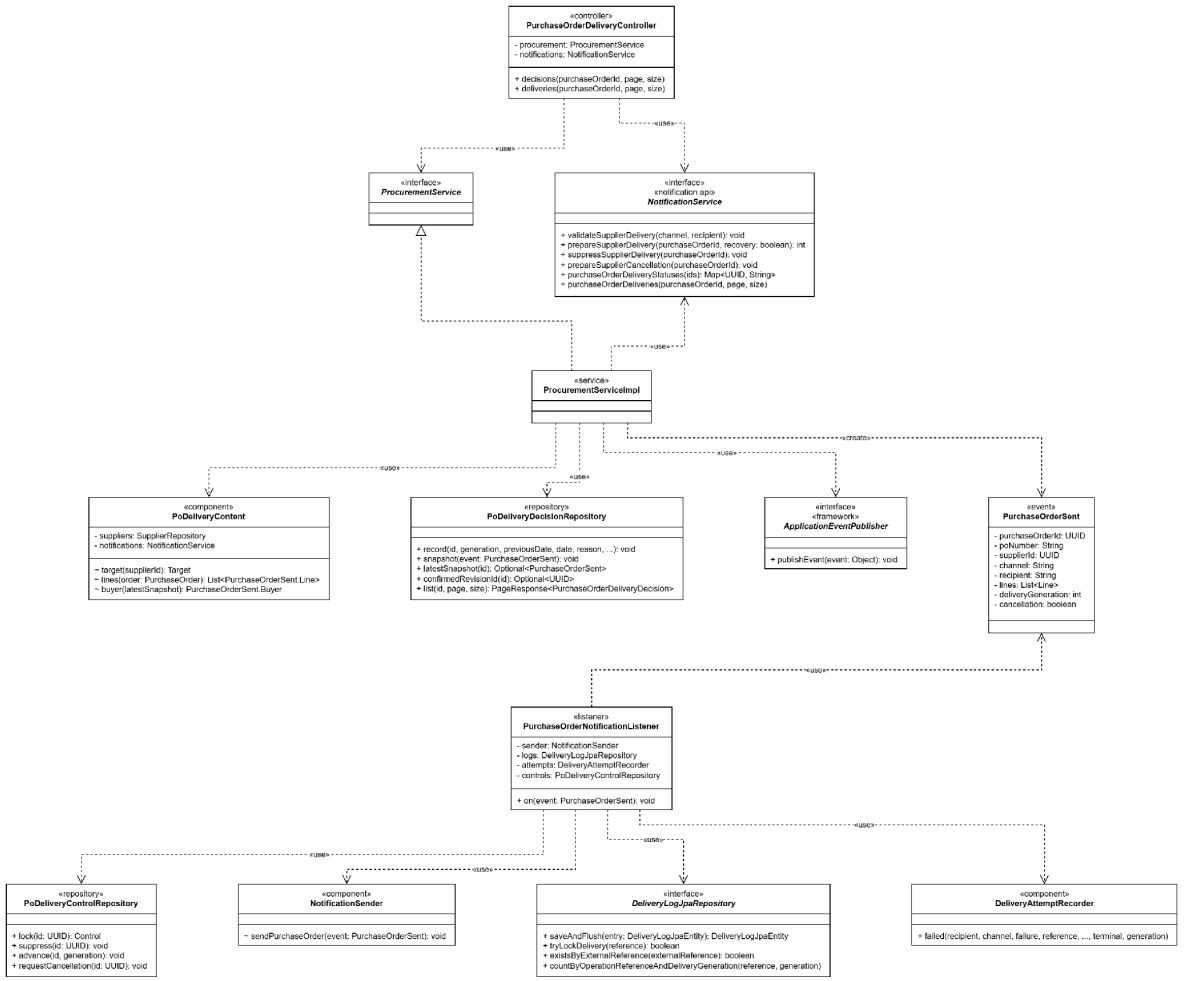


Figure 224. Procurement Class Diagram (3/5): Sending Orders to Suppliers

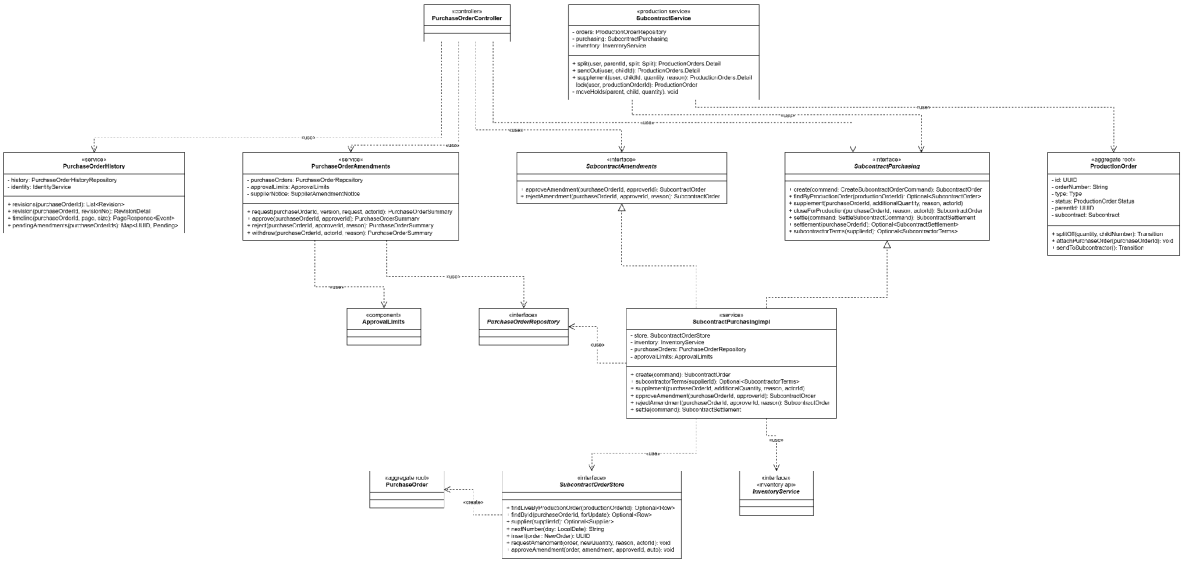


Figure 225. Procurement Class Diagram (4/5): Amendments and Subcontract Orders

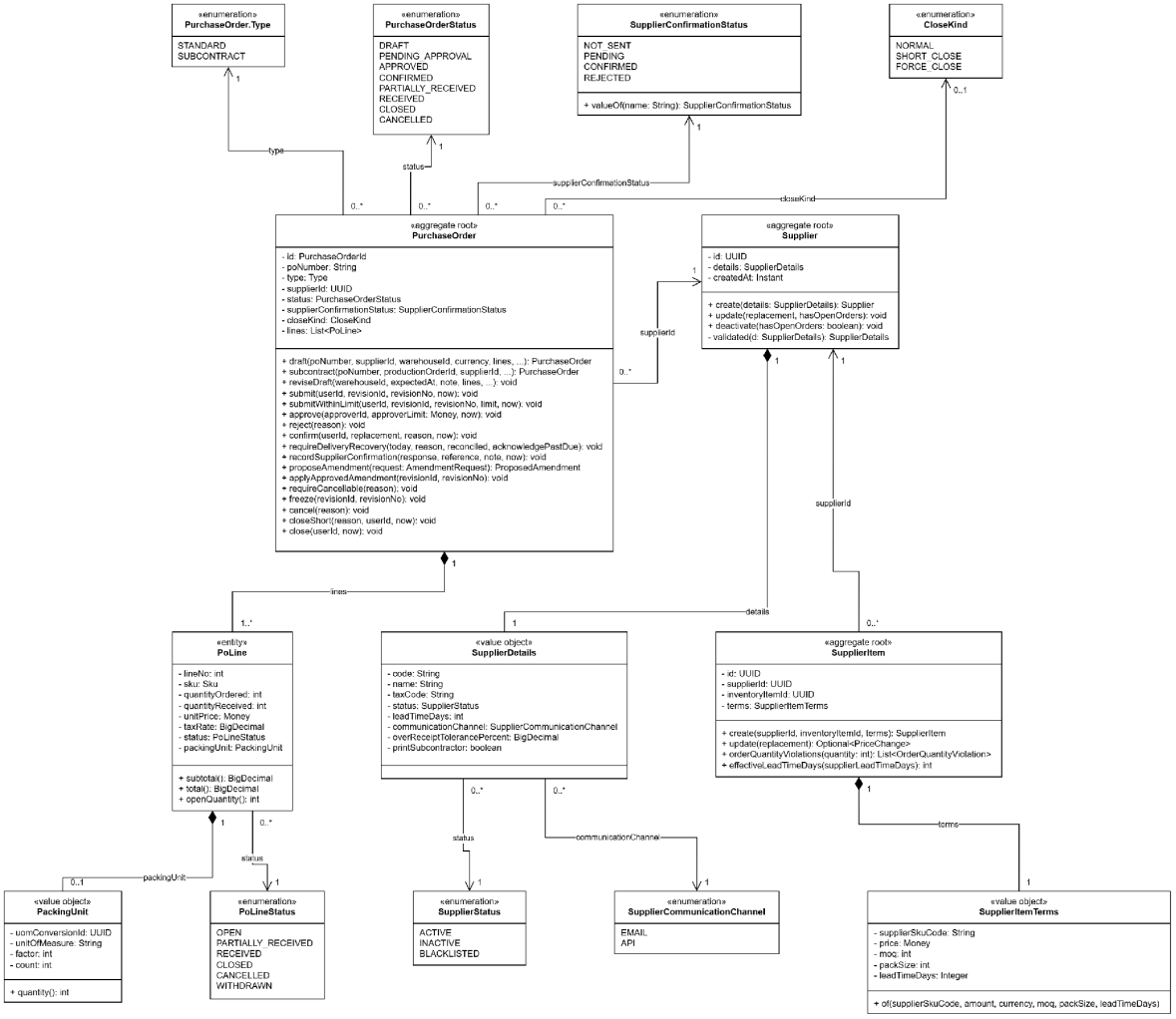


Figure 226. Procurement Class Diagram (5/5): Domain Model

3.14.2 Class Specification

PurchaseOrderController

No	Method	Description
1	create(request)	POST /api/v1/purchase-orders: Creates a DRAFT standard PO for an ACTIVE supplier; every line must be listed in the supplier catalog and meet MOQ and pack multiple (409 SUPPLIER_TERMS_VIOLATED); "5 CARTON" is sized to base units.
2	update(purchaseOrderId, request)	PUT /api/v1/purchase-orders/{purchaseOrderId}: Revises a DRAFT standard PO (lines, date, warehouse, note) with the same catalog checks; 409 PURCHASE_ORDER_NOT_EDITABLE or OPTIMISTIC_LOCK.
3	findOne(purchaseOrderId)	GET /api/v1/purchase-orders/{purchaseOrderId}: Returns one PO with its lines and delivery status; 404 when unknown.

4	list(page, size, supplierId, warehouseId, status, search, sort)	GET /api/v1/purchase-orders: Lists POs, paginated and filtered, within the caller's warehouse scope.
5	submit(purchaseOrderId, user)	POST /api/v1/purchase-orders/{purchaseOrderId}/submission: Freezes the draft as a revision and sends it for approval; auto-approved when the submitter's own limit covers the total and the 7-day auto-approved sum.
6	approve(purchaseOrderId, user)	POST /api/v1/purchase-orders/{purchaseOrderId}/approval: Approves a PENDING_APPROVAL PO; refused for the submitter (409 SELF_APPROVAL_NOT_ALLOWED) or above the approver's limit (409 PO_APPROVAL_LIMIT_EXCEEDED).
7	reject(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/rejection: Sends a PENDING_APPROVAL PO back to DRAFT with a reason.
8	amend(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/amendments: Amends an APPROVED or CONFIRMED standard PO (the header as it should read and every kept line, with the version read). It is approved at once within the requester's approval limit, otherwise it waits for another approver (409 PO_AMENDMENT_NOT_ALLOWED once receiving has started, 409 PO_AMENDMENT_ALREADY_PENDING, 400 PO_AMENDMENT_NO_CHANGE).
9	approveAmendment(purchaseOrderId, user)	POST /api/v1/purchase-orders/{purchaseOrderId}/amendment/approval: Approves the quantity amendment waiting on a SUBCONTRACT PO (four eyes, approver's limit).
10	rejectAmendment(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/amendment/rejection: Rejects the waiting SUBCONTRACT amendment with a reason.
11	withdrawAmendment(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/amendment/withdrawal: Withdraws the amendment still waiting on an approved standard PO, with a reason; anyone who may update the order may do it, the requester included.

12	subcontractSettlement(purchaseOrderId)	GET /api/v1/purchase-orders/{purchaseOrderId}/subcontract-settlement: Returns the settlement of a SUBCONTRACT PO (BR-PRD-17); 404 while not settled.
13	confirm(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/confirmation (alias /sending): Confirms an APPROVED PO (delivery date required) and queues it for e-mail / API delivery to the supplier; supplier response becomes PENDING.
14	recoverDelivery(purchaseOrderId, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/delivery-recovery: Resends a PO whose delivery failed terminally, with a reason and reconciled=true; never changes the PO status.
15	recordSupplierConfirmation(purchaseOrderId, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/supplier-confirmation: Records the supplier's CONFIRMED or REJECTED answer (a rejection needs a note and is refused after receipt) and stops further delivery.
16	cancel(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/cancellation: Cancels a DRAFT, PENDING_APPROVAL, APPROVED or CONFIRMED PO with a reason; refused while a receipt is DRAFT or PENDING_APPROVAL; a CONFIRMED PO sends a cancellation notice.
17	closeShort(purchaseOrderId, user, request)	POST /api/v1/purchase-orders/{purchaseOrderId}/closure-short: Closes a PARTIALLY_RECEIVED PO short, or force-closes a CONFIRMED one (no receipt in progress), with a reason.
18	close(purchaseOrderId, user)	POST /api/v1/purchase-orders/{purchaseOrderId}/closure: Closes a fully RECEIVED PO.
19	statusDashboard()	GET /api/v1/purchase-orders/reports/status-dashboard: Counts POs per status; refused to warehouse-bound callers (403 OUT_OF_DATA_SCOPE).
20	supplierSpend(page, size, supplierId, currency, expectedAtFrom, expectedAtTo)	GET /api/v1/purchase-orders/reports/supplier-spend: Sums spend per supplier and currency, highest first; company-wide callers only.

Table 84. PurchaseOrderController Class Specification

PurchaseOrderDeliveryController

No	Method	Description
1	decisions(purchaseOrderId, page, size)	GET /api/v1/purchase-orders/{purchaseOrderId}/delivery-decisions: Lists who authorised the first delivery or a recovery, with date changes and reconciliation reasons.
2	deliveries(purchaseOrderId, page, size)	GET /api/v1/purchase-orders/{purchaseOrderId}/deliveries: Lists the delivery attempts (SENT / FAILED, terminal); transport acceptance is not supplier confirmation.

Table 85. PurchaseOrderDeliveryController Class Specification

SupplierController

No	Method	Description
1	create(request)	POST /api/v1/suppliers: Creates a supplier profile; code and tax code unique (409), delivery channel's e-mail / API endpoint validated.
2	update(supplierId, request)	PUT /api/v1/suppliers/{supplierId}: Updates contacts, status and commercial defaults; code is immutable; leaving ACTIVE is refused while the supplier has open POs.
3	deactivate(supplierId)	DELETE /api/v1/suppliers/{supplierId}: Sets the supplier INACTIVE (kept for history); 409 SUPPLIER_HAS_OPEN_PURCHASE_ORDERS.
4	findOne(supplierId)	GET /api/v1/suppliers/{supplierId}: Returns one supplier profile.
5	list(page, size, search, status, printSubcontractor, sort)	GET /api/v1/suppliers: Searches suppliers by text and status; printSubcontractor=true lists print subcontractors.
6	performance(supplierId)	GET /api/v1/suppliers/{supplierId}/performance: Computes on-time delivery and QC acceptance rates from POs, receipts and inspections.

Table 86. SupplierController Class Specification

SupplierItemController

No	Method	Description
1	list(supplierId, page, size, sort)	GET /api/v1/suppliers/{supplierId}/items: Lists the SKUs a supplier sells with price, MOQ, pack size and lead time.

2	create(supplierId, request)	POST /api/v1/suppliers/{supplierId}/items: Lists a purchasable SKU for an ACTIVE supplier; optionally makes it the item's preferred supplier.
3	update(supplierId, supplierItemId, request)	PUT /api/v1/suppliers/{supplierId}/items/{supplierItemId}: Replaces a catalog row's terms (version required); a new price opens a new price period.
4	delete(supplierId, supplierItemId, version)	DELETE /api/v1/suppliers/{supplierId}/items/{supplierItemId}: Removes a catalog row unless an open PO still buys it (409 SUPPLIER_ITEM_IN_USE).
5	suppliersOf(sku)	GET /api/v1/procurement/skus/{sku}/suppliers: Lists every supplier that sells a SKU, preferred first, then by name.

Table 87. SupplierItemController Class Specification

3.14.3 Manage Suppliers & Supplier Catalog (Suppliers) Sequence Diagram

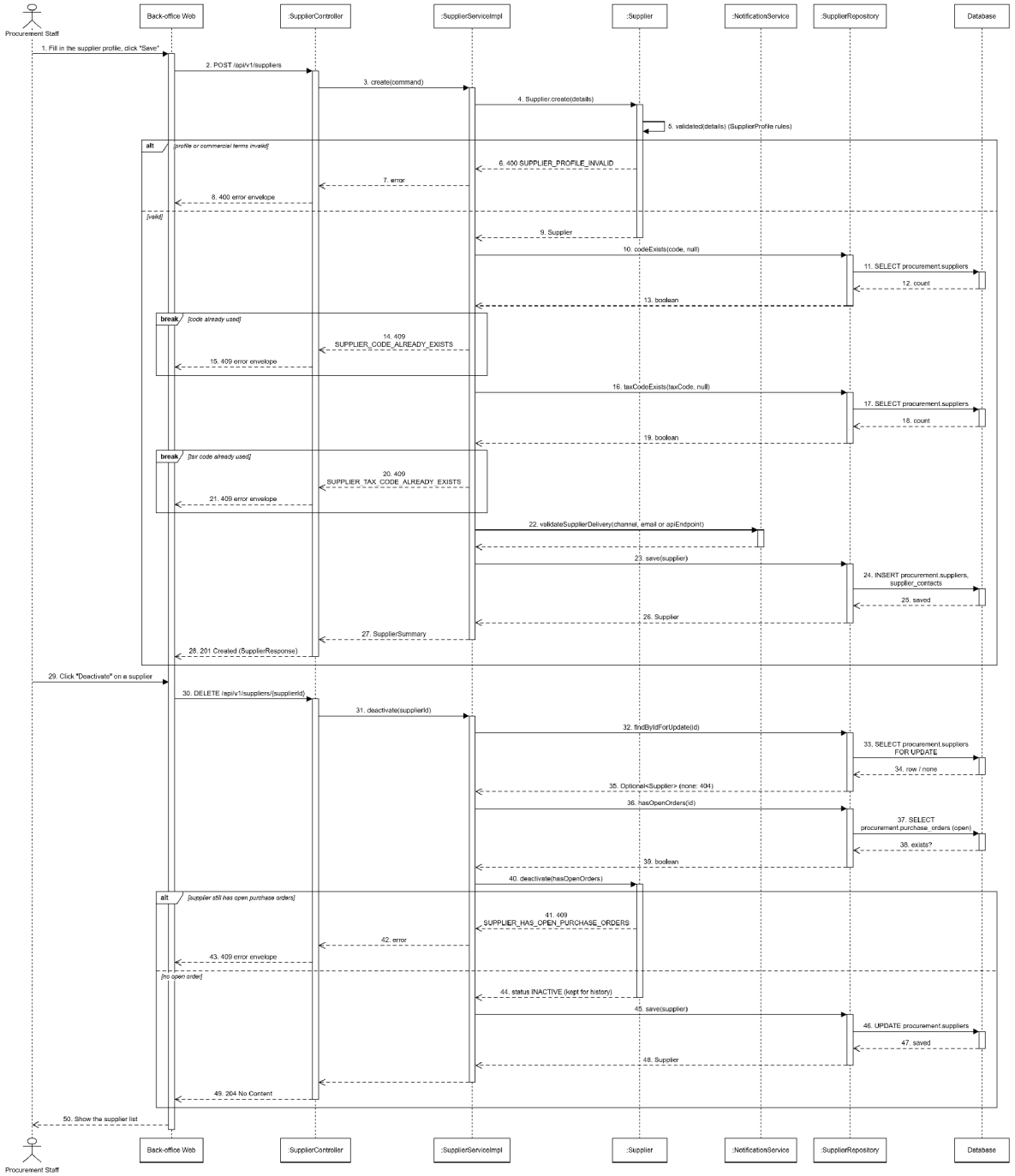


Figure 227. Manage Suppliers & Supplier Catalog (Suppliers) Sequence Diagram

3.14.4 Manage Suppliers & Supplier Catalog (Catalog) Sequence Diagram

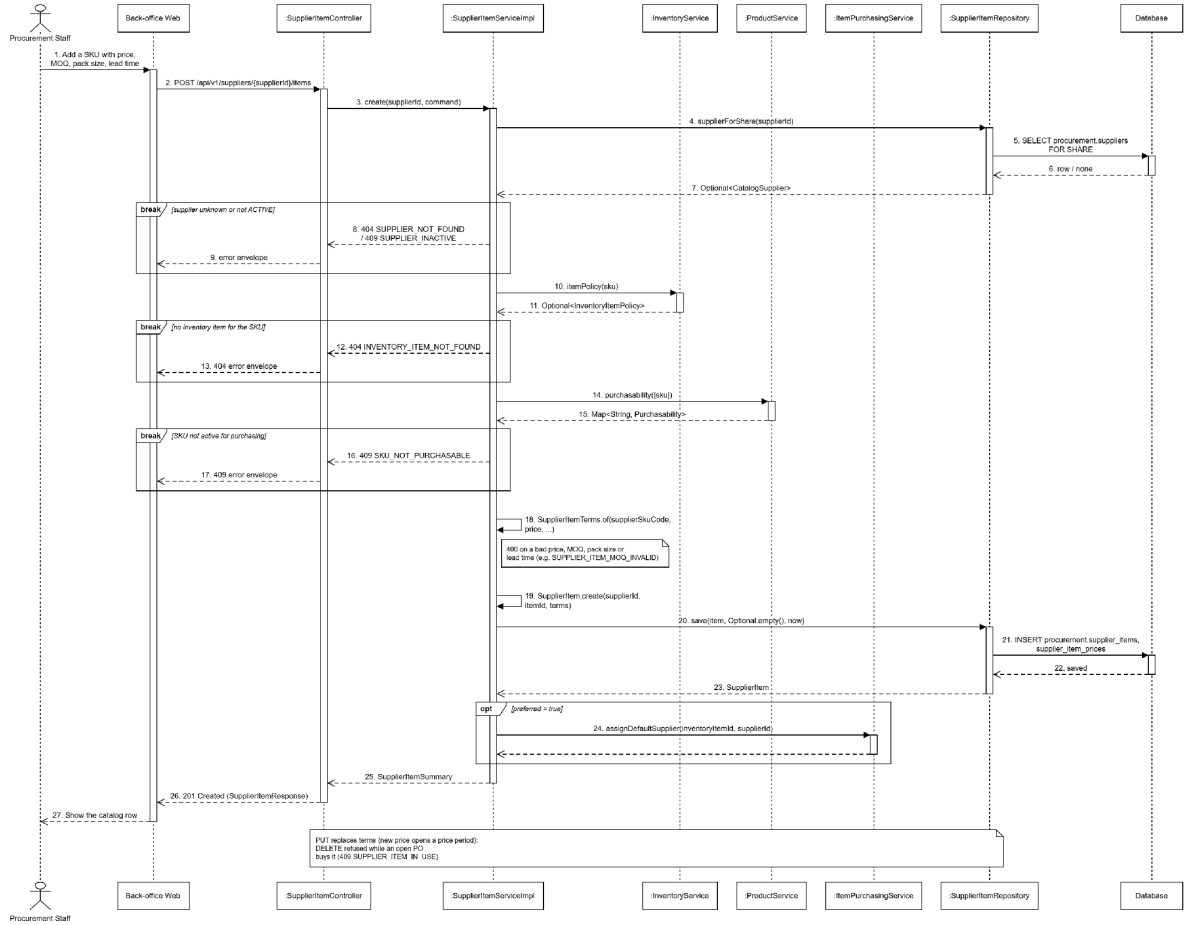


Figure 228. Manage Suppliers & Supplier Catalog (Catalog) Sequence Diagram

3.14.5 Create / Edit Purchase Order Sequence Diagram

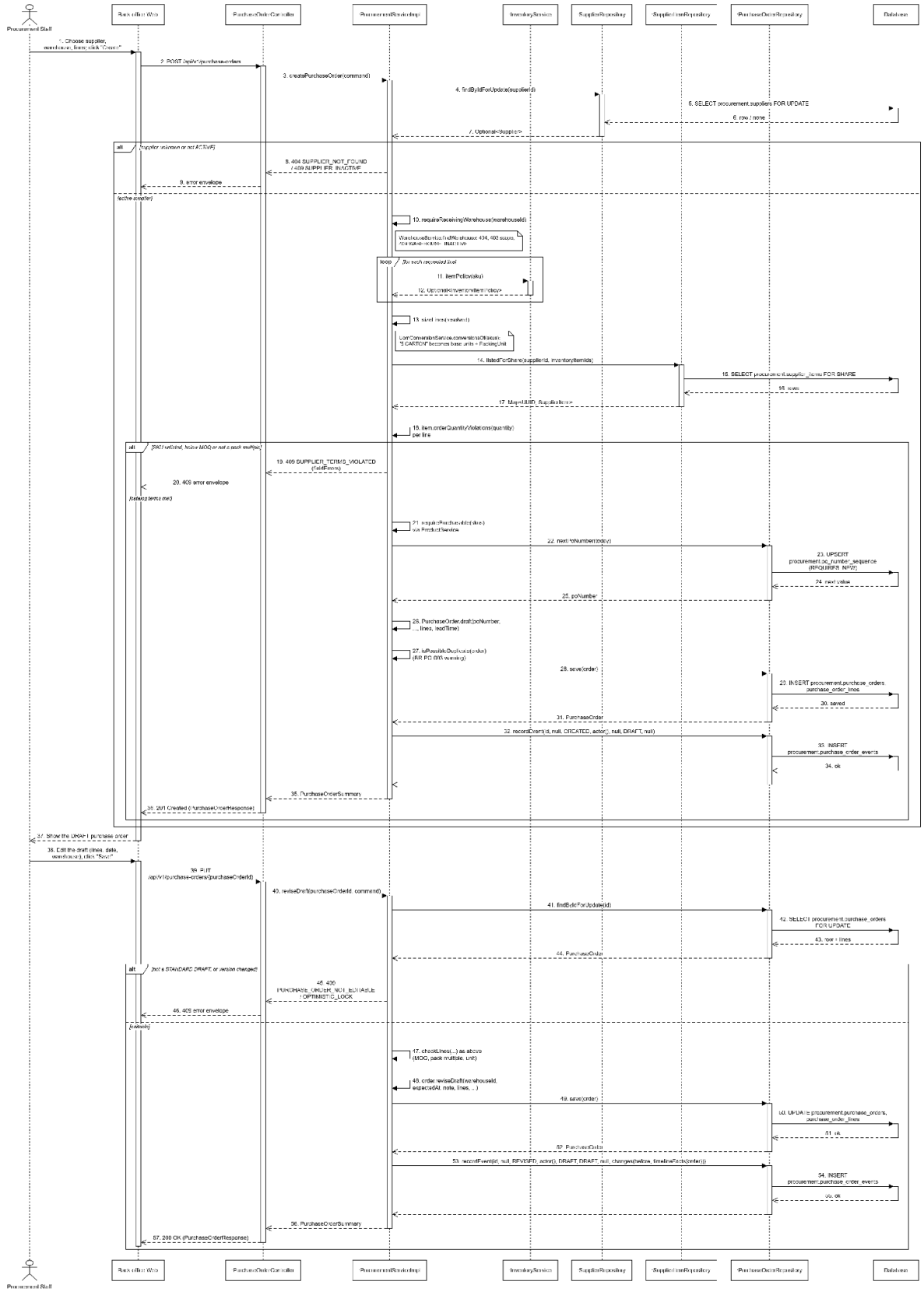


Figure 229. Create / Edit Purchase Order Sequence Diagram

3.14.6 Submit PO for Approval Sequence Diagram

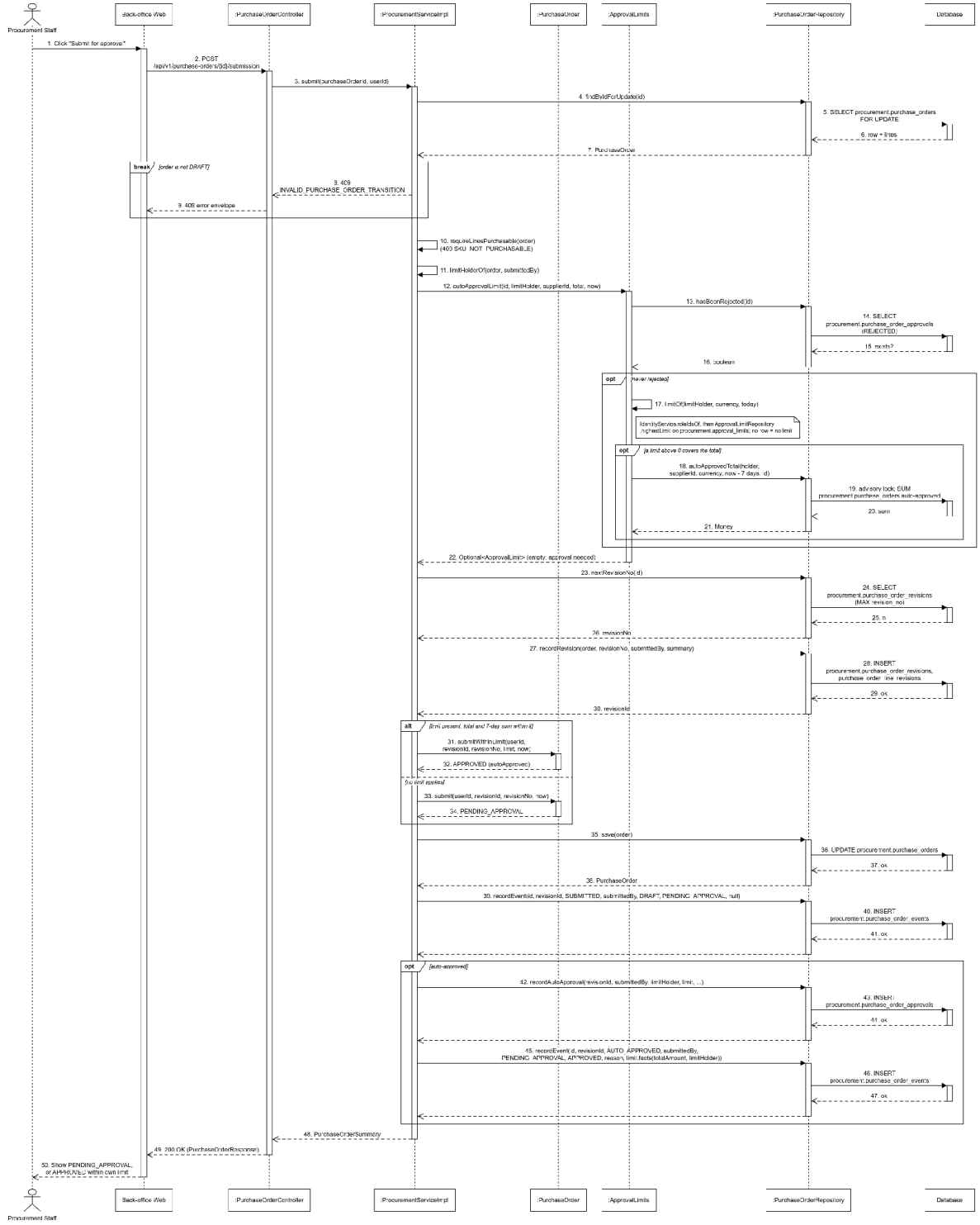


Figure 230. Submit PO for Approval Sequence Diagram

3.14.7 Approve / Reject Purchase Order Sequence Diagram

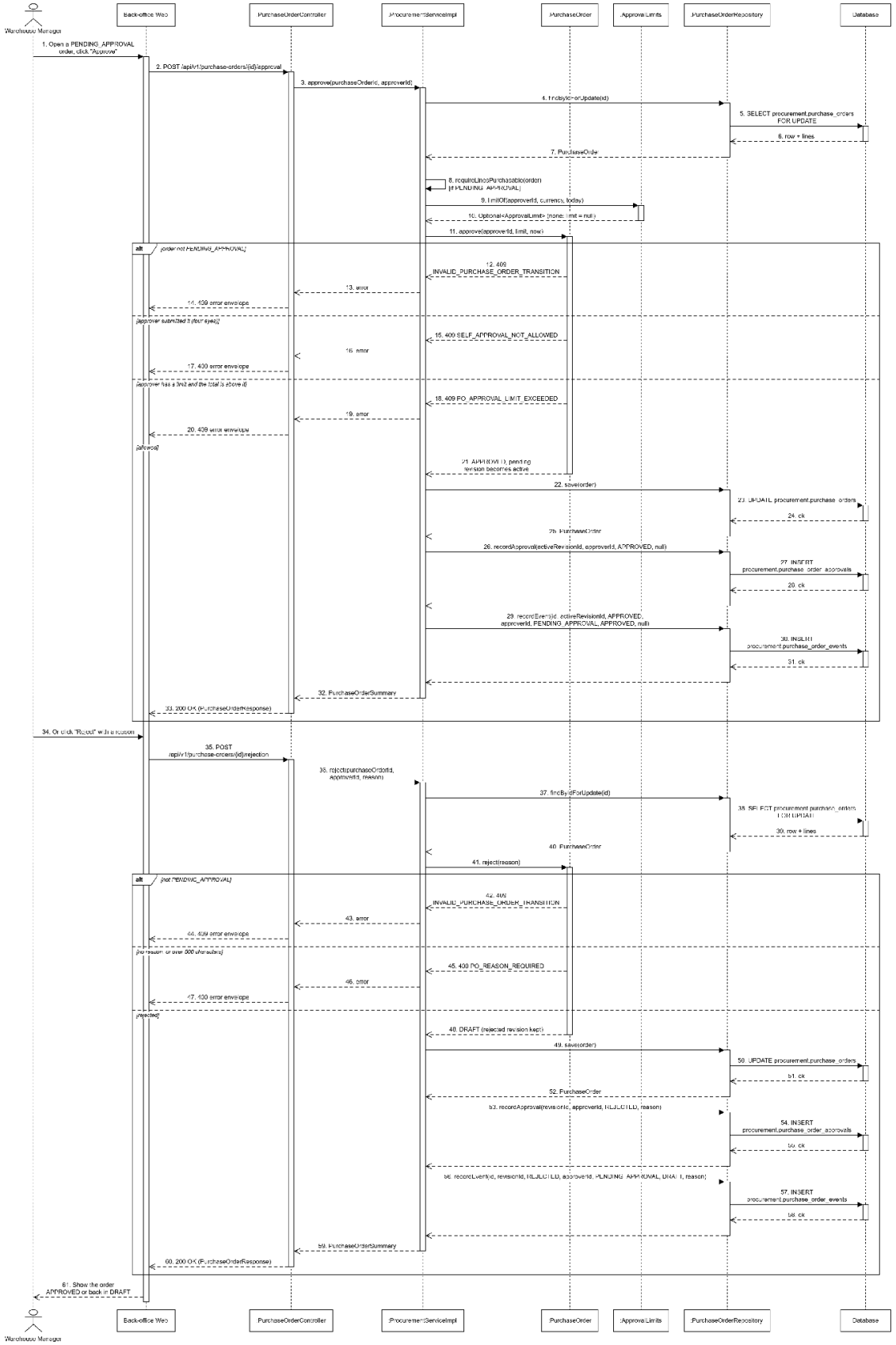


Figure 231. Approve / Reject Purchase Order Sequence Diagram

3.14.8 Confirm & Send PO to Supplier Sequence Diagram

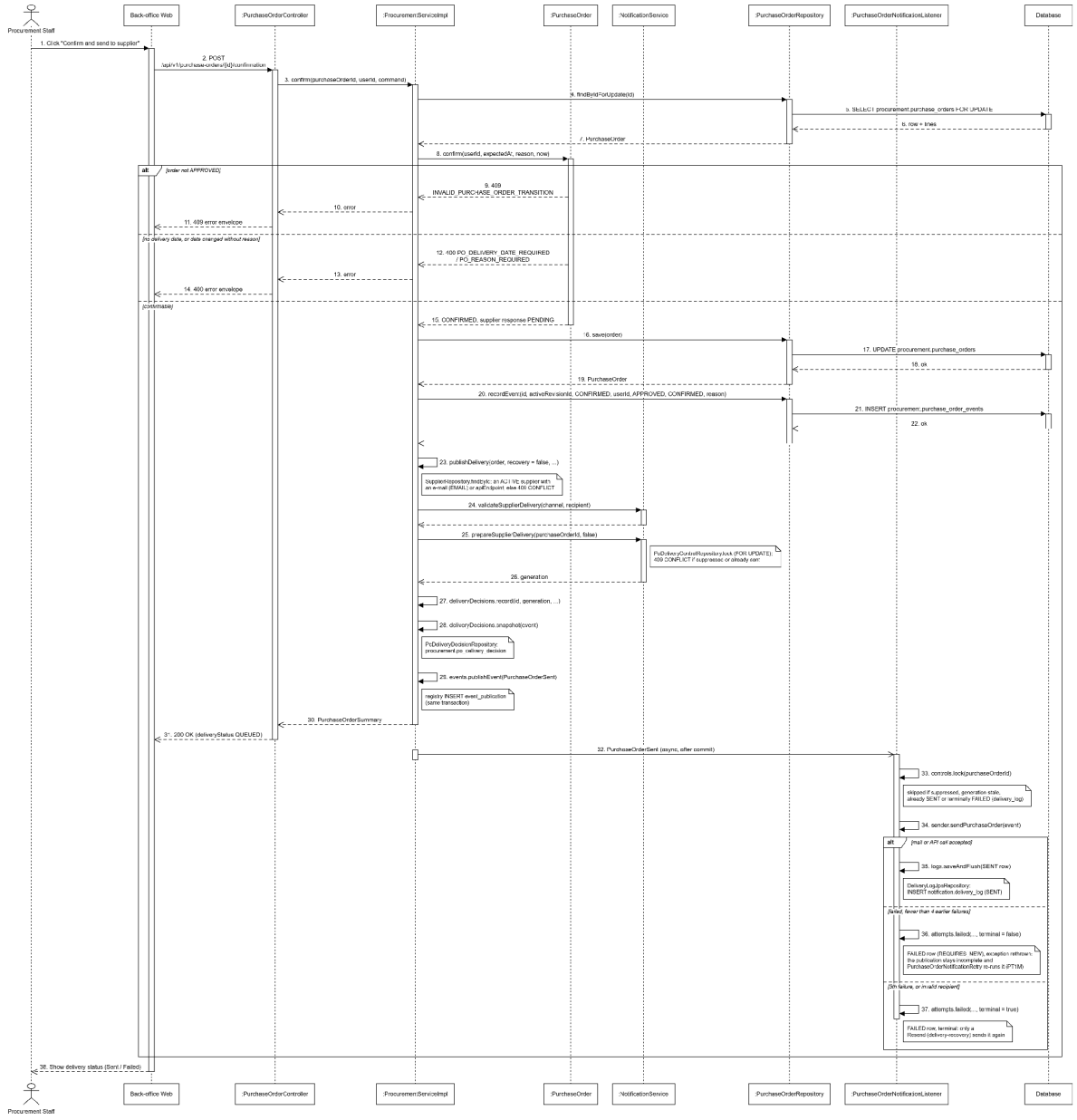


Figure 232. Confirm & Send PO to Supplier Sequence Diagram

3.14.9 Resend Failed PO Sequence Diagram

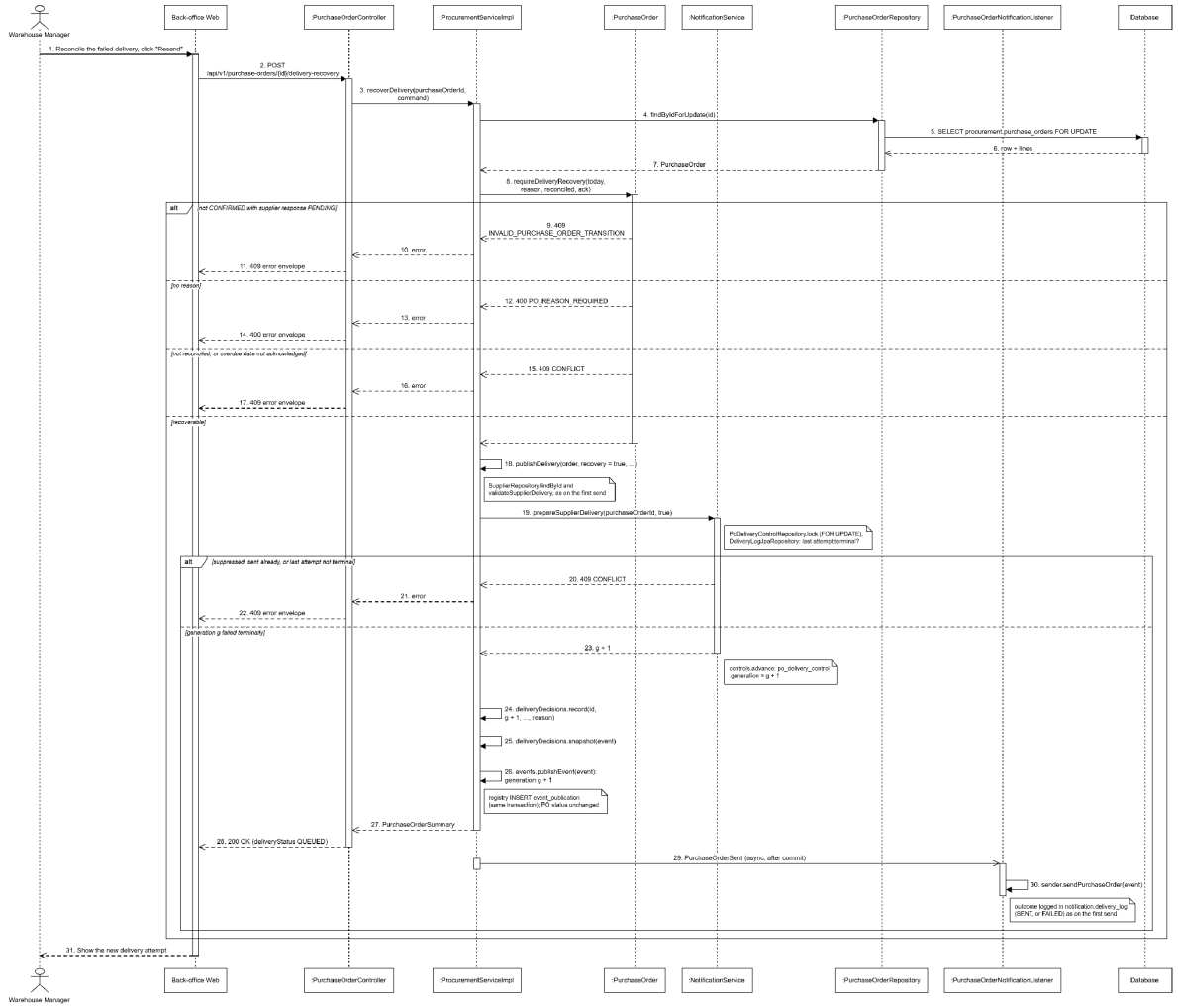


Figure 233. Resend Failed PO Sequence Diagram

3.14.10 Record Supplier Response Sequence Diagram

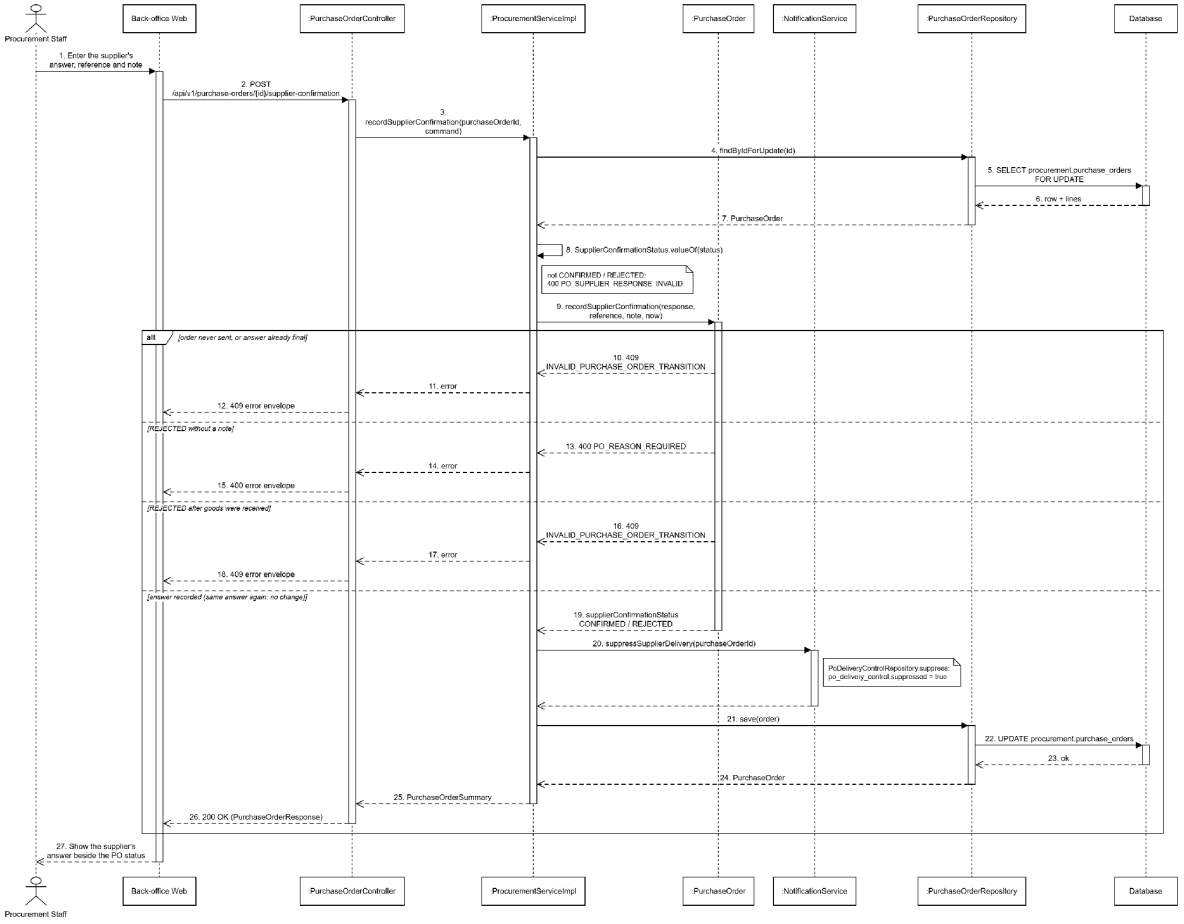


Figure 234. Record Supplier Response Sequence Diagram

3.14.11 Close / Cancel PO (Cancel) Sequence Diagram

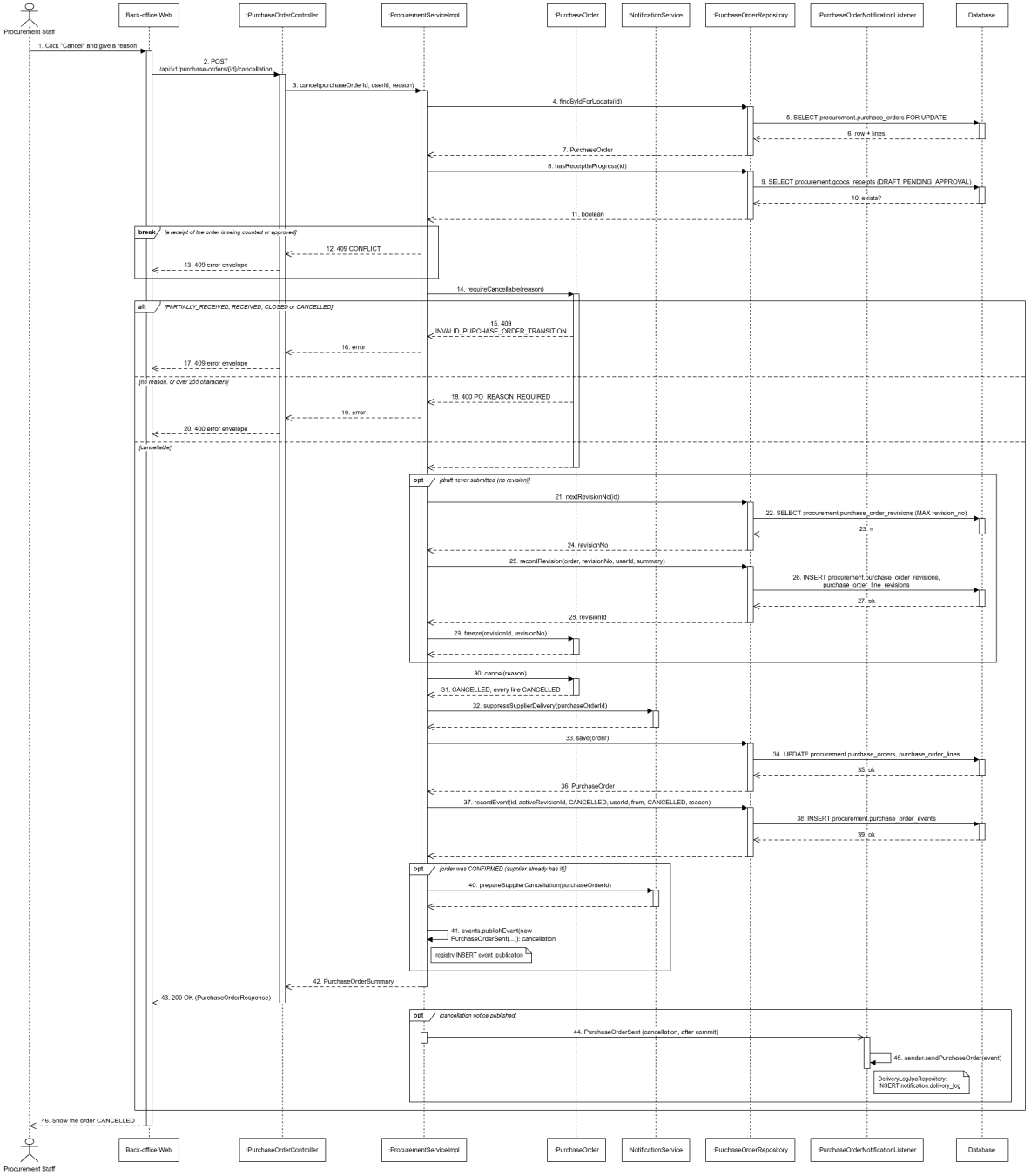


Figure 235. Close / Cancel PO (Cancel) Sequence Diagram

3.14.12 Close / Cancel PO (Close) Sequence Diagram

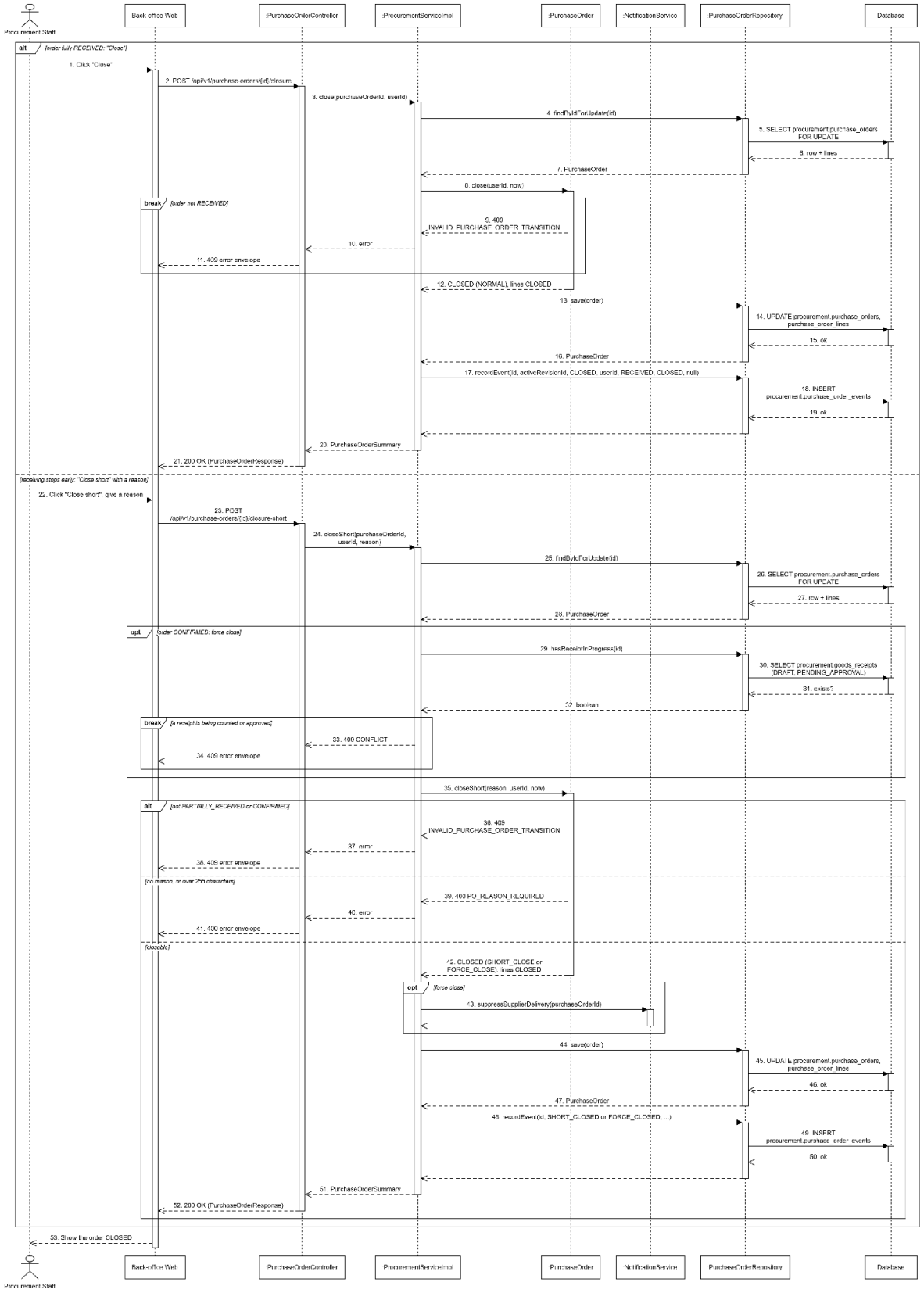


Figure 236. Close / Cancel PO (Close) Sequence Diagram

3.14.13 Create Subcontract PO (Procurement Side) Sequence Diagram

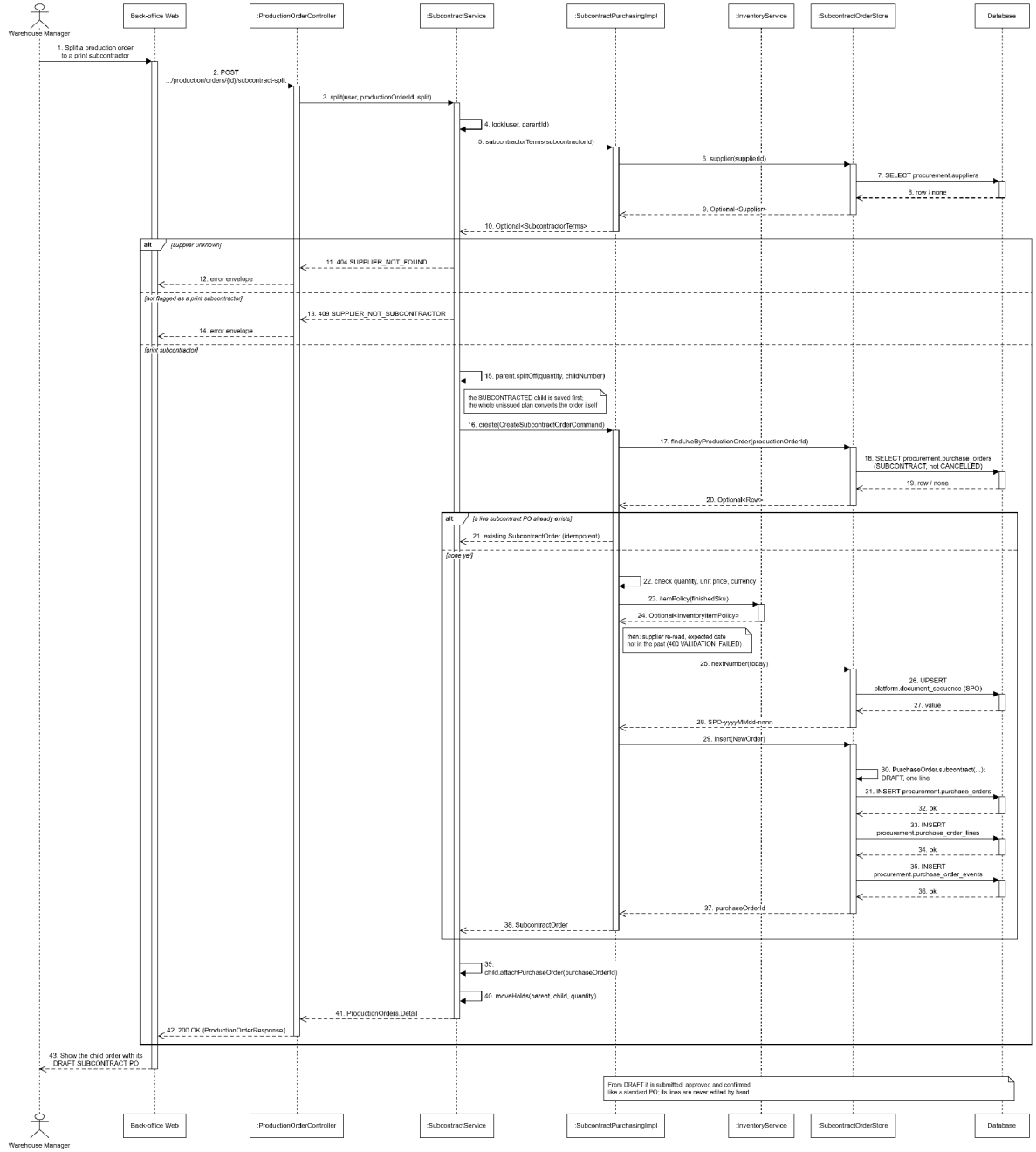


Figure 237. Create Subcontract PO (Procurement Side) Sequence Diagram

3.14.14 View Purchasing Reports Sequence Diagram

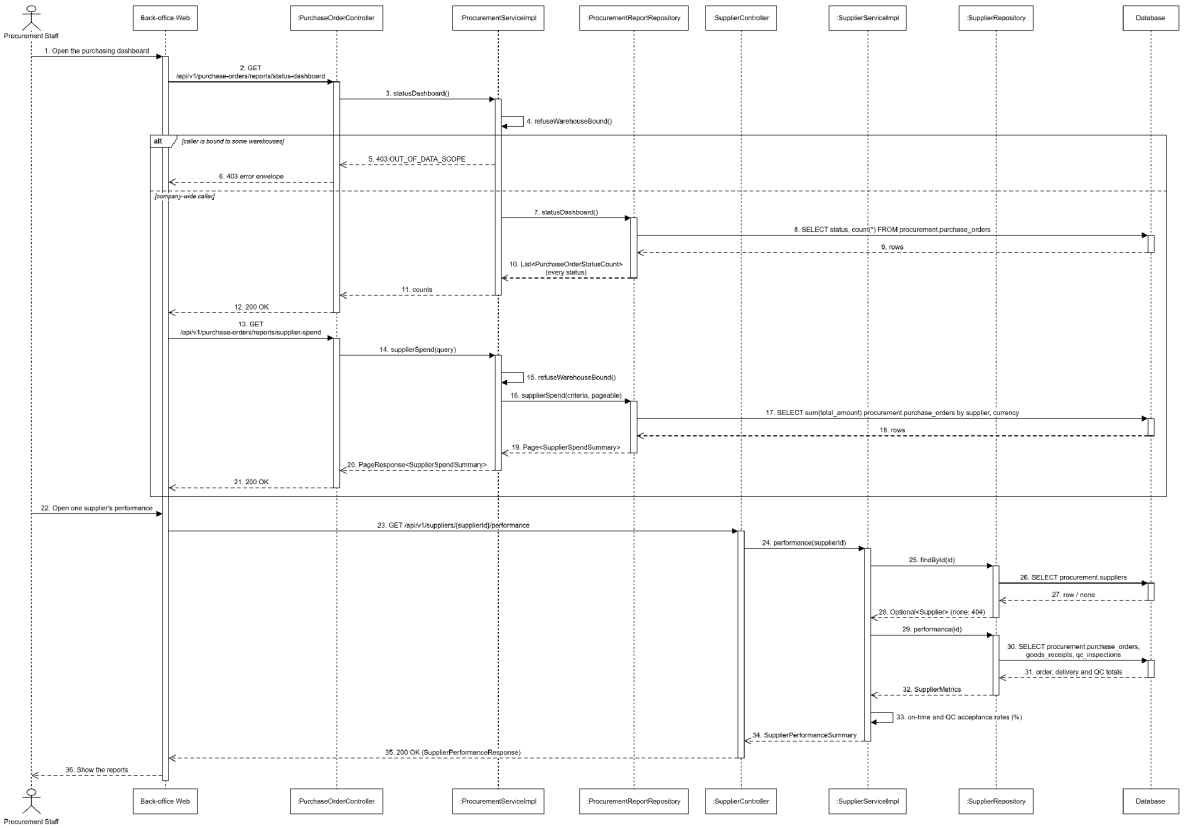


Figure 238. View Purchasing Reports Sequence Diagram

3.14.15 Purchase Order State Diagram

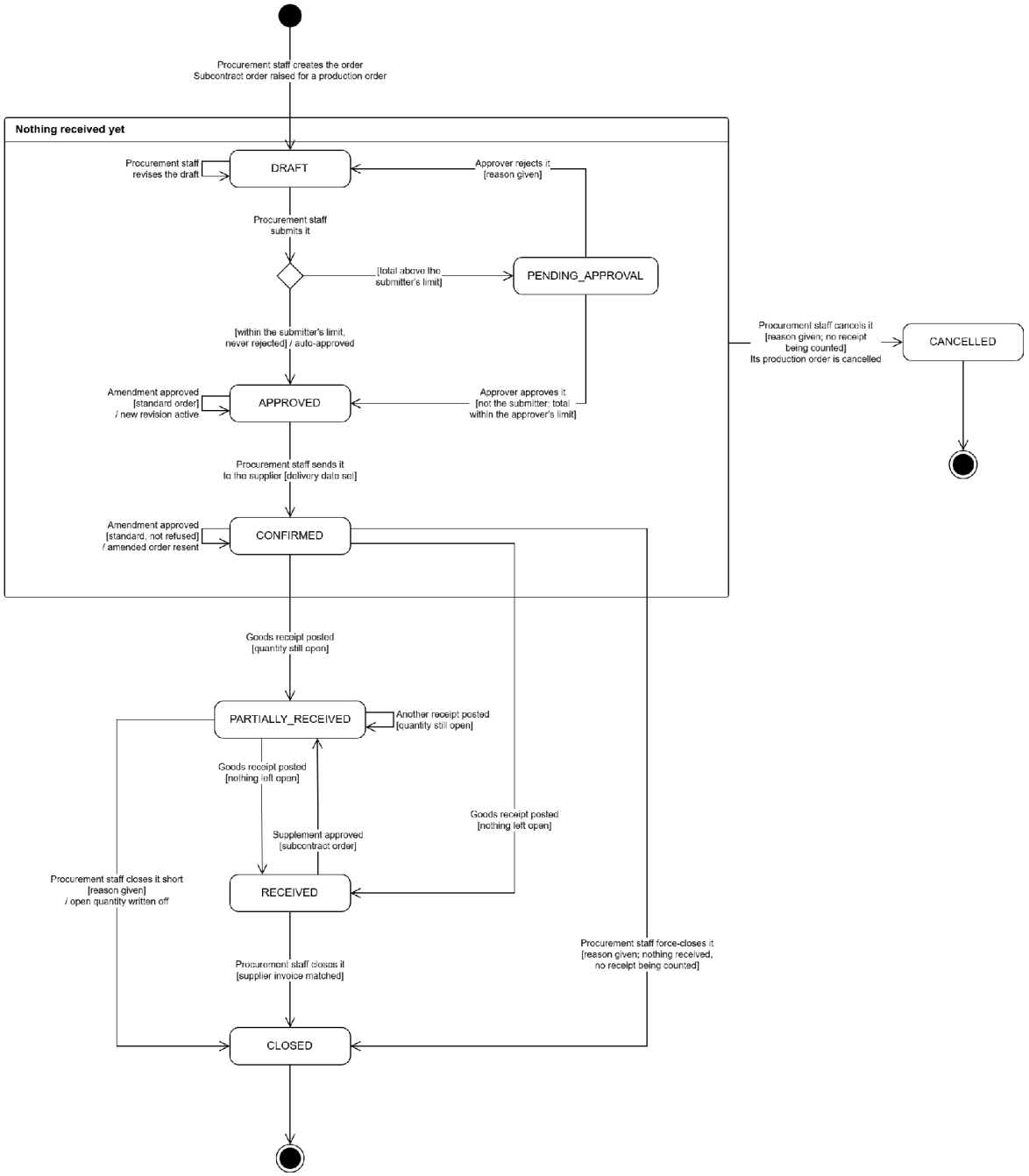


Figure 239. Purchase Order State Diagram

3.14.16 Supplier Confirmation State Diagram

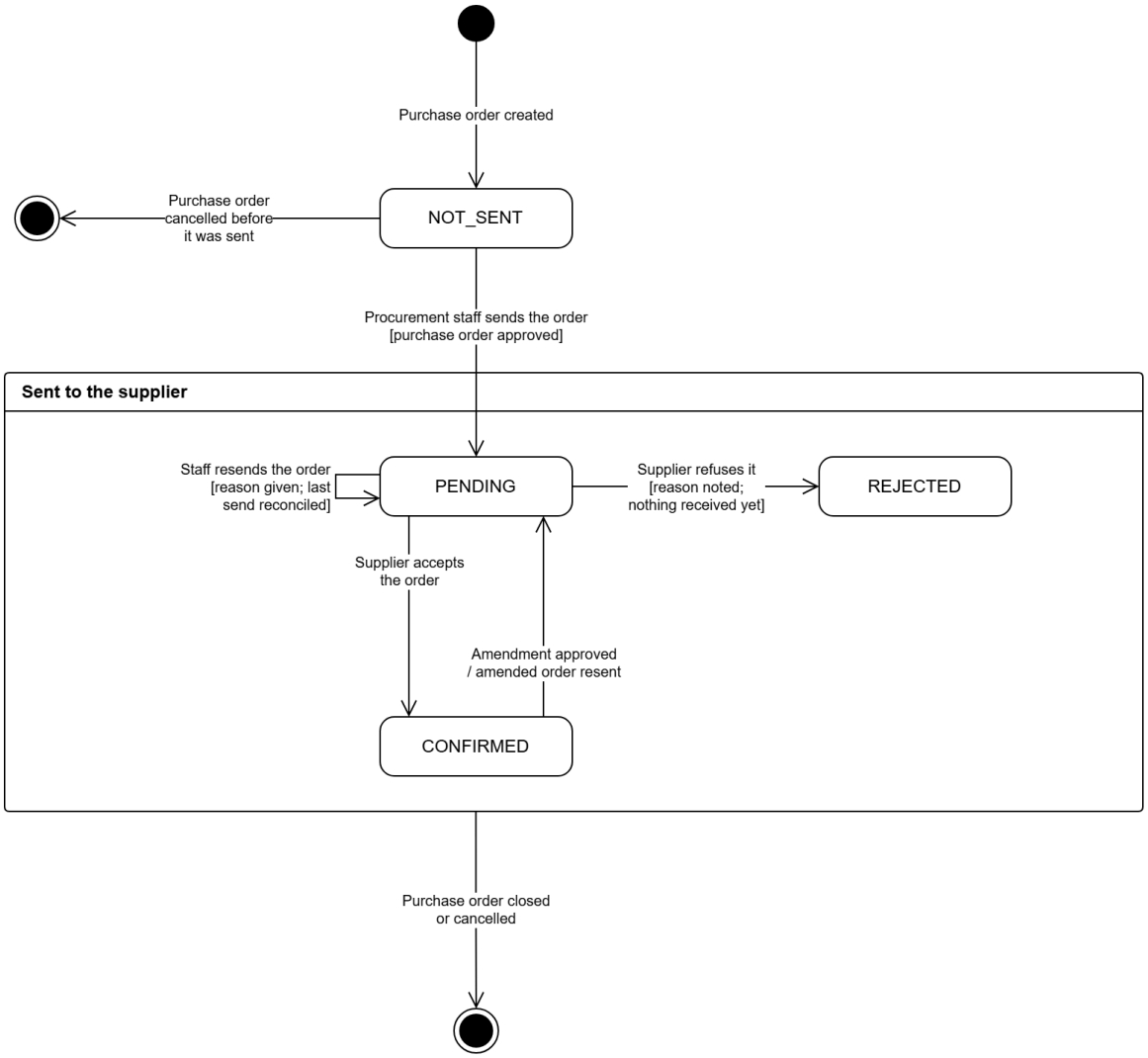


Figure 240. Supplier Confirmation State Diagram

3.14.17 Supplier State Diagram

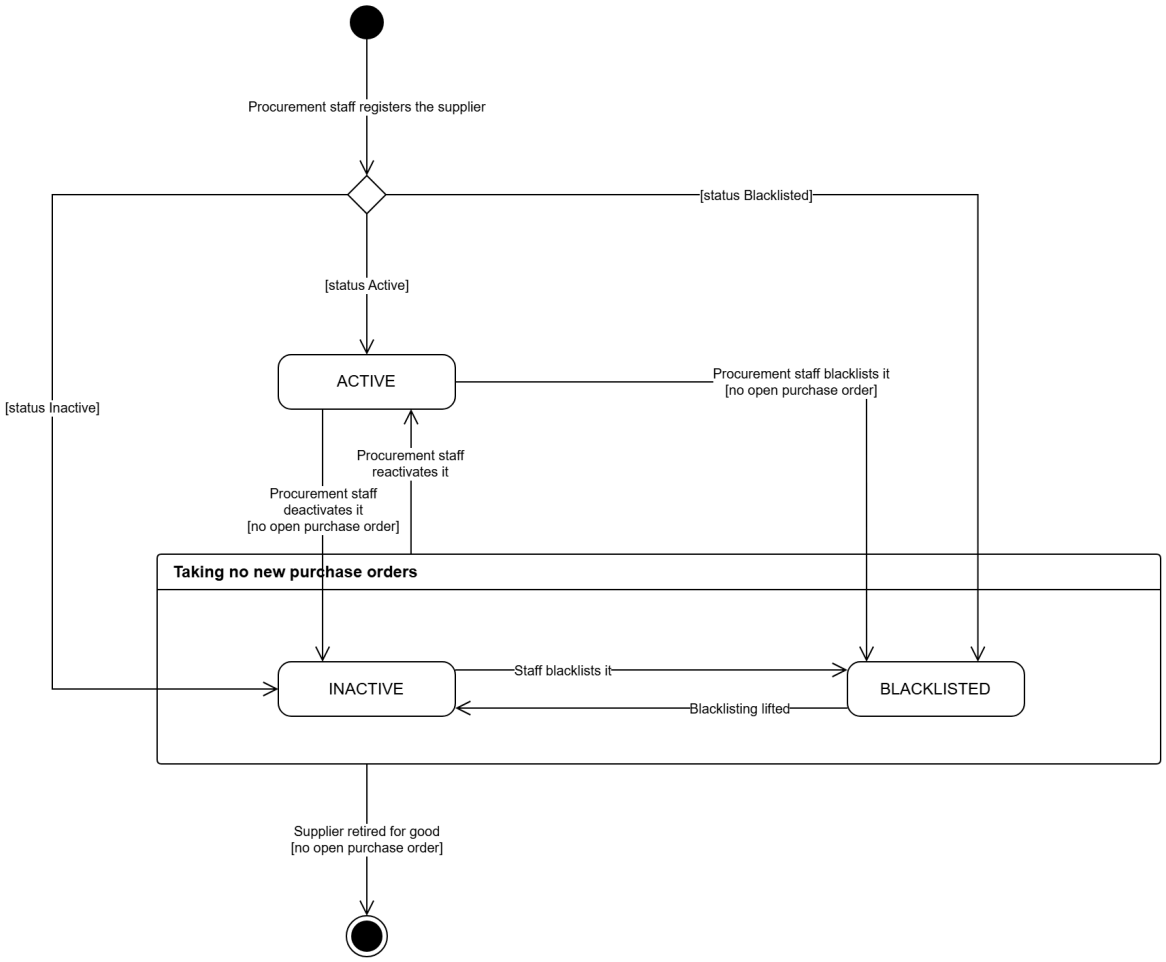


Figure 241. Supplier State Diagram

3.15 Goods Receipt, QC & Putaway

3.15.1 Class Diagram

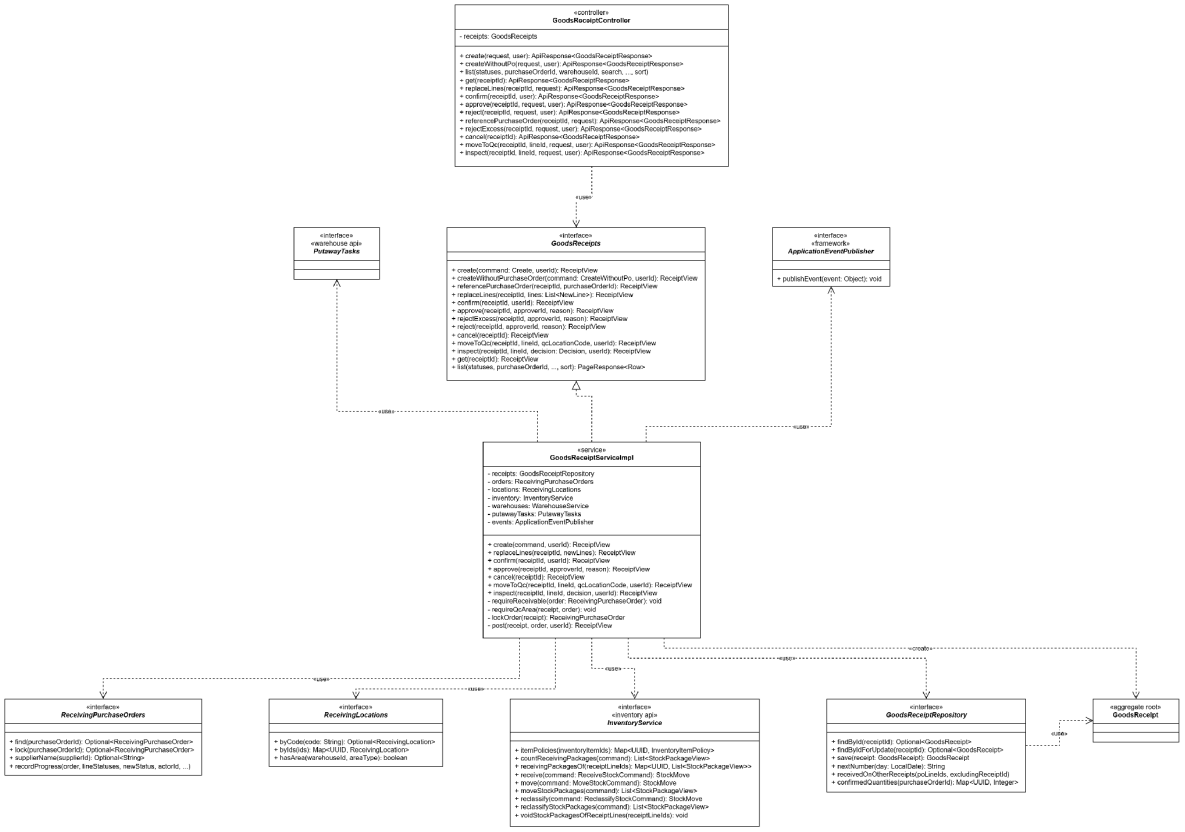


Figure 242. Goods Receipt, QC & Putaway Class Diagram (1/4): Receiving and QC

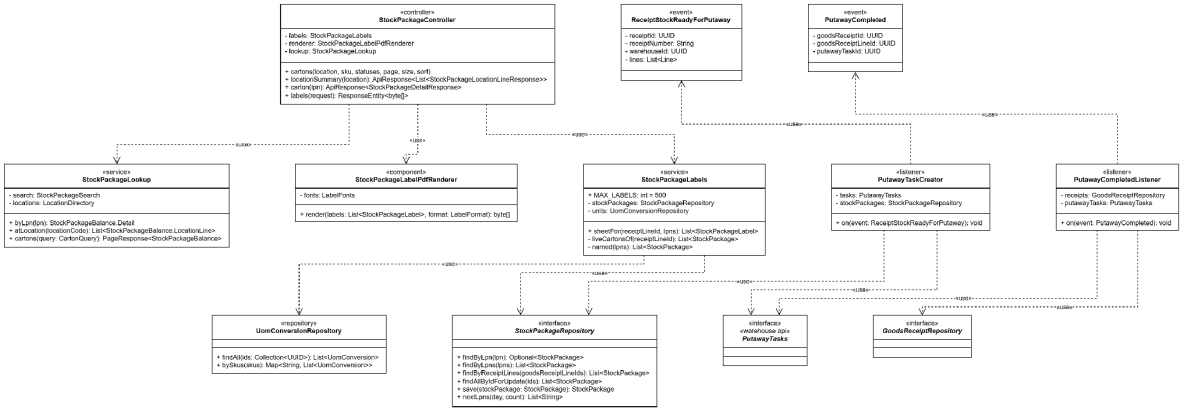


Figure 243. Goods Receipt, QC & Putaway Class Diagram (2/4): LPN Cartons and Receipt Events

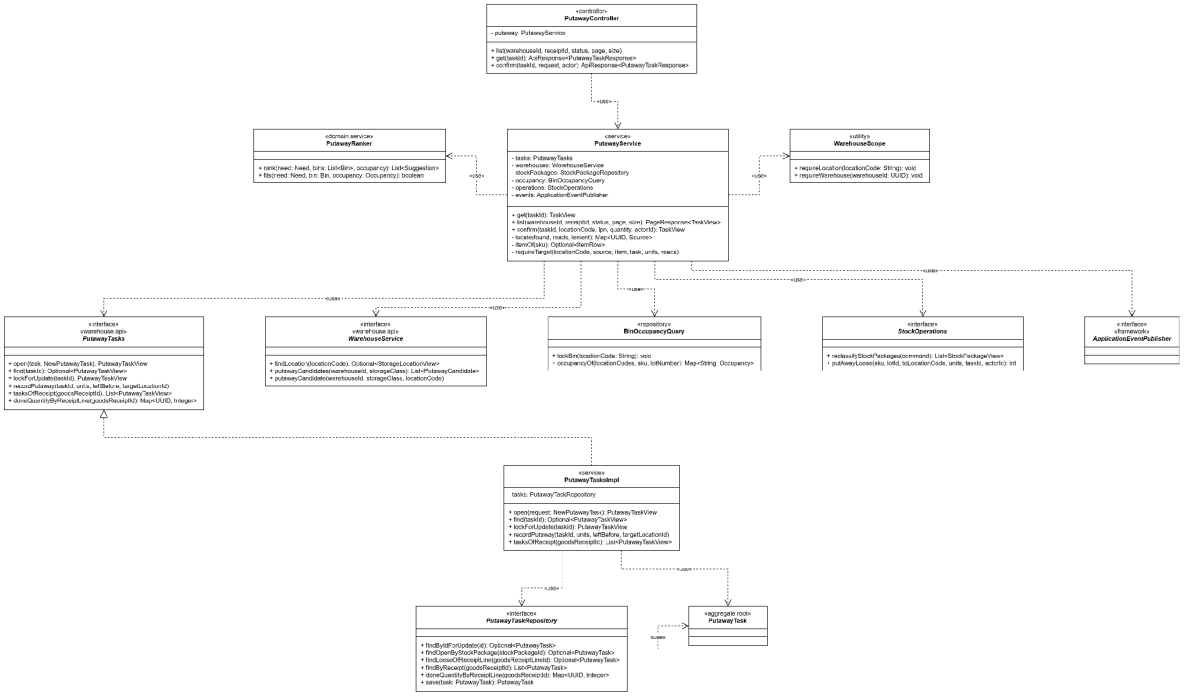


Figure 244. Goods Receipt, QC & Putaway Class Diagram (3/4): Putaway

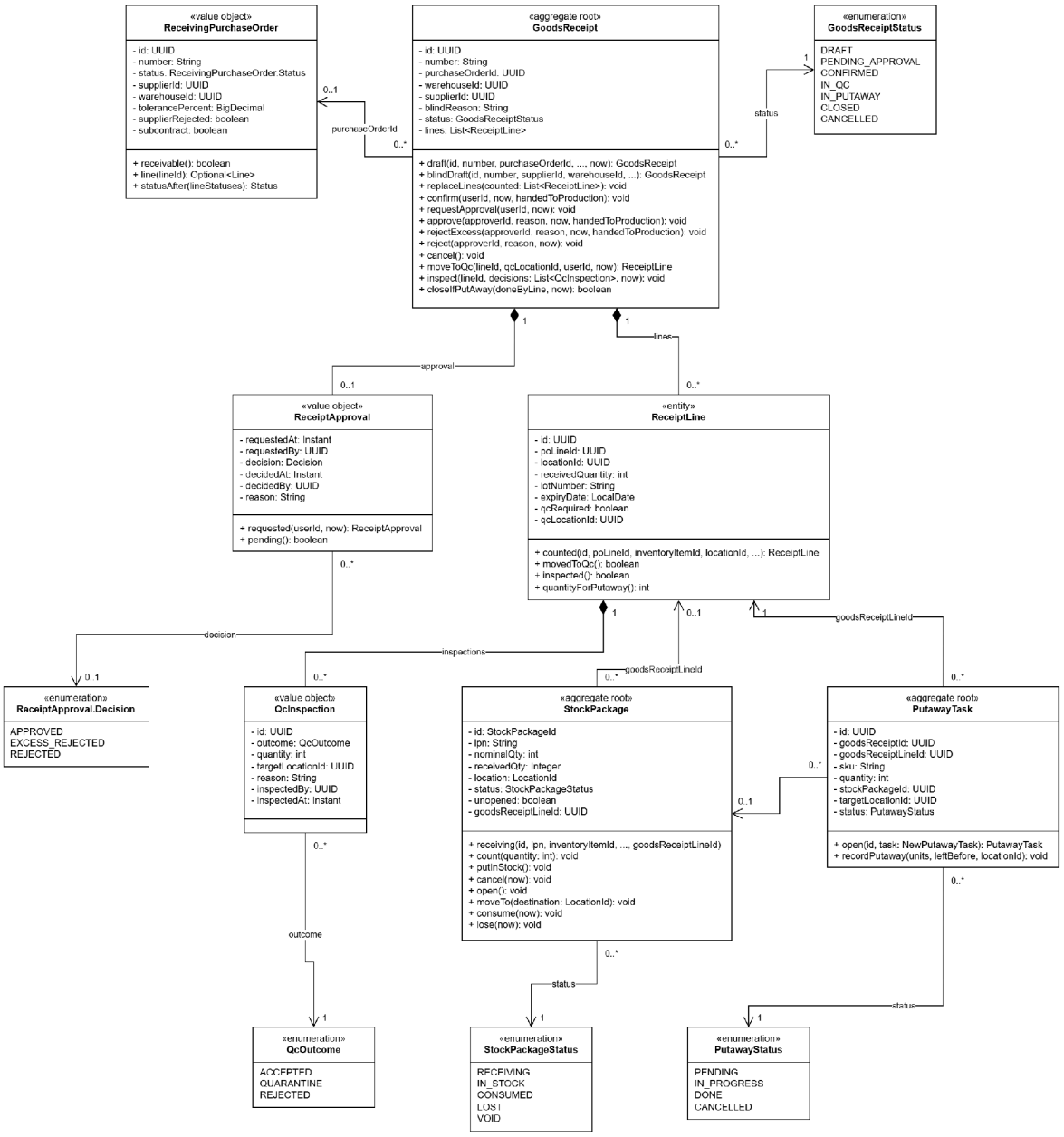


Figure 245. Goods Receipt, QC & Putaway Class Diagram (4/4): Domain Model

3.15.2 Class Specification

GoodsReceiptController

No	Method	Description
1	create(request, user)	POST /api/v1/goods-receipts: Starts a DRAFT receipt against a CONFIRMED or PARTIALLY_RECEIVED PO the supplier did not reject (409 PURCHASE_ORDER_NOT_RECEIVABLE, BR-01).
2	createWithoutPo(request, user)	POST /api/v1/goods-receipts/without-po: Starts a DRAFT receipt with no PO for a supplier and warehouse; a reason is required (BR-RCP-002).

3	list(statuses, purchaseOrderId, warehouseId, search, receivedFrom, receivedTo, withoutPo, page, size, sort)	GET /api/v1/goods-receipts: Lists receipts, newest first, with filters, within the caller's warehouse scope.
4	get(receiptId)	GET /api/v1/goods-receipts/{receiptId}: Returns one receipt with its lines, cartons and QC results.
5	replaceLines(receiptId, request)	PUT /api/v1/goods-receipts/{receiptId}/lines: Replaces the count of a DRAFT receipt (PO line open, RECEIVING area, lot / expiry BR-03) and labels, recounts or voids the lines' cartons (LPN).
6	confirm(receiptId, user)	POST /api/v1/goods-receipts/{receiptId}/confirmation: Confirms the count: stock is received INBOUND, the PO progresses, the receipt goes to IN_QC or IN_PUTAWAY; without a PO or over tolerance (BR-02) it waits in PENDING_APPROVAL.
7	approve(receiptId, request, user)	POST /api/v1/goods-receipts/{receiptId}/approval: Posts a PENDING_APPROVAL receipt whole; the decider is neither the counter nor the requester, and gives a reason.
8	reject(receiptId, request, user)	POST /api/v1/goods-receipts/{receiptId}/rejection: Cancels a PENDING_APPROVAL receipt that has no PO; its cartons are voided.
9	referencePurchaseOrder(receiptId, request)	PUT /api/v1/goods-receipts/{receiptId}/referenced-purchase-order: Attaches a PO of the same supplier to a receipt without one, as a reference only.
10	rejectExcess(receiptId, request, user)	POST /api/v1/goods-receipts/{receiptId}/excess-rejection: Cuts a PENDING_APPROVAL receipt back to the tolerance and posts the rest; cancels it when nothing is left.
11	cancel(receiptId)	POST /api/v1/goods-receipts/{receiptId}/cancellation: Cancels a DRAFT or PENDING_APPROVAL receipt and voids its cartons; a posted one is corrected by a stock adjustment (BR-05).
12	moveToQc(receiptId, lineId, request, user)	POST /api/v1/goods-receipts/{receiptId}/lines/{lineId}/qc-transfer: Moves a QC-required line's goods (cartons whole, loose units) to a QUALITY_CONTROL area.

13	inspect(receiptId, lineId, request, user)	POST /api/v1/goods-receipts/{receiptId}/lines/{lineId}/inspection: Records accepted / quarantined / rejected parts that add up to the line (BR-08); non-accepted parts need a reason and a QUARANTINE area; accepted units become ready for putaway.

Table 88. GoodsReceiptController Class Specification

PutawayController

No	Method	Description
1	list(warehouseId, receiptId, status, page, size)	GET /api/v1/putaway-tasks: Lists putaway tasks by warehouse, receipt and status, within the caller's warehouse scope.
2	get(taskId)	GET /api/v1/putaway-tasks/{taskId}: Returns one task with what is left, where it stands, and up to 10 ranked bin suggestions.
3	confirm(taskId, request, actor)	POST /api/v1/putaway-tasks/{taskId}/confirmations: Puts a carton (by LPN, whole) or a loose quantity into a scanned bin that is a usable putaway target of the right storage class with room (BR-SLT-001/003).

Table 89. PutawayController Class Specification

StockPackageController

No	Method	Description
1	cartons(location, sku, statuses, page, size, sort)	GET /api/v1/inventory/stock-packages: Lists cartons (LPN) by location, SKU and status from the stock_package_balance view.
2	locationSummary(location)	GET /api/v1/inventory/stock-packages/location-summary: Summarises what one bin or area holds, per SKU, lot and stock status: unopened cartons, opened cartons and loose units; 404 LOCATION_NOT_FOUND.
3	carton(lpn)	GET /api/v1/inventory/stock-packages/{lpn}: Returns one carton with its contents; 404 STOCK_PACKAGE_NOT_FOUND.
4	labels(request)	POST /api/v1/inventory/stock-packages/labels: Prints carton labels (LPN text, QR, Code 128) as a PDF for a receipt line's live cartons or listed LPNs; at most 500.

Table 90. StockPackageController Class Specification

3.15.3 Receive Goods against PO (Draft and Count) Sequence Diagram

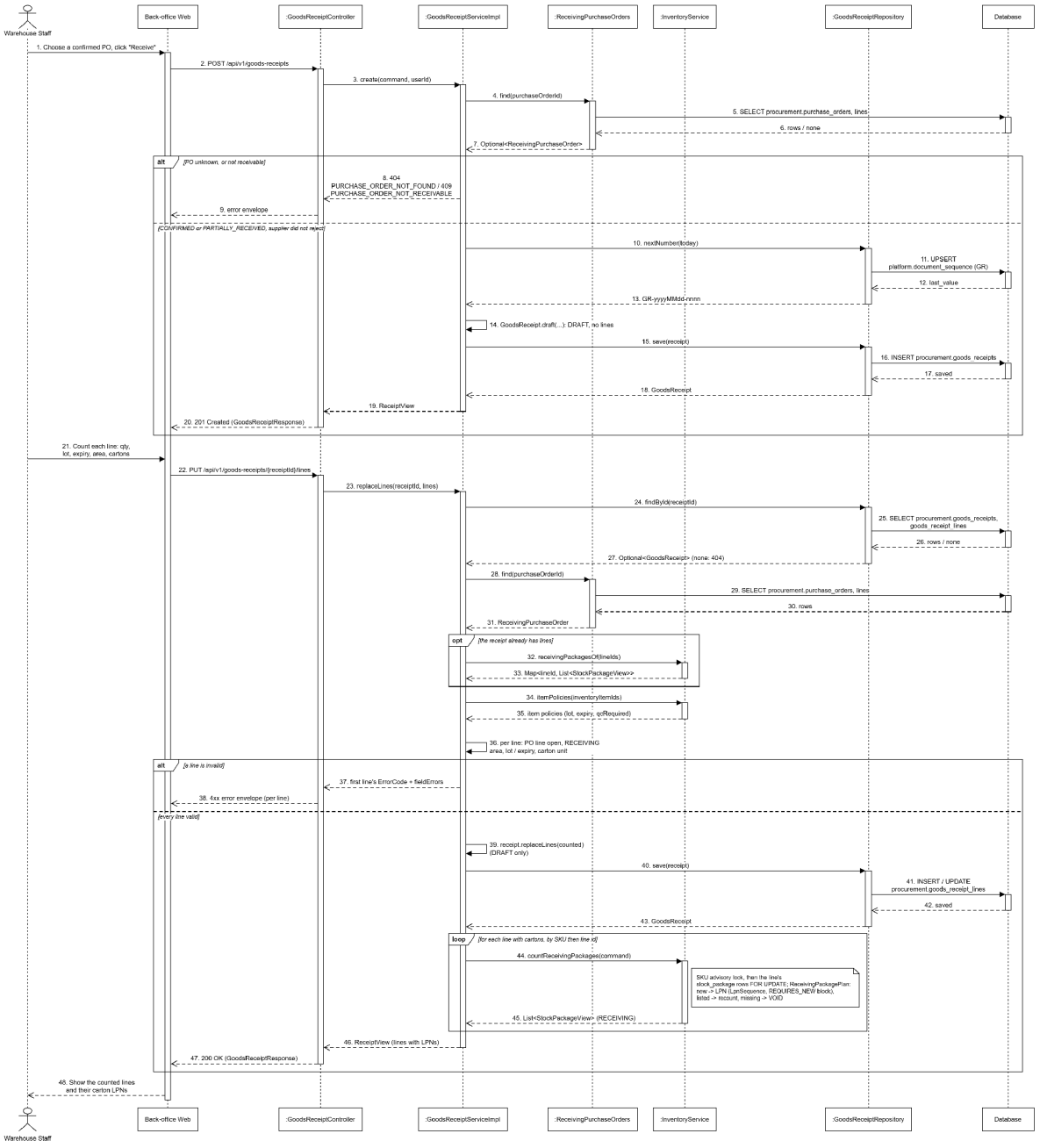


Figure 246. Receive Goods against PO (Draft and Count) Sequence Diagram

3.15.4 Receive Goods against PO (Confirm) Sequence Diagram

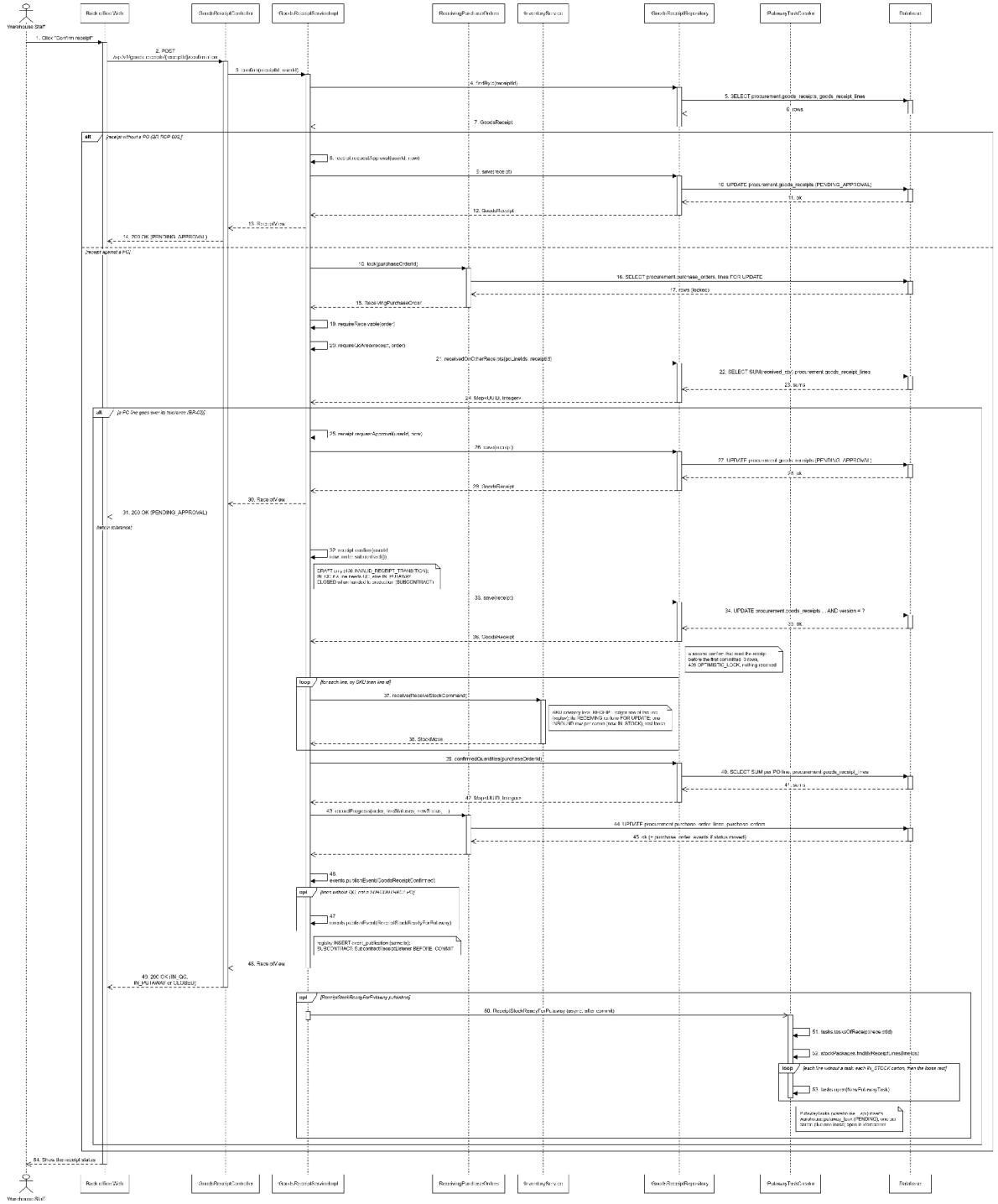


Figure 247. Receive Goods against PO (Confirm) Sequence Diagram

3.15.5 Label & Print LPN Cartons Sequence Diagram

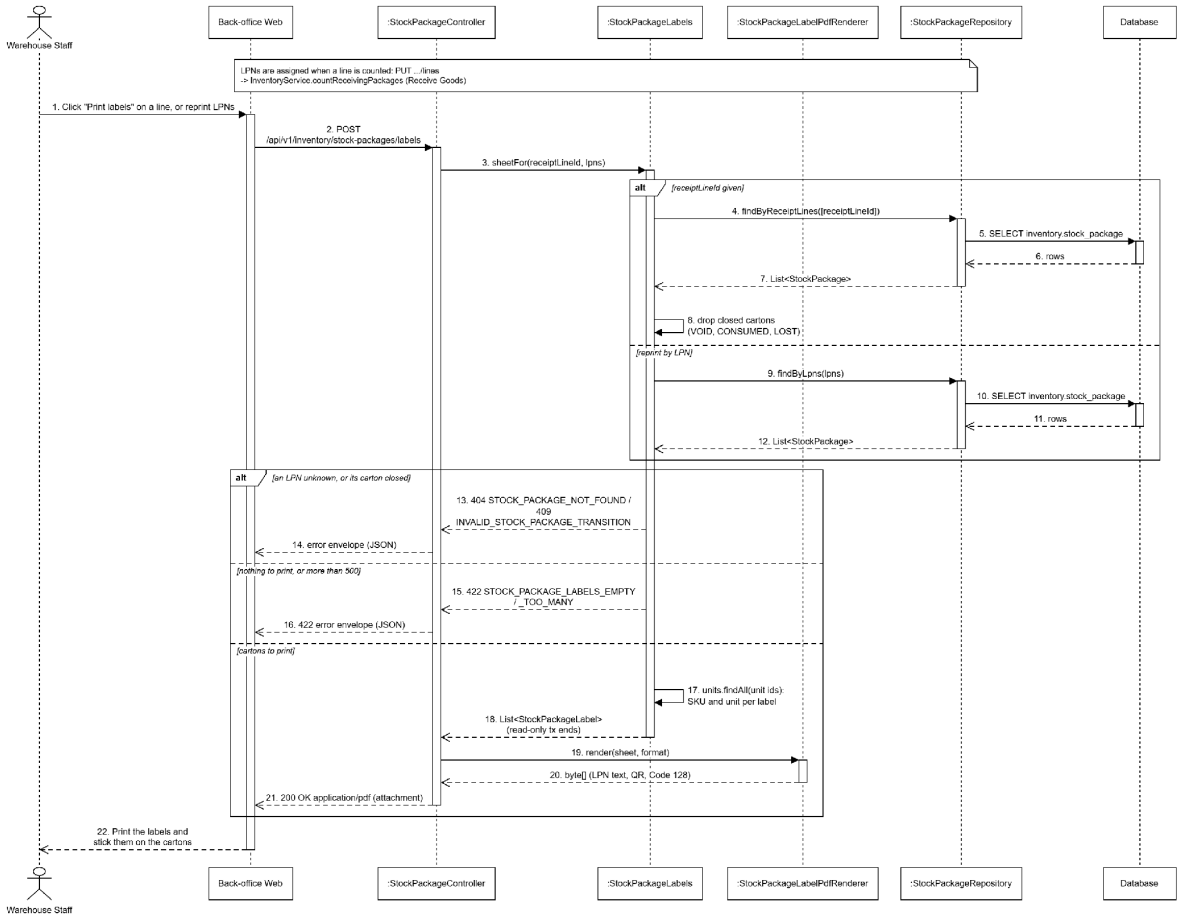


Figure 248. Label & Print LPN Cartons Sequence Diagram

3.15.6 Move Goods to QC Area Sequence Diagram

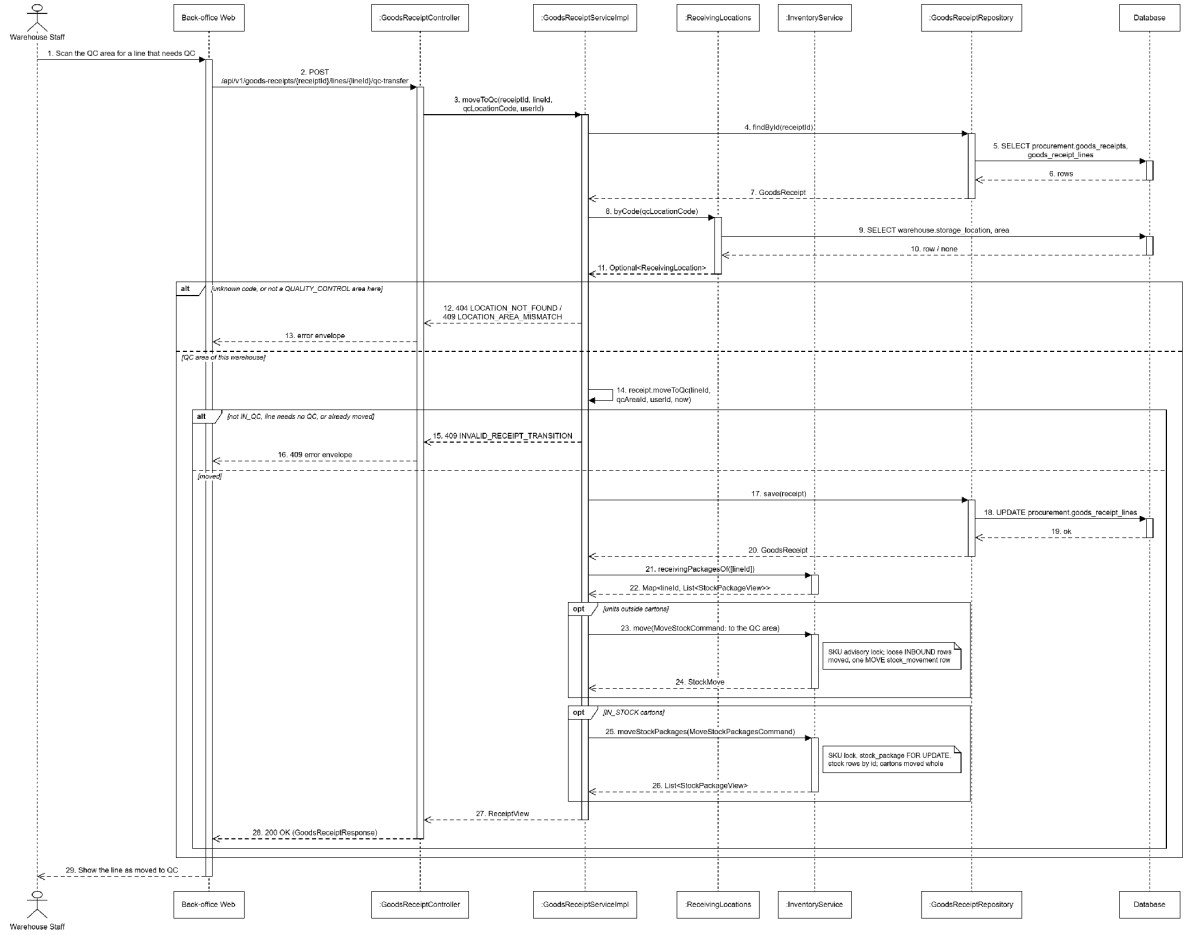


Figure 249. Move Goods to QC Area Sequence Diagram

3.15.7 Record Inbound QC Result Sequence Diagram

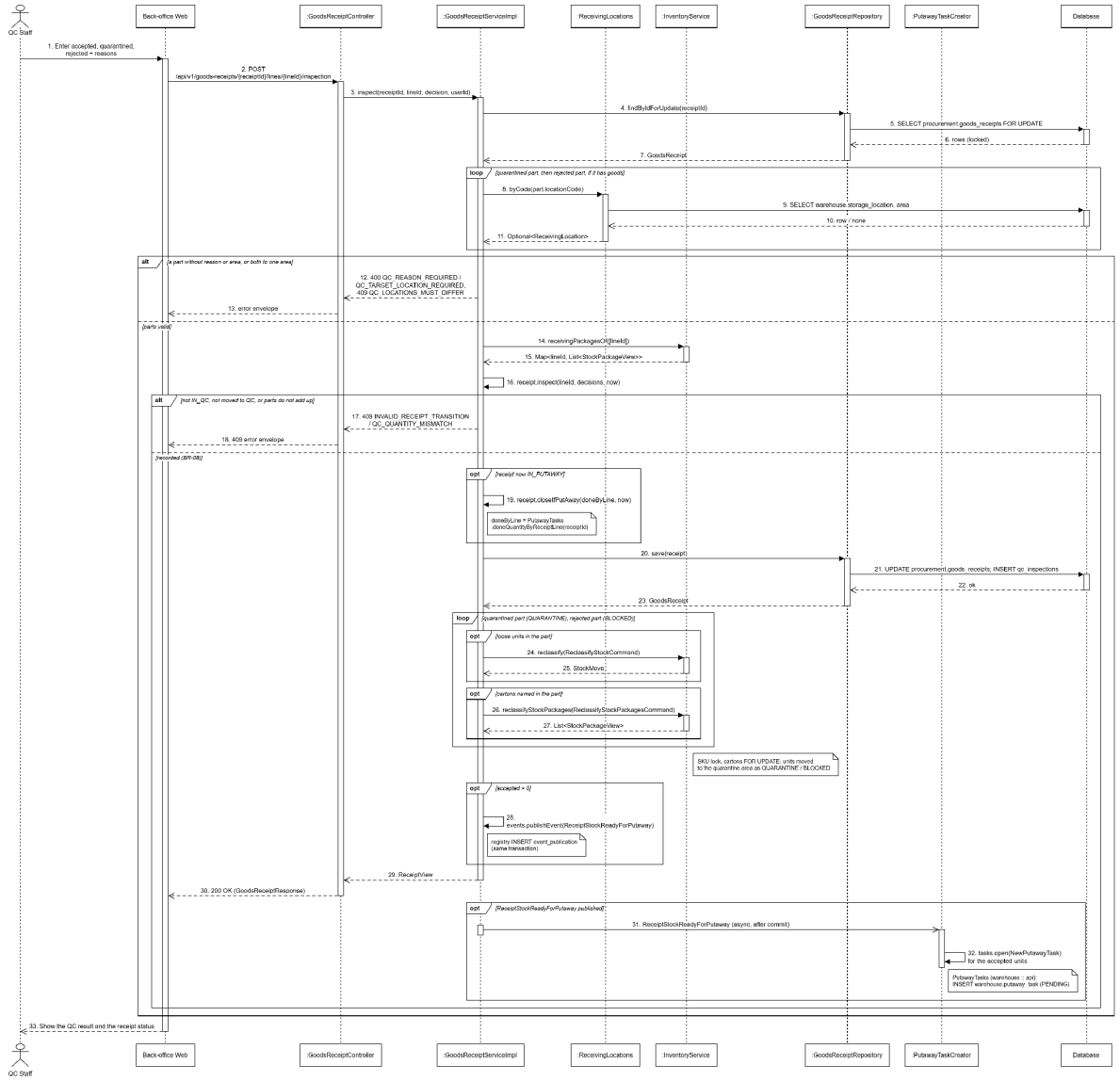


Figure 250. Record Inbound QC Result Sequence Diagram

3.15.8 Get Bin Suggestions Sequence Diagram

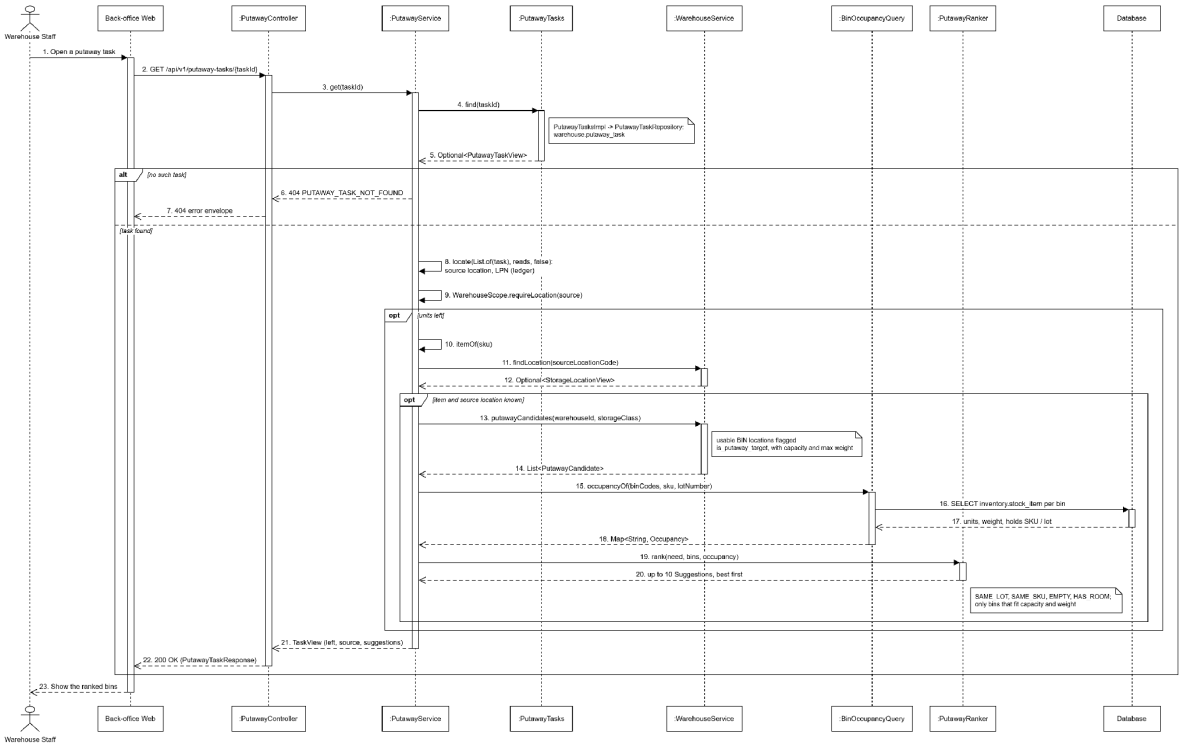


Figure 251. Get Bin Suggestions Sequence Diagram

3.15.9 Put Away Goods Sequence Diagram

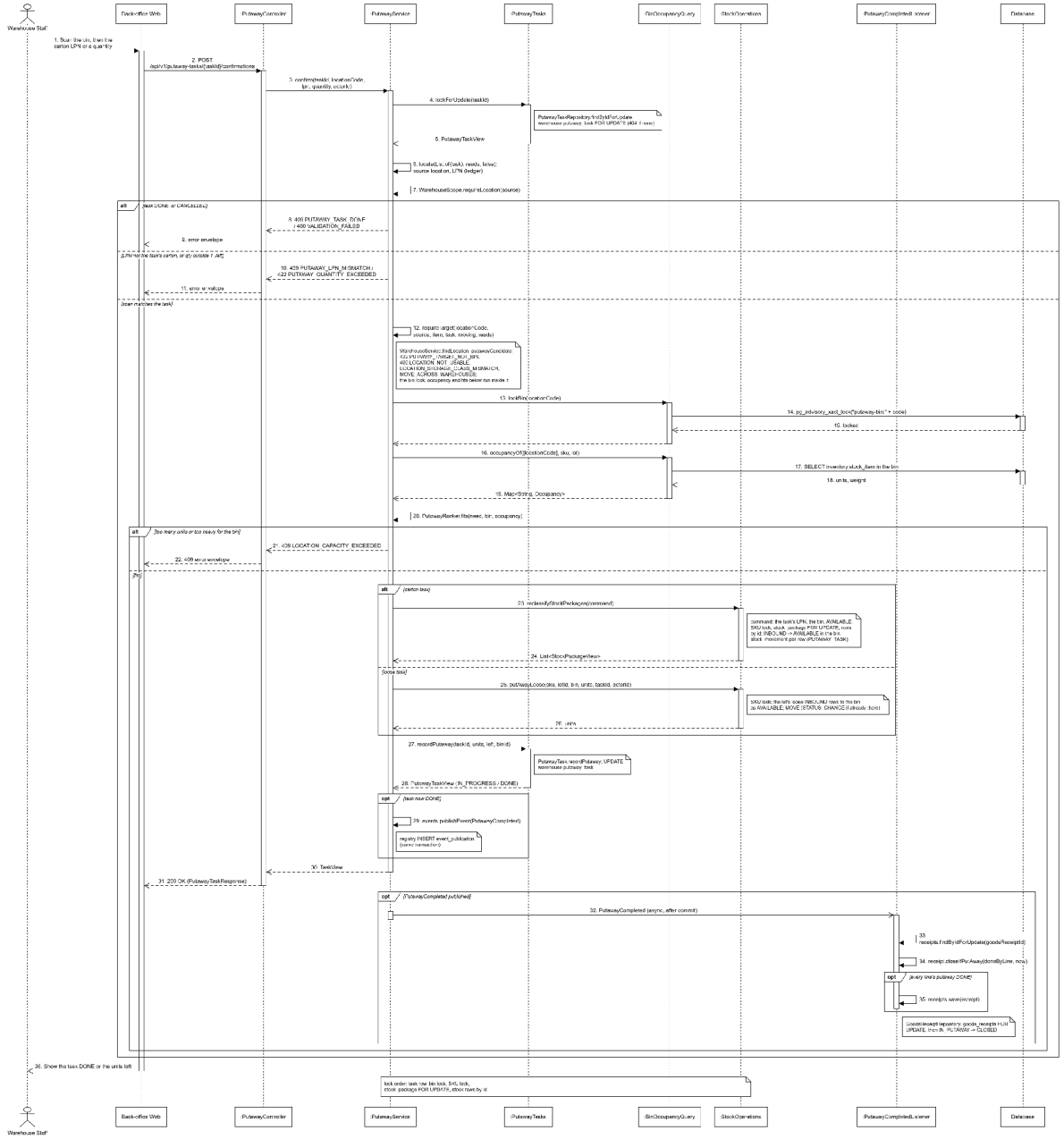


Figure 252. Put Away Goods Sequence Diagram

3.15.10 Goods Receipt State Diagram

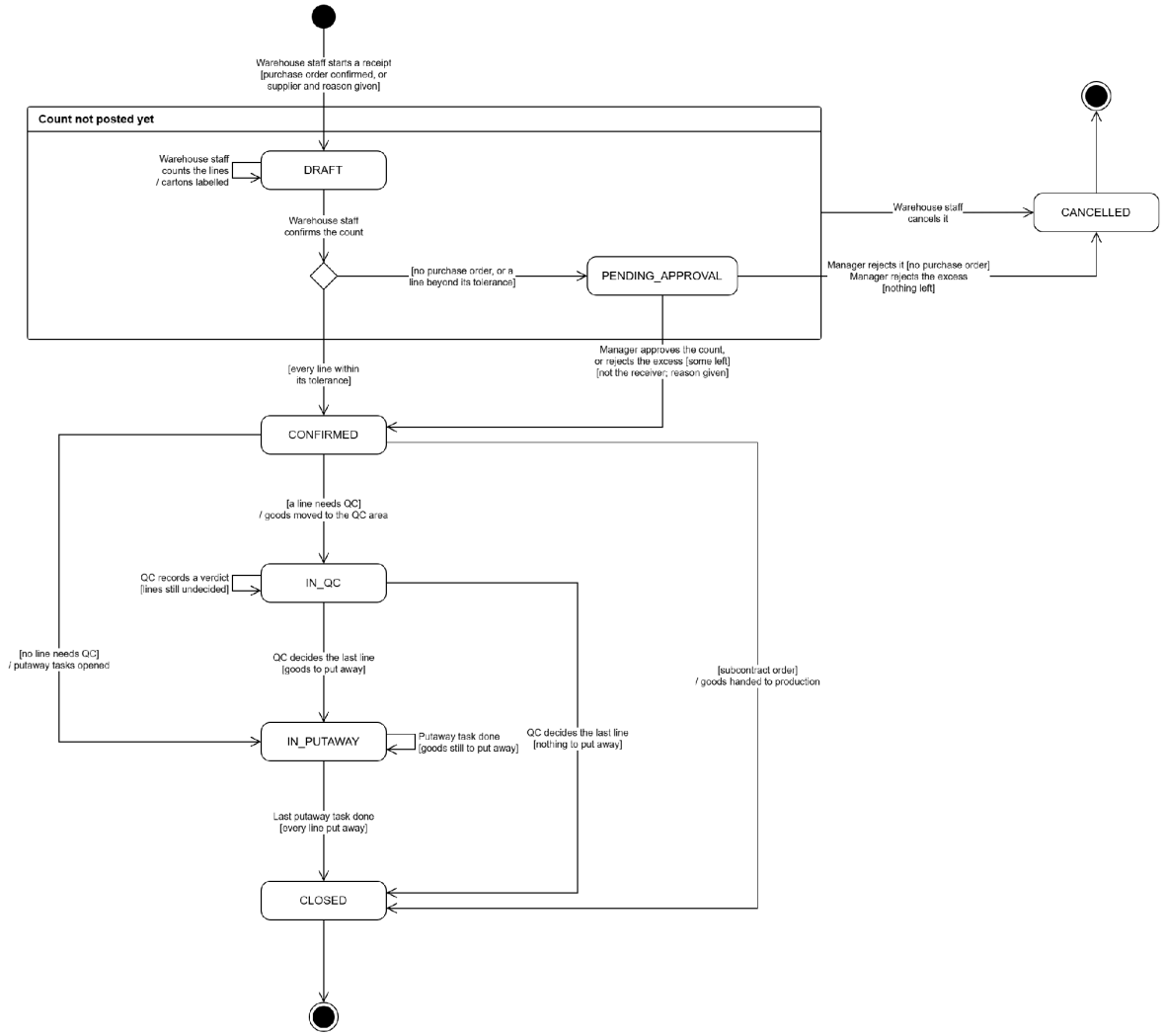


Figure 253. Goods Receipt State Diagram

3.15.11 Putaway Task State Diagram

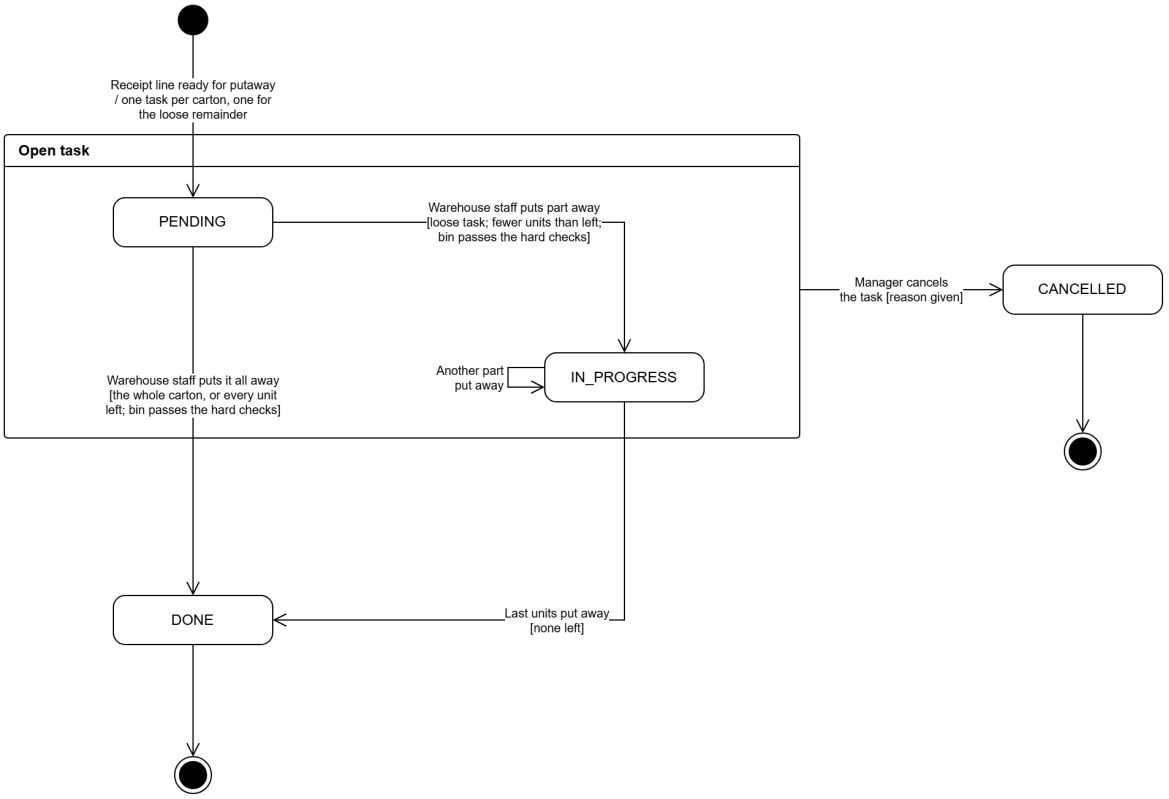


Figure 254. Putaway Task State Diagram

3.15.12 Stock Package (Carton) State Diagram

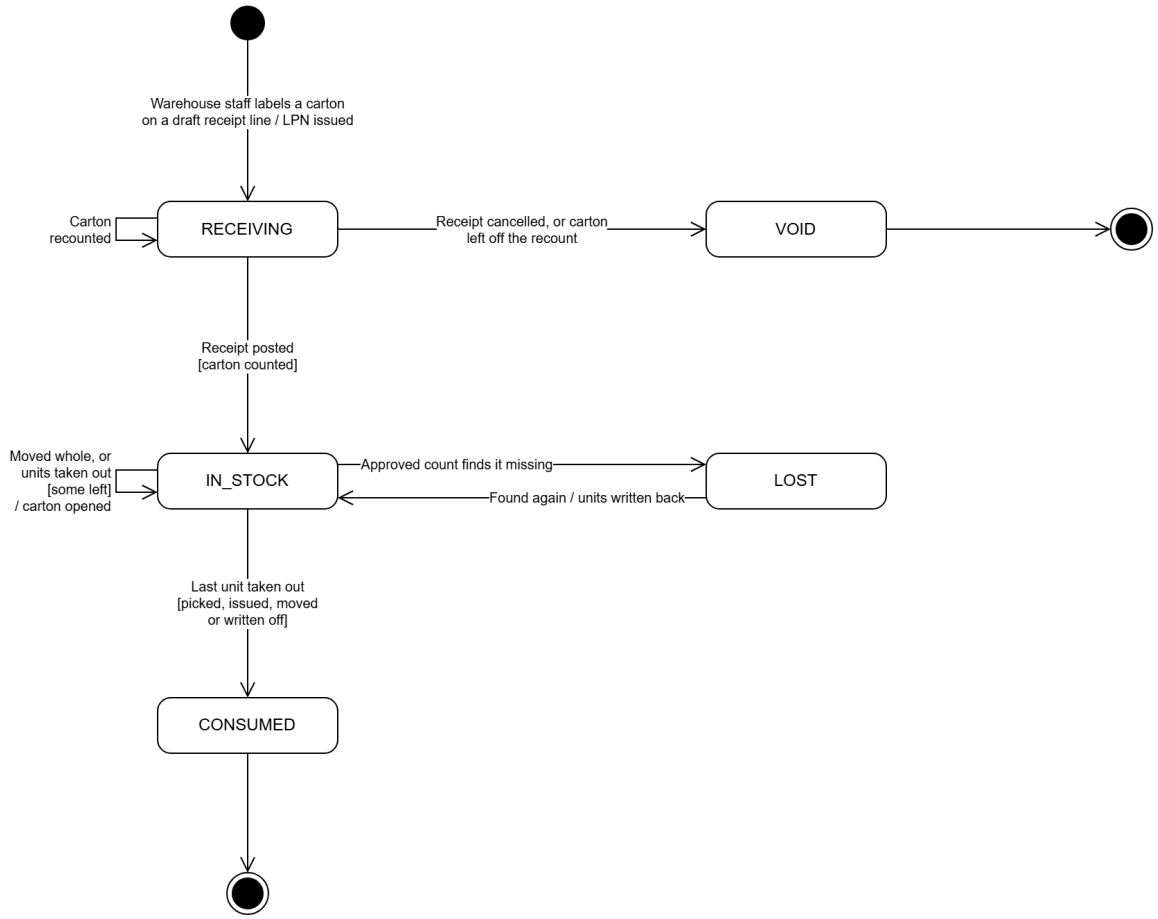


Figure 255. Stock Package (Carton) State Diagram

3.16 Inventory Management

3.16.1 Class Diagram

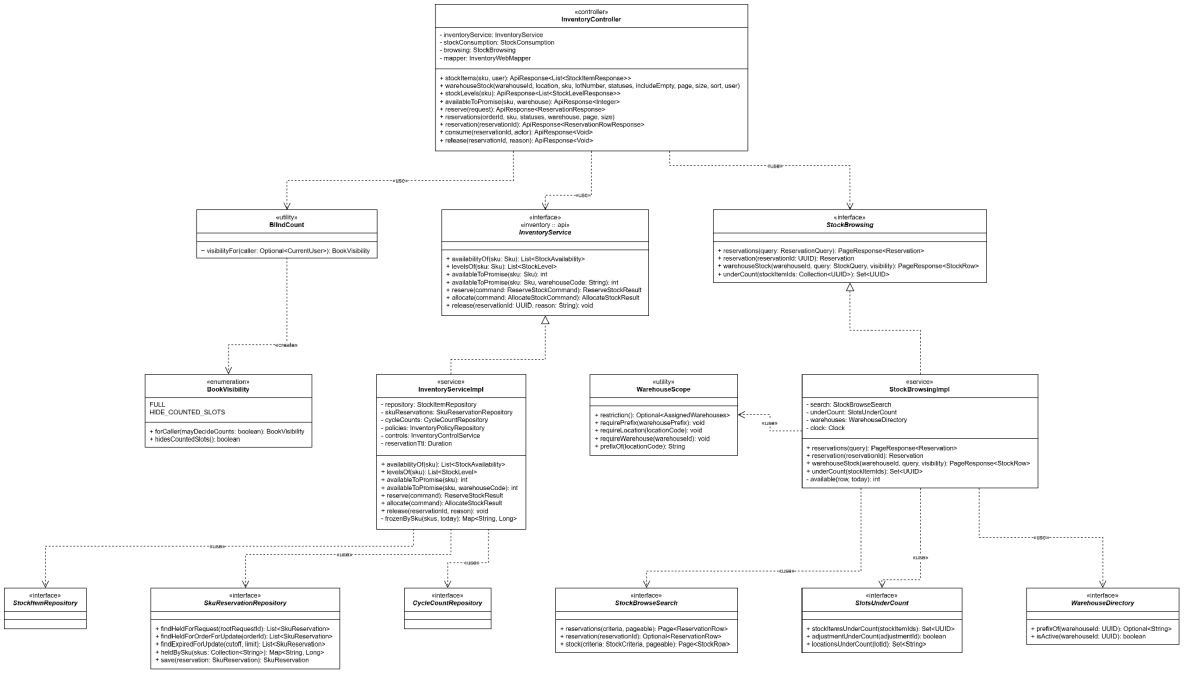


Figure 256. Inventory Management Class Diagram (1/7): Stock Lookup

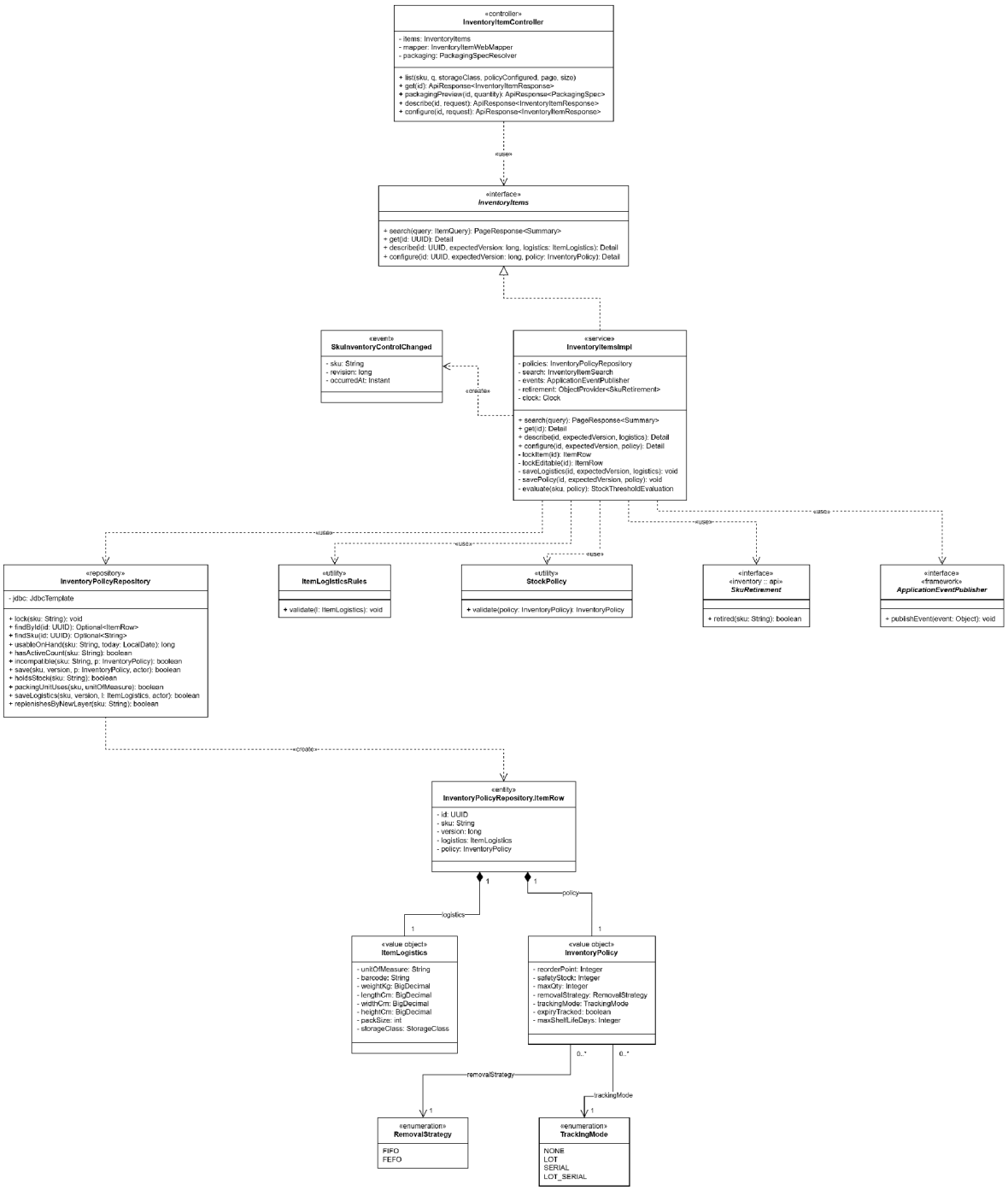


Figure 257. Inventory Management Class Diagram (2/7): Item Logistics and Stock Policy

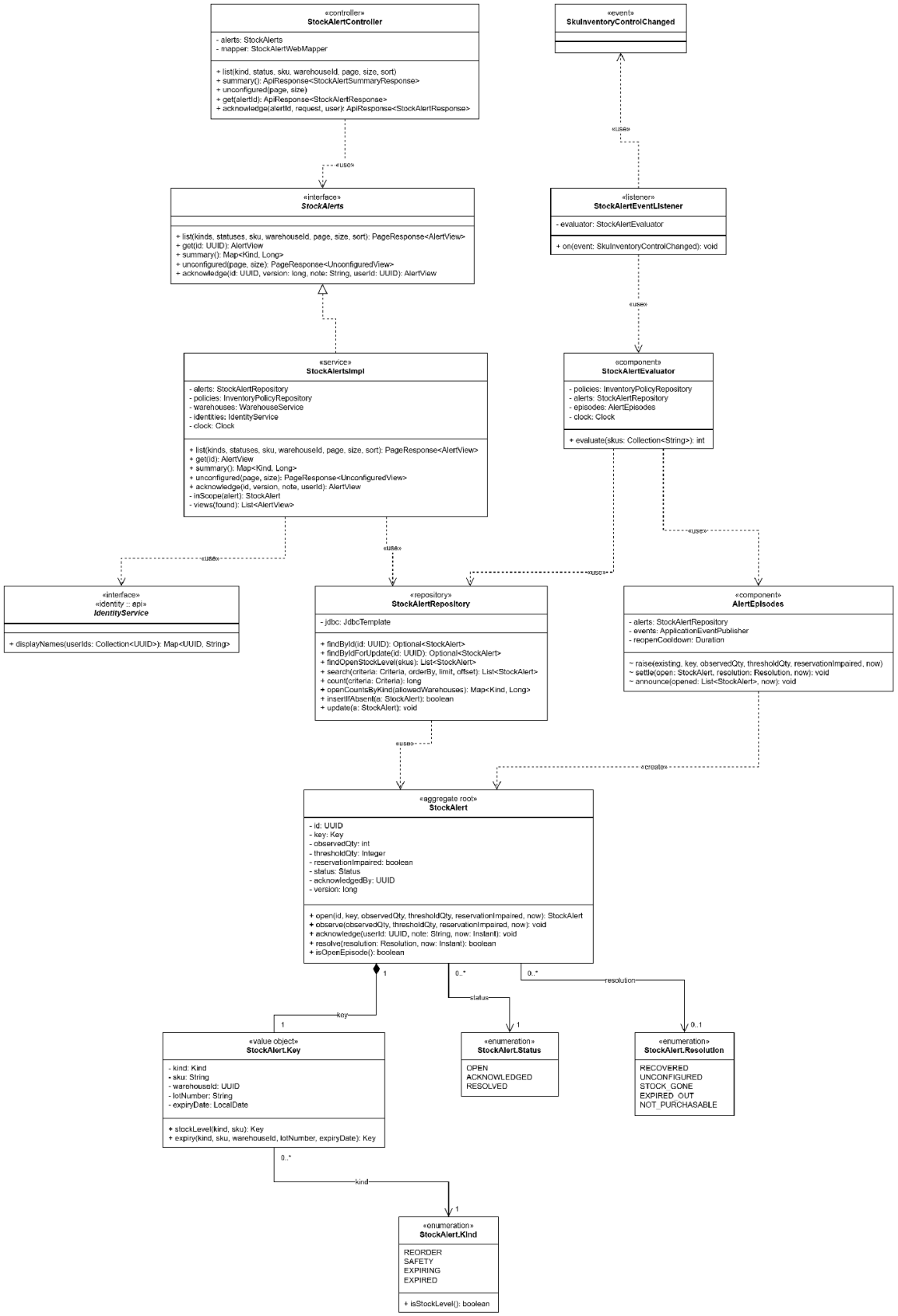


Figure 258. Inventory Management Class Diagram (3/7): Stock Alerts

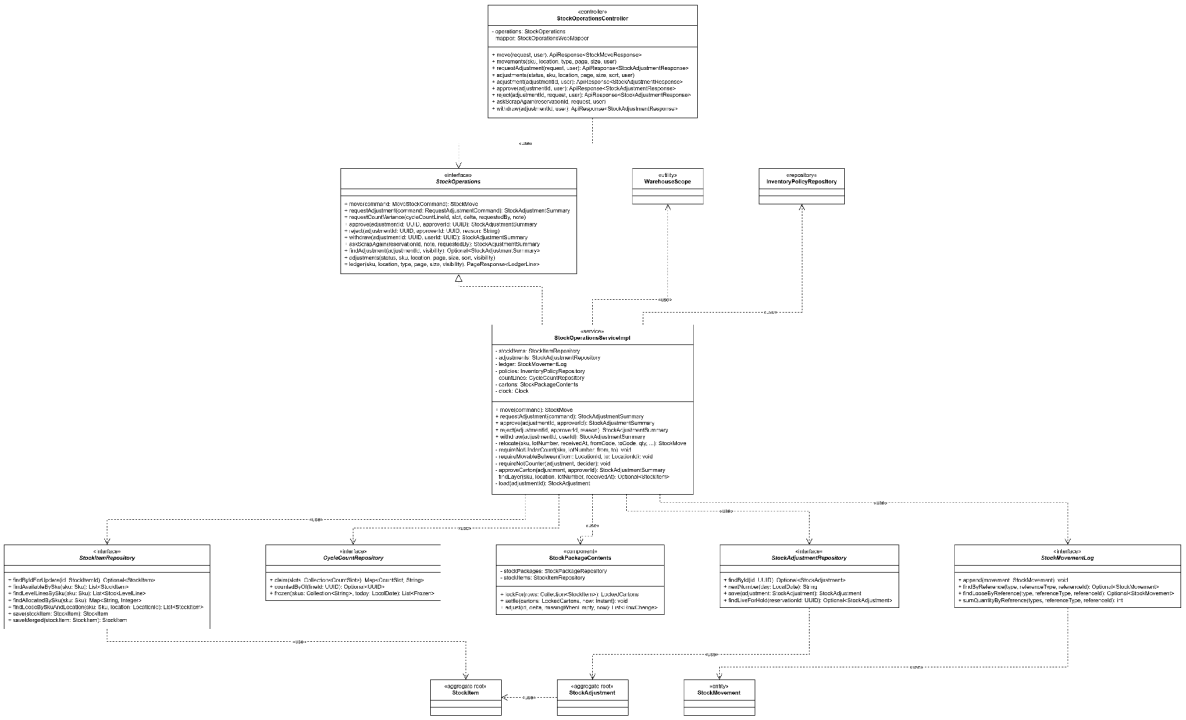


Figure 259. Inventory Management Class Diagram (4/7): Stock Moves and Adjustments

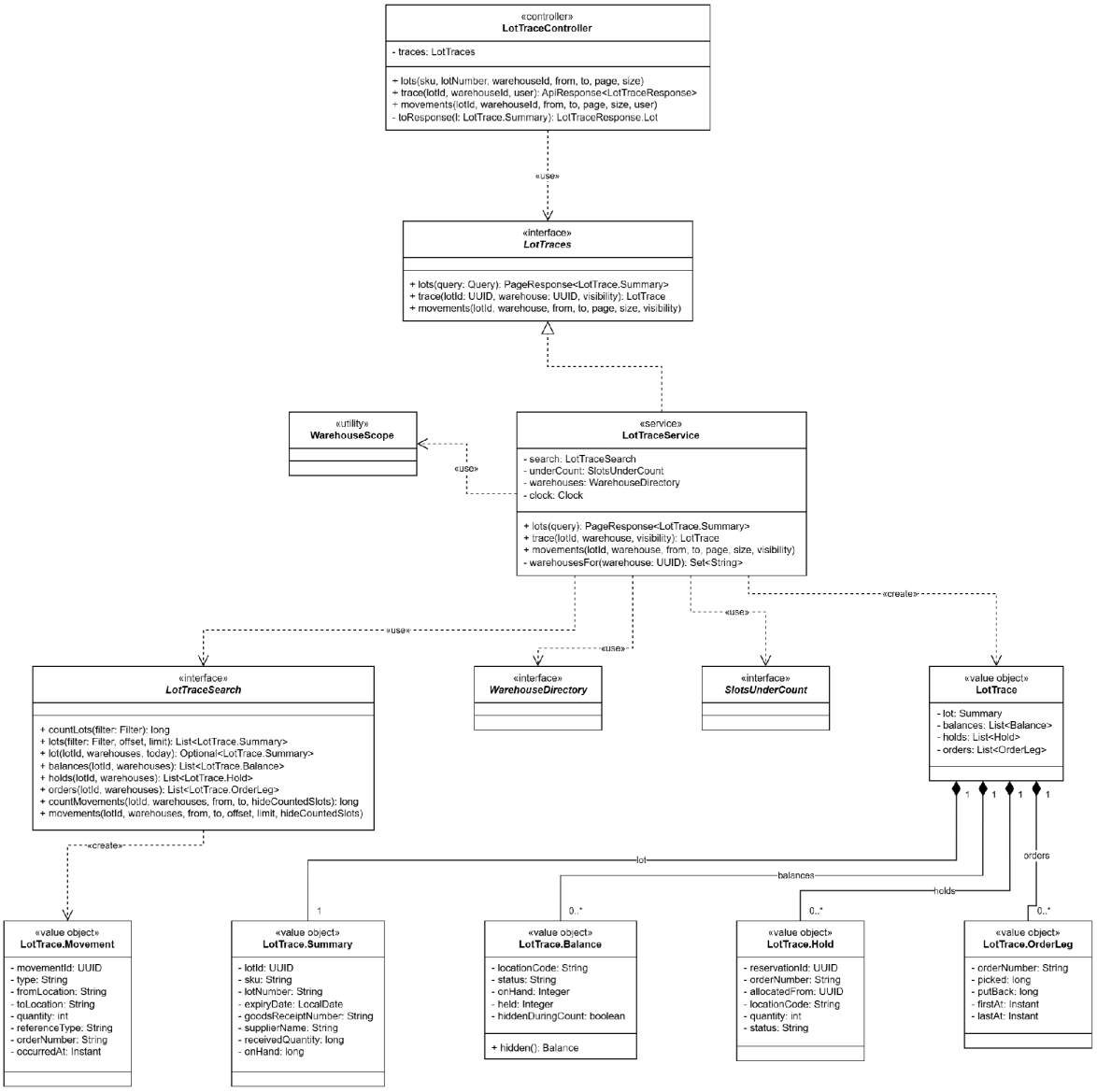


Figure 260. Inventory Management Class Diagram (5/7): Lot Trace

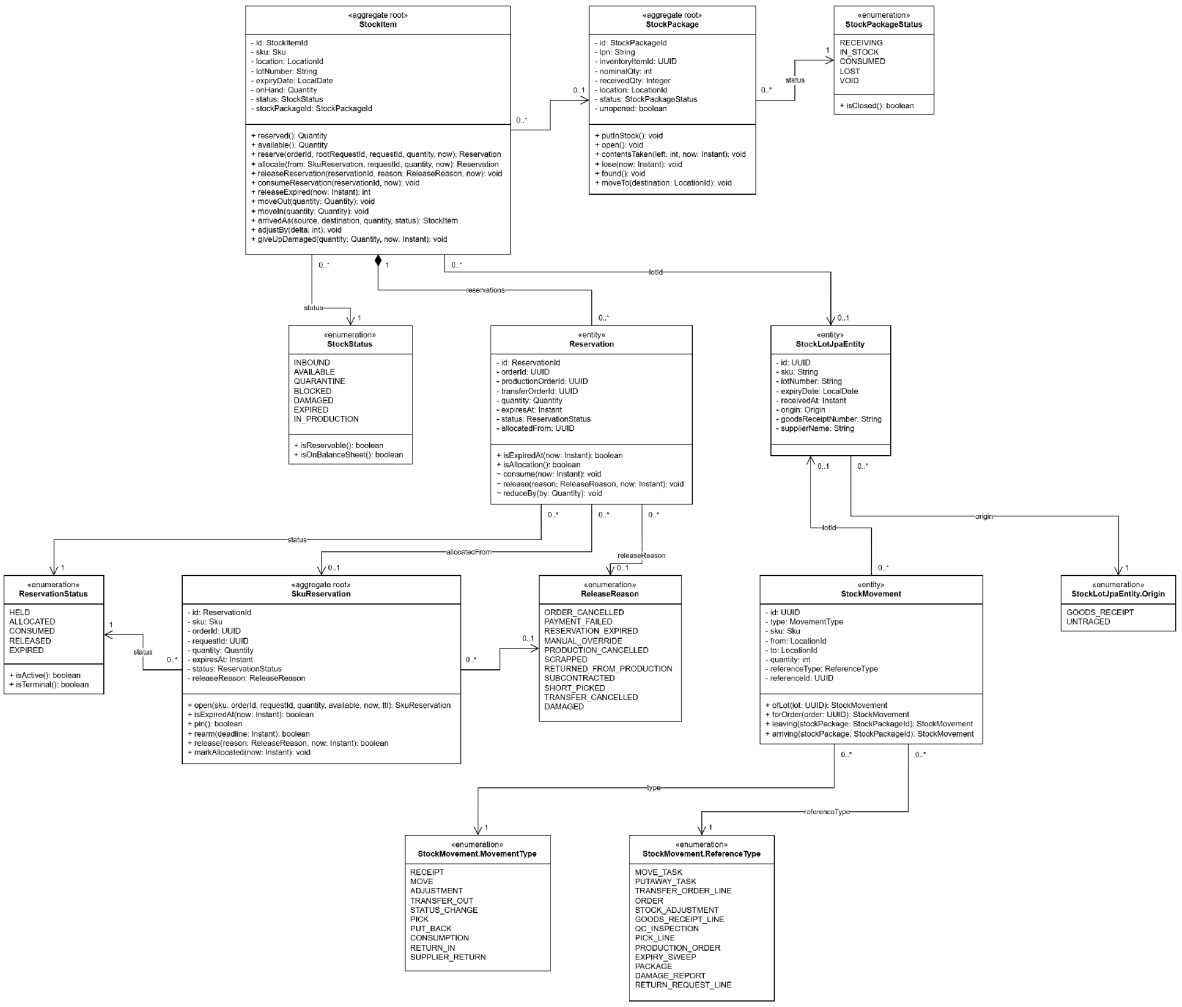


Figure 261. Inventory Management Class Diagram (6/7): Domain Model - Stock, Holds, Cartons and Ledger

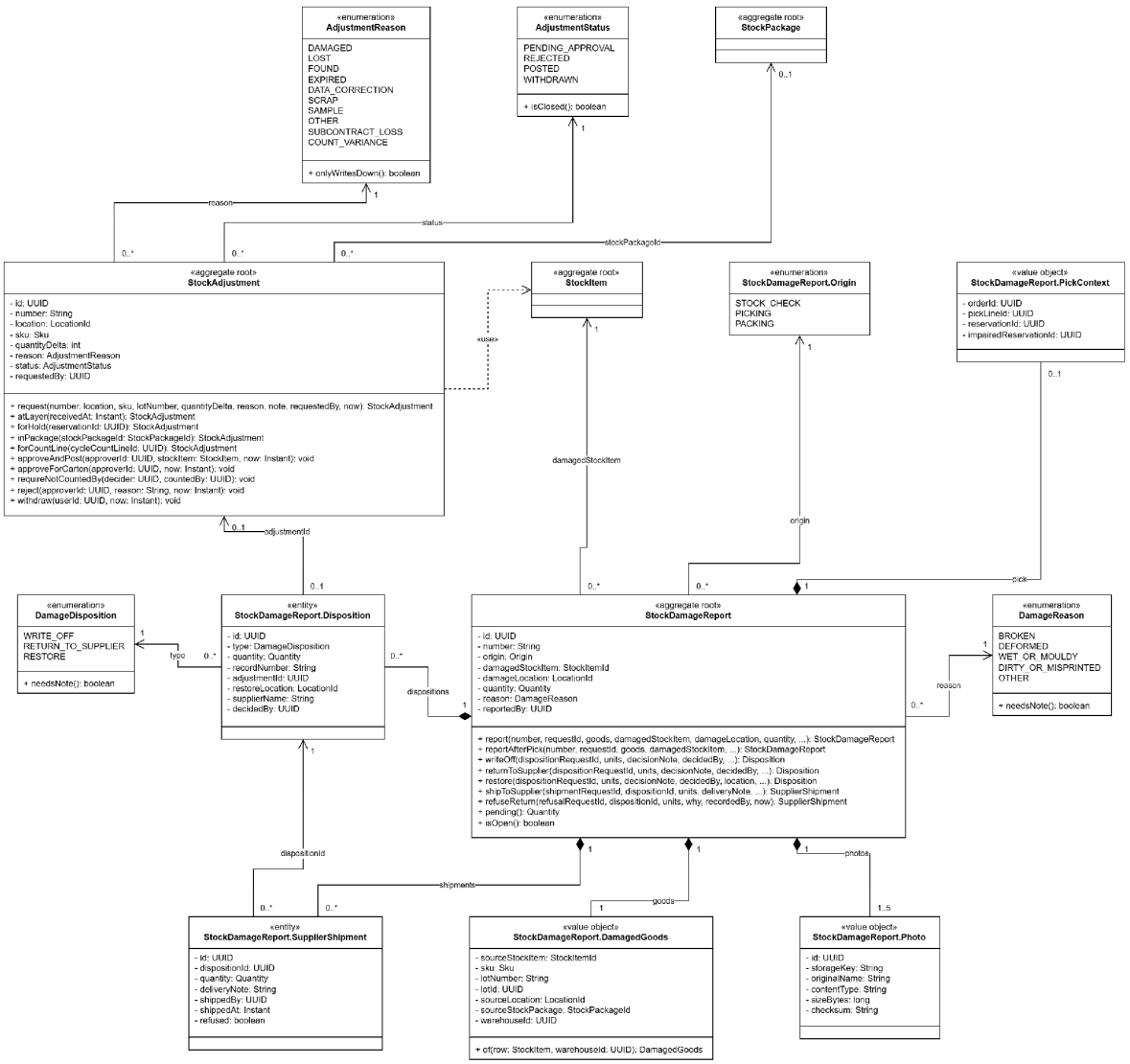


Figure 262. Inventory Management Class Diagram (7/7): Domain Model - Adjustments and Damage Reports

3.16.2 Class Specification

InventoryController

No	Method	Description
1	stockItems(sku, user)	GET /api/v1/inventory/stock-items: Sellable, unexpired stock rows of one SKU by location in removal-strategy order (FEFO/FIFO), limited to the caller's warehouses; for a caller without cycle-count APPROVE the quantities of rows under an open count are hidden. (:96)

2	warehouseStock(warehouseId, location, sku, lotNumber, statuses, includeEmpty, page, size, sort, user)	GET /api/v1/inventory/warehouses/{warehouseId}/stock-items: Every stock row of one warehouse in any status, paged and filtered by location prefix, SKU, lot and status; another warehouse is 403 OUT_OF_DATA_SCOPE, a location outside it 400; counted slots hidden from a blind counter. (:115)
3	stockLevels(sku)	GET /api/v1/inventory/stock-levels: On hand, available, reserved and ATP of one SKU per warehouse; ATP excludes stock frozen by a short-pick count and is capped at the system-wide ATP after live SKU reservations. (:135)
4	availableToPromise(sku, warehouse)	GET /api/v1/inventory/stock-items/atp: Available to promise of one SKU, system-wide (unheld sellable units minus live SKU reservations) or for one warehouse code. (:142)
5	reserve(request)	POST /api/v1/inventory/reservations: Opens a SKU reservation (no location) for an order line, 201; idempotent on requestId; above ATP 409 INSUFFICIENT_STOCK. Checkout does not use it: order calls InventoryService.reserve in-process. (:162)
6	reservations(orderId, sku, statuses, warehouse, page, size)	GET /api/v1/inventory/reservations: Holds newest first, filtered by order, SKU, status and warehouse. (:172)
7	reservation(reservationId)	GET /api/v1/inventory/reservations/{reservationId}: One hold; 404 RESERVATION_NOT_FOUND, 403 OUT_OF_DATA_SCOPE outside the caller's warehouses. (:187)
8	consume(reservationId, actor)	POST /api/v1/inventory/reservations/{reservationId}/consumption: Operations-only deduction of a hold outside a pick task (PickConsumptionService.consume): hold CONSUMED, on_hand falls, one PICK ledger row, StockDeducted published; a hold no longer HELD is 409 RESERVATION_NOT_ACTIVE. (:210)
9	release(reservationId, reason)	DELETE /api/v1/inventory/reservations/{reservationId}: Releases a SKU reservation or a stock-item hold (RELEASED); a closed hold is left as it is, a hold in the print shop or in packing is not released by this path. (:220)

Table 91. InventoryController Class Specification

InventoryItemController

No	Method	Description

1	list(sku, q, storageClass, policyConfigured, page, size)	GET /api/v1/inventory/items: Inventory items in SKU order; sku is exact, q matches SKU or barcode, filters by storage class and whether a policy is set. (:64)
2	get(id)	GET /api/v1/inventory/items/{id}: One item: logistics, stock policy and the reorder/safety-stock evaluation against usable on-hand; 404 INVENTORY_ITEM_NOT_FOUND. (:79)
3	packagingPreview(id, quantity)	GET /api/v1/inventory/items/{id}/packaging-preview: Advisory: how a base-unit quantity would be packed (carton, full cartons, remainder, warnings); status MISSING when no size and weight are known; nothing stored. (:89)
4	describe(id, request)	PUT /api/v1/inventory/items/{id}/logistics: Replaces unit, barcode, sizes and weight under the item's advisory lock; stale version 409 OPTIMISTIC_LOCK, retired SKU 409 VARIANT_OBSOLETE, unit change while stock is held 409 INVENTORY_POLICY_STOCK_CONFLICT, duplicate barcode 409 CONFLICT. (:96)
5	configure(id, request)	PUT /api/v1/inventory/items/{id}/policy: Replaces reorder point, safety stock, max, removal strategy and tracking; invalid 400 INVENTORY_POLICY_INVALID, tracking change under an open count 409 COUNT_POLICY_CONFLICT; publishes SkuInventoryControlChanged so StockAlertEventListener re-evaluates alerts after commit. (:104)

Table 92. InventoryItemController Class Specification

StockAlertController

No	Method	Description
1	list(kind, status, sku, warehouseId, page, size, sort)	GET /api/v1/inventory/alerts: Alerts newest first by kind, status, SKU and warehouse; a warehouse-limited user sees only expiry alerts of their warehouses, another warehouse is 403 OUT_OF_DATA_SCOPE. (:65)
2	summary()	GET /api/v1/inventory/alerts/summary: Number of open alerts per kind, for the cards above the list. (:80)
3	unconfigured(page, size)	GET /api/v1/inventory/alerts/unconfigured: SKUs that hold stock but have no reorder point; 403 for a warehouse-limited caller. (:89)
4	get(alertId)	GET /api/v1/inventory/alerts/{alertId}: One alert; 404 ALERT_NOT_FOUND. (:99)

5	acknowledge(alertId, request, user)	POST /api/v1/inventory/alerts/{alertId}/acknowledgement: OPEN → ACKNOWLEDGED with an optional note; the condition stays open until it clears; stale version 409 OPTIMISTIC_LOCK, not OPEN 409 INVALID_ALERT_TRANSITION. (:106)

Table 93. StockAlertController Class Specification

StockOperationsController

No	Method	Description
1	move(request, user)	POST /api/v1/inventory/movements: Moves unreserved loose units between two locations of one warehouse (201), idempotent on requestId; slot under a count 409 STOCK_SLOT_UNDER_COUNT, other warehouse 409 MOVE_ACROSS_WAREHOUSES, unusable target 409 LOCATION_NOT_USABLE, too few free units 409 INSUFFICIENT_STOCK. (:77)
2	movements(sku, location, type, page, size, user)	GET /api/v1/inventory/movements: The stock ledger newest first; location matches either side of a line; counted slots hidden from a blind counter. (:87)
3	requestAdjustment(request, user)	POST /api/v1/inventory/adjustments: Asks for a correction with a reason (201, PENDING_APPROVAL); stock is unchanged until another person approves; COUNT_VARIANCE is refused (400), unknown stock 404 STOCK_ITEM_NOT_FOUND. (:103)
4	adjustments(status, sku, location, page, size, sort, user)	GET /api/v1/inventory/adjustments: Adjustments newest first; status=PENDING_APPROVAL is the approver's queue. (:113)
5	adjustment(adjustmentId, user)	GET /api/v1/inventory/adjustments/{adjustmentId}: One adjustment; unknown or outside the caller's warehouses 404 STOCK_ADJUSTMENT_NOT_FOUND. (:129)
6	approve(adjustmentId, user)	POST /api/v1/inventory/adjustments/{adjustmentId}/approval: Approves and posts (POSTED, ADJUSTMENT ledger line); the requester or the count line's counter is 409 ADJUSTMENT_SELF_APPROVAL, decided 409 INVALID_ADJUSTMENT_TRANSITION, below held units 409 INSUFFICIENT_STOCK. (:140)
7	reject(adjustmentId, request, user)	POST /api/v1/inventory/adjustments/{adjustmentId}/rejection: Rejects with a reason (REJECTED), stock unchanged; same four-eyes rule as approval. (:148)

8	askScrapAgain(reservationId, request, user)	POST /api/v1/inventory/reservations/{reservationId}/scrap-request: Asks again (201) to scrap a cancelled order's printed goods whose write-off was rejected or withdrawn (BR-PRD-10); decided by the same approval. (:162)
9	withdraw(adjustmentId, user)	POST /api/v1/inventory/adjustments/{adjustmentId}/withdrawal: The requester takes back a pending adjustment (WITHDRAWN); someone else 403 ADJUSTMENT_NOT_REQUESTER, a count variance 409 COUNT_VARIANCE_NOT_WITHDRAWABLE. (:171)

Table 94. StockOperationsController Class Specification

LotTraceController

No	Method	Description
1	lots(sku, lotNumber, warehouseId, from, to, page, size)	GET /api/v1/inventory/lots: Lots most recently received first, by SKU, lot number, warehouse and the days they moved; another warehouse 403. (:47)
2	trace(lotId, warehouseId, user)	GET /api/v1/inventory/lots/{lotId}: One lot traced: origin (receipt, PO, supplier), balances by location, holds and allocations, customer orders it went to; 404 NOT_FOUND outside the caller's warehouses. (:62)
3	movements(lotId, warehouseId, from, to, page, size, user)	GET /api/v1/inventory/lots/{lotId}/movements: The lot's ledger lines oldest first, paged and date-filtered. (:80)

Table 95. LotTraceController Class Specification

3.16.3 Look Up Stock Sequence Diagram

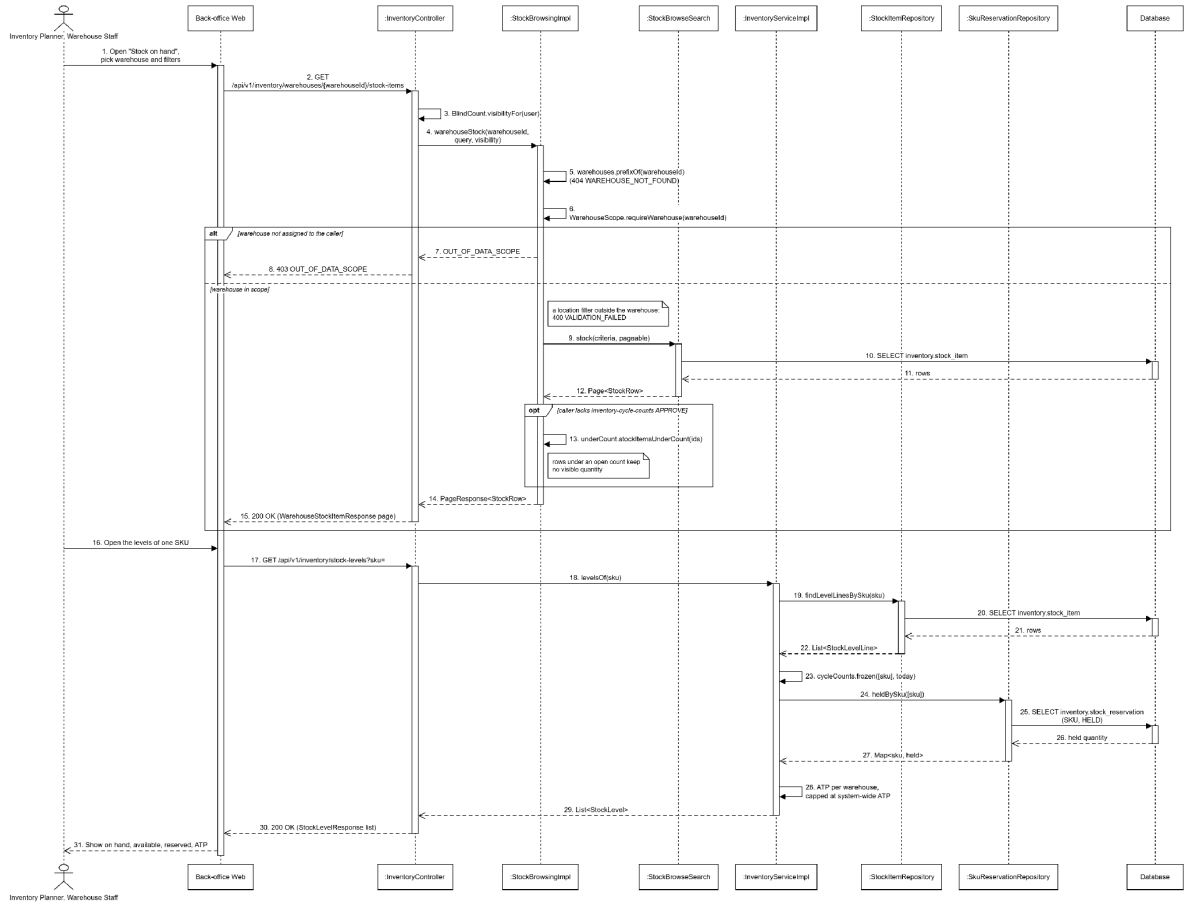


Figure 263. Look Up Stock Sequence Diagram

3.16.4 Maintain Inventory Item & Stock Policy (Logistics) Sequence Diagram

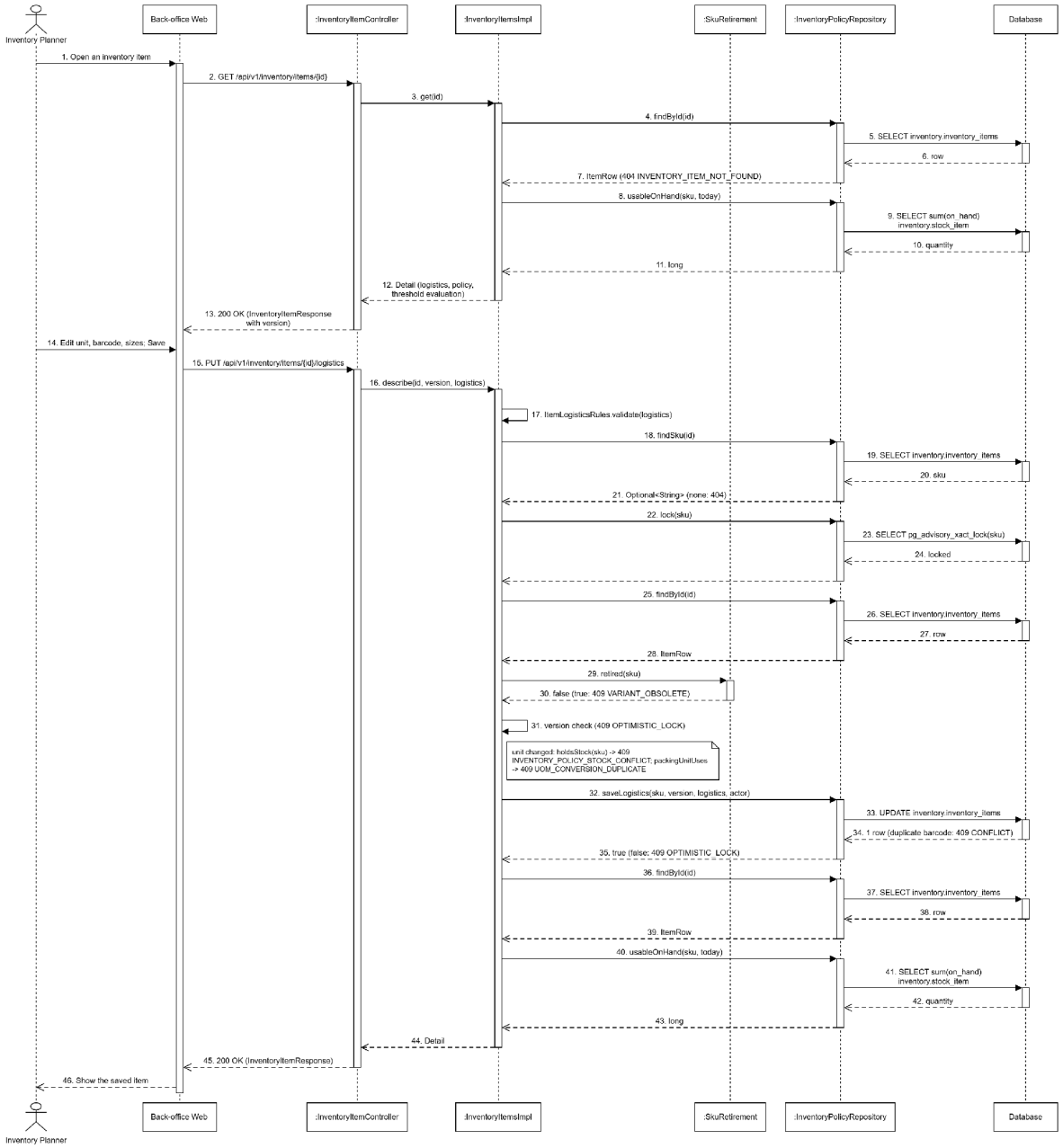


Figure 264. Maintain Inventory Item & Stock Policy (Logistics) Sequence Diagram

3.16.5 Maintain Inventory Item & Stock Policy (Stock Policy) Sequence Diagram

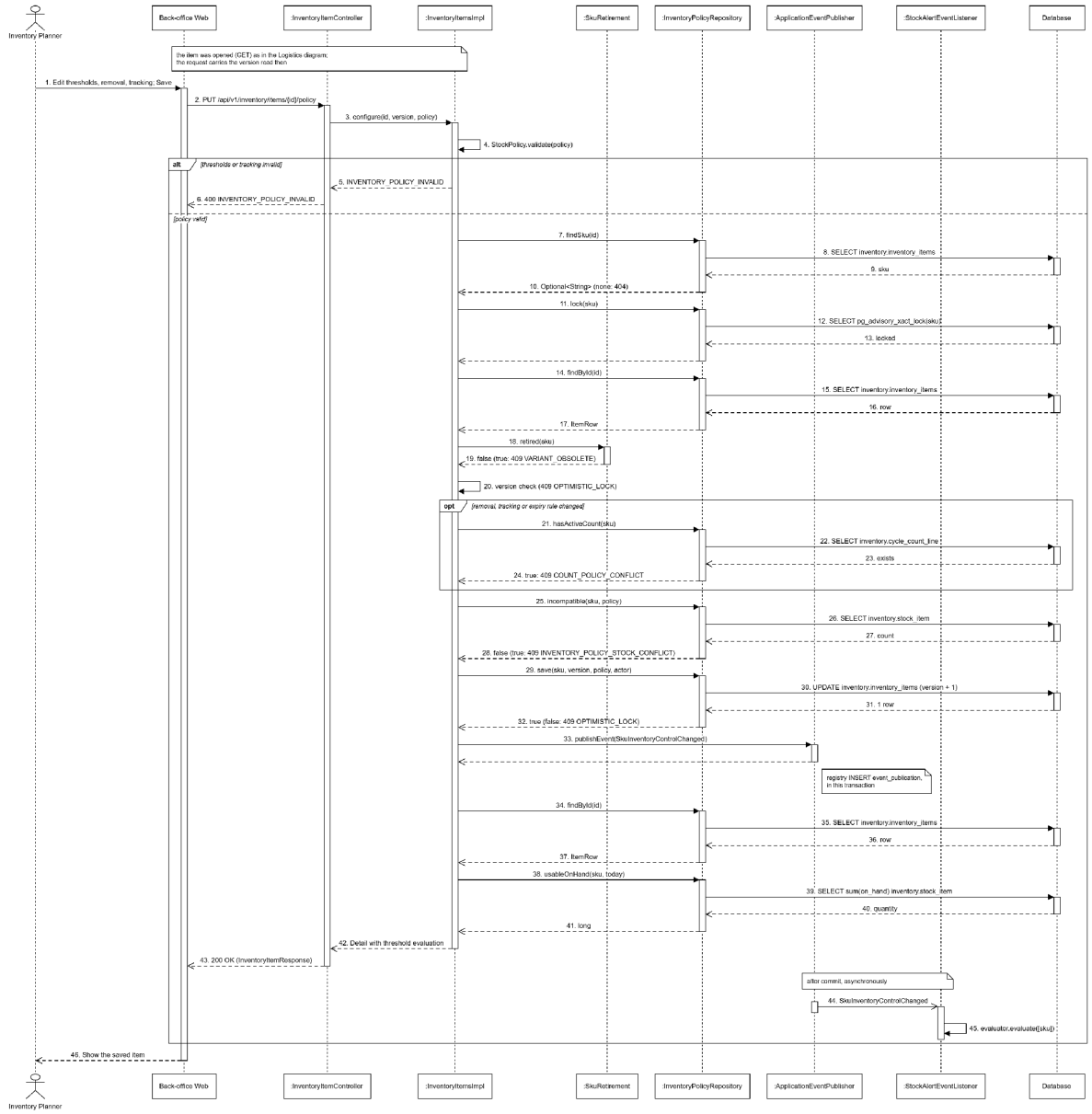


Figure 265. Maintain Inventory Item & Stock Policy (Stock Policy) Sequence Diagram

3.16.6 Acknowledge Stock Alert Sequence Diagram

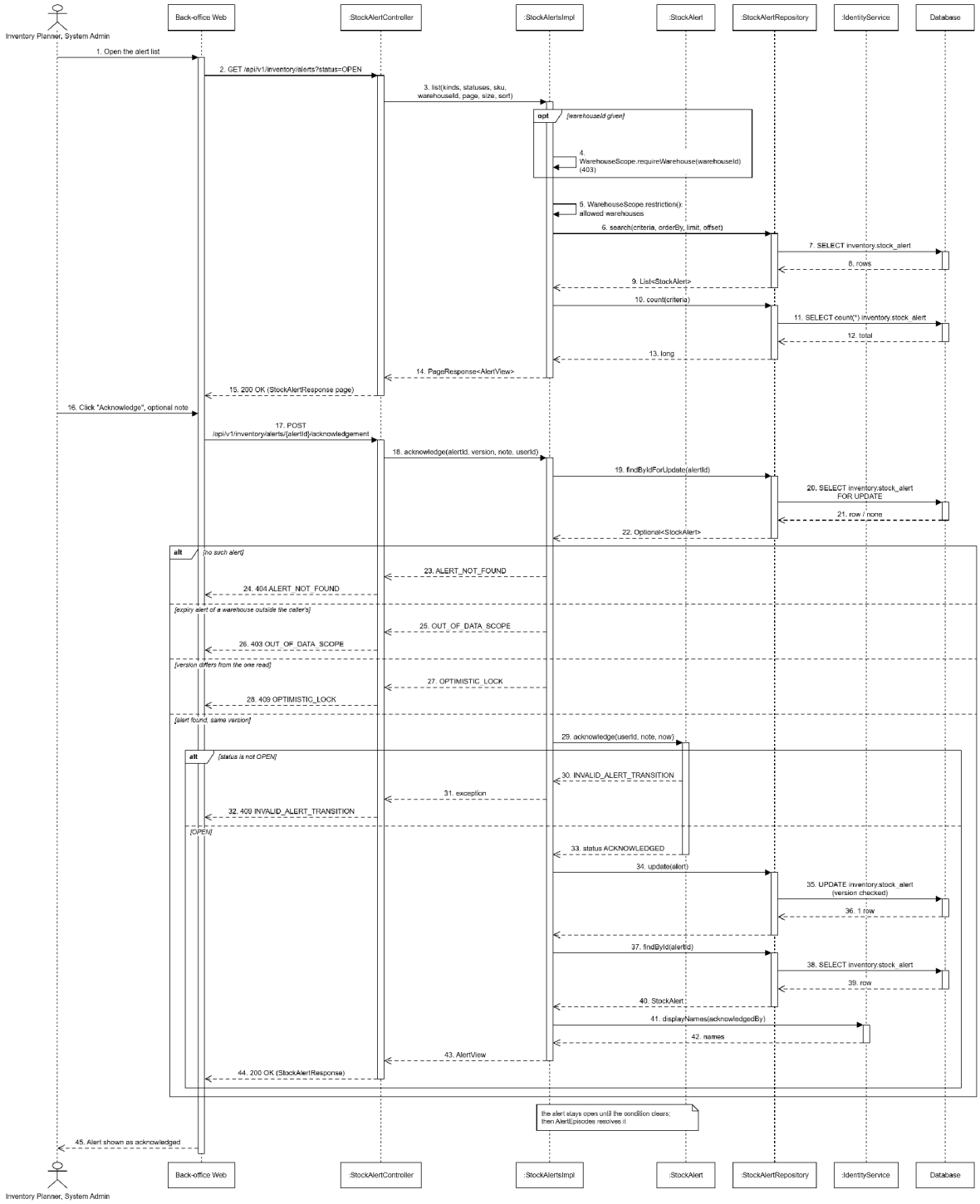


Figure 266. Acknowledge Stock Alert Sequence Diagram

3.16.7 Request Inventory Adjustment Sequence Diagram

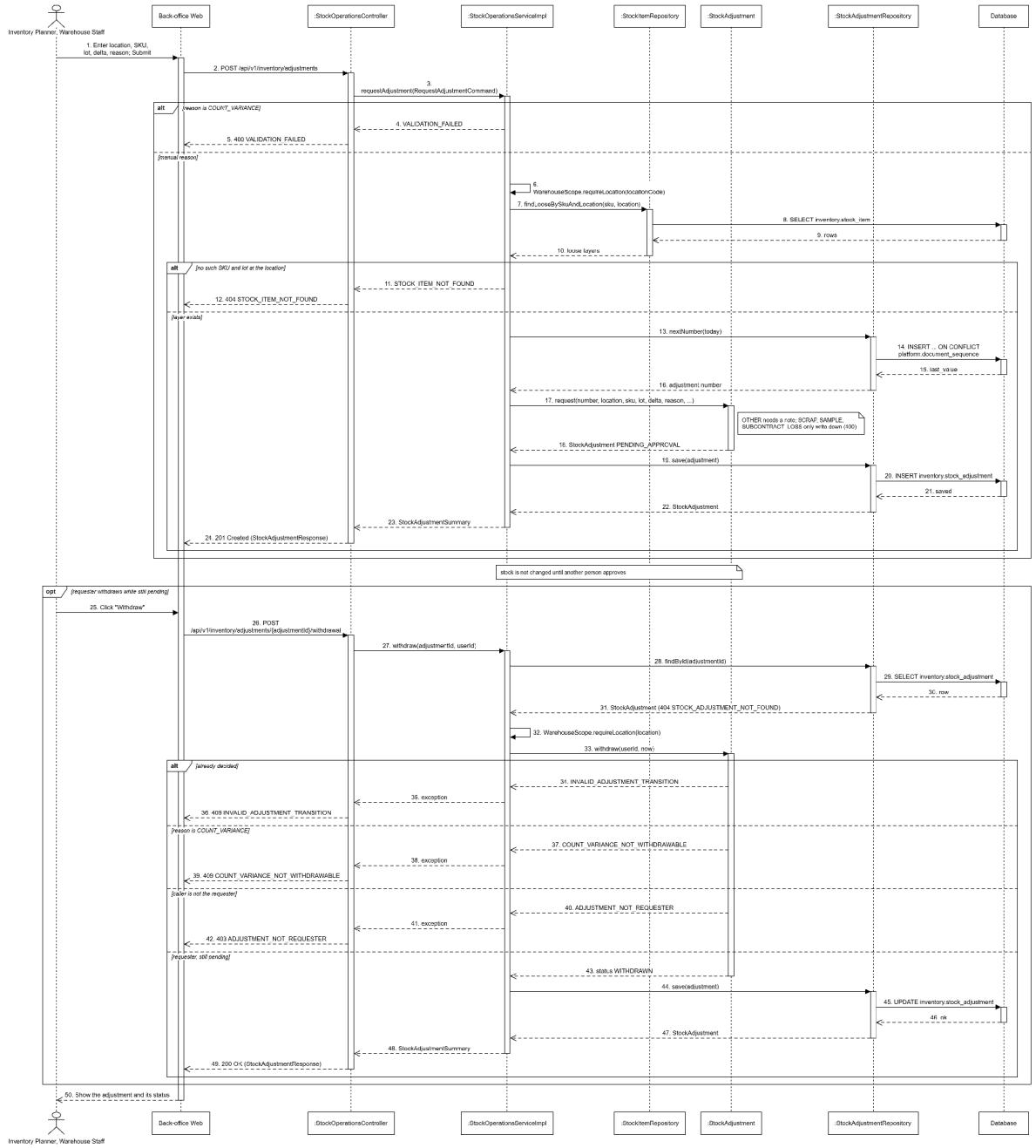


Figure 267. Request Inventory Adjustment Sequence Diagram

3.16.8 Review Count Variances & Adjustments (Approve or Reject) Sequence Diagram

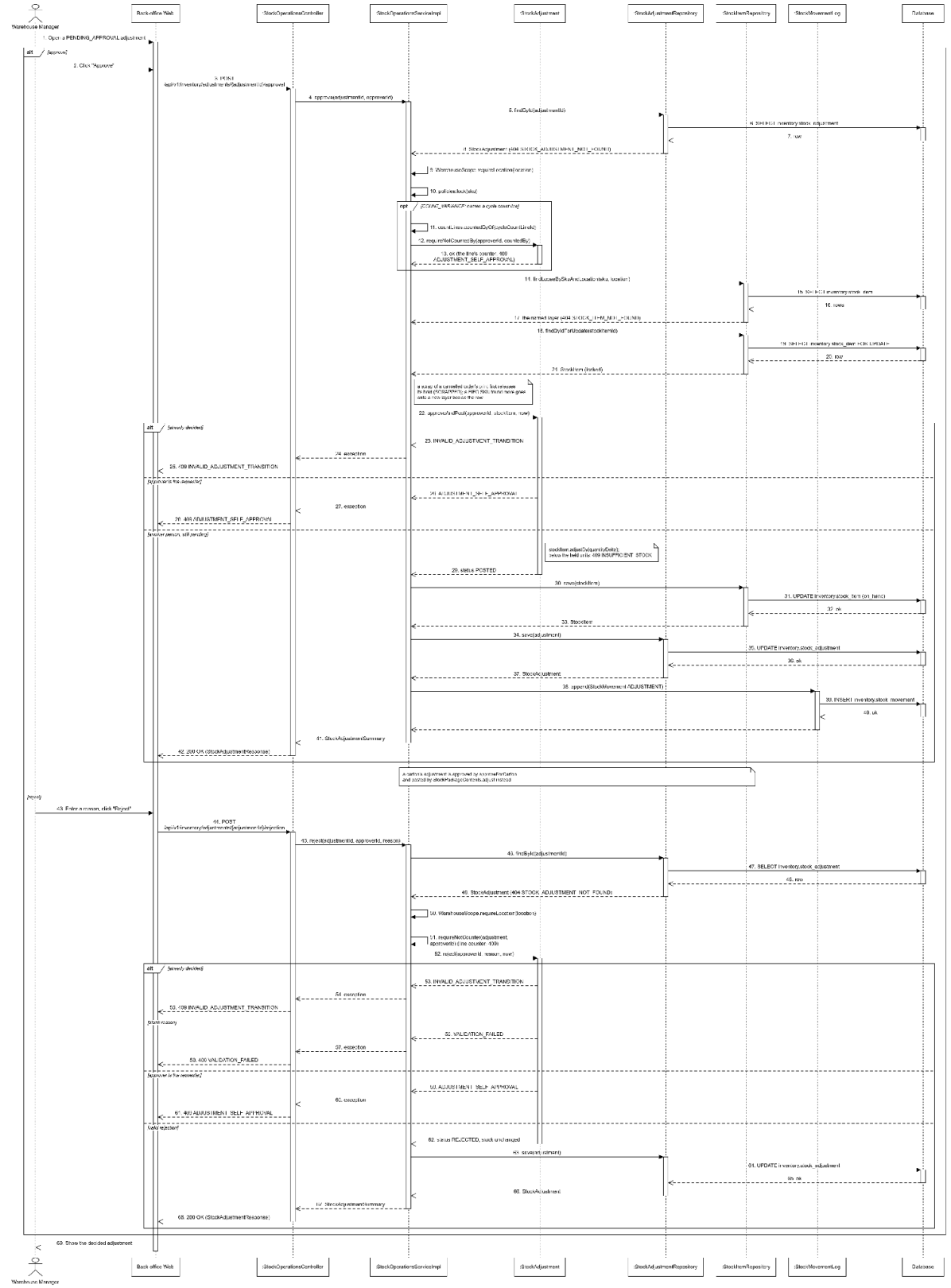


Figure 268. Review Count Variances & Adjustments (Approve or Reject) Sequence Diagram

3.16.9 Move Stock Within Warehouse Sequence Diagram

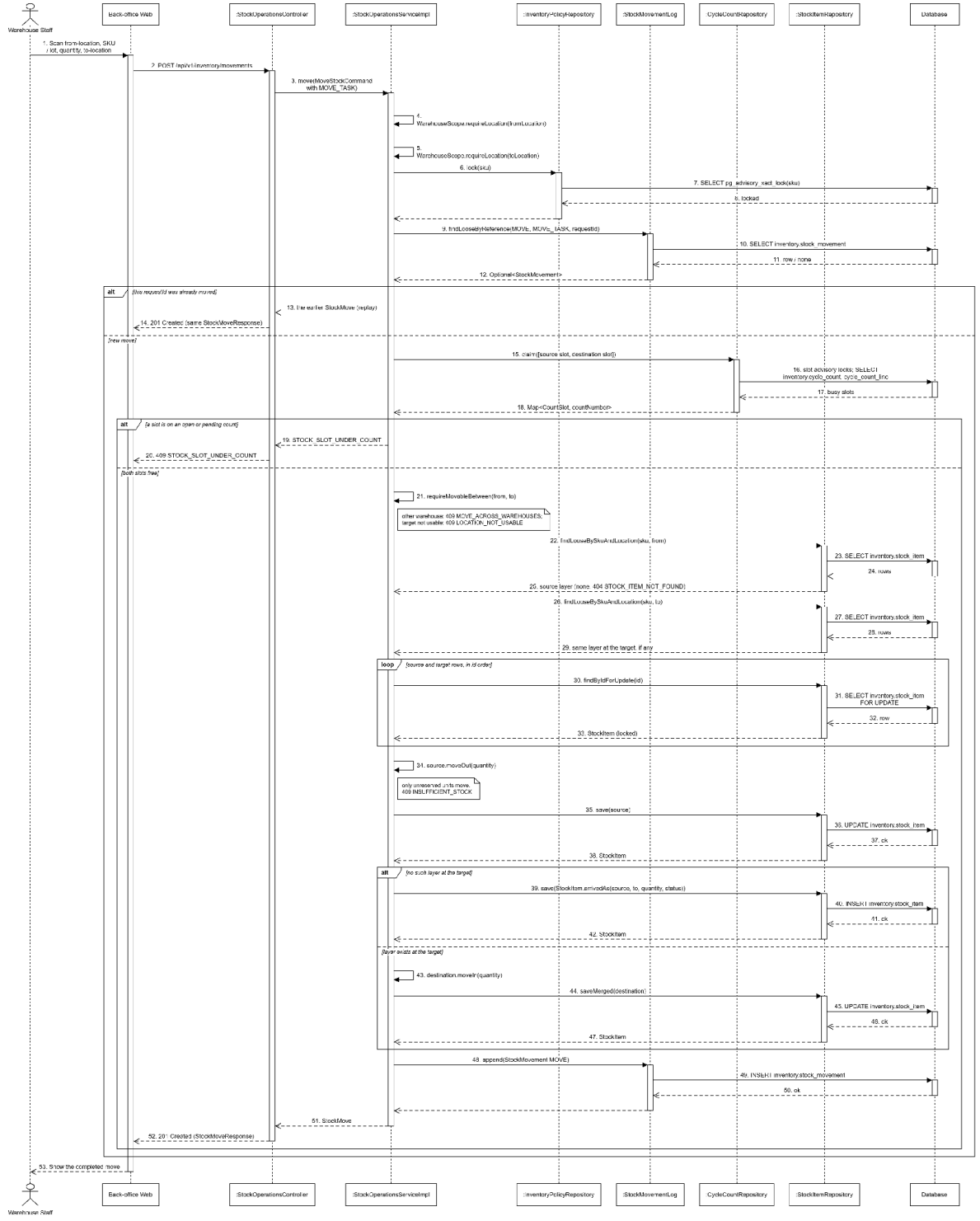


Figure 269. Move Stock Within Warehouse Sequence Diagram

3.16.10 Trace Lot Sequence Diagram

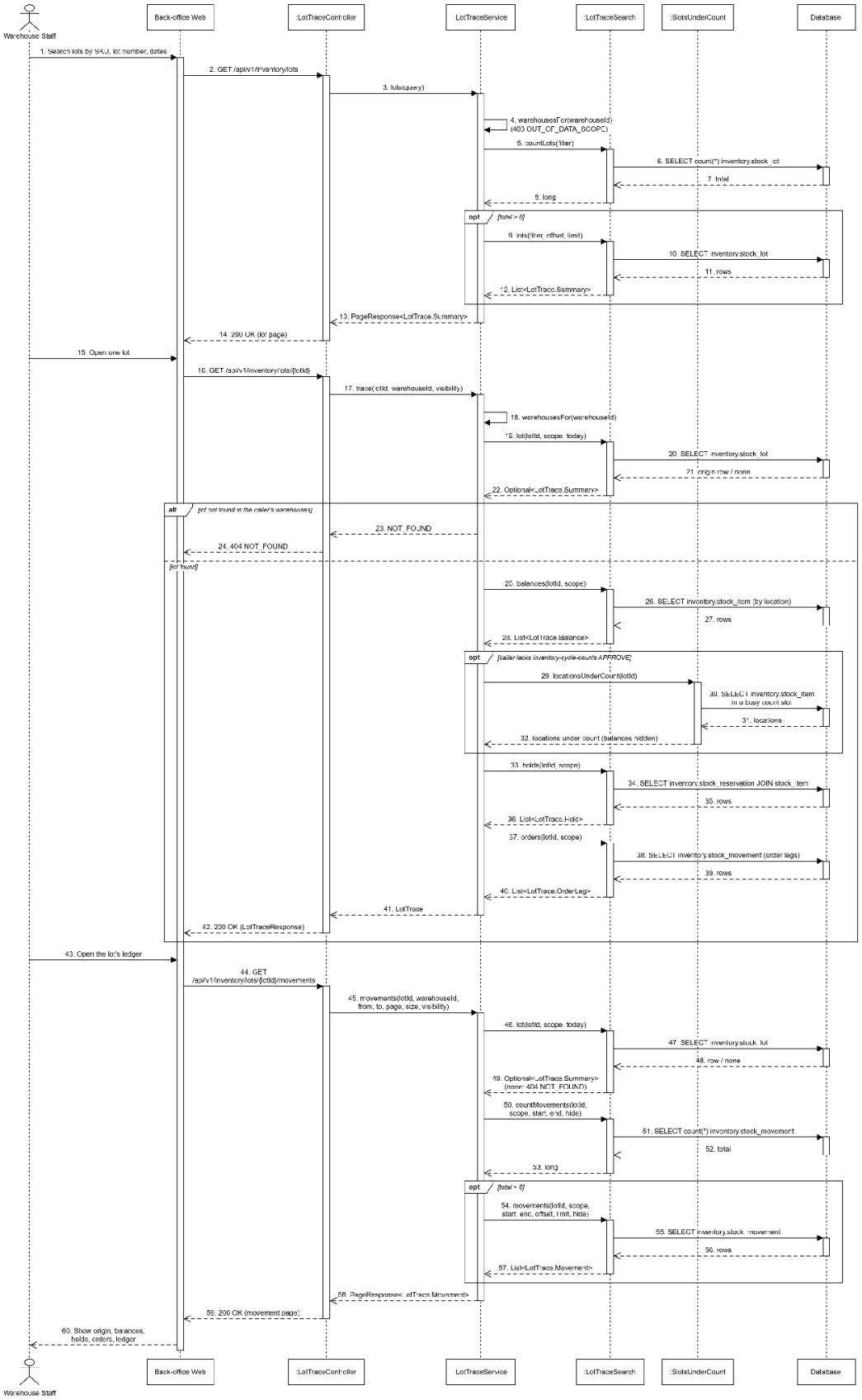


Figure 270. Trace Lot Sequence Diagram

3.16.11 Stock Adjustment State Diagram

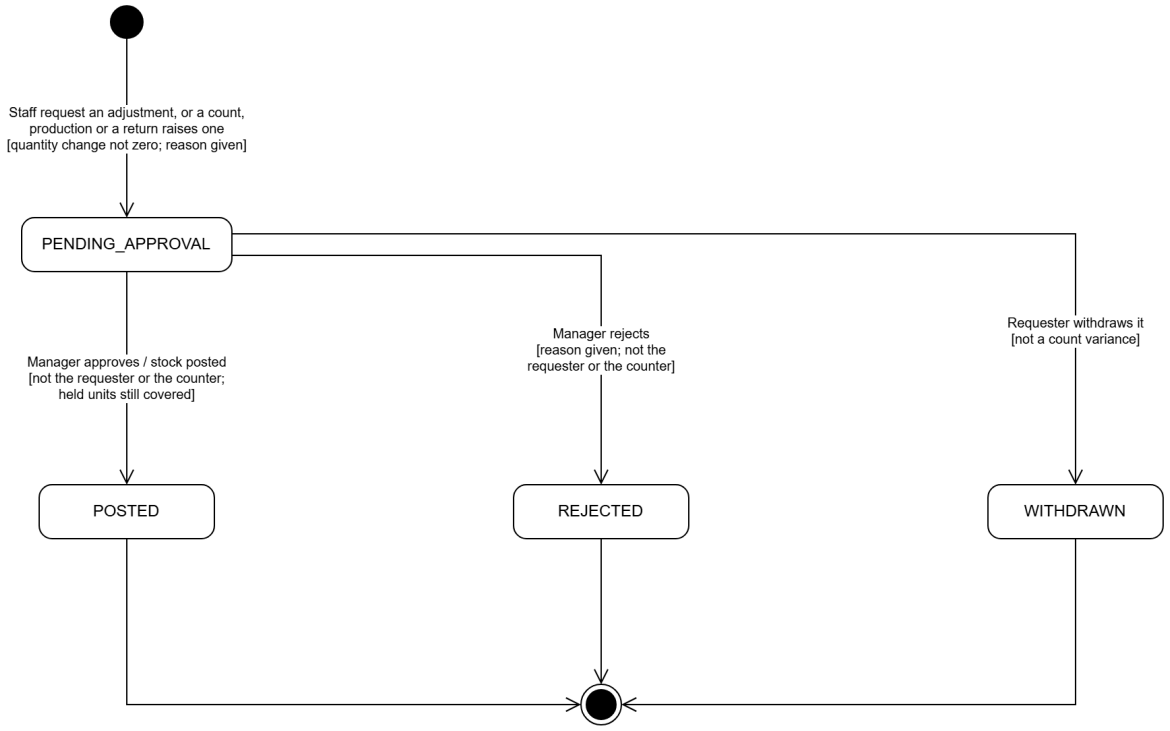


Figure 271. Stock Adjustment State Diagram

3.16.12 Stock Reservation State Diagram

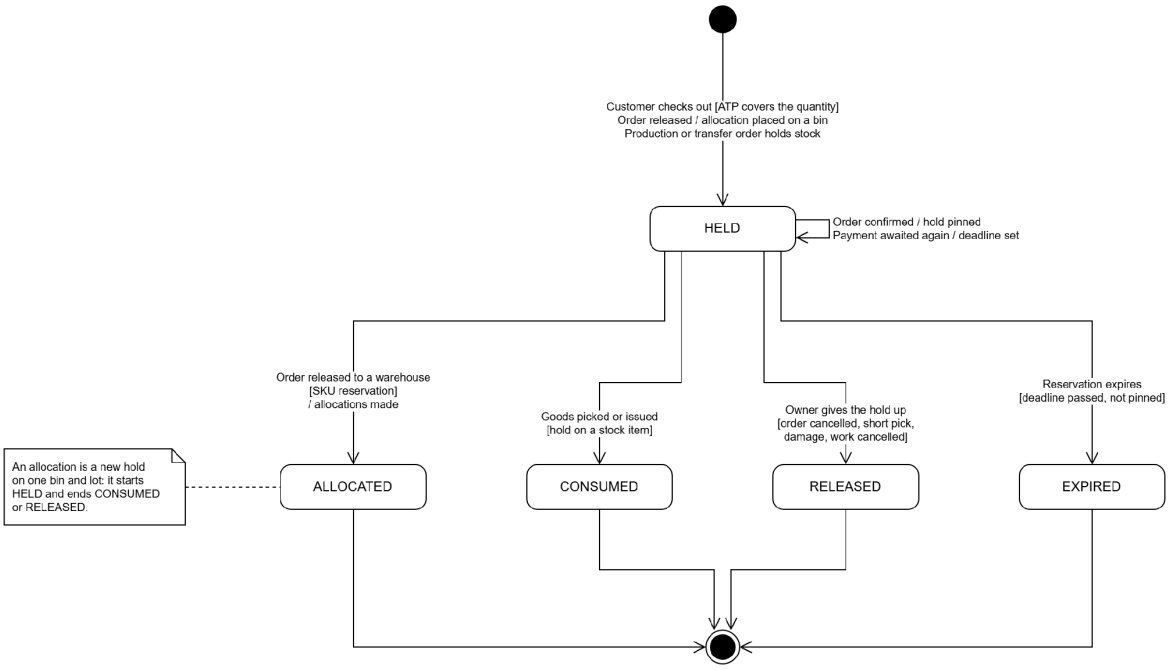


Figure 272. Stock Reservation State Diagram

3.16.13 Stock Item Status State Diagram

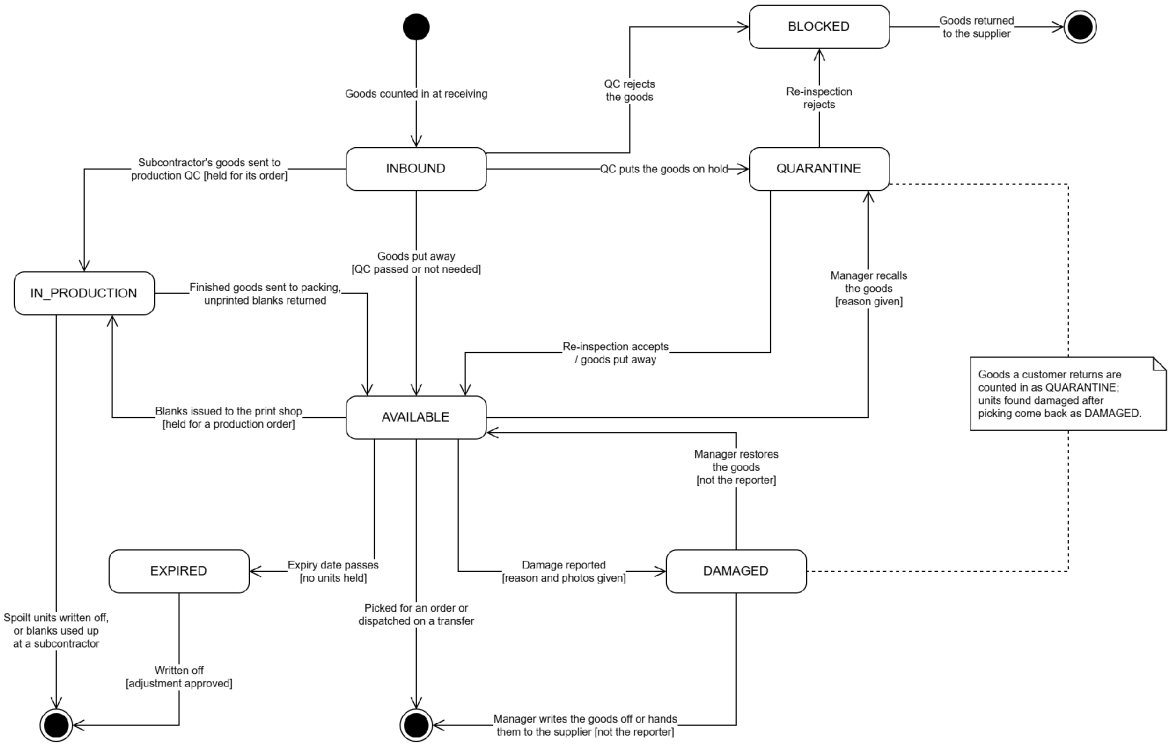


Figure 273. Stock Item Status State Diagram

3.16.14 Stock Alert State Diagram

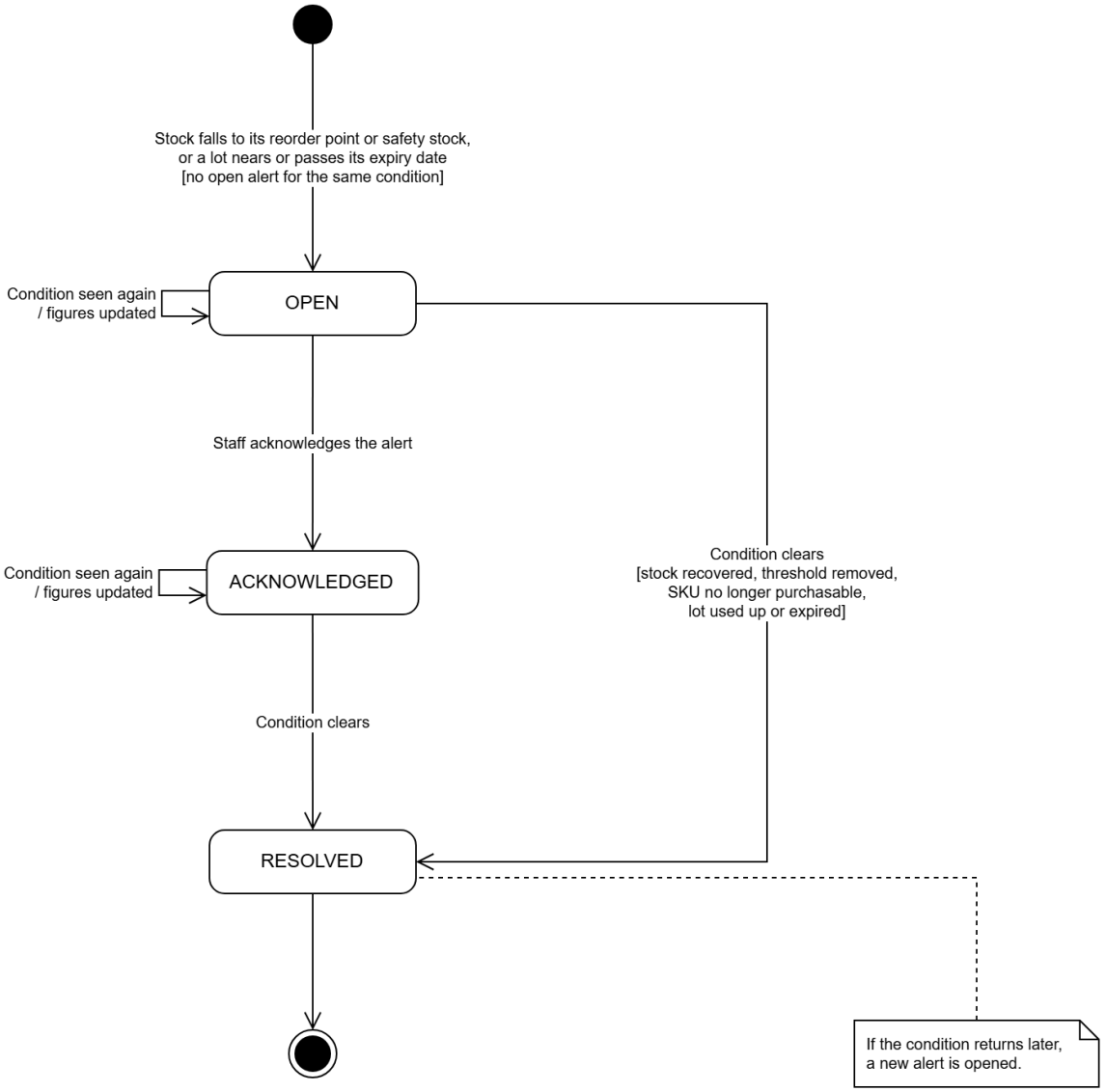


Figure 274. Stock Alert State Diagram

3.17 Transfer Orders

3.17.1 Class Diagram

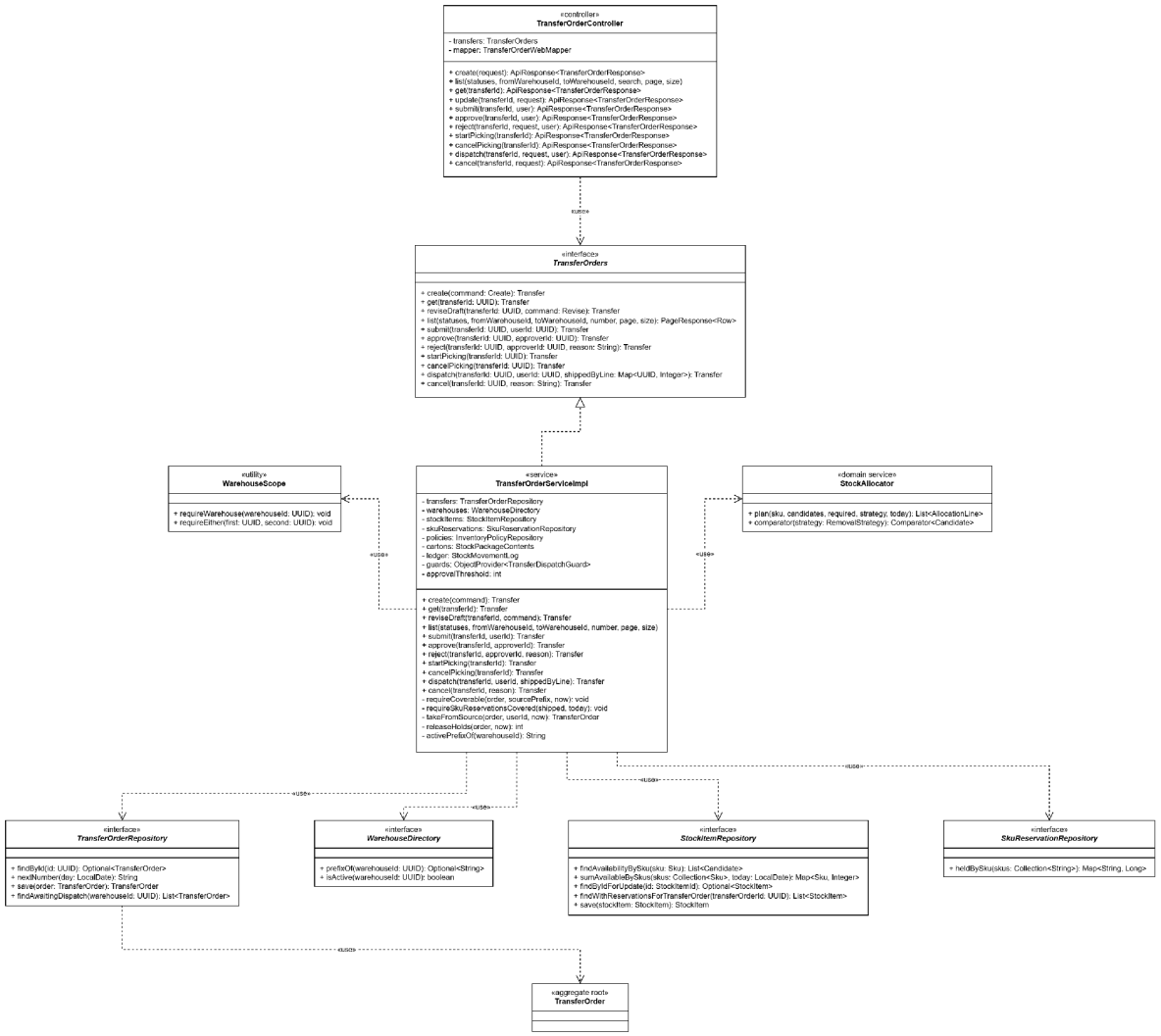


Figure 275. Transfer Orders Class Diagram (1/3): Creating, Submitting and Approving Transfers

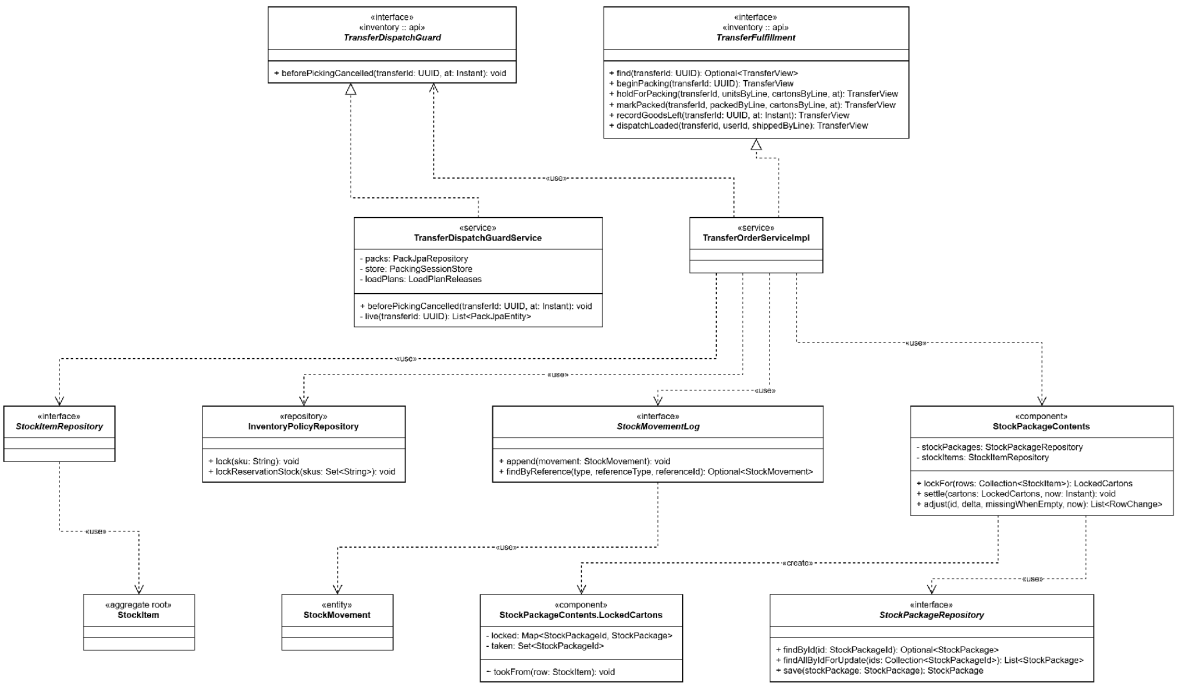


Figure 276. Transfer Orders Class Diagram (2/3): Picking, Cancelling Picking and Dispatch

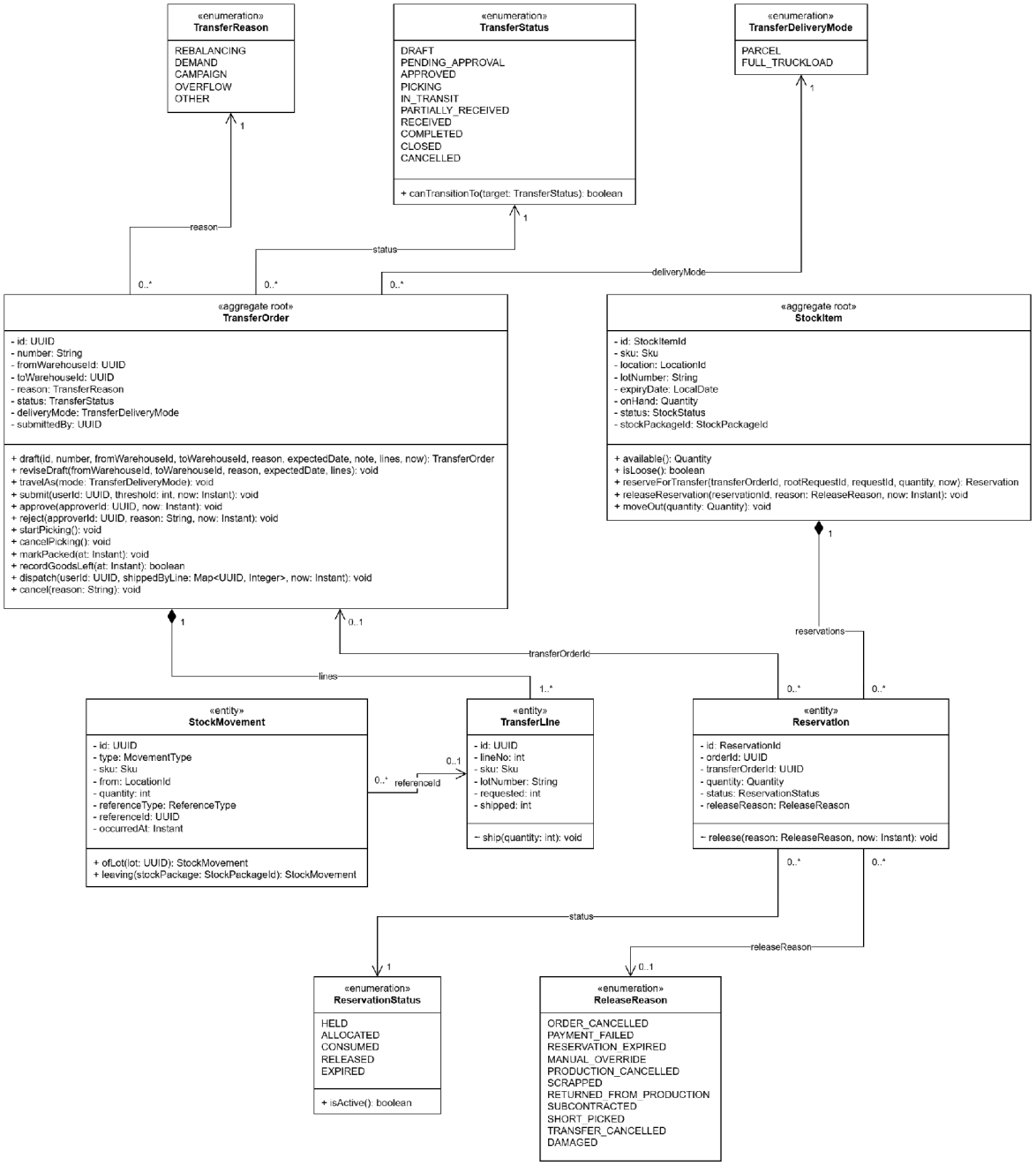


Figure 277. Transfer Orders Class Diagram (3/3): Domain Model

3.17.2 Class Specification

TransferOrderController

No	Method	Description
1	create(request)	POST /api/v1/inventory/transfer-orders: Creates a DRAFT transfer between two ACTIVE warehouses (201); every line must be coverable now from unheld sellable stock at the source minus SKU reservations (409 INSUFFICIENT_STOCK); same warehouse or a repeated SKU and lot 400. (:70)

2	list(statuses, fromWarehouseId, toWarehouseId, search, page, size)	GET /api/v1/inventory/transfer-orders: Transfers newest first by status, warehouses and number. (:80)
3	get(transferId)	GET /api/v1/inventory/transfer-orders/{transferId}: One transfer with its lines and last rejection; 404 TRANSFER_ORDER_NOT_FOUND, 403 unless the caller has one of its two warehouses. (:94)
4	update(transferId, request)	PUT /api/v1/inventory/transfer-orders/{transferId}: Replaces a DRAFT's warehouses, reason, date, lines and delivery mode; outside DRAFT 409 TRANSFER_ORDER_NOT_EDITABLE, stale version 409 OPTIMISTIC_LOCK, 409 WAREHOUSE_INACTIVE, 409 INSUFFICIENT_STOCK. (:104)
5	submit(transferId, user)	POST /api/v1/inventory/transfer-orders/{transferId}/submission: DRAFT → APPROVED at or under the threshold (100 units by default) or → PENDING_APPROVAL above it. (:116)
6	approve(transferId, user)	POST /api/v1/inventory/transfer-orders/{transferId}/approval: PENDING_APPROVAL → APPROVED; the submitter is 409 TRANSFER_SELF_APPROVAL. (:124)
7	reject(transferId, request, user)	POST /api/v1/inventory/transfer-orders/{transferId}/rejection: Back to DRAFT with a reason, kept as rejectionReason/rejectedBy/rejectedAt; same self-decision rule. (:134)
8	startPicking(transferId)	POST /api/v1/inventory/transfer-orders/{transferId}/picking: APPROVED → PICKING, by staff of the source warehouse only. (:143)
9	cancelPicking(transferId)	POST /api/v1/inventory/transfer-orders/{transferId}/picking-cancellation: PICKING → APPROVED; fulfillment voids its parcels and cancels its load plan in the same transaction and its stock holds are released; goods already on a vehicle 409 TRANSFER_GOODS_LEFT, a vehicle being loaded 409 LOAD_PLAN_LOADING_STARTED. (:152)

10	dispatch(transferId, request, user)	POST /api/v1/inventory/transfer-orders/{transferId}/dispatch : Parcel dispatch: shipped units per line leave the source by FIFO/FEFO (TRANSFER_OUT ledger lines) and the transfer is IN_TRANSIT; a full-truckload transfer is 409 TRANSFER_DISPATCH_BY_LOAD_PLAN. (:160)
11	cancel(transferId, request)	POST /api/v1/inventory/transfer-orders/{transferId}/cancellation: Cancels with a reason from DRAFT or APPROVED only (BR-04); any other status 409 INVALID_TRANSFER_TRANSITION. (:171)

Table 96. TransferOrderController Class Specification

3.17.3 Create / Submit Transfer Order Sequence Diagram

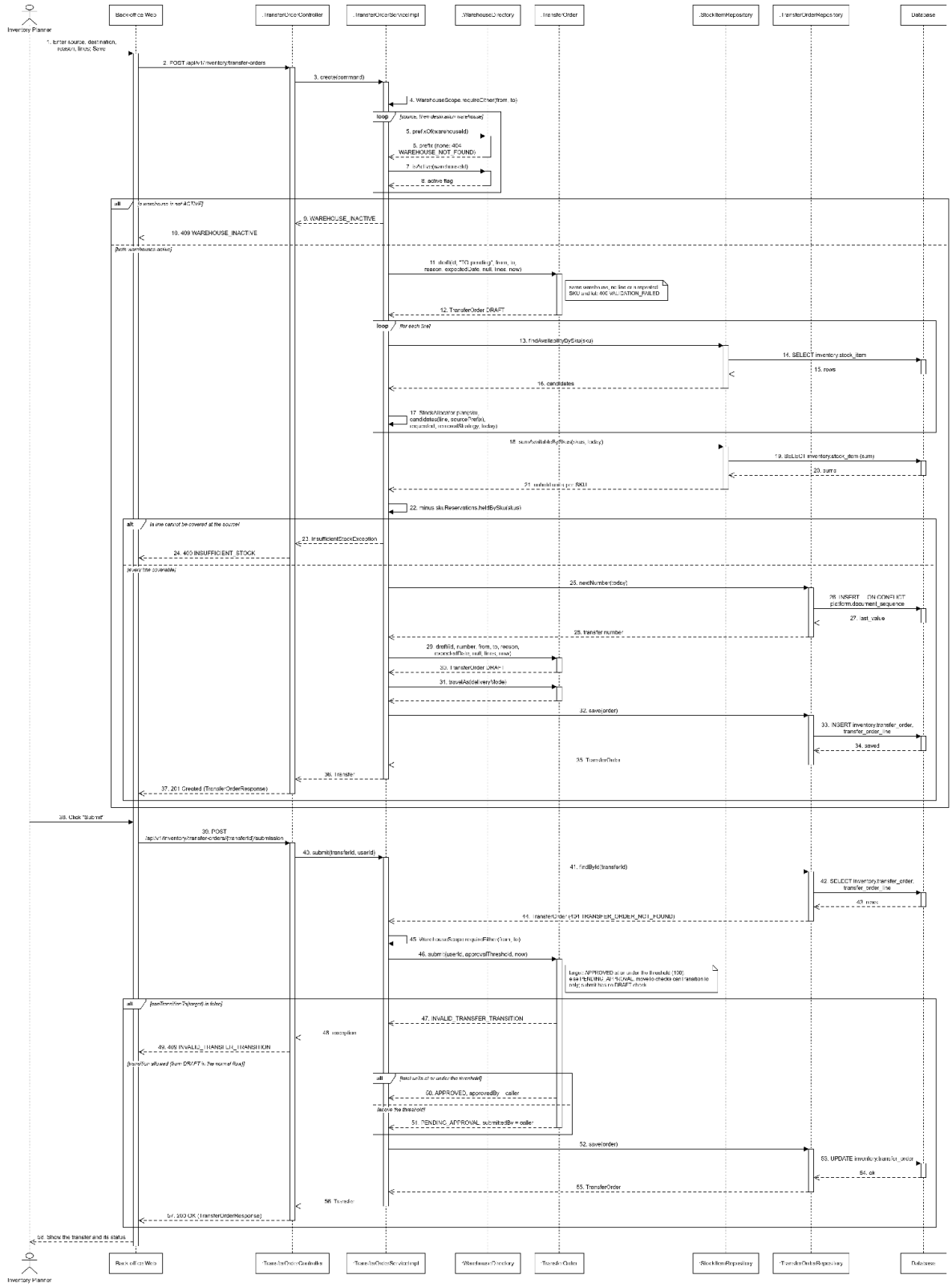


Figure 278. Create / Submit Transfer Order Sequence Diagram

3.17.4 Approve / Reject Transfer Order Sequence Diagram

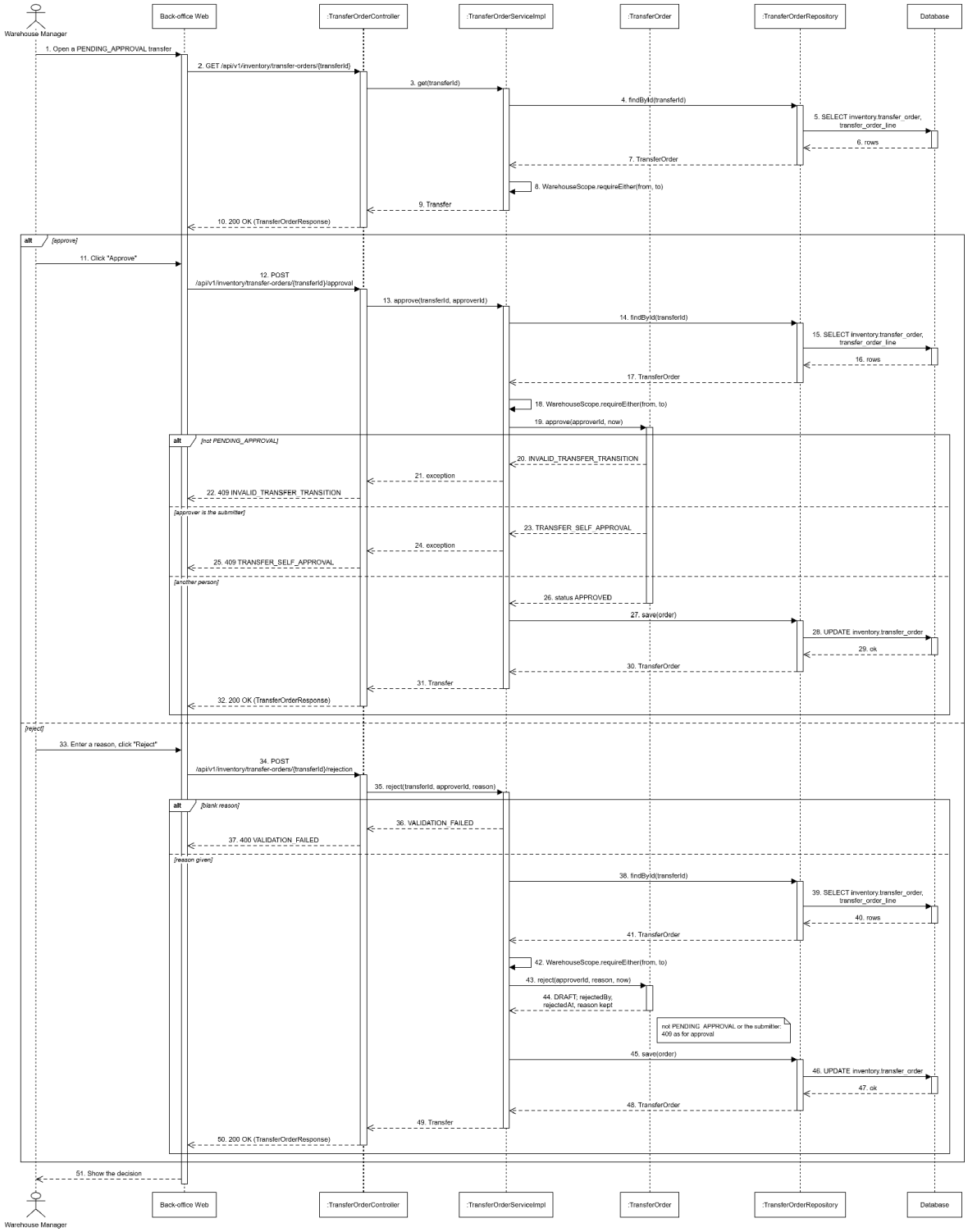


Figure 279. Approve / Reject Transfer Order Sequence Diagram

3.17.5 Cancel Transfer Picking Sequence Diagram

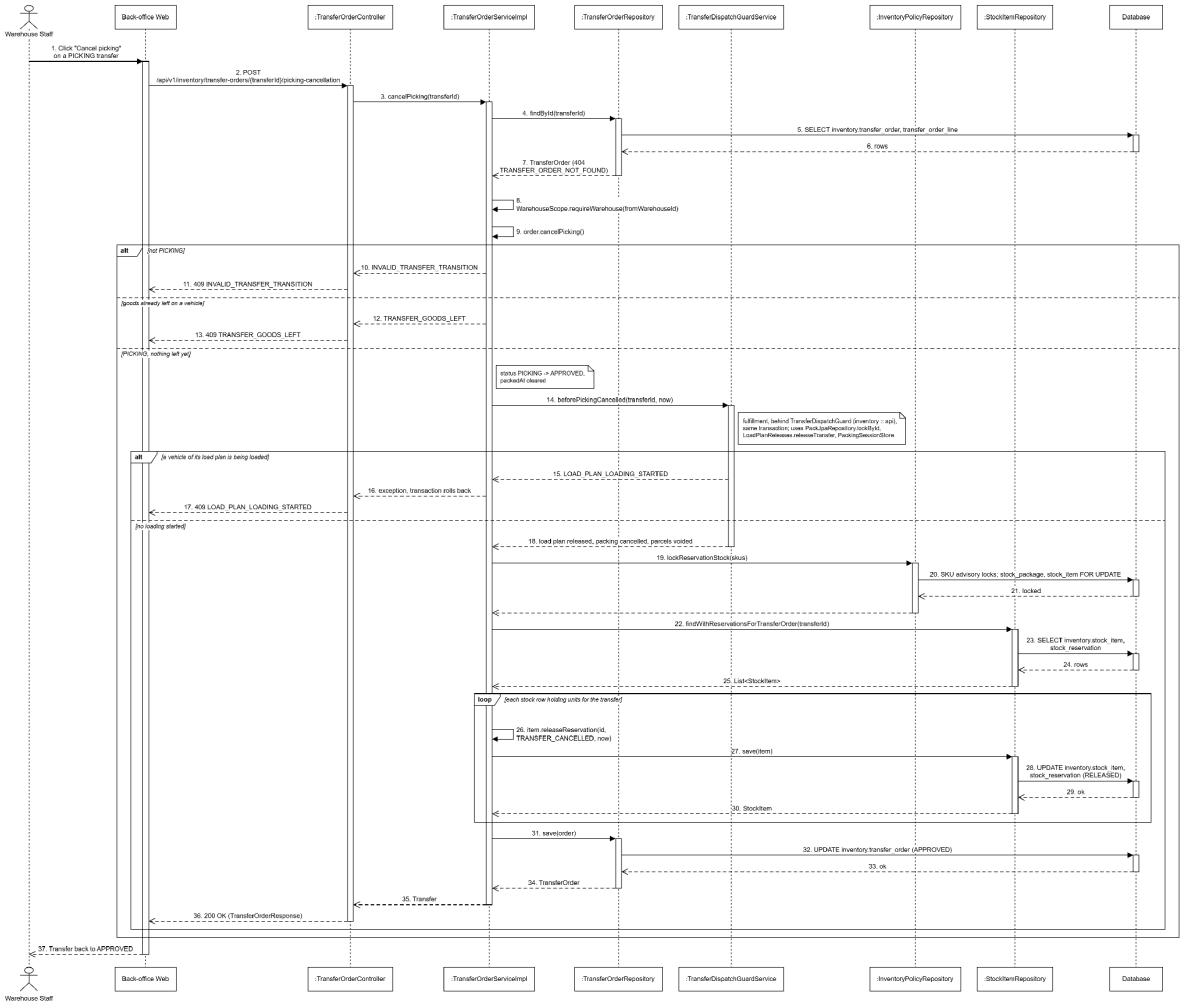


Figure 280. Cancel Transfer Picking Sequence Diagram

3.17.6 Pack & Dispatch Transfer Order (Parcel Mode: Pick and Dispatch, No Packing) Sequence Diagram

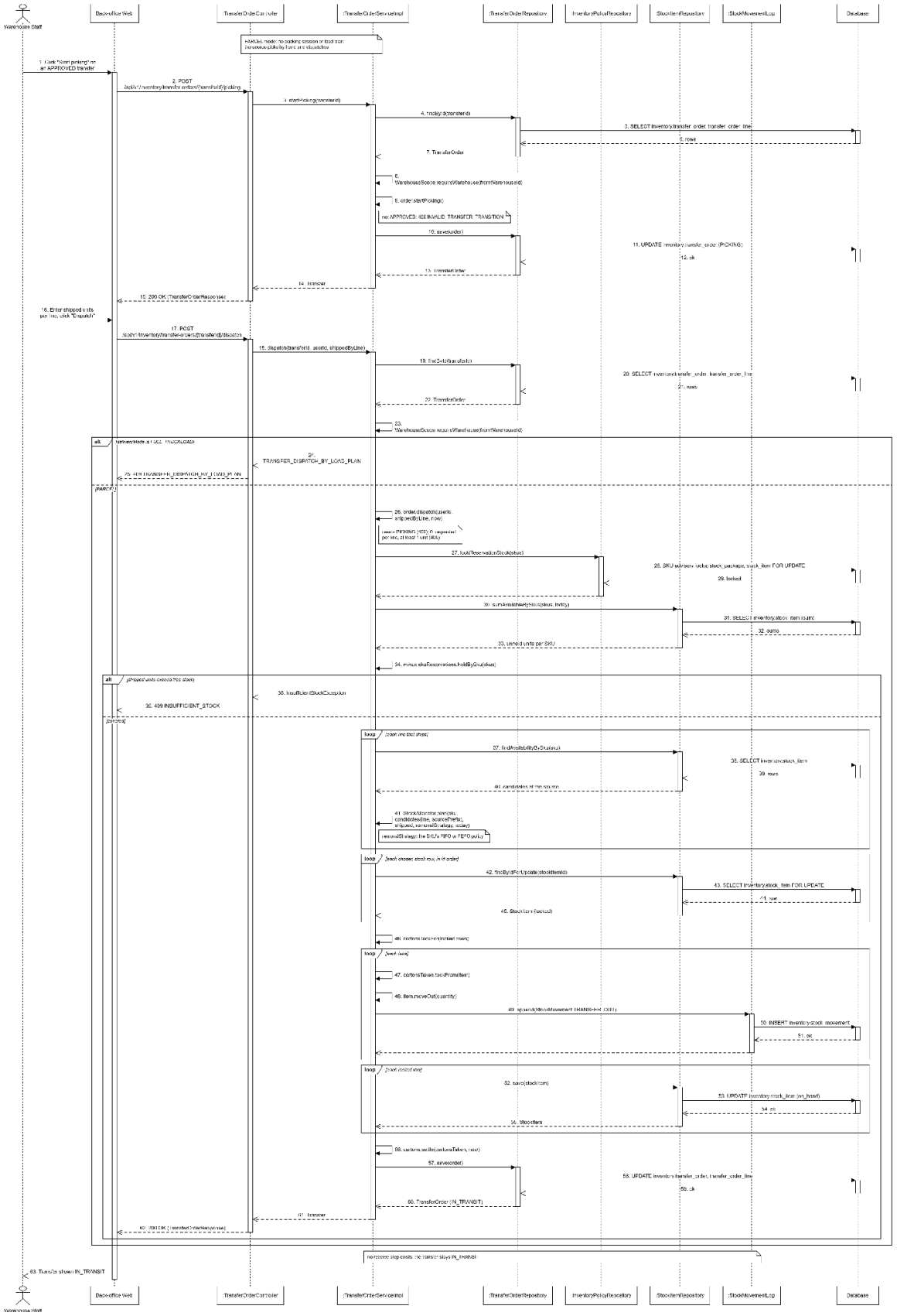


Figure 281. Pack & Dispatch Transfer Order (Parcel Mode: Pick and Dispatch, No Packing) Sequence Diagram

3.17.7 Transfer Order State Diagram

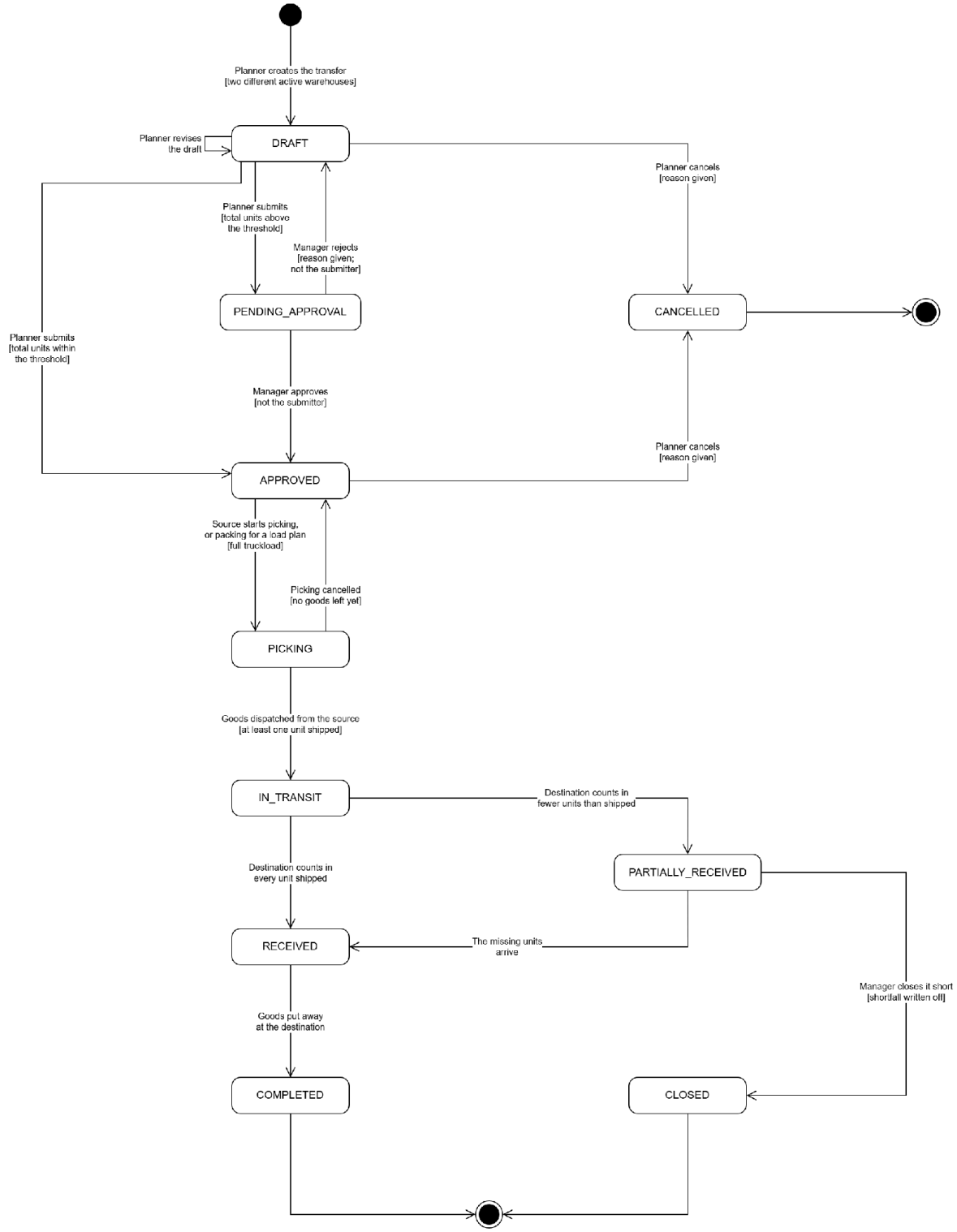


Figure 282. Transfer Order State Diagram

3.18 Cycle Count

3.18.1 Class Diagram

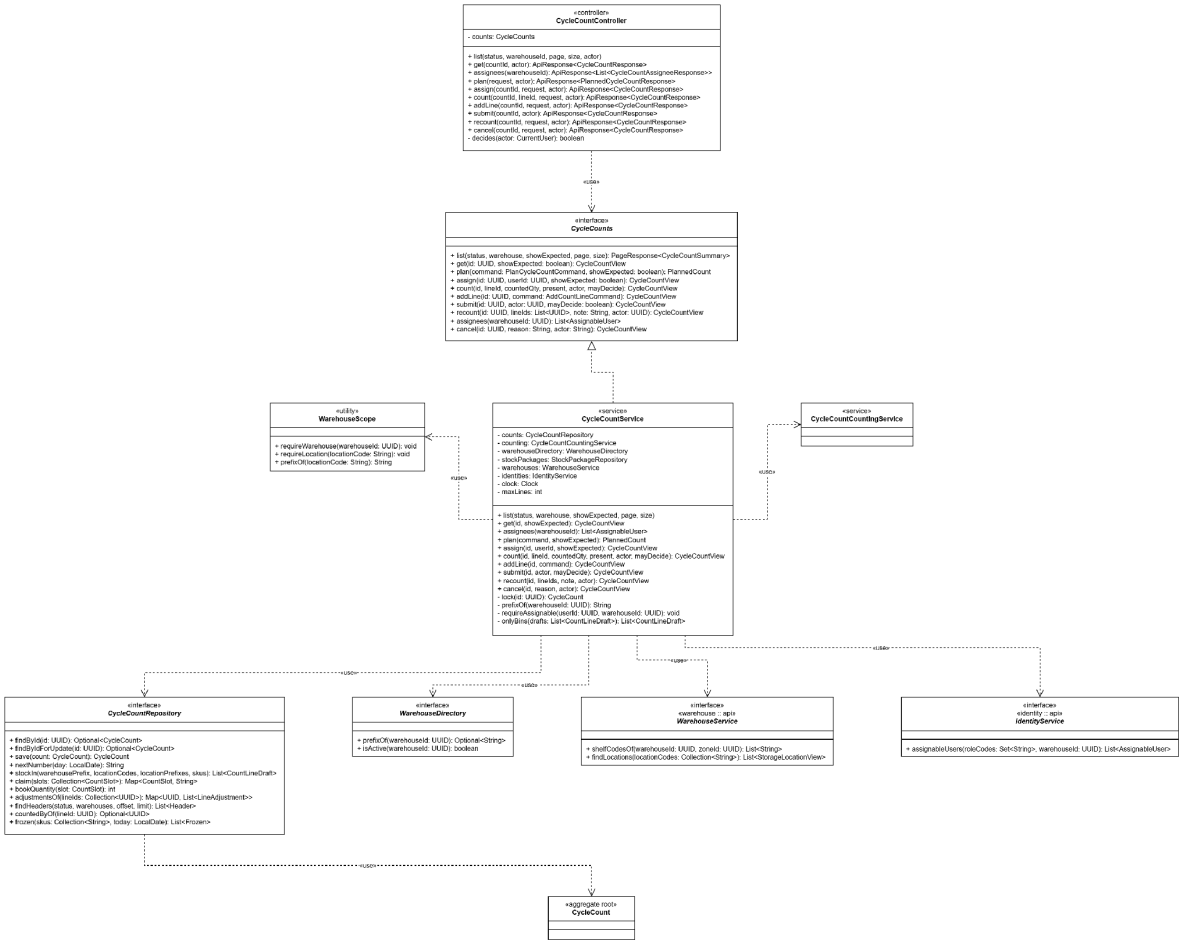


Figure 283. Cycle Count Class Diagram (1/3): Planning, Assigning, Recounting and Cancelling

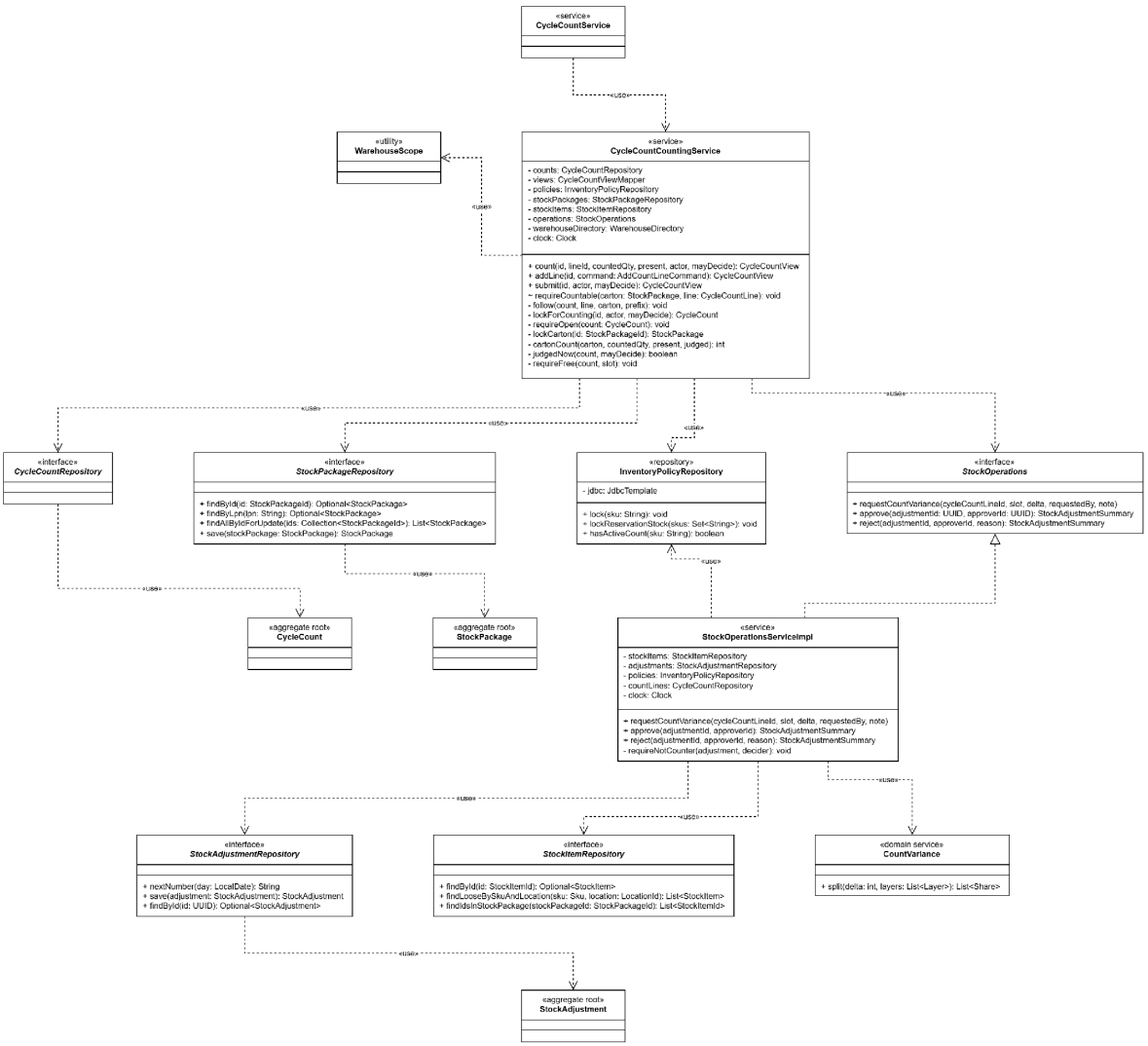


Figure 284. Cycle Count Class Diagram (2/3): Counting and Submitting Variances

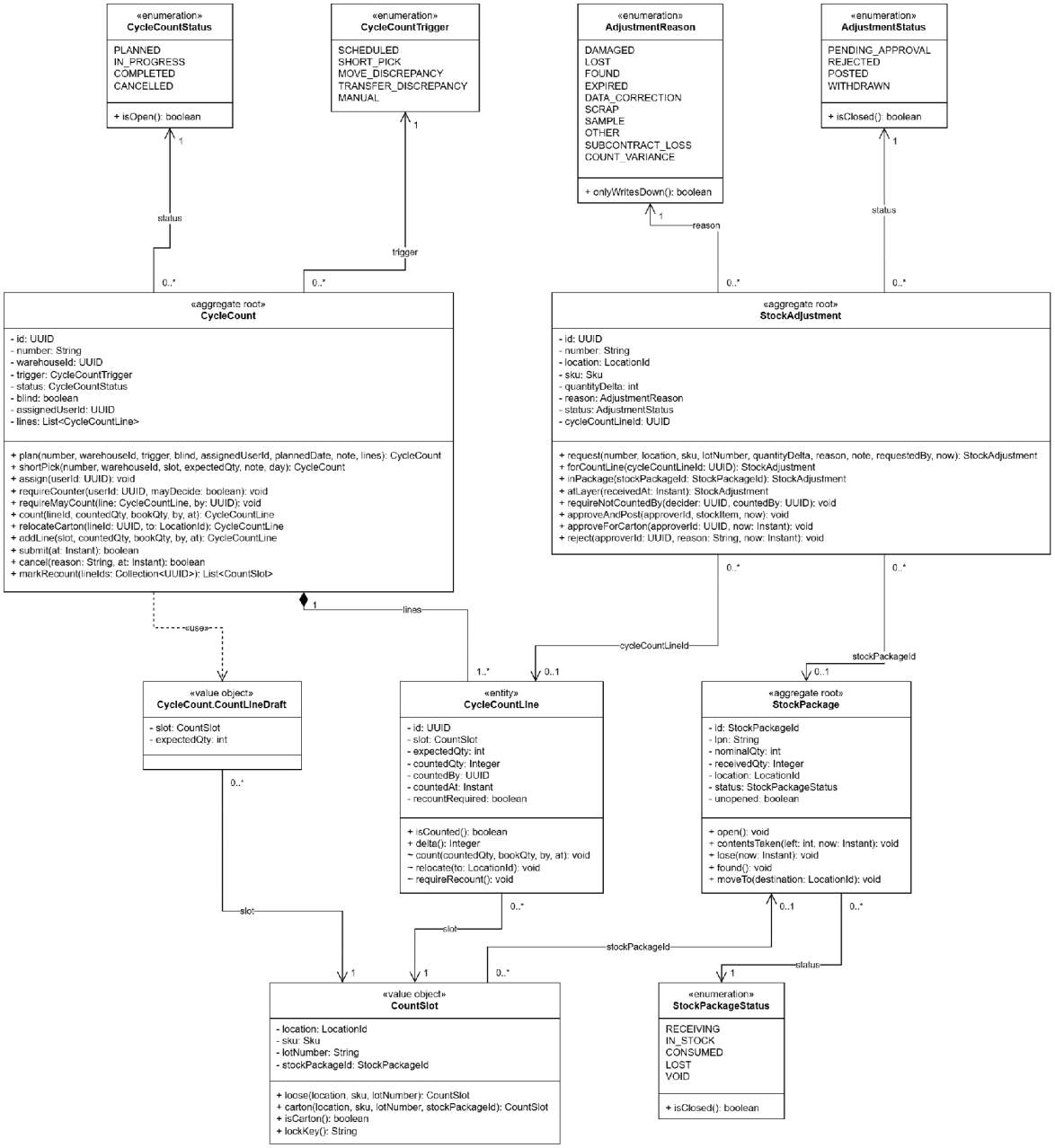


Figure 285. Cycle Count Class Diagram (3/3): Domain Model

3.18.2 Class Specification

CycleCountController

No	Method	Description
1	list(status, warehouseId, page, size, actor)	GET /api/v1/inventory/cycle-counts: Counts newest first without lines, by status and warehouse; book figures only for a holder of cycle-count APPROVE. (:74)
2	get(countId, actor)	GET /api/v1/inventory/cycle-counts/{countId}: One count with its lines; blind for a counter (no expected quantities); 404 NOT_FOUND. (:87)

3	assignees(warehouseId)	GET /api/v1/inventory/cycle-counts/assignees: Active warehouse staff and managers of that warehouse a count may be handed to. (:95)
4	plan(request, actor)	POST /api/v1/inventory/cycle-counts: Plans a MANUAL count over zones, shelves, bins and SKUs (201), blind by default; slots already on an open count are skipped and listed; too many slots 422 CYCLE_COUNT_TOO_LARGE, nothing to count 422 CYCLE_COUNT_EMPTY_SCOPE. (:105)
5	assign(countId, request, actor)	PUT /api/v1/inventory/cycle-counts/{countId}/assignment: Hands a PLANNED count to a staff member or manager of its warehouse; not PLANNED 409 CONFLICT, ineligible user 400. (:120)
6	count(countId, lineId, request, actor)	PUT /api/v1/inventory/cycle-counts/{countId}/lines/{lineId} /count: Records a line's figure (units, or present for a sealed carton); the book quantity is re-read now; PLANNED becomes IN_PROGRESS; another's count 403 CYCLE_COUNT_NOT_ASSIGNEE, closed count 409 CYCLE_COUNT_NOT_OPEN. (:129)
7	addLine(countId, request, actor)	POST /api/v1/inventory/cycle-counts/{countId}/lines: Adds a slot found on the shelf (bin + carton LPN, or bin + SKU + lot) already counted; a slot on another count 409 CYCLE_COUNT_SLOT_BUSY. (:139)
8	submit(countId, actor)	POST /api/v1/inventory/cycle-counts/{countId}/submission: Completes a fully counted count and raises one pending COUNT_VARIANCE adjustment per differing loose layer or carton, all in one transaction; uncounted line 409 CYCLE_COUNT_LINE_NOT_COUNTED, stale stock 409 CYCLE_COUNT_STALE; repeat submit is a no-op. (:150)
9	recount(countId, request, actor)	POST /api/v1/inventory/cycle-counts/{countId}/recounts: Manager sends lines whose variance was rejected (none pending) to a new blind count (201); otherwise 409 CONFLICT, slot busy 409 CYCLE_COUNT_SLOT_BUSY. (:158)
10	cancel(countId, request, actor)	POST /api/v1/inventory/cycle-counts/{countId}/cancellation: Calls off an open count with a reason (CANCELLED); a COMPLETED count 409 CYCLE_COUNT_NOT_OPEN; a SHORT_PICK slot becomes sellable again. (:170)

Table 97. CycleCountController Class Specification

3.18.3 Create Cycle Count Sequence Diagram

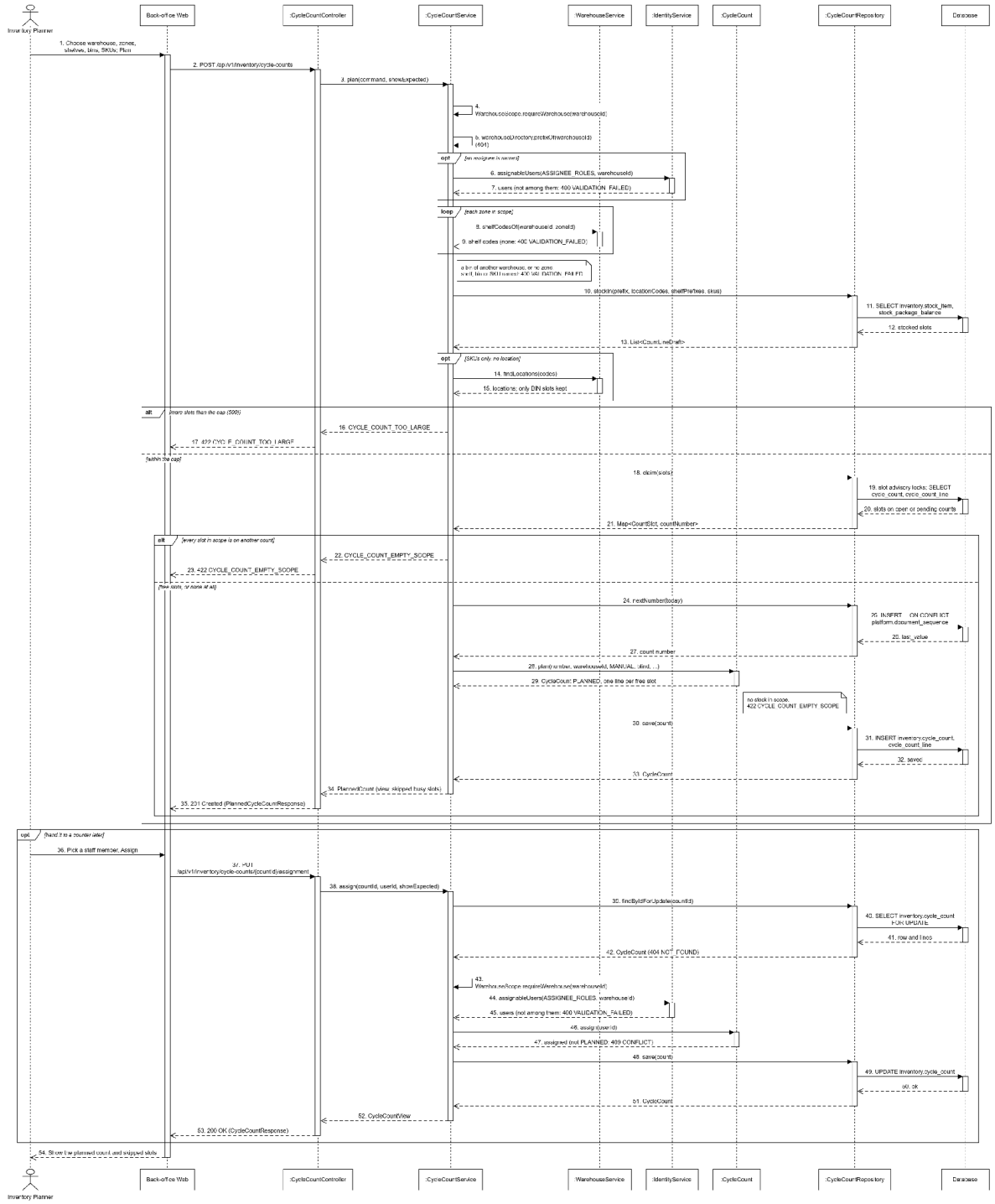


Figure 286. Create Cycle Count Sequence Diagram

3.18.4 Perform Cycle Count (Count Lines) Sequence Diagram

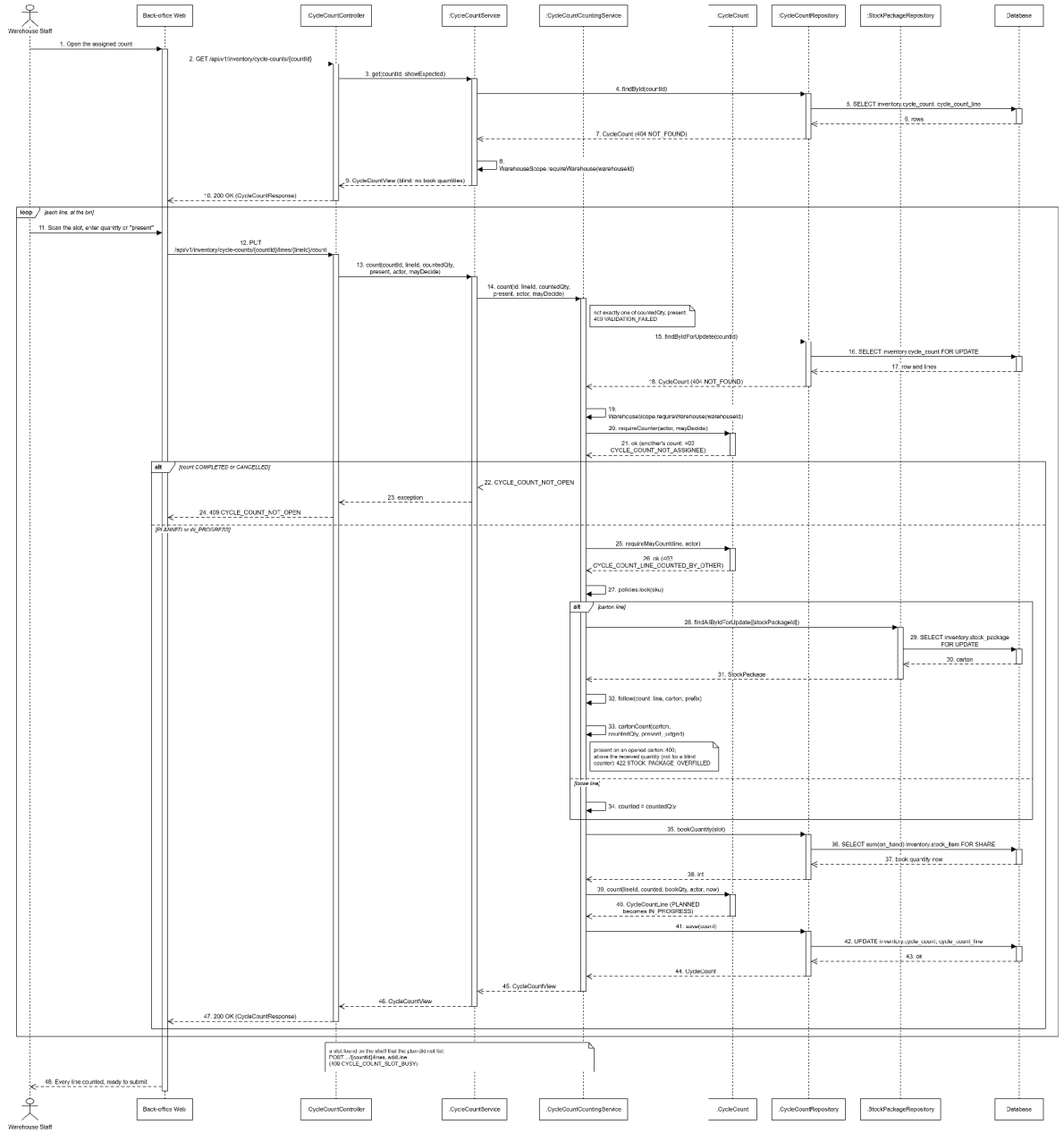


Figure 287. Perform Cycle Count (Count Lines) Sequence Diagram

3.18.5 Perform Cycle Count (Submit) Sequence Diagram

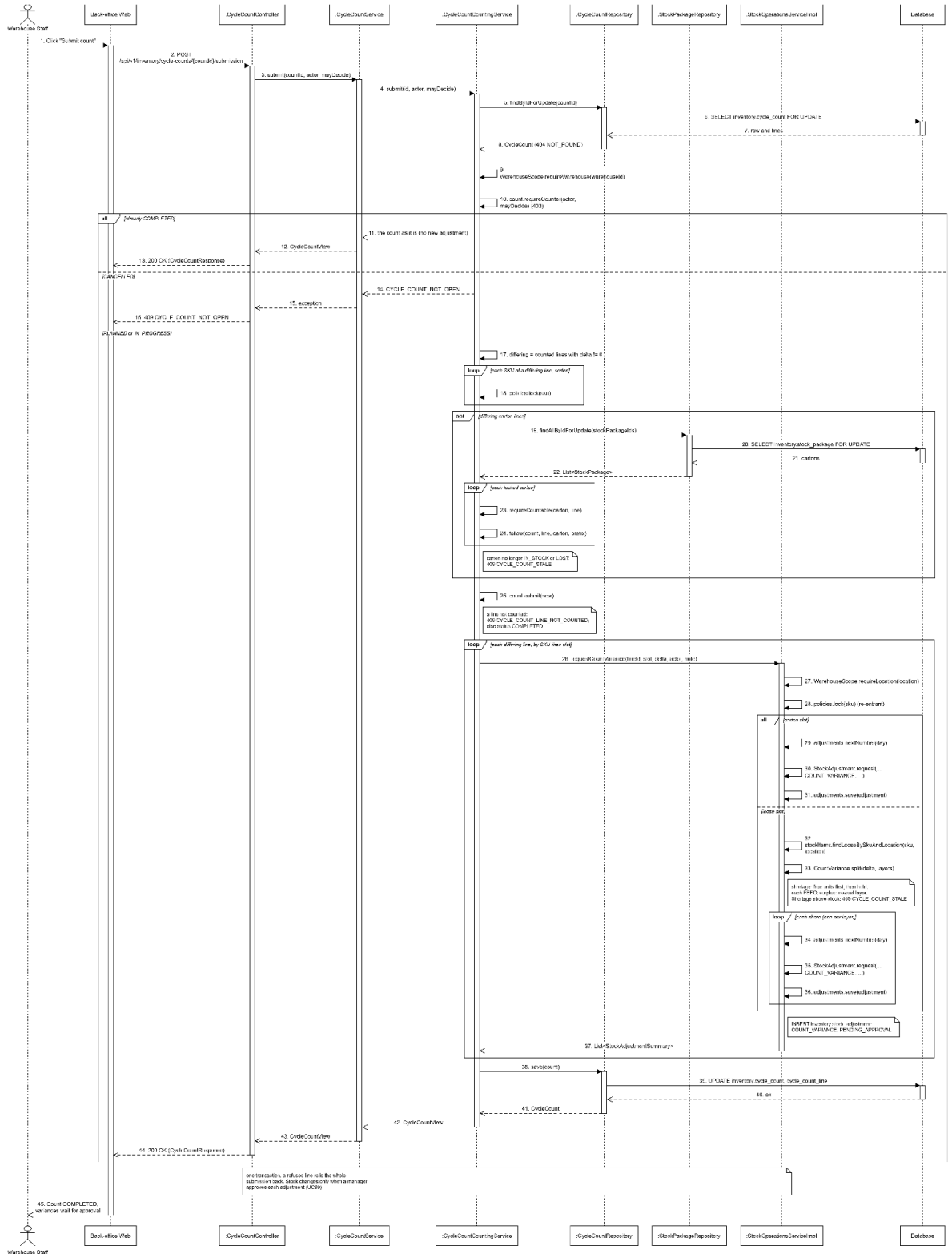


Figure 288. Perform Cycle Count (Submit) Sequence Diagram

3.18.6 Review Count Variances & Adjustments (Recount or Cancel Count) Sequence Diagram

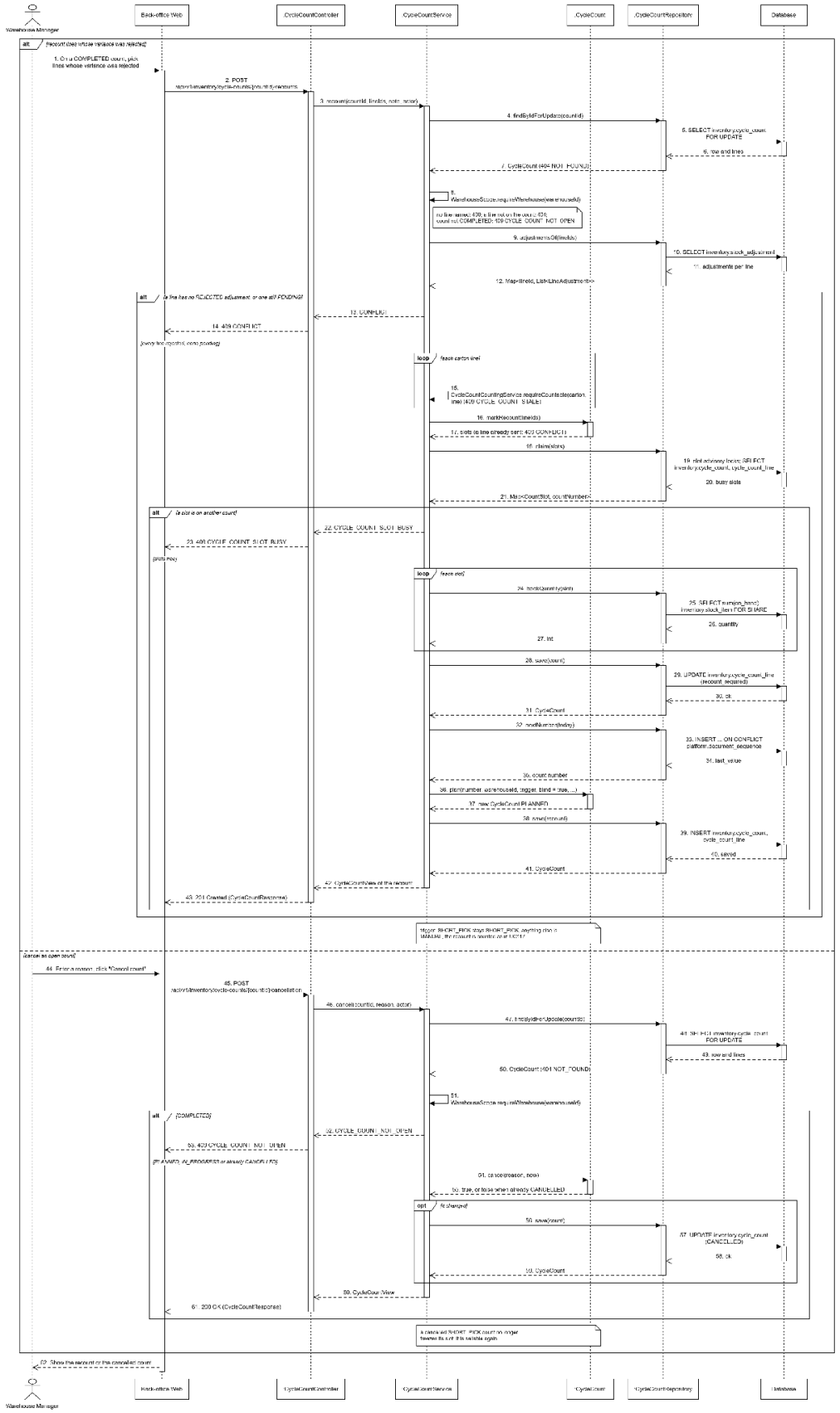


Figure 289. Review Count Variances & Adjustments (Recount or Cancel Count) Sequence Diagram

3.18.7 Cycle Count State Diagram

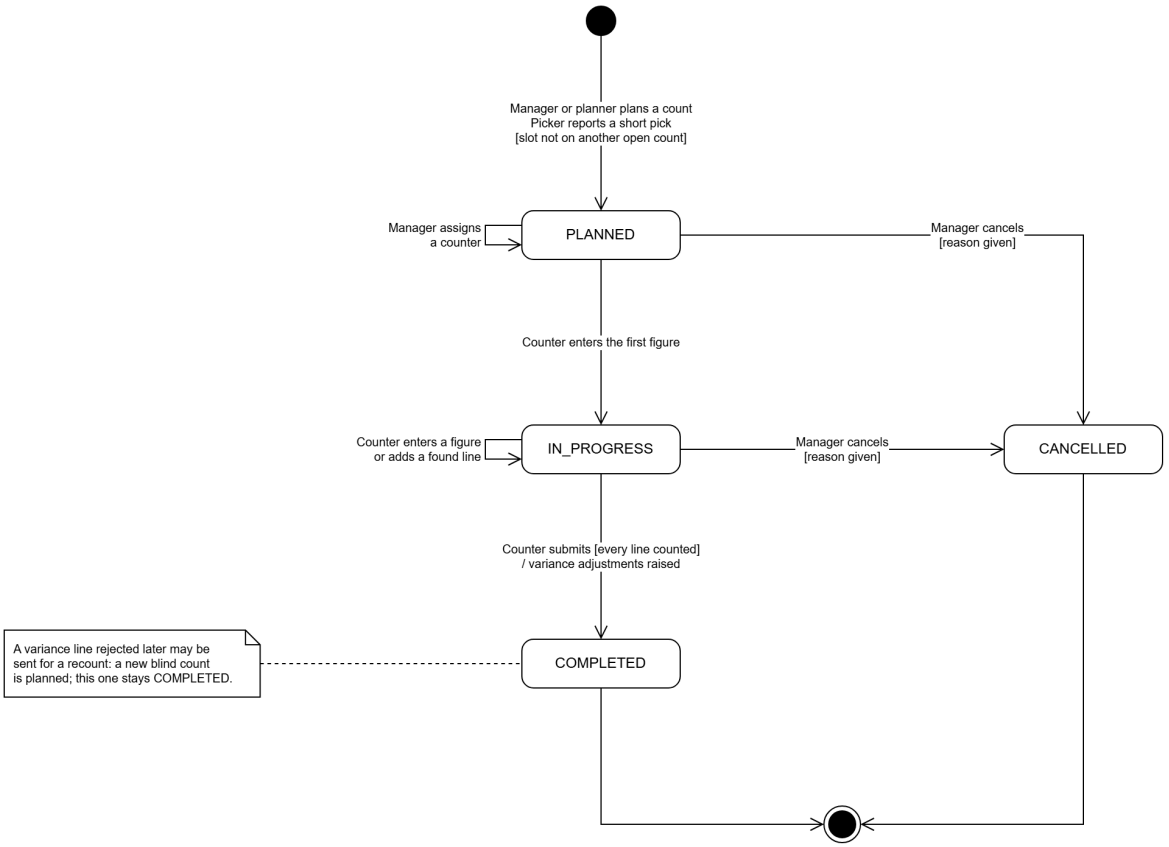


Figure 290. Cycle Count State Diagram

3.19 Picking & Packing

3.19.1 Class Diagram

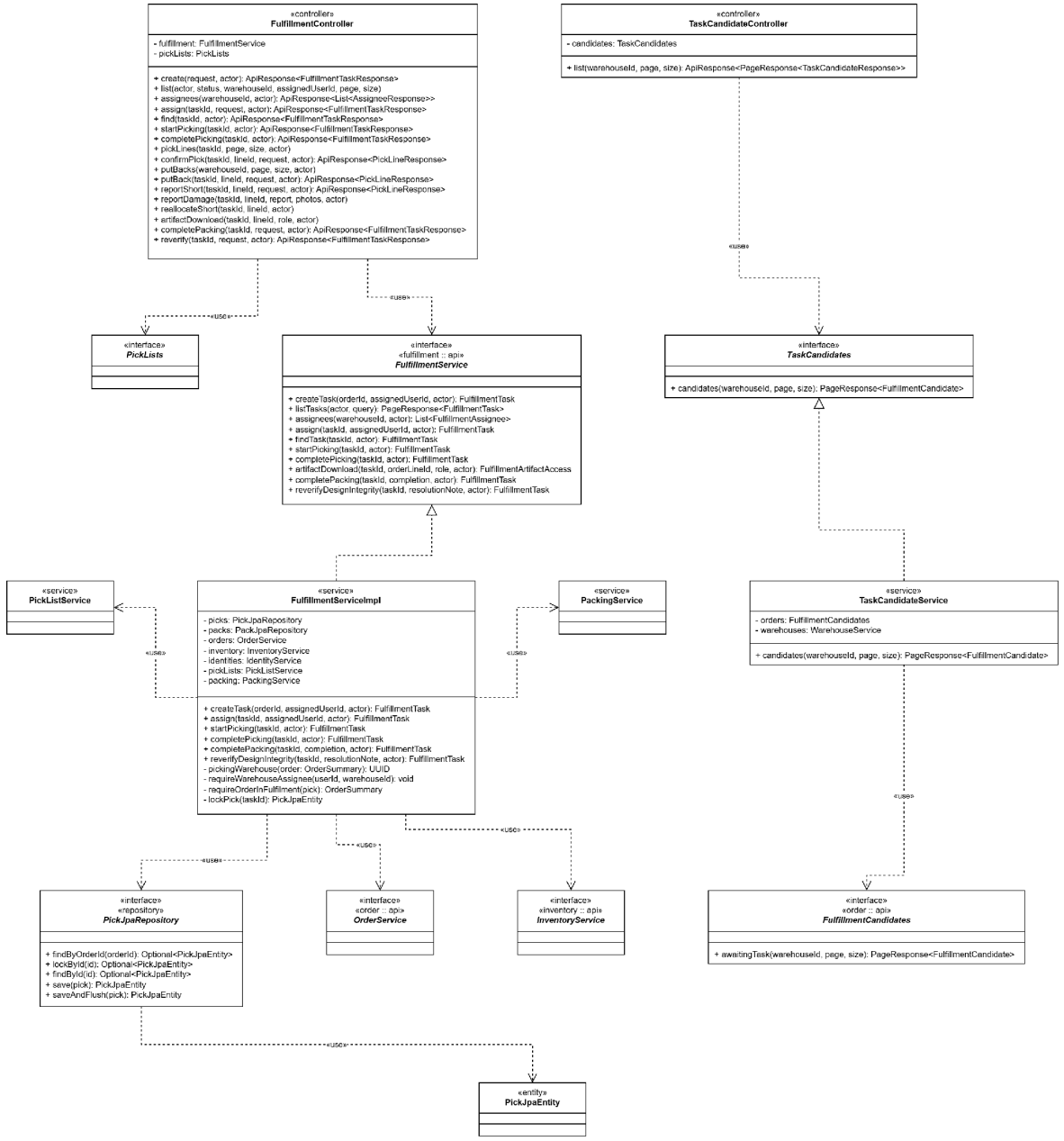


Figure 291. Picking and Packing Class Diagram (1/4): Pick Tasks

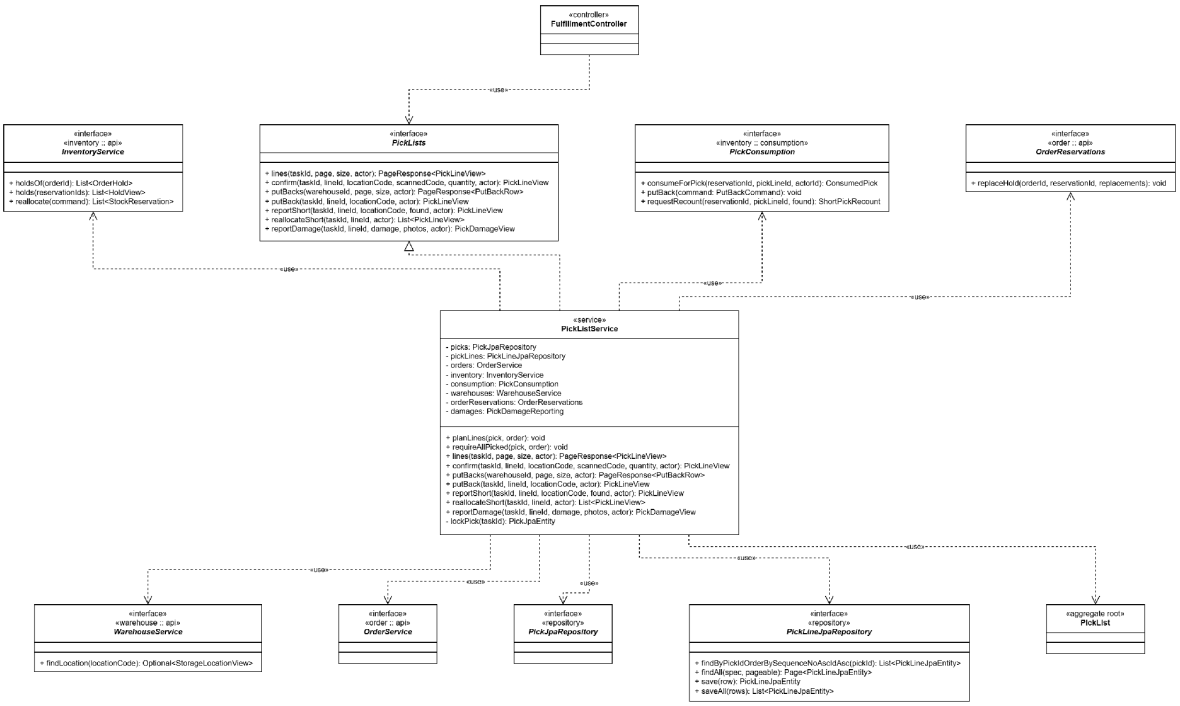


Figure 292. Picking and Packing Class Diagram (2/4): Pick Lines, Short Picks and Put-backs

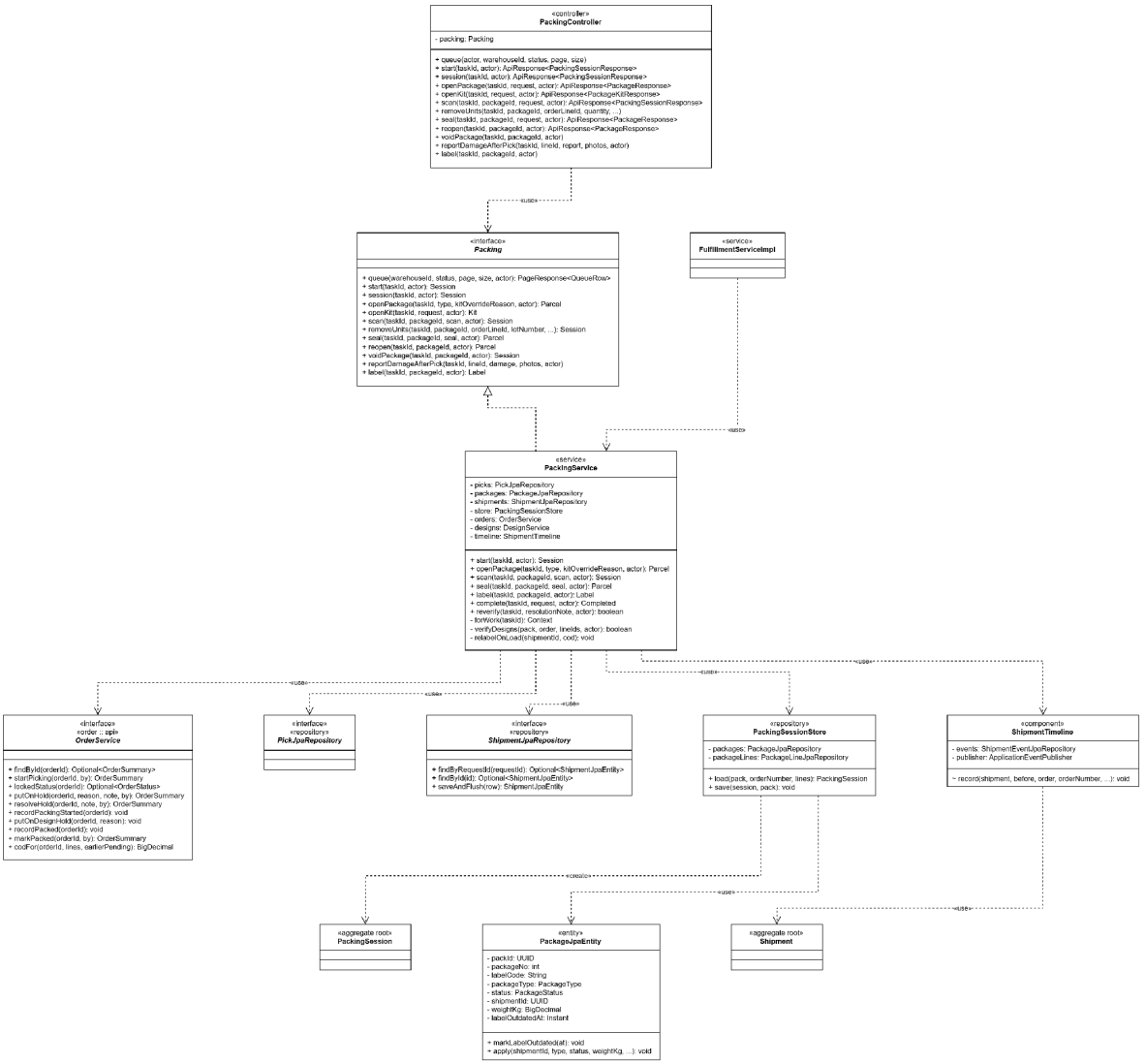


Figure 293. Picking and Packing Class Diagram (3/4): Packing, Labels and Shipment

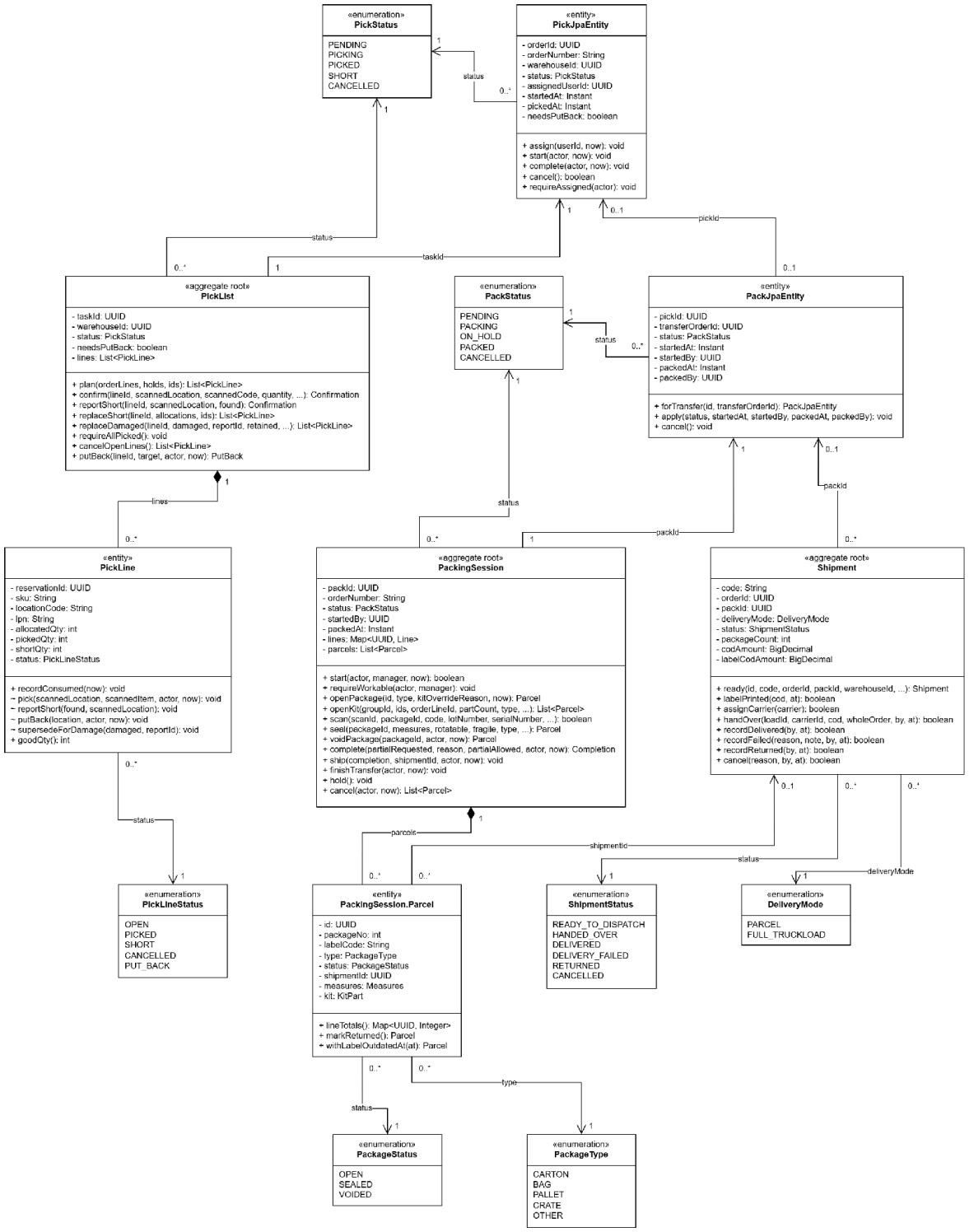


Figure 294. Picking and Packing Class Diagram (4/4): Domain Model

3.19.2 Class Specification

FulfillmentController

No	Method	Description

1	create(request, actor)	POST /api/v1/fulfillment/tasks: Opens the pick task of a released order (READY_TO_FULFILL, or IN_PRODUCTION with stock lines) for an active warehouse staff member or manager of its warehouse, plans one pick line per hold and moves the order to PICKING; an order that already has a task gets that task back (409 PICK_TASK_STATE_CONFLICT for other statuses, 409 FULFILLMENT_HOLDS_OUTSIDE_WAREHOUSE when its stock is held elsewhere).
2	list(actor, status, warehouseId, assignedUserId, page, size)	GET /api/v1/fulfillment/tasks: Pages the tasks the caller may see, newest first, limited to the caller's warehouses and narrowed by status, warehouse and assignee.
3	assignees(warehouseId, actor)	GET /api/v1/fulfillment/assignees: Lists the active warehouse staff and managers of one warehouse a task can be given to (not paged; a manager only for their own warehouse).
4	assign(taskId, request, actor)	PUT /api/v1/fulfillment/tasks/{taskId}/assignment: Gives a task to another eligible user of its warehouse; only while the task is PENDING (409 PICK_TASK_STATE_CONFLICT), a non-eligible user is 400 VALIDATION_FAILED.
5	find(taskId, actor)	GET /api/v1/fulfillment/tasks/{taskId}: Returns one task with its order lines when the caller may see it.
6	startPicking(taskId, actor)	POST /api/v1/fulfillment/tasks/{taskId}/picking-start: The assignee starts picking (PENDING -> PICKING) and the lines are planned if the task has none; starting a PICKING task again changes nothing.
7	completePicking(taskId, actor)	POST /api/v1/fulfillment/tasks/{taskId}/picking-completion: The assignee closes picking (PICKING -> PICKED) only when every line is PICKED or CANCELLED (409 PICK_LINES_INCOMPLETE); a retry on a PICKED task returns it unchanged.
8	pickLines(taskId, page, size, actor)	GET /api/v1/fulfillment/tasks/{taskId}/pick-lines: Pages the task's pick lines in walking order with each SKU's barcode, for the assignee and the warehouse's managers, QC and coordinators.

9	confirmPick(taskId, lineId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/confirmation: Confirms one line by scan - bin, then SKU/barcode (or the carton LPN for a whole-carton line), then the whole allocated quantity - and deducts its hold in the same transaction through inventory :: consumption; only the assignee; the same scan again returns the line unchanged (409 PICK_LOCATION_MISMATCH, PICK_ITEM_MISMATCH, PICK_QUANTITY_SHORT, PICK_QUANTITY_EXCEEDS_ALLOCATION).
10	putBacks(warehouseId, page, size, actor)	GET /api/v1/fulfillment/put-backs: Pages the PICKED lines of tasks cancelled after picking started (needsPutBack), limited to the caller's warehouses: the put-back queue.
11	putBack(taskId, lineId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/put-back: Puts everything a line of a cancelled order picked back onto an active bin of the task's warehouse as AVAILABLE stock (one PUT_BACK ledger row); any staff of that warehouse; 409 PUT_BACK_NOT_REQUIRED / PUT_BACK_LOCATION_INVALID.
12	reportShort(taskId, lineId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/short: The assignee reports fewer units than allocated at the line's bin: the line turns SHORT with nothing deducted, the order goes ON_HOLD (INVENTORY) and the slot is recounted (a count already holding it is reused, otherwise a SHORT_PICK count is opened); the same report again returns the line unchanged.
13	reportDamage(taskId, lineId, report, photos, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/damage (multipart, 1-5 photos): The picker reports allocated units found damaged at the bin. They go to quarantine as DAMAGED; the line is replaced by a line for the good units and lines for replacement stock, or by a SHORT line that puts the order ON_HOLD (INVENTORY) when no stock replaces them. The same report again returns what it made.

14	reallocateShort(taskId, lineId, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/reallocation: The coordinator / warehouse manager reallocates a SHORT line's quantity to other pickable bins of the warehouse (FEFO, never the short bin), replacing it with new OPEN lines and resolving the INVENTORY hold when no line is SHORT; 409 INSUFFICIENT_STOCK changes nothing.
15	artifactDownload(taskId, lineId, role, actor)	GET /api/v1/fulfillment/tasks/{taskId}/lines/{lineId}/design-artifacts/{role}/download-url: Returns a short-lived download URL for one artifact of a print line's confirmed design (Cache-Control no-store); 404 when the line has no confirmed design, 400 UNSUPPORTED_PARAMETER for an unknown role.
16	completePacking(taskId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packing-completion: Completes packing: the sealed packages not shipped yet become one READY_TO_DISPATCH shipment and the order is marked packed; without partial everything must be packed (409 PACKING_INCOMPLETE); a printed design that no longer matches answers 200 with packing ON_HOLD and no shipment.
17	reverify(taskId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/design-integrity-reverification: QC or a warehouse manager re-checks held packing: when every print line matches its design again the design hold is resolved and packing is back in progress (409 PICK_TASK_STATE_CONFLICT when not on hold).

Table 98. FulfillmentController Class Specification

PackingController

No	Method	Description
1	queue(actor, warehouseId, status, page, size)	GET /api/v1/fulfillment/packing-queue: Pages picked orders waiting at the packing table, oldest pick first, limited to the caller's warehouses.
2	start(taskId, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packing-start: A packer takes a picked order (pack PENDING -> PACKING); taking it again returns it as it is; 409 PACKING_NOT_READY before picking ends, PACKING_TAKEN while another packer has it (a manager may take over).

3	session(taskId, actor)	GET /api/v1/fulfillment/tasks/{taskId}/packing-session: Returns the packing session: per line what the order still wants and what was picked, and the packages.
4	openPackage(taskId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packages: Opens a new empty package labelled PKG-<order>-<n> and locks the order's delivery mode (LP-1); each call opens another.
5	openKit(taskId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/package-kits: Opens the N packages of a kit for one unit of an item that ships as N > 1 boxes (422 KIT_PART_COUNT_MISMATCH otherwise).
6	scan(taskId, packageId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}/scans: Scans units (SKU or barcode, lot or serial) into an open package; the same scanId again changes nothing; 409 PACK_ITEM_NOT_IN_ORDER, PACK_QUANTITY_EXCEEDS_ORDER / _PICKED, PACK_LOT_MISMATCH, PACK_SERIAL_MISMATCH.
7	removeUnits(taskId, packageId, orderLineId, quantity, lotNumber, serialNumber, scanId, actor)	DELETE /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}/lines/{orderLineId}: Takes units of an order line out of an open package again; idempotent by scanId.
8	reportDamageAfterPick(taskId, lineId, report, photos, actor)	POST /api/v1/fulfillment/tasks/{taskId}/pick-lines/{lineId}/damage-after-pick (multipart, 1-5 photos): Units found damaged at packing go back to stock as DAMAGED in quarantine and the task returns to picking for what replaces them, or a SHORT line puts the order ON_HOLD (INVENTORY); packing resumes once they are picked again.
9	seal(taskId, packageId, request, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}/seal: Seals a package with its measured weight and size (OPEN -> SEALED); a missing measure is 400 VALIDATION_FAILED, an empty package 409 PACKAGE_EMPTY.
10	reopen(taskId, packageId, actor)	POST /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}/reopening: Opens a sealed package again (SEALED -> OPEN) while it is in no shipment (409 PACKAGE_SHIPPED) and no load plan (409 PACKAGE_IN_LOAD_PLAN).

11	voidPackage(taskId, packageId, actor)	DELETE /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}: Takes a package apart before it shipped (-> VOIDED, its goods count as unpacked again); a kit part voids the whole kit.
12	label(taskId, packageId, actor)	GET /api/v1/fulfillment/tasks/{taskId}/packages/{packageId}/label: Returns what the internal label of a sealed package prints (n of N, recipient, COD, size, flags, contents) for the front end to render; printing records the label COD on a READY_TO_DISPATCH shipment; 409 PACK_STATE_CONFLICT when not sealed.

Table 99. PackingController Class Specification

TaskCandidateController

No	Method	Description
1	list(warehouseId, page, size)	GET /api/v1/fulfillment/task-candidates: Pages the orders a fulfillment task may be opened for now (READY_TO_FULFILL), oldest release first, warehouse-scoped.

Table 100. TaskCandidateController Class Specification

3.19.3 Create & Assign Pick Task Sequence Diagram

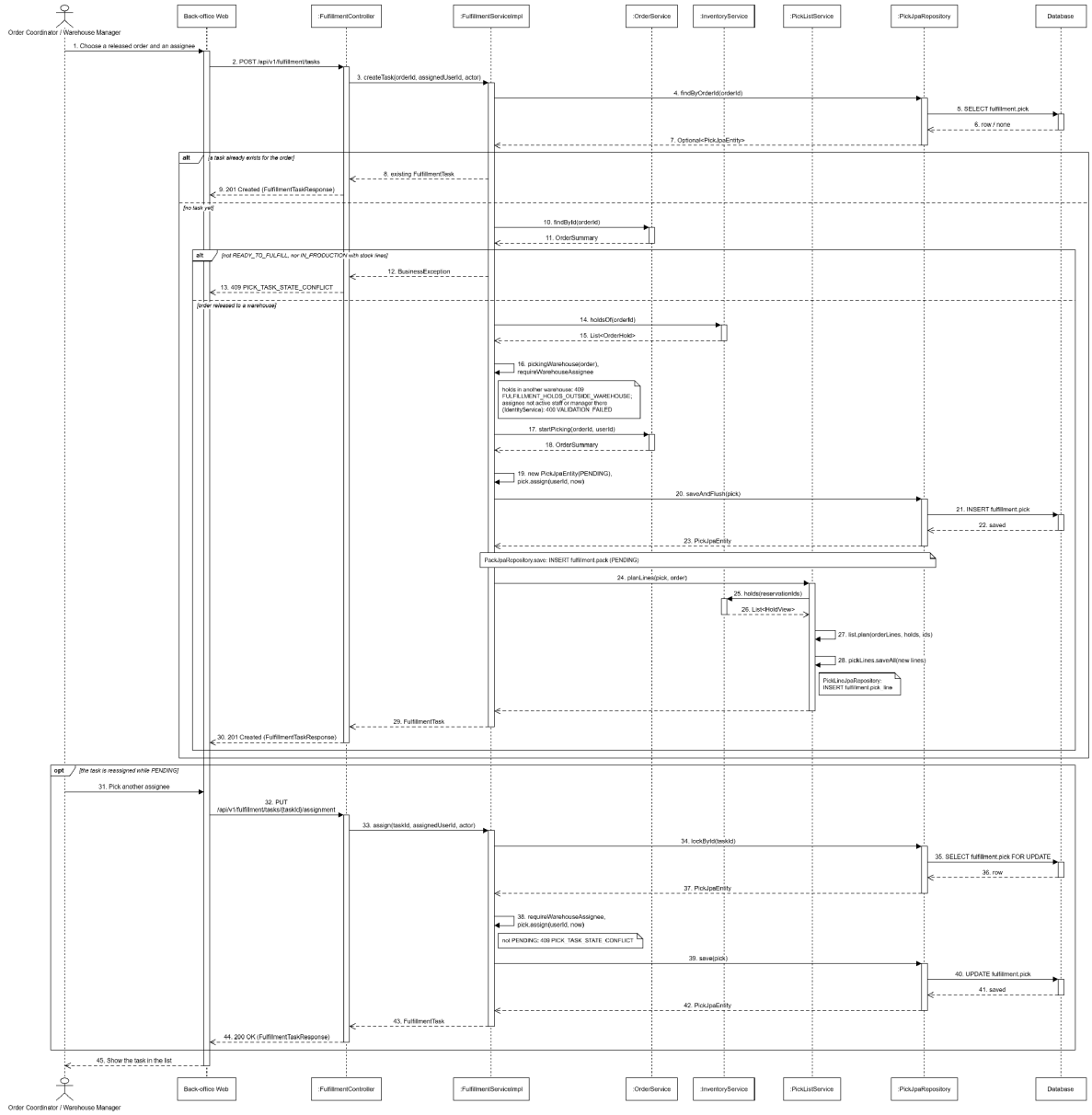


Figure 295. Create & Assign Pick Task Sequence Diagram

3.19.4 Pick Goods Sequence Diagram

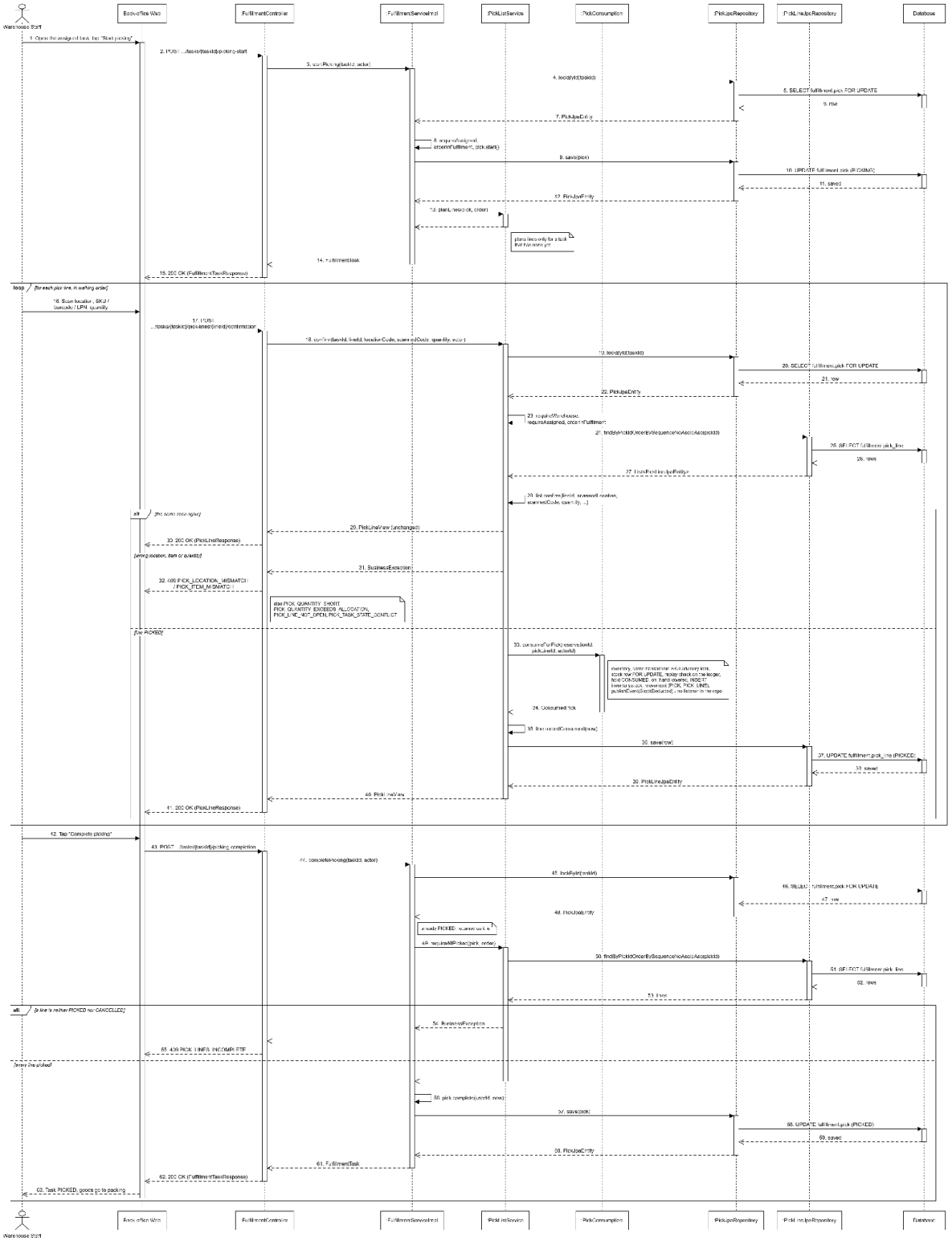


Figure 296. Pick Goods Sequence Diagram

3.19.5 Report Short Pick Sequence Diagram

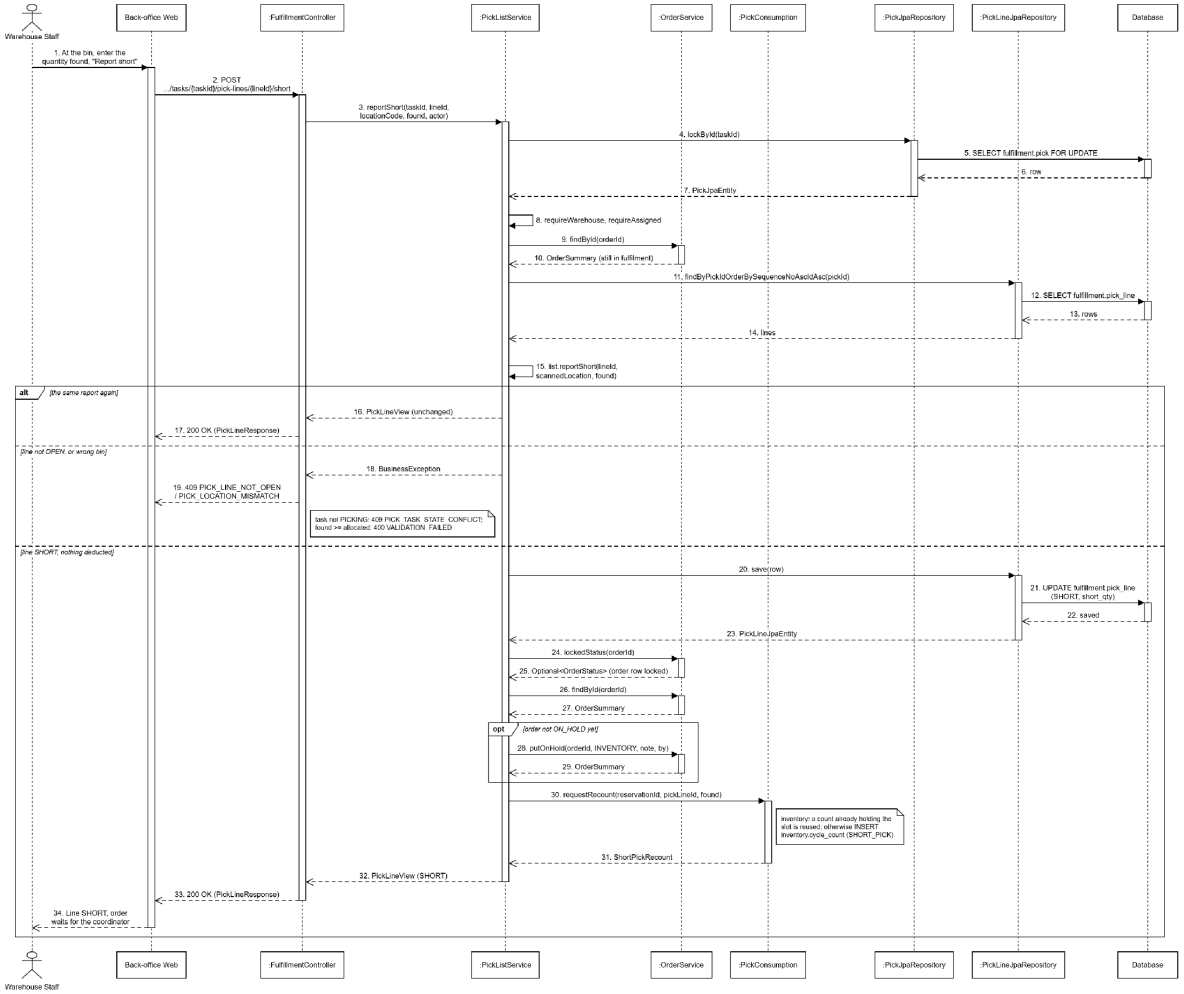


Figure 297. Report Short Pick Sequence Diagram

3.19.6 Reallocate Short-picked Line Sequence Diagram

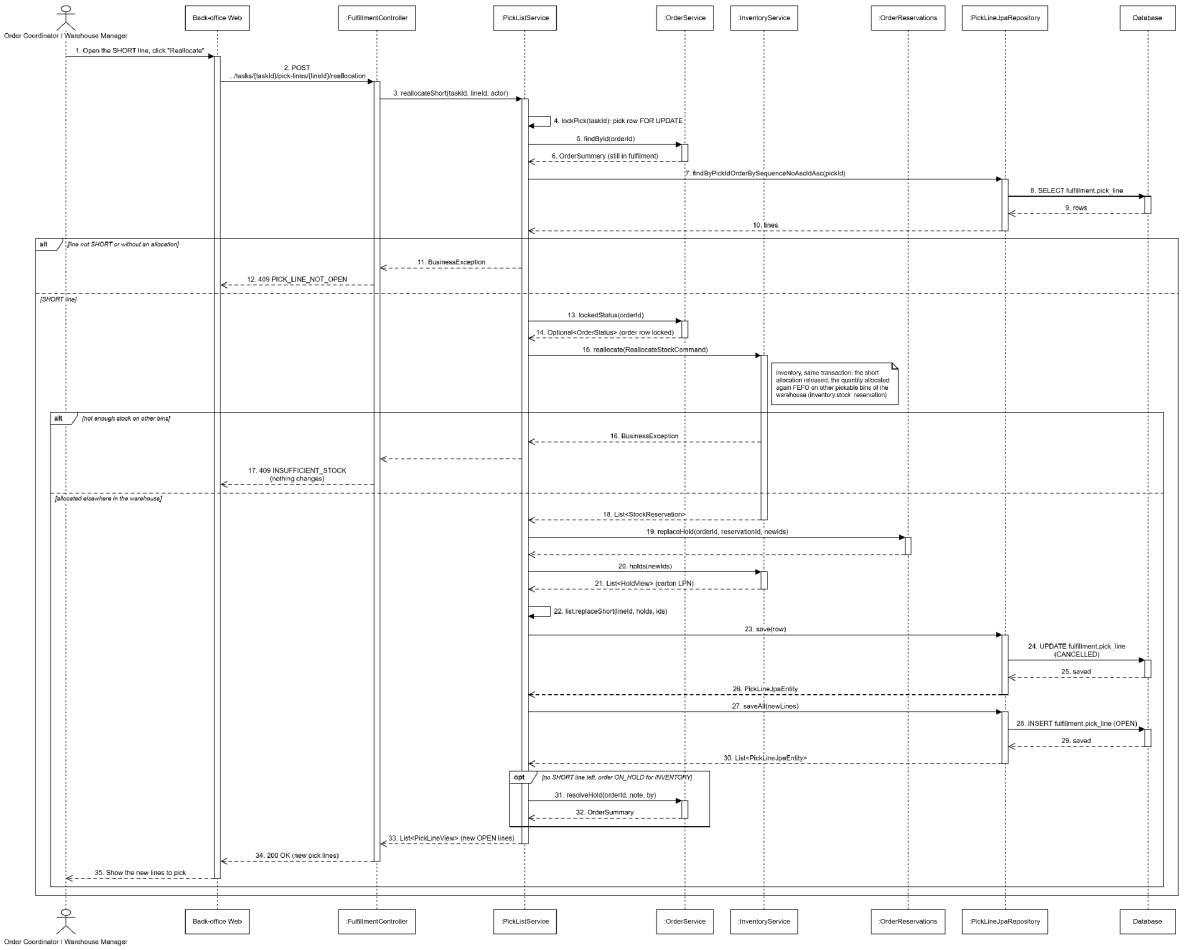


Figure 298. Reallocate Short-picked Line Sequence Diagram

3.19.7 Put Back Goods of Cancelled Order Sequence Diagram

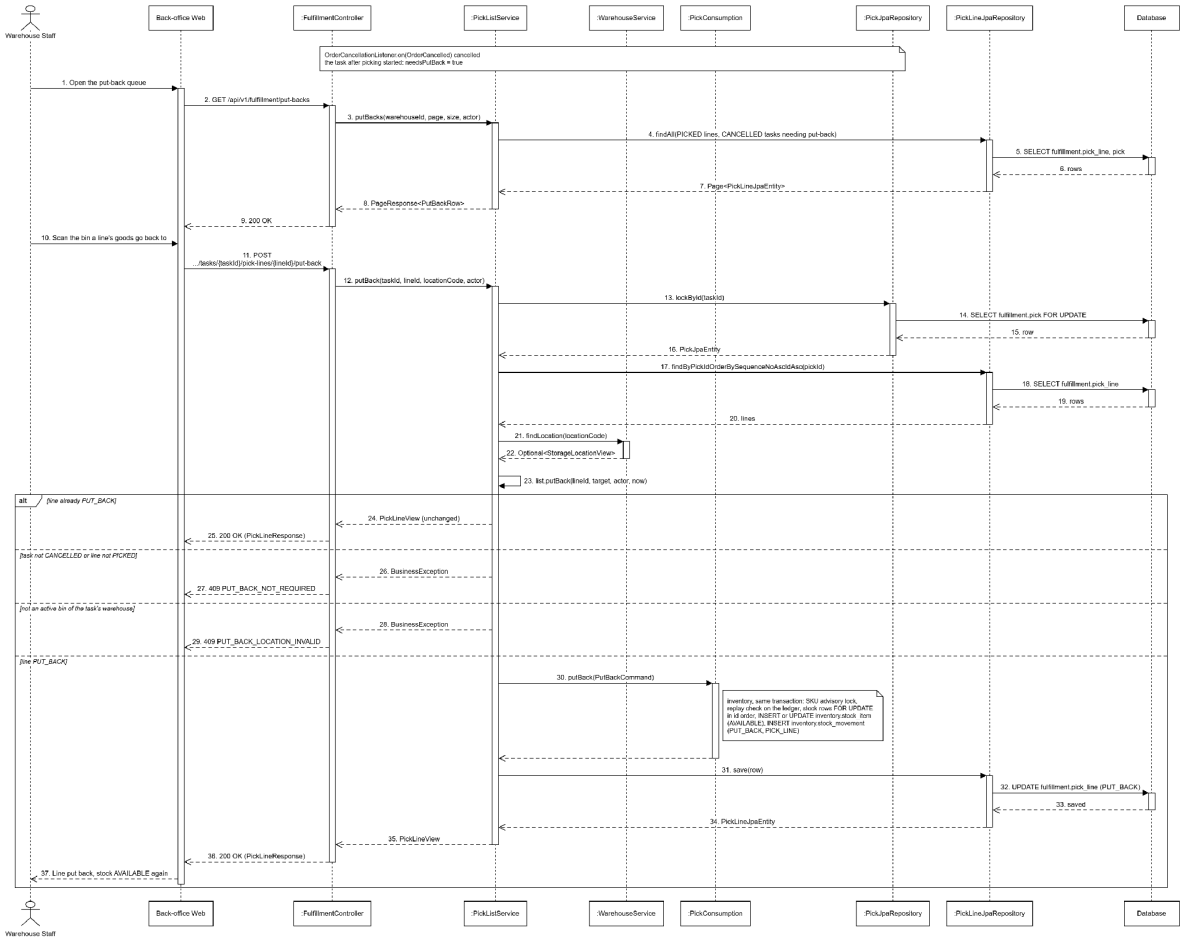


Figure 299. Put Back Goods of Cancelled Order Sequence Diagram

3.19.8 Print Package Labels Sequence Diagram

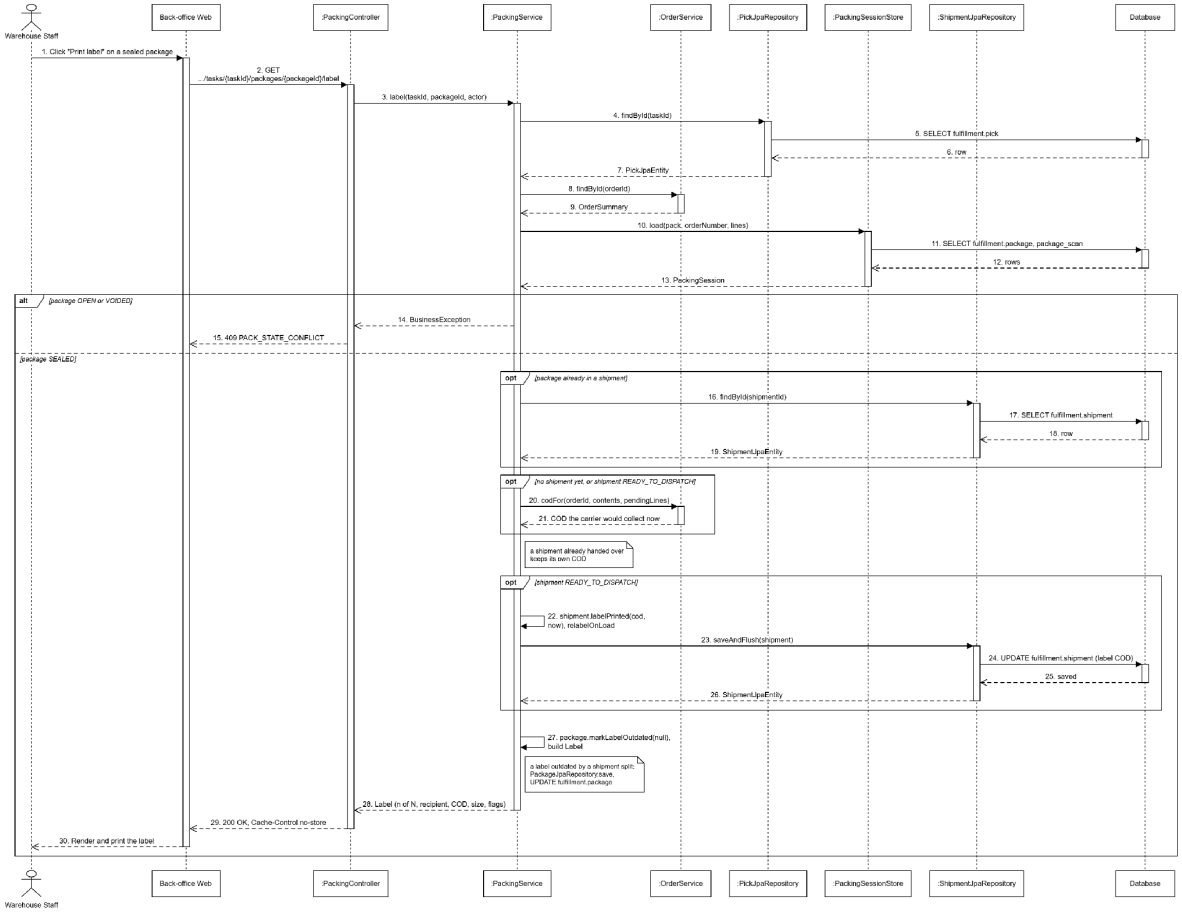


Figure 300. Print Package Labels Sequence Diagram

3.19.9 Pack Order & Complete Packing (Pack Parcels) Sequence Diagram

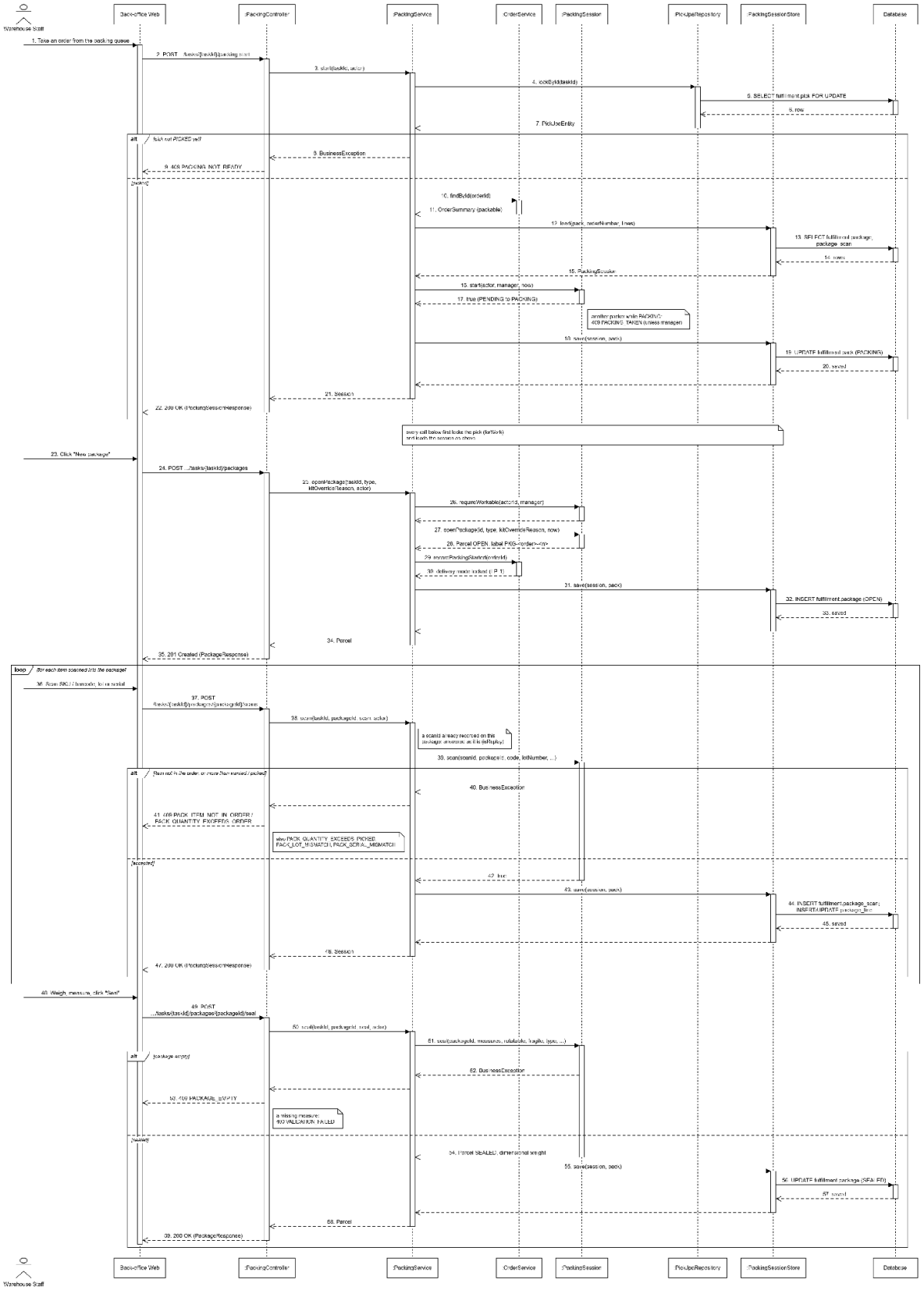


Figure 301. Pack Order & Complete Packing (Pack Parcels) Sequence Diagram

3.19.10 Pack Order & Complete Packing (Complete Packing) Sequence Diagram

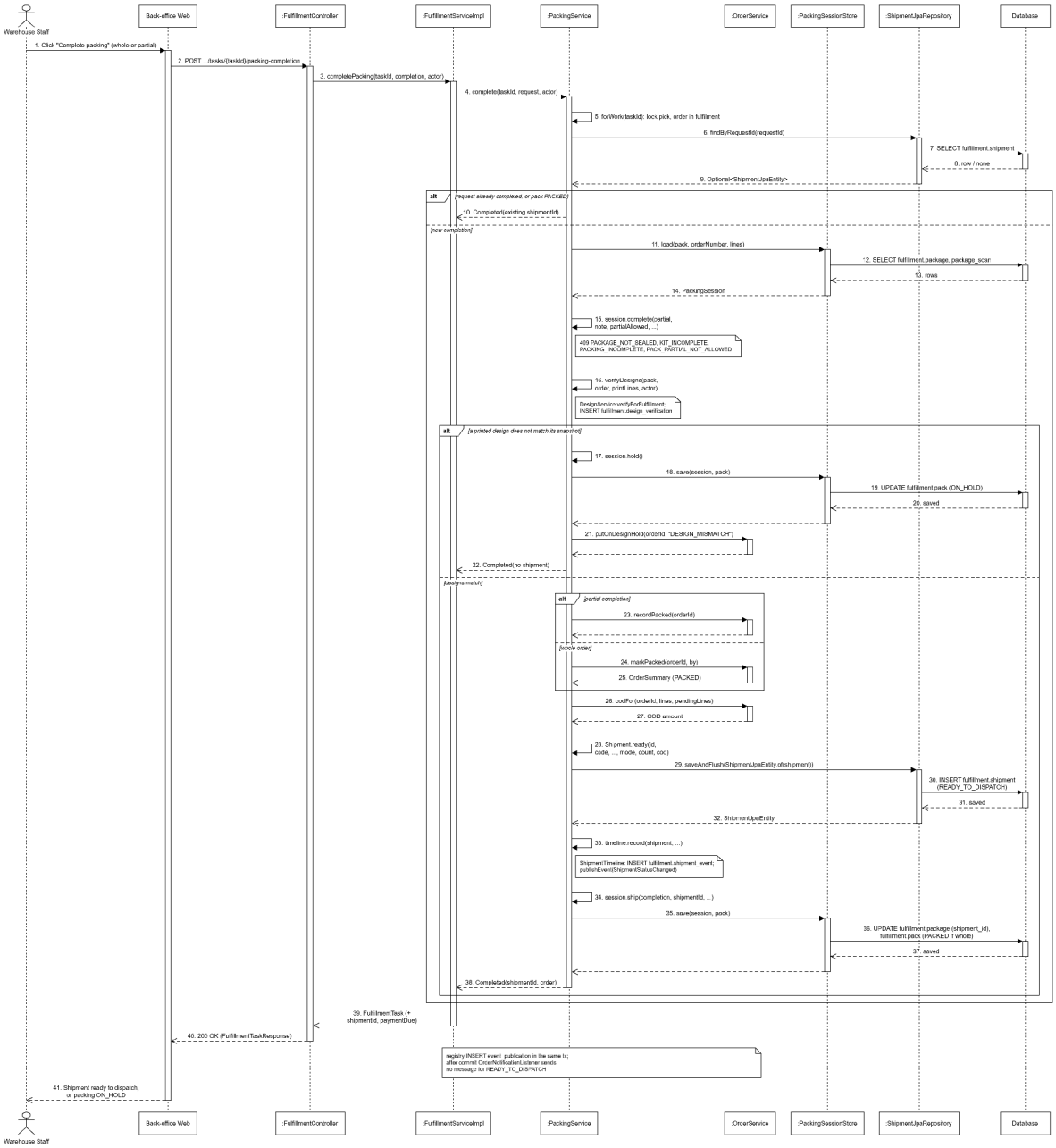


Figure 302. Pack Order & Complete Packing (Complete Packing) Sequence Diagram

3.19.11 Pick Task State Diagram

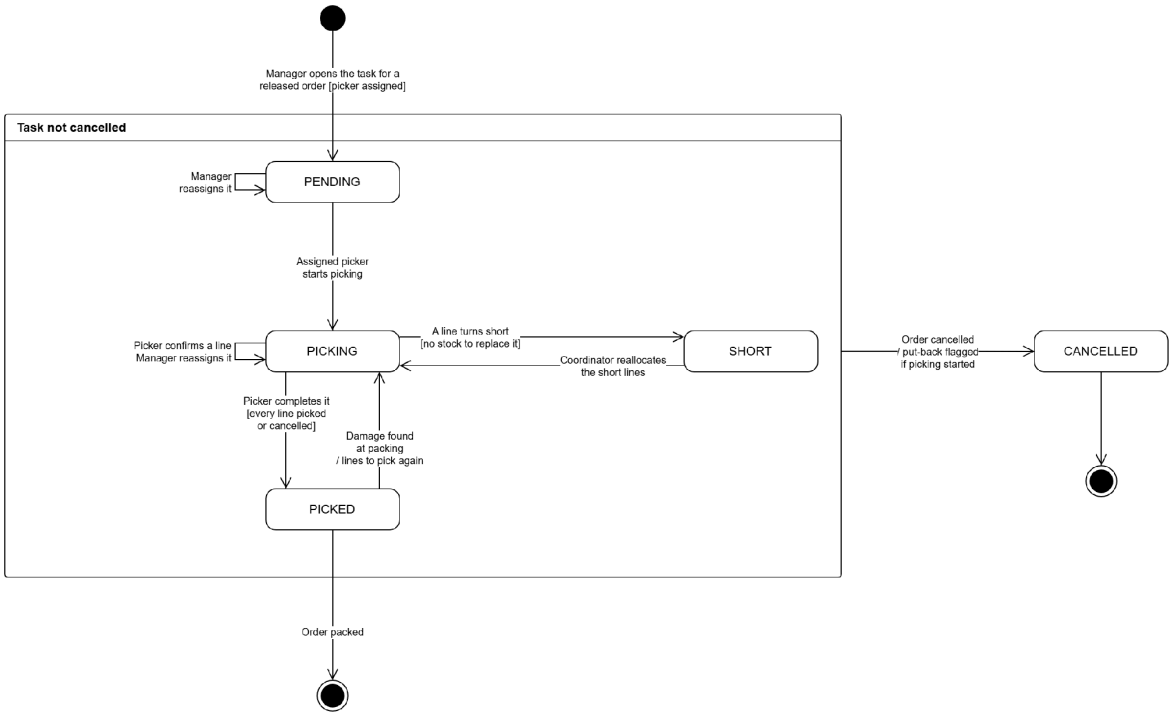


Figure 303. Pick Task State Diagram

3.19.12 Pick Line State Diagram

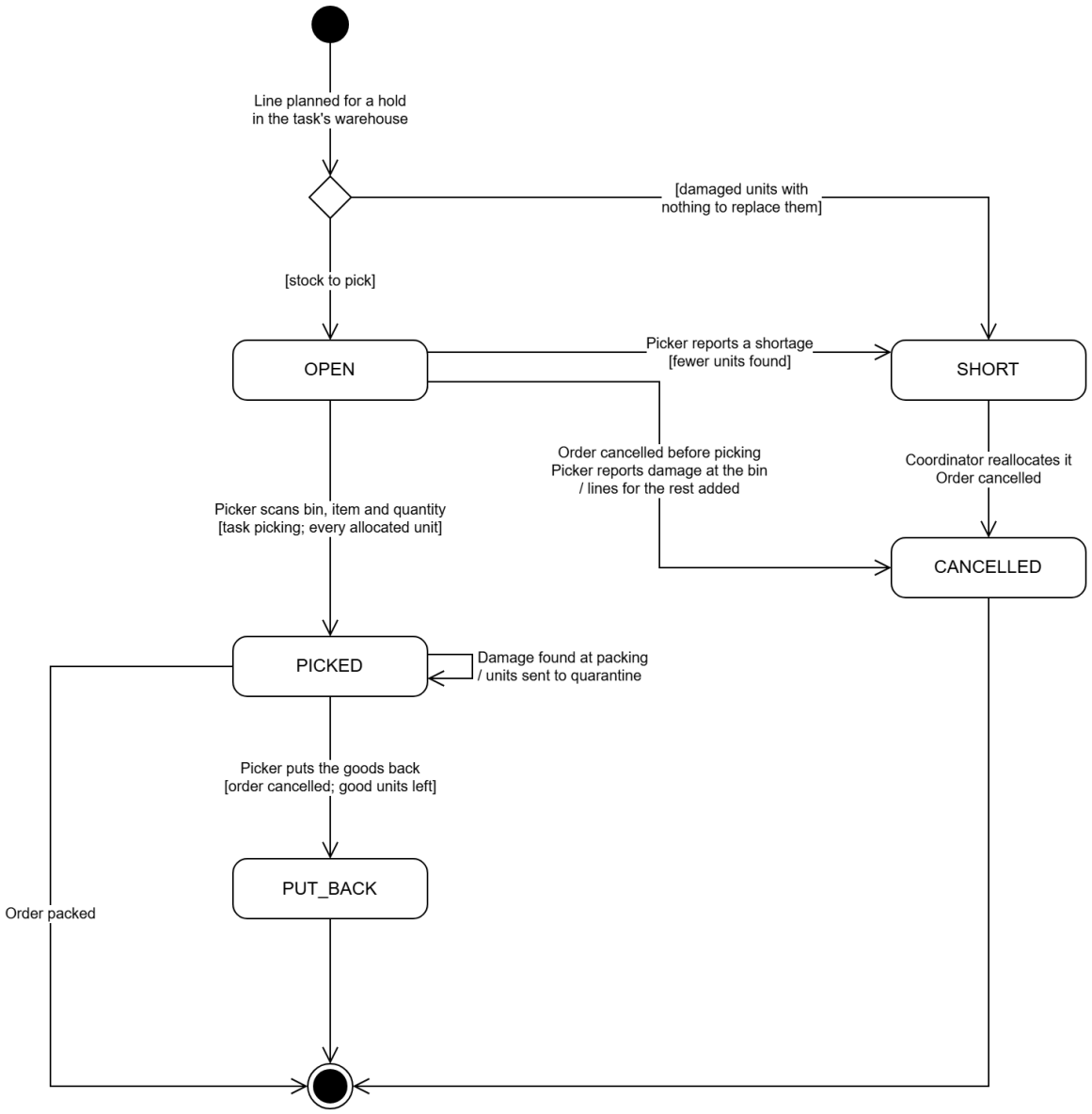


Figure 304. Pick Line State Diagram

3.19.13 Packing Session State Diagram

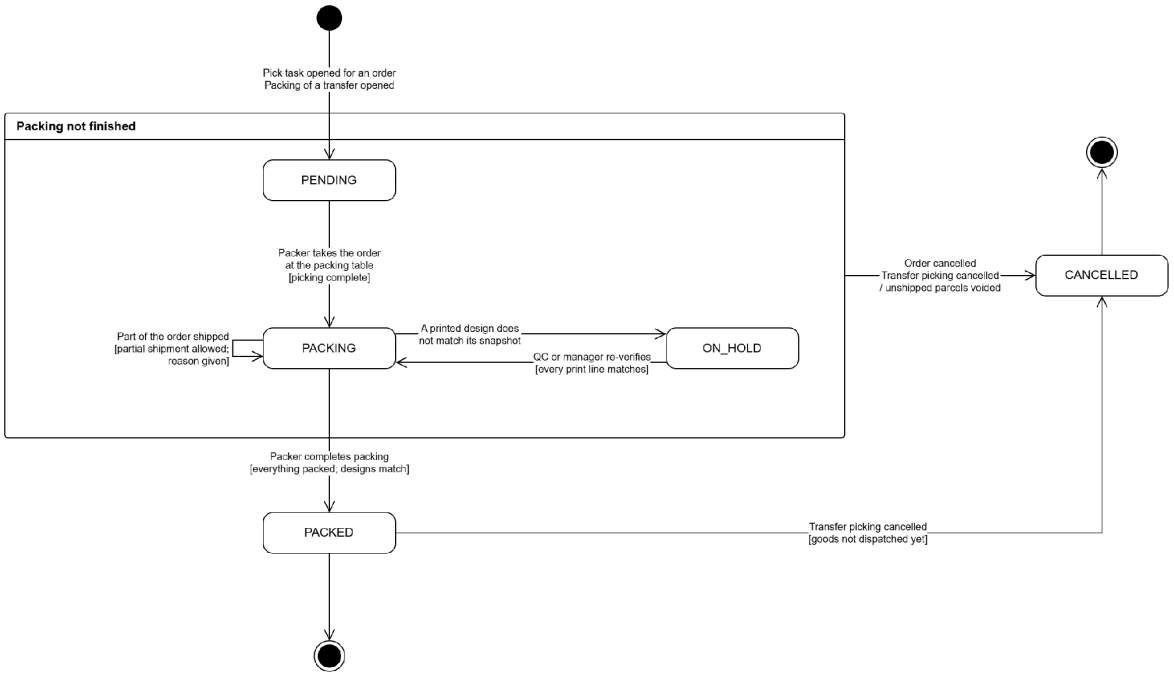


Figure 305. Packing Session State Diagram

3.19.14 Package State Diagram

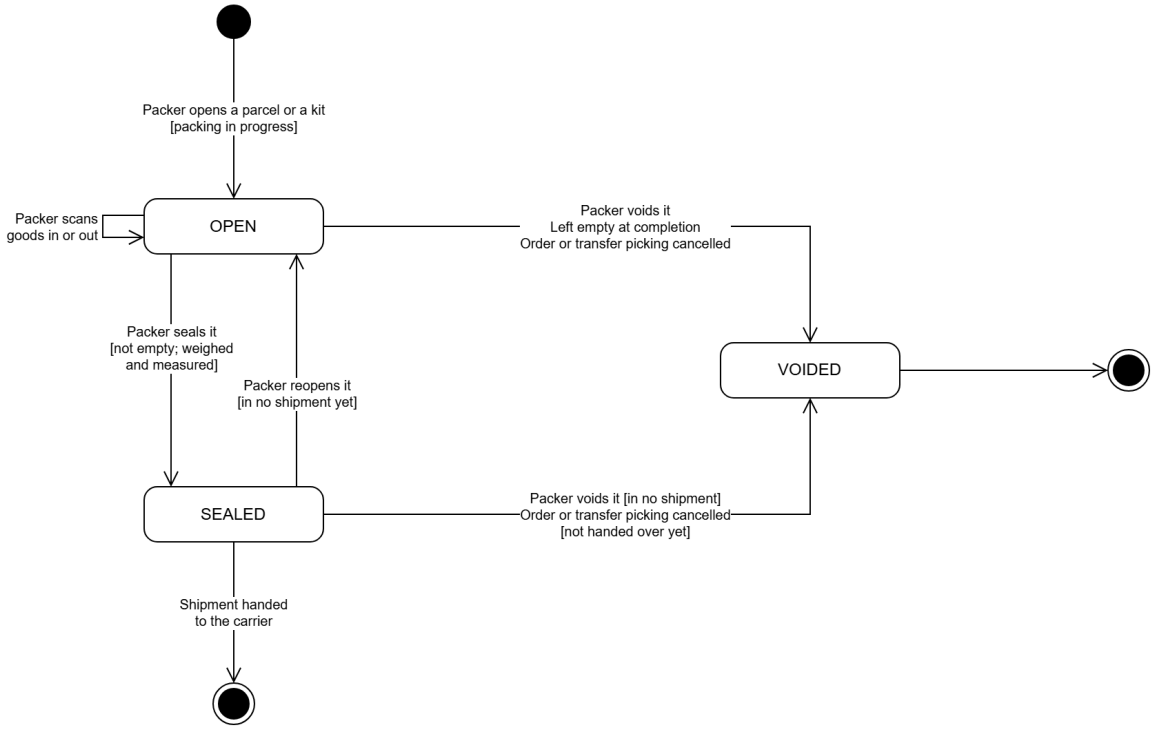


Figure 306. Package State Diagram

3.20 Shipping & Delivery

3.20.1 Class Diagram

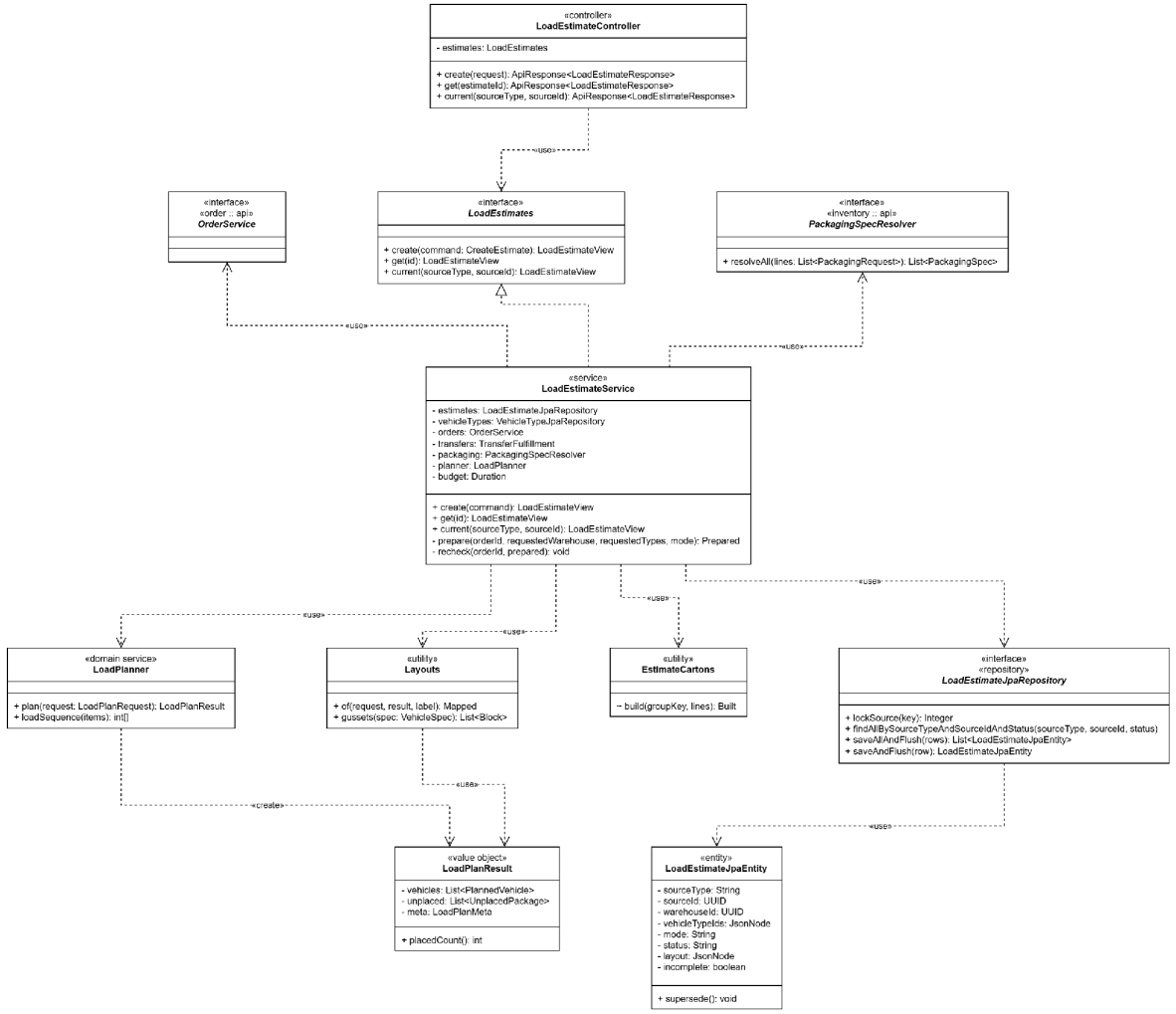


Figure 307. Shipping and Delivery Class Diagram (1/6): Load Estimates

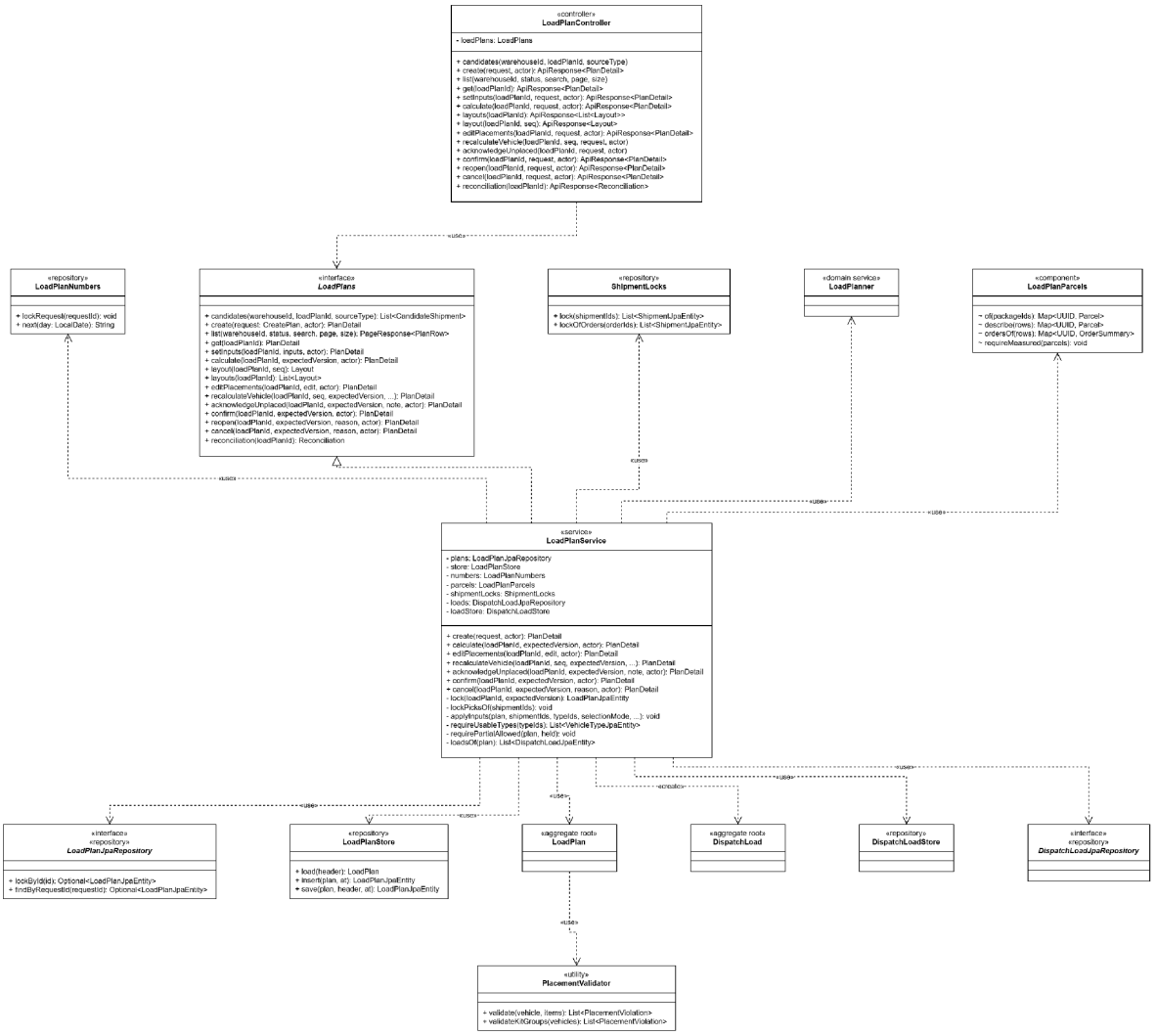


Figure 308. Shipping and Delivery Class Diagram (2/6): Load Plans

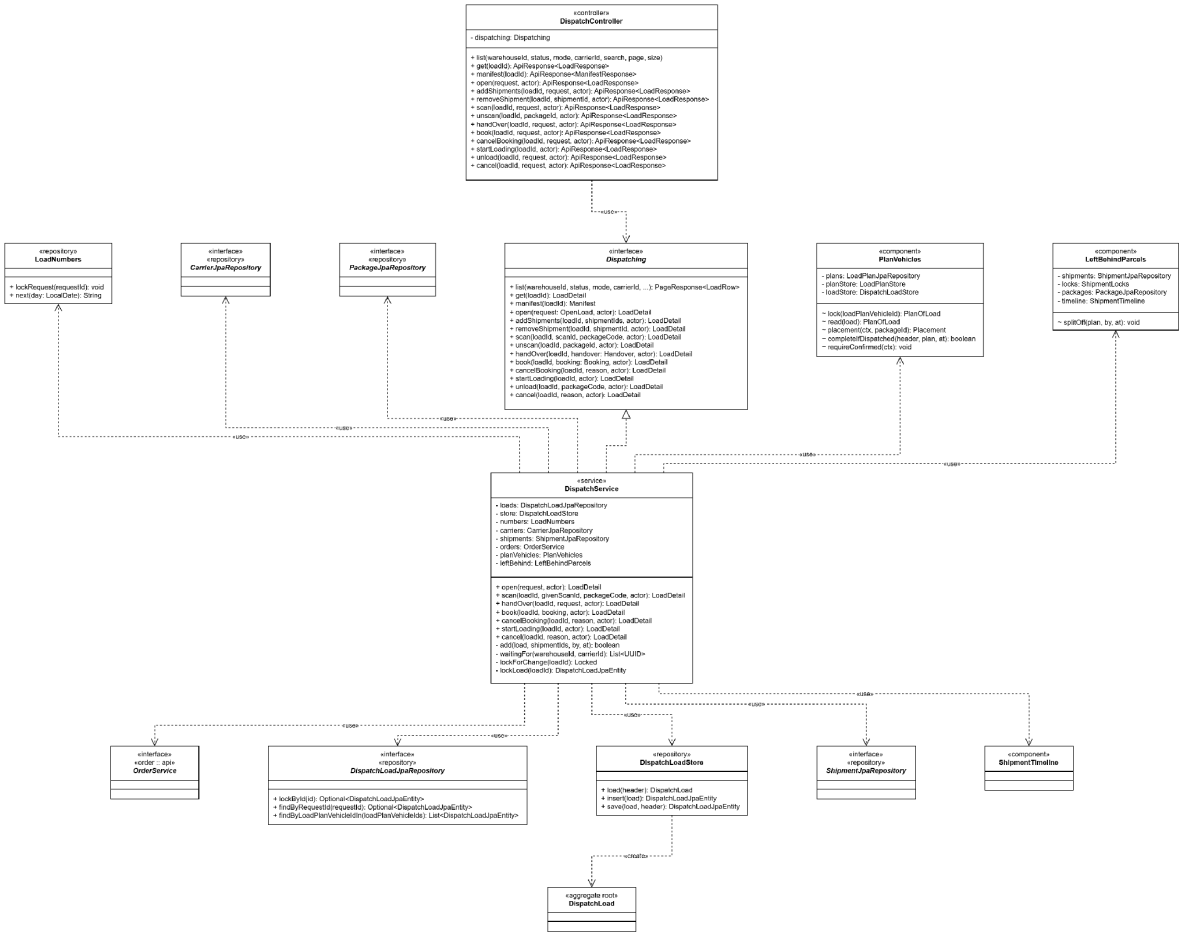


Figure 309. Shipping and Delivery Class Diagram (3/6): Dispatch Loads and Handover

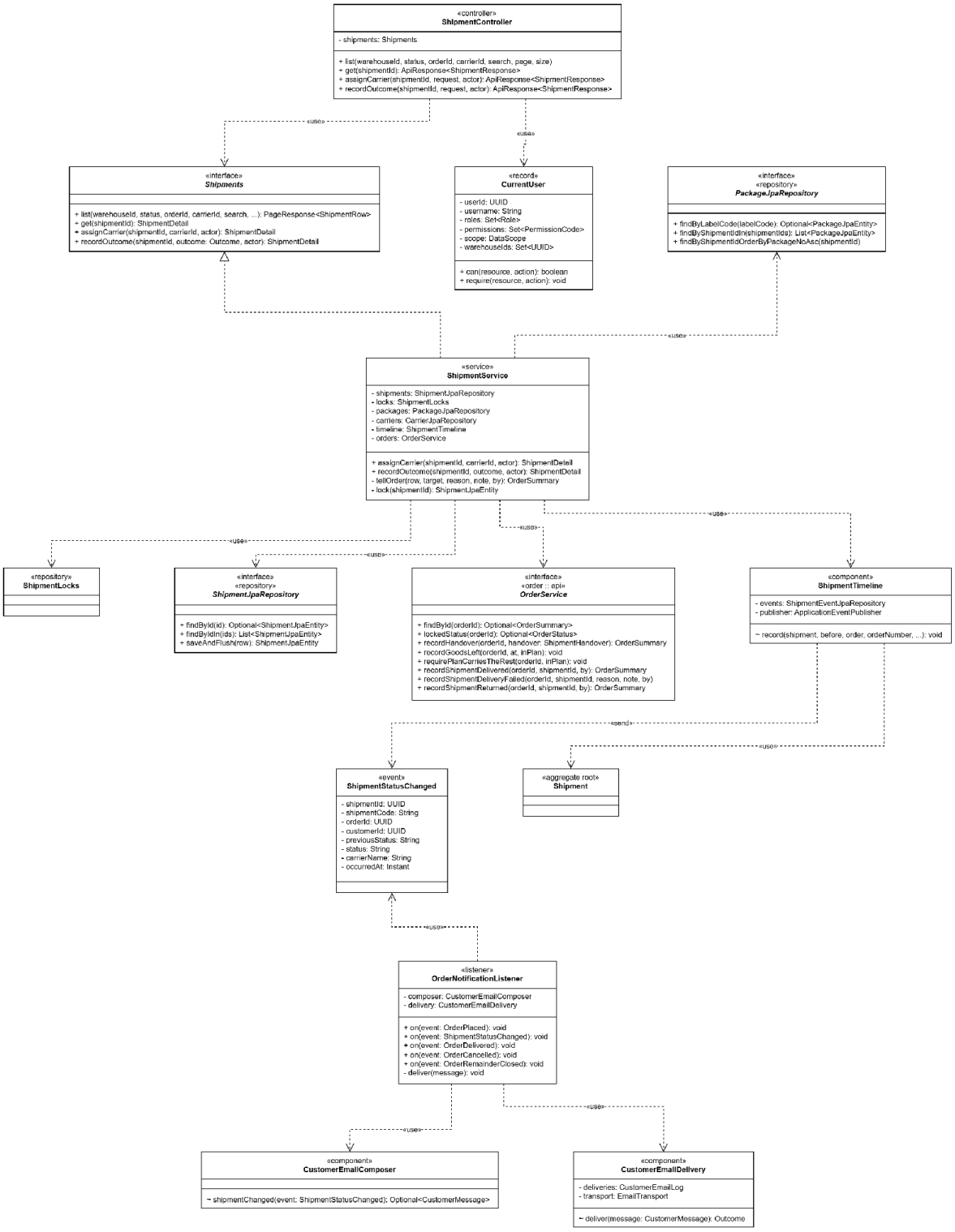


Figure 310. Shipping and Delivery Class Diagram (4/6): Shipments, Delivery Outcomes and Notification

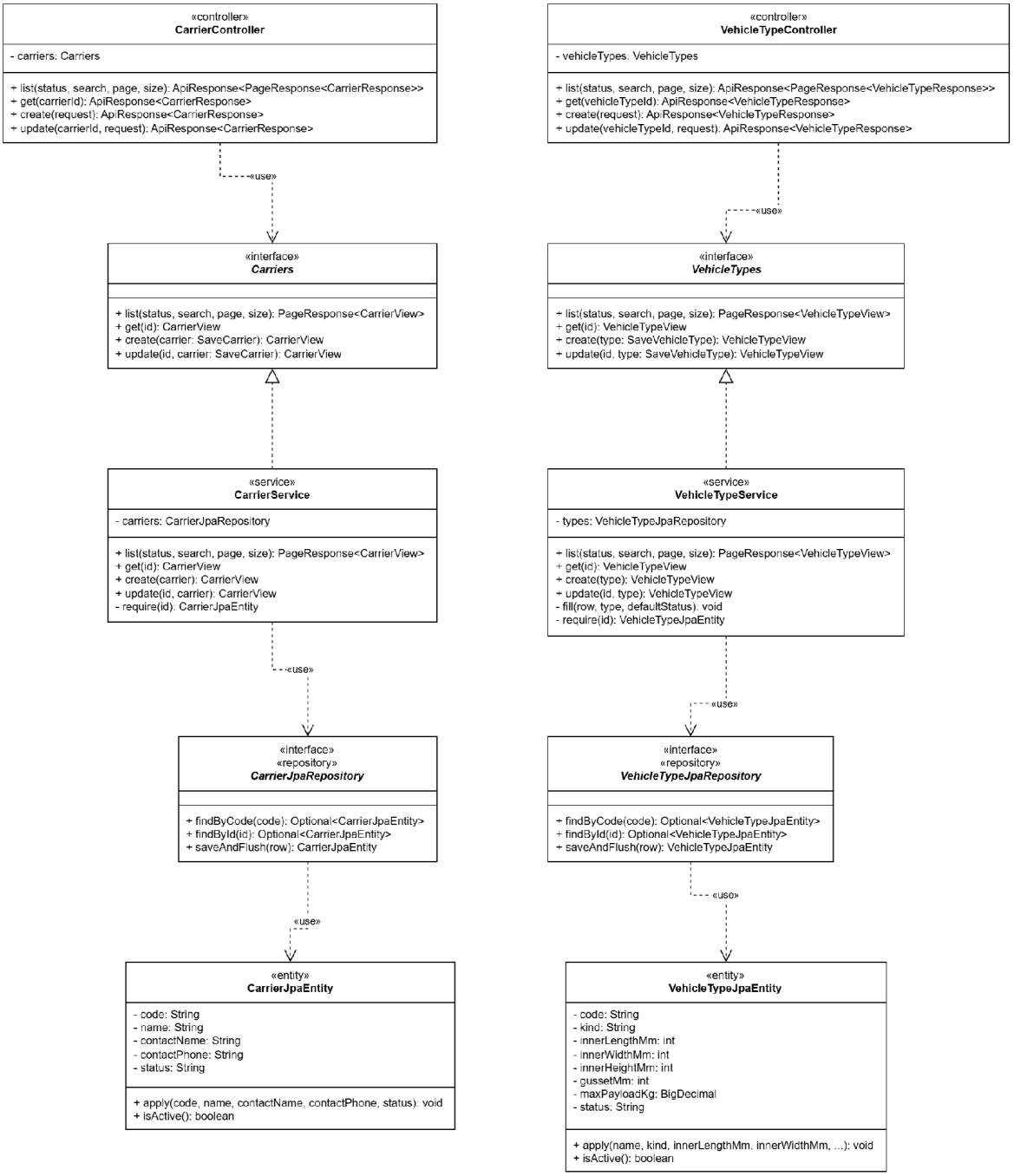


Figure 311. Shipping and Delivery Class Diagram (5/6): Carriers and Vehicle Types

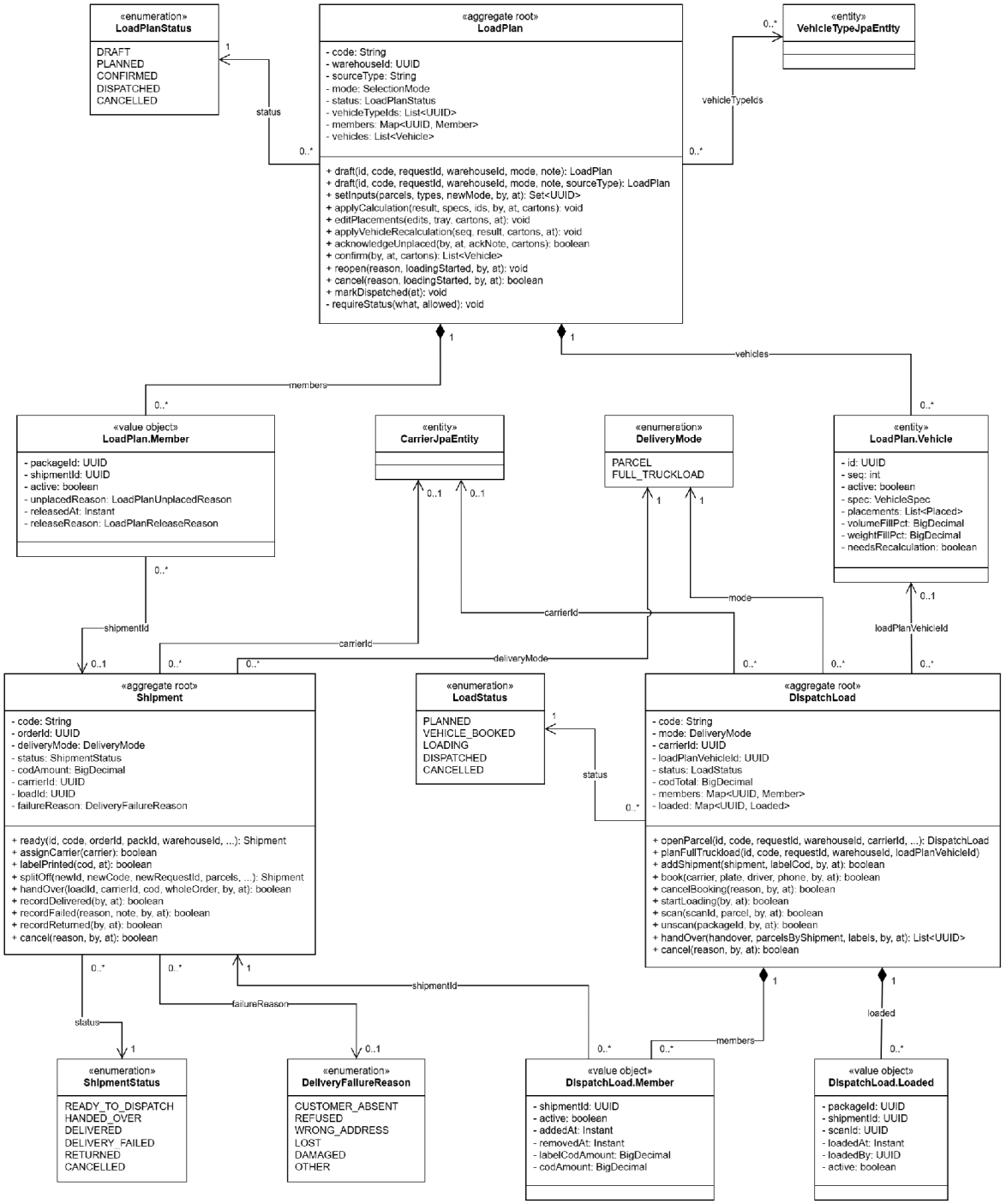


Figure 312. Shipping and Delivery Class Diagram (6/6): Domain Model

3.20.2 Class Specification

LoadEstimateController

No	Method	Description

1	create(request)	POST /api/v1/fulfillment/load-estimates: Estimates (advisory, synchronous within the solver budget) how an order's remaining units - or a transfer order's lines - load into the chosen vehicle types and makes it the source's CURRENT estimate (earlier ones SUPERSEDED); lines without size or weight are listed as missing; 409 LOAD_ESTIMATE_SOURCE_NOT_ELIGIBLE, 422 LOAD_ESTIMATE_PACKAGING_MISSING.
2	get(estimateId)	GET /api/v1/fulfillment/load-estimates/{estimateId}: Returns one estimate, current or superseded; 404 LOAD_ESTIMATE_NOT_FOUND.
3	current(sourceType, sourceId)	GET /api/v1/fulfillment/load-estimates?sourceType=&sourceId=: Returns the current estimate of a source; 404 LOAD_ESTIMATE_NOT_FOUND when it has none.

Table 101. LoadEstimateController Class Specification

LoadPlanController

No	Method	Description
1	candidates(warehouseId, loadPlanId, sourceType)	GET /api/v1/fulfillment/load-plan-candidates: Lists the warehouse's full-truckload shipments waiting for dispatch, whole, with the packages no other live plan holds (sourceType=TRANSFER: the packed full-truckload transfers leaving it).
2	create(request, actor)	POST /api/v1/fulfillment/load-plans: Starts a DRAFT load plan, optionally with whole shipments (or one packed transfer), vehicle types and mode; idempotent by requestId; 409 LOAD_PLAN_PACKAGE_TAKEN, SHIPMENT_STATE_CONFLICT, SHIPMENT_NOT_FOR_LOAD, VEHICLE_TYPE_INACTIVE; 422 LOAD_PLAN_PACKAGE_UNMEASURED.
3	list(warehouseId, status, search, page, size)	GET /api/v1/fulfillment/load-plans: Pages load plans of the caller's warehouses, newest first, by status and code.
4	get(loadPlanId)	GET /api/v1/fulfillment/load-plans/{loadPlanId}: Returns a plan: shipments, vehicle types, vehicles (fill, dispatch load) and each package's vehicle or unplaced reason; 404 LOAD_PLAN_NOT_FOUND.
5	setInputs(loadPlanId, request, actor)	PUT /api/v1/fulfillment/load-plans/{loadPlanId}/inputs: Replaces the shipments, vehicle types and mode; a PLANNED plan whose inputs change goes back to DRAFT; 409 OPTIMISTIC_LOCK, INVALID_LOAD_PLAN_TRANSITION, LOAD_PLAN_PACKAGE_TAKEN.

6	calculate(loadPlanId, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/calculation: Runs the 3D load planner over the whole plan (synchronous, solver budget) and stores vehicles and placements (-> PLANNED); 409 LOAD_PLAN_EMPTY, VEHICLE_TYPE_INACTIVE.
7	layouts(loadPlanId)	GET /api/v1/fulfillment/load-plans/{loadPlanId}/layouts: Returns the 3D layout of every vehicle of the plan, in seq order.
8	layout(loadPlanId, seq)	GET /api/v1/fulfillment/load-plans/{loadPlanId}/vehicles/{seq}/layout: Returns one vehicle's layout for the 3D viewer; 404 LOAD_PLAN_VEHICLE_NOT_FOUND.
9	editPlacements(loadPlanId, request, actor)	PUT /api/v1/fulfillment/load-plans/{loadPlanId}/placements: Replaces the whole plan's placements by hand, all or nothing: every vehicle must pass the loading rules and kits stay together, else 422 PLACEMENT_RULE_VIOLATION naming each box and rule; only PLANNED.
10	recalculateVehicle(loadPlanId, seq, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/vehicles/{seq}/recalculation: Re-solves one vehicle with its packages plus optional packages from the tray; what does not fit goes to the tray; only PLANNED (409 INVALID_LOAD_PLAN_TRANSITION).
11	acknowledgeUnplaced(loadPlanId, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/unplaced-acknowledgement: Accepts that the unplaced packages stay at the dispatch area (BR-13); refused for an order that may not ship in parts (409 ORDER_PARTIAL_SHIPMENT_NOT_ALLOWED) or a split kit (422 PLACEMENT_RULE_VIOLATION).
12	confirm(loadPlanId, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/confirmation: Confirms a PLANNED plan: locks layout and packages, releases unplaced ones to the dispatch area and opens one PLANNED dispatch load per vehicle; 409 LOAD_PLAN_UNPLACED_NOT_ACKNOWLEDGED, LOAD_PLAN_EMPTY, SHIPMENT_STATE_CONFLICT, ORDER_PARTIAL_SHIPMENT_NOT_ALLOWED.

13	reopen(loadPlanId, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/reopening: Takes a CONFIRMED plan back to PLANNED with a reason while no vehicle loads (BR-14), cancelling its PLANNED and VEHICLE_BOOKED loads; 409 LOAD_PLAN_LOADING_STARTED.
14	cancel(loadPlanId, request, actor)	POST /api/v1/fulfillment/load-plans/{loadPlanId}/cancellation: Cancels a DRAFT, PLANNED or CONFIRMED plan with a reason while no vehicle loads: packages back to the dispatch area, its loads cancelled; 409 LOAD_PLAN_LOADING_STARTED.
15	reconciliation(loadPlanId)	GET /api/v1/fulfillment/load-plans/{loadPlanId}/reconciliation: Returns the piece check (packages in = placed + unplaced) for the plan, per shipment and per order.

Table 102. LoadPlanController Class Specification

DispatchController

No	Method	Description
1	list(warehouseId, status, mode, carrierId, search, page, size)	GET /api/v1/fulfillment/loads: Pages loads of the caller's warehouses, newest first, by status, mode (PARCEL / FULL_TRUCKLOAD, else 400 UNSUPPORTED_PARAMETER), carrier and code / plate / plan code.
2	get(loadId)	GET /api/v1/fulfillment/loads/{loadId}: Returns a load with its manifest: shipments, parcels and which are on the vehicle; 404 LOAD_NOT_FOUND.
3	manifest(loadId)	GET /api/v1/fulfillment/loads/{loadId}/manifest: Returns the manifest to print: shipments, recipients, parcels, COD per shipment and in total.
4	open(request, actor)	POST /api/v1/fulfillment/loads: The carrier's driver arrived: opens a parcel load (LOADING) for an active carrier with the chosen shipments, or those waiting for that carrier; idempotent by requestId; 404 CARRIER_NOT_FOUND, 409 CARRIER_INACTIVE.
5	addShipments(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/shipments: Adds READY_TO_DISPATCH shipments of this warehouse and carrier to an open manifest; 409 SHIPMENT_ON_ANOTHER_LOAD, SHIPMENT_OTHER_CARRIER, SHIPMENT_NOT_FOR_LOAD, SHIPMENT_STATE_CONFLICT.

6	removeShipment(loadId, shipmentId, actor)	DELETE /api/v1/fulfillment/loads/{loadId}/shipments/{shipmentId}: Takes a shipment and its scanned parcels off the manifest before the handover.
7	scan(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/scans: Scans a sealed parcel onto the vehicle by its label code (a plan's vehicle takes only the parcels placed on it); the same scanId again gets the same answer; 404 PACKAGE_NOT_FOUND; 409 PACKAGE_NOT_IN_LOAD, PACKAGE_ALREADY_LOADED, PACKAGE_NOT_READY, PACKAGE_OTHER_CARRIER, LABEL_OUTDATED.
8	unscan(loadId, packageId, actor)	DELETE /api/v1/fulfillment/loads/{loadId}/scans/{packageId}: Takes a scanned parcel off the vehicle again before the handover.
9	handOver(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/handover: Records both signatures: the load becomes DISPATCHED, its shipments HANDED_OVER with the COD the order fixes and the orders PARTIALLY_FULFILLED / SHIPPED; 400 HANDOVER_NOT_SIGNED; 409 LOAD_EMPTY, LOAD_NOT_FULLY_LOADED, LOAD_UNLOAD_PENDING, OPTIMISTIC_LOCK, ORDER_BALANCE_NOT_PAID, ORDER_INVALID_TRANSITION; signing again returns it as it is.
10	book(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/booking: Records the vehicle booked with a carrier (outside the system) for a CONFIRMED plan's vehicle (PLANNED -> VEHICLE_BOOKED, again replaces the booking); 404 CARRIER_NOT_FOUND; 409 INVALID_LOAD_TRANSITION, INVALID_LOAD_PLAN_TRANSITION, CARRIER_INACTIVE, SHIPMENT_OTHER_CARRIER.
11	cancelBooking(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/booking-cancellation : The booking fell through: back to PLANNED with the reason, to book again; 409 INVALID_LOAD_TRANSITION once loading started.
12	startLoading(loadId, actor)	POST /api/v1/fulfillment/loads/{loadId}/loading-start: The booked vehicle arrived: VEHICLE_BOOKED -> LOADING; from now the plan cannot be reopened or cancelled (BR-14) and its left-behind parcels split off into a new shipment; 409 INVALID_LOAD_TRANSITION, ORDER_PARTIAL_SHIPMENT_NOT_ALLOWED.

13	unload(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/unloads: Records a parcel of a cancelled order scanned off the vehicle; with nothing left to carry the vehicle is called off; 404 PACKAGE_NOT_FOUND, 409 PACKAGE_NOT_IN_LOAD.
14	cancel(loadId, request, actor)	POST /api/v1/fulfillment/loads/{loadId}/cancellation: Calls an open parcel load off with a reason: its shipments and parcels go back to the dispatch area untouched; a plan's vehicle is refused (409 LOAD_MANAGED_BY_PLAN), a load that left too (409 INVALID_LOAD_TRANSITION).

Table 103. DispatchController Class Specification

ShipmentController

No	Method	Description
1	list(warehouseId, status, orderId, carrierId, search, page, size)	GET /api/v1/fulfillment/shipments: Pages shipments of the caller's warehouses, newest first, by warehouse, status, order, carrier and code.
2	get(shipmentId)	GET /api/v1/fulfillment/shipments/{shipmentId}: Returns a shipment with its parcels and status timeline; 404 SHIPMENT_NOT_FOUND.
3	assignCarrier(shipmentId, request, actor)	PUT /api/v1/fulfillment/shipments/{shipmentId}/carrier: Chooses (null clears) the carrier of a shipment waiting for dispatch; 409 SHIPMENT_ON_ANOTHER_LOAD, SHIPMENT_STATE_CONFLICT, CARRIER_INACTIVE.
4	recordOutcome(shipmentId, request, actor)	POST /api/v1/fulfillment/shipments/{shipmentId}/delivery-outcome: Records what the carrier reported - DELIVERED, DELIVERY_FAILED (reason; OTHER needs a note) or RETURNED - and the order follows (a credit order bills a delivered shipment); DELIVERED / DELIVERY_FAILED also need fulfillment-delivery-outcomes:APPROVE (403); 409 SHIPMENT_STATE_CONFLICT.

Table 104. ShipmentController Class Specification

CarrierController

No	Method	Description
1	list(status, search, page, size)	GET /api/v1/fulfillment/carriers: Pages carriers by name, filtered by ACTIVE / INACTIVE and code or name.

2	get(carrierId)	GET /api/v1/fulfillment/carriers/{carrierId}: Returns one carrier; 404 CARRIER_NOT_FOUND.
3	create(request)	POST /api/v1/fulfillment/carriers: Declares a carrier (ACTIVE unless stated); 409 CARRIER_CODE_EXISTS.
4	update(carrierId, request)	PUT /api/v1/fulfillment/carriers/{carrierId}: Changes a carrier with the version read (409 OPTIMISTIC_LOCK, CARRIER_CODE_EXISTS); an INACTIVE carrier takes no new load.

Table 105. CarrierController Class Specification

VehicleTypeController

No	Method	Description
1	list(status, search, page, size)	GET /api/v1/fulfillment/vehicle-types: Pages vehicle types by name, filtered by status and code or name.
2	get(vehicleTypeId)	GET /api/v1/fulfillment/vehicle-types/{vehicleTypeId}: Returns one vehicle type; 404 VEHICLE_TYPE_NOT_FOUND.
3	create(request)	POST /api/v1/fulfillment/vehicle-types: Declares a vehicle type (truck, pickup or container: inner dimensions and gusset clearance in mm, payload in kg); 409 VEHICLE_TYPE_CODE_EXISTS, 400 VALIDATION_FAILED for impossible measures.
4	update(vehicleTypeId, request)	PUT /api/v1/fulfillment/vehicle-types/{vehicleTypeId}: Changes a vehicle type with the version read; the code never changes (400); status INACTIVE retires it; 404 VEHICLE_TYPE_NOT_FOUND, 409 OPTIMISTIC_LOCK.

Table 106. VehicleTypeController Class Specification

3.20.3 Estimate Shipment Load Sequence Diagram

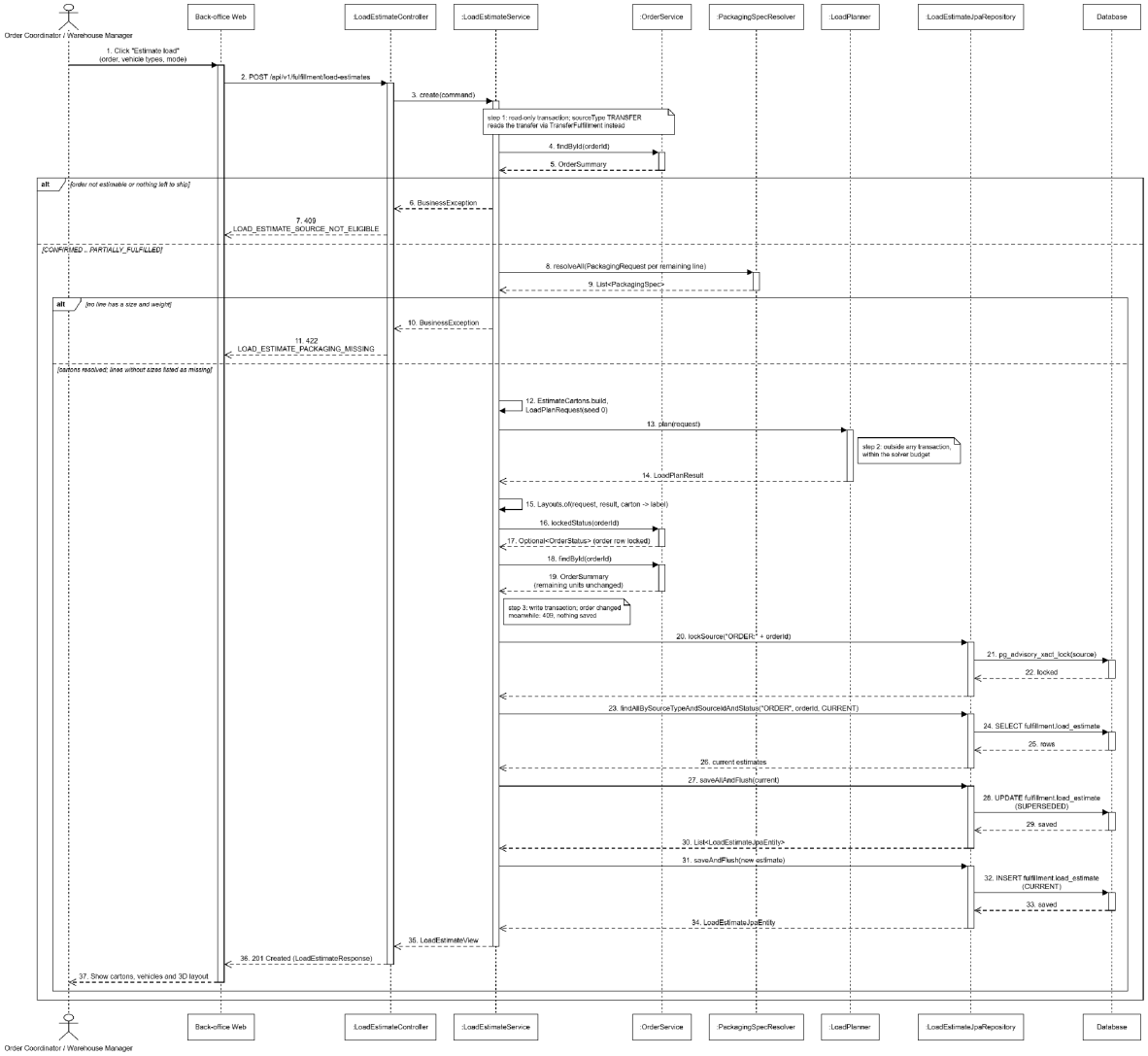


Figure 313. Estimate Shipment Load Sequence Diagram

3.20.4 Plan & Confirm Truck Load (Calculate 3D Plan) Sequence Diagram

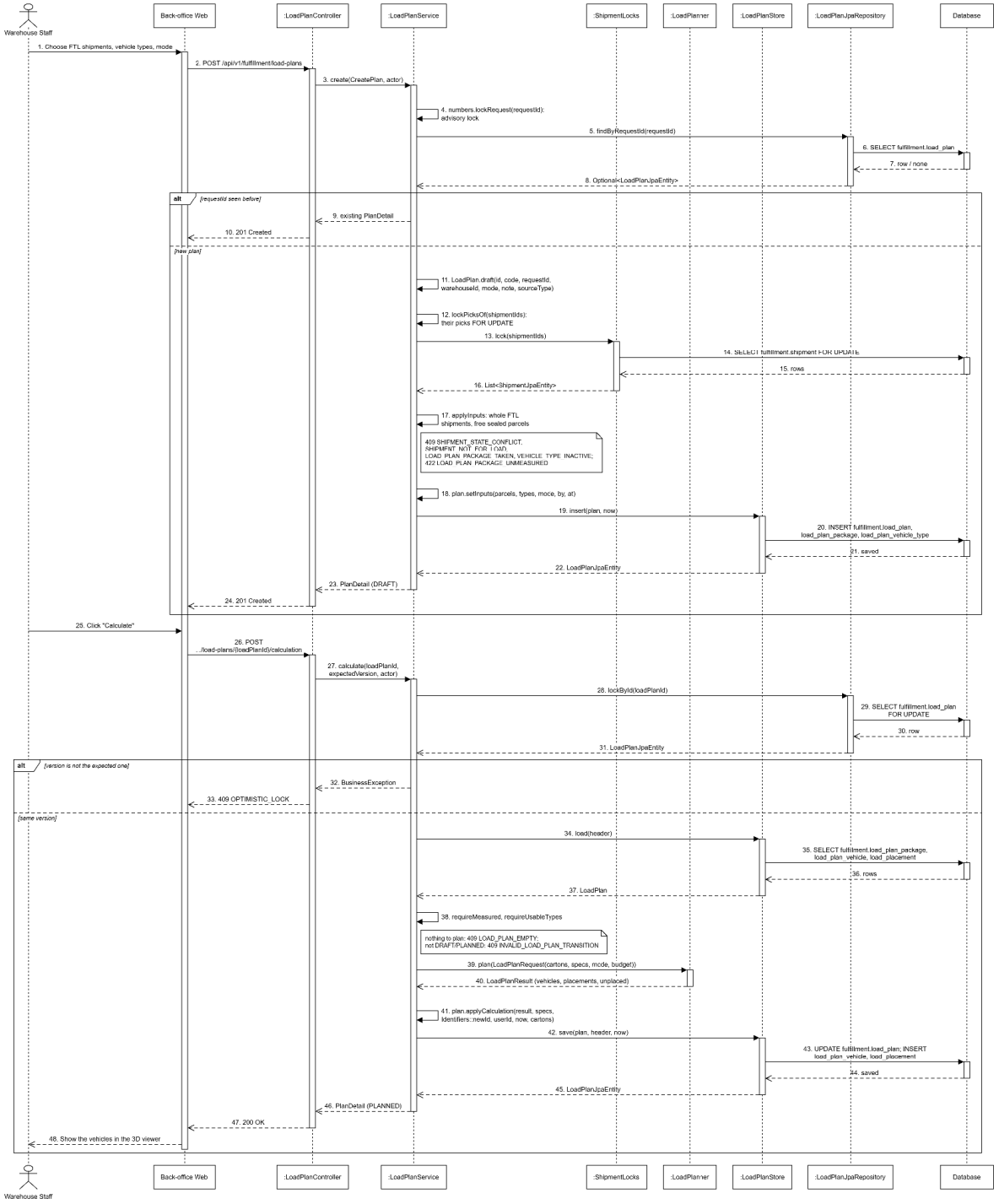


Figure 314. Plan & Confirm Truck Load (Calculate 3D Plan) Sequence Diagram

3.20.5 Plan & Confirm Truck Load (Confirm 3D Plan) Sequence Diagram

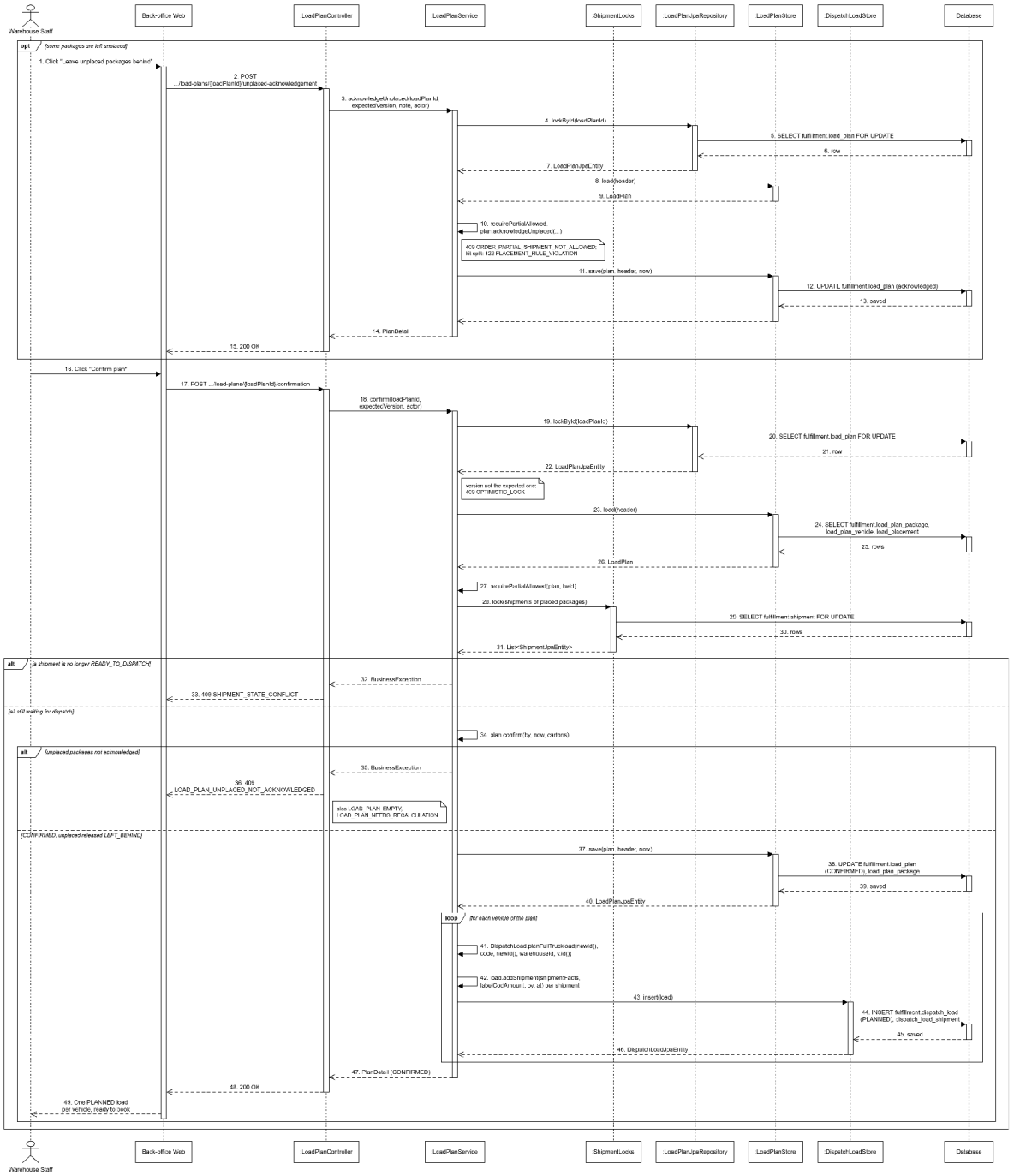


Figure 315. Plan & Confirm Truck Load (Confirm 3D Plan) Sequence Diagram

3.20.6 Adjust Load Plan Manually Sequence Diagram

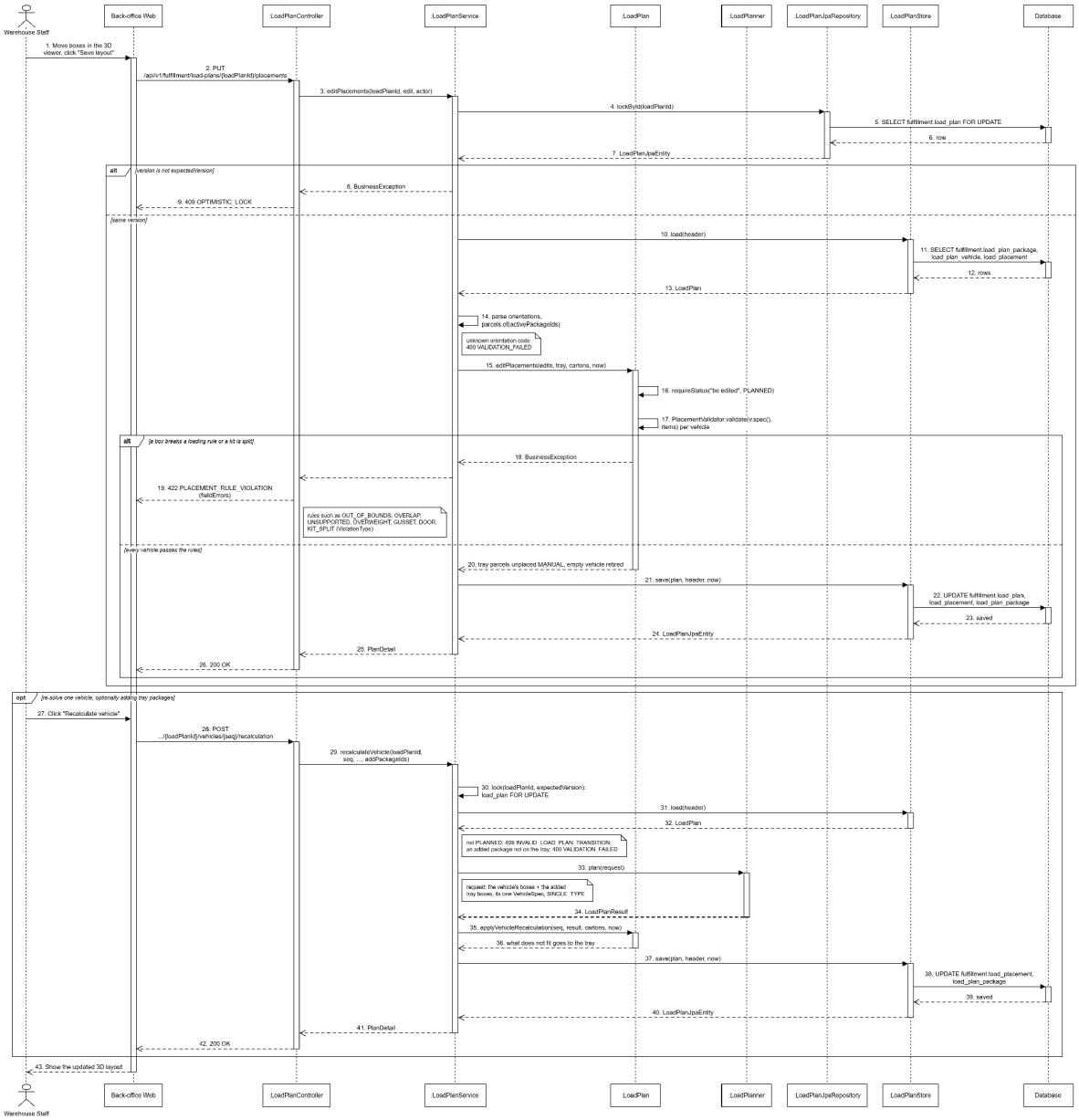


Figure 316. Adjust Load Plan Manually Sequence Diagram

3.20.7 Book Vehicle & Dispatch Truckload (Book and Start Loading) Sequence Diagram

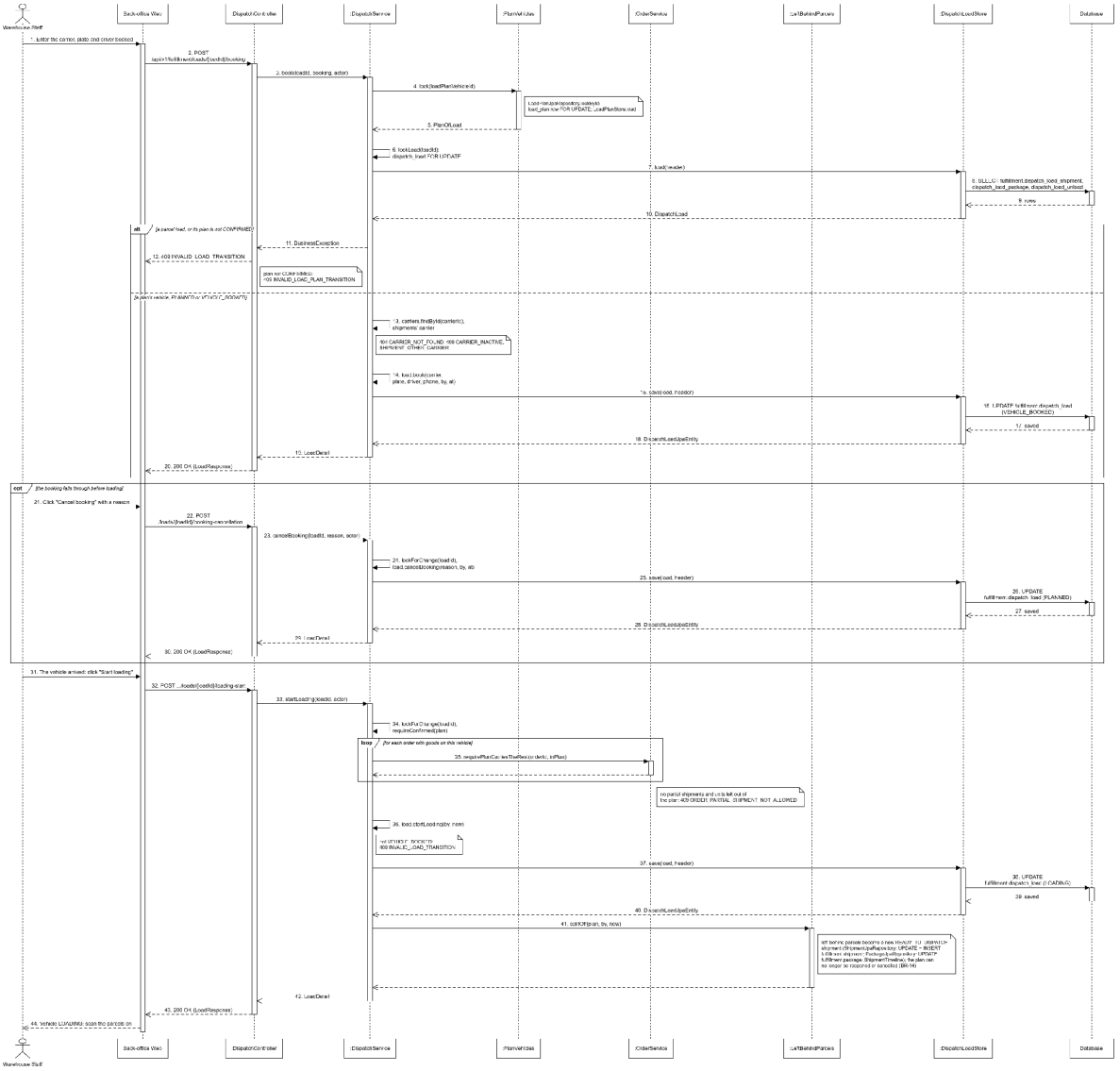


Figure 317. Book Vehicle & Dispatch Truckload (Book and Start Loading) Sequence Diagram

3.20.8 Book Vehicle & Dispatch Truckload (Load and Hand Over) Sequence Diagram

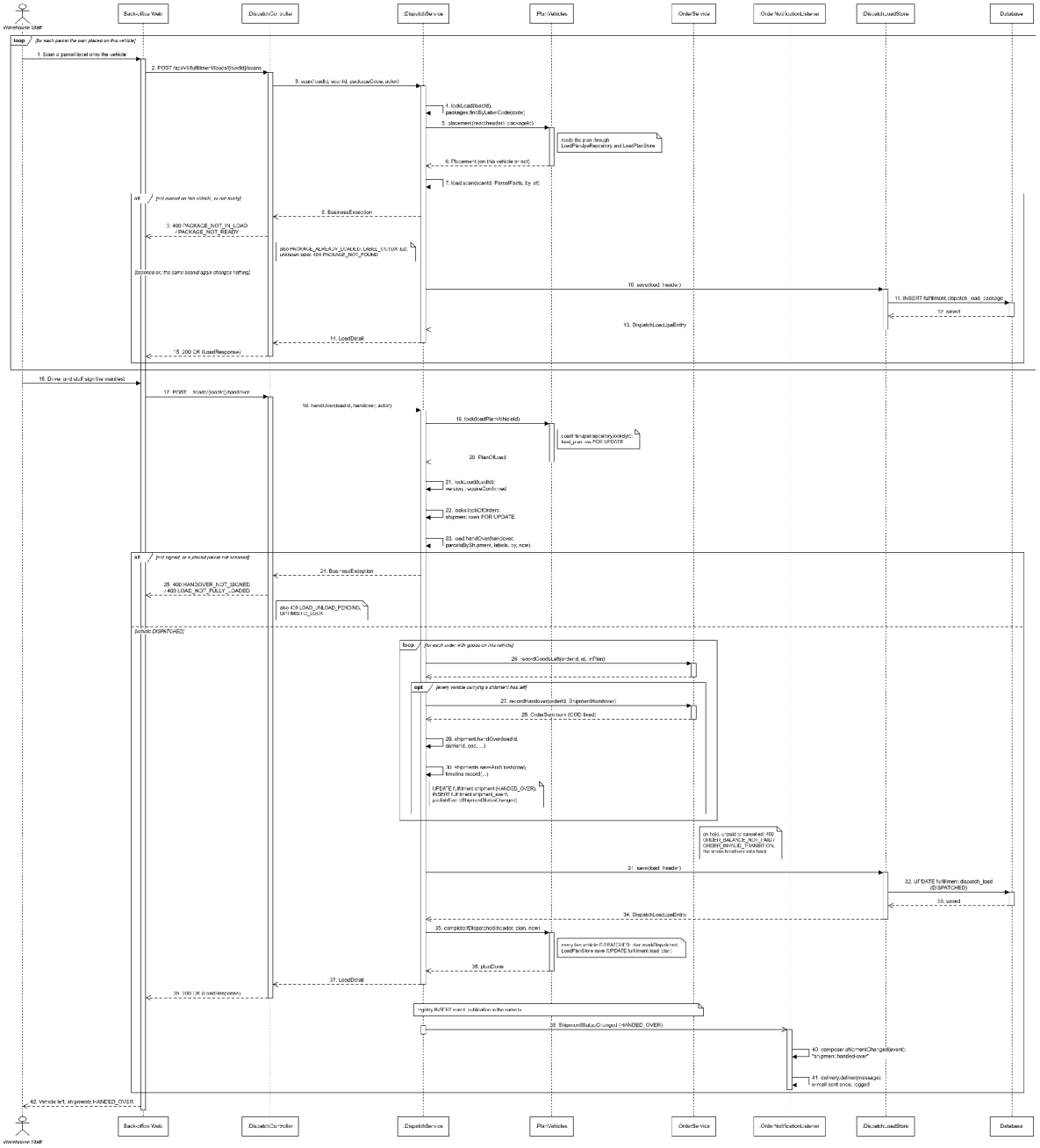


Figure 318. Book Vehicle & Dispatch Truckload (Load and Hand Over) Sequence Diagram

3.20.9 Cancel Dispatch Trip Sequence Diagram

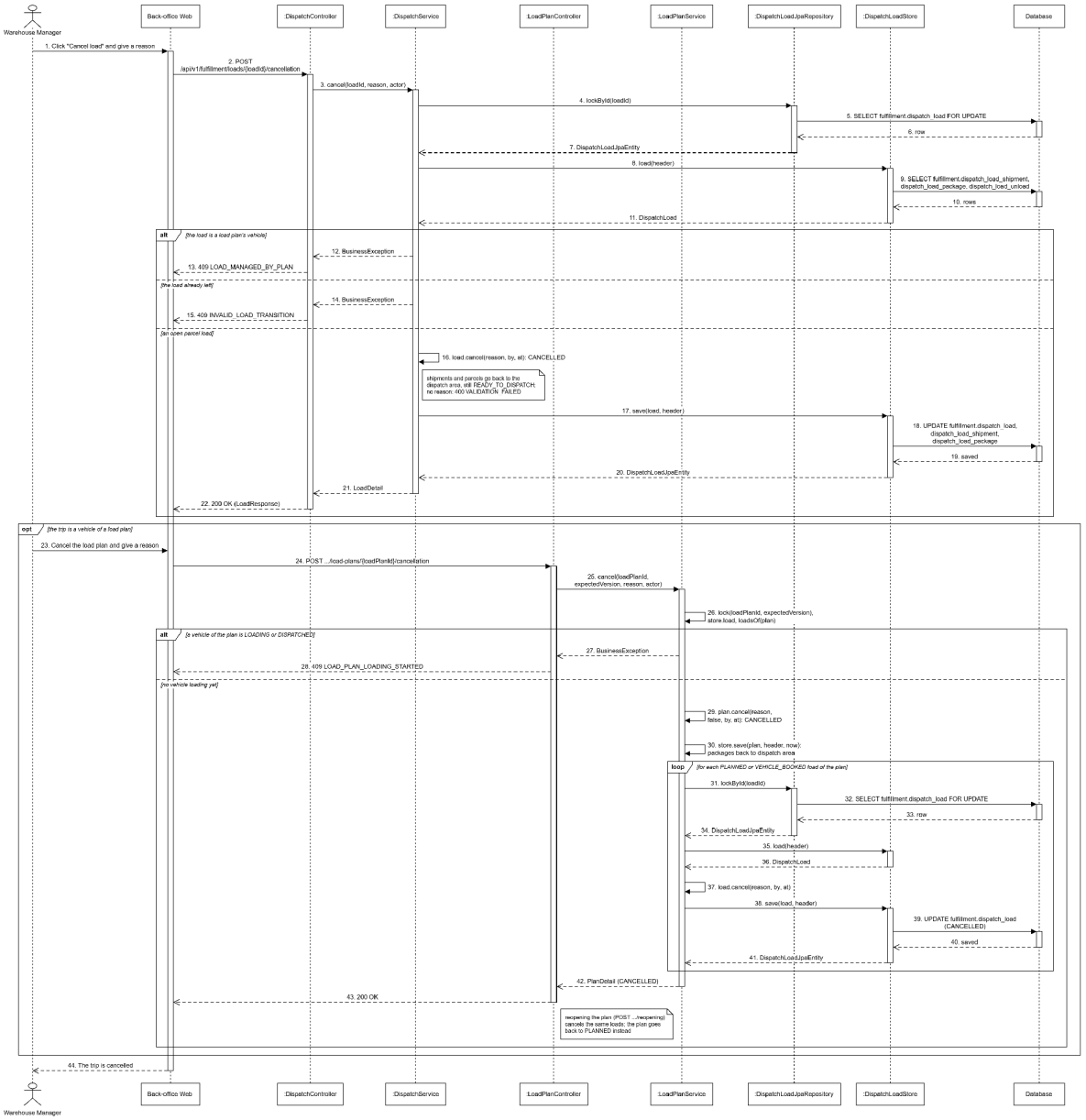


Figure 319. Cancel Dispatch Trip Sequence Diagram

3.20.10 Hand Over Parcels to Carrier Sequence Diagram

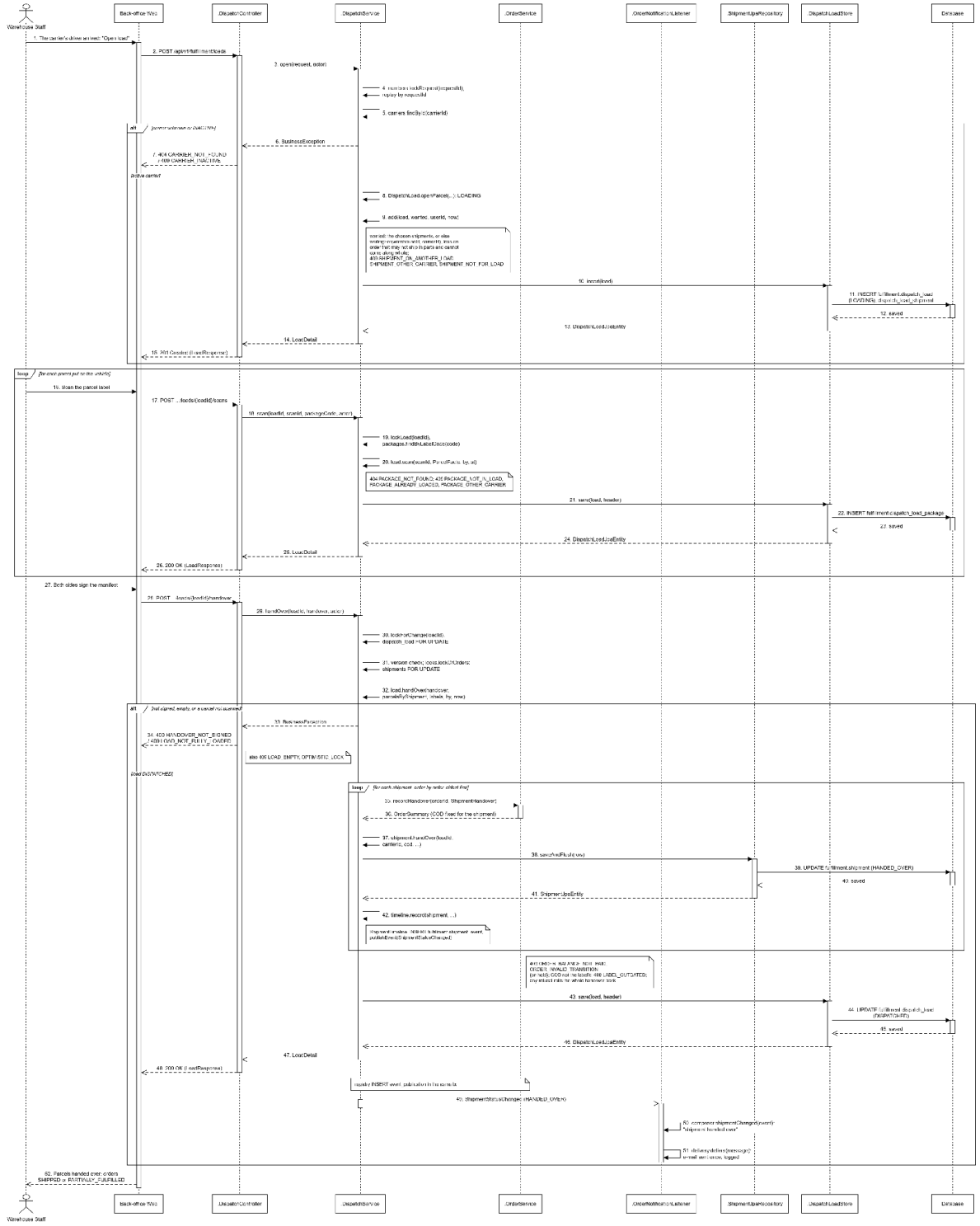


Figure 320. Hand Over Parcels to Carrier Sequence Diagram

3.20.11 Record Delivery Result and Returned Shipment Sequence Diagram

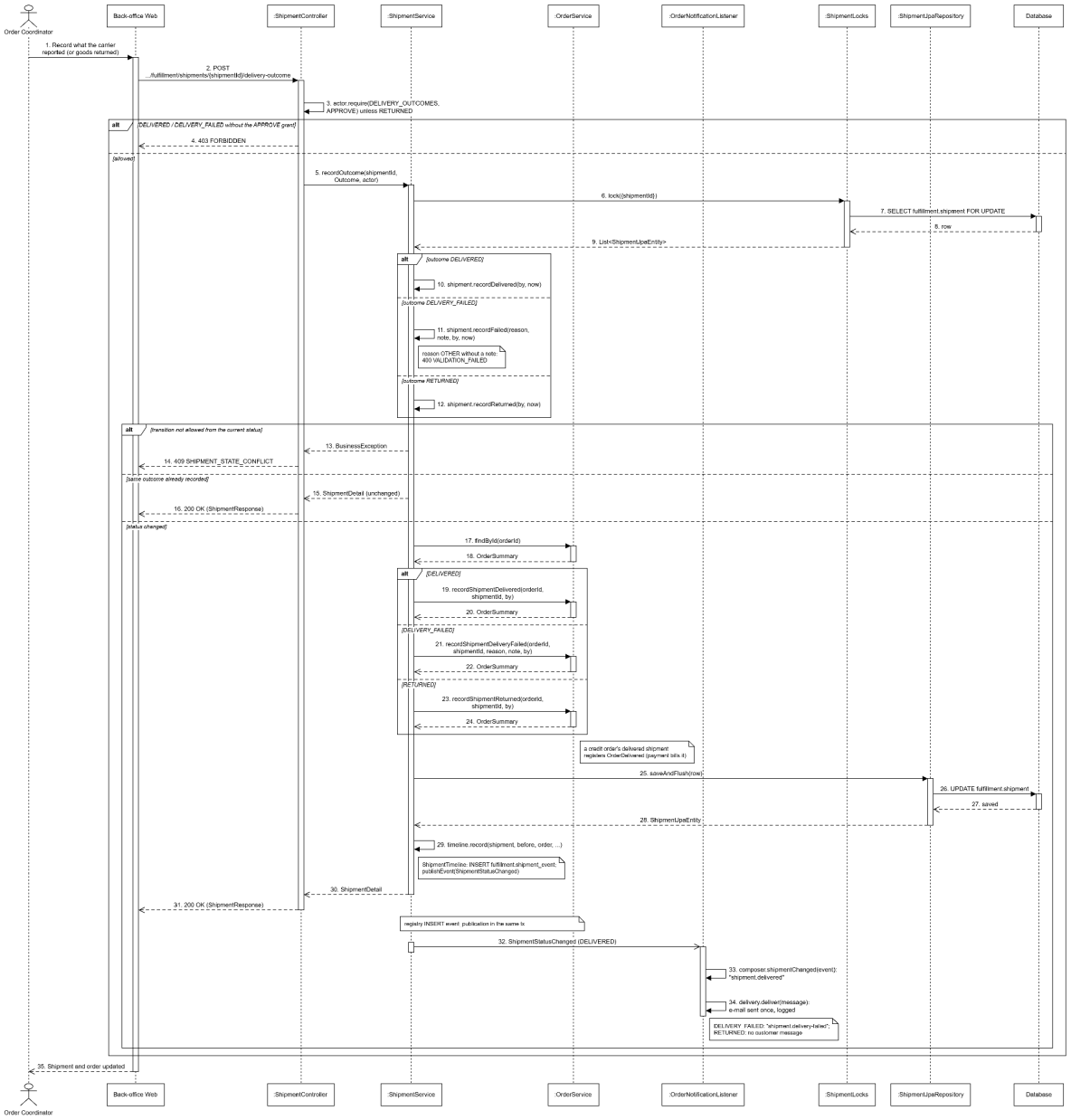


Figure 321. Record Delivery Result and Returned Shipment Sequence Diagram

3.20.12 Manage Carriers & Vehicle Types Sequence Diagram

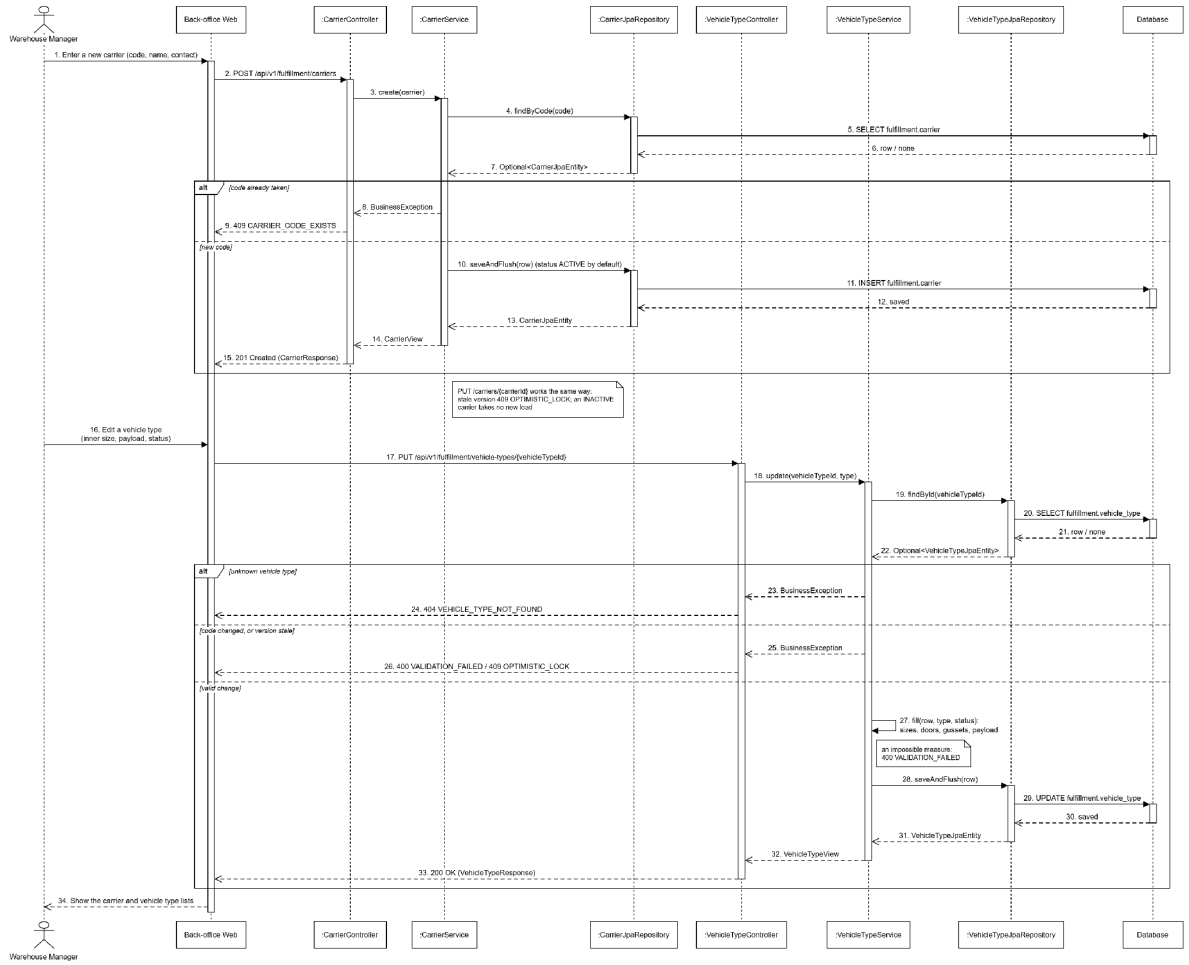


Figure 322. Manage Carriers & Vehicle Types Sequence Diagram

3.20.13 Shipment State Diagram

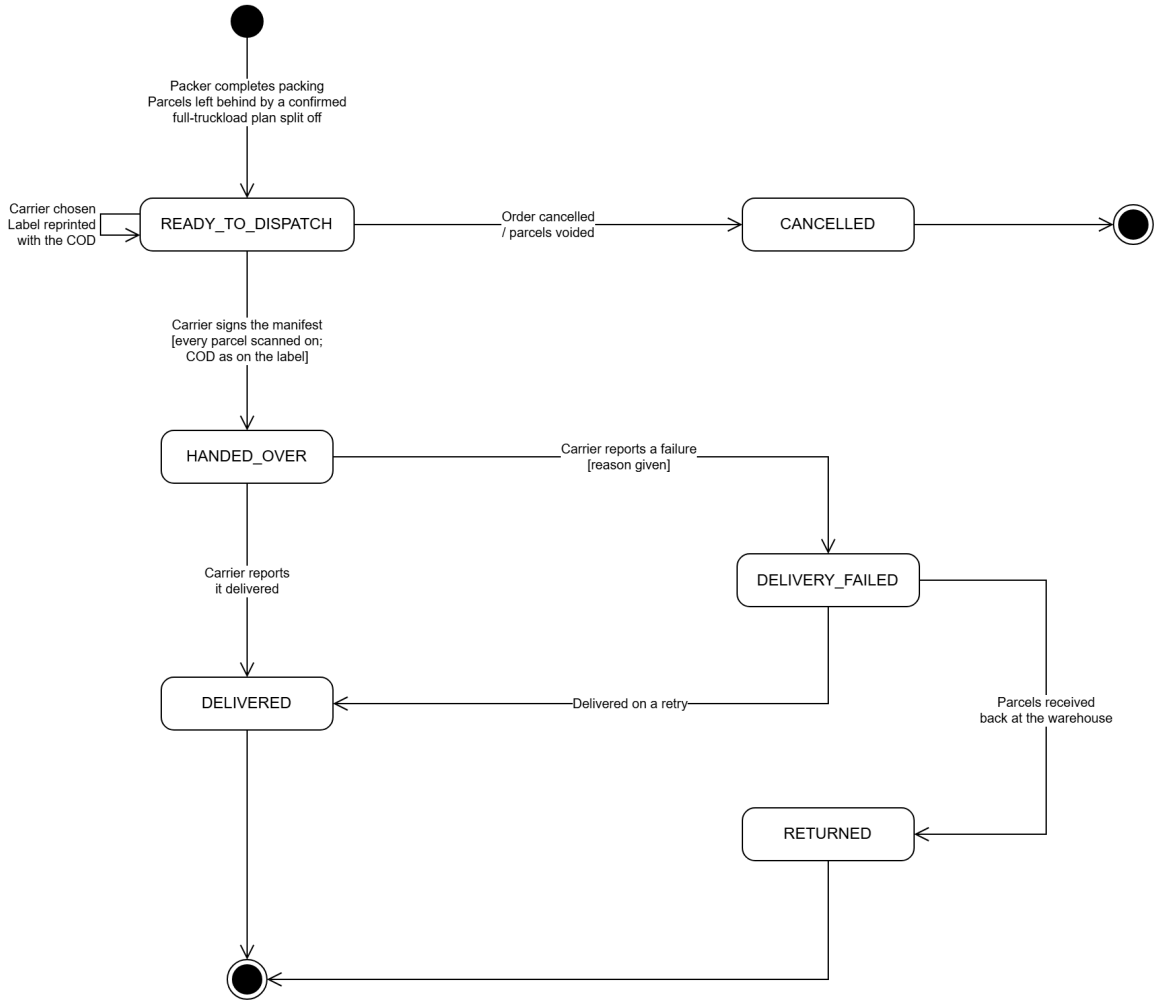


Figure 323. Shipment State Diagram

3.20.14 Load Plan State Diagram

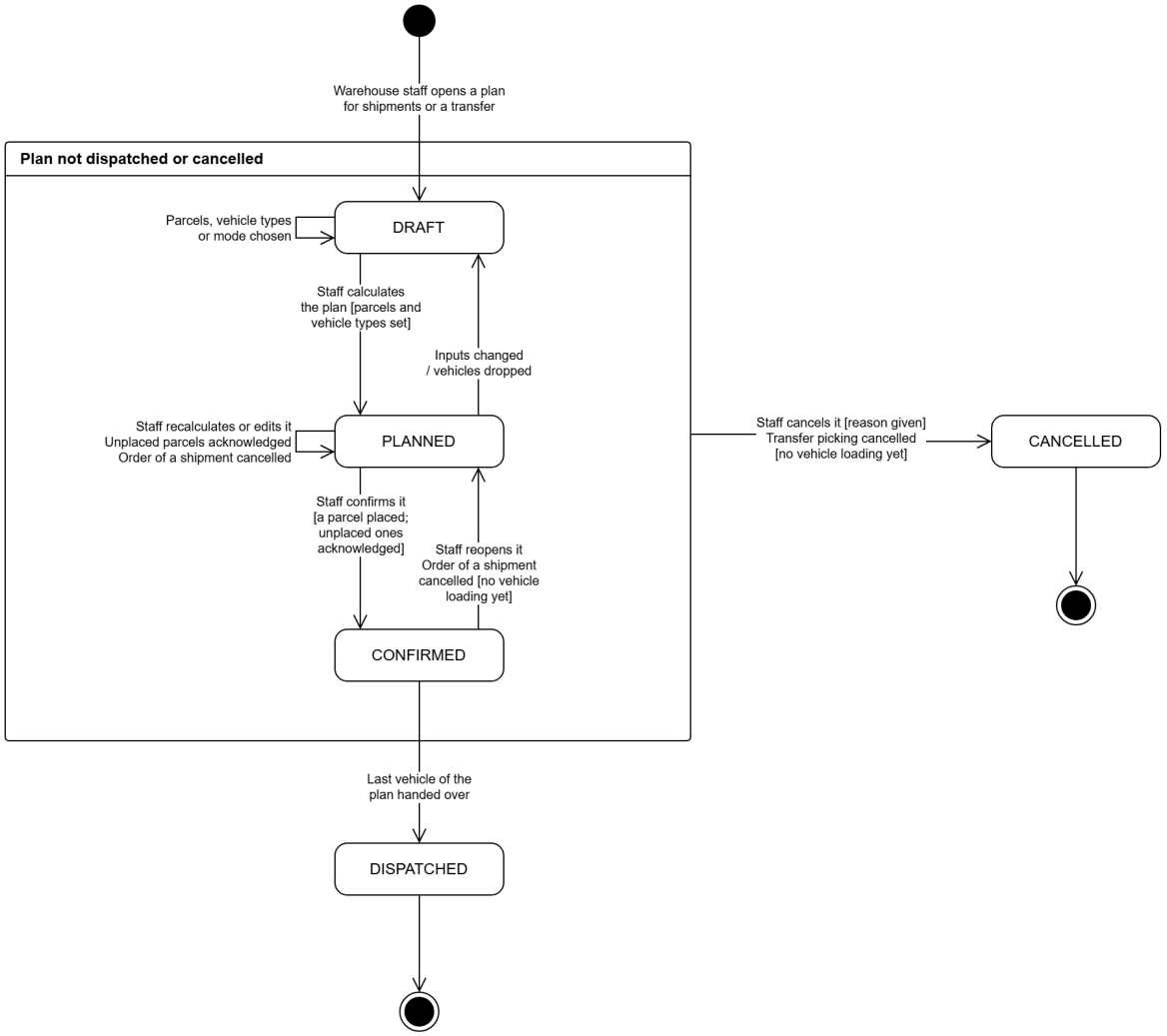


Figure 324. Load Plan State Diagram

3.20.15 Dispatch Load State Diagram

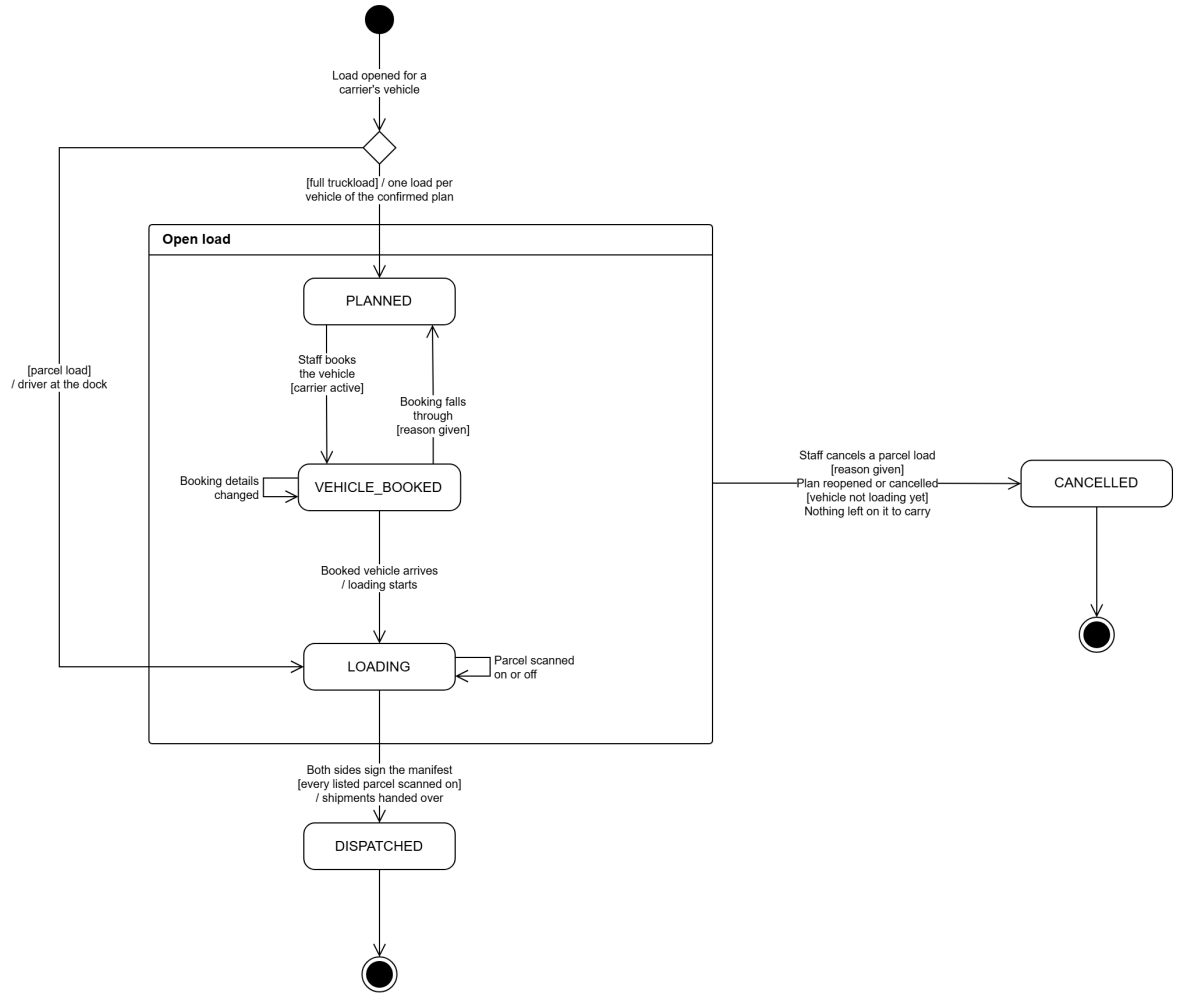


Figure 325. Dispatch Load State Diagram

3.21 Transfer Order Packing & Dispatch

3.21.1 Class Diagram

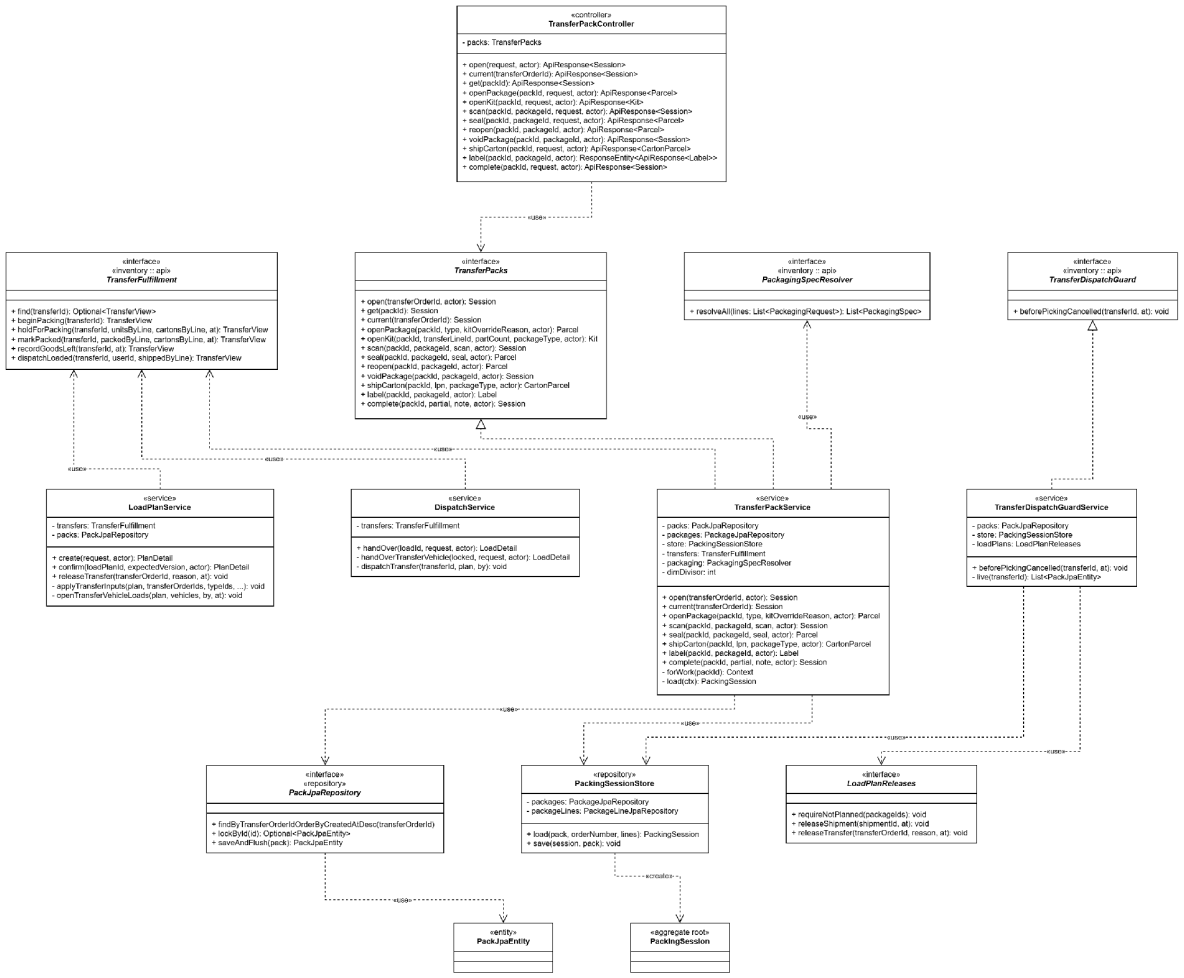


Figure 326. Transfer Order Packing and Dispatch Class Diagram (1/2): Packing and Dispatch

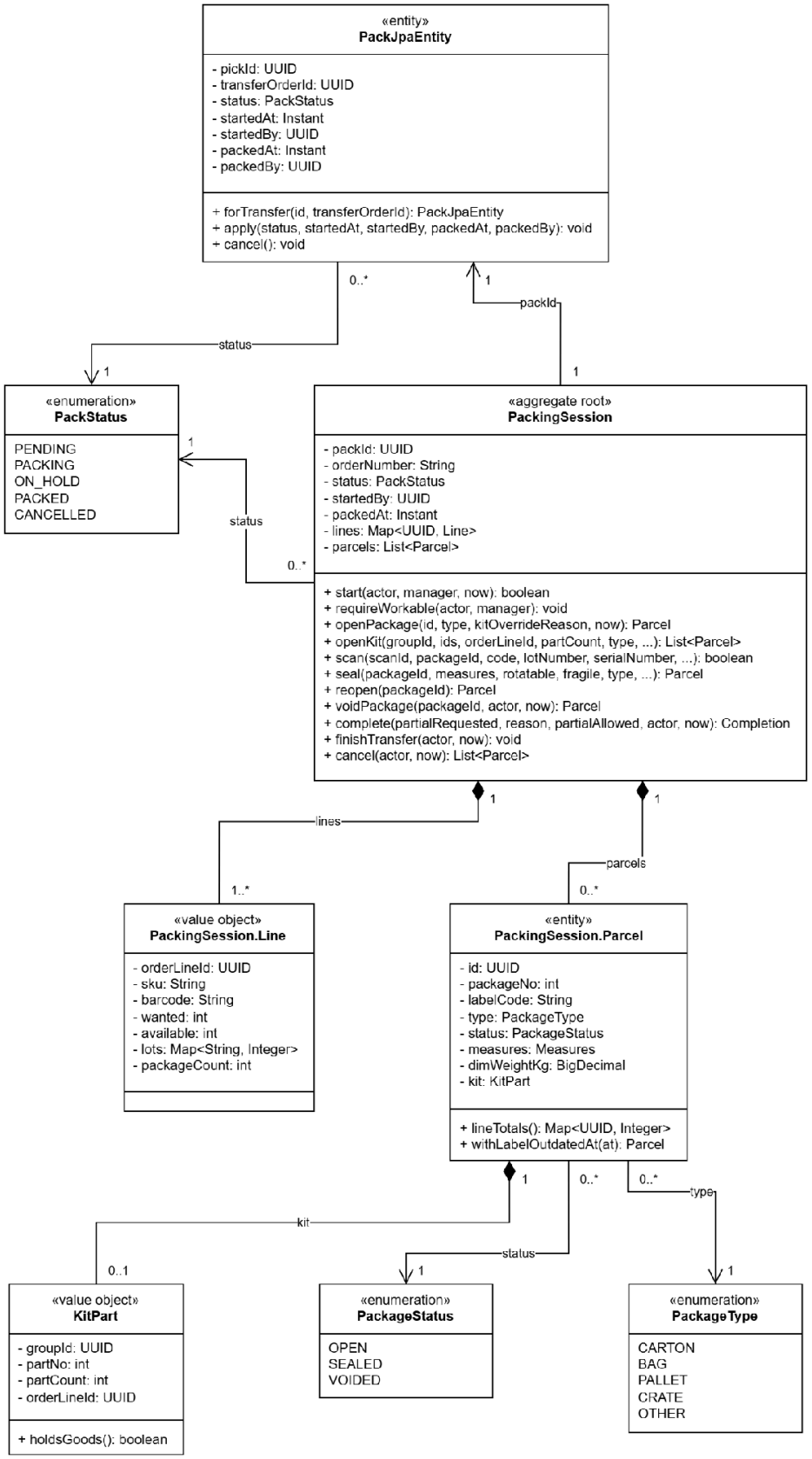


Figure 327. Transfer Order Packing and Dispatch Class Diagram (2/2): Domain Model

3.21.2 Class Specification

TransferPackController

No	Method	Description
1	open(request, actor)	POST /api/v1/fulfillment/transfer-packs: Opens (or returns) the packing session of an APPROVED or PICKING full-truckload transfer; APPROVED starts picking and every requested unit is held at the source; 409 TRANSFER_NOT_FULL_TRUCKLOAD, INVALID_TRANSFER_TRANSITION, PACKING_TAKEN.
2	current(transferOrderId)	GET /api/v1/fulfillment/transfer-packs?transferOrderId=: Returns the transfer's live packing session, else its latest, without starting anything; 404 when it was never packed.
3	reopen(packId, packageId, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/packages/{packageId}/reopening: Opens a sealed transfer package again before packing completes.
4	label(packId, packageId, actor)	GET /api/v1/fulfillment/transfer-packs/{packId}/packages/{packageId}/label: Returns the internal label of a sealed transfer package (transfer, n of N, from / to warehouse, weight, size, flags, contents, kit part); printing makes an outdated label current again.
5	get(packId)	GET /api/v1/fulfillment/transfer-packs/{packId}: Returns a transfer's packing session: lines (requested, packed, remaining, kit suggested) and packages.
6	openPackage(packId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/packages: Opens a new empty transfer package (kitOverrideReason for an item that ships as several boxes).
7	openKit(packId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/package-kits: Opens the N packages of a kit for one unit of a transfer line whose item ships as N > 1 boxes; 422 KIT_PART_COUNT_MISMATCH.

8	scan(packId, packageId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/packages/{packageId}/scans: Scans units into an open transfer package (lot when the line names one); the same scanId again changes nothing; 409 PACK_ITEM_NOT_IN_ORDER, PACK_QUANTITY_EXCEEDS_ORDER, PACK_LOT_MISMATCH.
9	seal(packId, packageId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/packages/{packageId}/seal: Seals a transfer package with its real weight and size (a whole stock carton too).
10	voidPackage(packId, packageId, actor)	DELETE /api/v1/fulfillment/transfer-packs/{packId}/packages/{packageId}: Takes a transfer package (a kit whole) apart before it left; its units count as unpacked again.
11	shipCarton(packId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/cartons: Ships an unopened stock carton as it is: one package holding all its units, with its packing unit's size and weight suggested; 409 STOCK_CARTON_NOT_SHIPPABLE.
12	complete(packId, request, actor)	POST /api/v1/fulfillment/transfer-packs/{packId}/completion: Completes transfer packing (-> PACKED): every package sealed, kits whole, less than requested only with partial and a note; the transfer is marked packed and is ready for a TRANSFER load plan; 409 PACKING_INCOMPLETE, PACKAGE_NOT_SEALED, KIT_INCOMPLETE.

Table 107. TransferPackController Class Specification

3.21.3 Pack & Dispatch Transfer Order Sequence Diagram

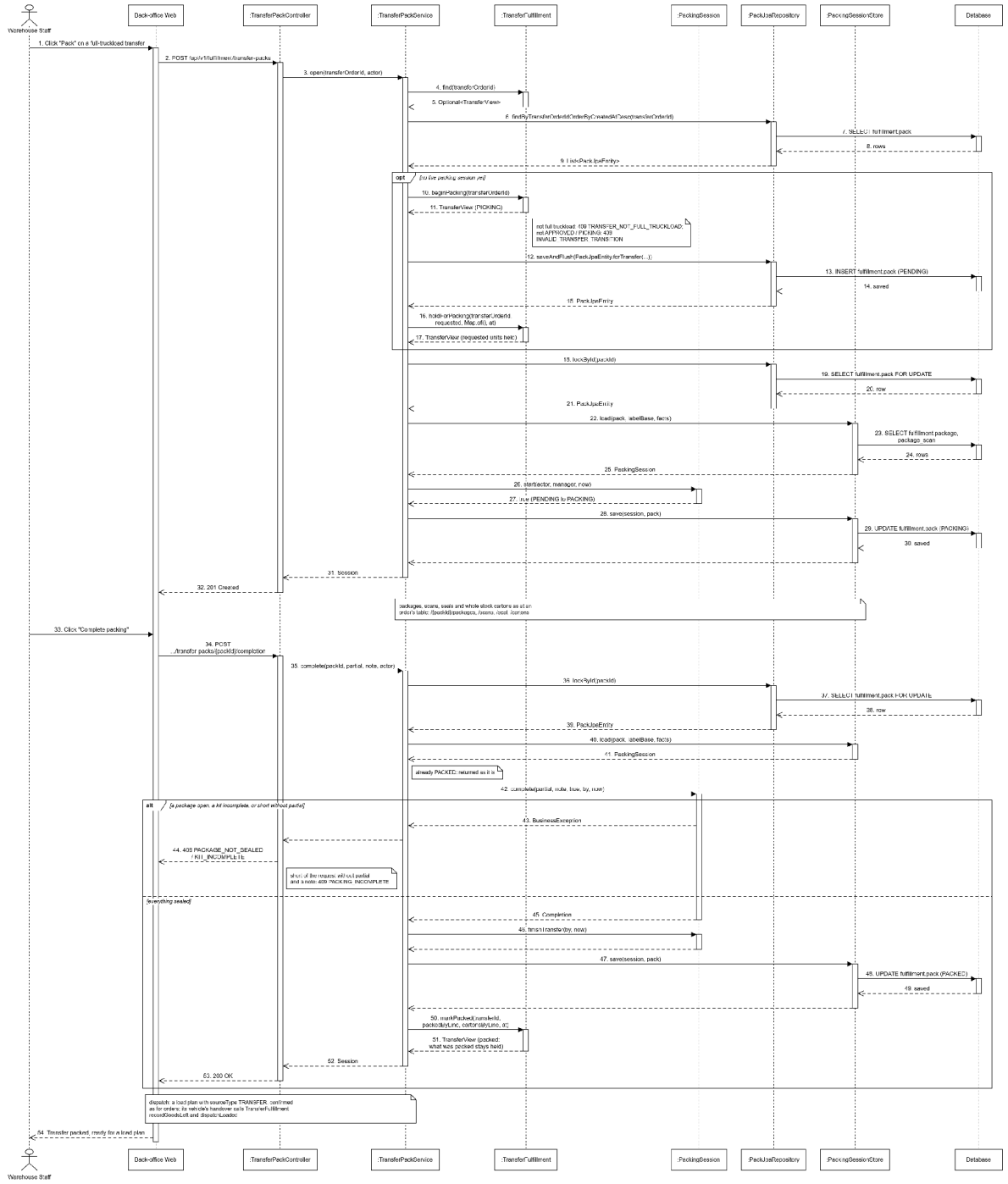


Figure 328. Pack & Dispatch Transfer Order Sequence Diagram

3.22 Reporting & Dashboards

3.22.1 Class Diagram

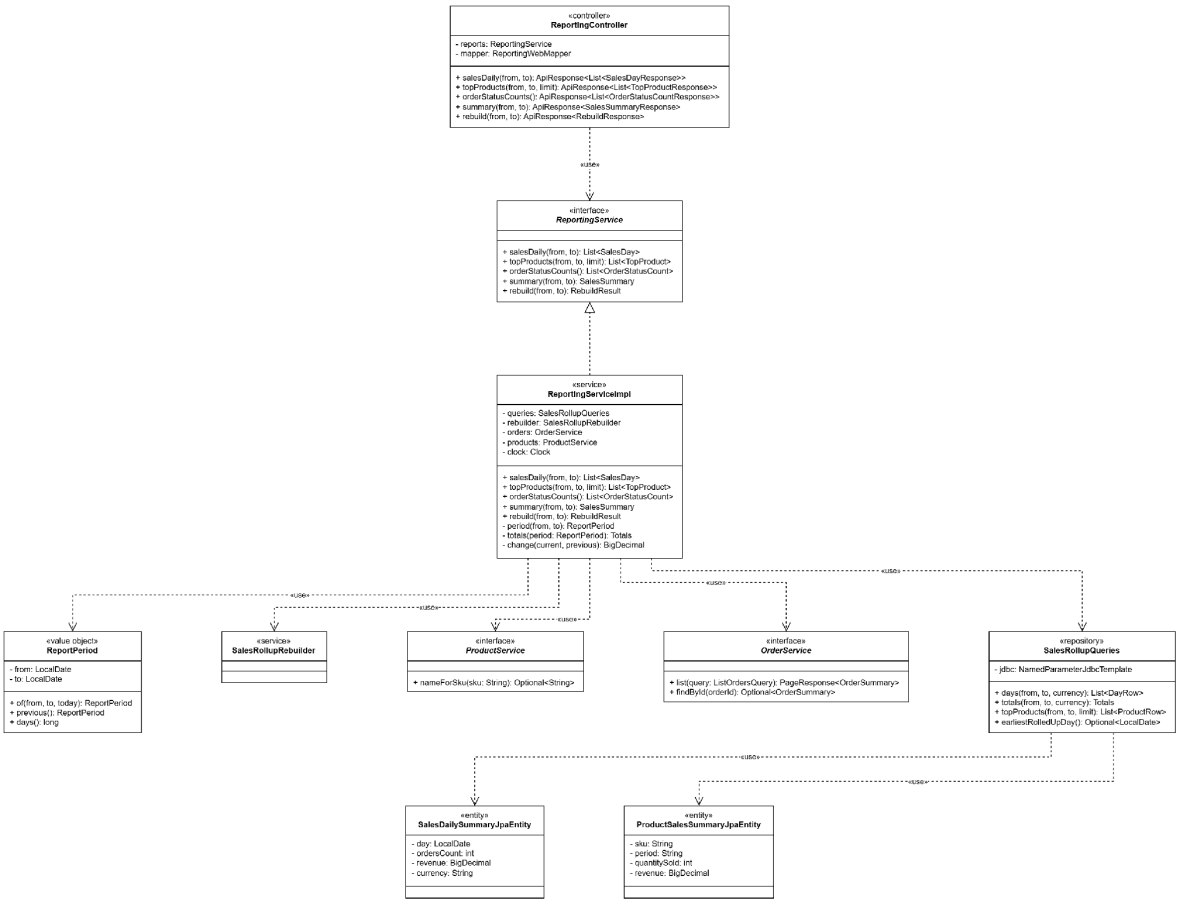


Figure 329. Reporting & Dashboards Class Diagram (1/3): Sales Dashboard

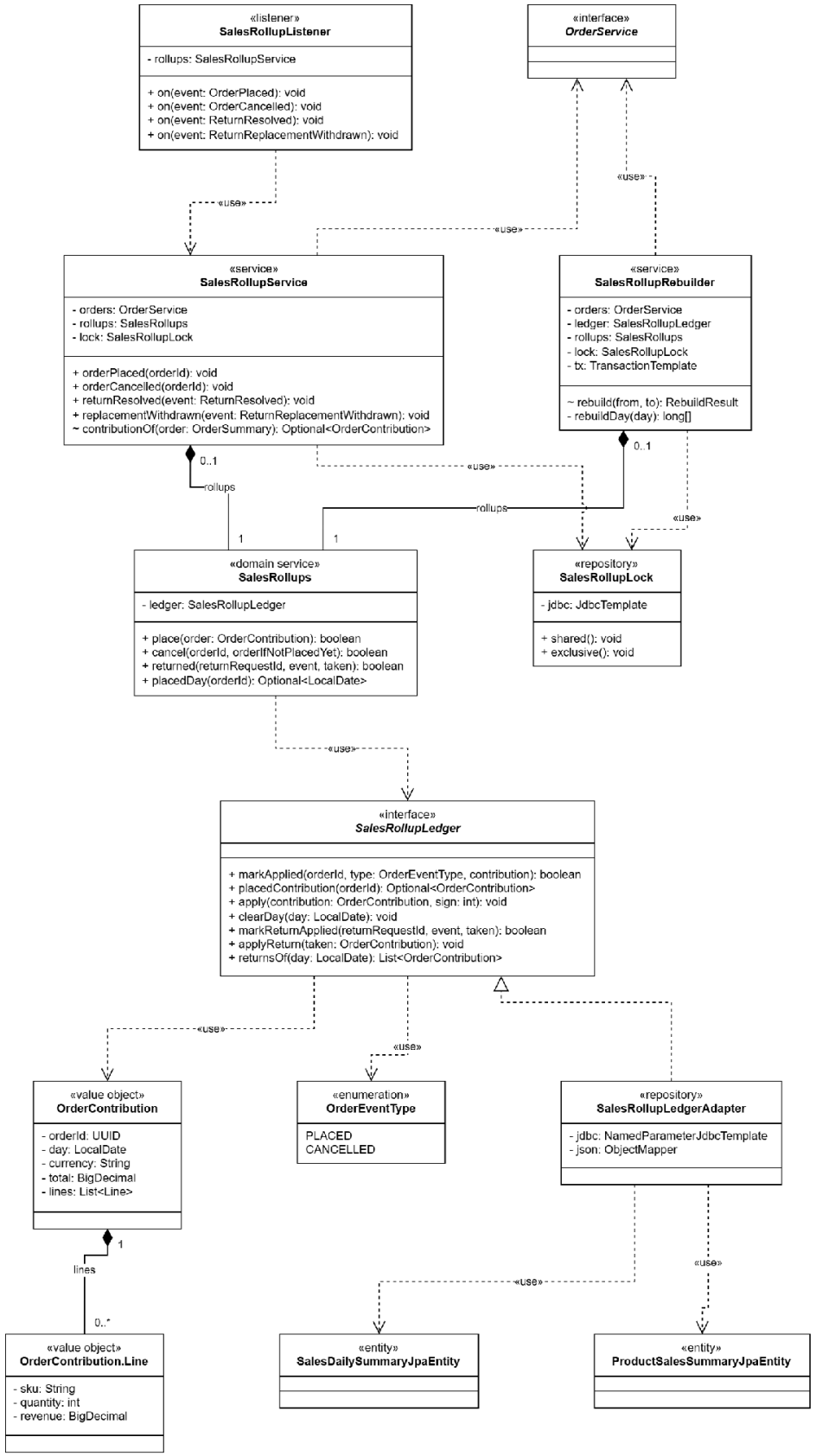


Figure 330. Reporting & Dashboards Class Diagram (2/3): Sales Rollups from Order Events

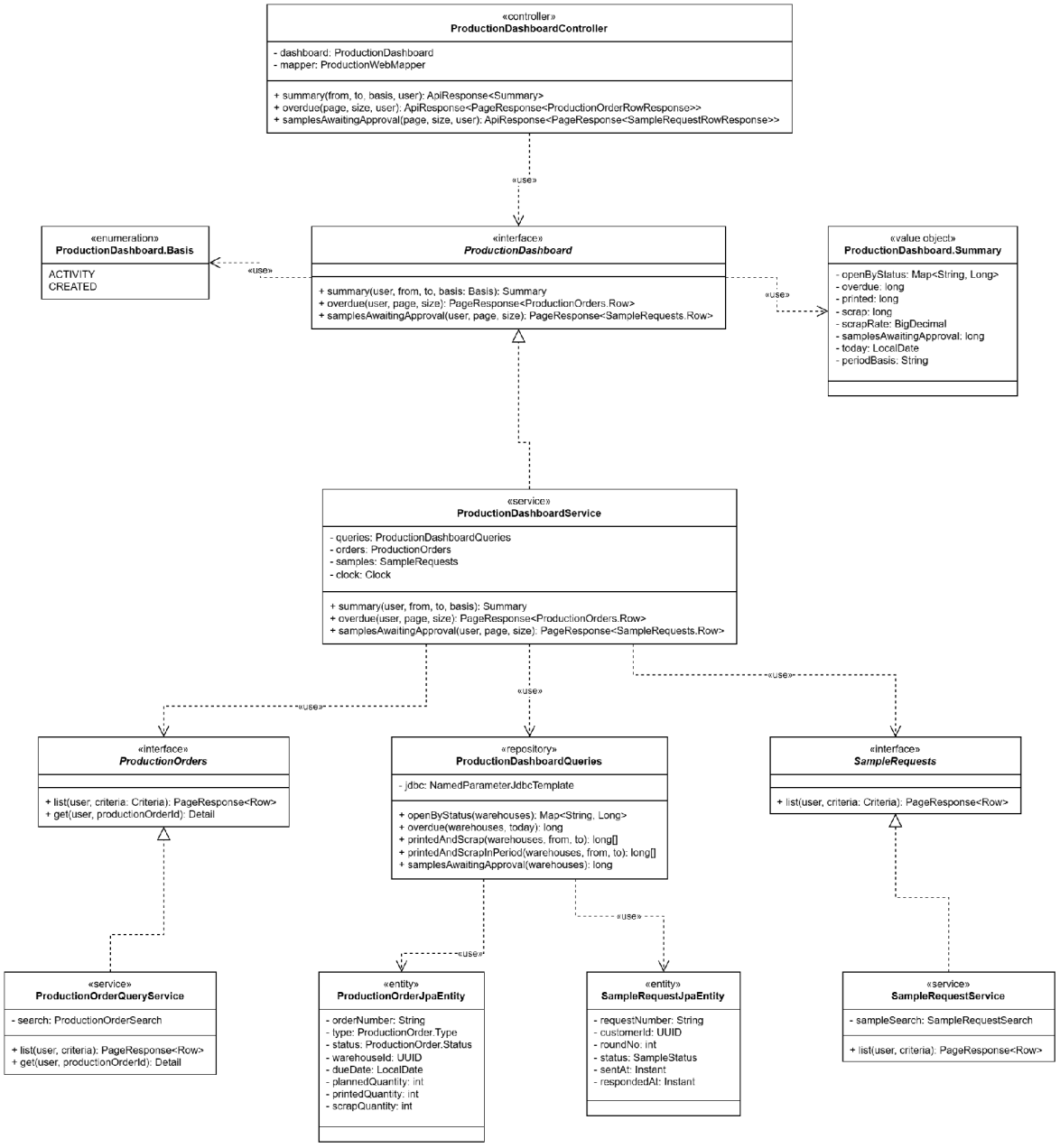


Figure 331. Reporting & Dashboards Class Diagram (3/3): Production Dashboard

3.22.2 Class Specification

ReportingController

No	Method	Description
1	salesDaily(from, to)	GET /api/v1/reports/sales-daily: Orders and revenue per day, one entry for every day of the range, zero-filled (reporting-reports:READ, scope ALL); range over 366 days or inverted is 400.
2	topProducts(from, to, limit)	GET /api/v1/reports/top-products: The best-selling SKUs of the range by revenue then units, with the product name when the product module knows the SKU; limit 1..100, default 10, else 400.

3	orderStatusCounts()	GET /api/v1/reports/order-status-counts: How many orders are in each OrderStatus right now, counted live through the order module (not date-bound).
4	summary(from, to)	GET /api/v1/reports/summary: Orders, revenue and average order value of the range against the previous period of the same length, with percent change (null when the previous is 0).
5	rebuild(from, to)	POST /api/v1/reports/rebuild: Recomputes the rollups of a range from the orders themselves, one transaction per day, idempotent (reporting-reports:EXPORT).

Table 108. ReportingController Class Specification

ProductionDashboardController

No	Method	Description
1	summary(from, to, basis, user)	GET /api/v1/production/dashboard: Open production orders by status, overdue count, samples awaiting the customer, and the scrap rate of the period by activity (default) or by orders created then; only the caller's warehouses; inverted period is 400.
2	overdue(page, size, user)	GET /api/v1/production/dashboard/overdue: Pages open production orders past their due date, most overdue first.
3	samplesAwaitingApproval(page, size, user)	GET /api/v1/production/dashboard/samples-awaiting-approval: Pages sample rounds SENT_TO_CUSTOMER and not answered, oldest first.

Table 109. ProductionDashboardController Class Specification

3.22.3 View Production Dashboard Sequence Diagram

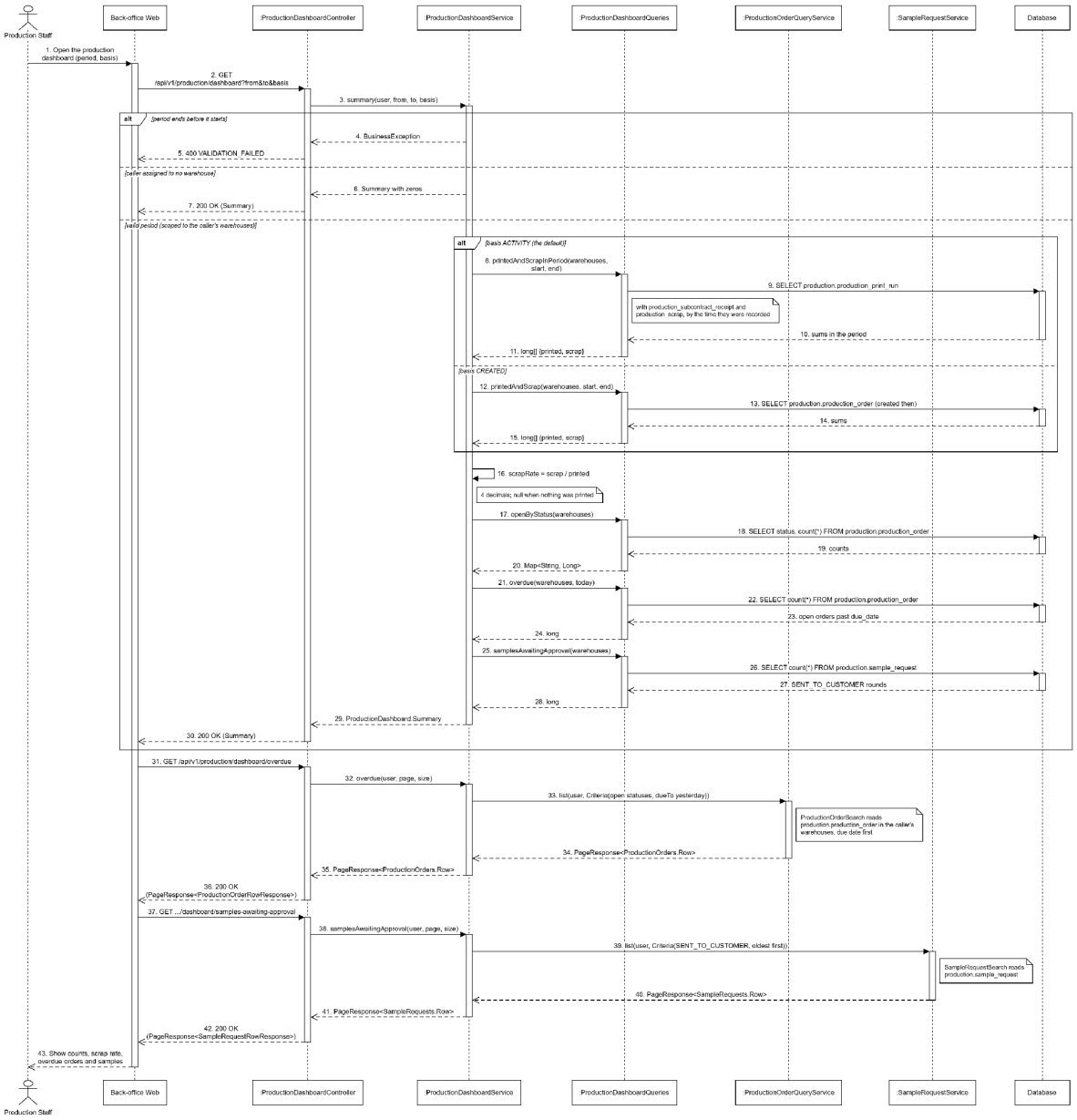


Figure 332. View Production Dashboard Sequence Diagram

3.22.4 View Sales Dashboard Sequence Diagram (1/2): Read the Figures

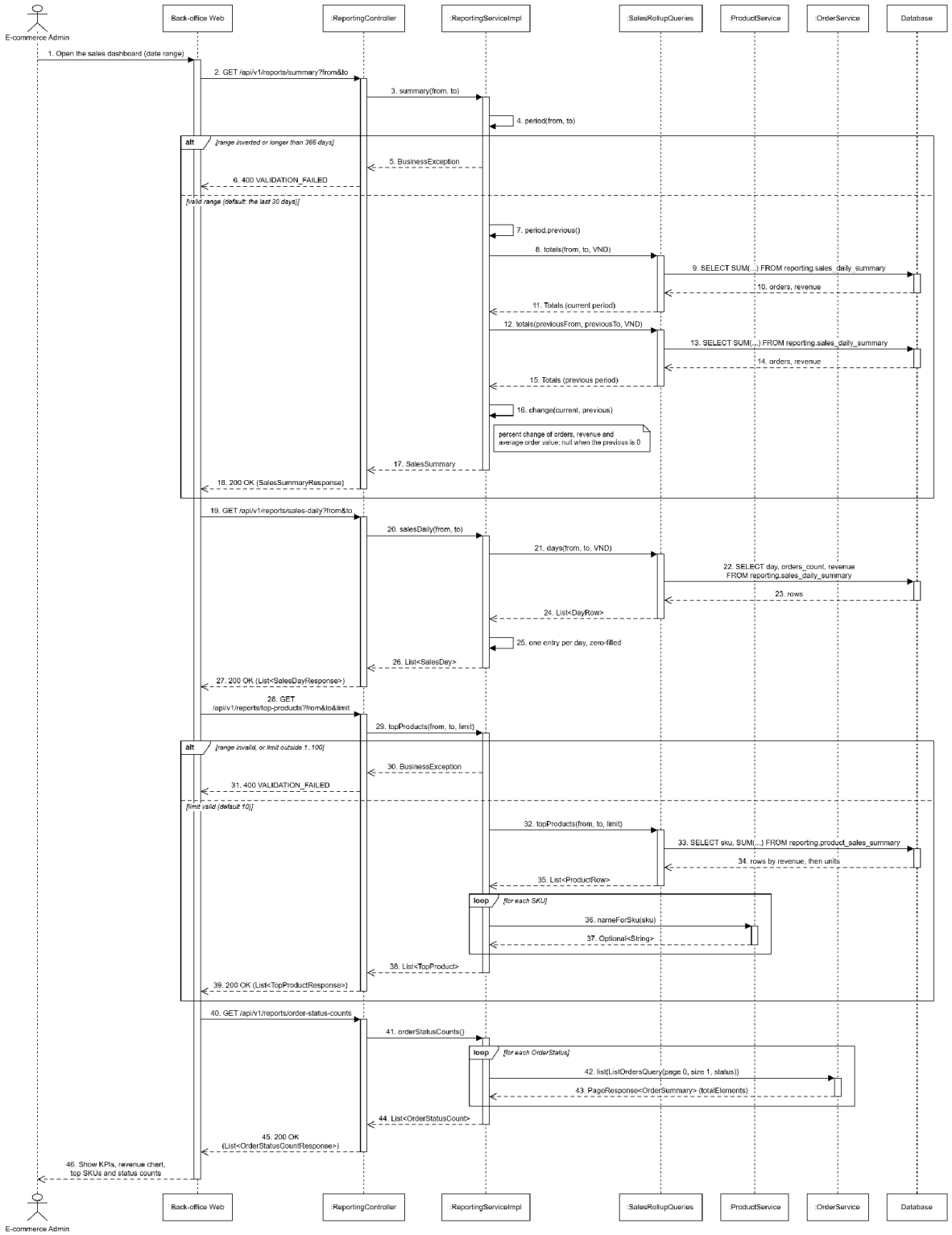


Figure 333. View Sales Dashboard Sequence Diagram (1/2): Read the Figures

3.22.5 View Sales Dashboard Sequence Diagram (2/2): Rollups Updated from Order Events

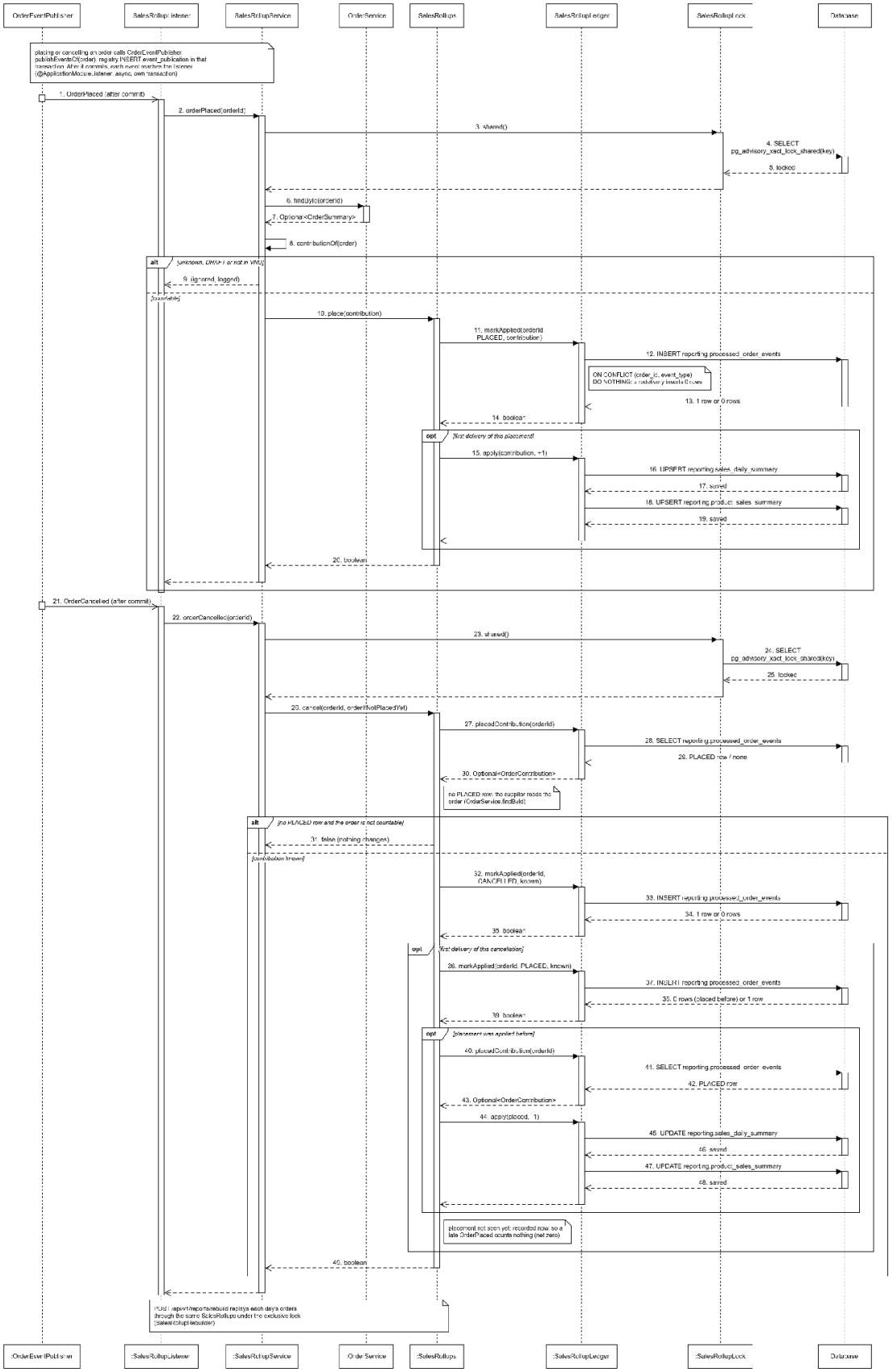


Figure 334. View Sales Dashboard Sequence Diagram (2/2): Rollups Updated from Order Events